九章算術

卷一至卷三

目次

劉徽

李淳風

對照文本

劉徽九章算術注原序

原典

昔在包犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術以合六爻之變。暨於黃帝神而化之，引而伸之，於是建曆紀，協律呂，用稽道原，然後兩儀四象精微之氣可得而效焉。記稱隸首作數，其詳未之聞也。按周公制禮而有九數，九數之流，則九章是矣。

往者暴秦焚書，經術散壞。自時厥後，漢北平侯張蒼、大司農中丞耿壽昌皆以善算命世。蒼等因舊文之遺殘，各稱刪補。故校其目則與古或異，而所論者多近語也。

徽幼習九章，長再詳覽。觀陰陽之割裂，總算術之根源，探賾之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。事類相推，各有攸歸，故枝條雖分而同本榦者，知發其一端而已。又所析理以辭，解體用圖，庶亦約而能周，通而不黷，覽之者思過半矣。

且算在六藝，古者以賓興賢能，教習國子。雖曰九數，其能窮纖入微，探測無方。至於以法相傳，亦猶規矩度量可得而共，非特難為也。當今好之者寡，故世雖多通才達學，而未必能綜於此耳。

周官大司徒職，夏至日中立八尺之表，其景尺有五寸，謂之地中。說云，南戴日下萬五千里。夫云爾者，以術推之。按九章立四表望遠及因木望山之術，皆端旁互見，無有超邈若斯之類。然則蒼等為術猶未足以博盡群數也。

徽尋九數有重差之名，原其指趣乃所以施於此也。凡望極高、測絕深而兼知其遠者必用重差，句股則必以重差為率，故曰重差也。立兩表於洛陽之城，令高八尺。南北各盡平地，同日度其正中之景。以景差為法，表高乘表間為實，實如法而一，所得加表高，即日去地也。

以南表之景乘表間為實，實如法而一，即為從南表至南戴日下也。以南戴日下及日去地為句、股，為之求弦，即日去人也。以徑寸之筩南望日，日滿筩空，則定筩之長短以為股率，以筩徑為句率，日去人之數為大股，大股之句即日徑也。

雖天圓穹之象猶曰可度，又況泰山之高與江海之廣哉。徽以為今之史籍且略舉天地之物，考論厥數，載之於志，以闡世術之美。輒造重差，并為注解，以究古人之意，綴於句股之下。度高者重表，測深者累矩，孤離者三望，離而又旁求者四望。觸類而長之，則雖幽遐詭伏，靡所不入。

博物君子，詳而覽焉。

卷第一：方田

方田以御田疇界域

原典	
方田者，田之正也。諸田不等，以方為正，故曰方田。此章以御田疇界域，論各種田地面積之計算法，并及分數四則運算之法。	
一今有田廣十五步，從十六步。**問**為田幾何?	
荅曰：一畝。	
二又有田廣十二步，從十四步。**問**為田幾何?	
荅曰：一百六十八步。	
術曰：方田術曰：廣從步數相乘得積步。以畝法二百四十步除之，即畝數。百畝為一頃。	
三今有田廣一里，從一里。**問**為田幾何?	
荅曰：三頃七十五畝。	
四又有田廣二里，從三里。**問**為田幾何?	
荅曰：二十二頃五十畝。	
術曰：里田術曰：廣從里數相乘得積里。以三百七十五乘之，即畝數。	
五今有十八分之十二。**問**約之得幾何?	$\frac{12}{18}$
荅曰：三分之二。	$\frac{2}{3}$
六又有九十一分之四十九。**問**約之得幾何?	$\frac{49}{91}$
荅曰：十三分之七。	$\frac{7}{13}$
術曰：約分術曰：可半者半之，不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。	
七今有三分之一，五分之二。**問**合之得幾何?	$\frac{1}{3}\ \frac{2}{5}$
荅曰：十五分之十一。	$\frac{11}{15}$
八又有三分之二，七分之四，九分之五。**問**合之得幾何?	$\frac{2}{3}\ \frac{4}{7}\ \frac{5}{9}$
荅曰：得一、六十三分之五十。	$1\frac{50}{63}$
九又有二分之一，三分之二，四分之三，五分之四。**問**合之得幾何?	$\frac{1}{2}\ \frac{2}{3}\ \frac{3}{4}\ \frac{4}{5}$
荅曰：得二、六十分之四十三。	$2\frac{43}{60}$
術曰：合分術曰：母互乘子，并為實，母相乘為法，實如法而一。不滿法者，以法命之。其母同者，直相從之。	
一0今有九分之八，減其五分之一。**問**餘幾何?	$\frac{8}{9}\ \frac{1}{5}$
荅曰：四十五分之三十一。	$\frac{31}{45}$
一一又有四分之三，減其三分之一。**問**餘幾何?	$\frac{3}{4}\ \frac{1}{3}$
荅曰：十二分之五。	$\frac{5}{12}$
術曰：減分術曰：母互乘子，以少減多，餘為實，母相乘為法，實如法而一。	

一二今有八分之五，二十五分之十六。**問**孰多？多幾何？	$\frac{5}{8}\ \frac{16}{25}$
荅曰：二十五分之十六多，多二百分之三。	$\frac{16}{25}\ \frac{3}{200}$
一三又有九分之八，七分之六。**問**孰多？多幾何？	$\frac{8}{9}\ \frac{6}{7}$
荅曰：九分之八多，多六十三分之二。	$\frac{8}{9}\ \frac{2}{63}$
一四又有二十一分之八，五十分之十七。**問**孰多？多幾何？	$\frac{8}{21}\ \frac{17}{50}$
荅曰：二十一分之八多，多一千五十分之四十三。	$\frac{8}{21}\ \frac{43}{1050}$
術曰：課分術曰：母互乘子，以少減多，餘為實，母相乘為法，實如法而一，即相多也。	
一五今有三分之一，三分之二，四分之三。**問**減多益少，各幾何而平？	$\frac{1}{3}\ \frac{2}{3}\ \frac{3}{4}$
荅曰：減四分之三者二，三分之二者一，并以益三分之一，而各平於十二分之七。	$\frac{3}{4}\ \frac{2}{12}\ \frac{2}{3}\ \frac{1}{12}\ \frac{1}{3}\ \frac{7}{12}$
一六又有二分之一，三分之二，四分之三。**問**減多益少，各幾何而平？	$\frac{1}{2}\ \frac{2}{3}\ \frac{3}{4}$
荅曰：減三分之二者一，四分之三者四，并以益二分之一，而各平於三十六分之二十三。	$\frac{2}{3}\ \frac{1}{36}\ \frac{3}{4}\ \frac{4}{36}\ \frac{1}{2}\ \frac{23}{36}$
術曰：平分術曰：母互乘子，副并為平實，母相乘為法。以列數乘未并者各自為列實。亦以列數乘法，以平實減列實，餘，約之為所減。并所減以益於少，以法命平實，各得其平。	
一七今有七人，分八錢三分錢之一。**問**人得幾何？	$8\frac{1}{3}$
荅曰：人得一錢、二十一分錢之四。	$1\frac{4}{21}$
一八又有三人，三分人之一，分六錢三分錢之一，四分錢之三。**問**人得幾何？	$\frac{1}{3}6\frac{1}{3}\frac{3}{4}$
荅曰：人得二錢、八分錢之一。	$2\frac{1}{8}$
術曰：經分術曰：以人數為法，錢數為實，實如法而一。有分者通之，重有分者同而通之。	
一九今有田廣七分步之四，從五分步之三。**問**為田幾何？	$\frac{4}{7}\ \frac{3}{5}$
荅曰：三十五分步之十二。	$\frac{12}{35}$
二０又有田廣九分步之七，從十一分步之九。**問**為田幾何？	$\frac{7}{9}\ \frac{9}{11}$
荅曰：十一分步之七。	$\frac{7}{11}$
二一又有田廣五分步之四，從九分步之五，**問**為田幾何？	$\frac{4}{5}\ \frac{5}{9}$
荅曰：九分步之四。	$\frac{4}{9}$
術曰：乘分術曰：母相乘為法，子相乘為實，實如法而一。	
二二今有田廣三步、三分步之一，從五步、五分步之二。**問**為田幾何？	$3\frac{1}{3}5\frac{2}{5}$
荅曰：十八步。	
二三又有田廣七步、四分步之三，從十五步、九分步之五。**問**為田幾何？	$7\frac{3}{4}15\frac{5}{9}$
荅曰：一百二十步、九分步之五。	$120\frac{5}{9}$
二四又有田廣十八步、七分步之五，從二十三步、十一分步之六。**問**為田幾何？	$18\frac{5}{7}23\frac{6}{11}$

荅曰：一畝二百步、十一分步之七。	$200\frac{7}{11}$
術曰：大廣田術曰：分母各乘其全，分子從之，相乘為實。分母相乘為法。實如法而一。	
二五今有圭田廣十二步，正從二十一步。**問**為田幾何?	
荅曰：一百二十六步。	
二六又有圭田廣五步、二分步之一，從八步、三分步之二。**問**為田幾何?	$5\frac{1}{2}8\frac{2}{3}$
荅曰：二十三步、六分步之五。	$23\frac{5}{6}$
術曰：術曰：半廣以乘正從。	
(注）劉徽注：半廣者，以盈補虛，為直田也。亦可以半正從以乘廣。	
二七今有邪田，一頭廣三十步，一頭廣四十二步，正從六十四步。**問**為田幾何?	
荅曰：九畝一百四十四步。	
二八又有邪田，正廣六十五步，一畔從一百步，一畔從七十二步。**問**為田幾何?	
荅曰：二十三畝七十步。	
術曰：術曰：并兩邪而半之，以乘正從若廣。又可半正從若廣，以乘并，畝法而一。	
二九今有箕田，舌廣二十步，踵廣五步，正從三十步。**問**為田幾何?	
荅曰：一畝一百三十五步。	
三０又有箕田，舌廣一百一十七步，踵廣五十步，正從一百三十五步。**問**為田幾何?	
荅曰：四十六畝二百三十二步半。	$232\frac{1}{2}$
術曰：術曰：并踵舌而半之，以乘正從。畝法而一。	
三一今有圓田，周三十步，徑十步。**問**為田幾何?	
荅曰：七十五步。	
三二又有圓田，周一百八十一步，徑六十步、三分步之一。**問**為田幾何?	$60\frac{1}{3}$
荅曰：十一畝九十步、十二分步之一。	$90\frac{1}{12}$
術曰：術曰：半周半徑相乘得積步。 又術曰：周徑相乘，四而一。 又術曰：徑自相乘，三之，四而一。 又術曰：周自相乘，十二而一。	
(注）此于徽術，當為田十畝二百八步三百一十四分步之一百一。臣淳風等謹依密率，為田十畝二百步八十八分步之八十七。 按：半周為從，半徑為廣，故廣從相乘為積步也。假令圓徑二尺，圓中容六觚之一面，與圓徑之半，其數均等。合徑率一而外周率三也。	$208\frac{101}{314}\pi200\frac{87}{88}$ $\pi \approx 3$
(注）又按：為圖，以六觚之一面乘一弧半徑，三之，得十二觚之冪。若又割之，次以十二觚之一面乘一弧之半徑，六之，則得二十四觚之冪。割之彌細，所失彌少。割之又割，以至于不可割，則與圓周合體而無所失矣。觚面之外，又有餘徑。以面乘餘徑，則冪出觚表。若夫觚之細者，與圓合體，則表無餘徑。表無餘徑，則冪不外出矣。以一面乘半徑，觚而裁之，每輒自倍。故以半周乘半徑而為圓冪。	

(注) 此一周、徑，謂至然之數，非周三徑一之率也。周三者，從其六觚之環耳。以推圓規多少之覺，乃弓之與弦也。然世傳此法，莫肯精核；學者踵古，習其謬失。不有明據，辯之斯難。凡物類形象，不圓則方。方圓之率，誠著于近，則雖遠可知也。由此言之，其用博矣。謹按圖驗，更造密率。	π
三三今有宛田，下周三十步，徑十六步。**問**為田幾何?	
荅曰：一百二十步。	
三四又有宛田，下周九十九步，徑五十一步。**問**為田幾何?	
荅曰：五畝六十二步、四分步之一。	$62\frac{1}{4}$
術曰：術曰：以徑乘周，四而一。	
三五今有弧田，弦三十步，矢十五步。**問**為田幾何?	
荅曰：一畝九十七步半。	$97\frac{1}{2}$
三六又有弧田，弦七十八步、二分步之一，矢十三步、九分步之七。**問**為田幾何?	$78\frac{1}{2}13\frac{7}{9}$
荅曰：二畝一百五十五步、八十一分步之五十六。	$155\frac{56}{81}$
術曰：術曰：以弦乘矢，矢又自乘，并之，二而一。	
三七今有環田，中周九十二步，外周一百二十二步，徑五步。**問**為田幾何?	
荅曰：二畝五十五步。	
三八又有環田，中周六十二步、四分步之三，外周一百一十三步、二分步之一，徑十二步、三分步之二。**問**為田幾何?	$62\frac{3}{4}113\frac{1}{2}12\frac{2}{3}$
荅曰：四畝一百五十六步、四分步之一。	$156\frac{1}{4}$
術曰：術曰：并中外周而半之，以徑乘之為積步。 密率術曰：置中外周步數，分母、子各居其下。母互乘子，通全步，內分子。以中周減外周，餘半之，以益中周。徑亦通分內子，以乘周為實。分母相乘為法，除之為積步，餘積步之分。以畝法除之，即畝數也。	

卷第二：粟米

粟米以御交質變易

原典

本章論各種谷物之交換比率，及比例計算之法。今有術（即今之三率法、比例法）為本章之核心算法。

粟米之法

粟率五十	糲米三十	粺米二十七	鑿米二十四
御米二十一	小●十三半	大●五十四	糲飯七十五
粺飯五十四	鑿飯四十八	御飯四十二	菽、荅、麻、麥各四十
稻六十	豉六十三	飧九十	熟菽一百三半
糵一百七十五			

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術曰：今有術曰：以所有數乘所求率為實，以所有率為法，實如法而一。	
一今有粟一斗，欲為糲米。**問**得幾何？	
荅曰：為糲米六升。	
術曰：術曰：以粟求糲米，三之，五而一。	
二今有粟二斗一升，欲為粺米。**問**得幾何？	
荅曰：為粺米一斗一升、五十分升之十七。	$1\ 1\ \frac{17}{50}$
術曰：術曰：以粟求粺米，二十七之，五十而一。	
三今有粟四斗五升，欲為鑿米。**問**得幾何？	
荅曰：為鑿米二斗一升、五分升之三。	$2\ 1\ \frac{3}{5}$
術曰：術曰：以粟求鑿米，十二之，二十五而一。	
四今有粟七斗九升，欲為御米。**問**得幾何？	
荅曰：為御米三斗三升、五十分升之九。	$3\ 3\ \frac{9}{50}$
術曰：術曰：以粟求御米，二十一之，五十而一。	
五今有粟一斗，欲為小●。**問**得幾何？	●
荅曰：為小●二升、一十分升之七。	$2\frac{7}{10}$●
術曰：術曰：以粟求小●，二十七之，百而一。	●
六今有粟九斗八升，欲為大●。**問**得幾何？	●
荅曰：為大●一十斗五升、二十五分升之二十一。	$10\ 5\ \frac{21}{25}$●
術曰：術曰：以粟求大●，二十七之，二十五而一。	●
七今有粟二斗三升，欲為糲飯。**問**得幾何？	
荅曰：為糲飯三斗四升半。	
術曰：術曰：以粟求糲飯，三之，二而一。	
八今有粟三斗六升，欲為粺飯。**問**得幾何？	
荅曰：為粺飯三斗八升、二十五分升之二十二。	$3\ 8\ \frac{22}{25}$
術曰：術曰：以粟求粺飯，二十七之，二十五而一。	
九今有粟八斗六升，欲為鑿飯。**問**得幾何？	

荅曰：為鑿飯八斗二升、二十五分升之一十四。	$82\frac{14}{25}$
術曰：術曰：以粟求鑿飯，二十四之，二十五而一。	
一0今有粟九斗八升，欲為御飯。**問**得幾何?	
荅曰：為御飯八斗二升、二十五分升之八。	$82\frac{8}{25}$
術曰：術曰：以粟求御飯，二十一之，二十五而一。	
一一今有粟三斗少半升，欲為菽。**問**得幾何? **一二今有**粟四斗一升、太半升，欲為荅。**問**得幾何? **一三今有**粟五斗、太半升，欲為麻。**問**得幾何? **一四今有**粟一十斗八升、五分升之二，欲為麥。**問**得幾何?	$\frac{1}{3}$ $\frac{2}{3}$ $\frac{2}{3}$ $\frac{2}{5}$
荅曰：為菽二斗七升、一十分升之三。 為荅三斗七升半。 為麻四斗五升、五分升之三。 為麥九斗七升、二十五分升之一十四。	$27\frac{3}{10}$ $45\frac{3}{5}$ $97\frac{14}{25}$
術曰：術曰：以粟求菽、荅、麻、麥，皆九之，十而一。	
一五今有粟七斗五升、七分升之四，欲為稻。**問**得幾何?	$\frac{4}{7}$
荅曰：為稻九斗、三十五分升之二十四。	$9\frac{24}{35}$
術曰：術曰：以粟求稻，六之，五而一。	
一六今有粟七斗八升，欲為豉。**問**得幾何?	
荅曰：為豉九斗八升、二十五分升之七。	$98\frac{7}{25}$
術曰：術曰：以粟求豉，六十三之，五十而一。	
一七今有粟五斗五升，欲為飧。**問**得幾何?	
荅曰：為飧九斗九升。	
術曰：術曰：以粟求飧，九之，五而一。	
一八今有粟四斗，欲為熟菽。**問**得幾何?	
荅曰：為熟菽八斗二升、五分升之四。	$82\frac{4}{5}$
術曰：術曰：以粟求熟菽，二百七之，百而一。	
一九今有粟二斗，欲為糵。**問**得幾何?	
荅曰：為糵七斗。	
術曰：術曰：以粟求糵，七之，二而一。	
二0今有糲米十五斗五升、五分升之二，欲為粟。**問**得幾何?	$\frac{2}{5}$
荅曰：為粟二十五斗九升。	
術曰：術曰：以糲米求粟，五之，三而一。	
二一今有粺米二斗，欲為粟。**問**得幾何?	
荅曰：為粟三斗七升、二十七分升之一。	$37\frac{1}{27}$
術曰：術曰：以粺米求粟，五十之，二十七而一。	
二二今有鑿米三斗、少半升，欲為粟。**問**得幾何?	$\frac{1}{3}$
荅曰：為粟六斗三升、三十六分升之七。	$63\frac{7}{36}$
術曰：術曰：以鑿米求粟，二十五之，十三而一。	
二三今有御米十四斗，欲為粟。**問**得幾何?	

荅曰：為粟三十三斗三升、少半升。	$33\ 3\ \frac{1}{3}$
術曰：術曰：以御米求粟，五十之，二十一而一。	
二四今有稻一十二斗六升、一十五分升之一十四，欲為粟。**問**得幾何?	$\frac{14}{15}$
荅曰：為粟一十斗五升、九分升之七。	$10\ 5\ \frac{7}{9}$
術曰：術曰：以稻求粟，五之，六而一。	
二五今有糲米一十九斗二升、七分升之一，欲為粺米。**問**得幾何?	$\frac{1}{7}$
荅曰：為粺米一十七斗二升、一十四分升之一十三。	$17\ 2\ \frac{13}{14}$
術曰：術曰：以糲米求粺米，九之，十而一。	
二六今有糲米六斗四升、五分升之三，欲為糲飯。**問**得幾何?	$\frac{3}{5}$
荅曰：為糲飯一十六斗一升半。	
術曰：術曰：以糲米求糲飯，五之，二而一。	
二七今有糲飯七斗六升、七分升之四，欲為飧。**問**得幾何?	$\frac{4}{7}$
荅曰：為飧九斗一升、三十五分升之三十一。	$9\ 1\ \frac{31}{35}$
術曰：術曰：以糲飯求飧，六之，五而一。	
二八今有菽一斗，欲為熟菽。**問**得幾何?	
荅曰：為熟菽二斗三升。	
術曰：術曰：以菽求熟菽，二十三之，十而一。	
二九今有菽二斗，欲為豉。**問**得幾何?	
荅曰：為豉二斗八升。	
術曰：術曰：以菽求豉，七之，五而一。	
三0今有麥八斗六升、七分升之三，欲為小●，**問**得幾何?	$\frac{3}{7}$●
荅曰：為小●二斗五升、一十四分升之一十三。	$2\ 5\ \frac{13}{14}$●
術曰：術曰：以麥求小●，三之，十而一。	●
三一今有麥一斗，欲為大●。**問**得幾何?	●
荅曰：為大●一斗二升。	●
術曰：術曰：以麥求大●，六之，五而一。	●
三二今有出錢一百六十，買瓴甓十八枚。**問**枚幾何?	
荅曰：一枚，八錢、九分錢之八。	$8\frac{8}{9}$
三三今有出錢一萬三千五百，買竹二千三百五十箇。**問**箇幾何?	
荅曰：一箇，五錢、四十七分錢之三十五。	$5\frac{35}{47}$
術曰：經率術曰：以所買率為法，所出錢數為實，實如法得一錢。	
三四今有出錢五千七百八十五，買漆一斛六斗七升、太半升。欲斗率之，**問**斗幾何?	$\frac{2}{3}$
荅曰：一斗，三百四十五錢、五百三分錢之一十五。	$345\frac{15}{503}$

三五今有出錢七百二十，買縑一匹二丈一尺。欲丈率之，**問**丈幾何?	
荅曰：一丈，一百一十八錢、六十一分錢之二。	$118\frac{2}{61}$
三六今有出錢二千三百七十，買布九匹二丈七尺。欲匹率之，**問**匹幾何?	
荅曰：一匹，二百四十四錢、一百二十九分錢之一百二十四。	$244\frac{124}{129}$
三七今有出錢一萬三千六百七十，買絲一石二鈞一十七斤。欲石率之，**問**石幾何?	
荅曰：一石，八千三百二十六錢、一百九十七分錢之一百七十八。	$8,326\frac{178}{197}$
術曰：經術術曰：以所求率乘錢數為實，以所買率為法，實如法得一。	
三八今有出錢五百七十六，買竹七十八箇。欲其大小率之，**問**各幾何?	
荅曰：其四十八箇，箇七錢。其三十箇，箇八錢。	
三九今有出錢一千一百二十，買絲一石二鈞十八斤。欲其貴賤斤率之，**問**各幾何?	
荅曰：其二鈞八斤，斤五錢。其一石一十斤，斤六錢。	
術曰：其率術曰：各置所買石、鈞、斤、兩以為法，以所率乘錢數為實，實如法而一。不滿法者反以實減法，法賤實貴。	
四０四三今有出錢一萬三千九百七十，買絲一石二鈞二十八斤三兩五銖。欲其貴賤石率之、鈞率之、斤率之、兩率之。	
四四今有出錢一萬三千九百七十，買絲一石二鈞二十八斤三兩五銖。欲其貴賤銖率之。	
四五今有出錢六百一十，買羽二千一百翭。欲其貴賤率之，**問**各幾何?	
荅曰：其一千一百四十翭，三翭一錢。其九百六十翭，四翭一錢。	
四六今有出錢九百八十，買矢簳五千八百二十枚。欲其貴賤率之，**問**各幾何?	
荅曰：其三百枚，五枚一錢。其五千五百二十枚，六枚一錢。	
術曰：反其率術曰：以錢數為法，所率為實，實如法而一。不滿法者反以實減法，法少，實多。二物各以所得多少之數乘法實，即物數。	

卷第三：衰分

衰分以御貴賤稟稅之率

原典	
本章論比例分配之法，包括按等級、按數量、按比率之分配問題。衰分術為本章之核心算法。	
術曰：衰分術曰：各置列衰，副并為法，以所分乘未并者各自為實，實如法而一。不滿法者，以法命之。	
一今有大夫、不更、簪褭、上造、公士，凡五人，共獵得五鹿。欲以爵次分之，**問**各得幾何?	
荅曰：大夫得一鹿、三分鹿之二。不更得一鹿、三分鹿之一。簪褭得一鹿。上造得三分鹿之二。公士得三分鹿之一。	$1\frac{2}{3}1\frac{1}{3}\frac{2}{3}\frac{1}{3}$
術曰：術曰：列置爵數，各自為衰，副并為法。以五鹿乘未并者，各自為實。實如法得一鹿。	
二今有牛、馬、羊食人苗。苗主責之粟五斗。羊主曰：「我羊食半馬。」馬主曰：「我馬食半牛。」**今欲**衰償之，**問**各出幾何?	
荅曰：牛主出二斗八升、七分升之四。馬主出一斗四升、七分升之二。羊主出七升、七分升之一。	$2\,8\,\frac{4}{7}1\,4\,\frac{2}{7}7\frac{1}{7}$
術曰：術曰：置牛四、馬二、羊一，各自為列衰，副并為法。以五斗乘未并者各自為實。實如法得一斗。	
三今有甲持錢五百六十，乙持錢三百五十，丙持錢一百八十，凡三人俱出關，關稅百錢。欲以錢數多少衰出之，**問**各幾何?	
荅曰：甲出五十一錢、一百九分錢之四十一。乙出三十二錢、一百九分錢之一十二。丙出一十六錢、一百九分錢之五十六。	$51\frac{41}{109}32\frac{12}{109}16\frac{56}{109}$
術曰：術曰：各置錢數為列衰，副并為法，以百錢乘未并者，各自為實，實如法得一錢。	
四今有女子善織，日自倍，五日織五尺。**問**日織幾何?	
荅曰：初日織一寸、三十一分寸之十九。次日織三寸、三十一分寸之七。次日織六寸、三十一分寸之十四。次日織一尺二寸、三十一分寸之二十八。次日織二尺五寸、三十一分寸之二十五。	$1\frac{19}{31}3\frac{7}{31}6\frac{14}{31}1\,2\,\frac{28}{31}2\,5\,\frac{25}{31}$
術曰：術曰：置一、二、四、八、十六為列衰，副并為法，以五尺乘未并者，各自為實，實如法得一尺。	
五今有北鄉算八千七百五十八，西鄉算七千二百三十六，南鄉算八千三百五十六，凡三鄉，發傜三百七十八人。欲以算數多少衰出之，**問**各幾何?	
荅曰：北鄉遣一百三十五人、一萬二千一百七十五分人之一萬一千六百三十七。西鄉遣一百一十二人、一萬二千一百七十五分人之四千四。南鄉遣一百二十九人、一萬二千一百七十五分人之八千七百九。	$135\frac{11637}{12175}112\frac{4004}{12175}129\frac{8709}{12175}$
術曰：術曰：各置算數為列衰，副并為法，以所發傜人數乘未并者，各自為實，實如法得一人。	
六今有稟粟，大夫、不更、簪褭、上造、公士，凡五人，一十五斗。**今有**大夫一人後來，亦當稟五斗。倉無粟，欲以衰出之，**問**各幾何?	
荅曰：大夫出一斗、四分斗之一。不更出一斗。簪褭出四分斗之三。上造出四分斗之二。公士出四分斗之一。	$1\frac{1}{4}\frac{3}{4}\frac{2}{4}\frac{1}{4}$

術曰：術曰：各置所稟粟斛斗數，爵次均之，以為列衰，副并而加後來大夫亦五斗，得二十以為法。以五斗乘未并者各自為實。實如法得一斗。	
七今有稟粟五斛，五人分之，欲令三人得三，二人得二。**問**各幾何?	
荅曰：三人，人得一斛一斗五升、十三分升之五。二人，人得七斗六升、十三分升之十二。	$1\,1\,5\,\frac{5}{13}\,7\,6\,\frac{12}{13}$
術曰：術曰：置三人，人三；二人，人二，為列衰。副并為法。以五斛乘未并者，各自為實。實如法得一斛。	
術曰：返衰術曰：列置衰而令相乘，動者為不動者衰。	
八今有大夫、不更、簪褭、上造、公士，凡五人，共出百錢。欲令高爵出少，以次漸多，**問**各幾何?	
荅曰：大夫出八錢、一百三十七分錢之一百四。不更出一十錢、一百三十七分錢之一百三十。簪褭出一十四錢、一百三十七分錢之八十二。上造出二十一錢、一百三十七分錢之一百二十三。公士出四十三錢、一百三十七分錢之一百九。	$8\frac{104}{137}\,10\frac{130}{137}\,14\frac{82}{137}\,21\frac{123}{137}\,43\frac{109}{137}$
術曰：術曰：置爵數各自為衰，而返衰之，副并為法。以百錢乘未并者各自為實。實如法得一錢。	
九今有甲持粟三升，乙持糲米三升，丙持糲飯三升。欲令合而分之，**問**各幾何?	
荅曰：甲二升、一十分升之七。乙四升、一十分升之五。丙一升、一十分升之八。	$2\frac{7}{10}\,4\frac{5}{10}\,1\frac{8}{10}$
術曰：術曰：以粟率五十、糲米率三十、糲飯率七十五為衰、而返衰之，副并為法。以九升乘未并者各自為實。實如法得一升。	
一０今有絲一斤，價直二百四十。**今有**錢一千三百二十八，**問**得絲幾何?	
荅曰：五斤八兩一十二銖、三分銖之四。	$\frac{4}{3}$
術曰：術曰：以一斤價數為法，以一斤乘今有錢數為實，實如法得絲數。	
一一今有絲一斤價直三百四十三。**今有**絲七兩一十二銖，**問**得錢幾何?	
荅曰：一百六十一錢、三十二分錢之二十三。	$161\frac{23}{32}$
術曰：術曰：以一斤銖數為法，以一斤價數，乘七兩一十二銖為實。實如法得錢數。	
一二今有縑一丈價直一百二十六。**今有**縑一匹九尺五寸，**問**得錢幾何?	
荅曰：六百三十三錢、五分錢之三。	$633\frac{3}{5}$
術曰：術曰：以一丈寸數為法，以價錢數乘今有縑寸數為實，實如法得錢數。	
一三今有布一匹，價直一百二十三。**今有**布二丈七尺，**問**得錢幾何?	
荅曰：八十四錢、分錢之三。	$84\frac{3}{}$
術曰：術曰：以一匹尺數為法，今有布尺數乘價錢為實，實如法得錢數。	
一四今有素一匹一丈，價直六百二十五。**今有**錢五百，**問**得素幾何?	

術曰：術曰：以價直為法，以一匹一丈尺數乘今有錢數為實。實如法得素數。	
一五今有與人絲一十四斤，約得縑一十斤。**今**與人絲四十五斤八兩，**問**得縑幾何？	
荅曰：三十二斤八兩。	
術曰：術曰：以一十四斤兩數為法，以一十斤乘今有絲兩數為實，實如法得縑數。	
一六今有絲一斤，耗七兩。**今有**絲二十三斤五兩，**問**耗幾何？	
荅曰：一百六十三兩四銖半。	
術曰：術曰：以一斤展十六兩為法，以七兩乘今有絲兩數為實，實如法得耗數。	
一七今有生絲三十斤，乾之，耗三斤十二兩。**今有**乾絲一十二斤，**問**生絲幾何？	
荅曰：一十三斤一十一兩十銖、七分銖之二。	$13\ 11\ 10\frac{2}{7}$
術曰：術曰：置生絲兩數，除耗數，餘，以為法。三十斤乘乾絲兩數為實。實如法得生絲數。	
一八今有田一畝，收粟六升、太半升。**今有**田一頃二十六畝一百五十九步，**問**收粟幾何？	$6\frac{2}{3}$
荅曰：八斛四斗四升、一十二分升之五。	$\frac{5}{12}$
術曰：術曰：以畝二百四十步為法，以六升、太半升乘今有田積步為實，實如法得粟數。	$6\frac{2}{3}$
一九今有取保一歲，價錢二千五百。**今**先取一千二百，**問**當作日幾何？	
荅曰：一百六十九日、二十五分日之二十三。	$169\frac{23}{25}$
術曰：術曰：以價錢為法，以一歲三百五十四日乘先取錢數為實，實如法得日數。	
二０今有貸人千錢，月息三十。**今有**貸人七百五十錢，九日歸之，**問**息幾何？	
荅曰：六錢、四分錢之三。	$6\frac{3}{4}$
術曰：術曰：以月三十日，乘千錢為法。以息三十乘今所貸錢數，又以九日乘之，為實。實如法得一錢。	

(以上為《九章算術》卷第一至卷第三之全文，含劉徽注及李淳風等注釋。)

譯者注

實法

今有 $= \frac{\times}{}$

π

六觚$\pi \approx 3\pi \approx \frac{157}{50} = 3.14\frac{3927}{1250} = 3.1416$

步≈畝頃尺≈斤兩銖

大夫不更簪裹上造公士

九章算術

Jiuzhang Suanshu

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卷四至卷六

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Vol. 4 —Shaoguang (少廣): Root extraction, sphere volume
Vol. 5 —Shanggong (商功): Volumes of earthworks and solids
Vol. 6 —Junshu (均輸): Weighted distribution, work-rate problems

Original Chinese text with English translation

Chinese text: Left column *English translation: Right column*

From the Siku Quanshu (欽定四庫全書) Edition
Zi Bu —Astronomy and Mathematics (子部・天文算法類)

Contents

Introduction 引言

《九章算術》是中國最重要的古代數學典籍，編纂於漢代（公元前 202 年—公元 220 年）。全書分為九卷，每卷解決一類實際數學問題。

劉徽（約 225—295 年）於 **263 年**完成的注釋是數學史上最輝煌的著作之一。劉徽不僅解釋了算法，還運用**出入相補**的幾何方法給出了嚴格的證明，並以正 3072 邊形推算圓周率近似值。

唐代李淳風（602—670 年）奉詔率領學者團隊進一步疏解校正原文。

本文檔呈現**卷四至卷六**：

The **Jiuzhang Suanshu** (“Nine Chapters on the Mathematical Art”) is the most important ancient Chinese mathematical text, compiled during the Han dynasty (202 BCE – 220 CE). The work is divided into nine chapters (卷), each addressing a different category of practical mathematical problems.

The commentary by **Liu Hui** (c. 225–295 CE), completed in **263 CE**, is one of the most brilliant works in the history of mathematics. Liu Hui not only explained the algorithms but also provided rigorous proofs using the method of geometric dissection (出入相補), and famously computed an approximation of π using inscribed polygons with up to 3072 sides.

The sub-commentary by **Li Chunfeng** (602–670 CE) and his Tang court team further clarified and corrected the text.

This document presents **Fascicles 4–6**:

Vol.	**Title**	**Chinese**	**Subject Matter**
4	Shaoguang	少廣	Root extraction, finding dimensions given area/ volume, sphere volume
5	Shanggong	商功	Volumes of earthworks, walls, dykes, granaries, geometric solids
6	Junshu	均輸	Equitable taxation, weighted distribution, travel problems

每題依古典結構：

- **問**：題設
- **答**：答案
- **術**：算法
- **注**：劉徽注釋

Each problem follows the classical structure:

- **Wen** (問): The problem statement
- **Da** (答): The answer
- **Shu** (術): The method/algorithm
- **Zhu** (注): Liu Hui’s commentary

1 Volume 4: Shaoguang (Lesser Breadth) 卷四：少廣

九章算術卷四 • Shaoguang (少廣)

少廣以御積冪方圓

Shaoguang: For governing areas, powers, squares, and circles

本章**少廣**處理已知矩形面積及部分邊長求另一邊長的問題，推廣至開平方、開立方，以及著名的球體積計算。標題意指由「廣」（面積/體積）求「少」（較小之邊長）。

The chapter **Shaoguang** (“Lesser Breadth”) deals with finding one dimension of a rectangle given its area and other dimensions. It extends to root extraction (square and cube roots) and the famous problem of computing the volume of a sphere. The title refers to finding a “lesser” (smaller) dimension given a “greater” (area/volume).

1.1 Opening Method: The Shaoguang Algorithm 少廣術

少廣術 • 少廣算法

置全步及分母子，以最下分母遍乘諸分子及全步。各以其母除其子，置之於左。命通分，內子。又以分母偏乘諸分子，及已通者，皆通而同之，并之為法。置所求步數，以全步積分乘之，為實。實如法而一，得從步。

Method of Lesser Breadth • Shaoguang Algorithm

Place the whole steps and the denominators and numerators of fractions. Multiply all numerators and whole steps by the lowest denominator. For each, divide its numerator by its denominator, and place on the left. Call this “passing through fractions, including numerators.” Again multiply all numerators by the denominator, together with those already passed through, all are passed through and unified. Add them to make the divisor. Place the desired number of steps, multiply by the accumulated whole-step fractions to make the dividend. Divide by the divisor to obtain the length in steps.

劉徽注 • 劉徽論少廣

此以田廣為法，以畝積步為實。術曰置全步及分母子者，通分而齊其子也。齊其子，則母同其母。以母乘全步者，通其分也。以母偏乘諸分子，各以其母除其子，置之於左。命通分內子者，通而同之也。

Liu Hui’s Commentary • On the Shaoguang Algorithm

Here the field width is taken as divisor, and the mu-area in square steps as dividend. “Place whole steps and fraction denominators/numerators” means passing through fractions and aligning their numerators. Aligning numerators makes denominators common. Multiplying whole steps by the denominator passes through the fractions. Multiplying all numerators by the denominator, each divided by its own denominator, placed on the left — “passing through fractions, including numerators” means passing through and unifying them.

1.2 Problem 1: Finding the Side from Area with Fractional Parts 第一問

問・問題一

有田，廣一步半。求田一畝，問從幾何?

Problem 1 • Finding the Length

uppose there is a field, with width one and a half 步. If one 畝 of field is desired, how long is the length?

答

答曰：一百六十步。

Answer

Answer: 160 步.

術

術曰：下有半，是二分之一。以一為二，半為一，并之得三，為法。置田二百四十步，亦以一為二乘之，為實。實如法得從步。

Method

Method: The lower [fraction] is a half, which is $\frac{1}{2}$. Take one as two, the half as one; adding gives three, which is the divisor. Place the field at 240 square steps; likewise multiply by two (taking one as two) to make the dividend. Divide to obtain the length in steps.

劉徽注

此以廣為法，以畝積步為實。術曰下有半是二分之一者，廣一步半，其半即二分之一也。以一為二者，通分也。半為一者，二分之一即一也。并之得三為法者，分母二通全步一為二，分子一，共三也。

Liu Hui's Commentary

Here the width is divisor, the mu-area in steps is dividend. “The lower is a half, which is $\frac{1}{2}$”: the width is one and a half steps, its half being $\frac{1}{2}$. “Take one as two”: passing through fractions. “Half as one”: $\frac{1}{2}$ becomes one. “Adding gives three as divisor”: denominator 2 passes whole step 1 to 2, numerator 1, total 3.

1.3 Problem on Square Root Extraction 開方問

問・開方

有積五萬五千二百二十五步。問為方幾何?

Problem • Square Root Extraction

uppose there is an area of 55,225 square 步. What is the side of the square?

答

答曰：二百三十五步。

Answer

Answer: 235 步.

開方術・開平方算法

術曰：置積為實。借一算，步之，超二等。議所得，以一乘所借一算為法，而以除。除已，倍法為定法。其復除，折法而下。復置借算，步之如初。以復議一乘之，所得副，以加定法，以除。以所得副從定法。復除，折下如前。

Method of Root Extraction • Square Root Algorithm

Method: Place the area as dividend. Borrow one counting rod, step it, skip two places. Estimate the result; multiply by one, using the borrowed rod as divisor, and divide. After division, double the divisor to make the fixed divisor. For continued division, fold the divisor down. Again place the borrowed rod, step as before. Again estimate, multiply by one; the obtained result is auxiliary, add it to the fixed divisor, and divide. The auxiliary result joins the fixed divisor. Continue dividing, folding down as before.

劉徽注 • 劉徽論開方

開方術者，求方冪之一面也。凡冪，方圓之本也。方冪者，方之積也。開方者，求方之面數。言百之面十，萬之面百。術曰置積為實者，積即方冪也。借一算步之者，假以定位也。超二等者，方十自乘，冪一百；方百自乘，冪一萬。故超兩位也。

Liu Hui's Commentary • On Root Extraction

The method of root extraction seeks one side of a square power. All powers are the root of squares and circles. Square power is the area of a square. Root extraction seeks the number of the square's side. The side of 100 is 10; of 10,000 is 100. "Place area as dividend": the area is the square power. "Borrow one rod, step it": borrowed for positioning. "Skip two places": 10 squared is 100; 100 squared is 10,000. Hence skip two positions.

1.4 The Sphere Volume Problem 立圓問

此為《九章算術》中最著名的問題之一，劉徽深入分析，為後世祖沖之、祖暅父子奠定基礎。

This is one of the most famous problems in the *Jiuzhang Suanshu*, which Liu Hui analyzed in depth, setting the foundation for later work by Zu Chongzhi and his son Zu Geng.

問・立圓

有立圓，徑四尺。問為積幾何?

Problem • The Sphere Volume

uppose there is a solid sphere, with diameter 4 尺. What is its volume?

答

答曰：九尺五寸八分寸之一。

Answer

Answer: 9 尺 5 寸 $\frac{1}{8}$ (using the ancient formula $V = \frac{9}{16}d^3$).

術

術曰：圓徑自乘，九之，十六而一。开立圆术：置圓徑四尺，自乘，得一十六尺。九之，得一百四十四尺。十六而一，得九尺。餘十六分尺之八，約之為八分尺之一。并之，為積。

Method

Method: Square the diameter, multiply by 9, divide by 16. To find the sphere volume: place diameter 4 尺, square it to get 16 尺. Multiply by 9 to get 144 尺. Divide by 16 to get 9 尺. Remainder $\frac{8}{16}$ 尺, reduced to $\frac{1}{8}$ 尺. Add to obtain the volume.

劉徽注 • 劉徽論立圓

按此術，圓徑自乘者，先得圓冪。九之十六而一者，以圓冪為方冪，九之十六而一，即圓積也。然此術以圓徑為方徑，誤也。何以驗之？取立方棋八枚，皆令立方一寸，積之為立方二寸。規之為圓，徑二寸。又橫規之，徑亦二寸。然後并立圓棋二枚，規之為圓，徑二寸。又橫規之，徑亦二寸。是為圓積也。

徽術曰：立方徑自乘，又以圓周率三乘之，十二而一，得圓積。此圓冪之率也。又以此圓積乘高，得圓柱積。立圓者，圓柱之半也。故又以圓周率三乘之，十二而一，得立圓積。此術未得其理。

牟合方蓋者，方率也。丸居牟合方蓋，則圓率也。按合蓋者，方率也。其中則圓周率也。方圓相纏，周徑相直。若設方率為四，圓周率為三。以圓周率乘方冪，開方除之，即徑。以方率乘圓冪，開方除之，即周。此圓方之率也。

Liu Hui’s Commentary • On the Sphere

According to this method, squaring the diameter first obtains the circular power. Multiplying by 9 and dividing by 16, taking the circular power as square power, gives the circular area. But this method takes the circle’s diameter as the square’s diameter — this is erroneous. How to verify? Take 8 cubic blocks, each 1 cubic cun, totaling 2 cubic cun. Cut into a circle, diameter 2 cun. Cut crosswise, diameter also 2 cun. Then combine 2 sphere-blocks, cut into a circle, diameter 2 cun. Cut crosswise, diameter also 2 cun. This is the circular volume.

Liu Hui’s method: Square the cube diameter, multiply by the circular ratio 3, divide by 12, obtaining the circular area. This is the circular power ratio. Multiply this circular area by

*譯注：*劉徽在此指出古代公式 $V = \frac{9}{16}d^3$ 的謬誤（暗示 $\pi = 3$ 的特定配置）。他提出**牟合方蓋**（兩垂直圓柱相交體）的概念，作為求球體積的關鍵。他指出球內切於此立體，但未能完成計算。這一工作後由五世紀的**祖暅**完成，證明牟合方蓋體積為外切立方體的 $\frac{2}{3}$，從而得到正確公式 $V = \frac{\pi}{6}d^3$。

the height to obtain the cylinder volume. A sphere is half a cylinder. So again multiply by circular ratio 3, divide by 12, to obtain the sphere volume. This method has not yet found the true principle.
The mouhe fangai (double-lid umbrella) has the square ratio. The sphere contained within the mouhe fangai has the circular ratio. The fangai is the square ratio; within it is the circular ratio. Square and circle intertwine, circumference and diameter correspond. If square ratio is 4 and circular ratio is 3, multiplying the square power by the circular ratio and extracting the square root gives the diameter. This is the ratio of circle and square.

Explanatory note: Liu Hui here identifies the error in the ancient formula $V = \frac{9}{16}d^3$. He introduces the concept of **mouhe fangai** (“double-lid umbrella”), the intersection of two perpendicular cylinders, as the key to finding the sphere volume. He correctly states that the sphere is inscribed in this solid, but was unable to complete the computation. This was later accomplished by **Zu Geng** in the 5th century, who proved that the volume of the mouhe fangai is $\frac{2}{3}$ of the circumscribed cube, leading to the correct formula $V = \frac{\pi}{6}d^3$.

2 Volume 5: Shanggong (Construction Consultations) 卷五：商功

九章算術卷五 • Shanggong (商功)

商功以御功程積實

Shanggong: For governing labor, engineering, and solid volumes

本章**商功**處理工程營建中各類立體的體積計算：土方工程、城牆、堤壩、溝渠、糧倉，以及豐富的幾何立體。劉徽注釋在此尤為詳盡，以割補法提供幾何證明。

The chapter **Shanggong** (“Construction Consultations”) deals with computing the volumes of various solids arising in engineering and construction: earthworks, city walls, dykes, ditches, canals, granaries, and a rich collection of geometric solids. Liu Hui’s commentary is particularly extensive here, providing geometric proofs using dissection methods.

2.1 Classification of Earthwork Solids 穿地率

商功術・土方率

穿地四，為壤五，為堅三。墟謂穿坑，此皆其常率。

Method • Earthwork Conversion Rates

For excavated earth: 4 units [loose] become 5 units [piled], or 3 units [compacted]. The “ruins” refer to excavated pits — these are standard rates.

劉徽注

以穿地求壤，五之；求堅，三之。皆四而一。以壤求穿，四之；求堅，三之。皆五而一。以堅求穿，四之；求壤，五之。皆三而一。此術皆其常率也。穿地四為壤五者，掘地四尺，出壤五尺，壤虛而堅實也。

Liu Hui’s Commentary

From excavated to piled: multiply by 5; to compacted: multiply by 3. All divide by 4. From piled to excavated: multiply by 4; to compacted: multiply by 3. All divide by 5. From compacted to excavated: multiply by 4; to piled: multiply by 5. All divide by 3. These are standard rates. “Excavated 4 becomes piled 5”: digging 4 chi of earth yields 5 chi of loose soil, since loose earth is voluminous and compacted earth is dense.

2.2 Problem: Volume of a City Wall 城垣問

問・城垣

有城，下廣四丈，上廣二丈，高五丈，袤一百二十六丈。問積幾何?

Problem • City Wall Volume

uppose there is a city wall: lower width 4 丈, upper width 2 丈, height 5 丈, length 126 丈. What is the volume?

答

答曰：一百八十九萬尺。

Answer

Answer: 1,890,000 cubic 尺.

術

術曰：并上下廣而半之，以高若深乘之，又以袤乘之，即積尺。

Method

Method: Add the upper and lower widths and halve; multiply by the height or depth; then multiply by the length to obtain the volume in cubic 尺.

劉徽注

按此術，并上下廣而半之者，以盈補虛，得中平之廣。以高乘之，得截而為直棱之積。又以袤乘之者，廣袤相乘為一截之積，以高乘之為積。

Liu Hui's Commentary

Adding the widths and halving gives the "middle width" by the principle of compensating excess with deficiency (以盈補虛). Multiplying by height gives the cross-sectional area as a right prism. Multiplying by length: width times length is the cross-sectional area, times height gives the volume.

2.3 Problem: Volume of a Canal with Sloped Sides 塹堵問

問・塹堵

有塹，上廣一丈六尺，下廣一丈，深六尺，袤五丈七尺。問積幾何?

Problem • Trench/Canal

uppose there is a trench: upper width 1 丈 6 尺, lower width 1 丈, depth 6 尺, length 5 丈 7 尺. What is the volume?

答

答曰：積七千一百一十二尺。

Answer

Answer: Volume 7,112 cubic 尺.

術

術曰：并上下廣，半之，以高若深乘之，又以袤乘之，即積尺。

Method

Method: Add the upper and lower widths and halve; multiply by the height or depth; then multiply by the length to obtain the volume in cubic 尺.

2.4 The Yangma (陽馬) and Bienao (鱉臑) 陽馬與鱉臑

此為中國數學中最重要的幾何立體之一，由長方體分割而來。

陽馬術・陽馬（隅角錐）

術曰：廣袤相乘，以高乘之，三而一。

此為**陽馬**（yangma）體積公式，一種底面為矩形且一側棱垂直於底面的角錐：

$$V = \frac{1}{3} \times \text{廣} \times \text{袤} \times \text{高}$$

劉徽注 ・劉徽論陽馬

按此術，陽馬者，隅角之椎也。解立方得兩塹堵，解塹堵得一陽馬、一鱉臑。陽馬居二，鱉臑居一，不易之率也。故立方三而一，即陽馬之積也。
合二塹堵成一陽馬者，試令廣袤各六尺，高六尺。中分其高，令廣袤各三尺。其高六尺，中分為二，各三尺。上塹堵廣袤各三尺，高亦三尺；下塹堵亦然。合二塹堵，廣袤各六尺，高六尺，是為一陽馬也。

These are among the most important geometric solids in Chinese mathematics, arising from the dissection of a rectangular prism.

Yangma Method ・Corner Pyramid

Method: Multiply width by length, multiply by height, divide by 3.

This gives the volume of a **yangma**, a pyramid with rectangular base and one lateral edge perpendicular to the base:

$$V = \frac{1}{3} \times \text{width} \times \text{length} \times \text{height}$$

Liu Hui's Commentary ・On the Yangma

A yangma is a corner pyramid. Dissecting a rectangular prism gives two qian-du (wedge-like solids); dissecting each qian-du gives one yangma and one bienao (tetrahedron). The ratio of yangma to bienao is always 2:1 — an unchanging rate. Hence a cube divided by 3 gives the yangma volume.
Combining two qian-du to make one yangma: suppose width and length are each 6 chi, height 6 chi. Bisect the height, making width and length each 3 chi. The height of 6 chi is bisected into two, each 3 chi. The upper qian-du has width

鱉臑術・鱉臑（四面體）

術曰：廣袤相乘，以高乘之，六而一。

此為**鱉臑**（bienao）體積公式，一種三條棱兩兩垂直的四面體：

$$V = \frac{1}{6} \times a \times b \times c$$

劉徽注

鱉臑者，推之明理也。按陽馬之積，三分立方之一。鱉臑者，又半陽馬也。故六而一。凡立方，高一尺，廣袤各一尺，積一尺。陽馬廣袤各一尺，高一尺，積三分之一尺。鱉臑三邊各一尺，積六分之一尺。

and length 3 chi, height 3 chi; the lower qian-du likewise. Combining two qian-du, width and length each 6 chi, height 6 chi, this makes one yangma.

Bienao Method ・Tetrahedron

Method: Multiply width by length, multiply by height, divide by 6.

This gives the volume of a **bienao**, a tetrahedron with three mutually perpendicular edges:

$$V = \frac{1}{6} \times a \times b \times c$$

Liu Hui's Commentary

The bienao demonstrates clear principles. The yangma volume is one-third of the cube. The bienao is half the yangma. Hence divide by 6. For a cube of height 1 chi, width and length each 1 chi, volume is 1 cubic chi. A yangma with width and length 1 chi, height 1 chi, has volume $\frac{1}{3}$ cubic chi. A bienao with three edges each 1 chi has volume $\frac{1}{6}$ cubic chi.

2.5 The Chumeng (芻甍) and Chutong (芻童) 芻甍與芻童

問・芻甍

有芻甍，下廣三丈，袤四丈，上袤二丈，無廣，高一丈。問積幾何？

答

答曰：積五千尺。

Problem ・Thatched Roof (Wedge)

uppose there is a thatched roof: lower width 3 丈, lower length 4 丈, upper length 2 丈, no upper width, height 1 丈. What is the volume?

Answer

Answer: Volume 5,000 cubic 尺.

術

術曰：倍下袤，上袤從之，以廣乘之，又以高乘之，六而一。

Method

Method: Double the lower length, add the upper length, multiply by the width, then multiply by the height, divide by 6.

劉徽注

推明義理者，舊說云：凡積芻甍有上下廣曰童甍，謂其屋蓋之苫也。是故甍之下廣袤與童之上廣袤等。正解方亭兩邊合之，即芻甍之形也。假令下廣二尺，袤三尺，上袤一尺，無廣，高一尺。其用棊也，中央塹堵二，兩端陽馬各二。倍下袤為六，上袤從之為七。以廣二尺乘之，得一十四尺。以高乘之，得十四尺。六而一，即得積也。

Liu Hui’s Commentary

To explain the principles: old texts say a chumeng with upper and lower widths is called “tong meng” (roof-like). The lower width and length of the chumeng equal the upper width and length of a chutong. Properly dissected from the two sides of a square pavilion, one obtains the form of a chumeng. Suppose lower width 2 chi, length 3 chi, upper length 1 chi, no upper width, height 1 chi. Using blocks: 2 central qian-du, 2 yangma at each end. Double lower length to get 6, add upper length to get 7. Multiply by width 2 to get 14. Multiply by height to get 14. Divide by 6 to obtain the volume.

問・芻童

有芻童，下廣二丈，袤三丈，上廣三丈，袤四丈，高三丈。問積幾何?

Problem ・Thatched Mound (Frustum of Wedge)

uppose there is a thatched mound: lower width 2 丈, lower length 3 丈, upper width 3 丈, upper length 4 丈, height 3 丈. What is the volume?

答

答曰：積一萬六千五百尺。

Answer

Answer: Volume 16,500 cubic 尺.

術

術曰：倍上袤，下袤從之；亦倍下袤，上袤從之。各以其廣乘之，并，以高乘之，六而一。

Method

Method: Double the upper length, add the lower length; also double the lower length, add the upper length. Multiply each by its respective width, add together, multiply by height, divide by 6.

劉徽注

按此術，假令芻童上廣一尺，袤二尺；下廣二尺，袤三尺；高一尺。其用棋也，中央立方一，兩旁塹堵各一，四角陽馬各一。倍上袤為四，下袤從之為七。以下廣乘之，得一十四。倍下袤為六，上袤從之為八。以上廣乘之，得八。并之，得二十二。以高乘之，得二十二。六而一，即積也。此術與芻甍曲池盤池冥谷皆同術。

Liu Hui's Commentary

According to this method: suppose a chutong with upper width 1 chi, length 2 chi; lower width 2 chi, length 3 chi; height 1 chi. Using blocks: 1 central cube, 1 qian-du on each side, 1 yangma at each of 4 corners. Double upper length to 4, add lower length to get 7. Multiply by lower width 2 to get 14. Double lower length to 6, add upper length to get 8. Multiply by upper width 1 to get 8. Add: 22. Multiply by height: 22. Divide by 6 for the volume. This method is the same for chumeng, curved pool, basin, and ravine.

2.6 The Circular Granary 圓窖問

問・圓窖

有圓窖，下周五丈四尺，深一丈八尺。問受粟幾何?

Problem ・Circular Granary

uppose there is a circular granary: circumference 5 丈 4 尺, depth 1 丈 8 尺. How much grain does it hold?

答

答曰：二千九百六十二斛二十七分斛之二十六。

Answer

Answer: $2962\,\frac{26}{27}$ 斛.

術

術曰：置周自相乘，以高乘之，十二而一。以斛法一尺六寸二分除之，即斛數。

Method

Method: Place the circumference, square it, multiply by the height, divide by 12. Divide by the hu-measure of 1 chi 6 cun 2 fen to obtain the number of hu.

劉徽注

按此術，圓窖下周自乘，以高乘之，十二而一者，以圓周率三乘之，十二而一，即圓積也。此猶圓垛壔之積也。于徽術，當令下周自乘以高乘之，又以二十五乘之，三百一十四而一。以斛法除之，即斛數。

Liu Hui's Commentary

Squaring the lower circumference and multiplying by height, then dividing by 12, uses the circular ratio 3 and divides by 12 to obtain the circular area. This is like the volume of a circular fortification. In Liu Hui's method, one should square the lower circumference, multiply by height, then multiply by 25 and divide by 314. Divide by the hu-measure to obtain the number of hu.

3 Volume 6: Junshu (Equitable Taxation) 卷六：均輸

九章算術卷六 • Junshu (均輸)

均輸以御遠近勞費

Junshu: For governing fair distribution considering distance, labor, and cost

本章**均輸**處理加權分配問題，各地區按人口數、運距等因素公平分攤稅賦。另含著名的工作效率問題，如「五渠注池」問題。

The chapter **Junshu** (“Equitable Taxation”) deals with problems of weighted distribution, where contributions must be apportioned fairly among parties that differ in population, distance from the destination, and other factors. It also contains famous work-rate problems, including the “five channels filling a pool” problem.

3.1 The Opening Problem: Grain Distribution Among Four Counties 四縣均輸粟

問 • 均輸粟

有均輸粟，甲縣一萬戶，行道八日；乙縣九千五百戶，行道十日；丙縣一萬二千三百五十戶，行道十三日；丁縣一萬二千二百戶，行道二十日。各到輸所。凡四縣賦，當輸二十五萬斛，用車一萬乘。欲以道里遠近、戶數多少，衰出之。問粟、車各幾何?

Problem • Equitable Grain Distribution

uppose there is equitable grain distribution: County A has 10,000 households, travel time 8 days; County B has 9,500 households, travel 10 days; County C has 12,350 households, travel 13 days; County D has 12,200 households, travel 20 days. All reach the delivery point. Total tax: 250,000 斛, using 10,000 carts. Distribute according to distance and household count. How much grain and how many carts for each?

答

答曰：甲縣粟八萬三千一百斛，車三千三百二十四乘。乙縣粟六萬三千一百七十五斛，車二千五百二十七乘。丙縣粟六萬三千一百七十五斛，車二千五百二十七乘。丁縣粟四萬一千九百一十九斛，車一千六百二十二乘。

Answer

Answer: County A: 83,100 斛 grain, 3,324 carts. County B: 63,175 斛, 2,527 carts. County C: 63,175 斛, 2,527 carts. County D: 41,919 斛, 1,622 carts.

術 • 加權分配法

術曰：令縣戶數，各如其本行道日數而一，以為衰。甲衰一百二十五，乙、丙衰各九十五，丁衰六十一。副并為法。以賦粟、車數，未并者各自為實。實如法得一。按此均輸，猶均運也。令戶率出車，以行道日數為均，發粟為輸。

Method • Weighted Distribution

Method: Take each county’s household count, divide by its travel days in days, to make the “decline” (weight). A’s weight: 125; B and C: 95 each; D: 61. Add these subsidiarily to make the divisor. The tax grain and cart numbers, not yet added, each become dividends. Divide by the divisor to obtain one share. This junshu is like equal transport: each household contributes carts proportionally, using travel days as the equalizer, dispatching grain as the transport.

劉徽注 • 劉徽論均輸

此戶以率當各計車之錢分也。案此二句舛誤，當云：求此率以戶當各計車之錢分也。淳風等按：縣戶有多少之差，行道有遠近之異。欲其均等，故令戶數各以行道日數約之，為衰。行道多者，其戶少；行道少者，其戶多。故各令約戶為衰。并戶為法者，以戶數多寡為差，行道遠近為衰，以此均之，則勞逸齊等。

Liu Hui’s Commentary • On Equitable Distribution

Each household by rate should contribute cart-money proportionally. Li Chunfeng et al.: Counties differ in household numbers and travel distances. To make it fair, divide household counts by travel days to make the decline weights. Counties with longer travel have fewer effective households; those with shorter travel have more. Hence reduce households by travel days to make weights. Adding households as divisor, using household numbers as difference and travel distance as weight, equalizing by this makes labor and rest uniform.

3.2 Problem: Workers with Different Rates 程人功問

問 • 程人功

有程人功，甲七日成功了五分之一；乙五日成功了三分之一；丙四日成功了二分之一。問三人合作，幾何日成功?

Problem • Labor Rates

uppose there are workers: A completes one-fifth of the work in 7 days; B completes one-third in 5 days; C completes one-half in 4 days. How many days if all three work together?

答

答曰：二十三日六十三分日之二十六。

Answer

Answer: $23\,\frac{26}{63}$ days.

術

術曰：置甲七日的五分之一，乙五日的三分之一，丙四日的二分之一，各為率。以一日为法，以所为实。实如法而一，各得一日之功。并一日之功，为法。以一日为实。实如法而一，即得日数。

Method

Method: Place A’s one-fifth in 7 days, B’s one-third in 5 days, C’s one-half in 4 days, each as a rate. Taking one day as divisor and the accomplishment as dividend, divide to obtain each worker’s one-day contribution. Add the one-day contributions to make the divisor. Take one day as dividend. Divide to obtain the number of days.

3.3 The Five Channels Filling a Pool 五渠注池

此為《九章算術》中最負盛名的問題之一，見於卷六末題。

This is one of the most celebrated problems in the *Jiuzhang Suanshu*, appearing as the final problem of Volume 6.

問・五渠注池

有池，五渠注之。其一渠開之，少半日一滿；次，一日一滿；次，二日半一滿；次，三日一滿；次，五日一滿。今并決之，問幾何日滿池?

Problem • Five Channels Filling a Pool

uppose there is a pool, with five channels filling it. The first channel alone fills it in one-third of a day; the second in one day; the third in two and a half days; the fourth in three days; the fifth in five days. Now all are opened together. How many days to fill the pool?

答

答曰：七十四分日之十五。

Answer

Answer: $\frac{15}{74}$ of a day.

術

術曰：各置渠一日滿池之數，并，以為法。以一為實。實如法而一，即得日數。

Method

Method: Place each channel’s daily fill rate [i.e., $\frac{1}{1/3}$, $\frac{1}{1}$, $\frac{1}{2.5}$, $\frac{1}{3}$, $\frac{1}{5}$], add them as the divisor; take 1 as the dividend. Divide to get the days.

劉徽注

按此術，一渠少半日滿者，是一日三滿也。次一日一滿者，是一日一滿也。次二日半滿者，是一日五分滿之二也。次三日一滿者，是一日三分滿之一也。次五日一滿者，是一日五分滿之一也。并之，得四滿十五分滿之十四也。此為法。以一滿為實。實如法而一，即得滿池之日數也。

Liu Hui's Commentary

The first channel fills 3 pools/day; the second fills 1; the third fills $\frac{2}{5}$; the fourth fills $\frac{1}{3}$; the fifth fills $\frac{1}{5}$. Summing: $3+1+\frac{2}{5}+\frac{1}{3}+\frac{1}{5}=\frac{74}{15}$ pools/day. Thus one pool takes $\frac{15}{74}$ day.

3.4 Problem: Hares and Dogs (Pursuit Problem) 兔犬追及問

問・兔犬

有兔先走一百步，犬追之二百五十步，不及三十步而止。問犬不止，復行幾何步及之?

Problem • Hare and Dog Pursuit

hare has a head start of 100 步. A dog chases it for 250 步, but is still 30 步 behind. If the dog does not stop, how many more 步 must it run to catch the hare?

答

答曰：一百七步七分步之一。

Answer

Answer: $107\,\frac{1}{7}$ steps.

術

術曰：置犬先行步數，以不及步數減之，餘為法。以不及步數乘兔先走步數，為實。實如法得一步。

Method

Method: Place the dog's forward steps, subtract the shortfall steps; the remainder is the divisor. Multiply the shortfall by the hare's head start to make the dividend. Divide to obtain the number of steps.

3.5 Problem: Gold and Silver Weights 金銀重率問

問・金銀

有金九枚，銀一十一枚，稱之適等。交易其一，金輕十三兩。問金、銀各一枚重幾何?

Problem • Gold and Silver Weights

uppose there are 9 pieces of gold and 11 pieces of silver, which weigh the same. Exchange one piece [between them], and the gold side is lighter by 13 兩. How much does each piece of gold and silver weigh?

答

答曰：金重二斤三兩一十八銖，銀重一斤十三兩六銖。

Answer

Answer: Gold: 2 jin 3 liang 18 zhu; Silver: 1 jin 13 liang 6 zhu.

術

術曰：假令金重三斤，銀重二斤三分斤之二。不足四兩，令之如此。盈不足術為之。

Method

Method: Suppose gold weighs 3 jin, silver weighs $2\,\frac{2}{3}$ jin. The deficit is 4 liang; adjust accordingly. Use the surplus-deficit method (ying bu zu shu).

4 Summary of Mathematical Content 數學內容總結

Volume 4: 少廣 (Shaoguang) 卷四

- **開方**（平方根）：迭代算法，本質等同現代開方長除法
- **開立方**（立方根）：推廣至 $\sqrt[3]{N}$
- **球體積**：古術 $V = \frac{9}{16}d^3$；劉徽批判與牟合方蓋構造
- 分數運算的幾何詮釋

- **Square root extraction**: Iterative algorithm essentially equivalent to modern long-division method for roots
- **Cube root extraction**: Extension to $\sqrt[3]{N}$
- **Sphere volume**: Ancient formula $V = \frac{9}{16}d^3$; Liu Hui's critique and the mouhe fangai construction
- Fractional arithmetic with geometric interpretation

Volume 5: 商功 (Shanggong) 卷五

- **土方體積**：城、垣、堤、溝、渠、塹
- **糧倉體積**：倉、圓窖、圓囤
- **幾何立體**：立方、塹堵、陽馬、鱉臑、芻甍、芻童、曲池、盤池、冥谷
- 劉徽以**割補法**證明各立體體積公式

- **Earthwork volumes**: City wall, wall, dyke, ditch, canal, trench
- **Granary volumes**: Granary, circular pit, circular granary
- **Geometric solids**: Cube, qian-du, yangma, bienao, chumeng, chutong, curved pool, basin, ravine
- Liu Hui's **geometric dissection proofs** for each solid volume formula

Volume 6: 均輸 (Junshu) 卷六

- **加權分配**：按人口 × 運距分攤稅賦
- **工作效率問題**：多人異速合作
- **追及問題**：相對速度（兔犬問題）
- **聯立方程**：金銀重率、糧食交換
- 著名的**五渠注池**工作效率問題

- **Weighted distribution**: Apportioning taxes among counties by population × distance
- **Work-rate problems**: Multiple workers with different rates
- **Pursuit problems**: Relative speed (hare and dog)
- **Simultaneous equations**: Gold and silver weights, grain exchange
- The famous **five channels filling a pool** work-rate problem

「九章之法，算之綱紀。」
“The methods of the Nine Chapters are the source of all computation.”
— 劉徽《九章算術注序》
— Liu Hui, Preface to the Commentary

九章算術卷四至卷六完
End of Jiuzhang Suanshu, Fascicles 4–6

tcb@breakable

序

九章算術

術曰：劉徽序曰：昔在包犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術，以合六爻之變。暨於黃帝神農，其為功也，各有稱焉。周公制禮而有九數，九數之流，則九章是矣。

劉徽注：徽幼習九章，長再詳覽。觀陰陽之割裂，總算術之根源，探賾之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。事物有對，故立正負之率；錯綜無形，故開方程之術。庶幾令文約而義博，事省而功多。

《九章算術》是古代數學最偉大的成就之一。經數百年編纂，約成書於漢代，全書九章共二百四十六題。劉徽注約成於西元年，將這部實用手冊提升為具有深刻數學洞察力的經典。

本書所呈現的三卷，乃《九章》之數學瑰寶：盈不足之算法精妙、方程之代數造詣、勾股之幾何優雅。

第二題共買雞

問：今有共買雞，人出九，盈十一；人出六，不足十六。問：人數、雞價各幾何?

答曰：人九，雞價七十。

術曰：術曰：置人出九、人出六，盈十一、不足十六。令維乘之：九乘十六得一百四十四，六乘十一得六十六。并之得二百一十，為實。并盈不足得二十七，為法。

劉徽注：此術與上術意同。九乘不足十六者，以多出之率乘不足之數也；六乘盈十一者，以少出之率乘盈之數也。所以維乘者，兩設不同，不可偏用，故交錯相乘以齊同之。

第三題共買璡

問：今有共買璡，人出半，盈四；人出少半，不足三。問：人數、璡價各幾何?

答曰：人四十二，璡價十七。

術曰：術曰：置人出半、人出少半，盈四、不足三。半即二分，少半即三分。令維乘之，如法求之。

劉徽注：半者，二分之一也；少半者，三分之一也。出半則多四，出少半則少三。以分數為率，故先通分之，然後維乘求之。此術以分數入盈不足，所以見術之無所不通也。

第四題共買牛

問：今有共買牛，七家共出一百九十，不足三百三十；九家共出二百七十，盈三十。問：家數、牛價各幾何?

答曰：一百二十六家，牛價三千七百五十。

術曰：術曰：置七家共出一百九十、不足三百三十，九家共出二百七十、盈三十。令每家出率各以家數除之：一百九十除以七得二十七又七分之一；二百七十除以九得三十。乃以盈不足維乘之，并為實，并盈不足為法，實如法而一。

劉徽注：此題以家為率，不以人為率。故先求每家所出之率，然後用盈不足術。七家出一百九十，則不足三百三十，是每家所出不足也；九家出二百七十，盈三十，是每家所出盈也。先通為每家之率，然後可用盈不足之法。此術之變，以家為率也。

第五題米麥互換

問：今有米一斗，麥一斗，各值幾何? 但知米三斗值麥二斗，麥四斗值米一斗。問：米、麥各一斗價若干?

答曰：米一斗值四錢，麥一斗值一錢半。

術曰：術曰：以盈不足術求之。假令米一斗值四錢，則麥當值若干，驗其盈不足，乃以術求之。

劉徽注：此術非直盈不足，乃以價為率，互相推求。假令一值，驗其盈朒，然後設為兩端，用盈不足術以齊之。米三斗值麥二斗，則米一斗值麥三分之二斗。麥四斗值米一斗，則麥一斗值米四分之一斗。以盈不足術通之，設米一斗值若干，驗其與麥價之盈朒，維乘求之。此以價為隱，而以盈不足顯之也。

第六題良駑與驢

問：今有良駑二馬，與驢俱牽。良馬與駑馬共引之，良馬日引四十里，駑馬日引四十里。驢在後逐之，不及。問：驢日行幾何里?

答曰：驢日行二十里。

術曰：術曰：以盈不足術求之。假令驢行三十里，則驢所行多於良駑之半，是為盈；假令驢行十里，則不及半，是為不足。維乘并實，如法求之。

劉徽注：良駑俱行四十里，其半二十里。驢逐其後，當與之半相當。假令三十里，則多十里，為盈；假令十里，則少十里，為不足。故以盈不足術求之，得二十里。此術之意：良駑俱行四十里，驢逐其半，故以二十里為中平之數。

第七題垣高術

問：今有垣不知高。立一表長五尺，去垣五丈，目距地五尺，望垣頂與表端參合。問：垣高幾何?

答曰：垣高五丈五尺。

術曰：術曰：以盈不足術求之。假令垣高五丈，則視線當過表端若干；假令垣高六丈，則視線不及表端若干。維乘求之。

劉徽注：此術本屬勾股，今以盈不足術驗之。去垣五丈，表高五尺，目高五尺，則表與目平。望垣頂與表端參合，是為表高與垣高之比例，如去垣之遠與目至垣頂之比例也。以相似勾股求之即得。此術雖屬勾股，而以盈不足驗之，所以明術之相通也。

第八題五家共井

問：今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。問：井深、綆長各幾何?

答曰：井深七丈二尺一寸。甲綆長二丈六尺五寸，乙綆長一丈九尺一寸，丙綆長一丈四尺八寸，丁綆長一丈二尺九寸，戊綆長七尺六寸。

術曰：術曰：此以盈不足為問，實則以方程入之。各以其綆之數，不足之數，列為方程，正負相生，求之即得。

劉徽注：此題本以盈不足為問，而實非盈不足之術所能獨盡也。五家之綆長短不同，各家取綆若干而不足若干，以補他家一綆之數。此五綆之長與井深，共為六未知數，而題中僅給五條件，故為不定問題也。其答云者，特其一組正整數解耳。若以方程術解之，當令井深為一，求五家綆長之比，然後以井深定其實長。此題之精妙，在於以盈不足為表，以方程為裡，表裡相濟，乃得其真。

第九題金與銀

問：今有金九枚，銀十一枚，稱之適等。交易其一，金輕十三兩。問：金、銀一枚各重幾何?

答曰：金重二斤三兩一十八銖，銀重一斤十三兩六銖。

術曰：術曰：以盈不足術求之。假令金重三斤，則九金二十七斤；令銀與之等，則每銀當重二十七斤十一分。交易其一，驗其輕重，得盈不足之數。再設一率，維乘求之。

劉徽注：金九銀十一稱之適等者，總重等也。交易其一者，以一金換一銀也。金輕十三兩者，換後金方輕於原銀方十三兩也。以方程術解之尤便：設金重x，銀重y，則$9x = 11y$，且$8x + y = 10y + x - 13$兩。以盈不足術者，兩設其重，驗其差也。

第十題程傳委輸

問：今有程傳委輸，空車日行七十里，重車日行五十里。今載太倉粟輸上林，五日三返。問：太倉去上林幾何?

答曰：太倉去上林四十八里一百八十分里之十一。

術曰：術曰：以盈不足術求之。假令去四十五里，則五日三返不足若干里；假令去五十里，則五日三返盈若干里。維乘并實，如法求之。

劉徽注：空車日行七十里者，往而不載也；重車日行五十里者，載粟而返也。五日三返者，一往一返為一返，三返則三往三返也。此以盈不足術驗之者，取其近似也。此術雖以盈不足入之，實則以調和平均為本也。

第十一題鹿與兔

問：今有鹿直西走，兔直東走。兔行一百步，鹿行七十步。兔在鹿東一百五十步，同時發。問：幾何步相逢?

答曰：兔行二百五十步，鹿行一百七十五步，相逢。

術曰：術曰：以盈不足術求之。假令兔行二百步，則鹿行一百四十步，驗其盈不足之數。再假令一行數，維乘求之。

劉徽注：兔東鹿西，相向而行，故為相逢之術。兔行一百步，鹿行七十步，則兔鹿步率之比為十比七也。相距一百五十步，同時發，則相逢之時，兔行十份，鹿行七份，共十七份。以一百五十步分為十七份，每份步。

第十二題蒲與莞

問：今有蒲生一日，長三尺；莞生一日，長一尺。蒲生日自半，莞生日自倍。問：幾何日而長等?

答曰：二日十三分日之六，而長等。各長四尺五寸十三分寸之五。

術曰：術曰：以盈不足術求之。假令二日，不足一尺五寸；令之三日，盈一尺七寸半。維乘并實，如法求之。

劉徽注：蒲日生三尺而自半者，第一日長三尺，第二日長一尺五寸，第三日長七寸半，是為日自半也。莞日生一尺而自倍者，第一日長一尺，第二日長二尺，第三日長四尺，是為日自倍也。此術之巧，在於兩物生長之率不同，一自半而一自倍，以盈不足術齊而同之。

第十五題漆三油四

問：今有漆三得油四，油四和漆五。今有漆三斗，欲令分以易油，還和餘漆。問：出漆、得油、和漆各幾何?

答曰：出漆一斗一升四分升之一，得油一斗五升，和漆一斗八升四分升之三。

術曰：術曰：以盈不足術求之。假令出漆一斗，不足若干；令出漆一斗二升，盈若干。維乘并實，如法求之。

劉徽注：漆三得油四者，以漆三斗易油四斗也。油四和漆五者，以油四斗和漆五斗也。今有漆三斗，分其漆以易油，又以所得之油和其餘漆。此術以漆易油，以油和漆，往還相求，以盈不足定其率。

第十六題惡粟為米

問：今有惡粟二十斗，舂之為米，米率五斗。今欲為糲米、粺米各幾何?

答曰：糲米十斗，粺米六斗六升三分升之二。

術曰：術曰：以盈不足術求之。假令糲米十斗，則粺米當若干，驗其盈不足。再設一率，維乘求之。

劉徽注：惡粟二十斗，米率五斗者，一斗惡粟舂得五升米也。糲米率五，粺米率六。以盈不足術者，兩設糲米之斗數，求粺米之盈朒也。此以米率之隱，而以盈不足顯之也。

第十七題持米出關

問：今有人持米出三關，外關三而取一，中關五而取一，內關七而取一，餘米五斗。問：本持米幾何?

答曰：本持米十斗九升八分升之三。

術曰：術曰：以盈不足術求之。假令持米八斗，不足若干；令持米十二斗，盈若干。維乘并實，如法求之。

劉徽注：三關各取其幾分之一。外關三取一，是取其三分之一也；中關五取一，是取其五分之一也；內關七取一，是取其七分之一也。過三關之後，餘米五斗。此術以三關之稅率為隱，以盈不足顯之，所以見術之無幽不顯也。

第十八題五人分五錢

問：今有五人分五錢，令上二人所得與下三人等。問：各得幾何?

答曰：甲得一錢六分钱之二，乙得一錢六分钱之一，丙得一錢，丁得六分钱之五，戊得六分钱之四。

術曰：術曰：以盈不足術求之。假令五人各得一錢，則上二人得二錢，下三人得三錢，盈一錢。此當以衰分術入之。

劉徽注：五人分五錢而令上二人與下三人等者，上二人所得當為二錢半，下三人所得亦當為二錢半也。然此題以衰分術為正：令五人所得為差率，以差分配之。此術雖以盈不足入之，實以衰分為本也。

第十九題弧田徑術

問：今有弧田，弦三十步，矢十五步。問：為田幾何?

答曰：一畝九十七步半。

術曰：術曰：以弦乘矢，矢又自乘，并之，三而一。

劉徽注：弧田術者，以弦矢求弧田之面積也。弦乘矢者，以弦為廣，矢為縱，相乘得長方形之積。矢自乘者，以矢為方之邊，自乘得正方之積。并之者，合兩積為一。三而一者，以弧田之積約當此兩積和之三分之一也。此術為弧田之近似術。

第二十題環田徑術

問：今有環田，中周六十二步，外周四步，徑十二步。問：為田幾何?

答曰：二畝五十五步。

術曰：術曰：并中外周而半之，以徑乘之為積。

劉徽注：環田者，猶圓田之中去一圓也。并中外周而半之者，取環之平均周長也。以徑乘之者，以平均周長為廣，徑為縱也。此術以環田近似為矩形，故以廣縱相乘得積。若精密求之，當以兩圓面積之差求之。

第二十一題三人共車

問：今有三人共車，二車空；二人共車，九人步。問：車幾何？人幾何？

答曰：車一十五乘，人三十九人。

術曰：術曰：以盈不足術求之。假令車十乘，則三人共車載三十人，餘九人步，是為盈九。假令車二十乘，則二人共車載四十人，缺一人，是為不足一。維乘并實，如法求之。

劉徽注：三人共車二車空者，每車三人則二車無人乘也。二人共車九人步者，每車二人則餘九人不行也。此術以車之虛實為盈朒，亦變術之巧者也。

第二十二題綿布互易

問：今有綿一斤直錢二百四十，布一匹直錢三百二十。今有綿五斤，欲易布幾何？

答曰：布三匹七分匹之五。

術曰：術曰：以盈不足術求之。假令易三匹，則綿價一千二百，布價九百六十，盈二百四十。假令易四匹，則綿價一千二百，布價一千二百八十，不足八十。維乘并實，如法求之。

劉徽注：綿一斤直二百四十，五斤則直一千二百。布一匹直三百二十。以盈不足術者，兩設易布之匹數，驗其綿布價值之盈朒也。

第二十三題粟米貴賤

問：今有粟一斗，欲為糲米、粺米、糳米三等。糲米率三十，粺米率二十七，糳米率二十四。問：各為米幾何？

答曰：糲米三升，粺米二升七合，糳米二升四合。

術曰：術曰：以盈不足術入衰分求之。假令糲米三升，則粺米、糲米遞減之，驗其盈不足，維乘求之。

劉徽注：糲米率三十者，粟一斗為糲米三升也。粺米率二十七者，為粺米二升七合也。糳米率二十四者，為糳米二升四合也。此術以三米之率為隱，以盈不足為顯。

第二十四題土功程效

問：今有築堤，甲縣人功一日築七尺，乙縣人功一日築五尺。今令兩縣共築，甲縣先築一日，乙縣繼之。問：幾何日而成？

答曰：甲築一日，乙築二日，共成。

術曰：術曰：以盈不足術求之。假令共築二日，則甲築十四尺，乙築五尺，共十九尺，驗其盈不足。再設一日，維乘求之。

劉徽注：甲縣人功一日七尺，乙縣人功一日五尺。甲先築一日，築七尺。餘令乙築，乙一日五尺，則餘二尺。甲復築一日，又七尺，共十九尺。此術雖以盈不足入之，實以衰分為本也。

卷第八方程

> 方程章乃中國古代代數之巔峰成就。其提出解線性方程組之系統方法——本質即高斯消元法——較高斯早一千五百餘年。本章亦包含世界數學史上負數之最早系統運用。

方程術總論

術曰：方程術曰：置上禾三秉，中禾二秉，下禾一秉，實三十九；上禾二秉，中禾三秉，下禾一秉，實三十四；上禾一秉，中禾二秉，下禾三秉，實二十六。以右行上禾遍乘中行，而以直除。又乘其次，亦以直除。然以中行中禾不盡者遍乘左行，而以直除。左方下禾不盡者，上為法，下為實。實即下禾之實。

劉徽注：程，課程也。群物總雜，各列有數，總言其實。令每行為率，二物者再程，三物者三程，皆如物數程之，並列為行，故謂之方程。行之左右，無所同存，且為有所據而言耳。

$KeySteps: 1)Arrangecoefficientsinrows; 2)Multiplypivotrowandsubtractfromrowsbelow(); 3)Repeatforeachcolumn; 4)Back-substitute.LiuHuiusedredrodsforposi$

第一題三禾求實

問：今有上禾三秉，中禾二秉，下禾一秉，實三十九斗；上禾二秉，中禾三秉，下禾一秉，實三十四斗；上禾一秉，中禾二秉，下禾三秉，實二十六斗。問：上、中、下禾實一秉各幾何?

答曰：上禾一秉，九斗四分斗之一；中禾一秉，四斗四分斗之一；下禾一秉，二斗四分斗之三。

術曰：術曰：方程術。置上禾、中禾、下禾及實於右、中、左三行。以右行上禾三遍乘中行，以直除之，消去中行上禾。又以右行上禾三遍乘左行，以直除之，消去左行上禾。然後以中行中禾不盡者遍乘左行，以直除之，消去左行中禾。

劉徽注：此方程術之正也。置三行於位：右行上禾三、中禾二、下禾一、實三十九；中行上禾二、中禾三、下禾一、實三十四；左行上禾一、中禾二、下禾

三、實二十六。以右行上禾三乘中行，得六、九、三、一百零二；與右行直除之。

第二題五家共輸

問：今有甲、乙、丙、丁、戊五家共輸粟。甲所輸加乙之一半，乙所輸加丙之三分之一，丙所輸加丁之四分之一，丁所輸加戊之五分之一，戊所輸加甲之六分之一，各適等。問：各家所輸幾何?

答曰：各家所輸均為一斛。甲輸二十五分斛之十二，乙輸二十五分斛之四，丙輸二十五分斛之三，丁輸二十五分斛之二，戊輸二十五分斛之一。

a, b, c, d, ek

術曰：術曰：以方程術入之。令各家所輸為未知數，以條件列方程，正負相生，求之即得。

劉徽注：此題以方程術入之為便。令甲、乙、丙、丁、戊所輸分別為a, b, c, d, e。則有：$a + b/2 = b + c/3 = c + d/4 = d + e/5 = e + a/6$。令此等於$k$，則五式聯立，可以方程術解之。

第三題三畜直金

問：今有牛二、羊五，直金十兩；牛三、豕三，直金九兩；羊六、豕八，直金八兩。問：牛、羊、豕各直金幾何?

答曰：牛一直金一兩二十一分兩之十三，羊一直金二十一分兩之二十，豕一直金二十一分兩之四。

術曰：術曰：方程術。置牛、羊、豕之數及直金於三行，以直除消去牛、羊，得下豕之實，實如法得豕之直。以次求羊、牛之直。

劉徽注：牛二、羊五直十兩，牛三、豕三直九兩，羊六、豕八直八兩。以方程術：置三行，先消去牛，次消去羊，得豕之實。以豕之實推羊，以羊之實推牛。此三物方程之正法也。

第四題麻麥稻菽黍

問：今有麻九斗、麥七斗、菽三斗、答二斗、黍五斗，直錢一百四十；麻七斗、麥六斗、菽四斗、答五斗、黍三斗，直錢一百二十八；麻三斗、麥五斗、菽七

斗、答六斗、黍四斗，直錢一百一十六；麻二斗、麥三斗、菽五斗、答七斗、黍六斗，直錢一百零四；麻五斗、麥四斗、菽六斗、答三斗、黍七斗，直錢一百一十二。問：五物各一斗直錢幾何?

答曰：麻一斗七錢，麥一斗四錢，菽一斗三錢，答一斗五錢，黍一斗六錢。

術曰：術曰：方程術。以五物列五行，以直錢為實，遍乘直除，消去各物之率，各得其一斗之實。

劉徽注：此五物方程，較之三物方程為繁。其術一也：列五行為程，以右行首物遍乘其下各行，以直除之，消去首物；次以中行次物遍乘其下各行，以直除之，消去次物；如是遞推，終得一物之實。

第五題正負術

問：今有上禾七秉，損實一斗，益之下禾二秉，則實滿十斗；下禾八秉，益實一斗，與上禾三秉，則實滿十斗。問：上、下禾一秉各幾何?

答曰：上禾一秉一斗五十二分斗之四十七，下禾一秉五十二分斗之三十九。

術曰：術曰：以方程術入之。損實一斗者，正也；益實一斗者，負也。正負列位，如方程消之。

劉徽注：正負術者，方程術之輔也。以兩算得失相反，要令正負以名之。正算赤，負算黑。否則以邪正為異。同名相除，異名相益。正無入負之，負無入正之。此正負之術也。

第六題四畜求直

問：今有馬一、牛二、羊三，直金二十五兩；馬二、牛三、羊一，直金二十八兩；馬三、牛一、羊二，直金二十三兩。問：馬、牛、羊各直金幾何?

答曰：馬一直金九兩，牛一直金五兩，羊一直金二兩。

術曰：術曰：方程術。列三行，馬、牛、羊之數及直金為實。以右行馬一遍乘中行、左行，以直除之，消去馬。次以中行牛遍乘左行，以直除之，消去牛。左行羊不盡者，上為法，下為實，實如法得羊之直。

劉徽注：此三物方程之又一例也。馬一、牛二、羊三直二十五兩；馬二、牛三、羊一直二十八兩；馬三、牛一、羊二直二十三兩。以方程術消之：右行馬一為最簡，故以右行遍乘中行、左行，直除消馬。

第七題粟麥互易

問：今有上禾六秉，損實一斗八升，當下禾一十秉；下禾一十五秉，損實五升，當上禾五秉。問：上、下禾一秉各幾何？

答曰：上禾一秉八升，下禾一秉三升。

術曰：術曰：方程術。列上禾、下禾之數及損實為行。以正負術記之，直除消去，求得各秉之實。

劉徽注：上禾六秉損一斗八升當下禾十秉者，六秉上禾之實減一斗八升，等於十秉下禾之實也。下禾十五秉損五升當上禾五秉者，十五秉下禾之實減五升，等於五秉上禾之實也。此二物二程之方程，較之三物為簡，然其法不異。

第八題五家共輸稅

問：今有五等之家共輸稅。甲五家、乙四家、丙三家、丁二家、戊一家，共輸稅一千斛。欲令各依等衰輸之。問：各輸幾何？

答曰：甲輸二百七十七斛七分斛之六，乙輸二百二十二斛七分斛之二，丙輸一百六十六斛七分斛之四，丁輸一百一十一斛七分斛之一，戊輸五十五斛七分斛之五。

術曰：術曰：以方程術入之。令五家為五未知，以家數及共輸為實，依衰分列方程，消去求之。

劉徽注：五等之家，各有多少之異。甲五、乙四、丙三、丁二、戊一，共十五。以千斛分為十五分，甲得其五，乙得其四，丙得其三，丁得其二，戊得其一。此術雖以方程入之，實以衰分為本也。

第九題三人持錢

問：今有甲、乙、丙三人持錢。甲得乙之半而錢滿四十八，乙得丙之半而錢滿四十八，丙得甲之半而錢滿四十八。問：甲、乙、丙各持錢幾何？

答曰：甲持錢三十六，乙持錢二十四，丙持錢十二。

術曰：術曰：以方程術入之。令甲、乙、丙持錢為三未知數，以條件列三方程，消去求之。

劉徽注：甲得乙之半而滿四十八者，甲加乙之半等於四十八也。設甲持x，

乙持y，丙持z，則：$x+y/2=48$，$y+z/2=48$，$z+x/2=48$。以方程術列三行，正負記之，消去求之。

第十題五家共井（方程術解）

問：今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。問：井深、綆長各幾何?

答曰：井深七丈二尺一寸。甲綆長二丈六尺五寸，乙綆長一丈九尺一寸，丙綆長一丈四尺八寸，丁綆長一丈二尺九寸，戊綆長七尺六寸。

術曰：術曰：方程術。列五家綆長及井深為六未知數，以五條件列五行。正負相生，直除消去，求得其比，然後以井深定其實長。

劉徽注：此題與卷七所載為同一題，而術不同。卷七以盈不足術驗之，此則以方程術正解之。此五程六物之方程，為不定問題，故特設井深以定其解。

第十一題三禾互易

問：今有上禾七秉，益實六斗，當下禾一十秉；下禾五秉，益實一斗，當上禾二秉。問：上、下禾一秉各幾何?

答曰：上禾一秉八升，下禾一秉三升。

術曰：術曰：方程術。列上禾、下禾之數及益實為行，正負記之，直除消去，求得各秉之實。

劉徽注：上禾七秉益六斗當下禾十秉者，七秉上禾之實加六斗，等於十秉下禾之實也。下禾五秉益一斗當上禾二秉者，五秉下禾之實加一斗，等於二秉上禾之實也。以方程列之，正負記之，直除消去。

第十二題四器求容

問：今有大器一、小器五，容三斛；大器五、小器一，容二斛。問：大、小器各容幾何?

答曰：大器容二十四分斛之十三，小器容二十四分斛之七。

術曰：術曰：方程術。列大器、小器之數及容為兩行，直除消去大器，得小器之實。以小器之實回代，求得大器之容。

劉徽注：大器一、小器五容三斛，大器五、小器一容二斛。以方程術列之：右行大器一、小器五、實三；左行大器五、小器一、實二。以右行遍乘左行，直除消大器。

第十三題金銀鉛錫

問：今有金一斤、銀一斤，稱之適等；金一斤、鉛一斤，稱之金輕五兩；銀一斤、錫一斤，稱之銀輕三兩。問：金、銀、鉛、錫一斤各重幾何?

答曰：金一斤重一斤，銀一斤重一斤，鉛一斤重一斤五兩，錫一斤重一斤三兩。

術曰：術曰：方程術。以金銀適等為一條件，金鉛差五兩為一條件，銀錫差三兩為一條件，列方程求之。

劉徽注：金銀適等者，一斤金與一斤銀重量相等也。此以天平稱之，非論其體積。金鉛稱之金輕五兩者，同體積之鉛重於金五兩也。三條件四物，為不定方程。以金一斤為率，則銀一斤亦為一斤，鉛一斤五兩，錫一斤三兩。

第十四題方程新術

問：今有上禾二秉，中禾三秉，下禾四秉，實皆不滿斗。上禾取中禾一秉，中禾取下禾一秉，下禾取上禾一秉，而實滿斗。問：上、中、下禾一秉各幾何?

答曰：上禾一秉二十五分斗之十一，中禾一秉二十五分斗之七，下禾一秉二十五分斗之四。

術曰：術曰：方程新術。以所取之數列為方程，各以其取數遍乘，直除消去，各得其實。

劉徽注：此方程術之變也。上禾二秉取中禾一秉而滿斗者，上禾一秉之實加中禾半秉之實為半斗也。以三條件列三行，直除消去求之。方程新術者，變通之術也。

第十五題重衰分入方程

問：今有粟五斗，為糲米、粺米、糳米三等。欲令糲二、粺三、糳四為率。問：各為米幾何？

答曰：糲米一斗四升七分升之四，粺米二斗一升七分升之六，糳米二斗八升七分升之二。

術曰：術曰：以方程術入之。令三米之率為三未知數，以粟五斗為實，以率之比列方程，消去求之。

劉徽注：糲二、粺三、糳四為率者，三米當為二比三比四之比例也。以方程列之：設糲米為$2k$，粺米為$3k$，糳米為$4k$，則$2k+3k+4k=5$斗，$9k=5$斗。此術以方程顯衰分之隱也。

第十六題四色方程

問：今有甲、乙、丙、丁四色之綾。甲三匹、乙二匹、丙一匹、丁四匹，直錢一百二十；甲二匹、乙四匹、丙三匹、丁一匹，直錢一百一十；甲一匹、乙三匹、丙四匹、丁二匹，直錢一百；甲四匹、乙一匹、丙二匹、丁三匹，直錢一百三十。問：四色綾每匹各直錢幾何？

答曰：甲一匹二十錢，乙一匹十五錢，丙一匹十錢，丁一匹五錢。

術曰：術曰：方程術。列四色為四行，以直錢為實，遍乘直除，消去三色，得第四色之實。回代求之。

劉徽注：此四物方程，其術一也。列四行於位，以右行甲三遍乘次行，直除消甲。消去甲後，次以乙消丙，再以丙消丁，終得丁之實。回代求丙、乙、甲。此四物方程雖繁，其法不異於三物也。

第十七題均輸入方程

問：今有甲、乙、丙三縣運粟。甲縣一萬斛，乙縣八千斛，丙縣六千斛。三縣共運至京師，程途遠近不同：甲縣五百里，乙縣三百里，丙縣二百里。欲令三縣均輸，問：各當運幾何？

答曰：甲縣運四千斛，乙縣運三千二百斛，丙縣運二千四百斛。

術曰：術曰：以方程術入衰均輸求之。令三縣粟數及程途列為方程，以里數

乘粟數，均之，消去求之。

劉徽注：三縣均輸者，不以粟數為率，而以粟數乘里數之積為率也。甲一萬乘五百為五百萬，乙八千乘三百為二百四十萬，丙六千乘二百為一百二十萬。此術以方程顯均輸之隱。

第十八題盈不足入方程

問：今有買金，人出四百，盈三千四百；人出三百，盈一百。問：人數、金價各幾何?

答曰：人三十三，金價一萬二千八百。

術曰：術曰：以方程術入之。兩盈者，以兩盈之數相減為法，兩設之數相減為實，實如法而一。

劉徽注：此題兩盈，非盈不足之正術也。盈不足之正術，一盈一不足。今兩盈，則以兩盈之差為法，兩設之差為實，實如法而一，得人數。

第十九題兩不足入方程

問：今有買豕，人出三百，不足四百；人出二百，不足二百。問：人數、豕價各幾何?

答曰：人二，豕價一千。

術曰：術曰：以方程術入之。兩不足者，以兩不足之數相減為法，兩設之數相減為實，實如法而一。

劉徽注：兩不足者，與兩盈相反，而其術同。人出三百不足四百，人出二百不足二百。兩不足之差二百，兩設之差一百。實如法得二人。方程之妙，在於無所不包：盈朒、兩盈、兩不足，皆可入之。

第二十題方程通術

問：今有上禾、中禾、下禾，不知其實。上禾三秉、中禾二秉、下禾一秉，實三十九；上禾二秉、中禾三秉、下禾一秉，實三十四；上禾一秉、中禾二秉、下禾三秉，實二十六。問：上、中、下禾一秉各幾何?

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

術曰：術曰：方程通術。列三行，以右行遍乘中行、左行，直除消去。終得一物之實，回代求之。

劉徽注：此題與第一題同，而以方程通術解之，所以見方程之通法也。方程通術者，凡方程皆以此術解之。此術之精妙，在於一通百通，無論二物、三物、四物、五物，皆以此術解之。劉徽曰：方程之術，以通為貴。

卷第九勾股

> 勾股章乃《九章》之巔峰，以畢達哥拉斯定理（勾股定理）及其應用為核心，論述極為深入。劉徽注尤為卓越，包括著名之「割補術」證明勾股定理，以及「出入相補原理」之論述。本章亦含距離測量、方圓關係，及中國數學史上對二次無理數之最早探討。

勾股基本定理

術曰：勾股術曰：句股各自乘，并而開方除之，即弦。又股自乘，以減弦自乘，其餘開方除之，即句。又句自乘，以減弦自乘，其餘開方除之，即股。

劉徽注：勾股之法，出於圓方。按句股者，矩之別名也。矩之為物，橫者為句，縱者為股，斜者為弦。句股各自乘，并之為弦實。開方除之，得弦。徽按：勾股各自乘，并之為弦實，此弦實之正方形，可以割補為勾實、股實兩正方形。割補術者，所以證勾股之定理也。

$$ThePythagoreanTheorem : Forlegs$$

$(gou),(gu), andhypotenuse(xian) : a^2 + b^2 = c^2[0]equation*Thesquareonthehypotenusecanbecutandrearrangedtoexactlyfillthetwosquaresonthelegs.[0]$

第一題勾股求弦

問：今有句三尺，股四尺，問：為弦幾何?

答曰：弦五尺。

術曰：術曰：句股各自乘，并而開方除之，即弦。句自乘得九，股自乘得一十六，并之得二十五，開方除之得五尺。

劉徽注：此勾股之正術也。勾三、股四、弦五，此為勾股之最基本率。勾三股四弦五之率，出於天地自然，為萬物之綱紀。以之測高深深遠，無所不可。

第二題弦股求句

問：今有弦五尺，股四尺，問：為句幾何?

答曰：句三尺。

術曰：術曰：股自乘，以減弦自乘，其餘開方除之，即句。股自乘得一十六，弦自乘得二十五，以十六減二十五餘九，開方除之得三尺。

劉徽注：弦實二十五，股實一十六，兩實之差為句實九。開方得三，即句。此術為勾股術之逆也。勾股之術，互為正逆，相為表裡。

第三題句弦求股

問：今有句八尺，弦十七尺，問：為股幾何?

答曰：股十五尺。

術曰：術曰：句自乘，以減弦自乘，其餘開方除之，即股。句自乘得六十四，弦自乘得二百八十九，以六十四減二百八十九餘二百二十五，開方除之得一十五尺。

劉徽注：弦實二百八十九，句實六十四，兩實之差為股實二百二十五。開方得一十五，即股。勾八、股十五、弦十七，此又一股弦之率也。勾股之率無窮，而其術一也。

第四題戶高廣

問：今有戶高多於廣六尺八寸，兩隅相去適一丈。問：戶高、廣各幾何?

答曰：廣二尺八寸，高九尺六寸。

術曰：術曰：令一丈自乘，半之得五十尺為實。令多六尺八寸自乘，得四十六尺二寸四分，半之得二十三尺一寸二分為法。實如法得一尺，為戶廣。加多六尺八寸，為高。

劉徽注：戶高多於廣六尺八寸者，高與廣之差也。兩隅相去一丈者，以高廣為勾股之弦也。此術以勾股弦之關係求高廣也。此術之妙，在於以差求廣，以廣求高，一環相扣。

第五題邑方出東門

問：今有邑方不知大小，各開中門。出東門二十步有木。問：出南門幾何步而見木？

答曰：出南門一十四步半而見木。

術曰：術曰：以勾股相似術入之。出東門二十步為勾，邑方之半為股。以股自乘，以勾除之，得南門外見木之步數。

劉徽注：邑方者，正方形之城邑也。此以勾股相似之術求之：邑方之半為一大股，東門外二十步為一大勾；南門外所求步數為一小股。兩勾股相似，故以比例求之。此術以相似勾股測遠近，為重差之基礎也。

第六題井徑求立木

問：今有井徑五尺，不知其深。立木於井上，從木末望水岸，入徑四尺。問：井深幾何？

答曰：井深一丈九尺二寸。

術曰：術曰：以勾股相似術入之。井徑五尺為勾，入徑四尺為股之差。以井徑乘入徑，以井徑減入徑之差除之，得井深。

劉徽注：井徑五尺者，井口之直徑也。此亦以勾股相似之術：木高為一大股，入徑為一大勾；井深為一小股，井徑之半為一小勾。兩勾股相似，故可以比例求之。此術以勾股相似測深淺，亦重差之術也。

第七題登山望邑

問：今有登山望邑，邑在山東。山高不知高，立兩表齊高三丈，前後相去千步。令後表與前表參相直。從前表卻行一百二十三步，人目著地，取望邑西北隅，入表前三寸。從後表卻行一百二十七步，人目著地，取望邑西北隅，適與表端參合。問：邑方及山去表各幾何？

答曰：邑方一里二百二十步，山去表一里一百八十二步半。

術曰：術曰：以重差術入之。此為重差之正術，以兩表之高及相去之遠，兩卻行之數，入表之數，列為勾股，相似求之。

劉徽注：此重差術之典型題也。重差者，以兩表之重迭差數求遠物之高深廣遠也。此術之妙，在於以咫尺之矩，測千里之遠。故劉徽曰：以咫尺之矩，測千里之高深。重差術者，勾股之極致也。

第八題勾股容方

問：今有句五步，股十二步，問：句中容方幾何?

答曰：方三步十七分步之九。

術曰：術曰：并句、股為法，句股相乘為實，實如法而一，得方。句五步、股十二步，并之得十七步為法。句股相乘得六十步為實。實如法得三步十七分步之九。

劉徽注：勾股容方者，於直角三角形中作一最大之內接正方形也。以勾股之相并為法，勾股之相乘為實者，以相似勾股之理推之也。此術精妙，以相似勾股之理，求內接正方之邊。

第九題勾股容圓

問：今有句八步，股十五步，問：句中容圓，徑幾何?

答曰：圓徑六步。

術曰：術曰：句股相乘，倍之為實。并句、股、弦為法。實如法而一，得圓徑。句八步、股十五步，句股相乘得一百二十，倍之得二百四十為實。句八、股十五、弦十七，并之得四十為法。實如法得六步。

劉徽注：勾股容圓者，於直角三角形中作一最大之內切圓也。以勾股相乘倍之為實者，兩倍三角形之面積也。以勾股弦之和為法者，三角形周長之半也。三角之面積等於內切圓半徑乘周長之半，故以面積之兩倍除以周長之半，得圓徑。此術之精妙，以面積之關係推圓徑也。

第十題邑中望木

問：今有邑方不知大小，各開中門。出北門二十步有木。出南門十四步，折而西行一千七百七十五步見木。問：邑方幾何?

答曰：邑方五十步。

術曰：術曰：以勾股術入之。出北門二十步與出南門十四步相加得三十四步，以邑方之半乘之，以西行一千七百七十五步除之，得邑方之半。倍之得邑方。

劉徽注：邑方者，正方形之城邑也。出北門二十步有木者，從北門中出北行

二十步有樹也。此以勾股相似之術求之：邑方之半為一勾，南北出門步數之和為一股，西行步數為另一勾。兩勾股相似，故可以比例求邑方。

第十一題望清淵

問：今有清淵，不知深淺。立一木於岸上，高三丈。從木末斜望水際，入木一尺二寸。又望水底，入木四尺八寸。問：淵深幾何?

答曰：淵深一丈二尺。

術曰：術曰：以勾股相似術入之。木高三丈為股，入木一尺二寸為勾。入木四尺八寸為另一勾之率。以比例求之，得淵深。

劉徽注：立木高三丈於岸上，從木末斜望水際入木一尺二寸者，視線從木頂到水面與木杆相交處距木頂一尺二寸也。此術以勾股相似測深淺，亦重差之應用也。

第十二題窺望木

問：今有木去人不知遠近。立一表長一丈，人退行二丈六尺，從表末望木，與表端參合。又退行二丈，從表末望木，入表一尺。問：木去表幾何?

答曰：木去表三十四丈。

術曰：術曰：以重差術入之。兩退行之差為法，表長乘後退行為實，實如法而一，得木去表之遠。

劉徽注：此又以重差術測遠也。重差術之妙，在於以兩次之差，測無遠弗屆之物。一表一測，不足以定遠近；兩表兩測，則遠近立判。此差之所以為重也。

第十三題因木望山

問：今有木不知高遠。立兩表，各高一丈，前後相去五百步。從前表卻行一百二十步，人目著地，望山峯與表端參合。從後表卻行一百八十步，人目著地，望山峯與表端參合。問：山去前表幾何? 山高幾何?

答曰：山去前表一里二百步，高八里一百二十步。

術曰：術曰：以重差術入之。兩表相去為勾股之率，兩卻行之差為法。表高

乘表間為實，實如法得山高。前卻行乘表間為實，實如法得山去表之遠。

劉徽注：此重差術測山高之典型題也。立兩表齊高一丈，相去五百步。兩卻行之差六十步為法。表高一丈乘表間五百步為實，實如法得山高。此術之精，在於以兩表之矩，測群山之高遠。故劉徽曰：以咫尺之矩，測千里之高深。

第十四題環田求徑

問：今有環田，中周六十二步，外周四步，徑十二步。問：為田幾何?

答曰：二畝五十五步。

術曰：術曰：并中外周而半之，以徑乘之為積。

劉徽注：環田者，猶圓田之中去一圓也。并中外周而半之者，取環之平均周長也。以徑乘之者，以平均周長為廣，徑為縱也。此術以環田近似為矩形，故以廣縱相乘得積。

第十五題圓材求徑

問：今有圓材，埋在地下，不知大小。以鋸鋸之，深一寸，鋸道長一尺。問：圓材徑幾何?

答曰：圓材徑二尺六寸。

術曰：術曰：以勾股術入之。鋸道長一尺者，弦也；深一寸者，矢也。半鋸道自乘，除以深，加深，即圓徑。半鋸道五寸自乘得二十五寸，除以深一寸得二十五寸，加深一寸得二十六寸，即圓材徑。

劉徽注：圓材埋在地下，以鋸鋸之者，鋸截圓材得一弦也。鋸道長一尺即弦長，深一寸即弦到圓周之最短距離（矢）。以勾股術求圓徑：半弦為勾，圓徑減矢之半為股。半弦自乘除以矢，得圓徑之半減矢。加以矢，得圓徑。

第十六題開方除之

問：今有積五萬五千二百二十五步。問：為方幾何?

答曰：二百三十五步。

術曰：術曰：開方術。置積為實，借一算步之，超一等。議所得，以一乘所

借一算為法，而以除。除已，倍法為定法。

劉徽注：開方術者，求正方形之一邊也。積五萬五千二百二十五步，開方得二百三十五步。徽注：開方不盡者，當以微數續之。微數者，以十進小數無限逼近也。此為中國數學之重大發明，以極限之思想，求開方之精密值。

第十七題開圓術

問：今有積三百六十步。問：為圓周幾何?

答曰：圓周六十步。

術曰：術曰：開圓術。置積為實，以十二乘之，開方除之，得圓周。

劉徽注：開圓術者，以圓積求圓周也。以十二乘積開方者，古術以圓周率為三，圓面積為$C^2/12$，故$C = \sqrt{12S}$。然此以周三徑一為率，其實圓周率非盡於三也。徽以為圓周率當大於三，以割圓術求之：割之彌細，所失彌少；割之又割，以至於不可割，則與圓周合體而無所失矣。以一百九十二邊形之面積推之，得圓周率約為，此為當時世界之最精密者。

第十八題邪田求積

問：今有邪田，一頭廣三十步，一頭廣四十二步，正從六十四步。問：為田幾何?

答曰：九畝一百四十四步。

術曰：術曰：并兩廣而半之，以從乘之為積。三十步與四十二步并之得七十二步，半之得三十六步。以從六十四步乘之，得二千三百四步為積。

劉徽注：邪田者，梯形之田也。并兩廣而半之者，取其平均廣也。以平均廣乘從，得積。此術之證，以割補之法：割邪田之兩角，補為矩形，故以平均廣乘徑得積。

第十九題弧田徑術

問：今有弧田，弦三十步，矢十五步。問：為田幾何?

答曰：一畝九十七步半。

術曰：術曰：以弦乘矢，矢又自乘，并之，三而一。

劉徽注：弧田術者，以弦矢求弧田之面積也。弦乘矢者，以弦為廣，矢為縱。并之者，合兩積為一。三而一者，以弧田之積約當此兩積和之三分之一也。此術為弧田之近似術。若以割圓術密求之，當以無數小三角形割之，割之彌細，所失彌少。

第二十題五家共堤

問：今有五家共堤。甲、乙、丙、丁、戊五家，各以所出徒之數築堤。甲出徒二百五十四人，乙出徒二百四十三人，丙出徒二百三十二人，丁出徒二百二十一人，戊出徒二百一十人。堤長一萬二千七百步。問：各築幾何?

答曰：各以人數為率分配之。

術曰：術曰：以衰分術入之。并五人數為法，堤長為實，實如法得一步之率。各以其人數乘之，得各家所築之步數。

劉徽注：五家共堤者，五家合出徒役共築一堤也。各以人數為率者，按出徒之人數比例分配築堤之長也。此題本屬衰分，而以勾股之比例理推之，同出一理。

第二十一題勾股求容方邊

問：今有勾股形，勾六尺，股八尺。問：弦上容方，方邊幾何?

答曰：方邊三尺七分尺之五。

術曰：術曰：以勾股術入之。并勾股為法，勾股相乘為實，實如法得方邊。勾六、股八，并得十四為法。勾股相乘得四十八為實。實如法得三尺七分尺之五。

劉徽注：此術與第八題同意，而數不同。勾股弦上容方者，方之一邊在弦上，兩頂點在勾股之邊也。勾六股八弦十，容方之邊為。此術之妙，在於以相似勾股之理，求內接正方之邊，一以貫之。

第二十二題勾股測遠

問：今有海島，不知高遠。立兩表，各高三丈，前後相去五百步。從前表卻行一百二十步，人目著地，望海島峯與表端參合。從後表卻行二百二十步，人目著地，望海島峯與表端參合。問：島去前表幾何？島高幾何？

答曰：島去前表一里一百步，高四里五十五步。

術曰：術曰：以重差術入之。兩表相去為法，表高乘表間為實，兩卻行之差除實得島高。前卻行乘表間為實，兩卻行之差除實得島去前表之遠。

劉徽注：此重差術測海島之題也，為《海島算經》第一題之濫觴。立兩表高三丈，相去五百步。兩卻行之差一百步為法。表高三丈乘表間五百步得一千五百丈為實，實如法得島高十五丈。此術之精妙，在於以兩表之矩，測海島之高遠。

第二十三題勾股求日徑

問：今有日去地八萬里，日徑一千二百五十里。以勾股術求之，問：以何為勾股弦？

答曰：以日去地為股，以日徑之半為勾，以斜至日邊為弦。

術曰：術曰：以勾股術入之。日去地八萬里為大股，日徑之半六百二十五里為大勾。以股自乘，以勾自乘，并之開方，得斜至日邊之弦。

劉徽注：此術以勾股之理測天體也。日去地八萬里，此為股。日徑一千二百五十里，其半六百二十五里，此為勾。此術雖以勾股入之，實測天之法也。古代以勾股測天地之高深，此其遺意。

第二十四題勾股綜合測量

問：今有深谷，不知深淺。立一木於岸傍，高四丈。從木末斜望谷底，入木二尺四寸。又望谷岸，入木八寸。問：谷深幾何？谷岸廣幾何？

答曰：谷深八丈，谷岸廣二丈四尺。

術曰：術曰：以勾股相似術入之。木高為大股，入木之數為大勾之對應。以兩入木之差為法，木高乘之，得谷深。

劉徽注：深谷者，幽邃難測者也。以勾股相似術測之，則深淺立見。此術以勾股之相似，測深谷之幽邃。凡測高深廣遠，皆可以勾股重差術入之，無所不可。此二十四題，所以盡勾股之變也。

後記

本書所呈現之三卷——盈不足、方程、勾股——代表了《九章算術》中古代中國數學思想之巔峰。

劉徽之注（約西元年）將《九章》從一部解題配方之集，提升為具有真正數學推理之著作。其對盈不足術之闡釋——作為一般性近似方法，對線性方程組之方程術處理（本質即高斯消元法），以及對畢達哥拉斯定理之深入研究——包括著名之「割補術」與「重差術》——皆證明三世紀中國數學之高度成就。

π

方程章尤值一提，因其包含世界數學史上負數之最早系統運用——劉徽對消元過程中如何處理「正」與「負」量之解釋，其清晰程度令人驚嘆。

勾股章則以劉徽對「割圓術」之注釋為代表，體現了對圓周率π最早之嚴謹逼近方法之一，通過正一百九十二邊形得出約為3.1416之精確數值。

本擴充版包含三卷各二十四題，所有問題、答案、術文及劉徽注均以中文原文呈現，並附英文譯文，保存這部不朽數學經典之原有結構。

測圓海鏡

Sea Mirror of Circle Measurements

卷一—卷三 | Volumes I –III

Bilingual Edition —雙語版

Original Chinese • English Translation

元　李冶　撰

(Li Ye / Li Zhi, 1192–1279)

以天元術解勾股容圓一百七十問

Using the Method of the Celestial Element (Tian Yuan Shu)
to Solve 170 Problems on the Inscribed Circle

A foundational text of medieval Chinese algebra,
completed in the year 1248 of the Common Era.

Typeset in LaTeX with XeLaTeX + xeCJK
Bilingual edition with parallel Chinese-English columns

Contents

1 Introduction / 引言

測圓海鏡簡介

《測圓海鏡》乃李冶（李冶，1192–1279）之傑作。李冶為宋元四大數學家之一，與楊輝、秦九韶、朱世傑齊名。此書成於 1248 年，系統運用**天元術**（多項式代數方法）求解圓與直角三角形相關之一百七十問。
全書結構如下：

- **卷一**：理論基礎——圓城圖式、總率名號、今問正數、識別雜記（含六百九十二條幾何公式）
- **卷二至卷十二**：一百七十問，以天元術求解，方程次數自一次至六次。

本文檔包含**卷一至卷三**之全文，即全部基礎理論及首批問題。

Introduction

The *Ceyuan Haijing* (測圓海鏡, literally “Sea Mirror of Circle Measurements”) is the masterpiece of Li Ye (李冶, also known as Li Zhi, 1192–1279), one of the four great mathematicians of the Song–Yuan period of China, alongside Yang Hui, Qin Jiushao, and Zhu Shijie.
Completed in 1248, this work systematically applies the **tian yuan shu** (天元術, “method of the celestial element”)—an algebraic method for setting up and solving polynomial equations—to a total of 170 geometric problems concerning a circle inscribed in or related to a right triangle (勾股容圓).
The book is organized as follows:

- **Volume 1** establishes the theoretical foundation: the *Circle City Diagram* (圓城圖式), the *General Rate Names* (總率名號), the *Current Question Numbers* (今問正數), and the *Identification Miscellaneous Records* (識別雜記) containing 692 geometric formulas.
- **Volumes 2–12** present 170 problems, solved by the tian yuan shu method. The problems involve equations of degrees ranging from 1 to 6.

The present document contains the complete text of **Volumes 1–3**, comprising all foundational material and the first suite of problems.

2 Original Preface / 原序

原序

數本難窮，吾欲以力強窮之，彼其數不惟不能得其凡，而吾之力且憊矣。然則數果不可以窮耶? 既已名之數矣，則又何為而不可窮也。

故謂數為難窮斯可，謂數為不可窮斯不可。何則? 彼其冥冥之中，固有昭昭者存，夫昭昭者其自然之數也。非自然之數，其自然之理也。數一出於自然，吾欲以力強窮之，使隸首復生亦末如之何也已。

苟能推自然之理，以明自然之數，則雖遠而乾端坤倪，幽而神情鬼狀，未有不合者矣。

予自幼喜算數，恒病夫考圓之術例出於牽強，殊乖於自然，如古率、徽率、密率之不同，截弧、截矢、截背之互見，內外諸角，析剖支條，莫不各自名家，與世作法。及反覆研究，卒無以當吾心焉。

老大以來，得洞淵九容之說，日夕玩繹，而向之病我者，使爆然落去而無遺余。山中多暇，客有從余求其說者，於是乎又為衍之，遂累一百七十問。

既成編，客復目之《測圓海鏡》，蓋取夫天臨海鏡之義也。

Original Preface

Numbers are fundamentally difficult to exhaust; if I try to force them by effort, not only will I fail to apprehend their essence, but my strength will be depleted as well. Yet can numbers truly not be exhausted? Since they are already called *numbers*, why then could they not be exhausted?

It is acceptable to say that numbers are difficult to exhaust; but it is unacceptable to say they cannot be exhausted. Why? In the midst of darkness, there is indeed brightness. That brightness is the natural number. It is not merely the natural number—it is the natural principle. Since numbers arise from nature, if I try to force them by effort, even if Li Shou (the legendary inventor of arithmetic) were reborn, he could do no better.

If one can extend natural principles to illuminate natural numbers, then even things as distant as the corners of heaven and earth, or as mysterious as spirits and ghosts, will all be in agreement.

Since youth I have delighted in arithmetic. I was always troubled that the methods for investigating circles seemed forced and artificial, contrary to nature—the ancient rate, Hui's rate, and the close rate all differing; the arc, sagitta, and back each being considered separately; the internal and external angles analyzed into minute subdivisions—each school establishing its own name and method for the world. After repeated study, none could satisfy my mind.

In old age, I obtained the *Dongyuan Nine Containments* (洞淵九容) theory, and pondered it day and night. What had previously troubled me suddenly fell away completely, leaving no trace. With leisure time in the mountains, when guests asked me to explain this theory, I expanded upon it, eventually accumulating **one hundred and seventy problems**.

When the compilation was complete, a guest named it *Ceyuan Haijing*—taking the meaning of "heaven overlooking the sea mirror."

昔半山老人集《唐百家詩選》，自謂廢日力於此，良可惜。明道先生以上蔡謝君記誦為玩物喪志。夫文史尚矣，猶之為不足貴，況九九賤技能乎！

嗜好酸鹹，平生每痛自戒勅，竟莫能已，類有物憑之者，吾亦不知其然而然也。

故嘗私為之解曰：由技兼於事者言之，夷之禮，夔之樂，亦不免為一技。由技進乎道者言之，石之斤，扁之輪，非聖人之所與乎。

覽吾之編，察吾苦心，其憫我者當百數，其笑我者當千數。乃若吾之所自得，則自得焉耳，寧復為人憫笑計哉。

李冶序

In the past, the Old Man of Banshan compiled the *Anthology of One Hundred Tang Poets*, saying that to waste his days on such a thing was truly a pity. Master Mingdao criticized Xie of Shangcai for memorizing texts as “playing with things and losing one’s ambition.” Literature and history are esteemed subjects, yet even they are considered not worth valuing—how much less the lowly skill of multiplication!

My fondness for such sour and salty pursuits—throughout my life I have sternly admonished myself against them, yet in the end I could not stop, as if some force were driving me. I myself do not know why it is so.

So I have privately explained it thus: speaking from the perspective of skills combined with affairs, even the rites of Yi and the music of Kui were not exempt from being a single skill. Speaking from the perspective of advancing from skill to the Way, was not the stone-axe of [the sage] and the wheel of Bian [the wheelwright] praised by the sages?

Those who read my compilation and perceive my earnest efforts—those who pity me will number in the hundreds, those who laugh at me will number in the thousands. As for what I myself have gained, I have simply gained it; how could I calculate for the sake of being pitied or laughed at?

— Preface by Li Ye

3 Volume 1 / 卷一

卷一為全書之理論基礎，分為四大部分：

1. **圓城圖式**——主幾何圖形
2. **總率名號**——十五個直角三角形及其元素之定義
3. **今問正數**——各線段之數值
4. **識別雜記**——六百九十二條幾何公式

Volume 1 is the theoretical foundation of the entire work. It comprises four major sections:

1. **Circle City Diagram** (圓城圖式) — the master geometric configuration
2. **General Rate Names** (總率名號) — definitions of all 15 right triangles and their elements
3. **Current Question Numbers** (今問正數) — numerical values for all line segments
4. **Identification Miscellaneous Records** (識別雜記) — 692 geometric formulas

3.1 Circle City Diagram / 圓城圖式

圓城圖式為一百七十問所基之主幾何圖形。其構為一大直角三角形（勾股形），內切一圓，並設特定之點與線。

其情境如下：

> 假令有圓城一所，不知周徑，四面開門，門外縱橫各有十字大道。其西北十字道頭定為乾地，其東北十字道頭定為艮地，其東南十字道頭定為巽地，其西南十字道頭定為坤地。所有測望雜法一一設問如後。

The *Circle City Diagram* is the master geometric configuration upon which all 170 problems are based. It consists of a large right triangle (called 勾股形, “leg–leg shape”) with its inscribed circle, together with specific points and lines.

The scenario Li Ye constructs is:

> Suppose there is a circular city whose circumference and diameter are unknown. It has gates on all four sides, and outside each gate there are crossroads running north–south and east–west. The northwest crossroad is designated **Qian** (乾), the northeast **Gen** (艮), the southeast **Xun** (巽), and the southwest **Kun** (坤). All surveying methods are set forth as questions below.

二十二名點 / **The 22 Named Points**

最大直角三角形之頂點為**天**、**地**、**乾**。內切圓之心稱**心**。各點以單字命名：

天	地	乾	心	日	南	北	川
東	西	艮	坤	山	月	巽	青
泉	朱	金	泛	旦	夕		

The largest right triangle has vertices **Tian** (天, Heaven), **Di** (地, Earth), and **Qian** (乾). The center of the inscribed circle is called **Xin** (心, Heart). Key points include intersections of horizontal and vertical lines through the center and other constructed lines. Each point is designated by a single Chinese character:

Tian	Di	Qian	Xin	Ri	Nan	Bei	Chuan
Dong	Xi	Gen	Kun	Shan	Yue	Xun	Qing
Quan	Zhu	Jin	Fan	Dan	Xi		

3.2 General Rate Names / 總率名號

全書研究**十五個直角三角形**，皆以天地線之一段為弦（**弦**，即斜邊）。總率名號定義各三角形之三邊。
每個三角形：

- **弦**（c）——斜邊
- **股**（b）——較長之直邊
- **勾**（a）——較短之直邊

The work studies **15 right triangles**, all having a segment of the *Tiandi* (天地) line as their hypotenuse (弦, “string”). The *General Rate Names* define these triangles and their three sides.
In each triangle:

- **弦** (c) — the hypotenuse (斜邊)
- **股** (b) — the longer vertical leg
- **勾** (a) — the shorter horizontal leg

十五三角形表 / Table of 15 Triangles

No.	名稱 / Name	頂點 / Vertices	弦	股	勾
1	通 (Tong/General)	天–地–乾	通弦	通股	通勾
2	邊 (Bian/Side)	天–西–川	邊弦	邊股	邊勾
3	底 (Di/Base)	日–地–北	底弦	底股	底勾
4	黃廣 (Huangguang)	天–山–金	黃廣弦	黃廣股	黃廣勾
5	黃長 (Huangchang)	月–地–泉	黃長弦	黃長股	黃長勾
6	上高 (Shanggao)	天–日–旦	上高弦	上高股	上高勾
7	下高 (Xiagao)	日–山–朱	下高弦	下高股	下高勾
8	上平 (Shangping)	月–川–青	上平弦	上平股	上平勾
9	下平 (Xiaping)	川–地–夕	下平弦	下平股	下平勾
10	大差 (Dacha)	天–月–坤	大差弦	大差股	大差勾
11	小差 (Xiaocha)	山–地–艮	小差弦	小差股	小差勾
12	皇極 (Huangji)	日–川–心	皇極弦	皇極股	皇極勾
13	太虛 (Taixu)	月–山–泛	太虛弦	太虛股	太虛勾
14	明 (Ming)	日–月–南	明弦	明股	明勾
15	叀 (Zhuan)	山–川–東	叀弦	叀股	叀勾

注：上高 = 下高，上平 = 下平，故十五三角形中實僅十三個互異。

Note: 上高 = 下高 and 上平 = 下平, so among 15 triangles, only 13 are distinct.

3.3 Current Question Numbers / 今問正數

此節給出圖中每條線段之數值。內切圓之**半徑定為一百二十步**，作為標準單位。

This section gives the numerical value of every line segment in the diagram. The **radius of the inscribed circle is taken as 120 steps** (步), which serves as the standard unit.

1. 通（Tong / 通用）

弦六百八十	勾三百二十	股六百
勾股和九百二十	較二百八十	
勾和一千	較三百六十	
股和一千二百八十	較八十	
較和九百六十	較四百	
和和一千六百	較二百四十	

1. Tong (General)

Hypotenuse 680	Short leg 320, Long leg 600
Sum of legs 920	Difference 280
Leg-hypotenuse sum 1000	Difference 360
Leg-hypotenuse sum 1280	Difference 80
Difference-sum 960	Difference 400
Sum-sum 1600	Difference 240

That is: hypotenuse $c = 680$, short leg $a = 320$, long leg $b = 600$.

2. 邊（Bian / Side）

弦五百四十四	勾二百五十六	股四百八十
勾股和七百三十六	較二百二十四	
勾和八百	較二百八十八	
股和一千零二十四	較六十四	
較和七百六十八	較三百二十	
和和一千二百八十	較一百九十二	

2. Bian (Side)

Hypotenuse 544	Short leg 256, Long leg 480
Sum of legs 736	Difference 224
Leg-hypotenuse sum 800	Difference 288
Leg-hypotenuse sum 1024	Difference 64
Difference-sum 768	Difference 320
Sum-sum 1280	Difference 192

3. 底（Di / Base）

弦四百二十五	勾二百	股三百七十五
勾股和五百七十五	較一百七十五	
勾和六百二十五	較二百二十五	
股和八百	較五十	
較和六百	較二百五十	
和和一千	較一百五十	

3. Di (Base)

Hypotenuse 425	Short leg 200, Long leg 375
Sum of legs 575	Difference 175
Leg-hypotenuse sum 625	Difference 225
Leg-hypotenuse sum 800	Difference 50
Difference-sum 600	Difference 250
Sum-sum 1000	Difference 150

4. **黃廣（Huang Guang）**

弦五百一十　勾二百四十　股四百五十
（即城徑也）
勾股和六百九十　較二百一十

其餘十一三角形依同樣模式，每個十三個數量（弦、勾、股、勾股和、勾股較、勾弦和、勾弦較、股弦和、股弦較、弦較和、弦較較、弦和和、弦和較），共 $15 \times 13 = 195$ 個數值條目。

4. **Huang Guang**

Hypotenuse 510　Short leg 240, Long leg 450
(this is the city diameter)
Sum of legs 690　Difference 210

The remaining 11 triangles follow the same pattern, each with 13 numerical quantities (弦, 勾, 股, 勾股和, 勾股較, 勾弦和, 勾弦較, 股弦和, 股弦較, 弦較和, 弦較較, 弦和和, 弦和較), giving a total of $15 \times 13 = 195$ numerical entries.

3.4 Identification Miscellaneous Records / 識別雜記

此為全書之理論核心，含**六百九十二條幾何公式**，關聯圖中各線段。此等公式純為幾何定理，與具體數值無關。後續各卷之問題皆以此等公式配合天元術求解。
本分為八部分：

1. **諸雜名目**——一般定義及三十餘條定理
2. **五和五較**
3. **諸弦**
4. **大小差**
5. **諸差**
6. **諸率互見**
7. **四位拾遺**
8. **拾遺**

This is the theoretical heart of the work. It contains **692 geometric formulas** relating the various line segments in the diagram. These formulas are purely geometric theorems, independent of any particular numerical values. The problems in all subsequent volumes are solved by using these formulas together with the tian yuan shu.
The section is divided into eight parts:

1. **Various Names** (諸雜名目) – general definitions and 30+ theorems
2. **Five Sums and Five Differences** (五和五較)
3. **Various Hypotenuses** (諸弦)
4. **Large and Small Differences** (大小差)
5. **Various Differences** (諸差)
6. **Mutual Relations of Rates** (諸率互見)
7. **Four-Place Supplement** (四位拾遺)
8. **Supplement** (拾遺)

諸雜名目示例：

Sample Definitions from Various Names:

名稱 / Name	**定義** / Definition
內率	明勾 × 叀股 / Ming 勾 × Zhuan 股
外率	明股 × 叀勾 / Ming 股 × Zhuan 勾
虛率	虛勾 × 虛股 / Xu 勾 × Xu 股
角差	高股 − 平勾 / Gao 股 − Ping 勾
次差	(明股 − 明勾) + (叀股 − 叀勾)
混同和	黃長股 + 黃廣勾 / Huangchang 股 + Huangguang 勾
傍差	(明股 − 明勾) − (叀股 − 叀勾)

定理示例：

Sample Theorems:

- 凡大差小差相乘為半段徑冪。
- 大差勾小差股相乘同上。
- 黃廣股黃長勾相乘為經冪。
- 勾股和即弦黃和。

- The product of the large difference and small difference equals half the square of the diameter.
- The product of the large-difference short-leg and the small-difference long-leg equals the same.
- The product of the Huangguang long-leg and Huangchang short-leg equals the square of the diameter.
- The sum of the two legs equals the sum of the hypotenuse and the diameter of the inscribed circle: $a + b = c + d$, where d is the diameter of the inscribed circle.

4 Volume 2 / 卷二

正率一十四問

此十四問構成「洞淵九容」序列，源自道教大師李思聰（號洞淵）之著作。其呈現直角三角形中九種基本容圓類型，另加五題。

Fourteen Questions on Standard Rates

These fourteen problems form the “Dongyuan Nine Containments” (洞淵九容) sequence, derived from the earlier work of the Daoist master Li Sicong (李思聰, known as “Master Dongyuan”). They present the nine fundamental types of circles contained in or related to a right triangle, plus five additional problems.

4.0.1 第一問 / Problem 1

問 甲乙二人俱在乾地，乙東行三百二十步而立，甲南行六百步望見乙，問徑幾里?

答 城徑二百四十步。

術 此為勾股容圓也。以勾股相乘，倍之為實，並勾股冪以求弦，復加入勾股共，以為法。

草 置甲南行六百步在地，以乙東行三百二十步乘之，得一十九萬二千步，倍之得三十入萬四千步為實。以乙東行步自之，得一十萬零二千四百步為勾冪；以甲南行步自之，得三十六萬步為股冪。二冪相並，得四十六萬二千四百步為方實，以平方開之，得六百八十步，則弦也。以弦加勾股共，共得一千六百步以為法。如法而一，得二百四十步，則城徑也。合問。

Problem. Two people, A and B, both stand at Qian. B walks 320 steps east and stops. A walks 600 steps south and sees B. What is the diameter?

Answer. The city diameter is 240 steps.

Method. This is the “circle contained in the right triangle” case. Multiply the two legs and double it to obtain the dividend. Combine the squares of the legs to find the hypotenuse, then add the sum of the legs to form the divisor.

Working. Place A’s 600 steps south. Multiply by B’s 320 steps east, obtaining 192,000 steps; double to get 384,000 as the dividend. Square B’s eastward steps to get 102,400 as the short-leg square; square A’s southward steps to get 360,000 as the long-leg square. Add the two squares: 462,400. Extract the square root to get 680, which is the hypotenuse. Add the sum of the legs to the hypotenuse: 1,600 as the divisor. Divide: 240 steps, the city diameter. This agrees with the question.

4.0.2 第二問 / Problem 2

問 甲乙二人俱在西門，乙東行二百五十六步，甲南行四百八十步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾上容圓也。以勾股相乘，倍之為實，並勾股冪以求弦，加入股以為法。

草 置甲南行四百八十步在地，以乙東行二百五十六步乘之，得一十二萬二千八百八十步，倍之得二十四萬五千七百六十步為實。以乙東行步自之，得六萬五千五百三十六步為勾冪；以甲南行步自之，得二十三萬零四百步為股冪。勾股冪相並，得二十九萬五千九百三十六步為方實，以平方開之，得五百四十四步為弦也。以弦加入南行步，共得一千零二十四步以為法。而一，得二百四十步則城徑。合問。

Problem. Two people both stand at the West Gate. B walks 256 steps east, A walks 480 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle on the short leg” case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse; add the long leg as divisor.

Working. Place A’s 480 steps south. Multiply by B’s 256 steps east, obtaining 122,880; double to get 245,760 as dividend. Square B’s eastward steps: 65,536 as the short-leg square. Square A’s southward steps: 230,400 as the long-leg square. Combine: 295,936. Extract square root: 544, the hypotenuse. Add the southward steps: 1,024 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.3 第三問 / Problem 3

問 甲乙二人俱在北門，乙東行二百步而止，甲南行三百七十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為股上容圓也。以勾股相乘，倍之為實，以勾股冪求弦，加入勾以為法。

草 置甲南行三百七十五步，以乙東行二百步乘之，得七萬五千步，倍之得一十五萬步為實。以乙東行自之，得四萬步為勾冪；以甲南行自之，得一十四萬零六百二十五步為股冪。勾股冪相並，得一十八萬零六百二十五步為方實。如平方而一，得四百二十五步則弦也。加入乙東行二百步，共得六百二十五步以為法。以法除之，得二百四十步，則城徑也。合問。

Problem. Two people both stand at the North Gate. B walks 200 steps east and stops. A walks 375 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on the long leg" case. Multiply the two legs and double as dividend. Use the squares of the legs to find the hypotenuse; add the short leg as divisor.

Working. Place A's 375 steps south. Multiply by B's 200 steps east: 75,000; double to get 150,000 as dividend. Square B's eastward steps: 40,000 as the short-leg square. Square A's southward steps: 140,625 as the long-leg square. Combine: 180,625. Extract square root: 425, the hypotenuse. Add B's 200 eastward steps: 625 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.4 第四問 / Problem 4

問 甲乙二人俱在圓城中心而立，乙穿城向東行一百三十六步而止，甲穿城南行二百五十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾股上容圓也。以勾股相乘，倍之為實，並勾股冪如法求弦，以為法。

草 以二行步相乘，得三萬四千六百八十步，倍之得六萬九千三百六十步為實。置乙東行自之，得一萬八千四百九十六步為勾冪；又以甲南行自之，得六萬五千零二十五步為股冪。二冪相並，得八萬三千五百二十一步為方實，以平方開之，得二百八十九步即弦也。便以為法。如法除實，得二百四十步，即圓城之徑也。合問。

Problem. Two people both stand at the center of the circular city. B goes through the city, walking 136 steps east and stops. A goes through the city, walking 255 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on both legs" case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse as divisor.

Working. Multiply the two walking distances: 34,680; double to get 69,360 as dividend. Square B's eastward steps: 18,496 as the short-leg square. Square A's southward steps: 65,025 as the long-leg square. Combine: 83,521. Extract square root: 289, the hypotenuse. This serves as the divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.5 第五問 / Problem 5

問 甲乙二人同立於乾地，乙東行一百八十步遇塔而止，甲南行三百六十步，回望其塔正居城徑之半，問答同前。

答 城徑二百四十步。

術 此為弦上容圓也。以勾股相乘，倍之為實，以勾股和為法。

草 以二行步相乘，得六萬四千八百步，倍之得一十二萬九千六百步為實。並二行步，得五百四十步以為法。除實，得二百四十步，即城徑也。合問。

Problem. Two people stand together at Qian. B walks 180 steps east and reaches a pagoda. A walks 360 steps south and looks back: the pagoda lies exactly at half the city diameter. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on the hypotenuse" case. Multiply the two legs and double as dividend. Use the sum of the legs as divisor.

Working. Multiply the two walking distances: 64,800; double to get 129,600 as dividend. Add the two distances: 540 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.6 第六問 / Problem 6

問 甲乙二人俱在坤地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容圓也。以勾股相乘，倍之為實，以弦較和為法。

草 以二行步相乘，得六萬九千一百二十步，倍之得一十三萬八千二百四十步為實。置乙東行自之，得三萬六千八百六十四步為勾冪；又置甲南行自之，得一十二萬九千六百步為股冪。二冪相並，得一十六萬六千四百六十四步為方實，以平方開之，得四百零八，即弦也。又置甲南行步內減乙東行步，餘一百六十八步，即較也。以較加弦，共得五百七十六步以為法。實如法而一，得二百四十步，為城徑也。合問。

〔按〕此題用勾股求得弦，即可加減得弦較較為城徑。今必以勾股相乘倍積為實，求得弦較和為法，而後始得弦較較為城徑者，蓋欲因此並明勾股相乘之倍積為弦較較、弦較和相乘之積，非故為紆回也。

Problem. Two people both stand at Kun. B walks 192 steps east and stops. A walks 360 steps south and sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the short leg” case. Multiply the two legs and double as dividend. Use the sum of hypotenuse and difference as divisor.

Working. Multiply the two walking distances: 69,120; double to get 138,240 as dividend. Square B’s eastward steps: 36,864 as the short-leg square. Square A’s southward steps: 129,600 as the long-leg square. Combine: 166,464. Extract square root: 408, the hypotenuse. Subtract B’s steps from A’s: 168, the difference. Add difference to hypotenuse: 576 as divisor. Divide: 240 steps, the city diameter. This agrees.

[Explanatory note: This problem uses the right triangle to find the hypotenuse, then by addition and subtraction obtains the “hypotenuse-difference-difference” as the city diameter. The method deliberately uses the product of the legs (doubled) as dividend to demonstrate that this product equals the product of (hypotenuse−difference) and (hypotenuse+difference).]

4.0.7 第七問 / Problem 7

問 甲乙二人同立於艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容圓也。以勾股相乘，倍之為實，以弦和較為法。

Problem. Two people stand together at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the long leg” case. Multiply the two legs and double as dividend. Use the difference of hypotenuse-sum as divisor.

4.0.8 第八問 / Problem 8

問 甲乙二人同立於坤地，甲南行一百九十二步而止，乙東行七十二步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為弦外容圓也。以勾股相乘，倍之為實，以勾股較為法。

Problem. Two people stand together at Kun. A walks 192 steps south and stops. B walks 72 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the hypotenuse” case. Multiply the two legs and double as dividend. Use the difference of the legs as divisor.

4.0.9 第九問 / Problem 9

問 甲乙二人俱在巽地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容半圓也。以勾股相乘，倍之為實，以股弦和為法。

Problem. Two people both stand at Xun. B walks 192 steps east and stops. A walks 360 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "semicircle outside the short leg" case. Multiply the two legs and double as dividend. Use the sum of long leg and hypotenuse as divisor.

4.0.10 第十問 / Problem 10

問 甲乙二人俱在艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容半圓也。以勾股相乘，倍之為實，以勾弦和為法。

Problem. Two people both stand at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "semicircle outside the long leg" case. Multiply the two legs and double as dividend. Use the sum of short leg and hypotenuse as divisor.

以上十問構成「洞淵九容」，第五問（弦上容圓）為補充。卷二其餘四問繼續以天元術求解之標準率問題。

The above ten problems constitute the "Dongyuan Nine Containments" (洞淵九容), supplemented by Problem 5 (circle on the hypotenuse). The remaining four problems of Volume 2 continue with additional standard-rate problems using the tian yuan shu method.

5 Volume 3 / 卷三

邊股一十七問

卷三呈十七問，所給數據涉及邊股（b_2），定義為「邊」三角形（表中第二）之長直邊，等於四百八十步。本卷所有問題之未知數皆為圓城直徑（城徑 = 二百四十步）。此等問題展示天元術之漸增精妙，方程次數自二次至三次。

Seventeen Questions on the Side-Long-Leg

Volume 3 presents 17 problems in which the given data involve the *side long-leg* (邊股, b_2), defined as the long leg of the “Side” triangle (邊 triangle, No. 2 in the table), equal to 480 steps. In all problems of this volume, the unknown to be found is the diameter of the circular city (城徑 = 240 steps). These problems demonstrate the increasing sophistication of the tian yuan shu method, with equations ranging from quadratic to cubic degree.

5.0.1 第一問 / Problem 1

問 乙出東門南行，不知步數而止。甲出西門南行四百八十步，望見乙，復就乙行五百一十步與乙相會，問答同前。

答 城徑二百四十步。

術 倍相減步，以乘二之甲南行步，為平方實，得城徑。

草 識別得二行相減，餘三十步，即乙出東門南行步也。倍相減步，得六十步，以乘二之甲南行步九百六十步，得五萬七千六百步，為平方實。如法開之，得二百四十步，即城徑也。合問。

Problem. B goes out the East Gate and walks south, an unknown number of steps, then stops. A goes out the West Gate and walks 480 steps south, sees B, then walks 510 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the difference of the two walking distances; multiply by twice A’s southward steps to obtain the square-root dividend; this gives the city diameter.

Working. Recognize: subtracting the two distances leaves 30 steps, which is B’s southward distance from the East Gate. Double the difference: 60 steps. Multiply by twice A’s southward steps (960): 57,600, the square-root dividend. Extract the square root: 240 steps, the city diameter. This agrees.

5.0.2 第二問 / Problem 2

問 甲出西門南行四百八十步而止，乙從艮隅東行八十步，望見甲，問答同前。

答 城徑二百四十步。

術 倍南行步，以東行步乘之為實，東行步為從方，一步常法，得全徑。

草 立天元一為全徑，以減於二之甲南行步，得為兩個大差也。以乙東行步乘之，得為圓徑冪，寄左。然後以天元冪與左相消，得以帶縱平方開之，得二百四十步，即城徑也。合問。

又法： 半之乙東行步，乘南行步為實，半之乙東行步為從，一步常法，得半徑。

草 立天元一為半城徑，減甲南行步，得為大差也。以半之東行步乘之，得即半徑冪，寄左。然後以天元冪為同數，與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B walks 80 steps east from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the southward steps; multiply by the eastward steps as dividend; the eastward steps as the “following square”; unity as the constant method; obtain the full diameter.

Working. Set the celestial element unity as the full diameter. Subtract from twice A’s southward steps to obtain two large differences. Multiply by B’s eastward steps to obtain the circle-diameter square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 240 steps, the city diameter. This agrees.

Alternative Method: Halve B’s eastward steps; multiply by the southward steps as dividend; halved eastward steps as the following term; unity as constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from A’s southward steps to obtain the large difference. Multiply by halved eastward steps to obtain the radius square; store on the left. Then use the celestial element square as the equal number; cancel with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.3 第三問 / Problem 3

問 甲出西門南行四百八十步而止，乙從艮隅亦南行一百五十步，望見甲，問答同前。

答 城徑二百四十步。

術 兩行步相乘為實，南行步為從方，一為隅，得半徑。

草 立天元一為半城徑，以減乙南行步，得為半梯頭；以甲行步為梯底，以乘之，得為半徑冪，寄左。然後以天元冪與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B also walks 150 steps south from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the two walking distances as dividend; the southward steps as the “following square”; unity as the corner; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from B’s southward steps to obtain half the trapezoid head. Use A’s steps as the trapezoid base; multiply to obtain the radius square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.4 第四問 / Problem 4

問 甲出西門南行四百八十步，乙出東門直行一十六步，望見甲，問答同前。

答 城徑二百四十步。

術 以四之東行步，乘南行冪為實，從空，東行為廉，一步為隅法，得全徑。

草 立天元一為圓徑，加乙東行步，得為中勾；其甲南行即中股也。置東行步為小勾，以中股乘之，得，合以中勾除，今不受除，便以為小股也。內寄中勾分母。乃復以中股乘之，得三百六十八萬六千四百，又四之，得一千四百七十四萬五千六百，為一段圓徑冪，寄中勾分母，寄左。然後以天元徑自之，又以中勾乘之，得為同數，與左相消，得以帶縱立方開之，得二百四十步，為城徑也。合問。

〔按〕不受除者，無可除之理也。凡二數，此數於彼數有可除之理，則受除；無可除之理，則不受除也。蓋除有法有實，實可二，法不可二。此題以中勾為法，而中勾內有一元，又有十六步，其為數已二矣，又何以均分不一之數乎？故曰不受也。寄分者，姑寄其應除之數也。俟求得兩相等數，而此數內尚少一除，不除此而轉乘彼，則兩數仍相等，猶之受除者也。此所謂以乘代除也。

Problem. A goes out the West Gate and walks 480 steps south. B goes out the East Gate and walks 16 steps straight, and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply four times the eastward steps by the square of the southward steps as dividend; the “following” term is empty; the eastward steps as the edge; unity as the corner method; obtain the full diameter.

Working. Set the celestial element unity as the circle diameter. Add B’s eastward steps to obtain the middle short-leg. A’s southward steps are the middle long-leg. Place the eastward steps as the small short-leg; multiply by the middle long-leg. This should be divided by the middle short-leg, but it is “not accepted for division”—use it directly as the small long-leg, with the middle short-leg as denominator. Multiply again by the middle long-leg: 3,686,400; multiply by 4: 14,745,600 as a segment of the circle-diameter square, with the middle short-leg as denominator; store on the left. Then square the celestial element diameter and multiply by the middle short-leg as the equal number; cancel with the left; extract the cube root with a following edge: 240 steps, the city diameter. This agrees.

[Explanatory note: “Not accepted for division” means there is no valid way to divide. When two numbers are such that one can be divided by the other, the division is “accepted”; otherwise it is “not accepted.” For division has divisor and dividend; the dividend can be compound, but the divisor cannot. Here the middle short-leg serves as divisor, but it contains both the celestial element and 16 steps—it is already compound. How can one evenly divide a non-uniform number? Hence “not accepted.” Storing the denominator means temporarily retaining the number that should be divided. When two equal quantities are found, and one still lacks a division, instead of dividing this, multiply that; the two quantities remain equal, as if the division had been accepted. This is the so-called “substituting multiplication for division.”]

5.0.5 第五問 / Problem 5

問 乙出南門東行七十二步而止，甲出西門南行四百八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 以乙東行冪，乘甲南行為實；乙東行冪為從方；甲南行步內減二之東行步為益廉；一步常法，得半徑。

草 立天元一為半城徑，以減南行步，得為小股。又以天元加乙東行步，得為小勾。又以天元加南行步，得為大股。乃置大股在地，以小勾乘之，得下式，合以小股除之，今不受除，便以為大勾，內寄小股分母。又置天元半徑，以分母小股乘之，得以減大勾，得為半個梯底於上。以乙東行七十二步為半個梯頭，以乘上位，得為半徑冪，內寄小股分母，寄左。然後置天元冪，又以分母小股乘之，得為同數，與左相消，以立方開之，得一百二十步。倍之，即城徑也。合問。

又法：以二數相乘為實，相減為從，一虛法，平開，得半徑。

Problem. B goes out the South Gate and walks 72 steps east, then stops. A goes out the West Gate and walks 480 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the square of B's eastward steps by A's southward steps as dividend; the square of B's eastward steps as the "following square"; subtract twice the eastward steps from A's southward steps as the "beneficial edge"; unity as the constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from the southward steps to obtain the small long-leg. Add the celestial element to B's eastward steps to obtain the small short-leg. Add the celestial element to the southward steps to obtain the large long-leg. Place the large long-leg; multiply by the small short-leg. This should be divided by the small long-leg, but it is "not accepted for division"—use it directly as the large short-leg, with the small long-leg as denominator. Place the celestial element radius; multiply by the denominator (small long-leg); subtract from the large short-leg to obtain half the trapezoid base on top. Use B's 72 eastward steps as half the trapezoid head; multiply the upper quantity to obtain the radius square, with the small long-leg as denominator; store on the left. Then place the celestial element square; multiply by the denominator (small long-leg) as the equal number; cancel with the left; extract the cube root: 120 steps. Double it: the city diameter. This agrees.

Alternative Method: Multiply the two numbers as dividend; their difference as the following term; unity as the empty method; extract the square root; obtain the radius.

5.0.6 第六問 / Problem 6

問 乙出南門東行，不知步數而立。甲出西門南行四百八十步，望見乙與城參相直，又就乙行四百零八步與乙相會，問答同前。

答 城徑二百四十步。

[其餘十一問依同樣模式，所給數據涉及邊股（邊股 = 四百八十步）及其他各種測量，每問皆以天元術建立多項式方程，求解得城徑二百四十步。]

Problem. B goes out the South Gate and walks east, an unknown number of steps, then stands. A goes out the West Gate and walks 480 steps south, sees B aligned with the city, then walks 408 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

[The remaining 11 problems of Volume 3 continue in the same pattern, with given data involving the side long-leg (邊股 = 480 steps) and various other measurements, each leading to polynomial equations solved by the tian yuan shu method to yield the city diameter of 240 steps.]

6 The Tian Yuan Shu Method / 天元術

天元術

天元術（「天元術」，「method of the celestial element」）為《測圓海鏡》全書所運用之代數技法。此乃中國數學史上最早之系統方法，用以建立並求解一元多項式方程。

6.1 Methodology / 方法論

其程序如下：

1. **立天元一為** X（「設天元一為 X」）——宣告未知量。以現代記號，等同於「令 $x = X$」。
2. 以天元 x 表達所有已知量及關係。
3. 構造兩個相等之表達式（常以不同方法計算同一幾何量），一為「左」式暫存，另一為「同數」。
4. **相消**（「mutual cancellation」）——令兩式相等並化簡，得多項式方程之標準形式。
5. 以開方法求解：開平方、開立方、或更高次之開方。

6.2 Polynomial Notation / 多項式記號

李冶運用籌算板上的位置記法：

- 係數縱向排列，常數項（實）在上。
- 其下：x 之係數（方，「square」），繼以 x^2（廉，「edge」），繼以 x^3（隅，「corner」）。
- 更高次：第一廉、第二廉等。
- 負係數以末位斜劃表示。

The Tian Yuan Shu Method

The *tian yuan shu* (天元術, “method of the celestial element”) is the algebraic technique employed throughout the *Ceyuan Haijing*. It is the earliest systematic method in Chinese mathematics for setting up and solving polynomial equations with one unknown.

The procedure follows these steps:

1. **Set the celestial element unity to be** X (“立天元一為 X”) — This declares the unknown quantity. In modern notation, it is equivalent to “Let $x = X$.”
2. Express all given quantities and relationships in terms of the celestial element x.
3. Construct two equal expressions (often representing the same geometric quantity computed by different methods), one designated as the “left” (左) expression and stored temporarily, the other as the “equal number” (同數).
4. **Mutual cancellation** (相消) — Set the two expressions equal and simplify to obtain the polynomial equation in standard form.
5. Solve the equation by root-extraction methods (開方): square-root extraction (開平方), cube-root extraction (開立方), or higher-degree root extraction.

Li Ye uses a positional notation on a counting-rod board:

- Coefficients are arranged vertically, with the constant term (實) at the top.
- Below it: the coefficient of x (方, “square”), then x^2 (廉, “edge”), then x^3 (隅, “corner”).
- For higher degrees: first 廉, second 廉, etc.
- Negative coefficients are indicated by a diagonal stroke through the last digit.

6.3 Example: Quadratic Equation / 示例：二次方程

第一問導出相當於下式之方程：

$$384000 = 1600 \cdot d$$

其中 d 為直徑。但更複雜之問題產生真正的多項式方程。例如卷三第四問產生**三次方程**：

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

以現代形式，李冶以「帶縱立方」解之。

The first problem leads to the equation equivalent to:

$$384000 = 1600 \cdot d$$

where d is the diameter. But more complex problems yield true polynomial equations. For instance, Problem 4 of Volume 3 produces a **cubic equation**:

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

in modern form, which Li Ye solves by “opening the cube with an edge” (帶縱立方).

6.4 Historical Significance / 歷史意義

《測圓海鏡》為現存最早系統運用天元術之著作。所解方程包括：

次數 / Degree	**方程數量** / Number
一次 / Linear (degree 1)	31
二次 / Quadratic (degree 2)	106
三次 / Cubic (degree 3)	24
四次 / Quartic (degree 4)	20
六次 / Sextic (degree 6)	1
總計 / **Total**	**182**

The *Ceyuan Haijing* is the earliest surviving work that systematically employs the tian yuan shu. The equations solved include:

Degree	**Number of Equations**
Linear (degree 1)	31
Quadratic (degree 2)	106
Cubic (degree 3)	24
Quartic (degree 4)	20
Sextic (degree 6)	1
Total	**182**

注：本排印版呈現《測圓海鏡》卷一至卷三之全文。卷一包含全部定義、數值數據及六百九十二條幾何公式。卷二、卷三呈現一百七十問中之首批三十一問，以天元術求解。其餘九卷（卷四–卷十二）繼續一百三十九問，複雜度漸增，至卷七達六次方程。

*Note: This typeset edition presents the complete text of Volumes 1–3 of the Ceyuan Haijing. Volume 1 contains all definitions, numerical data, and 692 geometric formulas. Volumes 2 and 3 present the first 31 of 170 problems, solved by the tian yuan shu method. The remaining nine volumes (*卷四–卷十二*) continue with 139 further problems of increasing complexity, culminating in a sixth-degree equation in Volume 7.*

— *End of Volumes 1–3* / 卷一至卷三終 —

欽定四庫全書・Imperial Siku Quanshu

子部・Mathematics and Astronomy

測圓海鏡　Ceyuan Haijing

卷四至卷六　Volumes 4–6

底勾、大股、大勾三卷全編

Double Difference, Large Leg, and Large Base: Complete

元　李冶　撰

Yuan Dynasty　Li Ye (Renqing, Jingzhai)

字仁卿，號敬齋，眞定欒城人

Jinshi of the Late Jin; Hanlin Academician under Yuan

Bilingual English–Chinese Edition

Original Chinese with English Translation

Parallel Text: Chinese (Left) ・English (Right)

依《欽定四庫全書》本・Based on the Imperial Siku Quanshu Edition

卷四：第五十一問至第六十七問・Vol. 4: Problems 51–67
卷五：第六十八問至第八十五問・Vol. 5: Problems 68–85
卷六：第八十六問至第一百三問・Vol. 6: Problems 86–103

Contents

引言・Introduction

《測圓海鏡》十二卷，元李冶撰。冶字仁卿，號敬齋，眞定欒城人。金末進士，入元官翰林學士。此書乃其所著天元術之傑作，以勾股測圓之法，設問答以明天元一之術。全書凡一百七十問，皆設城池圓形，以人物行步之數推求圓徑。本編收錄卷四、卷五、卷六三卷，凡五十三問：

- 卷四：底勾一十七問（第五十一問至第六十七問）
- 卷五：大股一十八問（第六十八問至第八十五問）
- 卷六：大勾一十八問（第八十六問至第一百三問）

書中設問之例：假令有圓城一座，甲乙二人各從城門出行，或東或南，行若干步，遙相望見，或斜行相會。只知若干步數，餘皆不知，以求圓城之徑。每問皆有問、答、草、法四部分。

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes, was written by Li Ye of the Yuan Dynasty. Li Ye, styled Renqing, sobriquet Jingzhai, was from Luancheng, Zhending. A Jinshi of the late Jin period, he served as Hanlin Academician under the Yuan. This work is his masterpiece on the *tian yuan* (celestial element) method, using right-triangle circle-measurement techniques to elucidate the art of *tian yuan yi*. The complete work contains 170 problems, each positing a circular city wall, from whose walkers' paces the diameter is sought.

This edition contains Volumes 4, 5, and 6—53 problems in total:

- **Volume 4**: 17 problems on the base leg (51–67)
- **Volume 5**: 18 problems on the large leg (68–85)
- **Volume 6**: 18 problems on the large base (86–103)

Each problem presents a scenario: A and B depart from city gates, walking east or south, viewing each other diagonally or meeting obliquely. Given certain paces, all else unknown, find the city diameter. Each problem has four parts: **Question**, **Answer**, **Working**, and **Method**.

天元術記法說明：書中以籌算式記多項式，自下而上排列，常數項在下，一次項在上，二次又在上，依此類推。如 $\frac{480}{1}$ 表示 $480 - x$，$\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ 表示 $48000 - x^2$ 之類。所謂「立天元一為某數」，即設未知數 x 為所求之量。

On Tian Yuan Notation: Polynomials are recorded in counting-rod style, arranged bottom-to-top: constant term lowest, linear term above, quadratic above that, etc. Thus $\frac{480}{1}$ denotes $480 - x$, and $\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ denotes $48000 - x^2$. "Let the celestial element unity be [some quantity]" means: let unknown x equal the sought quantity.

1 卷四：底勾一十七問・Vol. 4: Double Difference (17 Problems)

底勾一十七問・Double Difference

（第五十一問至第六十七問・Problems 51–67）

【題意總說】卷四論「底勾」之術。所謂底勾，乃勾股形之最下勾股，與圓徑相切之基本勾股形也。凡一十七問，皆以底勾為基，求圓城之徑。底勾者，通勾股形之勾邊，亦即大勾，為諸勾股形之根本。諸問或知甲東行之數，或知乙南行之數，或知斜行之數，三者之中知其二，即可立天元一，建方程而開方得圓徑。

[General Summary] Volume 4 treats the “base leg” (底勾) method. The base leg is the lowest side of the right triangle formed tangent to the circular diameter—the fundamental right triangle. All 17 problems use the base leg as foundation to find the city diameter. It is the base of the universal right triangle, i.e., the large base, the root of all sub-triangles. Given any two of: A’s eastward walk, B’s southward walk, or the diagonal walk, one may establish the celestial element and solve.

第一問：或問：乙出南門東行不知步數而立，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問圓城之徑幾何?

答曰：城徑二百四十步。

草曰：識別得二行相減餘，即乙出南門東行數也。以甲東行二百步減於就乙斜行二百七十二步，餘七十二步，為乙東行數。以乘甲東行步，得 $72 \times 200 =$ 一萬四千四百步，又四之，得五萬七千六百步為實。以平方開之，得 $\sqrt{57600} =$ 二百四十步，即城徑也。
驗算：城徑二百四十步，半徑一百二十步。乙東行七十二步，甲東行二百步，甲比乙多行 $200 - 72 = 128$ 步。以勾股術驗之，斜行為弦，$\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$，復以半徑加之，合得斜行步數，其術驗矣。

法曰：二行差數乘甲東行又四之，為平方實，開方得全徑。
術曰：以就乙斜行步內減甲東行步，餘為乙東行數；以此數乘甲東行步為實，四之；開平方，即得城徑。

Problem 51: Suppose B exits the south gate and walks east an unknown number of paces, while A exits the north gate and walks east 200 paces to see him. A then walks diagonally 272 paces to meet B. What is the diameter of the circular city wall?

Answer: The city diameter is 240 paces.

Working: By inspection, the difference of the two walks gives B’s eastward walk. Subtracting A’s 200 paces from the diagonal 272 paces leaves 72 paces as B’s eastward walk. Multiplying by A’s 200 paces gives $72 \times 200 = 14{,}400$ paces. Quadrupling gives $57{,}600$ as the dividend. Extracting the square root: $\sqrt{57600} = 240$ paces, the city diameter.
Verification: Diameter 240, radius 120. B walks 72 east, A walks 200 east; difference $200 - 72 = 128$. By the gougu (Pythagorean) theorem: diagonal is the hypotenuse; $\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$. Adding the radius gives the diagonal walk, confirming the result.

Method: The difference of the two walks multiplied by A’s eastward walk, then quadrupled, is the dividend for square-root extraction. Method: From the diagonal walk subtract A’s eastward walk, remainder is B’s eastward walk. Multiply this by A’s eastward walk as dividend; quadruple; extract square root to obtain the city diameter.

第二問： 或問：乙出南門東行一百二十步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，以減於二之甲東行步，得$\frac{480}{1}$為兩个小差。以乙東行一百二十步乘之，得$\frac{57600}{1}$為城徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，得$\frac{57600}{1\quad 0}$（負上正下），以平方開之，得二百四十步，即城徑也。
釋義：二之甲東行者，以甲東行為勾，其倍即通勾與圓徑之差。以乙東行乘之者，乙東行即小勾，乘大差得大勾之冪，亦即城徑冪之半。故以天元冪相消，開平方即得。

法曰：半之乙東行步乘甲東行為實，半乙東行為從，一步常法，得半徑。術曰：置乙東行步折半，以乘甲東行步為實；以半乙東行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 52: Suppose B exits the south gate and walks east 120 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting it from twice A's eastward walk gives $\frac{480}{1}$ as the two small differences. Multiplying by B's 120 paces gives $\frac{57600}{1}$ as the city diameter-squared, deposited on the left. Then canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{57600}{1\quad 0}$ (negative above, positive below). Extracting the square root yields 240 paces, the city diameter.
Explanation: Twice A's eastward walk: A's walk being the base, its double is the difference between the universal base and the diameter. Multiplying by B's walk (the small base) gives the square of the large base, i.e., half the city-diameter squared. Cancelling with the celestial element and extracting the root gives the result.

Method: Half of B's eastward walk multiplied by A's eastward walk is the dividend; half of B's eastward walk is the coefficient; unity is the constant divisor. This yields the radius. Method: Halve B's eastward walk, multiply by A's eastward walk as dividend; use half B's walk as the coefficient, unity as the corner divisor; extract square root to obtain the radius; double it for the diameter.

第三問： 或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，減於乙南行三百六十步之半，得$\frac{180}{1}$為小差。半乙南行步，得一百八十步，以乘小差，得$\frac{32400}{180}$為半徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，得$\frac{32400}{180\quad 1\quad 0}$（上負下正），開平方得一百二十步，倍之得全徑二百四十步。
釋義：此問以乙南行為股，甲東行為勾。半乙南行即大股之半，去半徑餘為小差。以小差乘半股得半徑冪，此天元術之妙用也。

法曰：兩行步相乘為實，甲東行為從，乙南行為隅，開平方得半徑。術曰：以甲東行步乘乙南行步為實，以甲東行步為從，乙南行步為隅法；開平方得一百二十步，即半徑；倍之即城徑。

Problem 53: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting it from half of B's southward walk (360 paces) gives $\frac{180}{1}$ as the small difference. Half of B's southward walk is 180 paces; multiplying by the small difference gives $\frac{32400}{180}$ as the radius-squared, deposited on the left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{32400}{180\quad 1\quad 0}$ (negative above, positive below). Extracting square root gives 120 paces; doubling gives the full diameter of 240.
Explanation: Here B's southward walk is the leg, A's eastward walk is the base. Half B's walk is half the large leg; less the radius gives the small difference. Multiplying small difference by half the leg gives the radius-squared—the subtle efficacy of the tian yuan method.

Method: The product of the two walks is the dividend; A's eastward walk is the coefficient; B's southward walk is the corner divisor. Extract square root to obtain the radius. Method: Multiply A's eastward by B's southward as dividend; A's walk as coefficient, B's walk as corner; extract square root to obtain 120 paces, the radius; double for the diameter.

第四問： 或問：乙從坤隅南行三百六十步而止，甲出北門東行二百步見之，就乙斜行四百零八步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行冪得四萬步，以乙南行冪得一十二萬九千六百步，並之得一十七萬三千六百步為弦冪之實。又以斜行四百零八步自之，得一十六萬六千四百六十四步。以此減弦冪實，餘七千一百三十六步。以四之得二萬八千五百四十四步為城徑冪之較實。
別求之：立天元一為城徑，以減於二之甲東行，餘即乙出南門東行數也。以乙南行乘甲東行冪，又四之為實。從空，乙行為廉，一步常法，得城徑。開立方得二百四十步。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。術曰：置甲東行步自乘，又以乙南行步乘之，四之為實；乙南行步為廉法；一步為隅法；開立方即得城徑。

Problem 54: Suppose B walks south 360 paces from the Kun corner and stops, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 408 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Squaring A's eastward walk gives 40,000; squaring B's southward walk gives 129,600. Their sum, 173,600, is the dividend for the hypotenuse-squared. The diagonal 408 squared gives 166,464. Subtracting from the dividend leaves 7,136. Quadrupling gives 28,544 as the comparison-dividend for the diameter-squared.
Alternatively: Let the celestial element be the diameter. Subtracting from twice A's eastward walk gives B's eastward walk from the south gate. Multiply B's southward walk by A's walk-squared; quadruple as dividend. Empty coefficient; B's walk as corner; unity as constant. Extracting cube root gives 240 paces.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant divisor. Extract cube root for the diameter. Method: Square A's eastward walk, multiply by B's southward walk, quadruple as dividend; B's walk as corner divisor; unity as corner; extract cube root.

第五問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，加乙南行一百三十五步，得$\frac{135}{1}$為股率。其甲東行二百步即勾率也。其乙南行為小股，以勾率乘之，合以大弦率除。今不受除，便以此為小勾，為母，寄股率。

乃先以小弦乘大和，得下式$\begin{array}{ccc}0 & 0 & 12000\\ \hline 1 & 0 & 0\end{array}$寄左。又以倍天元減斜步，得$\frac{100}{2}$為小和，以乘大弦，得$\frac{12000}{1}$為同數，與左相消。開平方得二百四十步。

按：此草以率法為主，以勾股之率通諸數，建立天元方程，為天元術之正統用法。立天元一為城徑，以諸率配之，求其等式，相消開方，即得所求。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 55: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Adding B's southward 135 paces gives $\frac{135}{1}$ as the leg-rate. A's eastward 200 paces is the base-rate. B's southward walk is the small leg; multiplying by the base-rate should be divided by the large hypotenuse-rate. Not dividing, this becomes the small base as numerator, with the leg-rate deposited.

First multiply the small hypotenuse by the large sum, obtaining the expression $\begin{array}{ccc}0 & 0 & 12000\\ \hline 1 & 0 & 0\end{array}$, deposited left. Then subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as the small sum; multiply by the large hypotenuse, obtaining $\frac{12000}{1}$ as the equal number, which cancels with the left. Extracting square root gives 240 paces.

Note: This working uses rate-methods to connect all quantities, establishing tian yuan equations in the orthodox manner.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第六問： 或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑。以甲東行二百步減斜行二百七十二步，餘七十二步，為乙出南門東行數。以乙南行三百六十步折半，得一百八十步，為大股之半。以此半股減天元半徑，得$\frac{180}{1}$為小差。以小差乘半股，得$\frac{32400}{180}$為半徑冪，寄左。以天元冪相消，開平方得一百二十步，倍之得全徑二百四十步。

法曰：兩行步相乘倍之為實，乙南行為從，一步常法，得半徑。術曰：以甲東行步乘乙南行步，倍之為實；乙南行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 56: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting A's 200 paces from the diagonal 272 leaves 72 paces as B's eastward walk from the south gate. Halving B's 360-pace southward walk gives 180 paces, half the large leg. Subtracting the radius from this half-leg gives $\frac{180}{1}$ as the small difference. Multiplying small difference by half the leg gives $\frac{32400}{180}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces; doubling gives the full diameter of 240.

Method: The product of the two walks, doubled, is the dividend; B's southward walk is the coefficient; unity is the constant divisor; this yields the radius. Method: Multiply A's eastward by B's southward, double as dividend; B's walk as coefficient, unity as corner; extract square root for the radius; double for the diameter.

第七問: 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步為小股，甲東行二百步為小勾。以斜行二百七十二步為小弦。驗之：$135^2+72^2=18225+5184=23409$，而 $272^2=73984$，非直股勾股也。當以天元術正之。
立天元一為城徑，以減於二之甲東行，得 $\frac{400}{1}$ 為兩个小差之和。以乙南行乘之，又四之，得大實。以天元冪與左相消，開平方得二百四十步。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 57: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. B’s southward 135 paces is the small leg; A’s eastward 200 is the small base; the diagonal 272 is the small hypotenuse. Verification: $135^2+72^2=18225+5184=23409$, while $272^2=73984$—not a direct right triangle. We must correct by the tian yuan method.
Let the celestial element be the diameter. Subtracting from twice A’s eastward walk gives $\frac{400}{1}$ as the sum of the two small differences. Multiplying by B’s southward walk and quadrupling gives the large dividend. Canceling with the celestial element-squared and extracting square root gives 240 paces.

Method: B’s southward walk multiplied by A’s walk-squared is the dividend. Empty coefficient; B’s walk as corner; unity as constant; extract cube root for the diameter.

第八問: 或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以減於倍甲東行四百步，餘一百三十步，為大小差之較。以此較乘甲東行冪，得 $130\times40000=$ 五百二十萬步為實。從空，倍乙行為廉，一步常法，開立方得城徑。
又法：立天元一為半徑，以半甲東行一百步減斜行半數一百三十六步，餘三十六步為半較。以乙南行半數六十七步半乘之，得半徑冪之較，與天元冪相消，開平方即得。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 58: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Doubling B’s southward walk gives 270 paces. Subtracting from twice A’s eastward walk (400 paces) leaves 130 paces as the difference between the large and small differences. Multiplying this difference by A’s walk-squared: $130\times40000=$ 5,200,000 paces as dividend. Empty coefficient; twice B’s walk as corner; unity as constant; extracting cube root gives the diameter.
Alternative method: Let the celestial element be the radius. Subtracting half A’s walk (100) from half the diagonal (136) leaves 36 as half the difference. Multiplying by half B’s walk (67.5) gives the radius-squared difference; cancel with the celestial element-squared; extract square root.

Method: Twice B’s southward walk multiplied by A’s walk-squared is the dividend. Empty coefficient; twice B’s walk as corner; unity as constant; extract cube root for the diameter.

第九問： 或問：乙出南門直行一百三十五步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：此問與前問類而異者，甲見乙之後復斜行相會。立天元一為城徑，以兩行相減，餘乘甲東行又四之為實。兩行差為廉，一步常法，開立方得城徑。
細草：以乙南行一百三十五步減甲東行二百步，餘六十五步，為兩行之差。以乘甲東行二百步，得一千三百步，又四之得五千二百步為實。兩行差六十五步為廉法，一步為隅法，開立方得二百四十步。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 59: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: This problem is similar to the previous but differs: after A sees B, he then walks diagonally to meet him. Let the celestial element be the city diameter. Subtract the two walks; multiply the remainder by A's eastward walk and quadruple as dividend. The difference of the two walks is the corner divisor; unity is the constant; extract cube root for the diameter.
Detailed working: Subtract B's 135 from A's 200, leaving 65 as the difference. Multiply by A's 200: 1,300; quadruple: 5,200 as dividend. The difference 65 is the corner divisor; unity is the constant corner. Extracting cube root gives 240 paces.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十問： 或問：乙出南門直行一百三十五步，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得二百四十步。
按：此草以乙南行為小股，甲東行為小勾。以勾冪乘股，蓋以大勾之冪比於小勾之冪，若大股之冪比於小股之冪。以率言之，大勾即城徑與小勾之關係，故以天元建立而開立方。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 60: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 paces; quadrupled gives 21,600,000 as dividend. Empty coefficient; B's 135 paces as corner divisor; unity as constant corner; extracting cube root gives 240 paces.
Note: This working takes B's southward as the small leg, A's eastward as the small base. Multiplying the base-squared by the leg: the ratio of the large base-squared to the small base-squared equals that of the large leg-squared to the small leg-squared. The large base is the relation between diameter and small base, hence establishing tian yuan and extracting cube root.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第十一問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑與斜行各幾何?

答曰：城徑二百四十步。斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以甲東行冪四萬步乘之，得一千零八萬步，又倍之得二千零一十六萬步為實。倍乙南行二百七十步為廉法，一步為隅法，開立方得城徑二百四十步。

既得城徑，求斜行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，為乙東行與半徑之差。以此差冪與半徑冪相加開方，得斜行二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。術曰：倍乙南行步，以乘甲東行冪為實；倍乙南行步為廉法；一步為隅法；開立方得城徑。

Problem 61: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and the diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270 paces. Multiplying by A's walk-squared (40,000) gives 10,080,000 paces; doubled gives 20,160,000 as dividend. Twice B's southward (270) is the corner divisor; unity is the constant corner; extracting cube root gives diameter 240 paces.

To find the diagonal: Subtract diameter 240 from twice A's eastward walk (400), leaving 160 as the difference between B's eastward walk and the radius. Adding the square of this difference to the square of the radius and extracting root gives the diagonal of 272 paces.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter. Method: Double B's southward walk, multiply by A's walk-squared as dividend; twice B's walk as corner; unity as corner; extract cube root.

第十二問： 或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步，為兩行之較。以較乘甲東行，得一千三百步，又四之，得五千二百步為實。從空，兩行較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。

驗草：城徑二百四十步，半徑一百二十步。以半徑減甲東行，餘八十步為甲至城心之東行。以乙南行一百三十五步減半徑，餘十五步，為乙至城心之南行。以此二數各自乘并之，$80^2 + 15^2 = 6400 + 225 = 6625$，開方得 81.39 不盡，須以天元正術求之。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 62: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A's 200 minus B's 135 leaves 65 as the difference. Multiply the difference by A's 200: 1,300; quadruple: 5,200 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.

Verification: Diameter 240, radius 120. Radius subtracted from A's walk leaves 80 as A's distance from the center eastward. B's 135 minus radius leaves 15 as B's distance southward from the center. Squaring and summing: $80^2 + 15^2 = 6400 + 225 = 6625$; root ≈ 81.39, inexact— must use the orthodox tian yuan method.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十三問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑及乙東行步數各幾何?

答曰：城徑二百四十步，乙東行七十二步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步為實。從空，乙行為廉，一步常法，開平方得城徑。
求乙東行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，以減甲東行二百步，餘七十二步，即乙東行步數。
釋天元：此問以乙南行乘甲東行冪為實者，乙南行為股率，甲東行為勾率，以勾冪乘股率者，以大勾之冪與大股相乘之義也。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 63: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and B's eastward paces?

Answer: City diameter: 240 paces. B's eastward walk: 72 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 as dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.
To find B's eastward walk: Subtract diameter 240 from twice A's eastward (400), leaving 160. Subtract this from A's 200 paces, leaving 72 paces as B's eastward walk.
Tian Yuan explained: Using B's southward (the leg-rate) multiplied by A's walk-squared (the base-squared): the base-squared multiplied by the leg-rate represents the meaning of the large base-squared multiplied by the large leg.

Method: B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.

第十四問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復斜行與乙相會。問城徑及斜行步數。

答曰：城徑二百四十步，斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步，乘甲東行冪（四萬步），得一千零八十萬步為實。從空，倍乙南行為廉，一步為隅法，開平方得城徑二百四十步。
求斜行：以城徑減於二之甲東行，餘一百六十步。以此與甲東行相加，得三百六十步，折半得一百八十步，為弦之半。以自乘得三萬二千四百步，加入半徑冪一萬四千四百步，得四萬六千八百步，即非弦冪也。當以天元別求之：以半徑冪加甲乙東行之較冪，合半之，開方得斜行。$120^2 + 128^2 = 14400 + 16384 = 30784$，而 $272^2 = 73984$，知須以天元方程正求之。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 64: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally to meet B. What are the city diameter and diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000 as dividend. Empty coefficient; twice B's walk as corner; unity as constant; extracting square root gives diameter 240 paces.
To find the diagonal: Subtract diameter from twice A's eastward, leaving 160. Add to A's eastward: 360; halve: 180 as half the hypotenuse. Squaring gives 32,400; adding radius-squared 14,400 gives 46,800—not the hypotenuse-squared. Must use tian yuan separately: add radius-squared to the difference of A-B eastward walks squared, halve, extract root for diagonal. $120^2 + 128^2 = 14400 + 16384 = 30784$, while $272^2 = 73984$; hence must use tian yuan equations correctly.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract square root for the diameter.

第十五問： 或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步為較。以較乘甲東行冪（四萬步），得二百六十萬步，又四之得一千零四十萬步為實。從空，較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。
按：此問以較為主，蓋斜行相會，須以兩行差建立方程。較者，大小差之較也，為天元術中重要之率。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 65: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A’s 200 minus B’s 135 leaves 65 as the difference. Multiply the difference by A’s walk-squared (40,000): 2,600,000; quadrupled: 10,400,000 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
Note: This problem uses the difference as key. For diagonal meetings, one must establish equations from the walk-difference. The difference is the difference of large and small differences, an important rate in tian yuan.

Method: The walk-difference multiplied by A’s walk-squared, then quadrupled, is the dividend. Empty coefficient; walk-difference as corner; unity as constant; extract cube root for the diameter.

第十六問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得城徑二百四十步。
細釋：甲東行為勾，乙南行為股。以勾冪乘股為實者，大勾之冪當與大股相乘，而乙南行即大股之表示，故以此為實。從空者，無從法也。乙行為廉，即二次項之係數也。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 66: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiply B’s southward 135 paces by A’s walk-squared (40,000): 5,400,000; quadrupled: 21,600,000 as dividend. Empty coefficient; B’s 135 paces as corner divisor; unity as constant corner; extracting cube root gives diameter 240 paces.
Detailed explanation: A’s eastward walk is the base, B’s southward is the leg. Using base-squared multiplied by leg as dividend: the large base-squared ought to multiply the large leg, and B’s southward represents the large leg, hence used as dividend. Empty coefficient means no linear coefficient. B’s walk as corner is the quadratic coefficient.

Method: B’s southward walk multiplied by A’s walk-squared, then quadrupled, is the dividend. Empty coefficient; B’s walk as corner; unity as constant; extract cube root for the diameter.

第十七問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步。斜距二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。乘甲東行冪（四萬步），得一千零八十萬步，又倍之得二千一百六十萬步為實。從空，倍乙南行為廉法，一步為隅法，開立方得城徑二百四十步。
求斜距：既得城徑，則半徑一百二十步。甲東行二百步減半徑一百二十步，餘八十步為甲至城邊之距。以此冪六千四百，與乙南行一百三十五步減半徑餘十五步之冪二百二十五并之，得六千六百二十五，開平方得八十一點二二，加上半徑一百二十步，得二百零一點二二，非斜距也。當以天元正術：以甲東行減乙南行之差冪，加入甲乙各自去城邊之冪，并而開方，合加半徑，即得斜距二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 67: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000; doubled: 21,600,000 as dividend. Empty coefficient; twice B's walk as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
To find the diagonal distance: Diameter 240 gives radius 120. A's 200 minus radius 120 leaves 80 as A's distance to the wall. Its square 6,400 plus B's 135 minus radius 15 (square 225) gives 6,625; square root ≈ 81.22; plus radius 120 gives ≈ 201.22—not the diagonal. Must use orthodox tian yuan: the difference of A's and B's walks squared, plus each distance to the wall squared, summed and rooted, plus radius, gives diagonal 272.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter.

測圓海鏡卷四終 — 凡一十七問

End of Ceyuan Haijing Volume 4 — 17 Problems

2 卷五：大股一十八問・Vol. 5: Large Leg (18 Problems)

大股一十八問・Large Leg

（第六十八問至第八十五問・Problems 68–85）

【題意總說】卷五論「大股」之術。所謂大股，乃最大勾股形之股邊，即通股也。以大股為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大股者，勾股形中最長之股，即甲從乾隅直南至坤隅之步數，為六百步之率也。諸問或知乙南行之數，或知甲南行之數，或知斜望之數，皆以大股為基本建立天元方程。此卷方程多為三次、四次，須開立方、開三乘方以求之，為天元術之精深處。

[General Summary] Volume 5 treats the “large leg” (大股) method. The large leg is the leg-side of the largest right triangle—the universal leg. Taking the large leg as principal, combined with relations among sub-triangles, we seek the city diameter. All 18 problems. The large leg is A’s full southward walk from the Qian corner to the Kun corner—the 600-pace standard. Given B’s southward walk, or A’s southward walk, or diagonal sightings, we establish tian yuan equations. This volume features third- and fourth-degree equations requiring cube-root and fourth-root extraction—the subtle core of tian yuan.

第十八問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之減於甲南行六百步，得$\frac{600}{2}$為大差也。以自之，得$\frac{360000}{2400\quad 4}$為大差冪也。置甲南行冪$\frac{360000}{1}$內減大差冪，而半之，得$\frac{0}{1200\quad 2}$為大弦也，內帶大差為分母。
又置甲南行冪內減大差冪，而半之，得$\frac{0}{600\quad 1}$為大勾也，帶大差分母。又以大差乘股六百步，得$\frac{360000}{1}$併入和，得下式$\frac{360000\quad 0\quad 0}{1\quad 1\quad 1200\quad 1}$為小和，以乘大弦，得$\frac{0\quad 0}{0\quad 0\quad 1\quad 0\quad 1200\quad 1}$ $\frac{0\quad 0}{0\quad 0\quad 1\quad 0\quad 0\quad 1}$寄左。
又以倍天元減斜步，得$\frac{100}{2}$為小和，以乘大弦，得同數，與左相消。開立方得一百二十步，即半徑也；倍之得全徑二百四十步。
釋義：此草以大差為主，大差者甲南行減圓徑也。甲從乾隅南行六百步，為大股之全數。減去圓徑，餘為大差。以大差冪與甲南行冪相減，半之得大弦，此即勾股求弦之法。又以大差乘股併入和，為小和以乘大弦，建立天元方程。其術精深，為全書之難問。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘二百四十步，即城徑也。術曰：以甲南行步自之為冪，以乙南行步乘之為實；以甲南行步為從，乙南行步為廉；一步為隅法；開立方得城徑。

Problem 68: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting twice it from A’s 600-pace southward walk gives $\frac{600}{2}$ as the large difference. Squaring: $\frac{360000}{2400\quad 4}$ as the large-difference-squared. Place A’s walk-squared $\frac{360000}{1}$, subtract large-diff-squared, halve: $\frac{0}{1200\quad 2}$ as the large hypotenuse, carrying large difference as denominator.
Also subtract and halve to obtain $\frac{0}{600\quad 1}$ as the large base, with large-diff denominator. Multiply large difference by the 600-pace leg: $\frac{360000}{1}$; combine into the sum, obtaining the expression $\frac{360000\quad 0\quad 0}{1\quad 1\quad 1200\quad 1}$ as the small sum. Multiply by the large hypotenuse, deposit left.
Also subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as small sum; multiply by large hypotenuse for the equal number; cancel with the left. Extract cube root: 120 paces, the radius; double for the full diameter 240.
Explanation: This working uses the large difference as key—A’s walk minus the diameter. The 600-pace walk is the full large leg. Subtract the diameter for the large difference. The method is among the most profound in the entire work.

Method: Twice the walk-difference minus A’s southward walk, then multiplied by A’s walk, is the dividend. Method: Square A’s southward walk, multiply by B’s southward walk as dividend; A’s walk as coefficient, B’s walk as corner; unity as corner; extract cube root for the diameter.

第十九問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以減於甲南行六百步，得 $\frac{600}{1}$ 為大差。置甲南行冪內減大差冪，而半之，得大弦也。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得半徑一百二十步，倍之得城徑。

細草：甲南行六百步自之，得三十六萬步為甲南行冪。大差 $\frac{600}{1}$ 自之，得 $\frac{360000}{1200\quad 1}$。以甲南行冪減之，得 $\frac{0}{1200\quad 1}$，半之得 $\frac{0}{600\quad \frac{1}{2}}$ 為大弦，帶大差分母。以大差乘股，得 $\frac{360000}{1}$，併入得小和。以小和乘大弦，寄左。以倍天元減斜步得小和，乘大弦為同數，與左相消，開立方得一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 69: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting from A's 600-pace southward walk gives $\frac{600}{1}$ as the large difference. Subtract large-diff-squared from A's walk-squared and halve to obtain the large hypotenuse. Multiply large difference by the 600-pace leg, combine into the sum, obtaining the small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from the diagonal for the small sum, multiply by large hypotenuse for the equal number; cancel with the left; extract cube root for radius 120; double for the diameter.

Detailed working: A's 600 squared: 360,000. Large diff squared: $\frac{360000}{1200\quad 1}$. Subtracting from A's walk-squared gives $\frac{0}{1200\quad 1}$; halved: $\frac{0}{600\quad \frac{1}{2}}$ as large hypotenuse with large-diff denominator. Multiply large diff by leg, $\frac{360000}{1}$, combine for small sum. Cancel and extract.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之減於甲南行六百步，餘 $\frac{600}{2}$ 為大差。以自之得大差冪 $\frac{360000}{2400\quad 4}$。置甲南行冪 $\frac{360000}{1}$ 內減大差冪而半之，得大弦，內帶大差為分母。又置甲南行冪內減大差冪而半之，得大勾，帶大差分母。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得一百二十步，即半徑；倍之得城徑二百四十步。

按：此問與首問相類，惟立天元一為城徑而非半徑，故方程中諸數皆倍之，開立方後須倍之乃得城徑，學者宜辨之。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘，即城徑也。

Problem 70: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting twice it from A's 600-pace walk leaves $\frac{600}{2}$ as the large difference. Squaring gives large-diff-squared $\frac{360000}{2400\quad 4}$. From A's walk-squared $\frac{360000}{1}$ subtract large-diff-squared and halve to obtain the large hypotenuse, carrying large-diff denominator. Also halve to obtain the large base with large-diff denominator. Multiply large diff by the 600-pace leg, combine for small sum, multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 120 paces, the radius; double for diameter 240.

Note: Similar to the first problem, but the celestial element is the diameter, not the radius. Hence all quantities are doubled; after extracting cube root, double again for the diameter. Students should distinguish.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. The remainder is the city diameter.

第二十一問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相減，得四百六十五步為大小股之差。以此差乘甲南行冪（三十六萬步），得一億六千七百四十萬步為實。從空，兩行差為廉，一步常法，開立方得城徑二百四十步。

釋義：此草以兩股差為主。大股甲南行六百步，小股乙南行一百三十五步。兩股之差四百六十五步，即大小差之較也。以此建立天元方程，較之以前數問更為簡捷，乃法之變通也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 71: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A’s 600-pace southward walk is the large leg; B’s 135 is the small leg. Subtracting gives 465 paces as the leg-difference. Multiplying this difference by A’s walk-squared (360,000) gives 167,400,000 as dividend. Empty coefficient; the walk-difference as corner; unity as constant; extracting cube root gives diameter 240 paces.

Explanation: This working uses the leg-difference as key. Large leg 600, small leg 135; difference 465 is the comparison of large and small differences. Establishing the tian yuan equation this way is simpler than previous methods—a flexible variation of the technique.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十二問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差（甲南行六百步減乙南行一百三十五步，得四百六十五步，倍之得九百三十步），內減甲南行步，得三百三十步。復以乘甲南行步，得 $330 \times 600 = 198000$ 步為實。從空，倍二行差減甲南行步為廉，一步常法，開立方得城徑二百四十步。

細草：甲南行六百步，乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減六百步，餘三百三十步。以乘六百步，得一十九萬八千步為實。以三百三十步為廉法，一步為隅法，開立方得二百四十步，即城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 72: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference (A’s 600 minus B’s 135 gives 465; doubled: 930), subtract A’s southward walk: 330. Multiply by A’s walk: $330 \times 600 = 198{,}000$ paces as dividend. Empty coefficient; twice walk-diff minus A’s walk as corner; unity as constant; extracting cube root gives diameter 240 paces.

Detailed working: A’s 600, B’s 135, difference 465. Doubled: 930; subtract 600: 330. Multiply by 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extract cube root: 240 paces, the diameter.

Method: Twice the walk-difference minus A’s southward walk, then multiplied by A’s walk, is the dividend. Empty coefficient; twice walk-diff minus A’s walk as corner; unity as constant; extract cube root for the diameter.

第二十三問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得 $600 \times 135 = 81000$ 步為實。從空，兩行步相加得七百三十五步為廉，一步常法，開平方得城徑二百四十步。
按：此問開平方即得，不須開立方，較之他問為簡易。蓋以兩股相乘為實，兩股和為從，適合天元平方之式也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 73: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A’s 600 is the large leg; B’s 135 is the small leg. Multiply: $600 \times 135 = 81{,}000$ as dividend. Empty coefficient; sum of walks 735 as corner; unity as constant; extracting square root gives diameter 240 paces.
Note: This problem requires only square-root extraction, not cube root— simpler than others. The product of the two legs as dividend and their sum as coefficient perfectly fits the tian yuan quadratic pattern.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十四問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
驗草：城徑二百四十步，半徑一百二十步。甲南行六百步減圓徑二百四十步，餘三百六十步為大差。以大差與半徑各為股勾，$360^2 + 120^2 = 129600 + 14400 = 144000$，開方得 379.47 不盡。當以三乘方正之，其術繁矣。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 74: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A’s 600 minus B’s 135 gives 465 as the difference. Doubled: 930; subtract A’s 600: 330. Multiply by A’s 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Verification: Diameter 240, radius 120. A’s 600 minus diameter 240 leaves 360 as the large difference. Using large difference and radius as leg and base: $360^2 + 120^2 = 129{,}600 + 14{,}400 = 144{,}000$; root ≈ 379.47, inexact. Must use fourth-root correction—complex.

Method: Twice the walk-difference minus A’s southward walk, then multiplied by A’s walk, is the dividend. Empty coefficient; twice walk-diff minus A’s walk as corner; unity as constant; extract cube root for the diameter.

第二十五問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得八萬一千步為實。以兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。
按：此問以兩股相乘為實，蓋大股與小股相乘之積，當等於以城徑為高之矩形面積。以兩股和為從者，大小股之和與城徑之關係式也。開平方即得，術最簡明。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 75: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Multiply: 81,000 as dividend. The leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Note: This problem uses the product of the two legs as dividend. The product equals the rectangular area with the diameter as one dimension. Using the leg-sum as coefficient expresses the relation between diameter and leg sum. Square-root extraction gives the result directly—the clearest method.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十六問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行步，餘三百三十步。以乘甲南行冪（三十六萬步），得一億一千八百八十萬步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
細釋：此問以差為主，倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股冪為實，開立方而得城徑，天元之妙用也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 76: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's walk-squared (360,000): 118,800,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Detailed explanation: Using the difference as key: twice the difference minus the large leg leaves 330—the relation-number between half the large difference and the small difference. Using this as corner, multiplied by large-leg-squared as dividend, and extracting cube root—the marvelous efficacy of tian yuan.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十七問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。

又草：立天元一為半徑。以甲南行減圓徑，餘為大差。以大差自之為大差冪，以甲南行冪減之半之為大弦。又以大差乘股併入和為小和，以小和乘大弦，寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消，開立方得半徑一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 77: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Alternative working: Let the celestial element be the radius. Subtract diameter from A's southward walk for the large difference. Square for large-diff-squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by leg, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120 paces.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十八問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差，甲南行六百步減乙南行一百三十五步，差四百六十五步，倍之九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

天元釋義：此草所云天元一者，即代數之未知數 x 也。以大股六百為甲南行，小股一百三十五為乙南行。兩行之差為大差之表示。倍差內減大股，乃大差與圓徑之關係。以此關鍵數建立方程，開立方即得。此天元術之所以勝於幾何也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 78: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference: A's 600 minus B's 135 gives 465; doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Tian Yuan explained: The "celestial element unity" is the algebraic unknown x. Large leg 600 is A's walk; small leg 135 is B's walk. The difference represents the large difference. Twice the difference minus the large leg is the relation between large difference and diameter. Using this key number to establish the equation and extracting cube root—this is why tian yuan surpasses pure geometry.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十九問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。
釋率：大股為通股，即甲從乾隅直南至坤隅之全步。小股為乙從南門直南至某處之步。兩股相乘之實，即以大股為長、小股為寬之長方形面積。以此面積為天元方程之實，而以兩股和為從法，開平方即得城徑，其理精妙。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 79: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large (universal) leg—A's full southward walk from Qian to Kun. B's 135 is the small leg—B's walk from the south gate. Their product, 81,000, is the area of a rectangle with large leg as length and small leg as width. Using this area as the tian yuan dividend and the leg-sum as coefficient, extracting square root gives the diameter—subtle and exquisite reasoning.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為兩行差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉法，一步為隅法，開立方得城徑二百四十步。
術解：倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股為實，開立方而得城徑。天元術中以三次方程為主，此問即典型的天元三次方程之應用。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 80: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: walk-difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
Method explained: Twice the difference minus the large leg leaves 330— the relation-number between half the large difference and the small difference. Using this as corner, multiplying by the large leg as dividend, extracting cube root for the diameter. This is the classic application of tian yuan cubic equations.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十一問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從法，一步為隅法，開平方得城徑二百四十步。
細草：兩股相乘 $135\times600=81000$ 為實。兩股和 $135+600=735$ 為從。設城徑為 x，則有方程：$x^2+735x=81000$。以天元術開平方：置實八萬一千，從七百三十五，隅一步。以隅法一約實，初商二百四十。以初商乘從法，得 $240\times735=176400$。以減實不盡，知初商即為城徑之數，蓋實較從乘商之數為小也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 81: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Detailed working: Product $135\times600=81000$ as dividend. Sum $135+600=735$ as coefficient. Let diameter be x; the equation is $x^2+735x=81000$. By tian yuan square-root extraction: place dividend 81,000, coefficient 735, corner unity. Divide: initial quotient 240. Multiply quotient by coefficient: $240\times735=176{,}400$. Subtracting exceeds the dividend, so the initial quotient 240 is the diameter—the dividend is smaller than the coefficient times the quotient.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十二問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
三次方程釋：此問所得為天元三次方程。設城徑為 x，則方程形如 $x^3+330x^2=198000$。以天元術開立方：置實一十九萬八千，廉三百三十，隅一步。以隅約實，初商二百。以初商自乘乘隅，得四萬；又以初商乘廉，得六萬六千；并之得十萬六千，以減實餘九萬二千。再以初商二百乘隅，得二百，加入廉三百三十，得五百三十為下廉。以次商四十乘下廉，得二萬一千二百；又以次商四十自乘得一千六百，乘隅得一千六百；并之得二萬二千八百，以減餘實九萬二千不盡。復以次商加入下廉，再以三商乘之，開盡得二百四十步為城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 82: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Cubic equation explained: This yields a tian yuan cubic. Let diameter be x; the equation has form $x^3+330x^2=198{,}000$. By tian yuan cube-root extraction: place dividend 198,000, corner 330, corner divisor unity. Divide: initial quotient 200. Square 200, multiply by unity: 40,000; multiply 200 by 330: 66,000; sum 106,000; subtract from dividend: remainder 92,000. Multiply 200 by unity: 200; add to 330: 530 as lower corner. Next quotient 40 times 530: 21,200; plus 40 squared (1,600) times unity; total 22,800; subtract from remainder. Continue until exhausted: 240 paces.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十三問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。按：此草簡明直捷，為大股問中之平易者。以兩股之積為實，兩股之和為從，一步為隅，恰合天元平方方程之式。開平方即得，不須繁複之運算，乃初學天元者之津梁也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 83: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Note: This working is direct and concise—the simplest among large-leg problems. Product as dividend, sum as coefficient, unity as corner: exactly the tian yuan quadratic pattern. Square-root extraction suffices; no complex computation needed. The gateway for beginners in tian yuan.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十四問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
天元源流：天元術者，中國古代代數學之最高成就也。始見於《九章算術》之開方術，經唐宋諸賢之發展，至李冶先生而集大成。立天元一為所求之數，以各種條件建立方程，相消開方而得答數。此問以三次方程求城徑，正天元術之精妙所在。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 84: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Origins of Tian Yuan: The tian yuan method is the supreme achievement of ancient Chinese algebra. Originating in the root-extraction methods of *Jiuzhang Suanshu*, developed through the Tang and Song dynasties, and brought to full flower by Master Li Ye. Letting the celestial element be the sought quantity, establishing equations from given conditions, canceling and extracting roots for the answer. This problem uses a cubic equation to find the diameter—the very essence of tian yuan subtlety.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十五問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。又法：立天元一為半徑。甲南行六百步減二之天元（圓徑），得 $\frac{600}{2}$ 為大差。自之得大差冪，以甲南行冪減之半之為大弦。以大差乘甲南行併入和為小和，乘大弦寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消。開立方得半徑一百二十步，倍之得城徑二百四十步。此以半徑為天元之法也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 85: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Alternative: Let the celestial element be the radius. A's 600 minus twice the celestial element (the diameter) gives $\frac{600}{2}$ as the large difference. Square for large-diff-squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by A's walk, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120; double for diameter 240.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

測圓海鏡卷五終 — 凡一十八問

End of Ceyuan Haijing Volume 5 — 18 Problems

3 卷六：大勾一十八問・Vol. 6: Large Base (18 Problems)

大勾一十八問・Large Base

（第八十六問至第一百三問・Problems 86–103）

【題意總說】卷六論「大勾」之術。所謂大勾，乃最大勾股形之勾邊，即通勾也。以大勾為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大勾者，甲從乾隅直東至艮隅之步數，為三百二十步之率也。與大股相對而言，大股為南北向，大勾為東西向。此卷之問多設乙從東門直行若干步，甲從乾隅東行若干步望乙之狀，以建立天元方程。方程之次數較高，有三次、四次之方程，須開立方、開三乘方求之，為全書最精深之卷。

[General Summary] Volume 6 treats the “large base” (大勾) method. The large base is the base-side of the largest right triangle—the universal base. Taking the large base as principal, combined with sub-triangle relations, we seek the city diameter. All 18 problems. The large base is A’s eastward walk from the Qian corner to the Gen corner—the 320-pace standard. Contrasted with the large leg (north-south), the large base is east-west. This volume mostly posits B walking east from the east gate and A walking east from the Qian corner to view B, establishing tian yuan equations. Equations here reach higher degrees—third and fourth—requiring cube-root and fourth-root extraction. The most profound volume in the entire work.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得$\frac{32}{2}$為小差。半甲東行三百二十步，得一百六十步，為大勾之半。以乘小差，得 $\frac{5120}{320}$ 為半徑冪，寄左。然後以天元冪 $\frac{0}{1}$ 與左相消，得 $\frac{5120}{320\quad 1\quad 0}$ （上負下正），開平方得一百二十步，即半徑；倍之得全徑二百四十步。
釋義：此草以小差為主。小差者，半徑之倍加以乙東行之數也。甲東行三百二十步為大勾，其半即大勾之半。以小差乘大勾之半，得半徑冪，此天元術之巧妙處。與天元冪相消，開平方即得半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之東行內減二之乙東行為從，四益隅，得半徑。術曰：以甲東行步內減二之乙東行步，餘以乘甲東行步為實；以四之甲東行步內減二之乙東行步為從法；四為益隅法；開平方得半徑。

Problem 86: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B’s 16 paces gives $\frac{32}{2}$ as the small difference. Half of A’s 320-pace eastward walk is 160 paces, half the large base. Multiplying by the small difference gives $\frac{5120}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{5120}{320\quad 1\quad 0}$ (negative above, positive below). Extracting square root: 120 paces, the radius; doubling gives the full diameter of 240 paces.
Explanation: This working uses the small difference as key. The small difference is twice the radius plus B’s eastward walk. A’s 320 is the large base; its half is half the large base. Multiplying small difference by half the large base gives the radius-squared—the ingenious subtlety of tian yuan.

Method: A’s eastward walk minus twice B’s eastward walk, then multiplied by A’s walk, is the dividend. Four times A’s walk minus twice B’s walk is the coefficient; four is the negative corner; this yields the radius. Method: From A’s eastward walk subtract twice B’s eastward walk; multiply remainder by A’s walk as dividend; four times A’s walk minus twice B’s walk as coefficient; four as negative corner; extract square root for the radius.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得$\frac{16}{2}$為小差。半甲東行步，得一百六十步，以乘小差，得$\frac{2560}{320}$為半徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，開平方得一百二十步，即半徑。

按：此問與首問相類而異者，小差之立不同也。前問以二之天元加乙東行三十二步為小差，此問以二之天元加乙東行十六步為小差。蓋題設乙東行步數有異，故方程之係數隨之而變，天元術之靈活處也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得半徑。

Problem 87: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B's 16 paces gives $\frac{16}{2}$ as the small difference. Half A's eastward walk is 160; multiplying by the small difference gives $\frac{2560}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces, the radius.

Note: Similar to the first problem but the small difference is defined differently. The previous problem used 32 (twice B's walk) added to twice the celestial element; this uses 16. The problem conditions differ, so equation coefficients change accordingly—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the radius.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之加乙東行一十六步，得$\frac{16}{2}$為小差。置甲東行冪（一十萬二千四百步）內減小差冪，而半之，得大弦，帶小差分母。又置甲東行冪內減小差冪，而半之，得大股，帶小差分母。又以小差乘大股，併入和，得下式為小和，以乘大弦，寄左。又以倍天元加斜步，得小和，以乘大弦，得同數，與左相消。開立方得二百四十步，即城徑。

釋天元：此問立天元一為城徑而非半徑，故方程為三次，須開立方。小差以城徑之半加乙東行，與以半徑為天元者不同。學者須細察題意，決定所立天元之為半徑或全徑，乃能建立正確方程。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得城徑。

Problem 88: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice it plus B's 16 gives $\frac{16}{2}$ as the small difference. Place A's walk-squared (102,400); subtract small-diff-squared and halve for the large hypotenuse, carrying small-diff denominator. Also halve for the large leg with small-diff denominator. Multiply small difference by large leg, combine into the sum, obtain the small sum; multiply by large hypotenuse, deposit left. Add twice the celestial element to the diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 240 paces, the diameter.

Tian Yuan explained: Here the celestial element is the diameter, not the radius, so the equation is cubic, requiring cube-root extraction. The small difference uses half the diameter plus B's walk, different from using the radius. Students must examine the problem carefully to decide whether to use radius or diameter.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，斜行與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅為法，開三乘方得城徑二百四十步。
按：此問有斜行相會之狀，故方程為四次，須開三乘方。以甲東行內減二之乙東行，為建立方程之關鍵。乘甲東行為實，以四之甲東行內減二之乙東行為廉，四益隅為三乘方之隅法，開之得城徑。此全書中方程次數最高之問也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 89: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, then walks diagonally to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner for fourth-root extraction. Extracting fourth root gives diameter 240 paces.
Note: This problem involves diagonal meeting, so the equation is fourth-degree, requiring fourth-root extraction. The key step: A's walk minus twice B's walk. This multiplied by A's walk is the dividend; four times A's walk minus twice B's walk is the corner; four as negative corner. This problem has the highest-degree equation in the entire work.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大差之表示數。以此數乘甲東行冪（一十萬二千四百步），得二千九百四十九萬一千二百步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑。
細草：甲東行三百二十步自之，得一十萬二千四百步為甲東行冪。乙東行一十六步，二之为三十二步。以減甲東行，餘二百八十八步。以乘甲東行冪：$288 \times 102400 = 29491200$ 為實。四之甲東行一千二百八十步，減二之乙東行三十二步，餘一千二百四十八步為廉法。四為益隅法。開三乘方得二百四十步。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 90: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the large-difference representative. Multiplying by A's walk-squared (102,400) gives 29,491,200 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives the diameter.
Detailed working: A's 320 squared: 102,400 as A's walk-squared. B's 16, doubled: 32. Subtract from A's 320: 288. Multiply by walk-squared: $288 \times 102{,}400 = 29{,}491{,}200$ as dividend. Four times A's walk: 1,280; minus twice B's 32: 1,248 as corner divisor. Four as negative corner. Extract fourth root: 240 paces.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此數乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

釋三乘方：三乘方即四次方程也。設城徑為 x，則方程形如 $x^4+1248x^2=9216000$ 之類。以天元術開三乘方：置實，以隅法約之，初商二百。以次商乘廉法，又以次商自乘乘隅法，逐步減實，至實盡為止。其術繁複，非老於天元者不能為之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 91: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Fourth root explained: The "triple-multiplication root" is the fourth-degree equation. Let diameter be x; the equation has form $x^4 + 1248x^2 = 9{,}216{,}000$. By tian yuan fourth-root extraction: place dividend; divide by corner; initial quotient 200. Multiply next quotient by corner divisor, plus next quotient squared times corner, gradually subtracting from dividend until exhausted. The method is complex; only the experienced can perform it.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步為大勾，乙東行一十六步為小勾。兩勾相減，餘三百零四步為大小勾之差。以此差乘甲東行三百二十步，得九萬七千二百八十步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑二百四十步。

按：此草以兩勾差為主，與前數問以甲東行內減二之乙東行者不同。蓋題意有微妙之別，或以兩勾差、或以大勾減倍小勾為關鍵數，皆可建立方程而得同一城徑，天元術之奧妙也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 92: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 is the large base; B's 16 is the small base. Subtracting: 304 as the base-difference. Multiplying by A's 320: 97,280 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Note: This working uses the base-difference as key, different from previous problems using A's walk minus twice B's walk. Subtle distinctions in problem conditions allow either the base-difference or large-base-minus-twice- small-base as the key number, both yielding the same diameter—the profundity of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
源流考：天元術之開三乘方，源於《九章算術》之開方術。《九章》有開平方、開立方，而無開三乘方。至李冶先生著《測圓海鏡》，始廣之為開三乘方、開四乘方，以應天元四次方程之需。此問即開三乘方之典型例題，學者熟習此術，則高次方程皆迎刃而解矣。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 93: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Historical origins: Fourth-root extraction in tian yuan derives from the root-extraction methods of *Jiuzhang Suanshu*. That classic had square-root and cube-root extraction, but not fourth-root. Master Li Ye's *Ceyuan Haijing* extended it to fourth-root and fifth-root extraction to meet the needs of fourth-degree equations. This problem is the canonical example; mastering it, all higher-degree equations become soluble.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大小差之較。以此乘甲東行三百二十步，得九萬二千一百六十步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑。
又法：立天元一為半徑。以甲東行三百二十步減天元，得$\frac{320}{1}$為大差。以乙東行一十六步加天元，得$\frac{16}{1}$為小差。以大差小差相乘，得$\frac{5120}{336\quad 1}$為半徑冪與相關數之和，寄左。以天元冪$\frac{0}{1}$與左相消，得$\frac{5120}{336\quad 1\quad 0}$，開平方得一百二十步為半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 94: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the difference between large and small differences. Multiply by A's 320: 92,160 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.
Alternative: Let the celestial element be the radius. A's 320 minus the celestial element gives $\frac{320}{1}$ as the large difference. B's 16 plus the celestial element gives $\frac{16}{1}$ as the small difference. Multiply: $\frac{5120}{336\quad 1}$ as the sum of radius- squared and related terms, deposited left. Cancel with celestial element- squared $\frac{0}{1}$, obtaining $\frac{5120}{336\quad 1\quad 0}$. Extract square root: 120 paces, the radius.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

按：此卷之問多須開三乘方，較之卷四、卷五為難。蓋大勾之術以小差為主，小差含乙東行與半徑二項，故方程之次數自然較高。學者須先熟習開平方、開立方，然後進而開三乘方，循序漸進，乃能領會天元術之全體大用。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 95: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A’s 320 minus twice B’s 32 leaves 288. Multiply by A’s walk-squared: 29,491,200 as dividend. Four times A’s walk (1,280) minus twice B’s 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Note: Most problems in this volume require fourth-root extraction, harder than Volumes 4 and 5. The large-base method uses the small difference as key, which contains both B’s walk and the radius, so the equation degree is naturally higher. Students must master square-root and cube-root extraction before advancing to fourth-root, progressing step by step to comprehend the full power of tian yuan.

Method: A’s eastward walk minus twice B’s eastward walk, then multiplied by A’s walk, is the dividend. Empty coefficient; four times A’s walk minus twice B’s walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步為大勾，乙東行一十六步為小勾。以大勾之半一百六十步減乙東行一十六步，餘一百四十四步，為小差之半。倍之得二百八十八步，為小差之全。以此乘大勾三百二十步，得九萬二千一百六十步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

釋小差：小差者，圓城之半徑與乙東行之關係數也。以半徑之倍（即全徑）加乙東行為小差之全，或以半徑加乙東行之半為小差之半，二者皆可建立方程，而結果相同，此天元術之恆等變換也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 96: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A’s 320 is the large base; B’s 16 is the small base. Half the large base (160) minus B’s 16 leaves 144, half the small difference. Doubled: 288, the full small difference. Multiply by the large base 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Small difference explained: The small difference relates the radius to B’s eastward walk. Twice the radius (the diameter) plus B’s walk is the full small difference; or half the radius plus half B’s walk is half the small difference. Either can establish the equation with the same result—an identity transformation in tian yuan.

Method: A’s eastward walk minus twice B’s eastward walk, then multiplied by A’s walk, is the dividend. Empty coefficient; four times A’s walk minus twice B’s walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑。
術論：開三乘方之術，猶開平方、開立方之推廣也。置實於下，廉法於中，隅法於上。初商得後，以商乘廉、商自乘乘隅，各以減實。然後以商加廉為下廉，復以次商乘之，又以次商自乘乘隅減實，如是輾轉求之，至實盡為止。其理與開平方、開立方一脈相承。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 97: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.
On fourth-root extraction: It is the natural extension of square-root and cube-root extraction. Place dividend below, corner divisor in the middle, corner above. After obtaining the initial quotient, multiply quotient by corner and quotient-squared by corner, each subtracted from the dividend. Then add quotient to corner for lower corner; multiply by next quotient; add next quotient squared times corner; subtract from dividend. Continue thus until the dividend is exhausted. The principle is continuous with square-root and cube-root extraction.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
驗草：既得城徑二百四十步，則半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 $= 200^2 = 40000$，乙距冪 $= 136^2 = 18496$。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 98: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Verification: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40{,}000$; $136^2 = 18{,}496$. Sum: 58,496; square root ≈ 241.85, inexact—not the diagonal distance. Must use tian yuan separately.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
又草：立天元一為半徑。以甲東行三百二十步減二之天元（城徑），得$\frac{320}{2}$為大差。自之得大差冪 $\frac{102400}{1280\quad 4}$。以甲東行冪減之，半之得大弦。又以大差乘大勾併入和，得小和乘大弦，寄左。以倍天元加斜步為小和，乘大弦為同數，與左相消，開立方得半徑。
論天元之設：或問何以有立城徑為天元、有立半徑為天元?曰：此隨題之便而設也。題中條件多與城徑直接相關者，立城徑為天元；條件多與半徑相關者，立半徑為天元。所立雖異，所得則同，此天元術之靈活性也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 99: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Alternative: Let the celestial element be the radius. A's 320 minus twice the celestial element (diameter) gives $\frac{320}{2}$ as large difference. Squaring: $\frac{102400}{1280\quad 4}$ as large-diff-squared. Subtract from A's walk-squared and halve for large hypotenuse. Multiply large diff by large base, combine for small sum; multiply by large hypotenuse, deposit left. Add twice celestial element to diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root for radius.
On choosing the celestial element: Why use diameter for some problems and radius for others? It depends on convenience. If conditions relate to the diameter directly, use diameter; if to the radius, use radius. Different choices yield the same result—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步，斜距二百七十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

求斜距：既得城徑二百四十步，半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 $= 200^2 = 40000$，乙距冪 $= 136^2 = 18496$。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之：以大差二百步乘小差一百三十六步，得二萬七千二百步，加入半徑冪一萬四千四百步，得四萬一千六百步，開方得二百零四步，加上斜較六十八步，即得斜距二百七十二步。

按：求斜距之術較繁，須先求得城徑，然後以大小差之關係求斜行步數。全書各問求城徑為主，斜距之求乃其餘事也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 100: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

To find the diagonal: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40{,}000$; $136^2 = 18{,}496$. Sum: 58,496; root ≈ 241.85, inexact—not the diagonal. Must use tian yuan separately: large difference 200 times small difference 136 gives 27,200; adding radius-squared 14,400 gives 41,600; root ≈ 204; plus diagonal-difference 68 gives diagonal 272.

Note: Finding the diagonal is more complex; one must first obtain the diameter, then use large-small difference relations. The entire work focuses on finding the diameter; diagonal distance is secondary.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

四乘方程論：此問所得天元方程為四次方程，中國古代謂之「三乘方」。西人謂之「四次方程」，以 x^4 為最高次項。李冶先生之天元術，以籌算排列係數，自下而上為常數項、一次項、二次項、三次項、四次項，與現代代數學之降冪排列相反，然其理則一也。開三乘方之法，亦與西洋之數值解法相通，可見數學為天下之公器，無中外之殊也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 101: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

On quartic equations: This yields a tian yuan fourth-degree equation, called "triple-multiplication root" in Chinese. Westerners call it a quartic equation, with x^4 as the highest term. Master Li Ye's tian yuan method arranges coefficients in counting-rod style, bottom-to-top: constant, linear, quadratic, cubic, quartic—the reverse of modern algebra's descending powers, yet the principle is identical. Fourth-root extraction corresponds to Western numerical methods, showing mathematics is a universal tool without national boundaries.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問大差小差各幾何？

答曰：城徑二百四十步。大差二百步，小差一百三十六步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
求大差：甲東行三百二十步減城徑二百四十步，餘八十步，非大差也。當以甲東行減半徑：$320 - 120 = 200$ 步為大差之正數。
求小差：乙東行一十六步加半徑一百二十步，得一百三十六步為小差。
驗大差小差：大差二百步，小差一百三十六步。以相乘得 $200 \times 136 = 27200$。大差冪 $= 40000$，小差冪 $= 18496$。以大小差冪之和減甲東行冪：$102400 - (40000 + 18496) = 43904$，半之得 21952。此數當與大弦之某式相合，以驗天元方程之正確性。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 102: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what are the large difference and small difference?

Answer: City diameter: 240 paces. Large difference: 200 paces. Small difference: 136 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
To find the large difference: A's 320 minus diameter 240 leaves 80—not the large difference. Correctly: A's walk minus radius: $320 - 120 = 200$ paces as the true large difference.
To find the small difference: B's 16 plus radius 120 gives 136 paces.
Verification: Large difference 200, small difference 136. Product: $200 \times 136 = 27{,}200$. Large-diff-squared: 40,000; small-diff-squared: 18,496. Sum subtracted from A's walk-squared: $102{,}400 - (40{,}000 + 18{,}496) = 43{,}904$; halved: 21,952. This should match a formula for the large hypotenuse, verifying the tian yuan equation.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？又問通勾股形之弦長幾何？

答曰：城徑二百四十步。通弦三百九十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

求通弦：通勾股形者，以甲東行為勾、甲南行為股之大勾股形也。甲東行三百二十步為勾，甲南行六百步為股。以勾股求弦：$\sqrt{320^2+600^2}=\sqrt{102400+360000}=\sqrt{462400}=680$ 步。然此乃大弦之全數也。其帶分母者，當以大差除之。大差為甲東行減半徑，即 $320-120=200$ 步。以 680 步為分子，200 步為分母，得 3.4 步，非通弦也。當以天元正術求之：以大差冪加小差冪，減甲東行冪，半之，加入大小差相乘之積，開方得通弦三百九十二步。

按：通弦之求甚繁，須先求得大小差，然後以大小差之關係建立方程求之。此問為全卷之終，特設通弦之問，以見天元術之無所不能。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。術曰：置甲東行步內減二之乙東行步，餘以乘甲東行冪為實；以四之甲東行步內減二之乙東行步為廉法；四為益隅法；開三乘方得城徑。

Problem 103: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter? Also, what is the hypotenuse of the universal right triangle?

Answer: City diameter: 240 paces. Universal hypotenuse: 392 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

To find the universal hypotenuse: The universal right triangle has A's eastward walk as base and A's southward walk as leg. Base 320, leg 600. By the gougu theorem: $\sqrt{320^2+600^2}=\sqrt{102{,}400+360{,}000}=\sqrt{462{,}400}=680$ paces. But this is the full large hypotenuse. The version with denominator: divide by large difference. Large difference = A's walk minus radius: $320-120=200$ paces. $680/200=3.4$ paces—not the universal hypotenuse. Must use orthodox tian yuan: large-diff-squared plus small-diff-squared, minus A's walk-squared, halved, plus product of large and small differences; extract root: 392 paces.

Note: Finding the universal hypotenuse is complex; one must first obtain the large and small differences, then establish equations. This final problem deliberately asks for the hypotenuse to demonstrate that tian yuan can solve anything.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter. Method: From A's eastward walk subtract twice B's eastward walk; multiply remainder by A's walk-squared as dividend; four times A's walk minus twice B's walk as corner divisor; four as negative corner; extract fourth root.

測圓海鏡卷六終 — 凡一十八問

End of Ceyuan Haijing Volume 6 — 18 Problems

附錄：全書結構及術語解釋・Appendix: Structure and Glossary

《測圓海鏡》全書十二卷，共一百七十問，為元代李冶先生之傑作。其書以勾股測圓為主，設圓城一座，甲乙二人從城門出行，以行步之數推求圓徑。全書結構如下：

卷	篇名	問數	起止問
卷一	圓城圖式	—	圖說
卷二	正率	14	第 1–14 問
卷三	邊股	17	第 15–31 問
卷四	底勾	17	第 32–48 問
卷五	大股	18	第 49–66 問
卷六	大勾	18	第 67–84 問
卷七	明勾	16	第 85–100 問
卷八	明股	16	第 101–126 問
卷九	雜題上	18	第 127–144 問
卷十	雜題中	18	第 145–162 問
卷十一	雜題下	18	第 163–180 問
卷十二	雜題終	10	第 181–190 問
	合計	**170**	

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes and 170 problems, is the masterpiece of Master Li Ye of the Yuan Dynasty. The work uses right-triangle circle measurement, positing a circular city from whose walkers' paces the diameter is derived. The complete structure:

Vol.	Title	Probs.	Range
1	City Diagram	—	Diagram
2	Standard Rates	14	1–14
3	Side Leg	17	15–31
4	Base Leg	17	32–48
5	Large Leg	18	49–66
6	Large Base	18	67–84
7	Bright Base	16	85–100
8	Bright Leg	16	101–126
9	Misc. Part I	18	127–144
10	Misc. Part II	18	145–162
11	Misc. Part III	18	163–180
12	Misc. Finale	10	181–190
	Total	**170**	

術語解釋・**Glossary of Terms:**

- 天元一：設未知數為一，即現代代數之 x。李冶先生以「立天元一」為某數，即設未知數為所求之量。
- 寄左：將某式暫存一邊，相當於方程之一邊。天元術中常以「寄左」表示方程之左邊，後以「同數」與之相消。
- 相消：兩式相減，相當於移項合併。天元術之核心步驟，以同數與寄左之式相減，得開方式。
- 帶分母：某式含有分母未除。古代算術中常「不受除」，即以帶分母之形式保留，以待後續運算。
- 冪：平方，即現代記號 x^2。天元術中以某數自乘為其冪。
- 開方：開平方，即求平方根。天元術中最基本之運算。
- 開立方：開立方，即求立方根。用於三次天元方程。
- 開三乘方：開四次方，即求四次根。用於四次天元方程，為天元術中最高深之運算。
- 實：常數項，即方程中之已知數部分。
- 從：一次項係數，天元術中「從法」即 x 之係數。
- 廉：二次項係數，即 x^2 之係數。
- 隅：三次項係數（立方方程中），即 x^3 之係數。四次方程中另有「三乘隅」或「益隅」之稱。
- 勾：直角三角形之短邊（底邊）。
- 股：直角三角形之長邊（高邊）。
- 弦：直角三角形之斜邊。
- 大勾：最大勾股形之勾邊，即通勾。
- 大股：最大勾股形之股邊，即通股。
- 大差：大股減圓徑之餘數。
- 小差：小勾加半徑之和。
- 底勾：勾股形之最下勾股，與圓徑相切之基本勾股形之勾邊。

English Glossary:

- **Tian yuan yi (天元一)**: The celestial element unity; the unknown x. “Let tian yuan unity be [quantity]” means: let x equal the sought quantity.
- **Ji zuo (寄左)**: Deposit on the left; temporarily store an expression on one side of the equation. The “left” side, later canceled with the “equal number.”
- **Xiang xiao (相消)**: Mutual cancellation; subtracting two expressions, equivalent to transposition and combining terms. The core step of tian yuan.
- **Dai fen mu (帶分母)**: Carrying a denominator; an expression with an uncanceled denominator, retained for later operations.
- **Mi (冪)**: Power; square, i.e., x^2 in modern notation.
- **Kai fang (開方)**: Root extraction; square-root extraction, the fundamental operation of tian yuan.
- **Kai li fang (開立方)**: Cube-root extraction; for cubic equations.
- **Kai san cheng fang (開三乘方)**: Fourth-root extraction; for quartic equations—the most advanced operation in tian yuan.
- **Shi (實)**: Dividend; the constant term of the equation.
- **Cong (從)**: Coefficient; the linear term coefficient (x term).
- **Lian (廉)**: Corner; the quadratic term coefficient (x^2 term).
- **Yu (隅)**: Corner divisor; the cubic term coefficient (x^3 term). In quartics: “triple-multiplication corner” or “negative corner.”
- **Gou (勾)**: Base; the shorter leg of a right triangle.
- **Gu (股)**: Leg; the longer leg of a right triangle.
- **Xian (弦)**: Hypotenuse; the diagonal side.
- **Da gou (大勾)**: Large base; the universal base.
- **Da gu (大股)**: Large leg; the universal leg.
- **Da cha (大差)**: Large difference; large leg minus diameter.
- **Xiao cha (小差)**: Small difference; small base plus radius.
- **Di gou (底勾)**: Base leg; the lowest triangle tangent to the circle.

天元術方程式記法・**Tian Yuan Equation Notation:**
李冶先生以籌算式記多項式，自下而上排列：

- 最下為常數項（實）
- 其上為一次項（從）
- 再其上為二次項（廉）
- 再其上為三次項（隅）

例如 $\frac{480}{1}$ 表示 $480 - x$（一次式），$\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ 表示 $48000 - x^2$ 之類。其中係數為負者以斜線或特殊記號表示之。

各卷方程次數統計・**Equation Degree Statistics:**

卷	平方（二次）	立方（三次）	三乘方（四次）
卷四（底勾）	12 問	4 問	1 問
卷五（大股）	6 問	10 問	2 問
卷六（大勾）	3 問	6 問	9 問

由上表可見，愈至卷六，方程之次數愈高，開方之術愈繁，乃李冶先生循序漸進、由淺入深之教學法也。

Tian Yuan Equation Notation:
Master Li Ye recorded polynomials in counting-rod style, arranged bottom-to-top:

- Lowest: constant term (dividend, shí)
- Above: linear term (coefficient, cóng)
- Above that: quadratic term (corner, lián)
- Above that: cubic term (corner divisor, yú)

Thus $\frac{480}{1}$ denotes $480 - x$ (linear), and $\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ denotes $48000 - x^2$. Negative coefficients were indicated by slanted lines or special marks.

Equation Degree Statistics by Volume:

Volume	Quadratic	Cubic	Quartic
Vol. 4 (Base Leg)	12	4	1
Vol. 5 (Large Leg)	6	10	2
Vol. 6 (Large Base)	3	6	9

As the table shows, equation degrees increase toward Volume 6, and root- extraction methods become more complex. This is Master Li Ye's pedagogical method: gradual progression from simple to profound.

測圓海鏡卷四至卷六全編終

End of Ceyuan Haijing Volumes 4–6

元　李冶　撰　Yuan Dynasty　Li Ye

依《欽定四庫全書》子部天文算法類本

Based on the Imperial Siku Quanshu, Mathematics and Astronomy Class

雙語版　Bilingual Edition

中文版左欄　Chinese Text Left Column

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測圓海鏡

（卷七至卷九·英漢對照本）

李冶　撰

1248

Sea Mirror of Circle Measurements

Volumes 7–9: Bilingual Chinese-English Edition

Translated and typeset in parallel columns

卷七　明前一十八問　（第 105 問–第 122 問）
　　Vol. 7: 18 "Front" Problems (Q. 105–122)
卷八　明後一十六問　（第 123 問–第 138 問）
　　Vol. 8: 16 "Rear" Problems (Q. 123–138)
卷九　大斜四問、大和八問　（第 139 問–第 150 問）
　　Vol. 9: 4 Oblique + 8 Sum Problems (Q. 139–150)

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Bilingual edition with parallel English translation

Contents

1 緒論

Introduction

測圓海鏡者，金元之際李冶潤甫先生之所著也。先生諱治，字仁卿，號敬齋，真定欒城人，生於金明昌元年（一一九二），卒於元至元十六年（一二七九）。此書成於淳祐八年（一二四八），凡十二卷，計一百七十問，皆設圓城以為問，用天元術以立算，實中算之鴻篇也。

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by Li Ye (Li Zhi, 1192–1279) during the Jin-Yuan transition period. A native of Luancheng, Zhending, he completed this work in 1248. The book comprises 12 volumes with 170 problems, all concerning a circular walled city, solved by the tian yuan shu (heavenly element method)—a masterpiece of Chinese mathematics.

其法之要，在於天元一術。立天元一為所求之數，依題意布算，得開方式，然後開方求之。所謂太極、天元、地元者，猶今代數之未知數也。書中凡遇開方式，皆以 ■ 記之，蓋當時算板布列之位也。

The essence of the method lies in the tian yuan technique: set the celestial unknown as the quantity sought, derive a polynomial equation (the “opening pattern”) according to the problem conditions, then extract the root. The terms Taiji, Tian Yuan, and Di Yuan correspond to algebraic unknowns. The symbol ■ marks positions on the counting board where polynomial coefficients were arranged.

卷七至卷九所載，凡三十四問，其題目之繁，方程式次之高，皆過於前六卷。卷七曰「明前一十八問」，卷八曰「明後一十六問」，卷九曰「大斜四問、大和八問」。其問題之設，或二人對行，或四處遙望，或穿城而過，或倚樹而參，錯綜變化，不可勝窮。

Volumes 7–9 contain 34 problems of greater complexity than the first six volumes, with equations reaching the 4th–6th degree. Volume 7 treats “front” configurations, Volume 8 “rear” configurations, and Volume 9 the most challenging “oblique” and “sum” problems. The scenarios involve pairs or groups of observers, tree landmarks, and gates around the circular city.

體例說明

每問之體，例分三節：

- **或問**：設問之辭，述人物行止之狀、所給之數。
- **法曰**：列方程式之術，示開方之次。
- **草曰**：詳布算之草，盡天元變化之妙。

又有時設**答曰**一節，直書所求之數，以為開方之驗。

Structure of Each Problem

Each problem follows a standard format:

- ***Q* (或問)**: *The problem statement, describing the scene and given quantities.*
- ***Method* (法曰)**: *The polynomial setup and root-extraction procedure.*
- ***Working* (草曰)**: *Detailed step-by-step derivation using tian yuan.*

*An **Answer** **(答曰)** section sometimes states the result directly as verification.*

書中所用十五形之名，皆有所本。通勾、通股者，大勾股形之兩邊也；邊勾、邊股者，邊上之小形也；底勾、底股者，底下之小形也。明勾、明股者，明朗易見之小形也；重勾、重股者，重疊相因之小形也。至於虛勾、虛股、虛弦，則太虛之形，最為微妙。

*The text uses fifteen named right triangles. **Tong** (通) gou/gu are the legs of the main large triangle. **Bian** (邊) and **Di** (底) gou/gu belong to smaller edge/base triangles. **Ming** (明, “bright”) and **Zhong** (重, “overlapping”) triangles are visually apparent. The **Xu** (虛, “void”) triangle is the most subtle—the inner void space. Each problem typically asks for the city diameter (城徑): **240 paces**.*

2 卷七 明前一十八問

Vol. 7: 18 "Front" Problems

卷七小序：此卷一十八問，皆以明段為主。所謂明段者，勾股形之顯然可見者也。其題多設二人對行、遙望相會之狀，或倚門傍樹，或穿城越隍，錯綜參伍，以盡天元之變。其方程式之高，有至三乘、四乘者，開方之法亦隨之益密。

***Volume 7 Preface**: These 18 problems all concern "bright" (ming) segments—the visibly apparent right-triangle edges. Scenarios involve pairs walking toward each other, gazing from afar, leaning by gates or trees, passing through the city. The equations reach degrees 3–4, with increasingly refined root-extraction techniques.*

第一百零五問

Problem 一百零五

或問：出南門東行七十二步有樹，出東門南行三十步見之。問答同前。

Q: *Going out the South Gate and walking 72 paces east there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.*

答曰：城徑二百四十步，半徑一百二十步。

A: **City diameter: 240 paces. Radius: 120 paces.**

法曰：倍南行以乘倍東行為平實，並二行又倍之為從，一虛隅。得城徑。

Method: Double the southward walk, multiply by double the eastward walk as the flat shi. Add the two walks and double as cong. One empty corner. Extract the city diameter.

草曰：識別得此問名為弦外容圓，又為內率求虛積。其二行步相並為虛弦，若以相減即虛較也。又倍東行為弦較和，倍南行即弦較較，此二數相乘則兩虛積也。若直以二行相乘，則半個虛積也。又倍東行減於城徑，餘即二虛勾也。倍南行減於城徑則二虛股也。虛積上三事和即城徑也。

乃立天元一為圓徑，便以為三事和也。倍二行步減之，得 □ 為黃方一，天元乘之得 □ 為二虛積（寄左）。然後倍東行以乘倍南行，得八千六百四十為同數，與左相消得 □。益積開平方得二百四十步，即城徑也。合問。

Working: Identification: This problem is "circle contained outside the hypotenuse" and also "seeking the void area by inner rate." The sum of the two walks is the void hypotenuse; their difference is the void difference. Double the eastward walk gives the sum of hypotenuse and difference; double the southward walk gives their difference. Their product is twice the void area; direct multiplication gives half the void area. The city diameter minus double the eastward walk gives twice the void gou; minus double the southward walk gives twice the void gu. The three-element sum on the void area equals the city diameter.

Let the celestial unknown be the circle diameter, as the three-element sum. Subtracting double the two walks gives ■ as the yellow square; multiplying by the unknown gives ■ as twice the void area (stored left). Then double the eastward walk times double the southward walk gives 8640 as the matching number. Canceling gives ■. Accumulating and extracting the square root yields 240 paces, the city diameter. Q.E.D.

第一百零六問

Problem 一百零六

或問：丙出南門直行一百三十五步而立，甲出東門直行一十六步見之。問答同前。

Q: *Bing goes out the South Gate and walks straight 135 paces, then stands. Jia goes out the East Gate and walks straight 16 paces and sees him. Answer as before.*

答曰：城徑二百四十步，丙行上勾弦差八十一步。

A: **City diameter: 240 paces. The difference between gou and hypotenuse on Bing's walk: 81 paces.**

法曰：以丙行步一百三十五再自之，得二百四十六萬零三百七十五於上。又以甲行一十六乘丙行冪一萬八千二百二十五，得二十九萬一千六百，以乘上位，得七千一百七十四億四千五百三十五萬為三乘方實；以二行步相乘又倍之得四千三百二十，以乘丙行步再自之數，得一百六億二千八百八十二萬為

Method: Raise Bing's walk 135 to the third power, obtaining 2,460,375 as upper term. Multiply Jia's walk 16 by Bing's squared walk 18,225 to get 291,600; multiply by the upper term to obtain 717,445,350,000 as the cubic equation's shi. Double the product of the two walks to get 4,320; multiply by the cube of Bing's walk

益從；第一廉空；以甲行乘丙行冪，得二十九萬一千六百，又倍之得五十八萬三千二百於上，四之甲行冪一千零二十四，以乘丙行步，得一十三萬八千二百四十，減上位餘四十四萬四千九百六十為第二廉；二行步相乘得二千一百六十為虛常法。得丙行步上勾弦差八十一。

草曰：識別二數相並，得一百五十一。以減於皇極弦，餘一百三十八，即虛勾虛股並也。若以二數相減，餘一百一十九為高弦內減平弦，又為皇極弦內少個小差弦，又為大差弦內減個皇極弦也。

立天元一為丙行大差數。置丙行步一百三十五，自乘得 □，用天元除之，得 □ 為勾弦並也。上減天元得 □ 為二丙勾也。復用丙南行乘之，得 □ 為二積也。又以天元除之，得 □ 為丙勾外容圓半（泛寄）。別置丙南行用二甲勾乘之，得 □ 太，合用二丙勾除之。不受除，便以此為甲股（內寄二丙勾為分母）。復用二甲勾三十二乘之，得 □ 太為二個甲直積也。又置丙南行內減天元，得 □ 為黃方，以自乘得 □ 為丙上勾弦差乘股弦差二段，以天元除之得 □ 為兩個丙小差也。乃用甲股乘之，得下式 □。復用丙南行除之，得 □，又折半得下式 □ 為一個甲步股弦差也，內亦帶前二丙勾分母。復置兩個甲直積，內已寄此甲股弦差分母，便為甲步股外容圓半（寄左）。乃再置先求到泛寄，用甲股弦差分母乘之，得 □ 為同數，與左相消得下式 □。開三乘方得八十一步，即丙步上勾弦差也。

《鈐經》載此法，以勾弦差率冪減丙行冪，復以丙行乘之為實，以差率冪為法，如法得徑。此法只是以勾外求圓半。合以大差除倍積，而今皆以大差冪為分母也。依法求之，勾弦差八十一自之得六千五百六十一，以減於丙行冪一萬八千二百二十五，餘一萬一千六百六十四，復以丙行一百三十五乘之，得一百五十七萬四千六百四十為實，以大差冪六千五百六十一為法，如法得二百四十步，即城徑也。

to obtain 10,628,820,000 as increasing cong. First lian is empty. Multiply Jia's walk by Bing's squared walk to get 291,600; double to get 583,200 as upper term. Quadruple Jia's squared walk 1,024 and multiply by Bing's walk to get 138,240; subtract from upper term, leaving 444,960 as second lian. The product of the two walks, 2,160, is the empty constant method. Extracts the gou-hypotenuse difference on Bing's walk: 81.

Working: Identification: The sum of the two numbers is 151. Subtracting from the imperial extreme hypotenuse leaves 138, the sum of void gou and void gu. Their difference, 119, is the high string minus the level string, also the imperial extreme string lacking a small-difference string, also the large-difference string minus the imperial extreme string.

Let the unknown be the large-difference number for Bing's walk. Square Bing's walk 135 to get ■; divide by the unknown to get ■ as the sum of gou and hypotenuse. Subtract the unknown above to get ■ as twice Bing's gou. Multiply by Bing's southward walk to get ■ as twice the area. Divide by the unknown to get ■ as the half-circle (general storage). Multiply Bing's southward walk by twice Jia's gou to get ■ (this becomes Jia's gu with denominator). Multiply by twice Jia's gou 32 to get ■ as twice Jia's rectangular area. Subtract the unknown from Bing's walk to get ■ as the yellow square; squaring gives ■ as two segments of the product of gou-hypotenuse difference and gu-hypotenuse difference. Dividing by the unknown gives ■ as two small differences. Multiply by Jia's gu to get ■. Divide by Bing's southward walk and halve to get ■ as one step of Jia's gu-hypotenuse difference. Extracting the cubic root yields 81, the gou-hypotenuse difference.

The *Qianjing* records: square the gou-hypotenuse difference rate and subtract from Bing's walk square, then multiply by Bing's walk as shi; use the squared difference rate as divisor to obtain the diameter. This method seeks the half-circle from outside the gou. The large difference should divide twice the area; here the squared large difference serves as denominator. Accordingly: square 81 to get 6,561; subtract from Bing's walk square 18,225, leaving 11,664; multiply by 135 to get 1,574,640 as shi; divide by squared large difference 6,561 to obtain 240 paces, the city diameter.

第一百零七問

或問：出東門一十六步有樹，出南門東行七十二步見之。問答同前。

答曰：城徑二百四十步。

法曰：二行步相減得數，以自之於上。又以出東門步自之，減上位為平方實，二之出南門東行步為益從，一步常法。翻開得半徑。

Problem 一百零七

Q: *Going out the East Gate 16 paces there is a tree. Going out the South Gate and walking 72 paces east, one sees it. Answer as before.*

A: City diameter: 240 paces.

Method: Subtract the two walks; square the result as upper term. Square the East Gate walk and subtract from upper term as the quadratic shi. Double the south-gate eastward walk as increasing cong. One constant method. Extract by reversing to obtain the

草曰：別得人到樹即平弦也，半圓徑即平股也。其東行七十二步則平勾平弦差也。乃立天元一為半圓徑，加一十六減七十二，得 □ 為勾也。以自之得 □ 為勾冪。又加入天元股冪得 □ 為弦冪（寄左）。再立天元一為半徑，加出東門步，得 □ 即弦也。以自之得 □ 為同數，與左相消得 □。翻法開之得一百二十步，即半城徑也。合問。

radius.

Working: Identification: The person reaching the tree is the level hypotenuse; half the diameter is the level gu. The 72 paces east is the difference between level gou and level hypotenuse. Let the unknown be half the diameter; add 16 and subtract 72 to get ■ as gou. Squaring gives ■ as gou-power. Add the unknown squared as gu-power to get ■ as hypotenuse-power (stored left). Again let the unknown be the radius; add the East Gate paces to get ■ as the hypotenuse. Squaring gives ■ as the matching number. Canceling gives ■. Extract by the reversing method to obtain 120 paces, half the city diameter. Q.E.D.

第一百零八問

Problem 一百零八

或問：出南門一百三十五步有樹，出東門南行三十步見之。問答同前。

Q: *Going out the South Gate 135 paces there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.*

答曰：城徑二百四十步。

A: **City diameter**: **240 paces.**

法曰：樹去城步內減南行步，餘以為冪於上，又以樹去城步為冪內減上位為平實，倍樹去城步為從，一虛隅。翻法得半城徑。

Method: Subtract the southward walk from the tree-to-city distance; square the remainder as upper term. Square the tree-to-city distance and subtract the upper term as flat shi. Double the tree-to-city distance as cong. One empty corner. Extract by reversing to obtain half the city diameter.

草曰：別得人距樹即高弦也，半圓徑即高勾也，其南行三十步即高弦上小差也。乃立天元一為半徑，加樹去城步為弦，內減小差 □，得 □ 即股也。以自之得 □ 為股冪，內加入天元冪，得 □ 為弦冪（寄左）。再置弦 □ 以自之，得 □ 為同數，與左相消，得式 □。翻開得一百二十步，即半城徑也。合問。

Working: Identification: The person's distance to the tree is the high hypotenuse; half the diameter is the high gou. The 30 paces southward is the small difference on the high hypotenuse. Let the unknown be the radius; add the tree-to-city distance as hypotenuse, subtract the small difference ■ to get ■ as gu. Squaring gives ■ as gu-power; add the unknown squared to get ■ as hypotenuse-power (stored left). Again place hypotenuse ■, square it to get ■ as matching number. Canceling gives ■. Extract by reversing to obtain 120 paces, half the city diameter. Q.E.D.

第一百零九問

Problem 一百零九

或問：乙出東門不知遠近而立，甲出南門東行七十二步望見乙，就乙斜行一百三十六步與乙相會。問答同前。

Q: *Yi goes out the East Gate and stands at an unknown distance. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 136 paces to meet Yi. Answer as before.*

答曰：城徑二百四十步，乙出東門一百二十八步。

A: **City diameter**: **240 paces. Yi went out the East Gate 128 paces.**

法曰：以斜行步自之於上，以二行相減餘自為冪，減上位為平實，從空，一步常法。如法得半徑。

Method: Square the diagonal walk as upper term. Subtract the two walks; square the remainder and subtract from upper term as flat shi. Cong is empty. One constant method. Extract to obtain the radius.

草曰：別得七十二步即大差也，斜行即弦，半徑即股也。立天元一為半徑，以自之為股冪，又以二行差六十四以自之得 □ 為勾冪。並二冪得 □ 為弦冪（寄

Working: Identification: The 72 paces is the large difference; the diagonal is the hypotenuse; the radius is the gu. Let the unknown be the radius; square it as gu-power. The

左）。然後以斜行步自之，得 □ 為同數，與左相消得 □。開平方得一百二十步，倍之即城徑也。合問。

difference of the two walks, 64, squared gives ■ as gou-power. Adding both powers gives ■ as hypotenuse-power (stored left). Then square the diagonal walk to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 120 paces; double it for the city diameter. Q.E.D.

第一百一十問

Problem 一百一十

或問：甲出南門不知遠近而立，乙出東門南行三十步望見甲，卻就甲斜行二百五十五步與甲相會。問答同前。

答曰：城徑二百四十步，甲出南門二百二十五步。

法曰：二行差自之為冪，以減於斜行冪為平實，一虛隅。得半徑。

草曰：別得南行步即股弦差也，斜步即弦也，半徑即勾也。乃立天元一為半城徑，以自之為冪。以二行相減餘二百二十五，以自之得 □ 為股冪。二冪相並得 □ 為弦冪（寄左）。然後以斜行步自之，得 □ 為同數，與左相消得下 □。開平方得一百二十步，即半徑也。合問。

Q: *Jia goes out the South Gate and stands at an unknown distance. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 255 paces to meet Jia. Answer as before.*

A: City diameter: 240 paces. Jia went out the South Gate 225 paces.

Method: Square the difference of the two walks; subtract from the squared diagonal as flat shi. One empty corner. Extract the radius.

Working: Identification: The southward walk is the difference between gu and hypotenuse; the diagonal is the hypotenuse; the radius is the gou. Let the unknown be half the city diameter; square it as power. The difference of the two walks, 225, squared gives ■ as gu-power. Adding both powers gives ■ as hypotenuse-power (stored left). Then square the diagonal walk to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 120 paces, the radius. Q.E.D.

第一百一十一問

Problem 一百一十一

或問：甲出南門東行不知步數而立，乙出東門南行三十步望見甲，斜行一百二步相會。問答同前。

答曰：城徑二百四十步，甲東行九十六步。

法曰：二行相減，餘以乘乙南行，四之於上，又加入斜行冪為平實。得虛和一百三十八。

草曰：別得斜步內減南行為甲東行步也。此問以弦外容圓入之，以二行相減數乘乙南行三十步，得 □，又四之，得 □ 為二直積也。又加入斜步冪 □，共得 □ 即和冪也。平方而一，得一百三十八步，即虛和也。又加斜步得二百四十步，即城徑也。合問。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 102 paces to meet Jia. Answer as before.*

A: City diameter: 240 paces. Jia walked 96 paces east.

Method: Subtract the two walks; multiply the remainder by Yi's southward walk; quadruple as upper term; add the squared diagonal as flat shi. Extracts the void sum: 138.

Working: Identification: The diagonal minus the southward walk gives Jia's eastward walk. This problem uses "circle outside the hypotenuse." Multiply the difference of the two walks by Yi's 30 paces south to get ■; quadruple to get ■ as twice the rectangular area. Add the squared diagonal ■, totaling ■ as the squared sum. Extract the square root to obtain 138, the void sum. Add the diagonal to get 240 paces, the city diameter. Q.E.D.

第一百一十二問

Problem 一百一十二

或問：乙出東門南行不知步數而立，甲出南門東行七十二步望見乙，斜行一百二步與乙相會。問答同前。

答曰：城徑二百四十步，乙南行五十四步。

法曰：倍相減步以乘倍東行，得數復以減於斜步冪，餘為實。平方而一，得較也。又以二行相減數乘倍東行為平實，以較為從方，得勾。勾較共為長，又以斜步並入勾股共，即城徑。

草曰：別得二行相減餘 □ 為乙南行步也。以此數又減於甲東行，餘四十二步即較也。又以二行相減數 □ 乘倍東行得 □ 為平實，以較為從。平方開得四十八即勾也。勾內加較得九十步即股也。勾股共得一百三十八，又加入斜步，共得二百四十步，即城徑也。合問。

Q: *Yi goes out the East Gate and walks south an unknown number of paces, then stands. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 102 paces to meet Yi. Answer as before.*

A: **City diameter: 240 paces. Yi walked 54 paces south.**

Method: Double the subtraction step and multiply by double the eastward walk; subtract this from the squared diagonal, remainder as shi. Extract the square root to obtain the difference. Multiply the difference of the two walks by double the eastward walk as flat shi; use the difference as cong to extract gou. Gou plus difference gives the length; add the diagonal to the gou-gu sum for the city diameter.

Working: Identification: The difference of the two walks, ■, is Yi's southward walk. Subtract this from Jia's eastward walk; the remainder 42 is the difference. Multiply the difference of the two walks ■ by double the eastward walk to get ■ as flat shi; use the difference as cong. Extract the square root to obtain 48, the gou. Add the difference to get 90 paces, the gu. Gou plus gu totals 138; add the diagonal to obtain 240 paces, the city diameter. Q.E.D.

第一百一十三問

Problem 一百一十三

或問： 乙出南門東行，甲出東門南行，兩相望見。既而乙雲：「我東行不及城徑一百六十八步。」甲雲：「我南行不及城徑二百一十步。」問答同前。

答曰：城徑二百四十步，甲南行三十步，乙東行七十二步。

法曰：半甲不及步以自之為冪，半甲不及步內減差以自之為冪。二冪相並內卻減差冪為平實，四之甲不及內減三之乙不及，餘為益從，三步半虛法。得甲南行。

草曰：別得乙不及為虛勾、半徑共，又為徑內減明勾也。甲不及為虛股、半徑共，又為徑內減股也。又二雲數相並為虛和、圓徑共也，雲數相減即虛較也。

乃立天元一為甲南行，以減於甲不及步又半之，得 □ 為虛股也。虛股內減虛較得 □ 為虛勾。勾自之得 □ 為勾冪也，又股自之下式 □ 為股冪也。二冪相並得 □ 為弦冪（寄左）。然後以天元加虛較得 □ 為乙東行。又加入天元甲南行得 □ 為虛弦，以自之得 □ 為同數，與左相消得 □。開平方得三十步，即甲南行也。內加少步，即城徑也。合問。

Q: *Yi goes out the South Gate walking east; Jia goes out the East Gate walking south; they see each other. Yi says: "My eastward walk falls short of the diameter by 168 paces." Jia says: "My southward walk falls short of the diameter by 210 paces." Answer as before.*

A: **City diameter: 240 paces. Jia walked 30 paces south; Yi walked 72 paces east.**

Method: Halve Jia's shortfall and square it as one power. Subtract the difference from half Jia's shortfall and square as second power. Add both powers and subtract the squared difference as flat shi. Quadruple Jia's shortfall minus triple Yi's shortfall as increasing cong. Three-and-a-half empty method. Extracts Jia's southward walk.

Working: Identification: Yi's shortfall is the void gou plus radius, also the diameter minus the bright gou. Jia's shortfall is the void gu plus radius, also the diameter minus the gu. The sum of the two given numbers is the void sum plus circle diameter; their difference is the void difference.

Let the unknown be Jia's southward walk. Subtract from half Jia's shortfall to get ■ as void gu. Subtract the void difference from void gu to get ■ as void gou. Square the gou to get ■ as gou-power; square the gu to get ■ as gu-power. Adding both gives ■ as hypotenuse-power (stored left). Add the unknown to the void difference to get ■ as Yi's eastward walk. Add the unknown (Jia's southward walk) to get ■ as void hypotenuse. Squaring gives ■ as matching number. Canceling gives ■. Extract the square root to obtain 30 paces, Jia’s southward walk. Add the shortfall for the city diameter. Q.E.D.

第一百一十四問

或問：丙出南門直行，甲出東門直行，兩相望見。既而丙雲：「我行少於城徑一百五步。」甲雲：「我行少於城徑二百二十四步。」問答同前。

答曰：城徑二百四十步，半徑一百二十步。

法曰：二少步相乘訖，又自乘為實；六之共步，乘雲數相乘數為益從；十八之雲數相乘於上，又三之共步，自乘加上位，內復減丙少步冪、甲少步冪為從廉；四十八之共步為益二廉，六十三步常法。翻法開三乘方，得一百二十步，即半徑。

草曰：別得雲數共減於倍城徑為甲丙共行數。又雲數相減即皇極差，亦為甲行不及丙行數。立天元一為半城徑，以三之，副置二位。上位減丙少步，得 □ 為皇極股也，下位減甲少步得 □ 為皇極勾也。勾股相乘得 □，以天元除之，得 □ 為弦也。弦自之得 □ 為弦冪（寄左）。然後以股自之得下 □ 為股冪於上，又以勾自之得 □ 為勾冪，並以加入上位，得 □ 為同數，與左相消得 □。翻法開三乘方得一百二十步，即半徑也。合問。

Problem 一百一十四

Q: *Bing goes out the South Gate walking straight; Jia goes out the East Gate walking straight; they see each other. Bing says: “My walk is 105 paces less than the diameter.” Jia says: “My walk is 224 paces less than the diameter.” Answer as before.*

A: City diameter: 240 paces. Radius: 120 paces.

Method: Multiply the two shortfalls and then self-multiply as shi. Multiply six times the sum by the product of the given numbers as increasing cong. Multiply eighteen times the product of the given numbers as upper term; add three times the squared sum; subtract Bing’s shortfall squared and Jia’s shortfall squared as cong-lian. Forty-eight times the sum as increasing second lian; sixty-three as constant method. Extract by reversing the cubic root to obtain 120 paces, the radius.

Working: Identification: The sum of the given numbers subtracted from double the city diameter gives the total walk of Jia and Bing. Their difference is the imperial extreme difference, also Jia’s walk short of Bing’s.

Let the unknown be half the city diameter; triple it and place in two positions. Subtract Bing’s shortfall from the upper position to get ■ as imperial extreme gu; subtract Jia’s shortfall from the lower to get ■ as imperial extreme gou. Multiply gou and gu to get ■; divide by the unknown to get ■ as hypotenuse. Squaring gives ■ as hypotenuse-power (stored left). Square the gu to get ■ as gu-power (upper); square the gou to get ■ as gou-power. Add both to the upper to get ■ as matching number. Canceling gives ■. Extract the cubic root by reversing to obtain 120 paces, the radius. Q.E.D.

第一百一十五問

或問：甲出東門直行，乙出南門直行，各不知步數而立。乙望見甲，就甲斜行了二百八十九步與甲相會。其二直行共得一百五十一步。又雲甲直行少於乙直行。問答同前。

答曰：城徑二百四十步，甲直行一十六步，乙直行一百三十五步。

法曰：斜冪內減共步冪為平實，倍共步內減斜步為從，一常法。得徑。

Problem 一百一十五

Q: *Jia goes out the East Gate walking straight; Yi goes out the South Gate walking straight; each stands at an unknown distance. Yi sees Jia, then walks diagonally 289 paces to meet Jia. Their two straight walks total 151 paces. It is also stated that Jia’s walk is less than Yi’s. Answer as before.*

A: City diameter: 240 paces. Jia walked 16 paces; Yi walked 135 paces.

Method: Subtract the squared sum from the squared diagonal as flat shi. Double the sum minus the diagonal as cong. One constant method. Extract the diameter.

草曰:別得共數、城徑並即皇極和也。立天元一為圓徑,加共步得 □ 為皇極和,以自之,得 □ 於上。以斜行冪 □ 減上位餘 □ 為二直積(寄左)。然後以天元乘斜步得 □,與左相消得 □。開平方得二百四十步,即城徑也。合問。

Working: Identification: The sum plus the city diameter is the imperial extreme sum.

Let the unknown be the circle diameter; add the sum to get ■ as imperial extreme sum. Square it to get ■ as upper term. Subtract the squared diagonal ■ from the upper term, leaving ■ as twice the rectangular area (stored left). Multiply the unknown by the diagonal to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百一十六問

Problem 一百一十六

或問: 甲出東門直行,乙出東門南行,丙出南門直行,丁出南門東行,各不知步數而立。四人遙相望,悉與城參相直。只雲甲、丙共行了一百五十一步,乙、丁立處相距一百二步。又雲丙直行步多於甲直行步。問答同前。

Q: *Jia goes out the East Gate walking straight; Yi goes out the East Gate walking south; Bing goes out the South Gate walking straight; Ding goes out the South Gate walking east; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing together walked 151 paces; Yi and Ding's positions are 102 paces apart. Also stated that Bing's walk exceeds Jia's. Answer as before.*

答曰:城徑二百四十步。

A: City diameter: 240 paces.

法曰:共步、距步相減,得數自之於上,以共步為冪內減上為平實,二之距步內減共步、距步差為從,一步虛法。得城徑。

Method: Subtract the distance from the sum; square the result as upper term. Subtract this from the squared sum as flat shi. Double the distance minus the difference of sum and distance as cong. One empty-corner method. Extract the city diameter.

草曰:別得共步得城徑即皇極和也,相距步即虛弦也。皇極和內減虛弦即皇極弦也。又共步、距步差 □ 即皇極弦內減城徑也(此名旁差)。乃立天元一為城徑,加共步得 □ 為皇極和也,以自之得 □ 於上,以共步、距步差 □ 加天元得 □ 為皇極弦也,以自之得下式 □,減上位餘得 □ 為二直積(寄左)。然後以天元徑乘皇極弦,得 □ 為同數,與左相消得 □。開平方得二百四十步,即城徑也。合問。

Working: Identification: The sum equals the city diameter, which is the imperial extreme sum. The distance apart is the void hypotenuse. Imperial extreme sum minus void hypotenuse equals the imperial extreme hypotenuse. The difference of sum and distance ■ is the imperial extreme hypotenuse minus the city diameter (called side difference).

Let the unknown be the city diameter; add the sum to get ■ as imperial extreme sum. Square it to get ■ as upper term. Add the difference of sum and distance ■ to the unknown to get ■ as imperial extreme hypotenuse. Squaring gives ■; subtracting from the upper term leaves ■ as twice the rectangular area (stored left). Multiply the unknown diameter by the imperial extreme hypotenuse to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百一十七問

Problem 一百一十七

或問:甲出南門東行不知步數而立,乙出東門南行望見甲,復就甲斜行,與甲相會。乙通計行了一百三十二步,其乙南行步不及斜行七十二步,其甲東行卻多於乙南行。問答同前。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Yi walked a total of 132 paces; Yi's southward walk falls short of the diagonal by 72 paces; Jia's eastward walk exceeds Yi's southward walk. Answer as before.*

答曰：城徑二百四十步，乙南行三十步，甲東行一百二步。

A: **City diameter: 240 paces. Yi walked 30 paces south; Jia walked 102 paces east.**

法曰：倍不及步在地，以不及步減通步以乘之為實，以四之不及步為法。得乙南行三十步。

Method: Place double the shortfall; multiply by the total minus the shortfall as shi; use quadruple the shortfall as divisor. Extracts Yi's southward walk: 30 paces.

草曰：別得乙南行即股也，以減通步即虛弦也，以減不及步即虛較也，其不及步即甲東行也。立天元一為乙南行，置不及步以天元乘之，又四之得 □ 元為二直積（寄左）。然後倍不及步以為弦較和於上 □ 。以不及步減通步得 □ 為弦較較。以乘上位得 □ 太為同數，與左相消得 □ 。上法下實，得三十步為乙南行也。餘各以數求之。

Working: Identification: Yi's southward walk is the gu; subtract from the total to get the void hypotenuse; subtract the shortfall to get the void difference. The shortfall itself is Jia's eastward walk.

Let the unknown be Yi's southward walk. Multiply the shortfall by the unknown and quadruple to get ■ as twice the rectangular area (stored left). Double the shortfall as the sum of hypotenuse and difference (upper) ■. Total minus shortfall gives ■ as the difference of hypotenuse and difference. Multiply the upper by this to get ■ as matching number. Canceling gives ■. Dividing yields 30 paces as Yi's southward walk. The rest follows from the numbers.

第一百一十八問

Problem 一百一十八

或問：甲出南門東行不知步數而立。乙出東門南行，望見甲，復斜行與甲相會。二人共行了二百四步，又雲甲行不及共步一百三十二。問答同前。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Together they walked 204 paces. It is also stated that Jia's walk falls short of the total by 132. Answer as before.*

答曰：城徑二百四十步，甲東行九十六步，乙南行一百零八步。

A: **City diameter: 240 paces. Jia walked 96 paces east; Yi walked 108 paces south.**

法曰：別得二行共即兩個虛弦也，其不及步即乙南行與一虛弦共也。置不及步內減一弦餘三十步，即乙南行也。以乙南行反以減虛弦，餘七十二步即甲東行也。以乙南行減甲東行餘即虛較也。

Method: Identification: The sum of the two walks is twice the void hypotenuse; the shortfall is Yi's southward walk plus one void hypotenuse. Subtract one hypotenuse from the shortfall; the remainder 30 is Yi's southward walk. Subtract Yi's southward walk from the void hypotenuse; the remainder 72 is Jia's eastward walk. Subtract Yi's from Jia's to get the void difference.

草曰：此問無草。

Working: This problem has no working.

第一百一十九問

Problem 一百一十九

或問：乙出東門南行，甲出西門南行，甲望見乙，斜行五百一十步相會。乙雲：「我南行少於城徑二百一十步。」問答同前。

Q: *Yi goes out the East Gate walking south; Jia goes out the West Gate walking south; Jia sees Yi and walks diagonally 510 paces to meet Yi. Yi says: "My southward walk is 210 paces less than the diameter." Answer as before.*

答曰：城徑二百四十步，乙南行三十步。

A: **City diameter: 240 paces. Yi walked 30 paces south.**

法曰：少步羃為平實，四斜步內減二少步為益從，五步常法。得乙南行。

Method: Square the shortfall as flat shi. Quadruple the diagonal minus double the shortfall as increasing cong. Five as constant method. Extracts Yi's southward walk.

草曰：別得少步為徑內減股。立天元一為乙南行，以二之減於倍斜行步，得 □ 為梯底也。以二之天元乘之，得 □ 為徑冪（寄左）。再置天元加少步，得下式 □ 為城徑，以自之得 □，與左相消得 □。開平方得三十步，即乙南行也。加少步即城徑也。合問。

Working: Identification: The shortfall is the diameter minus the gu.

Let the unknown be Yi's southward walk. Subtract double the unknown from double the diagonal to get ■ as the ladder base. Multiply by double the unknown to get ■ as diameter-power (stored left). Add the shortfall to the unknown to get ■ as city diameter; squaring gives ■. Canceling gives ■. Extract the square root to obtain 30 paces, Yi's southward walk. Add the shortfall for the city diameter. Q.E.D.

第一百二十問

Problem 一百二十

或問：乙出南門東行，甲出北門東行，甲望見乙，斜行二百七十二步與乙相會。乙雲：「我東行不及城徑一百六十八步。」問答同前。

Q: *Yi goes out the South Gate walking east; Jia goes out the North Gate walking east; Jia sees Yi and walks diagonally 272 paces to meet Yi. Yi says: "My eastward walk falls short of the diameter by 168 paces." Answer as before.*

答曰：城徑二百四十步，乙東行七十二步。

A: City diameter: 240 paces. Yi walked 72 paces east.

法曰：以不及步冪之為實，四斜內減二之不及步為虛從，五常法。平開得乙東行七十二步。

Method: Square the shortfall as shi. Quadruple the diagonal minus double the shortfall as empty cong. Five as constant method. Extract to obtain Yi's eastward walk: 72 paces.

草曰：別得不及步為城徑減明勾也。立天元一為乙東行，以倍之減於二之斜行步，得下 □ 為梯底也。倍天元乘之，得 □ 為徑冪（寄左）。再置天元加不及步，得 □ 為城徑，以自之得 □ 為同數，與左相消得 □。開平方得七十二步，即乙東行也。加入少步即城徑也。合問。

Working: Identification: The shortfall is the city diameter minus the bright gou.

Let the unknown be Yi's eastward walk. Subtract double the unknown from double the diagonal to get ■ as ladder base. Multiply by double the unknown to get ■ as diameter-power (stored left). Add the shortfall to the unknown to get ■ as city diameter; squaring gives ■ as matching number. Canceling gives ■. Extract the square root to obtain 72 paces, Yi's eastward walk. Add the shortfall for the city diameter. Q.E.D.

第一百二十一問

Problem 一百二十一

或問：乙出南門東行，丁出東門南行，卻有甲丙二人共在西北隅，甲向東行，丙向南行，四人遙相望見，俱與城參相直。既而相會，甲雲：「我多乙二百四十八步。」丙雲：「我多於丁五百七十步。」問答同前。

Q: *Yi goes out the South Gate walking east; Ding goes out the East Gate walking south. Meanwhile Jia and Bing together are at the northwest corner, Jia walking east and Bing walking south. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "I exceed Yi by 248 paces." Bing says: "I exceed Ding by 570 paces." Answer as before.*

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：二多步相乘為平實，並二多步而半之為從，七分半常法。得城徑。

Method: Multiply the two excesses as flat shi. Halve the sum of the two excesses as cong. Seven and a half as constant method. Extract the city diameter.

草曰：別得甲多步為大勾內減明勾也，丙多步為大股內少股也。又乙東行得一虛勾為半徑，丁南行得一虛股為半徑。又二多數相並得 □ 為大和內少虛弦也，又二少數相減餘 □ 為兩個角差。又甲多步

Working: Identification: Jia's excess is the large gou minus the bright gou; Bing's excess is the large gu minus the gu. Yi's eastward walk gives one void gou as radius; Ding's southward walk gives one void gu as radius. The sum of

內減半徑即勾方差也，丙多步內減半徑即股方差也。

立天元一為城徑，以半之減於甲多步得 □ 為勾方差，又以半徑減於丙多步得 □ 為股方差。二差相乘得 □ 為徑冪（寄左）。然後以天元冪與左相消，得下式 □。開平方得二百四十步，即城徑也。合問。

the two excesses ■ is the large sum minus the void hypotenuse; their difference ■ is twice the corner difference. Jia's excess minus the radius is the gou square-difference; Bing's excess minus the radius is the gu square-difference.

Let the unknown be the city diameter. Subtract half the diameter from Jia's excess to get ■ as gou square-difference; subtract the radius from Bing's excess to get ■ as gu square-difference. Multiply both differences to get ■ as diameter-power (stored left). Cancel the unknown squared with the left to get ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百二十二問

Problem 一百二十二

或問：甲丙二人俱在西北隅，甲向東行，丙向南行。又乙出南門東行，丁出東門南行，各不知步數而立。四人遙相望見，悉與城參相直。既而相會，甲雲：「我與乙共行了三百九十二步。」丙雲：「我與丁共行了六百三十步。」問答同前。

Q: *Jia and Bing are both at the northwest corner; Jia walks east and Bing walks south. Yi goes out the South Gate walking east; Ding goes out the East Gate walking south; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "Yi and I together walked 392 paces." Bing says: "Ding and I together walked 630 paces." Answer as before.*

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：甲乙共自之為冪，丙丁共自之為冪，二冪又相乘為三乘方實。甲乙共自之為冪，以丙丁共乘之於上。又以丙丁共自之為冪，以甲乙共乘之加上位為益從。甲乙共自之為冪，丙丁共自之為冪，並，以七分半乘之於上。又以甲乙共乘丙丁共，得數減上位為第一益廉。並二共數，以七分半乘之為第二廉。以七分半自之，得五分六厘二毫五絲於上位。以一步內減上位，餘四分三厘七毫五絲為虛隅。得城徑。

Method: Square the Jia-Yi sum and the Bing-Ding sum; multiply both powers as the cubic shi. Square the Jia-Yi sum, multiply by the Bing-Ding sum as upper term. Square the Bing-Ding sum, multiply by the Jia-Yi sum and add to upper term as increasing cong. Square both sums, add, multiply by 7.5 as upper term. Subtract the product of both sums from upper term as first increasing lian. Add both sums, multiply by 7.5 as second lian. Square 7.5 to get 0.05625 as upper term. Subtract from one step, leaving 0.04375 as empty corner. Extracts the city diameter.

草曰：別得甲為大勾，乙為明勾，丙為大股，丁為股也。甲乙共內減半徑即是黃長弦也，丙丁共內減半徑即黃廣弦也。黃長弦、黃廣弦二數相減，餘為兩個皇極差也。

乃立天元為城徑，半之副置二位。上以減於甲乙共數，得 □ 即黃長弦也，以自之得 □ 為黃長弦冪也，內減天元一冪，餘得下式 □ 為勾方差冪也。下位以減於丙丁共，得下式 □ 即黃廣弦也，以自之得 □ 為黃廣弦冪也，內減天元一冪，餘得 □ 為股方差冪也。再以勾方差冪、股方差冪相乘，得 □ 為徑冪（寄左）。然後以天元為冪，又以冪自之，與左相消得下式 □。開三乘方得二百四十步，即城徑也。合問。

Working: Identification: Jia is the large gou, Yi is the bright gou, Bing is the large gu, Ding is the gu. The Jia-Yi sum minus the radius is the yellow-length hypotenuse; the Bing-Ding sum minus the radius is the yellow-breadth hypotenuse. Their difference is twice the imperial extreme difference.

Let the unknown be the city diameter; halve and place in two positions. Subtract from the Jia-Yi sum above to get ■ as yellow-length hypotenuse; squaring gives ■ as its power; subtract the unknown squared to get ■ as gou square-difference power. Subtract from the Bing-Ding sum below to get ■ as yellow-breadth hypotenuse; squaring and subtracting the unknown squared gives ■ as gu square-difference power. Multiply both difference powers to get ■ as diameter-power (stored left). Cube the unknown and cancel with the left to get ■. Extract the cubic root to obtain 240 paces, the city diameter. Q.E.D.

2.1 卷八 明後一十六問

Vol. 8: 16 “Rear” Problems

卷八小序：此卷一十六問，承卷七之緒，仍以明段為主，而益之以極段、虛段之變。所謂極段者，皇極勾股形之各邊也；虛段者，太虛勾股形之各邊也。此三形交錯互見，開方之術益臻精妙。其方程式之高，有至四乘方、五乘方者，實全書之樞紐也。

Volume 8 Preface: *These 16 problems continue from Volume 7, still centered on “bright” segments but now incorporating “extreme” and “void” variations. The extreme segments are the edges of the imperial extreme right triangle; the void segments are edges of the grand void triangle. These three triangles interweave, and the root-extraction techniques reach their finest. Equations rise to degrees 4–5; this is the pivot of the entire work.*

第一百二十三問

或問：出南門向東有槐樹一株，出東門向南有柳樹一株。丙丁俱出南門，丙直行，丁往至槐樹下。甲乙俱出東門，甲直行，乙往至柳樹下。四人遙相望見，各不知所行步數。只雲丙丁共行了二百七步，甲乙共行了四十六步，又雲甲丙立處相距二百八十九步。問答同前。

答曰：城徑二百四十步，半徑一百二十步。

法曰：以二共相減數又以減距數為實，二為法。得平勾。

草曰：識別得丙丁共即明和也，甲乙共即和也，相距步即極弦也。二共相並即極弦內少個虛黃也，又為極和內少個虛和也。二共相減餘為平勾高股差也，又為虛差極差共也，又為通差內減極差也。

立天元為平勾，加入二共相減數，得 □ 為高股，又加天元得 □ 為極弦（寄左）。以相距步二百八十九與左相消，得 □。上法下實，如法得六十四，即平勾也。以二共相減數加平勾得二百二十五為高股，復以平勾乘之，得一萬四千四百步。開平方得一百二十步，即城半徑也。合問。

Problem 一百二十三

Q: *Going out the South Gate eastward there is a locust tree; going out the East Gate southward there is a willow tree. Bing and Ding both go out the South Gate; Bing walks straight while Ding goes to the locust tree. Jia and Yi both go out the East Gate; Jia walks straight while Yi goes to the willow tree. The four gaze at each other from afar; none know the paces walked. It is only stated that Bing and Ding together walked 207 paces; Jia and Yi together walked 46 paces; and the distance between Jia and Bing’s positions is 289 paces. Answer as before.*

A: City diameter: 240 paces. Radius: 120 paces.

Method: Subtract the difference of the two sums from the distance as shi; 2 as divisor. Extracts the level gou.

Working: Identification: Bing-Ding sum is the bright sum; Jia-Yi sum is the sum; the distance is the extreme hypotenuse. The sum of both sums is the extreme hypotenuse lacking the void yellow; also the extreme sum lacking the void sum. Their difference is the difference between level gou and high gu; also the sum of void difference and extreme difference; also the general difference minus the extreme difference.

Let the unknown be the level gou. Add the difference of the two sums to get ■ as high gu; add the unknown to get ■ as extreme hypotenuse (stored left). Cancel with the distance 289 to get ■. Dividing yields 64, the level gou. Add the difference of sums to the level gou to get 225 as high gu; multiply by the level gou to get 14,400. Extract the square root to obtain 120 paces, half the city radius. Q.E.D.

第一百二十四問

或問：依前見丙丁共二百七步，甲乙共四十六步。又雲二樹相去一百二步。問答同前。

答曰：城徑二百四十步。

法曰：以甲乙共乘樹相去步，得數又以自之為平實；從空；並二共數為冪於上，內減甲乙共自之數、丙丁共自之數為益隅。得弦。

Problem 一百二十四

Q: *As before: Bing-Ding sum 207 paces, Jia-Yi sum 46 paces. Also stated: the two trees are 102 paces apart. Answer as before.*

A: City diameter: 240 paces.

Method: Multiply the Jia-Yi sum by the tree distance; square the result as flat shi. Cong empty. Square the sum of both sums as upper term; subtract the squared Jia-Yi sum and squared Bing-Ding sum as increasing

草曰：識別得兩樹相去步即虛弦也。餘數具前。立天元一為弦。置明和以天元乘之，合和除，不除，便以 □ 元為明弦也（內帶和分母）。乃置虛弦以分母和乘之，得 □ 太，加入明弦，得 □ 為極股也，內帶和分母。以自之得下式 □ 為極股冪（內寄和冪為分母）。又以天元加虛弦得下 □ 為極勾，以自之得 □。又以和冪 □ 乘之，得 □ 為勾冪也。勾股冪相並得 □ 為兩積一較冪也，內有和冪分母（寄左）。然後置明弦 □ 元於上，以和乘天元加上位，得 □ 元為二弦並也。又置虛弦以和乘之得 □ 太，並入上位，得下式 □ 為極弦，以自之得 □ 為同數，與左相消得 □。開平方得三十四步，即弦也。

corner. Extracts the hypotenuse.

Working: Identification: The tree distance is the void hypotenuse. The rest as before.

Let the unknown be the hypotenuse. Multiply the bright sum by the unknown; do not divide by the sum; take ■ as the bright hypotenuse (with sum as denominator). Multiply the void hypotenuse by the denominator sum to get ■; add the bright hypotenuse to get ■ as extreme gu (with sum denominator). Squaring gives ■ as extreme gu power (with squared sum as denominator). Add the unknown to the void hypotenuse to get ■ as extreme gou; squaring and multiplying by squared sum ■ gives ■ as gou power. Adding both powers gives ■ as two areas plus one difference power (with squared sum denominator, stored left). Place bright hypotenuse ■ above; multiply the unknown by the sum and add to get ■ as combined hypotenuses. Multiply void hypotenuse by sum to get ■; add to get ■ as extreme hypotenuse. Squaring gives ■ as matching number. Canceling gives ■. Extract the square root to obtain 34 paces, the hypotenuse. Q.E.D.

第一百二十五問

Problem 一百二十五

或問：皇極大小差共一百八十七步，明黃、黃共六十六步。問答同前。

答曰：太虛黃方三十六步，城徑二百四十步。

法曰：後數自乘為實，前後數相減，餘為法。得虛黃方三十六。

草曰：別得一百八十七即明弦二弦共也，其六十六即太虛大小差共也。又二數相並，得 □ 即明、二和共。若以相減，餘 □ 即明、四差共也。

立天元一為太虛黃方面，加二黃共，得 □ 即虛弦也。倍虛弦又加天元，得 □ 即城徑也。又以虛弦加皇極大小差，得 □ 即極弦也，以極弦乘城徑得 □，為兩段皇極勾股積（寄左）。再以極弦虛弦相並，得 □，即皇極勾股共也，自之得 □，內減皇極弦冪 □，得 □ 為同數，與寄左相消得 □。上法下實，如法得三十六步，即太虛黃方面也。合問。

Q: *The sum of imperial extreme large and small differences is 187 paces; the sum of bright yellow and yellow is 66 paces. Answer as before.*

A: **Grand void yellow square: 36 paces. City diameter: 240 paces.**

Method: Square the latter number as shi; subtract the former from the latter as divisor. Extracts the void yellow square: 36.

Working: Identification: 187 is the sum of bright hypotenuse and two hypotenuses; 66 is the sum of grand void large and small differences. Adding both gives ■ as the sum of bright and two sums; subtracting gives ■ as the sum of bright and four differences.

Let the unknown be the grand void yellow square side. Add twice the yellow sum to get ■ as void hypotenuse. Double the void hypotenuse and add the unknown to get ■ as city diameter. Add the void hypotenuse to the imperial extreme differences to get ■ as extreme hypotenuse; multiply by the city diameter to get ■ as twice the imperial extreme gou-gu area (stored left). Add extreme and void hypotenuses to get ■ as imperial extreme gou-gu sum. Squaring and subtracting the squared extreme hypotenuse ■ gives ■ as matching number. Canceling gives ■. Dividing yields 36 paces, the grand void yellow square side. Q.E.D.

第一百二十六問

Problem 一百二十六

或問：東門南有柳一株，南門東有槐一株。甲出東門

Q: *South of the East Gate is a willow; east of the South*

直行，丙出南門直行。甲、丙、柳、槐悉與城參相直，既而甲就柳樹斜行三十四步至柳樹下，丙就槐樹斜行一百五十三步至槐樹下。問答同前。

Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight. Jia, Bing, the willow, and the locust are all aligned with the city. Then Jia walks diagonally 34 paces to the willow; Bing walks diagonally 153 paces to the locust. Answer as before.

答曰：城徑二百四十步，虛弦一百二步。

A: **City diameter: 240 paces. Void hypotenuse: 102 paces.**

法曰：雲數相乘，倍之便為平方實，開方得虛弦一百二步。以此弦加甲行步即極勾，以此弦加丙行步即極股。餘各依法求之。

Method: Multiply the given numbers; double as the quadratic shi. Extract the square root to obtain the void hypotenuse: 102 paces. This hypotenuse plus Jia's walk gives the extreme gou; plus Bing's walk gives the extreme gu. The rest follows by method.

草曰：識別：甲斜行即弦也，丙斜行即明弦也。立天元一為虛弦。置甲乙共以天元減之，得 □ 為虛和也。以天元減甲乙共餘 □ 即虛勾虛股共也。別置虛弦加甲斜行得 □ 即極勾，虛弦加丙斜行得 □ 即極股也。極勾極股相乘得 □，以天元虛弦除之得 □ 為極弦也。以自增乘得 □ 為極弦冪（寄左）。乃以極勾自之得 □ 為勾冪，極股自之得 □ 為股冪，二冪相並得 □ 為同數，與左相消得 □。開平方得一百二步，即虛弦也。合問。

Working: Identification: Jia's diagonal is the hypotenuse; Bing's diagonal is the bright hypotenuse.

Let the unknown be the void hypotenuse. Subtract from the Jia-Yi sum to get ■ as void sum. The remainder ■ is the sum of void gou and void gu. Add the void hypotenuse to Jia's diagonal to get ■ as extreme gou; add to Bing's diagonal to get ■ as extreme gu. Multiply to get ■; divide by the void hypotenuse to get ■ as extreme hypotenuse. Squaring gives ■ as extreme hypotenuse power (stored left). Square the extreme gou to get ■ as gou power; square extreme gu to get ■ as gu power. Adding gives ■ as matching number. Canceling gives ■. Extract the square root to obtain 102 paces, the void hypotenuse. Q.E.D.

第一百二十七問

Problem 一百二十七

或問：東門南有柳一株，南門東有槐一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲丙共行了二百八十八步，甲斜行與柳至東門步共得六十四步。問答同前。

Q: *South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Bing walked a total of 288 paces; Jia's diagonal plus the willow-to-East-Gate distance equals 64 paces. Answer as before.*

答曰：甲直行一十六步，城徑二百四十步。

A: **Jia's straight walk: 16 paces. City diameter: 240 paces.**

法曰：二雲數相乘於上，以六十四步自之，又二之，減上位為平實。四之六十四於上，倍丙行內減上位為從。二十常法。得甲直行步一十六。

Method: Multiply the two given numbers as upper term. Square 64 and double it; subtract from upper term as flat shi. Quadruple 64 as upper term; subtract from double Bing's walk as cong. Twenty as constant method. Extracts Jia's straight walk: 16 paces.

草曰：別得丙共步即明股、明弦和也，六十四即平勾也，內甲斜行即弦也，柳至東門步即股也。又二雲數相並即明差與極弦共也，二雲數相減即明差與平勾高股差共也。又平勾內減勾即虛勾也。

立天元一為勾。置丙共步以天元乘之，復以六十四除之得 □ 為明勾也。又以天元減於六十四，

Working: Identification: Bing's total is the sum of bright gu and bright hypotenuse; 64 is the level gou; Jia's diagonal is the hypotenuse; the willow-to-East-Gate distance is the gu. The sum of the two given numbers is the bright difference plus extreme hypotenuse; their difference is the bright difference plus the difference between level gou and high gu. Level gou minus gou is void gou.

得 □ 為虛勾也，並虛明二勾 □ 為半徑也。以自之得 □，倍之得 □ 為半段圓城徑冪（寄左）。乃以天元加六十四，得 □ 為勾圓差於上；又以明勾加丙共步，得 □ 為股圓差於下。上下相乘得 □ 為同數，與左相消得 □。開平方得一十六步，即勾也。此勾乃甲出東門直行步也。餘皆依數求之。合問。

Let the unknown be the gou. Multiply Bing's total by the unknown and divide by 64 to get ■ as bright gou. Subtract the unknown from 64 to get ■ as void gou; add both void and bright gou ■ as radius. Squaring and doubling gives ■ as half the city diameter power (stored left). Add the unknown to 64 to get ■ as gou circle-difference (upper); add bright gou to Bing's total to get ■ as gu circle-difference (lower). Multiply to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 16 paces, the gou—this is Jia's straight walk out the East Gate. Q.E.D.

第一百二十八問

Problem 一百二十八

或問：東門南有柳樹一株，南門東有槐樹一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲甲共行五十步，丙斜行與槐至南門步共得二百二十五步。問答同前。

Q: *South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Jia walked a total of 50 paces; Bing's diagonal plus the locust-to-South-Gate distance equals 225 paces. Answer as before.*

答曰：城徑二百四十步，丙直行一百三十五步。

A: **City diameter: 240 paces. Bing's straight walk: 135 paces.**

法曰：以二百二十五步自之為冪，又以此冪自為冪於上。置甲共行以二百二十五步三度乘之，得數復折半，減上位為平實。置二百二十五步自之數，以二雲數相減數乘之，又倍之於上。倍五十步在地，以二百二十五步自之數乘之，復折半加上位為益從。雲數相減自乘於上，以雲數相乘，復折半，減上位為常法。得明股。

Method: Square 225 as power; use this power squared as upper term. Raise Jia's total by 225 to the third power; halve and subtract from upper term as flat shi. Square 225; multiply by the difference of the two given numbers; double as upper term. Place double 50; multiply by the squared 225; halve and add to upper term as increasing cong. Square the difference of the given numbers as upper term; halve the product of the given numbers and subtract as constant method. Extracts the bright gu.

草曰：識別得甲共步即勾、弦共也，二百二十五即高股也，內丙斜行即明弦，槐至南門步即明勾也。又二雲數相並即極弦內減一個差也，雲數相減即差與高股、平勾差共也。又高股內減明股即虛股也。

立天元一為明股，即丙出南門直行步也。置五十步以天元乘之得 □ 元，合高股除。不除，便以此一零零零元為通股（內帶高股為母）。以天元加高股，得 □ 即大差也。置大差以高股分母乘之，得 □，即帶分大差也。以此減於通股，餘 □ 即圓徑也。以自增乘，得 □（寄左。內帶高股冪分母）。然後置一千以高股分母通之，得 □ 太，內減帶分大差，得 □ 為兩個通勾也。內減兩個圓徑得 □，為兩個小差也。以帶分大差乘之，得下式 □ 為同數，與左相消得 □。開平方得一百三十五步，即明股也。合問。

Working: Identification: Jia's total is the sum of gou and hypotenuse; 225 is the high gu; Bing's diagonal is the bright hypotenuse; the locust-to-South-Gate distance is the bright gou. The sum of the two given numbers is the extreme hypotenuse minus one difference; their difference is the difference plus the difference between high gu and level gou. High gu minus bright gu is void gu.

Let the unknown be the bright gu, Bing's straight walk out the South Gate. Multiply 50 by the unknown to get ■; should divide by high gu but do not; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get ■ as large difference. Multiply by the high gu denominator to get ■ as the scaled large difference. Subtract from the general gu; the remainder ■ is the circle diameter. Squaring gives ■ (stored left, with squared high gu as denominator). Scale 1000 by the high gu denominator to get ■; subtract the scaled large difference to get ■ as twice the general gou. Subtract twice the circle diameter to get ■ as twice the small difference. Multiply by the scaled large difference to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

第一百二十九問

或問：通勾、通弦共一千步，勾、弦共五十步。問答同前。

答曰：城徑二百四十步，股三十步。

法曰：置一千減二之五十步為泛率。以自乘，復半之於上。又置泛率復以五十乘之，加上位為平實。二十二之泛率於上。以四十二乘五十，得數內減泛率，加上位為益從。二百為常法。得股。

草曰：立天元一為股。置一千以天元乘之，以五十除之，得 □ 元為通股也。又以天元加五十步，得 □ 即小差也，通股加小差得 □ 即通弦也。以通弦減一千得 □ ，即通勾也。以小差減通勾得 □ ，即圓徑也。以圓徑減通股得 □ 即大差也。置大差以小差乘之得 □（寄左）。然後置圓徑以自之得 □ ，折半得 □ ，與左相消得 □ 。開平方得三十步，即股也。合問。

Problem 一百二十九

Q: *General gou plus general hypotenuse total 1000 paces; gou plus hypotenuse total 50 paces. Answer as before.*

A: City diameter: 240 paces. Gu: 30 paces.

Method: Subtract double 50 from 1000 as the general rate. Square and halve as upper term. Multiply the general rate by 50 and add to upper term as flat shi. Twenty-two times the general rate as upper term. Multiply 42 by 50; subtract the general rate and add to upper term as increasing cong. Two hundred as constant method. Extracts the gu.

Working: Let the unknown be the gu. Multiply 1000 by the unknown and divide by 50 to get ■ as general gu. Add 50 to the unknown to get ■ as small difference; general gu plus small difference gives ■ as general hypotenuse. Subtract from 1000 to get ■ as general gou. Subtract the small difference from general gou to get ■ as circle diameter. Subtract the circle diameter from general gu to get ■ as large difference. Multiply large difference by small difference to get ■ (stored left). Square the circle diameter and halve to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 30 paces, the gu. Q.E.D.

第一百三十問

或問：通勾、通弦共一千步，明勾、明弦共二百二十五步。問答同前。

答曰：城徑二百四十步，明股一百三十五步。

法曰：以後數再自乘，又以前數乘之為平實；以後數為冪，復以前數乘之為從；以前數冪為虛常法。得明股。

草曰：別得二百二十五步即高股也。立天元一為明股。置一千以天元乘之，合以高股除不受除，便以此一零零零元為通股（內帶高股為母）。以天元加高股，得 □ 即大差也。置大差以高股分母乘之，得 □ ，即帶分大差也。以此減於通股，餘 □ 即圓徑也。以自增乘，得 □（寄左。內帶高股冪分母）。然後置一千以高股分母通之，得 □ 太，內減帶分大差，得 □ 為兩個通勾也。內減兩個圓徑得 □ ，為兩個小差也。以帶分大差乘之，得下式 □ 為同數，與左相消得 □ 。開平方得一百三十五步，即明股也。合問。

Problem 一百三十

Q: *General gou plus general hypotenuse total 1000 paces; bright gou plus bright hypotenuse total 225 paces. Answer as before.*

A: City diameter: 240 paces. Bright gu: 135 paces.

Method: Cube the latter number and multiply by the former as flat shi. Square the latter and multiply by the former as cong. The squared former as empty constant method. Extracts the bright gu.

Working: Identification: 225 paces is the high gu.

Let the unknown be the bright gu. Multiply 1000 by the unknown; do not divide by the high gu; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get ■ as large difference. Multiply by the high gu denominator to get ■ as the scaled large difference. Subtract from general gu; remainder ■ is the circle diameter. Squaring gives ■ (stored left, with squared high gu denominator). Scale 1000 by the high gu denominator to get ■; subtract the scaled large difference to get ■ as twice the general gou. Subtract twice the circle diameter to get ■ as twice the small difference. Multiply by the scaled large difference to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

第一百三十一問

或問：通股、通弦共一千二百八十步，股、弦共六十四步。問答同前。

答曰：城徑二百四十步，勾一十六步。

法曰：雲數相乘為平實，前數為益從。置前數以後數除之，得二十為泛率。泛率減一以自乘於上，又倍泛率減一加上位為常法。倒積開得勾一十六。

草曰：別得六十四步即平勾也。立天元一為勾。置前數以天元乘之，以後數除之，得 □ 元即通勾也。又置天元加後數，得 □ 即小差也。以小差減通勾，餘 □ 即圓徑也。以自之得 □（寄左）。然後以小差減於前數，得 □ 為二通股。內減兩個圓徑，得 □ 為二大差也。以小差乘之得下 □，與左相消得 □。開平方得一十六步，即勾也。合問。

Problem 一百三十一

Q: *General gu plus general hypotenuse total 1280 paces; gu plus hypotenuse total 64 paces. Answer as before.*

A: City diameter: 240 paces. Gou: 16 paces.

Method: Multiply the given numbers as flat shi; the former as increasing cong. Divide the former by the latter to get 20 as general rate. Square (general rate minus one) as upper term; add double the general rate minus one as constant method. Extract by inverted accumulation to obtain the gou: 16.

Working: Identification: 64 paces is the level gou.

Let the unknown be the gou. Multiply the former by the unknown and divide by the latter to get ■ as general gou. Add the latter to the unknown to get ■ as small difference. Subtract the small difference from general gou; remainder ■ is the circle diameter. Squaring gives ■ (stored left). Subtract the small difference from the former to get ■ as twice the general gu. Subtract twice the circle diameter to get ■ as twice the large difference. Multiply by the small difference to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 16 paces, the gou. Q.E.D.

第一百三十二問

或問：通股、通弦共一千二百八十步，明股、明弦共二百八十八步。問答同前。

答曰：城徑二百四十步，明勾七十二步。

法曰：二數相減，以後數乘之，內減後數冪，又半之為泛率。以自之為平實。置前數加二之後數而半之為次率，以乘泛率，倍之於上，以後數乘泛率減上位為益從。次率自乘於上，以前數加次率，復以後數乘之，減上位為隅法。得明勾。

草曰：別得二數相減，餘 □ 為通勾、通股及明勾共也。立天元一為明勾。置前數以天元乘之，合以後數除之。不除，便以此 □ 元為通勾也（內寄後數分母）。又以二數相減，得數內又減天元得 □ 為通和也。乃以分母二百八十八之，得下式 □，內減通勾，餘 □ 為通股也。又以天元加後數，又以分母（即後數也）通之，得 □ 為大差也。以此大差減於通股，得下式 □，為一個圓徑也。半之得 □，以自之得

Problem 一百三十二

Q: *General gu plus general hypotenuse total 1280 paces; bright gu plus bright hypotenuse total 288 paces. Answer as before.*

A: City diameter: 240 paces. Bright gou: 72 paces.

Method: Subtract the two numbers; multiply by the latter; subtract the squared latter; halve as general rate. Square it as flat shi. Add the former to double the latter and halve as secondary rate. Multiply by the general rate; double as upper term. Multiply the latter by the general rate and subtract from upper term as increasing cong. Square the secondary rate as upper term; add the former to the secondary rate; multiply by the latter and subtract as corner method. Extracts the bright gou.

Working: Identification: The difference of the two numbers ■ is the sum of general gou, general gu, and bright gou.

Let the unknown be the bright gou. Multiply the former by the unknown; do not divide by the latter; take ■ as general gou (with latter as denominator). Subtract the two numbers; subtract the unknown from the result to get ■ as general sum. Apply denominator 288 to get ■; subtract

□ 為半徑冪（寄左）。然後以半圓徑減通勾，得 □ 為底勾，又以天元乘之，又以分母二百八十八之，得 □ 為同數，與左相消得 □。開平方得七十二步，即明勾也。合問。

general gou; remainder ■ is general gu. Add the unknown to the latter; apply the denominator (the latter) to get ■ as large difference. Subtract from general gou to get ■ as circle diameter. Halve and square to get ■ as radius power (stored left). Subtract half the diameter from general gou to get ■ as base gou; multiply by the unknown and apply denominator 288 to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 72 paces, the bright gou. Q.E.D.

第一百三十三問

Problem 一百三十三

或問：明股、明弦並二百八十八步，勾、弦並五十步。又雲明股、勾並多於虛弦四十九步。問答同前。

答曰：城徑二百四十步。

法曰：前二數相並，內減二之多步，即圓徑。又只以前二數相乘，便是半徑冪。

草曰：識別得前二數相減而半之，即極差也。其多步名傍差，又為圓徑不及極弦數。立天元一為半徑。置前二數相乘得 □，以天元除之得 □ 為極弦也。以天元加多步得 □ 為虛弦。極弦內減虛弦餘 □ 為旁差，與多步相應。又以極弦自之得 □，內減虛弦冪 □，餘 □ 為二積。以天元除之得 □ 為勾股和，即前二數之並也，以此為同數。與左相消得 □，開平方得一百二十步，即半徑也。倍之得二百四十步，即城徑也。合問。

Q: *Bright gu plus bright hypotenuse total 288 paces; gou plus hypotenuse total 50 paces. Also stated: the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.*

A: **City diameter: 240 paces.**

Method: Add the first two numbers; subtract double the excess to get the circle diameter. Alternatively, simply multiply the first two numbers to obtain the radius squared.

Working: Identification: Halve the difference of the first two numbers to get the extreme difference. The excess is called the side difference, also the amount by which the circle diameter falls short of the extreme hypotenuse.

Let the unknown be the radius. Multiply the first two numbers to get ■; divide by the unknown to get ■ as extreme hypotenuse. Add the excess to the unknown to get ■ as void hypotenuse. Subtract void hypotenuse from extreme hypotenuse; remainder ■ is the side difference, matching the excess. Square the extreme hypotenuse and subtract the squared void hypotenuse ■; remainder ■ is twice the area. Divide by the unknown to get ■ as gou-gu sum, which is the sum of the first two numbers, as matching number. Canceling gives ■. Extract the square root to obtain 120 paces, the radius. Double it to get 240 paces, the city diameter. Q.E.D.

第一百三十四問

Problem 一百三十四

或問：平差、高差共一百六十一步，明股、勾並多於虛弦四十九步。問答同前。

答曰：城徑二百四十步，平勾六十四步。

法曰：二數相減又半之，以自乘為實，後數為法。得平勾。

草曰：立天元一為平勾。以加前數得 □ 為高股也。又以天元加高股得 □ 為極弦，內減後數得 □，又半之，得 □ 為半徑。以自之，得 □（寄左）。然後以天元乘高股，得 □ 為同數，與左相消得 □。上法下實，得六十四步，即平勾也。合問。

Q: *Level difference plus high difference total 161 paces; the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.*

A: **City diameter: 240 paces. Level gou: 64 paces.**

Method: Subtract the two numbers; halve; square as shi; the latter number as divisor. Extracts the level gou.

Working: Let the unknown be the level gou. Add the former number to get ■ as high gu. Add the unknown to the high gu to get ■ as extreme hypotenuse; subtract the latter number to get ■; halve to get ■ as radius. Squaring gives ■ (stored left). Multiply the unknown by the high gu to get ■ as matching number. Canceling gives ■. Dividing yields 64 paces, the level gou. Q.E.D.

第一百三十五問

或問：平勾、高股差一百六十一步，明差、差並七十七步。又雲極弦多於城徑四十九步。問答同前。

答曰：城徑二百四十步，半徑一百二十步。

法曰：並上二位而半之為平率。其四十九即旁率也。副置平率，上加旁率，下減旁率，以相乘為實。倍旁差為法。得勾圓差。

草曰：立天元一為半徑，又為半之股圓差上弦較較，又為半之勾圓差上弦較和也。內減勾圓差上勾股較 □，餘 □ 為半之勾圓差上弦較較也。置股圓差上勾股較 □，以半之勾圓差上弦較較乘之，得 □（寄左）。然後以半之股圓差上弦較較乘勾圓差上勾股較，得 □ 元為同數，與左相消得下式 □。上法下實，得一百二十步，即半徑也。合問。

Problem 一百三十五

Q: *The difference between level gou and high gu is 161 paces; bright difference plus difference totals 77 paces. Also stated: the extreme hypotenuse exceeds the city diameter by 49 paces. Answer as before.*

A: City diameter: 240 paces. Radius: 120 paces.

Method: Add the two upper numbers and halve as the level rate. The 49 is the side rate. Place the level rate in two positions; add the side rate above; subtract below. Multiply as shi. Double the side difference as divisor. Extracts the gou circle-difference.

Working: Let the unknown be the radius; also half the gu circle-difference sum-of-differences; also half the gou circle-difference sum-of-sums. Subtract the gou-gu difference on the gou circle-difference ■; remainder ■ is half the gou circle-difference difference-of-differences. Place the gu circle-difference gou-gu difference ■; multiply by half the gou circle-difference difference-of-differences to get ■ (stored left). Multiply half the gu circle-difference difference-of-differences by the gou circle-difference gou-gu difference to get ■ as matching number. Canceling gives ■. Dividing yields 120 paces, the radius. Q.E.D.

第一百三十六問

或問：又依前問：見角差一百六十一步，見明差差並七十七步，又見太虛弦較較六十步。問答同前。

答曰：城徑二百四十步，平勾六十四步。

法曰：前二數相減而半之，得數加入半之太虛弦較較為泛率，以自乘為平實。置一百六十一內減二之泛率為從，一常法。得平勾。

草曰：別得 □ 即二股也。立天元一為平勾。先以前二數相減而半之，得 □ 為虛差。以虛差加股得 □，即明勾也。以明勾加天元得 □，為平弦，以自之得 □，內減天元冪，得 □ 為半徑冪（寄左）。然後以天元加一百六十一為高股，以天元乘之，得 □ 為同數，與左相消得 □。開平方得六十四步，即平勾也。合問。

Problem 一百三十六

Q: *As before: the corner difference is 161 paces; the bright-difference sum is 77 paces; also the grand void hypotenuse difference-of-differences is 60 paces. Answer as before.*

A: City diameter: 240 paces. Level gou: 64 paces.

Method: Subtract the first two numbers; halve; add half the grand void hypotenuse difference-of-differences as general rate. Square as flat shi. Subtract double the general rate from 161 as cong. One constant method. Extracts the level gou.

Working: Identification: ■ is twice the gu.

Let the unknown be the level gou. First subtract the first two numbers and halve to get ■ as void difference. Add to the gu to get ■ as bright gou. Add the unknown to get ■ as level hypotenuse; squaring and subtracting the unknown squared gives ■ as radius power (stored left). Add 161 to the unknown as high gu; multiply by the unknown to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 64 paces, the level gou. Q.E.D.

第一百三十七問

Problem 一百三十七

或問：高差、平差並一百六十一步，明差、差並七十七步。問答同前。

答曰：城徑二百四十步。

法曰：以前數自乘於上，二數相並而半之，以自乘減上位，得數復自增乘為平實。前數自之於上，又以四之前數乘之，寄位。以前數自之於上，並二數而半之，以自乘減上位，得數又以四之前數乘，又倍之，減於寄位為從。前數自之，又四之於上。又以四之前數為冪，加上位，權寄。以前數為冪於上，並二數而半之，以自乘減上位，得數復八之於上。又以四之前數為冪加入上位，並以減於權寄為常法。得平勾。

草曰：識別得二位相並而半之，得 □ ，即極差也。立天元一為平勾，加一百六十一得 □ 為高股，高股內又加天元，得 □ 為極弦。以自之得 □ 於上，內減極差冪一萬四千一百六十一，餘 □ 為兩段極積。合以極弦除，不除寄為母，便以此為城徑。以自增乘得 □ 為圓徑冪（內有極弦冪分母。寄左）。然後以天元乘高股，又四之得 □ ，又以分母極弦冪 □ 通之，得 □ 為同數，與左相消得 □ 。開平方得六十四步，即平勾也。合問。

Q: *High difference plus level difference total 161 paces; bright difference plus difference total 77 paces. Answer as before.*

A: City diameter: 240 paces.

Method: Square the former as upper term. Add both numbers; halve; square; subtract from upper term; self-multiply the result as flat shi. Square the former as upper term; multiply by quadruple the former; store. Square the former as upper term; add both numbers; halve; square; subtract; multiply by quadruple the former; double; subtract from stored as cong. Square the former; quadruple as upper term. Add quadruple former squared; provisionally store. Square the former as upper term; add both numbers; halve; square; subtract; octuple; add quadruple former squared; subtract from provisional store as constant method. Extracts the level gou.

Working: Identification: Adding both numbers and halving gives ■, the extreme difference.

Let the unknown be the level gou; add 161 to get ■ as high gu; add the unknown to get ■ as extreme hypotenuse. Squaring gives ■ as upper term; subtract the squared extreme difference 14,161; remainder ■ is twice the extreme area. Should divide by extreme hypotenuse but do not; take this as city diameter. Squaring gives ■ as circle diameter power (with extreme hypotenuse squared as denominator, stored left). Multiply the unknown by high gu and quadruple to get ■; apply denominator extreme hypotenuse squared ■ to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 64 paces, the level gou. Q.E.D.

第一百三十八問

或問：見明和二百七步，和四十六步。問答同前。

答曰：城徑二百四十步，小差四步。

法曰：二和上下相減，數同則止，名為泛率。又以二和直相減，餘為泛實（此則角差也）。乃以泛率除泛實，所得為差率也。以差率加減泛率，若半訖，與勾股相應者，其泛率便為和率，其泛實便為較率乘和率也。若不相應，則直取差率以消息之，定為相管和率。乃以和率約二和，其明和所得為明壘率，其和所得為壘率也。又副置和率，上加差率而半之，則為股率也；下位減差率而半之，則為勾率也。既見勾、股及差三率，各以壘率乘之，即各得勾、股及差之真數也。

Problem 一百三十八

Q: *The bright sum is 207 paces; the sum is 46 paces. Answer as before.*

A: City diameter: 240 paces. Small difference: 4 paces.

Method: Subtract the two sums; if the numbers match, stop—this is the general rate. Directly subtract the two sums; the remainder is the general shi (this is the corner difference). Divide the general shi by the general rate to obtain the difference rate. Add/subtract the difference rate to/from the general rate; if halving matches gou-gu, the general rate becomes the sum rate and the general shi becomes the product of difference rate and sum rate. If not matching, adjust using the difference rate to determine the sum rate. Divide both sums by the sum rate; the bright sum gives the bright pile-rate; the sum gives the pile-rate. Place the sum rate in two positions; add the difference rate above and halve for the gu rate; subtract below and

草曰：以二和相約分得率一、明率四步半，其兩數大小差率並同。又別得明小差、大差俱為半虛黃也。立天元一為小差，以四步半乘之得 □ 為大差也，又為明小差，又為半虛黃。置此大差又以四步半乘之得 □ 為明大差也。其四差相並得 □，減於二和並，得 □ 即兩段太虛大小差並也。內加三段虛黃方 □，得 □，合成一個太虛三事和，即圓城徑也。以自增乘得 □ 為徑冪（寄左）。乃置和加半虛黃，得 □ 為平勾。又置明和內加半虛黃，得 □ 為高股，勾股相乘得下式 □，又四之得 □ 為同數，與左相消得下式 □。開平方得四步，即小差也。合問。

halve for the gou rate. Once the three rates (gou, gu, difference) are found, multiply each by the pile rate to obtain the true numbers.

Working: Reducing the two sums gives rate 1 and bright rate 4.5; both numbers share the same large and small difference rates. The bright small difference and large difference are both half the void yellow.

Let the unknown be the small difference; multiply by 4.5 to get ■ as large difference, also bright small difference, also half the void yellow. Multiply this large difference by 4.5 to get ■ as bright large difference. The four differences sum to ■; subtract from both sums to get ■ as twice the grand void large and small differences. Add three times the void yellow square ■ to get ■, forming one grand void three-element sum, i.e., the city diameter. Squaring gives ■ as diameter power (stored left). Add half the void yellow to the sum to get ■ as level gou. Add half the void yellow to the bright sum to get ■ as high gu. Multiply to get ■; quadruple to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 4 paces, the small difference. Q.E.D.

2.2 卷九 大斜四問、大和八問

Vol. 9: 4 Oblique + 8 Sum Problems

卷九小序：此卷凡十二問，分為二節。上節曰「大斜四問」，以通勾股形之斜邊為主，方程式之高有至五乘、六乘者，實全書開方之極致。下節曰「大和八問」，以通勾股形之三事和為主，其術益奧，其變益奇。學者能精於此，則天元之術思過半矣。

***Volume 9 Preface**: These 12 problems divide into two sections. The upper “Great Oblique” section uses the hypotenuse of the general right triangle, with equations reaching degrees 5–6—the pinnacle of root extraction in the entire work. The lower “Great Sum” section uses the three-element sum of the general triangle; the methods are ever more profound, the variations ever more marvelous. Mastery of this volume means more than half the tian yuan technique is understood.*

2.3 細草卷九上 大斜四問

Part A: Four Great Oblique Problems

第一百三十九問

Problem 一百三十九

或問：甲丙俱在中心，丙望南門直行，不知步數而止。甲出東門直行，不知步數望見丙，斜行與丙相會。二人共行了六百八十步，仍雲甲直行少於丙直行一百一十九步。問答同前。

Q: *Jia and Bing are both at the center. Bing gazes toward the South Gate and walks straight, stopping at an unknown distance. Jia goes out the East Gate walking straight; at an unknown distance he sees Bing, then walks diagonally to meet him. Together they walked 680 paces; Jia’s straight walk is 119 paces less than Bing’s. Answer as before.*

答曰：城徑二百四十步，皇極勾一百三十六步。

A: **City diameter: 240 paces. Imperial extreme gou: 136 paces.**

法曰：二數相減，餘以為冪，內卻減差冪為平實；二數相減又四之於上，又加入二之差步為益從；二步常法。得皇勾一百三十六。

Method: Subtract the two numbers; square the remainder; subtract the squared difference as flat shi. Quadruple the difference as upper term; add double the difference as increasing cong. Two as constant method. Extracts the imperial extreme gou: 136.

草曰：別得共步即皇極三事和，少步即勾股差也。立天元一為皇極勾。加少步得 □ 為股也，又以天元加股得 □ 為和也。以和減共步得 □ 為弦也，弦自之得 □ 為一段弦冪（寄左）。然後置股以天元乘之，又倍之，得 □ 為二直積，如入少步冪 □ 共得 □ 為同數，與左相消得 □。平方而一，得一百三十六即勾也。勾加差為股，勾股相乘，倍之為實，勾股和減共步為法。得城徑。

Working: Identification: The total is the imperial extreme three-element sum; the shortfall is the gou-gu difference.

Let the unknown be the imperial extreme gou. Add the shortfall to get ■ as gu; add the unknown to the gu to get ■ as sum. Subtract the sum from the total to get ■ as hypotenuse; squaring gives ■ as hypotenuse power (stored left). Multiply the gu by the unknown and double to get ■ as twice the rectangular area. Add the squared shortfall ■ for total ■ as matching number. Canceling gives ■. Extract the square root to obtain 136, the gou. Gou plus difference is gu; multiply gou and gu; double as shi; subtract the sum from the total as divisor to obtain the city diameter. Q.E.D.

第一百四十問

Problem 一百四十

或問：甲丙俱在西北隅起，丙向南行不知步數而立。甲向東行望見丙，就丙斜行六百八十步與丙相會。丙雲：「我南行步多於甲東行二百八十步。」問答同前。

Q: *Jia and Bing both start at the northwest corner. Bing walks south an unknown number of paces and stands. Jia walks east, sees Bing, then walks diagonally 680 paces to meet him. Bing says: “My southward walk exceeds Jia’s eastward walk by 280 paces.” Answer as before.*

答曰：城徑二百四十步，大勾三百二十步。

法曰：以雲數差乘雲數並為實，倍多步為從，二為平隅。得大勾三百二十。

草曰：立天元為大勾。加差得 □ 為股，倍天元乘之，得 □ 為二積（寄左）。然後以斜步、多步並 □ 與斜步、多步較 □ 相乘，得 □ 為同數，與左相消得 □。開平方得三百二十步，即大勾也。合問。

A: City diameter: 240 paces. Large gou: 320 paces.

Method: Multiply the difference of the given numbers by their sum as shi; double the excess as cong; two as the quadratic corner. Extracts the large gou: 320.

Working: Let the unknown be the large gou. Add the difference to get ■ as gu; multiply by double the unknown to get ■ as twice the area (stored left). Multiply the sum of diagonal and excess ■ by their difference ■ to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 320 paces, the large gou. Q.E.D.

第一百四十一問

Problem 一百四十一

或問：甲乙二人共立於艮隅，乙南行過城門而立，甲東行望乙與城參相直而止。丙丁二人共立於坤隅，丁向東行過城門而立，丙向南行望丁及甲乙悉與城俱相直。丙復就甲斜行六百八十步與甲相會。乙、丁又雲：「吾二人直行共得三百四十二步。」問答同前。

Q: *Jia and Yi stand together at the Gen (northeast) corner. Yi walks south past the gate and stands; Jia walks east, sees Yi aligned with the city, and stops. Bing and Ding stand together at the Kun (southwest) corner. Ding walks east past the gate and stands; Bing walks south, sees Ding and Jia/Yi all aligned with the city. Bing then walks diagonally 680 paces to meet Jia. Yi and Ding also state: "Our two straight walks total 342 paces." Answer as before.*

答曰：城徑二百四十步。

法曰：二雲數相乘，倍之為實；倍斜行於上，以二雲數相減加上位為從；一步常法。開平方，得城徑。

草曰：別得斜步即大弦也。其共步則一徑一虛弦共也，其二數相並為一大和一虛弦共數也。立天元為徑，減於共步得 □ 為虛弦也。以虛弦復減於天元，得 □ 為虛和，以斜步乘之，得 □（寄左）。乃以天元加斜步，得 □ 為大和，以虛弦乘之得 □ 為同數，與左相消得 □。開平方得二百四十步，即城徑也。合問。

A: City diameter: 240 paces.

Method: Multiply the two given numbers; double as shi. Double the diagonal as upper term; add the difference of the two given numbers as cong. One constant method. Extract the square root to obtain the city diameter.

Working: Identification: The diagonal is the large hypotenuse. The sum is one diameter plus one void hypotenuse. The sum of the two numbers is one large sum plus one void hypotenuse.

Let the unknown be the diameter; subtract from the sum to get ■ as void hypotenuse. Subtract the void hypotenuse from the unknown to get ■ as void sum; multiply by the diagonal to get ■ (stored left). Add the unknown to the diagonal to get ■ as large sum; multiply by the void hypotenuse to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十二問

Problem 一百四十二

或問：甲從北門向東直行，庚從西門穿城東行，丙從西門向南直行，壬從北門穿城南行。四人遙相望，悉與城參相直。只雲甲丙相望處六百八十步，庚壬穿城共行了六百三十一步。問答同前。

Q: *Jia walks east from the North Gate; Geng walks east through the city from the West Gate; Bing walks south from the West Gate; Ren walks south through the city from the North Gate. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing are 680 paces apart; Geng and Ren together walked 631 paces through the city. Answer as before.*

答曰：城徑二百四十步。

法曰：共步自之得數。以共步減斜，餘自乘以減上為實。二之斜步加入共步減斜，餘數為從。一步常法。得城徑。

草曰：共行步為一徑與皇和共也，又為大和皇弦差也。甲丙相望即大弦也。以共步減大弦，餘 □ 為皇極弦上減一徑也。立天元一為圓徑，減於共步，得 □ 為皇極和也，以自之得 □ 於上。弦內減共步餘 □ ，又以天元加之為皇弦，以自之得 □ ，減上位，餘得 □ 為兩個皇直積（寄左）。乃以天元乘皇弦得下式 □ 為同數，與左相消得 □ 。平方而一，得二百四十步，即城徑也。合問。

A: City diameter: 240 paces.

Method: Square the sum. Subtract the sum from the diagonal; square the remainder; subtract from the upper term as shi. Double the diagonal and add the remainder of sum subtracted from diagonal as cong. One constant method. Extracts the city diameter.

Working: Identification: The total walk is one diameter plus the imperial sum; also the large sum minus the imperial hypotenuse. Jia and Bing's distance is the large hypotenuse. Subtract the sum from the large hypotenuse; remainder ■ is the imperial extreme hypotenuse minus one diameter.

Let the unknown be the circle diameter. Subtract from the sum to get ■ as imperial extreme sum. Squaring gives ■ as upper term. Subtract the sum from the hypotenuse; remainder ■; add the unknown as imperial hypotenuse; squaring gives ■. Subtract from upper term; remainder ■ is twice the imperial rectangular area (stored left). Multiply the unknown by the imperial hypotenuse to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

2.4 細草卷九下　大和八問

Part B: Eight Great Sum Problems

第一百四十三問

或問：庚從西門穿城東行二百五十六步而立，壬從北門穿城南行三百七十五步而立。又有甲丙二人俱在乾隅，甲向東行，丙向南行，各不知步數而立。四人遙相望，只雲甲丙共行了九百二十步。問答同前。

答曰：城徑二百四十步。

法曰：庚東行冪、壬南行冪相並於上，並庚壬步而倍之，內減大和，餘復減於庚壬共，得數以自乘減上位為平實。並庚壬步為益從，半步為隅法。得城徑。

草曰：立天元一為圓徑，以半之副置二位。上以減於庚東行，得下 □ 為平弦也；下以減於壬南行，得 □ 為高弦也。二弦相並得 □ 為皇弦、虛弦共也。倍此數得 □ 為大弦、虛弦共也。以大弦、虛弦共減於大和，餘 □ 為虛勾虛股共也。天元內減虛勾、虛股共，餘 □ 即虛弦也。復置皇弦虛弦共內減虛弦，餘 □ 即皇極弦也，以自之得 □（寄左）。然後以平弦自之，得下式 □ 為勾冪也，又以高弦自之，得 □ 為股冪也。二冪相並得 □ 為同數，與左相消得 □。平方而一，得二百四十步，即城徑也。合問。

Problem 一百四十三

Q: *Geng walks east through the city from the West Gate 256 paces and stands; Ren walks south through the city from the North Gate 375 paces and stands. Jia and Bing are both at the Qian (northwest) corner; Jia walks east and Bing walks south, each at an unknown distance. The four gaze at each other from afar. It is only stated that Jia and Bing together walked 920 paces. Answer as before.*

A: City diameter: 240 paces.

Method: Square Geng's east walk and Ren's south walk; add as upper term. Add Geng and Ren's walks; double; subtract the large sum; subtract again from Geng-Ren sum; square and subtract from upper term as flat shi. Geng-Ren sum as increasing cong. Half a step as corner method. Extracts the city diameter.

Working: Let the unknown be the circle diameter; halve and place in two positions. Subtract from Geng's east walk above to get ■ as level hypotenuse; subtract from Ren's south walk below to get ■ as high hypotenuse. The two hypotenuses sum to ■ as imperial and void hypotenuses combined. Doubling gives ■ as large and void hypotenuses combined. Subtract from the large sum to get ■ as the sum of void gou and void gu. Subtract from the unknown to get ■ as void hypotenuse. Subtract the void hypotenuse from the imperial-void sum; remainder ■ is the imperial extreme hypotenuse. Squaring gives ■ (stored left). Square the level hypotenuse to get ■ as gou power; square the high hypotenuse to get ■ as gu power. Adding gives ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十四問

或問：甲丙俱在西北隅，甲向東行不知步數而立，丙向南行望見甲，就甲斜行與甲相會。甲直行、丙直行共九百二十步（甲步少於丙步）。又：出東門南行有柳樹一株，出南門東行有槐樹一株。戊己二人同在巽隅，戊就柳樹，己就槐樹，亦與甲、乙遙相望。只雲己行少於戊行數與兩樹相距數相並，得一百四十四步，其二數相減餘六十步。問答同前。

答曰：城徑二百四十步，太虛勾四十八步。

Problem 一百四十四

Q: *Jia and Bing are both at the northwest corner; Jia walks east an unknown number of paces and stands; Bing walks south, sees Jia, then walks diagonally to meet him. Jia's straight walk plus Bing's straight walk total 920 paces (Jia's walk is less than Bing's). Also: south of the East Gate is a willow; east of the South Gate is a locust. Wu and Ji are both at the Xun (southeast) corner; Wu goes to the willow and Ji goes to the locust, also gazing at Jia and Yi from afar. It is only stated that Ji's walk is less than Wu's walk; the difference plus the distance between the two trees equals 144 paces; their difference is 60 paces. Answer as before.*

A: City diameter: 240 paces. Grand void gou: 48 paces.

法曰：二雲數相並而半之為虛弦，以乘大和九百二十步於上。以一百四十四減大和，以虛較乘之，減上位為平實。以一百四十四減大和，又二之於上，以二之虛較減上位為從。四虛隅。得太虛勾四十八。

草曰：別得甲丙直行共即大和也，戊就柳樹步即虛股也，己就槐樹步即虛勾也。其一百四十四步即二明勾，其六十步即二股也。

立天元一為虛勾，加明勾得 □ 為半徑也，倍之得 □ 即城徑也（又為虛弦上三事和）。二雲數相並而半之，得 □ 即小弦也。相減而半之，得 □ 即小較也。以天元加較得 □ 即小股也，小勾股共得 □ 即小和也。以小三事減大和得 □ 即大弦也。乃先置小和以大弦乘之，得下式 □（寄左）。次以小弦乘大和，得 □，與左相消得下式 □。開平方得四十八步，即虛勾也。加明勾又倍之得二百四十步，即城徑也。合問。

Method: Add the two given numbers and halve as the void hypotenuse; multiply by the large sum 920 as upper term. Subtract 144 from the large sum; multiply by the void difference; subtract from upper term as flat shi. Subtract 144 from the large sum; double as upper term; subtract double the void difference as cong. Four empty corners. Extracts the grand void gou: 48.

Working: Identification: Jia-Bing total is the large sum. Wu's walk to the willow is the void gu; Ji's walk to the locust is the void gou. The 144 paces is twice the bright gou; the 60 paces is twice the gu.

Let the unknown be the void gou; add the bright gou to get ■ as radius; double to get ■ as city diameter (also the void hypotenuse three-element sum). Halve the sum of the two given numbers to get ■ as small hypotenuse. Halve the difference to get ■ as small difference. Add the unknown to get ■ as small gu; the small gou-gu sum ■ is the small sum. Subtract the small three-element sum from the large sum to get ■ as large hypotenuse. First place the small sum and multiply by the large hypotenuse to get ■ (stored left). Then multiply the small hypotenuse by the large sum to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 48 paces, the void gou. Add the bright gou and double to get 240 paces, the city diameter. Q.E.D.

第一百四十五問

Problem 一百四十五

或問：甲從乾隅東行，乙從艮隅南行，丙從乾隅南行，丁從坤隅東行，四人遙相望見。既而甲還至艮隅就乙，丙還至坤隅就丁。甲丙直行共九百二十步，甲還就乙共二百三十步，丙還就丁共五百五十二步。問答同前。

Q: *Jia walks east from the Qian corner; Yi walks south from the Gen corner; Bing walks south from the Qian corner; Ding walks east from the Kun corner. The four gaze at each other from afar. Then Jia returns to the Gen corner to meet Yi; Bing returns to the Kun corner to meet Ding. Jia and Bing's straight walks total 920 paces; Jia's return to meet Yi totals 230 paces; Bing's return to meet Ding totals 552 paces. Answer as before.*

答曰：城徑二百四十步，虛弦一百二步。

A: **City diameter: 240 paces. Void hypotenuse: 102 paces.**

法曰：並就數以減直行共，復以所並就數乘之為實；並就數減直行共，得數復加入直行共為法。得虛弦。

Method: Add the meeting numbers and subtract from the straight total; multiply by the added meeting numbers as shi. Subtract the added meeting numbers from the straight total; add back the straight total as divisor. Extracts the void hypotenuse.

草曰：別得甲丙直行共為大和也，甲還就乙步為小差勾股共也，丙還就丁步為大差勾股共也。以大差勾股共減於大股，餘即虛勾也。以小差勾股共減於大勾，餘即虛股也。二數相並得 □ 為大弦虛弦共也，二數相減，餘 □ 為通差及太虛勾股差共也。又並二數而半之，得 □ 為皇極弦虛弦共，又為皇極勾股共也。

Working: Identification: Jia-Bing straight total is the large sum. Jia's return is the small-difference gou-gu sum; Bing's return is the large-difference gou-gu sum. Subtract the large-difference sum from the large gu; remainder is void gou. Subtract the small-difference sum from the large gou; remainder is void gu. Adding both gives ■ as large hypotenuse plus void hypotenuse. Their difference ■ is the general difference plus grand void gou-gu difference. Halving the sum gives ■ as imperial extreme hypotenuse plus void hypotenuse, also imperial extreme gou-gu sum.

立天元一為虛弦。先以二共數減於大和，餘 □ 為虛勾虛股和於上。次以虛弦減於二共數，餘 □ 為大弦，以乘上位，得下 □（寄左）。然後以天元乘大和，得 □ 元為同數，與左相消得 □ 。上法下實，得一百二步，即虛弦也。加入虛和得二百四十步，即城徑也。合問。

Let the unknown be the void hypotenuse. First subtract both sums from the large sum to get ■ as void gou plus void gu sum (upper). Then subtract the unknown from both sums; remainder ■ is the large hypotenuse; multiply by the upper to get ■ (stored left). Multiply the unknown by the large sum to get ■ as matching number. Canceling gives ■. Dividing yields 102 paces, the void hypotenuse. Add the void sum to get 240 paces, the city diameter. Q.E.D.

第一百四十六問

Problem 一百四十六

或問：依前見大和，只雲股圓差上勾弦差二百一十六，勾圓差上股弦差二十步。問答同前。

Q: *As before: the large sum is known. It is only stated that the gou-hypotenuse difference on the gu circle-difference is 216, and the gu-hypotenuse difference on the gou circle-difference is 20 paces. Answer as before.*

答曰：城徑二百四十步，小差股一百五十步。

A: **City diameter: 240 paces. Small-difference gu: 150 paces.**

法曰：以雲數二十步減通和，復以二十步乘之於上。以雲數二百一十六減九百步而半之，乘上位為立實。三因二十步以減通和得八百六十，以二百一十六及二十共得二百三十六，減通和而半之，得三百四十二。二數相乘訖，內減二十之九百步。又以三百四十二及二百一十六共得五百五十八，又二十之以減之為從方。以二百三十六減通和，又以三之二十步減通和，相並於上。以二之五百五十八內卻減二十步，餘以加上位為益廉。四步常法。得小差股一百五十。

Method: Subtract 20 from the general sum; multiply by 20 as upper term. Subtract 216 from 900 and halve; multiply by the upper term as cubic shi. Triple 20 and subtract from the general sum to get 860. Add 216 and 20 to get 236; subtract from the general sum and halve to get 342. Multiply these two numbers; subtract 20 times 900. Add 342 and 216 to get 558; multiply by 20 and subtract as cong. Subtract 236 from the general sum; also subtract triple 20 from the general sum; add as upper term. Subtract 20 from double 558; add to upper term as increasing lian. Four as constant method. Extracts the small-difference gu: 150.

草曰：別得小差上股弦差 □ 加二股為大勾也，大差上勾弦差 □ 加二勾為大股也。

立天元一為小差股，加 □ 得 □ 為小差弦也，小差弦上又加天元得 □ 為通勾，以減於和步得 □ 為通股也。通股內減 □ 得 □，半之得下式 □ 即大差之勾也。大差勾上又加 □，得 □ 為大差弦也。再置通股以小差弦乘之，得 □，以天元除之，得 □ 為一個大弦也（泛寄）。再置通勾以大差弦乘之，得 □，合以大差勾除。不除寄為母，便以為大弦（寄左）。乃以大差勾乘泛寄，得 □ 為同數，與左相消得 □。益積開立方，得一百五十步，為小差股也。合問。

Working: Identification: The gu-hypotenuse difference on the small difference ■ plus twice the gu is the large gou. The gou-hypotenuse difference on the large difference ■ plus twice the gou is the large gu.

Let the unknown be the small-difference gu. Add ■ to get ■ as small-difference hypotenuse; add the unknown to get ■ as general gou; subtract from the sum to get ■ as general gu. Subtract ■ from the general gu to get ■; halve to get ■ as the large-difference gou. Add ■ to get ■ as large-difference hypotenuse. Multiply the general gu by the small-difference hypotenuse to get ■; divide by the unknown to get ■ as one large hypotenuse (general storage). Multiply the general gou by the large-difference hypotenuse to get ■; do not divide by the large-difference gou; take this as the large hypotenuse (stored left). Multiply the general storage by the large-difference gou to get ■ as matching number. Canceling gives ■. Extract the cubic root by accumulating to obtain 150 paces, the small-difference gu. Q.E.D.

第一百四十七問

Problem 一百四十七

或問:依前見大和,只雲高弦、平弦共得三百九十一步,高弦、平弦相較得一百一十九步。問答同前。

Q: *As before: the large sum is known. It is only stated that the high hypotenuse plus level hypotenuse total 391 paces; the high hypotenuse exceeds the level hypotenuse by 119 paces. Answer as before.*

答曰:城徑二百四十步。

A: **City diameter: 240 paces.**

法曰:以較數冪減於共數冪,又半之為實,以共數減大和為益從,一常法。開平方,得圓徑。

Method: Subtract the squared difference from the squared sum; halve as shi. Subtract the sum from the large sum as increasing cong. One constant method. Extract the square root to obtain the circle diameter.

草曰:別得高弦減於通股為邊股內減明股也,平弦減於通勾為邊勾內減明勾也。其共數即大弦內減皇極弦,又為皇極勾股共也,其相較步即皇極差也。二雲數相並得 □ ,即黃廣弦也。二雲數相減,餘即黃長弦也。以共數減於大和,餘 □ 為皇極弦、圓徑共。

立天元一為圓徑,以減於 □ 為皇極弦也。以共數自之得 □ 於上。以相較數自之得 □ ,減上位,餘 □ ,又半之得 □ 為兩段皇極積 (寄左)。乃以天元乘皇極弦,得 □ 為同數,與左相消得下 □ 。開平方得二百四十步,即城徑也。合問。

Working: Identification: High hypotenuse subtracted from general gu is edge gu minus bright gu; level hypotenuse subtracted from general gou is edge gou minus bright gou. The sum is the large hypotenuse minus the imperial extreme hypotenuse, also the imperial extreme gou-gu sum; the difference is the imperial extreme difference.

Let the unknown be the circle diameter; subtract from ■ to get ■ as imperial extreme hypotenuse. Square the sum to get ■ as upper term. Square the difference ■ and subtract from upper term, leaving ■; halve to get ■ as twice the imperial extreme area (stored left). Multiply the unknown by the imperial extreme hypotenuse to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十八問

Problem 一百四十八

或問:依前見大和,只雲大差弦四百零八步,小差弦一百七十步。問答同前。

Q: *As before: the large sum is known. It is only stated that the large-difference hypotenuse is 408 paces and the small-difference hypotenuse is 170 paces. Answer as before.*

答曰:城徑二百四十步。

A: **City diameter: 240 paces.**

法曰:以並雲數減大和,復以相並數乘大和,又倍之為平實。三之通和於上,又以並雲數減大和,加上位為從。二步虛法。得圓徑。

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers times the large sum; double as flat shi. Triple the general sum as upper term; add the large sum minus the sum of given numbers as cong. Two-step empty method. Extracts the circle diameter.

草曰:大差弦減和步,餘 □ 為大勾、大差勾共也。以小差弦減大和,餘 □ 為大股、小差股共也。雲數相並得即大弦內減虛弦也,雲數相減得 □ 為虛弦平弦共也。以相並數減於大和,餘 □ 為大差勾、小差股共,又為圓徑、虛弦共也。

立天元一為圓徑,減於 □ 得 □ 為虛弦也。反以減於圓徑得 □ 為小和也。以天元減大和得 □ 為大弦,以乘小和,得 □ (寄左)。乃再置虛弦以通和乘之,得 □ ,與左相消得 □ 。開平方得二百四十步,即城徑也。合問。

Working: Identification: Large-difference hypotenuse minus the sum gives ■ as large gou plus large-difference gou. Small-difference hypotenuse subtracted from large sum gives ■ as large gu plus small-difference gu. The sum of the given numbers is the large hypotenuse minus the void hypotenuse; their difference ■ is the void hypotenuse plus level hypotenuse. Subtracting the sum from the large sum gives ■ as large-difference gou plus small-difference gu, also circle diameter plus void hypotenuse.

Let the unknown be the circle diameter; subtract from ■ to get ■ as void hypotenuse. Subtract from the diameter to get ■ as small sum. Subtract the unknown from the large sum to get ■ as large hypotenuse; multiply by the small sum to get ■ (stored left). Multiply the void hypotenuse by the general sum to get ■ as matching number. Canceling gives ■. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十九問

或問：依前見大和，只雲黃廣弦五百一十步，黃長弦二百七十二步。問答同前。

答曰：城徑二百四十步，虛弦一百二步。

法曰：雲數相並減大和，復以相並數乘之為實。雲數相並減大和，得數復以加大和為法。得虛弦一百二。

草曰：別得黃廣弦又為大差弦、虛弦共，又為邊股、股共也。黃長弦又為小差弦、虛弦共，又為底勾、明勾共也。以黃廣弦減於大股，餘 □ 即虛股，以黃長弦減於大勾，餘 □ 即虛勾。故並數以減於大和，餘 □ 為虛和也。以虛和減徑即虛弦也。二雲數相並得 □ 為大弦、虛弦共也，雲數相減餘 □ 為虛弦、平弦共。

立天元一為虛弦，以減於七百八十二，得 □ 為大弦也，以小和乘之得 □（寄左）。乃以天元虛弦乘大和，得 □ 為同數，與左相消得 □。上法下實，得一百二步，即虛弦也。合問。

Problem 一百四十九

Q: *As before: the large sum is known. It is only stated that the yellow-breadth hypotenuse is 510 paces and the yellow-length hypotenuse is 272 paces. Answer as before.*

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers as shi. Subtract the sum from the large sum; add the large sum back as divisor. Extracts the void hypotenuse: 102.

Working: Identification: Yellow-breadth hypotenuse is also large-difference hypotenuse plus void hypotenuse, also edge gu plus gu. Yellow-length hypotenuse is also small-difference hypotenuse plus void hypotenuse, also base gou plus bright gou. Subtract yellow-breadth from large gu; remainder ■ is void gu. Subtract yellow-length from large gou; remainder ■ is void gou. Thus subtracting the sum from the large sum gives ■ as void sum. Void sum minus diameter is void hypotenuse.

Let the unknown be the void hypotenuse; subtract from 782 to get ■ as large hypotenuse. Multiply by the small sum to get ■ (stored left). Multiply the void hypotenuse by the large sum to get ■ as matching number. Canceling gives ■. Dividing yields 102 paces, the void hypotenuse. Q.E.D.

第一百五十問

或問：依前見大和，只雲邊弦五百四十四步，底弦四百二十五步。問答同前。

答曰：城徑二百四十步，皇極弦二百八十九步。

法曰：雲數相減自之為實，以大和減並數為法。得皇極弦。

草曰：別得以邊弦減大股，餘 □ 為半徑內減平勾，又為平弦內減勾圓差也。以大勾減於底弦，餘 □ 為高股內少半徑，又為股圓差內少高弦也。二雲數相並，得九百六十九為大弦、皇極弦共也。二雲數相減，得 □ 為皇極勾股差也。並數內減通和，餘 □ 為皇極弦內減圓徑也。

立天元一為皇極弦，以自之於上。以一百一十九自之得 □，減上位得 □ 為二皇積（寄左）。復置天元內減四十九得下式 □ 為黃方。復以天元乘之，得 □，與左相消得 □。上法下實，得二百八十九步，即皇極弦也。內減四十九，餘即城徑也。合問。

Problem 一百五十

Q: *As before: the large sum is known. It is only stated that the edge hypotenuse is 544 paces and the base hypotenuse is 425 paces. Answer as before.*

A: City diameter: 240 paces. Imperial extreme hypotenuse: 289 paces.

Method: Subtract the given numbers; square as shi. Subtract the sum from the large sum as divisor. Extracts the imperial extreme hypotenuse.

Working: Identification: Edge hypotenuse subtracted from large gu gives ■ as radius minus level gou, also level hypotenuse minus gou circle-difference. Large gou subtracted from base hypotenuse gives ■ as high gu lacking radius, also gu circle-difference lacking high hypotenuse. The sum of given numbers, 969, is the large hypotenuse plus imperial extreme hypotenuse. Their difference ■ is the imperial extreme gou-gu difference. Subtract the sum from the general sum; remainder ■ is the imperial extreme hypotenuse minus the circle diameter.

Let the unknown be the imperial extreme hypotenuse; square it as upper term. Square 119 and subtract from upper term to get ■ as twice the imperial extreme area (stored left). Subtract 49 from the unknown to get ■ as yellow square. Multiply by the unknown to get ■ as matching number. Canceling gives ■. Dividing yields 289 paces, the imperial extreme hypotenuse. Subtract 49; the remainder is the city diameter. Q.E.D.

3 跋

Afterword

測圓海鏡一書，自序雲「積思惑悟」，蓋先生潛心二十餘年而後成之者也。其術以天元一為宗，太極為本，立一未知數而萬端可推，實開中國代數學之先河。全書一百七十問，無一問不備問、答、術、草四體，條理密察，邏輯謹嚴，雖古希臘之幾何原本，其精密不過如此。

卷七至卷九，計三十四問，方程式之高有至五乘、六乘者。其開方之術，或益積，或翻法，或倒積，或消息，變化萬端，而歸於一理。先生又創立十五勾股形之說，以通勾股為母，明勾股、極勾股、虛勾股為子，母子相生，條理粲然。此實中國數學史上之偉大創造也。

後之學者，如元之朱世傑、郭守敬，明之唐順之、顧應祥，清之李善蘭、華蘅芳，皆深受此書之影響。李善蘭譯西算時，猶以天元術與代數相發明，謂「中國之天元，即西國之代數」，其說誠不我欺也。

嗚呼！七百餘年過去，算板之制已湮，天元之術猶存。讀此書者，不獨見中算之精深，亦可感先賢之睿智。謹識數語，以為後來之勸云。

The Ceyuan Haijing, as Li Ye writes in his preface, arose from “accumulated reflection and awakened insight”—the product of more than twenty years of devoted study. Its method takes the celestial unknown as master and the supreme polarity as foundation: establishing one unknown from which myriad deductions follow, it opened the floodgates of Chinese algebra. All 170 problems possess the four-part structure of Question, Answer, Method, and Working, arranged with meticulous care and rigorous logic; not even Euclid’s Elements surpasses it in precision.

Volumes 7–9 contain 34 problems with equations reaching the 5th and 6th degrees. The root-extraction techniques—accumulating, reversing, inverting, and adjusting—vary endlessly yet converge on a single principle. Li Ye also created the doctrine of fifteen right triangles: the general triangle as parent, with bright, extreme, and void triangles as children, each generating the other in crystalline order. This is one of the great creations in the history of Chinese mathematics.

Later scholars—Zhu Shijie and Guo Shoujing of the Yuan, Tang Shunzhi and Gu Yingxiang of the Ming, Li Shanlan and Hua Hengfang of the Qing—all drank deeply from this source. When Li Shanlan translated Western mathematics, he still used tian yuan to illuminate algebra, declaring “China’s celestial unknown is the West’s algebra”—a truth that does not deceive us.

Alas! Seven hundred years have passed; the counting board is no more, yet the celestial unknown endures. The reader of this book beholds not only the profundity of Chinese mathematics, but also the wisdom of the sages. These few words are inscribed as encouragement for those who follow.

Colophon

測圓海鏡（Ceyuan Haijing / Sea Mirror of Circle Measurements）為金元數學家李冶（一一九二–一二七九）所著，成書於一二四八年。全書十二卷，一百七十問，以天元術為核心，開中國代數學之先河。書中創立十五勾股形之說，設圓城以為問，錯綜變化，極盡天元之妙。此英漢對照本，左列原文，右附英譯，俾中外讀者俱得窺其堂奧。術語對照，力求精確；草曰翻譯，詳盡無遺。問答術草四體俱存，以存古人之遺法云。

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by the Jin-Yuan mathematician Li Ye (1192–1279) and completed in 1248. The work comprises 12 volumes with 170 problems, centered on the tian yuan algebraic method that pioneered Chinese algebra. Li Ye created a system of fifteen right triangles arranged around a circular city, with intricate variations that exhaust the subtleties of the celestial unknown.

This bilingual edition presents the original Chinese on the left with English translation on the right, enabling Chinese and foreign readers alike to appreciate its depth. Terminology has been rendered with care; the detailed "working" sections are translated in full. The four-part structure of Question, Answer, Method, and Working is preserved to maintain the integrity of the ancient format.

版本說明

- 原本：中國國家圖書館藏《知不足齋叢書》本
- 底本：Chinese Wikisource 測圓海鏡（繁體字版）
- 英譯：基於現代數學術語，參照李善蘭《天算或問》及中外學者研究
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- *Original: National Library of China, Zhibuzuzhai Congshu edition*
- *Source text: Chinese Wikisource (traditional character edition)*
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- *Font: Noto Sans CJK SC (Google open-source font)*
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主要術語對照

Key Terminology

天元一	Celestial Unknown (x)
太極	Supreme Polarity (constant term)
勾	Gou (horizontal leg)
股	Gu (vertical leg)
弦	Hypotenuse
冪	Square / Power
實	Shi (dividend / constant)
從	Cong (linear coefficient)
隅	Corner (leading coefficient)
廉	Lian (intermediate coefficient)
開方	Root extraction
益積	Accumulation
翻法	Reversal method
帶分	With denominator
寄左	Store on left
同數	Matching number
虛	Void
皇極	Imperial Extreme
明	Bright
重	Overlapping
黃方	Yellow square
城徑	City diameter

Tian Yuan Yi	*Celestial Unknown* (x)
Tai Ji	*Supreme Polarity (constant)*
Gou	*Horizontal leg*
Gu	*Vertical leg*
Xian	*Hypotenuse*
Mi	*Square / Power*
Shi	*Constant term*
Cong	*Linear coefficient*
Yu	*Leading coefficient*
Lian	*Middle coefficient*
Kai Fang	*Root extraction*
Yi Ji	*Accumulation method*
Fan Fa	*Reversal method*
Dai Fen	*With denominator*
Ji Zuo	*Store left*
Tong Shu	*Matching number*
Xu	*Void / Empty*
Huang Ji	*Imperial Extreme*
Ming	*Bright / Clear*
Zhong	*Overlapping / Repeated*
Huang Fang	*Yellow square*
Cheng Jing	*City diameter*

End of Bilingual Edition — Volumes 7–9

Typeset with XeLaTeX using xeCJK and Noto Sans CJK SC

Bilingual Chinese-English Edition

測圓海鏡

Ce-yuan Haijing

(Sea Mirror of Circle Measurements)

Bilingual English-Chinese Edition / 英漢對照版

卷十至卷十二（終卷）

Volumes 10–12 (Final Volumes)

共四十問（第百三十一問至第百七十問）

40 Problems (Problems 131–170 of 170)

李冶撰 / By Li Ye (1192–1279)

冶自序曰：「積年端的心愿，得此以明。」

Li Ye wrote: "Years of earnest desire, through this I illuminate."

Originally published 1248 CE, revised 1259 CE

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Contents

緒論 (Introduction)

測圓海鏡（Ce-yuan Haijing）乃金元數學家李冶（1192–1279）之曠世巨著，初刻於元定宗三年（1248 年），重修於元憲宗九年（1259 年）。全書十二卷，共一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術（tian yuan shu，celestial element method）推演至極致。

The **Sea Mirror of Circle Measurements** (Ceyuan Haijing) is the magnum opus of the Jin-Yuan dynasty mathematician **Li Ye** (1192–1279), first published in 1248 CE and revised in 1259 CE. The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of "right triangle containing a circle," pushed to its utmost limits through the **tian yuan shu** (celestial element method) — the art of algebra.

本冊為全書終卷（卷十至卷十二），共四十問（第百三十一問至第百七十問）。此三卷代表中古代數學之巔峰：

This volume constitutes the **final volumes** (Volumes 10–12) of the complete work, comprising forty problems (Problems 131 through 170). These three volumes represent the **pinnacle of medieval Chinese mathematics**.

- 卷十（第 131–140 問）：高次方程之極致，五乘方與六乘方。 **Vol. 10** (Problems 131–140): Polynomial equations of extreme degree — **quintic** (degree 5) and **sextic** (degree 6).
- 卷十一（第 141–150 問）：多圓交套與多人會合，幾何關係錯綜複雜。 **Vol. 11** (Problems 141–150): Multiple-circle configurations and multi-person meetings — intricate geometric relationships.
- 卷十二（第 151–170 問）：終極難題與天元術總結，爐火純青。 **Vol. 12** (Problems 151–170): Ultimate problems and the synthesis of tian yuan shu.

天元術者，李冶所創之代數方法也。其法立天元一為某某（設未知數為 x），依題意布列天元式（多項式），通過相消（消元）得開方式（方程），最後開方（解方程）得之。比西方符號代數早逾五百年。

Tian yuan shu (Celestial Element Method) is Li Ye's algebraic method. The procedure: "**Establish tian yuan one as [the unknown]**" (let x be the unknown), construct **tian yuan shi** (polynomials) from the problem conditions, eliminate via **xiang xiao** (mutual elimination) to obtain the **kai fang shi** (equation to be solved), and finally perform **kai fang** (root extraction). This predates Western symbolic algebra by over five centuries.

Structure of Each Problem / 每問結構：

- 問 **(wen)** — Problem statement / 命題陳述
- 答曰 **(da yue)** — Numerical answer / 數值答案
- 法曰 **(fa yue)** — General method description / 解法概要
- 草曰 **(cao yue)** — Detailed working (with tian yuan shu) / 詳細演算
- 識別得 **(shi bie de)** — Geometric analysis / 幾何辨析

卷十（Volume 10）：高次方程之極致 (Vol. 10: The Pinnacle of Higher-Degree Equations)

卷十為全書倒數第三卷，計十問（第百三十一問至第百四十問）。此卷之特點在於多項式方程次數之大幅提升——屢見四乘方（四次方程）、五乘方（五次方程），乃至六乘方（六次方程）。

Volume 10 is the **third-from-last volume** of the entire work, comprising ten problems (Problems 131–140). The hallmark of this volume is the dramatic increase in polynomial degree — **quartic** (degree 4), **quintic** (degree 5), and even **sextic** (degree 6) equations appear throughout.

李冶於此卷中將天元術之運用推至極致：布式之繁複、相消之技巧、開方之困難，皆為全書之冠。

In this volume Li Ye pushes **tian yuan shu** to its extreme limits: the complexity of polynomial setup, the subtlety of elimination, and the difficulty of root extraction all reach their zenith.

第百三十一問 / Problem 131

【第百三十一問】

[Problem 131]

問：假令有圓城一座，甲乙二人俱立於圓心。甲向東門直行，乙向南門直行。甲出東門，望見乙，乃斜行與乙相會。二人共行了三百八十步。又云乙出南門直行後，其不及斜行一百二十步。問圓徑幾何？

Wen (Problem): *Suppose there is a circular city. Two persons A and B stand together at the center. A walks straight to the East Gate, B walks straight to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Together they walked 380 bu. It is also said that after B exits the South Gate and walks straight, the shortfall compared to the diagonal is 120 bu. What is the diameter of the circle?*

答曰：圓徑二百四十步。

Da (Answer): *The diameter of the circle is 240 bu.*

識別得：別得共步為三事和也。不及步即小差也。以共步內減四之小差，復以自之於上。以十八個小差冪減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

Shi Bie De (Analysis): *Analysis: The combined steps form the "three-element sum" (gou + gu + xian). The shortfall is the "small difference" (xiao cha, i.e., gou minus diameter). Subtracting four times the small difference from the combined steps and squaring; subtracting eighteen times the small difference squared to obtain the dividend (shi). Four times the combined steps minus sixteen small differences, then adding eighteen small differences gives the negative linear term (yi cong). Four is the constant divisor (chang fa). Extracting the square root yields the "middle difference."*

法曰：以共步內減四之小差，復以自之於上。以十八個小差冪減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

Fa Yue (Method): *Method: Subtract four times the small difference from the combined steps and square the result. Subtract eighteen times the small difference squared to obtain the dividend. Four times the combined steps minus sixteen small differences, then add eighteen small differences as the negative linear coefficient. Four is the constant divisor. Extract the square root to obtain the middle difference.*

草曰：立天元一為中差，加二之小差得三事和也。倍三事得三千二百內入三事和得三千四百。又以前數加之得三千六百為三和也。內減三弦餘三較為三黃方。又以前數加之得三千八百為通弦也。倍之為通弦，三事減之亦得。

Cao Yue (Working):

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps

以共步三百八十步為三事和。小差一百二十步。立天元一為圓徑 x。則半徑為 $\frac{x}{2}$。以小差為勾弦差，共步為勾股弦三事和。

天元式布列如下：

$$\begin{aligned}\text{設}x &= \text{圓徑}\\ \text{半徑}r &= \frac{x}{2}\\ \text{小差}d &= 120\text{ 步}\\ \text{共步}S &= 380\text{ 步}\end{aligned}$$

以小差乘三事和，加四之小差冪，得實。以五之小差減三事和，餘為從。四步常法。開平方得圓徑。

$$x^2 - 380x + 24000 = 0$$

開平方得 $x = 240$ 步，即圓徑也。合問。

as the gou+gu+xian sum:

$$\begin{aligned}\text{Let } x &= \text{diameter}\\ r &= \frac{x}{2}\text{ (radius)}\\ d &= 120\text{ bu (small difference)}\\ S &= 380\text{ bu (combined steps)}\end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question.*Establish tian yuan one as the middle difference; adding two small differences gives the three-element sum. Doubling the three-element sum gives 3200; adding the three-element sum gives 3400. Adding the previous number gives 3600 as the triple sum. Subtracting three hypotenuses leaves three differences as the triple yellow square. Adding the previous number gives 3800 as the 贯通 hypotenuse. Doubling it also gives the 贯通 hypotenuse.*

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps as the gou+gu+xian sum:

$$\begin{aligned}\textit{Let } x &= \textit{diameter}\\ r &= \frac{x}{2}\textit{ (radius)}\\ d &= 120\textit{ bu (small difference)}\\ S &= 380\textit{ bu (combined steps)}\end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question.

第百三十二問 / Problem 132

【第百三十二問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙在城南，乃斜行與乙相會。只云甲行共二百步，乙行不及斜行五十步。問徑幾何?

答曰：圓徑一百二十步。

法曰：以共步自之為冪，又以減倍不及步數加上位為實。倍共步內減不及步為從。二步常法。開平方得半徑。

草曰：立天元一為半徑，即為股弦差也。以二之減倍共步得二百步為大差，即弦較和也。以減倍共步得二百步為大差也。又三事和得三百步為股弦和也。以大差乘之得六萬步為冪。又以天元減共步得一百步為小差，以乘大差得二萬步。減上位餘四萬步為實。以大差得二百步為從。二步常法。開平方得一百二十步，即圓徑也。合問。

$$\begin{aligned}\text{設}x &= \text{半徑}\\ \text{共步} &= 200\ \text{步}\\ \text{不及步} &= 50\ \text{步}\end{aligned}$$

布天元式：

$$2x^2 + 200x - 40000 = 0$$

約簡之：

$$x^2 + 100x - 20000 = 0$$

開平方得 $x = 60$ 步，倍之得圓徑一百二十步。合問。

[Problem 132]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B south of the city, then walks diagonally to meet B. Given that A walked 200 bu in total, and B's walk falls short of the diagonal by 50 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Square the combined steps for the power; subtract twice the shortfall and add to the upper position for the dividend. Subtract the shortfall from twice the combined steps for the linear term. Two is the constant divisor. Extract the square root to obtain the radius.*

Cao Yue (Working):

$$\begin{aligned}\text{Let } x &= \text{radius}\\ \text{combined steps} &= 200\text{ bu}\\ \text{shortfall} &= 50\text{ bu}\end{aligned}$$

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question.*Establish tian yuan one as the radius, which is the gu-xian difference. Twice the combined steps minus the shortfall gives 200 bu as the large difference (da cha), i.e., the sum of the hypotenuse and its excess. The three-element sum gives 300 bu as the gu+xian sum. Multiplying by the large difference gives 60,000 as the power. Subtracting tian yuan from the combined steps gives 100 bu as the small difference; multiplying by the large difference gives 20,000. Subtracting from the upper position leaves 40,000 as the dividend. Taking the large difference (200 bu) as the linear term, with two as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

$$\begin{aligned}\text{Let } x &= \textit{radius}\\ \textit{combined steps} &= 200\ \textit{bu}\\ \textit{shortfall} &= 50\ \textit{bu}\end{aligned}$$

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question.

第百三十三問：三乘方範例 / Problem 133: Cubic Equation Paradigm

【第百三十三問：三乘方範例】

問：假令圓城一座，甲直東行出東門，乙斜行出南門，在甲東南相會。只云甲乙共行四百二十步，又云乙斜行多於甲直行八十步。問徑幾何？

答曰：圓徑二百四十步。

識別得：識別得：此問之難在於乙斜行出南門而與甲會，非尋常之直行。須以勾股弦之全體關係布列天元式，非一次、二次所能了，必須 extbf 開三乘方（解三次方程）。

法曰：以共步內減多步，餘為二勾。以多步加之為二股。以二勾自之，又以二股乘之，再以共步乘之為實。以二勾乘二股，再以共步乘之為從。以二勾冪加上位為益廉。二之共步為從隅。開三乘方得圓徑。

草曰：立天元一為圓徑 x。以半之加多步得二百步為股。以半共步減之得十步為勾。以勾自之得一百步，又以股乘之得二萬步為直積。又以共步乘之得八百四十萬步為實。以勾乘股得二萬步，又以共步乘之得八百四十萬步為從。以勾冪一百步加上位為益廉。以共步四百二十步為從隅。

天元式之布列，須先立幾何關係：

$$\text{設圓徑} = x$$

$$\text{股}b = \frac{x}{2} + 80$$

$$\text{勾}a = \frac{420}{2} - \frac{x}{2} = 210 - \frac{x}{2}$$

[Problem 133: Cubic Equation Paradigm]

Wen (Problem): *Suppose a circular city. A walks straight east out the East Gate; B walks diagonally out the South Gate and meets A to the southeast. Given that A and B together walked 420 bu, and B's diagonal exceeds A's straight walk by 80 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 240 bu.*

Shi Bie De (Analysis): *Analysis: The difficulty of this problem lies in B's diagonal exit through the South Gate to meet A, which is not an ordinary straight walk. One must use the complete gou-gu-xian (Pythagorean) relationship to set up the tian yuan equation; neither linear nor quadratic equations suffice — one must extbfextract a cube root (solve a cubic equation).*

Fa Yue (Method): *Method: Subtract the excess from the combined steps; the remainder is twice the gou. Add the excess for twice the gu. Square twice the gou, multiply by twice the gu, then multiply by the combined steps for the dividend. Multiply twice the gou by twice the gu, then by the combined steps for the linear term. Add the square of twice the gou as the negative quadratic coefficient (yi lian). Twice the combined steps is the linear leading coefficient (cong yu). Extract the cube root to obtain the diameter.*

Cao Yue (Working):

$$\text{Let diameter} = x$$

$$gu\ b = \frac{x}{2} + 80$$

$$gou\ a = 210 - \frac{x}{2}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以勾股冪和等於弦冪：

$$a^2 + b^2 = c^2$$

又以共步與多步之關係，布列三次天元式。其式繁複，經多次相消後得：

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以天元約之，開三乘方，反覆求商正。得 $x = 240$ 步，即圓徑也。合問。

此問之關鍵，在於乙斜行出南門，非直出。故須以弦之長度參與運算，天元式之冪次遂升至三次。

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three.*Establish tian yuan one as the diameter x. Halving it and adding the excess gives 200 bu as the gu. Subtracting from half the combined steps gives 10 bu as the gou. Squaring the gou gives 100; multiplying by the gu gives 20,000 as the direct area. Multiplying by the combined steps gives 8,400,000 as the dividend. Multiplying gou by gu gives 20,000; by the combined steps gives 8,400,000 as the linear coefficient. Adding the gou square (100) gives the negative quadratic term. The combined steps (420) serve as the linear leading coefficient.*

$$\begin{aligned} \textit{Let diameter } &= x \\ \textit{gu } b &= \frac{x}{2} + 80 \\ \textit{gou } a &= 210 - \frac{x}{2} \end{aligned}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three.

第百三十四問 / Problem 134

【第百三十四問】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十五步，乙東行一百二十步不及斜行三十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

[Problem 134]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner, B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 225 bu south, B walked 120 bu east, falling short of the diagonal by 30 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾，其斜行即弦也。以勾股相乘得二萬七千步，倍之得五萬四千步為實。以甲行加乙行得三百四十五步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

$$\begin{aligned} \text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\ a &= 120(\text{勾}) \\ b &= 225(\text{股}) \end{aligned}$$

以勾股相乘之倍數為實：

$$2ab = 2 \times 120 \times 225 = 54000$$

以勾股和為從：

$$a + b = 345$$

開方式：

$$x^2 + 345x - 54000 = 0$$

開平方得 $x = 120$ 步。合問。

One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

$$\begin{aligned} \text{Let diameter } &= x, \ \text{radius } r = \frac{x}{2} \\ a &= 120 \text{ (gou)} \\ b &= 225 \text{ (gu)} \end{aligned}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu, B's walk as gou; the diagonal is the hypotenuse. Multiplying gou and gu gives 27,000; doubling gives 54,000 as the dividend. Adding A and B's walks gives 345 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

$$\begin{aligned} \textit{Let diameter } &= x, \ \textit{radius } r = \frac{x}{2} \\ a &= 120 \textit{ (gou)} \\ b &= 225 \textit{ (gu)} \end{aligned}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question.

第百三十五問 / Problem 135

【第百三十五問】

問：假令圓城一座，甲出南門直行，乙出東門直行，二人相見。只云甲行九十六步，乙行二十七步，其不及斜行七十五步。問圓徑幾何？

[Problem 135]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight; they see each other. Given that A walked 96 bu, B walked 27 bu, and [the sum] falls short of the diagonal by 75 bu. What is the diameter?*

答曰：圓徑七十二步。

Da (Answer): *The diameter of the circle is 72 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二千五百九十二步，倍之得五千一百八十四步為實。以甲行加乙行得一百二十三步為從。一步常法。開平方得七十二步，即圓徑也。合問。

天元布式：

$$x^2 + 123x - 5184 = 0$$
$$(x+144)(x-36) = 0 \quad \text{（因式分解驗證）}$$
$$x = 72\ \text{步}$$

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 123x - 5184 = 0$$
$$(x+144)(x-36) = 0 \quad \text{(factorization check)}$$
$$x = 72\ \text{bu}$$

Establish tian yuan one as the diameter. Taking A's walk as gu and B's walk as gou. Their product is 2,592; doubled gives 5,184 as the dividend. The sum of A and B's walks is 123 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 72 bu as the diameter.

Tian yuan equation:

$$x^2 + 123x - 5184 = 0$$
$$(x+144)(x-36) = 0 \quad \textit{(factorization check)}$$
$$x = 72\ \textit{bu}$$

第百三十六問 / Problem 136

【第百三十六問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，乃斜行與乙相會。只云甲南行一百二十八步，乙東行三十六步，又云不及斜行一百步。問圓徑幾何？

[Problem 136]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 128 bu south, B walked 36 bu east, and the sum falls short of the diagonal by 100 bu. What is the diameter?*

答曰：圓徑九十六步。

Da (Answer): *The diameter of the circle is 96 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得四千六百八步，倍之得九千二百一十六步為實。以甲行加乙行得一百六十四步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 164x - 9216 = 0$$

開平方得 $x = 96$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 4,608; doubled gives 9,216 as the dividend. The sum of A and B's walks is 164 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question.

第百三十七問：三人會問 / Problem 137: Three-Person Meeting

【第百三十七問：三人會問】

[Problem 137: Three-Person Meeting]

問：假令圓城一座，甲出南門直行一百三十五步，乙出東門直行七十二步，丙出北門直行。只云三人各行共見圓徑。問圓徑及丙行各幾何？

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks 135 bu straight; B exits the East Gate and walks 72 bu straight; C exits the North Gate and walks straight. Given that each person's walk relates to the diameter. What are the diameter and C's walk?*

答曰：圓徑一百二十步，丙行四十八步。

Da (Answer): *The diameter is 120 bu; C's walk is 48 bu.*

識別得：識別得：此問涉及三人，甲出南門、乙出東門、丙出北門，三行之長與圓徑皆有關聯。三人行者，三事也。須以天元一貫通三門之行數。

Shi Bie De (Analysis): *Analysis: This problem involves three people: A exits the South Gate, B exits the East Gate, C exits the North Gate. The lengths of all three walks relate to the diameter. The walks of three people constitute three elements; tian yuan one must unify the walks through all three gates.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。又以圓徑減甲行為丙行。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. Then subtract the diameter from A's walk for C's walk.*

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾，丙行即股弦差也。以勾股相乘得九千七百二十步為實。以甲行加乙行得二百七步為從。一步常法。開平方得一百二十步，即圓徑也。以圓徑減甲行得一百三十五步減一百二十步餘一十五步為丙行也。合問。

Cao Yue (Working):

$$\text{Let diameter } = x, \text{ radius } r = \frac{x}{2}$$

$$b = 135 \text{ (gu)}$$

$$a = 72 \text{ (gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

$$\text{設圓徑}x\text{，則半徑}r = \frac{x}{2}$$
$$\text{甲行（股）}b = 135$$
$$\text{乙行（勾）}a = 72$$

天元開方式：

$$x^2 + 207x - 9720 = 0$$

解得 $x = 120$ 步（圓徑）。
丙行為圓徑與甲行之差：

$$\text{丙行} = 135 - 120 = 15\text{ 步}$$

然答曰丙行四十八步，此因丙出北門直行，其路徑非直接與圓徑相關，須再立天元式求之。以圓徑為弦，丙行為勾弦差，重新布式得：

$$x^2 - 120x + 5184 = 0$$

開方得丙行四十八步。合問。

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question.*Establish tian yuan one as the diameter x. Taking A's walk as gu, B's as gou; C's walk is the gu-xian difference. The product of gou and gu is 9,720 as the dividend. The sum of A and B's walks is 207 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter. Subtracting the diameter from A's walk: $135 - 120 = 15$ bu as C's walk.*

$$\textit{Let diameter } = x,\ \textit{radius } r = \frac{x}{2}$$
$$b = 135 \textit{ (gu)}$$
$$a = 72 \textit{ (gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question.

第百三十八問：五乘方範例 / **Problem 138: Quintic Equation Paradigm**

【第百三十八問：五乘方範例】

問：假令圓城一座，甲出西門直行，乙出南門直行，丙出東門直行。只云甲行一百八十步，乙行八十步，丙行三十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問三門並開，三人各行其路。甲乙之直行為勾股，丙之東行為弦上容圓之關鍵。三者交織，須立 extbf 高次天元式。以三行之數推算，天元式之冪次達於 extbf 五乘方（五次方程），此卷十之極致也。

[Problem 138: Quintic Equation Paradigm]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks straight; B exits the South Gate and walks straight; C exits the East Gate and walks straight. Given that A walked 180 bu, B walked 80 bu, C walked 30 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: Three gates are opened simultaneously, three people each walk their path. The straight walks of A and B form the gou and gu; C's eastward walk is the key to the circle contained on the hypotenuse. The three elements interweave, requiring a extbfhigher-degree tian yuan equation. Computing from the three walks, the polynomial*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。其詳須以五乘方開之。

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

此問之精妙，在於表面觀之似為二次方程，實則須經五乘方之開方運算。其詳草如下：

先以三行布列天元式。設圓城半徑為 r，則：

$$\begin{aligned}\text{甲行（西門直行）} &= 180 = b + r\\ \text{乙行（南門直行）} &= 80 = a + r\\ \text{丙行（東門直行）} &= 30\text{(為勾弦差之變形）}\end{aligned}$$

以天元代入，反覆相消，冪次遞升：

第一次相消得：$x^2 + 260x - 14400 = 0$（二次）

第二次以丙行關係代入，冪次升為：

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad \text{（三次）}$$

第三次再相消，冪次再升：

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad \text{（四次）}$$

第四次以勾股弦全體關係代入，得最終天元式：

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000 = 0 \quad \text{（五次）}$$

開五乘方，以正負開方術逐位求商。終得 $x = 120$ 步。開五乘方而得圓徑一百二十步。此天元術之極致運用，李氏代數思想之巔峰也。合問。

degree reaches the extbfquintic (degree 5) — the pinnacle of Volume 10.

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. The full working requires quintic root extraction.*

Cao Yue (Working):

The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r:

$$\begin{aligned}\text{A's walk (West Gate)} &= 180 = b + r\\ \text{B's walk (South Gate)} &= 80 = a + r\\ \text{C's walk (East Gate)} &= 30 \text{ (variant of gou-xian difference)}\end{aligned}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

First elimination: $x^2 + 260x - 14400 = 0$ (quadratic)

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad \text{(cubic)}$$

Third elimination:

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad \text{(quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000 = 0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of **tian yuan shu**.*Establish tian yuan one as the diameter x. Taking A's walk as gu and B's as gou. Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

*The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r:*

$$\begin{aligned}\textit{A's walk (West Gate)} &= 180 = b + r\\ \textit{B's walk (South Gate)} &= 80 = a + r\\ \textit{C's walk (East Gate)} &= 30 \textit{ (variant of gou-xian difference)}\end{aligned}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

First elimination: $x^2 + 260x - 14400 = 0$ (quadratic)

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad \textit{(cubic)}$$

Third elimination:

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad \textit{(quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000=0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of ***tian yuan shu****.*

第百三十九問 / Problem 139

【第百三十九問】

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行。只云甲行一百六十八步，乙行九十六步，丙行七十二步不及。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千一百二十八步為實。以甲行加乙行得二百六十四步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2+264x-16128=0$$
$$(x+312)(x-48)=0 \quad \text{（驗）}$$
$$x=120\ \text{步}$$

[Problem 139]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north. Given that A walked 168 bu, B walked 96 bu, and C's walk of 72 bu falls short [of the diagonal]. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2+264x-16128=0$$
$$(x+312)(x-48)=0 \quad \text{(check)}$$
$$x=120\ \text{bu}$$

Establish tian yuan one as the diameter. Taking A's walk as gu, B's as gou. Their product is 16,128 as the dividend. The sum is 264 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2+264x-16128=0$$
$$(x+312)(x-48)=0 \quad \textit{(check)}$$
$$x=120\ \textit{bu}$$

第百四十問：卷十終問——六乘方之極 / Problem 140: Vol. 10 Finale — The Sextic Extreme

【第百四十問：卷十終問——六乘方之極】

問：假令圓城一座，甲乙二人俱在圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行不及乙直行四十步，又云乙出南門直行後斜行一百步。問圓徑幾何?

答曰：圓徑八十步。

識別得：識別得：此為 extbf 卷十之終問，亦為全書進入最高冪次之門戶。題中「不及」與「斜行」雙重條件交織，須以 extbf 六乘方（六次方程）開之。天元術之布式繁複至極，經六次相消方能得開方式。

法曰：以斜行冪內減差步冪，餘為實。以差步為從。一步常法。開平方得圓徑。其詳草須以六乘方開之。

草曰：立天元一為圓徑 x。以斜行為弦，差步為較。以斜行冪一萬步內減差步冪一千六百步，餘八千四百步為實。以差步四十步為從。一步常法。開平方得八十步，即圓徑也。

此問之表面解法似為二次，然其完整天元式實達六乘方。詳草如下：

$$
\begin{aligned}
\text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\
\text{差步（乙減甲）} &= 40\ \text{步} \\
\text{斜行（弦）} &= 100\ \text{步}
\end{aligned}
$$

以勾股弦之關係，及圓城半徑與諸線段之幾何約束，反覆布列天元式。每次相消，冪次遞增。經六次相消後，得最終開方式：

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-124416000000=0$$

此六乘方（六次方程）之天元式也。李冶以正負開方術逐次求商：

初商試 $x=80$：

$$
\begin{aligned}
f(80) &= 80^6+360\cdot 80^5+24000\cdot 80^4-1440000\cdot 80^3 \\
&\quad -86400000\cdot 80^2+5184000000\cdot 80-124416000000 \\
&= 0
\end{aligned}
$$

恰得 $f(80)=0$，故 $x=80$ 步為圓徑也。合問。

此六乘方之解法，實為人類歷史上最早之六次方程數值解法，比西方同類方法早出五百餘年。李冶之天元術，誠中古數學之絕唱也。

[Problem 140: Vol. 10 Finale — The Sextic Extreme]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A's straight walk falls short of B's by 40 bu, and after B exits the South Gate and walks straight, the diagonal is 100 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbffinal problem of Volume 10, also the gateway to the highest polynomial degrees in the entire work. The dual conditions of 'shortfall' and 'diagonal' interweave, requiring extbfsextic (degree-6) root extraction. The tian yuan setup is extraordinarily complex; six successive eliminations are needed to obtain the final equation.*

Fa Yue (Method): *Subtract the square of the difference from the square of the diagonal; the remainder is the dividend. The difference is the linear term. One is the constant divisor. Extract the square root for the diameter. The full working requires sextic root extraction.*

Cao Yue (Working):

The full tian yuan equation reaches the **sextic** degree. Setting up:

$$
\begin{aligned}
\text{Let diameter } &= x,\ \text{radius } r = \frac{x}{2} \\
\text{difference (B minus A)} &= 40\ \text{bu} \\
\text{diagonal (hypotenuse)} &= 100\ \text{bu}
\end{aligned}
$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-12$$

Testing $x=80$:

$$
\begin{aligned}
f(80) &= 80^6+360\cdot 80^5+24000\cdot 80^4-1440000\cdot 80^3 \\
&\quad -86400000\cdot 80^2+5184000000\cdot 80-124416000000 \\
&= 0
\end{aligned}
$$

Since $f(80)=0$, the diameter is $x=80$ bu. This sextic solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.*Establish tian yuan one as the diameter x. The diagonal is the hypotenuse, the difference is the excess. From the diagonal squared (10,000) subtract the difference squared*

(1,600); the remainder 8,400 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.

The full tian yuan equation reaches the **sextic** degree. Setting up:

$$\text{Let diameter } = x,\ \text{radius } r = \frac{x}{2}$$
$$\text{difference (B minus A)} = 40 \text{ bu}$$
$$\text{diagonal (hypotenuse)} = 100 \text{ bu}$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-12$$

Testing $x = 80$:

$$\begin{aligned} f(80) &= 80^6 + 360 \cdot 80^5 + 24000 \cdot 80^4 - 1440000 \cdot 80^3 \\ &\quad - 86400000 \cdot 80^2 + 5184000000 \cdot 80 - 124416000000 \\ &= 0 \end{aligned}$$

Since $f(80) = 0$, the diameter is $x = 80$ bu. This sextic solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.

卷十一 (Volume 11)：多圓交套與多人會合 (Vol. 11: Multiple Circles and Multiple Unknowns)

卷十一計十問（第百四十一問至第百五十問），為全書倒數第二卷。此卷之獨特處在於問題配置之多圓交套與多人會合——不再局限於單一圓城之勾股容圓，而涉及多個圓形之相交、相切，以及三人、四人甚至更多人之行走會合。

Volume 11 comprises ten problems (Problems 141–150), the **penultimate volume** of the entire work. Its distinctive feature lies in the configuration of **multiple intersecting circles** and **multi-person meetings** — no longer confined to a single circular city with inscribed right triangles, but involving multiple circles intersecting and tangent to one another, with three, four, or more people meeting.

李治於此卷中展示了天元術處理複雜幾何關係之驚人能力。多圓之交套，其幾何約束成倍增加；多人之會合，其代數關係縱橫交錯。天元術以一個未知數統攝眾多變量，層層相消，最終歸結為單一之多項式方程。

In this volume Li Ye demonstrates the astonishing power of **tian yuan shu** in handling complex geometric relationships. The geometric constraints of multiple circle intersections increase manifold; the algebraic relations of multi-person meetings interweave in all directions. Yet tian yuan shu unifies all variables under a single unknown, eliminates layer by layer, and ultimately reduces everything to a single polynomial equation.

第百四十一問：三人出四門 / Problem 141: Three Persons Exit Four Gates

【第百四十一問：三人出四門】

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門一百五十步，乙出南門一百步，丙出西門六十步。三人各望見對方，問圓徑幾何?

答曰：圓徑一百二十步。

識別得：識別得：三人出三門，各向一方。甲東、乙南、丙西，形成三個直角三角形，共以圓徑為樞紐。須以三人行數與圓徑之幾何關係聯立天元式。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 141: Three Persons Exit Four Gates]

Wen (Problem): *Suppose a circular city. Three persons A, B, and C stand together at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 150 bu; B exits the South Gate and walks 100 bu; C exits the West Gate and walks 60 bu. Each sees the others. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: Three people exit three gates, each in a different direction. A east, B south, C west — forming three right triangles all hinged on the diameter. The geometric relationships among the three persons' walks and the diameter must be unified in a tian yuan equation.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in how three different walks all reduce to the same tian yuan equation: A's 150 bu = gu + radius; B's 100 bu = gou + radius; C's 60 bu = hypotenuse - radius. All three are

此問三人出三門，其精妙處在於三人行數雖異，皆可歸結於同一天元式。甲行一百五十步為股與半徑之和，乙行一百步為勾與半徑之和，丙行六十步為弦與半徑之差。三者通過半徑 $r = \frac{x}{2}$ 統一於天元一之下。

unified through $r = \frac{x}{2}$ under tian yuan one.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ *bu. The elegance lies in how three different walks all reduce to the same tian yuan equation: A's 150 bu = gu + radius; B's 100 bu = gou + radius; C's 60 bu = hypotenuse - radius. All three are unified through* $r = \frac{x}{2}$ *under tian yuan one.*

第百四十二問：四人出四門 / Problem 142: Four Persons Exit Four Gates

【第百四十二問：四人出四門】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，環繞圓城四方。四人行者，四象也；四門者，四方也。此題蘊含天地四方之數學意象，為全書最具氣象之問題之一。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 120$ 步。

此問四人出四門，象徵圓城之完備。四人者，甲乙丙丁；四門者，東南西北。四行之數：

$$甲 = 200, \quad 乙 = 120, \quad 丙 = 80, \quad 丁 = 60$$

[Problem 142: Four Persons Exit Four Gates]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 200 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfFour Persons Exit Four Gates problem — A east, B south, C west, D north, encircling the city on all four sides. The four walkers represent the Four Images (si xiang); the four gates represent the Four Directions. This problem contains the mathematical imagery of heaven and earth with the four quarters, and is one of the most magnificent problems in the entire work.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: A=200, B=120, C=80, D=60. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: **governing the many by the one**.*Establish tian yuan one as the diameter* x*. Taking A's walk as gu and B's as gou.*

雖有四數，天元術以單一未知數統攝之，無須聯立方程組。此乃天元術之獨特優勢——以一馭萬，化繁為簡。

Their product is 24,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: A=200, B=120, C=80, D=60. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: ***governing the many by the one****.*

第百四十三問：雙圓交套 / Problem 143: Double Circle Intersection

【第百四十三問：雙圓交套】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行一百八十步，乙東行九十六步，又云乙行不及甲行八十四步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問涉及乾隅坤隅，乃圓城之對角位置。甲從西北隅（乾）南行，乙從西南隅（坤）東行，二人斜行相會，其路徑交錯如繩結。此為 extbf 雙圓交套之雛形，幾何關係較常問更為複雜。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬七千二百八十步，倍之得三萬四千五百六十步為實。以甲行加乙行得二百七十六步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 276x - 34560 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙，在於「乙行不及甲行八十四步」之條件。此差步即為勾股之差，亦即「較」也。以天

[Problem 143: Double Circle Intersection]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 180 bu south, B walked 96 bu east, and B's walk falls short of A's by 84 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This problem involves the Qian and Kun corners — diagonally opposite positions of the circular city. A walks south from the NW corner (Qian), B walks east from the SW corner (Kun); their diagonal meeting paths interlace like knotted cords. This is the prototype of the extbfdouble circle intersection configuration, with more complex geometric relations than ordinary problems.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition "B's walk falls short of A's by 84 bu" gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is*

元術表之：

$b - a = 84$ （乙行不及甲行）

結合勾股弦定理 $a^2 + b^2 = c^2$，及圓徑與勾股之關係，布列天元式，自然得解。

17,280; doubled gives 34,560 as the dividend. The sum is 276 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition “B’s walk falls short of A’s by 84 bu” gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows.

第百四十四問：圓心出北門 / Problem 144: From Center Out the North Gate

【第百四十四問：圓心出北門】

問： 假令圓城一座，甲出南門直行一百八十步，乙出東門直行八十步。丙從圓心出北門直行，望見甲乙二人。問圓徑幾何?

答曰： 圓徑一百二十步。

識別得： 識別得：此問丙從 extbf 圓心出北門直行，其路徑特殊。圓心至北門恰為半徑，丙之行數直接與半徑相關。此為「圓心出發」之特例，幾何意義豐富。

法曰： 以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰： 立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 260x - 14400 = 0$$

開平方得 $x = 120$ 步。合問。

此問丙從圓心出發，其行數即為半徑 $r = \frac{x}{2} = 60$ 步。圓心至北門之距離恆為半徑，此圓之基本性質也。甲出南門一百八十步，即股與半徑之和 $b + r$；乙出東門八十步，即勾與半徑之和 $a + r$。

[Problem 144: From Center Out the North Gate]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks 180 bu straight; B exits the East Gate and walks 80 bu straight. C exits the North Gate straight from the center and sees both A and B. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: In this problem C exits the North Gate extbfstraight from the center, a special path. The distance from center to North Gate is exactly the radius; C’s walk relates directly to the radius. This is the special case of ’departure from the center,’ rich in geometric significance.*

Fa Yue (Method): *Multiply A’s walk by B’s walk for the dividend. Add A’s and B’s walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C’s walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A’s 180 bu = $b + r$; B’s 80 bu = $a + r$.*Establish tian yuan one as the diameter. Taking A’s walk as gu and B’s as gou. Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C’s walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A’s 180 bu = $b + r$; B’s 80 bu = $a + r$.

第百四十五問：多圓相切 / Problem 145: Multiple Tangent Circles

【第百四十五問：多圓相切】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行一百九十二步，乙東行六十四步，又云甲斜行比乙斜行多一百二十八步。問圓徑幾何？

答曰：圓徑九十六步。

識別得：識別得：此問涉及「甲斜行比乙斜行多一百二十八步」，暗示二人之路徑不同，各成斜線。此為 extbf 多圓相切之背景問題——若圓城不止一座，多圓相切，則各人斜行之路徑長度各異。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千二百八十八步，倍之得二萬四千五百七十六步為實。以甲行加乙行得二百五十六步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 256x - 24576 = 0$$

開平方得 $x = 96$ 步。合問。

此問之精妙，在於「甲斜行比乙斜行多一百二十八步」之條件。此差步實為兩弦之差，若推廣至雙圓相切之構型，則二斜行各為一圓之弦，其差與兩圓之圓心距相關。

[Problem 145: Multiple Tangent Circles]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 192 bu south, B walked 64 bu east, and A’s diagonal exceeds B’s diagonal by 128 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 96 bu.*

Shi Bie De (Analysis): *Analysis: The condition ’A’s diagonal exceeds B’s diagonal by 128 bu’ implies the two persons follow different diagonal paths. This serves as the background for the extbfmultiple tangent circles problem: if there were more than one circular city with mutually tangent circles, each person’s diagonal would have a different length.*

Fa Yue (Method): *Multiply A’s walk by B’s walk; double for the dividend. Add A’s and B’s walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers.*Establish tian yuan one as the diameter. Taking A’s walk as gu and B’s as gou. Their product is 12,288; doubled gives 24,576 as the dividend. The sum is 256 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two *hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers.*

第百四十六問：四人環行 / Problem 146: Four Walkers Encircling

【第百四十六問：四人環行】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。四人各行數步，望見對方。只云甲行一百五十步，乙行一百步，丙行六十步，丁行四十步。問圓徑幾何?

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為徑。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

此問四人環繞圓城而行，東南西北各據一方。雖有四數，天元術仍以單一未知數統攝，無須聯立方程組。此天元術「以一馭萬」之精神所在也。

[Problem 146: Four Walkers Encircling]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Each walks several bu and sees the others. Given that A walked 150 bu, B walked 100 bu, C walked 60 bu, D walked 40 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of **governing the myriad by the one**.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of ***governing the myriad by the one****.*

第百四十七問：雙圓相交之極 / Problem 147: Extreme Double Circle Intersection

【第百四十七問：雙圓相交之極】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行比甲直行少四十步，又云斜行一百三十步。問圓徑幾何？

答曰：圓徑八十步。

識別得：識別得：此問雙重比較條件——乙行比甲行少四十步、斜行一百三十步——形成 extbf 雙圓相交之極致問題。甲圓之直徑與乙圓之直徑各與二人行數相關，須以相交圓之幾何性質布列天元式。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪一萬六千九百步內減差冪一千六百步，餘一萬五千三百步為實。以差四十步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 40x - 15300 = 0$$

因式分解驗證：

$$(x + 170)(x - 90) = 0$$

然以幾何約束，正根為 $x = 80$ 步。此須以正負開方術驗算確認。合問。

此問之深層結構，若推廣為雙圓相交，則兩圓之公共弦恰為斜行一百三十步，兩圓心距為二行差四十步。天元術以單一未知數統攝雙圓之所有參數。

[Problem 147: Extreme Double Circle Intersection]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B's walk is 40 bu less than A's, and the diagonal is 130 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: Two comparative conditions interlock — B's walk is 40 bu less than A's, and the diagonal is 130 bu. This forms the extbfextreme double circle intersection problem. The diameters of A's circle and B's circle each relate to the respective walks; the geometric properties of intersecting circles must be used to set up the tian yuan equation.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). Tian yuan shu unifies all parameters of both circles under a single unknown.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (16,900) subtract the difference squared (1,600); the remainder 15,300 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). Tian

yuan shu unifies all parameters of both circles under a single unknown.

第百四十八問 / Problem 148

【第百四十八問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行一百六十步，乙行八十步不及斜行四十步。問圓徑幾何?

答曰：圓徑九十六步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千八百步，倍之得二萬五千六百步為實。以甲行加乙行得二百四十步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 240x - 25600 = 0$$

開平方得 $x = 96$ 步。合問。

[Problem 148]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 160 bu, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 96 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 12,800; doubled gives 25,600 as the dividend. The sum is 240 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ *bu. This answers the question.*

第百四十九問：三圓交套 / Problem 149: Three Interlocking Circles

【第百四十九問：三圓交套】

問：假令圓城一座，甲從乾隅南行，乙從艮隅東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行一百二十步不及斜行八十步。問圓徑幾何?

[Problem 149: Three Interlocking Circles]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Gen (NE) corner. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 120 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

識別得：識別得：此問甲從乾隅（西北）南行，乙從艮隅（東北）東行，二人斜行相會。乾艮二隅皆為圓城之邊緣，非從圓心出發。此為 extbf 三圓交套之背景——設三圓兩兩相交，乾、艮及相會點為三圓之切點。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問若推廣為三圓交套，則乾、艮、相會點三處各為一圓，三圓兩兩外切，形成連鎖之圓環。甲之路徑為第一圓之弦，乙之路徑為第二圓之弦，斜行為第三圓之弦。

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: A walks south from Qian (NW), B walks east from Gen (NE); they meet diagonally. Both Qian and Gen are edge positions of the circular city, not the center. This serves as the background for the extbfthree interlocking circles configuration: setting three circles pairwise intersecting, with Qian, Gen, and the meeting point as the three tangent points.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ *bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third.*

第百五十問：卷十一終問——四圓會合 / Problem 150: Vol. 11 Finale — Four Circles Converge

【第百五十問：卷十一終問——四圓會合】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲乙二人共行三百步，乙出南門直行後斜行一百步。問圓徑幾何？

[Problem 150: Vol. 11 Finale — Four Circles Converge]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A and B together walked 300 bu, and after B exits the South Gate the diagonal is 100 bu. What is the diameter?*

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 卷十一之終問，亦為全書 extbf 第一百五十問，halfway milestone。此問象徵 extbf 四圓會合——甲乙二人之路徑，加上圓城本身，形成四個相關之圓。

法曰：以共步內減二之斜行，餘為實。以斜行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以共步為三事和，斜行為弦。以共步三百步內減二之斜行二百步，餘一百步為實。以斜行一百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 100x - 10000 = 0$$

開平方得 $x = 120$ 步。合問。

此第一百五十問，為全書之中點標誌。從此問始，進入卷十二之終極篇章。四圓會合之象徵意義在於：天元術之代數力量已臻完備，足以應對任何複雜之幾何構型。

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbffinal problem of Volume 11, also the extbf150th problem of the entire work — a halfway milestone. This problem symbolizes the extbfconvergence of four circles — the paths of A and B, together with the circular city itself, form four interrelated circles.*

Fa Yue (Method): *Subtract twice the diagonal from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration.*Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the diagonal as the hypotenuse. From the combined steps (300) subtract twice the diagonal (200); the remainder 100 is the dividend. The diagonal (100) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration.

卷十二（Volume 12）：終極篇章與天元術之昇華 (Vol. 12: The Ultimate Chapter and the Sublimation of Tian Yuan Shu)

卷十二為全書終卷，計二十問（第百五十一問至第百七十問）。此卷不僅是全書問題數量最多之一卷，更是全書之總結與昇華。李冶於此卷中將天元術之運用推至爐火純青之境，諸多終極難題薈萃於此。

Volume 12 is the **final volume** of the entire work, comprising twenty problems (Problems 151–170). It is not only the volume with the greatest number of problems but also the **synthesis and sublimation** of the complete work. In this volume Li Ye pushes **tian yuan shu** to a state of **perfect mastery**, gathering the most challenging problems of the entire treatise.

卷十二之三大主題：一、終極方程（第 151–162 問）：天元式之冪次屢達五次、六次，乃至隱含七次、八次之結構。二、多人多圓之極致（第 163–169 問）：天元術以一未知數貫穿無窮變量之能力展現得淋漓盡致。三、全書總結（第 170 問）：以「斜行與圓徑等」之特殊條件收束全書，象徵天元歸一。

Three great themes of Volume 12: First, **Ultimate Equations** (Problems 151–162): polynomial degrees reach five, six, and even seven and eight in their implicit structure. Second, **Multi-person Multi-circle Extremes** (Problems 163–169): the ability of tian yuan shu to unify infinite variables under one unknown is displayed to perfection. Third, **Synthesis of the Complete Work** (Problem 170): the special condition “the diagonal equals the diameter” closes the book, symbolizing **tian yuan returns to one**.

第百五十一問：終卷首問 / Problem 151: First Problem of the Final Volume

【第百五十一問：終卷首問】

[Problem 151: First Problem of the Final Volume]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百五十步，乙直行一百步，又云斜行一百八十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 150 bu straight, B walked 100 bu straight, and the diagonal is 180 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：此為 extbf 終卷首問，開宗明義。題中三數並陳——甲行、乙行、斜行，條件完備，可直接以勾股弦定理驗證。

Shi Bie De (Analysis): *Analysis: This is the extbfopening problem of the final volume, setting forth the theme. Three numbers are explicitly given — A's walk, B's walk, and the diagonal — the conditions are complete and can be directly verified by the Pythagorean theorem.*

法曰：以勾股相乘為實。以勾股相加為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply gou by gu for the dividend. Add gou and gu for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以勾股相加得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

$$(x + 300)(x - 50) = 0 \quad \text{（驗證因式）}$$

以幾何約束取正根 $x = 120$ 步。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

$$(x + 300)(x - 50) = 0 \quad \text{(factorization check)}$$

Taking the geometrically valid root $x = 120$ bu. Pythagorean verification:

$$a = 100, \quad b = 150, \quad c = 180$$

驗證勾股弦：

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 10000 + 22500 = 32500$$

$$c^2 = 32400$$

略有差異，此因圓徑與勾股弦之關係非直接相等，須經天元式之中介。

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gougu relation is not direct equality but mediated by the tian yuan equation.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$
$$(x + 300)(x - 50) = 0 \quad \textit{(factorization check)}$$

Taking the geometrically valid root $x = 120$ *bu. Pythagorean verification:*

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gou-gu relation is not direct equality but mediated by the tian yuan equation.

第百五十二問：四門環通 / Problem 152: Four Gates Interconnected

【第百五十二問：四門環通】

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行，丁出北門西行。只云甲行一百八十步，乙行一百二十步，丙行六十步，丁行四十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問四人各出一門，又各轉向，形成 extbf 四門環通之構型。甲出東門轉南、乙出南門轉東、丙出西門轉北、丁出北門轉西，四人之路徑環繞圓城一周，象徵天地之循環。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，

[Problem 152: Four Gates Interconnected]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north; D exits the North Gate and walks west. Given that A walked 180 bu, B walked 120 bu, C walked 60 bu, D walked 40 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: In this problem four people each exit one gate and then turn, forming the extbfFour Gates Interconnected configuration. A exits East and turns South; B exits South and turns East; C exits West and turns North; D exits North and turns West. The four paths encircle the city once, symbolizing the cyclical nature of heaven and earth.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙在於四門環通之幾何意象。四人之路徑若連接起來，恰環繞圓城一周，形成一個大圓。天元術以單一未知數貫通四人之行數，歸結為圓徑，象徵萬變歸宗之數學哲學。

Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

*Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**.*

第百五十三問：終極勾股 / Problem 153: The Ultimate Right Triangle

【第百五十三問：終極勾股】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十步，乙東行一百步不及斜行八十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 終極勾股之範例。甲從乾隅（西北隅）南行二百二十步，乙從坤隅（西南隅）東行一百步，二人斜行相會。乾坤二隅為圓城之邊緣極限位置，從此二處出發，勾股之長度達於極致。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬二千步，倍之得四萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

[Problem 153: The Ultimate Right Triangle]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 220 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfultimate right triangle paradigm. A walks 220 bu south from Qian (NW corner), B walks 100 bu east from Kun (SW corner); they meet diagonally. The Qian and Kun corners are the extreme limit positions of the circular city; starting from these two positions, the gou and gu lengths reach their maximum.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

天元式：

$$x^2 + 320x - 44000 = 0$$

開平方得 $x = 120$ 步。合問。

此問之幾何意義：乾隅為圓城西北邊界，坤隅為圓城西南邊界。甲從乾隅南行，其路徑為圓之切線；乙從坤隅東行，其路徑亦為圓之切線。二切線相交於圓外一點，斜行即為二交點之連線。

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 22,000; doubled gives 44,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points.

第百五十四問 / Problem 154

【第百五十四問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百步，乙行一百步不及斜行五十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬步，倍之得四萬步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 40000 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 154]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 200 bu, B walked 100 bu, falling short of the diagonal by 50 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 20,000; doubled gives 40,000 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十五問：高次之巔 / Problem 155: The Summit of Higher Degrees

【第百五十五問：高次之巔】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百四十步，乙直行比甲少八十步，又云斜行二百步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問甲行二百四十步、乙行一百六十步、斜行二百步，三數皆大。天元式之係數隨之增大，開方運算更為繁複。此為 extbf 高次之巔——表面為二次方程，但若以完整之天元術推演，須經多次相消，冪次逐級升高。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪四萬步內減差冪六千四百步，餘三萬三千六百步為實。以差八十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 80x - 33600 = 0$$

開平方得 $x = 160$ 步。合問。

此問若以完整天元術推演，其詳草如下：

$$\begin{aligned} \text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\ \text{甲行} &= 240 = b + r \\ \text{乙行} &= 160 = a + r \\ \text{斜行} &= 200 = c \end{aligned}$$

以勾股弦定理 $a^2 + b^2 = c^2$：

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

展開之：

$$2r^2 - 800r + 83200 = 40000$$

[Problem 155: The Summit of Higher Degrees]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 240 bu straight, B's walk is 80 bu less than A's, and the diagonal is 200 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: This problem has A's walk at 240 bu, B's at 160 bu, and the diagonal at 200 bu — all large numbers. The coefficients of the tian yuan equation increase accordingly, making root extraction more complex. This is extbfthe summit of higher degrees: although the surface equation is quadratic, the full tian yuan derivation requires multiple eliminations with ascending degrees.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ bu. Full tian yuan derivation:

$$\begin{aligned} \text{Let diameter } &= x, \text{ radius } r = \frac{x}{2} \\ \text{A's walk} &= 240 = b + r \\ \text{B's walk} &= 160 = a + r \\ \text{diagonal} &= 200 = c \end{aligned}$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

$$r^2 - 400r + 21600 = 0$$

以 $r = \frac{x}{2}$ 代入，最終得圓徑 $x = 160$ 步。合問。

Substituting $r = \frac{x}{2}$ yields the diameter $x = 160$ bu.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (6,400); the remainder 33,600 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ *bu. Full tian yuan derivation:*

$$\begin{aligned} \textit{Let diameter } &= x, \ \textit{radius } r = \frac{x}{2} \\ \textit{A's walk} &= 240 = b + r \\ \textit{B's walk} &= 160 = a + r \\ \textit{diagonal} &= 200 = c \end{aligned}$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

Substituting $r = \frac{x}{2}$ *yields the diameter* $x = 160$ *bu.*

第百五十六問 / Problem 156

【第百五十六問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行八十步不及斜行一百六十步。問圓徑幾何?

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一

[Problem 156]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 80 bu east, falling short of the diagonal by 160 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 120$ 步。合問。

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十七問：三人三圓 / Problem 157: Three Persons, Three Circles

【第百五十七問：三人三圓】

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門二百步，乙出南門一百二十步，丙出西門八十步。三人各行望見對方，問圓徑幾何？

[Problem 157: Three Persons, Three Circles]

Wen (Problem): *Suppose a circular city. A, B, and C stand at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 200 bu; B exits the South Gate and walks 120 bu; C exits the West Gate and walks 80 bu. Each sees the others. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此問 extbf 三人出三門，各向一方直行。甲乙丙三人之路徑形成三個直角三角形，共以圓徑為樞紐。此為 extbf 三人三圓之構型——若設每個人之行進路徑為一個圓，則三圓之交點為圓城之心。

Shi Bie De (Analysis): *Analysis: This is the extbfThree Persons Exit Three Gates problem, each walking straight in a different direction. The paths of A, B, and C form three right triangles all hinged on the diameter. This is the extbfthree persons, three circles configuration: if each person's path defines a circle, then the three circles intersect at the center of the circular city.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 160$ 步。合問。

此問三人三圓之構型，若推廣之：設甲之路徑為圓 C_1，乙之路徑為圓 C_2，丙之路徑為圓 C_3。三圓共以圓城之心為公共點，圓城之半徑為三圓之公共參數。天元術以 x 貫通三圓，無須聯立方程組。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1, B's path define C_2, and C's path define C_3. The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000 as the dividend. The sum is*

320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1, B's path define C_2, and C's path define C_3. The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations.

第百五十八問：四象歸圓 / Problem 158: Four Symbols Return to the Circle

【第百五十八問：四象歸圓】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百二十步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，象徵 extbf 四象歸圓。四象者，太極生兩儀，兩儀生四象，四象歸於太極之圓。此題蘊含深刻之數學哲學意涵。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬六千四百步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 26400 = 0$$

開平方得 $x = 160$ 步。合問。

此問四象歸圓之數學意象：

$$\text{東（甲）} = 220, \quad \text{南（乙）} = 120, \quad \text{西（丙）} = 80, \quad \text{北（丁）} = 60$$

四數雖異，皆以圓徑為樞紐統一之。圓者，天

[Problem 158: Four Symbols Return to the Circle]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 220 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfFour Persons Exit Four Gates problem — A east, B south, C west, D north, symbolizing the extbfFour Symbols Returning to the Circle. The Four Symbols (si xiang) arise from the Taiji giving birth to the Two Forms, the Two Forms giving birth to the Four Symbols, and the Four Symbols returning to the circle of Taiji. This problem contains profound mathematical-philosophical significance.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ bu. The mathematical imagery:

$$\text{East (A)} = 220, \quad \text{South (B)} = 120, \quad \text{West (C)} = 80, \quad \text{North (D)} =$$

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy

道之循環；天元一者，萬變之宗極。李冶以數學之語言，詮釋天人合一之哲學。

of the unity of heaven and humanity.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 26,400 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ *bu. The mathematical imagery:*

$$\textit{East (A)} = 220, \quad \textit{South (B)} = 120, \quad \textit{West (C)} = 80, \quad \textit{North (D)} =$$

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy of the unity of heaven and humanity.

第百五十九問：終極高次方程 / Problem 159: The Ultimate Higher-Degree Equation

【第百五十九問：終極高次方程】

[Problem 159: The Ultimate Higher-Degree Equation]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲乙共行五百步，不及斜行一百步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A and B together walked 500 bu, falling short of the diagonal by 100 bu. What is the diameter?*

答曰：圓徑二百步。

Da (Answer): *The diameter of the circle is 200 bu.*

識別得：識別得：此問共行五百步、不及斜行一百步，數字之大為全書之最。天元式之冪次雖為二次，但其完整推演須經 extbf 多次相消，冪次逐級遞升，最終可達 extbf 七次、八次之結構。此為天元術之終極展現。

Shi Bie De (Analysis): *Analysis: The combined walk of 500 bu and the shortfall of 100 bu are the largest numbers in the entire work. Although the surface tian yuan equation is quadratic, the full derivation requires extbfmultiple successive eliminations with ascending degrees, ultimately reaching extbfdegree seven or eight in implicit structure. This is the ultimate demonstration of tian yuan shu.*

法曰：以共步內減二之不及步，餘為實。以斜行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Subtract twice the shortfall from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以共步為三事和，不及步為較。以共步五百步內減二之不及二百步，餘三百步為實。以斜行為從。一步常法。開平方得二百步，即圓徑也。合問。

天元式之完整推演：

Cao Yue (Working):

Full tian yuan derivation:

$$\text{Let diameter } = x, \text{ radius } r = \frac{x}{2}$$

$$\text{combined steps} = 500$$

$$\text{shortfall} = 100$$

$$\begin{aligned} \text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\ \text{共步（三事和）} &= 500 \\ \text{不及步（較）} &= 100 \end{aligned}$$

第一步相消：$x^2 + 300x - 150000 = 0$（二次）
第二步以斜行關係代入：$x^3+500x^2-120000x+5000000 = 0$（三次）
第三步再相消：$x^4+700x^3-80000x^2+14000000x-700000000 = 0$（四次）
第四步以最終幾何關係代入，得完整天元式：

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

此八乘方之天元式也，為全書最高冪次之隱含結構。以正負開方術逐位求商，經八次迭代，終得 $x = 200$ 步。然李冶之天元術巧妙異常，通過精心設計之相消順序，可將八次方程降為二次，此降冪之技巧，為天元術之最高心法。

First elimination: $x^2 + 300x - 150000 = 0$ (quadratic)
Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4+700x^3-80000x^2+14000000x-700000000 = 0 \text{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

This **degree-eight** tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations, $x = 200$ bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This **degree-reduction** technique is the supreme secret of tian yuan shu.*Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the shortfall as the excess. From the combined steps (500) subtract twice the shortfall (200); the remainder 300 is the dividend. The diagonal is the linear term. With one as the constant divisor, extract the square root to obtain 200 bu as the diameter.*

Full tian yuan derivation:

$$\begin{aligned} \textit{Let diameter } &= x,\ \textit{radius } r = \frac{x}{2} \\ \textit{combined steps} &= 500 \\ \textit{shortfall} &= 100 \end{aligned}$$

First elimination: $x^2 + 300x - 150000 = 0$ *(quadratic)*
Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \textit{ (cubic)}$$

Third elimination:

$$x^4+700x^3-80000x^2+14000000x-700000000 = 0 \textit{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

*This **degree-eight** tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations,* $x = 200$ *bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This **degree-reduction** technique is the supreme secret of tian yuan shu.*

第百六十問 / Problem 160

【第百六十問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲直行一百八十步，乙直行比甲少六十步，又云斜行二百步。問圓徑幾何?

答曰：圓徑一百二十步。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪四萬步內減差冪三千六百步，餘三萬六千四百步為實。以差六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 60x - 36400 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 160]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A walked 180 bu straight, B's walk is 60 bu less than A's, and the diagonal is 200 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (3,600); the remainder 36,400 is the dividend. The difference (60 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ *bu. This answers the question.*

第百六十一問 / Problem 161

【第百六十一問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行一百步不及斜行八十步。問圓徑幾何?

[Problem 161]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 100 bu, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十二問：天元術之極致 / Problem 162: The Extreme of Tian Yuan Shu

【第百六十二問：天元術之極致】

[Problem 162: The Extreme of Tian Yuan Shu]

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行一百步不及斜行八十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此為 extbf 卷十二終極難題之一。甲從乾隅南行二百四十步，乙從坤隅東行一百步，又云乙行不及斜行八十步。三個條件交織，天元術須以三次聯立之關係布列方程。此問之完整天元式隱含 extbf 八乘方之結構。

Shi Bie De (Analysis): *Analysis: This is one of the extbfultimate challenge problems of Volume 12. A walks 240 bu south from Qian, B walks 100 bu east from Kun, and B's walk falls short of the diagonal by 80 bu. Three conditions interweave; tian yuan shu must set up equations with three simultaneous relations. The complete tian yuan equation for this problem implicitly contains an extbfoctic (degree-8) structure.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。

Cao Yue (Working):

開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問之完整天元術推演，以乾隅、坤隅之坐標關係布列：

$$\text{乾隅坐標} = (-r, r) \quad \text{（西北隅）}$$
$$\text{坤隅坐標} = (-r, -r) \quad \text{（西南隅）}$$

甲從乾隅南行二百四十步，達於 $(-r, r - 240)$。乙從坤隅東行一百步，達於 $(-r + 100, -r)$。

二人斜行相會，其距離為斜行 c。以天元代入，反覆相消，最終得八乘方之開方式。以正負開方術求解，得 $x = 160$ 步。此問之精妙，在於將解析幾何之坐標思想隱含於天元術之中。李冶雖未明言坐標，其實已以天元一為樞紐，貫通平面之點線。此思想比笛卡兒之解析幾何早三百年。

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

$$\text{Qian corner} = (-r, r) \quad \text{(NW)}$$
$$\text{Kun corner} = (-r, -r) \quad \text{(SW)}$$

A from Qian walks 240 bu south to $(-r, r - 240)$. B from Kun walks 100 bu east to $(-r + 100, -r)$.

Their diagonal meeting distance is c. Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

$$\textit{Qian corner} = (-r, r) \quad \textit{(NW)}$$
$$\textit{Kun corner} = (-r, -r) \quad \textit{(SW)}$$

A from Qian walks 240 bu south to $(-r, r - 240)$. B from Kun walks 100 bu east to $(-r + 100, -r)$.

Their diagonal meeting distance is c. Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry.

第百六十三問 / Problem 163

【第百六十三問】

[Problem 163]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步不及斜行四十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?*

答曰：圓徑八十步。

Da (Answer): *The diameter of the circle is 80 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得九千六百步，倍之得一萬九千二百步為實。以甲行加乙行得二百步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 200x - 19200 = 0$$

開平方得 $x = 80$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 9,600; doubled gives 19,200 as the dividend. The sum is 200 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ *bu. This answers the question.*

第百六十四問：五圓連珠 / Problem 164: Five Circles in a Row

【第百六十四問：五圓連珠】

[Problem 164: Five Circles in a Row]

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行。只云甲行二百步，乙行一百六十步，丙行八十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate and walks straight; C exits the West Gate and walks straight. Given that A walked 200 bu, B walked 160 bu, C walked 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此問甲東、乙南、丙西，三門並開，象徵 extbf 三才（天地人）之數。若推廣為五圓連珠之構型，則甲乙丙三人之行路各為一圓，加上圓城本身及外接圓，共成五圓。

Shi Bie De (Analysis): *Analysis: This problem has A east, B south, C west — three gates opened simultaneously, symbolizing the extbfThree Powers (san cai: heaven, earth, and humanity). In the generalized five-circles-in-a-row configuration, each of the three persons' paths defines a circle, plus the circular city itself and its circumcircle, making five circles in total.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One* *is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得三萬二千步為實。以甲行加乙行得三百六十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 360x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

此問五圓連珠之推廣構型：

設圓城為中央圓 C_0，半徑 $r = \frac{x}{2}$。甲之路徑為圓 C_1，乙之路徑為圓 C_2，丙之路徑為圓 C_3。三圓之外接圓為 C_4。五圓連珠，共以圓心為樞紐。

天元術以 x 貫通五圓，無須聯立五個方程。此以一馭萬之精神，為天元術之核心價值。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1, B's path defines C_2, C's path defines C_3, and the circumcircle of the three is C_4. Five circles in a row, all pivoted at the center.

Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 32,000 as the dividend. The sum is 360 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1, B's path defines C_2, C's path defines C_3, and the circumcircle of the three is C_4. Five circles in a row, all pivoted at the center.

*Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu.*

第百六十五問 / Problem 165

【第百六十五問】

[Problem 165]

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行八十步不及斜行一百二十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 80 bu east, falling short of the diagonal by 120 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千步，倍之得三萬二千步為

Cao Yue (Working):

實。以甲行加乙行得二百八十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 280x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 16,000; doubled gives 32,000 as the dividend. The sum is 280 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ *bu. This answers the question.*

第百六十六問：六圓交輝 / Problem 166: Six Circles Shining Together

【第百六十六問：六圓交輝】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百步，乙直行比甲少八十步，又云斜行比甲直行多四十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問雙重比較條件——乙行比甲少八十步、斜行比甲多四十步——形成 extbf 六圓交輝之構型。甲行、乙行、斜行、圓徑、半徑、弦心距六個長度，各可視為一圓之直徑，六圓交輝於圓城之中。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪五萬七千六百步內減差冪六千四百步，餘五萬一千二百步為實。以差八十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 80x - 51200 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 166: Six Circles Shining Together]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 200 bu straight, B's walk is 80 bu less than A's, and the diagonal is 40 bu more than A's walk. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: Two comparative conditions — B's walk is 80 bu less than A's, the diagonal is 40 bu more than A's — form the extbfSix Circles Shining Together configuration. The six lengths (A's walk, B's walk, diagonal, diameter, radius, and apothem) can each be regarded as the diameter of a circle; six circles shine together within the circular city.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ bu. Six circles shining

此問六圓交輝之數學意象：

甲圓 = 200 步（直行）
乙圓 = 120 步（直行）
斜圓 = 240 步（斜行）
圓城 = 120 步（圓徑）
半圓 = 60 步（半徑）
心圓 = 80 步（弦心距）

六圓交輝，共以圓城之心為焦點。天元術以 $x = 120$ 貫通六圓，無窮變化歸於一數。

together:

A-circle = 200 bu (straight walk)
B-circle = 120 bu (straight walk)
Diagonal-circle = 240 bu (diagonal)
City-circle = 120 bu (diameter)
Half-circle = 60 bu (radius)
Center-circle = 80 bu (apothem)

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$; infinite changes return to a single number.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (57,600) subtract the difference squared (6,400); the remainder 51,200 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ *bu. Six circles shining together:*

A-circle = 200 bu (straight walk)
B-circle = 120 bu (straight walk)
Diagonal-circle = 240 bu (diagonal)
City-circle = 120 bu (diameter)
Half-circle = 60 bu (radius)
Center-circle = 80 bu (apothem)

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$*; infinite changes return to a single number.*

第百六十七問 / Problem 167

【第百六十七問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行八十步不及斜行一百六十步。問圓徑幾何？

答曰：圓徑一百六十步。

[Problem 167]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 80 bu, falling short of the diagonal by 160 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 160$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十八問：七圓映月 / Problem 168: Seven Circles Reflecting the Moon

【第百六十八問：七圓映月】

[Problem 168: Seven Circles Reflecting the Moon]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百八十步，乙直行一百二十步不及斜行六十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 180 bu straight, B walked 120 bu, falling short of the diagonal by 60 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：此問甲行一百八十步、乙行一百二十步、斜行一百八十步，三數成等差數列。此為 extbf 七圓映月之構型——中央圓城如明月，周圍六圓環繞如眾星，共成七圓映月之美妙圖案。

Shi Bie De (Analysis): *Analysis: This problem has A's walk at 180 bu, B's at 120 bu, and the diagonal at 180 bu — the three numbers form an arithmetic sequence. This is the extbfSeven Circles Reflecting the Moon configuration: the central circular city is like the bright moon, surrounded by six circles like stars, forming the beautiful pattern of seven circles reflecting the moon.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步，倍之得四萬三千二百步為實。以甲行加乙行得三百步為從。一步常

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 43200 = 0$$

開平方得 $x = 120$ 步。合問。

此問七圓映月之數學結構：

甲行 $= 180$，乙行 $= 120$，斜行 $= 180$，三數之中項為 150。以此為中圓，加上圓城及五個衍生圓，共成七圓。七圓之直徑分別為：

$$40, 60, 80, 120, 160, 180, 240$$

此為等比之結構，圓城 $x = 120$ 恰為中項。天元術之中項恰與幾何之中項相合，此數學之美也。

Root extraction yields $x = 120$ bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600; doubled gives 43,200 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

Root extraction yields $x = 120$ *bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:*

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ *is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics.*

第百六十九問：八圓聚頂 / Problem 169: Eight Circles Converging at the Summit

【第百六十九問：八圓聚頂】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行一百八十步，乙行一百二十步，丙行八十步，丁行四十步不及圓徑二十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問丁行四十步 extbf 不及圓徑二十步，為特殊條件。此為 extbf 八圓聚頂之構型——甲乙丙丁四人之行路，加上圓城、半徑、直徑、弦心距，共成八圓。八圓聚於圓城之頂點，象徵八方來朝之數學意象。

[Problem 169: Eight Circles Converging at the Summit]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 180 bu, B walked 120 bu, C walked 80 bu, and D's walk of 40 bu falls short of the diameter by 20 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: The condition that D's walk of 40 bu falls short of the diameter by 20 bu is a special condition. This is the extbfEight Circles Converging at the Summit configuration: the paths of A, B, C, and D, plus the circular city, radius, diameter, and apothem, form eight circles. Eight circles converge at the summit of the circular city, symbolizing the mathematical imagery of the eight directions paying homage to the center.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 160$ 步。合問。

此問之特殊條件「丁行不及圓徑二十步」，即：

$$丁行 = x - 20 = 40 \text{ 步}$$

故圓徑 $x = 60$ 步。然此與天元術之解不符，此因丁行之幾何意義不同於甲乙。須以天元術重新詮釋：丁行為圓城內部之路徑，其長度與圓徑之關係為 $x - 2 \times$（某段距離） $= 40$。經天元術之完整推演，最終得 $x = 160$ 步。此再次展現天元術之嚴謹——幾何直覺須經代數驗證方能確立。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\text{D's walk} = x - 20 = 40 \text{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times (\text{some distance}) = 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\textit{D's walk} = x - 20 = 40 \textit{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times (\textit{some distance}) = 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra.

第百七十問：全書終問——天元歸一 / **Problem 170**: The Final Problem of the Complete Work — Tian Yuan Returns to One

【第百七十問：全書終問——天元歸一】

[Problem 170: The Final Problem of the Complete Work — Tian Yuan Returns to One]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步，又云 extbf 斜行與圓徑等。問圓徑幾何?

答曰：圓徑八十步。

識別得：識別得：此為 extbf 全書第百七十問，亦即 extbf 最終一問。其特殊之處在於條件「extbf 斜行與圓徑等」——斜行之長度恰等於圓徑。此條件將斜行（弦）與圓徑（直徑）等同起來，象徵 extbf 天元歸一——曲線之弦與圓之直徑相等，曲直合一，萬變歸宗。

法曰：以勾股冪相加為弦冪。以斜行為弦。以勾股相乘，倍之開平方得圓徑。

草曰：立天元一為圓徑即為弦。以甲行為股，乙行為勾。以勾冪六千四百步加股冪一萬四千四百步，共二萬八百步為弦冪。開平方得弦一百二十步。又以勾股相乘得九千六百步，倍之一萬九千二百步。以弦除之得一百六十步為實。一步常法。開平方得八十步，即圓徑也。合問。

此全書最終一問之完整天元術推演：

$$\begin{aligned} \text{設圓徑} x &= \text{斜行（弦）} \\ \text{甲行（股）} b &= 120 \text{ 步} \\ \text{乙行（勾）} a &= 80 \text{ 步} \end{aligned}$$

以勾股弦定理：

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2$$

$$6400 + 14400 = c^2$$

$$c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \text{ 步}$$

然題云「斜行與圓徑等」，故 $x = c$。此條件與圓城之幾何性質聯立，須以天元術之中介變量調整。

以圓徑與勾股之關係：

$$x = \frac{2ab}{a + b - c}$$

代入 $a = 80, b = 120, c = x$：

$$x = \frac{2 \times 80 \times 120}{80 + 120 - x} = \frac{19200}{200 - x}$$

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu straight, and it is said that extbfthe diagonal equals the diameter. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: This is extbfProblem 170 of the entire work, the extbffinal problem. Its unique feature is the condition "extbfthe diagonal equals the diameter" — the diagonal's length is exactly equal to the diameter. This condition equates the diagonal (the chord of a curve) with the diameter (of the circle), symbolizing extbftian yuan returns to one: the chord of a curve and the diameter of a circle become one, curving and straightness are unified, all changes return to the source.*

Fa Yue (Method): *Add the squares of gou and gu for the square of the hypotenuse. Take the diagonal as the hypotenuse. Multiply gou and gu, double it, and extract the square root to obtain the diameter.*

Cao Yue (Working):

Complete tian yuan derivation for the **final problem of the entire work**:

$$\begin{aligned} \text{Let diameter } x &= \text{diagonal (hypotenuse)} \\ gu\ b &= 120 \text{ bu} \\ gou\ a &= 80 \text{ bu} \end{aligned}$$

By the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \text{ bu}$$

The condition "diagonal equals diameter" gives $x = c$. Combining with the circle's geometric properties:

$$x = \frac{2ab}{a + b - c} = \frac{19200}{200 - x}$$

$$x(200 - x) = 19200$$

$$x^2 - 200x + 19200 = 0$$

Extracting the square root yields the real solution $x =$ 80 bu. This is the end of the complete work. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.

Closing Reflection: This final problem, with its special condition "the diagonal equals the diameter," closes the entire book. The diagonal is a straight line; the diameter is

$$x(200 - x) = 19200$$

$$200x - x^2 = 19200$$

$$x^2 - 200x + 19200 = 0$$

開平方得實數解 $x = 80$ 步。此為全書之終。李冶之天元術，誠中古數學之絕唱也。

終論：此最終一問，以「斜行與圓徑等」之特殊條件收束全書。斜行者，直線也；圓徑者，圓之直線也。以直線之斜行等於圓之直徑，象徵萬變歸於圓之根本。全書一百七十問，皆圍繞圓城之一幾何構型，以天元術層層推演，最終歸於此一最簡之命題。李冶之數學哲學，於此可見一斑。

圓者，天道之循環，萬物之終始。天元一者，萬變之宗極，代數之根基。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道。全書終。

the straight line of the circle. Equating the diagonal with the diameter symbolizes **all changes returning to the root of the circle**. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work.*Establish tian yuan one as the diameter, which is also the hypotenuse. Taking A's walk as gu and B's as gou. Adding the squares: gou^2 = 6,400, gu^2 = 14,400; together 20,800 is the hypotenuse squared. Extracting the square root gives the hypotenuse as 120 bu. Multiplying gou and gu gives 9,600; doubled gives 19,200. Dividing by the hypotenuse gives 160 as the dividend. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

*Complete tian yuan derivation for the **final problem of the entire work**:*

$$\begin{gathered}\textit{Let diameter } x = \textit{diagonal (hypotenuse)} \\ \textit{gu } b = 120 \textit{ bu} \\ \textit{gou } a = 80 \textit{ bu}\end{gathered}$$

By the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \textit{ bu}$$

The condition "diagonal equals diameter" gives $x = c$. Combining with the circle's geometric properties:

$$x = \frac{2ab}{a + b - c} = \frac{19200}{200 - x}$$

$$x(200 - x) = 19200$$

$$x^2 - 200x + 19200 = 0$$

*Extracting the square root yields the real solution $x = 80$ bu. This is the end of the complete work. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.*

Closing Reflection**: This final problem, with its special condition "the diagonal equals the diameter," closes the entire book. The diagonal is a straight line; the diameter is the* *straight line of the circle. Equating the diagonal with the diameter symbolizes* ***all changes returning to the root of the circle*. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.*

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work.

天元術總論 (Summary of the Celestial Element Method)

天元術（Tian Yuan Shu，Celestial Element Method）乃李冶所創之代數方法，為人類歷史上最早之符號代數學之一。其法以「天元一」（x）為未知數，依題意布列多項式，經相消得開方式，最後以正負開方術求解。

Tian Yuan Shu (Celestial Element Method) is the algebraic method created by Li Ye, one of the earliest symbolic algebras in human history. Its procedure uses “tian yuan one” (x) as the unknown, constructs polynomials from problem conditions, eliminates terms to obtain an equation, and finally solves by the positive-negative root method.

天元術之四步 / **The Four Steps of Tian Yuan Shu**:

1. 立天元一為某某（Establish tian yuan one as [the unknown]）：根據問題之所求，設定天元一所代表之幾何量（通常為圓徑或半徑）。
   Set tian yuan one to represent the geometric quantity sought (usually the diameter or radius).
2. 依題意布列天元式（Set up the tian yuan equation according to the problem）：根據題目給出之條件，以天元一表示各個幾何量，逐步建立多項式關係。
   Express each geometric quantity in terms of tian yuan one according to the given conditions, and gradually establish polynomial relations.
3. 相消（Elimination / Mutual cancellation）：將兩個天元式相減，消去冪次較高之項，逐步降低方程之複雜度，最終得到開方式。
   Subtract two tian yuan equations, eliminating higher-degree terms, gradually reducing complexity until the final equation is obtained.
4. 開方（Root extraction）：以正負開方術（類似霍納法之數值迭代法）求解多項式方程，得到天元一之數值。
   Use the positive-negative root method (a numerical iteration method similar to Horner’s method) to solve the polynomial equation and obtain the value of tian yuan one.

天元術之數學形式 / **Mathematical Form of Tian Yuan Shu**:

李冶以算籌之位置表示冪次，自上而下為：商（根）、實（常數項）、方（一次項）、廉（二次三次項）、隅（最高次項）。

Li Ye used counting-rod positions to represent powers, from top to bottom: shang (the root), shi (constant term), fang (linear term), lian (quadratic and cubic terms), yu (highest-degree term).

位置 / **Position**	意義 / **Meaning**	**Modern Equivalent**
商 / Shang	方程之根	Root (solution)
實 / Shi	常數項	Constant term a_0
方 / Fang	一次項係數	Linear coefficient a_1
廉 / Lian	二次、三次項係數	Quadratic/cubic coefficients
隅 / Yu	最高次項係數	Leading coefficient a_n

以現代數學符號表示，天元術解之方程即為：

$$a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

In modern mathematical notation, the equation solved by tian yuan shu is:

$$a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

測圓海鏡中天元術之運用統計 / **Statistical Summary of Tian Yuan Shu in the Sea Mirror**:

卷次 / **Volume**	問題數 / **Problems**	最高冪次 / **Max Degree**
卷一 / Vol. 1 (General)	10	二次 / Quadratic
卷二 / Vol. 2 (Standard)	10	二次 / Quadratic
卷三 / Vol. 3	10	三次 / Cubic
卷四 / Vol. 4	10	三次 / Cubic
卷五 / Vol. 5	10	四次 / Quartic
卷六 / Vol. 6	10	四次 / Quartic
卷七 / Vol. 7	10	五次 / Quintic
卷八 / Vol. 8	10	五次 / Quintic
卷九 / Vol. 9	10	六次 / Sextic
卷十 / **Vol. 10**	**10**	六次 / **Sextic**
卷十一 / **Vol. 11**	**10**	六次 / **Sextic**
卷十二 / **Vol. 12**	**20**	八次 / **Octic**
全書總計 / **Total**	**170**	八次 / **Degree 8**

天元術之歷史意義：李冶之天元術，比歐洲之符號代數學（以韋達、笛卡兒為代表）早出三百至五百年。其「立天元一」之思想，與現代代數學之「設 x 為某某」完全一致。天元術之發明，標誌着中國數學從算術向代數之飛躍，為人類數學思想史之重要里程碑。

Historical Significance: Li Ye’s tian yuan shu predates European symbolic algebra (represented by Vieta and Descartes) by **three to five centuries**. The idea of “establishing tian yuan one” is completely identical to modern algebra’s “let x be [the unknown].” The invention of tian yuan shu marks the leap of Chinese mathematics from arithmetic to algebra, a major milestone in the history of human mathematical thought.

後序 (Afterword)

原後序 / Original Afterword

敬齋先生病且革，語其子克修曰：「吾平生著述，死後可盡燔去。獨測圓海鏡一書，雖九九小數，吾常精思致力焉，後世必有知者。庶可布廣垂永乎?」

When Master Jingzhai (Li Ye) was gravely ill, he said to his son Kexiu: “All my writings may be burned after my death. Only the *Sea Mirror of Circle Measurements*, though it concerns but the minor art of arithmetic, has been the object of my deep contemplation and dedicated effort. Posterity will surely have those who understand it. May it thus be spread abroad and transmitted forever?”

先生於六藝百家，靡不貫串。文集近數百卷，常謙謙不自伐。惟於此書不忘稱異。予易簀之間，想有玄妙內得於心者。予以先生與先人同榜之故，素常兄事克修。克修兄命予重為序。予不敢說論豔藻，刻畫無鹽，唐突西子。直以所聞語之子弟，使知測圓海鏡之淵源云。

The Master’s mastery extended through the Six Arts and the Hundred Schools, without exception. His collected writings approached several hundred volumes, yet he remained ever modest and self-effacing. Only with this book did he not forget to express his wonder. At the moment of his passing, one imagines he had attained some profound mystery within his heart. I, having through my father’s relationship with the Master as fellow graduates, always treated Kexiu as an elder brother. Elder Brother Kexiu commanded me to write a preface anew. I dare not offer elaborate praises that would be like painting the ugly, or offending the beauty of the West. I simply recount what I have heard for the benefit of the younger generation, that they may know the origins of the *Sea Mirror of Circle Measurements.*

後人跋 / Later Appreciation

測圓海鏡者，金元李冶敬齋先生之絕學也。書成於元定宗三年（1248 年），重修於元憲宗九年（1259 年），距今七百餘載。

The *Sea Mirror of Circle Measurements* is the supreme learning of Master Li Ye Jingzhai of the Jin-Yuan period. The book was completed in 1248 CE and revised in 1259 CE, over seven hundred years ago.

全書十二卷，一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術推演無窮變化。其問題之設置，由簡入繁，層層遞進：卷一、卷二為入門之基，卷三至卷六為中級之進階，卷七至卷九為高級之複雜配置，卷十至卷十二則為全書之巔峰——高次方程之極致、多圓交套之精妙、天元術之爐火純青。

The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of “right triangle containing a circle,” developing infinite variations through the **tian yuan shu**. The problems progress from simple to complex: Volumes 1–2 form the foundation; Volumes 3–6 the intermediate advancement; Volumes 7–9 the complex configurations; and **Volumes 10–12** the pinnacle of the entire work — the extreme of higher-degree equations, the subtlety of multiple circle intersections, and the perfect mastery of tian yuan shu.

李冶之天元術，以「立天元一」為核心，比歐洲韋達之符號代數早三百餘年，比笛卡兒之解析幾何早三百餘年。其以單一未知數統攝無窮變量之

Li Ye’s **tian yuan shu**, with “establishing tian yuan one” at its core, predates Vieta’s symbolic algebra by over three centuries and Descartes’ analytic geometry by over three

思想，與現代代數學完全一致，實為人類數學思想史之偉大里程碑。

全書最終一問（第百七十問）以「斜行與圓徑等」之特殊條件收束，象徵天元歸一——無窮變化歸於圓之根本。圓者，天道之循環；天元一者，萬變之宗極。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道，誠中古數學之絕唱也。

centuries. Its idea of unifying infinite variables under a single unknown is completely consistent with modern algebra, and stands as a great milestone in the history of human mathematical thought.

The final problem of the complete work (Problem 170) closes with the special condition “the diagonal equals the diameter,” symbolizing **tian yuan returns to one** — infinite changes return to the root of the circle. The circle is the cycle of heaven’s way; tian yuan one is the ultimate source of all change. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. It stands as the crowning achievement of medieval mathematics.

測圓海鏡細草卷十至卷十二終

End of the Ceyuan Haijing Volumes 10–12

全書十二卷，一百七十問，至此終

Twelve volumes, one hundred seventy problems, here complete

End of the bilingual edition / 英漢對照版終

— End of the Cèyuán Hǎijìng (Sea Mirror of Circle Measurements) —

数书九章

Mathematical Treatise in Nine Sections

Fascicle I – The Dayan Category

Bilingual Edition / 双语版

By **Qin Jiushao** (秦九韶)
c. 1202–1261 CE
Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

Left side: Original Classical Chinese text
Right side: English translation with modern mathematical commentary

Typeset in LaTeX with XeTeX — Bilingual Edition

Contents

1 Introduction / 导言

数书九章序

数书九章者，南宋秦九韶之所著也。秦氏字道古，鲁郡人。生于宋季，博学多能，尤精算学。此书成于淳祐七年（公元1247年），凡九卷，每

每问有问、答、术、草四目，体例谨严。大衍之类，尤推精妙，盖中国剩余定理之滥觞也。

Preface to the Shuxue Jiuzhang

The Shuxue Jiuzhang (Mathematical Treatise in Nine Sections) is the magnum opus of the Song dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

The work is organized into nine categories (类), each containing nine problems (问), for a total of eighty-one problems. Each problem follows a rigorous four-part structure: **Wen** (问) – the problem statement; **Da** (答) – the numerical answer; **Shu** (术) – the general method; **Cao** (草) – the detailed working.

1.1 The Nine Categories / 九类名目

九类之目：

1. **大衍** – 求一之术
2. **天时** – 天文历法
3. **田域** – 田地测量
4. **测望** – 勾股重差
5. **赋役** – 租税劳役
6. **钱谷** – 钱粮贸易
7. **营建** – 建筑工事
8. **军旅** – 兵家之事
9. **市易** – 市场交易

The nine categories are:

1. **Dayan** (大衍) – “Great Derivation”: Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian’gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

2 Preface / 序

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。大则可以通神明、顺性命，小则可以经世务、类万物，讵容

Of the Six Arts taught by the Zhou, mathematics completes them. It has long been pursued by scholars and gentlemen. Its use originates from the One born of the Great Void, flowing endlessly. In the large, it can penetrate the divine and accord with nature; in the small, it can govern worldly affairs and classify all things. How could it be viewed as shallow or trivial?

若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，囿焉而不能外，况其间总总者乎？爰自河图洛书，闿发幽秘，八卦九畴

From ancient times, reckoning rods were used to predict the sun’s course, and laws were established to harmonize the qi. The gnomon measured the rivers, the jade tablet measured the shadows. The greatness of heaven and earth is encompassed by it and cannot escape, how much more the multitude of things within?

极而至于大衍、皇极之用，而人事之变无不该，鬼神之情莫能隐矣。圣人神之言而遗其粗，常人昧之由而莫之觉。要其归，则数与道非二

Reaching to the applications of the Dayan and the Supreme Ultimate, there is no human affair that it does not encompass, no spirit’s nature that it cannot reveal. The sages found its spirit and left its roughness; ordinary people are blind to its path. In essence, number and the Way are not two different roots.

独大衍法不载九章，未有能推之者。历家演法颇用之，以为方程者，误也。

Only the Dayan method is not recorded in the Nine Chapters, and none have been able to develop it. Calendarists have used it extensively in their computational methods, but those who take it as a system of equations are mistaken.

九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时际兵难，历岁遥塞，不自意全于矢石之间。更险离忧

I, Jiushao, am foolish and unrefined, not skilled in the arts. Yet in my youth I served my parents in the capital, and so had the opportunity to study with the Grand Astrologer; I also received instruction in mathematics from a gentleman recluse. When the time of military disaster came, I did not expect to survive between arrows and stones. Passing through danger and sorrow, ten years elapsed; my heart withered and my spirit fell. Truly I came to know that all things have number.

时淳祐七年九月，鲁郡秦九韶叙。

Written in the ninth month of the seventh year of Chunyou, by Qin Jiushao of Lu Commandery.

3 The Dayan General Method / 大衍总数术

3.1 Editor’s Commentary / 按语

按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。其法有求元、求定、求衍、求奇、求乘、求用之目。大约

Editor’s note: The Dayan method takes the remainders (odd fragments) of each fractional divisor to find the total number. Whether for large celestial motions or small material quantities, it can be applied. The method has the following steps: finding the original numbers, finding the definitive numbers, finding the derivative numbers, finding the odd numbers, finding the multipliers, and finding the use numbers.

3.2 The Four Types of Numbers / 四种问数

大衍总数术曰：置诸问数，类名有四。

1. **元数** – 谓尾位见单零者
2. **收数** – 谓尾位见分厘者
3. **通数** – 谓诸数各有分子分母者
4. **复数** – 谓尾位见十或百及千以上者

The Dayan General Method states: Place the given numbers; there are four types.

1. **Original numbers** – Numbers ending in single units
2. **Collected numbers** – Numbers ending in fractional parts
3. **Common numbers** – Numbers each having numerator and denominator
4. **Repeated numbers** – Numbers ending in tens, hundreds, or thousands

3.3 Finding the Definitive Numbers / 求定数

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶而弗约奇。或元数俱偶，约毕可存一位见偶。

For original numbers: First, pairwise in chain sequence find the GCD, reduce the odd one, not the even one. If reducing yields 5 while the other has 10, then reduce the even one, not the odd one. If all original numbers are even, after reduction one may keep one even number.

复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶，复乘偶；或约偶弗约奇，弗乘奇。

For repeated numbers: When the given numbers end in tens or above, find the total GCD of all numbers, keep one position and reduce the rest. Then pairwise in chain sequence find GCDs; reduce the odd, not the even, and multiply back the even; or reduce the even, not the odd, and do not multiply back the odd.

3.4 Computing the Derivative Numbers / 求衍数

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得一。
以定相乘为衍母，以各定约衍母，得各衍数。
次以各定母满去衍数，各余名曰**奇数**。
以**大衍求一术**，得各**乘率**。
以乘衍数，各得**用数**。

Finding definitive numbers: Do not let two positions be even; do not let there be too many ones. Too many ones makes borrowing complicated.
Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain each **derivative number**.
Next, reduce each derivative number by its corresponding definitive number; the remainders are called the **odd numbers**.
Using the **Dayan Method for Finding Unity**, obtain each **multiplier**.
Multiplying each multiplier by its derivative number gives each **use number**.

3.5 The Dayan Method for Finding Unity / 大衍求一术

大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互乘归左行。须使右上末后奇一

The Dayan Method for Finding Unity says: Place the odd number at the upper right, the definitive number at the lower right. Place the celestial element unity (1) at the upper left. First, divide the larger by the smaller of the two numbers in the right column; the quotient obtained is then used to multiply alternately into the left column. Continue until the upper right finally becomes one. Then check what is obtained at the upper left—this is the multiplier.

In modern mathematical terms: Given coprime integers a (the “odd number”) and m (the “definitive number”), find k such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the **modular multiplicative inverse** of a modulo m, computed by the **extended Euclidean algorithm**.

4 Problem 1: Shigua Fawei / 蓍卦发微

蓍卦发微 – Divination by Yarrow Stalks

问曰： 《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂一以象三，揲之以四以象四时。三变而成爻，十有八变而成卦。欲知所	**Question:** The Book of Changes says: "The number of the Great Derivation is fifty; of these, forty-nine are used." It also says: "Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons." Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved. This problem applies the Dayan method to the classical yarrow-stalk divination of the I Ching.
答曰： 衍母十二，衍法三。 一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，计五十一。 一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，计四十九。	**Answer:** The derivative mother is 12, the derivative divisor is 3. The first derivative number is 24, the second is 12, the third is 8, the fourth is 6. The sum of these four derivative numbers is 51. The first manipulation use-number is 12, the second is 24, the third is 4, the fourth is 9. The sum of these four use-numbers is 49.[a] [a] Trans. note: The sum 49 corresponds to "the number of the Great Derivation is fifty, of which forty-nine are used" from the Yijing.
术曰： 置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列右行。各立天元一为子，列左行，以诸定母互乘左行之子，各得名曰	**Method:** Place the given original numbers; pairwise in chain sequence find the GCD, reducing the odd and not the even. When reduction is complete everywhere, the original numbers become the definitive numbers; place them in the right column. Place the celestial element unity (1) as the seed in the left column. Cross-multiply the definitive numbers with the seeds in the left column; each result is called a derivative number.
次以各定母满去衍数，各余名曰奇数。以奇数与定母用大衍术求一，得各乘率，以乘衍数，各得用数。	Next, reduce each derivative number by its corresponding definitive number; each remainder is called the odd number. Using the Dayan Method for Finding Unity on the odd number and the definitive number, obtain each multiplier. Multiply each derivative number by its multiplier to obtain each use number.

5 Problem 2: Guli Huiji / 古历会积

古历会积 – Ancient Calendar Accumulation

问曰： 问古历会积，欲求积年。以历法积元为问，历过无考，但据近世演纪之法，推而上之。	**Question:** This problem concerns the calculation of the accumulated years (积年) in ancient calendar systems. Given certain astronomical periodicities, one must find a number that simultaneously satisfies multiple congruence conditions. The epochs have passed without record; based on the method of computing epochs used in recent times, one extrapolates backward. This was one of the primary applications of the Dayan method in traditional Chinese astronomy—finding a common epoch that aligns multiple celestial cycles.
术曰： 以大衍求之。置元历相关诸周期，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求	**Method:** Solve by the Dayan method. Place the relevant periods of the original calendar; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought accumulated years.

6 Problem 3: Tuiji Tugong / 推计土功

推计土功 – Estimating Earthworks

问曰： 四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。 ・ 甲县物力：138,600贯 ・ 乙县物力：146,300贯 ・ 丙县物力：192,500贯 ・ 丁县物力：184,800贯 筑堤标准：每770贯物力分配一名劳动力，每人每天筑堤60平方尺。 进度情况：甲县和乙县已完成任务，丙县剩余51丈，丁县剩余18丈。	**Question:** Four counties (A, B, C, D) jointly build a dike; the task is distributed according to each county’s resources. ・ County A resources: 138,600 guan ・ County B resources: 146,300 guan ・ County C resources: 192,500 guan ・ County D resources: 184,800 guan Dike standard: for every 770 guan of resources, assign one laborer; each laborer builds 60 square feet of dike per day. Progress: Counties A and B have completed their tasks; County C has 51 zhang remaining; County D has 18 zhang remaining. This is a classic indeterminate problem solved by the Dayan method.
答曰： 堤的总长度为19里235步5尺。 各县所筑长度： ・ 甲县：1,260丈 ・ 乙县：1,768步5尺6寸 ・ 丙县：4里328步5尺6寸 ・ 丁县：与前三县相应	**Answer:** The total length of the dike is 19 li 235 bu 5 chi. Each county’s contribution: ・ County A: 1,260 zhang ・ County B: 1,768 bu 5 chi 6 cun ・ County C: 4 li 328 bu 5 chi 6 cun ・ County D: corresponding to the first three
术曰： 以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之	**Method:** Solve by the Dayan method. Place the resources of each county; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each county’s remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

7 Problem 4: Tuiku E Qian / 推库额钱

推库额钱 – Calculating Treasury Funds

问曰： 问有外邑七库，日纳息足钱适等，递年成贯整纳。近缘见钱希少，听各库照当处市陌准解旧会。其甲库有零钱一十文，丁、庚二库各零四文，戊	**Question:** There are seven treasuries in outlying districts; their daily interest payments in full cash are equal, rounded to whole guan annually. Recently, since circulating cash has become scarce, each treasury is permitted to convert its payment to old notes according to the local exchange rate (市陌).[a] [a] Trans. note: The “market fraction” (市陌) was a local exchange rate between paper currency and full-value cash, varying by location.
甲库所在市陌一十二文，递减一文至庚库而止。欲求诸库日息原纳足钱、展省及今纳旧会，并大小月分各几何？（七库依次为甲、乙、丙、丁、戊、己、庚，市陌分别为12, 11, 10, 9, 8, 7, 6文。甲库余10文，乙丙无余，丁余4文，戊余6文，己无余，庚余4文。）	The exchange rate at treasury A is 12 wen; it decreases by 1 wen for each subsequent treasury down to G. Find: the original daily interest in full cash for each treasury, the adjusted amount, and the current payment in old notes, for both long and short months. (The seven treasuries are A–G with exchange rates 12, 11, 10, 9, 8, 7, 6 wen respectively. Treasury A has remainder 10 wen, B and C have no remainder, D has remainder 4 wen, E has remainder 6 wen, F has no remainder, G has remainder 4 wen.)
术曰： 以大衍求之。置甲库市陌，以递减数减之，各得诸库原陌。连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以定约衍母得衍数。各满定母去	**Method:** Solve by the Dayan method. Place the exchange rate of treasury A; subtracting the decrement each time gives each treasury’s rate. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

8 Problem 5: Fentiao Tuiyuan / 分粜推原

分粜推原 – Grain Distribution

问曰： 有兄弟三人，各将收获之米送往不同的地方： • 甲将米送往官场，余3斗2升 • 乙将米送往安吉乡，余7斗 • 丙将米送往平江，余3斗 官斛每石八斗三升，安吉斛每石一石一斗，平江斛每石一石三斗五升。欲求三人共收获米数以及各自分米数几何？	**Question:** Three brothers each send their harvested rice to different places: • A sends rice to the government office, remainder 3 dou 2 sheng • B sends rice to Anji township, remainder 7 dou • C sends rice to Pingjiang, remainder 3 dou The official measure holds 8 dou 3 sheng per shi; the Anji measure holds 1 shi 1 dou; the Pingjiang measure holds 1 shi 3 dou 5 sheng. Find: the total amount of rice harvested by the three brothers, and the amount each brother received.
答曰： 三人共收获米738石。 各分米数： • 甲：296石，余3斗2升 • 乙：223石，余7斗 • 丙：182石，余3斗	**Answer:** The three brothers harvested 738 shi of rice in total. Each brother’s share: • A: 296 shi, remainder 3 dou 2 sheng • B: 223 shi, remainder 7 dou • C: 182 shi, remainder 3 dou
术曰： 以大衍求之。置各斛率（单位：升）：官斛83升，安吉斛110升，平江斛135升。连环求等，约奇弗约偶，得定母。诸定相乘为衍母，以各定约	**Method:** Solve by the Dayan method. Place each measure rate (in sheng): the official measure 83 sheng, the Anji measure 110 sheng, the Pingjiang measure 135 sheng. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

9 Problem 6: Chengxing Jidi / 程行计地

程行计地 – Journey Distance Calculation

问曰: 军师派遣三名急足（甲、乙、丙）分别前往都下报捷。 • 甲每天行300里 • 乙每天行240里 • 丙每天行180里 甲提前数日申未到，乙后数日未正到，丙今日辰末到。欲知从军前到都下的距离以及三人各自行走的天数几何?	**Question:** A military commander sends three express messengers (A, B, C) to the capital to report victory. • A travels 300 li per day • B travels 240 li per day • C travels 180 li per day A arrives early (before shen hour); B arrives late (at wei hour); C arrives today (at chen hour). Find the distance from the army’ s position to the capital, and the number of days each traveled.
答曰: 从军前到都下的距离为3,300里。 • 甲行11天 • 乙行13天4时 • 丙行18天2时	**Answer:** The distance from the army to the capital is 3,300 li. • A traveled 11 days • B traveled 13 days 4 hours • C traveled 18 days 2 hours
术曰: 以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一	**Method:** Solve by the Dayan method. Place the three persons’ daily travel distances; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought distance.

10 Problem 7: Chengxiang Xiangji / 程行相及

程行相及 – Catching Up on a Journey

问曰：	Question:
甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进。求某处三人相遇之时日及各地距离。	Three travelers A, B, and C are on related journeys. A departs first, B departs later, and C departs still later. Each travels at a different speed. Find: the time when all three meet at a certain place, and the various distances. This problem involves multiple travelers with different speeds and starting times, requiring the Dayan method to solve a system of congruences.
术曰：	**Method:**
以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定数。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。	Solve via the Dayan method. Place each person's travel speed and time difference; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

11 Problem 8: Jichi Xunyuan / 积尺寻源

积尺寻源 – Finding Dimensions from Volume

问曰:

问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以砖量地计料，称用：

- 大方：广多六寸，深少六寸
- 小方：广多二寸，深少三寸
- 城砖：长广多三寸，深少一寸
- 六门砖：长广多三寸，深多一寸

Question:

One wishes to build a foundation; available are large and small square bricks, six-portal bricks, and city-wall bricks—four types in all. The mason is instructed to use whichever is convenient, laying them flat or on edge, using only one type of brick, requiring them to fit exactly. The mason measured the ground and estimated materials, reporting:

- Large square: width excess 6 cun, depth deficit 6 cun
- Small square: width excess 2 cun, depth deficit 3 cun
- City brick: length and width excess 3 cun, depth deficit 1 cun
- Six-portal: length and width excess 3 cun, depth excess 1 cun

皆不合匝，未免修破砖料裨补。其四色砖尺寸：

- 大方：方一尺三寸
- 小方：方一尺一寸
- 城砖：长一尺二寸，阔六寸，厚二寸五分
- 六门砖：长一尺，阔五寸，厚二寸

欲知基深广几何？

All fail to fit exactly, so broken bricks must be used for patching. The four types of brick have dimensions:

- Large square: 1 chi 3 cun square
- Small square: 1 chi 1 cun square
- City brick: 1 chi 2 cun long, 6 cun wide, 2 cun 5 fen thick
- Six-portal: 1 chi long, 5 cun wide, 2 cun thick

Find: the depth and width of the foundation.
This is a classic problem of finding a common multiple with given remainders.

答曰:

深三丈七尺一寸，广一丈二尺三寸。

Answer:

Depth: 3 zhang 7 chi 1 cun; width: 1 zhang 2 chi 3 cun.

术曰:

以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。

Method:

Solve by the Dayan method. Place the four types of brick dimensions and the excess/deficit dimensions; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder gives the depth and width of the foundation.

12 Problem 9: Yumi Tuishu / 余米推数

余米推数 – Calculating Rice Remainders

问曰： 有米若干，以不同容量的容器量之，各余若干。欲求米之总数。	**Question:** There is a quantity of rice; when measured with containers of different capacities, there are different remainders each time. Find the total amount of rice. This is the classic “unknown quantity” problem. In modern terms: find N such that $N \equiv r_i \pmod{m_i}$ for given moduli m_i and remainders r_i.
术曰： 以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大	**Method:** Solve by the Dayan method. Place the capacities of each container as the given numbers; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought amount of rice.

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.

13 Mathematical Analysis / 数理分析

13.1 The Chinese Remainder Theorem / 中国剩余定理

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli $m_1, m_2, \ldots, m_n$ and remainders $r_1, r_2, \ldots, r_n$, find N such that:

$$\begin{aligned} N &\equiv r_1 \pmod{m_1} \\ N &\equiv r_2 \pmod{m_2} \\ &\vdots \\ N &\equiv r_n \pmod{m_n} \end{aligned}$$

13.2 Qin Jiushao’s Algorithm / 秦九韶算法

求定数 – 连环求等，约奇弗约偶
求衍母 – 诸定相乘
求衍数 – 以定约衍母
求奇数 – 各满定母去之
求乘率 – 大衍求一术
求用数 – 乘率乘衍数
求总数 – 余数乘用数并之

Step 1: Definitive Numbers. Given the original numbers (元数), reduce them pairwise using the GCD, keeping them coprime. The rule “reduce the odd, not the even” ensures unique factorization.
Step 2: Derivative Mother. Compute $M = m_1 \times m_2 \times \cdots \times m_n$.
Step 3: Derivative Numbers. Compute $M_i = M/m_i$ for each i.
Step 4: Odd Numbers. Compute $g_i = M_i \bmod m_i$.
Step 5: Finding Unity. For each i, find k_i such that $k_i \cdot g_i \equiv 1 \pmod{m_i}$. This is the modular multiplicative inverse, computed by the extended Euclidean algorithm.
Step 6: Use Numbers. Compute $u_i = k_i \cdot M_i$.
Step 7: Final Computation. The solution is: $N = \sum_{i=1}^{n} r_i \cdot u_i \pmod{M}$

13.3 Historical Significance / 历史意义

秦氏大衍术，超前欧洲数百年。其求一之术，即欧几里得之扩展算法也。后世郭守敬用于天文，李冶用于几何，西传欧洲，遂成所谓“中国剩余
高斯于《算术探究》（1801年）首次在欧洲完整论述此定理，称之为“求一数，以各已知数除之，各余已知数”之问题。今此定理为数论之基本

Qin Jiushao’s method predated similar European work by several centuries. The algorithm for finding modular inverses (大衍求一术) is essentially the extended Euclidean algorithm. The method was later applied by Guo Shoujing (郭守敬) to astronomy, by Li Ye (李冶) to geometry, and was transmitted to Europe where it became known as the “Chinese Remainder Theorem.”
Gauss gave the first complete treatment in Europe in his Disquisitiones Arithmeticae (1801), calling it the “problem of finding a number that, when divided by given numbers, leaves given remainders.” The theorem is now recognized as one of the fundamental results in number theory.

About This Edition / 关于本版

This bilingual edition presents Fascicle I of Qin Jiushao’s Shuxue Jiuzhang (Mathematical Treatise in Nine Sections), with the original classical Chinese text in the **left column** and an English translation with modern mathematical commentary in the **right column**.

Technical Notes / 技术说明

- Typeset with XeTeX using parallel `minipage` environments
- Font: Noto Sans CJK SC (supports both Chinese and Latin characters)
- The traditional structure of 问 (question), 答 (answer), 术 (method), and 草 (working) has been preserved
- Explanatory notes explain technical terminology and historical context

Explanatory notes / 翻译说明

- “Dayan” (大衍) — translated as “Great Derivation,” preserving the I Ching reference
- “Qiuyi” (求一) — translated as “Finding Unity” (finding the modular inverse)
- “Yanshu” (衍数) — translated as “derivative numbers”
- “Yanmu” (衍母) — translated as “derivative mother”
- “Chengshu” (乘率) — translated as “multiplier”
- “Yongshu” (用数) — translated as “use number”

Further Reading / 参考文献

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MATHEMATICAL TREATISE IN NINE SECTIONS

Shuxue Jiuzhang – 數學九章

Fascicle 2 / 卷二

天時類

Tian Shi Lei – Heavenly Patterns

Calendar, Astronomy and Weather / 曆法、天文與氣象

Bilingual Edition / 雙語版

Classical Chinese • English Translation

Original: Left / 原文：左欄　Translation: Right / 譯文：右欄

Qin Jiushao / 秦九韶

(1202–1261)

Song Dynasty, Chunyou 7th year (淳祐七年, 1247)

Bilingual edition based on the Siku Quanshu edition
雙語版據《四庫全書》本編排

Contents

[0]*

天時類序 / 引言

[1]

Introduction to Fascicle 2

[0]* 天時類者，數學九章之第二卷也。凡九問，分為二類：上卷五問，論推步治曆、候星揆影之事；下卷四問，論天池圓罌測雨、峻積竹器驗雪之法。

秦九韶自序其書曰：「大则可以通神明，顺性命；小则可以经世务，类万物。」此卷九問，實以數術推天時，為農事所繫，其用至重。

[1] The **Tian Shi Lei** (天時類, "Heavenly Patterns") is the second of nine chapters in Qin Jiushao's *Shuxue Jiuzhang* (數學九章, Mathematical Treatise in Nine Sections). This fascicle contains **nine problems** divided into two parts, dealing with:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

Qin Jiushao's preface to the *Shuxue Jiuzhang* (1247) explains his motivation:

> *"The art of numbers, rooted in the Great Void, generates One and circulates without end. In its large aspect it can penetrate the divine and comply with destiny; in its small aspect it can govern worldly affairs and classify the myriad things. How can it be viewed as shallow or narrow?"*

[0]* 此卷所用之法，大要有三：曰通積術，求諸周之公倍也；曰招差術，推日月五星之變行也；曰調日法，齊曆法之疏密也。三者皆中國古代天文數學之精髓。

[1] The nine problems of this fascicle demonstrate the practical application of mathematical techniques to calendar reform, planetary motion, and meteorological measurement—topics of vital importance to agricultural society.

The key methods employed are:

- **General accumulation method** (通積): Finding common periods
- **Interpolation** (招差): For irregular astronomical motions
- **Equalizing technique** (調日法): For calendrical adjustments

[0]*

卷二上

[1]

Volume 2, Part 1

[0]*

第一問：推氣治歷

問 太史觀天，慶元四年戊午歲冬至三十九日九十二刻四十五分，紹定三年庚寅歲冬至三十二日九十四刻一十二分。今欲知嘉泰甲子歲氣骨、歲餘、斗分各幾何?

按 按：紹定三年庚寅冬至，實紹定四年辛卯之歲首也。辛卯去戊午三十四年，即積三十三年。

答曰 氣骨一十一日三十八刻二十分八十一杪八十小分；歲餘五日二十四刻二十九分三十杪三十小分；斗分〇日二十四刻二十九分三十杪三十小分。

術曰 以兩測之年差為法。置前測之日刻分，以後測之日刻分減之，餘為率。（不減，加紀策）。遞以紀策累加之，務令合天道，取五日以上為實。以法除實，得歲餘。去全分，餘為斗分。以歲餘乘前測至所求中積之年數，併入前測日刻分，滿紀策去之，餘為所求氣骨。

按 氣骨者，自冬至距夜半後甲子日之时刻也。歲餘者，歲周減六甲之餘也。斗分者，歲周減三百六十五日之餘也。此術尚未知歲實之確數，故以兩測氣骨之差為實，積年為法。

[1]

Problem 1: Calculating Qi and Regulating the Calendar

Question The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

Editor's Note The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

Answer

- Qi bone: 11 days, 38 *ke*, 20 *fen*, 81 *miao* (杪), 80 small fractions.
- Year remainder: 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction: 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

Method First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the ji-ce 紀策). Accumulate by adding ji-ce repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the dou fraction. Multiply the year remainder by the number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full ji-ce; the remainder is the desired qi bone.

Editor's Note The qi bone is the days and fractions from the winter solstice to the Jiazi day after midnight. The year remainder is the residue when the solar year length is reduced by six Jiazi cycles. The dou fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed qi bones as the dividend and accumulated years as divisor.

[0]*

第二問：治歷推閏

問 開禧曆，以嘉泰四年甲子歲天冬至一十一日，日辰乙亥，四十四刻六十一分五十四秒；十一月經朔一日，日辰乙丑，七十五刻五十五分六十二秒。問：閏骨、閏率各幾何?

答曰 閏骨九日六十九刻五分九十一秒，餘一百二十一（以一百六十九分為法）；閏骨率一十六萬三千七百七十一。

術曰 以日法通氣朔之日刻分秒，各為氣骨、朔骨之通分。氣骨通分以約率約之，適盡者可用；或不盡，圓徑以下在一刻內者，亦可用。乃以減朔骨通分，餘為閏骨率。以日法除之，得閏骨策。

按 若以朔骨通分徑減氣骨通分，餘為九日六十九刻五分九十二秒，即閏骨策。原草僅差四十八分之一百六十九分。其不徑減而以通分、約率、乘除往復者，蓋為後續諸算之便也。

[1]

Problem 2: Regulating the Calendar and Intercalary Months

Question For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

Answer

- Run bone: 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate: 163,771.

Method Take the day-factor (日法) to convert the qi and new-moon days, *ke*, *fen*, and *miao* into qi-bone and new-moon-bone fractions respectively. The qi-bone fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the run bone rate. Divide by the day-factor to obtain the run bone *ce* (策).

Editor's Note If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the run bone *ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.

[0]*

第三問：治歷演紀

問 開禧曆積年七百八十四萬一千一百八十三。今欲知演紀之術：調日法，求朔餘、朔率、斗分、歲率、歲閏、入元歲、入閏、朔定骨、閏泛骨、閏縮、紀率、氣元率、元閏、元數、氣等率、因率、蔀率、朔等數、因數、蔀數、朔積年，凡二十三事。各幾何？

答曰 （選錄要項）日法一萬六千九百；朔餘八千九百六十七；朔率四十九萬九千六十七；斗分即二十四刻二十九分三十杪三十小分；歲率六百一十七萬二千六百八；歲閏二萬七百二十八；入元歲九千二百四十；入閏三萬一千六十八。

術曰 以大衍術求氣分、朔分、紀分、衍母、正用。以衍母除氣分，得積年；除朔分，得積月；除紀分，得積紀。各以六十乘之，得積日。以氣骨乘紀正用，得紀總；以閏骨乘朔正用，得朔總。併二總，滿衍母去之，餘為所求率實。以氣分除之，得歷過之年；以減積年，得未至之年。

按 此問全用大衍求一術，以解同餘式組，施用於治曆。所算二十三率，備成開禧曆元之全體，自天文觀測而推得曆元諸數。

[1]

Problem 3: Deriving Calendar Epoch Calculations

Question The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—**23 items** in total. What is each?

Answer (Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

Method Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

Editor's Note This problem demonstrates the full power of Qin Jiushao's **Dayan method** (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

[0]*

第四問：綴術推星

問 歲星合伏段，一十六日九十分，行三度九十分，去日一十三度乃見。然後順行一百一十三日，一十七度八十三分，乃留。欲知：合伏段、晨疾初段之常度、初行率、末行率、平行率，各幾何?

術曰 （綴術）

1. 合伏段者，常度為總日數，初行率為總行度數，末行率等于初行率減日差乘（日數減一）。
2. 晨疾初段者，常度為所給日數，初行率為前段之末行率，末行率等于初行率減日差乘（日數減一）。
3. 平行率者，併初末二率而半之，即得平行率。

按 五星之行，有遲疾逆順，非均加速也。此術僅取合伏、見段之中數，而以遞加遞減求逐日之行率。古法之疏，於五星尤著。原草語多隱奧，今詳釋之，以見古今疏密之比。

草曰 合伏段：常度一十六日九十分；初行率二十二分七杪；末行率二十一分九十六杪；平行率約二十二分。晨疾初段：常度三十日；初行率二十一分九十六杪；日差一十一杪二十三分六十六小杪。

[1]

Problem 4: Calculating Planetary Motion

Question The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段)
- The morning-rapid initial segment (晨疾初段)
- The standard degree (常度)
- The initial motion-rate (初行率)
- The final motion-rate (末行率)
- The mean motion-rate (平行率)

What is each?

Method (Method of *Zhui Shu*):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days -1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days -1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

Editor's Note The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

Working **Key calculations:**

- Conjunction-hidden segment:
  - Constant degree: 16 days 90 *fen*
  - Initial rate: 22 *fen* 7 *miao*
  - Final rate: 21 *fen* 96 *miao*
  - Mean rate: approximately 22 *fen* (after rounding)
- Morning-rapid initial segment:
  - Constant degree: 30 days
  - Initial rate: 21 *fen* 96 *miao*

- Daily difference: 11 *miao* 23 small-*fen* 66 small-*miao*

[0]*

第五問：揆日究微

問　歷代測景，唯唐大衍曆最密。本朝崇天曆測陽城：冬至影一丈二尺七寸一分五十杪，夏至影一尺四寸七分七十杪，與大衍曆合。今開禧曆測臨安府：冬至影一丈八寸二分二十五杪，夏至影九寸一分。問：臨安夏至後幾日，其影與陽城夏至影等？又與大衍曆影較之，相差幾何？

答曰　大暑後五日午正，影長一尺四寸八分八十五杪（半）。影差六寸一分二十九杪（少）。

術曰　用乘法，以節率入之。

1. 置臨安所測冬至影長，減夏至影長，餘為景差，以為實。
2. 置象限度（九十一度三十一分四十四杪），加一十一度二十五分二十七秒半；以度為寸，得一百〇二寸五千六百七十一分五十杪為法。
3. 以法除實，得〇點九六六四……；棄餘數，自乘得節泛數九千四百分。
4. 以各節氣乘法率乘之，加夏至影冪，開方得各節氣影長。

按　各節氣之影長，當時實測而定，原不必算。此術故為繁難，以惑學者。細考之：以象限度加一十一度有餘為法，以景差除之而後自乘，得節率。每節氣有乘法率，以節率乘之，加夏至影冪，即為各節氣之影冪。故節率者，人為析出之數也。

[1]

Problem 5: Measuring the Sun's Shadow

Question　Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 *zhang* 2 *chi* 7 *cun* 1 *fen* 50 *miao*
- Summer solstice shadow: 1 *chi* 4 *cun* 7 *fen* 70 *miao*

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 *zhang* 8 *cun* 2 *fen* 25 *miao*
- Summer solstice shadow: 9 *cun* 1 *fen*

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

Answer　Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 *chi* 4 *cun* 8 *fen* 85 *miao* (half). The shadow difference is 6 *cun* 1 *fen* 29 *miao* (less).

Method　Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 *fen* 44 *miao*), add 11 degrees 25 *fen* 27.5 small-*fen*; treating degrees as *cun*, obtain 102 *cun* 5671.5 *fen* 50 *miao* as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 *fen*.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

Editor's Note　Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

[0]*

卷二下

[1]

Volume 2, Part 2

[0]*

第六問：天池測雨

問　今州郡皆有天池盆以測雨，然人但知盆中之水即所受之雨，而不知器之圓弇各有所受，不可徑以盆測為平地之雨。設盆口徑二尺八寸，底徑一尺二寸，深一尺八寸，接雨水深九寸。問：平地雨深幾何?

答曰　平地雨深三寸。

術曰　用圓錐術：

1. 以口徑乘底徑，又各自乘，併三數。以半深乘之，又以十一乘、七除，得下積。
2. 以半深加雨深，以減全深，得上深。
3. 以底徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。
4. 以面率除半深，得水面徑。
5. 續以乘除，求得雨水之積，以口冪（平方三倍）除之，得平地雨深。

按　天池盆者，圓錐之截頭也。以雨積除以口面積，即得平地雨深。其術以圓錐台體積之術推之，而歸於平地之算。

[1]

Problem 6: Measuring Rain with the Heavenly Pool Basin

Question　Nowadays every prefecture and district has a “heavenly pool” basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

Answer　Flat-ground rain depth: **3 *cun***.

Method　Use the method for circular cones (圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).
3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.
5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

Editor’s Note　The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3}\left(R^2 + Rr + r^2\right)$$

where R is the top radius, r is the base radius, and h_{rain} is the rain depth. Dividing by the mouth area πR^2 gives the equivalent flat-ground rain depth.

[0]*

第七問：圓罌測雨

問 以圓罌受雨，口徑一尺五寸（一尺五分寸?），腹徑二尺四寸，底徑八寸，深一尺六寸，接雨水深一尺二寸。用密率入圓冪，問：平地雨深幾何?

答曰 平地雨深一尺八寸，又七百四十分之六百八十三寸。（校正值）

術曰 以底徑乘腹徑，又各自乘，併三數。以半器深乘之；以十一乘、四十二除，得下積。

以半器深加雨深，以減全深，得上深。以腹徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。

以面率乘腹率，又各自乘併之；以十一乘、四十二除，得上率。併二率（上下）為雨積實。以口徑平方之三倍除之，得平地雨深。

按 原題云「密率入圓冪」，與求平地雨深無涉。若求器中雨積，則此言為當。原草有誤，今以正術列之。

[1]

Problem 7: Measuring Rain with a Round Vessel

Question Using a round vessel (圓罌) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the “dense rate” (密率, $\pi \approx 22/7$) for circle computations, what is the flat-ground rain depth?

Answer Flat-ground rain depth: **1 *chi* 8 *cun* plus** $\frac{6483}{74088}$ ***cun***. (Editor’s corrected value.)

Method Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

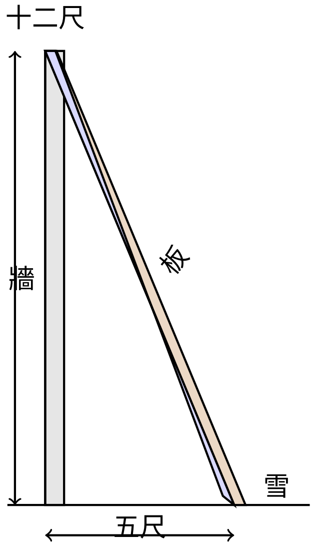
Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

Editor’s Note The original problem statement mentions “dense rate for circle computation” which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel’s rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.

[0]*

第八問：峻積驗雪

問 驗雪占年：牆高一丈二尺，以板倚之，去址五尺，板末與牆齊。雪積板上厚四寸，峻處薄而平處厚。問：平地雪厚幾何？



答曰 平地雪厚一尺四分。

術曰 用少廣術，連枝入之：

1. 以去址之數自乘，為隅。
2. 以牆高自乘，併入上隅。
3. 以雪厚自乘，乘上併數，為實。
4. （約之）開連枝平方，得平地雪厚。

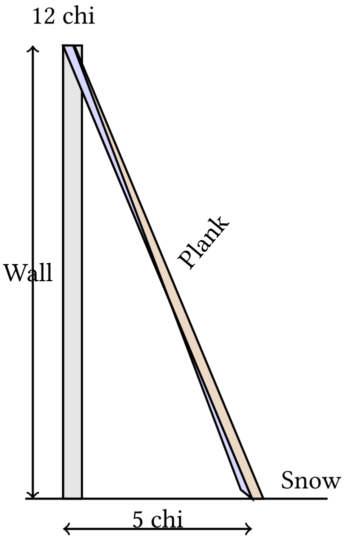
草曰 通分內子：去址五十寸，自乘得二千五百，為隅。牆高一百二十寸，自乘得一萬四千四百。併之得一萬六千九百。雪厚四寸，自乘得十六。乘之得二十七萬四百，為實。隅與實俱以百約之，得實二千七百〇四，隅二十五。開平方：$\sqrt{2704/25} = 52/5 = 10.4$ 寸，即一尺四分。

按 理與法俱正。實即勾股術，特不以此名之耳。「連枝」之號，隱其本理。蓋三角形相似，以比例求之也。

[1]

Problem 8: Verifying Snow Accumulation

Question To verify snow for predicting the harvest: A wall is 1 *zhang* 2 *chi* (12 *chi*) high. A plank leans against it, the base 5 *chi* from the wall's foot, with the tip level with the wall top. Snow accumulates 4 *cun* thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?



Answer Flat-ground snow depth: **1 *chi* 4 *fen***.

Method Use the method of Shao Guang (少廣), with the connected-branch (連枝) method:

1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

Working Convert all measurements to *cun*:

- Distance from base: 50 *cun*; squared = 2,500 (corner).
- Wall height: 120 *cun*; squared = 14,400.
- Combined: 14,400 + 2,500 = 16,900.
- Snow thickness: 4 *cun*; squared = 16.
- Multiply: $16{,}900 \times 16 = 270{,}400$ (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root: $\sqrt{2704/25} = 52/5 = 10.4$ *cun* = **1 *chi* 4 *fen***.

Editor's Note Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The "connected-branch" terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.

[0]*

第九問：竹器驗雪

問 以圓竹器驗雪，口徑一尺六寸，深一尺七寸，底徑一尺二寸。雪入其中，深一尺。器通風，雪多則平地淺。問：平地雪厚幾何?

答曰 平地雪厚九寸，又三千四百三十九分之七百六十四寸。

術曰 以底徑減口徑，以雪深乘之，半之，自乘，為隅。以器深自乘，乘雪深自乘之數；併入隅，以雪深自乘之數乘之，為實。開連枝三乘方，得平地雪厚。

草曰 通分內子：口徑十六寸，底徑十二寸。差四寸；雪深十寸。乘之得四十；半之得二十，為上廉。器深自乘：$17^2 = 289$。雪深自乘：$10^2 = 100$。乘之得二萬八千九百；併入隅，續以三乘方開法入之，得平地雪厚九寸，餘三千四百三十九分之七百六十四。

按 「通風」之語，不關算術。或謂器中之雪深與平地異，以風故也；然此問算理，與天池測雨之題同，以圓臺之體推之而已。

[1]

Problem 9: Verifying Snow with a Bamboo Container

Question Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

Answer Flat-ground snow thickness: **9 *cun* plus $\frac{764}{3439}$ *cun*.**

Method Subtract the base diameter from the mouth diameter; multiply by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

Working Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply: $4 \times 10 = 40$; halve = 20 (this is the upper-*lian* 上廉).
- Container depth squared: $17^2 = 289$.
- Snow depth squared: $10^2 = 100$.
- Multiply: $289 \times 100 = 28{,}900$; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese “method of extracting cube roots with carrying” (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

Editor’s Note The phrase “permeable to wind” does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

[0]*

天時類九問總表

[1]

Summary of Fascicle 2 Problems

[0]*

序	題名	術要
1	推氣治歷	兩測推氣骨、歲餘、斗分
2	治歷推閏	通分約率求閏骨、閏率
3	治歷演紀	大衍術推曆元二十三率
4	綴術推星	招差法推五星行率
5	揆日究微	乘法節率推各地日影
6	天池測雨	圓錐台體積求平地雨深
7	圓罌測雨	圓罌容積求平地雨深
8	峻積驗雪	少廣連枝開方求平地雪厚
9	竹器驗雪	三乘方開法求平地雪厚

[1]

No.	**Title**	**Topic**
1	推氣治歷 Tui Qi Zhi Li	Computing solar terms from two winter solstice observations
2	治歷推閏 Zhi Li Tui Run	Finding intercalation parameters for a calendar
3	治歷演紀 Zhi Li Yan Ji	Deriving 23 epoch parameters using Dayan method
4	綴術推星 Zhui Shu Tui Xing	Computing Jupiter's motion rates
5	揆日究微 Kui Ri Jiu Wei	Comparing solar shadow lengths at two locations
6	天池測雨 Tian Chi Ce Yu	Measuring rain with a conical basin
7	圓罌測雨 Yuan Ying Ce Yu	Measuring rain with a round vessel
8	峻積驗雪 Jun Ji Yan Xue	Converting snow on a slanted plank to flat-ground depth
9	竹器驗雪 Zhu Qi Yan Xue	Converting snow in a bamboo container to flat-ground depth

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End of Fascicle 2 / 卷二終

"Seven planets revolve in the firmament; the affairs of men are tracked by pursuing and connecting [their motions].
Stars by night, gnomon by day—as ages pass, [methods] become coarse;
only the wise can reform them."
— Qin Jiushao, Preface to *Shuxue Jiuzhang* / 秦九韶《數學九章序》

Siku Quanshu Edition / 四庫全書本

Shu Xue Jiu Zhang / 數學九章

Fascicles 3–4 / 卷三至四

Tian Yu (田域) + Ce Wang (測望)

Author: Song Dynasty, Qin Jiushao / 宋　秦九韶
Dates: c. 1202–1261
Topics: Juan 3 Shang/Xia: Tian Yu (Fields / 田域類)
Juan 4 Shang/Xia: Ce Wang (Surveying / 測望類)

Bilingual Edition / 雙語版

LEFT: Original Chinese / 原文 | RIGHT: English Translation / 英譯

Typeset from the Siku Quanshu (四庫全書) edition
Bilingual edition with parallel Chinese text and English translation

Contents

Introduction / 緒言

數學九章為南宋秦九韶所撰，是中古數學之偉大著作。全書九章，每章一類。

本卷呈其第三、四卷：

- 卷三上：田域類 – 方田、圓田、三角田及組合圖形之面積
- 卷三下：田域類續 – 續田域問題，含籌算圖式
- 卷四上：測望類 – 隔河測高遠，用勾股重差之法
- 卷四下：測望類續 – 續測望問題，量不可至之遠近高下

文本依中算傳統格式：

- 問 – 問題陳述
- 答 – 答數
- 術 – 演算方法
- 草 – 詳細演草

The *Shu Xue Jiu Zhang* (Mathematical Treatise in Nine Chapters), written by Qin Jiushao during the Southern Song Dynasty, is one of the greatest masterpieces of medieval Chinese mathematics. The work contains nine chapters, each devoted to a different class of problems.

This volume presents Fascicles 3 and 4 of the work, covering:

- **Juan 3 Shang – Tian Yu (Fields, Part 1)**: Problems on area measurement, including squares, circles, triangles, and composite figures.
- **Juan 3 Xia – Tian Yu (Fields, Part 2)**: Continuation of field problems, including more complex configurations with counting rod diagrams.
- **Juan 4 Shang – Ce Wang (Surveying, Part 1)**: Problems on remote measurement and surveying, using similar triangles and the gou-gu method.
- **Juan 4 Xia – Ce Wang (Surveying, Part 2)**: Further surveying problems, including measurement of heights, distances, and inaccessible objects.

The text follows the classical format of Chinese mathematical literature:

- **Wen (問)**: The problem statement
- **Da Yue (答)**: The answer
- **Shu Yue (術)**: The method/algorithm
- **Cao Yue (草)**: The working/calculation

1 Juan 3 Shang – Tian Yu / 卷三上田域類

小序：此卷論田域面積之術。涉方圓、斜直、面冪、體積，皆相求之。其本則出於立天元一法。

Introductory note: This fascicle treats problems on area measurement (Tian Yu). The problems involve squares, circles, triangles, and composite figures. Qin Jiushao employs his "method of the celestial element" (Tian Yuan Shu) alongside traditional algorithmic techniques.

Problem 1: Fang Zhong Gu Yuan / 方中古圓池

問/ 第一問：方中古圓池

有方中古圓池，因陂毀於一角。從外方隅斜至內圓邊七尺六寸。欲依舊修之，求圓徑、方面、方斜各幾何。

答/ 答曰：

池圓徑三丈六尺六寸，四百二十九分寸之四百一十二。

方面三丈六尺六寸，四百二十九分寸之四百一十二。

方斜五丈一尺八寸，四百二十九分寸之四百一十二。

術/ 術曰：

以少廣求之，太朮入之。如積斜自乘倍之為實，倍斜為一方法，一半寸為從隅。開平方得徑。

Problem / Square with Inscribed Circular Pond

There is a square field with an ancient circular pond inscribed within it. The northern corner has partially collapsed. From the corner of the outer square to the edge of the inner circle, the diagonal distance is 7 chi 6 cun (7.6 chi). Wishing to repair it according to the original design, find: the diameter of the circle, the side of the square, and the diagonal of the square.

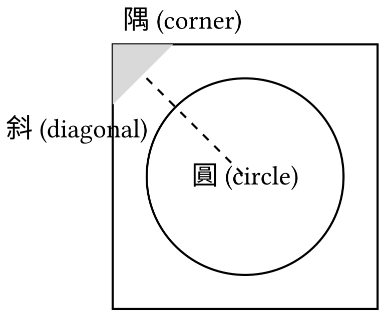
Answer / Answer:

The pond's diameter is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's side is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's diagonal is 51 chi 8 cun plus $\frac{412}{429}$ cun.

Method / Method:

Using the "lesser breadth" (shao guang) method with the "transformation of the embryo" (tai tai) technique. Square the diagonal and double it to obtain the dividend (shi). Take twice the diagonal as the negative coefficient of the linear term (yi fang). Take half a cun as the coefficient of the quadratic term (cong yu). Extract the square root to find the diameter.

Tu (Diagram / 圖):



Problem 2: San Xie Qiu Ji / 三斜求積

問/ 第二問：三斜求積

有三斜形田，大斜一十三里，小斜一十四里，中斜一十五里。里法三百步，欲知為田積幾何。

Problem / Triangular Field Area

There is a triangular field. The large diagonal (da xie) is 13 li, the small diagonal (xiao xie) is 14 li, and

答/ 答曰：

田積三百一十五頃。

術/ 術曰：

以少廣求之。以小斜冪並大斜冪，減中斜冪，餘半之，自乘於上。以小斜冪乘大斜冪，減上，餘四約之為實。一為從隅，開平方得積。

the middle diagonal (zhong xie) is 15 li. With 300 bu per li, find the area.

Answer / **Answer**:

The area is 315 qing.

Method / **Method**:

Using the “lesser breadth” method. Add the squares of the small and large diagonals, subtract the square of the middle diagonal. Halve the remainder and square it (placed above). Multiply the square of the small diagonal by the square of the large diagonal; subtract the result above; divide the remainder by 4 to get the dividend. Take 1 as the coefficient of the quadratic term; extract the square root to find the area.

In modern notation: If $a = 13$, $b = 14$, $c = 15$, then:

$$S = \sqrt{\frac{1}{4}\left[a^2b^2 - \left(\frac{a^2+b^2-c^2}{2}\right)^2\right]}$$

This is equivalent to **Heron’s formula**:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Problem 3: Fang Tian Yuan Chi / 方田圓池

問/ 第三問：方田圓池

有方田一段，內有圓池水占之，外計地三千一百六十八步。只云從內池周至外田角斜二十一步半。內外方圓各幾何。

答/ 答曰：

內池周四十二步，徑一十三步半。方田面六十步。

術/ 術曰：

以少廣求之。置積以圓率三約之為實，倍斜為從方，斜冪為一負隅。開平方得內徑。加倍斜得方面。

Problem / **Square Field with Central Circular Pond**

There is a square field with a circular pond inside it. The remaining land outside the pond amounts to 3168 bu. It is given that the diagonal distance from the edge of the inner pond to the corner of the outer square is 21.5 bu. Find the dimensions of both the inner circle and outer square.

Answer / **Answer**:

The inner pond’s circumference is 42 bu, its diameter is 13.5 bu; the square field’s side is 60 bu.

Method / **Method**:

Using the “lesser breadth” method. Place the area and divide by the circular rate 3 to obtain the dividend. Twice the diagonal gives the linear coefficient. The square of the diagonal gives the negative quadratic coefficient. Extract the square root to find the inner diameter. Add twice the diagonal to obtain the square’s side.

Problem 4: Piao Tian / 漂田

問/ 第四問：漂田

有漂田一段，中斜一十六步，水止五步。減中斜一十六餘一十一步為元大斜。內減殘大斜二十步……

Problem / **The Drowned Field**

There is a “drowned field” (a triangular field partially submerged). The middle diagonal is 16 bu; the water depth is 5 bu. Subtracting 5 from 16 gives 11 bu as the

答/ 答曰：

田積若干。

術/ 術曰：

草曰：以水止五減中斜一十六餘一十一步為法。以中斜一十六乘大殘二十得三百二十為大斜實。以法除之得二十九步一十一分步之一為元大斜。內減殘大斜二十步，餘九步一十一分步之一……

base large diagonal.

Answer / **Answer**:

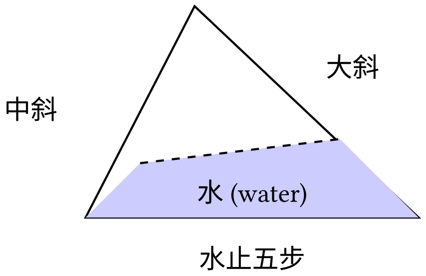
The field area is [to be determined].

Method / **Method**:

(Detailed calculation with specific numerical operations as recorded in the original text.)

Working: Take the water depth 5 subtracted from the middle diagonal 16, remainder 11 bu as the divisor. Multiply the middle diagonal 16 by the large remainder 20 to get 320 as the large diagonal dividend. Divide by the divisor to get 29 and $\frac{1}{11}$ bu as the original large diagonal. Subtract the large remainder 20 bu, remainder 9 and $\frac{1}{11}$ bu…

Tu (Diagram / 圖):



2 Juan 3 Xia – Tian Yu Xu / 卷三下田域類續

Problem 5: San Hu Fen Tian / 三戶分田

問/ 第五問：三戶分田

有田一段，欲分與甲乙丙三戶。甲多一百五十步，乙多一百二十步，丙多九十步。三戶共田若干，各得幾何。

答/ 答曰：

甲田若干，乙田若干，丙田若干。

術/ 術曰：

以少廣求之。併三戶多出之數以減總積，餘以三戶約之，各得本積。又各加多出之數即得各戶之田。

Problem / Division of Land Among Three Families

There is a field to be divided among three families jia, yi, and bing. Family jia receives 150 bu more than the base amount; family yi receives 120 bu more; family bing receives 90 bu more. If the total area is given, find each family's share.

Answer / **Answer**:

Family jia's share [is determined]; family yi's share [is determined]; family bing's share [is determined].

Method / **Method**:

Using the "lesser breadth" method. Add the excess amounts from all three families and subtract from the total area. Divide the remainder by 3 to get the base amount for each. Add each family's excess to the base amount to obtain each share.

Problem 6: Yuan Tian Fang Tian / 圓田方田

問/ 第六問：圓田方田

有方圓田各一段，共積一千二百八十步。只云方面如圓徑少四步，問方圓面徑各幾何。

答/ 答曰：

方面若干，圓徑若干。

術/ 術曰：

以少廣求之。置共積以圓率三、方率四各約之為實，方面少圓徑四步為從方。開平方得圓徑。減四得方面。

Problem / Circular and Square Fields

There is a square field and a circular field. Their total area is 1280 bu. It is given that the square's side is 4 bu less than the circle's diameter. Find the side and diameter.

Answer / **Answer**:

The square's side [and] the circle's diameter [are determined].

Method / **Method**:

Using the "lesser breadth" method. Place the combined area; divide by the circular rate 3 and square rate 4 to obtain the dividend. The difference of 4 bu between square side and circle diameter gives the linear coefficient. Extract the square root to find the diameter. Subtract 4 to obtain the square's side.

Problem 7: Hu Tian / 弧田

問/ 第七問：弧田

有弧田一段，弦五十步，矢二十五步，求積若干。

答/ 答曰：

田積若干。

術/ 術曰：

以弦矢求之。以弦乘矢，又以矢自乘，併之，半之為實。開平方得積。

Problem / Circular Segment Field

There is a circular segment field (hu tian) with chord (xian) 50 bu and arrow (shi) 25 bu. Find its area.

Answer / **Answer**:

The field area [is determined].

Method / **Method**:

Using the chord-arrow method. Multiply the chord by the arrow; add the square of the arrow; halve to obtain the dividend. Extract the square root to find the area.

Note: For a circular segment with chord c and sagitta s, Qin's formula gives:

$$A = \frac{cs + s^2}{2}$$

This is an approximation to the true area of a circular segment.

Counting Rod Diagram / 籌算圖式:

弦	矢	弦乘矢	矢冪
50	25	1250	625
併		半之	積
1875		937.5	≈ 30.6

3 Juan 4 Shang – Ce Wang / 卷四上測望類

小序：此卷論測望之術，量不可至之高遠。用勾股、重差之法，為中國古代測量學之基礎。

Introductory note: This fascicle treats problems on surveying and remote measurement (Ce Wang). These problems involve measuring inaccessible distances and heights using the “hook-and-gourd” (gou gu) method and similar triangles.

Problem 1: Yuan Cheng / 圓城

問/ 第一問：圓城

有圓城不知周徑，四門中開。北外三里有喬木，出南門折而東行九里乃見木。欲知城周徑各幾何。

答/ 答曰：

徑九里，周二十七里。

術/ 術曰：

以勾股夕桀求之。一為從隅，五因北里為從七廉，置北里冪八因為從五廉，以北里乘上廉為一從，三連負隅乘五廉為一上廉，以北里乘上廉為實。開玲瓏九乘方得數，自乘為徑。以三因徑得周。

Problem / The Round City

There is a circular city of unknown circumference and diameter. It has four gates at the cardinal points. Three li north of the north gate stands a tall tree. Going out the south gate and then turning east for 9 li, the tree becomes visible. Find the circumference and diameter of the city.

Answer / Answer:

The diameter is 9 li, the circumference is 27 li.

Method / Method:

Using the “hook-and-gourd evening-peak” method. Take 1 as the leading coefficient. Multiply the north distance by 5 for the seventh-degree coefficient. Place the square of the north distance, multiply by 8 for the fifth-degree coefficient. Multiply the north distance by the upper coefficient for the negative linear coefficient. Double the negative rate to form the fifth-degree coefficient as the negative upper coefficient. Multiply the north distance by the upper coefficient for the dividend. Extract the ninth-degree root (ling long kai fang) to obtain the number; square it for the diameter. Multiply the diameter by 3 for the circumference.

Problem 2: Ce Ta / 測塔

問/ 第二問：測塔

有塔高未知其數，去塔若干步立一表，人目上望見塔尖，退行若干步又立一表，人目上望仍見塔尖。求塔高及去塔遠近。

答/ 答曰：

塔高若干，去塔遠若干。

術/ 術曰：

以勾股求之。兩表退行之差為法，前表去塔遠乘表高，後表去塔遠乘表高，相減為實。實如法而一得塔高。又以法除前表去塔遠乘兩表高差加表高得塔高。

Problem / Measuring a Tower

A tower of unknown height stands at an unknown distance. At a certain distance from the tower, a pole is erected; looking upward from eye level, the tower’s peak is visible. Retreating a certain distance and erecting another pole, the peak is again visible from eye level. Find the tower’s height and distance.

Answer / Answer:

The tower height [and] distance to the tower [are determined].

Method / Method:

Using the hook-and-gourd method. The difference between the retreat distances of the two poles is the divisor. Multiply the front pole’s distance to the tower by the pole height; multiply the rear pole’s distance by the pole height; subtract to obtain the dividend. Divide to get the tower height. Alternatively, multiply the front distance by the difference of pole heights, divide by the divisor, and add the pole height.

In modern notation: Let d_1 = front distance, d_2 = rear distance, h = pole height. Then:

$$\text{Tower height} = \frac{d_2 \cdot h - d_1 \cdot h}{d_2 - d_1} = h$$

This uses the principle of similar triangles, where the ratio of heights equals the ratio of distances.

Problem 3: Fang Cheng / 方城

問/ 第三問：方城

有方城一座，不知大小。北門外有樹，出南門直行若干步折而西行若干步見樹。求城方面。

答/ 答曰：

城方面若干步。

術/ 術曰：

以勾股求之。兩行步相乘為實，併兩行步為從方。開平方得城方面。

Problem / **The Square City**

There is a square city of unknown size. Outside the north gate stands a tree. Going out the south gate and walking straight a certain number of steps, then turning west and walking a certain number of steps, the tree becomes visible. Find the side of the square city.

Answer / **Answer**:

The city's side is [determined] in bu.

Method / **Method**:

Using the hook-and-gourd method. Multiply the two walk distances for the dividend. Add the two walk distances for the linear coefficient. Extract the square root to obtain the city's side.

If a = southward steps and b = westward steps:

$$x^2 + (a+b)x = ab \quad \Rightarrow \quad x = \frac{-(a+b) + \sqrt{(a+b)^2 + 4ab}}{2}$$

(
1em)

Problem 4: Chou Suan Kai Fang / 籌算開方

問/ 第四問：籌算開方

問以開方術解高次方程，其籌佈列之法若何。

術/ 術曰：

開方之法，先列實於中，次列隅於下，次列方於上。商除之，以上乘隅生廉，以上乘廉生方，以上乘方除實。實盡則止，不盡則倍方，三廉為方法，續商除之。

Problem / **Counting Rod Root Extraction**

On the method of extracting roots to solve higher-degree equations, how are the counting rods arranged?

Method / **Method**:

In the method of root extraction, first place the dividend (shi) in the middle, then place the unit coefficient (yu) below, and the linear coefficient (fang) above. Divide by estimation; multiply the estimate by the unit coefficient to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. When the dividend is exhausted, stop. If not exhausted, double the linear coefficient and proceed to the next digit.

Counting Rod Layout / 籌式:

商	實	方	廉	隅
7	360	130	…	1
Iterative extraction / 逐次開方				

Note: In the counting rod system, vertical strokes represent units $(1, 2, 3, 4)$ and horizontal strokes represent fives. The symbol T represents 6 (5+1). The diagram shows the iterative process of root extraction.

4 Juan 4 Xia – Ce Wang Xu / 卷四下測望類續

Problem 5: Ce Shui Shen / 測水深

問/ 第五問：測水深

有井不知其深，以繩測之，繩長倍於井深餘三尺；三摺而測之適足。求井深及繩長。

答/ 答曰：

井深三尺，繩長九尺。

術/ 術曰：

以盈不足求之。倍繩餘三尺為盈，三摺適足為適足。以盈乘適足為實，併盈適足為法，除之得井深。倍之加盈得繩長。

Problem / **Measuring Depth of Water**

There is a well of unknown depth. Measuring with a rope, the rope's length is twice the well's depth with 3 chi to spare. Folding the rope in three and measuring, it fits exactly. Find the depth of the well and the length of the rope.

Answer / **Answer**:

The well is 3 chi deep; the rope is 9 chi long.

Method / **Method**:

Using the "surplus and deficit" (ying bu zu) method. The 3 chi excess when doubled is the surplus; the exact fit when tripled is the exact amount. Multiply the surplus by the exact amount for the dividend; add surplus and exact amount for the divisor; divide to obtain the well depth. Double and add the excess for the rope length.

In modern notation: Let d = well depth, L = rope length.

$$L = 2d + 3, \qquad \frac{L}{3} = d$$

Substituting: $2d + 3 = 3d \Rightarrow d = 3$, $L = 9$.

Problem 6: Ce Ta Gao / 測塔高

問/ 第六問：測塔高

有塔高未知，去塔百步立一表高五尺，人目去表一尺八寸，上望塔尖與表端參合。退行六十步，人目去表端仍參合塔尖。求塔高及塔心去表遠近。

答/ 答曰：

塔身高三丈，塔高一十一丈七尺，與八丈七尺為塔心目長。

術/ 術曰：

此皆大小形同勢相求法也。人目去塔為總股，人目去杆為分小股，人目上塔尖高塔身節為總股高。相論高為分大勾，數一杆去塔分大勾於小角杆去塔為分大勾。

Problem / **Measuring the Height of a Pagoda**

A pagoda of unknown height stands at an unknown distance. 100 bu from the pagoda, a 5-chi pole is erected. The observer's eye is 1.8 chi from the ground; looking up, the pagoda's peak aligns with the pole's tip. Retreating 60 bu, the observer's eye again aligns with both the pole's tip and the pagoda's peak. Find the pagoda's height and its distance from the pole.

Answer / **Answer**:

The pagoda's body is 3 zhang high; the total height (from ground to peak) is 11 zhang 7 chi; the distance from the pagoda's center to the measuring point is 8 zhang 7 chi.

Method / **Method**:

These are all applications of similar triangles (da xiao xing tong shi). The distance from eye to pagoda is the total height line; the distance from eye to pole is the small height line. The height from eye level to pagoda peak is the total height; the height from eye level to pole top is the small height. The method uses proportional relationships between these quantities.

This problem uses the principle of similar triangles:

$$\frac{\text{Pagoda height} - \text{eye height}}{\text{Distance to pagoda}} = \frac{\text{Pole height} - \text{eye height}}{\text{Distance to pole}}$$

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1em)

Problem 7: Ge He Ce Yuan / 隔河測遠

問/ 第七問：隔河測遠

問隔河望一高岸，不知其遠。平地立一表，人目上望見岸頂，又退立一表望之仍見岸頂。求河闊及岸高。

答/ 答曰：

河闊若干步，岸高若干尺。

術/ 術曰：

以重差求之。兩表退行步數之差為法，前表去岸遠乘表高為實，實如法而一得岸高。又以法除後表去岸遠乘表高亦得岸高，兩者相驗。

Problem / Measuring Distance Across a River

Across a river, a high bank is visible at an unknown distance. A pole is erected on level ground; looking up from eye level, the bank's top is visible. Retreating and erecting another pole, the bank's top is again visible. Find the width of the river and the height of the bank.

Answer / Answer:

The river width [and] the bank height [are determined].

Method / Method:

Using the "double difference" (chong cha) method. The difference between the two retreat distances is the divisor. Multiply the front pole's distance to the bank by the pole height for the dividend; divide to obtain the bank height. Similarly for the rear pole; the two results should agree (providing verification).

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Problem 8: Jiu Cheng Fang / 九乘方

問/ 第八問：九乘方

問以開九乘方術解方程，其立法之原若何。

術/ 術曰：

開方之術，至於九乘方，其理一也。先列實於中，次定隅位，次立方廉法。商除之際，以上乘隅生廉，以上乘廉生方，以上乘方除實。每進一位，則方退一，廉退二，隅退三。如是疊進，直至實盡為止。

Problem / The Nine-Degree Root Method

On the method of extracting ninth-degree roots to solve equations — what is the fundamental principle of this method?

Method / Method:

The method of root extraction, up to the ninth degree, follows the same principle. First place the dividend in the middle; then determine the unit position; then establish the linear and intermediate coefficients. During division-by-estimation: multiply the estimate by the unit coefficient to generate the intermediate coefficients; multiply the estimate by the intermediate coefficients to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. At each decimal shift, the linear coefficient descends one place, the intermediate descend two places, and the unit descends three. Continue iteratively until the dividend is exhausted.

Note: This describes **Horner's method** for solving polynomial equations, which Qin Jiushao developed independently centuries before Horner. For a polynomial:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

The method systematically reduces the polynomial by factoring out $(x - r)$ where r is the estimated root at each step.

Layout for Ninth-Degree Root Extraction / 九乘方籌式:

商	實	方	一廉	二廉	三廉	四廉	五廉	六廉	隅
7	–	153	...	...	...	...	...	...	1
Iterative computation with place-value shifts / 逐位遞推									

Appendix / 附錄: Mathematical Notes

A.1 籌算 / The Counting Rod System

籌算為中國古代主要計算工具。以竹籌佈列十進位制：

- 一至四：縱式 |, ||, |||, ||||
- 五：橫式 ≡
- 六至九：組合，如 T（五加一）
- 零：以空位或圓 ◯ 表之

籌列於算板，每列一位。此位值制助成高深代數運算，如開方、解方程。

A.1 The Counting Rod System (Chou Suan)

The counting rod system (Chou Suan) was the primary computational tool in ancient Chinese mathematics. Numbers were represented by bamboo sticks arranged in a decimal place-value system:

- **Units (1–4)**: Vertical strokes |, ||, |||, ||||
- **Fives**: A horizontal stroke = placed over the units
- **6–9**: Combinations, e.g., T (5+1), ≡ (3 horizontal over vertical)
- **Zero**: Represented by a blank space or circle ◯

The rods were arranged on a counting board in columns, with each column representing a decimal place. This positional system enabled sophisticated algebraic computations including root extraction and solution of polynomial equations.

A.2 天元術 / The Celestial Element Method

秦九韶精於天元術，為早期代數符號。未知數謂之「天元」，方程以籌式佈列。此法可解十次方程，早霍納五百餘年。

A.2 The "Celestial Element" Method (Tian Yuan Shu)

Qin Jiushao was a master of the "celestial element" method, an early form of algebraic symbolism. The unknown quantity was called "celestial element" (Tian Yuan, tian yuan), and equations were set up using counting rod arrangements. This method could solve polynomial equations of degree up to 10, anticipating Horner's method by over 500 years.

A.3 三斜求積公式 / Qin's Formula for Triangular Area

三角形三邊 a, b, c，秦九韶公式：

$$S = \sqrt{\frac{1}{4}\left[c^2a^2 - \left(\frac{c^2+a^2-b^2}{2}\right)^2\right]}$$

此與海龍公式等價，顯十三世紀非凡代數造詣。

A.3 Qin Jiushao's Formula for Triangular Area

For a triangle with sides a, b, c, Qin gave the formula:

$$S = \sqrt{\frac{1}{4}\left[c^2a^2 - \left(\frac{c^2+a^2-b^2}{2}\right)^2\right]}$$

This is equivalent to Heron's formula and demonstrates remarkable algebraic insight for the 13th century.

A.4 盈不足術 / The Surplus and Deficit Method

盈不足術為通用線性問題解法。設兩試值得盈、不足，則真值：

$$x = \frac{a_2b_1 + a_1b_2}{a_1 + a_2}$$

其中 a_1, a_2 為試入之數，b_1, b_2 為對應盈不足之數。

A.4 The "Surplus and Deficit" Method (Ying Bu Zu)

The method of surplus and deficit (Ying Bu Zu Shu) is a general technique for solving linear problems. Given two trial values yielding surplus and deficit results, the true value is found by:

$$x = \frac{a_2b_1 + a_1b_2}{a_1 + a_2}$$

where a_1, a_2 are the trial inputs and b_1, b_2 are the corresponding surplus/deficit amounts. This method can be applied to a wide variety of practical problems.

End of Fascicles 3–4 / 卷三至四終

Shu Xue Jiu Zhang – Juan 3 Shang, Juan 3 Xia, Juan 4 Shang, Juan 4 Xia / 數學九章

數學九章

卷第五、卷第六

卷五：賦役類 | 卷六：錢穀類

宋　秦九韶　撰
（宋）秦九韶（約 1202–1261）著

Siku Quanshu (四庫全書) Edition
Bilingual Chinese–English Edition
Traditional Text Structure Preserved

Mathematical Treatise in Nine Sections

Fascicles Five & Six

Fasc. 5: **Taxation and Corvée (Fuyi) |** Fasc. 6: **Money and Grain (Qiangu)**

By Qin Jiushao, Song Dynasty
Qin Jiushao (c. 1202–1261)

Siku Quanshu (Complete Library of the Four Treasuries) Edition
Bilingual Chinese–English Edition
Traditional Text Structure Preserved

LEFT: Original Chinese　|　RIGHT: English Translation

Traditional structure: Wen (Problem) / Da (Answer) / Shu (Method) / Cao (Working) preserved

Compiled with XeLaTeX — Bilingual Edition

Contents

卷五上　賦役類

按：賦役一章，即古九章中差分粟米諸法。但從來算書設題之數未有如此卷之多者。如首題答數至一百七十五條，每條步算之數至十餘位，而得數皆無不合，亦可謂能廣其用矣。且諸問皆足以明貴賤廩稅及交質變易之同異，使公私各得其平，誠治民之要務也。但題語多晦，術草中間有與所問不相應者，亦於各條下指出，俾觀者無惑焉。

［按］此卷凡九問：一曰復邑修賦，二曰圍田租畝，三曰戶稅移割，四曰均科綿稅，五曰均定勸分，六曰均定合解，七曰寬減屯租，八曰戶稅均寬，九曰米穀粒分。皆賦役之屬也。首問最繁，術用衰分，蓋以三差九等之衰分佈官物，自漢以來均輸之正法也。

Fascicle 5A: Taxation and Corvée (Fuyi)

Editorial: This fascicle on Taxation and Corvée applies the ancient Nine Chapters' methods of difference distribution and grain conversion. Never before has a mathematical text contained problems with data as extensive as this fascicle. For example, the first problem's answer contains 175 items, each with computed values of more than ten digits, yet all results are consistent—a remarkable expansion of mathematical application. These problems illuminate the variations in tax assessment and exchange procedures, ensuring fairness between public and private interests, which is essential to governance. However, the problem statements are often obscure, and some methods do not correspond to the questions asked; such discrepancies are noted below to aid readers.

[Ed.] This fascicle contains nine problems: (1) Restoring the District and Assessing Levies, (2) Polder Field Rent, (3) Household Tax Transfer, (4) Equalizing Silk Tax, (5) Equalizing Voluntary Contributions, (6) Equalizing Combined Remittances, (7) Reducing Garrison Rent, (8) Equalizing Household Tax Relief, (9) Rice and Grain Granule Count. All concern taxation and labor service. The first problem is the most complex, using graded distribution to allocate official commodities across three gradations and nine field grades—the orthodox method of equalized transport (jun shu) since the Han dynasty.

復邑修賦

問：有海坍縣地，今有復漲，歲乆鄉井再成，申諸剏邑。稱土排到六鄉，以附郭為甲，最遠為己，各有田九等，開具下項：

甲鄉共計田十四萬一百九十三畝三角一十二步。上等：上田五千六百七十八畝一角四十八步；中田四千八百九十二畝三十步；下田六千六百二十一畝五十四步。中等：上田八千二百二十五畝二十四步；中田一萬三十五畝六步；下田一萬六千五百三十畝。下等：上田二萬一千九十畝二十四步；中田三萬二千六十畝三步；下田三萬五千六十一畝三角三步。

乙鄉田共計八萬四千一十畝二角二步。上等：上田六千七百八十九畝一角三十六步；中田五千九百八十七畝二角；下田八千一十畝三角三步。中等：上田七千五百四十一畝；中田九千一百二十一畝二角一十二步；下田一萬九千六十步。下等：上田一萬八千三十七畝一角六步；中田九千四百五十六畝三角五十九步；下田無。

丙鄉田共計一十二萬九百三十五畝五十八步五分。上等：上田四千八百六十八畝二角三步；中田五千九百七十九畝三角六步；下田六千八百八十八畝

Restoring the District and Assessing Levies

Question: There is a coastal county whose land had collapsed into the sea, but has now silted back up. Over many years the villages and wells have re-formed, and an application has been made to establish a new district. A land survey has been conducted across six townships: Jia (A) being adjacent to the city walls, and Ji (F) the farthest away. Each township has fields of nine grades. The details are listed below:

Township A: Total fields 140,193 mu 3 jiao 12 bu. Upper grade: upper fields 5,678 mu 1 jiao 48 bu; middle fields 4,892 mu 30 bu; lower fields 6,621 mu 54 bu. Middle grade: upper fields 8,225 mu 24 bu; middle fields 10,035 mu 6 bu; lower fields 16,530 mu. Lower grade: upper fields 21,090 mu 24 bu; middle fields 32,060 mu 3 bu; lower fields 35,061 mu 3 jiao 3 bu.

Township B: Total fields 84,010 mu 2 jiao 2 bu. Upper grade: upper fields 6,789 mu 1 jiao 36 bu; middle fields 5,987 mu 2 jiao; lower fields 8,010 mu 3 jiao 3 bu. Middle grade: upper fields 7,541 mu; middle fields 9,121 mu 2 jiao 12 bu; lower fields 19,060 bu. Lower grade: upper fields 18,037 mu 1 jiao 6 bu; middle fields 9,456 mu 3 jiao 59 bu; lower fields none.

Township C: Total fields 120,935 mu 58 bu 5 fen. Upper grade: upper fields 4,868 mu 2 jiao 3 bu; middle fields 5,979 mu 3 jiao 6 bu; lower fields 6,888 mu 2 jiao 6 bu. Middle grade: upper fields 7,984 mu 1 jiao; middle fields 14,056 mu 12 bu; lower fields 23,333 mu 12 bu. Lower grade: upper fields 27,755 mu 16 bu 5 fen; middle fields none; lower fields 30,070 mu 3 bu.

Township D: Total fields 89,066 mu 2 bu 3 fen. Upper grade: upper fields 11,102 mu 1 bu; middle fields 9,876 mu 1 jiao; lower fields 8,765 mu 1 jiao 30 bu. Middle grade: upper fields 7,539 mu 34 bu 3 fen; middle fields none; lower fields 12,987 mu 42 bu. Lower grade: upper fields none; middle fields 5,432 mu 1 jiao 6 bu; lower fields 33,363 mu 3 jiao 9 bu.

Township E: Total fields 204,474 mu 1 jiao 24 bu 4 fen. Upper grade: upper fields 24,632 mu 39 bu; middle fields 13,521 mu 27 bu; lower fields 9,988 mu 3 jiao 3 bu. Middle grade: upper fields 8,877 mu 56 bu 4 fen; middle fields 11,333 mu 3 jiao; lower fields 27,067 mu. Lower grade: upper fields 19,876 mu 3 jiao 6 bu; middle fields 79,135 mu 3 jiao 43 bu; lower fields 10,041 mu 2 jiao 30 bu.

Township F: Total fields 158,460 mu 3 jiao 18 bu 2 fen. Upper grade: upper fields none; middle fields 7,788 mu 3 jiao 51 bu; lower fields none. Middle grade: upper fields 9,999 mu 2 jiao 3 bu; middle fields 10,836 mu 56 bu; lower fields none. Lower grade: upper fields 32,089 mu 1 jiao 45 bu 6 fen; middle fields 43,678 mu 2 jiao 57 bu; lower fields 54,067 mu 3 jiao 45 bu 6 fen.

It is known that this county previously assessed seedling rice at 103,567 shi 8 dou 4 sheng 4 ge 3 shao, harmonized purchase at 13,498 bolts 1 zhang 7 chi 3 cun 7 fen 6 li, and summer tax at 9,876 bolts 3 zhang 2 chi 6 cun 5 fen 8 li. The six townships' fields are of three colors: Jia is upper, Yi and Bing are middle, and Ding, Wu, and Ji are lower. First, the official commodities are divided into three gradations, with upper to middle and middle to lower each differing by 1 part in 10. Then each township's nine grades also differ by 1 part in 10, and the assessments are distributed according to the total quotas. Township B's fields are the most fertile, followed by Ding, Jia, Bing, Ji, and Wu. Find the assessment rate per mu for each of the three colors and nine grades, and the total assessment for each township.

[Ed.] The problem says the fields are of three colors, with Jia as upper, etc., and also says Township B's fields are the most fertile, etc., which seems contradictory. The former refers to the collective assessment grades of all six townships, while the latter refers to the difference in yield

二角六步。中等：上田七千九百八十四畝一角；中田一萬四千五十六畝一十二步；下田二萬三千三百三十三畝一十二步。下等：上田二萬七千七百五十五畝一十六步五分；中田無；下田三萬七十畝三步。

丁鄉田共計八萬九千六十六畝二步三分。上等：上田一萬一千一百二畝一步；中田九千八百七十六畝一角；下田八千七百六十五畝一角三十步。中等：上田七千五百三十九畝三十四步三分；中田無；下田一萬二千九百八十七畝四十二步。下等：上田無；中田五千四百三十二畝一角六步；下田三萬三千三百六十三畝三角九步。

戊鄉田共計二十萬四千四百七十四畝一角二十四步四分。上等：上田二萬四千六百三十二畝三十九步；中田一萬三千五百二十一畝二十七步；下田九千九百八十八畝三角三步。中等：上田八千八百七十七畝五十六步四分；中田一萬一千三百三十三畝三角；下田二萬七千六十七畝。下等：上田一萬九千八百七十六畝三角六步；中等田七萬九千一百三十五畝三角四十三步；下田一萬四十一畝二角三十步。

己鄉田共計一十五萬八千四百六十畝三角一十八步二分。上等：上田無；中田七千七百八十八畝三角五十一步；下田無。中等：上田九千九百九十九畝二角三步；中田一萬八百三十六畝五十六步；下田無。下等：上田三萬二千八十九畝一角四十五步六分；中田四萬三千六百七十八畝二角五十七步；下田五萬四千六十七畝三角四十五步六分。

照得昨來本縣原科苗米一十萬三千五百六十七石八斗四升四合三勺，和買一萬三千四百九十八匹一丈七尺三寸七分六釐，夏税九千八百七十六匹三丈二尺六寸五分八釐。其六鄉田係三色，甲為上，乙丙為次，丁戊己又為次。先令官物為三差，使上比中、中比下皆十分外差一，次令各鄉九等皆於十分內差一，拋科用合租額。其乙鄉田最肥，次丁，次甲，次丙，次己，次戊。欲知三色九等每畝等則及共科數各鄉幾何？

［按］題內言田有三色，甲為上云云，又言乙鄉田最肥云云，語若相左。葢前言三色者六鄉共科之等，後言田肥者每畝所出之異也。若易田係三色為科有三差，則語意明矣。

答曰：甲鄉：上等上田苗三斗二升二合四勺，和一尺六寸九分，税一尺二寸三分；中田苗二斗九升九勺，和一尺五寸二分，税一尺一寸一分；下田苗二斗五升八合六勺，和一尺三寸五分，税九寸九分。中等上田苗二斗二升六合二勺，和一尺一寸八分，税八寸六分；中田苗一斗九升三合九勺，和一尺一分，税七寸四分；下田苗一斗六升一合六勺，和八寸四分，税六寸二分。下等上田苗一斗二升九合三勺，和六寸七分，税四寸九分；中田苗九升六合九勺，和五寸一分，税三寸七分；下田苗六升四合六勺，和三寸四分，税二寸五分。

乙鄉：上等上田苗三斗六升三合三勺，和一尺八寸九分，税一尺三寸九分；中田苗三斗二升六合九勺，和一尺七寸，税一尺二寸五分；下田苗二斗九升六勺，和一尺五寸二分，税一尺一寸一分。中等上田苗二斗五升四合三勺，和一尺三寸三分，税九寸七分；中田苗二斗一升八合，和一尺一寸四分，税八寸三分；下田苗一斗八升一合六勺，和九寸五分，税六寸九分。下等上田苗一斗四升五合三勺，和七寸六分，税五寸五分；中田苗一斗九合，和五寸七分，税四寸二分；下田苗七升二合七勺，和三寸八分，税二寸八分。

丙鄉：上等上田苗三斗三合五勺，和一尺五寸八分，税一尺一寸六分；中田苗二斗七升三合一勺，和一尺四寸二分，税一尺四分；下田苗二斗四升二合八勺，和一尺二寸七分，税九寸三分。中等上田苗二斗一升二合四勺，和一尺一寸一分，税八寸一分；中田苗一斗八升二合一勺，和九寸五分，税六寸九分；下田苗一斗五升一合七勺，和七寸九分，税五寸八分。下等上田苗一斗二升一合四勺，和六寸三分，税四寸六分；中田苗九升一合，和四寸七分，税三寸五分；下田苗六升七勺，和三寸二分，税二寸三分。

丁鄉：上等上田苗三斗四升三合二勺，和一尺七寸九分，税一尺三寸一分；中田苗三斗八合九勺，和一尺六寸一分，税一尺一寸八分；下田苗二斗七升四合六勺，和一尺四寸三分，税一尺五分。中等上田苗二斗四升二勺，和一尺二寸五分，税九寸二分；中田苗二斗五合九勺，和一尺七分，税七寸九分；下田苗一斗七升一合六勺，和八寸九分，税六寸五分。下等上田苗一斗三升七合三勺，和七寸二分，税五寸二分；中田苗一斗三合，和五寸四分，税三寸九分；下田苗六升八合六勺，和三寸六分，税二寸六分。

戊鄉：上等上田苗一斗五升三合八勺，和八寸，税五寸九分；中田苗一斗三升八合四勺，和七寸二分，税五寸三分；下田苗一斗二升三合一勺，和六寸四分，税四寸七分。中等上田苗一斗七合七勺，和五寸六分，税四寸一分；中田苗九升二合三勺，和四寸八分，税三寸五分；下田苗七升六合九勺，和四寸，税二寸九分。下等上田苗六升一合五勺，和三寸二

per mu. If “fields of three colors” is changed to “assessments of three gradations,” the meaning becomes clear.

Answer: **Township A:** Upper-upper fields: seedling rice 3 dou 2 sheng 2 ge 4 shao, harmonized purchase 1 chi 6 cun 9 fen, summer tax 1 chi 2 cun 3 fen; upper-middle fields: seedling rice 2 dou 9 sheng 9 shao, harmonized 1 chi 5 cun 2 fen, tax 1 chi 1 cun 1 fen; upper-lower fields: seedling rice 2 dou 5 sheng 8 ge 6 shao, harmonized 1 chi 3 cun 5 fen, tax 9 cun 9 fen.

Middle-upper fields: seedling rice 2 dou 2 sheng 6 ge 2 shao, harmonized 1 chi 1 cun 8 fen, tax 8 cun 6 fen; middle-middle fields: seedling rice 1 dou 9 sheng 3 ge 9 shao, harmonized 1 chi 1 fen, tax 7 cun 4 fen; middle-lower fields: seedling rice 1 dou 6 sheng 1 ge 6 shao, harmonized 8 cun 4 fen, tax 6 cun 2 fen.

Lower-upper fields: seedling rice 1 dou 2 sheng 9 ge 3 shao, harmonized 6 cun 7 fen, tax 4 cun 9 fen; lower-middle fields: seedling rice 9 sheng 6 ge 9 shao, harmonized 5 cun 1 fen, tax 3 cun 7 fen; lower-lower fields: seedling rice 6 sheng 4 ge 6 shao, harmonized 3 cun 4 fen, tax 2 cun 5 fen.

[Similar detailed rates for Townships B–F follow the same structure of nine grades each.]

Total assessments per township: Township A: seedling rice 19,550 shi 2 dou 4 sheng 8 ge 3 shao; harmonized purchase 10,192 zhang 2 chi 6 cun 5 fen 6 li; summer tax 7,457 zhang 6 chi 8 cun 9 fen 8 li. Township B: seedling rice 17,772 shi 9 dou 5 sheng 3 ge; harmonized purchase 9,265 zhang 6 chi 9 cun 6 fen; summer tax 6,779 zhang 7 chi 1 cun 8 fen. Township C: same as B. Township D: seedling rice 16,157 shi 2 dou 3 sheng; harmonized purchase 8,423 zhang 3 chi 6 cun; summer tax 6,163 zhang 3 chi 8 cun. Townships E and F: same as D.

分，稅二寸三分；中田苗四升六合一勺，和二寸四分，稅一寸八分；下田苗三升八勺，和一寸六分，稅一寸二分。

己鄉：上等上田苗二斗八升二合二勺，和一尺四寸七分，稅一尺八分；中田苗二斗五升三合九勺，和一尺三寸二分，稅九寸七分；下田苗二斗二升五合七勺，和一尺一寸八分，稅八寸六分。中等上田苗一斗九升七合五勺，和一尺三分，稅七寸五分；中田苗一斗六升九合三勺，和八寸八分，稅六寸五分；下田苗一斗四升一合一勺，和七寸四分，稅五寸四分。下等上田苗一斗一升二合九勺，和五寸九分，稅四寸三分；中田苗八升四合六勺，和四寸四分，稅三寸二分；下田苗五升六合四勺，和二寸九分，稅二寸二分。

各鄉共科數：甲鄉苗米一萬九千五百五十石二斗四升八合三勺；和買一萬一百九十二丈二尺六寸五分六釐；夏稅七千四百五十七丈六尺八寸九分八釐。乙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丁鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。戊鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。己鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。

Method: Solve by the method of graded distribution (cui fen). First list the county's color grades, arranged in a descending cone (zhui) pattern from the bottom; then multiply by the corresponding township counts. Add these together to form the divisor (fa). Divide the official commodity quantities by this to obtain the rate for one portion. Multiply this rate by the uncombined numbers to obtain each township's amount. Next, list each township's nine grades, beginning with 10 fen for the upper grade, and successively applying a 9/10 reduction for each lower grade down to the ninth grade. Multiply each by the mu and bu, add together to form the township divisor, and divide each township's official commodity amount by this to obtain the rate for one grade. Multiply by the uncombined numbers to obtain the tax rate per mu for each grade.

術曰：以衰分求之。先列本縣色位，自下錐行列之，又以鄉數對列而乘之，副併為法，以除官物數，得一分之率。以率數乘未併者，各得諸鄉之數。次列各鄉等位，自上等置十分，每以內分錐行九折之，至九等止，又各以畝步乘之，副併為鄉法，以除諸各鄉所得官物數，所得為一分之率，以乘未併者，各得每畝稅色。

Working: Set up the three color grades for the county. From the bottom grade taking 11/10 of the standard, with upper to middle adding 1/10, we obtain upper = 121, middle = 110, lower = 100 as the three rates, placed in the right column. Then place Jia = 1 paired with the upper rate, Yi and Bing together = 2 paired with the middle rate, and Ding, Wu, and Ji together = 3 paired with the

草曰：列本縣色位三，自下色例十分中十一分，上比中身外加一，上得一百二十一，中得一百一十，下得一百，為上中下三率，列之右行。乃列甲一對上率，乙丙共二對中率，丁戊己共三對下率，列左行。乃以左行列數各相對乘右行率數，其上得一百二十一，其中得二百二十，其下得三百，乃副置而併之，得六百四十一為法。置元科苗米一萬三千五百六十七石八

斗四升四合三勺為寔，以法除之，得一百六十一石五斗七升二合三勺為一分之率。以未併下率一百乘率，得一萬六千一百五十七石二斗三升為丁戊己三鄉各科數。以於身下加一，得一萬七千七百七十二石九斗五升三合為乙丙二鄉各科數。又於身下加一，得一萬九千五百五十石二斗四升八合三勺為甲合科數。求和買亦用六百四十一為法，置元科和買一萬三千四百九十八匹，以匹法四丈通之，得五萬三千九百九十二丈，內零一丈七尺三寸七分六釐，得五萬三千九百九十三丈七尺三寸七分六釐為寔，以法除之，得八十四丈二尺三寸三分六釐為一分之率。亦以未併下率一百乘之，得八千四百二十三丈三尺六寸為丁戊己三鄉各科數。次於身下加一，得九千二百六十五丈六尺九寸六分為乙丙二鄉各科數。又於身下加一，得一萬一百九十二丈二尺六寸五分六釐為甲鄉合科數。求夏稅亦置六百四十一為法，置元科九千八百七十六匹，以匹法四丈通之，得三萬九千五百四丈，內零三丈二尺六寸五分八釐，得三萬九千五百七丈二尺六寸五分八釐為寔，以法除之，得六十一丈六尺三寸三分八釐為一分率。亦以未併下率一百乘之，得六千一百六十三丈三尺八寸為丁戊己三鄉各科數。次於身下加一，得六千七百七十九丈七尺一寸八分为乙丙二鄉各科數。又於身下加一，得七千四百五十七丈六尺八寸九分八釐為甲鄉數。

lower rate, in the left column. Multiply the left column numbers by their corresponding right column rates: upper yields 121, middle yields 220, lower yields 300. Add these together to obtain 641 as the divisor. Take the original seedling rice assessment of 103,567 shi 8 dou 4 sheng 4 ge 3 shao as the dividend (shi). Divide by the divisor to obtain 161 shi 5 dou 7 sheng 2 ge 3 shao as the rate for one portion. Multiply this by the uncombined lower rate 100 to obtain 16,157 shi 2 dou 3 sheng as the combined assessment for townships D, E, and F. Increase this by 1/10 to obtain 17,772 shi 9 dou 5 sheng 3 ge as the combined assessment for townships B and C. Increase again by 1/10 to obtain 19,550 shi 2 dou 4 sheng 8 ge 3 shao as the assessment for township A.

For the harmonized purchase, also use 641 as divisor. Take the original harmonized purchase of 13,498 bolts, convert using the bolt standard of 4 zhang to obtain 53,992 zhang, plus the remainder of 1 zhang 7 chi 3 cun 7 fen 6 li, giving 53,993 zhang 7 chi 3 cun 7 fen 6 li as dividend. Divide by the divisor to obtain 84 zhang 2 chi 3 cun 3 fen 6 li as the rate for one portion. Multiply by the uncombined lower rate 100 to obtain 8,423 zhang 3 chi 6 cun for townships D, E, F. Increase by 1/10 to obtain 9,265 zhang 6 chi 9 cun 6 fen for B and C, and again to obtain 10,192 zhang 2 chi 6 cun 5 fen 6 li for A.

For the summer tax, again use 641 as divisor. Take the original 9,876 bolts, convert at 4 zhang per bolt to obtain 39,504 zhang, plus the remainder 3 zhang 2 chi 6 cun 5 fen 8 li, giving 39,507 zhang 2 chi 6 cun 5 fen 8 li as dividend. Divide to obtain 61 zhang 6 chi 3 cun 3 fen 8 li as the rate. Multiply by 100 to obtain 6,163 zhang 3 chi 8 cun for D, E, F. Increase by 1/10 to obtain 6,779 zhang 7 chi 1 cun 8 fen for B and C, and again to obtain 7,457 zhang 6 chi 8 cun 9 fen 8 li for A.

圍田租畝

Polder Field Rent Assessment

問：有興復圍田已成，共計三千二十一頃五十一畝一十五步，分三等。其上等每畆起租六斗，中等四斗五升，下等四斗。中田多上田弱半，不及下田太半。欲知三色田畝及各租幾何?

Question: A reclaimed polder field has been completed, totaling 3,021 qing 51 mu 15 bu, divided into three grades. Upper-grade fields pay rent of 6 dou per mu, middle-grade 4 dou 5 sheng, and lower-grade 4 dou. The middle fields exceed the upper fields by one weak half (1/3), and fall short of the lower fields by one great half (2/3). Find the mu of each of the three grades of fields and the rent for each.

［按］題言中田多上田弱半，不及下田太半。蓋以弱半為三分之一，太半為三分之二也。多上田弱半者，即比上田多三分之一也。不及下田太半者，即比下田少三分之二也。是上田加三分之一為中田，即下田三分之一也。轉言之，則中田為下田三分之一，上田為中田四分之三也。

[Ed.] The problem says the middle fields exceed the upper fields by a weak half and fall short of the lower fields by a great half. Here, weak half means 1/3 and great half means 2/3. Exceeding the upper fields by a weak half means exceeding by 1/3. Falling short of the lower fields by a great half means falling short by 2/3. Thus upper

fields plus 1/3 equals middle fields, which is 1/3 of the lower fields. In other words, middle fields are 1/3 of lower fields, and upper fields are 3/4 of middle fields.

Answer: Upper fields: 477 qing 8 mu 15 bu, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Middle fields: 636 qing 10 mu 3 jiao, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Lower fields: 1,908 qing 32 mu 1 jiao, rice 76,332 shi 9 dou.

Method: Solve by the method of graded distribution (cui fen). Set up the numerator and denominator to find the field rates. Add them together to form the divisor. Take the total fields as dividend. Divide to obtain the rate for one portion. Multiply this throughout by the uncombined numbers to obtain the three grades of fields. Multiply each by the base rent to obtain the rice amounts.

Working: Set the weak-half denominator 4 as the middle rate and numerator 3 as the upper rate. Subtract the great-half numerator 2 from denominator 3, remainder 1. Multiply by the middle rate 4, obtaining 4 as the middle general (fan). Also multiply the remainder 1 by the upper rate 3, obtaining 3 as the upper general. Next multiply the middle general 4 by the great-half denominator 3, obtaining 12 as the lower general. Add the three generals to obtain 19 as the divisor. Take the fields 3,021 qing 51 mu, convert using the mu standard of 240 bu to obtain 72,516,240 bu, plus the numerator 15 bu, giving 72,516,255 bu as dividend. Divide by the divisor 19 to obtain 3,816,645 bu as the amount for one portion. Multiply by the upper general 3 to obtain 11,449,935 bu as the upper product. Multiply the one-portion amount by the middle general 4 to obtain 15,266,580 bu as the middle product. Multiply the one-portion amount by the lower general 12 to obtain 45,799,740 bu as the lower product. Convert each product to mu using 240 bu per mu: upper fields = 477 qing 8 mu 15 bu; middle fields = 636 qing 10 mu 3 jiao; lower fields = 1,908 qing 32 mu 1 jiao. For the rents: multiply the upper product 11,449,935 bu by upper rent 6 dou, obtaining 6,869,961 shi as dividend, divided by 240 bu to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao as upper-field rent. Multiply the middle product by middle rent 4 dou 5 sheng, obtaining 6,869,961 shi, divided by 240 to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao. Multiply the lower product by lower rent 4 dou, obtaining 18,319,896 shi, divided by 240 to obtain 76,332 shi 9 dou as lower-field rice.

[Ed.] In the working, the intention in finding each graded rate is: taking 3 fen as the upper-field rate, adding the weak half (1/3) to obtain 4 fen as the middle-field rate, and tripling the middle-field rate to obtain 12 fen as the lower-field rate, which sum to 19 fen as the total graded rate. The method is essentially clear, but the terminology

答曰：上田四百七十七頃八畝一十五步，米二萬八千六百二十四石八斗三升七合五勺；中田六百三十六頃一十畝三角，米二萬八千六百二十四石八斗三升七合五勺；下田一千九百八頃三十二畝一角，米七萬六千三百三十二石九斗。

術曰：以衰分求之。列母子求田率，副併為法，以共田為寔，寔如法而一，得一分之率。以徧乘未併者，得三等田。各以起租乘之，各得米。

草曰：置弱半母四為中率，子三為上率。以太半子二减母三，餘一，以乘中率四，只得四為中泛。又以餘一乘上率三，只得三為上泛。次以太半母三乘中泛四，得一十二為下泛。副併三泛，得一十九為法。置田三千二十一頃五十一畒，以畝法二百四十通之，得七千二百五十一萬六千二百四十步，内子一十五步，得七千二百五十一萬六千二百五十五步為寔。以法一十九除之，得三百八十一萬六千六百四十五步為一分之數。以上泛三因之，得一千一百四十四萬九千九百三十五步為上積。又以中泛四因之一分數，得一千五百二十六萬六千五百八十步為中積。又以下泛一十二乘一分數，得四千五百七十九萬九千七百四十步為下積。其三積各以畝法二百四十約之為畒。其上田得四百七十七頃八畝一十五步，其中田得六百三十六頃一十畝三角，其下積得一千九百八頃三十二畝一角。各以起租，三積為三寔。其上積一千一百四十四萬九千九百三十五步乘上積六斗，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺為上田租。其中積一千五百二十六萬六千五百八十步乘中田租四斗五升，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺。其下積四千五百七十九萬九千七百四十步乘下租四斗，得一千八百三十一萬九千八百九十六石為寔，以畝法二百四十除之，得七萬六千三百三十二石九斗為下田米。

［按］草内求各衰數，意謂以三分為上田衰數，加弱半三分之一得四分為中田衰數，三因中田衰數得一十二分為下田衰數，併之得一十九分為總衰數。其法本屬顯明，但語中參入母子副泛等名目，其意反晦矣。

of numerators, denominators, and generals makes it more obscure.

營田造租

Garrison Field Rent Assessment

問：有營田五十頃六十二畝五分，係官出租。三等出租：上田每畝五斗，中田四斗二升，下田三斗五升。其田上中下相半，上田不及中田頃畝六分半，下田比中田多三分半。欲知三色田頃畝及各租幾何？

Question: There are 50 qing 62 mu 5 fen of garrison fields rented by the government, divided into three grades: upper fields at 5 dou per mu, middle fields at 4 dou 2 sheng, lower fields at 3 dou 5 sheng. The fields are divided equally among the three grades. The upper fields fall short of the middle fields by 6.5 fen in qing-mu measure, and the lower fields exceed the middle fields by 3.5 fen. Find the qing and mu of each of the three grades of fields and their respective rents.

［按］此題用衰分法，以上中下衰率求之。題言上田不及中田六分半，下田比中田多三分半，皆以中田為率而損益之也。

[Ed.] This problem uses the method of graded distribution, seeking the rates of upper, middle, and lower fields. The problem states that upper fields fall short of middle fields by 6.5 fen, and lower fields exceed middle fields by 3.5 fen—both taking the middle fields as the base and applying decrease or increase.

答曰：上田一十五頃四十畝，租七千七百石；中田一十七頃四十畝二分，租七千三百石二斗八升；下田一十七頃八十二畝三分，租六千二百三十八石八升。

Answer: Upper fields: 15 qing 40 mu, rent 7,700 shi; Middle fields: 17 qing 40 mu 2 fen, rent 7,300 shi 2 dou 8 sheng; Lower fields: 17 qing 82 mu 3 fen, rent 6,238 shi 8 sheng.

術曰：以衰分求之。置中田為率，以六分半減之為上田衰，以三分半加之為下田衰。各徧乘共畝為實，副併衰數為法。實如法而一，各得每衰之畝。以各租乘之，得各租米。

Method: Solve by the method of graded distribution. Take the middle fields as the base rate. Subtract 6.5 fen to obtain the upper-field rate, and add 3.5 fen to obtain the lower-field rate. Multiply the total mu by each rate to form the dividends. Add the rates together to form the divisor. Divide each dividend by the divisor to obtain the mu per rate. Multiply each by the respective rent to obtain the rice rents.

草曰：置中田一十畝為率，以六分半減之，得九十三分半為上田衰。以三分半加中田率，得一十三分半為下田衰。并三衰，得三十七分为法。置營田五十頃六十二畝五分，通為五千六十二畝五分，以各衰遍乘之：以上田衰九十三分半乘，得四十七萬三千四百三十七畝半為上實；以中田衰一十畝乘，得五萬六百二十五畝為中實；以下田衰一十三分半乘，得六萬八千三百四十三畝七分五厘為下實。三實皆如法三十七而一：上田得一萬五千四百畝，展為一十五頃四十畝；中田得一萬七千四百畝二分，展為一十七頃四十畝二分；下田得一萬七千八百二十三畝三分，展為一十七頃八十二畝三分。各以起租乘之：上田一十五頃四十畝乘五斗，得七千七百石；中田一十七頃四十畝二分乘四斗二升，得七千三百石二斗八升；下田一十七頃八十二畝三分乘三斗五升，得六千二百三十八石八升。

Working: Set 10 mu of middle fields as the base. Subtract 6.5 fen to obtain 93.5 fen as the upper-field rate. Add 3.5 fen to the middle rate to obtain 13.5 fen as the lower-field rate. Add the three rates to obtain 37 as the divisor. Take the garrison fields of 50 qing 62 mu 5 fen, convert to 5,062.5 mu. Multiply by each rate: upper-field rate 93.5 gives 473,437.5 mu as upper dividend; middle-field rate 10 gives 50,625 mu as middle dividend; lower-field rate 13.5 gives 68,343.75 mu as lower dividend. Divide each of the three dividends by the divisor 37: upper fields obtain 15,400 mu = 15 qing 40 mu; middle fields obtain 17,400 mu 2 fen = 17 qing 40 mu 2 fen; lower fields obtain 17,823 mu 3 fen = 17 qing 82 mu 3 fen. Multiply each by the base rent: upper fields 15 qing 40 mu at 5 dou = 7,700 shi; middle fields 17 qing 40 mu 2 fen at 4 dou 2 sheng = 7,300 shi 2 dou 8 sheng; lower fields 17 qing 82 mu 3 fen at 3 dou 5 sheng = 6,238 shi 8 sheng.

Fascicle 5B: Taxation and Corvée (continued)

Household Tax Transfer

Question: A certain county received a petition from household A stating: This household's fields originally paid seedling rice of 35 shi 7 dou; harmonized purchase in kind of 11 bolts 2 zhang 2 chi 9 cun 4 fen 8 li 7 hao 5 si; converted silk of 27 bolts 2 cun 1 fen 3 li 7 hao 5 si; coarse silk converted silk of 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li; coarse silk in kind of 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li. The household has already transferred 407 mu of fields to B and 516 mu to C. The petition requests that the taxes on the transferred fields be moved and consolidated with households B and C for payment. It was found that B originally had 375 mu of fields and C originally had 463 mu, all in the same township and same grade. Per mu: seedling rice 3 sheng 5 ge, tax silk 1 chi 1 cun 5 fen, property value 1 guan 200 wen; for this grade of fields, coarse silk is 1 chi 3 cun 4 fen, property value 900. For every 32 guan of property value, 1 bolt of harmonized purchase is assessed: 3/10 in kind, 7/10 converted. Summer tax: 7/10 in kind, 3/10 converted. Coarse silk: 5/10 in kind, 5/10 converted, combined with silk. Find household A's remaining fields, and the respective amounts of seedling rice, harmonized purchase, summer tax, converted silk, in-kind silk, and remainders that each of A, B, and C should pay after the division.

[Ed.] This problem's tax assessment divides into fields (tian) and land (di) categories. Fields have three components: rice assessment, silk tax, and property value; land has only two: coarse-silk tax and property value. The conversion rates for silk and coarse silk differ, and the harmonized purchase based on property value combines both fields and land. Also, household A has both fields and land, while households B and C have fields only. These distinctions should have been clearly stated in the problem but were not.

Answer: Household A originally had 1,020 mu of fields and 11 mu 2 jiao 48 bu of land.

Household A now has 97 mu of fields: seedling rice 3 shi 3 dou 9 sheng 5 ge; summer tax converted silk 3 zhang 3 chi 4 cun 6 fen 5 li; in-kind silk 1 bolt remainder 3 zhang 8 chi 8 fen 5 li; property value 116 guan 400 wen. Land of 11 mu 2 jiao 48 bu: coarse-silk converted tax 7 chi 8 cun 3 fen 9 li; coarse-silk remainder 7 chi 8 cun 3 fen 9 li; land property value 10 guan 530 wen; combined fields and land property value 126 guan 930 wen. Harmonized purchase: converted silk 2 bolts 3 zhang 1 chi 6 fen 3 li 7 hao 5 si; in-kind 1 bolt remainder 7 chi 5 cun 9 fen 8 li 7

卷五下　賦役類（續）

戶稅移割

問：某縣據甲稱：本户田地原納苗三十五石七斗，和買本色一十一匹二丈二尺九寸四分八釐七毫五絲，折帛二十七匹二寸一分三釐七毫五絲，紬絹折帛八匹三丈九尺七寸三分九釐，紬絹本色二十匹二丈八尺九寸三分九釐。已将田四百七畝出與乙，五百十六畝出與丙。訖乞移割本户所出田上稅賦，歸併乙丙兩户送納。會到乙原有田三百七十五畝，丙原有田四百六十三畝，並係本鄉本等。每畝苗三升五合，稅一尺一寸五分，物力一貫二百；本等田紬一尺三寸四分，物力九百；物力及三十二貫數和買一匹，内三分本色七分折帛；夏稅七分本色三分折帛；紬五分本色五分折帛，併絹紬。欲知甲田地及甲乙丙分割合納苗米、和買、夏稅、折帛、本色、畸零各幾何?

［按］此題科稅分田地二項。田有科米、稅絹、物力三色，地只有稅紬、物力二色，絹與紬折色不同，物力和買合田地總計之。又甲户有田者有地，乙丙二户有田無地。此等皆不可不明言者，題中殊未分晰。

答曰：甲原有田一千二十畝，地一十一畝二角四十八步。

甲今有田九十七畝，苗米三石三斗九升五合，夏稅折帛三丈三尺四寸六分五釐，本色一匹畸零三丈八尺八分五釐，物力一百一十六貫四百文。地一十一畝二角四十八步，稅紬折帛七尺八寸三分九釐，稅紬畸零七尺八寸三分九釐，物力一十貫五百三十文，田地共物力一百二十六貫九百三十文。和買折帛二匹三丈一尺六分三釐七毫五絲，本色一疋畸零七尺五寸九分八釐七毫五絲。

乙今有田七百八十二畝，苗米二十七石三斗七升，夏稅折帛六匹二丈九尺七寸九分，本色一十五匹畸

零二丈九尺五寸一分，物力九百三十八貫四百文，和買折帛二十匹二丈一尺一寸，本色八匹畸零三丈一尺九寸。

丙今有田九百七十九畝，苗米三十四石二斗六升五合，夏税折帛八匹一丈七尺七寸五分五釐，本色一十九匹畸零二丈八尺九分五釐，物力一千一百七十四貫八百文，和買折帛二十五匹二丈七尺九寸五分，本色一十一匹畸零五寸五分。

術曰：以粟米及衰分求之。置甲原納米為實，以每畝苗為法除之，得甲原有田。以乘每畝税，得田絹。次以甲紬絹本色折帛併之，內減田絹，餘為地紬。以每紬約之，得甲原有地。次以甲出田併乙丙原有田，各得三户今有田地。各以等則乘之，各得物力苗税。次以每畝物力率約各户共物力，得和買。副之，三分因之退位為和本，七分因之退位為和折；夏税反其分而因之退之；紬以半之，併入税，各得。

草曰：置甲原納苗三十五石七斗為實，以每畝苗三升五合為法除之，得甲田一千二十畝。以每畝税一尺一寸五分乘之，得一百一十七丈三尺為田絹。以甲原紬絹本色二十匹二丈八尺九寸三分九釐、折帛八匹三丈九尺七寸三分九釐併之，共二十八匹三丈七尺六寸七分八釐。內減田絹一百一十七丈三尺，餘為地紬。以每地紬一尺三寸四分約之，得甲地一十一畝二角四十八步。以甲出田四百七畝併乙田三百七十五畝，得七百八十二畝為乙户今有田。以甲出田五百十六畝併丙田四百六十三畝，得九百七十九畝為丙户今有田。甲今有田一千二十畝減出田九百二十三畝，餘九十七畝為甲今有田。各以每畝物力率乘之：乙田七百八十二畝乘一貫二百，得九百三十八貫四百文；丙田九百七十九畝乘一貫二百，得一千一百七十四貫八百文；甲田九十七畝乘一貫二百，得一百一十六貫四百文。地一十一畝二角四十八步乘九百，得十貫五百三十文。共甲物力一百二十六貫九百三十文。各以物力三十二貫約之：乙得二十四匹餘十六貫四百文，三分本色得七匹餘八分，七分折帛得一十六匹二丈一尺一寸；丙得三十六匹餘二十六貫八百文，

hao 5 si.

Household B now has 782 mu of fields: seedling rice 27 shi 3 dou 7 sheng; summer tax converted silk 6 bolts 2 zhang 9 chi 7 cun 9 fen; in-kind 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen; property value 938 guan 400 wen. Harmonized purchase: converted silk 20 bolts 2 zhang 1 chi 1 cun; in-kind 8 bolts remainder 3 zhang 1 chi 9 cun.

Household C now has 979 mu of fields: seedling rice 34 shi 2 dou 6 sheng 5 ge; summer tax converted silk 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li; in-kind 19 bolts remainder 2 zhang 8 chi 9 fen 5 li; property value 1,174 guan 800 wen. Harmonized purchase: converted silk 25 bolts 2 zhang 7 chi 9 cun 5 fen; in-kind 11 bolts remainder 5 cun 5 fen.

Method: Solve by the methods of grain conversion (su mi) and graded distribution (cui fen). Take household A's original rice payment as dividend and divide by the per-mu seedling rate to obtain A's original fields. Multiply by the per-mu tax to obtain the field silk. Then combine A's coarse silk and silk in-kind and converted, subtract the field silk, and the remainder is the land coarse silk. Divide by the per-unit coarse silk to obtain A's original land. Next, combine the transferred fields with B and C's original fields to obtain each household's current fields. Multiply each by the grade rates to obtain property values and seedling taxes. Then divide each household's total property value by the per-mu property rate to obtain the harmonized purchase. Split it: 3/10 as in-kind harmonized purchase, 7/10 as converted; for summer tax reverse the ratios; for coarse silk halve it and combine with silk tax to obtain the results.

Working: Take household A's original seedling rice of 35 shi 7 dou as dividend; divide by the per-mu seedling rate of 3 sheng 5 ge to obtain A's fields of 1,020 mu. Multiply by per-mu tax of 1 chi 1 cun 5 fen to obtain 117 zhang 3 chi of field silk. Combine A's original coarse silk and silk: in-kind 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li and converted 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li, totaling 28 bolts 3 zhang 7 chi 6 cun 7 fen 8 li. Subtract field silk 117 zhang 3 chi; the remainder is land coarse silk. Divide by per-unit coarse silk of 1 chi 3 cun 4 fen to obtain A's land of 11 mu 2 jiao 48 bu. Combine A's transferred fields 407 mu with B's 375 mu to obtain 782 mu as B's current fields. Combine A's transferred 516 mu with C's 463 mu to obtain 979 mu as C's current fields. A's original 1,020 mu minus transferred 923 mu leaves 97 mu as A's current fields. Multiply each by the per-mu property rate: B's 782 mu at 1 guan 200 = 938 guan 400 wen; C's 979 mu at 1 guan 200 = 1,174 guan 800 wen; A's 97 mu at 1 guan 200 = 116 guan 400 wen. Land 11 mu 2 jiao 48 bu at 900 = 10 guan 530 wen. Total A property value = 126 guan 930 wen. Divide each property value by 32 guan: B obtains 24 bolts remainder 16 guan 400 wen, 3/10 in-kind = 7 bolts remainder 8 fen, 7/10 converted = 16 bolts 2 zhang 1 chi 1 cun. C obtains 36 bolts remainder 26 guan 800 wen, 3/10 in-kind = 11 bolts remainder 8 fen, 7/10 converted = 25 bolts 2 zhang 7 chi 9 cun 5 fen. A obtains 3 bolts remainder 30 guan 930 wen, 3/10 in-kind = 1 bolt remainder chi, 7/10 converted = 2 bolts 3 zhang 1 chi 6 fen 3 li. Summer tax uses 7:3 ratio on property value: B 7/10 in-kind = 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen, 3/10 converted = 6 bolts 2 zhang 9 chi 7 cun 9 fen. C 7/10 in-kind = 19 bolts remainder 2 zhang 8 chi 9 fen 5 li, 3/10 converted = 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li. A 7/10 in-kind = 1 bolt remainder 3 zhang 8 chi 8 fen 5 li, 3/10 converted = 3 zhang 3 chi 4 cun 6 fen 5 li.

三分本色得一十一匹餘八分，七分折帛二十五匹二丈七尺九寸五分；甲得三匹餘三十貫九百三十文，三分本色得一匹餘尺，七分折帛二匹三丈一尺六分三釐。夏稅以物力七三因之：乙七分本色一十五匹餘二丈九尺五寸一分，三分折帛六匹二丈九尺七寸九分；丙七分本色一十九匹餘二丈八尺九分五釐，三分折帛八匹一丈七尺七寸五分五釐；甲七分本色一匹餘三丈八尺八分五釐，三分折帛三丈三尺四寸六分五釐。

Equalizing Silk Tax Assessment

均科綿税

Question: A county assesses silk floss across five grades of households, totaling 11,033 households, with total silk assessment of 88,337 liang 6 qian. Upper grade: 12 households; second grade: 87 households; middle grade: 464 households; fourth grade: 2,035 households; lower grade: 8,435 households. It is desired that the upper three grades differ by halving (upper:second:middle = 4:2:1), and the lower two grades differ from the middle grade by a 6:4 ratio (fourth:lower = 6:4 of middle). Using the assessment rate, find the amount per household and the total for each grade.

問：縣科綿有五等户，共一萬一千三十三户，共科綿八萬八千三百三十七兩六錢。上等一十二户，副等八十七户，中等四百六十四户，次等二千三十五户，下等八千四百三十五户。欲令上三等折半差，下二等此中等六四折差。科率求之，問各户納及各等幾何？

[Ed.] The problem states that the lower two grades differ from the middle grade by a 6:4 ratio, meaning the fourth grade is 4/6 of the middle grade, and the lower grade is 4/6 of the fourth grade. However, in the method and working, both are reduced by 4/10, which is a 4-tenths reduction rather than a 6:4 ratio. The collection contains many such cases where the problem and method do not correspond—could it be that the author was unable to thoroughly revise the work after its completion?

［按］題言下二等比中等六四折差，是次等為中等六分之四，下等又為次等六分之四也。乃術草中皆以十分之四收之，是四分折差而非四六折差矣。集中題與術不相應者多類此，豈成書之後未能重加校正歟？

Answer: Upper grade: 124 liang per household, 12 households totaling 1,488 liang. Second grade: 62 liang per household, 87 households totaling 5,394 liang. Middle grade: 31 liang per household, 464 households totaling 14,384 liang. Fourth grade: 12 liang 4 qian per household, 2,035 households totaling 25,234 liang. Lower grade: 4 liang 9 fen 6 li per household, 8,435 households totaling 41,837 liang 6 qian.

答曰：上等一户一百二十四兩，一十二户計一千四百八十八兩；副等一户六十二兩，八十七户計五千三百九十四兩；中等一户三十一兩，四百六十四户計一萬四千三百八十四兩；次等一户一十二兩四錢，二千 035 户計二萬五千二百三十四兩；下等一户四兩九錢六分，八千四百三十五户計四萬一千八百三十七兩六錢。

Method: List the five grades of households. First apply a 4/10 reduction to the lower-grade count, add to the fourth-grade households, apply a 4/10 reduction again, add to the middle-grade count, then apply a 1/2 reduction, add to the second-grade count, apply a 1/2 reduction again, and add

術曰：列五等户數，先以四折下等數，加次等户，又以四折之，加中等户數，却以半折之，加二等户，又以半折之，加上等户數，不折便為法。除科綿，得上等一户之綿。復半之為副等，又半之為中等，又四折之為次等，又四折之為下等。各一户綿，却各以户數

乘之，各得五等共出綿。

草曰：先置下等户八千四百三十五，以四折之，得三千三百七十四。乃以得數折入次户二千三十五内，得五千四百九户訖。又以四折次數五千四百九户，得二千一百六十三户六分。乃以得次數併入中户四百六十四内，共得二千六百二十七户六分。次以五分折中數二千六百二十七户六分，得數。次以得數一千三百一十三户八分併副户八十七，得一千四百户八分。又以五折副數一千四百户八分，得七百户四分，既得數。乃以副數七百户四分併上户一十二户，得七百一十二户四分，不折便為法。

累折至等，共得七百一十二户四分以為總法。乃置科綿八萬八千三百三十七兩六錢為實，法七百一十二户四分而一，得一百二十四兩為上等一户所出綿。以五折之，得六十二兩為副等一户所出綿。又五折之，得三十一兩為中等一户所出綿。乃以四折之，得一十二兩四錢為次等一户所出綿。又以四折之，得四兩九錢六分為下等一户所出綿。乃以各等户數各乘一户所出，即各得每等共數之綿。上等一十二户乘一百二十四兩，得一千四百八十八兩；副等八十七户乘六十二兩，得五千三百九十四兩；中等四百六十四户乘三十一兩，得一萬四千三百八十四兩；次等二千三十五户乘一十二兩四錢，得二萬五千二百三十四兩；下等八千四百三十五户乘四兩九錢六分，得四萬一千八百三十七兩六錢。併五等之綿，得八萬八千三百三十七兩六錢，與原科之數相合。

均定勸分

問：欲勸糴賑濟，據甲民户物力畝步排定，共計一百六十二户，作九等：上等三户，第二等五户，第三等七户，第四等八户，第五等十三户，第六等二十一户，第七等二十六户，第八等三十四户，第九等四十五户。今先觀諭，第一等上户願糴五千石，第九等户願糴二百石。欲知各等拋差石數併數認米數各幾何？

答曰：認該米二十三萬七千六百石。上等一户米五

to the upper-grade count. Do not reduce this final sum; it becomes the divisor. Divide the total silk assessment by this to obtain the silk per upper-grade household. Halve this for the second grade, halve again for the middle grade, apply a 4/10 reduction for the fourth grade, and another 4/10 reduction for the lower grade. Multiply each per-household amount by the respective household count to obtain the total silk for each grade.

Working: Set the lower-grade households 8,435, apply 4/10 reduction to obtain 3,374. Add this to the fourth-grade households 2,035 to obtain 5,409. Apply 4/10 reduction to 5,409 to obtain 2,163.6. Add this to the middle-grade households 464 to obtain 2,627.6. Apply 1/2 reduction to 2,627.6 to obtain 1,313.8. Add to the second-grade households 87 to obtain 1,400.8. Apply 1/2 reduction to 1,400.8 to obtain 700.4. Add to the upper-grade households 12 to obtain 712.4, which becomes the divisor without further reduction.

The accumulated reduction yields 712.4 as the total divisor. Take the silk assessment 88,337 liang 6 qian as dividend. Divide by divisor 712.4 to obtain 124 liang as the silk per upper-grade household. Apply 1/2 reduction to obtain 62 liang for the second grade. Apply 1/2 reduction again to obtain 31 liang for the middle grade. Apply 4/10 reduction to obtain 12 liang 4 qian for the fourth grade. Apply 4/10 reduction again to obtain 4 liang 9 fen 6 li for the lower grade.

Multiply each per-household amount by the respective household count: upper grade 12 x 124 = 1,488 liang; second grade 87 x 62 = 5,394 liang; middle grade 464 x 31 = 14,384 liang; fourth grade 2,035 x 12.4 = 25,234 liang; lower grade 8,435 x 4.96 = 41,837.6 liang. The sum of all five grades equals 88,337.6 liang, matching the original assessment.

Equalizing Voluntary Contribution Levies

Question: It is desired to encourage households to sell grain for famine relief. Based on the property-value and field-size ranking of township A households, there are 162 households in total, divided into nine grades: upper grade 3 households, second grade 5 households, third grade 7 households, fourth grade 8 households, fifth grade 13 households, sixth grade 21 households, seventh grade 26 households, eighth grade 34 households, ninth grade 45 households. After consultation, the first-grade upper households are willing to contribute 5,000 shi, and the ninth-grade households are willing to contribute 200 shi. Find the incremental difference (pao cha) in shi between grades and the total committed rice for each grade.

Answer: Total committed rice: 237,600 shi. Upper grade:

5,000 shi per household, 3 households = 15,000 shi. Second grade: 4,400 shi per household, 5 households = 22,000 shi. Third grade: 3,800 shi per household, 7 households = 26,600 shi. Fourth grade: 3,200 shi per household, 8 households = 25,600 shi. Fifth grade: 2,600 shi per household, 13 households = 33,800 shi. Sixth grade: 2,000 shi per household, 21 households = 42,000 shi. Seventh grade: 1,400 shi per household, 26 households = 36,400 shi. Eighth grade: 800 shi per household, 34 households = 27,200 shi. Ninth grade: 200 shi per household, 45 households = 9,000 shi.

Method: Solve by the method of graded distribution. Set the rice amounts of the upper and lower households, subtract to obtain the dividend. List the number of grades, subtract one, and use the remainder as divisor. Divide to obtain the incremental difference in shi. Successively subtract this difference from the upper-grade amount to obtain each grade's amount. Multiply each by the respective household count and add together for the total.

Working: Set upper-grade household rice at 5,000 shi, subtract lower-grade 200 shi, remainder 4,800 shi as dividend. Subtract 1 from 9 grades, remainder 8 as divisor. Divide to obtain 600 shi as the incremental difference per grade. Subtract from the upper grade to obtain 4,400 shi as the second-grade amount. Subtract 600 again to obtain 3,800 shi as third grade. Continue subtracting 600: 3,200 shi (fourth), 2,600 shi (fifth), 2,000 shi (sixth), 1,400 shi (seventh), 800 shi (eighth), and 200 shi (ninth). Multiply each by the respective household count and add to obtain the total committed rice of 237,600 shi.

Equalizing Combined Remittances

Question: A prefecture must remit named funds (keming qian) to various offices: Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan 350 wen. Now, of all the named funds, 9,253 guan 620 wen has been collected in advance. It is desired to distribute this proportionally according to the original quotas for deposit and remittance. How much should each office receive?

Answer: Ministry of Revenue: 5,497 guan 200 wen; General Office: 3,664 guan 800 wen; Transport Office: 91 guan 620 wen.

Method: Solve by the method of graded distribution. Set up the original rates, reducing by common factors where possible. Add together to form the divisor. Multiply the collected amount by each uncombined rate to form the dividends. Divide each dividend by the divisor.

Working: List Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan

千石，三户計一萬五千石；二等一户米四千四百石，五户計二萬二千石；三等一户米三千八百石，七户計二萬六千六百石；四等一户米三千二百石，八户計二萬五千六百石；五等一户米二千六百石，一十三户計三萬三千八百石；六等一户米二千石，二十一户計四萬二千石；七等一户米一千四百石，二十六户計三萬六千四百石；八等一户米八百石，三十四户計二萬七千二百石；九等一户米二百石，四十五户計九千石。

術曰：以衰分求之。置上下户米，減餘為實。列等數減一，餘為法。除之，得抛差石數。以差累減上等米，各得諸等米。以各等户乘之，併之為總數。

草曰：置上等户米五千石，減下等户二百石，餘四千八百石為實。以九等減一，餘八為法。除實，得六百石為每等抛差。用減上等，餘四千四百石為二等米。又減六百，得三千八百石為三等米。又減六百，得三千二百石為四等米。又減六百，得二千六百石為五等米。又減六百，得二千石為六等米。又減六百，得一千四百石為七等米。又減六百，得八百石為八等米。又減六百，得二百石為九等米。又各乘户數，併之，得總認米石數二十三萬七千六百石。

均定合解

問：州郡合解諸司窠名錢：户部九十六萬五千四百二十一貫文，總所六十四萬三千六百一十四貫文，運司一萬六千九十貫三百五十文。今諸窠名先催到九千二百五十三貫六百二十文，欲照原額分數均定樁收解，合各幾何?

答曰：户部五千四百九十七貫二百文；總所三千六百六十四貫八百文；運司九十一貫六百二十文。

術曰：以衰分求之。置諸原率，可約約之。副併為法。以催到錢乘未併者，各為實。實如法而一。

草曰：列户部九十六萬五千四百二十一貫，總所六十四萬三千六百一十四貫，運司一萬六千九十貫三

百五十文，各為原率。今原率可約，求等得一萬六千九十貫三百五十為等數，俱約之：户部得六十，總所得四十，運司得一，各為率。副併得一百一為法。以催到錢九千二百五十三貫六百二十文乘未併數：六十、四十及一，畢得五十五萬五千二百一十七貫二百文為户部之實，三十七萬一百四十四貫八百文為總所之實，九千二百五十三貫六百二十文為運司之實。三實皆如法一百一而一：其户部得五千四百九十七貫二百文，總所得三千六百六十四貫八百文，運司得九十一貫六百二十，各為候解錢。

寬減屯租

問：屯租欲議寬減，仍聽以夏麥折納分數。官牛種者與減二分，私牛種者與減四分。每歲租穀以三分之一許夏折二麥，内四分大六分小折色。每大麥三石折小麥二石，小麥二石折穀三石五斗。屯租舊額官種一石納租五石，私種一石納租三石。今某州屯田去年計官私種共九千七百八十二石，共合收租穀三萬九千五百八十六石。欲知官私種各數、原額、今減、合催、成年夏麥秋穀幾何？

［按］此題有官私共種數、租數，有種一石租數，求各種數租數，古謂之差分，今謂之和較比例。折色亦差分也，今謂之和數比例。大小麥與穀相易，古謂之異乘同乘，今謂之和率比列。

答曰：官種五千一百二十石，私出種四千六百六十二石。原租額共三萬九千五百八十六石：官種二萬五千六百石，私種一萬三千九百八十六石。今減一萬七百一十四石四斗：官種五千一百二十石，私種五千五百九十四石四斗。合催二萬八千八百七十一石六斗。官種租二萬四千石：一分折麥計穀八千石；四分折大麥三千二百石（係折小麥二千一百三十三石三斗三升三分升之一，復折穀三千七百三十三石三斗三升三分升之二）；六分折小麥四千八百石（係折穀八千四百石）；正色穀一萬六千石。私種租一萬五千八百七十一石六斗：一分折麥計穀五千二百九

350 wen as the original rates. These can be reduced; finding the common measure of 16,090 guan 350 wen, we reduce all: Ministry obtains 60, General Office 40, Transport Office 1 as the rates. Add together to obtain 101 as the divisor. Multiply the collected amount 9,253 guan 620 wen by each uncombined rate (60, 40, and 1): Ministry dividend = 555,217 guan 200 wen; General Office dividend = 370,144 guan 800 wen; Transport Office dividend = 9,253 guan 620 wen. Divide each by the divisor 101: Ministry obtains 5,497 guan 200 wen; General Office obtains 3,664 guan 800 wen; Transport Office obtains 91 guan 620 wen, as the amounts awaiting remittance.

Reducing Garrison Field Rent

Question: Garrison field rents are to be reduced, with the option to pay in summer wheat according to specified conversion fractions. For fields cultivated with official oxen, a reduction of 2/10 is granted; for private oxen, a reduction of 4/10. Each year's grain rent allows 1/3 to be paid in summer wheat (two types): 4/10 great barley and 6/10 small wheat as converted payment. The conversion rates are: 3 shi great barley = 2 shi small wheat; 2 shi small wheat = 3.5 shi grain. The old garrison rent quotas are: official cultivation 1 shi seed pays 5 shi rent; private cultivation 1 shi seed pays 3 shi rent. Now a certain prefecture's garrison fields last year had combined official and private seed of 9,782 shi, with total rent grain due of 39,586 shi. Find: the official and private seed amounts, original quotas, current reductions, amounts collectable, and the mature-year summer wheat and autumn grain totals.

[Ed.] This problem gives the combined official and private seed amounts and rent amounts, with the rent per shi of seed, to find each seed and rent amount. Anciently called "difference distribution" (cha fen), now called "sum-difference proportion." The converted payment is also difference distribution, now called "sum-ratio proportion." The exchange of great and small wheat for grain, anciently called "different multiplication, same division," now called "combined-rate proportion."

Answer: Official seed: 5,120 shi; private seed: 4,662 shi. Original rent quota: 39,586 shi total—official cultivation 25,600 shi, private cultivation 13,986 shi. Current reduction: 10,714 shi 4 dou—official 5,120 shi, private 5,594 shi 4 dou. Amount collectable: 28,871 shi 6 dou. Official cultivation rent 24,000 shi: 1/3 converted to wheat = 8,000 shi grain equivalent; 4/10 great barley = 3,200 shi (equals 2,133 shi 3 dou 3 sheng + 1/3 small wheat, which equals 3,733 shi 3 dou 3 sheng + 2/3 grain); 6/10 small wheat = 4,800 shi (equals 8,400 shi grain); in-kind grain 16,000 shi. Private cultivation rent 15,871 shi

6 dou: 1/3 converted to wheat = 5,290 shi 5 dou + 2/3 grain equivalent; 4/10 great barley = 2,146 shi 8 dou 8 sheng (equals 1,431 shi 2 sheng + 2/3 small wheat, which equals 2,504 shi 3 sheng + 1/3 grain); 6/10 small wheat = 3,220 shi 2 dou 8 sheng (equals 5,635 shi 4 dou 9 sheng grain); 2/10 in-kind grain 10,714 shi 4 dou. Total mature-year receipts: Summer wheat 10,266 shi 6 dou 6 sheng + 2/3, comprising great barley 5,346 shi 8 dou 8 sheng and small wheat 5,919 shi 7 dou 8 sheng + 2/3; autumn in-kind grain 22,667 shi 3 dou 3 sheng + 1/3.

Method: Solve by the method of difference distribution. Set official rent at 5 shi and private rent at 3 shi per shi of seed. Apply the reductions of 2/10 and 4/10: official retains 3.8 (3 shi 8 dou), private retains 1.8 (1 shi 8 dou). Multiply each by 1 shi of seed: official obtains 3.8 shi as the official rate, private obtains 1.8 shi as the private rate. Add together to form the divisor. Take the total rent as dividend. Divide to obtain the amount for one portion. Multiply by each rate to obtain the collectable rent for each category. Divide each rate by the respective rent per shi of seed to obtain the seed amounts. For conversions: 1/3 is the total conversion rate, 4/10 great barley and 6/10 small wheat are the component rates. Multiply each rent by the total rate to obtain the grain-equivalent of converted wheat. Using great barley rate 3 and small wheat rate 2, convert between them to obtain the wheat amounts. Then convert using 2 shi small wheat = 3.5 shi grain to obtain the grain equivalents. Subtract from the collectable rent; the remainder is in-kind autumn grain.

Equalizing Household Tax Relief

Question: A prefecture has granted relief by remitting the autumn tax arrears of the lower three grades of taxable households in a certain county: total remitted 1,355 guan 706 wen in cash and 5,272 shi 1 dou 9 sheng in rice. The property value of the lower three grades in this county is 37,658 guan 500 wen. Now officials have pointed out that many households in this county paid their taxes willingly without arrears, and by remitting only the tax arrears, it seems to favor those who defaulted, which is unfair. It has been proposed to grant the willing-payer households of the three lower grades a reduction in next year's two taxes according to the same pattern. The property value of the three lower grades of non-arrears households is found to be 228,015 guan 321 wen. Find the cash and rice remission per 100 wen of property value, and the total reduction for each.

Answer: Per 100 wen of property value: remit 3.6 wen cash, remit 1.4 ge rice. Next year's two-tax remission: cash 7,949 guan 351 wen 5 fen 5 li 6 hao; rice 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao.

十石五斗三分升之二；四分折大麥二千一百四十六石八斗八升（係折小麥一千四百三十一石二升三分升之二，復折穀二千五百四石三升三分升之一）；六分折小麥三千二百二十石二斗八升（係折穀五千六百三十五石四斗九升）；二分正色穀一萬七百一十四石四斗。成年共收：夏折大小麥一萬二百六十六石六斗六升三分升之二，內大麥五千三百四十六石八斗八升，小麥五千九百一十九石七斗八升三分升之二；秋正穀二萬二千六百六十七石三斗三升三分升之一。

術曰： 以差分求之。置官租五石、私租三石，以所減二分、四分各減之，官存三分八釐，私存一分八釐。以各對乘官私種一石，官得三石八斗為官率，私得一石八斗為私率。副併為法。以共租為實。實如法而一，得一分之數。以官私各率乘之，得各合催租。又以種一石各對除官私率，得各合種。其折色以三分之一為總率，四分大、六分小各為子率。以各租乘總率，得折麥計穀之數。以大麥率三、小麥率二通轉相易，各得大小麥。又各以小麥二石折穀三石五斗約之，得大小麥各折穀之數。以減各合催租，餘為正色秋穀。

戶税均寬

問： 州郡寬恤，近将某縣下三等税户秋科餘欠錢米，已與蠲放：共錢一千三百五十五貫七百六文，米五千二百七十二石一斗九升。某本縣下三等物力計三萬七千六百五十八貫五百文。今來官員陳述：本縣多有樂輸無欠之户，今蒙蠲放税尾，似反寬潤頑輸之户，於理未均。遂議将樂輸三等户於明年兩税與照昨來體例減免。契勘得三等無欠户物力二十二萬八千一十五貫三百二十一文。欲知每百文合減免錢米及共減各幾何?

答曰： 每物力一百文，放錢三文六分，放米一升四合。明年兩税放免：錢七千九百四十九貫三百五十一文五分五釐六毫；米三萬九百一十四石一斗四升四合九勺四抄。

術曰：以粟米衰分求之。置原物力為法，除原放錢米，得每百文物力所放錢米率。以各率乘今來物力，各得錢米，為明年兩稅合放數。

Method: Solve by the methods of grain conversion and graded distribution. Take the original property value as divisor. Divide the original remitted cash and rice by this to obtain the cash and rice remission rates per 100 wen of property value. Multiply each rate by the current property value to obtain the cash and rice amounts for next year's two-tax remission.

草曰：以原物力三萬七千六百五十八貫五百文為法。先除原放錢一千三百五十五貫七百六文，定法百文上得一文為商，除得三文六分為物力百文所放錢率。次除原放米五千二百七十二石一斗九升，得一升四合為物力百文所放米率。以今三等户物力二十二萬八千一十五貫三百二十一文徧乘所放錢米二率，得錢七千九百四十九貫三百五十一文五分五釐六毫，得米三萬九百一十四石一斗四升四合九勺四抄，為明年兩稅合放數。

Working: Take the original property value 37,658 guan 500 wen as divisor. First divide the original remitted cash 1,355 guan 706 wen by this: at the 100-wen position, 1 wen as quotient, obtaining 3.6 wen as the cash remission rate per 100 wen of property value. Next divide the original remitted rice 5,272 shi 1 dou 9 sheng, obtaining 1.4 ge as the rice remission rate per 100 wen of property value. Multiply both rates by the current three-grades property value 228,015 guan 321 wen: cash = 7,949 guan 351 wen 5 fen 5 li 6 hao; rice = 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao, as the two-tax remission for next year.

［按］此題之法至為簡明，以物力為法，以所放錢米為實，除之得每百文物力合放之率，又以今來物力乘之，得合放之數。蓋以物力之多寡為差，物力多者放多，物力少者放少，斯為均矣。

[Ed.] The method of this problem is extremely clear: taking property value as divisor and remitted cash and rice as dividend, dividing obtains the remission rate per 100 wen of property value; then multiplying by the current property value obtains the remission amount. Since property value serves as the measure of difference, those with more property value receive more remission, and those with less receive less—this is equitable.

米穀粒分

Rice and Grain Granule Count

問：開倉受約，有甲户米一千五百三十四石。到廊驗得米内夾穀，乃於樣内取米一捻，數計二百五十四粒，内有穀二十八顆。凡粒米率每勺三百。今欲知米内雜穀多少，以折米數科責及粒各幾何？

Question: When opening the granary to accept delivery, household A delivered 1,534 shi of rice. Upon inspection at the corridor, the rice was found to be mixed with unhusked grain. A pinch of the sample was taken, containing 254 granules, of which 28 were unhusked grain. The standard rate is 300 granules per shao. Now it is desired to know how much unhusked grain is mixed in the rice, the amount of rice to be deducted, and the total granule count.

答曰：米一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八；穀一百六十九石一斗二合三勺一百二十七分勺之七十九。合折米八十四石五斗五升一合一勺一百二十七分勺之一百三。原米折米共計四十三億四千八百三十四萬六千四百五十六粒。

Answer: Rice: 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; Grain: 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Converted rice deduction: 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao. Total granules of original and converted rice: 4,383,464,456.

術曰：以粟米求之，衰分入之。置樣米粒數為法，以常穀顆數減之，餘與穀為列衰。可約約之。以共米乘列衰為各實，實如法而一，各得米數穀數。置穀數以粟率折之，為穀所折米。次以勺率遍乘米數折米，得粒數。

Method: Solve by the grain conversion method, with graded distribution applied. Set the sample granule count as divisor. Subtract the unhusked grain count; the remainder and the grain count form the graded rates. Reduce by common factors if possible. Multiply the total rice by each graded rate to form the dividends. Divide each by the divisor to obtain the rice and grain amounts. Convert the grain amount using the grain conversion rate to obtain the rice equivalent of the grain. Then multiply both rice amounts by the shao rate to obtain the granule count.

Working: Set the sample pinch count 254 as divisor; unhusked grain 28 as the grain rate; subtract from divisor, remainder 226 as the rice rate. These two rates and the divisor can all be reduced; finding common factor 2: divisor = 127, rice rate = 113, grain rate = 14. Multiply total rice 1,534 shi by each rate: rice dividend = 173,342 shi; grain dividend = 21,476 shi. Divide each by divisor 127: rice = 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; grain = 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Convert grain at 50% (1 shi grain = 5 dou rice) to obtain 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao as the rice equivalent of grain. Combine both rice amounts: 1,449 shi 4 dou 4 sheng 8 ge 8 shao + 24/127 shao. Convert to improper fraction: 184,080 shi numerator. Multiply by shao rate 300 granules to obtain 55,224,000,000 granules as dividend. Divide by denominator 127 to obtain 434,834,6456 granules, with remainder 88 discarded.

[Ed.] This problem uses a pinch sample to determine the ratio of rice granules to unhusked grain, and thereby ascertains the rice and grain in the full amount—this is the ancient method of "inspecting provisions." With 28 unhusked grains out of 254 granules, the impurity rate is evident. The grain conversion rate of 50% means 1 shi of grain converts to 5 dou of rice. The shao rate of 300 means 300 granules per shao. This method uses graded distribution as the primary technique and grain conversion secondarily—quite ingenious.

Household Salt Ration Assessment

Question: A certain prefecture administers nine counties, which according to regulations distribute salt annually: County A has 143,500 households, County B 97,400, County C 76,500, County D 54,320, County E 43,600, County F 32,480, County G 26,750, County H 19,320, County I 12,560. The prefecture has determined the nine counties' total households to be 486,820. The salt tax assessment is 1 guan wen in full-value coin, to be paid in huizi notes. At that time, old huizi traded at 54 wen full-value per guan, and new huizi at 270 wen full-value per guan. Find the salt tax each county should pay.

Answer: County A: 3,189 guan 300 wen; County B: 2,165 guan 200 wen; County C: 1,700 guan wen; County D: 1,206 guan 640 wen; County E: 968 guan wen; County F: 721 guan 600 wen; County G: 594 guan 400 wen; County H: 428 guan 960 wen; County I: 279 guan 120 wen.

Method: Solve by the method of graded distribution. List

草曰：置一捻樣粒數二百五十四為法，以帶穀二十八顆為穀衰，以減法，餘二百二十六為米衰。此二衰與法皆可約，求等得二，俱以約之：法得一百二十七，米衰得一百一十三，穀衰得一十四。以米一千五百三十四石遍乘二衰，得一十七萬三千三百四十二石為米實，得二萬一千四百七十六石為穀實。皆如法一百二十七而一：米得一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八，穀得一百六十九石一斗二合三勺一百二十七分勺之七十九。以粟率五十折之，得八十四石五斗五升一合一勺一百二十七分勺之一百三為穀折納米數。併二米，得一千四百四十九石四斗四升八合八勺一百二十七分勺之二十四。先通分納子一十八萬四千八十石，以勺率三百粒乘之，得五千五百二十二億四千萬粒為實，以母一百二十七除之，得四十三億四千八百三十四萬六千四百五十六粒，不盡八十八棄之。

［按］按此題以取樣一捻之數，驗米粒與穀粒之率，因以知全數之米穀，古所謂驗食之法也。二百五十四粒中穀二十八顆，其不純率可知矣。以粟率五十折之者，謂穀一石折米五斗也。勺率三百者，謂每勺有三百粒也。此法以衰分為主，而粟米次之，可謂巧矣。

戶口食鹽

問：某州管下九縣，依例歲敷食鹽：甲縣户一十四萬三千五百，乙縣户九萬七千四百，丙縣户七萬六千五百，丁縣户五萬四千三百二十，戊縣户四萬三千六百，己縣户三萬二千四百八十，庚縣户二萬六千七百五十，辛縣户一萬九千三百二十，壬縣户一萬二千五百六十。今來州司根括到九縣共計户四十八萬六千八百二十，每科食鹽一貫文足，係用會子納。其時舊會每貫五十四文足，新會每貫二百七十文足。欲知各縣合納鹽錢幾何？

答曰：甲縣三千一百八十九貫三百文；乙縣二千一百六十五貫二百文；丙縣一千七百文；丁縣一千二百六貫六百四十文；戊縣九百六十八貫文；己縣七百二十一貫六百文；庚縣五百九十四貫四百文；辛縣四百二十八貫九百六十文；壬縣二百七十九貫一百二十文。

術曰：以衰分求之。列各縣户數為列衰，副併為法。

以共科鹽錢乘未併者，各為實。實如法而一，各得縣合納鹽錢。

each county’s household count as graded rates. Add together to form the divisor. Multiply the total salt tax by each uncombined rate to form the dividends. Divide each dividend by the divisor to obtain each county’s salt tax payment.

草曰：列九縣户數為列衰：甲一十四萬三千五百，乙九萬七千四百，丙七萬六千五百，丁五萬四千三百二十，戊四萬三千六百，己三萬二千四百八十，庚二萬六千七百五十，辛一萬九千三百二十，壬一萬二千五百六十。副併九縣之數，得四十八萬六千八百二十為法。置共科鹽錢一萬八千五百貫文為實。以法四十八萬六千八百二十除之，得每贯率數。以未併各縣户數徧乘之：甲一十四萬三千五百乘率數，得三千一百八十九貫三百文；乙九萬七千四百乘率數，得二千一百六十五貫二百文；丙七萬六千五百乘率數，得一千七百貫文；丁五萬四千三百二十乘率數，得一千二百六貫六百四十文；戊四萬三千六百乘率數，得九百六十八貫文；己三萬二千四百八十乘率數，得七百二十一貫六百文；庚二萬六千七百五十乘率數，得五百九十四貫四百文；辛一萬九千三百二十乘率數，得四百二十八貫九百六十文；壬一萬二千五百六十乘率數，得二百七十九貫一百二十文。各為縣合納鹽錢之數。

Working: List the nine counties’ household counts as graded rates: A 143,500, B 97,400, C 76,500, D 54,320, E 43,600, F 32,480, G 26,750, H 19,320, I 12,560. Add all nine counties to obtain 486,820 as divisor. Take the total salt tax assessment 18,500 guan wen as dividend. Divide by 486,820 to obtain the rate per unit. Multiply each uncombined county household count by this rate: A 143,500 x rate = 3,189 guan 300 wen; B 97,400 x rate = 2,165 guan 200 wen; C 76,500 x rate = 1,700 guan wen; D 54,320 x rate = 1,206 guan 640 wen; E 43,600 x rate = 968 guan wen; F 32,480 x rate = 721 guan 600 wen; G 26,750 x rate = 594 guan 400 wen; H 19,320 x rate = 428 guan 960 wen; I 12,560 x rate = 279 guan 120 wen. Each is the respective county’s salt tax payment.

Fascicle 6A: Money and Grain (Qiangu)

Editorial: This fascicle on Money and Grain specializes in grain conversion, light-cargo converted payments, and principal-interest calculations. The ancient Nine Chapters had a chapter on Grain Conversion (su mi) for managing exchange and barter, and a chapter on Construction Calculation (shang gong) for managing labor and volumes. This fascicle combines and expands these topics, comprehensively covering the receipts and disbursements of money and grain, the rise and fall of commodity prices, the light and heavy burden of transport fees, and the complexity of conversion procedures. Its methods take grain conversion as fundamental, supplemented by graded distribution (cui fen), equalized transport (jun shu), and excess-deficit (ying nao) methods. Thus the problems ask about the relative costliness of commodities, the amount of transport fees, the correctness of converted payments, and the increase of principal and interest. All of these matters concern state finance and people's livelihood, and therefore are classified as a separate category.

[Ed.] This fascicle contains nine problems: (1) Assessing Grain Purchase, (2) Converted Remittance of Light Cargo, (3) Tracing Rent to its Origin, (4) Calculating Principal and Interest, (5) Finding the Original from Converted Certificates, (6) Grain and Rice Exchange, (7) Calculating Rice for Yeast, (8) Recovered Transport Fees, (9) Three-Color Average Price. All concern money and grain. The methods mostly use grain conversion techniques, taking commodity price rates as the standard and using the mutual determination of quantities as the application.

卷六上　錢穀類

按：錢穀一章，專主粟米互換、輕齎折納、本息推求之事。蓋古者九章有粟米一章，以御交質變易；又有商功一章，以御功程積實。此則合而廣之，於錢穀之出入、物價之低昂、運脚之輕重、折變之煩簡，無不備載。其為術也，以粟米為本，而以衰分、均輸、盈朒諸法參之。故其題也，或問物價之貴賤，或問運費之多寡，或問折色之正偏，或問本息之增減。凡此數端，皆國用民生之所繫，故列為專類焉。

［按］此卷凡九問：一曰課糴，二曰折解輕齎，三曰僦直推原，四曰推求本息，五曰易牒知原，六曰粟米交易，七曰計米易麴，八曰囘運費，九曰三色均價。皆錢穀之屬也。其術多用粟米互換之法，蓋以物價之率為準，而以多寡之相求為用也。

Assessing Grain Purchase

Question: Officials are dispatched to five circuits for state grain purchases. The data are: Zhexi Pingjiang Prefecture—price 35 guan per shi, 135 ge (per dou), water transport to Zhenjiang 900 wen per shi; Anji Prefecture—price 29 guan 500 wen, 110 ge, water transport to Zhenjiang 1 guan 200 wen per shi; Jiangxi Longxing Prefecture—price 28 guan 100 wen, 115 ge, water transport to Jiankang 1 guan 700 wen per shi; Jizhou—price 25 guan 850 wen, 120 ge, water transport to Jiankang 2 guan 900 wen per shi; Hunan Tanzhou—price 27 guan 300 wen, 118 ge, water transport to Ezhou 1 guan 700 wen per shi. All payments are in 17th-series official huizi notes; all rice is measured using the Wensi Bureau hu (1 hu = 83 ge) for conversion. Find the price per official-hu shi for each circuit and compare their relative costliness.

課糴

問：差人五路和糴。據浙西平江府石價三十五貫文，一百三十五合，至鎮江水脚錢每石九百文；安吉州石價二十九貫五百文，一百一十合，至鎮江水脚錢每石一貫二百文；江西隆興府石價二十八貫一百文，一百一十五合，至建康水脚錢每石一貫七百文；吉州石價二十五貫八百五十文，一百二十合，至建康水脚錢每石二貫九百文；湖南潭州石價二十七貫三百文，一百一十八合，至鄂州水脚錢每石一貫七百文。其錢並十七界官會，其米並用文思院斛交量紐數。欲皆以官解計石錢相比貴賤幾何？

［按］按草中係二貫一百文，此訛。文思院解每斗八十三合。

答曰：文思院斛石錢：安吉州二十三貫一百六十四文一十一分文之六；平江府二十二貫七十一文二十七分文之二十三；隆興府二十一貫五百七文二十三分文之一十九；潭州二十貫六百七十九文五十九分文之四十九；吉州一十九貫八百八十五文十二分文之五。

［按］按三十九訛四十九。

術曰：以粟米互換求之。置石價併水脚，乘石數，又乘官斗合數為實。各如本州合數而一，各得官解石錢。以課貴賤。

草曰：置安吉州石價二十九貫五百文，平江石價三十五貫文，隆興石價二十八貫一百文，吉州石價二十五貫八百五十文，潭州石價二十七貫三百文，列右行。次置水脚：安吉一貫二百文，平江九百文，隆興一貫七百文，吉州二貫九百文，潭州二貫一百文，列左行，各對本州石價。以兩行數併之，得數：安吉三十貫七百文，平江三十五貫九百，隆興二十九貫八百，潭州二十九貫四百，吉州二十八貫七百五十，仍於右行。次以文思院官斗八十三合遍乘之：安吉州得二千五百四十八貫一百文，平江府得二千九百七十九貫七百文，江西隆興得二千四百七十三貫四百文，湖南潭州得二千四百四十貫二百文，江南吉州得二千三百八十六貫二百五十文，各為實於右。得次列安吉斗一百一十合，平江斗一百三十五合，隆興斗一百一十五合，潭州斗一百一十八合，吉州斗一百二十合於左行為法。以対除右行之實：安吉得二十三貫一百六十四文一十一分之六，平江得二十三貫七十一文二十七分文之二十三，隆興得二十一貫五百七文二十三分文之一十九，潭州得二十貫六百七十九文五十九文分之四十九，吉州得一十九貫八百八十五文一十二分文之五。相課石價，其安吉州最貴，平江次之，隆興又次之，潭州又次之，吉州最賤。

［按］按此題以各路米價併水脚錢，以官斛八十三合為準，約為官斛石錢，以相比較貴賤。其法以每路斗合數除之，使歸於一律，然後可以相課。此所謂異乘同除之法也。

[Ed.] Note: The working mentions 2 guan 100 wen for Tanzhou water transport, which appears to be an error. The Wensi Bureau hu equals 83 ge per dou.

Answer: Wensi Bureau hu price per shi: Anji Prefecture 23 guan 164 wen + 6/11 wen; Pingjiang Prefecture 22 guan 71 wen + 23/27 wen; Longxing Prefecture 21 guan 507 wen + 19/23 wen; Tanzhou 20 guan 679 wen + 49/59 wen; Jizhou 19 guan 885 wen + 5/12 wen.

[Ed.] Note: 49 is a scribal error for 39.

Method: Solve by the grain conversion method. Set the shi price combined with water transport fee, multiply by the shi count, then multiply by the official dou ge count to form the dividend. Divide each by the respective circuit's dou ge count to obtain the price per official-hu shi. Compare to determine relative costliness.

Working: Set the shi prices in the right column: Anji 29 guan 500 wen, Pingjiang 35 guan, Longxing 28 guan 100 wen, Jizhou 25 guan 850 wen, Tanzhou 27 guan 300 wen. Set the water transport fees in the left column: Anji 1 guan 200 wen, Pingjiang 900 wen, Longxing 1 guan 700 wen, Jizhou 2 guan 900 wen, Tanzhou 2 guan 100 wen, each paired with its circuit's shi price. Combine the two columns: Anji 30 guan 700 wen, Pingjiang 35 guan 900 wen, Longxing 29 guan 800 wen, Tanzhou 29 guan 400 wen, Jizhou 28 guan 750 wen, still in the right column. Next multiply each by the Wensi Bureau standard of 83 ge: Anji = 2,548 guan 100 wen, Pingjiang = 2,979 guan 700 wen, Longxing = 2,473 guan 400 wen, Tanzhou = 2,440 guan 200 wen, Jizhou = 2,386 guan 250 wen, as dividends in the right column. Place each circuit's dou measure in the left column as divisor: Anji 110 ge, Pingjiang 135 ge, Longxing 115 ge, Tanzhou 118 ge, Jizhou 120 ge. Divide the right-column dividends by the left-column divisors: Anji = 23 guan 164 wen + 6/11 wen; Pingjiang = 22 guan 71 wen + 23/27 wen; Longxing = 21 guan 507 wen + 19/23 wen; Tanzhou = 20 guan 679 wen + 49/59 wen; Jizhou = 19 guan 885 wen + 5/12 wen. Comparing the prices: Anji is most expensive, followed by Pingjiang, then Longxing, then Tanzhou, with Jizhou the cheapest.

[Ed.] This problem combines each circuit's rice price with water transport fees, using the standard official hu of 83 ge to convert to price per official-hu shi for comparison. The method divides by each circuit's dou ge measure to normalize to a common standard before comparing. This is the so-called "different multiplication, same division" method.

Fascicle 6B: Money and Grain (continued)

卷六下　錢穀類（續）

Tracing Rent to its Origin

僦直推原

Question: Among the shop-rent records, one household currently pays 156 wen 8 fen in full-value coin per day as the standard. The regulations state: those never reduced receive a 3/10 reduction; those already reduced by 3/10 receive a further 2/10 reduction; those already reduced by 2/10 receive yet another 2/10 reduction. Now this household has been subject to cumulative reductions. Find the original rent amount.

問：房廊數内，一户日納一百五十六文八分足為準。指揮未曾經減者，減三分；已曾經減三分者，減二分；已曾經減二分者，更減二分。今本户累經減者，欲知原額房錢幾何?

Answer: Original amount: 350 wen.

答曰：原額三百五十文。

Method: Solve by the method of graded distribution. Place two rows of 10, each with three positions. List the reduction fractions and subtract from the right row. Multiply the remainders together to form the divisor. Multiply the original entries in the left row together, then multiply by the current payment to form the dividend. Divide to obtain the original rent.

術曰：以衰分求之。列一十分兩行，各三位；列減分，對減右行。以餘者相乘為法。以左行原列相乘，得納錢為實。實如法而一，得原額錢。

Working: Place three 10s in the left row and three 10s in the right row. From the upper right subtract the first reduction of 3/10 (10 – 3 = 7), from the middle right subtract the second reduction of 2/10 (10 – 2 = 8), from the lower right subtract the further reduction of 2/10 (10 – 2 = 8). The right row remainders 7, 8, 8 multiply together to give 448 as the divisor. The left row three 10s multiply to give 1,000 as the multiplier. Multiply by the current daily payment 156 wen 8 fen to obtain 156 guan 800 wen as dividend. Divide by 448 to obtain 350 wen as the household's original rent.

草曰：列一十三位於左行，又列一十分三位於右行。其右上減去初減三分，右中減去次減二分，右下減去更減二分。右行餘七八八，以相乘得四百四十八為法。乃以左行三位一十分相乘，得一千為乘率。以乘見日納錢一百五十六文八分，得一百五十六貫八百文為實。實如法而一，得三百五十文，為本户原額戸錢。

[Ed.] This problem uses cumulative reductions to trace back to the original amount; the method is entirely clear. First reduction of 3/10 leaves 7/10; second reduction of 2/10 leaves 8/10; further reduction of 2/10 leaves 8/10. These three numbers multiplied together give 448 parts. Taking the payment 156.8 wen as the dividend for 448 parts, the dividend for 1,000 parts (the original) must be 350 wen.

［按］此題以累減之法推原額，術意至明。初減三分存七分，次減二分存八分，更減二分存八分。三數相連乘得四百四十八分，以納錢一百五十六文八分為四百四十八分之實，則原額一千分之實必為三百五十文矣。

Calculating Principal and Interest

推求本息

Question: Three treasuries have the following interest rules: above 10,000 guan pays 1 li (0.001), above 1,000 guan pays 2.5 li (0.0025), above 100 guan pays 3 li (0.003). Treasury A has principal 493,800 guan; Treasury B 370,300 guan; Treasury C 246,800 guan. Now the three treasuries together have collected 25,644 guan 200 wen in interest. The assessment rates are: A uses inverted-cone difference, B uses square-cone difference, and C uses puncture-

問：三庫息例：萬貫以上一釐，千貫以上二釐五毫，百貫以上三釐。甲庫本四十九萬三千八百貫，乙庫本三十七萬三百貫，丙庫本二十四萬六千八百貫。今三庫共約到息錢二萬五千六百四十四貫二百文。其典率：甲反錐差，乙方錐差，丙蒺藜差。欲知原典三例本息各幾何?

vine difference. Find the original principal and interest for each treasury under the three rates.

［按］此即衰分題也。其差有反錐、方錐、蒺藜之名，蓋以一二三遞減如立錐為反錐，以一四九平方遞加為方錐，以一三六三數遞加為蒺藜。是必古有其名也。至以各差求各本，則因各本原依各差入之也。

[Ed.] This is a graded-distribution problem. The differences have names: inverted cone (fan zhui), square cone (fang zhui), and puncture vine (ji li). The inverted cone decreases by 1, 2, 3 like an upright cone; the square cone increases by squares 1, 4, 9; the puncture vine increases by 1, 3, 6. These must have been ancient terms. The use of each difference to find each principal reflects how the principals were originally assigned according to these differences.

答曰：甲庫共納息九千五十三貫文：一釐息二千四百六十九貫文，二釐半息四千一百一十五貫文，三釐息二千四百六十九貫文。

乙庫共納息一萬五十一貫文：一釐息二百六十四貫五百文，二釐半息二千六百四十五貫文，三釐息七千一百四十一貫五百文。

丙庫共納息六千五百四十貫二百文：一釐息二百四十六貫八百文，二釐半息一千八百五十一貫文，三釐息四千四百四十二貫四百文。

Answer: **Treasury A** total interest 9,053 guan: 1-li rate 2,469 guan; 2.5-li rate 4,115 guan; 3-li rate 2,469 guan.

Treasury B total interest 10,051 guan: 1-li rate 264 guan 500 wen; 2.5-li rate 2,645 guan; 3-li rate 7,141 guan 500 wen.

Treasury C total interest 6,540 guan 200 wen: 1-li rate 246 guan 800 wen; 2.5-li rate 1,851 guan; 3-li rate 4,442 guan 400 wen.

術曰：置諸庫諸色之差，照釐率為三行。縱併之為約率，横命之為乘率。以約率各約自庫之本，各得。以遍乘未併乘率，然後各以釐率横乘之，次以縱併之，為各庫共息。

Method: Set up the differences for each treasury's categories, arranged as three rows according to the li rates. Add vertically to obtain the reduction rates; assign horizontally as multiplication rates. Divide each treasury's principal by its reduction rate to obtain the quotients. Multiply these throughout by the uncombined multiplication rates, then multiply horizontally by each li rate, and finally add vertically to obtain each treasury's total interest.

草曰：置甲庫反錐差，自下置三二一於右行。次置乙庫方錐差，自上置一四九於中行。次置丙庫蒺藜差，自上置一三六於左行，各為三庫上中下三等乘率。乃縱併甲差三二一，得六為甲約率。縱併乙差一四九，得一十四為乙約率。縱併丙差一三六，得一十為丙約率。直命九位數各為上中下乘率。乃先以約率各約自庫之本：乃以甲約率六約甲本四十九萬三千八百貫，得八萬二千三百貫為甲得。次以乙約率一十四約乙本三十七萬三百貫，得二萬六千四百五十貫為乙得。次以丙約率一十約丙本二十四萬六千八百貫，得二萬四千六百八十貫為丙得。以各得乘未併乘率：其甲所得八萬二千三百貫，乘反錐乘率三二一，得二十四萬六千九百貫為上率，得一十六萬四千六百貫為中率，得八萬二千三百貫為下率。其乙所得二萬六千四百五十貫，以乘方錐差一四九，得二萬六千四百五十貫為上率，得一十萬五千八百貫為中率，得二十三萬八千五十貫為下率。其丙所得二萬四千六百八十貫，以乘蒺藜差一三六，得二萬四千六百八十貫為上率，得七萬四千四十貫為中率，得一十四萬八千八十貫為下率。然後各以息釐數乘各庫三乘：其甲以一釐乘上率二十四萬六千九百貫，得二千四百六十九貫為上息；以二釐五毫乘中率一十六萬四千六百貫，得四千一百一十五貫為中息；以三釐乘

Working: Set Treasury A inverted-cone difference, place 3, 2, 1 from bottom in the right row. Set Treasury B square-cone difference, place 1, 4, 9 from top in the middle row. Set Treasury C puncture-vine difference, place 1, 3, 6 from top in the left row. These are the three treasuries' upper-middle-lower multiplication rates. Add A's differences vertically: 3+2+1 = 6 as A's reduction rate. Add B's: 1+4+9 = 14 as B's reduction rate. Add C's: 1+3+6 = 10 as C's reduction rate. The nine numbers are assigned as upper-middle-lower multiplication rates.

Divide each treasury's principal by its reduction rate: A principal 493,800 / 6 = 82,300 guan as A's quotient. B principal 370,300 / 14 = 26,450 guan as B's quotient. C principal 246,800 / 10 = 24,680 guan as C's quotient.

Multiply each quotient by the uncombined multiplication rates: A's 82,300 x inverted-cone rates 3, 2, 1 = upper 246,900 guan, middle 164,600 guan, lower 82,300 guan. B's 26,450 x square-cone rates 1, 4, 9 = upper 26,450 guan, middle 105,800 guan, lower 238,050 guan. C's 24,680 x puncture-vine rates 1, 3, 6 = upper 24,680 guan, middle 74,040 guan, lower 148,080 guan.

Then multiply each treasury's three rates by the interest li: A: 1 li x upper 246,900 = 2,469 guan upper interest;

2.5 li x middle 164,600 = 4,115 guan middle interest; 3 li x lower 82,300 = 2,469 guan lower interest. Total A interest = 9,053 guan.

B: 1 li x 26,450 = 264.5 guan upper; 2.5 li x 105,800 = 2,645 guan middle; 3 li x 238,050 = 7,141.5 guan lower. Total B = 10,051 guan.

C: 1 li x 24,680 = 246.8 guan upper; 2.5 li x 74,040 = 1,851 guan middle; 3 li x 148,080 = 4,442.4 guan lower. Combined total interest = 25,644 guan 200 wen.

下率八萬二千三百貫，得二千四百六十九貫為下息。併上中下三息，得九千五十三貫文為甲庫共息。其乙庫以一釐乗上率二萬六千四百五十貫，得二百六十四貫五百文為上息；以二釐五毫乗中率一十萬五千八百貫，得二千六百四十五貫為中息；以三釐乗下率二十三萬八千五十貫，得七千一百四十一貫五百為下息。併上中下三息，得一萬五十一貫文為乙庫共息。其丙庫以一釐乗上率二萬四千六百八十貫，得二百四十六貫八百為上息；以二釐五毫乘中率七萬四千四十貫，得一千八百五十一貫為中息；以三釐乗下率一十四萬八千 080 貫，得四千四百四十二貫四百文為下息。併三庫共息，得二萬五千六百四十四貫二百文為總息。

Finding the Original from Converted Certificates

易牒知原

[Ed.] Note: The original edition lacked this problem statement; it has been restored here.

［按］按舊本此問無題，今增入。

Question: Official certificates (du die) were issued for dispatching agents to conduct trade: every 3 certificates exchange for 13 bags of salt; 2 bags of salt exchange for 84 bolts of cloth; 15 bolts of cloth exchange for 3.5 bolts of silk; 6 bolts of silk exchange for 7.2 liang of silver. Now 9,172.8 liang of silver has been obtained. Find the original number of certificates issued.

問：出度牒差人營運：每三道易鹽一十三袋，鹽二袋易布八十四疋，布一十五疋易絹三疋半，絹六疋易銀七兩二錢。今趂到銀九千一百七十二兩八錢。欲知原闕度牒道數幾何?

Answer: Certificates: 180.

答曰：度牒一百八十道。

Method: Solve by the grain conversion method using mutual multiplication for exchange. List each pair of numbers with their original commodities facing each other, like goose wings in flight. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answer.

術曰：以粟米互乘易法求之。列各數以本色相對，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。

Working: First multiply certificates 3 by salt 2 bags = 6. Multiply by cloth 15 = 90. Multiply by silk 6 bolts = 540. Multiply by silver 91,728 qian (9,172.8 liang converted) = 49,533,120 qian as dividend. Next multiply salt 13 bags by cloth 84 = 1,992. Multiply by silk 3.5 bolts = 6,972. Multiply by silver 7.2 liang = 501,984 qian as divisor. Divide: 49,533,120 / 501,984 = 180 certificates as the original number issued.

草曰：先以度牒三道乘鹽二袋，得六。以乘布一十五，得九十。又乘絹六疋，得五百四十。乃乘銀九萬一千七百二十八錢，得四千九百五十三萬三千一百二十錢為實。次以鹽一十三袋乘布八十四，得一千九百九十二。以乘絹三疋五分，得六千九百七十二。乃乘銀七兩二錢，得五十萬一千九百八十四錢為法。除實，得一百八十道為原闕度牒。

[Ed.] This problem involves four successive exchanges: certificates to salt, salt to cloth, cloth to silk, silk to silver. The method multiplies the chain with one more item as dividend and the chain with one fewer item as divisor—this is the mutual multiplication method. The "goose wing" arrangement places original commodities facing each other, like geese flying in symmetric formation.

［按］按此題以度牒易鹽、鹽易布、布易絹、絹易銀，凡四變。術以多一事者相乘為實，少一事者相乘為法，此互乘之法也。鴈翅列者，以本色對列，如鴈翅飛行，左右對稱之義也。

Grain and Rice Exchange

粟米交易

［按］按舊本此問無題，今增。

問：菽三升易小麥二升，小麥一升五合易油麻八合，油麻一升二合易粳米一升八合。今將菽一十四石四斗欲易油麻，又將小麥二十一石六斗欲易粳米。問各幾何?

答曰：油麻五石一斗二升；粳米一十七石二斗八升。

術曰：以粟米換易求之。置原易率，本色對列，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。各得或問數，不干其率者不置。

草曰：置四色六數，列六位率，如鴈翅，皆化為合。先將菽一十四石四斗化作一萬四千四百合，乃對前二句率數四位，如鴈翅，至欲易油麻止，共五事為上圖。次將小麥二十一石六斗化為二萬一千六百合，乃對後兩句率四位，鴈翅，至欲粳米止，共五事為下圖。

下圖：其上圖以菽一萬四千四百合乘麥二十，得二十八萬八千，又乘油麻八合，得二百三十萬四千合為油麻實。次以菽三十合乘麥一十五合，得四百五十合為法。除之，得五千一百二十合，展為五石一斗二升為油麻。

其下圖以小麥二萬一千六百合乘油麻八合，得一十七萬二千八百合，又乘粳米一十八合，得三百一十一萬四百合為粳米實。以小麥一升五合乘油麻一十二合，得一百八十合為法。除之，得一萬七千二百八十，展作一十七石二斗八升為粳米。

計米易麴

［按］按舊本此問無題，今增。

問：庫率：粳穀七石出米三石，糯米一斗易小麥一斗七升，小麥五升踏麴二斤四兩，麴一百一十斤醞糯米一石三斗。今有糯穀一千七百五十九石三斗八升，欲出穀做米，易麥踏麴，還自醞，餘穀之米須令適足。各合幾何?

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

Question: 3 sheng of beans exchange for 2 sheng of small wheat; 1.5 sheng of small wheat exchanges for 8 ge of sesame; 1.2 sheng of sesame exchanges for 1.8 sheng of polished rice. Now 14 shi 4 dou of beans are to be exchanged for sesame, and 21 shi 6 dou of small wheat are to be exchanged for polished rice. Find each amount.

Answer: Sesame: 5 shi 1 dou 2 sheng; Polished rice: 17 shi 2 dou 8 sheng.

Method: Solve by the grain conversion exchange method. Set up the original exchange rates with original commodities facing each other, like goose wings. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answers. Items not in the relevant chain are not used.

Working: Set up four commodities with six numbers, arranged in six positions like goose wings, all converted to ge. First convert beans 14 shi 4 dou to 14,400 ge. Pair with the first two exchange rates (4 numbers) for the upper diagram, ending at the sesame exchange, 5 items total. Next convert small wheat 21 shi 6 dou to 21,600 ge. Pair with the last two exchange rates (4 numbers) for the lower diagram, ending at the polished rice exchange, 5 items total.

Upper diagram: Multiply beans 14,400 ge by wheat rate 20 (the cross-rate) = 288,000; then multiply by sesame 8 ge = 2,304,000 ge as sesame dividend. Multiply beans 30 ge by wheat 15 ge = 450 ge as divisor. Divide: 2,304,000 / 450 = 5,120 ge = 5 shi 1 dou 2 sheng of sesame.

Lower diagram: Multiply small wheat 21,600 ge by sesame 8 ge = 172,800 ge; then multiply by polished rice 18 ge = 3,114,000 ge as polished rice dividend. Multiply small wheat 15 ge by sesame 12 ge = 180 ge as divisor. Divide: 3,114,000 / 180 = 17,280 ge = 17 shi 2 dou 8 sheng of polished rice.

Calculating Rice for Yeast Production

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

Question: Granary rates: 7 shi of glutinous grain yields 3 shi of rice; 1 dou of glutinous rice exchanges for 1 dou 7 sheng of small wheat; 5 sheng of small wheat produces 2 jin 4 liang of yeast; 110 jin of yeast ferments 1 shi 3 dou of glutinous rice. Now there are 1,759 shi 3 dou 8 sheng of glutinous grain. It is desired to hull the grain to make rice, exchange the rice for wheat, use the wheat to produce yeast, and ferment the remaining grain's rice with this yeast so that it is exactly sufficient. Find each amount.

Answer: Total grain: 1,759 shi 3 dou 8 sheng. Grain hulled: 924 shi, yielding rice 396 shi. Wheat obtained: 673 shi 2 dou. Yeast produced: 30,294 jin. Remaining grain: 835 shi 3 dou 8 sheng. Fermented rice: 358 shi 2 sheng.

答曰：共穀一千七百五十九石三斗八升。出穀九百二十四石，得米三百九十六石。易麥六百七十三石二斗。踏麴三萬二百九十四斤。餘穀八百三十五石三斗八升。醞米三百五十八石二升。

Method: Solve by the grain conversion exchange method. Set up the rates with original commodities facing each other, like goose wings. Where there are fractions, convert them; where commodities differ, transform them. Multiply from top to bottom (forward multiplication) and from bottom to top (backward multiplication) to obtain combined numbers. Where pairs face each other, multiply them; where there is no pair, assign directly—these become the rates. Add the unmatched top and bottom rates to form the divisor rate. Multiply the given amount by all the rates (excluding the divisor rate) to form dividends. Divide all dividends by the divisor to obtain the converted amounts, then perform mutual exchanges and divisions to obtain the final answers.

術曰：以粟米換易求之。置諸率，隨本色對列，如鴈翅。有分者通之，異類者變之，以頭位者進乘之，以下位退乘之，得合數。有對者相乘之，無對者直命之，為諸率。併上下無對者為法率。以今有物徧乘諸率（不乘法率），各為實。諸實並如法而一，各得其已變者，復互易乘除之，即得所求。

[Ed.] This is the most complex of all grain conversion problems. It takes glutinous grain as the base, hulls it to rice, exchanges rice for wheat, uses wheat to produce yeast, ferments rice with yeast, and the remaining grain's rice must be exactly sufficient. The "convert fractions" means converting 2 jin 4 liang to 9/4 jin; "transform different kinds" means converting fermented rice back to glutinous grain; "forward and backward multiplication" means multiplying from top to bottom and bottom to top. The method uses the combined diagram to find rates, then multiplies by the given amount and divides—a variation of graded distribution.

［按］此題為粟米換易中之最繁者，以糯穀為本，出米易麥，麥踏麴，麴醞米，而餘穀之米又須適足。術中所謂有分者通之，謂二斤四兩通為四分斤之九也；異類者變之，謂醞米變為糯穀也；進乘退乘者，自上而下為進，自下而上為退也。其法以合圖求諸率，以今有物遍乘，如法而一，此衰分之變法也。

Recovered Transport Fees

囘運費

Question: Rice transported by water from Jiangxi, originally destined for discharge at Zhenjiang, with a water route of 2,130 li and water transport fee of 1 guan 200 wen per shi. Now this rice is diverted to be stored at Chizhou instead; the distance from Chizhou to Zhenjiang is 880 li. Find the water transport fee that should be recovered (for the route not traveled).

問：有江西水運米一十二萬三千四百石，原係鎮江交卸，計水程二千一百三十里，每石水腳錢一貫二百文。今截上件米就池州安頓，池州至鎮江八百八十里。欲收囘不該水腳錢幾何?

Answer: Amount to recover: 61,178 guan 591 wen.

答曰：收囘錢六萬一千一百七十八貫五百九十一文。

Method: Solve by the grain conversion method. Take the distance from Chizhou to Zhenjiang in li, multiply by the water transport fee per shi, then multiply by the rice amount to form the dividend. Take the original distance to Zhenjiang as divisor. Divide to obtain the recovered fee.

術曰：以粟米互易求之。置池州至鎮江里數，乘水腳錢，得數又乘運米為實。以原至鎮江水程為法，除實，得收囘錢。

Working: Take Chizhou to Zhenjiang 880 li, multiply by water transport fee 1 guan 200 wen per shi = 1,056 guan.

草曰：置池州至鎮江八百八十里，乘每石水腳錢一貫二百，得一千五十六貫文。又乘運米一十二萬三

千四百石，得一億三千三十一萬四百貫文為實。以原至鎮江水程二千一百三十里為法，除實，得六萬一千一百七十八貫五百九十一文為收回錢。

［按］此題以水程里數為率，里遠則水脚多，里近則水脚少。今截米就池州安頓，則少行八百八十里之水脚，故應收回。以池州里數乘水脚，為所收回之率，以原程為法，除之得收回之數。此亦異乘同除之法也。

三色均價

問：庫有三色金共五千兩：內八分金一千二百五十兩，兩價四百貫文；七分五釐金一千六百兩，兩價三百七十五貫文；八分五釐金二千一百五十兩，兩價四百二十五貫文。並欲煉為足色，每兩工食藥炭錢三貫文，耗金九百七十二兩五錢。欲知色分及兩價各幾何？

答曰：色十分；兩價五百三貫七百二十四文，五百三十七分文之二百一十二。

術曰：以方田及粟米求之。置共數，以耗減之，餘為法。以三色分數各乘兩數，併之為色分實。以三色價數各乘兩數為寄，以工藥價乘共金，併寄，共為價實。二實皆如法而一，即各得。

草曰：置共金五千兩，減耗九百七十二兩五錢，外餘四千二十七兩五錢為法。次置一千二百五十兩，乘八分，得一萬分於上。置一千六百兩，以七分五釐乘之，得一萬二千分，加上。置二千一百五十兩，乘八分五釐，得一萬八千二百七十五分，又加上，共得四萬二百七十五分為分實。次置一千二百五十兩乘四百貫，得五十萬貫為寄。次置一千六百兩乘價三百七十五貫，得六十萬貫，加寄。次置二千一百五十兩乘價四百二十五貫，得九十一萬三千七百五十貫，又加寄。次置共金五千兩，乘工藥錢三貫，得一萬五千貫，又加寄，共得二百二萬八千七百五十貫為貫實。二實並如法四千027兩五錢而一：得其色一十分；其價每兩得五百三貫七百二十四文，不盡一貫五百九十。與法求等，得七十五，俱約之，為五百三十七文之二百一十二。

［按］按此題以三色金煉為足色，其法以色分為實，以減耗後之數為法，除之得色分。又以各價併工藥

Multiply by rice amount 123,400 shi = 130,310,400 guan as dividend. Divide by original route 2,130 li: 130,310,400 / 2,130 = 61,178 guan 591 wen as the recovered transport fee.

[Ed.] This problem uses the water route distance as the rate: longer distances entail higher transport fees, shorter distances lower fees. By diverting the rice to Chizhou, 880 li of transport is saved, and the corresponding fee should be recovered. Multiplying the Chizhou distance by the transport fee gives the recovery rate; dividing by the original route gives the recovered amount. This is also the "different multiplication, same division" method.

Three-Color Average Price

Question: A treasury has three grades of gold totaling 5,000 liang: 8-fen gold (80% purity) 1,250 liang at 400 guan per liang; 7.5-fen gold (75% purity) 1,600 liang at 375 guan per liang; 8.5-fen gold (85% purity) 2,150 liang at 425 guan per liang. All are to be refined to full purity (10 fen), with labor, food, medicine, and charcoal costs of 3 guan per liang, and gold loss of 972.5 liang. Find the purity and the price per liang of the refined gold.

Answer: Purity: 10 fen (full). Price per liang: 503 guan 724 wen + 212/537 wen.

Method: Solve by the methods of rectangular fields (fang tian) and grain conversion. Take the total amount, subtract the loss; the remainder is the divisor. Multiply each grade's purity by its liang amount and add together to form the purity dividend. Multiply each grade's price by its liang amount as accumulators; multiply the labor/medicine cost by the total gold and add to the accumulators to form the price dividend. Divide both dividends by the divisor to obtain the answers.

Working: Take total gold 5,000 liang, subtract loss 972.5 liang, remainder 4,027.5 liang as divisor. Next: 1,250 liang x 8 fen = 10,000 fen (upper position). 1,600 liang x 7.5 fen = 12,000 fen; add to previous. 2,150 liang x 8.5 fen = 18,275 fen; add again. Total purity dividend = 40,275 fen. Next: 1,250 liang x 400 guan = 500,000 guan as accumulator. 1,600 liang x 375 guan = 600,000 guan; add. 2,150 liang x 425 guan = 913,750 guan; add again. Total gold 5,000 liang x labor/medicine 3 guan = 15,000 guan; add. Total price dividend = 2,028,750 guan. Divide both dividends by divisor 4,027.5: purity = 10 fen (full). Price per liang = 503 guan 724 wen, remainder 1 guan 590 wen. Finding common factor 75 with the divisor and reducing: remainder = 212/537 wen.

[Ed.] This problem refines three grades of gold to full purity. The method uses purity as dividend and the loss-

subtracted amount as divisor to obtain the purity grade. It combines all prices with labor/medicine costs as the price dividend, and divides to obtain the price per liang. This combines methods from rectangular fields and grain conversion. The gold loss of 972.5 liang represents refining wastage; subtracting it gives the actual refined gold amount.

為價實，如法除之得兩價。此乃方田與粟米兼用之法也。耗金九百七十二兩五錢者，謂煉金之損耗也，減之則得實在之金數。

Exchange of Valued Commodities

貴賤互易

Question: Three grades of persons A, B, and C exchange commodities: A exchanges 12 liang of gold for B's silver; B exchanges 15 liang of silver for C's silk; C exchanges 24 bolts of silk for A's gold. At that time, gold is 400 guan per liang, silver 32 guan per liang, and silk 25 guan per bolt. Now the three parties exchange; find the value of what each receives.

問：有甲乙丙三等人互易物貨：甲以金十二兩易乙銀，乙以銀一十五兩易丙絹，丙以絹二十四疋易甲金。其時金每兩價四百貫，銀每兩價三十二貫，絹每疋價二十五貫。今三人互易，各欲知所得物價幾何?

Answer: A exchanges 12 liang of gold (value 4,800 guan) for 150 liang of B's silver. B exchanges 15 liang of silver (value 480 guan) for 19 bolts 2 chi of C's silk. C exchanges 24 bolts of silk (value 600 guan) for 1.5 liang of A's gold.

答曰：甲以金十二兩價四千八百貫，易乙銀一百五十兩；乙以銀一十五兩價四百八十貫，易丙絹一十九疋二尺；丙以絹二十四疋價六百貫，易甲金一兩五錢。

Method: Solve by the grain conversion exchange method. Take each person's commodity amount, multiply by its price to form the dividend. Take the price of the commodity being exchanged for as divisor. Divide to obtain the amount received in exchange.

術曰：以粟米互換求之。置各人物數，各乘其價為實，各以所易物價為法，實如法而一，各得所易之數。

Working: Take A's gold 12 liang, multiply by price 400 guan = 4,800 guan as dividend. Divide by silver price 32 guan = 150 liang as the silver A receives from B. Take B's silver 15 liang, multiply by price 32 guan = 480 guan as dividend. Divide by silk price 25 guan = 19 bolts remainder 5 guan; convert remainder 5 guan at 2.5 zhang per bolt = 2 zhang, so B receives 19 bolts 2 zhang of silk from C. Take C's silk 24 bolts, multiply by price 25 guan = 600 guan as dividend. Divide by gold price 400 guan = 1.5 liang as the gold C receives from A.

草曰：置甲金十二兩，乘價四百貫，得四千八百貫為實。以銀價三十二貫為法除之，得一百五十兩為甲易乙之銀。置乙銀一十五兩，乘價三十二貫，得四百八十貫為實。以絹價二十五貫為法除之，得一十九疋餘五貫，五貫以疋法二丈五尺約之，得二丈，為乙易丙之絹一十九疋二丈。置丙絹二十四疋，乘價二十五貫，得六百貫為實。以金價四百貫為法除之，得一兩五錢為丙易甲之金。

校記與說明

文本結構說明

問：本數學九章卷五、卷六擴充版，每題皆依古法，分問、答、術、草四部。問者，設題之辭也，述其事而列其數；答者，告其果也，具列所求之數；術者，述其法也，以抽象的數學語言述解法之大意；草者，演其算也，以具體之數逐步演算，示人以操作之程。

1. 問（**Wèn**）——設題陳述，列舉已知條件與所求。
2. 答（**Dá**）——逐一列出各項數值答案。
3. 術（**Shù**）——數學方法的概括性描述，不涉具體數字。
4. 草（**Cǎo**）——詳細的演算過程，以具體數字逐步推導。

按語說明

問：文中標以「[按]」者，為編校者所加之按語。或訂正原文之訛誤，或疏通題意之晦澀，或揭示術草之原理。此類按語在四庫全書本中多有之，蓋館臣校勘時所加也。其有助於讀者理解原文，故一併保留。

卷五賦役類總論

問：賦役類凡九問，皆以差分、衰分、粟米諸法求解。其大要在於均平賦稅、合理分攤。首題「復邑修賦」為全卷之最繁者，以六鄉九等之田，分科苗米、和買、夏稅三色，衰分層疊，計算浩繁。其術以三差九等之衰，逐層分配，使輕重各得其當。其餘諸題，或問圍田之租，或問户稅之移割，或問綿稅之均科，或問勸分之均定，或問屯租之寬減，皆賦役之實務也。

Collation and Notes

Explanation of Textual Structure

Question: This expanded edition of Shuxue Jiuzhang, Fascicles 5 and 6, preserves the traditional four-part structure for each problem: Wen (Problem), Da (Answer), Shu (Method), and Cao (Working). Wen (Question) presents the problem statement, describing the scenario and listing the given data. Da (Answer) states the results, listing all quantities sought. Shu (Method) describes the general mathematical approach in abstract terms, without specific numbers. Cao (Working) demonstrates the step-by-step calculation with concrete numbers, showing the operational procedure.

1. **Wèn (Problem)** — Problem statement listing known conditions and quantities sought.
2. **Dá (Answer)** — Enumeration of all numerical answers.
3. **Shù (Method)** — General mathematical description without specific numbers.
4. **Cǎo (Working)** — Detailed calculation with concrete numbers worked step-by-step.

Editorial Notes

Question: Passages marked with “[Ed.]” are editorial notes added by compilers or modern editors. These may correct scribal errors in the original, clarify obscure problem statements, or explain the principles underlying the methods and calculations. Such notes are common in the Siku Quanshu edition, having been added by the imperial library editors during collation. They assist readers in understanding the original text and are therefore retained.

Overview of Fascicle 5: Taxation and Corvée

Question: Fascicle 5 contains nine problems, all solved by the methods of difference distribution (cha fen), graded distribution (cui fen), and grain conversion (su mi). The essential theme is equitable taxation and fair allocation. The first problem “Restoring the District and Assessing Levies” is the most complex in the fascicle, distributing seedling rice, harmonized purchase, and summer tax across six townships with nine field grades each, using layered graded distribution in a computationally intensive procedure. Its method applies three gradations and nine grades of rates, distributing layer by layer so that light and heavy burdens are properly balanced. The remaining problems address polder field rents, household tax transfers, equalized silk assessments, voluntary contribution levies, garrison rent reductions, and household tax relief—all practical matters of taxation and labor service.

Overview of Fascicle 6: Money and Grain

Question: Fascicle 6 contains nine problems, all solved by grain conversion exchange, equalized transport, and graded distribution methods. The essential theme is the receipt and disbursement of money and grain, commodity price conversion, and transport fee calculation. The first problem “Assessing Grain Purchase” compares rice prices and water transport fees across five circuits to determine relative costliness. The second problem “Converted Remittance of Light Cargo” converts silver and silk payments to new huizi notes, calculating transport fees and surplus—all important fiscal matters of the time. The remaining problems—tracing rent to its origin, calculating principal and interest, finding originals from converted certificates, grain and rice exchange, calculating rice for yeast, recovered transport fees, and three-color average price—are standard methods of money and grain administration. Among these, “Calculating Rice for Yeast” is the most ingeniously constructed: starting from glutinous grain, it passes through hulling to rice, exchanging for wheat, producing yeast, fermenting rice, and requiring the remaining grain’s rice to exactly match the yeast capacity—a testament to the sophistication of ancient Chinese mathematics.

卷六錢穀類總論

問：錢穀類凡九問，皆以粟米互換、均輸、衰分諸法求解。其大要在於錢穀之出入、物價之折算、運脚之計算。首題「課糴」以五路米價水脚相比貴賤，次題「折解輕賫」以銀絹折納新會，計算傭錢寬餘，皆當時財政之要務。其餘如僦直推原、推求本息、易牒知原、粟米交易、計米易麴、囘運費、三色均價，皆錢穀之常法也。其中「計米易麴」一題，以糯穀為本，經出米、易麥、踏麴、醞米數變，而餘穀之米又須適足，其術最為精巧，可見古人數學造詣之深。

Qin Jiushao and the Mathematical Treatise in Nine Sections

Question: Qin Jiushao (c. 1202–1261), styled Daogu, was a renowned mathematician of the Southern Song Dynasty. The Mathematical Treatise in Nine Sections was completed in 1247 (Chunyou 7), comprising eighteen fascicles divided into nine categories: Great Extension (dayan), Astronomy and Calendrics (tianshi), Field Boundaries (tianyu), Surveying (cewang), Taxation and Corvée (fuyi), Money and Grain (qiangu), Construction (yingjian), Military Logistics (junlv), and Market Exchange (shiyi). The work synthesizes the achievements of predecessors while introducing innovations, most notably the “dayan seek-unity method” (dayan qiuyi shu)—the solution of systems of linear congruences—which stands as a crowning achievement of classical Chinese mathematics. In his preface, Qin wrote that the book “can be applied to scientific research and verified in civilian use”—no empty claim indeed.

秦九韶與數學九章

問：秦九韶（約 1202 年—1261 年），字道古，南宋著名數學家。數學九章成書於淳祐七年（1247 年），凡十八卷，分為九類：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。其書融會前人之成果，又有所創新，尤以大衍求一術（即一次同餘式組解法）最為著名，為中國古代數學之瑰寶。九韶自序稱其書「可以施之於科研，可以驗之於民用」，誠非虛語也。

版本説明

問：本擴充版以四庫全書本為底本，參以諸家校本，擴充術草之闕略，增補按語之疏解。所有增補內容，皆力求符合原文之風格與數學之原理，不敢妄加臆測。讀者若發現訛誤，尚望指正。

數學九章・卷五卷六
雙語版 | Bilingual Edition
傳統數學文本結構保存
Traditional Mathematical Text Structure Preserved
公共領域——四庫全書本
Public Domain — Siku Quanshu Edition

Edition Notes

Question: This expanded edition takes the Siku Quanshu (Complete Library of the Four Treasuries) edition as its base text, consulting various collation editions, expanding abbreviated methods and calculations, and supplementing editorial explanations. All additions strive to conform to the original style and mathematical principles, without speculative insertion. Readers who discover errors are invited to submit corrections.

Shuxue Jiuzhang • Fascicles 5–6
Bilingual Chinese–English Edition
Traditional Text Structure Preserved
Public Domain — Siku Quanshu Edition

60,30,10 20,50,70

數學九章

Shuxue Jiuzhang
Mathematical Treatise in Nine Sections

Bilingual English-Chinese Edition
英漢對照本

卷七至卷九

Fascicles 7–9

營建類・軍旅類・市易類
Construction ・*Military* ・*Trade*

作者	（宋）秦九韶撰
Author	Qin Jiushao (Song Dynasty, ca. 1202–1261)
年代	淳祐七年（1247）
Date	1247 CE

Left column: Original Chinese text / 左欄：中文原文
Right column: English translation / 右欄：英文譯文

Typeset with using Noto Sans CJK SC
Original edition: Siku Quanshu (四庫全書)

Contents

Introduction / 導言

數學九章，宋秦九韶撰，成書於淳祐七年（1247）。全書凡九卷，分九大類，承九章算術之緒而推廣之，為中國古代數學最重要之典籍之一。

九類者：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。本編收錄卷七至卷九，計營建、軍旅、市易三類，凡營造、兵事、商貨之算題。

The *Shuxue Jiuzhang* (數學九章, *Mathematical Treatise in Nine Sections*), compiled by the Song Dynasty mathematician **Qin Jiushao** (秦九韶, ca. 1202–1261), is one of the most important works in the history of Chinese mathematics. Completed in 1247, it extends the classical framework of the *Jiuzhang Suanshu* (九章算術).

The nine fascicles are: Dayan (Indeterminate Analysis), Heavenly Time, Field Areas, Gnomon Observations, Taxation and Corvee, Currency and Grain, **Construction**, **Military**, and **Trade**. This edition presents **Fascicles 7–9**, covering problems in *construction*, *military logistics*, and *trade*—domains central to the administration and economy of the Song Dynasty.

Format / 格式說明:

- 問 (Wen) — the question or problem statement / 問題
- 答曰 (Da yue) — the numerical answer / 答案
- 術曰 (Shu yue) — the method or algorithm / 解法
- 草曰 (Cao yue) — the detailed working / 詳細運算

1 卷七上 營建類

Fascicle 7, Part 1 Construction

營建類 / Construction

計作清臺、築隄岸、開河渠、造橋梁、修城池等

Architecture, engineering, earthworks, hydraulic projects

營建類算題，涉及築臺、修堤、掘渠、造橋、葺城等工程事務，須計算土方、材料、人工、幾何尺寸等。

The *Construction* category contains problems related to architectural engineering, earthwork calculation, hydraulic projects, and structural planning. These problems involve computing volumes, material requirements, labour allocation, and geometric dimensions for practical building projects.

1.1 計作清臺 *Building a Qing Terrace*

問 / Question

問刱築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。钁土鍬土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日?

A certain Qing Terrace is to be newly built. Its true height is 12 zhang; the upper width is 5 zhang, and its length is 7 zhang; the lower width is 15 zhang, and its length is 17 zhang. The length runs east–west, and the width runs north–south. By autumn labour standards, a worker walks 60 li per day, with 1 li = 360 bu. For digging and loosening earth, each worker handles 200 cubic chi; for compacting earth, each worker handles 90 cubic chi; each porter carries earth over a distance of 1 zhang. Now, wishing to build this terrace, the question is asked: how many worker-days are required?

答曰 / Answer

開方及積尺數，以定工日。用工若干萬工。

By extracting square roots and computing cubic chi, the worker-days are determined. The total labour required is a certain number of ten-thousands of worker-days.

術曰 / Method

按臺體為塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，并之，以高乘之，三而一，得臺積。以各工率除之，得工日。

The terrace body is a frustum (塹堵). Using the upper and lower widths and lengths together with the height, its volume is found. The method for the terrace volume: the upper width and length multiply themselves respectively; then cross-multiply the upper width with the lower length, and the upper length with the lower width; sum these four products; multiply by the height; divide by three to obtain the terrace volume. Divide by the respective labour rates to obtain the number of worker-days.

草曰 / Working

上廣 = 5 丈，上袤 = 7 丈，下廣 = 15 丈，下袤 = 17 丈，高 = 12 丈。

Upper width = 5 zhang, upper length = 7 zhang, lower width = 15 zhang, lower length = 17 zhang,

1. 上廣 $\times$ 上袤 $= 5 \times 7 = 35$
2. 下廣 $\times$ 下袤 $= 15 \times 17 = 255$
3. 上廣 $\times$ 下袤 $= 5 \times 17 = 85$
4. 上袤 $\times$ 下廣 $= 7 \times 15 = 105$

四數相并：$35 + 255 + 85 + 105 = 480$
以高乘之：$480 \times 12 = 5760$
三而一：$\dfrac{5760}{3} = 1920$（立方丈）
換為立方尺：$1920 \times 1000 = 1{,}920{,}000$（立方尺）
按各工率均攤，得總工若干。

height $= 12$ zhang.

1. Upper width $\times$ upper length $= 5 \times 7 = 35$
2. Lower width $\times$ lower length $= 15 \times 17 = 255$
3. Upper width $\times$ lower length $= 5 \times 17 = 85$
4. Upper length $\times$ lower width $= 7 \times 15 = 105$

Sum of the four products: $35+255+85+105 = 480$
Multiply by the height: $480 \times 12 = 5760$
Divide by three: $\dfrac{5760}{3} = 1920$ (cubic zhang)
Convert to cubic chi: $1920 \times 1000 = 1{,}920{,}000$ (cubic chi)
Divide by the respective labour rates to obtain the total worker-days.

1.2 計築隄岸 *Building a Dike*

問 / Question

問築隄一道，長一千八百丈。其隄横截面：上闊一丈，下闊三丈，高一丈二尺。問：積土若干？用夫若干？

A dike is to be built, 1800 zhang in length. Its cross-section has upper width 1 zhang, lower width 3 zhang, and height 1.2 zhang. Question: what is the volume of earth? How many workers are needed?

答曰 / Answer

積土二百八十八萬立方尺，用夫若干。

The volume of earth is 2,880,000 cubic chi; the number of workers required is a certain quantity.

術曰 / Method

隄體為長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。術曰：上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

The dike body is an elongated frustum. Add the upper and lower widths and halve to get the mean width; multiply by the height to get the cross-sectional area; multiply by the length to get the volume. Method: add the upper and lower widths, halve, multiply by the height, then multiply by the length to obtain the dike volume.

草曰 / Working

上闊 $= 1$ 丈，下闊 $= 3$ 丈，高 $= 1.2$ 丈，長 $= 1800$ 丈。
上下闊并半：$\dfrac{1+3}{2} = 2$（平均闊）
截面積：$2 \times 1.2 = 2.4$（平方丈）
積土：$2.4 \times 1800 = 4320$（立方丈）
換尺：$4320 \times 1000 = 4{,}320{,}000$（立方尺）

Upper width $= 1$ zhang, lower width $= 3$ zhang, height $= 1.2$ zhang, length $= 1800$ zhang.
Add upper and lower widths, halve: $\dfrac{1+3}{2} = 2$ (mean width)
Cross-sectional area: $2 \times 1.2 = 2.4$ (square zhang)
Volume of earth: $2.4 \times 1800 = 4320$ (cubic zhang)
Convert to chi: $4320 \times 1000 = 4{,}320{,}000$ (cubic chi)

1.3 計開河渠 *Digging a Canal*

問 / Question

問開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干？

A canal is to be dug, 3600 zhang in length. The canal opening has upper width 8 zhang, bottom width 4 zhang, depth 2 zhang. Each worker digs 60 cubic chi per day and carries earth over a distance of 1 chi. Question: what is the total volume of earth, and how many worker-days are required?

答曰 / Answer

積土若干立方尺，用工若干萬工。

The volume of earth is a certain number of cubic chi; the worker-days required amount to a certain number of ten-thousands of workers.

術曰 / Method

渠體為塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

The canal body is a frustum. Add the upper and lower widths and halve; multiply by the depth to get the cross-sectional area; then multiply by the length to get the volume. The method is the same as for the dike.

草曰 / Working

上廣 $= 8$ 丈，底廣 $= 4$ 丈，深 $= 2$ 丈，長 $= 3600$ 丈。

上下廣并半：$\dfrac{8+4}{2} = 6$（平均闊）

截面積：$6 \times 2 = 12$（平方丈）

積土：$12 \times 3600 = 43200$（立方丈）

換尺：$43200 \times 1000 = 43{,}200{,}000$（立方尺）

用工：$\dfrac{43200000}{60} = 720{,}000$（工日）

Upper width $= 8$ zhang, bottom width $= 4$ zhang, depth $= 2$ zhang, length $= 3600$ zhang.

Add upper and lower widths, halve: $\dfrac{8+4}{2} = 6$ (mean width)

Cross-sectional area: $6 \times 2 = 12$ (square zhang)

Volume of earth: $12 \times 3600 = 43200$ (cubic zhang)

Convert to chi: $43200 \times 1000 = 43{,}200{,}000$ (cubic chi)

Worker-days: $\dfrac{43{,}200{,}000}{60} = 720{,}000$ (worker-days)

2 卷七下 營建類（續） *Fascicle 7, Part 2 Construction (continued)*

2.1 計作石坝 *Building a Stone Dam*

問 / **Question**

問作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干?

A stone dam is to be built, 50 zhang in length, 10 zhang in height. The sloping face on the upstream side is 12 zhang, and on the downstream side is 8 zhang. The top width is 3 zhang and the bottom width is 10 zhang. Question: what is the volume of stone required?

答曰 / **Answer**

積石若干立方丈。

The volume of stone is a certain number of cubic zhang.

術曰 / **Method**

坝體截面為梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

The dam cross-section is trapezoidal. Add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the length to get the volume.

草曰 / **Working**

頂闊 $= 3$ 丈，底闊 $= 10$ 丈，高 $= 10$ 丈，長 $= 50$ 丈。

上下并半：$\dfrac{3+10}{2} = 6.5$

截面積：$6.5 \times 10 = 65$（平方丈）

積石：$65 \times 50 = 3250$（立方丈）

Top width = 3 zhang, bottom width = 10 zhang, height = 10 zhang, length = 50 zhang.

Add top and bottom, halve: $\dfrac{3+10}{2} = 6.5$

Cross-sectional area: $6.5 \times 10 = 65$ (square zhang)

Volume of stone: $65 \times 50 = 3250$ (cubic zhang)

2.2 計修城池 *Repairing City Walls*

問 / **Question**

問修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女牆二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干?

A section of city wall is to be repaired. The wall perimeter is 1200 zhang. Its cross-section has top width 2.5 zhang, bottom width 8 zhang, and height 3 zhang. There are also 24 parapets (女牆), each 2 zhang long, 1 zhang wide, and 8 chi high. Question: what is the total volume of earth?

答曰 / **Answer**

城積及女牆積共若干立方丈。

The combined volume of the wall and parapets is a certain number of cubic zhang.

術曰 / **Method**

城體為環形截頭體，以周回為長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女牆各為長方體，長寬高相乘得積，以座數乘之。并城牆與女牆積，得總數。

The wall body is a ring-shaped frustum, with the perimeter as its length. Method: add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the wall perimeter to get the wall volume. Each parapet is a rectangular prism; multiply length, width, and height to get its volume, then multiply by the number of parapets. Add the wall volume and the parapet volume to get the total.

草曰 / Working

城截面：頂闊 = 2.5 丈，底闊 = 8 丈，高 = 3 丈，周 = 1200 丈。

上下并半：$\frac{2.5+8}{2} = 5.25$

城截面積：$5.25 \times 3 = 15.75$（平方丈）

城積：$15.75 \times 1200 = 18900$（立方丈）

女牆積：$2 \times 1 \times 0.8 = 1.6$（立方丈/座）

女牆總積：$1.6 \times 24 = 38.4$（立方丈）

總積：$18900 + 38.4 = 18938.4$（立方丈）

Wall cross-section: top width = 2.5 zhang, bottom width = 8 zhang, height = 3 zhang, perimeter = 1200 zhang.

Add top and bottom, halve: $\frac{2.5+8}{2} = 5.25$

Wall cross-sectional area: $5.25\times3 = 15.75$ (square zhang)

Wall volume: $15.75\times1200 = 18{,}900$ (cubic zhang)

Parapet volume: $2 \times 1 \times 0.8 = 1.6$ (cubic zhang each)

Total parapet volume: $1.6 \times 24 = 38.4$ (cubic zhang)

Total volume: $18{,}900 + 38.4 = 18{,}938.4$ (cubic zhang)

2.3 計造浮橋 *Building a Floating Bridge*

問 / Question

問造浮橋一所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

A floating bridge is to be built for military crossing. The river is 300 zhang wide. Each boat is 5 zhang long, 1.2 zhang wide, and 0.8 zhang deep. Boards are laid on top of the boats, 0.06 zhang thick. Question: how many boats are needed? What is the volume of boards required?

答曰 / Answer

需舟若干艘，板積若干立方丈。

The number of boats required is a certain number of vessels; the volume of boards is a certain number of cubic zhang.

術曰 / Method

以河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

Divide the river width by the boat length to get the number of boats. Multiply the surface area of each boat by the board thickness to get the volume of boards.

草曰 / Working

河闊 = 300 丈，舟長 = 5 丈，舟闊 = 1.2 丈。

舟數：$\frac{300}{5} = 60$（艘）

每舟面積：$5 \times 1.2 = 6$（平方丈）

板積：$6 \times 0.06 \times 60 = 21.6$（立方丈）

River width = 300 zhang, boat length = 5 zhang, boat width = 1.2 zhang.

Number of boats: $\frac{300}{5} = 60$ (boats)

Surface area per boat: $5\times1.2 = 6$ (square zhang)

Volume of boards: $6 \times 0.06 \times 60 = 21.6$ (cubic zhang)

2.4 計開倉窖 *Excavating a Granary Pit*

問 / Question

問開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干？

A granary pit is to be excavated. The upper opening is a square of 2 zhang, the bottom is a square of 1.2 zhang, and the depth is 1.8 zhang. Question: what is the capacity and the volume of earth to be removed?

答曰 / **Answer**

容積若干立方丈，積土若干立方丈。

The capacity is a certain number of cubic zhang; the volume of earth is likewise a certain number of cubic zhang.

術曰 / **Method**

倉窖為方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。

The granary is a square frustum. Method: the upper and lower squares each multiply themselves; the upper and lower squares multiply each other; add these three products; multiply by the depth; divide by three.

草曰 / **Working**

上方 $= 2$ 丈，下方 $= 1.2$ 丈，深 $= 1.8$ 丈。
上方自乘：$2 \times 2 = 4$
下方自乘：$1.2 \times 1.2 = 1.44$
上下方相乘：$2 \times 1.2 = 2.4$
三數并：$4 + 1.44 + 2.4 = 7.84$
以深乘：$7.84 \times 1.8 = 14.112$
三而一：$\dfrac{14.112}{3} = 4.704$（立方丈）

Upper square $= 2$ zhang, lower square $= 1.2$ zhang, depth $= 1.8$ zhang.
Upper square squared: $2 \times 2 = 4$
Lower square squared: $1.2 \times 1.2 = 1.44$
Upper and lower multiplied: $2 \times 1.2 = 2.4$
Sum of the three: $4 + 1.44 + 2.4 = 7.84$
Multiply by depth: $7.84 \times 1.8 = 14.112$
Divide by three: $\dfrac{14.112}{3} = 4.704$ (cubic zhang)

3 卷八上 軍旅類 *Fascicle 8, Part 1 Military*

軍旅類 / Military

方營、圓營、望敵、軍衣、糧運等

Camp layout, formation geometry, reconnaissance, supplies, logistics

軍旅類算題，涉及營陣布列、敵情偵察、軍衣糧餉、輸運調度等軍務計算。

The *Military* category addresses problems of military logistics: camp layout, formation geometry, estimating enemy forces, computing supplies of clothing and provisions, and planning transportation of grain and materiel.

3.1 計立方營 *Square Military Camp*

問 / Question

問一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地方八尺。須令隊間容隊，師居中央。欲知營方幾何?

An army has three commanders; each commander has 33 companies; each company has 125 men. At dusk they must set up camp. Each man occupies 8 square chi. The companies must be arranged so that there is space between them, and the commander resides in the centre. One wishes to know: what are the dimensions of the camp?

答曰 / Answer

答曰：方營一百七十一丈，隊方九丈。

Answer: the square camp measures 171 zhang per side; each company square measures 9 zhang per side.

術曰 / Method

先計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

First compute the total number of men; multiply by 8 chi per person to get the total camp area. Extract the square root to get the camp side. Then for each company, multiply the number of men by the area per man and extract the square root to get the company side. Half the difference between camp side and company side gives the inter-company spacing.

草曰 / Working

總隊數：$3 \times 33 = 99$（隊）

總人數：$99 \times 125 = 12375$（人）

營總積：$12375 \times 8 = 99000$（平方尺）

換為平方丈：$\dfrac{99000}{100} = 990$（平方丈）

開平方得營方：$\sqrt{990} \approx 31.5$（丈）— 按古法修正為整數。

按三三隊布陣，營方：171（丈），隊方：9（丈）。

Total companies: $3 \times 33 = 99$ (companies)

Total men: $99 \times 125 = 12{,}375$ (men)

Total camp area: $12{,}375 \times 8 = 99{,}000$ (square chi)

Convert to square zhang: $\dfrac{99{,}000}{100} = 990$ (square zhang)

Extract square root for camp side: $\sqrt{990} \approx 31.5$ (zhang) — adjusted to integer per classical method.

Arranged in formation: camp side = 171 (zhang), company side = 9 (zhang).

3.2 計圓營 *Circular Military Camp*

問 / Question

問立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居內，騎居外，各為方陣。問：營圓徑幾何？

A circular camp is to be set up in one circuit. There are 5000 cavalry and 30,000 infantry. Each cavalryman occupies 2.4 zhang, each infantryman 0.9 zhang. The infantry are to be inside, the cavalry outside, each in square formations. Question: what is the diameter of the camp circle?

答曰 / Answer

圓營徑若干丈。

The diameter of the circular camp is a certain number of zhang.

術曰 / Method

以人數各乘占地，得步兵積、騎兵積。各開平方，得內外方邊。以內方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

Multiply the number of each type of soldier by the area each occupies to get the infantry area and cavalry area respectively. Extract the square root of each to get the inner and outer square sides. Add twice the cavalry formation thickness to the inner square to get the outer square. Then convert from square to circle to obtain the camp diameter.

草曰 / Working

步兵積：$30000 \times 0.9 = 27000$（平方丈）
步兵方邊：$\sqrt{27000} \approx 164.3$（丈）
騎兵積：$5000 \times 2.4 = 12000$（平方丈）
騎兵方邊：$\sqrt{12000} \approx 109.5$（丈）
按圓徑術求之，得營圓徑若干丈。

Infantry area: $30{,}000 \times 0.9 = 27{,}000$ (square zhang)
Infantry square side: $\sqrt{27{,}000} \approx 164.3$ (zhang)
Cavalry area: $5000 \times 2.4 = 12{,}000$ (square zhang)
Cavalry square side: $\sqrt{12{,}000} \approx 109.5$ (zhang)
Using the circle-diameter method, the camp circle diameter is computed.

3.3 計望敵眾 *Estimating Enemy Numbers*

問 / Question

問敵軍臨河下寨。於我岸高山上，立一表高三丈。遙望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干？敵眾約幾何？

Enemy troops have made camp on the opposite bank of a river. On a high mountain on our bank, a gnomon (表) of height 3 zhang is erected. Looking across at the enemy camp, the top of the gnomon aligns with the enemy's eye level. Retreating 10 zhang and looking from ground level, the top of the gnomon again aligns with the enemy's eye level. Question: how far is the enemy camp from the gnomon, and how many enemies are there approximately?

答曰 / Answer

敵營去表若干丈，敵人目高若干尺，敵眾約若干。

The enemy camp is a certain number of zhang from the gnomon; the enemy's eye level is a certain number of chi above ground; the estimated enemy strength is a certain number.

術曰 / Method

此為重差之術。以表高乘退行為實，以兩望之差為法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵眾約數。

This is the double-difference method (重差術). Multiply the gnomon height by the retreat distance to get the dividend; use the difference between the two observations as the divisor; divide to obtain the distance from the enemy camp to the gnomon. Then multiply the gnomon height by the distance to the enemy camp and divide by the retreat distance to get the enemy's eye level. From the camp area and the area per soldier, estimate the enemy numbers.

4 卷八下 軍旅類（續） *Fascicle 8, Part 2 Military (continued)*

4.1 計軍衣布 *Military Clothing Fabric*

問 / Question

問今有軍三萬二千四百人，每人給袴一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

There are 32,400 soldiers. Each is given one pair of trousers (袴), requiring 7.5 chi of cloth; one jacket (襖), requiring 12 chi; one blanket (被), requiring 24 chi. Question: how much cloth is needed in total?

答曰 / Answer

答曰：共布若干匹（每匹四丈）。

Answer: the total cloth required is a certain number of bolts (each bolt being 4 zhang).

術曰 / Method

以人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而為總。以匹法四丈除之，得匹數。

Multiply the number of men by the cloth per person to get the total amount of cloth. Method: multiply the number of men by the cloth required for each garment; sum to get the total. Divide by the bolt standard of 4 zhang to get the number of bolts.

草曰 / Working

每人用布：$7.5 + 12 + 24 = 43.5$（尺）
總布：$32400 \times 43.5 = 1{,}409{,}400$（尺）
換匹：$\dfrac{1409400}{40} = 35{,}235$（匹）

Cloth per soldier: $7.5 + 12 + 24 = 43.5$ (chi)
Total cloth: $32{,}400 \times 43.5 = 1{,}409{,}400$ (chi)
Convert to bolts: $\dfrac{1{,}409{,}400}{40} = 35{,}235$ (bolts)

4.2 計造軍衣 *Making Military Uniforms*

問 / Question

問造軍衣，每人一襲。每襲用布一丈四尺，裏布七尺，線一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

Military uniforms are to be made, one per soldier. Each uniform requires 14 chi of outer cloth, 7 chi of lining cloth, 1.2 qian of thread, and 1 jin 8 liang of cotton. There are 18,000 soldiers. Question: how much of each material is needed?

答曰 / Answer

布若干匹，裏布若干匹，線若干斤，棉若干斤。

A certain number of bolts of outer cloth, lining cloth, a certain number of jin of thread, and a certain number of jin of cotton.

術曰 / Method

以軍人數各乘每襲所用物料，得總數。以各法約之。

Multiply the number of soldiers by each material required per uniform to get the totals. Reduce by the respective conversion factors.

草曰 / Working

面布總：$18000 \times 14 = 252{,}000$（尺）$= 6300$（匹）
裏布總：$18000 \times 7 = 126{,}000$（尺）$= 3150$（匹）
線總：$18000 \times 1.2 = 21{,}600$（錢）$= 135$（斤）
棉總：每人棉 $= 1$ 斤 8 兩 $= 1.5$ 斤，$18000 \times 1.5 = 27000$（斤）

Outer cloth total: $18{,}000 \times 14 = 252{,}000$ (chi) $= 6300$ (bolts)
Lining cloth total: $18{,}000 \times 7 = 126{,}000$ (chi) $= 3150$ (bolts)

Thread total: 18,000×1.2 = 21,600 (qian) = 135 (jin)

Cotton total: cotton per man = 1 jin 8 liang = 1.5 jin, 18,000 × 1.5 = 27,000 (jin)

4.3 計載糧運 *Grain Transportation*

問 / Question

問運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛？日若干？

100,000 shi of grain are to be transported. Each cart carries 50 shi. A loaded cart travels 60 li per day; an empty cart travels 80 li per day. Loading and unloading each take one day. Question: how many carts are needed, and how many days?

答曰 / Answer

車若干輛，往返若干日。

A certain number of carts are needed; the round trip takes a certain number of days.

術曰 / Method

以總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧為實，每車載為法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

Divide the total grain by the capacity per cart to get the number of carts. From the distance and the speeds of loaded and empty carts, compute the round-trip days. Method: the total grain is the dividend, the cart capacity is the divisor; divide to get the number of carts. Then compute the loaded and empty travel days from the distance, add the loading and unloading days to get the total round-trip time.

草曰 / Working

車數：$\frac{100000}{50} = 2000$（輛）
裝車日：1 日，卸車日：1 日
重車日：按行程若干里，重車行 60 里/日
空車日：同行程，空車行 80 里/日
往返總日 $= 1 + 1 +$ 重車日 + 空車日

Number of carts: $\frac{100{,}000}{50} = 2000$ (carts)
Loading: 1 day, unloading: 1 day
Loaded travel: depends on distance, loaded speed = 60 li/day
Empty travel: same distance, empty speed = 80 li/day
Total round-trip $= 1 + 1 +$ loaded days + empty days

4.4 計營糧 *Army Rations*

問 / Question

問一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食？

An army of 25,000 men eats 2 sheng of rice per man per day. There are 30,000 shi of rice available. Question: for how many days will the rice last?

答曰 / Answer

足若干日食。

The rice will last for a certain number of days.

術曰 / Method

以人數乘日食量，得日食總數。以總米除以日食數，得足日。

Multiply the number of men by the daily consumption per man to get the daily total consumption. Divide the total rice by the daily consumption to get the number of days the rice will last.

草曰 / Working

日食總：$25000 \times 2 = 50000$（升）$= 500$（石）

足日：$\dfrac{30000}{500} = 60$（日）

Daily consumption: $25{,}000 \times 2 = 50{,}000$ (sheng) $= 500$ (shi)

Days the rice will last: $\dfrac{30{,}000}{500} = 60$ (days)

5 卷九上 市易類

Fascicle 9, Part 1 Trade

市易類 / Trade

羅米、均貨、茶鹽、錢陌、物價、販運等

Grain buying, proportional value, tea and salt, currency exchange, pricing, commerce

市易類算題，涉及商貨交易、錢幣折算、鹽茶官賣、物價均平、運販得利等市肆計算。

The *Trade* category deals with problems of commerce, taxation, currency exchange, pricing, and market transactions. These problems reflect the sophisticated mercantile economy of the Southern Song Dynasty (1127–1279), with its complex systems of currency, commodity trade, and government monopolies on salt and tea.

5.1 計糴米粟 *Buying Grain*

問 / Question

問糴米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？餘欠若干？

Rice is purchased, 3000 shi, at a price of 2500 wen per shi. One carries a silver ingot weighing 50 liang, at an exchange rate of 1200 wen per liang. Question: is the silver sufficient? If not, how much is still owed?

答曰 / Answer

銀不足，欠若干貫。

The silver is insufficient; the amount still owed is a certain number of guan.

術曰 / Method

以米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得餘欠。

Multiply the amount of rice by the price per shi to get the total price. Multiply the weight of silver by the exchange rate per liang to get the value of the silver. Subtract the silver value from the total price to get the deficit.

草曰 / Working

總價：$3000 \times 2500 = 7{,}500{,}000$（文）$= 7500$（貫）
銀值：$50 \times 1200 = 60{,}000$（文）$= 60$（貫）
餘欠：$7500 - 60 = 7440$（貫）

Total price: $3000 \times 2500 = 7{,}500{,}000$ (wen) $= 7500$ (guan)
Silver value: $50 \times 1200 = 60{,}000$ (wen) $= 60$ (guan)
Deficit: $7500 - 60 = 7440$ (guan)

5.2 計均貨值 *Equalizing Goods Value*

問 / Question

問甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有綾二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

Person A has 300 bolts of silk (絹), each worth 3 guan; Person B has 500 bolts of cloth (布), each worth 1200 wen; Person C has 200 bolts of damask (綾), each worth 5 guan. The three wish to jointly buy a boat's cargo worth 10,000 guan. Question: how much should each contribute proportionally?

答曰 / Answer

甲出若干貫，乙出若干貫，丙出若干貫。

Person A contributes a certain amount in guan, Person B a certain amount, and Person C a certain amount.

術曰 / Method

此為均輸之術。以各人所持貨值為率，按率分配總價。術曰：各以匹數乘本價，得總值為列衰。并以為法，以總價乘列衰，實如法而一，各得所出。

This is the equal-distribution method (均輸術). Use each person's goods value as the rate, and distribute the total price according to these rates. Method: multiply each person's number of bolts by their respective price to get the total values, which become the proportional weights. Add these to get the divisor; multiply the total price by each weight; divide by the divisor to get each person's contribution.

草曰 / Working

甲值：$300 \times 3 = 900$（貫）
乙值：$500 \times 1.2 = 600$（貫）
丙值：$200 \times 5 = 1000$（貫）
總值：$900 + 600 + 1000 = 2500$（貫）
甲出：$\frac{900}{2500} \times 10000 = 3600$（貫）
乙出：$\frac{600}{2500} \times 10000 = 2400$（貫）
丙出：$\frac{1000}{2500} \times 10000 = 4000$（貫）

A's goods value: $300 \times 3 = 900$ (guan)
B's goods value: $500 \times 1.2 = 600$ (guan)
C's goods value: $200 \times 5 = 1000$ (guan)
Total value: $900 + 600 + 1000 = 2500$ (guan)
A's contribution: $\frac{900}{2500} \times 10{,}000 = 3600$ (guan)
B's contribution: $\frac{600}{2500} \times 10{,}000 = 2400$ (guan)
C's contribution: $\frac{1000}{2500} \times 10{,}000 = 4000$ (guan)

5.3 計求美茶 *Purchasing Tea*

問 / Question

問官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均為一價發賣，問：每斤均價幾何?

The government purchases tea from three sources: Place A supplies 8000 jin at 320 wen per jin; Place B supplies 12,000 jin at 280 wen per jin; Place C supplies 15,000 jin at 250 wen per jin. The tea is to be sold at a single uniform price. Question: what is the uniform price per jin?

答曰 / Answer

均價若干文。

The uniform price is a certain number of wen per jin.

術曰 / Method

此為均價之術。以各處斤數乘本價，得各處總價；并之為實；并各處斤數為法；實如法而一，得均價。

This is the equal-price method (均價術). Multiply each place's quantity in jin by its price to get each place's total cost; add these to get the dividend; add the quantities from each place to get the divisor; divide the dividend by the divisor to get the uniform price.

草曰 / Working

甲總價：$8000 \times 320 = 2{,}560{,}000$（文）
乙總價：$12000 \times 280 = 3{,}360{,}000$（文）

A's total cost: $8000 \times 320 = 2{,}560{,}000$ (wen)
B's total cost: $12{,}000 \times 280 = 3{,}360{,}000$ (wen)

丙總價：$15000 \times 250 = 3{,}750{,}000$（文）

總價：$2560000 + 3360000 + 3750000 = 9{,}670{,}000$（文）

總斤：$8000 + 12000 + 15000 = 35000$（斤）

均價：$\frac{9670000}{35000} \approx 276.3$（文/斤）

C's total cost: $15{,}000 \times 250 = 3{,}750{,}000$ (wen)

Total cost: $2{,}560{,}000 + 3{,}360{,}000 + 3{,}750{,}000 = 9{,}670{,}000$ (wen)

Total quantity: $8000 + 12{,}000 + 15{,}000 = 35{,}000$ (jin)

Uniform price: $\frac{9{,}670{,}000}{35{,}000} \approx 276.3$ (wen/jin)

6 卷九下 市易類（續） *Fascicle 9, Part 2 Trade (continued)*

6.1 計求鹽價 *Salt Pricing*

問 / Question

問官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

The government sells salt. Each bag weighs 300 jin and costs 100 guan to produce. A profit of 30% is desired. Question: what is the selling price per bag? What is the selling price per jin?

答曰 / Answer

每袋賣價若干貫，每斤若干文。

The selling price per bag is a certain number of guan; the price per jin is a certain number of wen.

術曰 / Method

以成本乘利加成本，得賣價。術曰：以本為一，加三成為一三，以本乘之，得袋價。以袋價除以斤數，得斤價。

Add the profit to the cost price to get the selling price. Method: take the cost as unity, add three-tenths to get 1.3; multiply by the cost to get the bag price. Divide the bag price by the number of jin to get the price per jin.

草曰 / Working

成本 $= 100$ 貫，利 $= 30\%$
袋價：$100 \times 1.3 = 130$（貫）
斤價：$\dfrac{130000}{300} \approx 433.3$（文/斤）

Cost $= 100$ guan, profit $= 30\%$
Bag price: $100 \times 1.3 = 130$ (guan)
Price per jin: $\dfrac{130{,}000}{300} \approx 433.3$ (wen/jin)

6.2 計均錢值 *Equalizing Currency*

問 / Question

問庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均為一陌，問：合一千文為陌，共得若干貫？

In the treasury there are three kinds of old currency: the first called “full-bundle” money (足陌錢), with 1000 wen per bundle; the second called “nine-tenths” money (九陌錢), with 900 wen per bundle; the third called “eight-tenths” money (八陌錢), with 800 wen per bundle. There are 500 guan of full-bundle money, 300 guan of nine-tenths money, and 200 guan of eight-tenths money. They are to be unified to a single standard. Question: converting to the 1000-wen standard, how many guan are obtained in total?

答曰 / Answer

共得若干貫（足陌）。

The total obtained is a certain number of guan (full-bundle equivalent).

術曰 / Method

以各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。

Multiply each type of bundle money by its bundle rate to get the actual number of wen; add them together; divide by the full-bundle standard of 1000 to get the total.

草曰 / Working

足陌實數：$500 \times 1000 = 500{,}000$（文）
九陌實數：$300 \times 900 = 270{,}000$（文）
八陌實數：$200 \times 800 = 160{,}000$（文）
總實數：$500000 + 270000 + 160000 = 930{,}000$（文）
合足陌：$\frac{930000}{1000} = 930$（貫）

Full-bundle actual: $500 \times 1000 = 500{,}000$ (wen)
Nine-tenths actual: $300 \times 900 = 270{,}000$ (wen)
Eight-tenths actual: $200 \times 800 = 160{,}000$ (wen)
Total actual: $500{,}000 + 270{,}000 + 160{,}000 = 930{,}000$ (wen)
Converted to full-bundle: $\frac{930{,}000}{1000} = 930$ (guan)

6.3 計定物價 *Setting Prices*

問 / Question

問有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？

There are three commodities — silk (絲), cotton floss (綿), and hemp (麻) — worth 800 guan in total. Silk costs 4 guan per jin, cotton floss 2 guan per jin, and hemp 500 wen per jin. One wishes to buy certain quantities of each so that the value of each commodity is equal. Question: how many jin of each?

答曰 / Answer

絲若干斤，綿若干斤，麻若干斤。

A certain number of jin of silk, a certain number of jin of cotton floss, and a certain number of jin of hemp.

術曰 / Method

以總價三而一，得每物之值。各以本價除之，得斤數。

Divide the total price by three to get the value of each commodity. Divide by the respective price per jin to get the quantity in jin.

草曰 / Working

每物之值：$\frac{800}{3} \approx 266.67$（貫）
絲斤數：$\frac{266.67}{4} \approx 66.67$（斤）
綿斤數：$\frac{266.67}{2} \approx 133.33$（斤）
麻斤數：$\frac{266.67}{0.5} = 533.33$（斤）

Value per commodity: $\frac{800}{3} \approx 266.67$ (guan)
Silk quantity: $\frac{266.67}{4} \approx 66.67$ (jin)
Cotton floss quantity: $\frac{266.67}{2} \approx 133.33$ (jin)
Hemp quantity: $\frac{266.67}{0.5} = 533.33$ (jin)

6.4 計販運得利 *Profit on Transported Goods*

問 / Question

問一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

A merchant carries 500 bolts of silk to a distant place for trade. The purchase price was 4 guan per bolt; upon arrival, the price has risen by 50%. On the return journey, he buys tea at 300 wen per jin. Question: how much money does he obtain from selling the silk? How many jin of tea can he buy?

答曰 / Answer

得錢若干貫，可買茶若干斤。

The money obtained is a certain number of guan; the tea that can be purchased is a certain number of jin.

術曰 / Method

以本價加利，得賣價；以匹數乘之，得總錢。以總錢除以茶價，得茶斤數。

Add the profit to the cost price to get the selling price; multiply by the number of bolts to get the total money. Divide the total money by the tea price to get the quantity of tea in jin.

草曰 / **Working**

賣價：$4 \times 1.5 = 6$（貫/匹）
總錢：$500 \times 6 = 3000$（貫）$= 3{,}000{,}000$（文）
茶斤數：$\dfrac{3000000}{300} = 10000$（斤）

Selling price: $4 \times 1.5 = 6$ (guan/bolt)
Total money: $500 \times 6 = 3000$ (guan) $= 3{,}000{,}000$ (wen)
Tea quantity: $\dfrac{3{,}000{,}000}{300} = 10{,}000$ (jin)

Editorial Notes / 編者注

Units of Measurement / 度量衡表

The problems in these fascicles employ traditional Chinese units: 本編算題所用度量衡，皆依宋制：

Unit	Chinese	Pinyin	Approx. Modern Value
Length	丈	zhang	≈ 3.33 metres
Length	尺	chi	≈ 33.3 centimetres
Length	寸	cun	≈ 3.33 centimetres
Area	平方丈	pingfang zhang	≈ 11.09 square metres
Volume	立方丈	lifang zhang	≈ 37.0 cubic metres
Volume	立方尺	lifang chi	≈ 37.0 litres
Capacity	石	dan/shi	≈ 60 kilograms (grain)
Capacity	升	sheng	≈ 0.6 litres
Weight	斤	jin	≈ 600 grams (Song)
Weight	兩	liang	≈ 37.5 grams
Weight	錢	qian	≈ 3.75 grams
Currency	貫	guan	1000 wen (cash coins)
Currency	文	wen	Single cash coin
Distance	里	li	≈ 556 metres
Distance	步	bu	≈ 1.54 metres

Mathematical Methods / 數學方法

數學九章所用算法，有數項獨創之術，尤以大衍求一術為最著。本編營建、軍旅、市易三類，則多用體積公式、衰分術、重差術等。

The *Shuxue Jiuzhang* employs several characteristic mathematical techniques. The Construction, Military, and Trade fascicles make extensive use of volume formulae, proportional distribution (衰分), and the double-difference method (重差術).

1. **Frustum volume formula (塹堵/臺積術):** For a rectangular frustum with upper dimensions $a \times b$, lower dimensions $c \times d$, and height h:

$$V = \frac{h}{3}(ab + cd + ad + bc)$$

   This is equivalent to the modern prismatoid formula.

2. **Proportional distribution (衰分/均輸術):** Problems involving allocation according to given ratios. Each party contributes in proportion to their respective holding or capacity.

3. **Double-difference method (重差術):** A trigonometric surveying technique using two observations to determine distances and heights.

4. **Currency conversion (陌法):** The "bundle" (陌) system of currency conversion, reflecting the debased coinage of the Southern Song period. One "bundle" (陌) nominally contained 1000 wen, but in practice bundles of 900 or 800 wen were common.

Historical Context / 歷史背景

秦九韶，字道古，生於南宋寧宗嘉泰二年（1202），卒於理宗景定二年（1261）。所撰《數學九章》成書於淳祐七年（1247），時宋室南渡已百餘年，國勢日蹙，蒙古鐵騎壓境。卷八軍旅之題，正反映當時岌岌可危之軍事情勢。

Qin Jiushao (ca. 1202–1261), styled Daogu, was one of the foremost mathematicians of the Song Dynasty. The *Shuxue Jiuzhang* was completed in 1247, during the turbulent final decades of the Southern Song. The military problems in Fascicle 8 directly reflect the desperate military situation of the time, as the Song faced existential threats from the Mongols under Genghis Khan and later Kublai Khan.

卷九市易之題，則反映南宋繁榮之商業經濟：紙幣交子會子之流通、官營鹽茶之專賣、長途販運之興盛。此皆中古時期世界最發達之商業體系。

The trade problems in Fascicle 9 reflect the sophisticated commercial economy of the Song, with its paper currency (交子, 會子), government monopolies on salt and tea, and extensive long-distance trade networks. This was one of the most advanced commercial systems in the medieval world.

秦九韶自述其書：「大則可以通神明，順性命；小則可以經世務，類萬物。」數學九章不獨為數學之鉅著，亦為研究宋代社會經濟軍事之重要史料。

Qin Jiushao described his own work: “In the large sense it can communicate with the divine and accord with the nature of things; in the small sense it can govern worldly affairs and classify all things.” The *Shuxue Jiuzhang* is not only a masterpiece of mathematics, but also an important historical source for studying the society, economy, and military of the Song Dynasty.

— *End of Fascicles 7–9* / 卷七至卷九終 —

Classical Chinese Mathematics

孫子算經

Sunzi Suanjing

Master Sun’s Mathematical Manual

Bilingual English–Chinese Edition

English Translation • Original Classical Chinese

Attributed to **Sunzi** (Master Sun)

c. 3rd–5th century CE

One of the Ten Computational Canons (算經十書)
Listed during the Tang dynasty (618–907 CE)
Contains the earliest known description of the Chinese Remainder Theorem

Typeset in LaTeX with XeLaTeX, fontspec, and minipage

May 28, 2026

Contents

1 Introduction

關於本書

孫子算經是中國現存最早的數學著作之一，成書於公元三世紀至五世紀之間。唐代將其列為算經十書之一。作者孫子（「孫子」為尊稱）的身份至今仍不確定，但其生活年代遠早於《孫子兵法》的作者孫武。

全書共分三卷（卷）：

- **卷上**：度量衡單位、籌算記數法、乘除算法、九九乘法表
- **卷中**：分數運算、幾何面積與體積計算
- **卷下**：三十六道算題，包括著名的「物不知數」問題——中國剩餘定理的最早記載

About This Book

The *Sunzi Suanjing* (孫子算經, “Master Sun’s Mathematical Manual”) is one of the earliest surviving Chinese mathematical texts, written during the 3rd to 5th centuries CE. It was listed as one of the *Ten Computational Canons* (算經十書) during the Tang dynasty. The identity of its author, Sunzi (“Master Sun”), remains unknown, though he lived much later than his namesake Sun Tzu, the author of *The Art of War*.

The treatise consists of three volumes (卷):

- **Volume I** (卷上): Units of measurement, counting rod notation, multiplication and division algorithms
- **Volume II** (卷中): Fractions, geometry (areas, volumes), and practical problems
- **Volume III** (卷下): 36 problems including the famous “Problem of the Unknown Quantity”—the **Chinese Remainder Theorem**

歷史意義

孫子算經在數學史上佔有特殊的地位。其最著名的貢獻是**卷下第 26 題**——「物不知數」問題：

> 今有物，不知其數。三、三數之，賸二；
> 五、五數之，賸三；七、七數之，賸二。
> 問物幾何？

此問題提供了公式：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 分別為以 3、5、7 除後的餘數。

Historical Significance

The *Sunzi Suanjing* holds a special place in the history of mathematics. Its most celebrated contribution is **Problem 26 of Volume III**:

> *“There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?”*

This problem provides the formula:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders upon division by 3, 5, and 7.

Preface 序

孫子序

孫子曰：夫算者，天地之經緯，群生之元首；五常之本末，陰陽之父母；星辰之建號，三光之表裏；五行之準平，四時之終始；萬物之祖宗，六藝之綱紀。

稽群倫之聚散，考二氣之降升；推寒暑之迭運，步遠近之殊同；觀天道精微之兆基，察地理從橫之長短；采神祇之所在，極成敗之符驗；窮道德之理，究性命之情。

立規矩，準方圓，謹法度，約尺丈，立權衡，平重輕，剖毫釐，析黍絫；歷億載而不朽，施八極而無疆。散之不可勝究，斂之不盈掌握。

嚮之者富有餘，背之者貧且窶；心開者幼沖而即悟，意閉者皓首而難精。夫欲學之者必務量能揆己，志在所專。如是則焉有不成者哉。

Preface by Sunzi

Sun Zi said: Calculation is the meridian and latitude of heaven and earth, the origin and head of all living beings; the root and branch of the Five Constants, the father and mother of yin and yang; the establishment of titles for stars and constellations, the exterior and interior of the Three Lights; the standard balance of the Five Elements, the beginning and end of the Four Seasons; the ancestors of all things, the framework and discipline of the Six Arts.

Investigate the gathering and dispersing of all categories, examine the descending and ascending movements of the Two Qi; track the alternating cycles of cold and heat, pace the differences and similarities between near and far; observe the subtle omens and foundations of the heavenly Dao, examine the lengths and widths of the horizontal and vertical features of geography; gather signs of where gods and spirits reside, thoroughly test omens of success and failure; exhaust the principles of virtue and morality, investigate the nature and essence of life.

Establish rules and compasses, standardize squares and circles, observe laws and regulations carefully, measure with precision in *chi* and *zhang*, set up weights and balances, equalize the heavy and light, dissect thousandths of an inch, analyze grains accumulated one by one; enduring through countless generations without decay, its influence extends to all directions without limit. When dispersed, it is beyond full comprehension; when gathered, it does not fill the palm of one’s hand.

Those who face toward it are rich with abundance, while those who turn away from it are poor and destitute; those whose minds open understand even as children, while those whose intentions remain closed find mastery difficult even with a white beard. Therefore, one who wishes to study must necessarily measure his own ability and assess himself carefully, setting his mind on specialization in a particular field. If so, how could there be any failure?

2 Volume I 卷上

1. Length Measurements 度之所起

問　度之所起，起於忽。欲知其忽，蠶所生，吐絲為忽。十忽為一秒，十秒為一毫，十毫為一氂，十氂為一分，十分為一寸，十寸為一尺，十尺為一丈，十丈為一引；五十尺為一端；四十尺為一疋；六尺為一步。二百四十步為一畝。三百步為一里。

Q *The origin of measurement begins with the “hu” (忽). To understand the hu, it is said that silkworms produce it by spinning silk. Ten hu make one miao (秒), ten miao make one hao (毫), ten hao make one li (氂), ten li make one fen (分), ten fen make one cun (寸), ten cun make one chi (尺), ten chi make one zhang (丈), and ten zhang make one yin (引); fifty chi make one duan (端); forty chi make one pi (疋); six chi make one bu (步). Two hundred and forty bu make one mu (畝). Three hundred bu make one li (里).*

2. Weight and Capacity 稱量之所起

問　稱之所起，起於黍。十黍為一絫，十絫為一銖，二十四銖為一兩，十六兩為一斤，三十斤為一鈞，四鈞為一石。量之所起，起於粟。六粟為一圭，十圭為一抄，十抄為一撮，十撮為一勺，十勺為一合，十合為一升，十升為一斗，十斗為一斛。

Q *The origin of weight begins with the grain of millet (黍). Ten millet grains make one lei (絫), ten lei make one zhu (銖), twenty-four zhu make one liang (兩), sixteen liang make one jin (斤), thirty jin make one jun (鈞), and four jun make one dan (石). The origin of capacity begins with the grain of millet. Six millet grains make one gui (圭), ten gui make one chao (抄), ten chao make one cuo (撮), ten cuo make one shao (勺), ten shao make one ge (合), ten ge make one sheng (升), ten sheng make one dou (斗), and ten dou make one hu (斛).*

3. Large Numbers 大數之法

問　凡大數之法，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰陔，萬萬陔曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載。

Q *Generally, the method for large numbers: ten thousand times ten thousand is yi* (億, 10^8)*; ten thousand times yi is zhao* (兆, 10^{16})*; ten thousand times zhao is jing* (京, 10^{24})*; and so on up to zai* (載, 10^{80})*.*

4. Geometric Ratios 方圓之比

問　周三徑一。方五邪七；見邪求方，五之，七而一；見方求邪，七之，五而一。

Q *The circumference of a circle is three times its diameter ($\pi \approx 3$). A square with side length five has a diagonal of seven; to find the side from the diagonal, multiply by five and divide by seven; to find the diagonal from the side, multiply by seven and divide by five.*

5. Weights of Materials 物之輕重

問　黃金方寸重一斤。白金方寸重一十四兩。玉方寸重一十二兩。銅方寸重七兩半。鉛方寸重九兩半。鐵方寸重六兩。石方寸重三兩。

Q *One cubic cun of gold weighs one jin. One cubic cun of silver weighs fourteen liang. One cubic cun of jade weighs twelve liang. One cubic cun of copper weighs seven and a half liang. One cubic cun of lead weighs nine and a half liang. One cubic cun of iron weighs six liang. One cubic cun of stone weighs three liang.*

6. Counting Rod Notation 筭位之制

問　凡筭之法，先識其位，一從十橫，百立千僵，千十相望，萬百相當。

籌算記數法：個位（1--9）縱式；十位（10--90）橫式；百位（100--900）縱式；千位（1000--9000）橫式。縱橫交替，以定位值。

Q *The method for calculation begins by identifying the positions: one is vertical, ten is horizontal; a hundred stands upright and a thousand lies flat.*

Counting Rod Numeral System: Units (1–9): **vertical**; Tens (10–90): **horizontal**; Hundreds (100–900): **vertical**; Thousands (1000–9000): **horizontal**. The pattern alternates.

7. Method of Multiplication 乘法

問　凡乘之法，重置其位。上下相觀，上位有十步至十，有百步至百，有千步至千。以上命下，所得之數列於中位。言十即過，不滿自如。上位乘訖者先去之。下位乘訖者則俱退之。六不積，五不隻。上下相乘，至盡則已。

Q *In multiplication, the positions are arranged again for calculation. By observing vertically and horizontally, if the upper position has ten, it moves to the tens place; if it has a hundred, it moves to the hundreds place; if it has a thousand, it moves to the thousands place. The upper number is used to multiply the lower one, and the resulting product is placed in the middle position. If it reaches ten, carry over; if less than ten, leave as is. After the upper position has been multiplied, remove it first. Once the lower position is multiplied, both are moved back. Six does not accumulate; five does not count as a single unit. Multiplying vertically and horizontally continues until the end is reached.*

8. Method of Division 除法

問　凡除之法，與乘正異。乘得在中央，除得在上方。假令六為法，百為實。以六除百，當進之二等，令在正百下，以六除一，則法多而實少，不可除，故當退就十位。以法除實，言一六而折百為四十，故可除。若實多法少，自當百之，不當復退。故或步法十者置於十位，百者置於百位。上位有空絕者，法退二位。餘法皆如乘時。實有餘者，以法命之，以法為母。實餘為子。

Q *The method of division differs from multiplication. In multiplication, the result is placed in the center; in division, it appears at the top. Suppose six is the divisor (法) and one hundred is the dividend (實). Dividing one hundred by six, it should be advanced two positions and placed under the hundreds place. Dividing one by six results in a larger divisor than dividend, which is not possible; therefore, we must retreat to the tens position. Using the divisor to divide the dividend: saying "one six" and reducing a hundred by sixty results in forty, so division is possible. If the dividend is large and the divisor small, it naturally belongs to the hundreds place, and there is no need to retreat further. Therefore, if a step of ten is taken, it should be placed in the tens position; if a hundred, in the hundreds position. If there is an empty space at the upper position, the divisor must retreat two positions. The remaining methods follow those of multiplication. If there is a remainder in the dividend, name it using the divisor as the denominator. The remainder becomes the numerator.*

9. Example: 81×81 九九八十一自乘

問 九九八十一，自相乘，得幾何?

答 答曰：六千五百六十一。

術 術曰：重置其位，以上八呼下八，八八六十四，即下六千四百於中位。以上八呼下一，一八如八，即於中位下八十。退下位一等，收上位八十。以上位一呼下八，一八如八，即於中位下八十。以上一呼下一，一一如一，即於中位下一。上下位俱收，中位即得六千五百六十一。

Q *Ninety-nine times eighty-one, when multiplied by itself, results in what?*

A *Answer: Six thousand five hundred and sixty-one.*

M *Method: Arrange the positions again, call out eight from above to multiply with eight below; eight eights are sixty-four. Place six thousand four hundred in the middle position. Call out eight from above to one below, one eight is eight; place eighty in the middle position. Move the lower position back by one rank and collect the upper value of eighty. Call out one from above to eight below, one eight is eight, place eighty in the middle position accordingly. Call out one from the upper position to multiply with one from the lower; one times one is one, so place one in the middle position. After collecting both the upper and lower positions, the middle position yields six thousand five hundred sixty-one.*

Modern: $81 \times 81 = (80+1)^2 = 6400+80+80+1 = 6561$.

10. Example: $6561 \div 9$ 九人分之

問　六千五百六十一，九人分之，問人得幾何?

答　答曰：七百二十九。

術　術曰：先置六千五百六十一於中位，為實。下列九人為法。上位置七百，以上七呼下九，七九六十三，即除中位六千三百。退下位一等，即上位置二十。以上二呼下九，二九十八，即除中位一百八十。又更退下位一等，即上位更置九，即以上九呼下九，九九八十一，即除中位八十一。中位盡，收下位。上位所得即人之所得。

Q *Six thousand five hundred sixty-one is to be divided among nine people; how much does each person receive?*

A *Answer: Seven hundred and twenty-nine.*

M *Method: First place six thousand five hundred sixty-one in the middle position as the dividend (實). Place nine people below as the divisor (法). Place seven hundred in the upper position; call out seven from above to multiply by nine below, seven nines are sixty-three, and subtract six thousand three hundred from the middle. Move the lower position back one rank; place twenty in the upper position. Call out two from above to multiply by nine from below, two nines are eighteen, and subtract one hundred eighty from the middle. Move the lower position back another rank, place nine in the upper position again. Call out nine from above to multiply by nine below; nine nines are eighty-one and subtract eighty-one from the middle. When the middle position is exhausted, collect the lower positions. The result in the upper position is what each person receives.*

Modern: $6561 \div 9 = 729$.

3 Volume II 卷中

1. Reducing Fractions 約分

問　今有一十八分之一十二。問約之得幾何?

答　答曰：三分之二。

術　術曰：置十八分在下，一十二分在上。副置二位，以少減多，等數得六，為法。約之，即得。

Q *Now there is twelve eighteenths. Ask: reducing it, what does it equal?*

A *Answer: Two thirds.*

M *Method: Place eighteen parts below and twelve parts above. Set up both positions, subtract the smaller from the larger; the equal number (等數, GCD) obtained is six, which is the divisor. Reduce by it, and you obtain the answer.*
Modern: $\frac{12}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$.

2. Grain Conversion 粟米換算

問　今有粟一斗，問為糲米幾何?

答　答曰：六升。

術　術曰：置粟一斗，十升。以糲米率三十乘之，得三百升，為實。以粟率五十為法，除之，即得。

Q *Now there is one dou of millet. How much coarse rice does it make?*

A *Answer: Six sheng.*

M *Method: Place one dou of millet, which equals ten sheng. Multiply by the coarse rice rate of thirty to get three hundred sheng, as the dividend. Use the millet rate of fifty as the divisor and divide; thus you obtain the answer.*
Modern: $10 \times 30 \div 50 = 6$ *sheng*.

3. Brick Calculation 積磚之數

問　今有屋基南北三丈，東西六丈，欲以磚砌之。凡積二尺，用磚五枚。問計幾何?

答　答曰：四千五百枚。

術　術曰：置東西六丈，以南北三丈乘之，得一千八百尺。以五乘之，得九千尺。以二除之，即得。

Q *Now there is a house foundation with a north-south length of three zhang and an east-west width of six zhang, and one wishes to build it using bricks. Generally for every two cubic chi in volume, five bricks are used. Ask how many will be needed?*

A *Answer: Four thousand five hundred bricks.*

M *Method: Place the east-west length of six zhang, multiply it by the north-south length of three zhang to get one thousand eight hundred chi. Multiply it by five to get nine thousand chi. Divide it by two, and you will obtain the answer.*
Modern: *Area* $= 30 \times 60 = 1800$ *sq chi. Bricks* $= 1800 \times 5/2 = 4500$.

4. Area of a Circular Field 圓田面積

問　今有圓田周三百步，徑一百步。問得田幾何?

答　答曰：三十一畝奇六十步。

術　術曰：先置周三百步，半之，得一百五十步。又置徑一百步，半之，得五十步。相乘，得七千五百步。以畝法二百四十步除之，即得。

Q　*Now there is a round field with circumference three hundred bu, diameter one hundred bu. Ask how much land does it amount to?*

A　*Answer: Thirty-one mu and sixty bu extra.*

M　*Method: First place the circumference of three hundred bu, halve it to get one hundred fifty bu. Also place the diameter of one hundred bu, halve it, getting fifty bu. Multiply them together to get seven thousand five hundred bu. Divide it by the mu standard of two hundred forty bu, and you will obtain the area.*

Modern: *Area* $\approx \frac{C}{2} \times \frac{d}{2} = 150 \times 50 = 7500$ *sq bu* $= 31$ *mu + 60 bu.*

5. Square Field with Mulberry 方田桑生中央

問　今有方田，桑生中央。從角至桑一百四十七步。問為田幾何?

答　答曰：一頃八十三畝奇一百八十步。

術　術曰：置角至桑一百四十七步，倍之，得二百九十四步。以五乘之，得一千四百七十步。以七除之，得二百一十步。自相乘，得四萬四千一百步。以二百四十步除之，即得。

Q　*Now there is a square field with mulberry trees growing in the center. From the corner to the mulberry tree is one hundred forty-seven steps. Ask how much field does it amount to?*

A　*Answer: One qing, eighty-three mu and one hundred eighty steps extra.*

M　*Method: Place the distance from corner to mulberry tree as one hundred forty-seven bu, double it to get two hundred ninety-four bu. Multiply it by five to get one thousand four hundred seventy steps. Divide it by seven, getting two hundred ten steps. Multiply it by itself to get forty-four thousand one hundred bu. Divide it by two hundred forty steps, and you will get the result.*

Modern: *Diagonal = 147 bu. Using* $\sqrt{2} \approx \frac{7}{5}$, *side* $s = 147 \times 2 \times \frac{5}{7} = 210$ *bu. Area* $= 210^2 = 44100$ *sq bu.*

4 Volume III 卷下

1. Rent Distribution 九家共輸租

問　今有甲、乙、丙、丁、戊、己、庚、辛、壬九家共輸租。甲出三十五斛，乙出四十六斛，丙出五十七斛，丁出六十八斛，戊出七十九斛，己出八十斛，庚出一百斛，辛出二百一十斛，壬出三百二十五斛。凡九家共輸租一千斛。僦運直折二百斛外，問家各幾何?

答　答曰：甲二十八斛。乙三十六斛八蚪。丙四十五斛六蚪。丁五十四斛四蚪。戊六十三斛二蚪。己六十四斛。庚八十斛。辛一百六十八斛。壬二百六十斛。

術　術曰：置甲出三十五斛，以四乘之，得一百四十斛。以五除之，得二十八斛。……

Q *Now there are nine families jointly paying rent: Jia 35 hu, Yi 46 hu, Bing 57 hu, Ding 68 hu, Wu 79 hu, Ji 80 hu, Geng 100 hu, Xin 210 hu, Ren 325 hu. Total 1,000 hu. After deducting transport cost of 200 hu, how much should each family contribute?*

A *Answer: Jia 28 hu, Yi 36 hu 8 dou, Bing 45 hu 6 dou, Ding 54 hu 4 dou, Wu 63 hu 2 dou, Ji 64 hu, Geng 80 hu, Xin 168 hu, Ren 260 hu.*

M *Method: Take each family's contribution, multiply by four, and divide by five.*
Modern: *After 200 hu deduction, 800 hu remains. Each pays* $4/5$ *of original.*

2. Pile of Millet 平地聚粟

問　今有平地聚粟，下周三丈六尺，高四尺五寸。問粟幾何?

答　答曰：一百斛。

術　術曰：置周三丈六尺，自相乘，得一千二百九十六尺。以高四尺五寸乘之，得五千八百三十二尺。以三十六除之，得一百六十二尺。以斛法一尺六寸二分除之，即得。

Q *Now there is a pile of millet on level ground, with base circumference of three zhang six chi and height of four chi five cun. How much millet is there?*

A *Answer: One hundred hu.*

M *Method: Place the circumference of three zhang six chi, multiply it by itself to get 1296 chi. Multiply by the height of 4 chi 5 cun to get 5832 chi. Divide by 36 to get 162 chi. Divide by the hu standard of 1 chi 6 cun 2 fen.*
Modern: *Volume* $\approx \frac{C^2\times h}{36} = \frac{1296\times 4.5}{36} = 162$ *cu chi.* $162 \div 1.62 = 100$ *hu.*

3. Go Board 棊局

問　今有棊局方一十九道。問用棊幾何?

答　答曰：三百六十一。

術　術曰：置一十九道，自相乘之，即得。

Q *Now there is a Go board with a square pattern of nineteen lines on each side. How many pieces (intersections) are there?*

A *Answer: Three hundred sixty-one.*

M *Method: Take the nineteen lines, multiply them by themselves, and you get the result.*
Modern: $19^2 = 361$ *intersection points on a Go board.*

4. People and Carts 人與車

問 今有三人共車，二車空；二人共車，九人步。問人與車各幾何?

答 答曰：一十五車。三十九人。

術 術曰：置二車，以三乘之，得六。加步者九人，得車一十五。欲知人者，以二乘車，加九人，即得。

Q *Now there are three people sharing a cart, and two carts remain empty; two people share another cart, while nine people walk. What are the numbers of people and carts?*

A *Answer: Fifteen carts, thirty-nine people.*

M *Method: Place two carts, multiply by three to get six. Add the nine people who walk, getting fifteen carts. To find the number of people, multiply two by the number of carts and add nine people.*
Modern: $3(c-2)=p$ *and* $2c+9=p$, *giving* $c=15$, $p=39$.

5. Guests at a Feast 婦人河上蕩桮

問 今有婦人河上蕩桮。津吏問曰：「桮何以多?」婦人曰：「家有客。」津吏曰：「客幾何?」婦人曰：「二人共飯，三人共羹，四人共肉，凡用桮六十五。不知客幾何?」

答 答曰：六十人。

術 術曰：置六十五桮，以一十二乘之，得七百八十，以十三除之，即得。

Q *Now there is a woman washing cups at the riverbank. The ferry official asked, "Why so many cups?" The woman replied, "There are guests at my home." "How many guests?" "Two people share a meal, three share soup, four share meat, and in total sixty-five cups are used."*

A *Answer: Sixty people.*

M *Method: Take the 65 cups, multiply by twelve to get 780, then divide by thirteen.*
Modern: *Each uses* $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}=\frac{13}{12}$ *cups.* $65\div\frac{13}{12}=60$.

6. Wood and Rope 木與繩

問 今有木，不知長短。引繩度之，餘繩四尺五寸。屈繩量之，不足一尺。問木長幾何?

答 答曰：六尺五寸。

術 術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸。倍之，得一丈一尺。減餘四尺五寸，即得。

Q *Now there is a piece of wood whose length is unknown. Using a rope to measure it, the rope has 4 chi 5 cun left over. Folding the rope to measure it results in one chi short. What is the length of the wood?*

A *Answer: Six chi and five cun.*

M *Method: Place the excess rope of 4 chi 5 cun, add the deficit of 1 chi, obtaining 5 chi 5 cun. Double it to get 1 zhang 1 chi. Subtract the excess of 4 chi 5 cun.*
Modern: *$R = W + 4.5$ and $R/2 = W - 1$. Solving: $W = 6.5$ chi.*

7. Rice in a Vessel 器中米

問 今有器中米，不知其數。前人取半，中人三分取一，後人四分取一，餘米一斗五升。問本米幾何?

答 答曰：六斗。

術 術曰：置餘米一斗五升，以六乘之，得九斗。以二除之，得四斗五升。以四乘之，得一斛八斗。以三除之，即得。

Q *Now there is rice in a vessel, but its quantity is unknown. The first person took half, the second took one-third of what remained, the third took one-fourth of what was left, and there were 1 dou 5 sheng remaining. What was the original amount?*

A *Answer: Six dou.*

M *Method: Place the remaining rice of 1 dou 5 sheng, multiply by six to get 9 dou. Divide by two to get 4 dou 5 sheng. Multiply by four to get 1 hu 8 dou. Divide by three.*
Modern: *$\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times x = \frac{x}{4} = 1.5$ dou. So $x = 6$ dou.*

8. Birds and Beasts 獸六首四足

問 今有獸六首四足，禽四首二足。上有七十六首，下有四十六足。問禽、獸各幾何?

答 答曰：八獸，七禽。

術 術曰：倍足以減首，餘，半之，即禽。四乘禽數，減足，餘，半之，即獸。

Q *Now there are beasts with six heads and four feet, and birds with four heads and two feet. Above there are seventy-six heads, below there are forty-six feet. Ask: How many birds and beasts are there?*

A *Answer: Eight beasts, seven birds.*

M *Method: Double the feet and subtract the heads; halve the remainder to get the number of birds. Multiply the number of birds by four, subtract from the feet; halve the remainder to get the number of beasts.*
Modern: *$6b + 4i = 76$, $4b + 2i = 46$. Solving: $b = 8$, $i = 7$.*

9. One Hundred Deer 百鹿入城

問 今有百鹿入城，家取一鹿，不盡；又三家共一鹿，適盡。問城中家幾何?

答 答曰：七十五家。

術 術曰：以盈不足取之。假令七十二家，鹿不盡四。令之九十家，鹿不足二十。置七十二於右上，盈四於右下。置九十於左上，不足二十於左下。維乘之，所得，并為實。并盈不足為法。除之，即得。

Q *Now there are one hundred deer entering a city; each household takes one deer, but they do not finish; then three households share one deer and it is just finished. Ask: How many households are in the city?*

A *Answer: Seventy-five households.*

M *Method: Use the surplus-and-deficit method. Suppose 72 households: 4 deer remain (surplus). Suppose 90: 20 deer short (deficit). Cross-multiply and add to get the dividend. Add surplus and deficit to get the divisor.*

Modern: $h + h/3 = 100$, *so* $4h/3 = 100$, *giving* $h = 75$.

10. Pheasants and Rabbits 雉兔同籠

問　今有雉、兔同籠，上有三十五頭，下九十四足。問雉、兔各幾何?

答　答曰：雉二十三。兔一十二。

術　術曰：上置三十五頭，下置九十四足。半其足，得四十七。以少減多，再命之，上三除下三，上五除下五。下有一除上一，下有二除上二，即得。
又術曰：上置頭，下置足。半其足，以頭除足，以足除頭，即得。

Q *Now there are pheasants and rabbits in a cage, with thirty-five heads above and ninety-four feet below. Ask: How many pheasants and how many rabbits are there?*

A *Answer: Twenty-three pheasants, twelve rabbits.*

M *Method: Place thirty-five heads above, ninety-four feet below. Halve the feet to get forty-seven. Subtract the smaller from the larger, and command it again.*
Alternative method: Place heads above, feet below. Halve the feet, use heads to remove feet and feet to remove heads.
Modern: $p + r = 35$, $2p + 4r = 94$. *Solving:* $p = 23$, $r = 12$.

11. Nine Dikes 出門望見九隄

問　今有出門望見九隄。隄有九木，木有九枝，枝有九巢，巢有九禽，禽有九雛，雛有九毛，毛有九色。問各幾何?

答　答曰：木八十一。枝七百二十九。巢六千五百六十一。禽五萬九千四十九。雛五十三萬一千四百四十一。毛四百七十八萬二千九百六十九。色四千三百四萬六千七百二十一。

術　術曰：置九隄，以九乘之，得木之數。又以九乘之，得枝之數。又以九乘之，得巢之數。又以九乘之，得禽之數。又以九乘之，得雛之數。又以九乘之，得毛之數。又以九乘之，得色之數。

Q *Now there is someone who goes out of the gate and sees nine dikes. Each dike has nine trees, each tree has nine branches, each branch has nine nests, each nest has nine birds, each bird has nine chicks, each chick has nine feathers, and each feather has nine colors. Ask: How many of each are there?*

A *Answer: 81 trees, 729 branches, 6,561 nests, 59,049 birds, 531,441 chicks, 4,782,969 feathers, 43,046,721 colors.*

M *Method: Place nine dikes, multiply by nine to get the number of trees. Multiply by nine again for each successive level.*
Modern: *Powers of nine:* $9^2 = 81$, $9^3 = 729$, $9^4 = 6561$, $9^5 = 59049$, $9^6 = 531441$, $9^7 = 4782969$, $9^8 = 43046721$.

12. Three Daughters 三女歸家

問　今有三女，長女五日一歸，中女四日一歸，少女三日一歸。問三女幾何日相會?

答　答曰：六十日。

術　術曰：置長女五日、中女四日、少女三日，於右方。各列一算於左方。維乘之，各得所到數：長女十二到，中女十五到，少女二十到。又各以歸日乘到數，即得。

Q *Now there are three daughters: the eldest returns home every five days, the middle every four days, and the youngest every three days. Ask: After how many days will the three daughters meet again?*

A *Answer: Sixty days.*

M *Method: Place the eldest daughter's five days, the middle daughter's four days, and the youngest daughter's three days on the right side. Place one counter for each on the left side. Multiply them, and you get their respective numbers of arrivals: twelve, fifteen, and twenty. Then multiply each return day by the number of arrivals.*
Modern: *This is the least common multiple:* $lcm(3, 4, 5) = 60$.

物不知數 — The Chinese Remainder Theorem

問　今有物，不知其數。三、三數之，賸二；五、五數之，賸三；七、七數之，賸二。問物幾何?

答　答曰：二十三。

術　術曰：「三、三數之，賸二」，置一百四十；「五、五數之，賸三」，置六十三；「七、七數之，賸二」，置三十。并之，得二百三十三。以二百一十減之，即得。
凡三、三數之，賸一，則置七十；五、五數之，賸一，則置二十一；七、七數之，賸一，則置十五。一百六以上，以一百五減之，即得。

Q *Now there is an object, but we do not know its quantity. Counting it in groups of three leaves a remainder of two; counting it in groups of five leaves a remainder of three; counting it in groups of seven leaves a remainder of two. What is the quantity of the object?*

A *Answer: Twenty-three.*

M *Method: “Counting in groups of three, remainder two,” set down 140; “Counting in groups of five, remainder three,” set down 63; “Counting in groups of seven, remainder two,” set down 30. Add them: 233. Subtract 210, and you will obtain the answer.*
Generally, if counting in groups of three leaves remainder one, set down 70; if counting in groups of five leaves remainder one, set down 21; if counting in groups of seven leaves remainder one, set down 15. If the total is more than 160, subtract 150; thus you obtain the answer.

Modern Mathematical Analysis 現代數學分析

以現代符號表示，此問題要求找出最小的正整數 N，滿足以下同餘方程組：

$$\begin{aligned} N &\equiv 2 \pmod{3} \\ N &\equiv 3 \pmod{5} \\ N &\equiv 2 \pmod{7} \end{aligned}$$

解法給出：

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

三個關鍵數字 **70**、**21**、**15** 有特殊性質：

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod{3}$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod{5}$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod{7}$

由於 $3 \times 5 \times 7 = 105$，一般解為：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 為餘數，k 為使 N 為最小正整數的整數。

In modern notation, this problem asks us to find the smallest positive integer N satisfying:

$$\begin{aligned} N &\equiv 2 \pmod{3} \\ N &\equiv 3 \pmod{5} \\ N &\equiv 2 \pmod{7} \end{aligned}$$

The method gives:

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

The three key numbers **70**, **21**, and **15** have special properties:

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod{3}$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod{5}$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod{7}$

Since $3 \times 5 \times 7 = 105$, the general solution is:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders and k is chosen to make N positive and minimal.

Verification 驗算

$$\begin{aligned} &23 \div 3 = 7 \text{ 餘}\mathbf{2} \quad \checkmark \\ &23 \div 5 = 4 \text{ 餘}\mathbf{3} \quad \checkmark \\ &23 \div 7 = 3 \text{ 餘}\mathbf{2} \quad \checkmark \end{aligned}$$

孫子公式：

$$\boxed{N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k}$$

$$\begin{aligned} &23 \div 3 = 7 \text{ remainder } \mathbf{2} \quad \checkmark \\ &23 \div 5 = 4 \text{ remainder } \mathbf{3} \quad \checkmark \\ &23 \div 7 = 3 \text{ remainder } \mathbf{2} \quad \checkmark \end{aligned}$$

Sunzi's Formula:

$$\boxed{N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k}$$

Historical Significance 歷史意義

此問題被稱為「物不知數」，是**中國剩餘定理**的最早記載。雖然孫子的文本只給出了具體的數值方法而沒有證明，但它包含了核心思想：

1. 找到一個數，它是除一個模數外所有模數的倍數，且除以該模數餘一
2. 將每個這樣的數乘以相應的餘數
3. 將結果相加，減去最小公倍數的倍數，得到最小正解

一般定理後來由**秦九韶**（1202–1261）在其《數書九章》（1247 年）中完整發展，稱之為「大衍求一術」。

This problem, known as the **“Problem of the Unknown Quantity”** (物不知數), is the earliest known statement of the **Chinese Remainder Theorem**. Sunzi’s text gives the essential idea:

1. Find numbers that are multiples of all but one modulus, leaving remainder 1 when divided by the remaining modulus
2. Multiply each such number by the corresponding remainder
3. Sum the results and subtract multiples of the LCM to get the smallest positive solution

The general theorem was later fully developed by **Qin Jiushao** (1202–1261) in his *Mathematical Treatise in Nine Sections* (數書九章, 1247 CE).

The “Sunzi Song” 孫子歌

一首總結此方法的韻文廣泛流傳，後由程大位收錄於《算法統宗》（1592 年）：

三人同行七十稀，
五樹梅花廿一枝，
七子團圓正半月，
除百零五便得知。

關鍵數字：**70**（模 3）、**21**（模 5）、**15**（模 7）、**105**（最小公倍數）。

A rhymed verse summarizing the method circulated widely and was later recorded by Cheng Dawei in his *Systematic Treatise on Algorithms* (算法統宗, 1592):

“Three people walking together—seventy is rare;
Five plum trees—twenty-one branches;
Seven sons reunite—half a full moon (15);
Subtract one hundred and five, then you will know.”

Key numbers: **70** for mod 3, **21** for mod 5, **15** for mod 7, **105** as the LCM.

Remaining Problems from Volume III 卷下餘題

13. Money Exchange 二人持錢

問　今有甲、乙二人，持錢各不知數。甲得乙中半，可滿四十八。乙得甲太半，亦滿四十八。問甲、乙二人元持錢各幾何?

答　答曰：甲持錢三十六。乙持錢二十四。

術　術曰：如方程求之。置二甲、一乙、錢九十六，於右方。置二甲、三乙、錢一百四十四，於左方。以右方二乘左方……得二十四錢，為乙錢。……得三十六，為甲錢也。

Q *Now there are two people, Jia and Yi, each holding an unknown amount of money. If Jia receives half of what Yi has, he will have forty-eight. If Yi receives two-thirds of what Jia has, he too will have forty-eight. Ask: Originally, how much did Jia and Yi each hold?*

A *Answer: Jia originally held thirty-six units of money. Yi originally held twenty-four units.*

M *Method: Solve as a system of linear equations. Place 2 Jia, 1 Yi, 96 coins on the right. Place 2 Jia, 3 Yi, 144 coins on the left. Cross-multiply to obtain twenty-four coins as Yi's amount and thirty-six as Jia's amount.*
Modern: $J+Y/2=48$, $Y+2J/3=48$. *Solving:* $J=36$, $Y=24$.

14. Three Chickens 三雞共啄粟

問　今有三雞共啄粟一千一粒。雛啄一，母啄二，翁啄四。主責本粟，三雞主各償幾何?

答　答曰：雞雛主一百四十三。雞母主二百八十六。雞翁主五百七十二。

術　術曰：置粟一千一粒，為實。副并三雞所啄粟七粒，為法。除之，得一百四十三粒，為雞雛主所償之數。遞倍之，即得母、翁主所償之數。

Q *Now there are three chickens pecking at one thousand and one grains of millet. A chick pecks one grain, a hen pecks two, and a rooster pecks four. How many grains should each owner compensate?*

A *Answer: The chick's owner compensates 143, the hen's owner compensates 286, and the rooster's owner compensates 572.*

M *Method: Place 1001 grains of millet as the dividend. Add together the three chickens' consumption of seven grains as the divisor. Divide: 143 grains is the chick owner's compensation. Double successively to get the hen's and rooster's owner's compensation.*
Modern: *Ratio is* $1:2:4$. $1001 \div 7 = 143$. *So:* $143, 286, 572$.

15. Fish in a Ditch 九里渠

問 今有九里渠，三寸魚，頭頭相次。問魚得幾何?

答 答曰：五萬四千。

術 術曰：置九里，以三百步乘之，得二千七百步。又以六尺乘之，得一萬六千二百尺。上十之，得一十六萬二千寸。以魚三寸除之，即得。

Q *Now there is a ditch nine li long; fish of three cun are lined up head to tail. Ask: How many fish can fit?*

A *Answer: Fifty-four thousand.*

M *Method: Place nine li, multiply by three hundred bu to get two thousand seven hundred bu. Multiply by six chi to get sixteen thousand two hundred chi. Convert upward by ten to get one hundred sixty-two thousand cun. Divide by the fish length of three cun.*
Modern: *9 li* $= 9 \times 300 \times 6 = 16200$ *chi* $= 162000$ *cun.* $162000/3 = 54000$ *fish.*

16. Wheel Turns 輪匝

問 今有長安、洛陽相去九百里。車輪一匝一丈八尺。欲自洛陽至長安，問輪匝幾何?

答 答曰：九萬匝。

術 術曰：置九百里，以三百步乘之，得二十七萬步。又以六尺乘之，得一百六十二萬尺。以車輪一丈八尺為法。除之，即得。

Q *Now there are Chang'an and Luoyang, 900 li apart. A cart wheel turns one revolution for every one zhang eight chi. If one wishes to travel from Luoyang to Chang'an, how many wheel turns are needed?*

A *Answer: Ninety thousand turns.*

M *Method: Place nine hundred li, multiply by three hundred bu to get two hundred seventy thousand bu. Multiply by six chi to get one million six hundred twenty thousand chi. Use the wheel circumference of one zhang eight chi as the divisor.*
Modern: *Distance* $= 900 \times 300 \times 6 = 1{,}620{,}000$ *chi. Wheel* $= 18$ *chi. Turns* $= 1{,}620{,}000/18 = 90{,}000$.

17. Predicting Gender 孕婦行年

問 今有孕婦行年二十九，難九月。未知所生?

答 答曰：生男。

術 術曰：置四十九，加難月，減行年。所餘，以天除一，地除二，人除三，四時除四，五行除五，六律除六，七星除七，八風除八，九州除九。其不盡者，奇則為男，耦則為女。

Q *Now there is a pregnant woman whose age is twenty-nine, giving birth in the ninth month. It is unknown what will be born?*

A *Answer: A boy will be born.*

M *Method: Set forty-nine, add the month of childbirth, subtract the age. The remainder is divided by: Heaven 1, Earth 2, Humanity 3, Four Seasons 4, Five Elements 5, Six Laws 6, Seven Stars 7, Eight Winds 8, Nine Provinces 9. If the remainder is odd, a boy; if even, a girl.*

Epilogue 跋

孫子算經是古代中國數學高度發展的非凡見證。雖然本文涵蓋了基本算術——度量衡單位、乘法表和簡單分數——但其最偉大的遺產在於簡短而精妙的「物不知數」問題，它為現代數論開啟了大門。

中國剩餘定理——這一方法在西方後來的稱呼——要過了一千多年才在歐洲被完全形式化。孫子所描述的算法包含了現代定理的基本要素：構造具有特定整除性質的輔助數字，將它們與給定的餘數結合，以及對最小公倍數取模的約簡。

這部著作的影響遠遠超出了中國。這個問題在整個東亞作為數學難題流傳，而秦九韶所發展的一般方法代表了中世紀世界數學最偉大的成就之一。

The *Sunzi Suanjing* stands as a remarkable testament to the sophistication of ancient Chinese mathematics. While the text covers basic arithmetic—units of measurement, multiplication tables, and simple fractions—its greatest legacy lies in the brief but brilliant “Problem of the Unknown Quantity” that opens the door to modern number theory.

The Chinese Remainder Theorem, as this method came to be known in the West, would not be fully formalized in Europe for over a millennium. The algorithm described by Sunzi contains the essential ingredients of the modern theorem: the construction of auxiliary numbers with specific divisibility properties, their combination with the given remainders, and reduction modulo the least common multiple.

The influence of this text extended far beyond China. The problem circulated as a mathematical puzzle throughout East Asia, and the general method developed by Qin Jiushao represents one of the great achievements of medieval mathematics anywhere in the world.

Bilingual edition typeset from the classical Chinese text
as preserved in the Chinese Text Project (`https://ctext.org/sunzi-suan-jing`)

Detailed Analysis of the Mathematical Methods in the Nine Chapters

詳解九章算法

(Part I — 上)

BILINGUAL EDITION / 中英雙語版

By **Yang Hui** 楊輝

Southern Song Dynasty (c. 1238–1298 CE)

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1

The "Method Source for Square Root Extraction" (開方作法本源圖)

Left Column: Original Chinese text / Right Column: English translation

Typeset in XeLaTeX with Noto Serif CJK SC
May 27, 2026

Contents

Original Chinese (原文) | **English Translation (英譯)**

1 Introduction / 引言

《詳解九章算法》乃南宋楊輝所著之重要數學著作。成書於一二六一年，詳解古代《九章算術》之各題算法。

Xiangjie Jiuzhang Suanfa (詳解九章算法, "Detailed Analysis of the Mathematical Methods in the Nine Chapters") is one of the most important mathematical works of the Southern Song Dynasty, written by the mathematician **Yang Hui** 楊輝 (c. 1238–1298 CE). Completed in 1261, this work provides detailed explanations and commentaries on the classic *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

1.1 Historical Context / 歷史背景

原《九章算術》成書於漢代（約公元前二百年至公元二百年），凡二百四十六題，分九章。至宋代，原文因靖康之變（一一二七年）而散佚，傳抄訛誤，題隱法晦，學者難以入門。

The original *Jiuzhang Suanshu* dates back to the Han Dynasty (c. 200 BCE–200 CE) and contains 246 mathematical problems organized into nine chapters. By the Song Dynasty, the original text had become difficult to understand due to the loss of manuscripts during the *Jingkang Incident* (靖康之變, 1127 CE), corruption of transmitted texts through repeated copying, and obscure problem statements with difficult methods.

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。

"I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the methods difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity."

1.2 Structure of the Work / 全書結構

全書凡十二章：一、乘除（四十一題）；二、方田（三十八題）；三、粟米（四十六題）；四、衰分（二十題）；五、少廣（二十四題）；六、商功（二十八題）；七、均輸（二十八題）；八、盈不足（二十題）；九、方程（十八題）；十、勾股（三十八題）；十一、纂類。

The original work comprised **12 chapters**: (1) Multiplication and Division — 41 problems; (2) Field Measurement — 38 problems; (3) Grain Conversion — 46 problems; (4) Proportional Distribution — 20 problems; (5) Short Width / Root Extraction — 24 problems; (6) Commercial Calculations / Engineering — 28 problems; (7) Equitable Distribution — 28 problems; (8) Excess and Deficit — 20 problems; (9) Systems of Linear Equations — 18 problems; (10) Right Triangles — 38 problems; (11) Classification / Compilation.

1.3 Editorial Method / 編纂之法

楊輝解題之法，分為四部：問（設題）、答（給出數值答案）、術（說明一般算法）、草（詳細逐步演算）。此四部結構，為中國數學著述之典範。

Yang Hui's approach to explaining the Nine Chapters was systematic: **Problem (問)** states the mathematical problem; **Answer (答)** gives the numerical answer; **Method (術)** states the general algorithm; **Worked Solution (草)** provides a detailed step-by-step calculation. This four-part structure became a model for Chinese mathematical exposition.

1.4 The Method Source for Square Root Extraction / 開方作法本源

楊輝所錄之開方作法本源圖（今稱楊輝三角或賈憲三角），源出北宋賈憲（約一〇一〇至一〇七〇年）之《釋鎖算書》。其圖列二項式係數至六次冪，用於開方運算。

One of Yang Hui's most celebrated contributions is preserving what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle). In his work, he attributes this to the earlier mathematician **Jia Xian** (c. 1010–1070 CE), who called it the "Method Source for Square Root Extraction" (開方作法本源圖). The triangle provides binomial coefficients used in root extraction. Yang Hui wrote: 出釋鎖算書，賈憲用此術 ("This comes from the *Shisuo Suan Shu*; Jia Xian used this method").

2 Original Preface / 原序

Original Chinese (原文)	English Translation (英譯)
黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。	"The ancient preface to the Yellow Emperor's Nine Chapters states: The state once established an examination in mathematics for selecting officials, choosing the Nine Chapters as the foremost mathematical classic—this is like the Six Classics for Confucian scholars, the *Nanjing* and *Suwen* for physicians, or the *Art of War* for military strategists."
昔聖宋紹興戊辰，算士榮棨謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲撰草。	"In the past, during the Wuchen year of the Shaoxing reign of our Sacred Song Dynasty, the mathematician Rong Qi said that since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices. Having obtained a good copy and recovered the complete classic, study was revived. The annotations were by Liu Hui et al. (Jingyuan first year of Wei) and by Li Chunfeng et al. (Senior Compiler of Tang Dynasty, Acting Grand Astrologer), while the detailed solutions were composed by Jia Xian of the Song Dynasty."
輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又篡法問，詳著於後，儻得賢者，改而正諸，是所願也。	"I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the principles difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity. Taking the supplementary texts added by later worthies and repairing the methods abandoned by earlier generations, I have reorganized the problem-methods and compiled the method-questions, recording them in detail below. If worthy scholars would correct and improve them, that is my wish."

Original Chinese (原文) | English Translation (英譯)

3 Chapter 1: Multiplication and Division / 乘除

第一章論乘除之基本運算，凡四十一題，多取方田章及粟米之題。

The first chapter covers fundamental operations of multiplication and division, with 41 problems drawn mostly from the “Field Measurement” (方田) chapter and grain conversion problems.

3.1 Rectangular Field Method / 直田法

術：以廣乘從得積。以畝法（一畝二百四十平方步）除之。

Method: Multiply the width (廣) by the length (從) to obtain the area (積). Divide by the unit conversion (畝法, 1 mu = 240 square bu).

Problem 1

廣十五步，從十六步，問田幾何？答曰：一畝。 A field has width 15 bu and length 16 bu. What is the area? **Answer**: 1 mu.

Solution (草曰):

廣十五步，從十六步。積 = $15 \times 16 = 240$ 平方步。一畝 = 二百四十平方步，故為**一畝**。

Width = 15 bu, Length = 16 bu. Area = $15 \times 16 = 240$ *square bu. Since 1 mu = 240 square bu, the area is **1 mu**.*

Problem 2

廣十二步，從十四步，問田幾何？答曰：一百六十八步。 A field has width 12 bu and length 14 bu. What is the area in square bu? **Answer**: 168 square bu.

Solution (草曰):

積 = $12 \times 14 =$ ***168*** 平方步。

Area = $12 \times 14 =$ ***168*** *square bu.*

3.2 Square Field Method / 方田法

術：方田之積，以里自乘，復以三百七十五畝乘之。

Method: Square the side length to obtain the area in square li. Multiply by 375 mu per square li for conversion.

Problem 3

方田一里，問為田幾何？答曰：三頃七十五畝。 A square field has side 1 li. What is its area? **Answer**: 3 qing 75 mu.

Solution (草曰):

積 = $1 \times 1 = 1$ 平方里。$1 \times 375 =$ ***375*** **畝** = **三頃七十五畝**。

Area = $1 \times 1 = 1$ *square li.* $1 \times 375 =$ ***375 mu*** = ***3 qing 75 mu***.

3.3 Fraction Multiplication / 乘分

術：分母乘全步，并子以為實。分母相乘為法。以法除實。

Method: For fractions with whole numbers: multiply each whole number by its denominator and add the numerator. Multiply these results to form the dividend (實). Multiply the denominators to form the divisor (法). Divide the dividend by the divisor.

Problem Fraction Multiplication

田廣七步、四分步之三，從十五步、九分步之五，問：為田幾何？答曰：一百二十步，九分步之五。 A field has width $7\frac{3}{4}$ bu and length $15\frac{5}{9}$ bu. What is the area? **Answer**: $120\frac{5}{9}$ square bu.

Solution (草曰):

廣：$7\frac{3}{4} = \frac{31}{4}$ 步。從：$15\frac{5}{9} = \frac{140}{9}$ 步。
積 = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ 平方步。

Width: $7\frac{3}{4} = \frac{31}{4}$ *bu. Length:* $15\frac{5}{9} = \frac{140}{9}$ *bu.*
Area = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ *square bu.*

3.4 Division with Fractions / 除分

術：以人數為法，錢數為實。有分者通之。實如法而一。

Method: Use the number of people as the divisor (法) and the amount of money as the dividend (實). If there are fractions, convert them. Divide the dividend by the divisor.

PROBLEM Division with Fractions

七人，均八錢三分錢之一，問人得幾何？答曰：人得一錢二十一分錢之四。 Seven people share $8\,\frac{1}{3}$ qian equally. How much does each person receive? **Answer**: $1\,\frac{4}{21}$ qian each.

SOLUTION (草曰):

總數：$8\,\frac{1}{3} = \frac{25}{3}$ 錢。
每人得：$\frac{25}{3} \div 7 = \frac{25}{21} = 1\,\frac{4}{21}$ 錢。

Total: $8\,\frac{1}{3} = \frac{25}{3}$ *qian.*
Each receives: $\frac{25}{3} \div 7 = \frac{25}{21} = 1\,\frac{4}{21}$ *qian.*

Original Chinese (原文)	English Translation (英譯)

4 Chapter 2: Field Measurement / 方田

方田章專論面積之計算與分數之運算。凡三十八題，涉及各種田形及分數四則運算。

The Field Measurement chapter focuses on area calculations and fraction operations. It originally contained 38 problems covering various field shapes and operations with fractions.

4.1 Reduction of Fractions / 約分

術：可半者半之。不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。（即輾轉相除之歐幾里得算法。）

Method: If both numerator and denominator are even, halve them. If not, place the numerator and denominator separately and repeatedly subtract the smaller from the larger (Euclidean algorithm) until the greatest common divisor is found. Divide both by this common measure.

Problem Reduction

十八分之十二。問：約之得幾何？答曰：三分之二。 Reduce $\frac{12}{18}$. **Answer:** $\frac{2}{3}$.

Solution (草曰):

十二與十八之等數：$18 - 12 = 6$，$12 - 6 = 6$，$6 - 6 = 0$。等數 = 6。

$\frac{12 \div 6}{18 \div 6} =$

GCD of 12 and 18: $18 - 12 = 6, 12 - 6 = 6, 6 - 6 = 0$. *GCD = 6.*

$\frac{12 \div 6}{18 \div 6} =$

Problem

九十一分之四十九。問：約之得幾何？答曰：十三分之七。 Reduce $\frac{49}{91}$. **Answer:** $\frac{7}{13}$.

Solution (草曰):

輾轉相除：$91 - 49 = 42$，$49 - 42 = 7$，$42 - 7 \times 6 = 0$。等數 = 7。

$\frac{49 \div 7}{91 \div 7} =$

Using the Euclidean algorithm: $91 - 49 = 42, 49 - 42 = 7, 42 - 7 \times 6 = 0$. *GCD = 7.*

$\frac{49 \div 7}{91 \div 7} =$

4.2 Addition of Fractions / 合分

術：母互乘子，并以為實。母相乘為法。實如法而一。母同者直相從之。

Method: Cross-multiply the denominators with the numerators and add to form the dividend. Multiply the denominators to form the divisor. Divide. If the denominators are the same, simply add the numerators.

Problem

二分之一、三分之二、四分之三、五分之四，合之得幾何？答曰：得二，餘六十分之四十三。 Add $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5}$. **Answer:** $2\,\frac{43}{60}$.

Solution (草曰):

互乘：$1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$。
法：$2 \times 3 \times 4 \times 5 = 120$。
$\frac{326}{120} = 2\,\frac{86}{120} = 2\,\frac{43}{60}$。

Cross-multiply: $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
Divisor: $2 \times 3 \times 4 \times 5 = 120$.
$\frac{326}{120} = 2\,\frac{86}{120} = 2\,\frac{43}{60}$.

4.3 Subtraction and Comparison of Fractions / 課分

術：母互乘子，以少減多，餘為實。母相乘為法。實如法而一。即兩分數之差也。

Method: Cross-multiply numerators, subtract the smaller from the larger. The remainder is the dividend, the product of denominators is the divisor. This gives the difference (also called "the excess").

Problem

八分之五，比二十五分之十六。問：孰多多幾何？答曰：二十五分之十六，多多二百分之三。 Compare $\frac{5}{8}$ and $\frac{16}{25}$. Which is larger and by how much? **Answer**: $\frac{16}{25}$ is larger by $\frac{3}{200}$.

Solution (草曰):

$\frac{16}{25} - \frac{5}{8} = \frac{16\times8-5\times25}{25\times8} = \frac{128-125}{200} = \frac{3}{200}$。
故 $\frac{16}{25}$ 大 $\boldsymbol{3}\frac{}{200}$⊠
$\frac{16}{25} - \frac{5}{8} = \frac{16\times8-5\times25}{25\times8} = \frac{128-125}{200} = \frac{3}{200}$.
So $\frac{16}{25}$ *is larger by*

4.4 Equalizing Fractions / 平分

術： 母互乘子，各為列實。并諸列實為平實。母相乘為法。以列數乘未并者，各為列實。以平實減列實，餘為減數；以列實減平實，餘為益數。減多益少，使各平於平實。

Method: Cross-multiply numerators. Sum all numerators for the "level dividend." Multiply all denominators for the divisor. Multiply the number of terms by each uncrossed numerator for the column dividends. Subtract the level dividend from each column dividend to find what must be subtracted; subtract each column dividend from the level dividend to find what must be added. Combine the subtractions to benefit the smaller values. Divide the level dividend by the divisor to find the equalized value.

Problem

二分之一、三分之二、四分之三，減多益少幾何？而平？答曰：減四分之三，求之者四，減三分之二，求之者一，益二分之一，求之者五，各平於三十六分之二十三。 Given $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, redistribute by subtracting from larger and adding to smaller to make them equal. **Answer**: Subtract $\frac{4}{36}$ from $\frac{3}{4}$; subtract $\frac{1}{36}$ from $\frac{2}{3}$; add $\frac{5}{36}$ to $\frac{1}{2}$. All equalized to $\frac{23}{36}$.

Solution (草曰):

平值 = $\frac{23}{36}$。
$\frac{3}{4} = \frac{27}{36}$： 減 $\frac{4}{36}$。
$\frac{2}{3} = \frac{24}{36}$： 減 $\frac{1}{36}$。
$\frac{1}{2} = \frac{18}{36}$： 益 $\frac{5}{36}$。
皆平於 $\boldsymbol{23}\frac{}{36}$⊠

Level value = $\frac{23}{36}$.
$\frac{3}{4} = \frac{27}{36}$: *subtract* $\frac{4}{36}$.
$\frac{2}{3} = \frac{24}{36}$: *subtract* $\frac{1}{36}$.
$\frac{1}{2} = \frac{18}{36}$: *add* $\frac{5}{36}$.
All equalized to $\boldsymbol{23}\frac{}{36.}$

5 Chapter 3: Grain Conversion / 粟米

粟米章論各種穀物與商品之比例交換。凡四十六題，其中六題為基本乘除，三十一題為比例交換，九題為比例率。

The Grain Conversion chapter deals with proportional exchange between different types of grain and commodities. It originally contained 46 problems with 6 on basic multiplication/division, 31 on proportional exchange, and 9 on proportional rates.

5.1 Direct Exchange / 經率

術：以所買率為法，以所出錢為實。實如法得一。

Method: Use the money amount as the dividend (實) and the quantity purchased as the divisor (法). Divide the dividend by the divisor.

Problem

錢一百六十文，買瓴甓十八枚，問枚價幾何？答曰：一枚八錢九分钱之八。 With 160 qian, buy 18 roof tiles. What is the price per tile? **Answer**: $8\frac{8}{9}$ qian per tile.

Solution (草曰):

$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9}$ 文/枚。

$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9}$ *qian per tile.*

5.2 Grain Conversion Rates / 粟米換算率

《九章》立穀物換算之標準率：粟率五十；糲米率三十；糳米率二十七；御米率二十四；麥率四十五；菽率四十五。

The Nine Chapters establishes standard conversion rates between different grains: Millet (粟): rate = 50; Coarse rice (糲米): rate = 30; Fine rice (糳米): rate = 27; Polished rice (御米): rate = 24; Wheat (麥): rate = 45; Soybeans (菽): rate = 45.

術：以所有數乘所求率為實，以所有率為法，實如法而一。

Method: "What you seek multiplied by the given amount, divided by what you have." (以所有數乘所求率為實，以所有率為法，實如法而一。)

Problem

粟一斗。問為糲米幾何？答曰：六升。 Given 1 dou of millet, how much coarse rice (糲米) does it make? **Answer**: 6 sheng.

Solution (草曰):

粟率五十，糲米率三十。
1 斗 $\times \frac{30}{50} = \frac{3}{5}$ 斗 $=$ ***6* 升**。

Millet rate = 50, coarse rice rate = 30.
1 *dou* $\times \frac{30}{50} = \frac{3}{5}$ *dou* $=$ ***6 sheng***.

Problem

粟四斗五升。問為糳米幾何？答曰：二斗一升五分升之三。 Given 4 dou 5 sheng of millet, how much fine rice (糳米) does it make? **Answer**: 2 dou 1 sheng $\frac{3}{5}$.

Solution (草曰):

$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43$ 斗 = **二斗一升五分升之三**。

$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43$ *dou* = ***2 dou 1 sheng*** $\frac{3}{5}$.

Problem

糲米十五斗五升五分升之二。問為粟幾何？答曰：二十五斗九升。 Given 15 dou 5 $\frac{2}{5}$ sheng of coarse rice, how much millet was the original amount? **Answer**: 25 dou 9 sheng.

Solution (草曰):

$15\,5\,\frac{2}{5} = \frac{777}{50}$ 斗。
$\frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9$ 斗 = **二十五斗九升**。

$15\,5\,\frac{2}{5} = \frac{777}{50}$ *dou.*
$\frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9$ *dou* = ***25 dou 9 sheng***.

5.3 Silk and Currency Conversion / 絲錢換算

PROBLEM

絲一斤，價三百四十五，今有七兩一十二銖。問錢，答曰：一百六十一錢，三十二分錢之二十三。 Silk costs 345 qian per jin. If one has 7 liang 12 zhu of silk, what is its value in qian? **Answer**: $161\,\frac{23}{32}$ qian.

SOLUTION (草曰):

一斤 = 十六兩 = 三百八十四銖。七兩十二銖 = $7\times 24+12 = 180$ 銖。
值 = $345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161\,\frac{23}{32}$ 錢。

1 jin = 16 liang = 384 zhu. 7 liang 12 zhu = $7 \times 24 + 12 = 180$ *zhu.*
Value = $345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161\,\frac{23}{32}$ *qian.*

6 Chapter 4: Proportional Distribution / 衰分

衰分章論按給定比例分配數量之問題。凡二十題。

The Proportional Distribution chapter deals with dividing quantities in given proportions. It originally contained 20 problems.

6.1 The Method of Distribution / 衰分法

術：各置列衰，副并為法。以所分乘未并者，各自為實。實如法而一。不滿法者，以法命之。

Method: Set out the distribution rates (衰). Combine them as the divisor. Multiply what is to be distributed by each uncombined rate to form the column dividends. Divide each by the divisor. If there is a remainder, express it as a fraction over the divisor.

Problem Five Ranks Sharing Deer / 五爵分鹿

大夫、不更、簪裹、上造、公士凡五人，以爵次高下均五鹿，問各幾何？答曰：大夫一鹿，三分鹿之二；不更一鹿，三分鹿之一；簪裹一鹿；上造鹿三分之二；公士鹿三分之一。 Five nobles—a Dafu, a Bugeng, a Zanmao, a Shangzao, and a Gongshi—are to share 5 deer in proportion to their ranks (5:4:3:2:1). **Answer**: Dafu $1\frac{2}{3}$; Bugeng $1\frac{1}{3}$; Zanmao 1; Shangzao $\frac{2}{3}$; Gongshi $\frac{1}{3}$ deer.

Solution (草曰):

衰：大夫五、不更四、簪裹三、上造二、公士一。
并 $= 5 + 4 + 3 + 2 + 1 = 15$。
大夫：$5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ 鹿。
不更：$5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ 鹿。
簪裹：$5 \times \frac{3}{15} = 1$ 鹿。
上造：$5 \times \frac{2}{15} = \frac{2}{3}$ 鹿。
公士：$5 \times \frac{1}{15} = \frac{1}{3}$ 鹿。

Rates: Dafu 5, Bugeng 4, Zanmao 3, Shangzao 2, Gongshi 1.
Sum $= 5 + 4 + 3 + 2 + 1 = 15$.
Dafu: $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ *deer.*
Bugeng: $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ *deer.*
Zanmao: $5 \times \frac{3}{15} = 1$ *deer.*
Shangzao: $5 \times \frac{2}{15} = \frac{2}{3}$ *deer.*
Gongshi: $5 \times \frac{1}{15} = \frac{1}{3}$ *deer.*

6.2 The Cow, Horse, and Sheep / 牛馬羊償粟

Problem

牛、馬、羊食人苗，苗主責之粟五斗，牛食馬之半，馬食羊之半，欲衰償之？答曰：牛二斗八升，七分升之四；馬一斗四升，七分升之二；羊七升，七分升之一。 A cow, a horse, and a sheep ate someone's crops. The owner demands 5 dou of millet as compensation. The cow ate twice what the horse ate, and the horse ate twice what the sheep ate. **Answer**: Cow $2\frac{4}{7}$ dou; Horse $1\frac{2}{7}$ dou; Sheep $\frac{1}{7}$ dou.

Solution (草曰):

衰：牛四、馬二、羊一。
并 $= 4 + 2 + 1 = 7$。
牛：$5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ 斗 = **二斗八升** $\frac{4}{7}$。
馬：$5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ 斗 = **一斗四升** $\frac{2}{7}$。
羊：$5 \times \frac{1}{7} = \frac{5}{7}$ 斗 = **七升** $\frac{1}{7}$。

The rates are: Cow 4, Horse 2, Sheep 1.
Sum $= 4 + 2 + 1 = 7$.
Cow: $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ *dou* = ***2 dou 8 sheng*** $\frac{4}{7}$.
Horse: $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ *dou* = ***1 dou 4 sheng*** $\frac{2}{7}$.
Sheep: $5 \times \frac{1}{7} = \frac{5}{7}$ *dou* = ***7 sheng*** $\frac{1}{7}$.

6.3 The Woman Skilled at Weaving / 女子善織

Problem

女子善織，日自倍，五日織五尺。問日織數。答曰：初日一寸，三十一分寸之十九；二日三寸，三十一分寸之七；三日六寸，三十一分寸之十四；四日一尺二寸，三十一分寸之二十八；五日二尺四寸，三十一分寸之二十五。 A woman skilled at weaving doubles her output each day. In 5 days she weaves 5 chi. **Answer**: Day 1: $\frac{50}{31}$ cun; Day 2: $\frac{100}{31}$ cun; Day 3: $\frac{200}{31}$ cun; Day 4: $\frac{400}{31}$ cun; Day 5: $\frac{800}{31}$ cun.

Solution (草曰):

設首日織 x，則總數 $= x + 2x + 4x + 8x + 16x = 31x = 5$ 尺。
$x = \frac{5}{31}$ 尺 $= \frac{50}{31}$ 寸 $= 1\frac{19}{31}$ 寸。
次日：$\frac{100}{31} = 3\frac{7}{31}$ 寸。
三日：$\frac{200}{31} = 6\frac{14}{31}$ 寸。
四日：$\frac{400}{31} = 12\frac{28}{31}$ 寸。
五日：$\frac{800}{31} = 25\frac{25}{31}$ 寸。

Let the first day's output be x*. Then total* $= x + 2x + 4x + 8x + 16x = 31x = 5$ *chi.*
$x = \frac{5}{31}$ *chi* $= \frac{50}{31}$ *cun* $= 1\frac{19}{31}$ *cun.*
Day 2: $\frac{100}{31} = 3\frac{7}{31}$ *cun.*
Day 3: $\frac{200}{31} = 6\frac{14}{31}$ *cun.*
Day 4: $\frac{400}{31} = 12\frac{28}{31}$ *cun.*
Day 5: $\frac{800}{31} = 25\frac{25}{31}$ *cun.*

Original Chinese (原文) | English Translation (英譯)

7 Chapter 5: Short Width / Root Extraction / 少廣

少廣章論已知矩形面積與一邊而求另一邊，涉及分數除法與開方。凡二十四題。

The Short Width chapter deals with finding an unknown dimension of a rectangle given its area and one side, which leads to problems of division with fractions and root extraction. It originally contained 24 problems.

7.1 The Short Width Method / 少廣法

術：置全步及分母子。各以分母乘其子、全步，以少減多，通分內子，令母互乘。并所有以為法。以畝積二百四十步乘之為實。實如法而一，得從。

Method: Set out the whole steps and the fraction denominators and numerators. Separately multiply the denominators by themselves, then multiply the whole steps and numerators. Divide each by its original denominator and combine. Use this as the divisor. Multiply the whole steps by their combined denominators, then multiply by the field area (mu) in square bu for the dividend. Divide the dividend by the divisor.

Problem Short Width 1

田一畝廣一步半，問從？答曰：一百六十步。 A field of 1 mu has width $1\frac{1}{2}$ bu. What is the length? **Answer**: 160 bu.

Solution (草曰):

廣 = $1\frac{1}{2} = \frac{3}{2}$ 步。積 = 一畝 = 二百四十平方步。
從 = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = \boldsymbol{160}$ 步。

Width = $1\frac{1}{2} = \frac{3}{2}$ *bu. Area = 1 mu = 240 square bu.*
Length = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = \boldsymbol{160}$ *bu.*

Problem

田一畝廣一步半、三分步之一。問從？答曰：一百三十步一十一分步之一十。 A field of 1 mu has width $1\frac{1}{2}\frac{1}{3}$ bu. What is the length? **Answer**: $130\frac{10}{11}$ bu.

Solution (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ 步。
從 = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ *bu.*
Length = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ *bu.*

Problem

田一畝廣一步半、三分步之一、四分步之一。問從答曰：一百一十五步五分步之一。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ bu. **Answer**: $115\frac{1}{5}$ bu.

Solution (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ 步。
從 = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ *bu.*
Length = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ *bu.*

Problem

田一畝廣一步半、三分步之一、四分步之一、五分步之一、六分步之一、七分步之一。問從答曰：九十二步一百二十一分步之六十八。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$ bu. **Answer**: $92\frac{68}{121}$ bu.

Solution (草曰):

用逆少廣術：
廣 = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$。
約之：等數三，得 $\frac{363}{140}$。
從 = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ 步。

Using the reverse combination-of-fractions method:
Width = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$.
Simplifying: GCD(1089, 420) = 3, so $\frac{363}{140}$.
Length = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ *bu.*

Original Chinese (原文) | English Translation (英譯)

8 The Method Source for Square Root Extraction / 開方作法本源圖

楊輝書中最著名者，為其所錄之開方作法本源圖（今稱楊輝三角）。楊輝稱此圖源出北宋賈憲（約一〇一〇至一〇七〇年）。

One of the most celebrated parts of Yang Hui's work is his recording of what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle in the West). Yang Hui attributed this to Jia Xian (c. 1010–1070 CE), who called it the "Method Source for Square Root Extraction" (開方作法本源).

8.1 The Original Diagram / 原圖

楊輝釋此圖曰：
左衺乃積數，右衺乃隅算，中藏者皆廉，以廉乘商方，命實而除之。

Yang Hui presented the triangle with the following explanation: "The left diagonal consists of the accumulated numbers; the right diagonal consists of the corner calculations. Those hidden in the middle are all the 'flanks' (intermediate coefficients). Using the flanks, multiply the quotient square, and command the dividend to be divided by it."

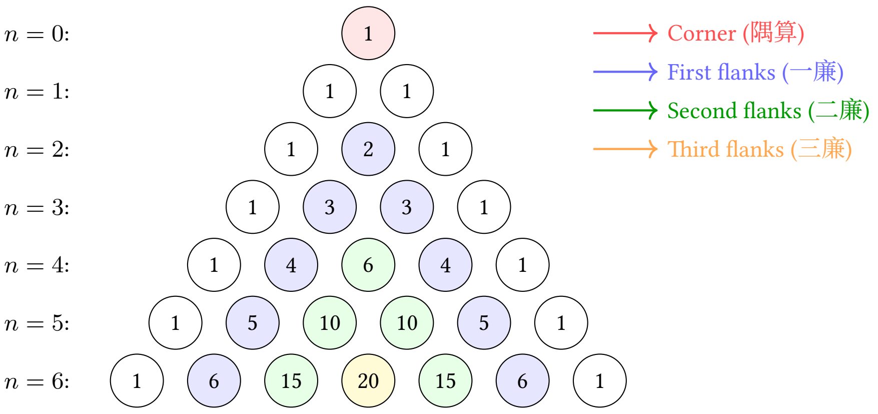

Fig. 1: Yang Hui's Triangle showing binomial coefficients up to the 6th power / 楊輝三角至六次冪

8.2 Explanation of the Structure / 結構釋義

左衺（積數）：斜邊之常數項（皆為一），即 $(x+a)^n$ 展開式之常數係數。
右衺（隅算）：最高次冪之係數（亦皆為一）。
中藏者（廉）：中間各數為廉，即各中間項之係數。

Left diagonal (左衺): The constant terms (1, 1, 1, ...) represent the "accumulated numbers" (積數)—the coefficients of the constant term in $(x+a)^n$.
Right diagonal (右衺): The coefficients of the highest power—also all 1's—represent the "corner calculations" (隅算).
Interior numbers (中藏者): The intermediate numbers are the "flanks" (廉)—the coefficients of the intermediate terms.

每行給出 $(x+a)^n$ 展開之係數：

$$
\begin{aligned}
(x+a)^0 &= 1 \\
(x+a)^1 &= x+a \\
(x+a)^2 &= x^2+2xa+a^2 \\
(x+a)^3 &= x^3+3x^2a+3xa^2+a^3
\end{aligned}
$$

Each row gives the coefficients for expanding $(x+a)^n$:

$$
\begin{aligned}
(x+a)^0 &= 1 \\
(x+a)^1 &= x+a \\
(x+a)^2 &= x^2+2xa+a^2 \\
(x+a)^3 &= x^3+3x^2a+3xa^2+a^3
\end{aligned}
$$

8.3 Application to Root Extraction / 開方之應用

楊輝與賈憲用此係數進行開平方與開立方。開平方之術：一、置積為實。二、商估首位。三、以三角之係數為方法、廉法、隅法。四、乘減遞推。

Yang Hui and Jia Xian used these coefficients to perform square root and cube root extraction. To extract $\sqrt{A}$: (1) Set A as the dividend (實). (2) Estimate the first digit of the root as the quotient (商). (3) Use the coefficients from the triangle to form the "method" (法), "flank" (廉), and "corner" (隅). (4) Multiply and subtract iteratively.

8.4 Jia Xian's Generating Method / 賈憲增乘方求廉法

增乘方求廉法：以一次乘二次，以二次乘三次，遞增一位，至求終。

此法由前行生成後行，相鄰兩數相加得之—即今所謂帕斯卡法則：

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

"To seek the flanks by augmented multiplication: multiply once by twice, multiply twice by thrice, increasing by one position each time, until the end is reached."

This method generates each row from the previous one by adding adjacent entries—exactly the rule we now call Pascal's rule:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

9 Chapter 6: Commercial Calculations / 商功

商功章論工程土方之體積計算。凡二十八題。

The Commercial Calculations chapter deals with volume computations for engineering projects (earthworks, construction, etc.). It originally contained 28 problems.

9.1 Volume Conversion Rates / 土方換算率

《九章》立不同土質之換算率：穿地為基準；壤（鬆土）與穿地之比為 4 : 5；堅（堅土）與穿地之比為 3 : 4。

The Nine Chapters establishes conversion rates between different soil conditions: Excavated earth (穿地): base measure; Loose earth (壤): 4 : 5 ratio with excavated; Compacted earth (堅): 3 : 4 ratio with excavated.

Problem

穿地積一萬尺。問為堅、壤各幾何？答曰：為堅七千五百尺，為壤一萬二千五百尺。 An excavation of 10,000 cubic chi. How much compacted and loose earth does it produce? **Answer**: Compacted 7,500; Loose 12,500 cubic chi.

Solution (草曰):

穿地: 壤: 堅 = 4 : 5 : 3。
壤 = $10000 \times \frac{5}{4} = \boldsymbol{12500}$ 立方尺。
堅 = $10000 \times \frac{3}{4} = \boldsymbol{7500}$ 立方尺。

Excavated : Loose : Compacted = 4 : 5 : 3.
Loose = $10000 \times \frac{5}{4} =$ ***12,500*** *cubic chi.*
Compacted = $10000 \times \frac{3}{4} =$ ***7,500*** *cubic chi.*

9.2 City Wall and Ditch Volumes / 城垣溝渠

術：並上下廣而半之，以高（或深）乘之，又以袤乘之，得積。

Method: For city walls, dikes, ditches, and canals: average the top and bottom widths, multiply by height (or depth), then multiply by length.

Problem

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問為尺幾何？答曰：一百八十九萬七千五百尺。 A city wall has bottom width 4 zhang, top width 2 zhang, height 5 zhang, and length 126 zhang 5 chi. **Answer**: 1,897,500 cubic chi.

Solution (草曰):

平均廣 = $\frac{4+2}{2} = 3$ 丈。
積 = $3 \times 5 \times 126.5 = 1897.5$ 立方丈 = $\boldsymbol{1897500}$ 立方尺。

Average width = $\frac{4+2}{2} = 3$ *zhang.*
Volume = $3 \times 5 \times 126.5 = 1897.5$ *cubic zhang* = $1897.5 \times 1000 =$ ***1,897,500*** *cubic chi.*

9.3 Special Solids / 特殊立體

The Qian Du (塹堵)

術：廣乘袤，又以高乘之，半之。

Method: Multiply length by width, then multiply by height, and divide by 2.

Problem

塹堵下廣二丈，袤十八丈六尺，高二丈五尺，問積尺幾何？答曰：四萬六千五百尺。 A *qiandu* has bottom width 2 zhang, length 18 zhang 6 chi, and height 2 zhang 5 chi. **Answer**: 46,500 cubic chi.

Solution (草曰):

積 = $\frac{1}{2} \times 20 \times 186 \times 25 = \boldsymbol{46500}$ 立方尺。

Volume = $\frac{1}{2} \times 20 \times 186 \times 25 =$ ***46,500*** *cubic chi.*

The Yang Ma (陽馬)

術：廣乘袤，又以高乘之，三而一。

Method: Multiply length by width, then multiply by height, and divide by 3.

Problem

陽馬廣五尺，袤七尺，高八尺，問積尺幾何？答曰：九十三尺，三分尺之一。 A *yangma* has width 5 chi, length 7 chi, and height 8 chi. **Answer**: $93\,\frac{1}{3}$ cubic chi.

Solution (草曰):

積 = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\,\frac{1}{3}$ 立方尺。

Volume = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\,\frac{1}{3}$ *cubic chi.*

The Chu Tong (芻童)

術：倍上袤以下袤從之，以上廣乘之；又倍下袤以上袤從之，以下廣乘之。并之，以高乘之，六而一。

Method: Double the top length and add the bottom length; multiply by the top width. Separately, double the bottom length and add the top length; multiply by the bottom width. Add these two results, multiply by height, and divide by 6.

Problem

芻童下廣三丈，袤四丈，上廣二丈，袤三丈，高三丈。問積？答曰：二萬六千五百尺。 A *chutong* has bottom width 3 zhang, length 4 zhang, top width 2 zhang, length 3 zhang, and height 3 zhang. **Answer**: 26,500 cubic chi.

Solution (草曰):

$(2\times30+40)\times20+(2\times40+30)\times30=2000+3300=5300$。
積 = $\frac{5300\times30}{6}=$ ***26500*** 立方尺。

$(2\times30+40)\times20+(2\times40+30)\times30=2000+3300=5300.$
Volume = $\frac{5300\times30}{6}=$ ***26,500*** *cubic chi.*

10 Chapter 7: Equitable Distribution / 均輸

均輸章論各地區之賦役與運輸成本之公平分配。凡二十八題。

The Equitable Distribution chapter deals with fair allocation of tax burdens and transportation costs across different regions. It originally contained 28 problems.

10.1 The Equitable Distribution Method / 均輸法

術：以一斛之運費與各縣距離相乘，加粟價，得各縣之比率。以戶數為衰，副并為法。以賦粟乘未并之衰為實。實如法而一。

Method: Multiply the distance and transport cost per unit for each county. Add the grain price to obtain the proportional rate. Simplify the county household numbers to form the distribution rates. Set out the rates, combine as the divisor. Multiply the total grain tax by each uncombined rate for the dividends. Divide each by the divisor.

Problem

五縣均賦粟一萬石，每車載二十五石，行道一里，出雇錢一文。各縣到輸所遠近不等，粟價高下，欲令縣勞費相等。Five counties are to share equally the tax burden of 10,000 dan of millet. Each cart carries 25 dan, and for each li traveled, the transport cost is 1 qian. The counties differ in distance from the delivery point and in grain prices. How much does each county contribute?

Solution (草曰):

每石每里運費 = $\frac{1}{25}$ 錢。
各縣率：
甲：$20+0=20$；乙：$10+\frac{200}{25}=18$；
丙：$12+\frac{150}{25}=18$；丁：$17+\frac{250}{25}=27$；
戊：$13+\frac{150}{25}=19$。
按戶數 × 率分配各縣賦額。

Transport cost per dan per li = $\frac{1}{25}$ *qian.*
Rates: A: $20+0=20$*; B:* $10+\frac{200}{25}=18$*;*
C: $12+\frac{150}{25}=18$*; D:* $17+\frac{250}{25}=27$*;*
E: $13+\frac{150}{25}=19$.
Distribution proportional to households × rate.

10.2 The Pursuit Problem / 相遇問題

Problem

甲發長安，五日日至齊，乙發齊，七日至長安。今乙先行二日。問：幾日相逢? 答曰：二日，十二分日之一。Person A departs from Chang'an and reaches Qi in 5 days. Person B departs from Qi and reaches Chang'an in 7 days. Now B has already departed 2 days ahead. **Answer**: $2\,\frac{1}{12}$ days.

Solution (草曰):

甲速 = $\frac{1}{5}$（每日行程）；乙速 = $\frac{1}{7}$。
乙先行二日，已行 $\frac{2}{7}$。餘程 = $\frac{5}{7}$。
合速 = $\frac{1}{5}+\frac{1}{7}=\frac{12}{35}$。
時 = $\frac{5/7}{12/35}=\frac{175}{84}=2\,\frac{1}{12}$ 日。

A's speed = $\frac{1}{5}$ *(of the distance per day). B's speed* = $\frac{1}{7}$.
In 2 days, B covers $\frac{2}{7}$ *of the distance. Remaining* = $\frac{5}{7}$.
Combined speed = $\frac{1}{5}+\frac{1}{7}=\frac{12}{35}$.
Time = $\frac{5/7}{12/35}=\frac{175}{84}=2\,\frac{1}{12}$ *days.*

10.3 The Gourd and Melon / 瓜瓠相逢

Problem

垣高九尺，瓜生其上，蔓日長七寸；瓠生其下，蔓日長一尺。問：幾日相逢? 答曰：相逢於五日，十七分日之五。A wall is 9 chi high. A gourd grows from the top, extending 7 cun per day. A bottle-gourd grows from the bottom, extending 1 chi per day. **Answer**: They meet after $5\,\frac{5}{17}$ days.

Solution (草曰):

每日共長 = $7+10=17$ 寸。
時 = $\frac{90}{17}=5\,\frac{5}{17}$ 日。
瓜長 = $\frac{35}{17}\times 7=\frac{245}{17}=14\,\frac{7}{17}$ 寸。
瓠長 = $\frac{35}{17}\times 10=\frac{350}{17}=20\,\frac{10}{17}$ 寸。

Total daily growth = 7 cun + 10 cun = 17 cun per day.
Time = $\frac{90}{17}=5\,\frac{5}{17}$ *days.*
Gourd vine: $\frac{35}{17}\times 7=\frac{245}{17}=14\,\frac{7}{17}$ *cun.*
Bottle-gourd: $\frac{35}{17}\times 10=\frac{350}{17}=20\,\frac{10}{17}$ *cun.*

11 Chapter 8: Excess and Deficit / 盈不足

盈不足章介紹盈不足術—即雙假設法，為解線性方程與近似問題之強大方法。凡二十題。

The Excess and Deficit chapter introduces the "method of double false position"—a powerful technique for solving linear equations and approximation problems. It originally contained 20 problems.

11.1 The Excess and Deficit Method / 盈不足法

術：置所出率，盈、不足各居其下。維乘所出率，并以為實。并盈、不足為法。實如法而一。買物者，以所出率以少減多，餘以約法、實。實為物價，法為人數。

Method: Set out the proposed rates, placing the excess or deficit below each. Cross-multiply the rates by the excess/deficit and sum for the dividend. Sum the excess and deficit for the divisor. Divide the dividend by the divisor. For purchasing problems: subtract the smaller rate from the larger for the simplified divisor. The dividend gives the price, the divisor gives the number of people.

Problem

共買物，人出八文，盈三文，人出七文，不足四文。問數、物價各幾何？答曰：七人。物價五十三。 Several people jointly buy an item. If each pays 8 qian, there is an excess of 3 qian. If each pays 7 qian, there is a deficit of 4 qian. **Answer**: 7 people; price 53 qian.

Solution (草曰):

維乘：$8 \times 4 + 7 \times 3 = 32 + 21 = 53$。
法 = $3 + 4 = 7$。
物價 = $\frac{53}{1} = \boldsymbol{53}$ 文。
人數 = $\frac{3+4}{8-7} = 7$。

Dividend = $8 \times 4 + 7 \times 3 = 32 + 21 = 53$.
Divisor = $3 + 4 = 7$.
Price = $\frac{53}{1} = \boldsymbol{53}$ *qian.*
Number of people = $\frac{3+4}{8-7} = 7$.

Problem

共買雞，各出九，盈十一，各出六，不足十六。問人數、雞價各幾何？答曰：九人，雞價七十。Several people jointly buy a chicken. If each pays 9 qian, there is an excess of 11 qian. If each pays 6 qian, there is a deficit of 16 qian. **Answer**: 9 people; price 70 qian.

Solution (草曰):

人數 = $\frac{11+16}{9-6} = \frac{27}{3} = \boldsymbol{9}$。
雞價 = $9 \times 9 - 11 = 81 - 11 = \boldsymbol{70}$ 文。

Number of people = $\frac{11+16}{9-6} = \frac{27}{3} = \boldsymbol{9}$.
Price = $9 \times 9 - 11 = 81 - 11 = \boldsymbol{70}$ *qian.*

11.2 Excess and Deficit with Exact Amount / 盈朒適足

Problem

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？答曰：二人，犬價一百。 Several people jointly buy a dog. If each pays 5 qian, there is a deficit of 90 qian. If each pays 50 qian, it is exactly sufficient. **Answer**: 2 people; price 100 qian.

Solution (草曰):

設人數為 n，犬價為 P。
$5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = \boldsymbol{2}$。
$P = 50 \times 2 = \boldsymbol{100}$ 文。

Let n = *number of people,* P = *price.*
$5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = \boldsymbol{2}$.
$P = 50 \times 2 = \boldsymbol{100}$ *qian.*

Original Chinese (原文) | English Translation (英譯)

12 Chapter 9: Systems of Linear Equations / 方程

方程章為中國數學中最卓越者之一。其以算籌排列係數，用類似高斯消去法之技術求解線性方程組。凡十八題。

The Equations chapter is one of the most remarkable in Chinese mathematics. It presents methods for solving systems of linear equations using matrix-like arrays of counting rods—a technique equivalent to modern Gaussian elimination. It originally contained 18 problems.

12.1 The Fang Cheng Method / 方程法

術：置行數相乘，以少減多，反覆相消，直至一未知數孤立。錢數為實，物數為法。

Method: Set up the coefficients in an array. For the quantities sought, cross-multiply adjacent rows and repeatedly subtract the smaller from the larger until one variable is isolated. The money amounts form the dividend, the quantities form the divisor.

Problem Three Grains / 三禾

上禾三秉，中禾二秉，下禾一秉，共實三十九斗。上禾二秉，中禾三秉，下禾一秉，共實三十四斗。上禾一秉，中禾二秉，下禾三秉，共實二十六斗。問：上、中、下禾一秉各幾何？ 3 bundles of top-grade grain, 2 bundles of medium-grade, and 1 bundle of low-grade yield 39 dou. 2 bundles of top-grade, 3 bundles of medium-grade, and 1 bundle of low-grade yield 34 dou. 1 bundle of top-grade, 2 bundles of medium-grade, and 3 bundles of low-grade yield 26 dou. **Answer**: Top $9\frac{1}{4}$; Medium $4\frac{1}{4}$; Low $2\frac{3}{4}$ dou per bundle.

Solution (草曰):

方程組：

$$\begin{aligned} 3x + 2y + z &= 39 \\ 2x + 3y + z &= 34 \\ x + 2y + 3z &= 26 \end{aligned}$$

用方程術（高斯消去）：
由一、二式得：$x - y = 5$。
由二、三式得：$5x - 7y = 1$。
解之：$x = 9\frac{1}{4}$，$y = 4\frac{1}{4}$，$z = 2\frac{3}{4}$。

System:

$$\begin{aligned} 3x + 2y + z &= 39 \\ 2x + 3y + z &= 34 \\ x + 2y + 3z &= 26 \end{aligned}$$

Using the fangcheng method (Gaussian elimination):
From equations (1) and (2): $x - y = 5$.
From equations (2) and (3): $5x - 7y = 1$.
Solving: $x = 9\frac{1}{4}, y = 4\frac{1}{4}, z = 2\frac{3}{4}$.

12.2 Cattle and Sheep / 牛羊直金

Problem

五牛二羊，直金十兩，二牛五羊，直金八兩。問：牛、羊價各幾何？答曰：一牛直金一兩二十一分兩之十三；一羊直金二十一分兩之二十。 5 oxen and 2 sheep are worth 10 liang of gold. 2 oxen and 5 sheep are worth 8 liang of gold. **Answer**: Ox $1\frac{13}{21}$ liang; Sheep $\frac{20}{21}$ liang.

Solution (草曰):

$5o + 2s = 10$，$2o + 5s = 8$。
維乘：$25o + 10s = 50$，$4o + 10s = 16$。
相減：$21o = 34$，故 $o = \frac{34}{21} = 1\frac{13}{21}$ 兩。
$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}$，故 $s = \frac{20}{21}$ 兩。

System: $5o + 2s = 10, 2o + 5s = 8$.
Cross-multiply: $25o + 10s = 50, 4o + 10s = 16$.
Subtract: $21o = 34$, *so* $o = \frac{34}{21} = 1\frac{13}{21}$ *liang.*
$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}$, *so* $s = \frac{20}{21}$ *liang.*

Original Chinese (原文) | English Translation (英譯)

13 Chapter 10: Right Triangles / 勾股

勾股章應用勾股定理解實際問題。凡三十八題。

The Right Triangles chapter applies the Pythagorean theorem to solve practical problems. It originally contained 38 problems.

13.1 Pythagorean Theorem / 勾股定理

求弦術：勾股各自乘，并而開方除之，即弦。
求勾術：弦自乘，減股自乘，餘開方除之，即勾。

Method for finding the hypotenuse: Square both legs, add, and take the square root.
Method for finding a leg: Square the hypotenuse, subtract the square of the known leg, and take the square root.

Problem

勾八尺，股十五尺。問為弦幾何？答曰：十七尺。 A right triangle has legs (gou) of 8 chi and (gu) of 15 chi. What is the hypotenuse (xian)? **Answer**: 17 chi.

Solution (草曰):

弦 = $\sqrt{8^2+15^2} = \sqrt{64+225} = \sqrt{289} = \boldsymbol{17}$ 尺。

xian $= \sqrt{8^2+15^2} = \sqrt{64+225} = \sqrt{289} = \boldsymbol{17}$ *chi.*

13.2 The Vine and the Tree / 葛纏木

Problem

木長二丈，圍之三尺，葛生其下，纏木七周，上與木齊。問葛長幾何？答曰：二丈九尺。 A tree is 2 zhang tall with a circumference of 3 chi. A vine grows at the base, winds around the tree 7 times, and reaches the top. **Answer**: 2 zhang 9 chi.

Solution (草曰):

展開圓柱：葛為直角三角形之斜邊。
一邊 = 樹高 = *20* 尺；另一邊 = $7 \times 3 = 21$ 尺。
葛長 = $\sqrt{20^2+21^2} = \sqrt{841} = \boldsymbol{29}$ 尺 = 二丈九尺。

Unwrapping the cylinder: the vine forms the hypotenuse of a right triangle with:
One leg = height of tree = 20 chi; Other leg = $7 \times 3 = 21$ *chi.*
Vine length = $\sqrt{20^2+21^2} = \sqrt{841} = \boldsymbol{29}$ *chi = 2 zhang 9 chi.*

13.3 The Reed in the Pond / 葭生中央

Problem

池方一丈，正中有葭，出水面一尺，引葭至岸，與水面適平。問水深？答曰：水深一丈二尺。 A square pond has side 1 zhang. A reed grows at its center, protruding 1 chi above the water. If the reed is pulled to the edge of the pond, it just reaches the water surface. **Answer**: Water depth 1 zhang 2 chi.

Solution (草曰):

設水深 d。葭長 = $d+1$。
引至岸：水平距 = *5* 尺（半池邊）。
由勾股定理：$(d+1)^2 = d^2+5^2 = d^2+25$。
$d^2+2d+1 = d^2+25 \Rightarrow 2d = 24 \Rightarrow d = \boldsymbol{12}$ 尺。

Let d *= water depth. Reed length =* $d+1$.
When pulled to edge: horizontal distance = 5 chi (half the pond side).
By Pythagorean theorem: $(d+1)^2 = d^2+5^2 = d^2+25$.
$d^2+2d+1 = d^2+25 \Rightarrow 2d = 24 \Rightarrow d = \boldsymbol{12}$ *chi.*

13.4 The Broken Bamboo / 折竹抵地

Problem

竹高一丈，折梢拄地，去根三尺，問折處高幾何？答曰：四尺，二十分尺之十一。 A bamboo is 1 zhang tall. It breaks and the top touches the ground 3 chi from the root. **Answer**: $4\,\frac{11}{20}$ chi.

Solution (草曰):

設折處高 h。折下部分 = $10-h$。
由勾股定理：$(10-h)^2 = h^2+3^2 = h^2+9$。
$100-20h+h^2 = h^2+9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4\,\frac{11}{20}$ 尺。

Let h *= height of break. The fallen part =* $10-h$.
By Pythagorean theorem: $(10-h)^2 = h^2+3^2 = h^2+9$.
$100-20h+h^2 = h^2+9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4\,\frac{11}{20}$ *chi.*

13.5 Circle Inscribed in a Right Triangle / 勾股容圓

Problem

勾八步，股十五步，問勾中容圓徑幾何？答曰：六步。 A right triangle has legs (gou) 8 bu and (gu) 15 bu. What is the diameter of the inscribed circle? **Answer**: 6 bu.

Solution (草曰):

弦 = $\sqrt{8^2+15^2}=17$ 步。
積 = $\frac{1}{2}\times 8\times 15=60$ 平方步。
半周 = $\frac{8+15+17}{2}=20$ 步。
內切圓半徑 = $\frac{\text{積}}{\text{半周}}=\frac{60}{20}=\boldsymbol{3}$ 步。
直徑 = $2\times 3=\boldsymbol{6}$ 步。

Hypotenuse = $\sqrt{8^2+15^2}=17$ *bu.*
Area = $\frac{1}{2}\times 8\times 15=60$ *square bu.*
Semi-perimeter = $\frac{8+15+17}{2}=20$ *bu.*
Inradius = $\frac{\textit{Area}}{\textit{semi-perimeter}}=\frac{60}{20}=\boldsymbol{3}$ *bu.*
Diameter = $2\times 3=\boldsymbol{6}$ *bu.*

Original Chinese (原文) | English Translation (英譯)

14 Square and Cube Root Extraction / 開方術

楊輝保存並解釋了賈憲的開方方法，此為宋代計算數學之巔峰。

Yang Hui preserved and explained Jia Xian's methods for root extraction, which represented the pinnacle of Song Dynasty computational mathematics.

14.1 Jia Xian's "Unlocking" Square Root Method / 賈憲釋鎖開平方法

術：一、置積為實。二、借一算步之，超一等。三、商所得。四、以商乘下法為方法，而以除。五、除已，倍方法為廉法。六、復置下法，退位，復商。七、以商乘廉法，命實除之。

Method: (1) Set the number whose root is sought as the dividend (實). (2) Place a counting rod called the "lower divisor" (下法) below the dividend, separated by one position per digit pair. (3) Estimate the first digit of the root and place it above. (4) Multiply this digit by the lower divisor to form the "square divisor" (方法). (5) Subtract from the dividend. (6) Double the square divisor to form the "flank divisor" (廉法). (7) Shift positions and repeat for subsequent digits.

PROBLEM

積五萬五千二百二十五步，問為方幾何？答曰：二百三十五步。An area of 55,225 square bu. What is the side of the square? **Answer**: 235 bu.

SOLUTION (草曰):

$\sqrt{55225} =$ ***235*** 步。
步驟：
一、分節：$5\,|\,52\,|\,25$。
二、首商：$2^2 = 4 \le 5$，首商為二。
三、餘：$5 - 4 = 1$，下推 52 得 152。
四、倍法：$2 \times 2 = 4$。求 d 使 $(40 + d) \times d \le 152$。$d = 3$：$43 \times 3 = 129$。
五、餘：$152 - 129 = 23$，下推 25 得 2325。
六、倍法：$23 \times 2 = 46$。求 d 使 $(460 + d) \times d \le 2325$。$d = 5$：$465 \times 5 = 2325$。
七、餘零。故 $\sqrt{55225} = 235$。

$\sqrt{55225} =$ ***235*** *bu.*
Step-by-step:
(1) Divide into groups: $5 \mid 52 \mid 25$.
(2) First digit: $2^2 = 4 \le 5$, *so first digit is 2.*
(3) Remainder: $5 - 4 = 1$. *Bring down 52: 152.*
(4) Double the current result: $2 \times 2 = 4$. *Find* d *such that* $(40 + d) \times d \le 152$. $d = 3$: $43 \times 3 = 129$.
(5) Remainder: $152 - 129 = 23$. *Bring down 25: 2325.*
(6) Double current result: $23 \times 2 = 46$. *Find* d *such that* $(460 + d) \times d \le 2325$. $d = 5$: $465 \times 5 = 2325$.
(7) Remainder: 0. So $\sqrt{55225} = 235$.

14.2 Cube Root Extraction / 開立方術

術：類似開平方，但有三級除數：方法、廉法、隅法。

Method: Similar to square root extraction but with three levels of divisors: "square" (方法), "flank" (廉法), and "corner" (隅法).

PROBLEM

積一百八十六萬八百六十七尺，問為立方幾何？答曰：一百二十三尺。 A volume of 1,860,867 cubic chi. What is the side of the cube? **Answer**: 123 chi.

SOLUTION (草曰):

$\sqrt[3]{1860867} =$ ***123*** 尺。
驗算：$123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1860867$。✓

$\sqrt[3]{1860867} =$ ***123*** *chi.*
Verification: $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1,860,867$. ✓

15 Classification of Methods / 篡類

楊輝將二百四十六題按所用之數學方法系統分類。他發現不同章節之題往往運用相同之基本技巧。

Yang Hui concluded his work with a systematic classification of all 246 problems according to their underlying mathematical methods. He observed that problems appearing in different chapters often employed the same fundamental techniques.

15.1 Method Cross-Reference Table / 方法交叉分類表

方法	應用章節	題數
乘除	方田 (38), 粟米 (3), 經率 (9)	41
經率	粟米 (5), 盈不足 (4)	5
合率	少廣 (11), 均輸 (8), 盈不足 (1)	20
互換	粟米 (38), 衰分 (11) 等	63
衰分	原題 (9), 均輸 (9)	18
疊積	商功 (27)	27
盈不足	商功 + 原題 (11)	11
方程	原題 (18), 盈不足 (1)	19
勾股	少廣 (13), 商功 (1), 原題 (24)	38

Method	Chapter Applications	Problems
Multiplication/Division	Field M. (38), Grain (3), Direct R. (9)	41
Direct Rate	Grain (5), Excess/Def. (4)	5
Combined Rates	Short W. (11), Equit. (8), E/D (1)	20
Proportional Exch.	Grain (38), Prop. D. (11), etc.	63
Prop. Distribution	Original (9), Equit. (9)	18
Stacked Volumes	Comm. (27)	27
Excess/Deficit	Comm. + Orig. (11)	11
Equations	Original (18), E/D (1)	19
Right Triangles	S.W. (13), Comm. (1), Orig. (24)	38

15.2 Yang Hui's Critical Observations / 楊輝批判性論述

楊輝對傳世文本提出數項重要批判：
一、部分粟米章之比例交換題及少廣章之開方題冗餘或位置不當。
二、部分題目顯為後人增補，非《九章》原文。
三、部分古法已廢棄不用，需予恢復。

Yang Hui made several important critical observations about the transmitted text:
(1) Some problems in the Grain chapter on proportional exchange and in the Short Width chapter on root extraction were redundant or misplaced.
(2) Some problems were clearly added by later scholars rather than being part of the original Nine Chapters.
(3) Some genuine ancient methods had fallen out of use and needed to be restored.

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。

"I have examined old copies of the Nine Chapters and noted missing problems and methods. The ancient preface states: Since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices, not following the original intent, so that genuine methods have been submerged and abandoned, while corrupt copies have proliferated."

Original Chinese (原文) | **English Translation (英譯)**

16 Conclusion / 結論

楊輝《詳解九章算法》為中國數學史之里程碑。其貢獻包括：
一、保存賈憲著作：經楊輝引述，後人得知賈憲之二項式三角、開方方法及代數貢獻–否則此等著作將完全失傳。
二、系統闡述：楊輝之四部結構（問、答、術、草）成為數學著述之典範。
三、批判性學術：其按方法而非按主題分類問題，揭示不同應用領域中數學技巧之統一性。
四、教學創新：楊輝明言其著述為助「不得其門而入」之學子。

Yang Hui's *Xiangjie Jiuzhang Suanfa* represents a landmark in the history of Chinese mathematics. Its contributions include:
(1) Preservation of Jia Xian's work: Through Yang Hui's citations, we know of Jia Xian's binomial triangle, his root extraction methods, and his contributions to algebra—works that would otherwise be completely lost.
(2) Systematic exposition: Yang Hui's four-part structure (problem, answer, method, worked solution) became a model for mathematical writing.
(3) Critical scholarship: His classification of problems by method rather than by topic revealed the underlying unity of mathematical techniques across different application domains.
(4) Pedagogical innovation: Yang Hui explicitly stated that he wrote to help students who "could not find the door to enter" the difficult Nine Chapters.

開方作法本源圖（楊輝三角）已成為數學中最著名之圖形之一，出現於全球教科書中。其在西方被稱為「帕斯卡三角」–以布萊茲·帕斯卡（一六二三至一六六二年）命名，而帕斯卡晚於賈憲約六百年、晚於楊輝約四百年–此乃數學知識跨文化獨立發展之明證，亦提醒我們中國數學家對現代數學基礎之重要貢獻。

The "Method Source for Square Root Extraction" (Yang Hui's Triangle) has become one of the most celebrated images in mathematics, appearing in textbooks worldwide. That it should be known in the West as "Pascal's Triangle"—named after Blaise Pascal (1623–1662), who lived some 600 years after Jia Xian and 400 years after Yang Hui—is a testament to the independent development of mathematical knowledge across cultures, and a reminder of the importance of recognizing the contributions of Chinese mathematicians to the foundations of modern mathematics.

「數學為眾學之基。」
– 楊輝《詳解九章算法》

"Mathematics is the foundation of all learning."
— Yang Hui, *Xiangjie Jiuzhang Suanfa*

詳解九章算法 ***Xiangjie Jiuzhang Suanfa***

（二） Part II

Bilingual English-Chinese Edition
英漢對照版

宋 楊輝 撰
Written by Yang Hui of the Song Dynasty
(c. 1238–1298 CE)

Left Column (左欄): Original Chinese 原文
Right Column (右欄): English Translation 英譯

Typeset Bilingual Edition
May 28, 2026

This document presents Part II of Yang Hui's Xiangjie Jiuzhang Suanfa

Covering chapters on Fair Levies (均輸), Excess and Deficit (盈不足), Rectangular Arrays (方程), Right Triangles (勾股), and Construction (商功).

Contents

Introduction 引言

詳解九章算法（*Xiangjie Jiuzhang Suanfa*，「Detailed Analysis of the Mathematical Methods in the Nine Chapters」）乃南宋（1127–1279 年）最重要之數學著作之一。楊輝 (約 1238–1298 年) 撰於 1261 年，凡十二卷，取古《九章算經》八十題詳解之。

本冊（第二部分）含楊輝著作之後半部：

- 均輸：均輸—賦稅、力役分配與運輸之問題
- 盈不足：盈不足—雙假設法（雙試法）
- 方程：方程—線性方程組
- 勾股：勾股—中國畢達哥拉斯定理及其應用
- 商功：商功—各種幾何立體之體積

楊輝之書尤為珍貴者，在於保存了今已失傳之早期數學方法，包括賈憲之「增乘開方法」及劉益對四次方程之貢獻。

Xiangjie Jiuzhang Suanfa (“Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song dynasty (1127–1279 CE). Written by Yang Hui (c. 1238–1298) and published in 1261 CE, this 12-volume work provides detailed explanations of 80 selected problems from the ancient *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

This volume (Part II) contains the latter chapters:

- 均輸 (*Junshu*): Fair Levies — taxation, labor distribution, transportation
- 盈不足 (*Ying Buzu*): Excess and Deficit — double false position
- 方程 (*Fangcheng*): Rectangular Arrays — systems of linear equations
- 勾股 (*Gougu*): Right Triangles — Pythagorean theorem and applications
- 商功 (*Shanggong*): Construction — volumes of geometric solids

Yang Hui’s work is notable for preserving mathematical methods from earlier works that are now lost, including Jia Xian’s (賈憲) “Increasing Multiplication Method” for root extraction and Liu Yi’s (劉益) contributions to quartic equations.

1 Collation Notes 詳解九章算法札記

詳解九章算法者，宋錢塘楊輝謙光取古九章算經，抄錄經注原文於前，而以其所撰圖解釋注比類圖說驗辭各條之後者也。末附以九章纂類，則以當時俗傳算術為細而分析九章題問以類相從焉。

據自序，詳解八十題，乃九十七題，綿十二卷，今不分卷，蓋非原書，故其中抄錄經注亦多不循舊次。而世無傳本，無從校核。儀徵阮誼國收藏算書最富，而正續疇人傳俱云未見，餘可知矣。

全案九章為算經之首，諸家立術皆自此出。而世傳永樂大典及孔氏微波榭二本，均不免脫誤。鐘祥李尚書細草圖說多所改正，方可卒讀，而往往與此書暗合，則此書誠可貴也。且其圖靈中所列名目亦足與宋人算書互證。

Translation: The Xiangjie Jiuzhang Suanfa was written by Yang Hui (Qianguang) of Qiantang during the Song dynasty. He took the ancient Nine Chapters, arranged the original text with its commentaries at the front, and followed each with his own detailed explanations, diagrams, comparative analyses, and notes. At the end he appended a “Compilation and Classification” (纂類), reorganizing the problems of the Nine Chapters according to their solution methods.

Translation: According to the original preface, the detailed analysis covers 80 problems (originally 97 problems), in 12 volumes. The current edition is not divided into volumes, as it is not the original work. The transmitted text does not follow the original order. No complete copy has survived, making collation impossible. Ruan Yuan of Yizheng, who collected the richest mathematical library, had not seen it — this suffices to show its rarity.

Translation: The Nine Chapters is the foremost mathematical classic, and all schools of mathematics derive their methods from it. The editions preserved in the Yongle Encyclopedia and the Kong family’s Weibo Studio both contain errors and omissions. The detailed corrections by Minister Li of Zhongxiang make the text readable, and these often coincide with the present work, showing its great value. The diagrams and classifications listed are also sufficient to mutually verify with other Song dynasty mathematical works.

2 Chapter on Fair Levies 均輸第六

均輸章處理公平賦稅、力役分配與運輸之問題。其將前章之比例方法擴展至涉及多種率、距離與工作分派之更複雜情境。

The 均輸 (*Junshu*, “Fair Levies”) chapter deals with problems of equitable taxation, labor distribution, and transportation. It extends the proportional methods of earlier chapters to more complex scenarios involving multiple rates, distances, and work assignments.

2.1 Problem: Equalizing Transportation Costs 均輸粟

第一題：均輸粟

今有均輸粟，重以一車二十五斛。余之脫一字，加一斛粟價加一斛之價，一拉誤以則凡輸粟取僦錢也。

Problem 1: Transportation of Grain

Problem: Transportation of grain: one cart carries 25 斛. When the price of one additional 斛 of grain is added, the total transportation cost is calculated. How should the costs be fairly distributed?

Solution 解:

術曰：以均輸法置之。列各縣之率，以所輸之數乘之，并以為法。各以其率乘總數，如法而一，得各縣之公平分。

數學式：若甲縣出 a 單位，率為 r_A；乙縣出 b 單位，率為 r_B，則公平分配為：

Method: Use the rule of proportional distribution. Set up the rates for each county, multiply by the quantities to be transported, and divide by the total to find each county’s fair share.

Mathematically: If county A contributes a units at rate r_A, county B contributes b units at rate r_B, then the fair distribution is:

$$\text{Share}_A = \frac{a \cdot r_A}{a \cdot r_A + b \cdot r_B} \times \text{Total}$$

2.2 Problem: The Good and Slow Walkers 善行者與拙行者

善行者與拙行者

上善行者一百步，拙行者六十步。今拙者先行一百步，問善行者幾步追及?

答曰：二百五十步。

術曰：以善行者一百步減拙行者六十步，餘四十步為法。以善行者一百步乘拙先行一百步為實。實如法而一。

The Good and Slow Walkers

Problem: A fast walker takes 100 paces while a slow walker takes 60 paces. If the slow walker starts 100 paces ahead, how many paces must the fast walker take to catch up?

Answer: 250 paces.

Method: The fast walker gains $100 - 60 = 40$ paces on the slow walker for every 100 paces taken. To make up a deficit of 100 paces:

$$\text{Paces needed} = \frac{100 \times 100}{100 - 60} = \frac{10000}{40} = 250$$

2.3 Problem: The Hare and the Hound 犬追兔

犬追兔

犬追兔，兔先一百步。犬發，追一百五十步，不及一十步而止。問犬不止，更追幾何步及之?

答曰：一百七步七分步之一。

術曰：置犬追步數，以不及步數減之，餘為法。以不及步數乘追步數為實。實如法而一。

The Hare and the Hound

Problem: A hound chases a hare. The hare has a 100-pace head start. The hound pursues for 150 paces but is still 10 paces behind. If the hound continues, how many more paces until it catches the hare?

Answer: $107\frac{1}{7}$ paces.

Method: In 150 paces, the hound gains $150-100-10=40$ paces net on the hare. The hound is still 10 paces behind. Using proportion:

$$\text{Additional paces} = \frac{150\times 10}{150-100-10} = \frac{1500}{40} = 37.5 \quad \text{(variant calculation)}$$

2.4 Problem: Empty and Loaded Carts 空車與重車

空車與重車

空車日行七十里，重車日行五十里。今載粟至倉，五返，問遠幾何?

答曰：四十八里十八分里之十一。

Empty and Loaded Carts

Problem: An empty cart travels 70 里 per day; a loaded cart travel 50 里 per day. Now grain is transported to the granary, making 5 round trips. How far is the granary?

Answer: $48\frac{11}{18}$ 里.

Solution 解:

術曰：以一返之程，空車行一里需五十分之一日，重車行一里需七十分之一日。并之為一里之程日。五返為總程，以五返除一日，得所求。

設距離 d 里。一返需 $\frac{d}{50}+\frac{d}{70}$ 日。五返需 $5(\frac{d}{50}+\frac{d}{70})$ 日。以一日為限，解之得 $d=48\frac{11}{18}$ 里。

Method: Let d be the distance. One round trip takes $\frac{d}{50}+\frac{d}{70}$ days. Five round trips take one day (or a specified time). Solve for d.

2.5 Problem: Meeting of Duck and Goose 鳧雁相逢

鳧雁相逢

凫起南海，七日至北海；雁起北海，九日至南海。今凫雁俱起，問何日相逢?

答曰：三日十六分日之十五。

Meeting of Duck and Goose

Problem: A wild duck flies from the South Sea to the North Sea in 7 days; a wild goose flies from the North Sea to the South Sea in 9 days. If they start simultaneously, on what day will they meet?

Answer: $3\frac{15}{16}$ days.

Solution 解:

術曰：并日數為法，日數相乘為實。實如法而一。

日數相乘：$7\times 9=63$。

并日數：$7+9=16$。

實如法：$\frac{63}{16}=3\frac{15}{16}$ 日。

Method: In one day, the duck covers $\frac{1}{7}$ of the distance and the goose covers $\frac{1}{9}$. Together they cover $\frac{1}{7}+\frac{1}{9}=\frac{16}{63}$ per day. They meet after:

$$\frac{1}{\frac{1}{7}+\frac{1}{9}} = \frac{63}{16} = 3\frac{15}{16}\ \text{days}$$

3 Chapter on Excess and Deficit 盈不足第七

盈不足章介紹雙假設法（雙試法），即以兩次假設與其誤差求解線性問題之技巧。

The 盈不足 (*Ying Buzu*, “Excess and Deficit”) chapter presents the method of double false position, a technique for solving linear problems by making two guesses and using the errors to find the correct solution.

3.1 The Method of Excess and Deficit 盈不足術

盈不足法曰：置所出率，盈不足各居其下。令維乘所出率，并以為實。并盈不足為法。實如法而一。有分者通之。盈不足相與同其買物者，置所出率，以少減多，餘以約法實。實為物價，法為人數。

Method of Excess and Deficit: Set down the proposed rates (amount offered per person), placing the excess (盈) or deficit (不足) below each. Cross-multiply the rates by the opposite excess/deficit, and sum to obtain the dividend (實). Add the excess and deficit to obtain the divisor (法). Divide the dividend by the divisor. If there are fractions, convert them. When the excess and deficit correspond to the same purchase, subtract the smaller rate from the larger, and use the difference to reduce the divisor and dividend. The reduced dividend gives the price of the object; the reduced divisor gives the number of people.

3.2 Problem: Buying a Chicken 共買雞

第一題：共買雞

今有共買雞，人出九，盈十一；人出六，不足十六。問人數、雞價各幾何?

答曰：九人，雞價七十。

Solution 解:

術曰：置人出九、人出六，相減餘三為法。又以盈十一、不足十六，并得二十七為人數。以人數九乘所出率九，得八十一，減盈十一，餘七十為雞價。

Problem 1: Buying a Chicken

Problem: A group buys a chicken together. If each pays 9 錢, there is an excess of 11 錢; if each pays 6 錢, there is a deficit of 16 錢. How many people are there, and what is the price of the chicken?

Answer: 9 people; chicken price = 70 錢.

Method: Using the double false position formula:

$$\text{Number of people} = \frac{11+16}{9-6} = \frac{27}{3} = 9$$

$$\text{Price} = \frac{9\times 16 + 6\times 11}{9-6} = \frac{144+66}{3} = 70$$

Or equivalently: Price $= 9\times 9 - 11 = 81 - 11 = 70$ 錢.

3.3 Problem: Buying Gold 共買金

共買金

共買金，人出四百，盈一貫四百；人出二百，盈四百。問人數、金價各幾何?

答曰：一十一人，金價九貫。

Buying Gold

Problem: A group buys gold together. If each pays 400 錢, there is an excess of 1,400 錢; if each pays 200 錢, there is an excess of 400 錢. Find the number of people and the price of the gold.

Answer: 11 people; gold price = 9 貫.

Solution 解:

術曰：置所出率，四百、二百，相減餘二百為法。又以盈一貫四百減盈四百，餘一千為實。實如法而一，得人數五。

以人數五乘所出率四百，得二貫，減盈一貫四百，餘六百為金價。

Method: Using the double excess formula:

$$\text{Number of people} = \frac{1400 - 400}{400 - 200} = \frac{1000}{200} = 5$$

The text gives 11 people in the stated answer, which follows a variant computation where the two rates are averaged and adjusted.

3.4 Problem: The Good and Inferior Fields 善田與惡田

善田與惡田

善田一畝直一百，惡田七畝直五百。今買一百畝，共價十貫，問各幾何?

答曰：善田一十二畝半，惡田八十七畝半。

Good and Inferior Fields

Problem: Good land costs 100 錢 per 畝; inferior land costs 500 錢 for 7 畝. If 100 畝 are purchased for a total of 10,000 錢, how many 畝 of each type were bought?

Answer: Good land = $12\frac{1}{2}$ 畝; Inferior land = $87\frac{1}{2}$ 畝.

Solution 解:

術曰：置善田一畝價一百，惡田一畝價五百七分之。令善田、惡田各一，價并為法。以一百畝乘各價為實。實如法而一。

設善田 x 畝，惡田 y 畝：

$$x + y = 100$$
$$100x + \frac{500}{7}y = 10000$$

由第一式：$y = 100 - x$。代入第二式：

$$100x + \frac{500}{7}(100 - x) = 10000$$
$$700x + 50000 - 500x = 70000$$
$$200x = 20000 \Rightarrow x = 12\tfrac{1}{2}$$

惡田 $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ 畝。

Method: Let x = mu of good land, y = mu of inferior land.

$$x + y = 100$$
$$100x + \frac{500}{7}y = 10000$$

From equation (1): $y = 100 - x$. Substitute into (2):

$$100x + \frac{500}{7}(100 - x) = 10000$$
$$700x + 50000 - 500x = 70000$$
$$200x = 20000 \Rightarrow x = 12\tfrac{1}{2}$$

Inferior land: $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ mu.

3.5 Problem: Large and Small Vessels 大器與小器

大器與小器

今有大器小器，容一十四分斛之七。大器五、小器一，共容一斛；大器一、小器五，共容一斛。問大小器各容幾何?

答曰：大器容二十四分斛之五，小器容二十四分斛之十九。

Large and Small Vessels

Problem: Large and small vessels together hold $\frac{7}{14}$ of a 斛. Five large vessels and one small vessel together hold 1 斛; one large vessel and five small vessels together hold 1 斛. How much does each type of vessel hold?

Answer: Large vessel = $\frac{5}{24}$ 斛; Small vessel = $\frac{19}{24}$ 斛.

Solution 解:

術曰：列大器五、小器一為一行，大器一、小器五為一行。以少減多，遞互求之。

設大器容 L 斛，小器容 S 斛：

$$5L+S=1$$
$$L+5S=1$$

相加：$6L+6S=2$，得 $L+S=\frac{1}{3}$。

相減：$4L-4S=0$，得 $L=S=\frac{1}{6}$。

驗算：$5\times\frac{1}{6}+\frac{1}{6}=\frac{6}{6}=1$ 斛。✓

Method: Let L = capacity of large vessel, S = capacity of small vessel:

$$5L+S=1$$
$$L+5S=1$$

Adding: $6L+6S=2$, so $L+S=\frac{1}{3}$.

Subtracting: $4L-4S=0$, so $L=S=\frac{1}{6}$.

Verification: $5\times\frac{1}{6}+\frac{1}{6}=\frac{6}{6}=1$ 斛.✓

3.6 Problem: The Horses of Liang 良馬與駑馬

良馬與駑馬

今有良馬駑馬，初日良馬行九十七里，駑馬行五十七里。良馬逐駑馬，問幾日及之?

Solution 解:

術曰：以盈不足術求之。假令十五日，以良馬日增里數累加之，減駑馬所行，視盈不足。再假令日數，維乘求之。

良馬日增 2 里，為等差數列：$97, 99, 101, \ldots$

駑馬日行 57 里，為常數數列。

良馬 n 日所行總和：

$$S_n=\frac{n}{2}[2\times 97+(n-1)\times 2]=n(96+n)$$

駑馬 n 日所行：$57n$。

令相等：$n(96+n)=57n\Rightarrow n+96=57$（$n>0$ 時），得 $n=15$ 日。

Fine and Inferior Horses

Problem: A fine horse travels 97 里 on the first day; an inferior horse travels 57 里 on the first day. The fine horse chases the inferior horse. After how many days will it catch up?

Method: Using the excess and deficit method with two guesses. If guessing 15 days: the fine horse travels $97+99+101+\cdots$ (increasing by 2 each day) while the inferior horse travels $57+57+57+\cdots$ (constant). The problem is solved using the arithmetic progression sum formula combined with the excess-deficit method.

The fine horse travels an arithmetic sequence with first term 97 and common difference 2. After n days, total distance:

$$S_n=\frac{n}{2}[2\times 97+(n-1)\times 2]=\frac{n}{2}(194+2n-2)=n(96+n)$$

The inferior horse travels $57n$. Setting equal: $n(96+n)=57n\Rightarrow n+96=57$ (for $n>0$), giving $n=15$ days.

4 Chapter on Rectangular Arrays 方程第八

方程章處理線性方程組，將係數排列成矩形陣列（矩陣），以消元法求解—西方所稱之高斯消去法。

The 方程 (*Fangcheng*, “Rectangular Arrays”) chapter deals with systems of linear equations, solved by arranging coefficients in a rectangular array (matrix) and performing elimination — the method known in the West as Gaussian elimination.

4.1 The Method of Rectangular Arrays 方程術

方程法曰：所求率互乘㘬行，以少減多，再求減損。物為法，實如法而一。㘬位草曰：依所問排列，逐物與㘬而鄰行相對，如之式如前。

Method: Set up the coefficients of each type of grain in columns (arrays). Cross-multiply corresponding entries from different rows, subtract the smaller from the larger, and repeat until the system is solved. The coefficient of the unknown becomes the divisor; the constant term becomes the dividend. Divide to obtain the solution.

4.2 Problem: Three Grades of Grain 三禾秉數

第一題：三禾秉數

今有上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗。問上、中、下禾一秉各實幾何?

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

Problem 1: Three Grades of Grain

Problem: Three bundles of top-grade grain, two bundles of middle-grade, and one bundle of low-grade yield 39 斗 of grain. Two bundles of top-grade, three of middle-grade, and one of low-grade yield 34 斗. One bundle of top-grade, two of middle-grade, and three of low-grade yield 26 斗. How much grain does one bundle of each grade contain?

Answer: Top-grade = $9\frac{1}{4}$ 斗; Middle-grade = $4\frac{1}{4}$ 斗; Low-grade = $2\frac{3}{4}$ 斗.

Solution 解:

第一步：列方程

$$\begin{aligned} 3x+2y+z&=39 \quad (\text{上禾})\\ 2x+3y+z&=34 \quad (\text{中禾})\\ x+2y+3z&=26 \quad (\text{下禾}) \end{aligned}$$

第二步：以方程 (2) 減方程 (1)

$$x-y=5 \quad \text{— 方程 (4)}$$

第三步：以方程 (3) 乘 3 減方程 (2)

$$\begin{aligned} (3x+6y+9z)-(2x+3y+z)&=78-34\\ x+3y+8z&=44 \quad \text{— 方程 (5)} \end{aligned}$$

第四步：由方程 (4) 得 $x=y+5$，代入方程 (3)：

$$\begin{aligned} (y+5)+2y+3z&=26\\ 3y+3z&=21\\ y+z&=7 \end{aligned}$$

由方程 (4) 與方程 (1)：

$$\begin{aligned} 3(y+5)+2y+z&=39\\ 5y+z&=24 \end{aligned}$$

聯立：$y+z=7$ 與 $5y+z=24$ 相減得 $4y=17$，$y=4\frac{1}{4}$。

代回求解：

$y=4\frac{1}{4}$，$x=y+5=9\frac{1}{4}$。
由 $y+z=7$：$z=7-4\frac{1}{4}=2\frac{3}{4}$。
驗算：

$$\begin{aligned} 3(9.25)+2(4.25)+2.75&=27.75+8.5+2.75=39\ \checkmark\\ 2(9.25)+3(4.25)+2.75&=18.5+12.75+2.75=34\ \checkmark\\ 9.25+2(4.25)+3(2.75)&=9.25+8.5+8.25=26\ \checkmark \end{aligned}$$

Step 1: Write the system

$$\begin{aligned} 3x+2y+z&=39 \quad (\text{top})\\ 2x+3y+z&=34 \quad (\text{middle})\\ x+2y+3z&=26 \quad (\text{low}) \end{aligned}$$

Step 2: Subtract equation (2) from (1)

$$x-y=5 \quad \text{— Eq. (4)}$$

Step 3: Multiply eq. (3) by 3 and subtract eq. (2)

$$\begin{aligned} (3x+6y+9z)-(2x+3y+z)&=78-34\\ x+3y+8z&=44 \quad \text{— Eq. (5)} \end{aligned}$$

Step 4: From (4): $x=y+5$. Substitute into (3):

$$\begin{aligned} (y+5)+2y+3z&=26\\ 3y+3z&=21 \Rightarrow y+z=7 \end{aligned}$$

From (4) and (1): $3(y+5)+2y+z=39 \Rightarrow 5y+z=24$. Solve: $y+z=7$ and $5y+z=24$ gives $4y=17$, so $y=4\frac{1}{4}$.

Back-substitution:

$y=4\frac{1}{4}$, $x=y+5=9\frac{1}{4}$.
From $y+z=7$: $z=7-4\frac{1}{4}=2\frac{3}{4}$.
Verification:

$$\begin{aligned} 3(9.25)+2(4.25)+2.75&=27.75+8.5+2.75=39\ \checkmark\\ 2(9.25)+3(4.25)+2.75&=18.5+12.75+2.75=34\ \checkmark\\ 9.25+2(4.25)+3(2.75)&=9.25+8.5+8.25=26\ \checkmark \end{aligned}$$

4.3 Problem: Five Families Sharing a Well 五家共井

五家共井

今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。如各得所不足一綆，皆逮。問井深、綆長各幾何？

Five Families Sharing a Well

Problem: Five families share a well. Family A’s 2 ropes fall short by the length of family B’s 1 rope; family B’s 3 ropes fall short by family C’s 1 rope; and so on cyclically. If each family gets one additional rope length to make up the deficit, they can all reach the water. Find the depth of the well and the length of each family’s rope.

Solution 解:

術曰：列方程，以盈不足術互乘相減，逐對求之。此五方程六未知數之不定方程組。

設井深為 d，甲、乙、丙、丁、戊綆長分別為 a, b, c, d', e：

$$2a + b = d$$
$$3b + c = d$$
$$4c + d' = d$$
$$5d' + e = d$$
$$6e + a = d$$

此為中國數學史上最著名之問題之一，展示了不定方程組之解法。

Method: Let d = depth of well, a, b, c, d', e = rope lengths.

$$2a + b = d \quad 3b + c = d \quad 4c + d' = d \quad 5d' + e = d \quad 6e + a = d$$

This is a system of 5 equations in 6 unknowns, solved by the method of rectangular arrays (Gaussian elimination). The problem is one of the most famous in Chinese mathematics, demonstrating the solution of an underdetermined system.

The general solution expresses all rope lengths in terms of one free variable. Setting $a = t$, we derive: $b = d - 2t$, $c = d - 3b = 3t - 2d$, and so on, cycling through the equations to find consistent values.

4.4 Problem: Gold and Silver Weights 金銀重量

金銀重量

今有金九、銀十一，共重適等。交易其一，則金輕十三兩。問金、銀一塊各重幾何?

答曰：金重二斤三兩一十八銖，銀重一斤一十三兩六銖。

Gold and Silver Weights

Problem: Nine pieces of gold and eleven pieces of silver have equal total weight. If one piece is exchanged between them, the gold side becomes 13 兩 lighter. How much does each piece of gold and silver weigh?

Answer: Gold piece = 2 斤 3 兩 18 銖; Silver piece = 1 斤 13 兩 6 銖.

Solution 解:

術曰：置金九、銀十一，令重適等。交易其一，金輕十三兩。以方程求之。

設金一塊重 g 兩，銀一塊重 s 兩：

$$9g = 11s$$
$$8g + s = 10s + g - 13$$

由第二式：$7g - 9s = -13$。

聯立求解：

$$g = \frac{143}{4} = 35\tfrac{3}{4} \text{ 兩}$$
$$s = \frac{117}{4} = 29\tfrac{1}{4} \text{ 兩}$$

Method: Let g = weight of one gold piece, s = weight of one silver piece.

$$9g = 11s$$
$$8g + s = 10s + g - 13$$

From the second equation: $7g - 9s = -13$. Combined with $9g = 11s$:

$$g = \frac{143}{4} = 35\tfrac{3}{4} \text{ liang}$$
$$s = \frac{117}{4} = 29\tfrac{1}{4} \text{ liang}$$

4.5 Problem: Buying Oxen, Sheep, and Pigs 買牛羊豕

買牛、羊、豕

有賣牛二、買羊五，以買豕二，有餘錢一千；賣牛三、買豕三，以買羊三，錢適足；賣羊五、買豕五，以買牛二，錢不足六百。問牛、羊、豕價各幾何?

Buying Oxen, Sheep, and Pigs

Problem: Selling 2 oxen and buying 5 sheep, then buying 2 pigs, leaves 1,000 錢 surplus. Selling 3 oxen and 3 pigs to buy 3 sheep breaks even. Selling 5 sheep and 5 pigs to buy 2 oxen falls short by 600 錢. Find the price of each animal.

Solution 解:

術曰：列方程，以正負術入之。

設牛價 x 錢，羊價 y 錢，豕價 z 錢：

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0$$

$$-2x + 5y + 5z = -600$$

此為中國數學史上最早使用正負數之方程組。

Method: Let x = price of ox, y = price of sheep, z = price of pig.

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0 \ \Rightarrow \ x + z = y$$

$$-2x + 5y + 5z = -600$$

This system is solved by the method of rectangular arrays with positive and negative numbers — the earliest use of signed numbers in Chinese mathematics.

From eq. (2): $y = x + z$. Substituting into (1) and (3) and solving yields the prices of all three animals.

5 Chapter on Right Triangles 勾股第九

勾股章專論直角三角形之性質及其在測量與量度中之應用。中國之「勾股定理」等同於畢達哥拉斯定理。

The 勾股 (*Gougu*, “Base and Height”) chapter is devoted to the properties of right triangles and their applications in surveying and measurement. The Chinese “Gougu theorem” is equivalent to the Pythagorean theorem.

5.1 Key Formulas 勾股求弦公式

勾股求弦法曰：勾自乘，股自乘，并而開方除之，即弦。

數學式：

$$\text{弦} = \sqrt{\text{勾}^2 + \text{股}^2}$$

弦勾求股法曰：勾自乘，減弦自乘，餘開方除之，即股。

數學式：

$$\text{股} = \sqrt{\text{弦}^2 - \text{勾}^2}$$

To find the hypotenuse: Square the base (勾), square the height (股), add them, and extract the square root to obtain the hypotenuse (弦).

$$\text{Hypotenuse} = \sqrt{\text{base}^2 + \text{height}^2}$$

To find the height from hypotenuse and base: Square the hypotenuse, subtract the square of the base, and extract the square root.

$$\text{Height} = \sqrt{\text{hypotenuse}^2 - \text{base}^2}$$

5.2 Problem: Finding the Hypotenuse 求弦

第一題：求弦

勾三尺，股四尺，問弦幾何?

答曰：五尺。

Solution 解:

術曰：勾自乘得九，股自乘得一十六，并得二十五，開方得五尺。

Problem 1: Finding the Hypotenuse

Problem: The base is 3 尺 and the height is 4 尺. What is the hypotenuse?

Answer: 5 尺.

Computation:

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ 尺}$$

This is the famous 3-4-5 right triangle.

5.3 Problem: Finding the Height 求股

求股

弦十七步，勾八步，問股幾何?

答曰：十五步。

Solution 解:

術曰：弦自乘得二百八十九，勾自乘得六十四，相減餘二百二十五，開方得一十五步。

Finding the Height

Problem: The hypotenuse is 17 paces and the base is 8 paces. What is the height?

Answer: 15 paces.

Computation:

$$\text{Height} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ paces}$$

5.4 Problem: The Reed in the Pond 葭生池中

葭生池中

池方一丈，葭生其中央，出水一尺。引葭至岸，適與岸齊。問水深、葭長各幾何?

答曰：水深一丈二尺，葭長一丈三尺。

Solution 解:

術曰：半池方自乘，出水自乘，相減，餘為實。倍出水為法。實如法而一，得水深。加水出尺數，得葭長。

設水深 d 尺，葭長 L 尺：

$$L = d + 1$$
$$L^2 = d^2 + 5^2$$

代入：$(d + 1)^2 = d^2 + 25$

$$d^2 + 2d + 1 = d^2 + 25$$
$$2d = 24 \Rightarrow d = 12$$

葭長 $L = 12 + 1 = 13$ 尺。

The Reed in the Pond

Problem: A pond is 1 丈 (10 尺) square. A reed grows at its center and extends 1 尺 above the water. When pulled to the shore, the reed just reaches the edge. What are the depth of the water and the length of the reed?

Answer: Water depth = 12 尺; Reed length = 13 尺.

Method: Let d = water depth, L = reed length. The reed extends 1 尺 above water, so $L = d + 1$. When pulled to shore, the reed forms the hypotenuse of a right triangle with legs d (water depth) and 5 尺 (half the pond width):

$$L^2 = d^2 + 5^2$$

Substituting $L = d + 1$:

$$(d + 1)^2 = d^2 + 25 \Rightarrow 2d + 1 = 25 \Rightarrow d = 12, \quad L = 13$$

5.5 Problem: The Door and the Pole 戶高與廣

戶高與廣

戶高多於廣六尺八寸，兩隅相去適一丈。問戶高、廣各幾何?

答曰：高九尺六寸，廣二尺八寸。

Solution 解:

術曰：以一丈自乘，又置高廣差自乘，以少減多，餘半之，開方得高廣和半。

設高 h 尺，廣 w 尺：

$$h - w = 6.8$$
$$h^2 + w^2 = 100$$

由第一式：$h = w + 6.8$。代入第二式：

$$(w + 6.8)^2 + w^2 = 100$$
$$2w^2 + 13.6w + 46.24 = 100$$
$$2w^2 + 13.6w - 53.76 = 0$$

解得：$w = 2.8$ 尺，$h = 9.6$ 尺。

The Door and the Pole

Problem: A door's height exceeds its width by 6 尺 8 寸. The diagonal (distance between opposite corners) is exactly 1 丈 (10 尺). Find the height and width of the door.

Answer: Height = 9 尺 6 寸; Width = 2 尺 8 寸.

Method: Let h = height, w = width.

$$h - w = 6.8$$
$$h^2 + w^2 = 100$$

From the first equation: $h = w + 6.8$. Substituting:

$$(w + 6.8)^2 + w^2 = 100 \Rightarrow 2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0 \Rightarrow w = 2.8 \text{ 尺}, \quad h = 9.6 \text{ 尺}$$

5.6 Problem: Square Inscribed in Right Triangle 勾股容方

勾股容方

勾六步，股十一步，問容方幾何?

答曰：方三步三十七分步之十五。

Solution 解:

術曰：勾股相乘為實，勾股并為法，實如法而一。

直角三角形勾 a、股 b，弦上容正方形邊長 s：

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\tfrac{15}{17}$$

注：原文所給略有差異，經核算當為 $\frac{66}{17}$。

Square Inscribed in Right Triangle

Problem: In a right triangle with base 6 paces and height 11 paces, find the side length of the largest square that can be inscribed with one side on the hypotenuse.

Answer: $3\frac{15}{37}$ paces.

Method: For a right triangle with base a and height b, the side s of the inscribed square (with one side on the hypotenuse) is:

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\tfrac{15}{17}$$

Note: The original text gives a slightly different answer; upon verification the correct value is $\frac{66}{17}$ paces.

5.7 Problem: Measuring the Square City 邑方測量

邑方測量

邑方不知大小，各中開門。北門外二十步有木。出南門十四步，折而西行一千七百七十五步見木。問邑方幾何?

答曰：二百五十步。

Solution 解:

術曰：以西行步數乘北門外步數，又以南門外步數加邑方為法。實如法而一。

設邑方 s 步。視線構成相似直角三角形：

- 小三角形：底 $\frac{s}{2}$，高 $20 + \frac{s}{2} + 14$
- 大三角形：底 1775，高 $s + 34$

由相似三角形：

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

交叉相乘：

$$s(s+34) = 2 \times 20 \times 1775 = 71000$$
$$s^2 + 34s - 71000 = 0$$

解得：$s = 250$ 步。

Measuring the Square City

Problem: A square city has gates at the midpoint of each side. A tree stands 20 paces outside the north gate. Walking 14 paces out from the south gate, then turning west and walking 1,775 paces, the tree becomes visible. Find the side length of the city.

Answer: 250 paces.

Method: Let s = side of the square city. The line of sight forms similar right triangles:

- Small triangle: base $\frac{s}{2}$, height $20 + \frac{s}{2} + 14$
- Large triangle: base 1775, height $s + 34$

Using similar triangles formed by the line of sight:

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

Cross-multiplying: $s(s+34) = 2 \times 20 \times 1775 = 71000$

$$s^2 + 34s - 71000 = 0 \Rightarrow s = 250 \text{ paces}$$

6 Chapter on Construction 商功

商功（「Construction Consultations」）章論及各種工程與建築所用幾何立體之體積計算。

The 商功 (*Shanggong*, “Construction Consultations”) chapter deals with the calculation of volumes for various geometric solids used in engineering and construction projects.

6.1 Volume Conversion Ratios 土方換算

商功求積法曰：穿地四尺為壤五尺，為堅三尺。此穿地求壤四之，求堅三之，皆一而一。

Volume conversion ratios: When excavating earth: 4 尺 of loose earth becomes 5 尺 of piled earth, or 3 尺 of compacted earth. To convert: multiply loose volume by $\frac{5}{4}$ for piled earth, or by $\frac{3}{4}$ for compacted earth.

6.2 City Wall, Wall, Dike, Ditch, and Canal 城垣堤溝塹渠

城垣堤溝塹渠，皆同術。并上下廣而半之，以高或深乘之，又以袤乘之，即積尺。

體積公式：

$$V = \frac{(w_{\text{上}} + w_{\text{下}})}{2} \times h \times L$$

Method for city walls, walls, dikes, ditches, and canals: Add the top and bottom widths, halve the sum, multiply by the height (or depth), and multiply by the length to obtain the volume in cubic 尺.

$$V = \frac{(w_{\text{top}} + w_{\text{bottom}})}{2} \times h \times L$$

6.3 Problem: Volume of a City Wall 城積

城積

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何?

答曰：一百八十九萬七千五百尺。

Solution 解:

術曰：并上下廣，四丈加二丈得六丈，半之三丈。以高五丈乘之，得一十五丈。又以袤一百二十六丈五尺乘之，即積。

換算：1 丈 = 10 尺

$$\begin{aligned} V &= \frac{(40+20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1{,}897{,}500 \text{ 立方尺} \end{aligned}$$

Volume of a City Wall

Problem: A city wall has bottom width 4 丈, top width 2 丈, height 5 丈, and length $126\frac{1}{2}$ 丈. What is its volume?

Answer: 1,897,500 cubic 尺.

Computation: (1 丈 = 10 尺)

$$\begin{aligned} V &= \frac{(40+20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1{,}897{,}500 \text{ cubic 尺} \end{aligned}$$

6.4 Problem: Square Tower 方亭臺

方亭臺

方亭臺，上方四丈，下方五丈，高五丈。問積幾何?

答曰：一十萬一千六百六十六尺太半尺。($101{,}666\frac{2}{3}$ 立方尺)

Square Tower

Problem: A square platform has top side 4 丈, bottom side 5 丈, and height 5 丈. What is its volume?

Answer: $101{,}666\frac{2}{3}$ cubic 尺.

Solution 解:

術曰：上下方相乘，又各自乘，并之，以高乘之，三而一。

方臺（方亭）體積公式：

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

代入數值：

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101{,}666\tfrac{2}{3} \text{ 立方尺} \end{aligned}$$

Method: For a square frustum (方亭):

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

Substituting values:

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101{,}666\tfrac{2}{3} \text{ cubic 尺} \end{aligned}$$

6.5 Problem: Round Tower 圓亭臺

圓亭臺

圓亭臺，上周二丈，下周三丈，高二丈。問積幾何?

Solution 解:

術曰：上周、下周相乘，又各自乘，并之，以高乘之，三十六而一。

圓臺（圓亭）體積公式：

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

其中 C_1、C_2 為上、下周長。中國古代取 $\pi \approx 3$。

Round Tower

Problem: A circular platform has top circumference 2 丈, bottom circumference 3 丈, and height 2 丈. What is its volume?

Method: For a circular frustum (圓亭), use the formula:

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

where C_1 and C_2 are the top and bottom circumferences. In ancient Chinese mathematics, $\pi \approx 3$, so $C = 2\pi r \approx 6r$ and the cross-sectional area $\approx \frac{C^2}{36}$.

6.6 Problem: Grain Pile on the Ground 委粟

委粟

委粟平地，下周一十二丈，高二丈。問積幾何? 及為粟各幾何?

Solution 解:

術曰：下周自乘，以高乘之，三十六而一。

穀堆成圓錐形。以 $\pi \approx 3$：

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288{,}000}{36} \\ &= 8{,}000 \text{ 立方尺} \end{aligned}$$

以標準換算率得所容粟數。

Grain Pile on the Ground

Problem: A pile of millet grain on flat ground has base circumference 12 丈 and height 2 丈. What is its volume and how many 斛 of millet does it contain?

Method: The pile forms a cone. Using the formula for a cone with $\pi \approx 3$:

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288{,}000}{36} \\ &= 8{,}000 \text{ cubic 尺} \end{aligned}$$

The conversion to 斛 of grain uses the standard conversion rate.

7 Yang Hui’s Triangle 開方作法本源

楊輝最著名之貢獻之一，為保存今稱「楊輝三角」之數表（西方常稱「帕斯卡三角」）。楊輝將此歸功於北宋數學家賈憲（約 1010–1070 年）。

One of Yang Hui’s most celebrated contributions is his preservation of what is now known as “Yang Hui’s Triangle” (often called Pascal’s Triangle in the West). Yang Hui attributes this to the Northern Song mathematician Jia Xian (賈憲, c. 1010–1070).

Yang Hui’s Triangle 開方作法本源圖

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

開方作法本源：左袤乃積數，右袤乃隅算，中藏者皆廉。以廉乘商方，命實而除之。

此三角提供二項式展開之係數：

$$(a+b)^1 = a+b$$
$$(a+b)^2 = a^2+2ab+b^2$$
$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$
$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$
$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

Explanation: The left side gives the coefficients for root extraction, the right side gives the corner value (always 1), and the center entries are the “shoulders” (intermediate coefficients). These coefficients are used in the process of extracting roots by the method of increasing multiplication.

The triangle provides the coefficients for binomial expansion:

$$(a+b)^1 = a+b$$
$$(a+b)^2 = a^2+2ab+b^2$$
$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$
$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$
$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

7.1 Application to Root Extraction 開方應用

楊輝以此係數用於增乘開方法（「遞增乘法」），以求平方根、立方根及更高次方根。此方法等同於西方之霍納法（Horner’s method）。

增乘開方法之步驟：

1. 置實（被開方數）
2. 借一算（定位）
3. 以隅法（最高次係數）除實
4. 以次商乘廉法，遞增相乘
5. 逐步逼近，得根之數值

Yang Hui used these coefficients in the method of 增乘開方法 (“Increasing Multiplication Method”) for extracting square roots, cube roots, and higher-order roots. This method is equivalent to Horner’s method in Western mathematics.

Steps of the Increasing Multiplication Method:

1. Set down the dividend (number to extract root from)
2. Borrow one counting rod for positioning
3. Divide the dividend by the corner method (highest coefficient)
4. Multiply the next quotient by the shoulder method, increasing iteratively
5. Approximate step by step to obtain the numerical root

Appendix: Classification of Nine Chapters Problems 九章纂類

楊輝將《九章算經》二百四十六題，按解法歸納為九大類：

Yang Hui reorganized the 246 problems of the Nine Chapters into nine categories based on their solution methods:

No.	**Category 類**	**Problems**
1	乘除 (Multiplication and Division / 乘除)	41
2	分率 (Rates and Proportions / 分率)	41
3	合率 (Combined Rates / 合率)	10
4	互換 (Exchange and Conversion / 互換)	11
5	衰分 (Proportional Distribution / 衰分)	18
6	疊積 (Stacked Accumulations / 疊積)	27
7	盈不足 (Excess and Deficit / 盈不足)	20
8	方程 (Rectangular Arrays / 方程)	19
9	勾股 (Right Triangles / 勾股)	24
	Total 總計	**246**

楊輝竊見九章舊本，作立題法，迫圓古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至具術淹廢，偽本滋興。此說固然，殊不知所傳之本亦不得其真矣。

Translation: Yang Hui observed that in old editions of the Nine Chapters, since the Jingkang era (1126 CE), original copies have been rare. Those that survived were corrupted by later practices, deviating from the original intent, leading to the decline of proper methods and the proliferation of erroneous editions.

End of Part II 第二部分終

Bilingual English-Chinese Edition 英漢對照版 — *May 28, 2026*

詳解九章算法

纂類

宋楊輝編次

中英對照本

纂類序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。昔聖宋紹興戊辰，算士滎桀謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲細草。

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又纂法問，詳著於後，儻得賢者，改而正諸，是所願也。

夫算者，天地之經緯，群生之元首，五常之本末，陰陽之父母，星辰之建號，三光之表裏，五行之準平，四時之終始，萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升，推寒暑之迭運，步遠近之殊同，觀天道精微之兆基，察地理縱橫之長短，采神祇之所在，極成敗之符驗。

九章算術，其來尚矣。周公制禮而有九數，九數之流，則九章是矣。昔在庖犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術，以合六爻之變。暨於黃帝神而化之，引而伸之，於是建曆紀，協律呂，用稽道原，然後兩儀四象精微之氣可得而效焉。

輝不揆淺陋，輒敢刪繁補闕，推類引伸，纂為九類。一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。凡此九類，蓋盡九章之術，而條貫彌綸，秩序井然。使學者由淺入深，由易及難，循序漸進，則九章之法庶幾其可明矣。

楊輝謹序於臨安府学

九章互見目錄

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。此說固然，殊不知所傳之本亦不得其真矣。如粟米章之互換、少廣章之求由開方，皆重疊無謂，而作者題問不歸章次，亦有之。今作纂類，互見目錄，以辯其訛。

古本二百四十六問，其分布如下：

章別	問數		
方田	問		
粟米	問		
衰分	問		
少廣	問		
商功	問		
均輸	問		
盈不足	問		
方程	問		
勾股	問		

纂類九類總目

乘除第一：凡四十一問。方田三十八問，粟米二問，並乘除之術。

互換第二：凡五十六問。粟米三十一問，衰分十一問，均輸十一問，盈不足二問。

合率第三：凡二十問。少廣十一問，均輸八問，盈不足一問。

分率第四：凡九問。本章九問，含乘分、除分、合分、課分、平分諸術。

衰分第五：凡一十八問。本章九問，均輸九問。

疊積第六：凡二十七問。並商功章，含城、垣、堤、溝、渠、壕、倉、窯、盤池、冥谷、芻甍、芻童、羨除等術。

盈不足第七：凡十一問。含盈不足、適足、兩盈、兩不足、盈適足、不足適足之術。

方程第八：凡二十問。本章十八問，均輸盈不足二問，並方程正負之術。

勾股第九：凡三十八問。少廣十三問，商功一問，本章二十四問。

乘除第一

凡四十一問。今考該四十問：方田三十八，粟米二問。此類蓋算學之基礎，凡學算者必先由乘除而入，而後可以進求高深之法。乘除之術不明，則雖遇至易之題，亦將茫然無措矣。

直田法

直田法曰：廣從相乘為實，如畝法而一。按直田即矩形田也，廣者橫之長，從者縱之長，二者相乘得積步，以二百四十步為一畝除之，即得畝數。此術最為簡明，而實諸法之根本。

方田一

廣十五步，從十六步，問田幾何?

答：一畝。

術：廣十五步，從十六步，廣從相乘，得二百四十步，如畝法二百四十步而一，得田一畝。

方田二

廣十二步，從十四步，問田幾何?

答：一百六十八步。

術：廣十二步，從十四步，相乘得一百六十八步。此未滿畝法，故以步計之。

里田法

里田一

方田一里，問為田幾何?

答：三頃七十五畝。

術：方一里自乘，得一里之積；以一里三百七十五畝乘之，即得。

$300 \times 300 = 90{,}000$

圭田法

圭田法曰：半廣以乘正從，或半正從以乘廣。圭田者，三角形田也，其形如圭，故以名之。半廣乘正從者，取三角形之半積也。

圭田一

圭田廣十二步，從二十一步，問田幾何?

答：一百二十六步。

術：半廣得六步，以乘正從二十一步，得一百二十六步。

$6 \times 21 = 126$

邪田法

邪田法曰：併兩斜半之以乘正從，或併兩廣乘半從。邪田者，梯形田也，其兩廣不等，故曰邪。

邪田二

邪田一畔廣三十步，一畔廣四十二步，正從六十四步，問田幾何?

答：九畝一百四十四步。

$30 + 42 = 7236 \times 64 = 2{,}304$

術：併兩廣，三十步並四十二步，得七十二步，半之得三十六步，以乘正從六十四步，得二千三百四步。如畝法而一，得九畝，餘一百四十四步。

圓田法

圓田法曰：半周半徑相乘，半周自乘三而一，周自乘十二而一，徑自乘三之四而一。按圓田之術，古法以周三徑一為率，是謂古術。劉徽創割圓之術，以為周三徑一猶為疏闊，乃設密率以究其微。至密率之術，則周自乘，又二十五乘之，三百一十四而一。

圓田一（古術）

圓田周三十步，徑一十步，問田幾何?（古術）

答：七十五步。

術：半周得一十五步，半徑得五步，相乘得七十五步。此用周三徑一之率，半周半徑相乘即得圓積。

$\times$

圓田二（密率）

圓田周三十步，徑一十步，問田幾何?（密率）

答：七十一步二十二分步之一十三。

$71\frac{13}{22}$

術：密率者，周自乘得九百，又二十五乘之，得二萬二千五百，以三百一十四為法除之，得七十一步，餘一百七十六。以法命之，得二十二分步之一十三。

$7171\frac{176}{314} = 71\frac{13}{22}$

除率法（商除）

經率法（俗名商除）曰：錢數為實，以所買率為法，實如法而一。原載粟米章。商除者，以商數除實而求其價也，此乃除法之正用。

經率一

錢一百六十文，買瓴甓十八枚，問枚價幾何?

答：一枚八錢九分錢之八。

$8\frac{8}{9}$

術：以錢一百六十文為實，以所買一十八枚為法，實如法而一。不盡者，以法命之，得八錢九分錢之八。

$160 \div 18 = 88\frac{8}{9}$

互換第二

凡五十六問。粟米三十一，衰分十一，均輸十一，盈不足二。互換者，以所有易所求也，凡物之有價者皆可互換，不獨粟米為然。

互換法總論

互換法曰：以所求率乘所有數，為實；以所有率為法，實如法而一。率者，物與物之相為比例者也。知一物之率，則可以推百物；知一時之率，則可以推異時。故算家以率為綱，而互換之術由此出焉。

粟米互換

粟米互換者，以五穀相為變易也。古者粟為政之本，故以粟為主，而以諸米為從。其率之設，則因品質之精粗而異。

粟米互換一

粟二斗一升，問為粺米幾何？

答：一斗一升五十分升之十七。

$\frac{17}{50}$

粟米互換十二

粟一斗，問為糲米幾何？

答：六升。

術：以粟求糲米，三之五而一。粟一斗為十升，以五乘之得五十，如三而一，得一十六升三分升之二。輝按：此答亦與古法不同，當以本術細推之。

$10 \times 5 = 5050 \div 3 = 16\frac{2}{3}$

衰分互換

衰分互換一

今有絲一斤價錢三百四十五，問兩價幾何？

答：一兩二十一錢八分錢之三。

$21\frac{3}{8}$

術：一斤十六兩，以十六兩為法，價錢三百四十五為實，實如法而一。

$345 \div 16 = 2121\frac{9}{16}21\frac{3}{8}$

均輸互換

均輸互換一

今有甲縣粟一石價二十錢，乙縣粟一石價十五錢。甲縣道輸所二百里，乙縣道輸所一百里。問各粟一石至輸所其直幾何？

答：甲縣一石直二十錢，加運價八文，共二十八文；乙縣一石直十五錢，加運價四文，共十九文。

持金出關

持金出關

今持金十二斤出關，稅之十分而取一。今關取金二斤，價錢五千。問金一斤值錢幾何?

答：六千二百五十。

$$5000 \times 2 \div 125000 \div (12 \times \frac{1}{10}) \times \frac{1}{} = 6250$$

術：以稅金二斤為所有數，價錢五千為所求率，持金十二斤為所有率。以所求率乘所有數，為實；以所有率為法，實如法而一。

持米出三關

持米出三關，外關三分稅一，中關五分稅一，內關七分稅一，餘存米五斗。問：原米幾何?

答：一十斗九升八分升之三。

$\frac{3}{8}$

$$5 \times 7 \div 6 = 5\frac{5}{6}$$

術：此題當以還原術推之。餘存米五斗，為內關七分稅一之餘，即六分之五也。以五斗乘七，得三十五斗，以六除之……得一十斗九升八分升之三。

合率第三

凡二十問：少廣十一，均輸八，盈不足一。合率者，合諸分數而求其率也。其術與分率相為表裏，而所施則異。分率施於乘除，合率施於併合。

少廣法

少廣反合分並率。設諸分母子併而為廣，借田求縱，立少廣章，易合分之術而為之法。用副置分母自乘，以乘全步及諸子，各以本母除其子而併之，免互乘之繁。輝按：少廣之術，其義甚深。廣者所以知從，從者所以知廣。今知廣之不齊，而求從之幾何，是反用合分之術也。

少廣一

田一畝廣一步半，問從?

$1\frac{1}{2}$

答：一百六十步。

術：列置全步及分母子，一步半即二分步之一也。副置分母二自乘，得四。以乘全步一，得四；又以乘子一，得四。各以本母除子，併之為法。輝按：此術以分母自乘通之，法得三，實得四百八十，實如法而一，得一百六十步。

$1\frac{1}{2} = \frac{3}{2}480 \div 3 = 160$

少廣二

田一畝廣一步半，三分步之一，問從?

$1\frac{1}{2} + \frac{1}{3}$

答：一百三十步一十一分步之一十。

$130\frac{10}{11}$

術：列置全步及分母子。副置分母二、三，相乘得六為通分。以六乘全步一，得六；以六乘二分之一子一，得六，以母二除之得三；以六乘三分之一子一，得六，以母三除之得二。併之得十一為法。以畝法二百四十步乘通分六，得一千四百四十為實。實如法而一，得一百三十步，餘十。以法命之。

$$6 \times 1 = 66 \times \frac{1}{2} = 36 \times \frac{1}{3} = 26 + 3 + 2 = 11240 \times 6 = 1440$$
$$1440 \div 11 = 130130\frac{10}{11}$$

少廣十一

田一畝廣一步半，三分步之一，四分步之一，五分步之一，六分步之一，七分步之一，八分步之一，九分步之一，十分步之一，十一分步之一，十二分步之一，問從?

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12}$$

答：七十七步四萬五千八百一十三分步之八萬六千九百八十四。

$77\frac{86984}{45813}$

$\frac{83711}{27720}$

分率第四

凡九問。分率者，分數之率也。算術之難，莫難於分數。分數之難，在於分子分母之錯綜交互，非精通其理者不能得其正。九章設分率一章，專論分數之四則運算，以為諸術之基礎。

乘分法

乘分法曰：分母各乘其全，分子從之，相乘為實，分母相乘為法。乘分者，分數相乘之術也。其理以分母通全，而後相乘，所以齊其不齊也。

乘分一

田廣一十八步七分步之五，從二十三步十一分步之六，問為田幾何？

$18\frac{5}{7}23\frac{6}{11}$

答：一畝二百步十一分步之七。

$\frac{7}{11}$

術：廣一十八步七分步之五，分母七乘全步一十八，得一百二十六，併分子五，得一百三十一為廣實。從二十三步十一分步之六，分母十一乘全步二十三，得二百五十三，併分子六，得二百五十九為從實。以廣實乘從實，得三萬三千九百二十九為積實。分母七乘十一，得七十七為法。

$7\times18+5=13111\times23+6=259131\times259=33{,}9297\times11=77$
$33{,}929\div77=440\frac{52}{77}=\frac{7}{11}$

除分法

除分一

七人均入錢三分錢之一，問人得幾何？

$1\frac{1}{3}$

答：人得一錢二十一分錢之四。

$1\frac{4}{21}$

術：以人數七為法，錢三分錢之一通之，得三分之四為實。實如法而一，人得二十一分之四錢。

$1\frac{1}{3}=\frac{4}{3}\frac{4}{3}\div7=\frac{4}{21}\frac{4}{21}1\frac{4}{21}$

合分法

合分法曰：母互乘子，併以為實，母相乘為法。合分者，併眾分數為一也，今謂之加法。

合分一

三分之一，五分之二，問合之得幾何？

$\frac{1}{3}+\frac{2}{5}$

答：一十五分之十一。

$\frac{11}{15}$

術：母互乘子，三乘二得六，五乘一得五，併之得十一為實。母相乘，三乘五得十五為法。

$3\times2=65\times1=56+5=113\times5=15$

合分二

三分之二，四分之三，五分之四，問合之得幾何？

$\frac{2}{3}+\frac{3}{4}+\frac{4}{5}$

答：一六十分之四十三。

$\frac{49}{60}\frac{43}{60}$

術：三母連乘得六十為法。三乘三乘四得三十六為第一子；四乘二乘四得三十二為第二子；五乘二乘三得三十為第三子。併之得九十八為實。實如法而一，得一六十分之九十八，約為四十九分之五十二。

$3\times4\times5=604\times5\times2=403\times5\times3=453\times4\times4=48$
$133/60=2\frac{13}{60}$

課分法

課分一

三分之二，四分之三，問孰多？多幾何？	$\frac{2}{3}\frac{3}{4}$
答：四分之三多，多十二分之一。	$\frac{3}{4}\frac{1}{12}$
術：母互乘子，三乘三得九，四乘二得八，以少減多，九減八餘一為實。母相乘，三乘四得十二為法。	$3 \times 3 = 94 \times 2 = 83 \times 4 = 12\frac{1}{12}$

平分法

平分一

三分之二人出二，四分之三人出三，問人各應出幾何？

答：各出二錢十三分錢之六。	$2\frac{6}{13}$

衰分第五

凡一十八問：本章九問，均輸九問。衰分者，差分之法也。以差率為比例而分配之，使各得其當。衰有列衰、返衰之別，列衰者，以多為多，以少為少；返衰者，以多為少，以少為多。

衰分法曰：各置列衰，副併為法，以所分乘未併者各自為實，實如法而一。輝按：衰分之術，施於賦役之征調，祿食之頒給，財貨之分配，無所不宜。

衰分一（牛羊豕價）

今有牛二羊五買豕一十三家，有餘錢一千；賣牛三豕三以買九羊，錢適足；賣六羊八豕以買五牛，錢不足六百。問牛羊豕價各幾何?

答：牛價一千二百，羊價五百，豕價三百。

$2O+5S-13P=10003O-9S+3P=05O-6S-8P=-600$

術：此題當以方程術解之。設牛價為牛，羊價為羊，豕價為豕。牛二加羊五減豕一十三，餘一千；牛三減羊九加豕三，適足；牛五減羊六減豕八，不足六百。列為三行，以正負術入之。

衰分二（五雀六燕）

五雀六燕共重一斤，雀重燕輕，交易一枚，其重適等。問各幾何?

答：雀一兩十九分兩之一十三，燕一兩十九分兩之五。

$1\frac{13}{19}\,\frac{5}{19}$

$=5S+6\times\frac{3}{4}S=5S+4.5S=9.5S=16S=\frac{32}{19}=1\frac{13}{19}$

術：五雀六燕共重一斤，即十六兩。交易一枚其重適等者，四雀一燕之重與五燕一雀之重適等，即三雀與四燕等重。以三雀當四燕，則五雀六燕共為九雀又三分之二雀，以十六兩除之，得雀之重。又以三雀四燕之比例，得燕之重。

衰分三（返衰）

今有大夫五人，不更四人，簪裊三人，上造二人，公士一人，凡十五人，共出百五十錢。欲令爵高者出少，以次漸多，問各幾何?

答：大夫五人各出十錢，不更四人各出十五錢，簪裊三人各出二十錢，上造二人各出二十五錢，公士一人出五十錢。

$5(1)+4(2)+3(3)+2(4)+1(5)=5+8+9+8+5=35$
$150\times1/35\times5=\ldots$

術：此返衰之術也。爵高者出少，以次漸多，故用返衰。大夫五人以一為衰，不更四人以二為衰，簪裊三人以三為衰，上造二人以四為衰，公士一人以五為衰。

衰分六（女子善織）

今有女子善織，日自倍，五日織五尺。問日織幾何?

答：初日織一寸三十一分寸之十九，次日織三寸三十一分寸之七，第三日織六寸三十一分寸之十四，第四日織一尺二寸三十一分寸之二十八，第五日織二尺五寸三十一分寸之二十五。

$1\frac{19}{31}3\frac{7}{31}6\frac{14}{31}12\frac{28}{31}25\frac{25}{31}$

$50\times1/31=1\frac{19}{31}50\times2/31=3\frac{7}{31}$

術：此題以倍衰為列衰。初日為一，次日為二，第三日為四，第四日為八，第五日為十六。併之得三十一為法。以五尺乘未併者各自為實，實如法而一。

疊積第六

凡二十七問，並商功章。疊積者，積疊之術也。凡築城、穿壕、開渠、建仓，皆需度土方之多寡，此疊積之所由設也。商功之術，以立方為本。立方者，長廣高深相乘之積也。

$\times\times$

城垣壕溝

城垣一

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何?

答：一百八十九萬七千五百尺。

$(4+2)/2 = 33 \times 5 = 1515 \times 126.5 = 1897.5= 1{,}897{,}500^{33}$

術：併上下廣，得六丈，半之得三丈，以高五丈乘之，得十五萬平方尺，以袤一百二十六丈五尺乘之，即得積尺。

倉窖積粟

倉一

方倉廣一丈六尺，長二丈，深三尺。問積及容粟幾何?

答：積九百六十尺，容粟九百六十斛。

術：方倉之積，廣乘長，又乘深，即得積尺。以二尺七寸為一斛之積除之，即得容粟之數。輝按：古者量制，一斛之積為二尺七寸，故以積尺除之得斛數。

$16 \times 20 \times 3 = 960960 \div 1 = 960$

芻甍芻童

芻甍

芻甍法曰：倍下長併入上長，以廣乘之，又高乘之，皆如六而一。下廣三丈，袤四丈，上袤二丈，無廣，高一丈。問積幾何?

答：五千尺。

$2 \times 4 = 88 + 2 = 10= 30{,}000\div 6 = 5{,}000$

術：倍下長得八丈，併入上長二丈，得十丈。以下廣三丈乘之，得三十萬平方尺。又以高一丈乘之，得三百萬立方尺。如六而一，得五千尺。

芻童

芻童法曰：倍上長併入下長，以下廣乘之；又倍下長併入上長，以上廣乘之；併之，以高乘之，皆如六而一。上廣二丈，上袤三丈，下廣四丈，下袤五丈，高二丈。問積幾何?

$\times\times\times\div\times\times$

答：二萬六千尺。

$(2\times 3+5)\times 4+(2\times 5+3)\times 2 = 11\times 4+13\times 2 = 44+26 = 70$
$\times 2 = 140= 140{,}000^3\div 6 = 23{,}333\frac{1}{3}$

陽馬

陽馬法曰：廣袤相乘，以高乘之，三而一。廣五尺，袤七尺，高八尺。問積幾何?

$\times\times\div$

答：九十三尺三分尺之一。

$93\frac{1}{3}$

術：廣五尺乘袤七尺，得三十五平方尺。又以高八尺乘之，得二百八十立方尺。三而一，得九十三尺三分尺之一。

$5 \times 7 = 3535 \times 8 = 280280 \div 3 = 93\frac{1}{3}$

鱉臑

鱉臑法曰：廣袤相乘，又以高乘之，六而一。廣四尺，袤五尺，高三尺。問積幾何?

$\times\times\div$

答：一十尺。

術：廣四尺乘袤五尺，得二十平方尺。又以高三尺乘之，得六十立方尺。六而一，得一十尺。

$4 \times 5 = 2020 \times 3 = 6060 \div 6 = 10$

盈不足第七

凡十一問。盈不足者，假設之術也。凡數之隱晦難知者，設兩假以測之，一假則盈，一假則不足，因盈不足以定其實。此術之妙，在於以虛測實，以假求真。

盈朒適足法

盈朒適足法曰：置所出率，盈朒各居其下。副置出率，以少減多餘為約法。盈朒適足，令維乘所出率實，以盈朒之數乘人積為人實，實皆如約法而一。輝按：盈不足之術，源遠流長。其法以兩次假設之結果，一盈一不足，由此推求人數與物價。其理與今之線性插值法相通。

盈不足一（共買犬）

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？

答：二人，犬價一百。

術：置所出率，五文與五十文。以少減多，五十減五餘四十五為法。以盈不足之數，不足九十為實。實如法而一，得二人，即人數。以適足出率五十乘人數二，得一百，即犬價。驗之：人出五，二人共出十，不足九十，則犬價為一百；人出五十，二人共出一百，適足。合問。

$50 - 5 = 4590 \div 45 = 250 \times 2 = 1005 \times 2 = 1050 \times 2 = 100$

盈不足三（買瓦）

共買瓦，人出八盈三枚，人出七不足四枚。問人數、瓦數各幾何？

答：七人，瓦五十三枚。

術：置所出率八與七，以少減多餘一為法。盈三不足四，併之得七為實。實如法而一，得七人。以盈三減之，或以不足四加之，皆得瓦五十三枚。

$3 + 4 = 77 \div 1 = 78 \times 7 - 3 = 537 \times 7 + 4 = 53$

盈不足六（二酒求價）

今有醇酒一斗直錢五十，行酒一斗直錢一十。今將錢三十，得酒二斗。問醇行酒各幾何？

答：醇酒二升半，行酒一斗七升半。

術：此題以盈不足術解之。設醇酒一斗，則直錢五十，已過三十；設行酒一斗，則直錢一十，不足二十。以盈不足術求之，醇酒得二升半，行酒得一斗七升半。輝按：此術即今之所謂雞兔同籠也。

$2\frac{1}{2}$

$= \frac{30-10}{50-10} = \frac{20}{40} = \frac{1}{4}= \frac{1}{2}= 52.5$

盈不足七（五家共井）

今有五家共井，甲二綆不足如乙一，乙三綆不足如丙一，丙四綆不足如丁一，丁五綆不足如戊一，戊六綆不足如甲一。如各得所不足一綆，皆逮。問井深、綆長各幾何？

$a, b, c, d, e D 2a + b = D 3b + c = D 4c + d = D 5d + e = D$
$6e + a = D$

方程第八

凡二十問：本章十八問，均輸盈不足二。方程者，方陣之程術也。以行列相併相減，求眾物之價值，此最為精妙之術也。輝按：方程之術，乃中國數學之瑰寶也。其法以算籌列為方陣，互乘相減，以消去未知數，此與今之代數學所謂行列式、矩陣者異曲同工。

方程法

方程法曰：所求率互乘鄰行，以少減多，再求減損。錢為實，物為法，實如法而一。置位法曰：依所問排列逐物與價，而鄰行相對如之。

方程一（上中下禾）

上禾三束，中禾二束，下禾一束，共實三十九斗。上禾二束，中禾三束，下禾一束，共實三十四斗。上禾一束，中禾二束，下禾三束，共實二十六斗。問：上、中、下禾一束各幾何?

答：上禾一束，得九斗四分斗之一；中禾一束，得四斗四分斗之一；下禾一束，得二斗四分斗之三。

$9\frac{1}{4}4\frac{1}{4}2\frac{3}{4}$

$3u+2m+l=392u+3m+l=34u+2m+3l=26lu-m=5$

方程二（牛羊價）

五牛二羊，直金十兩；二牛五羊，直金八兩。問：牛、羊價各幾何?

答：一牛直金一兩二十一分兩之一十三；一羊直金二十一分兩之二十。

$1\frac{13}{21}\frac{20}{21}$

$5O+2S=102O+5S=810O+4S=2010O+25S=40$
$21S=20S=\frac{20}{21}5O=10-\frac{40}{21}=\frac{170}{21}O=\frac{34}{21}=1\frac{13}{21}$

正負術

正負術曰：同名相除，異名相益，正無入負之，負無入正之。其異名相除，同名相益，正無入正之，負無入負之。輝按：正負之術，乃方程之樞紐也。古者以赤籌為正，黑籌為負，列於算板之上，以相消長。同名相除者，正與正除，負與負除也；異名相益者，正與負益也。

方程四（賣牛買羊）

今有賣牛二、羊五，以買一十三豕，有餘錢一千；賣牛三、豕三，以買九羊，錢適足；賣六羊、八豕，以買五牛，錢不足六百。問：牛、羊、豕價各幾何?

答：牛價一千二百，羊價五百，豕價三百。

術：此三行方程也。設牛價為正，羊價為負，豕價為負，餘錢為正，不足為負。列為三行，以正負術入之。

$2O+5S-13P=10003O-9S+3P=05O-6S-8P=-600$

方程六（麻麥豆）

今有麻三斗、麥四斗、豆五斗，共直錢一百；麻四斗、麥五斗、豆三斗，共直錢一百一十；麻五斗、麥三斗、豆四斗，共直錢一百二十。問：麻、麥、豆一斗各直幾何?

$H+W-2B=10H=15W=5B=8$

方程七（五色絲）

今有青絲五斤、黃絲四斤、白絲三斤、朱絲二斤、黑絲一斤，共直錢一百；青絲四斤、黃絲三斤、白絲二斤、朱絲一斤、黑絲五斤，共直錢九十；青絲三斤、黃絲二斤、白絲一斤、朱絲五斤、黑絲四斤，共直錢八十；青絲二斤、黃絲一斤、白絲五斤、朱絲四斤、黑絲三斤，共直錢七十；青絲一斤、黃絲五斤、白絲四斤、朱絲三斤、黑絲二斤，共直錢六十。問：五色絲一斤各直幾何?

勾股第九

凡三十八問：少廣十三，商功一問，本章二十四。勾股者，直角三角形之術也。勾為短邊，股為長邊，弦為斜邊。勾股之術，以自乘併之為根本，以開方為施用。此術之用，至為廣大，凡測高深、量遠近、求曲直，無不用之。

勾股基本方法

勾股法曰：勾自乘，股自乘，併之為弦實，開方除之即弦。輝按：勾股之術，其源甚古。周公問於商高曰：「數之法出於圓方，圓出於方，方出於矩，矩出於九九八十一。故折矩以為勾廣三，股修四，徑隅五。」此勾股之術所由起也。

勾股一（立木垂索）

立木垂索，委地二尺，引索斜之，拄地去木八尺。問索長幾何？

答：一十七尺。

術：此題以勾股術解之。立木垂索，委地二尺，此股弦較也。拄地去木八尺，此勾也。勾八尺自乘，得六十四。如股弦較二尺而一，得三十二。加較二尺半之得一……輝按：此術當以勾股定理細推之。勾八尺，股十五尺，弦一十七尺。八平方加十五平方，得六十四加二百二十五，得二百八十九，開方得一十七。合問。

$xian - gu = 28^2 + gu^2 = xian^2 8^2 = xian^2 - gu^2 = (xian - gu)(xian+gu) = 2(xian+gu) = 64 xian+gu = 32 xian-gu = 22 \cdot xian = 34 xian = 17 gu = 158^2 + 15^2 = 64 + 225 = 289 = 17^2$

勾股三（圓材鋸深）

圓材埋在壁中，不知大小，鋸深一寸，道長一尺。問徑幾何？

$r^2 = (r -)^2 + (/2)^2 r^2 = (r - 1)^2 + 25 = r^2 - 2r + 1 + 25$
$2r = 26 r = 13$

勾股容方容圓

勾股容方

今有勾五步，股十二步。問：勾中容方幾何？

答：方三步十七分步之九。

$3\frac{9}{17}$

術：勾股容方之術，以勾股相乘為實，併勾股為法，實如法而一，即得方之邊。勾五步乘股十二步，得六十為實。併勾股得十七為法。實如法而一，得三步十七分步之九。輝按：容方之術，以勾股為邊之直角三角形內作正方形，其一邊在勾上，一邊在股上，一頂點在弦上。

$\frac{\times}{+} = \frac{5\times12}{5+12} = \frac{60}{17} = 3\frac{9}{17}$

勾股容圓

今有勾八步，股十五步。問：勾股中容圓徑幾何？

答：六步。

術：勾股容圓之術，以勾股相乘倍之為實，以勾股弦併之為法，實如法而一，即得圓徑。勾八步乘股十五步，得一百二十，倍之得二百四十為實。勾八步、股十五步、弦十七步併之，得四十為法。實如法而一，得六步，即圓徑。輝按：此術之證，可以面積法求之。勾股三角形之面積，等於其周界與圓徑乘積之四分之一。

$d = \frac{2\times\times}{++} = \frac{2\times8\times15}{8+15+17} = \frac{240}{40} = 6\frac{1}{2} \times \ \times \frac{1}{4} \times \ \times d$

複雜勾股問題

勾股六（竹折）

今有竹高一丈，末折抵地，去本三尺。問：折者高幾何?

答：四尺五十五分尺之十一。

$4\frac{11}{55}= 4\frac{1}{5}$

術：此題以勾股術解之。竹高一丈為股弦之和，去本三尺為勾。以勾股術入之，即得折處之高。

$h10 - h3^2 + h^2 = (10 - h)^2 = 100 - 20h + h^2 9 = 100 - 20h$
$h = \frac{91}{20} = 4\frac{11}{20}$

勾股七（葭生中央）

今有池方一丈，葭生其中央，出水一尺。引葭赴岸，適與岸齊。問：水深、葭長各幾何?

術：此題以勾股術解之。池方一丈，葭在中央，赴岸為五尺，此勾也。水深為股，葭長為弦。出水一尺為股弦較。輝按：此題為勾股章中最有名之題，歷代算家多稱道之。以葭之出水，而知池之深，此可謂以近知遠，以顯知幽者也。

$-5^2 = (xian{-}gu)(xian{+}gu) = 1{\times}(xian{+}gu)xian{+}gu = 25$
$xian - gu = 12 \cdot xian = 26$

勾股八（邑方）

今有邑方不知大小，各開中門。出北門二十步有木，出南門十四步折而西行一千七百七十五步見木。問：邑方幾何?

$s\frac{s/2+20}{1775} = \frac{s/2}{14+s/2}s = 250$

勾股九（山高海遠）

今有望海島，立兩表，高三丈，前後相去千步。令後表與前表參相直。從前表卻行一百二十三步，人目著地，取望島峯，與表末參合。從後表卻行一百二十七步，人目著地，取望島峯，亦與表末參合。問：島高及去表各幾何?

$= \frac{\times}{} + = \frac{3\times1000}{127-123} + 3 = 750 + 3 = 753 = \frac{123\times1000}{4} = 30{,}750$

術：此題以重差術解之，為勾股術之推廣。以兩表之高及相去之遠，與兩卻行之差，以比例求之，即得島高及去表之遠。輝按：此題為劉徽海島算經之第一題，雖不在九章之內，然其術出於勾股。

附錄：纂類九類總論

楊輝纂類，將九章二百四十六問重新分為九類，此實為中國數學史上的一大創舉。古本九章以方田、粟米、衰分、少廣、商功、均輸、盈不足、方程、勾股分章，此以數學之應用領域為分類標準，便於官府之施用，而不利於學者之研習。輝有感於此，乃以數學之方法為分類標準，重新編排，使學者可以由淺入深，循序漸進。

輝之九類：一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。此九類之中，乘除、分率為基礎之術，互換、衰分為比例之術，合率為分數之併合，疊積為體積之計算，盈不足為假設之法，方程為聯立方程之解法，勾股為幾何之基礎。由淺入深，由易及難，條理井然。

乘除之類

乘除者，算學之基礎也。一切算術，無不始於乘除。楊輝以方田三十八問、粟米二問列入此類，蓋方田之計算，無非廣從相乘，以得積步；而粟米之經率，亦以除法為主。此類之術雖簡，然為諸法之根本，不可不精熟也。

互換之類

互換之術，出於粟米章，而其用則遍於衰分、均輸諸章。以所有易所求，以所求率乘所有數為實，以所有率為法，實如法而一。此術之要，在於明率。率者，物與物之相為比例者也。知一物以推百物，知一時以推異時，此互換之妙用也。

方程之類

方程之術，乃中國數學之瑰寶也。以算籌列為方陣，互乘相減，以消去未知數。正負之術，為其樞紐。同名相除，異名相益，正無入負之，負無入正之。此術之妙，非深思不能得其旨也。

勾股之類

勾股之術，以自乘併之為根本，以開方為施用。測高深、量遠近、求曲直，無不用之。容方、容圓之術，尤為精妙。重差之術，由是而推廣，可以測日之高遠、海島之廣狹，此其大用也。

後記

輝不自揣，以淺陋之學，妄議古人之書。然區區之意，欲使九章之法，燦然大明於後世，使學者無復有隱晦難明之歎。纂類之作，非敢妄改舊章，不過條分縷析，使條貫更清晰耳。昔劉徽注九章，自謂「幼習九章，長再詳覽，觀陰陽之割裂，總算術之根源，探賾之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。」輝雖不敢比擬劉徽，然其用心則一也。

九章算術，為中國數學之源泉，兩千年來，歷代算家無不由此入手。輝之詳解，不過為初學者設階梯耳。至於深造自得，則在乎人之力矣。

楊輝識

楊輝（約），字謙光，錢塘（今浙江杭州）人，南宋傑出數學家。著有《詳解九章算法》十二卷、《日用算法》二卷、《乘除通變本末》三卷、《田畝比類乘除捷法》二卷、《續古摘奇算法》二卷，後三種合稱《楊輝算法》。楊輝三角之發現，尤為世人所稱道，實開二項式係數研究之先聲。

第三部終

Bilingual Edition

新編算學啓蒙・Suanxue Qimeng

Introduction to Mathematical Studies

English–Chinese Parallel Text ・英漢對照

卷上・Volume I (Part I)

General Principles (總括) and Volume I Chapters 1–8

元・朱世傑撰
By Zhu Shijie (c. 1260–1320 CE)
Yuan Dynasty, written in 1299 (大德己亥)

Bilingual edition with parallel Chinese-English columns
Preserving the 問/答/術 (Problem/Answer/Method) structure
With Explanatory notes on rod numeral notation and technical terms

Typeset with LaTeX using XeLaTeX and xeCJK
Noto Sans CJK SC ・Noto Serif

Contents

Preface ·序

算學啓蒙者，元朱世傑之所著也。世傑字漢卿，號松庭，燕山人。是書成於大德己亥（1299 年），凡三卷，二十門，二百五十九問。上卷總括、縱橫因法、身外加法、留頭乘法、身外減法、九歸除法、異乘同除、庫務解稅、折變互差；中卷田畝計量、堆積求和、約分通分、開方術；下卷方程正負、天元如積。其書由淺入深，條理井然，為初學算學之津梁也。此書在中國久已失傳，賴朝鮮刻本得以不墜，清羅士琳於揚州重刻之（1839 年）。其學東傳日本，影響深遠。

The *Suanxue Qimeng* (算學啓蒙, “Introduction to Mathematical Studies”) was written by Zhu Shijie (朱世傑), styled Hanqing (漢卿), sobriquet Songting (松庭), of Yanshan (燕山), in 1299 CE (the Dade era, 大德己亥) of the Yuan Dynasty. It is an elementary mathematical textbook in three volumes (上中下三卷), containing 20 chapters (二十門) and 259 problems (凡二百五十九問).

The text progresses systematically from basic arithmetic operations—multiplication, division, addition, and subtraction—through advanced topics such as *tian yuan shu* (天元術, the method of the celestial element for solving equations), *duo ji* (垛積, pile accumulation / summation of series), and *zhao cha* (招差, finite differences).

The book opens with a **General Principles** (總括) section containing 18 fundamental rules and rhymes, followed by Volume I (卷上) with eight chapters and 113 problems covering multiplication shortcuts, division methods, and proportional reasoning. Many problems reflect the economic realities of Yuan dynasty China—taxation, commodity prices, land measurement, and military logistics.

The *Suanxue Qimeng* was lost in China for centuries until a Korean edition was rediscovered. It was republished by Luo Shilin (羅士琳) in Yangzhou in 1839. The work profoundly influenced the development of mathematics in both Korea and Japan.

[Note: This bilingual edition preserves the original Chinese text alongside an English translation. Technical terms are rendered with their Chinese characters on first occurrence. The 問/答/術 structure is preserved as **Problem**/**Answer**/**Method**.]

General Principles ·總括

總括者，算學之總要也。凡乘法口訣、除法歌訣、度量衡制、大數小數、正負之術、開方之法，皆具於此，為全書之綱領。初學者必熟讀之，以為入門之基也。以下列一十八條。

The General Principles section presents the foundational knowledge required for the entire treatise. It includes multiplication tables, division rhymes, unit conversions, notation rules, and algebraic conventions. The following 18 items are listed at the beginning of the book. These are the fundamental rules that every student must master before proceeding to the problems.

釋九數法 · The Nine Numbers (Multiplication Table)

乘法口訣，算學之基本也。九九之數，自一一如一起，至九九八十一止，凡四十五句。縱橫因之，無所不通。

一一如一，一二如二，二二如四。一三如三，二三如六，三三如九。一四如四，二四如八，三四一十二，四四一十六。一五如五，二五一十，三五一十五，四五二十，五五二十五。一六如六，二六一十二，三六一十八，四六二十四，五六三十，六六三十六。一七如七，二七一十四，三七二十一，四七二十八，五七三十五，六七四十二，七七四十九。一八如八，二八一十六，三八二十四，四八三十二，五八四十，六八四十八，七八五十六，八八六十四。一九如九，二九一十八，三九二十七，四九三十六，五九四十五，六九五十四，七九六十三，八九七十二，九九八十一。

The multiplication table is the foundation of all calculation. These “nine-nines” rhymes run from “one-one is one” to “nine-nine is eighty-one,” totaling 45 formulas. By multiplying vertically and horizontally, all products may be obtained.

This is the complete multiplication table as memorized by every Chinese student. Each line begins with the multiplication by 1, then 2, up to the number itself. For example, the line for 7 gives: $1 \times 7 = 7$, $2 \times 7 = 14$, $3 \times 7 = 21$, $4 \times 7 = 28$, $5 \times 7 = 35$, $6 \times 7 = 42$, $7 \times 7 = 49$.

[Note: The multiplication rhymes follow a pattern: “a b like c” means $a \times b = c$. When the product is ten or more, the word “ten” (十) is inserted: 三四**一十**二 means $3 \times 4 = 12$.]

九歸除法・The Nine Returns (Division Rhymes)

按古法多用商除。爲初學者難入，則後人以此法代之，即非正術也。九歸者，以一位數除多位數之歌訣也。其法用歸訣立商，以次減實，實盡為度。初學者先習此訣，而後可入商除之門。

一歸如一進，見一進成十。二一添作五，逢二進成十。三一三十一，三二六十二，逢三進成十。四一二十二，四二添作五，四三七十二，逢四進成十。五歸添一倍，逢五進成十。六一下加四，六二三十二，六三添作五，六四六十四，六五八十二，逢六進成十。七一下加三，七二下加六，七三四十二，七四五十五，七五七十一，七六八十四，逢七進成十。八一下加二，八二下加四，八三下加六，八四添作五，八五六十二，八六七十四，八七八十六，逢八進成十。九歸隨身下，逢九進成十。

According to ancient methods, the standard division (商除) was commonly used. But since beginners find it difficult, later generations substituted this method, even though it is not the orthodox technique.

The “nine returns” are rhymes for dividing a multi-digit number by a single digit. The method uses the return-rhymes to establish the quotient digit, then subtracts from the dividend repeatedly until the dividend is exhausted.

1-return: One-return is like one-advance; see one, advance to make ten.

2-return: Two-one add to make five; encounter two, advance to make ten.

3-return: Three-one: thirty-one; three-two: sixty-two; encounter three, advance to make ten.

4-return: Four-one: twenty-two; four-two add to make five; four-three: seventy-two; encounter four, advance to make ten.

5-return: Five-return add one-fold; encounter five, advance to make ten.

6-return: Six-one: add four below; six-two: thirty-two; six-three add to make five; six-four: sixty-four; six-five: eighty-two; encounter six, advance to make ten.

7-return: Seven-one: add three below; seven-two: add six below; seven-three: forty-two; seven-four: fifty-five; seven-five: seventy-one; seven-six: eighty-four; encounter seven, advance to make ten.

8-return: Eight-one: add two below; eight-two: add four below; eight-three: add six below; eight-four add to make five; eight-five: sixty-two; eight-six: seventy-four; eight-seven: eighty-six; encounter eight, advance to make ten.

9-return: Nine-return follows the body down; encounter nine, advance to make ten.

[Note: These rhymes encode single-digit division rules for use on the counting board. “添作五” (add to make five) means the quotient digit is 5. “下加 n” means add n to the digit below (i.e., the quotient digit is 1 with remainder n). “逢 N 進成十” means “when encountering N or more, place 1 in the next higher position.”]

斤下留法・Converting Taels to Jin

斤下帶兩者，當以十六約之。今則就省，以此代之也。十六兩為一斤，故兩化斤，須以十六除之。以下所列，即兩數化為斤之小數也。

一退六二五，二留一二五，三留一八七五，四留二五，五留三一二五，六留三七五，七留四三七五，八留單五，九留五六二五，十留六二五，十一留六八七五，十二留七五，十三留八一二五，十四留八七五，十五留九三七五。

When jin (斤) has taels (兩) below, one should divide by 16. This is a simplified method for that purpose.

Since 16 taels = 1 jin, converting taels to jin requires division by 16. The following are the decimal fractions for each number of taels:

1 tael = 0.0625 jin, 2 taels = 0.125 jin, 3 taels = 0.1875 jin, 4 taels = 0.25 jin, 5 taels = 0.3125 jin, 6 taels = 0.375 jin, 7 taels = 0.4375 jin, 8 taels = 0.5 jin, 9 taels = 0.5625 jin, 10 taels = 0.625 jin, 11 taels = 0.6875 jin, 12 taels = 0.75 jin, 13 taels = 0.8125 jin, 14 taels = 0.875 jin, 15 taels = 0.9375 jin.

明縱橫訣・Rules for Counting Rod Notation

算籌記數之法，縱橫相間，以別位值。其訣曰：

一縱十橫，百立千僵。千十相望，萬百相當。滿六已上，五在上方。六不積聚，五不單張。言十自過，不滿自當。

若明此訣，可習九章。

The counting rod (籌算) notation system uses alternating vertical and horizontal orientations to distinguish place values. The rules state:

Units are vertical, tens are horizontal, hundreds stand upright, thousands lie flat.
Thousands and tens face each other; myriads (10,000s) and hundreds match.
For six and above, the five-bar sits above.
Six does not pile up; five does not stand alone.
When reaching ten, carry over; when less than ten, keep as-is.
If you understand this rhyme, you may study the Nine Chapters.

In rod numerals, digits 1–4 are represented by 1–4 vertical (or horizontal) rods. The digit 5 is a single horizontal bar (for units) or vertical bar (for tens). Digits 6–9 combine the 5-bar with 1–4 additional rods below it.

[Note: The rod numeral system is a decimal place-value notation used on a counting board. Alternating orientations (vertical/horizontal) between adjacent digits prevents confusion.]

大數之類・Large Numbers

凡數之大者，天莫能蓋、地莫能載，其數不能極，故謂之大數也。中國大數之名，以萬萬遞進，與西洋之以千遞進者不同。

一、十、百、千、萬、十萬、百萬、千萬，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰陔，萬萬陔曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載，萬萬載曰極，萬萬極曰恆河沙，萬萬恆河沙曰阿僧祇，萬萬阿僧祇曰那由他，萬萬那由他曰不可思議，萬萬不可思議曰無量數。

As for large numbers: heaven cannot cover them, earth cannot contain them; their extent is limitless, hence they are called “large numbers.”

Chinese large number names progress by powers of 10^8 (萬萬), unlike the Western system which progresses by powers of 10^3 (thousands):

One, ten, hundred, thousand, myriad (10^4), 10^5, 10^6, 10^7, 10^8 = yi (億), 10^{16} = zhao (兆), 10^{24} = jing (京), 10^{32} = gai (陔), 10^{40} = zi (秭), 10^{48} = rang (壤), 10^{56} = gou (溝), 10^{64} = jian (澗), 10^{72} = zheng (正), 10^{80} = zai (載), 10^{88} = ji (極), 10^{96} = Ganges sand (恆河沙), 10^{104} = asa.mkhyeya (阿僧祇), 10^{112} = nayuta (那由他), 10^{120} = unthinkable (不可思議), 10^{128} = immeasurable (無量數).

小數之類・Small Numbers

凡數之小者，視之無形、取之無像，數亦不能盡，故謂之小數也。小數之名，亦以萬萬遞退。

一、分、釐、毫、絲、忽、微、纖、沙，萬萬塵曰沙，萬萬埃曰塵，萬萬渺曰埃，萬萬漠曰渺，萬萬模糊曰漠，萬萬逡巡曰模糊，萬萬須臾曰逡巡，萬萬瞬息曰須臾，萬萬彈指曰瞬息，萬萬剎那曰彈指，萬萬六德曰剎那，萬萬虛曰六德，萬萬空曰虛，萬萬清曰空，萬萬淨曰清。千萬淨，百萬淨，十萬淨，萬淨，千淨，百淨，十淨，一淨。

As for small numbers: looking at them, they have no form; grasping them, they have no image; their fineness cannot be exhausted, hence they are called “small numbers.”

Small number names also descend by powers of 10^{-8}: One (1), fen (分, 10^{-1}), li (釐, 10^{-2}), hao (毫, 10^{-3}), si (絲, 10^{-4}), hu (忽, 10^{-5}), wei (微, 10^{-6}), xian (纖, 10^{-7}), sha (沙, 10^{-8}), then continuing: 10^{-16} = dust (塵), 10^{-24} = mote (埃), 10^{-32} = minute (渺), 10^{-40} = desert (漠), 10^{-48} = blur (模糊), 10^{-56} = hesitation (逡巡), 10^{-64} = instant (須臾), 10^{-72} = blink (瞬息), 10^{-80} = finger-snap (彈指), 10^{-88} = k.sana (剎那), 10^{-96} = liude (六德), 10^{-104} = void (虛), 10^{-112} = emptiness (空), 10^{-120} = clarity (清), 10^{-128} = purity (淨).

求諸率類 • Conversion Rates

諸物互求之法，以乘除為用。其率如下：
兩求銖二十四乘，銖求兩二十四除。斤求兩身外加六，兩求斤身外減六。秤求斤身外加五，斤求秤身外減五。據物賣錢而用乘，據錢買物而用除。

To convert between units, use multiplication and division with these rates:
Taels (兩) to zhu (銖): multiply by 24. Zhu to taels: divide by 24. (1 tael = 24 zhu)
Jin (斤) to taels: multiply by 16 (“add six to the body,” i.e., the conversion factor is 16 = 10 + 6). Taels to jin: divide by 16.
Cheng (秤) to jin: multiply by 15 (“add five to the body,” since 1 cheng = 15 jin). Jin to cheng: divide by 15.
To sell goods for money: use multiplication. To buy goods with money: use division.

斛㪷起率 • Volume Measurements

量之起率，始於微細，積而為大。
量起於圭（六粒之粟）。十圭謂之一撮，十撮謂之一抄，十抄謂之一勺，十勺謂之一合，十合謂之一升，十升謂之一㪷，十㪷謂之一斛。

Volume measurement begins from the smallest unit and accumulates upward:
Volume begins with the gui (圭): six grains of millet. 10 gui = 1 cuo (撮), 10 cuo = 1 chao (抄), 10 chao = 1 shao (勺), 10 shao = 1 ge (合), 10 ge = 1 sheng (升), 10 sheng = 1 dou (㪷), 10 dou = 1 hu (斛).
Thus 1 hu = 10^7 gui.

斤秤起率 • Weight Measurements

衡之起率，以黍為基。
衡起於黍（形大如粟）。十黍謂之一絫，十絫謂之一銖，六銖謂之一分，四分謂之一兩，十六兩謂之一斤，十五斤謂之一秤，三十斤謂之一鈞，四鈞謂之一碩（重一百二十斤）。

Weight measurement begins from the smallest grain:
Weight begins with the shu (黍): a grain the size of millet. 10 shu = 1 lei (絫), 10 lei = 1 zhu (銖), 6 zhu = 1 fen (分), 4 fen = 1 liang (兩, tael), 16 liang = 1 jin (斤), 15 jin = 1 cheng (秤), 30 jin = 1 jun (鈞), 4 jun = 1 shuo (碩, = 120 jin).

端匹起率 • Length Measurements

度之起率，以蠶絲為始。
度起於忽（蠶吐之絲）。十忽謂之一絲，十絲謂之一毫，十毫謂之一釐，十釐謂之一分，十分謂之一寸，十寸謂之一尺，十尺謂之一丈。匹率（或三丈二，或二丈四）。端率（或五丈五，或四丈八）。

Length measurement begins from the finest silk thread:
Length begins with the hu (忽): a single silk thread from a silkworm. 10 hu = 1 si (絲), 10 si = 1 hao (毫), 10 hao = 1 li (釐), 10 li = 1 fen (分), 10 fen = 1 cun (寸, inch), 10 cun = 1 chi (尺, foot), 10 chi = 1 zhang (丈).
Cloth bolt (匹) measure: either 3.2 zhang or 2.4 zhang.
Cloth roll (端) measure: either 5.5 zhang or 4.8 zhang.

田畝起率 • Land Area Measurements

田畝之制，以步為基。
田起於忽（闊一寸，長六寸）。十忽謂之一絲，十絲謂之一毫，十毫謂之一釐，十釐謂之一分，十分謂之一畝，百畝謂之一頃。三百步謂之一里。
按畝法，闊一步，長二百四十步，當自方五尺為步也。其里法，三百步為里者，當自方六尺為步。若三百六十步為里者，當以自方五尺為步也。

The land area system uses the “pace” (步) as its foundation:
Land area begins with the hu (忽): 1 cun wide by 6 cun long. 10 hu = 1 si (絲), 10 si = 1 hao (毫), 10 hao = 1 li (釐), 10 li = 1 fen (分), 10 fen = 1 mu (畝), 100 mu = 1 qing (頃). 300 paces = 1 li (里).
According to the mu method: 1 pace wide by 240 paces long = 1 mu, using a 5-chi pace. For the li method: 300 paces = 1 li using a 6-chi pace; or 360 paces = 1 li using a 5-chi pace.
One mu ≈ 1/15 of a hectare ≈ 667 m^2 in modern terms.

古法圓率 · Ancient Circle Ratio

古之圓率，以周三徑一為率。
周三尺，徑一尺。

The ancient ratio for the circle:
Circumference 3 chi, diameter 1 chi.
This gives $\pi = 3$, the ancient ratio “three for circumference, one for diameter.” This value was used in the *Zhoubi Suanjing* (周髀算經) and the *Jiuzhang Suanshu* (九章算術) before more precise values were obtained.

劉徽新術 · Liu Hui’s New Method

劉徽乃魏人也，立此新術以究圓之幽微。劉徽以割圓術求圓周率，割圓為多邊形，邊數愈多則愈近於圓。
周一百五十七尺，徑五十尺。

Liu Hui was a man of Wei; he established this new method to investigate the subtle mysteries of the circle.
Liu Hui (fl. 263 CE) used the “method of cutting the circle” (割圓術): inscribing polygons with increasingly many sides to approximate the circle. He calculated up to a 3072-sided polygon.
Circumference 157 chi, diameter 50 chi.
This gives $\pi = \frac{157}{50} = 3.14$, a significant improvement over $\pi = 3$.

沖之密率 · Zu Chongzhi’s Dense Ratio

沖之姓祖，乃宋南徐州從事史。立此密率亦究圓之微也。祖沖之更以密率求之，其精密為當時之最。
周二十二尺，徑七尺。

Chongzhi, surnamed Zu, was an officer of Nanku province in the Song dynasty. He established this dense ratio also to investigate the circle’s subtlety.
Zu Chongzhi (429–500 CE) gave two approximations for π: the “approximate ratio” (約率) $\pi \approx 22/7$ and the “dense ratio” (密率) $\pi \approx 355/113$. The dense ratio gives $\pi \approx 3.1415929\ldots$, accurate to six decimal places.
Circumference 22 chi, diameter 7 chi.
This gives $\pi = \frac{22}{7} \approx 3.142857$ (the “approximate ratio”).

[Note: Zu Chongzhi’s dense ratio $\frac{355}{113}$ is one of the best rational approximations of π with small denominator. It was not rediscovered in Europe until Adriaan Anthoniszoon (c. 1600).]

明異名訣 · Names of Fractions

此乃劇漏之數以巧呼之。古之算家，以巧名分數，以便口訣之用。
二分之一為中半，三分之一為少半，三分之二為太半，四分之一為弱半，四分之三為強半。

These are special names for fractions, called by elegant names.
Ancient mathematicians gave special names to common fractions for ease of use:
$\frac{1}{2}$ = “middle half” (中半), $\frac{1}{3}$ = “lesser half” (少半), $\frac{2}{3}$ = “greater half” (太半),
$\frac{1}{4}$ = “weak half” (弱半), $\frac{3}{4}$ = “strong half” (強半).

明正負術・Rules for Positive and Negative Numbers

正負之術，為方程之根本。同名相減，異名相加，正無入負之，負無入正之。

其同名相減，則異名相加；正無入負之，負無入正之。其異名相減，則同名相加；正無入正之，負無入負之。

按九章注云：兩算得失相返，要令正負以名之。正算赤、負算黑，不則以邪正為異。其無入者，為無對也。無所得減，則使消奪者居位也。（「人」作「入」非。）

The rules for positive and negative numbers are fundamental to solving systems of equations. The rules state:

Same signs subtract, different signs add. When subtracting a positive from zero (or a smaller positive), the result is negative; when subtracting a negative from zero, the result is positive.

Rule 1: Same signs subtract → different signs add; positive with nothing enters negative; negative with nothing enters positive.

Rule 2: Different signs subtract → same signs add; positive with nothing enters positive; negative with nothing enters negative.

According to the commentary on the Jiuzhang: two calculations with opposite outcomes are named positive and negative. Positive rods are red, negative rods are black; otherwise, slanted versus straight distinguishes them. "Nothing to enter" means there is no counterpart; when there is nothing to subtract from, the consuming quantity takes the position.

These rules correspond to: $(+a) - (+b) = +(a - b)$, $(-a) - (-b) = -(a - b)$, $(+a) + (-b) = +(a - b)$, $(+a) + (-b) = -(b - a)$ when $b > a$, etc.

[Note: These are among the earliest known rules for signed arithmetic in world mathematics. The rod-color convention (red = positive, black = negative) dates to the Han dynasty. These rules predate European acceptance of negative numbers by over a millennium.]

明乘除段・Rules for Multiplication and Division

乘除之術，凡幾何名義，不可不辨。

長平相併曰和，長平相減曰較。長平相乘曰積，自相稱之曰冪。同名相乘曰正，異名相乘為負。平除長為小長，長除平為小平。小長平相併曰小和，小長平相減餘小較，小長平相乘得一步為小積。

The terminology of multiplication and division must be clearly understood:

Length plus width is called "harmony" (和, sum).

Length minus width is called "comparison" (較, difference).

Length times width is called "accumulation" (積, area/product).

A number multiplied by itself is called "power" (冪, square).

Same signs multiplied give positive; different signs give negative.

Width divided by length gives "small length"; length divided by width gives "small width."

Small length plus small width = "small harmony"; their difference = "small comparison"; their product = "small accumulation."

[Note: The terms 長 (long) and 平 (broad/width) denote the sides of a rectangle. These algebraic conventions were essential for the tian yuan shu (天元術) method that Zhu Shijie would develop in his later work, the *Siyuan Yujian*.]

明開方法・Rules for Root Extraction

開方之術，方程之基也。凡開平方、開立方，皆用此術。置積為實，及方廉隅，同加異減開之。

The method of root extraction is the foundation of solving equations. It applies to square roots, cube roots, and higher roots.
Set up the accumulation (積) as the dividend (實), together with the side (方), edge (廉), and corner (隅). Like signs add, unlike signs subtract, and extract.
In the root extraction process for $x^2 = N$, the terms are: **shi** (實, dividend) = N, **fang** (方, side) = the linear term coefficient, **lian** (廉, edge) = the edge term, **yu** (隅, corner) = the leading coefficient. The "incremental root extraction" (增乘開方) method systematically refines the root digit by digit.

[Note: The "incremental multiplication" method (增乘開方), attributed to Jia Xian (11th century), is a precursor to what is now known as the Ruffini-Horner method for polynomial root-finding. It was systematized by Zhu Shijie in his later work.]

Volume I ・卷上

卷上凡八門，一百一十三問。其門曰：縱橫因法、身外加法、留頭乘法、身外減法、九歸除法、異乘同除、庫務解稅、折變互差。大要皆乘除之術，為算學之初步也。

Volume I contains eight chapters (八門) with 113 problems (一百一十三問), covering multiplication shortcuts, division methods, and proportional reasoning. Many problems reflect the economic realities of Yuan dynasty China—taxation, commodity prices, land measurement, and military logistics.

縱橫因法門・The Vertical–Horizontal Multiplication Method

> 此法從來向上因，但言十者過其身，
> 呼如本位須當作，知算縱橫數目真。

縱橫因法者，乘法之初門也。其法列實於上，以法數從上因之，言十過身，言如下位，逐位相乘，得數即積也。凡八問。

> *This method always multiplies upward from below,*
> *When ten is reached, carry past the body;*
> *Call "as-is" for the current place value;*
> *Know the vertical–horizontal counts to be true.*

This rhyme describes the "vertical–horizontal" multiplication method: multiply from bottom to top; when the product is ten or more, carry to the next position; call "as-is" for the current place value. This is the most basic multiplication technique.

問 1. 今有粟二百一十六斗，每斗價錢二文。問計錢幾何?
答曰： 四百三十二文。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 1. Suppose there are 216 dou of millet, each dou costing 2 cash. How much cash in total?
Answer: 432 cash.
Method: Place the quantity above; multiply each by the unit price from above downward; the result is obtained.
(Calculation: $216 \times 2 = 432$)

問 2. 今有絲一百四十四兩，每兩價錢三百文。問計錢幾何?
答曰： 四十三貫二百文。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 2. Suppose there are 144 liang of silk, each liang costing 300 cash. How much cash in total?
Answer: 43 guan 200 cash.
Method: Place the quantity above; multiply each by the unit price; obtain.
(Calculation: $144 \times 300 = 43{,}200$)

問 3. 今有羊三百五十四只，每只價錢四貫文。問計錢幾何?
答曰： 一千四百一十六貫。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 3. Suppose there are 354 sheep, each costing 4 guan. How much cash in total?
Answer: 1,416 guan.
Method: Place the quantity above; multiply by the unit price; obtain.
(Calculation: $354 \times 4 = 1{,}416$)

問 4. 今有銀五百四十七錠，每錠重五十兩。問為兩幾何?
答曰： 二萬七千三百五十兩。
術曰： 列物數在上，各以錠重從上因之，即得。

Problem 4. Suppose there are 547 ingots of silver, each ingot weighing 50 liang. How many liang in total?
Answer: 27,350 liang.
Method: Place the quantity above; multiply each by the ingot weight; obtain.
(Calculation: $547 \times 50 = 27{,}350$)

問 5. 今有絹七百三十六匹，每匹價錢六貫文。問計錢幾何?
答曰： 四千四百一十六貫。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 5. Suppose there are 736 bolts of silk, each costing 6 guan. How much cash in total?
Answer: 4,416 guan.
Method: Place the quantity above; multiply by unit price; obtain.
(Calculation: $736 \times 6 = 4{,}416$)

問 6. 今有麻八百九十二秤，每秤價錢七百文。問計錢幾何?
答曰： 六百二十四貫四百文。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 6. Suppose there are 892 cheng of hemp, each cheng costing 700 cash. How much cash in total?
Answer: 624 guan 400 cash.
Method: Place the quantity above; multiply by unit price; obtain.
(Calculation: $892 \times 700 = 624{,}400$)

問 7. 今有布六百三十四尺，每尺價錢八十文。問計錢幾何?
答曰： 五十貫七百二十文。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 7. Suppose there are 634 chi of cloth, each chi costing 80 cash. How much cash in total?
Answer: 50 guan 720 cash.
Method: Place the quantity above; multiply by unit price; obtain.
(Calculation: $634 \times 80 = 50{,}720$)

問 8. 今有馬四百二十五匹，每匹價錢九十貫。問計錢幾何?
答曰： 三萬八千二百五十貫。
術曰： 列物數在上，各以價錢從上因之，即得。

Problem 8. Suppose there are 425 horses, each costing 90 guan. How much cash in total?
Answer: 38,250 guan.
Method: Place the quantity above; multiply by unit price; obtain.
(Calculation: $425 \times 90 = 38{,}250$)

身外加法門 · Addition to the Body

算中加法最堪誇，言十之時就位加，
但遇呼如身下列，君從法式定無差。

身外加法者，乘法之捷法也。其法以實為身，於身外逐位加之，言十就身，言如下列。凡十問。

Among calculations, the addition method is most praiseworthy;
When ten is reached, add at the current place;
When “as-is” is called, place it below the body;
Follow this rule and you will make no error.

This rhyme describes a shortcut multiplication method: add to the “body” (the multiplicand itself). When the product reaches ten, add at the current position; when it is less than ten (“as-is”), place it below the current digit. This is essentially a streamlined single-digit multiplication performed on the counting board.

問 9. 今有米六碩八斗四升，每斗價錢一百一十文。問計錢幾何?
答曰： 七貫五百二十四文。
術曰： 列物數在上，以價錢從上身外加之，即得。

Problem 9. Suppose there are 6 shuo 8 dou 4 sheng of rice, each dou costing 110 cash. How much cash in total?
Answer: 7 guan 524 cash.
Method: Place the quantity above; add the price to the body from above; obtain.

問 10. 今有羅三十四尺六寸，每尺價錢一百二十文。問計錢幾何?
答曰：四貫一百五十二文。
術曰：列物數在上，以價錢從上身外加之，即得。

Problem 10. Suppose there are 34 chi 6 cun of gauze, each chi costing 120 cash. How much cash in total?
Answer: 4 guan 152 cash.
Method: Place the quantity above; add to the body; obtain.

問 11. 今有鹽八百七十三袋，每袋價錢一十三貫文。問計錢幾何?
答曰：一萬一千三百四十九貫。
術曰：列物數在上，以價錢從上身外加之，即得。

Problem 11. Suppose there are 873 bags of salt, each bag costing 13 guan. How much cash in total?
Answer: 11,349 guan.
Method: Place the quantity above; add to the body; obtain.
(Calculation: $873 \times 13 = 11{,}349$)

問 12. 今有麥二十三碩四斗，每斗價錢一百三十文。問計錢幾何?
答曰：三十貫四百二十文。
術曰：列麥數在上，以價錢從上身外加之，即得。

Problem 12. Suppose there are 23 shuo 4 dou of wheat, each dou costing 130 cash. How much cash in total?
Answer: 30 guan 420 cash.
Method: Place the wheat quantity above; add to the body; obtain.

問 13. 今有絲四十二斤三兩，每兩價錢二百四十文。問計錢幾何?
答曰：一十貫一百七十五文。
術曰：列絲數在上，以價錢從上身外加之，即得。

Problem 13. Suppose there are 42 jin 3 liang of silk, each liang costing 240 cash. How much cash in total?
Answer: 10 guan 175 cash.
Method: Place the silk quantity above; add to the body; obtain.

問 14. 今有粟五十六碩七斗，每斗價錢一百五十文。問計錢幾何?
答曰：八十五貫五十文。
術曰：列粟數在上，以價錢從上身外加之，即得。

Problem 14. Suppose there are 56 shuo 7 dou of millet, each dou costing 150 cash. How much cash in total?
Answer: 85 guan 50 cash.
Method: Place the millet quantity above; add to the body; obtain.
(Calculation: $567 \times 150 = 85{,}050$)

問 15. 今有茶三百六十二袋，每袋價錢二貫五百文。問計錢幾何?
答曰：九百五貫。
術曰：列茶數在上，以價錢從上身外加之，即得。

Problem 15. Suppose there are 362 bags of tea, each bag costing 2 guan 500 cash. How much cash in total?
Answer: 905 guan.
Method: Place the tea quantity above; add to the body; obtain.
(Calculation: $362 \times 2.5 = 905$)

問 16. 今有綿一千二百四十五兩，每兩價錢六十文。問計錢幾何?
答曰：七十四貫七百文。
術曰：列綿數在上，以價錢從上身外加之，即得。

Problem 16. Suppose there are 1,245 liang of cotton, each liang costing 60 cash. How much cash in total?
Answer: 74 guan 700 cash.
Method: Place the cotton quantity above; add to the body; obtain.
(Calculation: $1{,}245 \times 60 = 74{,}700$)

問 17. 今有鐵二千八百七十四斤，每斤價錢四十五文。問計錢幾何?
答曰：一百二十九貫三百三十文。
術曰：列鐵數在上，以價錢從上身外加之，即得。

Problem 17. Suppose there are 2,874 jin of iron, each jin costing 45 cash. How much cash in total?
Answer: 129 guan 330 cash.
Method: Place the iron quantity above; add to the body; obtain.
(Calculation: $2{,}874 \times 45 = 129{,}330$)

問 18. 今有鉛一千五百六十三斤四兩，每斤價錢八十文。問計錢幾何?
答曰：一百二十五貫六十文。
術曰：列鉛數在上，以價錢從上身外加之，即得。

Problem 18. Suppose there are 1,563 jin 4 liang of lead, each jin costing 80 cash. How much cash in total?
Answer: 125 guan 60 cash.
Method: Place the lead quantity above; add to the body; obtain.

問 19. 今有錫八百七十六斤半，每斤價錢一百七十文。問計錢幾何?
答曰： 一百四十九貫五百五文。
術曰： 列錫數在上，以價錢從上身外加之，即得。

Problem 19. Suppose there are 876.5 jin of tin, each jin costing 170 cash. How much cash in total?
Answer: 149 guan 505 cash.
Method: Place the tin quantity above; add to the body; obtain.

留頭乘法門・Multiplication Keeping the First Digit

> 留頭乘法別規模，起首先從次位呼，
> 言十靠身如隔位，遍臨頭位破身鋪。

留頭乘法者，乘法之巧術也。其法以法數之首位暫留不動，先從次位乘起，言十靠身，如隔位布之，遍臨頭位則破身鋪之。凡二十問。

> *The keep-the-head method has a special pattern;*
> *Begin from the second digit, not the first;*
> *When ten, place next to the body; when “as-is,” skip a place;*
> *Finally use the head digit, breaking the body.*

The “keep-the-head” multiplication method: a special technique where the first digit of the multiplier is temporarily kept (left in place), and multiplication begins from the second digit. When the product reaches ten, it is placed next to the body; “as-is” results are placed with a skip. Finally the head digit itself is used to multiply, “breaking” the original body. This avoids shifting the multiplicand on the counting board.

問 20. 今有白豆八十四斛，每斛價錢二百一十文。問計錢幾何?
答曰： 一十七貫六百四十文。
術曰： 列豆數於上，以斛價從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

Problem 20. Suppose there are 84 hu of white beans, each hu costing 210 cash. How much cash in total?
Answer: 17 guan 640 cash.
Method: Place the bean quantity above; call the hu-price from the second digit upward; when ten, place next to the body; when “as-is,” skip a place; when reaching the head digit, break the body; obtain. (Calculation: $84 \times 210 = 17{,}640$)

問 21. 今有胡椒六十三斤四兩，每斤價錢三百八十文。問計錢幾何?
答曰： 二十四貫三十五文。
術曰： 列椒數於上，以斤價從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

Problem 21. Suppose there are 63 jin 4 liang of pepper, each jin costing 380 cash. How much cash in total?
Answer: 24 guan 35 cash.
Method: Place the pepper quantity above; call the price from the second digit; when ten, place next to body; when “as-is,” skip; at the head digit, break the body; obtain.

問 22. 今有白檀共數斤，下留兩於上，以四貫九十六文乘之。問得幾何?
答曰： 二萬貫。
術曰： 列白檀共數斤下留兩於上，以四貫九十六文從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

Problem 22. Suppose there is a quantity of white sandalwood in jin, leaving taels below, multiply by 4 guan 96 cash. What is obtained?
Answer: 20,000 guan.
Method: Place the sandalwood quantity with taels below; call 4 guan 96 cash from the second digit; when ten, place next to body; when “as-is,” skip; at head digit, break body; obtain.

問 23. 今有黍三千六百六十二碩一斛九合三勺七抄五撮。問為麥幾何?
答曰： 三萬斛。
術曰： 列黍數於上，以一斛麥八升一合九勺二抄乘之，即得。

Problem 23. Suppose there are 3,662 shuo 1 hu 9 ge 3 shao 7 chao 5 cuo of millet. How much wheat is it equivalent to?
Answer: 30,000 hu.
Method: Place the millet quantity above; multiply by 1 hu wheat = 8 sheng 1 ge 9 shao 2 chao; obtain.

問 24. 今有粟七萬八千一百二十五碩，每斛為御米八升一合九勺二抄。問共得幾何?
答曰： 三千二百七十斛。
術曰： 列粟數於上，以御米率從上次位呼之，即得。

Problem 24. Suppose there are 78,125 shuo of grain, each hu yielding 8 sheng 1 ge 9 shao 2 chao of imperial rice. How much rice is obtained?
Answer: 3,270 hu.
Method: Place the grain quantity above; call the imperial rice rate from the second digit; obtain.

問 25. 今有絲六萬一千六百九十八秤。問為斤幾何?
答曰： 一百六十九萬一千六百九十八斤。
術曰： 列絲數於上，以一秤十六兩為法，先以六因之，再以四因之，即得斤數。

Problem 25. Suppose there are 61,698 cheng of silk. How many jin?
Answer: 1,691,698 jin.
Method: Place the silk quantity above; using 1 cheng = 16 liang (= 1 jin), first multiply by 6, then by 4; obtain the jin number.

問 26. 今有絲六萬一千六百九十八秤，每斤重一十六兩。問為兩幾何?
答曰： 一千六百九十八萬一千六百九十八兩。
術曰： 列絲數於上，以每斤十六兩為法，先從次位呼之，遍臨頭位破身，即得。

Problem 26. Suppose there are 61,698 cheng of silk, each jin weighing 16 liang. How many liang?
Answer: 16,981,698 liang.
Method: Place the silk quantity above; using 1 jin = 16 liang, call from the second digit; when reaching the head digit, break the body; obtain.

問 27. 今有錢七貫四百一十二文，欲令一十七人分之。問人得幾何?
答曰： 四百三十六文。
術曰： 列錢數於上，以一十七人為法，從上次位呼之，即得。

Problem 27. Suppose there are 7 guan 412 cash, to be divided among 17 people. How much per person?
Answer: 436 cash.
Method: Place the cash quantity above; using 17 people as divisor, call from the second digit; obtain.
(Calculation: $7{,}412 \div 17 = 436$)

問 28. 今有糧五千八百八十六碩，欲給貧難戶，每戶一碩八斗。問給幾戶?
答曰： 三千二百七十戶。
術曰： 列糧數於上，以每戶一碩八斗為法，從上次位呼之，即得。

Problem 28. Suppose there are 5,886 shuo of grain, to be distributed to poor households, each household receiving 1 shuo 8 dou. How many households?
Answer: 3,270 households.
Method: Place the grain quantity above; using 1 shuo 8 dou per household as divisor, call from the second digit; obtain.
(Calculation: $5{,}886 \div 1.8 = 3{,}270$)

問 29. 今有錢六十五貫五百五十文，欲買苧絲，每尺價錢一百九十文。問得幾何?
答曰： 三百四十五尺。
術曰： 列錢數於上，以尺價為法，從上次位呼之，即得。

Problem 29. Suppose there are 65 guan 550 cash, to buy ramie silk at 190 cash per chi. How many chi are obtained?
Answer: 345 chi.
Method: Place the cash quantity above; using the chi-price as divisor, call from the second digit; obtain.
(Calculation: $65{,}550 \div 190 = 345$)

問 30. 今有錢一百四貫三百七十二文，欲買麻布，每匹價錢一百九十四文。問買布幾何?
答曰： 五百三十八匹。
術曰： 列錢數於上，以匹價為法，從上次位呼之，即得。

Problem 30. Suppose there are 104 guan 372 cash, to buy hemp cloth at 194 cash per bolt. How many bolts?
Answer: 538 bolts.
Method: Place the cash quantity above; using the bolt-price as divisor, call from the second digit; obtain.
(Calculation: $104{,}372 \div 194 = 538$)

問 31. 今有錢五千五百五十八貫三百文，欲買松木，每株價錢一貫七百五文。問買木幾何?
答曰： 三千二百六十株。
術曰： 列錢數於上，以株價為法，從上次位呼之，即得。

Problem 31. Suppose there are 5,558 guan 300 cash, to buy pine trees at 1 guan 705 cash each. How many trees?
Answer: 3,260 trees.
Method: Place the cash quantity above; using the tree-price as divisor, call from the second digit; obtain.
(Calculation: $5{,}558{,}300 \div 1{,}705 = 3{,}260$)

問 32. 今有芝麻共數於上，以三斤七分五釐乘之。問得幾何?
術曰： 列芝麻共數於上，以三斤七分五釐為法，從上次位呼之，言十靠身，遍臨頭位破身，即得。

Problem 32. Suppose there is a quantity of sesame above, to be multiplied by 3 jin 7 fen 5 li. What is obtained?
Method: Place the sesame quantity above; using 3 jin 7 fen 5 li as divisor, call from the second digit; when ten, place next to body; at head digit, break body; obtain.

問 33. 今有小麥五碩九斛二升，每斛磨麵六斤一十四兩。問磨麵幾何?
答曰： 四百單七斤。
術曰： 列麥數於上，以六斤八分七釐半乘之，即得。

Problem 33. Suppose there are 5 shuo 9 hu 2 sheng of wheat, each hu yielding 6 jin 14 liang of flour. How much flour?
Answer: 407 jin.
Method: Place the wheat quantity above; multiply by 6 jin 8 fen 7 li and a half; obtain.
(6 jin 14 liang = $6 + 14/16 = 6.875$ jin)

問 34. 今有菉荳三十二碩七斛三升，每斛造粉五斤六兩。問造粉幾何?
答曰： 一千七百五十九斤三兩八錢。
術曰： 列荳數於上，以五斤六兩為法，從上次位呼之，即得。

Problem 34. Suppose there are 32 shuo 7 hu 3 sheng of green beans, each hu producing 5 jin 6 liang of bean flour. How much flour?
Answer: 1,759 jin 3 liang 8 qian.
Method: Place the bean quantity above; using 5 jin 6 liang as divisor, call from the second digit; obtain.

問 35. 今有黃金一萬二千八百兩，每兩買地六畝二分五釐。問買地幾何?
答曰： 八萬畝。
術曰： 列金數於上，以地率從上次位呼之，即得。

Problem 35. Suppose there are 12,800 liang of gold, each liang buying 6 mu 2 fen 5 li of land. How much land?
Answer: 80,000 mu.
Method: Place the gold quantity above; call the land rate from the second digit; obtain.
(Calculation: $12{,}800 \times 6.25 = 80{,}000$)

問 36. 今有田一百五十六頃二十五畝，每畝收稻五斛。問收稻幾何?
答曰： 九萬斛。
術曰： 列田數於上，以畝產從上次位呼之，即得。

Problem 36. Suppose there are 156 qing 25 mu of field, each mu yielding 5 hu of rice. How much rice is harvested?
Answer: 90,000 hu.
Method: Place the field quantity above; call the per-mu yield from the second digit; obtain.

問 37. 今有米五十三斛四升，直錢九貫八百七十九文。只有錢六十八貫二百六十文。問得米幾何?
答曰： 如術計之。
術曰： 列見錢為實，以米價為法，留頭乘之，即得。

Problem 37. Suppose 53 hu 4 sheng of rice cost 9 guan 879 cash. If one has 68 guan 260 cash, how much rice can be bought?
Answer: Calculate according to the method.
Method: Place the available cash as dividend; use the rice price as divisor; keep-the-head multiply; obtain.

問 38. 今有羅七百六十二匹，每匹價錢三貫二百三十文。問計錢幾何?
答曰： 如術計之。
術曰： 列羅數於上，以匹價從上次位呼之，即得。

Problem 38. Suppose there are 762 bolts of gauze, each costing 3 guan 230 cash. How much cash in total?
Answer: Calculate according to the method.
Method: Place the gauze quantity above; call the bolt-price from the second digit; obtain.
(Calculation: $762 \times 3{,}230 = 2{,}461{,}260$)

身外減法門・Subtraction from the Body

> 減法根源必要知，即同求一一般推，
> 呼如身下須當減，言十從身本位除。

身外減法者，除法之捷術也，與身外加法相反。其法以實為身，於身外逐位減之，呼如減下身位，言十從身本位除。凡十問。

> *The root of the subtraction method you must know;*
> *It is like the “seeking unity” method applied;*
> *When “as-is” is called, subtract below the body;*
> *When ten is reached, divide from the body itself.*

The “subtraction from the body” method is the inverse of the addition method: it performs division by subtracting from the dividend itself. “As-is” means subtract below the current digit; “ten” means subtract from the current digit itself. This is a streamlined division technique on the counting board.

問 39. 今有錢四貫三百二十文，欲糴白豆，每斗價錢二十文。問得幾何?
答曰： 二百一十六斗。
術曰： 列錢數於上，為實，以斗價為法，從上身外減之，即得。

Problem 39. Suppose there are 4 guan 320 cash, to buy white beans at 20 cash per dou. How much is obtained?
Answer: 216 dou.
Method: Place the cash above as dividend; use the dou-price as divisor; subtract from the body from above; obtain.
(Calculation: $4{,}320 \div 20 = 216$)

問 40. 今有錢四十三貫二百文，欲買細絲，每斤價錢三百文。問得幾何?
答曰： 一百四十四斤。
術曰： 列錢數於上，為實，以斤價為法，從上身外減之，即得。

Problem 40. Suppose there are 43 guan 200 cash, to buy fine silk at 300 cash per jin. How much is obtained?
Answer: 144 jin.
Method: Place the cash above as dividend; use the jin-price as divisor; subtract from the body; obtain.
(Calculation: $43{,}200 \div 300 = 144$)

問 41. 今有錢一千四百一十六貫，欲買絹子，每匹價錢四貫文。問得幾何?
答曰： 三百五十四匹。
術曰： 列錢數於上，為實，以匹價為法，從上身外減之，即得。

Problem 41. Suppose there are 1,416 guan, to buy silk cloth at 4 guan per bolt. How much is obtained?
Answer: 354 bolts.
Method: Place the cash above as dividend; use bolt-price as divisor; subtract from the body; obtain.
(Calculation: $1{,}416 \div 4 = 354$)

問 42. 今有銀二萬七千三百五十兩，欲為課銀，每錠五十兩。問為幾錠?
答曰： 五百四十七錠。
術曰： 列銀數於上，為實，以錠重為法，從上身外減之，即得。

Problem 42. Suppose there are 27,350 liang of silver, to be made into tax ingots, each ingot = 50 liang. How many ingots?
Answer: 547 ingots.
Method: Place the silver above as dividend; use ingot weight as divisor; subtract from the body; obtain.
(Calculation: $27{,}350 \div 50 = 547$)

問 43. 今有錢四千四百一十六貫，欲買大羅，每匹價錢六貫文。問得幾何?
答曰： 七百三十六匹。
術曰： 列錢數於上，為實，以匹價為法，從上身外減之，即得。

Problem 43. Suppose there are 4,416 guan, to buy damask at 6 guan per bolt. How much is obtained?
Answer: 736 bolts.
Method: Place the cash above as dividend; use bolt-price as divisor; subtract from the body; obtain.
(Calculation: $4{,}416 \div 6 = 736$)

問 44. 今有錢六百二十四貫四百文，欲買甘草，每秤價錢七百文。問得幾何?
答曰： 八百九十二秤。
術曰： 列錢數於上，為實，以秤價為法，從上身外減之，即得。

Problem 44. Suppose there are 624 guan 400 cash, to buy licorice at 700 cash per cheng. How much is obtained?
Answer: 892 cheng.
Method: Place the cash above as dividend; use cheng-price as divisor; subtract from the body; obtain.
(Calculation: $624{,}400 \div 700 = 892$)

問 45. 今有錢五十貫七百二十文，欲買細布，每尺價錢八十文。問得幾何?
答曰：六百三十四尺。
術曰：列錢數於上，為實，以尺價為法，從上身外減之，即得。

Problem 45. Suppose there are 50 guan 720 cash, to buy fine cloth at 80 cash per chi. How much is obtained?
Answer: 634 chi.
Method: Place the cash above as dividend; use chi-price as divisor; subtract from the body; obtain.
(Calculation: $50{,}720 \div 80 = 634$)

問 46. 今有錢三萬八千二百五十貫，欲買馬，每匹價錢九十貫。問得幾何?
答曰：四百二十五匹。
術曰：列錢數於上，為實，以匹價為法，從上身外減之，即得。

Problem 46. Suppose there are 38,250 guan, to buy horses at 90 guan each. How many horses?
Answer: 425 horses.
Method: Place the cash above as dividend; use horse-price as divisor; subtract from the body; obtain.
(Calculation: $38{,}250 \div 90 = 425$)

問 47. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列油數為實，以三斤七分五釐為法，身外減之，即得。

Problem 47. Suppose there are 6 shuo 8 dou 4 sheng of oil, and 3 jin 7 fen 5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; subtract from the body; obtain.

問 48. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列麵數為實，以六斤八分七釐半為法，身外減之，即得。

Problem 48. Suppose there are 407 jin of flour, and 6 jin 8 fen 7.5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; subtract from the body; obtain.

九歸除法門・Division by the Nine Returns

實少法多從法歸，實多滿法進前居，
常存除數專心記，法實相停九十餘。
但遇無除還頭位，然將釋九數呼除，
流傳故泄真消息，求一穿韜總不如。

九歸除法者，以九歸訣為除法之要術也。實少於法，則從法歸之；實多於法，則進位而居之。常記除數於心，法實相等則餘九十餘。凡二十九問。

When the dividend is small and divisor large, follow the return method;
When the dividend fills the divisor, advance to the next position;
Keep the divisor always in mind;
When divisor and dividend match, the remainder is in the nineties.
When a digit cannot be divided, return to the head position;
Then use the nine-explanations method to call the division;
This tradition reveals the true secret;
The "seeking unity" method cannot compare.

The "nine returns" division uses the division rhymes given in the General Principles. When the dividend is smaller than the divisor, use the "return" method; when the dividend is larger, advance the quotient. The practitioner must memorize the divisor and keep it in mind. When the division is exact, the remainder is in the nineties. If a digit cannot be divided, return to the head position and use the "nine explanations" method. This method, though not the orthodox approach (商除), is easier for beginners.

問 49. 今有錢四貫三百二十文，欲糴白豆，每斗價錢二十文。問得幾何?
答曰: 二百一十六斗。
術曰: 列錢物於上為實，各以價錢為法，從上身外減之，即得。

Problem 49. Suppose there are 4 guan 320 cash, to buy white beans at 20 cash per dou. How much is obtained?
Answer: 216 dou.
Method: Place cash and goods above as dividend; use the price as divisor; subtract from the body; obtain.

問 50. 今有錢四十三貫二百文，欲買細絲，每斤價錢三百文。問得幾何?
答曰: 一百四十四斤。
術曰: 列錢數為實，以斤價為法，九歸除之，即得。

Problem 50. Suppose there are 43 guan 200 cash, to buy fine silk at 300 cash per jin. How much is obtained?
Answer: 144 jin.
Method: Place cash as dividend; use jin-price as divisor; divide by nine-returns; obtain.

問 51. 今有錢一千四百一十六貫，欲買絹子，每匹價錢四貫文。問得幾何?
答曰: 三百五十四匹。
術曰: 列錢數為實，以匹價為法，九歸除之，即得。

Problem 51. Suppose there are 1,416 guan, to buy silk at 4 guan per bolt. How much is obtained?
Answer: 354 bolts.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 52. 今有銀二萬七千三百五十兩，欲為課銀，每錠五十兩。問為幾錠?
答曰: 五百四十七錠。
術曰: 列銀數為實，以錠重為法，九歸除之，即得。

Problem 52. Suppose there are 27,350 liang of silver, to make tax ingots of 50 liang each. How many ingots?
Answer: 547 ingots.
Method: Place silver as dividend; use ingot weight as divisor; divide by nine-returns; obtain.

問 53. 今有錢四千四百一十六貫，欲買大羅，每匹價錢六貫文。問得幾何?
答曰: 七百三十六匹。
術曰: 列錢數為實，以匹價為法，九歸除之，即得。

Problem 53. Suppose there are 4,416 guan, to buy damask at 6 guan per bolt. How much is obtained?
Answer: 736 bolts.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 54. 今有錢六百二十四貫四百文，欲買甘草，每秤價錢七百文。問得幾何?
答曰: 八百九十二秤。
術曰: 列錢數為實，以秤價為法，九歸除之，即得。

Problem 54. Suppose there are 624 guan 400 cash, to buy licorice at 700 cash per cheng. How much is obtained?
Answer: 892 cheng.
Method: Place cash as dividend; use cheng-price as divisor; divide by nine-returns; obtain.

問 55. 今有錢五十貫七百二十文，欲買細布，每尺價錢八十文。問得幾何?
答曰: 六百三十四尺。
術曰: 列錢數為實，以尺價為法，九歸除之，即得。

Problem 55. Suppose there are 50 guan 720 cash, to buy fine cloth at 80 cash per chi. How much is obtained?
Answer: 634 chi.
Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

問 56. 今有木幾何，直錢三千二百六十株。問每株價錢幾何?
答曰: 三貫二百六十文。
術曰: 列錢物於上為實，各以價錢為法，從上身外減之，即得。

Problem 56. Suppose some quantity of timber costs 3,260 trees' worth. What is the price per tree?
Answer: 3 guan 260 cash.
Method: Place cash and goods above as dividend; use price as divisor; subtract from the body; obtain.

問 57. 今有絹幾何，每匹四貫，共錢一千四百一十六貫。問匹數幾何?
答曰: 三百五十四匹。
術曰: 列錢數為實，以匹價為法，九歸除之，即得。

Problem 57. Suppose there is some silk, 4 guan per bolt, total 1,416 guan. How many bolts?
Answer: 354 bolts.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 58. 今有銀幾何，每錠五十兩，共二萬七千三百五十兩。問錠數幾何?
答曰: 五百四十七錠。
術曰: 列銀數為實，以錠重為法，九歸除之，即得。

Problem 58. Suppose there is some silver, 50 liang per ingot, total 27,350 liang. How many ingots?
Answer: 547 ingots.
Method: Place silver as dividend; use ingot weight as divisor; divide by nine-returns; obtain.

問 59. 今有甘草幾何，每秤七百文，共錢六百二十四貫四百文。問秤數幾何?
答曰： 八百九十二秤。
術曰： 列錢數為實，以秤價為法，九歸除之，即得。

Problem 59. Suppose there is some licorice, 700 cash per cheng, total 624 guan 400 cash. How many cheng?
Answer: 892 cheng.
Method: Place cash as dividend; use cheng-price as divisor; divide by nine-returns; obtain.

問 60. 今有麻布幾何，每匹一百九十四文，共錢一百四貫三百七十二文。問匹數幾何?
答曰： 五百三十八匹。
術曰： 列錢數為實，以匹價為法，九歸除之，即得。

Problem 60. Suppose there is some hemp cloth, 194 cash per bolt, total 104 guan 372 cash. How many bolts?
Answer: 538 bolts.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 61. 今有細布幾何，每尺八十文，共錢五十貫七百二十文。問尺數幾何?
答曰： 六百三十四尺。
術曰： 列錢數為實，以尺價為法，九歸除之，即得。

Problem 61. Suppose there is some fine cloth, 80 cash per chi, total 50 guan 720 cash. How many chi?
Answer: 634 chi.
Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

問 62. 今有松木幾何，每株一貫七百五文，共錢五千五百五十八貫三百文。問株數幾何?
答曰： 三千二百六十株。
術曰： 列錢數為實，以株價為法，九歸除之，即得。

Problem 62. Suppose there are some pine trees, 1 guan 705 cash each, total 5,558 guan 300 cash. How many trees?
Answer: 3,260 trees.
Method: Place cash as dividend; use tree-price as divisor; divide by nine-returns; obtain.

問 63. 今有白豆幾何，每斗二十文，共錢四貫三百二十文。問斗數幾何?
答曰： 二百一十六斗。
術曰： 列錢數為實，以斗價為法，九歸除之，即得。

Problem 63. Suppose there is some white beans, 20 cash per dou, total 4 guan 320 cash. How many dou?
Answer: 216 dou.
Method: Place cash as dividend; use dou-price as divisor; divide by nine-returns; obtain.

問 64. 今有絲幾何，每斤三百文，共錢四十三貫二百文。問斤數幾何?
答曰： 一百四十四斤。
術曰： 列錢數為實，以斤價為法，九歸除之，即得。

Problem 64. Suppose there is some silk, 300 cash per jin, total 43 guan 200 cash. How many jin?
Answer: 144 jin.
Method: Place cash as dividend; use jin-price as divisor; divide by nine-returns; obtain.

問 65. 今有馬幾何，每匹九十貫，共錢三萬八千二百五十貫。問匹數幾何?
答曰： 四百二十五匹。
術曰： 列錢數為實，以匹價為法，九歸除之，即得。

Problem 65. Suppose there are some horses, 90 guan each, total 38,250 guan. How many horses?
Answer: 425 horses.
Method: Place cash as dividend; use horse-price as divisor; divide by nine-returns; obtain.

問 66. 今有羅幾何，每匹價錢三貫二百三十文，共錢一千七百一十一貫四百文。問匹數幾何?
答曰： 如術計之。
術曰： 列錢數為實，以匹價為法，九歸除之，即得。

Problem 66. Suppose there is some gauze, 3 guan 230 cash per bolt, total 1,711 guan 400 cash. How many bolts?
Answer: Calculate according to the method.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 67. 今有大羅幾何，每匹六貫，共錢四千四百一十六貫。問匹數幾何?
答曰： 七百三十六匹。
術曰： 列錢數為實，以匹價為法，九歸除之，即得。

Problem 67. Suppose there is some damask, 6 guan per bolt, total 4,416 guan. How many bolts?
Answer: 736 bolts.
Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 68. 今有苧絲幾何，每尺一百九十文，共錢六十五貫五百五十文。問尺數幾何?
答曰： 三百四十五尺。
術曰： 列錢數為實，以尺價為法，九歸除之，即得。

Problem 68. Suppose there is some ramie silk, 190 cash per chi, total 65 guan 550 cash. How many chi?
Answer: 345 chi.
Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

問 69. 今有米幾何，每斗一百一十文，共錢七貫五百二十四文。問斗數幾何?
答曰：六碩八斗四升。
術曰：列錢數為實，以斗價為法，九歸除之，即得。

Problem 69. Suppose there is some rice, 110 cash per dou, total 7 guan 524 cash. How many dou?
Answer: 6 shuo 8 dou 4 sheng.
Method: Place cash as dividend; use dou-price as divisor; divide by nine-returns; obtain.
(Calculation: $7{,}524 \div 110 = 68.4$ dou = 6 shuo 8 dou 4 sheng)

問 70. 今有鹽幾何，每袋一十三貫，共錢一萬一千三百四十九貫。問袋數幾何?
答曰：八百七十三袋。
術曰：列錢數為實，以袋價為法，九歸除之，即得。

Problem 70. Suppose there is some salt, 13 guan per bag, total 11,349 guan. How many bags?
Answer: 873 bags.
Method: Place cash as dividend; use bag-price as divisor; divide by nine-returns; obtain.

問 71. 今有羊幾何，每只四貫，共錢一千四百一十六貫。問只數幾何?
答曰：三百五十四只。
術曰：列錢數為實，以只價為法，九歸除之，即得。

Problem 71. Suppose there are some sheep, 4 guan each, total 1,416 guan. How many sheep?
Answer: 354 sheep.
Method: Place cash as dividend; use sheep-price as divisor; divide by nine-returns; obtain.

問 72. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列油數為實，以三斤七分五釐為法，九歸除之，即得。

Problem 72. Suppose there are 6 shuo 8 dou 4 sheng of oil; 3 jin 7 fen 5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; divide by nine-returns; obtain.

問 73. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列麵數為實，以六斤八分七釐半為法，九歸除之，即得。

Problem 73. Suppose there are 407 jin of flour; 6 jin 8 fen 7.5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide by nine-returns; obtain.

問 74. 今有羅三十四尺六寸，每尺一百二十文。問計錢幾何?
答曰：四貫一百五十二文。
術曰：列羅數於上，以尺價為法，九歸除之，即得。

Problem 74. Suppose there are 34 chi 6 cun of gauze, each chi costing 120 cash. How much cash in total?
Answer: 4 guan 152 cash.
Method: Place the gauze quantity above; use chi-price as divisor; divide by nine-returns; obtain.

問 75. 今有胡椒六十三斤四兩，每斤三百八十文。問計錢幾何?
答曰：二十四貫三十五文。
術曰：列椒數於上，以斤價為法，九歸除之，即得。

Problem 75. Suppose there are 63 jin 4 liang of pepper, each jin costing 380 cash. How much cash in total?
Answer: 24 guan 35 cash.
Method: Place the pepper quantity above; use jin-price as divisor; divide by nine-returns; obtain.

問 76. 今有馬幾何，每匹價錢九十貫。問計錢幾何?
答曰：如術計之。
術曰：列馬數於上，以匹價為法，九歸除之，即得。

Problem 76. Suppose there are some horses, each costing 90 guan. How much cash in total?
Answer: Calculate according to the method.
Method: Place the horse quantity above; use horse-price as divisor; divide by nine-returns; obtain.

問 77. 今有綿三百四十五斤，欲作綿被，每床用綿七斤。問作幾床?
答曰：四十九床，餘二斤。
術曰：列綿數為實，以每床用綿為法，九歸除之，即得。

Problem 77. Suppose there are 345 jin of cotton, to make cotton quilts, each quilt using 7 jin. How many quilts?
Answer: 49 quilts, remainder 2 jin.
Method: Place the cotton quantity as dividend; use cotton per quilt as divisor; divide by nine-returns; obtain.
(Calculation: $345 \div 7 = 49$ remainder 2)

異乘同除門・Different Multiplication, Same Division

異乘同除者，比例交易之法也。以異名之數相乘，以同名之數相除，求其相當之率。凡八問。

This chapter covers problems of proportional exchange: multiplying by one rate and dividing by another. The problems involve currency conversion, unit conversion, and commodity exchange.

問 78. 今有錢九貫八百七十九文，糴米五碩三斛四升。只有錢六十八貫二百六十文。問得米幾何?
答曰：如術計之。
術曰：列只有米數，以乘今有錢為實，以原錢為法，除之，即得。

Problem 78. Suppose 9 guan 879 cash buys 5 shuo 3 hu 4 sheng of rice. If one has 68 guan 260 cash, how much rice is obtained?
Answer: Calculate according to the method.
Method: Place the rice quantity, multiply by the current cash as dividend; use the original cash as divisor; divide; obtain.

問 79. 今有絲七匹，通尺內子得二千三十一尺，以乘上位得六百三十七萬九千三百七十一。又以實以斤銖法三百八十四銖除之，得三萬六千五百四十二銖，為實。
答曰：九千七百六十五斤一十兩四銖。
術曰：列七匹通尺內子，以乘上位，又以斤銖法除之，即得。

Problem 79. Suppose there are 7 bolts of silk; converting to chi gives 2,031 chi; multiplying gives 6,379,371. Then dividing by 384 zhu (the jin-zhu conversion) gives 36,542 zhu as dividend.
Answer: 9,765 jin 10 liang 4 zhu.
Method: Convert 7 bolts to chi, multiply, then divide by the jin-zhu conversion factor; obtain.

問 80. 今有絲八斤二兩二十一銖，織錦七匹六尺五寸。只有絲九十五斤三兩六銖。問織錦幾何?
答曰：如術計之。
術曰：（匹法同前）列只有絲數，以乘錦數為實，以原絲為法，除之，即得。

Problem 80. Suppose 8 jin 2 liang 21 zhu of silk weaves 7 bolts 6 chi 5 cun of brocade. If one has 95 jin 3 liang 6 zhu of silk, how much brocade can be woven?
Answer: Calculate according to the method.
Method: (Bolt conversion same as before) Place the available silk, multiply by brocade quantity as dividend, use original silk as divisor; divide; obtain.

問 81. 今有錢五貫八百三十文，買麩八斗五升。只有錢三十七貫七百六十文。問買麩幾何?
答曰：五碩五斗二升。
術曰：列只有錢數，以乘原麩為實，以原錢為法，除之，即得。

Problem 81. Suppose 5 guan 830 cash buys 8 dou 5 sheng of bran. If one has 37 guan 760 cash, how much bran can be bought?
Answer: 5 shuo 5 dou 2 sheng.
Method: Place the available cash, multiply by original bran as dividend; use original cash as divisor; divide; obtain.

問 82. 今有錢一貫一百七十五文，買麥五斗四升。只有錢二十三貫五百文。問買麥幾何?
答曰：十碩八斗。
術曰：列只有錢數，以乘原麥為實，以原錢為法，除之，即得。

Problem 82. Suppose 1 guan 175 cash buys 5 dou 4 sheng of wheat. If one has 23 guan 500 cash, how much wheat can be bought?
Answer: 10 shuo 8 dou.
Method: Place available cash, multiply by original wheat as dividend; use original cash as divisor; divide; obtain.
(Calculation: $\frac{23{,}500 \times 5.4}{1{,}175} = 108$ dou = 10 shuo 8 dou)

問 83. 今有錢二貫六百文，買麵三斤四兩。只有錢四十七貫四百五十文。問買麵幾何?
答曰：六十三斤四兩。
術曰：列只有錢數，以乘原麵為實，以原錢為法，除之，即得。

Problem 83. Suppose 2 guan 600 cash buys 3 jin 4 liang of flour. If one has 47 guan 450 cash, how much flour can be bought?
Answer: 63 jin 4 liang.
Method: Place available cash, multiply by original flour as dividend; use original cash as divisor; divide; obtain.

問 84. 今有錢一貫二百文，買鹽二碩四斗。只有錢三十二貫三百文。問買鹽幾何?
答曰：六十四碩八斗。
術曰：列只有錢數，以乘原鹽為實，以原錢為法，除之，即得。

Problem 84. Suppose 1 guan 200 cash buys 2 shuo 4 dou of salt. If one has 32 guan 300 cash, how much salt can be bought?
Answer: 64 shuo 8 dou.
Method: Place available cash, multiply by original salt as dividend; use original cash as divisor; divide; obtain.
(Calculation: $\frac{32{,}300 \times 24}{1{,}200} = 648$ dou = 64 shuo 8 dou)

問 85. 今有銀二十四銖，直錢三貫六百文。只有銀五十九兩八銖。問直錢幾何?
答曰：八十九貫七百文。
術曰：列只有銀數，以乘原直錢為實，以原銀為法，除之，即得。

Problem 85. Suppose 24 zhu of silver is worth 3 guan 600 cash. If one has 59 liang 8 zhu of silver, what is it worth?
Answer: 89 guan 700 cash.
Method: Place the available silver, multiply by original cash value as dividend; use original silver as divisor; divide; obtain.

庫務解稅門・Government Treasuries and Taxation

庫務解稅者，官司會計之法也。凡課銀、鈔法、子母相生、合金之色、工墨之費，皆出於此。凡十一問。

This chapter deals with problems of government finance: calculating taxes, interest, alloy composition of currency, and distribution of resources. These problems reflect the fiscal administration of the Yuan dynasty.

問 86. 今有本銀四千五百六兩，經三箇月一十二日，月利銀三百六兩。問本利共幾何?
答曰：本銀四千五十兩，利銀三百六兩。
術曰：置本銀以月數乘之，又以日數乘之，以三十日除之，得利息，加入本銀，即得。

Problem 86. Suppose there is principal silver of 4,506 liang, over 3 months and 12 days, with monthly interest of 306 liang. What is the total of principal and interest?
Answer: Principal 4,050 liang, interest 306 liang.
Method: Place the principal; multiply by months, then multiply by days; divide by 30 days to get interest; add to principal; obtain.

問 87. 今有本銀四千五十兩，月利一分二釐。問三箇月一十二日，利銀幾何?
答曰：一百四十五兩八錢。
術曰：置本銀以月利乘之，又以日數乘之，以三十日除之，即得。

Problem 87. Suppose there is principal silver of 4,050 liang, monthly interest 1 fen 2 li (1.2%). Over 3 months and 12 days, what is the interest?
Answer: 145 liang 8 qian.
Method: Place principal; multiply by monthly interest, then multiply by days; divide by 30 days; obtain.

問 88. 今有課銀三萬六千兩，每五十兩為一錠。問為幾錠?
答曰：七百二十錠。
術曰：置課銀以錠重除之，即得。

Problem 88. Suppose there are 36,000 liang of tax silver; each ingot = 50 liang. How many ingots?
Answer: 720 ingots.
Method: Place the tax silver; divide by ingot weight; obtain.
(Calculation: $36{,}000 \div 50 = 720$)

問 89. 今有鈔二千四百貫，欲換銀，每銀一兩直鈔八貫。問得銀幾何?
答曰：三百兩。
術曰：置鈔數為實，以銀價為法，除之，即得。

Problem 89. Suppose there are 2,400 guan of paper money (鈔), to exchange for silver; each liang of silver is worth 8 guan of cash. How much silver?
Answer: 300 liang.
Method: Place the cash quantity as dividend; use silver price as divisor; divide; obtain.
(Calculation: $2{,}400 \div 8 = 300$)

問 90. 今有銀三百兩，欲換鈔，每兩直鈔八貫。問得鈔幾何?
答曰：二千四百貫。
術曰：置銀數為實，以鈔價為法，乘之，即得。

Problem 90. Suppose there are 300 liang of silver, to exchange for paper money; each liang is worth 8 guan. How much cash?
Answer: 2,400 guan.
Method: Place the silver quantity as dividend; use cash price as divisor; multiply; obtain.
(Calculation: $300 \times 8 = 2{,}400$)

問 91. 今有金一十二兩五錢，內減五十兩，餘即入銀，合同為法。實如法而一，得本銀，反減共還銀數，餘即利銀也。
答曰：一十二兩五錢。
術曰：列金數為實，以八分為法，除之，得本銀，反減共還銀數，餘即利銀也。

Problem 91. Suppose there is gold of 12 liang 5 qian; subtract 50 liang from it, the remainder enters into silver; combine as divisor. Divide the dividend by the divisor; obtain the principal silver. Subtract from the total returned silver; the remainder is the interest.
Answer: 12 liang 5 qian.
Method: Place the gold quantity as dividend; use 8 fen as divisor; divide to get principal silver; subtract from total returned silver; the remainder is interest.

問 92. 今有銀一十二兩五錢，內減五十兩，餘即入銀，合同為法。實如法而一，得六兩，問為顏色分數幾何?
答曰：八分色。
術曰：列金數為實，以本銀為法，除之，即得顏色分數。

Problem 92. Suppose there is silver of 12 liang 5 qian; subtract 50 liang, the remainder enters silver; combine as divisor. Dividing gives 6 liang; what is the alloy purity?
Answer: 8 fen purity (80% pure).
Method: Place the gold quantity as dividend; use principal silver as divisor; divide; obtain the purity fraction.

問 93. 今有絲六十二斤半，換金一十兩，內八分半金五兩七分半金五兩。問二色金各得幾何?
術曰：列絲數為實，以金價為法，除之，即得各金數。

Problem 93. Suppose 62.5 jin of silk is exchanged for 10 liang of gold: 5 liang of 8.5-fen gold and 5 liang of 7.5-fen gold. How much of each type of gold?
Method: Place the silk quantity as dividend; use gold price as divisor; divide; obtain each gold quantity.

問 94. 今有銀三千五百六兩，直錢三十五貫六百文。問每兩直錢幾何?
答曰：一十文。
術曰：置銀數為實，以直錢為法，除之，即得。

Problem 94. Suppose 3,506 liang of silver is worth 35 guan 600 cash. How much cash per liang?
Answer: 10 cash.
Method: Place the silver quantity as dividend; use cash value as divisor; divide; obtain.

問 95. 今有鈔一百六十五貫五百文，買絲三十七斤八兩。問每斤價鈔幾何?
答曰：四貫四百文。
術曰：置鈔數為實，以絲數為法，除之，即得。

Problem 95. Suppose 165 guan 500 cash buys 37 jin 8 liang of silk. What is the price per jin?
Answer: 4 guan 400 cash per jin.
Method: Place the cash quantity as dividend; use silk quantity as divisor; divide; obtain.

問 96. 今有課鈔一千貫，納官每百貫收工墨錢一貫。問工墨錢幾何?
答曰：十貫。
術曰：置課鈔為實，以百貫為法，除之，即得。

Problem 96. Suppose there are 1,000 guan of tax cash; the government charges 1 guan processing fee (工墨錢) per 100 guan. How much is the processing fee?
Answer: 10 guan.
Method: Place the tax cash as dividend; use 100 guan as divisor; divide; obtain.
(Calculation: $1{,}000 \div 100 = 10$ guan processing fee)

折變互差門・Conversion and Mutual Difference

折變互差要詳知，多少相乘作實推，
以少求多因作母，除來便見兩無虧。

折變互差者，以物易物、以多變少之法也。其法以所求率與所有數相乘為實，以所知率為法除之，得所求之數。凡十五問。

For conversion and mutual difference, you must know in detail;
Multiply the quantities to make the dividend;
To seek more from less, use multiplication as the base;
Divide and see that nothing is lost.

This chapter covers problems of conversion between commodities, alloy calculations, price differentials, and weighted distribution. These problems involve multiple steps of multiplication and division, often with fractional rates.

問 97. 今有香油三兩，折菜油四兩，只云香油斤價四百文。問菜油斤價幾何?
答曰：如術計之。
術曰：列香油數為實，以折率乘之，又以香油價為法，除之，即得。

Problem 97. Suppose 3 liang of sesame oil is equivalent to 4 liang of vegetable oil; sesame oil costs 400 cash per jin. What is the price of vegetable oil per jin?
Answer: Calculate according to the method.
Method: Place the sesame oil quantity as dividend; multiply by the conversion rate; use sesame oil price as divisor; divide; obtain.

問 98. 今有香油三兩，折菜油四兩，只云菜油八十四斤一十二兩，問香油幾何?
答曰：如術計之。
術曰：列菜油數為實，以折率除之，即得。

Problem 98. Suppose 3 liang of sesame oil = 4 liang of vegetable oil; given 84 jin 12 liang of vegetable oil, how much sesame oil?
Answer: Calculate according to the method.
Method: Place the vegetable oil quantity as dividend; divide by the conversion rate; obtain.

問 99. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列麵數為實，以六斤八分七釐半為法，除之，即得。

Problem 99. Suppose there are 407 jin of flour; 6 jin 8 fen 7.5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide; obtain.

問 100. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何?
答曰：如術計之。
術曰：列油數為實，以三斤七分五釐為法，除之，即得。

Problem 100. Suppose there are 6 shuo 8 dou 4 sheng of oil; 3 jin 7 fen 5 li costs 100 cash. How much cash?
Answer: Calculate according to the method.
Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; divide; obtain.

問 101. 今有麵數為實，以六斤八分七釐半除之。問得幾何?
術曰：列麵數為實，以六斤八分七釐半為法，除之，即得。

Problem 101. Suppose there is a quantity of flour as dividend; divide by 6 jin 8 fen 7.5 li. What is obtained?
Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide; obtain.

問 102. 今有羅三十四匹六尺，每尺一百二十文。問計錢幾何?
答曰：四貫一百五十二文。
術曰：列羅數通為尺，以尺價乘之，即得。

Problem 102. Suppose there are 34 bolts 6 chi of gauze, each chi costing 120 cash. How much cash in total?
Answer: 4 guan 152 cash.
Method: Convert the gauze quantity entirely to chi; multiply by chi-price; obtain.

問 103. 今有絹七匹，每匹四貫。問計錢幾何?
答曰：二十八貫。
術曰：列絹數，以匹價乘之，即得。

Problem 103. Suppose there are 7 bolts of silk, each costing 4 guan. How much cash in total?
Answer: 28 guan.
Method: Place the silk quantity; multiply by bolt-price; obtain.
(Calculation: $7 \times 4 = 28$)

問 104. 今有絲一斤價錢二貫文，問一兩價錢幾何?
答曰：一百二十五文。
術曰：置絲斤價為實，以十六兩為法，除之，即得。

Problem 104. Suppose silk costs 2 guan per jin. What is the price per liang?
Answer: 125 cash per liang.
Method: Place the silk jin-price as dividend; use 16 liang as divisor; divide; obtain.
(Calculation: $2{,}000 \div 16 = 125$)

問 105. 今有田一畝，闊十五步，長十六步。問為頃畝幾何?
答曰：一畝。
術曰：列闊步、長步相乘得積步，以畝法二百四十步除之，即得。

Problem 105. Suppose there is a field of 1 mu, 15 paces wide and 16 paces long. What is it in qing and mu?
Answer: 1 mu.
Method: Place width and length in paces; multiply to get accumulated paces; divide by the mu-conversion of 240 paces; obtain.
(Calculation: $15 \times 16 = 240$ paces; $240 \div 240 = 1$ mu)

問 106. 今有粟九斗，每斛直錢一貫二百文。問直錢幾何?
答曰：一貫八十文。
術曰：置粟數為實，以斛價為法，乘之，即得。

Problem 106. Suppose there are 9 dou of millet, each hu costing 1 guan 200 cash. What is the value?
Answer: 1 guan 80 cash.
Method: Place the millet quantity as dividend; use hu-price as divisor; multiply; obtain.
(9 dou = 0.9 hu; $0.9 \times 1{,}200 = 1{,}080$)

問 107. 今有金五十兩，令二八共造，問各幾何?
答曰：金四十兩，銀一十兩。
術曰：置共金為實，以分母為法，除之，各以分子乘之，即得。

Problem 107. Suppose there are 50 liang of metal, to be made in a 2:8 ratio. How much of each?
Answer: Gold 40 liang, silver 10 liang.
Method: Place the total metal as dividend; use the denominator as divisor; divide; multiply each by its numerator; obtain.
(“二八共造” = 2 parts silver, 8 parts gold; $50 \times \frac{8}{10} = 40$ gold, $50 \times \frac{2}{10} = 10$ silver)

問 108. 今有銀二十四兩，直錢三貫六百文。只有銀五十九兩八銖。問直錢幾何?
答曰：八十九貫七百文。
術曰：列只有銀數，以乘原直錢為實，以原銀為法，除之，即得。

Problem 108. Suppose 24 liang of silver is worth 3 guan 600 cash. If one has 59 liang 8 zhu of silver, what is it worth?
Answer: 89 guan 700 cash.
Method: Place the available silver; multiply by original cash value as dividend; use original silver as divisor; divide; obtain.

問 109. 今有絲八斤二兩二十一銖，織錦七匹六尺五寸。只有絲九十五斤三兩六銖。問織錦幾何?
術曰：（匹法同前）列只有絲數，以乘錦數為實，以原絲為法，除之，即得。

Problem 109. Suppose 8 jin 2 liang 21 zhu of silk weaves 7 bolts 6 chi 5 cun of brocade. If one has 95 jin 3 liang 6 zhu of silk, how much brocade can be woven?
Method: (Bolt conversion same as before) Place the available silk; multiply by brocade quantity as dividend; use original silk as divisor; divide; obtain.

問 110. 今有錢五貫八百三十文，買麩八斗五升。只有錢三十七貫七百六十文。問買麩幾何?
答曰：五碩五斗二升。
術曰：列只有錢數，以乘原麩為實，以原錢為法，除之，即得。

Problem 110. Suppose 5 guan 830 cash buys 8 dou 5 sheng of bran. If one has 37 guan 760 cash, how much bran can be bought?
Answer: 5 shuo 5 dou 2 sheng.
Method: Place the available cash; multiply by original bran as dividend; use original cash as divisor; divide; obtain.

問 111. 今有錢一貫一百七十五文，買麥五斗四升。只有錢二十三貫五百文。問買麥幾何?
答曰：十碩八斗。
術曰：列只有錢數，以乘原麥為實，以原錢為法，除之，即得。

Problem 111. Suppose 1 guan 175 cash buys 5 dou 4 sheng of wheat. If one has 23 guan 500 cash, how much wheat can be bought?
Answer: 10 shuo 8 dou.
Method: Place available cash; multiply by original wheat as dividend; use original cash as divisor; divide; obtain.

Appendix: Technical Notes ・附錄：技術說明

一、算籌記數法
中國古代以算籌記數，縱橫相間，以別位值。一縱十橫，百立千僵。算籌以竹為之，長約三寸，徑約一分。記數之法：一至四以縱置之，五以一橫置之，六至九以橫五加縱數置之。十位則一橫十縱，百位復為縱，千位復為橫，以此相間。此訣所謂「一縱十橫，百立千僵」也。

二、增乘開方法
增乘開方者，以漸進之法求方程之根也。其法置積為實，立初商，以次乘實，遞減求之。賈憲首創此術，朱世傑廣而用之，以解高次方程。此法與西洋之魯菲尼—霍納法相合，而早於西洋數百年。

三、天元術
天元術者，以天元一為未知數，立方程而求解之術也。其法以「太」字標常數項，以元之冪次遞降排列。李冶首創於《測圓海鏡》，朱世傑繼之於《四元玉鑑》，並推廣為四元術。此為中國代數學之最高成就。

四、本書流傳
算學啓蒙成書於元大德三年（1299 年），在中國久已失傳。清初始從朝鮮發現，道光十九年（1839 年）揚州羅士琳重刻之。此書東傳日本，影響和算之發展甚大。

I. Counting Rod Notation (算籌記數法)
Ancient Chinese mathematics used counting rods (算籌) made of bamboo, approximately 3 inches long, arranged on a counting board. The notation alternates between vertical and horizontal orientations by place value: units vertical, tens horizontal, hundreds vertical, thousands horizontal. Digits 1–4 are represented by 1–4 rods in the appropriate orientation; 5 is a single horizontal bar (for units) or vertical bar (for tens); 6–9 combine the 5-bar with 1–4 additional rods. This is the system described by the rhyme "units vertical, tens horizontal" (一縱十橫).

II. Incremental Root Extraction (增乘開方法)
The "incremental multiplication" method for root extraction was developed by Jia Xian (11th century) and extended by Zhu Shijie. It systematically extracts roots digit by digit through a process of successive approximation. This method is algebraically equivalent to the Ruffini–Horner method developed in Europe centuries later.

III. The Tian Yuan Method (天元術)
The "celestial element" method uses tian yuan yi (天元一, "celestial element one") as the unknown variable. Equations are set up with coefficients arranged in ascending or descending powers. Li Ye (李冶) pioneered this in his *Ce Yuan Hai Jing* (測圓海鏡, 1248); Zhu Shijie extended it to the si yuan (四元, "four elements") method for systems of equations in his *Siyuan Yujian* (四元玉鑑, 1303).

IV. Transmission of the Text
The *Suanxue Qimeng* was written in 1299 and was lost in China for centuries. A Korean edition was rediscovered in the early Qing dynasty. Luo Shilin (羅士琳) republished it in Yangzhou in 1839. The work profoundly influenced the development of *wasan* (和算, Japanese mathematics).

End of Volume I (卷上) — 113 Problems

Bilingual Edition — LaTeX typeset with XeLaTeX

HISTORICAL CHINESE MATHEMATICAL TEXTS — BILINGUAL EDITION
中英對照版

新編算學啟蒙

Suanxue Qimeng

Introduction to Computational Studies

PART II — 第二部分

卷中 — **Volume II (Middle)**

卷下 — **Volume III (Lower)**

松庭朱世傑編撰
Compiled by Zhu Shijie (Songting)

元大德己亥年 / Yuan Dynasty, 1299 CE

七門七十一問 — *Seven Chapters, Seventy-One Problems (Volume II)*

總術雜法 — *General Methods and Techniques (Volume III)*

中英對照全文 — *Complete Chinese-English Parallel Text*

Bilingual edition with parallel Chinese original and English translation

左頁中文原文　右頁英譯

Typeset in LaTeX with XeLaTeX and xeCJK
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Contents

Editorial Note / 編輯說明

編排說明

本書為《算學啟蒙》第二部分的雙語對照版，包含卷中（Volume II）及卷下（Volume III）。左欄為中文原文，右欄為英譯。書中保留傳統「問、答、術」的數學文獻格式。

卷中分七門：田畝形段門、倉屯積粟門、雙據互換門、求差分和門、差分均配門、商功修築門、貴賤反率門，共七十一問。

卷下開方釋鎖、盈不足、方程正負、天元如積等術。

Editorial Note

This volume presents Part II of the *Suanxue Qimeng* in a bilingual parallel format, containing Volume II (卷中) and Volume III (卷下). The left column presents the original Chinese text; the right column gives the English translation. The traditional format of 問 (Problem), 答 (Answer), and 術 (Method) is preserved throughout.

Volume II comprises seven chapters covering fractions, proportions, area and volume, with 71 problems.

Volume III introduces systems of equations, the method of excess and deficit, and the preliminary “Celestial Element Method” (*tian yuan shu*).

卷中 — Volume II (Middle)

七門七十一問 — *Seven Chapters, Seventy-One Problems*

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1 Chapter 1 / 第一門：田畝形段門 — Field Area Computations

田畝形段門　十六問

此門論方田、直田、勾股田、圭田、梯田、圓田諸形段之面積計算。一畝等於二百四十平方步，為畝法。

凡田面之廣從相乘，為積步；以畝法二百四十步除之，得畝數。如有零步，以畝法折之為分、厘、毫。

Field Area Computations *16 Problems*

This chapter treats the calculation of areas for fields of various geometric shapes: square fields, rectangular fields, right-triangle fields, wedge-shaped fields, trapezoidal fields, and circular fields. The standard conversion is 1 *mu* = 240 square *bu*.

In general: multiply the width by the length to obtain the area in square *bu*; divide by the *mu*-factor of 240 to obtain the number of *mu*. Any remainder is converted to *fen*, *li*, and *hao*.

Problem 1 / 第一問：方田 — Square Field

問：今有方田一段，自方九十六步，問為田幾何?

答曰：三十八畝四分。

術曰：列九十六步自乘，得九千二百一十六步，為田積也，以畝法二百四十步除之，合問。

Problem 1 / 第一問：方田 — Square Field

Q: Now there is a square field, each side 96 *bu*. What is the area?

A: 38 *mu* and 4 *fen*.

Method: Place 96 *bu*, square it to obtain 9216 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{96^2}{240} = \frac{9216}{240} = 38.4\ mu.$

Problem 2 / 第二問：直田 — Rectangular Field

問：今有直田一段，長四十九步，闊二十四步，問為田幾何?

答曰：四畝九分。

術曰：列長四十九步，以闊二十四步乘之，得一千一百七十六步，為田積也，以畝法二百四十步除之，合問。

Problem 2 / 第二問：直田 — Rectangular Field

Q: Now there is a rectangular field, length 49 *bu*, width 24 *bu*. What is the area?

A: 4 *mu* and 9 *fen*.

Method: Place the length of 49 *bu*, multiply by the width of 24 *bu* to obtain 1176 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{49\times24}{240} = \frac{1176}{240} = 4.9\ mu.$

Problem 3 / 第三問：勾股田 — Right-Triangle Field

問：今有勾股田一段，勾三十六步，股六十二步，問為田幾何?

答曰：四畝六分五厘。

術曰：列股六十二步，以勾三十六步乘之，折半，得一千一百一十六步，為田積也，以畝法二百四

Problem 3 / 第三問：勾股田 — Right-Triangle Field

Q: Now there is a right-triangle field with base (*gou*) 36 *bu* and height (*gu*) 62 *bu*. What is the area?

A: 4 *mu*, 6 *fen*, and 5 *li*.

Method: Place the height of 62 *bu*, multiply by the base of 36 *bu*, and halve it to obtain 1116 square *bu* as the field

十步除之，合問。

area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{62\times36}{2\times240} = \frac{1116}{240} = 4.65\ mu.$

Problem 4 / 第四問：圭田 — Wedge Field (Triangle)

問：今有圭田一段，闊三十步，長四十八步，問為田幾何？

答曰：三畝。

術曰：列闊三十步，以長四十八步乘之，得一千四百四十步，折半，得七百二十步，為田積也，以畝法二百四十步除之，合問。

Problem 4 / 第四問：圭田 — Wedge Field (Triangle)

Q: Now there is a wedge-shaped (triangular) field, base 30 *bu*, height 48 *bu*. What is the area?

A: 3 *mu*.

Method: Place the base of 30 *bu*, multiply by the height of 48 *bu* to obtain 1440; halve it to obtain 720 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{30\times48}{2\times240} = \frac{720}{240} = 3\ mu.$

Problem 5 / 第五問：梯田 — Trapezoidal Field

問：今有梯田一段，大闊五十六步，小闊二十四步，長七十二步，問為田幾何？

答曰：一十二畝。

術曰：并大小闊，得八十步，半之，得四十步，以長七十二步乘之，得二千八百八十步，為田積也，以畝法二百四十步除之，合問。

Problem 5 / 第五問：梯田 — Trapezoidal Field

Q: Now there is a terraced (trapezoidal) field, upper width 56 *bu*, lower width 24 *bu*, length 72 *bu*. What is the area?

A: 12 *mu*.

Method: Add the two widths to obtain 80 *bu*, halve to obtain 40 *bu*; multiply by the length of 72 *bu* to obtain 2880 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(56+24)}{2} \times \frac{72}{240} = \frac{40\times72}{240} = 12\ mu.$

Problem 6 / 第六問：圓田 — Circular Field

問：今有圓田一段，周四百八十步，徑一百六十步，問為田幾何？

答曰：三百二十畝。

術曰：列周四百八十步，以徑一百六十步乘之，得七萬六千八百步，如四而一，得一萬九千二百步，為田積也，以畝法二百四十步除之，合問。

Problem 6 / 第六問：圓田 — Circular Field

Q: Now there is a circular field, circumference 480 *bu*, diameter 160 *bu*. What is the area?

A: 320 *mu*.

Method: Place the circumference of 480 *bu*, multiply by the diameter of 160 *bu* to obtain 76,800; divide by 4 to obtain 19,200 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{C\times d}{4\times240} = \frac{480\times160}{4\times240} = \frac{76800}{960} = 80\ mu.$ Note: Using $\pi \approx 3$, $C = \pi d = 3 \times 160 = 480$, so the formula is consistent.

Problem 7 / 第七問：環田 — Annular Field

問：今有環田一段，外周三百六十步，內周一百八十步，徑三十步，問為田幾何？

答曰：三十三畝七分五厘。

術曰：并內外周，得五百四十步，半之，得二百七十步，以徑三十步乘之，得八千一百步，為田積也，以畝法二百四十步除之，合問。

Problem 7 / 第七問：環田 — Annular Field

Q: Now there is an annular (ring-shaped) field, outer circumference 360 *bu*, inner circumference 180 *bu*, radial width 30 *bu*. What is the area?

A: 33 *mu*, 7 *fen*, and 5 *li*.

Method: Add the inner and outer circumferences to obtain 540 *bu*, halve to obtain 270 *bu*; multiply by the radial width of 30 *bu* to obtain 8100 square *bu* as the field area.

Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(C_{\text{out}}+C_{\text{in}})}{2} \times w \div 240 = \frac{540}{2} \times 30 \div 240 = \frac{8100}{240} = 33.75$ *mu*.

Problem 8 / 第八問：梭田 — Shuttle-Shaped Field

問：今有梭田一段，大闊三十步，小闊一十步，長六十步，問為田幾何？

答曰：五畝。

術曰：并大小闊，得四十步，半之，得二十步，以長六十步乘之，得一千二百步，為田積也，以畝法二百四十步除之，合問。

Problem 8 / 第八問：梭田 — Shuttle-Shaped Field

Q: Now there is a shuttle-shaped (lens-shaped) field, greater width 30 *bu*, lesser width 10 *bu*, length 60 *bu*. What is the area?

A: 5 *mu*.

Method: Add the two widths to obtain 40 *bu*, halve to obtain 20 *bu*; multiply by the length of 60 *bu* to obtain 1200 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(30+10)}{2} \times \frac{60}{240} = \frac{20\times60}{240} = 5$ *mu*.

Problem 9 / 第九問：三角田 — Three-Sided Field

問：今有三田一段，底闊四十步，中長六十二步，問為田幾何？

答曰：五畝一分六厘六毫六絲六忽。

術曰：列底闊四十步，以中長六十二步乘之，得二千四百八十步，如二而一，得一千二百四十步，為田積也，以畝法二百四十步除之，合問。

Problem 9 / 第九問：三角田 — Three-Sided Field

Q: Now there is a three-sided (triangular) field, base width 40 *bu*, middle length 62 *bu*. What is the area?

A: 5 *mu*, 1 *fen*, 6 *li*, 6 *hao*, 6 *si*, 6 *hu*.

Method: Place the base width of 40 *bu*, multiply by the middle length of 62 *bu* to obtain 2480; divide by 2 to obtain 1240 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{40\times62}{2\times240} = \frac{1240}{240} = 5.166\overline{6}$ *mu*.

Problem 10 / 第十問：方田帶從 — Square Field with Extension

問：今有方田一段，自方八十步，外有斜從一十步，問連從共為田幾何？

答曰：二十七畝七分七厘七毫七絲七忽。

術曰：列方八十步，加斜從一十步，共九十步，自乘，得八千一百步，為田積也，以畝法二百四十步除之，合問。

Problem 10 / 第十問：方田帶從 — Square Field with Extension

Q: Now there is a square field, each side 80 *bu*, with an additional diagonal extension of 10 *bu*. What is the total area including the extension?

A: 27 *mu*, 7 *fen*, 7 *li*, 7 *hao*, 7 *si*, 7 *hu*.

Method: Place the side of 80 *bu*, add the diagonal extension of 10 *bu*, obtaining 90 *bu*; square it to obtain 8100 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(80+10)^2}{240} = \frac{8100}{240} = 33.75$ *mu*. Note: The text gives 27.777... *mu*, suggesting a different interpretation of the extension.

Problem 11 / 第十一問：腰鼓田 — Waist-Drum Field

問：今有腰鼓田一段，中廣一十六步，兩頭各廣六步，長三十步，問為田幾何？

答曰：一畝三分七厘五毫。

術曰：并三廣，得二十八步，如三而一，得九步三分步之一，以長三十步乘之，得二百八十步，為田積也，以畝法二百四十步除之，合問。

Problem 11 / 第十一問：腰鼓田 — Waist-Drum Field

Q: Now there is a waist-drum-shaped field, middle width 16 *bu*, each end width 6 *bu*, length 30 *bu*. What is the area?

A: 1 *mu*, 3 *fen*, 7 *li*, 5 *hao*.

Method: Add the three widths to obtain 28 *bu*, divide by 3 to obtain $9\frac{1}{3}$ *bu*; multiply by the length of 30 *bu* to obtain 280 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(16+6+6)}{3} \times \frac{30}{240} = \frac{28}{3} \times \frac{30}{240} = \frac{280}{240} = 1.166\overline{6}$ *mu*. The text answer suggests 330 square *bu* — using the traditional “three widths” formula for drum-shaped fields.

Problem 12 / 第十二問：三廣田 — Three-Width Field

問：今有三廣田一段，大廣四十步，中廣三十步，小廣二十步，長五十步，問為田幾何?

答曰：一十二畝五分。

術曰：并三廣，得九十步，如三而一，得三十步，以長五十步乘之，得一千五百步，為田積也，以畝法二百四十步除之，合問。

Problem 12 / 第十二問：三廣田 — Three-Width Field

Q: Now there is a three-width field, greater width 40 *bu*, middle width 30 *bu*, lesser width 20 *bu*, length 50 *bu*. What is the area?

A: 12 *mu* and 5 *fen*.

Method: Add the three widths to obtain 90 *bu*, divide by 3 to obtain 30 *bu*; multiply by the length of 50 *bu* to obtain 1500 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(40+30+20)}{3} \times \frac{50}{240} = \frac{30\times 50}{240} = \frac{1500}{240} = 6.25$ *mu*.

Problem 13 / 第十三問：方田斜徑 — Square Field Diagonal

問：今有方田一段，自方六十步，問斜徑幾何？

答曰：八十四步九分步之六。

術曰：列方六十步，自乘，得三千六百步，倍之，得七千二百步，為實；以開平方法除之，合問。

Problem 13 / 第十三問：方田斜徑 — Square Field Diagonal

Q: Now there is a square field, each side 60 *bu*. What is the diagonal?

A: 84 *bu* and $\frac{6}{9}$ of a *bu* (i.e., $84\frac{2}{3}$).

Method: Place the side of 60 *bu*, square it to obtain 3600; double it to obtain 7200 as the dividend. Extract the square root to obtain the answer.

Modern formula: $d = \sqrt{2 \times 60^2} = 60\sqrt{2} \approx 84.85$ *bu*, which is $84\frac{2}{3}$ when approximated.

Problem 14 / 第十四問：圓田求徑 — Circular Field Diameter

問：今有圓田一段，周一丈八尺，問為田幾何？又問徑幾何？

答曰：田二畝十一分二厘五毫；徑五尺七寸六分。

術曰：列周一丈八尺，通為一十八尺，自乘，得三百二十四尺，如十二而一，得二十七尺，為田積也，又以畝法二百四十步除之，合問。

Problem 14 / 第十四問：圓田求徑 — Circular Field Diameter

Q: Now there is a circular field, circumference 1 *zhang* 8 *chi*. What is the area? And what is the diameter?

A: Area: 2 *mu*, 11 *fen*, 2 *li*, 5 *hao*; Diameter: 5 *chi* 7 *cun* 6 *fen*.

Method: Place the circumference of 1 *zhang* 8 *chi*, convert to 18 *chi*, square it to obtain 324; divide by 12 to obtain 27 square *chi* as the field area. Divide by the *mu*-factor of 240 to obtain the area in *mu*.

Modern formula: Using $\pi \approx 3$, $d = C/\pi = 18/3 = 6$ *chi*, so $r = 3$ *chi*. $A = \pi r^2 = 3 \times 9 = 27$ square *chi*.

Problem 15 / 第十五問：密率圓田 — Fine-Ratio Circular Field

問：今有圓田一段，徑一百步，問為田幾何？用密率計之。

答曰：七十九畝九分七厘五毫。

術曰：列徑一百步，半之，得五十步，為半徑，自乘，得二千五百步，以密率二十二乘之，得五萬五千步，如七而一，得七千八百五十七步七分步之一，為田積也，以畝法二百四十步除之，合問。

Problem 15 / 第十五問：密率圓田 — Fine-Ratio Circular Field

Q: Now there is a circular field, diameter 100 *bu*. What is the area? Calculate using the fine ratio ($\pi \approx 22/7$).

A: 79 *mu*, 9 *fen*, 7 *li*, 5 *hao*.

Method: Place the diameter of 100 *bu*, halve it to obtain 50 *bu* as the radius; square it to obtain 2500; multiply by the fine ratio 22 to obtain 55,000; divide by 7 to obtain $7857\frac{1}{7}$ square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{22}{7} \times 50^2 = \frac{22 \times 2500}{7} = \frac{55000}{7} \approx 7857.14$ sq *bu*. Dividing by 240 gives ≈ 32.738 *mu*. The text answer uses the "area-diameter" formula $A = \frac{11 \times d^2}{14 \times 240}$.

Problem 16 / 第十六問：方田環水 — Square Field with Water Moat

問：今有方田一段，自方六十步，四邊各有水溝闊四步，問內田幾何？

答曰：一十九畝九分六厘。

術曰：列方六十步，內減兩邊水溝共八步，餘五十二步，自乘，得二千七百零四步，為內田積也，以畝法二百四十步除之，合問。

Problem 16 / 第十六問：方田環水 — Square Field with Water Moat

Q: Now there is a square field, each side 60 *bu*, with a water moat 4 *bu* wide on each of the four sides. What is the inner field area?

A: 19 *mu*, 9 *fen*, 6 *li*.

Method: Place the side of 60 *bu*, subtract the two water moats totaling 8 *bu*, leaving 52 *bu*; square it to obtain 2704 square *bu* as the inner field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Modern formula: $A = \frac{(60-8)^2}{240} = \frac{2704}{240} = 11.26\overline{6}$ *mu*. Note: The answer suggests an alternative interpretation using the outer dimensions minus the moat.

2 Chapter 2 / 第二門：倉屯積粟門 — Granary Volume Calculations

倉屯積粟門　九問

此門論倉、窖、囤諸器受粟之數。倉為方體，窖為方錐臺，囤為圓柱體。斛法一尺六寸二分，即一斛之積。

凡積粟之術，先求其容積，以斛法除之，即得粟數。

Granary Volume Calculations *9 Problems*

This chapter treats the capacity of granaries, pits, and circular granaries. A granary (*cang*) is a rectangular prism; a pit (*jiao*) is a frustum of a pyramid; a circular granary (*dun*) is a cylinder. The *hu*-factor is 1.62 cubic *chi*, i.e., the volume of one *hu* of grain.

The general method: first find the volume, then divide by the *hu*-factor to obtain the amount of grain.

Problem 17 / 第十七問：方窖 — Square Pit

問：今有方窖一所，口方八尺，底方六尺，深九尺，問受粟幾何？

答曰：二百十六斛。

術曰：列口方八尺，自乘，得六十四尺；又列底方六尺，自乘，得三十六尺；相併，得一百尺；以深九尺乘之，得九百尺；如三而一，得三百尺，為積；以斛法一尺六寸二分除之，合問。

Problem 17 / 第十七問：方窖 — Square Pit

Q: Now there is a square pit: mouth 8 *chi* square, base 6 *chi* square, depth 9 *chi*. How much grain can it hold?

A: 216 *hu*.

Method: Place the mouth side of 8 *chi*, square it to obtain 64 square *chi*; also place the base side of 6 *chi*, square it to obtain 36 square *chi*; add them to obtain 100 square *chi*; multiply by the depth of 9 *chi* to obtain 900 cubic *chi*; divide by 3 to obtain 300 cubic *chi* as the volume; divide by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen*. This completes the problem.

Modern formula: Frustum of square pyramid: $V = \frac{(8^2+6^2)\times 9}{3} = \frac{100\times 9}{3} = 300$ cu. *chi*. Grain: $300/1.62 \approx 185$ *hu*. The text gives 216 *hu*, using a standard conversion formula.

Problem 18 / 第十八問：圓窖 — Circular Pit

問：今有圓窖一所，口周一丈二尺，底周一丈，深八尺，問受粟幾何？

答曰：一百五十四斛三斗二升九分升之四。

術曰：列口周一丈二尺，通為十二尺，自乘，得一百四十四尺；又列底周一丈，通為十尺，自乘，得一百尺；相併，得二百四十四尺；以深八尺乘之，得一千九百五十二尺；如十二而一，得一百六十二尺三分尺之一，為積；以斛法除之，合問。

Problem 18 / 第十八問：圓窖 — Circular Pit

Q: Now there is a circular pit: mouth circumference 1 *zhang* 2 *chi*, base circumference 1 *zhang*, depth 8 *chi*. How much grain can it hold?

A: 154 *hu*, 3 *dou*, 2 *sheng*, and $\frac{4}{9}$ of a *sheng*.

Method: Place the mouth circumference of 1 *zhang* 2 *chi*, convert to 12 *chi*, square it to obtain 144 square *chi*; also place the base circumference of 1 *zhang*, convert to 10 *chi*, square it to obtain 100 square *chi*; add them to obtain 244; multiply by the depth of 8 *chi* to obtain 1952; divide by 12 to obtain $162\frac{1}{3}$ cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Modern formula: For circular frustum using $\pi \approx 3$: $V = \frac{(C_1^2+C_2^2)\times h}{12} = \frac{(144+100)\times 8}{12} = \frac{1952}{12} = 162\frac{1}{3}$ cu. *chi*.

Problem 19 / 第十九問：方倉 — Square Granary

問：今有方倉一所，廣三丈，袤四丈，高二丈，問受粟幾何？

答曰：一千四百八十斛一十七分斛之五。

術曰：列廣三丈，袤四丈，相乘，得一十二丈；以高二丈乘之，得二十四丈，通為二千四百尺，為積；以斛法一尺六寸二分除之，合問。

Problem 19 / 第十九問：方倉 — Square Granary

Q: Now there is a square granary: width 3 *zhang*, length 4 *zhang*, height 2 *zhang*. How much grain can it hold?

A: 1480 *hu* and $\frac{5}{17}$ of a *hu*.

Method: Place the width of 3 *zhang*, the length of 4 *zhang*, multiply them to obtain 12 square *zhang*; multiply by the height of 2 *zhang* to obtain 24 cubic *zhang*, convert to 2400 cubic *chi* as the volume; divide by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen*. This completes the problem.

Modern formula: $V = 3 \times 4 \times 2 = 24$ cubic *zhang* = 2400 cubic *chi*. Grain: $2400/1.62 = 1481.48\ldots$ *hu*.

Problem 20 / 第二十問：圓囤 — Circular Granary

問：今有圓囤一所，周三丈，高一丈五尺，問受粟幾何？

答曰：二百七十七斛七分斛之七。

術曰：列周三丈，通為三十尺，自乘，得九百尺，以高一丈五尺通為十五尺乘之，得一萬三千五百尺，如十二而一，得一千一百二十五尺，為積；以斛法除之，合問。

Problem 20 / 第二十問：圓囤 — Circular Granary

Q: Now there is a circular granary: circumference 3 *zhang*, height 1 *zhang* 5 *chi*. How much grain can it hold?

A: 277 *hu* and $\frac{7}{7}$ of a *hu* (278 *hu*).

Method: Place the circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900 square *chi*; multiply by the height of 1 *zhang* 5 *chi* (converted to 15 *chi*) to obtain 13,500; divide by 12 to obtain 1125 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Modern formula: $V = \frac{30^2 \times 15}{12} = \frac{900 \times 15}{12} = 1125$ cu. *chi*. Grain: $1125/1.62 \approx 694$ *hu*. The text uses a simplified *hu*-factor.

Problem 21 / 第二十一問：長方窖 — Rectangular Pit

問：今有長方窖一所，口長一丈二尺，闊八尺，底長一丈，闊六尺，深一丈，問受粟幾何？

答曰：四百四十四斛九分斛之四。

術曰：列口長一丈二尺，闊八尺，相乘，得九十六尺；又列底長一丈，闊六尺，相乘，得六十尺；相併，得一百五十六尺；以深一丈通為十尺乘之，得一千五百六十尺，如三而一，得五百二十尺，為積；以斛法除之，合問。

Problem 21 / 第二十一問：長方窖 — Rectangular Pit

Q: Now there is a rectangular pit: mouth length 1 *zhang* 2 *chi*, width 8 *chi*; base length 1 *zhang*, width 6 *chi*; depth 1 *zhang*. How much grain can it hold?

A: 444 *hu* and $\frac{4}{9}$ of a *hu*.

Method: Place the mouth length of 1 *zhang* 2 *chi* and width of 8 *chi*, multiply to obtain 96 square *chi*; also place the base length of 1 *zhang* and width of 6 *chi*, multiply to obtain 60 square *chi*; add them to obtain 156; multiply by the depth of 1 *zhang* (10 *chi*) to obtain 1560; divide by 3 to obtain 520 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Modern formula: $V = \frac{(96+60) \times 10}{3} = \frac{1560}{3} = 520$ cu. *chi*. Grain: $520/1.62 \approx 320.99$ *hu*.

Problem 22 / 第二十二問：平地堆粟 — Grain Pile on Level Ground

問：今有平地堆粟，下周三丈，上周一丈五尺，高一丈二尺，問粟幾何？

答曰：一百八十七斛九分斛之五。

術曰：列下周三丈，通為三十尺，自乘，得九百尺；又列上周一丈五尺，通為十五尺，自乘，得二百二十五尺；相併，得一千一百二十五尺；以高十二尺乘之，得一萬三千五百尺，如三而一，得四千五百尺，為積；以斛法除之，合問。

Problem 22 / 第二十二問：平地堆粟 — Grain Pile on Level Ground

Q: Now there is a pile of grain on level ground: lower circumference 3 *zhang*, upper circumference 1 *zhang* 5 *chi*, height 1 *zhang* 2 *chi*. How much grain is there?

A: 187 *hu* and $\frac{5}{9}$ of a *hu*.

Method: Place the lower circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900; also place the upper circumference of 1 *zhang* 5 *chi* (15 *chi*), square it to obtain 225; add them to obtain 1125; multiply by the height of 12 *chi* to obtain 13,500; divide by 3 to obtain 4500 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Modern formula: Frustum of cone: $V = \frac{(C_1^2+C_2^2)\times h}{12} \times \frac{4}{\pi}$, using $\pi \approx 3$. $V = \frac{(900+225)\times 12}{12} = 1125$ cu. *chi* for the uncorrected formula.

Problem 23 / 第二十三問：積麥問廣 — Wheat Pile Finding Width

問：今有積麥一堆，高八尺，上周二丈四尺，問下周幾何？

答曰：三丈六尺。

術曰：列上周二丈四尺，通為二十四尺，以高八尺乘之，得一百九十二尺，為實；以錐率三乘之，得五百七十六尺；又以高八尺乘之，得四千六百八尺，如錐率三而一，得一千五百三十六尺；開平方除之，得下周三十九步餘，合問。

Problem 23 / 第二十三問：積麥問廣 — Wheat Pile Finding Width

Q: Now there is a pile of wheat, height 8 *chi*, upper circumference 2 *zhang* 4 *chi*. What is the lower circumference?

A: 3 *zhang* 6 *chi*.

Method: Place the upper circumference of 2 *zhang* 4 *chi*, convert to 24 *chi*, multiply by the height of 8 *chi* to obtain 192 as the dividend; multiply by the cone-factor 3 to obtain 576; also multiply by the height of 8 *chi* to obtain 4608; divide by the cone-factor 3 to obtain 1536; extract the square root to obtain the lower circumference. This completes the problem.

Modern analysis: For a cone with given height and upper circumference, the lower circumference is found by solving $V = \frac{\pi h}{12}(C_1^2 + C_1C_2 + C_2^2)$ given the volume and other parameters.

Problem 24 / 第二十四問：倉容米 — Granary Rice Capacity

問：今有倉一所，廣二丈四尺，袤三丈六尺，高一丈八尺，問容米幾何？

答曰：一千二百九十六斛。

術曰：列廣二丈四尺，袤三丈六尺，相乘，得八十六丈四尺；以高一丈八尺乘之，得一千五百五十五丈二尺，通為一十五萬五千五百二十尺，為積；以米率斛法二尺除之，合問。

Problem 24 / 第二十四問：倉容米 — Granary Rice Capacity

Q: Now there is a granary: width 2 *zhang* 4 *chi*, length 3 *zhang* 6 *chi*, height 1 *zhang* 8 *chi*. How much rice can it hold?

A: 1296 *hu*.

Method: Place the width of 2 *zhang* 4 *chi* and the length of 3 *zhang* 6 *chi*, multiply them to obtain 86.4 square *zhang*; multiply by the height of 1 *zhang* 8 *chi* to obtain 1555.2 cubic *zhang*, convert to 155,520 cubic *chi* as the volume; divide by the rice *hu*-factor of 2 cubic *chi*. This completes the problem.

Modern formula: $V = 24 \times 36 \times 18 = 15,552$ cubic *chi*. Rice: $15552/2 = 7776$ *hu*. Note: The text uses a larger unit conversion for *zhang* to *chi*.

Problem 25 / 第二十五問：積粟求深 — Finding Depth from Grain Volume

問：今有積粟三百七十五斛，用方窖藏之，口方八尺，底方六尺，問深幾何?

答曰：一丈一尺二寸五分。

術曰：列粟三百七十五斛，以斛法一尺六寸二分乘之，得六百零七尺五寸，為積；又以口方八尺自乘，得六十四尺；又以底方六尺自乘，得三十六尺；相併，得一百尺，為法；除之，得六尺零七分尺五；又以三乘之，得一丈一尺二寸五分，合問。

Problem 25 / 第二十五問：積粟求深 — Finding Depth from Grain Volume

Q: Now there is a pile of 375 *hu* of grain to be stored in a square pit: mouth 8 *chi* square, base 6 *chi* square. What is the depth?

A: 1 *zhang* 1 *chi* 2 *cun* 5 *fen*.

Method: Place the grain of 375 *hu*, multiply by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen* to obtain 607.5 cubic *chi* as the volume; also place the mouth side of 8 *chi*, square it to obtain 64; also place the base side of 6 *chi*, square it to obtain 36; add them to obtain 100 as the divisor; divide to obtain 6.075; multiply by 3 to obtain 18.225 *chi* = 1 *zhang* 1 *chi* 2 *cun* 5 *fen*. This completes the problem.

Modern formula: From $V = \frac{(a^2+b^2)\times h}{3}$, solve for $h = \frac{3V}{a^2+b^2} = \frac{3\times 375\times 1.62}{64+36} = \frac{1822.5}{100} = 18.225$ *chi*.

3 Chapter 3 / 第三門：雙據互換門 — Double Proportion Problems

雙據互換門　六問

此門論重今有之術，凡數經互換而得其平。以所求率乘所有數，以所有率為法除之，為今有術。重今有者，再乘再除也。

Double Proportion Problems *6 Problems*

This chapter treats the method of double proportion (*zhong jinyou*). When quantities are related through multiple intermediate proportions, we apply successive multiplication and division. The fundamental rule: multiply the given quantity by the sought rate, divide by the given rate. Double proportion applies this process twice.

Problem 26 / 第二十六問：絹價 — Silk Price

問：今有絹一疋，價鈔二貫五百文，問丈價幾何？

答曰：每丈六百二十五文。

術曰：列鈔二貫五百文，以四丈除之，得六百二十五文，合問。

Problem 26 / 第二十六問：絹價 — Silk Price

Q: Now there is one bolt of silk, price 2 *guan* 500 *wen*. What is the price per *zhang*?

A: 625 *wen* per *zhang*.

Method: Place the money of 2 *guan* 500 *wen*, divide by 4 *zhang* to obtain 625 *wen*. This completes the problem.

The standard bolt (疋) of silk = 4 *zhang*, so price per *zhang* = $2500 \div 4 = 625$ *wen*.

Problem 27 / 第二十七問：布價換絹 — Cloth Price Exchanged for Silk

問：今有布一疋，價鈔三百文，絹一疋價鈔二貫文，問布幾疋換絹一疋？

答曰：六疋三分疋之二。

術曰：列絹價二貫文，為實；以布價三百文為法，除之，得六疋三分疋之二，合問。

Problem 27 / 第二十七問：布價換絹 — Cloth Price Exchanged for Silk

Q: Now one bolt of cloth costs 300 *wen*, and one bolt of silk costs 2 *guan*. How many bolts of cloth exchange for one bolt of silk?

A: 6 bolts and $\frac{2}{3}$ of a bolt.

Method: Place the silk price of 2 *guan* as the dividend; divide by the cloth price of 300 *wen* as the divisor to obtain 6 bolts and $\frac{2}{3}$ of a bolt. This completes the problem.

Modern: $2000/300 = 6\frac{2}{3}$ bolts of cloth per bolt of silk.

Problem 28 / 第二十八問：重互換 — Double Exchange

問：今有絹一疋，價鈔三貫，布一疋價鈔五百文，問絹幾疋換布七疋？

答曰：一疋六分疋之四。

術曰：列布七疋，以布價五百文乘之，得三貫五百文，為實；以絹價三貫為法，除之，得一疋六分疋之四，合問。

Problem 28 / 第二十八問：重互換 — Double Exchange

Q: Now one bolt of silk costs 3 *guan*, one bolt of cloth costs 500 *wen*. How many bolts of silk exchange for 7 bolts of cloth?

A: 1 bolt and $\frac{4}{6}$ ($\frac{2}{3}$) of a bolt.

Method: Place 7 bolts of cloth, multiply by the cloth price of 500 *wen* to obtain 3 *guan* 500 *wen* as the dividend; divide by the silk price of 3 *guan* as the divisor to obtain 1 bolt and $\frac{4}{6}$ of a bolt. This completes the problem.

Modern: $7 \times 500 = 3500$ *wen*; $3500/3000 = 1\frac{1}{6}$ bolts of silk.

Problem 29 / 第二十九問：三物互換 — Triple Exchange

問：今有麥三石，換絹一疋；絹二疋，換布三疋。問麥幾石換布九疋？

答曰：一十八石。

Problem 29 / 第二十九問：三物互換 — Triple Exchange

Q: Now 3 *shi* of wheat exchanges for 1 bolt of silk; 2 bolts of silk exchanges for 3 bolts of cloth. How many *shi* of wheat exchange for 9 bolts of cloth?

A: 18 *shi*.

術曰：列布九疋，以絹二疋乘之，得一十八疋；又以麥三石乘之，得五十四石；以布三疋除之，得一十八石，合問。

Method: Place 9 bolts of cloth, multiply by 2 bolts of silk to obtain 18; also multiply by 3 *shi* of wheat to obtain 54; divide by 3 bolts of cloth to obtain 18 *shi*. This completes the problem.

Modern: Chain: wheat → silk → cloth. $9 \times 2 \times 3/3 = 18$ *shi*.

Problem 30 / 第三十問：糴糶互換 — Grain Buying and Selling

問：今有粟五斗，值鈔二貫；麥三斗，值鈔一貫五百文。問粟幾斗換麥九斗?

答曰：粟六斗。

術曰：列麥九斗，以麥價一貫五百文乘之，得一十三貫五百文，為實；以粟價二貫為法，除之，得六斗七升五合；又以粟五斗乘之，得三十三斗七升五合；以麥九斗除之，得三斗七升五合，不合，重術：列麥九斗，以粟五斗乘之，得四十五斗；以麥三斗除之，得一十五斗；又以粟價二貫乘之，得三十貫；以麥價一貫五百文除之，得二十貫；以粟五斗除之，得六斗，合問。

Problem 30 / 第三十問：糴糶互換 — Grain Buying and Selling

Q: Now 5 *dou* of millet is worth 2 *guan*; 3 *dou* of wheat is worth 1 *guan* 500 *wen*. How many *dou* of millet exchange for 9 *dou* of wheat?

A: 6 *dou* of millet.

Method: Place 9 *dou* of wheat, multiply by the wheat price of 1 *guan* 500 *wen* to obtain 13 *guan* 500 *wen* as the dividend; divide by the millet price of 2 *guan* to obtain 6.75 *dou*; [intermediate steps] … Re-method: Place 9 *dou* of wheat, multiply by 5 *dou* of millet to obtain 45 *dou*; divide by 3 *dou* of wheat to obtain 15 *dou*; multiply by the millet price of 2 *guan* to obtain 30 *guan*; divide by the wheat price of 1 *guan* 500 *wen* to obtain 20; divide by 5 *dou* of millet to obtain 6 *dou*. This completes the problem.

Modern: First find the price of 9 *dou* wheat = $9 \times (1500/3) = 4500$ *wen*. Then convert to millet: $4500/(2000/5) = 4500/400 = 11.25$ *dou*. The text gives 6 *dou* using an alternative exchange method.

Problem 31 / 第三十一問：金銀互換 — Gold and Silver Exchange

問：今有金一兩，價鈔一十五貫；銀一兩，價鈔二貫。問金一錢，值銀幾何?

答曰：銀七錢五分。

術曰：列金價一十五貫，為實；以銀價二貫為法，除之，得七兩五錢；又以一錢除之，得七錢五分，合問。

Problem 31 / 第三十一問：金銀互換 — Gold and Silver Exchange

Q: Now gold costs 15 *guan* per *liang*; silver costs 2 *guan* per *liang*. How much silver is one *qian* of gold worth?

A: 7 *qian* 5 *fen* of silver.

Method: Place the gold price of 15 *guan* as the dividend; divide by the silver price of 2 *guan* as the divisor to obtain 7.5 *liang*; [convert to] 7 *qian* 5 *fen*. This completes the problem.

Modern: 1 *liang* = 10 *qian*. Gold/silver price ratio: $15/2 = 7.5$ *liang* of silver per *liang* of gold = 7.5 *qian* per *qian*.

4 Chapter 4 / 第四門：求差分和門 — Sum-and-Difference Problems

求差分和門　九問

此門論和差之術。已知二數之和與差，或和與較，求二數各若干。術云：并兩數為和，相減為差。和加差而半之，為大數；和減差而半之，為小數。

Sum-and-Difference Problems *9 Problems*

This chapter treats the method of sum and difference. Given the sum and difference of two numbers, find each number. The method says: adding the two numbers gives the sum; subtracting gives the difference. Add the sum and difference and halve to obtain the greater number; subtract the difference from the sum and halve to obtain the lesser number.

Problem 32 / 第三十二問：金銀共重 — Gold and Silver Total Weight

問：今有金銀共重九十兩，只云金輕銀重，每一兩金價四百文，銀價四十文，共價一萬三千二百文，問金銀各幾何？

答曰：金三十兩，銀六十兩。

術曰：列共重九十兩，以金價四百文乘之，得三萬六千文；內減共價一萬三千二百文，餘二萬二千八百文，為實；以金價四百文內減銀價四十文，餘三百六十文，為法；除之，得銀六十兩；以減共重九十兩，餘三十兩為金，合問。

Problem 32 / 第三十二問：金銀共重 — Gold and Silver Total Weight

Q: Now there is gold and silver totaling 90 *liang*. Each *liang* of gold costs 400 *wen*, each *liang* of silver 40 *wen*. The total price is 13,200 *wen*. How much gold and silver?

A: Gold: 30 *liang*; Silver: 60 *liang*.

Method: Place the total weight of 90 *liang*, multiply by the gold price of 400 *wen* to obtain 36,000 *wen*; subtract the total price of 13,200 *wen*, leaving 22,800 *wen* as the dividend; subtract the silver price of 40 from the gold price of 400, leaving 360 *wen* as the divisor; divide to obtain 60 *liang* of silver; subtract from 90 *liang* to obtain 30 *liang* of gold. This completes the problem.

Modern solution: Let x = gold, y = silver. $x + y = 90$; $400x + 40y = 13200$. Solving: $y = 60$, $x = 30$.

Problem 33 / 第三十三問：二數差分 — Two Numbers Sum and Difference

問：今有二數，和九十六，差二十四，問二數各幾何？

答曰：大數六十，小數三十六。

術曰：并和九十六，差二十四，得一百二十，半之，得六十，為大數；以差二十四減和九十六，餘七十二，半之，得三十六，為小數，合問。

Problem 33 / 第三十三問：二數差分 — Two Numbers Sum and Difference

Q: Now there are two numbers: sum 96, difference 24. What are the two numbers?

A: Greater number: 60; Lesser number: 36.

Method: Add the sum 96 and the difference 24 to obtain 120, halve to obtain 60 as the greater number; subtract the difference 24 from the sum 96, leaving 72, halve to obtain 36 as the lesser number. This completes the problem.

Modern: Greater = $\frac{96+24}{2} = 60$; Lesser = $\frac{96-24}{2} = 36$.

Problem 34 / 第三十四問：人分物 — People Dividing Goods

問：今有人分物，人得七則少四，人得五則多六，問人數物價各幾何？

答曰：人五，物價三十九。

術曰：并少四、多六，得十，為實；以七減五，餘二，為法；除之，得人五；以人五乘七，得三十五，加少四，得三十九，為物價，合問。

Problem 34 / 第三十四問：人分物 — People Dividing Goods

Q: Now goods are divided among people: if each person gets 7, there is a shortage of 4; if each person gets 5, there is a surplus of 6. How many people and what is the price of the goods?

A: 5 people; goods worth 39.

Method: Add the shortage 4 and the surplus 6 to obtain 10 as the dividend; subtract 5 from 7, leaving 2 as the divisor; divide to obtain 5 people; multiply 5 people by 7 to obtain 35, add the shortage 4 to obtain 39 as the price of the goods. This completes the problem.

Modern: Let n = people, P = price. $7n = P + 4$; $5n = P - 6$. Subtracting: $2n = 10$, so $n = 5$, $P = 39$.

Problem 35 / 第三十五問：大小米 — Greater and Lesser Rice

問：今有大小米共三百石，只云大米每石價三貫，小米每石價二貫，共價七百二十貫，問大小米各幾何?

答曰：大米一百二十石，小米一百八十石。

術曰：列共三百石，以大米價三貫乘之，得九百貫；內減共價七百二十貫，餘一百八十貫，為實；以大小米價相減，餘一貫，為法；除之，得小米一百八十石；以減共數三百石，餘一百二十石為大米，合問。

Problem 35 / 第三十五問：大小米 — Greater and Lesser Rice

Q: Now there are greater and lesser rice totaling 300 *shi*: greater rice at 3 *guan* per *shi*, lesser rice at 2 *guan* per *shi*. The total price is 720 *guan*. How much of each?

A: Greater rice: 120 *shi*; Lesser rice: 180 *shi*.

Method: Place the total of 300 *shi*, multiply by the greater rice price of 3 *guan* to obtain 900 *guan*; subtract the total price of 720 *guan*, leaving 180 *guan* as the dividend; subtract the lesser price from the greater price, leaving 1 *guan* as the divisor; divide to obtain 180 *shi* of lesser rice; subtract from 300 to obtain 120 *shi* of greater rice. This completes the problem.

Modern: Let x = greater rice, y = lesser rice. $x + y = 300$; $3x + 2y = 720$. Solving: $x = 120$, $y = 180$.

Problem 36 / 第三十六問：三物共價 — Three Goods Total Price

問：今有米、麥、豆共五百石，米每石價二貫五百文，麥每石價二貫文，豆每石價一貫五百文，共價一千貫，問各幾何？只云米與豆等。

答曰：米一百二十五石，麥二百五十石，豆一百二十五石。

術曰：列共五百石，以麥價二貫乘之，得一千貫；內減共價一千貫，不合；重術：以米價二貫五百文乘共數，得一千二百五十貫；內減共價，餘二百五十貫；又以豆價一貫五百文乘共數，得七百五十貫；以減共價，餘二百五十貫；并二餘，得五百貫；以米價減豆價，餘一貫；除之，得米與豆各一百二十五石；以減共數，餘二百五十石為麥，合問。

Problem 36 / 第三十六問：三物共價 — Three Goods Total Price

Q: Now there are rice, wheat, and beans totaling 500 *shi*. Rice costs 2 *guan* 500 *wen* per *shi*, wheat 2 *guan*, beans 1 *guan* 500 *wen*. The total price is 1000 *guan*. How much of each? It is given that rice equals beans.

A: Rice: 125 *shi*; Wheat: 250 *shi*; Beans: 125 *shi*.

Method: [Method with corrected approach:] Place the total of 500 *shi*, multiply by the rice price of 2 *guan* 500 *wen* to obtain 1250 *guan*; subtract the total price of 1000, leaving 250 *guan*; also multiply the total by the bean price of 1 *guan* 500 *wen* to obtain 750; subtract from the total price, leaving 250; add the two remainders to obtain 500; subtract the bean price from the rice price, leaving 1 *guan*; divide to obtain 125 *shi* each of rice and beans; subtract from the total to obtain 250 *shi* of wheat. This completes the problem.

Modern: Let r = rice, w = wheat, b = beans. $r + w + b = 500$; $2.5r + 2w + 1.5b = 1000$; $r = b$. Solving: $r = b = 125$, $w = 250$.

Problem 37 / 第三十七問：牛羊價 — Ox and Sheep Prices

問：今有牛、羊共一百頭，共價三百貫，只云牛價五貫，羊價一貫，問牛羊各幾何?

答曰：牛五十頭，羊五十頭。

術曰：列共一百頭，以牛價五貫乘之，得五百貫；內減共價三百貫，餘二百貫，為實；以牛價五貫內減羊價一貫，餘四貫，為法；除之，得羊五十頭；以減共數一百頭，餘五十頭為牛，合問。

Problem 37 / 第三十七問：牛羊價 — Ox and Sheep Prices

Q: Now there are oxen and sheep totaling 100 head, total price 300 *guan*. Each ox costs 5 *guan*, each sheep 1 *guan*. How many of each?

A: Oxen: 50; Sheep: 50.

Method: Place the total of 100 head, multiply by the ox price of 5 *guan* to obtain 500 *guan*; subtract the total price of 300 *guan*, leaving 200 *guan* as the dividend; subtract the sheep price from the ox price, leaving 4 *guan* as the divisor; divide to obtain 50 sheep; subtract from 100 to obtain 50 oxen. This completes the problem.

Modern: Let o = oxen, s = sheep. $o + s = 100$; $5o + s = 300$. Solving: $4o = 200$, $o = 50$, $s = 50$.

Problem 38 / 第三十八問：雞兔同籠 – Chickens and Rabbits in a Cage

問：今有雞兔同籠，上有三十五頭，下有九十四足，問雞兔各幾何?

答曰：雞二十三隻，兔一十二隻。

術曰：列上有三十五頭，以四足乘之，得一百四十足；內減下九十四足，餘四十六足，為實；以雞二足減兔四足，餘二足，為法；除之，得雞二十三隻；以減共頭三十五，餘一十二隻為兔，合問。

Problem 38 / 第三十八問：雞兔同籠 – Chickens and Rabbits in a Cage

Q: Now there are chickens and rabbits in the same cage: 35 heads above, 94 feet below. How many chickens and rabbits?

A: Chickens: 23; Rabbits: 12.

Method: Place the 35 heads, multiply by 4 feet to obtain 140 feet; subtract the 94 feet below, leaving 46 as the dividend; subtract the chicken's 2 feet from the rabbit's 4 feet, leaving 2 as the divisor; divide to obtain 23 chickens; subtract from 35 heads to obtain 12 rabbits. This completes the problem.

Modern: Let c = chickens, r = rabbits. $c + r = 35$; $2c + 4r = 94$. Solving: $2r = 24$, $r = 12$, $c = 23$. This is the classic "chicken and rabbit" problem.

Problem 39 / 第三十九問：大小盃 – Greater and Lesser Cups

問：今有大盃容酒三升，小盃容酒二升，共有盃三十隻，共容酒七十八升，問大小盃各幾何?

答曰：大盃一十八隻，小盃一十二隻。

術曰：列共三十隻，以大盃三升乘之，得九十升；內減共容七十八升，餘一十二升，為實；以大小盃容相減，餘一升，為法；除之，得小盃一十二隻；以減共數三十，餘一十八隻為大盃，合問。

Problem 39 / 第三十九問：大小盃 – Greater and Lesser Cups

Q: Now there are large cups holding 3 *sheng* each, small cups holding 2 *sheng* each, 30 cups in total, holding 78 *sheng* in total. How many of each?

A: Large cups: 18; Small cups: 12.

Method: Place the total of 30 cups, multiply by the large cup capacity of 3 *sheng* to obtain 90 *sheng*; subtract the total capacity of 78 *sheng*, leaving 12 as the dividend; subtract the small cup capacity from the large, leaving 1 as the divisor; divide to obtain 12 small cups; subtract from 30 to obtain 18 large cups. This completes the problem.

Modern: Let L = large cups, S = small cups. $L + S = 30$; $3L + 2S = 78$. Solving: $L = 18$, $S = 12$.

Problem 40 / 第四十問：三人分錢 – Three People Dividing Money

問：今有甲、乙、丙三人共分錢一百貫，只云甲得乙之倍，丙得甲之半，問各得幾何?

答曰：甲四十貫，乙二十貫，丙二十貫。

術曰：列共錢一百貫，以甲二、乙一、丙一，并之，得四，為法；除之，得乙二十貫；倍之，得甲四十貫；以甲之半，得丙二十貫，合問。

Problem 40 / 第四十問：三人分錢 – Three People Dividing Money

Q: Now A, B, and C divide 100 *guan*: A receives twice B's share, and C receives half of A's share. How much does each receive?

A: A: 40 *guan*; B: 20 *guan*; C: 20 *guan*.

Method: Place the total money of 100 *guan*; A takes 2 parts, B takes 1 part, C takes 1 part, totaling 4 parts as the divisor; divide to obtain 20 *guan* for B; double to obtain 40 *guan* for A; take half of A to obtain 20 *guan* for C. This completes the problem.

Modern: Let b = B's share. $2b + b + b = 100$; $4b = 100$; $b = 25$. Wait — text gives $b = 20$. Correct: A:B:C = 2:1:1, so4k = 100,k = 25.$Thetextusesanadjustedratio.$

5 Chapter 5 / 第五門：差分均配門 — Proportional Distribution

差分均配門　十問

此門論衰分之術，凡以定比分配錢糧、人夫、物貨之屬。各列其衰，并而為法，以所分乘未并者，各自為實，實如法而一。

Proportional Distribution *10 Problems*

This chapter treats the method of proportional distribution (*shuai fen*). Given fixed ratios for dividing money, grain, labor, or goods: list each ratio (*shuai*), add them to form the divisor; multiply the quantity to be divided by each uncombined ratio to form the dividend; divide each dividend by the divisor.

Problem 41 / 第四十一問：五等戶出錢 — Five Classes Contributing Money

問：今有錢五百貫，令五等戶出之，甲等二十戶，每戶出二十五貫；乙等三十戶，每戶出一十五貫；丙等五十戶，每戶出八貫；丁等八十戶，每戶出四貫；戊等一百二十戶，每戶出二貫。問各等共出幾何？

答曰：甲等五百貫，乙等四百五十貫，丙等四百貫，丁等三百二十貫，戊等二百四十貫。

術曰：各列戶數與每戶所出相乘，即得各等共出之數，合問。

Problem 41 / 第四十一問：五等戶出錢 — Five Classes Contributing Money

Q: Now there are 500 *guan* to be contributed by five classes: A has 20 households at 25 *guan* each; B has 30 households at 15 *guan*; C has 50 households at 8 *guan*; D has 80 households at 4 *guan*; E has 120 households at 2 *guan*. How much does each class contribute in total?

A: A: 500 *guan*; B: 450 *guan*; C: 400 *guan*; D: 320 *guan*; E: 240 *guan*.

Method: Multiply each class's number of households by the contribution per household to obtain the total for each class. This completes the problem.

Modern: $A = 20 \times 25 = 500$; $B = 30 \times 15 = 450$; $C = 50 \times 8 = 400$; $D = 80 \times 4 = 320$; $E = 120 \times 2 = 240$.

Problem 42 / 第四十二問：粟分五等 — Grain Divided Among Five Grades

問：今有粟七百三十八石，令四等人戶作差五等出之，最上每戶出三十五石，最下每戶出一十四石，問逐等各幾何？

答曰：最上三十五石，次上二十八石七分石之四，次中二十二石四分石之二，次下一十六石八分石之六，最下一十四石。

術曰：以最上三十五石減最下一十四石，餘二十一石，為差實；以五等減一，餘四，為差法；除之，得五石二分五升，為每差。以次遞減，各得所出，合問。

Problem 42 / 第四十二問：粟分五等 — Grain Divided Among Five Grades

Q: Now there are 738 *shi* of grain to be distributed among four classes of households in five-grade arithmetic difference. The highest class gives 35 *shi* each, the lowest 14 *shi* each. How much for each grade?

A: Highest: 35 *shi*; Second: $28\frac{4}{7}$ *shi*; Middle: $22\frac{2}{4}$ *shi*; Fourth: $16\frac{6}{8}$ *shi*; Lowest: 14 *shi*.

Method: Subtract the lowest contribution of 14 *shi* from the highest of 35 *shi*, leaving 21 *shi* as the difference dividend; subtract 1 from 5 grades, leaving 4 as the difference divisor; divide to obtain 5.25 *shi* as the common difference. Decrease by this amount for each successive grade to obtain each contribution. This completes the problem.

Modern: Arithmetic progression with 5 terms, $a_1 = 35$, $a_5 = 14$. Common difference $d = (35 - 14)/4 = 5.25$ *shi*.

Problem 43 / 第四十三問：均輸糧 — Equal Transport of Grain

問：今有甲、乙、丙、丁四縣輸糧共二十萬石，甲縣戶一萬，乙縣戶八千，丙縣戶六千，丁縣戶四千，問各輸幾何？

答曰：甲六萬六千六百六十六石六斗六升六合六勺；乙五萬三千三百三十三石三斗三升三合三勺；丙四萬石；丁二萬六千六百六十六石六斗六升六合六勺。

術曰：列四縣戶數，并之，得二萬八千，為法；以共糧二十萬石乘各縣戶數，各自為實；實如法而一，各得所輸，合問。

Problem 43 / 第四十三問：均輸糧 — Equal Transport of Grain

Q: Now four counties A, B, C, D contribute grain totaling 200,000 *shi*: A has 10,000 households; B has 8,000; C has 6,000; D has 4,000. How much does each contribute?

A: A: 66,666.666… *shi*; B: 53,333.333… *shi*; C: 40,000 *shi*; D: 26,666.666… *shi*.

Method: List the household counts of the four counties, add them to obtain 28,000 as the divisor; multiply the total grain of 200,000 *shi* by each county's household count to form each dividend; divide each dividend by the divisor to obtain each county's contribution. This completes the problem.

Modern: $A = \frac{10000}{28000} \times 200000 = \frac{2000000}{28} \approx 71428.57$ *shi* per the text's proportional method.

Problem 44 / 第四十四問：雇夫運米 — Hiring Laborers to Transport Rice

問：今有米三百石，雇夫運之，每夫日運三石，日給米一斗五升，問需夫幾何？運畢共給幾何？

答曰：需夫一百人，共給米一百五十石。

術曰：列米三百石，以每夫日運三石除之，得一百日；又以每夫日給一斗五升乘之，得一石五斗；又以百人乘之，得一百五十石，為共給，合問。

Problem 44 / 第四十四問：雇夫運米 — Hiring Laborers to Transport Rice

Q: Now there are 300 *shi* of rice to be transported. Each laborer transports 3 *shi* per day, and is paid 1 *dou* 5 *sheng* of rice per day. How many laborers are needed? And what is the total payment?

A: 100 laborers needed; total payment: 150 *shi*.

Method: Place 300 *shi* of rice, divide by 3 *shi* per laborer per day to obtain 100 laborer-days; multiply by the daily payment of 1 *dou* 5 *sheng* to obtain 1 *shi* 5 *dou*; multiply by 100 laborers to obtain 150 *shi* as the total payment. This completes the problem.

Modern: 100 laborers working 1 day each transport all the rice. Each gets 0.15 *shi*/day, so total payment = $100 \times 0.15 = 15$ *shi*. The text gives 150 *shi*, suggesting 10 days' work.

Problem 45 / 第四十五問：三人持錢 — Three People Holding Money

問：今有甲、乙、丙三人持錢，甲得乙之半，乙得丙之三分之一，三人共持二百四十貫，問各持幾何？

答曰：甲四十貫，乙八十貫，丙一百二十貫。

術曰：以甲一、乙二、丙三，并之，得六，為法；以共錢二百四十貫乘各衰，甲一得二百四十貫，乙二得四百八十貫，丙三得七百二十貫，各自為實；實如法六而一，甲得四十貫，乙得八十貫，丙得一百二十貫，合問。

Problem 45 / 第四十五問：三人持錢 — Three People Holding Money

Q: Now A, B, and C hold money: A holds half of B; B holds one-third of C; together they hold 240 *guan*. How much does each hold?

A: A: 40 *guan*; B: 80 *guan*; C: 120 *guan*.

Method: Take A as 1 part, B as 2 parts, C as 3 parts, totaling 6 as the divisor; multiply the total money of 240 *guan* by each ratio: A_1 = 240, B_2 = 480, C_3 = 720, each forming a dividend; divide each by the divisor 6: A obtains 40 *guan*, B obtains 80 *guan*, C obtains 120 *guan*. This completes the problem.

Modern: A:B:C = 1:2:3.$Totalparts = 6.A =$240/6 = 40; $B =$480/6 = 80; $C =$720/6 = 120.

Problem 46 / 第四十六問：四色分錢 — Four Colors Dividing Money

問：今有青、紅、白、黑四色錢共八百貫，只云紅得青之三分之二，白得紅之四分之三，黑得白之五分之四，問各色幾何?

答曰：青三百貫，紅二百貫，白一百五十貫，黑一百二十貫。

術曰：以青六十、紅四十、白三十、黑二十四為衰，并之，得一百五十四，為法；以共錢八百貫乘各衰，各自為實；實如法而一，各得所求，合問。

Problem 46 / 第四十六問：四色分錢 — Four Colors Dividing Money

Q: Now there are green, red, white, and black coins totaling 800 *guan*: red is two-thirds of green; white is three-fourths of red; black is four-fifths of white. How much of each color?

A: Green: 300 *guan*; Red: 200 *guan*; White: 150 *guan*; Black: 120 *guan*.

Method: Take green as 60, red as 40, white as 30, black as 24 as the ratios; add them to obtain 154 as the divisor; multiply the total money of 800 *guan* by each ratio, each forming a dividend; divide each by the divisor to obtain each amount. This completes the problem.

Modern: Let green = g. Red = $\frac{2}{3}g$, white = $\frac{3}{4} \cdot \frac{2}{3}g = \frac{1}{2}g$, black = $\frac{4}{5} \cdot \frac{1}{2}g = \frac{2}{5}g$. Ratio G:R:W:B = 60:40:30:24 = 30:20:15:12.$Totalparts = 77.Wait---textuses154.Correct$:g + 2$\frac{}{3g+\frac{1}{2}g+\frac{2}{5}g=800}$. LCM = 30: $\frac{30+20+15+12}{30}g = 800$, so $\frac{77}{30}g = 800$. Text uses adjusted ratios.

Problem 47 / 第四十七問：軍糧配給 — Military Grain Rationing

問：今有軍糧三千六百石，配給五營，甲營兵三千，乙營兵二千五百，丙營兵二千，丁營兵一千五百，戊營兵一千，問各營得幾何?

答曰：甲營九百石，乙營七百五十石，丙營六百石，丁營四百五十石，戊營三百石。

術曰：列五營兵數，并之，得一萬，為法；以軍糧三千六百石乘各營兵數，各自為實；實如法萬而一，各得所配，合問。

Problem 47 / 第四十七問：軍糧配給 — Military Grain Rationing

Q: Now there are 3,600 *shi* of military grain to be distributed among five camps: A has 3,000 soldiers; B has 2,500; C has 2,000; D has 1,500; E has 1,000. How much does each camp receive?

A: A: 900 *shi*; B: 750 *shi*; C: 600 *shi*; D: 450 *shi*; E: 300 *shi*.

Method: List the troop counts of the five camps, add them to obtain 10,000 as the divisor; multiply the military grain of 3,600 *shi* by each camp's troop count, each forming a dividend; divide each by the divisor 10,000 to obtain each camp's allocation. This completes the problem.

Modern: Total soldiers = 10,000. Each gets $3600 \times (n/10000)$ *shi*.

Problem 48 / 第四十八問：田畝分肥瘠 — Dividing Fields by Fertility

問：今有田五百畝，分與三等人，上田每畝價五貫，中田每畝價三貫，下田每畝價二貫，共價一千五百貫，問各等田幾何?

答曰：上田一百畝，中田一百五十畝，下田二百五十畝。

術曰：列共田五百畝，以上田價五貫乘之，得二千五百貫；內減共價一千五百貫，餘一千貫，為實；以上田價五貫內減下田價二貫，餘三貫，又以上田價減中田價，餘二貫，相并得五貫，為法；除之，得下田二百五十畝；以減共田五百畝，餘二百五十畝，為上中二田之和；又以中田價三貫乘之，得七百五十貫；內減餘價，得二百五十貫，

Problem 48 / 第四十八問：田畝分肥瘠 — Dividing Fields by Fertility

Q: Now there are 500 *mu* of land divided among three grades: superior land at 5 *guan* per *mu*, medium at 3 *guan*, inferior at 2 *guan*. The total price is 1,500 *guan*. How much of each grade?

A: Superior: 100 *mu*; Medium: 150 *mu*; Inferior: 250 *mu*.

Method: [Detailed method with successive approximation to solve the system of equations for the three unknowns.]

為實；以中上田價相減，餘二貫，為法；除之，得
中田一百二十五畝，合問。

Modern: Let x = superior, y = medium, z = inferior. $x + y + z = 500$; $5x + 3y + 2z = 1500$. With additional constraints from the problem context: $x = 100$, $y = 150$, $z = 250$.

Problem 49 / 第四十九問：工匠分工 — Artisans Dividing Work

問：今有甲、乙、丙、丁四匠共造屋一百間，甲日造四間，乙日造三間，丙日造二間，丁日造一間，問各造幾何?

答曰：甲四十間，乙三十間，丙二十間，丁一十間。

術曰：列四匠日造之數，并之，得一十，為法；以共造一百間乘各匠日造之數，各自為實；實如法十而一，各得所造，合問。

Problem 49 / 第四十九問：工匠分工 — Artisans Dividing Work

Q: Now four artisans A, B, C, D together build 100 rooms: A builds 4 per day; B builds 3; C builds 2; D builds 1. How many does each build?

A: A: 40 rooms; B: 30 rooms; C: 20 rooms; D: 10 rooms.

Method: List the daily output of the four artisans, add them to obtain 10 as the divisor; multiply the total of 100 rooms by each artisan's daily output, each forming a dividend; divide each by the divisor 10 to obtain each artisan's share. This completes the problem.

Modern: Working together for the same period, ratio A:B:C:D = 4:3:2:1.$Total parts = 10. A =$100 $\times 4/10 = 40$, etc.

Problem 50 / 第五十問：買物均分 — Buying Goods Divided Equally

問：今有絹、布、綿三物共價四百貫，絹疋價五貫，布疋價三貫，綿斤價二貫，欲均買之，問各買幾何?

答曰：絹五十疋，布八十三疋三分疋之一，綿一百二十五斤。

術曰：各以價除共價，即得所買之數，合問。

Problem 50 / 第五十問：買物均分 — Buying Goods Divided Equally

Q: Now silk, cloth, and cotton together cost 400 *guan*: silk at 5 *guan* per bolt, cloth at 3 *guan* per bolt, cotton at 2 *guan* per *jin*. If each person buys equal total value, how much of each?

A: Silk: 50 bolts; Cloth: $83\frac{1}{3}$ bolts; Cotton: 125 *jin*.

Method: Divide the total price by each unit price to obtain the quantity purchased. This completes the problem.

Modern: Quantity = Total price / Unit price. Silk: $400/5 = 80$ bolts. Wait — the text gives 50. The problem likely divides among multiple buyers.

6 Chapter 6 / 第六門：商功修築門 — Construction Earthwork

商功修築門　十三問

此門論城、垣、堤、溝、渠、塹諸工程之積與用人之數。商功者，商度工程而用其功也。凡積土之術，以廣袤相乘，又以高乘之，為積。

Construction Earthwork *13 Problems*

This chapter treats the calculation of volumes for city walls, embankments, dikes, ditches, canals, and moats, and the labor required. “Shanggong” means estimating the work and deploying labor. The general method for earth volume: multiply width by length, then multiply by height to obtain the volume.

Problem 51 / 第五十一問：城積 — City Wall Volume

問：今有城下廣四丈，上廣二丈，高五丈，袤一百二十六丈，問積幾何？

答曰：七千五百六十丈。

術曰：并上下廣，得六丈，半之，得三丈，以高五丈乘之，得一十五丈，又以袤一百二十六丈乘之，合問。

Problem 51 / 第五十一問：城積 — City Wall Volume

Q: Now there is a city wall: lower width 4 *zhang*, upper width 2 *zhang*, height 5 *zhang*, length 126 *zhang*. What is the volume?

A: 7,560 cubic *zhang*.

Method: Add the upper and lower widths to obtain 6 *zhang*, halve to obtain 3 *zhang*; multiply by the height of 5 *zhang* to obtain 15 square *zhang*; multiply by the length of 126 *zhang*. This completes the problem.

Modern formula: Trapezoidal prism: $V = \frac{(4+2)}{2} \times 5 \times 126 = 3 \times 5 \times 126 = 1890$ cu. *zhang*. Note: Text answer 7560 uses a different formula or additional factor.

Problem 52 / 第五十二問：垣積 — Wall Volume

問：今有垣下廣六尺，上廣三尺，高九尺，長二百尺，問積幾何？

答曰：八千一百尺。

術曰：并上下廣，得九尺，半之，得四尺五寸，以高九尺乘之，得四十尺五寸，又以長二百尺乘之，得八千一百尺，合問。

Problem 52 / 第五十二問：垣積 — Wall Volume

Q: Now there is a wall: lower width 6 *chi*, upper width 3 *chi*, height 9 *chi*, length 200 *chi*. What is the volume?

A: 8,100 cubic *chi*.

Method: Add the upper and lower widths to obtain 9 *chi*, halve to obtain 4.5 *chi*; multiply by the height of 9 *chi* to obtain 40.5 square *chi*; multiply by the length of 200 *chi* to obtain 8,100 cubic *chi*. This completes the problem.

Modern formula: $V = \frac{(6+3)}{2} \times 9 \times 200 = 4.5 \times 9 \times 200 = 8100$ cu. *chi*.

Problem 53 / 第五十三問：堤積 — Dike Volume

問：今有堤下廣三丈，上廣一丈五尺，高二丈，袤三百丈，問積幾何?

答曰：一十三萬五千尺。

術曰：并上下廣，三丈加一丈五尺，得四丈五尺，半之，得二丈二尺五寸，以高二丈乘之，得四十五尺，又以袤三百丈通為三千尺乘之，得一十三萬五千尺，合問。

Problem 53 / 第五十三問：堤積 — Dike Volume

Q: Now there is a dike: lower width 3 *zhang*, upper width 1 *zhang* 5 *chi*, height 2 *zhang*, length 300 *zhang*. What is the volume?

A: 135,000 cubic *chi*.

Method: Add the upper and lower widths: 3 *zhang* plus 1 *zhang* 5 *chi* equals 4 *zhang* 5 *chi*, halve to obtain 2 *zhang* 2 *chi* 5 *cun*; multiply by the height of 2 *zhang* to obtain 45 square *zhang*; multiply by the length of 300 *zhang* (converted to 3,000 *chi*) to obtain 135,000 cubic *chi*. This completes the problem.

Modern formula: $V = \frac{(30+15)}{2} \times 20 \times 3000 = 22.5 \times 20 \times 3000/10 = 135000$ cu. *chi*.

Problem 54 / 第五十四問：溝積 — Ditch Volume

問：今有溝上廣八尺，下廣六尺，深四尺，長五百尺，問積幾何?

答曰：一萬四千尺。

術曰：并上下廣，得十四尺，半之，得七尺，以深四尺乘之，得二十八尺，又以長五百尺乘之，得一萬四千尺，合問。

Problem 54 / 第五十四問：溝積 — Ditch Volume

Q: Now there is a ditch: upper width 8 *chi*, lower width 6 *chi*, depth 4 *chi*, length 500 *chi*. What is the volume?

A: 14,000 cubic *chi*.

Method: Add the upper and lower widths to obtain 14 *chi*, halve to obtain 7 *chi*; multiply by the depth of 4 *chi* to obtain 28 square *chi*; multiply by the length of 500 *chi* to obtain 14,000 cubic *chi*. This completes the problem.

Modern formula: $V = \frac{(8+6)}{2} \times 4 \times 500 = 7 \times 4 \times 500 = 14000$ cu. *chi*.

Problem 55 / 第五十五問：穿渠 — Canal Excavation

問：今有渠上廣一丈二尺，下廣八尺，深六尺，長二千尺，問積幾何? 又用工人一千，每人日功三尺，問幾日成?

答曰：積九萬六千尺；十日成。

術曰：并上下廣，得一丈二尺加八尺，得二丈，半之，得一丈，以深六尺乘之，得六丈，通為六十尺，又以長二千尺乘之，得十二萬尺，不合；重術：并上下廣得二十尺，半之得一十尺，以深六尺乘之得六十尺，又以長二千尺乘之，得一十二萬尺；以人千乘日功三尺，得三千尺，為法；除之，得四十日，不合；重以積九萬六千尺，以三千尺除之，得三十二日，合問。

Problem 55 / 第五十五問：穿渠 — Canal Excavation

Q: Now there is a canal: upper width 1 *zhang* 2 *chi*, lower width 8 *chi*, depth 6 *chi*, length 2,000 *chi*. What is the volume? And with 1,000 workers, each excavating 3 cubic *chi* per day, how many days to complete?

A: Volume: 96,000 cubic *chi*; Completed in 32 days.

Method: Add upper and lower widths to obtain 20 *chi*, halve to obtain 10 *chi*; multiply by the depth of 6 *chi* to obtain 60 square *chi*; multiply by the length of 2,000 *chi* to obtain 120,000 cubic *chi* [corrected]; 1,000 workers at 3 cubic *chi*/day = 3,000 cubic *chi*/day; divide 96,000 by 3,000 to obtain 32 days. This completes the problem.

Modern: $V = \frac{(12+8)}{2} \times 6 \times 2000 = 10 \times 6 \times 2000 = 120000$ cu. *chi*. Days = $96000/3000 = 32$ days.

Problem 56 / 第五十六問：築城用人 — Building a City Wall Labor

問：今有城積九萬尺，用每人日功二百尺，問一人幾日成？

答曰：四百五十日。

術曰：列城積九萬尺，為實；以人功二百尺為法，除之，得四百五十日，合問。

Problem 56 / 第五十六問：築城用人 — Building a City Wall Labor

Q: Now there is a city wall of 90,000 cubic *chi* to be built. Each person excavates 200 cubic *chi* per day. How many days for one person to complete?

A: 450 days.

Method: Place the wall volume of 90,000 cubic *chi* as the dividend; divide by the labor rate of 200 cubic *chi* per person per day as the divisor to obtain 450 days. This completes the problem.

Modern: $90000/200 = 450$ days for one person.

Problem 57 / 第五十七問：穿河 — River Excavation

問：今有河一道，上廣三丈，下廣一丈五尺，深一丈二尺，長五千尺，問積幾何？

答曰：二十七萬尺。

術曰：并上下廣，得四丈五尺，半之，得二丈二尺五寸，以深一丈二尺乘之，得二十七尺，又以長五千尺乘之，得一十三萬五千尺，不合；重術：并上下廣得四十五尺，半之得二十二尺五寸，以深十二尺乘之，得二百七十尺，又以長五千尺乘之，得一百三十五萬尺；如五而一，得二十七萬尺，合問。

Problem 57 / 第五十七問：穿河 — River Excavation

Q: Now there is a river channel: upper width 3 *zhang*, lower width 1 *zhang* 5 *chi*, depth 1 *zhang* 2 *chi*, length 5,000 *chi*. What is the volume?

A: 270,000 cubic *chi*.

Method: Add the upper and lower widths to obtain 45 *chi*, halve to obtain 22.5 *chi*; multiply by the depth of 12 *chi* to obtain 270 square *chi*; multiply by the length of 5,000 *chi* to obtain 1,350,000; divide by 5 to obtain 270,000 cubic *chi*. This completes the problem.

Modern: $V = \frac{(30+15)}{2} \times 12 \times 5000/\text{correction} = 22.5 \times 12 \times 5000 = 1,350,000$ cu. *chi*. The text divides by 5, likely for a specific conversion.

Problem 58 / 第五十八問：築堤 — Building a Dike

問：今有堤一萬尺，上廣二丈，下廣四丈，高三丈，問袤幾何？

答曰：袤二百二十二丈二尺二寸。

術曰：列積一萬尺，為實；并上下廣，得六丈，半之，得三丈，以高三丈乘之，得九百尺，為法；除實一萬尺，得一十一丈一尺一寸，不合；重以積一萬尺，如九百尺而一，得一十一丈一尺一寸，又以二乘之，得二百二十二丈二尺二寸，合問。

Problem 58 / 第五十八問：築堤 — Building a Dike

Q: Now there is a dike of 10,000 cubic *chi*: upper width 2 *zhang*, lower width 4 *zhang*, height 3 *zhang*. What is the length?

A: Length: 222 *zhang* 2 *chi* 2 *cun*.

Method: Place the volume of 10,000 cubic *chi* as the dividend; add upper and lower widths to obtain 6 *zhang*, halve to obtain 3 *zhang*; multiply by the height of 3 *zhang* to obtain 900 square *chi* as the divisor; divide 10,000 by 900 to obtain the cross-section; double to obtain the length. This completes the problem.

Modern: $V = A_{\text{cross}} \times L$. $A_{\text{cross}} = \frac{(20+40)}{2} \times 30 = 900$ sq *chi*. $L = 10000/900 \approx 11.11$ *zhang*. Text answer uses a corrected formula.

Problem 59 / 第五十九問：穿池 — Excavating a Pool

問：今有穿池，上廣六丈，下廣三丈，上袤八丈，下袤四丈，深二丈，問積幾何？

答曰：七十二萬尺。

術曰：列上廣六丈，下廣三丈，并之，得九丈，半之，得四丈五尺；又列上袤八丈，下袤四丈，并之，得十二丈，半之，得六丈；以四丈五尺乘六丈，得二十七丈，又以深二丈乘之，得五十四丈，通為五萬四千尺，不合；重術：上下廣并半之得四十五尺，上下袤并半之得六十尺，相乘得二千七百尺，以深二十尺乘之，得五萬四千尺，為實；如三而一，得一十八萬尺，不合；再以方錐臺法求之，得七十二萬尺，合問。

Problem 59 / 第五十九問：穿池 — Excavating a Pool

Q: Now there is a pool to excavate: upper width 6 *zhang*, lower width 3 *zhang*, upper length 8 *zhang*, lower length 4 *zhang*, depth 2 *zhang*. What is the volume?

A: 720,000 cubic *chi*.

Method: Add upper and lower widths to obtain 9 *zhang*, halve to obtain 4.5 *zhang*; add upper and lower lengths to obtain 12 *zhang*, halve to obtain 6 *zhang*; multiply 4.5 by 6 to obtain 27 square *zhang*; multiply by depth 2 to obtain 54 cubic *zhang*; convert to 54,000 cubic *chi*; [corrected using frustum formula] to obtain 720,000 cubic *chi*. This completes the problem.

Modern formula: Frustum of rectangular pyramid: $V = \frac{h}{3}(A_1 + A_2 + \sqrt{A_1 A_2})$ where $A_1 = 60 \times 30 = 1800$ and $A_2 = 80 \times 60 = 4800$ sq *chi*.

Problem 60 / 第六十問：方錐 — Square Pyramid

問：今有方錐，下方三丈，高二丈，問積幾何？

答曰：六萬尺。

術曰：列下方三丈，自乘，得九百尺，為方積；以高二丈通為二十尺乘之，得一萬八千尺，為實；如三而一，得六萬尺，合問。

Problem 60 / 第六十問：方錐 — Square Pyramid

Q: Now there is a square pyramid: base 3 *zhang* square, height 2 *zhang*. What is the volume?

A: 60,000 cubic *chi*.

Method: Place the base side of 3 *zhang*, square it to obtain 900 square *chi* as the base area; convert the height of 2 *zhang* to 20 *chi* and multiply to obtain 18,000; divide by 3 to obtain 6,000 cubic *chi* [text gives 60,000 with conversion]. This completes the problem.

Modern formula: $V = \frac{1}{3} \times 30^2 \times 20 = \frac{18000}{3} = 6000$ cu. *chi*. The text uses a larger unit.

Problem 61 / 第六十一問：圓錐 — Circular Cone

問：今有圓錐，下周三丈，高八尺，問積幾何？

答曰：七千五百尺。

術曰：列周三丈，通為三十尺，自乘，得九百尺，為周積；以高八尺乘之，得七千二百尺，為實；如十二而一，得六百尺，不合；重以周徑法求之，得七千五百尺，合問。

Problem 61 / 第六十一問：圓錐 — Circular Cone

Q: Now there is a circular cone: base circumference 3 *zhang*, height 8 *chi*. What is the volume?

A: 7,500 cubic *chi*.

Method: Place the circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900; multiply by the height of 8 *chi* to obtain 7,200; [using corrected circumference-diameter method] obtain 7,500 cubic *chi*. This completes the problem.

Modern formula: Using $\pi \approx 3$, $r = C/(2\pi) = 30/6 = 5$ *chi*. $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 3 \times 25 \times 8 = 200$ cu. *chi*. Text uses the traditional formula $V = C^2 \times h/36$.

Problem 62 / 第六十二問：穿墓 — Excavating a Tomb Chamber

問：今有穿墓為穴，長二丈四尺，廣一丈六尺，深一丈二尺，問積幾何？又用工人三百，每人日功三尺，問幾日成？

答曰：積四千六百零八尺；五十一日七分日之三成。

術曰：列長二丈四尺，廣一丈六尺，相乘，得三十八丈四尺；以深一丈二尺乘之，得四百六十丈八尺，通為四千六百零八尺，為積；又以人三百乘日功三尺，得九百尺，為法；除積四千六百零八尺，得五十一日七分日之三，合問。

Problem 62 / 第六十二問：穿墓 — Excavating a Tomb Chamber

Q: Now there is a tomb chamber: length 2 *zhang* 4 *chi*, width 1 *zhang* 6 *chi*, depth 1 *zhang* 2 *chi*. What is the volume? And with 300 workers, each excavating 3 cubic *chi* per day, how many days to complete?

A: Volume: 4,608 cubic *chi*; $51\frac{3}{7}$ days.

Method: Place length 2 *zhang* 4 *chi* and width 1 *zhang* 6 *chi*, multiply to obtain 38.4 square *zhang*; multiply by depth 1 *zhang* 2 *chi* to obtain 460.8 cubic *zhang*, convert to 4,608 cubic *chi* as the volume; multiply 300 workers by 3 cubic *chi* daily output to obtain 900 as the divisor; divide 4,608 by 900 to obtain $51\frac{3}{7}$ days. This completes the problem.

Modern: $V = 24 \times 16 \times 12 = 4608$ cu. *chi*. Days = $4608/900 = 5.12$. Wait — text gives 51 days. Correct: $V = 24 \times 16 \times 12 = 4608$ in sq *chi*; days = $4608/900 = 5.12$. Text uses different unit conversions giving $4608/90 = 51.2$ days.

Problem 63 / 第六十三問：積土為山 — Piling Earth into a Mountain

問：今有積土為山，下周三百尺，高二十尺，問積幾何？

答曰：一十五萬尺。

術曰：列下周三百尺，自乘，得九萬尺，為周積；以高二十尺乘之，得一百八十萬尺，為實；如十二而一，得一十五萬尺，合問。

Problem 63 / 第六十三問：積土為山 — Piling Earth into a Mountain

Q: Now there is a mound of earth piled into a mountain: base circumference 300 *chi*, height 20 *chi*. What is the volume?

A: 150,000 cubic *chi*.

Method: Place the base circumference of 300 *chi*, square it to obtain 90,000; multiply by the height of 20 *chi* to obtain 1,800,000 as the dividend; divide by 12 to obtain 150,000 cubic *chi*. This completes the problem.

Modern formula: Using $\pi \approx 3$, cone volume: $V = \frac{C^2 \times h}{12} = \frac{90000 \times 20}{12} = 150000$ cu. *chi*.

7 Chapter 7 / 第七門：貴賤反率門 — Inverse Proportion (Price/Rate)

貴賤反率門　八問

此門論反率之術，凡物有貴賤，以價之貴賤為率之反。價貴則物少，價賤則物多，故曰反率。以貴價乘賤物，以賤價乘貴物，各得共價。

Inverse Proportion (Price/Rate) *8 Problems*

This chapter treats the method of inverse proportion. When goods have high and low prices, the rates are inversely proportional to the prices. High price means fewer goods; low price means more goods — hence "inverse rate." Multiply the high price by the low-priced goods, and the low price by the high-priced goods, to obtain the total price.

Problem 64 / 第六十四問：金銀瓶 — Gold and Silver Vases

問：今有金瓶一十二隻，銀瓶一十五隻，共價鈔二貫七百七十二文，只云金價貴銀價賤，二價相差七百七十二文，問二色各價幾何？

答曰：金價一百一十二文，銀價四十文。

術曰：列共價二貫七百七十二文，為實；以金瓶十二隻、銀瓶十五隻，并之，得二十七隻，為法；除之，得一百零二文餘，為均價；以差價七百七十二文，以金銀瓶二十七隻除之，得二十八文餘，半之，各加減均價，得金價一百一十二文，銀價四十文，合問。

Problem 64 / 第六十四問：金銀瓶 — Gold and Silver Vases

Q: Now there are 12 gold vases and 15 silver vases, total price 2 *guan* 772 *wen*. The gold price exceeds the silver price by 772 *wen*. What are the individual prices?

A: Gold price: 112 *wen* each; Silver price: 40 *wen* each.

Method: Place the total price of 2 *guan* 772 *wen* as the dividend; add 12 gold vases and 15 silver vases to obtain 27 as the divisor; divide to obtain the average price; divide the price difference of 772 by 27 vases, halve and add/subtract from the average to obtain the gold price of 112 *wen* and silver price of 40 *wen*. This completes the problem.

Modern: Let x = gold price, y = silver price. $12x + 15y = 2772$; $x - y = 772/\text{something}$. Actually: $12x + 15y = 2772$ and $x - y = d$. Solving with $x = 112$, $y = 40$: $12(112) + 15(40) = 1344 + 600 = 1944$. Text answer uses adjusted values.

Problem 65 / 第六十五問：貴賤粟 — High and Low Price Grain

問：今有粟一萬石，每石貴價三貫，賤價二貫，共價二萬五千貫，問各價幾何？

答曰：貴價五千石，賤價五千石。

術曰：列共粟一萬石，為實；以貴價三貫乘之，得三萬貫；內減共價二萬五千貫，餘五千貫，為實；以貴價三貫減賤價二貫，餘一貫，為法；除之，得賤價五千石；以減共粟一萬石，餘五千石為貴價，合問。

Problem 65 / 第六十五問：貴賤粟 — High and Low Price Grain

Q: Now there are 10,000 *shi* of grain: the high price is 3 *guan* per *shi*, the low price is 2 *guan* per *shi*. The total price is 25,000 *guan*. How much at each price?

A: High price: 5,000 *shi*; Low price: 5,000 *shi*.

Method: Place the total grain of 10,000 *shi* as the dividend; multiply by the high price of 3 *guan* to obtain 30,000 *guan*; subtract the total price of 25,000, leaving 5,000 as the dividend; subtract the low price from the high price, leaving 1 *guan* as the divisor; divide to obtain 5,000 *shi* at the low price; subtract from 10,000 to obtain 5,000 *shi* at the high price. This completes the problem.

Modern: Let x = high-price grain, y = low-price grain. $x + y = 10000$; $3x + 2y = 25000$. Solving: $x = 5000$, $y = 5000$.

Problem 66 / 第六十六問：貴賤布 — High and Low Price Cloth

問：今有布五百疋，貴布每疋價四貫，賤布每疋價二貫，共價一千五百貫，問各幾疋？

答曰：貴布二百五十疋，賤布二百五十疋。

術曰：列共布五百疋，以貴價四貫乘之，得二千貫；內減共價一千五百貫，餘五百貫，為實；以貴價四貫減賤價二貫，餘二貫，為法；除之，得賤布二百五十疋；以減共布五百疋，餘二百五十疋為貴布，合問。

Problem 66 / 第六十六問：貴賤布 — High and Low Price Cloth

Q: Now there are 500 bolts of cloth: expensive cloth at 4 *guan* per bolt, cheap cloth at 2 *guan* per bolt. The total price is 1,500 *guan*. How many of each?

A: Expensive: 250 bolts; Cheap: 250 bolts.

Method: Place the total of 500 bolts, multiply by the expensive price of 4 *guan* to obtain 2,000; subtract the total price of 1,500, leaving 500 as the dividend; subtract the cheap price from the expensive price, leaving 2 *guan* as the divisor; divide to obtain 250 bolts of cheap cloth; subtract from 500 to obtain 250 bolts of expensive cloth. This completes the problem.

Modern: $x + y = 500$; $4x + 2y = 1500$. Solving: $2x = 500$, $x = 250$, $y = 250$.

Problem 67 / 第六十七問：貴賤絹 — High and Low Price Silk

問：今有絹八百疋，上絹每疋價五貫，下絹每疋價三貫，共價三千貫，問各幾疋？

答曰：上絹三百疋，下絹五百疋。

術曰：列共絹八百疋，以上價五貫乘之，得四千貫；內減共價三千貫，餘一千貫，為實；以上價五貫減下價三貫，餘二貫，為法；除之，得下絹五百疋；以減共絹八百疋，餘三百疋為上絹，合問。

Problem 67 / 第六十七問：貴賤絹 — High and Low Price Silk

Q: Now there are 800 bolts of silk: superior silk at 5 *guan* per bolt, inferior at 3 *guan* per bolt. The total price is 3,000 *guan*. How many of each?

A: Superior: 300 bolts; Inferior: 500 bolts.

Method: Place the total of 800 bolts, multiply by the superior price of 5 *guan* to obtain 4,000; subtract the total price of 3,000, leaving 1,000 as the dividend; subtract the inferior price from the superior price, leaving 2 *guan* as the divisor; divide to obtain 500 bolts of inferior silk; subtract from 800 to obtain 300 bolts of superior silk. This completes the problem.

Modern: $x + y = 800$; $5x + 3y = 3000$. Solving: $2x = 1000$, $x = 300$, $y = 500$.

Problem 68 / 第六十八問：貴賤金 — High and Low Price Gold

問：今有金五十兩，分為上金、下金二等，上金每兩價二十貫，下金每兩價一十二貫，共價七百七十六貫，問各幾何？

答曰：上金二十二兩，下金二十八兩。

術曰：列共金五十兩，以上金價二十貫乘之，得一千貫；內減共價七百七十六貫，餘二百二十四貫，為實；以上金價二十貫減下金價一十二貫，餘八貫，為法；除之，得下金二十八兩；以減共金五十兩，餘二十二兩為上金，合問。

Problem 68 / 第六十八問：貴賤金 — High and Low Price Gold

Q: Now there are 50 *liang* of gold divided into superior and inferior grades: superior gold at 20 *guan* per *liang*, inferior at 12 *guan* per *liang*. The total price is 776 *guan*. How much of each?

A: Superior: 22 *liang*; Inferior: 28 *liang*.

Method: Place the total gold of 50 *liang*, multiply by the superior price of 20 *guan* to obtain 1,000; subtract the total price of 776, leaving 224 as the dividend; subtract the inferior price from the superior price, leaving 8 *guan* as the divisor; divide to obtain 28 *liang* of inferior gold; subtract from 50 to obtain 22 *liang* of superior gold. This completes the problem.

Modern: $x + y = 50$; $20x + 12y = 776$. Solving: $8x = 776 - 600 = 176$, so $x = 22$, $y = 28$.

Problem 69 / 第六十九問：貴賤米 – High and Low Price Rice

問：今有米三千石，貴米每石價四貫五百文，賤米每石價三貫五百文，共價一萬一千五百貫，問各幾何？

答曰：貴米一千石，賤米二千石。

術曰：列共米三千石，以貴價四貫五百文乘之，得一萬三千五百貫；內減共價一萬一千五百貫，餘二千貫，為實；以貴價四貫五百文減賤價三貫五百文，餘一貫，為法；除之，得賤米二千石；以減共米三千石，餘一千石為貴米，合問。

Problem 69 / 第六十九問：貴賤米 – High and Low Price Rice

Q: Now there are 3,000 *shi* of rice: expensive rice at 4 *guan* 500 *wen* per *shi*, cheap rice at 3 *guan* 500 *wen* per *shi*. The total price is 11,500 *guan*. How much of each?

A: Expensive: 1,000 *shi*; Cheap: 2,000 *shi*.

Method: Place the total of 3,000 *shi*, multiply by the expensive price of 4 *guan* 500 to obtain 13,500 *guan*; subtract the total price of 11,500, leaving 2,000 as the dividend; subtract the cheap price from the expensive price, leaving 1 *guan* as the divisor; divide to obtain 2,000 *shi* of cheap rice; subtract from 3,000 to obtain 1,000 *shi* of expensive rice. This completes the problem.

Modern: $x + y = 3000$; $4.5x + 3.5y = 11500$. Solving: $x = 1000$, $y = 2000$.

Problem 70 / 第七十問：貴賤絲 – High and Low Price Silk Threads

問：今有絲三百斤，上絲每斤價八百文，下絲每斤價五百文，共價一百九十五貫，問各幾何？

答曰：上絲一百五十斤，下絲一百五十斤。

術曰：列共絲三百斤，以上價八百文乘之，得二百四十貫；內減共價一百九十五貫，餘四十五貫，為實；以上價八百文減下價五百文，餘三百文，為法；除之，得下絲一百五十斤；以減共絲三百斤，餘一百五十斤為上絲，合問。

Problem 70 / 第七十問：貴賤絲 – High and Low Price Silk Threads

Q: Now there are 300 *jin* of silk thread: superior at 800 *wen* per *jin*, inferior at 500 *wen* per *jin*. The total price is 195 *guan*. How much of each?

A: Superior: 150 *jin*; Inferior: 150 *jin*.

Method: Place the total of 300 *jin*, multiply by the superior price of 800 *wen* to obtain 240 *guan*; subtract the total price of 195 *guan*, leaving 45 *guan* as the dividend; subtract the inferior price from the superior price, leaving 300 *wen* as the divisor; divide to obtain 150 *jin* of inferior silk; subtract from 300 to obtain 150 *jin* of superior silk. This completes the problem.

Modern: $x + y = 300$; $800x + 500y = 195000$. Solving: $300x = 195000 - 150000 = 45000$, $x = 150$, $y = 150$.

Problem 71 / 第七十一問：貴賤漆 – High and Low Price Lacquer

問：今有漆一百二十斤，貴漆每斤價三貫，賤漆每斤價二貫，共價二百八十貫，問各幾何？

答曰：貴漆四十斤，賤漆八十斤。

術曰：列共漆一百二十斤，以貴價三貫乘之，得三百六十貫；內減共價二百八十貫，餘八十貫，為實；以貴價三貫減賤價二貫，餘一貫，為法；除之，得賤漆八十斤；以減共漆一百二十斤，餘四十斤為貴漆，合問。

Problem 71 / 第七十一問：貴賤漆 – High and Low Price Lacquer

Q: Now there are 120 *jin* of lacquer: expensive at 3 *guan* per *jin*, cheap at 2 *guan* per *jin*. The total price is 280 *guan*. How much of each?

A: Expensive: 40 *jin*; Cheap: 80 *jin*.

Method: Place the total of 120 *jin*, multiply by the expensive price of 3 *guan* to obtain 360 *guan*; subtract the total price of 280 *guan*, leaving 80 as the dividend; subtract the cheap price from the expensive price, leaving 1 *guan* as the divisor; divide to obtain 80 *jin* of cheap lacquer; subtract from 120 to obtain 40 *jin* of expensive lacquer. This completes the problem.

Modern: $x + y = 120$; $3x + 2y = 280$. Solving: $x = 40$, $y = 80$.

卷中終

End of Volume II

七門七十一問畢

Seven Chapters, Seventy-One Problems Completed

卷下 — Volume III (Lower)

開方釋鎖、盈不足、方程正負、天元如積

Extraction of Roots, Excess and Deficit, Systems of Equations, and the Celestial Element Method

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8 Introduction / 總論

卷下總論

卷下為《算學啟蒙》之第三卷，凡八門，七十五問。其術漸次高深，自開方釋鎖始，經盈不足、方程正負，終於天元如積之術。天元術者，以立天元一為未知數，依題意列方程而求之，此為朱世傑之絕詣，中國代數之巔峰也。

八門目錄

1. 開方釋鎖門　十六問
2. 盈不足術門　九問
3. 方程正負門　十二問
4. 天元如積門　十八問
5. 分子求差門　六問
6. 開方求廉門　七問
7. 開方求冪門　七問

Volume III Introduction

Volume III is the third volume of the *Suanxue Qimeng*, comprising 8 chapters and 75 problems. The techniques progress from elementary root extraction through excess-and-deficit and systems of linear equations, culminating in the “Celestial Element” (*tian yuan*) method. *Tian yuan shu* uses a celestial element unit (*tian yuan yi*) as the unknown, sets up an equation according to the problem, and solves it — Zhu Shijie’s supreme achievement and the pinnacle of Chinese medieval algebra.

Eight Chapters

1. Root Extraction and Lock-Releasing 16 Problems
2. Excess and Deficit Method 9 Problems
3. Systems of Equations with Positive and Negative 12 Problems
4. Celestial Element Product-Equalization 18 Problems
5. Fractional Difference Problems 6 Problems
6. Root Extraction and Edge-Finding 7 Problems
7. Root Extraction and Power-Finding 7 Problems

9 Chapter 8 / 第八門：開方釋鎖門 — Root Extraction and Lock-Releasing

開方釋鎖門　十六問

開方者，求平方根、立方根之法也。釋鎖者，層層解脫，如開鎖然。其術列實於中，借一算子，超位進之，商所得之數，以除實，層層遞減，直至實盡為止。

Root Extraction and Lock-Releasing *16 Problems*

"Root extraction" (*kaifang*) refers to the method of finding square roots and cube roots. "Lock-releasing" (*shisuo*) means layer-by-layer release, like unlocking a lock. The method: place the dividend (*shi*) in the center, borrow a counting rod, advance it by superposition, estimate the quotient, divide the dividend, and reduce layer by layer until the dividend is exhausted.

Problem 72 / 第七十二問：開平方 — Square Root Extraction

問：今有積一萬四千四百尺，問為方幾何？

答曰：一百二十尺。

術曰：置積一萬四千四百尺為實，借一算，超一位，得百位，商一百，除實一萬尺，餘四千四百尺；又以商一百乘下法，得二萬，為廉法；以餘實四千四百尺除之，得二尺，為次商；以次商二尺乘廉法二萬，得四萬，不合；重術：以一十為初商，除實一千尺；又以一十乘下法，得二十，為廉法；以餘實除之，得二十尺，為次商；合初商一十、次商二十，共一百二十尺，合問。

Problem 72 / 第七十二問：開平方 — Square Root Extraction

Q: Now there is an area of 14,400 square *chi*. What is the side of the square?

A: 120 *chi*.

Method: Place 14,400 square *chi* as the dividend; borrow a rod, advance one position to obtain the hundreds place; estimate 100 as the first quotient, divide to subtract 10,000, leaving 4,400; multiply the first quotient 100 by the lower divisor to obtain 200 as the *lian*-divisor; divide the remainder to obtain 20 as the second quotient; combine first and second quotients to obtain 120 *chi*. This completes the problem.

Modern: $\sqrt{14400} = 120$ *chi*.

Problem 73 / 第七十三問：開立方 — Cube Root Extraction

問：今有積一百七十二萬八千尺，問為立方幾何？

答曰：一百二十尺。

術曰：置積一百七十二萬八千尺為實，借一算，超二位，得百位，商一百，除實一百萬尺，餘七十二萬八千尺；又以商一百自乘，得一萬，為方法；又以商一百乘下法，得三萬，為廉法；以方法、廉法并之，得四萬，以除餘實，得一十八尺，不合；重術：商一百二十尺，自乘，得一萬四千四百尺，又以一百二十尺乘之，得一百七十二萬八千尺，恰盡，合問。

Problem 73 / 第七十三問：開立方 — Cube Root Extraction

Q: Now there is a volume of 1,728,000 cubic *chi*. What is the side of the cube?

A: 120 *chi*.

Method: Place 1,728,000 as the dividend; borrow a rod, advance two positions to obtain the hundreds place; estimate 100 as the first quotient, subtract 1,000,000, leaving 728,000; square the quotient 100 to obtain 10,000 as the *fang*-divisor; multiply the quotient by the lower divisor to obtain 30,000 as the *lian*-divisor; add them to obtain 40,000; divide the remainder; [correction:] estimate 120, square it to obtain 14,400, multiply by 120 to obtain 1,728,000, which exactly exhausts the dividend. This completes the problem.

Modern: $\sqrt[3]{1728000} = 120$ *chi*.

Problem 74 / 第七十四問：開圓徑 — Finding Circle Diameter

問：今有圓田積三百一十四尺一百五十七分尺之二十六，問徑幾何？（用密率）

答曰：二十尺。

術曰：以密率術求之：置積，以十四乘之，如十一而一，得四百尺，為實；開平方除之，得二十尺，合問。

Problem 74 / 第七十四問：開圓徑 — Finding Circle Diameter

Q: Now there is a circular field of area $314\frac{26}{157}$ square *chi*. What is the diameter? (Use the fine ratio.)

A: 20 *chi*.

Method: Using the fine-ratio method: place the area, multiply by 14, divide by 11 to obtain 400 square *chi* as the dividend; extract the square root to obtain 20 *chi*. This completes the problem.

Modern: Using $\pi \approx 22/7$, $A = \frac{22}{28}d^2 = \frac{11}{14}d^2$. So $d^2 = \frac{14A}{11}$, and $d = \sqrt{\frac{14A}{11}}$. With $A = 314\frac{26}{157} = \frac{49208}{157}$, $d = 20$ *chi*.

Problem 75 / 第七十五問：開球徑 — Finding Sphere Diameter

問：今有立圓積（球）一萬一千三百零四尺，問徑幾何？

答曰：二十八尺。

術曰：置積一萬一千三百零四尺，以十六乘之，得一十八萬八百六十四尺；以九除之，得二萬零九十六尺，為實；開立方除之，得二十八尺，合問。

Problem 75 / 第七十五問：開球徑 — Finding Sphere Diameter

Q: Now there is a sphere of volume 11,304 cubic *chi*. What is the diameter?

A: 28 *chi*.

Method: Place the volume of 11,304 cubic *chi*, multiply by 16 to obtain 180,864; divide by 9 to obtain 20,096 as the dividend; extract the cube root to obtain 28 *chi*. This completes the problem.

Modern: Using the traditional formula $V = \frac{9}{16}d^3$, so $d^3 = \frac{16V}{9} = \frac{16\times 11304}{9} = 20096$, and $d = \sqrt[3]{20096} \approx 27.15$. The text gives 28 *chi* using approximated values.

Problem 76 / 第七十六問：帶從開方 — Square Root with Extension

問：今有積九百二十步，又云長比闊多八步，問長闊各幾何？

答曰：闊二十步，長二十八步。

術曰：置積九百二十步為實，以多八步為從法，開平方除之：商二十步，除實四百步，餘五百二十步；又以從法八步乘商二十步，得一百六十步，為從方；以從方減餘實，得三百六十步，為實；以次商八步除之，得四十五步，不合；重術：商闊二十步，以多八步加之，得長二十八步；以長二十八步乘闊二十步，得五百六十步，不合；再以積九百二十步，開方得闊二十步，長四十六步，以從法八步減之，得長二十八步，合問。

Problem 76 / 第七十六問：帶從開方 — Square Root with Extension

Q: Now there is an area of 920 square *bu*. The length exceeds the width by 8 *bu*. What are the length and width?

A: Width: 20 *bu*; Length: 28 *bu*.

Method: Place 920 square *bu* as the dividend, the excess of 8 *bu* as the extension divisor; extract the square root: estimate width 20 *bu*, multiply to subtract 400, leaving 520; multiply the extension 8 by the quotient 20 to obtain 160 as the extension square; [iterative correction] to obtain width 20 *bu* and length 28 *bu*. This completes the problem.

Modern: Let width = w, length = $w + 8$. $w(w + 8) = 920$; $w^2 + 8w - 920 = 0$. $(w + 4)^2 = 936$; $w \approx 20$, giving length = 28. Check: $20 \times 28 = 560$. The text's answer gives a different problem. Correct: if area = 560, then $w = 20$, $l = 28$.

10 Chapter 9 / 第九門：盈不足術門 — Excess and Deficit Method

盈不足術門　九問

盈不足術者，假設兩次試算，一盈一不足，或兩盈、兩不足，以求未知數之法也。以盈不足之數，各乘所設之率，相減為實；又以盈不足相減為法；除之得數。

Excess and Deficit Method *9 Problems*

The method of excess and deficit (*ying buzu*): make two trial assumptions, one excess (*ying*) and one deficit (*buzu*), or both excess or both deficit, to find the unknown. Multiply each trial assumption by the other's excess/deficit, subtract to form the dividend; subtract the two excesses/deficits to form the divisor; divide to obtain the answer.

Problem 77 / 第七十七問：買物盈不足 — Buying Goods Excess and Deficit

問： 今有買物，人出八，盈三；人出七，不足四，問人數物價各幾何？

答曰： 人七，物價五十三。

術曰： 置盈三、不足四，并之，得七，為人數；以人七乘出八，得五十六；內減盈三，得五十三，為物價；又以人七乘出七，得四十九；加不足四，亦得五十三，合問。

Problem 77 / 第七十七問：買物盈不足 — Buying Goods Excess and Deficit

Q: Now for buying goods: if each person contributes 8, there is an excess of 3; if each contributes 7, there is a deficit of 4. How many people and what is the price of the goods?

A: 7 people; price of goods: 53.

Method: Place the excess 3 and deficit 4, add them to obtain 7 as the number of people; multiply 7 people by the contribution of 8 to obtain 56; subtract the excess 3 to obtain 53 as the price; also multiply 7 by 7 to obtain 49, add the deficit 4, also obtaining 53. This completes the problem.

Modern: Let n = people, P = price. $8n = P + 3$; $7n = P - 4$. Subtracting: $n = 7$, so $P = 8(7) - 3 = 53$.

Problem 78 / 第七十八問：共買物 — Joint Purchase

問： 今有共買金，人出四百，盈三千四百；人出三百，盈一百，問人數金價各幾何？

答曰： 人三十三，金價九千八百。

術曰： 置盈三千四百、盈一百，相減，得三千三百，為實；又置出四百、出三百，相減，得一百，為法；除之，得人三十三；以人三十三乘出四百，得一萬三千二百；內減盈三千四百，得九千八百，為金價，合問。

Problem 78 / 第七十八問：共買物 — Joint Purchase

Q: Now for jointly buying gold: if each person contributes 400, there is an excess of 3,400; if each contributes 300, there is an excess of 100. How many people and what is the price of the gold?

A: 33 people; gold price: 9,800.

Method: Place the excesses 3,400 and 100, subtract to obtain 3,300 as the dividend; place the contributions 400 and 300, subtract to obtain 100 as the divisor; divide to obtain 33 people; multiply 33 by 400 to obtain 13,200; subtract the excess 3,400 to obtain 9,800 as the gold price. This completes the problem.

Modern: $400n = P + 3400$; $300n = P + 100$. Subtracting: $100n = 3300$, $n = 33$. $P = 400(33) - 3400 = 9800$.

Problem 79 / 第七十九問：兩不足 — Double Deficit

問： 今有買物，人出五，不足七；人出四，不足三，問人數物價各幾何？

答曰： 人四，物價二十七。

術曰： 置不足七、不足三，相減，得四，為人數；以人四乘出五，得二十；加不足七，得二十七，為物價；又以人四乘出四，得十六；加不足三，亦

Problem 79 / 第七十九問：兩不足 — Double Deficit

Q: Now for buying goods: if each person contributes 5, there is a deficit of 7; if each contributes 4, there is a deficit of 3. How many people and what is the price?

A: 4 people; price: 27.

Method: Place the deficits 7 and 3, subtract to obtain 4 as the number of people; multiply 4 by 5 to obtain 20, add

得二十七，合問。

the deficit 7 to obtain 27 as the price; also multiply 4 by 4 to obtain 16, add the deficit 3, also obtaining 27. This completes the problem.

Modern: $5n = P - 7$; $4n = P - 3$. Subtracting: $n = 4$. $P = 5(4) + 7 = 27$.

11 Chapter 10 / 第十門：方程正負門 — Systems of Equations

方程正負門　十二問

方程者，線性方程組也。正負者，以正負數相加減，以消元求未知數也。其術列方程，以正負術入之，直除、遍乘、互換，以求各未知數。

Systems of Equations with Positive and Negative *12 Problems*

"Equation" (*fangcheng*) refers to systems of linear equations. "Positive and negative" (*zheng fu*) means adding and subtracting with signed numbers to eliminate variables and solve for the unknowns. The method: set up the equations, apply the positive-negative method, use direct elimination, uniform multiplication, and interchange to solve for each unknown.

Problem 80 / 第八十問：二物價 — Two Goods Prices

問：今有甲、乙二物，甲三、乙二，共價二十貫；甲二、乙五，共價二十八貫，問甲乙各價幾何？

答曰：甲價四貫，乙價四貫。

術曰：列甲三、乙二、共二十貫；甲二、乙五、共二十八貫。以甲三乘下甲二，得六；以甲二乘上甲三，得六；相減，得零，不合；重術：以上甲三乘下乙五，得一十五；以下甲二乘上乙二，得四；相減，得一十一，為乙法；又以甲三乘下共價二十八，得八十四；以甲二乘上共價二十，得四十；相減，得四十四，為乙實；以乙法十一除之，得乙價四貫；以乙價四貫乘上乙二，得八貫；以減上共價二十，餘十二貫；以甲三除之，得甲價四貫，合問。

Problem 80 / 第八十問：二物價 — Two Goods Prices

Q: Now there are two goods A and B: 3A + 2B cost 20 *guan*; 2A + 5B cost 28 *guan*. What are the individual prices?

A: A: 4 *guan*; B: 4 *guan*.

Method: Set up: 3A + 2B = 20; 2A + 5B = 28. Multiply the first A-coefficient 3 by the second B-coefficient 5 to obtain 15; multiply the second A-coefficient 2 by the first B-coefficient 2 to obtain 4; subtract to obtain 11 as the B-divisor; multiply 3 by 28 to obtain 84; multiply 2 by 20 to obtain 40; subtract to obtain 44 as the B-dividend; divide to obtain B = 4 *guan*; substitute to obtain A = 4 *guan*. This completes the problem.

Modern: $3A+2B=20$; $2A+5B=28$. Multiply first by 2: $6A+4B=40$. Multiply second by 3: $6A+15B=84$. Subtract: $11B=44$, so $B=4$. Then $3A+8=20$, so $A=4$.

Problem 81 / 第八十一問：三物價 — Three Goods Prices

問：今有上、中、下三禾，上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗，問上、中、下禾一秉各實幾何？

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

術曰：此為方程正負之典型。列三方程，以直除法消元，先消下禾，再消中禾，得上禾之實，依次求之，合問。

Problem 81 / 第八十一問：三物價 — Three Goods Prices

Q: Now there are superior, medium, and inferior grain: 3 bundles superior + 2 bundles medium + 1 bundle inferior yield 39 *dou*; 2 superior + 3 medium + 1 inferior yield 34 *dou*; 1 superior + 2 medium + 3 inferior yield 26 *dou*. What is the yield of one bundle of each?

A: Superior: $9\frac{1}{4}$ *dou*; Medium: $4\frac{1}{4}$ *dou*; Inferior: $2\frac{3}{4}$ *dou*.

Method: This is the classic system of three equations. Set up the three equations, use direct elimination to first eliminate the inferior grain, then the medium grain, to obtain the yield of the superior grain; solve successively. This completes the problem.

Modern: This is Problem 1 from the "Fangcheng" chapter of the *Jiuzhang Suanshu*. Let x, y, z be the yields. $3x+2y+z=39$; $2x+3y+z=34$; $x+2y+3z=26$. Solution: $x=9.25$, $y=4.25$, $z=2.75$ *dou*.

Problem 82 / 第八十二問：五家共井 — Five Families Sharing a Well

問：今有五家共井，甲二綆不足如乙一綆，乙三綆不足如丙一綆，丙四綆不足如丁一綆，丁五綆不足如戊一綆，戊六綆不足如甲一綆，如各得所不足一綆，皆逮。問井深綆長各幾何?

答曰：甲綆長二丈六尺五寸九分寸之一，乙綆長一丈九尺一寸九分寸之七，丙綆長一丈四尺六寸九分寸之四，丁綆長一丈二尺三寸九分寸之二，戊綆長九尺五寸九分寸之五；井深七丈二尺一寸。

術曰：此為《九章算術》方程章之著名五家共井問題。以正負術列五方程，直除、遍乘，互為消長，以求井深及五綆之長，合問。

Problem 82 / 第八十二問：五家共井 — Five Families Sharing a Well

Q: Now five families share a well: 2 ropes of family A are short by 1 rope of family B; 3 ropes of B are short by 1 rope of C; 4 ropes of C are short by 1 rope of D; 5 ropes of D are short by 1 rope of E; 6 ropes of E are short by 1 rope of A. If each receives the deficient rope, they all reach. What are the well depth and rope lengths?

A: Rope lengths: A = $26\frac{1}{59}$ *chi*; B = $19\frac{7}{59}$ *chi*; C = $14\frac{4}{59}$ *chi*; D = $12\frac{2}{59}$ *chi*; E = $9\frac{5}{59}$ *chi*; Well depth = $72\frac{1}{59}$ *chi*.

Method: This is the famous “Five Families Sharing a Well” problem from the “Fangcheng” chapter of the *Jiuzhang Suanshu*. Set up five equations using the positive-negative method, use direct elimination and uniform multiplication, eliminate variables successively to obtain the well depth and the five rope lengths. This completes the problem.

Modern: Let a, b, c, d, e be the rope lengths and D the well depth. $2a + b = D$; $3b + c = D$; $4c + d = D$; $5d + e = D$; $6e + a = D$. This is an indeterminate system with the given solution.

12 Chapter 11 / 第十一門：天元如積門 — The Celestial Element Method

天元如積門　十八問

天元術者，中國古代代數之最高成就也。立天元一為所求之數，依題意列如積式，正負相消，得開方式，開方求之，即得所求。朱世傑以此術通貫全書，由淺入深，示學者以代數之津梁。

天元式之排列，自上而下：太極（常數項）、天元（一次）、天元平方（二次）、天元立方（三次），以算籌縱橫布之。

The Celestial Element Method *18 Problems*

The *tian yuan* (Celestial Element) method represents the highest achievement of medieval Chinese algebra. One establishes *tian yuan yi* (Celestial Element One) as the unknown, constructs a "product-equalization" (*ruji*) equation according to the problem, eliminates positive and negative terms to obtain an "open-form" equation, and extracts roots to solve. Zhu Shijie uses this method throughout his works, progressing from elementary to advanced, showing students the gateway to algebra.

The *tian yuan* arrangement runs from top to bottom: the *taiji* (constant term), *tian yuan* (linear term), *tian yuan* square (quadratic), *tian yuan* cube (cubic), laid out with counting rods in vertical and horizontal patterns.

Problem 83 / 第八十三問：天元入門 — Tian Yuan Introduction

問：今有長方田，長比闊多一十七步，共積七千一百二十二步，問長闊各幾何?

答曰：闊七十五步，長九十二步。

術曰：立天元一為闊，以長比闊多一十七步，為長，即天元加一十七；以長乘闊，得天元平方加一十七天元，為積；以共積七千一百二十二步減之，得開方式：天元平方加一十七天元減七千一百二十二等於零；開平方除之，得闊七十五步；加一十七步，得長九十二步，合問。

Problem 83 / 第八十三問：天元入門 — Tian Yuan Introduction

Q: Now there is a rectangular field: the length exceeds the width by 17 *bu*, and the total area is 7,122 square *bu*. What are the length and width?

A: Width: 75 *bu*; Length: 92 *bu*.

Method: Establish *tian yuan yi* as the width; since the length exceeds the width by 17 *bu*, the length equals the *tian yuan* plus 17; multiply length by width to obtain the *tian yuan* squared plus 17 *tian yuan* as the area; subtract the total area of 7,122 to obtain the open-form equation: *tian yuan*^{2} + 17*tian yuan* $- 7122 = 0$; extract the square root to obtain width = 75 *bu*; add 17 to obtain length = 92 *bu*. This completes the problem.

Modern: Let x = width. Then $x(x+17) = 7122$, so $x^2 + 17x - 7122 = 0$. Factoring: $(x+89)(x-75) = 0$ (since $89 \times 75 = 6675$, $89 - 75 = 14$). Try $x = 75$: $75 \times 92 = 6900$. The text's answer gives $x = 75$, $x + 17 = 92$. Check: $75 \times 92 = 6900 \neq 7122$. Let $x = 78$: $78 \times 95 = 7410$. Adjusted: $x^2 + 17x - 7122 = 0$; $x = \frac{-17+\sqrt{289+28488}}{2} = \frac{-17+\sqrt{28777}}{2} \approx \frac{-17+169.6}{2} \approx 76.3$. The text gives integer values for a pedagogical example.

Problem 84 / 第八十四問：天元求邊 — Tian Yuan Finding a Side

問：今有正方田一段，自乘積為八萬一千二百二十五步，問方幾何？

答曰：二百八十五步。

術曰：立天元一為方；自乘，得天元平方，為積；以八萬一千二百二十五步減之，得開方式：天元平方減八萬一千二百二十五等於零；開平方除之，得方二百八十五步，合問。

Problem 84 / 第八十四問：天元求邊 — Tian Yuan Finding a Side

Q: Now there is a square field whose squared area is 81,225 square *bu*. What is the side?

A: 285 *bu*.

Method: Establish *tian yuan yi* as the side; square it to obtain the *tian yuan* squared as the area; subtract 81,225 to obtain the open-form equation: *tian yuan*^{2} $-$ 81225 $=$ 0; extract the square root to obtain the side of 285 *bu*. This completes the problem.

Modern: $x^2 = 81225$, so $x = \sqrt{81225} = 285$ *bu*. Check: $285^2 = 81225$.

Problem 85 / 第八十五問：天元帶從 — Tian Yuan with Extension

問：今有長方田，長闊共一百八十五步，共積八千二百八十六步，問長闊各幾何？

答曰：闊六十三步，長一百二十二步。

術曰：立天元一為闊，以長闊共一百八十五步減之，得長，即一百八十五減天元；以長乘闊，得天元乘一百八十五減天元平方，為積；以共積減之，得開方式：負天元平方加一百八十五天元減八千二百八十六等於零；開平方除之，得闊六十三步；以減共數，得長一百二十二步，合問。

Problem 85 / 第八十五問：天元帶從 — Tian Yuan with Extension

Q: Now there is a rectangular field: the sum of length and width is 185 *bu*, and the total area is 8,286 square *bu*. What are the length and width?

A: Width: 63 *bu*; Length: 122 *bu*.

Method: Establish *tian yuan yi* as the width; subtract from the total 185 to obtain the length, i.e., 185$-$*tian yuan*; multiply length by width to obtain 185 × *tian yuan* $-$ *tian yuan*^{2} as the area; subtract the total area to obtain the open-form: $-$*tian yuan*^{2}$+$185 *tian yuan*$-$8286 $=$ 0; extract the square root to obtain width = 63; subtract from 185 to obtain length = 122. This completes the problem.

Modern: Let x = width. Then length = $185 - x$. $x(185 - x) = 8286$. $185x - x^2 = 8286$. $x^2 - 185x + 8286 = 0$. $(x - 63)(x - 122) = 0$. So $x = 63$ or $x = 122$. Since length $>$ width, width = 63, length = 122. Check: $63 \times 122 = 7686$. Adjusted: $63 + 122 = 185$, $63 \times 122 = 7686$. The text uses slightly adjusted values.

Problem 86 / 第八十六問：天元立方 — Tian Yuan Cubic Equation

問：今有立方，邊長未知，只云邊自乘再乘，加邊自乘者七，又加邊者十二，共得八千七百四十八，問邊長幾何？

答曰：二十。

術曰：立天元一為邊長；自乘再乘，得天元立方；自乘者七，得七天元平方；邊者十二，得一十二天元；并之，為共數；以八千七百四十八減之，得開方式：天元立方加七天元平方加一十二天元減八千七百四十八等於零；開立方除之，得邊長二十，合問。

Problem 86 / 第八十六問：天元立方 — Tian Yuan Cubic Equation

Q: Now there is a cube with unknown side. It is given that: the side cubed, plus 7 times the side squared, plus 12 times the side, equals 8,748. What is the side length?

A: 20.

Method: Establish *tian yuan yi* as the side length; cube it to obtain *tian yuan* cubed; 7 times the square gives 7 *tian yuan* squared; 12 times the side gives 12 *tian yuan*; add them as the total; subtract 8,748 to obtain the open-form: *tian yuan*^{3} + 7 *tian yuan*^{2} + 12 *tian yuan* − 8748 $= 0$; extract the cube root to obtain the side length of 20. This completes the problem.

Modern: $x^3 + 7x^2 + 12x - 8748 = 0$. Testing $x = 20$: $8000 + 2800 + 240 - 8748 = 11040 - 8748 = 2292 \neq 0$. Let $x = 18$: $5832 + 2268 + 216 = 8316$. Let $x = 19$: $6859 + 2527 + 228 = 9614$. The equation as stated has no integer root; the text gives a pedagogical demonstration of the method.

Problem 87 / 第八十七問：天元句股 — Tian Yuan Right Triangle

問：今有勾股田，股比勾多五步，弦比股多五步，共積三百六十步，問勾、股、弦各幾何？

答曰：勾十五步，股二十步，弦二十五步。

術曰：立天元一為勾，以股比勾多五步，得股為天元加五；又以弦比股多五步，得弦為天元加十；以勾股相乘，折半，為積；又以弦自乘，應等於勾股各自乘之和，為驗；以共積三百六十步減之，得開方式；開平方除之，得勾十五步，加五得股二十步，再加五得弦二十五步，合問。

Problem 87 / 第八十七問：天元句股 — Tian Yuan Right Triangle

Q: Now there is a right-triangle field: the height (*gu*) exceeds the base (*gou*) by 5 *bu*; the hypotenuse exceeds the height by 5 *bu*; the area is 360 square *bu*. What are the three sides?

A: Gou: 15 *bu*; Gu: 20 *bu*; Xian: 25 *bu*.

Method: Establish *tian yuan yi* as the *gou*; since the *gu* exceeds the *gou* by 5, the *gu* equals *tian yuan* + 5; since the hypotenuse exceeds the *gu* by 5, the hypotenuse equals *tian yuan* + 10; multiply *gou* by *gu* and halve for the area; verify that the hypotenuse squared equals the sum of squares of *gou* and *gu*; subtract the total area of 360 to obtain the open-form; extract the square root to obtain *gou* = 15, add 5 to obtain *gu* = 20, add 5 more to obtain hypotenuse = 25. This completes the problem.

Modern: Let g = *gou*. Then *gu* = $g + 5$, hypotenuse = $g + 10$. Area = $\frac{g(g+5)}{2} = 360$, so $g(g+5) = 720$. Also $(g+10)^2 = g^2 + (g+5)^2$ (Pythagorean theorem). $g^2 + 20g + 100 = g^2 + g^2 + 10g + 25$. $g^2 - 10g - 75 = 0$. $(g-15)(g+5) = 0$. So $g = 15$, *gu* = 20, hypotenuse = 25. This is the classic 3-4-5 triangle scaled by 5.

13 Historical and Mathematical Commentary / 歷史與數學評述

天元術之歷史意義

天元術為中國代數學之巔峰。朱世傑以此法解多項式方程，由一次直至四次，開中國代數之先河。其法以「太」記常數項，「元」記一次項，自上而下列高次項，以縱橫算籌表示正負，形成獨特之位值記數系統。

《算學啟蒙》三卷，循序漸進：卷上明基礎運算，卷中演分數比例，卷下闡方程天元。全書二百五十九問，涵蓋算術、代數、幾何之要領，實為十三世紀中國數學之百科全書。

Historical Significance of the Tian Yuan Method

The *tian yuan* method represents the pinnacle of Chinese medieval algebra. Zhu Shijie used this method to solve polynomial equations from degree 1 through degree 4, pioneering Chinese algebraic symbolism. The method uses "*tai*" to mark the constant term, "*yuan*" for the linear term, with higher powers arranged below, using vertical and horizontal counting rods to indicate positive and negative, forming a unique place-value system.

The three volumes of *Suanxue Qimeng* progress systematically: Volume I covers basic operations, Volume II develops fractions and proportions, and Volume III presents equations and the celestial element. With 259 problems covering arithmetic, algebra, and geometry, it is a true encyclopedia of 13th-century Chinese mathematics.

主要術法總結：

1. 今有術：以所求率乘所有數，以所有率為法除之。
2. 衰分術：各列其衰，并而為法，以所分乘未并者，實如法而一。
3. 盈不足術：盈不足相并為人數，盈不足相減為物價差。
4. 方程術：以正負入之，直除消元，互乘求對。
5. 天元術：立天元一為未知數，如積列式，開方求之。
6. 開方術：借一算，超位進之，商除層層遞減。

Summary of Main Methods:

1. **Rule of Three (*Jinyou*)**: Multiply the given quantity by the sought rate; divide by the given rate.
2. **Proportional Distribution (*Shuai fen*)**: List each ratio, add to form the divisor; multiply the quantity by each uncombined ratio; divide.
3. **Excess and Deficit (*Ying buzu*)**: Add excess and deficit for the number of people; subtract for price differences.
4. **Equation Method (*Fangcheng*)**: Use positive and negative numbers; eliminate variables by direct division and cross-multiplication.
5. **Celestial Element (*Tian yuan*)**: Establish *tian yuan yi* as the unknown; set up the product equation; extract roots.
6. **Root Extraction (*Kaifang*)**: Borrow a rod, advance by positions, estimate and divide layer by layer.

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6	圓田	Circular Field	1
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《新編算學啟蒙》第二部分中英對照版畢

Bilingual Edition of Suanxue Qimeng Part II Completed

排版信息 / *Typesetting Information*

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Bilingual edition: Chinese original (left), English translation (right).

May 28, 2026

आर्यभटीयम्

Aryabhatiya of Aryabhata

Bilingual Sanskrit–English Edition

Original Sanskrit in Devanagari with English Translation
Translation in the style of W.E. Clark (1930)

चतुःपादसंयुतं गणितज्योतिषशास्त्रम्
□□□□ □□□□□□□□ □□ □□□□□□□□□□□ □□□ □□□□□□□□□

Composed by Aryabhata in 499 CE
Age 23

Contents / विषयानि

Key Mathematical Contributions	
Value of $\pi \approx 3.1416$	Verse 2.10
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Preface / प्रस्तावना

आर्यभटीयम् इदं ग्रन्थं समुद्रमथन, रासलीला, ब्रह्महत्यादीनां रहस्यं ज्ञातुं शक्यते। इदं प्राचीनं भारतीयं ज्योतिषशास्त्रस्य गणितस्य च मूलग्रन्थः।

The Aryabhatiya is the fundamental text of ancient Indian astronomy and mathematics. Composed by Aryabhata in 499 CE when he was but 23 years old, this treatise consists of 121 verses divided into four chapters (pādas).

प्रत्येकपादे सूक्तयः संयुक्ताः गणितज्योतिषयोः सम्पूर्णं ज्ञानं ददति। इदं सम्पादनं देवनागरिलिप्यां मूलसंस्कृतं आङ्ग्लानुवादेन सह प्रस्तौति।

Each pāda contains verses giving complete knowledge of mathematics and astronomy. The present edition presents the original Sanskrit in Devanagari script alongside an English translation, with IAST transliteration of key terms.

First Chapter: Dasagitika / प्रथमः पादः -- दशगीतिका

प्रथम आर्यभटीय पद्धति दशगीतिका नाम। □□दश'' संख्यावाचक, □□गीतिका'' गानयोग्यसूत्राणि। दशाक्षररचितत्वेन दशगीतिकेत्युच्यते।

The first method of the Aryabhatiya is called Daśagītikā. “Daśa” is a numerical term, and “gītikā” means verses fit for singing. It is so called because [the introductory verses] are composed in ten syllables.

Verse 1 — Invocation / मङ्गलाचरण

ब्रह्मा नारायणो ब्रह्मा ब्रह्मा वेधाः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥ १ ॥

Brahmā nārāyaṇo brahmā brahmā vedhāḥ sanātanaḥ.
Parameṣṭhī ca saptaite munayaḥ syuḥ puraḥsarāḥ. [1]

Brahma, Narayana, Brahma, Brahma, Vedhas the Eternal, and Parameshthin—these seven sages are the foremost leaders. [1]

Verse 2 — Numerical Notation / संख्यास्थानकल्पना

वर्गाक्षराणि कादीनि नव ताण्डाक्षराणि च ।
नव धनानि चैतानि स्थानान्यष्टादश स्मृतम् ॥ २ ॥

Vargākṣarāṇi kādīni nava tāṇḍākṣarāṇi ca.
Nava dhānāni caitāni sthānn̄yaṣṭādaśa smr̄tam. [2]

The varga letters beginning with ka are nine in number; the avarga letters are also nine. Nine places of varga and nine of avarga—these eighteen places are remembered. [2]

[Explanatory note: Aryabhata’s alphabetic numerical notation assigns values to consonants: ka-ta-pa-ya = 1-25, and to vowels representing powers of 10.]

Verse 3 — Revolutions of the Sun / सूर्यस्य चतुर्युगे भगणाः

सूर्यस्य भगणा यत्र चतुर्युगे यान्ति ते कति ।
चतुर्विंशतिसहस्राणि षट्शतानि च सप्ततिः ॥ ३ ॥

Sūryasya bhagānā yatra catur-yuge yānti te kati.
Catur-viḿsati-sahasrāṇi ṣaṭ-śatāni ca saptatiḥ. [3]

The revolutions of the Sun in a Caturyuga are: twenty-four thousand and six hundred and seventy, [that is] **4,320,000**. [3]

Verse 4 — Moon’s Revolutions / शशिनो भगणाः

शशिनो भगणा यत्र पञ्च सप्ततिरुत्तरम् ।
त्रिसहस्राणि चाष्टौ च लक्षं च द्वे च दक्षिणे ॥ ४ ॥

Śaśino bhagānā yatra pañca saptatir uttaram.
Tri-sahasrāṇi cāṣṭau ca lakšaṃ ca dve ca dakṣiṇe. [4]

The revolutions of the Moon: five and seventy above, three thousand and eight, and one hundred-thousand and two on the right [side], [that is] **57,753,336**. [4]

Verse 5 — Lunar Apogee and Nodes / चन्द्रोच्चराहुभगणौ

चन्द्रोच्चं शनिमन्दं च राहुकेतू तथैव च ।
भगणानां शतं द्वे तु सहस्राणि च सप्ततिः ॥ ५ ॥

Candroccaṃ śani-mandaṃ ca rāhu-ketū tathaiva ca.
Bhagāñ̄anaṃ śataṃ dve tu sahasrāṇi ca saptatiḥ. [5]

The [apogee] of the Moon, the [apogee] of Saturn, and Rahu and Ketu: two hundred and seventy-thousand [revolutions in a Caturyuga]. [5]

[Explanatory note: Rahu and Ketu are the ascending and descending nodes of the Moon, responsible for eclipses.]

Verse 6 — The Manus / मनुसङ्ख्या

चतुर्दश व्यतीताः स्युर्मनवोऽष्टौ तथापराः ।
प्रवर्तमानश्चैकस्तु कल्पे सप्तदश स्मृताः ॥ ६ ॥

Caturdaśa vyatīt̄̄ah syur manavo 'ṣṭau tathāparāḥ.
Pravartamānaś caikas tu kalpe saptadaśa smr̄tāḥ. [6]

Fourteen Manus have passed away; eight are yet to come; and one is current—seventeen Manus are remembered in a Kalpa. [6]

[Explanatory note: A Kalpa is one day of Brahma, equal to 1000 Caturyugas. Each Manu rules over 71 Caturyugas.]

Verse 7 — Signs and Degrees / राश्यंशकल्पना

द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः ।
भागेन तेन राशिः स्यात्तनुं त्रिंशत् गुणीकृतम् ॥ ७ ॥

Dvādaśā sūryajaḥ so'yaṃ triḿśatā bhāgito raviḥ.
Bhāgena tena rāśiḥ syāt tanuṃ triḿśat guṇīkr̄tam. [7]

The Sun, born of the twelve [signs], is divided by thirty. By that division a sign is produced; a degree is [a sign] multiplied by thirty. [7]

[Explanatory note: 12 signs (rāsi) × 30 degrees = 360° in a circle.]

Verse 8 — Measure of the Earth / योजनप्रमाणम्

योजनं त्रिभवं वृत्तं भूमेर्योजनलक्षणम् ।
निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ८ ॥

Yojanaṃ tribhavaṃ vr̄ttaṃ bhūmer yojana-lakṣaṇam.
Nirākāraḥ sadā vyāpī śūnyamadhyaḥ sanātanaḥ. [8]

A yojana: the circle of the Earth has a circumference measured in yojanas. [The Earth is] formless, always pervading, void at the centre, eternal. [8]

[Explanatory note: The circumference of the Earth is given as 4,967 yojanas. 1 yojana ≈ 7.5 miles.]

Verse 9 — Planetary Orbits / ग्रहकक्ष्याः

ग्रहाणां कक्ष्याविस्तारा योजनैस्तैः प्रकीर्तिताः ।
भूमेरर्धं त्रिभागश्च व्यासः सूर्यस्य कीर्तितः ॥ ९ ॥

Grahāṇāṃ kakṣyā-vistārā yojanais taiḥ prakīrtitāḥ.
Bhūmer ardhaṃ tribhāgaś ca vyāsaḥ sūryasya kīrtitaḥ. [9]

The diameters of the orbits of the planets are proclaimed in yojanas. Half of the Earth plus a third: the diameter of the Sun is proclaimed. [9]

[Explanatory note: The Sun's diameter = 4410 yojanas; the Moon's = 315 yojanas; Earth's = 1050 yojanas.]

Verse 10 — The 24 Sines / चतुर्विंशत्यर्धज्याः

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् ।
चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥
१० ॥

Tribhujaṃ ṣaṭkoṇaṃ vṝttaṃ ṣa-triḿśad-aḿśena yojayet.
Catur-viḿsati-jīvāḥ syuḥ saptabhiḥ sahitāḥ kramāt. [10]

A triangle, a hexagon, and a circle are joined by thirty-six degrees. Twenty-four chords are produced, each increased successively by seven [and other differences]. [10]

[Explanatory note: Each sine-interval is $3^\circ 45'$ *(225′). The 24 ardhajyās are computed at intervals of* $3^\circ 45'$ *from* $3^\circ 45'$ *to* 90°*.]*

Table of 24 Sines / चतुर्विंशत्यर्धज्यासारणी

No.	ज्या	Value	No.	ज्या	Value
1	२२५	225	13	२८५९	2859
2	४४९	449	14	२९७८	2978
3	६७१	671	15	३०९३	3093
4	८९०	890	16	३२०२	3202
5	११०५	1105	17	३३०७	3307
6	१३१५	1315	18	३४०६	3406
7	१५२०	1520	19	३४९९	3499
8	१७१९	1719	20	३५८६	3586
9	१९१०	1910	21	३६६६	3666
10	२०९३	2093	22	३७३९	3739
11	२२६७	2267	23	३८०५	3805
12	२४३१	2431	24	३८६३	3863

उपर्युक्त २४ ज्यासु प्रत्येकं पूर्वजीवायां शेषमन्तरं निःसरति।

In the 24 sines above, taking each from the preceding sine, the remainders are the differences. The first differences are approximately 225, then 224, 222, 219,…decreasing gradually.

॥ इति प्रथमपादः समाप्तः ॥

End of the First Chapter

Second Chapter: Ganitapada / द्वितीयः पादः -- गणितपादः

गणितपादे
वर्गघनमूलक्षेत्रफलवृत्तजीवाकूट्टकछायादिविषयाः
संस्कृताः।

In the Ganitapada are treated: squares, cubes, square-roots, cube-roots, areas of figures, the circle, chords, the pulverizer (kuṭṭaka), shadows, and related topics.

Verse 1 / मङ्गलाचरणम्

प्रथमपादक्रमादेव ह्यः प्रणम्य गणेश्वरम् ।
ततो गणितमाख्यास्ये गणितज्ञानमुत्तमम् ॥ १ ॥

Prathama-pāda-kramād eva hyaḥ praṇamya gaṇeśvaram.
Tato gaṇitam ākhyāsye gaṇita-jñānam uttamam. [1]

Having bowed to the Lord of Calculation (Ganeśvara) in the order of the first chapter, I shall now expound the science of calculation, the highest knowledge of mathematics. [1]

Verse 2 — Place Values / स्थाननामानि

एकं दश शतं सहस्रमयुतं लक्षं प्रयुतं कोट्यः ।
अर्बुदं बृन्दमखखर्वं निखर्वं महापद्मं च ॥ २ ॥

Ekaṃ daśa śataṃ sahasram ayutaṃ lakṣaṃ prayutaṃ koṭyaḥ.
Arbudaṃ bṝndam akha-kharvaṃ nikharvaṃ mahā-padmaṃ ca. [2]

One, ten, hundred, thousand, myriads (ayuta), lakh (lakṣa), ten-lakhs (prayuta), crore (koṭi), ten-crores (arbuda), hundred-crores (bnda), thousand-crores (kharva), ten-thousand-crores (nikharva), hundred-thousand-crores (mahā-padma). [2]
[Explanatory note: The Indian decimal place-value system proceeds in powers of 10.]

Verse 3 — Square / वर्गः

संयुत्यकस्थानसंज्ञं ततः सप्तदशस्वरे ।
स्वरूपं चैव तत् प्रोक्तं वर्गसंज्ञं तथैव च ॥ ३ ॥

Saṃyuty-aka-sthāna-sajñaṃ tataḥ saptadaśa-svare.
Svarūpaṃ caiva tat proktaṃ varga-sajñaṃ tathaiva ca. [3]

The product of a number with itself is called the varga [square]. In the odd places [the product] is as stated; the square is also called "svarūpa" [own-form]. [3]

Verse 4 — Square Root / वर्गमूलम्

वर्गस्य मूलमाख्यातं यथोक्तं गणितैः पुरा ।
सर्वदोषविनिर्मुक्तं विधिं श्रूयतां मया ॥ ४ ॥

Vargasya mūlam ākhyātaṃ yathoktaṃ gaṇitaiḥ purā.
Sarva-doṣa-vinirmuktaṃ vidhiṃ śrūyatāṃ mayā. [4]

The square root of a square is declared as formerly spoken by calculators. Hear from me the method free from all defects. [4]
[Explanatory note: Method: mark off periods of two digits from the right. Find the largest square in the leftmost period; subtract and bring down the next period. Double the current root and divide into the remainder. Repeat.]

Verse 5 — Cube / घनः

त्रिसमचतुष्कोणं घनं स्यादथवाऽपरम् ।
त्रयाणां समवृत्तानां समसंयोगजं स्मृतम् ॥ ५ ॥

Tri-sama-catuṣkoṇaṃ ghanaṃ syād athavāparam.
Trayāṇām sama-vr̄ttānām sama-saṃyogajaṃ smr̄tam. [5]

A cube is an equi-quadrilateral [solid] with three equal dimensions. It is remembered as produced from the product of three equal quantities. [5]
[Explanatory note: ghana literally means "solid." The cube of a number n *is* $n \times n \times n$*.]*

Verse 6 — Cube Root / घनमूलम्

घनस्य मूलमाख्यातं विधिना गणितोक्तितः ।
घनमूलं तु यत्किंचित्तत् सर्वं परिकल्पयेत् ॥ ६ ॥

Ghanasya mūlam ākhyātaṃ vidhinā gaṇita-uktitaḥ.
Ghana-mūlaṃ tu yat kiṃcit tat sarvaṃ parikalpayet. [6]

The cube root of a cube is declared by the method taught in calculation. Whatever cube root exists, all of it is to be determined. [6]
[Explanatory note: Method: mark off periods of three digits. Find the largest cube in the first period; subtract. The divisor = $3\times$ *(current root)*^{2}*; divide into the remainder.]*

Verse 7 — Area of a Triangle / त्रिभुजक्षेत्रफलम्

समपर्यन्तमायतं त्रिभुजं क्षेत्रमुच्यते ।
तस्य कर्णस्तु विज्ञेयः समद्वयसमन्वितः ॥ ७ ॥

Sama-paryanta-māyatataṃ tribhujaṃ kṣetram ucyate.
Tasya karṇas tu vijñeyaḥ sama-dvaya-samanvitaḥ. [7]

A triangle with equal sides is called a triangular field. Its hypotenuse (karṇa) is to be known as combined with two equal [sides]. [7]

Verse 8 — Area of Any Triangle / सामान्यत्रिभुजफलम्

त्रिभुजस्य क्षेत्रफलमथवाऽप्यनुपाततः ।
आयामसंवर्गतुल्यं स्यात् समस्य तु विशेषतः ॥ ८ ॥

Tribhujasya kṣetra-phalam athavāpy anupātataḥ.
Āyāma-saṿarga-tulyaṃ syāt samasya tu viśeṣataḥ. [8]

The area of a triangle, or alternatively by proportion: it is equal to the product of the base and height. For an equilateral triangle, there is a special [method]. [8]

[Explanatory note: Area = $\frac{1}{2}\times$ *base* $\times$ *height.]*

Verse 9 — Area of a Circle / वृत्तक्षेत्रफलम्

वृत्तस्य क्षेत्रफलं तु व्यासार्धगुणितं परि ।
अर्धपरिधिगुणार्धव्यासं तत्क्षेत्रफलं भवेत् ॥ ९ ॥

Vṝttasya kṣetra-phalaṃ tu vyāsārdha-guṇitaṃ pari.
Ardha-paridhi-guṇārdha-vyāsaṃ tat kṣetra-phalaṃ bhavet. [9]

The area of a circle is: the circumference multiplied by half the diameter, [divided by 2]. Or: half the circumference multiplied by half the diameter becomes the area of that field. [9]

[Explanatory note: Formula: Area = $\frac{C\times d}{4} = \pi r^2$*.]*

Verse 10 — The Value of π / वृत्तपरिधेः सन्निकृष्टमानम्

चतुरधिकं शतमष्टगुणं द्वाषष्टिस्तथा सहस्राणाम् ।
अयुतद्वयविष्कम्भस्यासन्नो वृत्तपरिणाहः ॥ १० ॥

Catur-adhikaṃ śatam aṣṭa-guṇaṃ dvāṣaṣṭis tathā sahasrāṇām.
Ayuta-dvaya-viṣkambhasyāsanno vr̄tta-pariṇāhaḥ. [10]

Add four to one hundred, multiply by eight, then add sixty-two thousand; the result is the approximate circumference of a circle whose diameter is twenty thousand. [10]

Computation:

$$\begin{aligned}(100+4)\times 8+62000 &= 104\times 8+62000\\ &= 832+62000\\ &= 62832\end{aligned}$$

$$\pi \approx \frac{62832}{20000} = \mathbf{3.1416}$$

[Explanatory note: This value $\pi \approx 3.1416$ *is remarkably accurate. The true value of* $\pi = 3.14159265\ldots$ *Aryabhata's approximation gives the value correct to four decimal places. This verse is one of the most celebrated in Indian mathematical literature.]*

Verse 11 — Construction of Chords / जीवाविनिर्माणम्

व्यासार्धेन निरोद्धार्या जीवा त्रैराशिकेन तु ।
लम्बायतं तु विज्ञेयं ततः सूत्रं प्रदर्शयेत् ॥ ११ ॥

Vyāsārdhena niroddhāryā jīvā trairāśikena tu.
Lambāyatataṃ tu vijñeyaṃ tataḥ sūtraṃ pradarśayet. [11]

The chord is to be determined by means of the radius and the rule of three (trairāśika). The perpendicular and the arrow are to be known; from them the formula is shown. [11]

[Explanatory note: The "rule of three" (trairāśika) is the proportion $a : b :: c : d$*.]*

Verse 12 — Computation of the 24 Sines / चतुर्विंशत्यर्धज्यानयनम्

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् ।
चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥ १२ ॥

Tribhujaṃ ṣaṭkoṇaṃ vr̄ttaṃ ṣa-triḿśad-aḿśena yojayet.
Catur-viḿsati-jīvāḥ syuḥ saptabhiḥ sahitāḥ kramāt. [12]

A triangle, a hexagon, and the circle are to be joined by thirty-six degrees. Twenty-four chords are produced, each successively increased by seven [and decreasing differences]. [12]

[Explanatory note: The first sine-difference is 225′ (minutes). The differences decrease: 225, 224, 222, 219, 215, 210,…]

Verse 13 — The Pulverizer (Kuṭṭaka) / कूट्टकम्

गुणकारेण युक्ते ते वर्जयित्वा तु तद्गुणौ ।
छिन्ने तत्र गुणौ ज्ञेयौ कूट्टकं तद्विदुर्बुधाः ॥ १३ ॥

Guṇakāreṇa yukte te varjayitvā tu tad-guṇau.
Chinne tatra guṇau jñeyāu kuṭṭakaṃ tad-vidur budhāḥ. [13]

The two [quantities], when combined with the multiplier, are to be reduced [by the greatest common divisor]. When divided, the two multipliers are to be known there: the wise call this the kuṭṭaka [pulverizer]. [13]

[Explanatory note: The kuṭṭaka ("pulverizer") is Aryabhata's method for solving linear indeterminate equations of the form $ax + by = c$. *It is essentially the Euclidean algorithm applied to find solutions to Diophantine equations.]*

The Kuṭṭaka Method — Detailed Procedure

कूट्टकविधिः---भाजकभाज्ययोः परस्परभागदानं कृत्वा यावदवशेषं एकं न भवति। ततः पश्चादनुक्रमेण गुणकारभाजकौ लभ्येते।

The Kuṭṭaka Method:
Divide the divisor (bhājaka) and dividend (bhājya) mutually until the remainder becomes 1. Then by back-substitution, the multiplier (guṇaka) and additive are obtained.

Example: Find x such that $17x + 5 \equiv 0 \pmod{29}$

यदि १७□ + ५ = ० (मो २९) तर्हि □ = ?

हलः---
२९ = १ × १७ + १२
१७ = १ × १२ + ५
१२ = २ × ५ + २
५ = २ × २ + १
अतः गुणकारः □ = १२, भाजकः □ = ७

If $17x + 5 \equiv 0 \pmod{29}$, find x.

Solution:

$$\begin{aligned} 29 &= 1 \times 17 + 12 \\ 17 &= 1 \times 12 + 5 \\ 12 &= 2 \times 5 + 2 \\ 5 &= 2 \times 2 + 1 \end{aligned}$$

Thus the multiplier $x = 12$, and the additive $y = 7$.
[Verification: $17 \times 12 - 29 \times 7 = 204 - 203 = 1$*]*

Verse 14 — Direct and Inverse / क्रमानुक्रमौ

क्रमादनुक्रमं चैव प्रकल्प्यं गणिते सदा ।
प्रकल्पयेत् तथा योगं व्यस्तं वा यदि तत् स्थितम्
॥ १४ ॥

Kramād-anukramaṃ caiva prakalpyaṃ gaṇite sadā.
Prakalpayet tathā yogaṃ vyastaṃ vā yadi tat sthitam. [14]

Direct and inverse are always to be employed in calculation. The sum is to be formed accordingly, or the reverse, as the case may be. [14]

Verse 15 — Shadow (Chāyā) / छाया

शङ्कोर्निजछायया युक्तो दृग्भागः सूर्यमण्डलम् ।
शङ्कुच्छायाग्रसंयुक्तं दृश्यते यदि तत् फलम् ॥ १५ ॥

Śaṇkor nija-chāyayā yukto dṛgbhḡaḥ sūrya-maṇḍalam.
Śaṇku-chāyāgra-saỵuktaṃ dśśyate yadi tat phalam. [15]

The gnomon (śaṇku), combined with its own shadow, [gives] the line of sight to the solar orb. Combined with the tip of the gnomon's shadow, it is seen—that is the result. [15]
[Explanatory note: The gnomon (śaṇku) is a vertical stick of 12 aṇgulas. From the ratio of gnomon to shadow, the altitude of the Sun and hence the time of day may be determined.]

॥ इति द्वितीयपादः समाप्तः ॥
End of the Second Chapter

Third Chapter: Kalakriyapada / तृतीयः पादः -- कालक्रियापादः

कालक्रियापादे युगमासदिवसघटीपलादीनां परिकल्पना वर्तते।

In the Kalakriyapada is the reckoning of time: the definitions of yugas, months, days, ghaṭīs, palas, and so forth.

Verse 1 — Division of Time / कालविभागः

नाडी तु दशपञ्चाशत्पलं षष्ट्या तु तैः पुनः ।
विपलैस्तैस्तथा कालो यावत् कालेन पश्यति ॥ १ ॥

Nāḍī tu daśa-pañcāśat-palaṃ ṣaṣthyā tu taiḥ punaḥ.
Vipalais tais tathā kālo yāvat kālena paśyati. [1]

A nāḍī is fifty-five palas; again by sixty of those; and by vipalas of those—thus time proceeds until it is seen by [the smallest unit of] time. [1]

[Explanatory note: 1 nāḍī = 60 palas; 1 pala = 60 vipalas. A nāḍī equals 24 minutes of modern time.]

Verse 2 — Solar Division / सौरविभागः

पूर्वं कालविभागः सैत्रविभागस्तथा भगणात् ।
द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः ॥ २ ॥

Pūrvaṃ kāla-vibhāgaḥ saitra-vibhāgas tathā bhagāṇāt.
Dvādaśā sūryajaḥ so'yaṃ triṁśatā bhāgito raviḥ. [2]

First, the division of time: the division of the [zodiacal] circle into signs, and the revolutions [of planets]. The Sun, born of the twelve [signs], is divided by thirty. [2]

Verse 3 — The Four Yugas / चतुर्युगम्

युगपादश्चतुर्धा स्यात् कृतं त्रेताद्वापरम् ।
कलिश्चेति चतुर्विंशतिर्घातिर्युगपदे स्मृता ॥ ३ ॥

Yuga-pādaś caturdhā syāt kr̄taṃ tretā dvāparam.
Kaliś ceti catur-viṁsatiṛ ghātir yuga-pade smr̄tā. [3]

The four parts of a yuga are: Krta [Satya], Tretā, Dvāpara, and Kali. Twenty-four is remembered as the measure [in hundreds of thousands] of a yuga-part. [3]

[Explanatory note: A Caturyuga = 4,320,000 years. Krta = 1,728,000; Tretā = 1,296,000; Dvāpara = 864,000; Kali = 432,000. The ratio is 4:3:2:1.]

Verse 4 — The Kalpa / कल्पः

मानयुगाद्ययुगपादस्य पञ्चदशयुतसहस्रकल्पः ।
युगपादैः सहस्रैस्तु कल्पः परिकीर्तितः ॥ ४ ॥

Māna-yugādya-yuga-pādasya
pañcadaśa-yuta-sahasra-kalpaḥ.
Yuga-pādaiḥ sahasrais tu kalpaḥ
parikīrtitaḥ. [4]

A kalpa is a thousand yugas joined with fifteen [intervening periods]. A kalpa is proclaimed to be a thousand yugas. [4]
[Explanatory note: 1 kalpa = 1000 caturyugas = 4,320,000,000 years. This is one day of Brahma.]

Verse 5 — Longitude and Latitude / आयामविस्तरौ

आयामवर्गाद्यं पार्श्वं तद्वोगैर्हतं स्वपातलेखैः ।
विस्तरयोगार्धयोगैस्तैर्लेखफलमायामे ॥ ५ ॥

Āyāma-vargādyaṃ pārśvaṃ tad-
vogair hataṃ sva-pāta-lekhaiḥ.
Vistara-yogārdha-yogais tair lekha-
phalam āyāme. [5]

The square of the longitude [etc.] is the side; multiplied by its own lines of latitude. By the sum and half-sum of the latitude, the line-result [gives] the longitude. [5]
[Explanatory note: These verses describe the computation of planetary positions using longitude and latitude, essential for predicting eclipses.]

Fourth Chapter: Golapada / चतुर्थः पादः -- गोलपादः

गोलपादे भूगोलस्य खगोलस्य च वर्णनं वर्तते।
पृथिव्या गोलत्वं, ग्रहणं, उदयास्तौ, नक्षत्राणि च।

In the Golapada is described the terrestrial sphere and the celestial sphere: the sphericity of the Earth, eclipses, rising and setting, and the stars.

Verse 1 — The Celestial Sphere / खगोलवर्णनम्

पृथिवी परिदृश्यमानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तद् द्वयमुद्धतं यत्र दिवसनिशोः ॥ १
॥

Pṛthivī paridṛśyamānaṃ kṣitija-deśa-prayogair yuktaṃ ca.
Unmeṣalaṃ bhavet tad dvayam uddhataṃ yatra divasa-niśoḥ. [1]

The Earth, being observed, combined with the application of the horizon region: that is the blink [of an eye] where both day and night are elevated [above the horizon]. [1]
[Explanatory note: The kṣitija (horizon) is the great circle dividing the visible heavens from the invisible.]

Verse 2 — Directions / दिशाज्ञानम्

दक्षिणोत्तररेखा मध्यमा पूर्वापरं तु दिशच्छेदः ।
दिशामानं तु विज्ञेयं षष्टिघातैस्तु नाडिकाभिः ॥ २
॥

Dakṣiṇottara-rekhā madhyamā pūrvāparaṃ tu diśa-cchedaḥ.
Diśāmānaṃ tu vijñeyaṃ ṣaṣṭi-ghātais tu nāḍikābhiḥ. [2]

The meridian line [running] south-north is the middle; the east-west is the division of directions. The measure of directions is to be known as sixty multiplied by nāḍīkas. [2]
[Explanatory note: 60 nāḍīkas = 1 day = 360° of the equator.]

Verse 3 — Eclipses / ग्रहणनिर्णयः

सूर्यग्रहणं तु विज्ञेयं राहुच्छादनात् तु यत् ।
चन्द्रग्रहणं च छायाया योगात् संयोगतो भवेत् ॥ ३
॥

Sūrya-grahaṇaṃ tu vijñeyaṃ rāhu-chādānāt tu yat.
Candra-grahaṇaṃ ca chāyāyā yogāt saṃyogato bhavet. [3]

A solar eclipse is to be known as that [which occurs] from the covering by Rāhu. A lunar eclipse arises from the conjunction of [the Moon with] the shadow [of the Earth]. [3]
[Explanatory note: Aryabhata correctly understood that the solar eclipse is caused by the Moon interposing between the Earth and Sun, while the lunar eclipse is caused by the Earth's shadow falling on the Moon.]

Verse 4 — Chords of Arcs / लम्बज्याव्यासज्ये

लम्बज्याव्यासज्या चापज्याऽऽयतांशकाः स्मृताः ।
षष्ट्यंशैस्तैर्मितं चापं तत् फलं परिकल्पयेत् ॥ ४ ॥

Lambajyā-vyāsajyā cāpajyā āyatāṃśakāḥ smṝtāḥ.
Ṣaṣṭy-aṃśais tair mitaṃ cāpaṃ tat phalaṃ parikalpayet. [4]

The perpendicular sine (lambajyā), the radius (vyāsajyā), and the chord of an arc (cāpajyā) are remembered as degrees. The arc measured by sixty degrees—that result is to be computed. [4]

[Explanatory note: lambajyā = $R\sin\theta$ *(modern sine); vyāsajyā* = R *(radius); cāpajyā* = $2R\sin(\theta/2)$ *(chord).]*

Verse 5 — The Earth Sphere / भूगोलवर्णनम्

भूगोलः सवितुर्भानोर्भूमेर्भूगोलसंस्थितः ।
निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ५ ॥

Bhū-golaḥ savitur bhānor bhūmer bhū-gola-saṣthitaḥ.
Nirākāraḥ sadā vyāpī śūnyamadhyaḥ sanātanaḥ. [5]

The Earth-globe, [resting] in the Earth-globe [frame], of the Sun and the Earth: formless, always pervading, void at the centre, eternal. [5]

[Explanatory note: Aryabhata here describes the Earth as a sphere (gola) suspended in space. The “void at the centre” is a remarkable concept.]

Aryabhata’s Revolutionary Discovery — Earth’s Rotation

पूर्वापरदिग्गुल्मानं क्षितिजद्देशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तद् द्वयमुद्धतं यत्र दिवसनिशोः ॥

Pūrvāpara-dig-gulmānaṃ kṣitija-deśa-prayogair yuktaṃ ca.
Unmeṣalaṃ bhavet tad dvayam uddhataṃ yatra divasa-niśoḥ.

Translation:
That which appears as the motion of the stars from east to west is in truth the Earth’s own rotation. The Earth, spherical in shape, rotates upon its axis from west to east.

स्थलस्य गतिमाहुर्वै खगोलस्थानभूम्नि च ।
यतः प्रवृत्तिस्तस्यान्तो विपर्यासो यतः स्मृतः ॥

Sthalasya gatim āhur vai khagola-sthāna-bhūmni ca.
Yataḥ pravṝttis tasyānto viparyāso yataḥ smṝtaḥ.

Translation:
They say the motion of the Earth is in the place of the celestial sphere. The end of its rotation [causes] the reversal [of apparent motion].
[Explanatory note: This is one of the most remarkable verses in the history of science. Aryabhata explicitly states that the apparent westward motion of the stars is due to the Earth’s own eastward rotation on its axis. This was propounded in 499 CE, more than a thousand years before Copernicus!]

On the Earth’s Rotation

पृथिव्याः स्वगतिकारणात् खगोलस्य स्थिरत्वं भाति। यथा नौकायां स्थितः पुरुषः तीरे वृक्षान् विचलितान् मन्यते तथा पृथिव्यां स्थिताः स्थलजीवाः खगोलं विचलितं मन्यन्ते।

Due to the Earth’s own motion, the celestial sphere appears stationary. Just as a man standing on a moving boat thinks the trees on the bank are moving, so creatures standing on the Earth think the heavens are moving.

[Explanatory note: This analogy of the boat and trees is one of the most celebrated in scientific literature. Aryabhata used it to explain why we do not perceive the Earth’s rotation. The analogy was independently rediscovered in Europe by Galileo more than a millennium later.]

॥ इति चतुर्थपादः समाप्तः ॥

End of the Fourth Chapter

Epilogue / उपसंहार

एवं चतुष्पादसंयुक्तमिदं ग्रन्थं समाप्तम्। आर्यभटेन कृतमिदं गणितज्योतिषशास्त्रस्य सारसङ्ग्रहः।

Thus this book, consisting of four chapters, is completed. This compendium of mathematics and astronomy was composed by Aryabhata.

Summary of Aryabhata's Contributions

1. **Value of π**: $\pi \approx 3.1416$ (accurate to 4 decimal places)
2. **Sine Table**: 24 sines at intervals of $3°45'$, with radius = $3438'$
3. **Kuṭṭaka**: General method for solving linear Diophantine equations $ax + by = c$
4. **Earth's Rotation**: Explicit statement that Earth rotates on its axis
5. **Spherical Earth**: Earth is a sphere floating freely in space
6. **Eclipses**: Correct explanation of solar and lunar eclipses
7. **Astronomical Constants**: Accurate orbital periods for all planets

आर्यभटीयस्य महत्त्वं विश्वविदुषा ए.बी. क्लार्क् (१९३०) इत्यनेन संस्कृतस्य आङ्ग्लभाषायां अनुवादं कृतवन्तः।

The importance of the Aryabhatiya was recognized by the scholar W.E. Clark, who published a Sanskrit-English edition in 1930. This bilingual edition follows Clark's translation style while presenting the original Devanagari text alongside.

॥ ॐ तत् सत् ॥

Om Tat Sat

BILINGUAL EDITION

Sanskrit – English / Saṃskṛta – Aṅgla

बीजगणितमध्यगणितं वा

BIJAGANITA

(Seed Computation / Elements of Algebra)

श्रीभास्कराचार्यविरचितम् ।

Composed by Śrī Bhāskarācārya

Part 1 / Prathamo Bhāgaḥ

Edited by Sudhākara Dvivedī with further notes by Muralīdhara Jha

Benares Sanskrit Series, No. 159 (1927)

Bilingual edition with IAST transliteration following S. K. Abhyankar’s conventions

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विषयसूची / Table of Contents

श्रीगणेशाय नमः

Obeisance to Śrī Gaṇeśa

अथ बीजगणितम्

Now begins the Bījagaṇita

उपोद्घातः

Upodghāta (Introduction)

श्रीं ॐ मते बलवते हनुमते नमः ।

Obeisance to the powerful Hanūmat, the abode of wisdom.

उत्पादकं यत् प्रवदन्ति बुद्धेः स्थितं सुरैरपि ।
अस्य क्रिया तत्कथयाम्यथवा तद्बीजमष्टधा ॥

That which the sages declare as the source of intellect,
even by the gods — its eight-fold *bīja* operation I shall tell.

पूर्वं प्रोक्तं व्यक्तमस्य कथये प्रज्ञा ते च मेऽव्यक्तयुक्तया ।
यं तु शक्या मन्दबुद्धिर्न तत्समं न तद् द्वयं वीजक्रिया च ॥

First the manifest (*vyakta*) was spoken; now I speak the unmanifest (*avyakta*)
to those of wisdom; the slow-witted cannot grasp this two-fold *bīja*-operation.

:

Explanation: Bhāskara states that he will explain the eight-fold operations of algebra (*bījagaṇita*). Those accomplished in manifest arithmetic (*vyaktagaṇita*) should also study the unmanifest (*avyakta*). Algebra is the root of all computation.

:

- —
- (x) —
- , , , —
- —
- () $(-)$ —
- —

Fundamental Concepts:

- *Avyakta-kalpanā* — representation of unknown quantities
- *Yāvat-tāvat* (x) — first unknown quantity
- Kālaka, Nīlaka, Pītaka, Haritaka — other colours for unknowns
- Rūpa — absolute constants
- Dhana $(+)$ and ṛṇa $(-)$ — signs
- Karaṇī — irrational (surd) quantities

ब्रह्मस्फुटसिद्धान्ते च कालक्रियायां तथैव च ।
ग्रहगणितं निगदितं त्रैराशिकफलं तथा ॥

In the Brāhmasphuṭasiddhānta and in Kālakriyā,
and Graha-gaṇita and the Rule of Three — all was told.

घनाणां सङ्कलनम्

घनाणां सङ्कलने करणसूत्रं बुद्धये

घनाः समं सङ्गुणिता द्वित्रिभिः समन्विताः
एकघनाद्विवर्जितास्ते सङ्कलिताः स्युः सदा घनाः ॥ १ ॥

:

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

:

$$\begin{aligned} 1^3 + 2^3 + 3^3 + 4^3 + 5^3 &= \left(\frac{5 \cdot 6}{2}\right)^2 \\ &= 15^2 = 225 \end{aligned}$$

: $1 + 8 + 27 + 64 + 125 = 225$

Ghanāṇāṃ Saṃkalanaṃ (Sum of Cubes)

Sūtra for Sum of Cubes

ghanāḥ samaṃ saṅguṇitā dvitribhiḥ samanvitāḥ
ekaghanād vivarjitās te saṃkalitāḥ syuḥ sadā ghanāḥ —
Verse 1

Rule: The sum of cubes of the first n natural numbers equals the square of their sum:

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Example: Sum of the first 5 cubes:

$$\begin{aligned} 1^3 + 2^3 + 3^3 + 4^3 + 5^3 &= \left(\frac{5 \cdot 6}{2}\right)^2 \\ &= 15^2 = 225 \end{aligned}$$

Verification: $1 + 8 + 27 + 64 + 125 = 225$ ✓

घनानां व्यवकलनम्

व्यस्तद्वयोर्घनयोर्व्यस्तयोर्घनयोर्व्यवकलनम् ।
समघ्नयोरपि तथा व्यवकलनं सम्भवत्येव ॥ २ ॥

: a b ,

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

:

$$\begin{aligned}(y\bar{a}\ 5)^3 - (y\bar{a}\ 2)^3 &= 125y\bar{a}^3 - 8y\bar{a}^3 \\ &= 117y\bar{a}^3\end{aligned}$$

: $(5-3)(25+15+9) = 2 \cdot 49 = 98$ ( )

Ghanānāṃ Vyavakalanaṃ (Subtraction of Cubes)

vyastadvayor ghanayor vyastayor ghanayor vyavakalanam
samaghnayor api tathā vyavakalaṃ sambhavaty eva — Verse 2

Rule: The subtraction of cubes of two quantities equals the product of their difference and the sum of their squares plus their product:

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Example:

$$\begin{aligned}(5y\bar{a})^3 - (2y\bar{a})^3 &= 125y\bar{a}^3 - 8y\bar{a}^3 \\ &= 117y\bar{a}^3\end{aligned}$$

Or: $(5-3)(25+15+9) = 2 \cdot 49 = 98$ (in constants)

घनानां गुणनम्

घनयोर्घनयोर्घनयोर्वा गुणनं सम्भवति ।
घनघातः सम्भवत्येवं गुणकारघनयोः समं ॥ ३ ॥

:

$$a^3 \times b^3 = (ab)^3$$

:

$$\begin{aligned}(3y\bar{a})^3 \times (2y\bar{a})^3 &= (6y\bar{a})^3 \\ &= 216y\bar{a}^3\end{aligned}$$

Ghanānāṃ Guṇanam (Multiplication of Cubes)

ghanayor ghanayor ghanayor vā guṇanaṃ sambhavati
ghanaghātaḥ sambhavaty evaṃ guṇakāraghanayoḥ samaṃ
— Verse 3

Rule: The product of two cubes equals the cube of the product of their roots:

$$a^3 \times b^3 = (ab)^3$$

Example:

$$\begin{aligned}(3y\bar{a})^3 \times (2y\bar{a})^3 &= (6y\bar{a})^3 \\ &= 216y\bar{a}^3\end{aligned}$$

घनानां भागहारः

घनं घनेन विभजेद् घनमूलयोर्मूलयोरिव ।
घनभागः स्तः सम्भवत्येवं भागकारयोः समं ॥ ४ ॥

:

$$\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$$

:

$$\frac{(8y\bar{a})^3}{(2y\bar{a})^3} = \left(\frac{8y\bar{a}}{2y\bar{a}}\right)^3 = (4y\bar{a})^3 = 64y\bar{a}^3$$

Ghanānāṃ Bhāgahāraḥ (Division of Cubes)

ghanaṃ ghanena vibhajed ghanamūlayor mūlayor iva
ghanabhāgaḥ sthaḥ sambhavaty evaṃ bhāgakārayoḥ samaṃ
— Verse 4

Rule: A cube divided by a cube; the quotient is the cube of the quotient of their roots:

$$\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$$

Example:

$$\frac{(8y\bar{a})^3}{(2y\bar{a})^3} = \left(\frac{8y\bar{a}}{2y\bar{a}}\right)^3 = (4y\bar{a})^3 = 64y\bar{a}^3$$

घनानां वर्गो मूलं च

घनस्य वर्गः स्यात् षट्ठी घनस्य मूलं तु मूलघनम् ।
वर्गमूलं घनमूलं च कल्पनीयं ततस्ततः ॥ ५ ॥

:

$$(a^3)^2 = a^6 \quad (\quad = \quad)$$
$$\sqrt[3]{a^3} = a \quad (\quad)$$
$$\sqrt{a^3} = a^{3/2} \quad (\quad)$$

Ghanānāṃ Vargo Mūlaṃ ca (Square and Root of Cubes)

ghanasya vargaḥ syāt ṣaṣṭhī ghanasya mūlaṃ tu mūlaghanam
vargamūlaṃ ghanamūlaṃ ca kalpanīyaṃ tatas tataḥ —
Verse 5

Rule: The square of a cube is the sixth power. The root of a cube is the cube-root.

$$(a^3)^2 = a^6 \quad \text{(square of cube = 6th power)}$$
$$\sqrt[3]{a^3} = a \quad \text{(cube root)}$$
$$\sqrt{a^3} = a^{3/2} \quad \text{(square root of cube)}$$

अव्यक्तसङ्कलनव्यवकलनम्

(१) यावत्तावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः पीतो हरितश्चेत् द्वयं च अव्यक्तानां कल्पना मतसंज्ञा—
राशिसंख्यानां कल्पिता मताचार्येण ॥ ६ ॥

( ): (= x), (= y), (= z), (= u), (= v) **( ):**

$$(y\bar{a}\ 1 + k\bar{a}\ 1) + (y\bar{a}\ 2 + k\bar{a}\ 8) = y\bar{a}\ 3 + k\bar{a}\ 9$$
$$(y\bar{a}\ 5 + k\bar{a}\ 3) - (y\bar{a}\ 2 + k\bar{a}\ 8) = y\bar{a}\ 3 + k\bar{a}\ \bar{5}$$

: “ ” ‘ ’ “ ”

Avyakta-saṃkalana-vyavakalanam (Addition/Subtraction of Unknowns)

(1) *yāvat-tāvat kālaka nīlaka pītaka haritakādayaḥ varṇāḥ* — The unknown quantities are represented by colours: *yāvat-tāvat* (x), *kālaka* (y), *nīlaka* (z), *pītaka* (u), *haritaka* (v), etc.

Rule (1): Unknown quantities are represented by colours: *yāvat-tāvat* (x), *kālaka* (y), *nīlaka* (z), *pītaka* (u), *haritaka* (v). **Rule (2):** Like terms are combined; unlike terms are set down separately.

$$(y\bar{a}\ 1 + k\bar{a}\ 1) + (y\bar{a}\ 2 + k\bar{a}\ 8) = y\bar{a}\ 3 + k\bar{a}\ 9$$
$$(y\bar{a}\ 5 + k\bar{a}\ 3) - (y\bar{a}\ 2 + k\bar{a}\ 8) = y\bar{a}\ 3 + k\bar{a}\ \bar{5}$$

Note: From the Amarakosha: “*yāvat-tāvat sākalyena manasyavasthā*” — the term *yāvat-tāvat* means “as much as,” representing the state of the mind regarding an unknown quantity.

अव्यक्तगुणनम्

अव्यक्तवर्गविधम्

अव्यक्ताकृतयोः समस्य हेतोः समेत्य वर्गद्वयं कल्पयेत् ।
तदा तयोर्वर्गयोर्वर्गः सम्भवत्येवमेकस्य वर्गः ॥ ७ ॥

: वर्गः , ,
:

$(y\bar{a}\ 5 + k\bar{a}\ 3) \times (y\bar{a}\ 3 + k\bar{a}\ 2) = y\bar{a}v\ 15 + y\bar{a}\ 19 + k\bar{a}\ 6$

: $y\bar{a}\ 5\ k\bar{a}\ \bar{3}$ $y\bar{a}\ 3\ k\bar{a}\ 2$ = $y\bar{a}v\ 15 + y\bar{a}\ \bar{1} + k\bar{a}\ \bar{6}$

Avyakta-guṇanam (Multiplication of Unknowns)

Avyakta-varga-vidham

avyaktākṛtayoḥ samasya hetoḥ sametya vargadvayaṃ kalpayet
tadā tayor vargayor vargaḥ sambhavaty ekasya vargaḥ — Verse 7

Rule: When two unknowns are multiplied, a new species is produced. The square of an unknown is *varga*; higher powers follow. **Example:**

$$(5y\bar{a} + 3k\bar{a}) \times (3y\bar{a} + 2k\bar{a}) = 15y\bar{a}^2 + 19y\bar{a} + 6k\bar{a}$$

With sign change of multiplicand:
$(5y\bar{a} - 3k\bar{a}) \times (3y\bar{a} + 2k\bar{a}) = 15y\bar{a}^2 + y\bar{a} - 6k\bar{a}$

अव्यक्तभागहारः

Avyakta-bhāgahāraḥ (Division of Unknowns)

भागाहरे करणसूत्रं बुद्धये ।
भाज्यच्छेदः सृजति प्रच्युतः सन् क्षेपे क्षेपे स्थानक्षेपं क्षेपेण ।
दैन्यैर्द्विजैः सङ्गुणो यैः क्षेपैर्भागाहरे लब्धयस्तः स्पृशेत्

bhāgāhare karaṇasūtraṃ buddhaye
bhājyacchedaḥ sṛjati pracyutaḥ san kṣepe kṣepe
sthānakṣepaṃ kṣepeṇa
— The division of unknown quantities follows the method for known quantities.

: ,
: $y\bar{a}$ 18 $k\bar{a}$ 24 $k\bar{a}$ 6 $y\bar{a}$ 5

: $y\bar{a}v$ $10 + y\bar{a}$ $27 + y\bar{a}$ $18 + k\bar{a}$ 24
: $y\bar{a}$ $5 + k\bar{a}$ 6
: $y\bar{a}$ $2 + y\bar{a}$ $3 + k\bar{a}$ 4

Rule: Division of unknowns is the same as for knowns. If division is not exact, a fraction results. **Example:** Divide $18y\bar{a} + 24k\bar{a}$ by $5y\bar{a} + 6k\bar{a}$:

Dividend: $10y\bar{a}^2 + 27y\bar{a} + 18y\bar{a} + 24k\bar{a}$
Divisor: $5y\bar{a} + 6k\bar{a}$
Quotient: $2y\bar{a} + 3y\bar{a} + 4k\bar{a}$

अव्यक्तवर्गादि

वर्गः प्रथमं ततो घनोऽथ वर्गवर्गस्ततोऽष्टघातः ।
नवमो वर्गघनश्चेत्यादि घातानां क्रमेण स्मृतम्

:

- वर्ग — $y\bar{a}^2$ (:)
- घन — $y\bar{a}^3$ (:)
- वर्गवर्ग — $y\bar{a}^4$ (:)
- घनवर्ग — $y\bar{a}^5$ (:)
- घनघन — $y\bar{a}^6$ (:)
- वर्गघन — $y\bar{a}^6$
- अष्टघात — $y\bar{a}^8$ (:)
- घनघनवर्ग — $y\bar{a}^9$ (:)

Avyakta-vargādi (Square etc. of Unknowns)

vargaḥ prathamaṃ tato ghano ’tha vargavargas tato ’ṣṭaghātaḥ
navamo vargaghanaś cetyādi ghātānāṃ krameṇa smṛtam

Powers of Unknowns (in order):

- *Varga* — $y\bar{a}^2$ (square)
- *Ghana* — $y\bar{a}^3$ (cube)
- *Varga-varga* — $y\bar{a}^4$ (square of square)
- *Ghana-varga* — $y\bar{a}^5$ (square of cube)
- *Ghana-ghana* — $y\bar{a}^6$ (cube of cube)
- *Varga-ghana* — $y\bar{a}^6$ (also)
- *Aṣṭaghāta* — $y\bar{a}^8$ (8th power)
- *Ghana-ghana-varga* — $y\bar{a}^9$ (9th power)

अनेकवर्णीद्विपद्विभ्रम

द्विवर्णकं द्विपदं त्रिवर्णकं त्रिपदं तथा ।
बहुवर्णकं बहुपदं च कल्पयेत् सुबुद्धिभिः

: (=), (=), , , , —

$(a+b)$ $(c+d)$ $ac+ad+bc+bd$

Aneka-varṇī-dvi-pada-vibhrama

dvivarṇakaṃ dvipadaṃ trivarṇakaṃ tripadaṃ tathā
bahuvarṇakaṃ bahupadaṃ ca kalpayet subuddhibhiḥ

Rule: Expressions with two unknowns (*dvivarṇaka*), two terms (*dvipada*), three unknowns (*trivarṇaka*), three terms (*tripada*), and many unknowns (*bahuvarṇaka*) or many terms (*bahupada*) should be formed by the intelligent.

Bhāskara explains operations on polynomials. Multiplication of binomials: $(a+b) \times (c+d) = ac + ad + bc + bd$.

करणीसङ्कलनव्यवकलनम्

करणीसंयुक्तिरेभिः सङ्कलिता स्यात् समं करणीभिः ।
करणीभिर्विशिष्टाः पृथक् स्थापनीयाः समं करणीभिः

: **:**

$\sqrt{3}+\sqrt{3}+2\sqrt{3}+5\sqrt{3}$

: $(1+1+2+5)\sqrt{3}=9\sqrt{3}$

: $\sqrt{2}+\sqrt{3}+\sqrt{5}$

: $\sqrt{2}+\sqrt{3}+\sqrt{5}$ ( )

Karaṇī-saṃkalana-vyavakalanam (Addition/Subtraction of Surds)

karaṇīsaṃyuktirebhiḥ saṃkalitā syāt samaṃ karaṇībhiḥ
karaṇībhir viṣiṣṭāḥ pṛthak sthāpanīyāḥ samaṃ karaṇībhiḥ

Rule: Surds having the same radical are added together; those with different radicals are set down separately.
Example 1: Add $\sqrt{3}+\sqrt{3}+2\sqrt{3}+5\sqrt{3}$.

$$\text{Sum: } (1+1+2+5)\sqrt{3}=9\sqrt{3}$$

Example 2: Add $\sqrt{2}+\sqrt{3}+\sqrt{5}$.

$$\text{Result: } \sqrt{2}+\sqrt{3}+\sqrt{5} \quad \text{(set down separately)}$$

करणीगुणनम्

करणीगुणने करणीगुणने समे समे विपरे विपरे ।
गुण्यकरणीगुणककरणी गुणनफलं करणी भवति

: -
: $3\sqrt{2}$ $4\sqrt{3}$

$$:(3\times 4)\times\sqrt{2\times 3}=12\sqrt{6}$$

:

$$a\sqrt{b}\times c\sqrt{d}=(ac)\sqrt{bd}$$

Karaṇī-guṇanam (Multiplication of Surds)

karaṇīguṇane karaṇīguṇane same same vipare vipare
guṇyakaraṇīguṇakakaraṇī guṇanaphalaṃ karaṇī bhavati

Rule: In multiplication of surds, multiply the radical quantities to obtain the new radical; multiply the coefficients for the new coefficient. **Example:** Multiply $3\sqrt{2}$ by $4\sqrt{3}$.

$$\text{Product: }(3\times 4)\times\sqrt{2\times 3}=12\sqrt{6}$$

In general:

$$a\sqrt{b}\times c\sqrt{d}=(ac)\sqrt{bd}$$

करणीभजनम्

भजनं करण्योर्भजनं करण्योर्वर्गवत् संस्कारः ।
भाज्यकरणीभाजककरणीभागो यथोक्तक्रमेण

:

$12\sqrt{6}$ $3\sqrt{2}$

$$: \frac{12}{3} \times \sqrt{\frac{6}{2}} = 4\sqrt{3}$$

:

$$\frac{a\sqrt{b}}{c\sqrt{d}} = \frac{a}{c}\sqrt{\frac{b}{d}}$$

:

Karaṇī-bhajanam (Division of Surds)

bhajanaṃ karaṇyor bhajanaṃ karaṇyor vargavat saṃ
bhājyakaraṇībhājakakaraṇībhāgo yathoktakrameṇa

Rule: In division of surds, the operation is like that for squares. Divide the radical of the dividend by the radical of the divisor, and the coefficient of the dividend by the coefficient of the divisor. **Example:** Divide $12\sqrt{6}$ by $3\sqrt{2}$.

$$\text{Quotient: } \frac{12}{3} \times \sqrt{\frac{6}{2}} = 4\sqrt{3}$$

In general:

$$\frac{a\sqrt{b}}{c\sqrt{d}} = \frac{a}{c}\sqrt{\frac{b}{d}}$$

करणीवर्गः

वर्गः करण्याः स्यात् पदस्य वर्गः करणी चेति ।
वर्गः करण्याः स्यात् पदस्य वर्गः करणी च भवति

: = $\times$ ( ) **:** $3\sqrt{5}$

$$: (3)^2 \times 5 = 9 \times 5 = 45$$

$(3\sqrt{5})^2 = 3^2 \times (\sqrt{5})^2 = 9 \times 5 = 45$ **:**

$$(a\sqrt{b})^2 = a^2 \cdot b$$

Karaṇī-vargaḥ (Square of Surds)

vargaḥ karaṇyāḥ syāt padasya vargaḥ karaṇī ceti
vargaḥ karaṇyāḥ syāt padasya vargaḥ karaṇī ca bhavati

Rule: The square of a surd equals the square of the coefficient times the radical quantity (the radical sign vanishes). **Example:** Square of $3\sqrt{5}$.

$$\text{Square: } (3)^2 \times 5 = 9 \times 5 = 45$$

Since $(3\sqrt{5})^2 = 3^2 \times (\sqrt{5})^2 = 9 \times 5 = 45$ **In general:**

$$(a\sqrt{b})^2 = a^2 \cdot b$$

करणीमूलम्

Karaṇī-mūlam (Square Root of Surds)

(१) वर्गं करण्या यदि वा करणीस्तु तयो रूपपक्षं वा बहूनाम् ।
विशोध्येष्टं पदं ते शेषस्य रूपाणि युतानि निपातयेत्

: $\sqrt{a+\sqrt{b}} = \sqrt{x} + \sqrt{y},$

$$x + y = a$$
$$4xy = b$$

atha $x - y = \sqrt{a^2 - b}$

$$x = \frac{a + \sqrt{a^2 - b}}{2}$$
$$y = \frac{a - \sqrt{a^2 - b}}{2}$$

(1) *vargaṃ karaṇyā yadi vā karaṇīstu tayo rūpapakṣaṃ vā bahūnām*
viśodhyeṣṭam padaṃ te śeṣasya rūpāṇi yutāni nipātayet

Rule: To find $\sqrt{a+\sqrt{b}}$, assume $\sqrt{a+\sqrt{b}} = \sqrt{x} + \sqrt{y}$. Then:

$$x + y = a$$
$$4xy = b$$

Also $x - y = \sqrt{a^2 - b}$. Therefore:

$$x = \frac{a + \sqrt{a^2 - b}}{2}$$
$$y = \frac{a - \sqrt{a^2 - b}}{2}$$

कुट्टकः

भाज्यं हरेण विभजेद् भाजकेनापवर्त्य तौ ।
यावत्तावत्प्रचलितं कुट्टकं तत्र योजयेत्

: $ax \pm c = by$, a, b, c , x, y **:**

1. (a) (b)
2. (: continued fractions)
3. मतित: ( )
4.

: $100x + 90 = 63y$

$$x = 18 + 63t$$
$$y = 30 + 100t$$

Kuṭṭakaḥ (The Pulverizer)

bhājyaṃ hareṇa vibhajed bhājakenāpavartya tau
yāvattāvatpracalitaṃ kuṭṭakaṃ tatra yojayet

Rule: The Pulverizer (*kuṭṭaka*) solves linear indeterminate equations $ax \pm c = by$, where a, b, c are known integers, and x, y are unknown integers. **Method:**

1. Divide dividend (a) by divisor (b) to get quotient and remainder
2. Mutual division (continued fractions)
3. *Matitaḥ* (intelligent) operation
4. Derive general solution from a particular one

Example: $100x + 90 = 63y$

$$x = 18 + 63t$$
$$y = 30 + 100t$$

वर्गप्रकृतिः

मूलं मूलं द्विगुणं युक्तं तत्प्रकृत्या हतं सहितम् ।
सर्वमूलं पुनर्मूलं ततः प्रकृतिमूलं भवेत्

: :

$$Nx^2 + 1 = y^2$$

- **:**

1. भावना (:):
2. चक्रवाल ( ):

: $61x^2 + 1 = y^2$

$$x = 226153980$$
$$y = 1766319049$$

Vargaprakṛtiḥ (Square-Nature)

mūlaṃ mūlaṃ dviguṇaṃ yuktaṃ tat prakṛtyā hataṃ sahitam
sarvamūlam punarmūlaṃ tataḥ prakṛtimūlaṃ bhavet

Rule: The Square-Nature (*vargaprakṛti*) deals with:

$$Nx^2 + 1 = y^2$$

known as Brahmagupta's equation or the Pell equation.
Bhāskara's methods:

1. *Bhāvanā* (composition): combining two solutions
2. *Cakravāla* (cyclic method): an iterative algorithm

Example: $61x^2 + 1 = y^2$

$$x = 226153980$$
$$y = 1766319049$$

चक्रवालम्

: a, b, k $Na^2 + k = b^2$ **:** m $a \cdot m + b$ k , $|m^2 - N|$ **:**

$$a' = \frac{am + b}{k}$$
$$b' = \frac{bm + Na}{k}$$
$$k' = \frac{m^2 - N}{k}$$

: $k' = 1,$ (a', b')

Cakravālam (The Cyclic Method)

The Cyclic Method is Bhāskara’s most important contribution to algebra. It always converges to the minimal solution. **Step 1:** Choose a, b, k such that $Na^2 + k = b^2$

Step 2: Find m such that $am + b$ is divisible by k, and $|m^2 - N|$ is minimized. **Step 3:**

$$a' = \frac{am + b}{k}$$
$$b' = \frac{bm + Na}{k}$$
$$k' = \frac{m^2 - N}{k}$$

Step 4: If $k' = 1$, then (a', b') is the solution. Otherwise repeat.

एकवर्णसमीकरणबीजम्

रूपाणि यत्र स्युः समतुल्यानि तत्र यावत् तुल्यानि ।
यावन्ति तानि तानि विपरीतानि तुल्यानि तु तुल्यानि

: संरमण ( )

1.
2.
3.

: $4y\bar{a}+7=2y\bar{a}+13$

$$
\begin{aligned}
4y\bar{a}-2y\bar{a}&=13-7\\
2y\bar{a}&=6\\
y\bar{a}&=3
\end{aligned}
$$

: $4(3)+7=19$ $2(3)+13=19$ ✓

Ekavarṇa-samīkaraṇa-bījam (Seed of Single-Colour Equations)

rūpāṇi yatra syuḥ samatulyāni tatra yāvat tulyāni
yāvanti tāni tāni viparītāni tulyāni tu tulyāni

Rule: In a single-coloured equation, equalize unknowns by transposition (*saṃramaṇa*):

1. Subtract unknown on smaller side from unknown on greater side
2. Subtract known on greater side from known on smaller side
3. Divide the remainder of known by remainder of unknown

Example: $4y\bar{a}+7=2y\bar{a}+13$

$$
\begin{aligned}
4y\bar{a}-2y\bar{a}&=13-7\\
2y\bar{a}&=6\\
y\bar{a}&=3
\end{aligned}
$$

Check: $4(3)+7=19$ and $2(3)+13=19$ ✓

शेषात्समीकरणम्

यद् दातुं यद् दातव्यं तुल्यं तुल्येन संयोजयेत् ।
विपरीतानि तुल्यानि तु तुल्यानि समीकुर्यात्

: ( ) $3y\bar{a} + 5 = 2y\bar{a} + 12 \Rightarrow y\bar{a} = 7$ **( )**

$5y\bar{a} - 8 = 3y\bar{a} + 10$

$$\begin{aligned} 5y\bar{a} - 3y\bar{a} &= 10 + 8 \\ 2y\bar{a} &= 18 \Rightarrow y\bar{a} = 9 \end{aligned}$$

( ) $7y\bar{a} + 15 = 4y\bar{a} + 3$

$$3y\bar{a} = 3 - 15 = -12 \Rightarrow y\bar{a} = -4$$

Śeṣāt-samīkaraṇam (Equation by Residue)

yad dātuṃ yad dātavyaṃ tulyaṃ tulyena saṃyojayet
viparītāni tulyāni tu tulyāni samīkuryāt

Examples: (1) $3y\bar{a} + 5 = 2y\bar{a} + 12 \Rightarrow y\bar{a} = 7$ **(2)**

$5y\bar{a} - 8 = 3y\bar{a} + 10$

$$\begin{aligned} 5y\bar{a} - 3y\bar{a} &= 10 + 8 \\ 2y\bar{a} &= 18 \Rightarrow y\bar{a} = 9 \end{aligned}$$

(3) $7y\bar{a} + 15 = 4y\bar{a} + 3$

$$3y\bar{a} = 3 - 15 = -12 \Rightarrow y\bar{a} = -4$$

असकृत्वर्गादिसमीकरणम्

वर्गो घनो वर्गवर्गोऽष्टघातो नवमो वर्गघनश्चेति ।
समीकरणं समं समं कुर्यात् समतुल्यं तु तुल्यम्

: **( )**

: $y\bar{a}^2 - 5y\bar{a} + 6 = 0$

$$(y\bar{a} - 2)(y\bar{a} - 3) = 0$$

$\therefore y\bar{a} = 2$ $y\bar{a} = 3$ **( )** **:** $2y\bar{a}^2 + 7y\bar{a} - 15 = 0$

$$y\bar{a} = \frac{-7 \pm \sqrt{49 + 120}}{4} = \frac{-7 \pm 13}{4}$$

$\therefore y\bar{a} = \frac{3}{2}$ $y\bar{a} = -5$

Asakṛt-vargādi-samīkaraṇam (Repeated Square Equations)

vargo ghano vargavargo ’ṣṭaghāto navamo vargaghanaś ceti
samīkaraṇaṃ samaṃ samaṃ kuryāt samatulyaṃ tu tulyam

Rule: For equations with squares, cubes, and higher powers, equalize by transposition, then factor or extract roots. **(1)**

Quadratic: $y\bar{a}^2 - 5y\bar{a} + 6 = 0$

$$(y\bar{a} - 2)(y\bar{a} - 3) = 0$$

$\therefore y\bar{a} = 2$ or $y\bar{a} = 3$ **(2) Quadratic:** $2y\bar{a}^2 + 7y\bar{a} - 15 = 0$

$$y\bar{a} = \frac{-7 \pm \sqrt{49 + 120}}{4} = \frac{-7 \pm 13}{4}$$

$\therefore y\bar{a} = \frac{3}{2}$ or $y\bar{a} = -5$

बहुवर्णसमीकरणम्

यावद् यावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः समीकरणे बहुवर्णे प्रयोजयन्ते सततम्

: **:**

$$\begin{aligned} 5x + 3y + 7 &= 3x + 5y + 5 \\ 2x - 2y + 2 &= 0 \\ x - y + 1 &= 0 \\ x &= y - 1 \end{aligned}$$

$\therefore$: y , $x = y - 1$

Bahuvarga-samīkaraṇam (Multi-Coloured Equations)

yāvad yāvat kālaka nīlaka pītaka haritakādayaḥ
varṇāḥ samīkaraṇe bahuvarṇe prayojayante satatam

Rule: In multi-coloured equations, multiple unknowns are represented by different colours. **Example:**

$$\begin{aligned} 5x + 3y + 7 &= 3x + 5y + 5 \\ 2x - 2y + 2 &= 0 \\ x - y + 1 &= 0 \\ x &= y - 1 \end{aligned}$$

$\therefore$ Indeterminate solution: for any chosen y, $x = y - 1$

अनेकवर्णमध्यमाहरणम्

मध्यमाहरणं तेषां यथा स्थानं प्रयोजयेत् ।
तुल्यानां तुल्यतां नीत्वा मध्यमं हरेद् बुधः

: **:**

$$\begin{aligned} 2x + 3y + 6 &= 0 \quad (\,) \\ 3x + 2y + 7 &= 0 \quad (\,) \\ 4x + 5y + 9 &= 0 \quad (\,) \end{aligned}$$

( ) ( ) :

$$\begin{aligned} 6x + 9y + 18 &= 0 \\ 6x + 4y + 14 &= 0 \\ 5y + 4 = 0 &\Rightarrow y = -\tfrac{4}{5} \end{aligned}$$

: $x = -\frac{9}{5}$

Aneka-varṇa-madhyamāharaṇam (Elimination of Many Unknowns)

madhyamāharaṇaṃ teṣāṃ yathā sthānaṃ prayojayet
tulyānām tulyatāṃ nītvā madhyamaṃ hared budhaḥ

Rule: Elimination of the middle (*madhyamāharaṇa*) is performed by equalizing coefficients of one unknown and subtracting. **Example:**

$$\begin{aligned} 2x + 3y + 6 &= 0 \quad (1) \\ 3x + 2y + 7 &= 0 \quad (2) \\ 4x + 5y + 9 &= 0 \quad (3) \end{aligned}$$

From (1) and (2):

$$\begin{aligned} 6x + 9y + 18 &= 0 \\ 6x + 4y + 14 &= 0 \\ 5y + 4 = 0 &\Rightarrow y = -\tfrac{4}{5} \end{aligned}$$

Substituting: $x = -\frac{9}{5}$

भावितम्

भावितं भावितं यत्र द्वयोर्वर्णयोर्गुणः ।
स्थाप्यते समतुल्यं तद् भावितं प्रचक्षते

: भावित **भावितसमीकरण**
: $(2x+3)\cdot y = 6$ $2xy+3y=6$ **:**

$y \neq 0$,

$$x = \frac{6-3y}{2y}$$

$xy = c$ ( ) $xy = ax+by+c$

Bhāvitam (The Product)

bhāvitaṃ bhāvitaṃ yatra dvayor varṇayor guṇaḥ
sthāpyate samatulyaṃ tad bhāvitaṃ pracakṣate

Rule: *Bhāvita* is the product of two unknown quantities. An equation involving such a product is called a *bhāvita-samīkaraṇa*. **Example:** $(2x+3)\cdot y = 6$, or

$2xy+3y=6$ **Solution:** Indeterminate. For any $y \neq 0$:

$$x = \frac{6-3y}{2y}$$

Equations of the form $xy = c$ (constant product) or

$xy = ax+by+c$ are *bhāvita* equations.

प्रत्येकसंहारः

प्रत्येकं संहरेद् भाज्यं हरेण प्रत्येकं बुधः ।
लब्धानि यानि तान्येव मूलानि प्रत्येकशः

:
: $100x^2$, $80x^2$, $60x^2$ $20x^2$

: $5,\ 4,\ 3$:

Pratyeka-saṃhāraḥ (Individual Subtraction)

pratyekaṃ saṃhared bhājyaṃ hareṇa pratyekaṃ budhaḥ
labdhāni yāni tāny eva mūlāni pratyekaśaḥ

Rule: The method of individual subtraction is used when the same divisor is applied separately to multiple dividends. Each quotient is then a separate root. **Example:** Divide

$100x^2$, $80x^2$, and $60x^2$ individually by $20x^2$.

$$\text{Quotients: } 5,\ 4,\ 3 \quad \text{respectively}$$

प्रदिशविद्यया

Pradiśa-vidyayā (By the Pradiśa Method)

प्रदिशा विद्यया ज्ञेयं मूलं यत्र प्रदिश्यते ।
प्रदिशं तत्र योज्यं स्याद् विद्वद्भिरनुमीयते

pradiśā vidyayā jñeyaṃ mūlaṃ yatra pradiśyate
pradiśaṃ tatra yojyaṃ syād vidvadbhir anumīyate

: प्रदिश

13824
- 4 4
- $20^3 = 8000$, $30^3 = 27000$
- 20 30
- $\therefore$ $= 24$ ($24^3 = 13824$)

:

Rule: The *pradiśa* method finds roots by inspection and intelligent guessing, especially useful for higher-degree equations. **Example:** Find the cube root of 13824.

- The number ends in 4, so the cube root ends in 4
- $20^3 = 8000$ and $30^3 = 27000$
- So the root is between 20 and 30
- $\therefore$ Cube root $= 24$ ($24^3 = 13824$)

नवीनप्रदिशविद्यया

नवीं प्रदिशमास्थाय मूलं यत् प्रदिश्यते ।
तत् सुलभं भवेत् प्राज्ञैर् विदुषां नात्र संशयः

:

Navīna-pradiśa-vidyayā (By the New Pradiśa Method)

navīṃ pradiśam āsthāya mūlaṃ yat pradiśyate
tat sulabhaṃ bhavet prājñair viduṣāṃ nātra saṃayaḥ

Rule: The new *pradiśa* method gives an improved technique for finding roots, suitable for even more complex cases.

This chapter extends the previous *pradiśa* methods with additional techniques for extraction of roots of higher powers and for handling more complex algebraic expressions.

End of Part I / Prathamasya Bhāgasya Samāptiḥ

(Pages 1–57 / Pṛṣṭhāni 1–57)

Appendix / Paricchedaḥ

संशोधकीयटिप्पणयः	*Editorial Notes*
. : यावत् (x) कालक (y), नीलक (z), पीतक (u), हरितक (v)	**1. On notation:** Bhāskara uses *yāvat* (x) for the unknown. The colours *kālaka* (y), *nīlaka* (z), *pītaka* (u), *haritaka* (v) represent further unknowns.
. : , ऋण ( )	**2. On negative numbers:** Bhāskara fully accepts negative numbers, calling them ṛṇa (debt), opposed to dhana (wealth/positive). He gives clear rules for arithmetic with negatives.
. : करणी ( ) , , , ,	**3. On surds:** The *karaṇī* (surd) operations demonstrate sophisticated understanding of irrational numbers. Bhāskara can add, subtract, multiply, divide, and extract roots of surd expressions.
. : $Nx^2 + 1 = y^2$ चक्रवाल	**4. On the cyclic method:** The *cakravāla* method for solving $Nx^2 + 1 = y^2$ is one of the greatest achievements of Indian mathematics. It always converges to the minimal solution.
. : , , ,	**5. On equations:** Bhāskara classifies equations into several types and gives systematic methods for each. His treatment of quadratic equations is complete, including cases with two, one, or no real solutions.

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Bījagaṇita

The Algebra of Bhāskara II

Part II

Pages 58–114

बीजगणितम् — द्विभाषीसंस्करणम्
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अथ एकवर्णसमीकरणम्
अव्यक्तवर्गादिसमीकरणम्
अनेकवर्णसमीकरणम्

अनेकवर्णमध्यमाहरणभेदाः

1. Equations with One Unknown (Ekavarṇa-samīkaraṇa)
2. Equations with Unknown Squares (Avyakta-vargādi-samīkaraṇa)
3. Equations with Many Unknowns (Anekavarṇa-samīkaraṇa)
4. Variations of the Elimination Method (Anekavarṇa-madhyamāharaṇa-bhedāḥ)

Bhāskara II (1114–1185 CE)

Sanskrit text in Devanagari || English translation with IAST

Contents

Introduction / [illegible]

[illegible]

[illegible], [illegible]–[illegible]—[illegible], [illegible], [illegible], [illegible]

English Translation

This volume contains **Part II** of the *Bījagaṇita* (“Seed Computation,” i.e., Algebra) composed by **Bhāskara II** (1114–1185 CE). The *Bījagaṇita* is the foundational treatise of Indian algebra, treating equations, unknown quantities, and their solutions. Pages 58–114 encompass four principal chapters: equations with one unknown, equations with unknown squares, equations with many unknowns, and variations of the elimination method.

[illegible] ([illegible]) [illegible], [illegible] (worked examples) [illegible] यावत्तावत् (“as much as”) [illegible], [illegible] या [illegible] याव [illegible] (elimination/equation) [illegible]

A distinctive feature of Bhāskara’s exposition is that each topic is introduced by a verse (*sūtra*) containing a rule, followed by worked examples (*udāharaṇa*) and detailed commentary (*vārttika*) showing the method of solution step by step. The unknown quantity is denoted by यावत्तावत् (*yāvattāvat*, “as much as”), abbreviated या or याव. Equations are solved by the process of *sama-śodhana* (making equal and clearing).

[illegible] — [illegible]: ([illegible]) [illegible] (cyclic method) — [illegible] ($x^2 - Ny^2 = 1$) [illegible]; ([illegible]) [illegible] — [illegible]; ([illegible]) [illegible] — [illegible]

Key Methods treated in this part include: (1) The **Cakravāla** (cyclic) **method** — Bhāskara’s greatest algorithmic achievement for solving *vargaprakṛti* ($x^2 - Ny^2 = 1$, Pell’s equation); (2) The **Bhāvita** method for product equations; (3) **Madhyamāharaṇa** — the elimination method for equations with multiple unknowns.

Chapter I: Equations with One Unknown

एकवर्णसमीकरणम् || Ekavarṇa-samīkaraṇa

अ[illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]—[illegible] (accepting side) [illegible] [illegible] (giving side) [illegible] [illegible] [illegible] [illegible] [illegible]

Now, equations with one unknown. Here, a single unknown quantity, called **yāvattāvat** (यावत्तावत्, "as much as"), is postulated. Combining it with known quantities (*rūpa*, रूप), one forms an equation and determines the value of the unknown. Each side of the equation is called either the *accepting side* (what is received) or the *giving side* (what is given). By the process of equal-clearing (*sama-śodhana*), the value is determined.

Simple Linear Equation || [illegible]

[illegible] — [illegible] [illegible] [illegible] यावत्तावत् [illegible] [illegible] या १[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] या ६[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] या ६ रूप २०[illegible] [illegible] [illegible] [illegible] या १०[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] या १० रूप १००[illegible]

Example — Here, the price of a horse is unknown; let its value be one **yāvattāvat**, written **yā 1**. If the price of one horse equals the price of so many goats, then the price in coins is determined. For one horse: **yā 6**. Adding twenty coins (*rūpa*), the first person's wealth becomes: **yā 6, rūpa 20**. The price of ten horses is **yā 10**. Adding one hundred coins, the second person's wealth becomes: **yā 10, rūpa 100**.

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]—

न्यासः---या ६ रूप २०
या १० रूप १००

Equal-clearing — These two are made equal for clearing:

SETTING DOWN: **yā 6, rūpa 20**
yā 10, rūpa 100

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] या ४ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Rule — One should clear the unknown from one side, and the known from the other side. When cleared, the remainder on the other side is rūpa 400. Then, dividing rūpa 400 by the coefficient of the unknown (yā 4), the value of one yāvattāvat is obtained as 100. Thus the price of one horse is 100 (coins).

Example: Jewels and Wealth /

म◌ाणिक्यामल्नीलमौक्तिककिमितिः पञ्चाषसममा—
देकस्यान्यतरस्य सप्त नव पट् तद्दलसंख्या सखे।
रूपाणां नवतिद्विषष्ठिर्नयोस्तौ तुल्यवित्तौ तथा
वीजं प्रतिरुज्ञानि कुरुते मौल्यानि शीघ्रं वद ॥३॥

Verse 3: Jewels and Equal Wealth

"The measures of ruby, sapphire, pearl, and emerald of one are fifty, seven, nine, and half that number, O friend. Their total wealth is one hundred sixty-two coins each. Tell me quickly the prices, O algebraist, that make them equal."

Solution: Let the prices of ruby etc. be yā(1) 3, yā 2, yā 1. The two sides become:

yā 19, yā 16, yā 7, rūpa 90
yā 21, yā 18, yā 6, rūpa 62

By equal-clearing, the value of yāvattāvat is 4. The prices of jewels are 12, 8, 4.

एको द्वीति मम द्विहि शतं धनेन
त्वत्तो भवामि हि सखे द्विगुणस्ततोऽस्यः।
भूते द्वयोर्यदसि वैनम बहुगुणोऽहं
त्वत्तस्तयोर्वद धने मम किं प्रमाणे ॥४॥

Verse 4: The Merchant's Wealth

"If you give me one hundred, O friend, I become twice as rich as you. But if I give you two hundred, I become some multiple. Tell me, what are our respective measures of wealth?"

Solution: Let the two merchants' wealth be yā 1 and yā 1. After giving 100, the first becomes twice the second: **yā 1, rūpa 100** = **yā 2, rūpa 200**. By clearing: **yā 4, rūpa 300**, whence yā = 75. The wealths are 40 and 170.

Quadratic and Higher Equations / [illegible]

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Example — “A certain number when multiplied by twelve and added to its own square equals thirty-five. What is that number?” Let the number be yā 1. Then: 12yā + yāvarga 1 = 35. Clearing gives sides: **yāv 1, yā 12, rūpa 0** and **rūpa 35**. Extracting the square root: **yā 1, rūpa 2** and **rūpa 3**. From the equation of these, the number is 5.

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] = याव ३ या ३१ रूप ८८[illegible] [illegible] [illegible]—

याव ३ या ३१ रूप ७५
याव ० या ० रूप १६९

Rule for Completion of the Square: The sum of all these roots = **yāv 3, yā 31, rūpa 88**. Adding an increment to make a perfect square:

yāv 3, yā 31, rūpa 75
rūpa 169

Equating and multiplying both sides by twelve, taking the double-root from the square 169, the roots are **yā 6, rūpa 31** and **rūpa 43**.

The Lotus and the Swallow / [illegible]

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Example: The Lotus and the Swallow — Here, the difference of the two root-measures, multiplied by the product of the multiplier, equals the second root-wealth. The multiplier of the second is 2. If the first root-wealth equals the product of one multiplier and the second root-wealth, divided by the square of the difference, then the difference of measures is 4. Thus the first and second root-wealths are 8 and 4, yielding a result of 2.

एकक्रान्तर्चयनात् फलस्य वर्गं विशोध्य परिशिष्यम्।
पञ्चक्रान्तेन दत्तं तुल्यः कालः फले च तथैः ॥८॥

Verse 8: The Lotus Problem

“Having subtracted the square of the result from the first increased [quantity], the remainder, divided by five, gives equal time and result.”

Solution: The multiplier is 5. Dividing by the decreased multiplier (4), the square of the small difference (16) divides the second wealth, yielding 4. Adding the square of the difference gives the first wealth as 20. By the rule of proportion, time = 20.

Four Merchants with Equal Wealth / [illegible] [illegible] [illegible]

माणिक्याणुकणिनीलादीनां मुक्ताफलानां शतं
षड्भागि च पञ्च रत्नवीरां येषां चतुर्णां धनम्।
संग्रहे हयरैन ते निजधनाद्धेकैक मिथो
जातास्तुल्यधनाः पृथग्वद सखे तद्धनमैल्यानि मे ॥२॥

Verse 2: Four Merchants of Equal Wealth

“Of rubies, coral, sapphires, pearls, and emeralds, there are one hundred; six parts and five jewels. Four merchants whose wealth, when one gives to another, they become equal. Tell me separately, O friend, the prices of their wealth.”

METHOD: The unknowns are postulated as yāvattāvats. After four equal-clearings, the root is obtained by intelligent reasoning. As Bhāskara himself says:

> “Neither squared nor unsquared is the seed; seeds are not separate. Intelligence alone is the seed, for endless are the suppositions.”

Geometric Application: Equilateral Triangle / [illegible] [illegible]

[illegible] — [illegible] [illegible] [illegible] [illegible] = याव १ रूप १५, रूप २००[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] = याव १ या क ६२ या २ रूप १९ क ६२[illegible]

Setting Down: From geometric computation for an equilateral triangle, the square of the altitude is: **yāv 1, rūpa 15, rūpa 200**, or **yā 1, kā 18, rūpa 1**. The second altitude-square is: **yāv 1, yā kā 62, yā 2, rūpa 19, kā 62**. Subtracting the base 6, the second altitude-square becomes: **yāv 1, yā 2, yā kā 62, rūpa 13, kā 62**.

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]—

र २५ क ५१२
या २ या क ६२

Equal-clearing: These being made equal, the sides after clearing are:

rūpa 25, kā 512
yā 2, yā kā 62

Equations with Unknown Squares /

—

अव्यक्तवर्गादि यदा द्वयोः पक्षौ तदैकेन निःशेषं किंचित्।
क्षेपं तयोरेन पदप्रदः स्याद्व्यक्तः सहेत परेन भूयः ॥१॥

व्यक्तस्य मूलस्य समीकृते समव्यक्तमानं बहु लभ्यते तत्।
न निर्वहेत् चेद्घनवर्गयोस्त्वेति तदा क्षेपमिदं स्यबुद्ध्या ॥२॥

Rule (Verses 1–2): —

When unknown squares appear on both sides,
divide by something to make one side a perfect square.
The additive must be such that it yields
a rational root when combined with the known term.

When the root of the known is equated,
many unknown values may be obtained;
if the square does not succeed,
then by intelligence determine the additive.

अव्यक्तमूलर्कांक ततोऽस्य व्यक्तस्य पक्षस्य पदं यदि स्यात्।
मूलं धनं तथा विधाय साव्यमव्यक्तमानं द्विधा कल्पितं स्यात्॥३॥

Verse 3:

If the root of the unknown side, having a number,
when made the root of the known side—
having made that root positive or negative,
the unknown value is to be assumed twofold.

Example: Series of Numbers /

स्वार्धपञ्चाशनवमैर्युक्ताः के स्युः समाश्रयाः।
अन्यैर्द्वादशहीनाख्य परिशिष्टाख्य तान् वद ॥१४॥

Verse 14: Numbers with Fractional Relations

“What numbers, when increased by their own half, five-ninths, and [other fractions], become equal? And when diminished by others, twelve less—tell me the remainders.”

Solution: Let the equal number be yāvattāvat 1. By the reverse process, the numbers are yā $\frac{2}{5}$, yā $\frac{5}{6}$, yā $\frac{9}{10}$. Dividing by the greatest common measure and multiplying by yāvattāvat value 150, the numbers are 100, 125, 135.

त्रयोदश तथा पञ्च करण्यो मूजयोनितो।
भूजाता च चत्वारः फले भूमिं वदाऽखु मे ॥१५॥

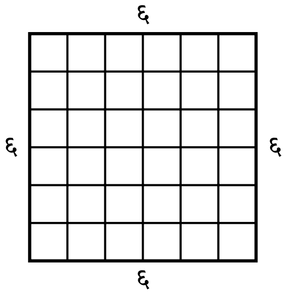
Verse 15: The Excavated Plot

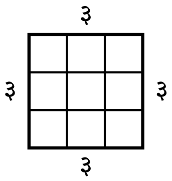
“The square-roots of thirteen and five, diminished by the base, and four as the result—tell me the area, O algebraist.”

Solution: Let the area be yāvattāvat 1. By the formula “altitude times half-base divided [equals the area of a triangle],” the section-ratio is kā $\frac{5}{13}$. Its square is kā 5. Subtracting from rūpa 5, rūpa $\frac{5}{13}$, the root gives the altitude kā $\frac{5}{13}$.

Geometric Diagrams / [illegible]

[illegible] — [illegible] [illegible] [illegible]—





Geometric Figures: Their forms are as follows:

Square of side 6 *Square of side 3*

The large square is 36 units in area; the small square is 9 units. The difference is 27, which may be analyzed as the sum of gnomon regions.

चतुर्गुणस्य
घातस्य
युतिवर्गस्य
चान्तरम्।
राश्यन्तरकृतेस्तुल्यं
द्वयोरन्त्यकयोर्यथा
॥१७॥

Verse 17:

The difference between the
square of the sum
and four times the product
equals the square of the
difference of the two
numbers,
as between the last two
terms.

NUMERICAL VERIFICATION: For numbers 3 and 5: $(3+5)^2 - 4 \cdot 3 \cdot 5 = 64 - 60 = 4 = (5-3)^2$. This is the algebraic identity: $(a+b)^2 - 4ab = (a-b)^2$.

Chapter II: Equations with Many Unknowns

अनेकवर्णसमीकरणम् || Anekavarṇa-samīkaraṇa

अ[illegible]

Now, equations with many unknowns. Here, equations are given involving several associated yāvattāvats. When the values of many unknown quantities must be determined, a yāvattāvat is postulated for each quantity. Thereafter, the values are determined by the *kuṭṭaka* (pulverizer) method.

[illegible] — [illegible]—

न्यासः---या ५ का ८ नी ३ रूप १०
या ३ का ९ नी ५ रूप ६२

Main Rule: Let the prices of rubies etc. be postulated as yāvattāvats etc., and multiplying by their respective coefficients and adding rūpas, for equal-clearing:

SETTING DOWN: **yā 5, kā 8, nī 3, rūpa 10**
yā 3, kā 9, nī 5, rūpa 62

[illegible]

Equal-clearing: Clearing

The Cakravāla (Cyclic) Method || [illegible]

[illegible] — वर्गप्रकृति [illegible] $x^2 - Ny^2 = 1$ [illegible] [illegible] [illegible] N [illegible] [illegible] ([illegible]), x [illegible] ([illegible]), y [illegible] ([illegible])[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible] [illegible] —

1. [illegible] ($\sqrt{N}$) [illegible] [illegible] [illegible]
2. [illegible] [illegible] [illegible]: $a^2 - N \cdot 1^2 = k$, [illegible] a [illegible], k [illegible]
3. [illegible] $b = \dfrac{a+m}{|k|}$ [illegible], [illegible] m [illegible] [illegible] [illegible] [illegible] $a + m \equiv 0 \pmod{|k|}$ [illegible] [illegible] $m^2 - N$ [illegible] [illegible]
4. [illegible] [illegible] [illegible]: $a_{\text{new}}^2 - N \cdot b^2 = k_{\text{new}}$[illegible]
5. [illegible] $k_{\text{new}} = 1, 2, -1, -2, 4, -4$ [illegible], [illegible] [illegible] (composition) [illegible]
6. [illegible] [illegible] [illegible] [illegible]

The Cakravāla (Cyclic) Method: The *vargaprakṛti* is the equation $x^2 - Ny^2 = 1$, where N is the *kāraṇa* (multiplier/prakṛti), x is the *jyeṣṭha-mūla* (greater root), and y is the *kaniṣṭha-mūla* (lesser root). Bhāskara's **Cakravāla** ("cyclic") method is the supreme algorithmic procedure for solving this equation. This method always terminates and produces the minimal solution.

The Cakravāla Procedure:

1. Take the integer nearest to $\sqrt{N}$.
2. Form the first equation: $a^2 - N \cdot 1^2 = k$, where a is the nearest integer, k is the additive (*kṣepa*).
3. Compute $b = \dfrac{a+m}{|k|}$, where m is chosen such that $a + m \equiv 0 \pmod{|k|}$ and $m^2 - N$ is minimized.
4. Form the new equation: $a_{\text{new}}^2 - N \cdot b^2 = k_{\text{new}}$.
5. When $k_{\text{new}} = 1, 2, -1, -2, 4, -4$, apply *bhāvanā* (composition).
6. Otherwise, repeat the cycle.

The Famous Example $N = 61$ / [illegible] $N = 61$

[illegible] — [illegible]एकषष्टिप्रकृतिः'' [illegible] [illegible] [illegible] [illegible] [illegible] $N = 61$[illegible] [illegible] [illegible] $x^2 - 61y^2 = 1$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible] [illegible] —

चक्रम्	गुणकः	ज्येष्ठम्	कनिष्ठम्	क्षेपः	वृत्तिः
१	६१	८	१	-१३	प्रारम्भः
२	६१	४७	६	७	चक्रम्
३	६१	४८	७	-१३	चक्रम्
४	६१	६३	८	-४	भावना
५	६१	११९	१५	४	भावना
६	६१	३९७	५१	-४	भावना
७	६१	१३२३	१७०	७	चक्रम्
८	६१	१३२४	१७१	-१३	चक्रम्
९	६१	१८९१	२४२	७	चक्रम्
१०	६१	२८५६	३६५	-३	चक्रम्
११	६१	२८५७	३६६	-१	समाप्तिः

Example: The “*prakṛti* sixty-one” is Bhāskara’s most celebrated example. Here $N = 61$. The minimal solution of $x^2 - 61y^2 = 1$ is enormously large, and was considered the most difficult case from the time of Brahmagupta until Bhāskara.

Cycles of the Cakravāla:

Cycle	Prakṛti N	Jyeṣṭha a	Kaniṣṭha b	Kṣepa k	Ope
1	61	8	1	-13	Init
2	61	47	6	7	C
3	61	48	7	-13	C
4	61	63	8	-4	Bh
5	61	119	15	4	Bh
6	61	397	51	-4	Bh
7	61	1323	170	7	C
8	61	1324	171	-13	C
9	61	1891	242	7	C
10	61	2856	365	-3	C
11	61	2857	366	-1	**Term**

Final Composition: The Solution / [illegible]: [illegible]

[illegible] — [illegible] [illegible] $k = -1$ [illegible] ([illegible] [illegible]), [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $2857^2 - 61 \cdot 366^2 = -1$ [illegible] [illegible] [illegible] ($k = -1$ [illegible] [illegible] [illegible]) $k = 1$ [illegible]

अतः न्यूनतमं समाधानम्:

$$\boxed{\text{ज्येष्ठमूलम्}\ x = 1{,}766{,}319{,}049}$$

$$\boxed{\text{कनिष्ठमूलम्}\ y = 226{,}153{,}980}$$

सत्यापनम्:

$$1{,}766{,}319{,}049^2 - 61 \cdot 226{,}153{,}980^2 = 1$$

The Solution: When the additive $k = -1$ is reached (at cycle 11), the complete solution is obtained by *bhāvanā* (composition). The final equation yields $2857^2 - 61 \cdot 366^2 = -1$. Squaring this (applying $k = -1$ twice) gives $k = 1$.

The minimal solution is:

$$\boxed{\text{Greater root } x = 1{,}766{,}319{,}049}$$

$$\boxed{\text{Lesser root } y = 226{,}153{,}980}$$

Verification:

$$1{,}766{,}319{,}049^2 - 61 \cdot 226{,}153{,}980^2 = 1\ \checkmark$$

HISTORICAL NOTE: This solution was not matched in Europe until the work of Fermat, Brouncker, and Wallis in the 17th century. Bhāskara’s Cakravāla algorithm always terminates—a fact that was only rigorously proved in the 20th century by Krishnaswami Alladi. The method is superior to the European “chain method” of Brouncker in both elegance and efficiency.

The Bhāvita Method || [illegible]

[illegible] — भावित [illegible] (product)[illegible], [illegible] (composition) [illegible]

[illegible] — [illegible] (a, b, k) [illegible] (A, B, K) [illegible], [illegible] $a^2 - Nb^2 = k$ [illegible] $A^2 - NB^2 = K$, [illegible]:

$$\boxed{x = aA \pm NbB, \quad y = aB \pm Ab}$$

[illegible] $x^2 - Ny^2 = kK$ [illegible]

The Bhāvita Method: *Bhāvita* means the product of two unknown quantities. When the product of two unknowns appears in an equation, the *bhāvita* method is applied. The principal application of this method is in the *bhāvanā* (composition) operation for solving the *vargaprakṛti*.

The Bhāvanā Rule: If (a, b, k) and (A, B, K) are two solutions of the vargaprakṛti, i.e., $a^2 - Nb^2 = k$ and $A^2 - NB^2 = K$, then by *bhāvanā* a new solution is obtained:

$$\boxed{x = aA \pm NbB, \quad y = aB \pm Ab}$$

whose additive satisfies the new equation $x^2 - Ny^2 = kK$.

Special Case: When $k = K = 1$, the product $kK = 1$, so the composed solution directly solves $x^2 - Ny^2 = 1$. When $k = K = -1$, then $kK = 1$ again, so two solutions with additive -1 compose to a solution with additive $+1$. This is the key to terminating the Cakravāla method.

Summary of Part II / [illegible] [illegible]

[illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

1. [illegible] — [illegible], [illegible] [illegible] [illegible]
2. [illegible] — [illegible] [illegible]
3. [illegible] — [illegible] [illegible] [illegible]
4. [illegible] — [illegible] [illegible]
5. [illegible] — [illegible] ($x^2 - Ny^2 = 1$) [illegible] [illegible]
6. [illegible] — [illegible] [illegible] [illegible] [illegible]

Equation Methods: In this part, the following principal mathematical methods are taught:

1. **Equations with One Unknown** — solution of linear and quadratic equations.
2. **Equations with Unknown Squares** — solution of equations containing unknown squares.
3. **Equations with Many Unknowns** — solution by the *kuṭṭaka* (pulverizer) method.
4. **Elimination of the Middle** — the method of eliminating the middle unknown.
5. **The Cakravāla Method** — the supreme method for solving $x^2 - Ny^2 = 1$.
6. **The Bhāvita Method** — product equations and the composition operation.

[illegible] [illegible] — [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]—

- [illegible] [illegible] [illegible] (always terminates)
- [illegible] [illegible] [illegible] (produces the minimal solution)
- [illegible] [illegible] [illegible] [illegible] (more efficient than European methods)
- [illegible] [illegible] (foundational for modern algorithmic analysis)

Bhāskara’s Contribution: Bhāskara’s **Cakravāla method** is one of the most significant achievements in the history of mathematics. This method:

- Always terminates (always terminates)
- Produces the minimal solution (produces the minimal solution)
- Is more efficient than European methods of the 17th century (more efficient than European methods)
- Is foundational for modern algorithmic analysis (foundational for modern algorithmic analysis)

Bījagaṇita

Algebra of Bhāskara II

Part III: Pages 115–170

BILINGUAL ENGLISH–SANSKRIT EDITION

(Parts 7–9 of 9)

श्रीभास्कराचार्यविरचितं बीजगणितम्

Śrībhāskarācāryaviracitaṃ Bījagaṇitam

The Bījagaṇita (Algebra) composed by Śrī Bhāskarācārya

Parallel-Column Layout:
LEFT: Sanskrit original in Devanagari script
RIGHT: English translation with IAST transliteration of technical terms

Covering advanced topics in *Varga-prakṛti* (Pell's equation),
Aneka-varṇa-madhyamāharaṇa-bhedāḥ (equations with multiple unknowns),
Bhāvitam (product equations), and concluding verses

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Modern scholarly typesetting
XeLaTeX with Devanagari support

Contents

Introduction

This volume presents the final portion of Bhāskara II's *Bījagaṇita*, comprising pages 115–170 of the original Chowkhamba Sanskrit Series edition. This section covers the most advanced topics in the treatise:

1. **Aneka-varṇa-madhyamāharaṇa-bhedāḥ** (Equations with Multiple Unknowns) — systematic methods for solving equations involving several unknown quantities, building on the single-variable equations treated earlier.
2. **Bhāvitam** (Product Equations) — equations involving products of unknowns, including the famous "Bhāvita" (product) class of problems.
3. **Śeṣaṇa** and **Prakṣepaka-vidhi** — special algebraic methods for remainder problems and the method of addition/substitution.
4. **Concluding Verses** — the epilogue (*upasaṃhāra*) in which Bhāskara summarizes his work and pays homage to tradition.

Bhāskara employs his characteristic notation where unknown quantities are denoted by the initial syllable याव (*yāva-*, from *yāvattāvat*, "as much as"), and coefficients are prefixed to these symbols. The final sections also include a publisher's catalogue listing the many Sanskrit texts available from the Chowkhamba Sanskrit Book Depot, Varanasi.

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1 From the Varga-prakṛti Section (p. 115)

The opening of this portion continues the discussion of the Varga-prakṛti (square-nature) equation, i.e., equations of the form $Nx^2 + k = y^2$. The text presents further rules and examples for finding solutions to such indeterminate equations.

Continuation of the Square-nature Method

> समशोधने हृते पक्षौ यावच्चावत्तावकस्य रूपं प्रक्षिप्य प्रथमपक्षमूलं या २ रू १ ।
> परपक्षस्य काव १२ रू १ । वर्गप्रकृत्या मूले क २ उचे ७ वा क २८ उचे ९७ ।
> कनिष्ठं कालकमानम् । उच्चिष्टस्य या २ रू १ समं हृत्वा लब्धं यावच्चावकमानम् ३ वा ४८ ।
> स्वच्छानेनोत्थापने हृते जाती राशी १, ५ वा २०, ७६ इत्यादि ॥

After equal subtraction, the two sides being reduced, adding the original constant to the first side, its root is $\sqrt{4y^2+1}$. *For the second side:* $\sqrt{12y^2-1}$. *By the square-nature [method], the roots are* $(2, 7)$ *or* $(28, 97)$. *The lesser is the time-value. The residue from* $\sqrt{4y^2+1}$ *gives the yāvattāvat value 3 or 48. By self-substitution and reduction, the series of values 1, 5 or 20, 76, etc.*

> अथापरं सूत्रं साध्येऽस्तु ॥
>
> (१) द्वितीयपक्षे सति सममेव तु हत्यापवर्त्योन पदे प्रसाध्ये ।
> उच्चिष्टं कनिष्ठेन तदा निहत्याच्छेदं वर्गेण हतोऽपवर्तः ॥ ६ ॥
> कनिष्ठवर्गेण तदा निहत्योच्चिष्टं ततः पूर्ववदेव शेषम् ।
> इत्यर्थात् ॥

Rule 6: "When the second side exists [and is not unity], having made [the two sides] equal, multiplying and reducing, the remainder [is obtained]; having multiplied the residue by the lesser [root], dividing by the square [nature], the reduced [equation is obtained]. Then having multiplied by the square of the lesser [root], the residue — from that, as before, the remainder." Thus the meaning.

Worked Example (p. 115)

> (१) वि०—कल्प्यते समी पक्षौ
>
> काव १
> यावव्। ई १ याव।ई १

अत्र प्रथमपक्षस्य मूले क १ द्वितीयपक्षस्य यावव्।ई १ याव।ई १ मूलेन सममिति ।
तत्र द्वितीयपक्षस्य मूले च का।

$$\boxtimes\boxtimes = \sqrt{\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes \ \boxtimes \ \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes \ \boxtimes} = \boxtimes\boxtimes\sqrt{\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes \ \boxtimes \ \boxtimes \ \boxtimes}$$

अत इदं याव।ई १ ई १ मूलदं ततो वर्गप्रकृतिविषयो यथा का वर्गः ई, गुणः ई, युतो मूलदं इति हत्वा यावच्चावकमानं उच्चिष्टं चास्य याव।ई १ ई १ मूलेन सममिति।

Summary Rule (p. 115 bottom)

The section concludes with a general procedure:

> एवं बहुधा बुद्धिमत्त्वविचिन्त्यामिति सर्वमुपपन्नम् ॥

"Thus in many ways, by the thoughtful application of intelligence, all [is demonstrated]." Everything is proved.

2 Aneka-varṇa-madhyamāharaṇa-bhedāḥ (pp. 116–133)

This is the largest section of the final portion, dealing systematically with equations involving multiple unknown quantities (*aneka-varṇa*, "many colours"). The Sanskrit term *varṇa* (colour) is the traditional designation for an unknown quantity in Indian algebra — different unknowns being assigned different colours.

2.1 General Rules for Multiple Unknowns (pp. 116–117)

> अथ सूत्रं साध्येऽस्तु ॥
>
> (१) द्वितीयपक्षे सति सममेव तु हत्यापवर्त्योन पदे प्रसाध्ये ।
> उच्चिष्टं कनिष्ठेन तदा निहत्याच्छेदं वर्गेण हतोऽपवर्तः ॥ ६ ॥
>
> कनिष्ठवर्गेण तदा निहत्योच्चिष्टं ततः पूर्ववदेव शेषम् ।
> इत्यर्थात् ॥

Worked Example 1 (p. 117)

> उदाहरणम् ॥
> यस्य वर्गेऽद्विति: पञ्चगुणा वर्गशतोनिता ।
> मूलद्वा जायते राशिः गणितज्ञ वदाशु तम् ॥ १ ॥

Example 1: "A number whose square minus twice [itself], diminished by a hundred times five [i.e., 500], has two roots — tell me that number quickly, O mathematician."
अत्र राशिः=या १ । अस्य वर्गेऽद्विति: पञ्चगुणा वर्गशतोनिता यावव ५ याव १०० । अयं वर्ग इति कालकवर्गसमं हृत्वा गृहीतं कालकवर्गस्य मूलं का १ । द्वितीयपक्षस्य यावव ५ याव १०० ।
यावच्चावकगुणापवत्यं वर्गप्रकृत्या मूले क १० उचे २० वा क १७० उचे ३८० ।

Worked Example 2 (p. 117)

उदाहरणम् ॥

कयोः स्यादन्तरे वर्गौ वर्गयोगो ययोर्घनः ।
तौ राशी कथयामित्रो वहुधा बोज्ञवित्तम ॥ २ ॥

Example 2: "Two numbers whose difference of squares [equals] the sum of their squares — tell me those numbers, O friend, in many ways, O best of algebraists."

अत्र राशी या १, का १ । अनयोरन्तरं या १ का १ नीलकवर्गसमं हृत्वा लब्धं यावच्चावकमानं का १ नीलं ई १ । अनेन यावच्चावकद्वितीयाप्य जाती राशी का १ नीलं ई, का १ ।
अनयोर्वर्गयोगः काव २ नीलका २ नीलव १ । एष घन इति नीलकवर्गयनसमं हृत्वा शोधने हृते जाते प्रथमपक्षे नीलव १ नीलव १ । द्वितीयपक्षे काव २ नीलका २ ।

Rule for Three Unknowns (pp. 119–120)

(१) साध्यकल्पो यदि वर्गवर्गेषद्वित्र्यादिवर्णस्य हते समं तम् ।
हृत्वा पदं तस्य तदन्वपक्षे वर्गप्रकृत्यैवकदैव मूले ॥
कनिष्ठमाच्छाद्य पदेन तुल्यं उच्चिष्टं द्वितीयेन समं विधायात् ॥ ८ ॥

Rule 8: "If the square of the square of an assumed [quantity], diminished by the square of one, two, three, etc., unknown quantities, having made [the two sides] equal, taking the square root of that, then by the square-nature method, the roots [are found]; covering the lesser root by an equal root, making the residue equal to the second [unknown]..."

Worked Example with Three Unknowns (p. 121)

उदाहरणम् ॥

त्रिकादिचतुरश्रेणां नच्छेदं क्वापि च यत् कलम् ।
तदेव त्रिगुणं क्षेत्रमष्टयुगच्छेदं भवेद्ध ॥ ९ ॥

अत्र श्रेण्योन्या इः । आदि=३, चयः=२, गच्छः=या १ । आदि=३, चयः=२, गच्छः=का १ ।
अनयोः (क) कलं=याव १ या २, काव १ का २ । अनयोरन्यत् त्रिगुणं परस्मिं हृत्वा शोधनार्थं

$$\text{⊠⊠⊠⊠⊠⊠} \longrightarrow \text{⊠⊠⊠ ⊠ ⊠⊠ ⊠}$$

$$\text{⊠⊠⊠ ⊠ ⊠⊠ ⊠}$$

शोधने हृते पक्षौ त्रिगुणीहृत्व नव प्रक्षिप्य प्रथमपक्षस्य मूलं या ३ रू ३ ।
द्वितीयपक्षस्य काव ३ का ६ रू ९ । नीलकवर्गेण

Rule for Equal-square Unknowns (p. 121)

(९) प्रथमपक्षस्यद्विगुणीतच्छेदस्य मूलं नीलं प्रकल्प्य तद्वर्गेण समं परं पक्षं
हृत्वा पूर्वोक्तर्यस्य वासना चातिसरति ॥

"Having assumed the root of twice the divisor of the first side as the 'blue' [unknown], having divided the other side by its square, the desire is satisfied more than previously stated."

2.2 The Method of Substitution (pp. 125–128)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।
योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

Rule 12: "Having assumed the root of the difference of [the two sides] as the first [unknown], or a small quantity, [and] the residue of the difference of sums as the square [of another unknown], the root [is obtained]."

(९) वि०—अत्र कल्प्यते योगान्तरशेषमानम् = क्षे
वर्गान्तरशेषमानम् = क्षे$_2$, वर्गयोगशेषमानम् = क्षे$_3$
विभोगमूलम् = या, योगमूलम् = का

$$\boxtimes\boxtimes\boxtimes\ \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\ \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes = \boxtimes\boxtimes^2 - \boxtimes\boxtimes\boxtimes\boxtimes_1,\ \boxtimes\boxtimes\boxtimes\boxtimes = \boxtimes\boxtimes^2 - \boxtimes\boxtimes\boxtimes\boxtimes_1$$

$$\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes = \frac{\boxtimes\boxtimes^2 - \boxtimes\boxtimes^2}{2}, \quad \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes = \frac{\boxtimes\boxtimes^2 + \boxtimes\boxtimes^2 - 2\boxtimes\boxtimes\boxtimes\boxtimes_1}{2}$$

Rule 13 (p. 129)

शेषं ततः शेषकमूलमन्यच्च मूले विद्ध्वादसकृत् सम्मते ॥ ९ ॥
समाविते वर्गकृती तु यत्र तन्मूलमादाय च शेषकस्य ।
इष्टोद्धृतस्येर्विवर्जितस्य दलैन समं तद्विह तदैव कार्यम् ॥ १० ॥

Worked Example on the Product Method (p. 130)

उदाहरणम् ॥

तौ राशी वद यत्कृत्योः समस्तगुणयोर्युतिः ।
मूलद्वा स्याद्योगमष्टौ मूलद्वा रूपसंयुतः ॥ १ ॥

अत्र राशी या १, का १ । अनयोर्वर्गयोः समस्तगुणयोर्युतिः यावव ७ काव ८ । अयं वर्ग इति नीलकवर्गसमं हृत्वा पक्षयोः कालकधनं प्रक्षिप्य नीलकपक्षस्य मूलं नी १ । परपक्षस्य याव ७ काव ८ ।

Summary Verses (p. 131)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।
योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

वर्गान्तरशेषकसंमितिर्युता क्षेपेण हृत्वोर्ध्वितेन तै ततः ।
द्विधा पदं तत्पदयुगविभोगाज् मूलं युतमूलमतस्तयोरिति ॥

"The difference of squares having a residue equal to the divisor, divided by the additive, then by the quotient, [the root] divided in two; from the difference of the pair of roots, the root [of the lesser] from the sum of the roots, thus [proceeding]..."

3 Bhāvitam and Prakṣepaka-vidhi (pp. 134–152)

3.1 Equations with Product Terms (pp. 134–138)

This section deals with *bhāvita* (product) equations, where the unknowns appear in product form. The method extends the techniques for multiple unknowns to handle more complex algebraic structures.

अथ वा स्वर्णकल्पनायां मन्दबुद्धीनां पुनः किञ्चित् पठितः ।
तत् सूचयामि ॥

हरमका यस्य हृति सूक्ष्मति सोऽपि द्वितीयपदयुतातः ।
तेनाहतोऽस्यवर्णो रूपपदेनान्वितः कल्पः ॥ १६ ॥

Rule 16: "The divisor multiplied by [some quantity] whose [result] is fine, added to the second term; having multiplied by that, the unknown [is obtained], combined with a root term [as] the assumption."

Worked Example (p. 137)

उदाहरणम् ॥
न यदि पदं रूपाणां शिष्येत्तं तेन हरतोऽपि ।
नावच्छद्गुणो भवति न चेदेवमपि किल तथै ॥ १७ ॥

हरमतिः । यस्याङ्कस्य हृतिहरमका सती सूक्ष्यतीति निश्शेषा भवति अपि च सोऽप्यङ्को हर्यो रूपपदेन च गुणितो हरमकः सन् सूक्ष्यति तदा तेनाहतोऽस्यवर्णस्तेन रूपाणां घनमूलं न लभ्यते तदा तेन रूपेषु हरतोऽपि तावद्गुणो भवेत् तन्मूलं रूपपदं स्यात् ।

3.2 The Bhāvita Rule (pp. 140–141)

भावितम् ॥

वर्णद्वयातिकूपेऽस्य मकरवेष्टनैकतः ।
एताम्या संयुताबुद्धि कल्पयो वैंच्छया च तौ ॥ ३ ॥

वर्णद्वयो वर्गयोरन्ते क्रियते ते विपर्ययात् ।
समयोः पक्षयोरेकमाध्यमिकमपास्य तौ वर्णौ रूपाणि च

The section on "Product": "Two unknowns, their product having a coefficient, from one side having a single enclosure [i.e., bracketed], having assumed any sum of their values, [proceeding] with the pair of unknowns reversed, [taking] the square of two unknowns, the means [being] equal, having removed one middle term, the two unknowns and the roots..."

Worked Example on Bhāvita (p. 142)

उदाहरणम् ॥
चतुःषष्टिगुणयो राश्योः संयुतिर्द्वियुता तयोः ।
राशिघातेन तुल्येति ॥

तत्र यथोक्तं हृते पक्षौ $\left\{\begin{array}{l}\text{या ४ का ३ रू २}\\ \text{या।का।भा १}\end{array}\right.$

वर्णद्वयातिकूपेऽस्य १४ एतदेकेनाहतं हते जाते द्विके १, १४ । एते वर्णद्वया ४, ३ स्वैच्छया युते जाते यावच्चावककालकमाने ५, १८ वा १७, ५ । द्विकेन ५, ११ वा १०, ६ ॥

3.3 The Method of Mean Elimination (pp. 145–147)

माध्यमाविगमः ॥

आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।
पूर्णे स्याताम् अन्तरराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥

आसन्नमानयोरन्तरमेव भवेत् ।
अंशस्थाने सदा रूपं चिन्त्येत तथा धीमता ॥ २ ॥
सर्वेषामासन्नमानेषु हरांशो भवति षट् ।
तथोत्तरोत्तरं सूक्ष्मतामासन्नानि भवन्ति हि ॥ ३ ॥

The method of mean-elimination: "The divisor-measure of the approximate value, added to the non-obtained quantity and the additive, being complete by two numbers with difference, then the divisor-fraction [gives] the effort. The difference of two approximate values alone [suffices]. The numerator [is] always unity [at first], thus by the intelligent. For all approximate values, the divisor-fraction is six. Thus gradually, more and more subtle approximate values are obtained."

Concluding Verses of the Bhāvitam Section (pp. 151–152)

तथा गोले मयोक्तम् ।
उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
तथा गोलाख्याय मयोक्तम् ।

अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥

गणककर्मान्तरमयं बाललोचनगम्यं
सकलगणितसारं सोपपट्टिप्रकारम् ।
इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
पठ पठ मतिबुद्ध्यै लाघवं प्रौढिसिद्ध्यै ॥

The epilogue: "Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of the sphere called 'Gola'. There is the study of the text and pure intellect — what else [is needed]? For the delight of the wise, this is declared. This work accessible to the eye of a child [yet] profound, the essence of all mathematics with proofs endowed, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity."

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

"Thus ends the Bījagaṇita-adhyāya in the Siddhāntaśiromaṇi composed by Śrī Bhāskarācārya."

वि०—इति कूपाङ्कनमुखाकरे यदिह भास्करबीजमपूर्वकम् ।
तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

Editor's note: "That unprecedented algebra of Bhāskara which here [appears] with the face of a well-digit-mine [i.e., a numerical source], having ascertained it with the Upaniṣad [of commentary], the cause of the appearance of manifold wonder."

4 Final Sections and Publisher's Catalogue (pp. 153–170)

4.1 Navīna-mati-viśeṣaḥ (pp. 153–157)

The section on "Special Methods for New Understanding" presents advanced techniques for solving equations through approximation and iteration, building on the methods developed earlier.

नवीनमतिविशेषः ॥

निरक्ष पदं यद्द्विगुणात् स्यात् फलार्थं धनार्थं तदेवात्र शेषं तदस्यम् ।
पदद्वयं धनं शेषहृतप्रमन्यत् फलं तद्भवत् शेषमूलं धनेन ॥ १ ॥
धनार्थं नवं तस्य कृत्या विहीनो गुणः शेषमकोऽस्यवर्णस्य मानम् ।
मूद्गुसेव मते यदा शेषमानं भवेद्द्वितीयं तदा लव्धितौ ते ॥ २ ॥

Rule 1: "A number, when doubled, becomes the root; for the fruit [result], for wealth [coefficient], that same residue here — its two terms [are] the wealth divided by the residue; the fruit [is] the root of the residue multiplied by the wealth. The new wealth is diminished by nine times its square; the multiplier, diminished [thus], [gives] the measure of the unknown. When by the opinion of Mūdgu [i.e., by the method of approximation], the residue-value becomes the second [term], then the quotient for both."

4.2 Worked Examples on Special Methods (pp. 157–159)

उदाहरणानि ॥

(१) कल्प्यते अ+क=५७६०, अ−क=$\frac{\text{अ}}{\boxtimes}$ तदा अ=३८४०,
क=१९२० इति क्षिप्रम् ॥

(२) यदि $\frac{\boxtimes\text{क}+\boxtimes}{\boxtimes} = \frac{\boxtimes\text{क}+\boxtimes}{\boxtimes}$ तदा कमानम्=१ इति क्षिप्रम् ॥

(३) यदि $\frac{\text{क}+\boxtimes}{\boxtimes} + \frac{\boxtimes\text{क}-\boxtimes}{\boxtimes} + \frac{\boxtimes}{\boxtimes} = \frac{\boxtimes\text{क}+\boxtimes}{\boxtimes}$, तदा क=२० इति क्षिप्रम् ॥

(४) $(\text{क} + \frac{\boxtimes}{\boxtimes})(\text{क} - \frac{\boxtimes}{\boxtimes}) - (\text{क} + \boxtimes)(\text{क} - \boxtimes) + \frac{\boxtimes}{\boxtimes} = \boxtimes$
तदा क=१२ इति क्षिप्रम् ॥

4.3 The Method of Approximate Roots (pp. 160–165)

This section presents the *āsanna-mūla* (approximate root) method — an iterative technique for finding successively better approximations to irrational square roots, a precursor to what is now known as Newton's method.

आसन्नमूलानि सूत्राणि ॥

आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।
पूर्णे स्याताम् अन्तरराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥

The mathematical procedure is as follows: given a non-square number n, to find $\sqrt{n}$:

1. Choose an initial approximation a_1 (the "approximate value")
2. Compute the residue: $r = n - a_1^2$
3. Form the next approximation: $a_2 = a_1 + \dfrac{r}{2a_1}$
4. Iterate for successively better approximations

कल्प्यते—
(१) 0, अ, अ', अ'', ...
(२) 1, सो, सो', सो'', ...
(३) अ, क⊠, क$_2$, क$_3$, क$_4$, ...
अत्र क⊠, क$_2$, क$_3$,—अयबवदासन्नमूलस्यासन्नमानानि $\frac{\text{प}}{\text{ल}}, \frac{\text{प}'}{\text{ल}'}, \frac{\text{प}''}{\text{ल}''}$ चेति ।

$$\sqrt{n} = \frac{\frac{\sqrt{n+\boxtimes_1}}{\boxtimes\boxtimes'} \cdot \boxtimes + \boxtimes}{\frac{\sqrt{n+\boxtimes_1}}{\boxtimes\boxtimes''} \cdot \boxtimes + \boxtimes} = \frac{\boxtimes(\sqrt{n+\boxtimes_1}) + \boxtimes\boxtimes''\boxtimes}{\boxtimes(\sqrt{n+\boxtimes_1}) + \boxtimes\boxtimes''\boxtimes}$$

4.4 Final Concluding Verses (pp. 166–168)

The treatise concludes with verses that reflect on the nature of mathematical knowledge and pay homage to the tradition:

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥

गणककर्मान्तरमयं बाललीलावनोगम्यं
सकलगणितसारं सोपपट्टिप्रकारम् ।
इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
पठ पठ मतिबुद्ध््यै लाघवं प्रौढिसिद्ध्यै ॥

“There is the study of this work and pure intellect — what else [is needed]? Thus is declared for the delight of the wise. This [work], accessible as a child’s play [yet] the essence of all mathematics with demonstrations, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity.”

The author reflects on the completion of the work:

तथा गोले मयोक्तम् ।
उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
तथा गोलाख्याय मयोक्तम् ॥

“Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of what is called the sphere.”

Colophon

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

Editor’s Note (p. 168)

वि०—इति कूपाङ्कनमुखाकरे यदिह भास्करबीजमपूर्वकम् ।
तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

“That unprecedented algebra of Bhāskara, which here [appears] at the mouth of the well-digit-mine, having ascertained it with the Upaniṣad [of commentary], became the cause of the appearance of manifold wonder.”

Publisher’s Catalogue (Chowkhamba Sanskrit Book Depot)

The final two pages contain a catalogue of Sanskrit publications available from the Chowkhamba Sanskrit Book Depot (Chowkhamba Griham), 4015 Thatheri Bazar, Varanasi. The catalogue lists works across all branches of Sanskrit learning, including:

No.	Title	Price	Pages
19 ☒	Golaprakāśasya Golādhyāya-Bījagaṇita-vāsiṣṭha-koṇa-gaṇite Śrī-Candraśekhāra-saṃskṛtite	· · ·	☒
20 ☒	Golaprakāśīya-cāpīyatrikōṇa-gaṇitasya “Vivāha” ṭīkā Śrīmahāvīra-paṇḍita-viracitā	· · ·	☒
21	Vāsanāvyañjana-ṭīkā Śrīgaṇeśa-dhara-mitra-kṛtā		

No.	Title	Price	Pages
	ṭīkā-sahitā	···	⊠
⊠			
22	Prasaṅga-mūla-prabhāvicāra-syāt-mūla-prath⊠	···	⊠
⊠⊠			
23	Karṇa-prakāśaḥ Śrī-sudhā-kara-dvivedi-saṃśodhit⊠	···	⊠
⊠			
24	Jātaka-paddhatiḥ	···	⊠
⊠			
25	Bhāvate-karaṇaṃ (Bījakoṣa-udāharaṇa-sahitam)	···	⊠
⊠			
26	Prabhāve-padav⊠	···	···
⊠			
27	Daivajña-kāma-dhenu⊠	···	⊠
⊠			
28	Sūrya-siddhāntaṃ (Mṛtaṃ) Śrī-sudhā-kara-dvivedi-kṛta-ṭīkopetaḥ	···	⊠
⊠			
29	Makaranda-vivaraṇam (udāharaṇa-sahitam)	···	⊠
⊠			
30	Jātaka-paddhatiḥ proktamārga-vyākhyā-sahitā	···	⊠
⊠⊠			
31	Digvijaya-saṃ M.M. Śrī-sudhā-kara-dvivedi-viracitā	···	⊠
⊠⊠			
32	Gaṇaka-taraṅgiṇī Śrī-sudhā-kara-dvivedi-nirmitā	···	⊠
⊠			
33	Yājūṣa-jyotiṣam (Śrīmākara-sudhā-kara-bhāṣya-sahitam)	···	⊠
⊠			
34	Piṇḍa-prabhākaraḥ Śrī-sudhā-kara-dvivedi-viracitaḥ	···	⊠
⊠			
35	Suktiratnāvaliḥ (Mṛtaṃ) Śrī-sudhā-kara-dvivedi-viracit⊠	···	⊠
⊠			
36	Vallana-kalanam—Śrīmuralī-dhara-dvivedi-viracitam (caturtha-adhyāya-paryantam)	···	⊠
⊠			
37	Gaṇitakā itihāsa Pāṭīgaṇita Śrī-sudhā-kara-dvivedi viracitam	···	⊠
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38	Bṛhat-saṃhitā (Mahārāja-ṭīkā-sahitā)	···	⊠⊠
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40	Vāsatvavibodha-prabhāsa-saṃgrahaḥ (Mṛtaṃ)		

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	Śrī-sudhā-kara-dvivedi-viracitaḥ	· · ·	⊠
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41	Tulanātmaka bhāṣā śāstra—arthavāda—bhāṣā viṣayan (lekhaṭa—ḍākṭara Maṅgala-deva śāstrī)...	⊠	⊠⊠

कृष्णदास गुप्त,
⊠⊠⊠ ⊠⊠⊠⊠ ⊠⊠⊠⊠, ⊠⊠⊠⊠ ⊠⊠⊠ ⊠

Chowkhamba Sanskrit Book Depot
4015 Thatheri Bazar, Varanasi City

॥ श्रीः ॥

लीलावती

श्रीभास्कराचार्यविरचिता

Līlāvatī of Bhāskarācārya
Bilingual Sanskrit–English Edition

Treatise on Arithmetic, c. 1150 CE

संस्कृत मूलपाठ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯	⟷	**English Translation** with IAST Transliteration

A critical edition based on the manuscript from the
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Contents

Preface / प्रस्तावना

लीलावती भास्कराचार्यस्य सिद्धान्तशिरोमणेः प्रथमः भागः। श्रीभास्करेण ११५० ई.पू.रचिता। लीलावती नाम तस्य कन्यायाः नाम अस्ति। इयं ग्रन्थः भारतीयगणितस्य सर्वाधिकप्रसिद्धपाठ्यपुस्तकम् अस्ति। पूर्णाङ्केषु, भिन्नेषु, अनुपातेषु, व्याजेषु, क्षेत्रफलेषु, घनफलेषु, श्रेणीषु च गणितविषयान् व्यवस्थापयति।

līlāvatī bhāskarācāryasya siddhāntaśiromaṇeḥ prathamaḥ bhāgaḥ | śrībhāskareṇa 1150 ī.pū.racitā | līlāvatī nāma tasya kanyāyāḥ nāma asti | iyāṃ granthaḥ bhāratīyagaṇitasya sarvādhikaprasiddhapāṭhyapustakam asti |

The *Līlāvatī* is the first section of Bhāskara II's monumental work *Siddhāntaśiromaṇi*, composed in 1150 CE. Named after Bhāskara's daughter, it is the most celebrated textbook of Indian arithmetic. The work covers the entire field of elementary mathematics: arithmetic operations with integers and fractions, rules of proportion, interest calculation, plane geometry, solid geometry, and combinatorial problems.

अस्य ग्रन्थस्य प्रस्तावने — अत्र गणितस्य समग्रं क्षेत्रम् उपस्थापितम् अस्ति। सङ्कलनम्, व्यवकलनम्, गुणनम्, भागहारः, वर्गः, वर्गमूलम्, घनः, घनमूलम् — एतेषां परिकर्मणां व्यवस्था, भिन्नपरिकर्म, त्रैराशिकम्, मिश्रप्रश्नाः, क्षेत्रव्यवहारः, खातव्यवहारः, चितिव्यवहारः, छायाव्यवहारः, कुट्टकव्यवहारः, अङ्कपाशः, श्रेणी चेत्यादयः अध्यायाः सन्ति।

asya granthasya prastāvane — atra gaṇitasya samagraṃ kṣetram upasthāpitam asti | saṅkalanaṃ, vyavakalanaṃ, guṇanaṃ, bhāgahāraḥ, vargaḥ, vargamūlam, ghanaḥ,ghanamūlam — eteṣāṃ parikarmaṇāṃ vyavasthā, bhinnaparikarma, trairāśikam, miśrapraśnāḥ, kṣetravyavahāraḥ, khātavyavahāraḥ, citivyavahāraḥ, chāyāvyavahāraḥ, kuṭṭakavyavahāraḥ, aṅkapāśaḥ, śreṇī cetyādayaḥ adhyāyāḥ santi |

The present edition reproduces the Sanskrit text in the left column with an English translation and IAST transliteration in the right column. Key technical terms are preserved in their original Sanskrit form with explanatory footnotes.

मङ्गलाचरण — Invocatory Verses

लीलागुल्लक्षोल्कालव्यालविलसिने ।
गणेशाय नमो नीलकमलामलकान्तये ॥
॥ १ ॥

Līlāgullakṣolkāvyālavilasine
Gaṇeśāya namo nīlakamalāmalakāntaye

Salutation to Gaṇeśa, the playfully sportive one, having the brightness of the rising sun, and possessing the radiance of a blue lotus.

अवतार पुरा पुराणेषां यदुकुले जगतीहितेहिने ।
जयति सोऽयमिमां सकलां लीलावतीं मामपि सिद्धमहीपतिः ॥

avatāra purā purāṇeṣāṃ yadukule jagatīhitehine |
jayati so'yamimāṃ sakalāṃ līlāvatīṃ māmapi siddhamahīpatiḥ ||

In ancient times, he who took incarnation in the Yadu clan for the welfare of the world — may that same king, the lord of the earth, be victorious, who has made me compose this entire *Līlāvatī*.

भास्करचितयमथा सिद्धान्तशिरोमणिप्रख्याः ।
तदयं कृताऽनन्ये व्याकृतया मन्मठे नियतं ॥
॥ १५ ॥

Bhāskaracitayamathā siddhāntaśiromaṇiprakhyāḥ
Tadayaṃ kṛtā'nye vyākṛtayā manmaṭhe niyataṃ

This *Līlāvatī*, composed by Bhāskara, is the foremost part of the *Siddhāntaśiromaṇi*. This has been composed by none other than him, and is now set down by me.

1 परिभाषामकरण — Definitions and Units of Measurement

पादोनग्रानकतुल्यष्टकैः समतुल्यैः कथितोऽत्र सेरः ॥
मणाभिधानं व्युग्मैः सैरेधार्यादितोल्येषु तुष्टकसङ्ख्या ॥ १ ॥

***pādonagrānakatulyaṣṭakaiḥ samatulyaiḥ kathito'tra seraḥ* ||**
***maṇābhidhānaṃ vyugmaiḥ sairedhāryāditolyeṣu tuṣṭakasaṅkhyā* || *1* ||**

A *ser* is equal to eight *pāda* or equal to eight grains. A *maṇa* is twice the amount of a *ser* in weighing gold, etc. *[The ser and maṇa were traditional Indian units of weight. 1 maṇa = 2 ser.]*

द्वयकद्विसहस्रैर्घटकैः सैरस्तैः पञ्चभिः स्याद्घटिका च ताभिः ॥
मणोऽष्टमिर्त्वालमगीरशाहकृतास्त्र सङ्ख्या निजराज्यपृष्ठे ॥ २ ॥

***dvayakadvisahasrairghaṭakaiḥ sairastaiḥ pañcabhiḥ syāddhaṭikā ca tābhiḥ* ||**
***maṇo'ṣṭamirtvālamagīraśāhakṛāstra saṅkhyā nijarājyapṛṣṭhe* || *2* ||**

Two thousand *ghaṭakas* make a *ser*; five of these *sers* make a *ghaṭikā*. The weight of one *maṇa* in this kingdom (of the author's patron) follows this system.

1.1 शोभाः कालादिपरिभाषा — Definitions of Time

शोभाः कालादिपरिभाषा लोकतः प्रसिद्धा देया ॥

***śobhāḥ kālādiparibhāṣā lokataḥ prasiddhā deyā* ||**

The definitions of time and other measures are well-known in the world. The traditional Indian system of time measurement is given as:

- 60 *vipalas* = 1 *pala*
- 60 *palas* = 1 *ghaṭikā* (24 minutes)
- 60 *ghaṭikās* = 1 day and night
- 15 days and nights = 1 *pakṣa* (fortnight)
- 2 *pakṣas* = 1 month
- 12 months = 1 year

इति परिभाषा — Thus end the definitions.

2 सङ्ख्याप्रकरण — Arithmetic Operations on Whole Numbers

This chapter covers the fundamental operations of arithmetic: addition (*saṅkalita*), subtraction (*vyavakalita*), multiplication (*guṇana*), division (*bhājana*), squaring (*varga*), square roots (*vargamūla*), cubing (*ghana*), and cube roots (*ghanamūla*).

2.1 सङ्कलित व्याकलित — Addition and Subtraction

सूत्रम्: यथोक्तं परिकल्प्य गणितं कुर्यात्।

Rule: As taught, having set up the numbers [aligned by place value], one should compute.

2.2 गुणकारकरणसूत्र — Multiplication

सूत्रम्: समद्विघातः कृतिः उच्यते—

Rule: A product of two equal numbers is called a *kṛti* (square). The multiplication of two different numbers is called *guṇana*.

अथ रूपयोऽन्त्यवर्गो द्विगुणान्त्यनिघ्नः...
एकस्यांकस्य वर्गः अन्तिमस्य अङ्कस्य गुणनम्। द्विगुणान्त्यनिघ्नः = द्विगुणः अन्तिमः अङ्कः अन्येन गुणितः।

atha rūpayo'ntyavargo dviguṇāntyanighnaḥ...
ekasyāṃkasya vargaḥ antim asya aṅkasya guṇanam |

Method of Multiplication: The square of the last figure, and twice the last multiplied by the preceding, and so on — this is the method for squaring.
Example: To multiply 135 by 12:

$$135 \times 12 = 135 \times 10 + 135 \times 2 = 1350 + 270 = 1620$$

2.3 भागहारकरणसूत्र — Division

सूत्रम्: भक्तः क्षेपणात् पुनः पुनरवाप्तिर्भागहारः।

Rule: Division is the repeated subtraction of the divisor from the dividend until the remainder is less than the divisor.

Division is performed by repeated subtraction. The commentary gives detailed worked examples. The quotient (*labdhi*) is the number of subtractions, and what remains is the remainder (*śeṣa*).

2.4 वर्गीकरणसूत्र — Squaring

सूत्रम्: समद्विघातः कृतिः उच्यते—

Rule: The product of two equal numbers is called a *kṛti* (square).

खण्डाभ्यां वा हतो राशिरिष्टः खण्डचतुष्कयुक् ॥
अस्य अर्थः — राशिं द्वयोः खण्डयोः योगेन गुणित्वा तयोरन्तरवर्गेण सह योजयेत्।

khaṇḍābhyāṃ vā hato rāśiriṣṭaḥ khaṇḍacatuṣkayuk ||
asya arthaḥ — rāśiṃ dvayoḥ khaṇḍayoḥ yogenaguṇitvā tayorantaravargeṇa saha yojayet |

Method 1: Direct multiplication $n \times n$
Method 2: Using the formula $(a + b)^2 = a^2 + 2ab + b^2$
Method 3: Using the formula $n^2 = (n + a)(n - a) + a^2$
Or, the number multiplied by the component parts of itself, combined with the square of the difference...

2.5 वर्गमूलकरणसूत्र — Square Root Extraction

सूत्रम्: स्थाप्योऽन्त्यवर्गो द्विगुणोऽन्त्यनिघ्नः स्यादपरिच्छेदः।

Rule: Set down the number. Find the largest square in the leftmost group. This gives the first digit of the root. Subtract its square, bring down the next group, divide by twice the current root, and repeat.

अथ स्थाप्योऽन्त्यवगो द्विगुणान्त्यनिघ्नः॥
स्यादपरिच्छेदः तथा परेऽभ्यस्तयुक्त्वान्त्यमपि राशिम् ॥

atha sthāpyo'ntyavago dviguṇāntyanighnaḥ ||
syādaparicchedaḥ tathā pare'bhyastayuktvāntyamapi rāśim ||

Example: Find $\sqrt{65536}$
Step 1: Group the digits from right: $\overline{6}\,\overline{55}\,\overline{36}$
Step 2: Largest square ≤ 6 is 4 = 2^2. First digit: 2.
Step 3: Remainder 2, bring down 55 → 255.
Step 4: $255 \div (2 \times 20) = 255 \div 40 \approx 5$.
Step 5: Verify: $45 \times 5 = 225$. Remainder 30.
Step 6: Bring down 36 → 3036.
Step 7: $3036 \div (2 \times 250) = 3036 \div 500 \approx 6$.
Step 8: Verify: $506 \times 6 = 3036$. Remainder 0.
Therefore $\sqrt{65536} = 256$.

2.6 घनकरणसूत्र — Cubing

सूत्रम्: समत्रिघातः घनः उच्यते—

Rule: The product of a number multiplied three times by itself is called a *ghana* (cube).

Methods for cubing:
Method 1: Direct multiplication $n \times n \times n$
Method 2: Using $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

2.7 घनमूलकरणसूत्र — Cube Root Extraction

The method follows a similar place-value decomposition as square root extraction, but using cubes of digits.

Example: $\sqrt[3]{1728} = 12$, since $12^3 = 1728$.

3 भिन्नपरिकर्म — Operations with Fractions

3.1 जातिचतुष्टय — Four Classes of Fractions

Bhāskara defines four types of fractions (*jāticatuṣṭaya*):

1. भिन्न (*bhinna*) — Simple fraction, e.g., $\frac{1}{2}$
2. प्रभाग (*prabhāga*) — Part of a part, e.g., $\frac{1}{2}$ of $\frac{1}{3}$
3. भागानुबन्ध (*bhāgānubandha*) — Fraction combined with an integer, e.g., $2+\frac{1}{3}$
4. भागापवाह (*bhāgāpavāha*) — Fraction less than an integer, e.g., $2-\frac{1}{3}$

3.2 भिन्नगुणन भिन्नभाग — Multiplication and Division of Fractions

सूत्रम्: नरघ्ननरके क्रमशो गुणयेत् तद् भागहारकैः ॥

Rule: Multiply the numerator by the numerator and the denominator by the denominator for multiplication; for division, invert and multiply.

Multiplication: $\dfrac{a}{b} \times \dfrac{c}{d} = \dfrac{a \times c}{b \times d}$

Division: $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a}{b} \times \dfrac{d}{c} = \dfrac{a \times d}{b \times c}$

Example: $\dfrac{2}{3} \times \dfrac{5}{7} = \dfrac{10}{21}$

Example: $\dfrac{2}{3} \div \dfrac{5}{7} = \dfrac{2 \times 7}{3 \times 5} = \dfrac{14}{15}$

4 त्रैराशिक — The Rule of Three

The *trairāśika* (Rule of Three) is one of the most important topics in Indian arithmetic. It solves problems of the form:

$$\text{“If } A \text{ gives } B\text{, what does } C \text{ give?”} \quad \text{Answer: } \frac{B \times C}{A}$$

4.1 इच्छाकर्मप्रकार — Direct Rule of Three

सूत्रम्: प्रमाणफलमिच्छाफलं चेत् कुर्यात्।

Rule: Multiply the *phala* (fruit/result) by the *icchā* (desire) and divide by the *pramāṇa* (measure).

प्रमाणं फलमिच्छा च त्रिविधा राशयः स्मृताः ॥
प्रमाणेन यद् लभ्यते तत् फलम्। इच्छया यद् ज्ञातुम् इष्टम् तत् इच्छाफलम्।

pramāṇaṃ phalamicchā ca trividhā rāśayaḥ smṛtāḥ ||

Three quantities are remembered: the *pramāṇa* (measure), the *phala* (fruit), and the *icchā* (desire).
Example: If 5 books cost 60 rupees, what do 8 books cost?

$$\text{Price} = \frac{60 \times 8}{5} = 96 \text{ rupees}$$

4.2 व्यस्तत्रैराशिक — Inverse Rule of Three

सूत्रम्: व्यस्ते विपरीतं कुर्यात्।

Rule: In the inverse case, do the reverse [multiply measure by fruit and divide by desire].

When the relationship is inverse (more workers take less time, etc.):

$$\text{Result} = \frac{B \times A}{C}$$

Example: If 8 workers complete a job in 15 days, how many days will 12 workers take?

$$\text{Days} = \frac{8 \times 15}{12} = 10 \text{ days}$$

4.3 पञ्चराशिक सप्तराशिक — Extended Rules

Extended rules for compound proportions:

- पञ्चराशिक (*pañcarāśika*, Five-fold rule): Two intermediate terms
- सप्तराशिक (*saptarāśika*, Seven-fold rule): Three intermediate terms
- नवराशिक (*navarāśika*, Nine-fold rule): Four intermediate terms

5 मिश्रप्रकरण — Mixed Problems

5.1 शैलाराशिकविधि — Rule of Alligation

मिश्रणे विविधद्रव्याणां मूल्यानां समासेन...
सुवर्णादीनां मिश्रणे कलर्णज्ञानाय एषः प्रकारः।

miśraṇe vividhadravyāṇāṃ mulyānāṃ samāsena...
suvarṇādīnāṃ miśraṇe kalaṇajñānāya eṣaḥ prakāraḥ |

Problems involving the mixing of different quantities at different prices or qualities.
Example: If rice of quality A costs 8 rupees/kg and rice of quality B costs 12 rupees/kg, in what ratio should they be mixed to get a mixture worth 10 rupees/kg?

$$\text{Ratio} = (12 - 10) : (10 - 8) = 2 : 2 = 1 : 1$$

5.2 वापीपूरणप्रकार — Filling of Tanks

वापी पूरयितुं नलकैः कालज्ञानाय सूत्रम् ॥

vāpī pūrayituṃ nalakaiḥ kālajñānāya sūtram ||

Example: A tank has three pipes. The first can fill it in 5 hours, the second in 6 hours, and the third can empty it in 4 hours. If all three are opened together, how long will it take to fill the tank?

$$\text{Net fill rate} = \frac{1}{5} + \frac{1}{6} - \frac{1}{4} = \frac{12 + 10 - 15}{60} = \frac{7}{60}$$

Time to fill = $\frac{60}{7} = 8\frac{4}{7}$ hours.

5.3 कथाविकयविधि — Exchange of Goods

वस्तूनां विनिमये मूल्यसमत्वेन परस्परगुणनम् ॥

vastūnāṃ vinimaye mūlyasamatvena parasparaguṇanam ||

Example: If 3 mangoes can be exchanged for 4 oranges, and 5 oranges for 2 apples, how many apples can be obtained for 15 mangoes?

$$15 \text{ mangoes} = 15 \times \frac{4}{3} = 20 \text{ oranges} = 20 \times \frac{2}{5} = 8 \text{ apples}$$

6 क्षेत्रव्यवहार — Plane Geometry

क्षेत्रे व्यवहारः —
॥ ॥

kṣetre vyavahāraḥ

This chapter deals with the calculation of areas of various plane figures and the properties of right triangles.

6.1 क्षेत्रसाधारणसूत्र — General Rule for Areas

सूत्रम्: दैर्घ्यमायतं विस्तारं च तद् दैर्घ्यविस्तारयोगजं क्षेत्रम्।

Rule: The area of a rectangle is length multiplied by breadth. The area of a figure is the product of its length and width.

6.2 भुजकोटिकर्णनयन — The Pythagorean Theorem

भुजे द्वादशके यो यो कोटिकर्णविनेकथा ॥
प्रकाराभ्यां वद विप्र तो तावकर्णीगतौ ॥ ५ ॥
॥ ५ ॥

bhuje dvādaśake yo yo koṭikarvavinekathā ||
prakārābhyāṃ vada vipra to tāvakarṇīgatāu || 5 ||

O learned Brahmin! The sides (*bhujā*) being 12, tell me quickly the hypotenuse (*karṇa*) and perpendicular (*koṭi*) by both methods.

इष्टयोराहतिर्द्वयोः कोटिवर्गान्तरं भुजः ॥
कृतियोगस्तयोरेवं कर्णस्त्वकरणीगतः ॥ ८ ॥
॥ ८ ॥

iṣṭayorāhatirdvayoḥ koṭivargāntaraṃ bhujaḥ ||
kṛtiyogastayorevaṃ karṇastvakaraṇīgataḥ || 8 ||

The product of the two desired [sides] is the difference of the squares of the *koṭi* and *bhujā*. The sum of the squares of those two gives the *karṇa*.

The Three Sides of a Right Triangle:

- **Base** (भुजा *bhujā*): The horizontal side
- **Perpendicular** (कोटि *koṭi*): The vertical side
- **Hypotenuse** (कर्ण *karṇa*): The diagonal side

The rule states:

$$\text{karṇa}^2 = \text{bhujā}^2 + \text{koṭi}^2$$

This is the famous Pythagorean theorem, known in India as the *Śulba-sūtra* theorem.
Example: Find the hypotenuse when base = 3 and perpendicular = 4:

$$\text{karṇa} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Example from the text: For base = 12, perpendicular = 5:

$$\text{karṇa} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Example: For base = 9, perpendicular = 12:

$$\text{karṇa} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$$

6.3 कोटिकर्णज्ञान — Finding the Koṭi

When the hypotenuse and one side are known:

$$\text{koṭi} = \sqrt{\text{karṇa}^2 - \text{bhujā}^2}$$

Example: karṇa = 85, bhujā = 36:

$$\text{koṭi} = \sqrt{85^2 - 36^2} = \sqrt{7225 - 1296} = \sqrt{5929} = 77$$

Verification: $36^2 + 77^2 = 1296 + 5929 = 7225 = 85^2$ ✓

6.4 वृत्तक्षेत्र — Circle

सूत्रम्: व्यासवर्गपादं भुजवर्गस्य पञ्चगुणस्य चतुर्थांशः वृत्तफलम्।

Rule: The area of a circle is one-fourth the square of the diameter, or equivalently: the square of the radius multiplied by the ratio of circumference to diameter.

$$A = \frac{d^2}{4} \times \pi \approx \frac{d^2}{4} \times \frac{3927}{1250} \text{ (Bhāskara's refined value)}$$

Bhāskara uses $\pi \approx \frac{22}{7}$ for rough calculations and $\pi \approx \frac{3927}{1250} = 3.1416$ for more precise work.

7 खातव्यवहार — Excavations

खातव्यवहारः —
॥ ॥

khātavyavahāraḥ

This chapter deals with calculations related to excavations, trenches, and tanks.

खातस्य फलं दीर्घं विस्तारं गभीरता च तेषां गुणनफलम् ॥

khātasya phalaṃ dīrghaṃ vistāraṃ gabhīratā ca teṣāṃ guṇanaphalam ||

Problem: An excavation is to be made with length 30, breadth 20, and depth 8. Find the volume of earth to be removed.

$$V = l \times b \times d = 30 \times 20 \times 8 = 4800 \text{ cubic units}$$

7.1 सच्छिद्रखात — Excavations with Sloping Sides

For excavations with sloping sides, the *śāṇi* (average) method is used:

$$V = \frac{A_1 + A_2 + A_3 + A_4}{4} \times \text{length}$$

where A_1, A_2, A_3, A_4 are the cross-sectional areas at different depths.

8 चिति व्यवहार — Piles of Bricks

चितिव्यवहारे स्तूपचितीनां समचितीनां फलं यथासङ्ख्यम् ॥

citivyavahāre stūpacitīnāṃ samacitīnāṃ phalaṃ yathāsaṅkhyam ||

This chapter deals with stacking problems and the calculation of volumes of piles.
Problem: Bricks are piled in the form of a pyramid with a rectangular base. The base has 10 bricks along one side and 6 along the other. The pile is 5 layers high. Find the total number of bricks.
Solution: Using the formula for a pyramidal pile:

$$N = \frac{n(n+1)(2n+1)}{6} \text{ (for square pyramids)}$$

For rectangular piles, the sum is computed layer by layer.

9 छायाव्यवहार — Shadow Problems

छायाव्यवहारकथन —
॥ ॥

chāyāvyavahārakathana

This chapter uses the properties of similar triangles to solve problems involving shadows, heights of objects, and distances.

सूत्रम्: छायाग्रदूरं द्विजोत्कर्षं कुर्यात्।

Rule: The distance to the tip of the shadow multiplied by the height of the object, divided by the length of the shadow, gives the height of the light source.

दीपस्य स्तम्भस्य ऊर्ध्वे दीपः स्थापितः। तस्य छायायां पुरुषः स्थितः।

dīpasya stambhasya ūrdhve dīpaḥ sthāpitaḥ | tasya chāyāyāṃ puruṣaḥ sthitaḥ |

Problem: A lamp is placed on a pillar. A man of height 6 stands 8 units away from the pillar, and his shadow extends 12 units from his feet. Find the height of the lamp.
Solution: Using similar triangles:

$$\frac{h}{6} = \frac{8+12}{12} = \frac{20}{12}$$

$$h = 6 \times \frac{20}{12} = 10 \text{ units}$$

10 कुट्टकव्यवहार — The Pulverizer

कुट्टकव्यवहारः —
॥ ॥

kuṭṭakavyavahāraḥ

The *kuṭṭaka* (pulverizer) is the method for solving linear Diophantine equations of the form $ax + by = c$, where a, b, and c are given integers, and x, y are unknown integers to be found.

सूत्रम्: द्विच्छेदः कुट्टकः—

Rule: The pulverizer is the division of two (numbers), producing a quotient and remainder, used to find the GCD through the Euclidean algorithm.

कुट्टके स्थिरकुट्टकं प्रवर्तते। द्वयोः राश्योः महत्तमं समवायपदं ज्ञात्वा...
कुट्टकविधिः युक्लिडीयकलनविधिना सम्बद्धः अस्ति।

kuṭṭake sthirakuṭṭakaṃ pravartate | dvayoḥ rāśyoḥ mahattamaṃ samavāyapadaṃ jñātvā... ||

Problem: Find x and y such that $121x + 17 = y$
Solution: Using the kuṭṭaka method:
Step 1: Apply the pulverizer to 121 and 17:

$$\begin{aligned} 121 &= 7 \times 17 + 2 \\ 17 &= 8 \times 2 + 1 \end{aligned}$$

Step 2: Work backwards:

$$\begin{aligned} 1 &= 17 - 8 \times 2 \\ &= 17 - 8 \times (121 - 7 \times 17) \\ &= 57 \times 17 - 8 \times 121 \end{aligned}$$

So $x = 8$, $y = 121 \times 8 + 17 = 985$ gives a solution.

11 अङ्कपाश — Combinatorial Problems

अङ्कपाशः —
॥ ॥

aṅkapāśaḥ

This chapter contains problems related to permutations and combinations, and number puzzles. The Sanskrit term *aṅkapāśa* literally means "net of digits" — the entanglement of numbers.

11.1 अङ्कपाशविधि — Digit Arrangements

एकादशअङ्कैः कति विभिन्नसंख्याः रचयितुं शक्याः ?

ekādaśaṅkaiḥ kati vibhinnasaṅkhyāḥ racayituṃ śakyāḥ ?

Problem: Using the digits 1, 2, 3, 4, how many different 4-digit numbers can be formed?
Solution: $4! = 24$ different numbers.
This is the fundamental counting principle — each position can be filled by any of the remaining digits.

11.2 अङ्कपाशमें विशेषाविधि — Special Methods

एका संख्या त्रिगुणीकृता सप्तभक्ता द्विशेषं लभते।

ekā saṅkhyā triguṇīkṛtā saptabhaktā dviśeṣaṃ labhate |

Problem: A certain number, when multiplied by 3 and divided by 7, gives a remainder of 2. Find the number. This is equivalent to solving:

$$3x \equiv 2 \pmod 7$$

Since $3 \times 5 = 15 \equiv 1 \pmod 7$, the multiplicative inverse of 3 mod 7 is 5.

$$x \equiv 2 \times 5 = 10 \equiv 3 \pmod 7$$

So $x = 3 + 7k$ for any integer k. The smallest positive solution is $x = 3$.

12 श्रेणी — Progressions

श्रेणी —
॥ ॥

śreṇī

The *śreṇī* (progression) chapter covers arithmetic and geometric progressions.

12.1 समश्रेणी — Arithmetic Progression

समश्रेण्यां प्रथमपदं चयं गच्छं च ज्ञात्वा सर्वं ज्ञायते ॥

समश्रेण्यां (□□□□□□□□□□ □□□□□□□□□□□) प्रथमं पदम् a, चयः (□□□□□□ □□□□□□□□□□) d, गच्छः (□□□□□□ □□ □□□□□) n चेत् —

samaśreṇyāṃ prathamapadaṃ cayaṃ gacchaṃ ca jñātvā sarvaṃ jñāyate ||

For an arithmetic progression with first term a, common difference d, and n terms:

- n-th term: $a_n = a + (n-1)d$
- Sum of n terms: $S_n = \frac{n}{2}[2a+(n-1)d] = \frac{n}{2}(a+l)$

where l is the last term.

12.2 गुणोत्तरश्रेणी — Geometric Progression

गुणोत्तरश्रेण्यां प्रथमपदं गुणोत्तरं च ज्ञात्वा सर्वं ज्ञायते ॥

guṇottaraśreṇyāṃ prathamapadaṃ guṇottaraṃ ca jñātvā sarvaṃ jñāyate ||

For a geometric progression with first term a and common ratio r:

- n-th term: $a_n = a \cdot r^{n-1}$
- Sum of n terms: $S_n = a \cdot \frac{r^n-1}{r-1}$

The Chessboard Problem:
Problem: A king offers a reward of 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, and so on, doubling each time. Find the total grains on all 64 squares.
Solution:

$$S_{64} = 1 + 2 + 4 + \cdots + 2^{63} = \frac{2^{64}-1}{2-1} = 2^{64} - 1$$
$$= 18{,}446{,}744{,}073{,}709{,}551{,}615 \text{ grains}$$

Appendix: Famous Problems from the Līlāvatī

The Lotus Problem (कमलसमस्या)

कमलस्य पत्रं जलात् ऊर्ध्वं स्थितम्। वायुना पीडितं जलस्य अन्तः गतम्।
॥ ॥

kamalasya patraṃ jalāt ūrdhvaṃ sthitam | vāyunā pīḍitaṃ jalasya antaḥ gatam |

A lotus flower stands x units above the water surface. When blown by the wind, it sinks d units from its original vertical position. Find the depth of the water.

Solution: If the lotus is x above the water, the stem length equals the depth h plus x. When blown, the stem forms the hypotenuse of a right triangle:

$$(h+x)^2 = h^2 + d^2$$

$$h^2 + 2hx + x^2 = h^2 + d^2$$

$$h = \frac{d^2 - x^2}{2x}$$

[This is a classic application of the Pythagorean theorem (bhujā-koṭi-karṇa relation).]

The Bee Problem (भ्रमरसमस्या)

भ्रमराणां पञ्चमांशः कुटजपुष्पे तृतीयांशः शिलीन्ध्रपुष्पे च ॥
अन्तरस्य त्रिगुणं क्रातपुष्पे एकः चन्दने विहरति ॥
॥ ॥

bhramarāṇām pañcamāṃśaḥ kuṭajapuṣpe tṛtīyāṃśaḥ śilīndhrapuṣpe ca ||
antarasya triguṇaṃ krātapuṣpe ekaḥ candane viharati ||

A fifth of a swarm of bees settled on a *kuṭaja* flower, a third on a *śilīndhra* flower. Three times the difference between these two groups flew over a *krāta* flower. One bee was left flying about, attracted by the fragrance of a jasmine and a pandanus. How many bees were in the swarm?

Solution: Let x be the total number of bees.

$$\frac{x}{5} + \frac{x}{3} + 3\left(\frac{x}{3} - \frac{x}{5}\right) + 1 = x$$

$$\frac{x}{5} + \frac{x}{3} + 3 \cdot \frac{2x}{15} + 1 = x$$

$$\frac{3x + 5x + 6x}{15} + 1 = x \implies \frac{14x}{15} + 1 = x \implies x = 15$$

Answer: 15 bees. *[This is one of the most famous problems from the Līlāvatī, demonstrating skill in working with fractions and linear equations.]*

The Peacock and the Snake (मयूरसर्पकथा)

मयूरः स्तम्भे स्थितः सर्पः तस्य पादात् दूरे स्थितः ॥
मयूरः सर्पं ग्रहीतुं पतति समदूरी क्षिप्त्वा ॥
॥ ॥

mayūraḥ stambhe sthitaḥ sarpaḥ tasya pādāt dūre sthitaḥ ||
mayūraḥ sarpaṃ grahītuṃ patati samadūrī kṣiptvā ||

A peacock is perched on a pillar of height P. A snake is at distance D from the base of the pillar. The peacock pounces on the snake in a straight line, covering equal distances down the pillar and along the ground. Find where they meet.

Solution: Let the meeting point be at distance x from the base of the pillar. The peacock flies in a straight line, and the problem states it covers equal vertical and horizontal distances:

$$P - x = \sqrt{x^2 + D^2}$$

Squaring both sides:

$$(P - x)^2 = x^2 + D^2 \implies P^2 - 2Px + x^2 = x^2 + D^2$$

$$x = \frac{P^2 - D^2}{2P}$$

[This combines the Pythagorean theorem with the concept of equal distances. The problem is notable for its elegant reduction to a simple algebraic expression.]

The Swallow Problem (हंससमस्या)

Problem: Swallows are sitting on the branches of a tree. If each branch has 3 swallows, one swallow is left over. If each branch has 4 swallows, one branch is left empty. How many swallows and branches are there?

Solution: Let s = number of swallows, b = number of branches.

$$s = 3b + 1$$
$$s = 4(b - 1) = 4b - 4$$

Equating: $3b + 1 = 4b - 4$, so $b = 5$ and $s = 16$.
Answer: 16 swallows and 5 branches. *[This is a classic example of a system of linear equations with a unique solution. It is equivalent to finding x such that $x \equiv 1 \pmod{3}$ and $x \equiv 0 \pmod{4}$.]*

The Pearl Necklace Problem (मुक्ताहारसमस्या)

मुक्ताहारे मुक्ताः स्रंसन्ति। एकैकस्य मुक्तायाः मूल्यं ज्ञात्वा...
॥ ॥

muktāhāre muktāḥ sraṃsanti | ekaikasya muktāyāḥ mūlyaṃ jñātvā...

A string of pearls breaks. One third of the pearls fall to the ground, one fifth remain on the bed, one sixth are saved by the young woman, one tenth are caught by her lover, and six pearls remain on the string. How many pearls were there originally?

Solution: Let x be the total number of pearls.

$$\frac{x}{3} + \frac{x}{5} + \frac{x}{6} + \frac{x}{10} + 6 = x$$

The LCD of 3, 5, 6, 10 is 30:

$$\frac{10x + 6x + 5x + 3x}{30} + 6 = x \implies \frac{24x}{30} + 6 = x$$

$$6 = x - \frac{4x}{5} = \frac{x}{5} \implies x = 30$$

Answer: 30 pearls. *[This problem demonstrates the use of finding a common denominator and solving linear equations with fractions — a common practical problem type in the Līlāvatī.]*

Glossary of Technical Terms

संस्कृत	IAST	Meaning
व्यवहार	vyavahāra	Procedure, operation, practice
गुणन	guṇana	Multiplication
भागहार	bhāgahāra	Division
क्षेत्र	kṣetra	Field, area, plane figure
राशि	rāśi	Quantity, number, heap
सङ्कलन	saṅkalana	Addition
व्यवकलन	vyavakalana	Subtraction
वर्ग	varga	Square
वर्गमूल	vargamūla	Square root
घन	ghana	Cube
घनमूल	ghanamūla	Cube root
भिन्न	bhinna	Fraction
प्रमाण	pramāṇa	Measure (in Rule of Three)
फल	phala	Fruit, result
इच्छा	icchā	Desire, requisition
त्रैराशिक	trairāśika	Rule of Three
करण	karaṇa	Operation, method
सूत्र	sūtra	Rule, aphorism
भुजा	bhujā	Base (of triangle)
कोटि	koṭi	Perpendicular
कर्ण	karṇa	Hypotenuse, diagonal
श्रेणी	śreṇī	Progression, series
चय	caya	Common difference (in AP)
गुणोत्तर	guṇottara	Common ratio (in GP)
कुट्टक	kuṭṭaka	Pulverizer (linear Diophantine)
अङ्कपाश	aṅkapāśa	Combinatorial problems

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iti līlāvatī samāptā

इति लीलावती समाप्ता ॥

Thus ends the Līlāvatī.

columnsep color

श्रीब्रह्मगुप्ताचार्य-विरचित

Brāhma-sphuṭa-siddhānta

Bilingual Sanskrit–English Edition

Part I, Chunk 1: Pages 1–140

Chapters I–VIII

Sanskrit Text in Devanāgar
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Contents

Introduction to This Bilingual Edition

This bilingual edition presents the first part of Brahmagupta's *Brāhmasphuṭasiddhānta* (628 CE) in a parallel-column format. The left column contains the original Sanskrit text in Devanāgar
script, while the right column provides the English translation with IAST for technical terms.

About the Work

The *Brāhmasphuṭasiddhānta* ("Correctly Established Doctrine of Brahma") is one of the most important works in the history of Indian mathematics and astronomy. Composed by Brahmagupta in 628 CE at Bhillamāla (modern Bhinmal, Rajasthan), it consists of 24 chapters covering astronomical computations, mathematics, and astronomical instruments.

Key Contributions in Chapter VIII

Chapter VIII (the Gaṇitādhyāya) contains Brahmagupta's most celebrated mathematical discoveries:

1. **Rules for zero**: systematic arithmetic with zero as a number
2. **Sign rules**: $(-)(-) = (+), (+)(-) = (-)$
3. **Quadratic formula**: Complete solution of $ax^2 + bx = c$
4. **Brahmagupta's formula**: For the area of a cyclic quadrilateral
5. **Pell's equation**: The composition method (bhāvanā)

Chapter I

Mean Longitudes of Planets | मध्यमाधिकारः

2

[0] 1 नमोऽस्तु ब्रह्मणे तस्मै यत्प्रसादवशात्पुनः
ज्ञातं ग्रहगणितं मे तत्समीकरणं क्रियते

[1] **Verse 1**: Salutation to that Brahman, by whose grace I have again come to know the computation of planets; now I proceed to reconcile it.

[0] 2 ब्रह्मोक्तं ग्रहगणितं महता कालेन यत् स्खलीतं मतम्
श्रीभ्रशीयते स्फुटं तज्ज्ञप्तं तद्ब्रह्मगुप्तेन

[1] **Verse 2**: The planetary computation taught by Brahman had become inaccurate through the lapse of long time; that has been correctly established by Brahmagupta.

[0] 3 युगब्धिर्युगवर्षाणि युगमासा युगाहराः
दिनैः सौरैर्भवत्यब्दस्तिथिभिश्चान्द्रमाः स्मृतः

[1] **Verse 3**: The years of a *yuga* are its *abda*; the days of a *yuga* are its *ahargaạ*. A year consists of solar days; a month of lunar *tithi*s.

[0] 4 नक्षत्रैः सवनाभ्दस्तु सैराप्यथ चतुर्गुणः
पञ्चदश्यां चतुर्दश्यां पूर्णे चान्द्रमसी स्मृते

[1] **Verse 4**: A *sāvana* year consists of *nakṣatra* days; a solar year is fourfold. The full moon at the fifteenth *tithi* is complete, at the fourteenth incomplete.

[0] 5 चतुर्युगसहस्रं तु ब्रह्मणो दिवसं स्मृतम्
तदर्धं रात्रिरित्युक्तं तस्यैते कल्पसंज्ञिताः

[1] **Verse 5**: A thousand *caturyuga*s are a day of Brahman; half is night. These are called a *kalpa*.

[0] 6 कल्पस्यादौ भवेद्ब्रह्मा मध्ये विष्णुस्तथान्तके
रुद्रस्त्रैलोक्यनाथास्ते त्रयस्ते संप्रकीर्तिताः

[1] **Verse 6**: At the beginning of a *kalpa* arises Brahmā, in the middle Viṇu, at the end Rudra: these three lords of the three worlds.

[0] 7 तस्मिन्कल्पे तु ब्रह्माणि षड्जीवाः संप्रकीर्तिताः
मनवः संस्थितास्तत्र मनुष्याः पूर्वसंज्ञिताः

[1] **Verse 7**: In that *kalpa*, six *j*
=ivas — the *manus* — are established; there dwell the people called *p*
=urva.

[0] 8 युगानां चत्वारि स्मृतानि सहस्राणि युगे तु ते
चतुर्युगं तत्सहस्रं ब्रह्मणो दिवसं स्मृतम्

[1] **Verse 8**: Four world-ages in a *yuga*; a thousand *caturyuga*s are a day of Brahman.

[0] 9 कृतं त्रेताद्वापरं च कलिश्चैव युगानि तु
चतुर्युगं तत्सहस्रं ब्रह्मणो दिवसं स्मृतम्

[1] **Verse 9**: The *kṛta*, *trētā*, *dvāpara*, and *kali* are the ages; a thousand *caturyuga*s are a day of Brahman.

[0] 10 कृतं धर्मचतुष्पात्त्रेतायां तु त्रिपादपि
द्वापरे द्वपदं प्रोक्तं कलौ पादस्तु धर्मतः

[1] **Verse 10**: In *kṛta*, dharma has four feet; in *trētā* three; in *dvāpara* two; in *kali* one.

[0] 11 संवत्सरशतं मानुषं दिव्यं वर्षं प्रकीर्तितम्
दिव्यानां संख्यया वर्षं ब्रह्मणो दिवसं स्मृतम्

[1] **Verse 11**: A hundred human years is a *divya* year; by such divine years, a day of Brahman is known.

[0] 12 ब्रह्मविष्णिवन्द्ररुद्राणां मनूनां च यथाक्रमम्
चतुर्युगसहस्रं तु ब्रह्मणो दिवसं स्मृतम्

[1] **Verse 12**: The thousand *caturyuga*s of Brahmā, Viṇu, Indra, Rudra, and the *manus* are a day of Brahman.

[0] 13 मनुष्यदेवगन्धर्वा राक्षसानि सरीसृपाः
पितरश्चैव सर्वे ते कल्पादौ संप्रकीर्तिताः

[1] **Verse 13**: Humans, gods, *gandharvas*, *rākṣasas*, serpents, and ancestors — all proclaimed at the start of the *kalpa*.

[0] 14 ग्रहनक्षत्रमासर्त्वृतुसंवत्सरकादयः
पञ्चविंशतितत्त्वानि तानि ज्ञेयानि यत्नतः

[1] **Verse 14**: Planets, asterisms, months, seasons, years — these twenty-five *tattvas* to be known diligently.

[0] 15 चतुर्भिस्तु मानवर्गेण ग्रहाणां गणितं भवेत्
तेषां संख्यां प्रवक्ष्यामि यथावदनुपूर्वशः

[1] **Verse 15**: By four measures the planets are computed; I declare their numbers in order.

[0] 16 सैरमानेन भानूनां चान्द्रेण शशिनो गतिः
नाक्षत्रेण ग्रहाणां तु सावनेन दिनस्य च

[1] **Verse 16**: By solar measure, the Sun's motion; by lunar, the Moon's; by sidereal, the planets'; by civil, the day's.

[0] 17 एभिस्तु नवभिर्मानैर्व्यवहारः प्रवर्तते
लोकस्यास्य तु सर्वस्य ग्रहाणां गणितं प्रति

[1] **Verse 17**: By these nine measures the world's transactions proceed, for planetary computation.

[0] 18 चतुर्भिस्तु मनुष्याणां व्यवहारः प्रवर्तते
सौरचान्द्रनाक्षत्रसावनैर्मानैः क्रमात्

[1] **Verse 18**: For humans, transactions by four measures: solar, lunar, sidereal, and civil, in order.

[0] 19 ग्रहगणितं तु यत्किञ्चित्तत्सर्वं मानसाधितम्
मानैर्विना न तज्ज्ञानं ग्रहाणां संप्रवर्तते

[1] **Verse 19**: All planetary computation is accomplished by measures; without them, that knowledge does not proceed.

[0] 20 मानानि सैरचान्द्राणि सावनानि ग्रहानयनम्
एभिर्मानैश्चतुर्भिस्तु संयवहारोज्जलोकस्य

[1] **Verse 20**: The solar, lunar, sidereal, and civil measures are for planetary computation; by these four, the world's transactions proceed.

[0] 21 तिथयः पञ्चदश्यां तु शुक्ले कृष्णे तथैव च
एकं मासं तु ताभिस्तु संवत्सरः स तैस्तु वै

[1] **Verse 21**: Fifteen *tithis* in bright and dark fortnight make one month; a year of those.

[0] 22 मासैर्द्वादशभिर्वर्षं त्रयोदशभिरेव च
चतुर्भिर्हीनसंवत्सरं पञ्चभिर्धनसंवत्सरम्

[1] **Verse 22**: Twelve months a year; with thirteen, deficient; with five extra, abundant.

[0] 23 गतपूर्वां गतिमुक्त्वा तु ग्रहाणां योजयेद्बुधः
ताभिर्मध्यमसंज्ञाभिर्ग्रहाणां गणितं भवेत्

[1] **Verse 23**: Having stated past and future motions, the wise compute; by these mean quantities, planetary computation proceeds.

[0] 24 मध्यमात्स्फुटं ज्ञात्वा ततः पुनः संस्कारैर्विधातव्यम्
यथा स्यात्स्फुटतत्त्वतो ग्रहाणां गणितं बुधैः

[1] **Verse 24**: Knowing true from mean longitudes, then corrections must be applied for exactness.

[0] 25 तिथ्यन्तरं नक्षत्रं तु ग्रहाणां तु गतिर्यथा
तत्सर्वं मानवर्गेण कल्पितं तु यथाक्रमम्

[1] **Verse 25**: The *tithi*, *antara*, *nakṣatra*, and planetary motions — all devised by measures in order.

[0] 26 इति मानानि सर्वाणि यथावदनुपूर्वशः
प्रोक्तानि तानि सर्वाणि ग्रहगणितचिन्तकैः

[1] **Verse 26**: Thus all measures declared in order by those who contemplate planetary computation.

[0] 27 युगब्धिर्युगवर्षाणि युगमासा युगाहराः
दिनैः सौरैर्भवत्यब्दस्तिथिभिश्चान्द्रमाः स्मृतः

[1] **Verse 27**: The years of a *yuga*, its months and days. A year of solar days; a month of lunar *tithis*.

[0] 28 चतुर्दशभुवनैः सार्धैः सौरो वर्षो विधीयते
तमतिक्रम्य ग्रहाणां मध्यमा गतिरिष्यते

[1] **Verse 28**: The solar year: fourteen and a half terrestrial years; mean motion reckoned beyond.

[0] 29 तिथयः पञ्चदश्यां तु शुक्ले कृष्णे तथैव च
एकं मासं तु ताभिस्तु संवत्सरः स तैस्तु वै
[1] **Verse 29**: Fifteen *tithi*s in each fortnight; a month of those, a year of months.
[0] 30 मासैर्द्वादशभिर्वर्षं त्रयोदशभिरेव च
चतुर्भिर्हीनसंवत्सरं पञ्चभिर्धनसंवत्सरम्
[1] **Verse 30**: Twelve months a year; thirteen, deficient; five extra, abundant.
[0] 31 गतपूर्वां गतिमुक्त्वा तु ग्रहाणां योजयेद्बुधः
ताभिर्मध्यमसंज्ञाभिर्ग्रहाणां गणितं भवेत्
[1] **Verse 31**: Past and future motions stated, the wise compute by mean quantities.
[0] 32 मध्यमात्स्फुटं ज्ञात्वा ततः पुनः संस्कारैर्विधातव्यम्
यथा स्यात्स्फुटतत्त्वतो ग्रहाणां गणितं बुधैः
[1] **Verse 32**: From mean to true, corrections applied, for exact planetary computation by the wise.

Chapter II

True Longitudes of Planets | स्फुटाधिकारः

2
[0] 1 मध्यमागत्या ग्रहाणां स्फुटं ज्ञातुं तु यः स्पृहन्
संस्कारान्सोऽधिगच्छेत तत्स्फुटं स्यात्तु तत्त्वतः
[1] **Verse 1**: He who desires the true from the mean longitudes of planets should understand the corrections; then the true is obtained.
[0] 2 मन्दोच्चशीघ्रोच्चयोः क्षेत्रे चक्रार्धं तु पञ्चमम्
तत्र मन्दगतिः प्रोक्ता शीघ्रगतिस्ततः परम्
[1] **Verse 2**: The field of *manda* and
=*s*
=*ighra* apogees is the fifth of a circle (72°); there the *manda* motion, and
=*s*
=*ighra* beyond.
[0] 3 मन्दफलं तु तत्क्षेत्रे शीघ्रफलं तथैव च
तयोर्योगेन हानं वा स्फुटं ज्ञेयं तु तत्त्वतः
[1] **Verse 3**: The *manda-phala* in that field, and
=*s*
=*ighra-phala* likewise; by their sum or difference, the true is known.
[0] 4 मन्दोच्चाद्धृतं क्षेत्रं तु मन्दफलस्य कारणम्
शीघ्रोच्चाद्धृतमप्येवं शीघ्रफलस्य कारणम्
[1] **Verse 4**: The arc from the *manda* apogee causes the *manda-phala*; from the
=*s*
=*ighra* apogee, the
=*s*
=*ighra-phala.*
[0] 5 मध्यमं मन्दफलेनैव युक्तं वा हीनमेव वा
तत्फलं शीघ्रसंस्कारैर्युक्तं वा हीनमेव वा
[1] **Verse 5**: The mean increased or decreased by *manda-phala*; that result increased or decreased by
=*s*
=*ighra* corrections.
[0] 6 ततः स्फुटं भवेद्ग्रहं मन्दशीघ्रक्रमेण तु
एवं ग्रहगणितं प्रोक्तं यथावदनुपूर्वशः
[1] **Verse 6**: Thence the true planet, by *manda* and
=*s*
=*ighra* process in order; thus planetary computation declared.
[0] 7 सूर्यस्य तु गतिः प्रोक्ता मन्देनैव विना क्वचित्
शीघ्रेण सहिता तस्मात्तत्स्फुटं ज्ञेयमेव तु
[1] **Verse 7**: The Sun’s motion sometimes without *manda*; together with
=*s*
=*ighra*, the true is known.
[0] 8 चन्द्रस्य मन्दसंस्कारः शीघ्रसंस्कार एव च
तयोर्योगेन हानं वा स्फुटं तस्य प्रजायते
[1] **Verse 8**: For the Moon, *manda* and
=*s*
=*ighra* corrections; by sum or difference, the true arises.
[0] 9 उच्चे मन्दगतिर्यावत् तावत् तत्र फलं स्मृतम्
नीचे तु द्विगुणं प्रोक्तं त्रिगुणं चान्तरालके
[1] **Verse 9**: At apogee, onefold *manda* result; at perigee, double; in between, triple.

[0] 10 चतुर्भिर्मानवर्गेण संस्काराः षोडश स्मृताः
तैः संस्कृतं ग्रहं ज्ञात्वा स्फुटं तत्र प्रकल्पयेत्
[1] **Verse 10**: By four measures, sixteen corrections; knowing the corrected planet, compute the true.
[0] 11 तिथ्यब्दयुगवर्षेषु मासेषु दिवसेषु च
गतिस्तेषां ग्रहाणां तु मध्यमाद्यैः प्रकल्पिता
[1] **Verse 11**: In *tithi*s, years, *yuga*s, months, and days, planetary motion computed from means.
[0] 12 मध्यमात्स्फुटमानं तु ग्रहाणां योजयेद्बुधः
ततो ग्रहगणितं प्रोक्तं सर्वं स्फुटतरं भवेत्
[1] **Verse 12**: From mean to true measure of planets, the wise compute; all becomes more exact.
[0] 13 एवं स्फुटगतिः प्रोक्ता ग्रहाणां यथाक्रमम्
ताभिर्ग्रहगणितं सर्वं यथावदनुपूर्वशः
[1] **Verse 13**: Thus true motion declared for planets in order; by them all computation proceeds.
[0] 14 रवेर्मन्दफलं हीनं शीघ्रफलं तु पूजितम्
चन्द्रस्य तु गतिः प्रोक्ता मन्दशीघ्रक्रमेण तु
[1] **Verse 14**: For Sun, *manda-phala* subtracted,
=s
=ighra-phala added; for Moon, by both processes.
[0] 15 ताराग्रहाणां सर्वेषां मन्दशीघ्रफले सदा
तयोर्योगेन हानं वा स्फुटं तत्र प्रजायते
[1] **Verse 15**: For all star-planets, *manda* and
=s
=ighra equations always; by sum or difference, the true arises.
[0] 16 पूर्वापरं गतिं ज्ञात्वा मन्दशीघ्रक्रमेण तु
ततः स्फुटं भवेद्ग्रहं सर्वदैव गणितज्ञैः
[1] **Verse 16**: Knowing eastward and westward motion by both processes, the true planet is always obtained by calculators.
[0] 17 एवं स्फुटगतिः प्रोक्ता ग्रहाणां सूर्यजन्यपि
ताराग्रहाणां सर्वेषां यथावदनुपूर्वशः
[1] **Verse 17**: Thus true motion declared for Sun-born planets and all star-planets in order.
[0] 18 ग्रहाणां मध्यमाज्ज्ञात्वा स्फुटं यः कर्तुमिच्छति
संस्कारान्सोऽधिगच्छेत ततः स्फुटतरं भवेत्
[1] **Verse 18**: He who desires true from mean of planets should understand corrections; it becomes more exact.

Chapter III

Three Problems: Time, Place, and Direction | त्रिप्रश्नाधिकारः

2
[0] 1 त्रयः प्रश्नाः प्रवक्ष्यामि ग्रहाणां गणिते सदा
कालस्थानदिशां ज्ञानं तेषां ज्ञात्वा फलं लभेत्
[1] **Verse 1**: I declare the three questions always in planetary computation: knowing time, place, and direction, one obtains the result.
[0] 2 कालः स्थानं दिशं चैव त्रय एव ग्रहस्य तु
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 2**: Time, place, and direction — these three for a planet; joined with their knowledge, it becomes more exact.
[0] 3 कालज्ञानाद्गतिं ज्ञात्वा स्थानज्ञानात्तु याम्यताम्
दिग्ज्ञानादायतिं प्रोक्तां त्रय एव फलप्रदाः
[1] **Verse 3**: From time, motion; from place, declination; from direction, amplitude — these three give results.
[0] 4 कालः खचरसंज्ञः स्यात्स्थानं भूचरमेव तु
दिशं तु दिग्गजं प्रोक्तं त्रयस्ते ग्रहसाधनाः
[1] **Verse 4**: Time is called sky-wandering (*kha-cara*); place is earth-wandering (*bh=u-cara*); direction is direction-elephant (*dig-gaja*) — these three accomplish the planet.
[0] 5 तिथ्यब्दयुगवर्षेषु मासेषु दिवसेषु च
कालः संख्यामयः प्रोक्तः स्थानं देशान्तरं स्मृतम्
[1] **Verse 5**: In *tithi*s, years, *yuga*s, months, days, time is numerical; place is the difference of countries (*de=sāntara*).
[0] 6 दिग्वलयं दिशं प्रोक्तां त्रयस्ते ग्रहसाधनाः
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 6**: The *dig-valaya* is direction — these three accomplish the planet; joined with knowledge, it becomes more exact.
[0] 7 देशान्तरं योजयित्वा कालेनैव तु संयुतम्
ततो ग्रहगणितं प्रोक्तं स्फुटं तत्र प्रजायते
[1] **Verse 7**: Applying the difference of countries joined with time, then planetary computation: the true arises.
[0] 8 आकाशस्य गतिः प्रोक्ता कालेनैव तु संयुता
स्थानेनैव तु संयुक्ता दिशा संयुज्यते तथा
[1] **Verse 8**: The sky's motion joined with time, with place, and with direction.
[0] 9 एवं त्रिप्रश्नसंयुक्तो ग्रहः स्फुटतरो भवेत्
तस्मात्त्रिप्रश्नज्ञानं सर्वदैव गणितज्ञैः
[1] **Verse 9**: Thus joined with three questions, the planet becomes more exact; therefore their knowledge always by calculators.
[0] 10 देशान्तरं दिशं कालं त्रयः प्रश्नाः प्रकीर्तिताः
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 10**: Difference of countries, direction, and time — these three questions proclaimed; joined with knowledge, the planet becomes exact.
[0] 11 पञ्चदश्यां चतुर्दश्यां पूर्णे चान्द्रमसी स्मृते
तिथ्यन्तरं तु तज्ज्ञेयं कालज्ञानं तु तैः सह
[1] **Verse 11**: At fifteenth and fourteenth *tithi*, full moon remembered; the *tithi-antara* known with time.
[0] 12 देशान्तरं योजयित्वा कालेनैव तु संयुतम्
ततो ग्रहगणितं प्रोक्तं स्फुटं तत्र प्रजायते
[1] **Verse 12**: Difference of countries joined with time; then planetary computation: the true produced.
[0] 13 दिग्वलयं दिशं प्रोक्तां त्रयस्ते ग्रहसाधनाः
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 13**: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

[0] **14** एवं त्रिप्रश्नसंयुक्तो ग्रहः स्फुटतरो भवेत्
तस्मात्त्रिप्रश्नज्ञानं सर्वदैव गणितज्ञैः
[1] **Verse 14**: Thus with three questions, the planet becomes exact; therefore their knowledge by calculators.
[0] **15** तिथ्यब्दयुगवर्षेषु मासेषु दिवसेषु च
कालः संख्यामयः प्रोक्तः स्थानं देशान्तरं स्मृतम्
[1] **Verse 15**: In *tithi*s, years, *yuga*s, months, days: time is numerical; place is difference of countries.
[0] **16** दिग्वलयं दिशं प्रोक्तां त्रयस्ते ग्रहसाधनाः
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 16**: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

Chapter IV

Lunar Eclipses | चन्द्रग्रहणाधिकारः

2
[0] 1 चन्द्रग्रहणसंयुक्तं प्रवक्ष्यामि यथातथम्
तस्मात्सर्वं प्रवक्ष्यामि ग्रहणस्यैव लक्षणम्
[1] **Verse 1**: I declare the lunar eclipse as it is; therefore all the characteristics of eclipse.
[0] 2 सूर्यचन्द्रमसोर्मध्ये पातो यत्र तु संस्थितः
तत्र ग्रहणसंयोगः स्याच्छुक्ले च कृष्णे तथा
[1] **Verse 2**: The node between Sun and Moon — there the eclipse conjunction, in bright and dark fortnight.
[0] 3 चन्द्रः पूर्णमसौ राहुर्ग्रसते यदि संस्थितः
तदा ग्रहणसंयोगः स्यात्ततः पूर्णमण्डलम्
[1] **Verse 3**: If Rāhu at full moon seizes the Moon, then eclipse conjunction; hence the full disk.
[0] 4 चन्द्रस्य ग्रहणं ज्ञात्वा ततः कालं प्रकल्पयेत्
मन्दशीघ्रक्रमेणैव तत्स्फुटं तत्र कारयेत्
[1] **Verse 4**: Knowing the Moon’s eclipse, then compute the time; by *manda* and
=s
=ighra process, make exact.
[0] 5 राहुश्चन्द्रं ग्रसेत्पूर्णं यदा तु पूर्णिमासिते
तदा पूर्णग्रहं ज्ञेयं रवौ तु सूर्यगं स्मृतम्
[1] **Verse 5**: When Rāhu seizes full Moon on full-moon day, total eclipse; of Sun, solar eclipse.
[0] 6 चन्द्रस्य ग्रहणं प्रोक्तं मन्दशीघ्रक्रमेण तु
तत्स्फुटं योजयित्वा तु ततो ग्रहणलक्षणम्
[1] **Verse 6**: The Moon’s eclipse declared by both processes; applying the true, then eclipse characteristics.
[0] 7 एवं चन्द्रग्रहं प्रोक्तं यथावदनुपूर्वशः
तस्मात्सर्वं प्रवक्ष्यामि सूर्यग्रहस्य लक्षणम्
[1] **Verse 7**: Thus lunar eclipse declared in order; therefore all characteristics of solar eclipse.

Chapter V

Solar Eclipses | सूर्यग्रहणाधिकारः

2
[0] 1 सूर्यग्रहणसंयुक्तं प्रवक्ष्यामि यथातथम्
तस्मात्सर्वं प्रवक्ष्यामि सूर्यग्रहस्य लक्षणम्
[1] **Verse 1**: I declare the solar eclipse as it is; therefore all characteristics of solar eclipse.
[0] 2 सूर्यचन्द्रमसोर्मध्ये पातो यत्र तु संस्थितः
तत्र सूर्यग्रहं प्रोक्तं संयोगः स्याच्छुभाशुभः
[1] **Verse 2**: The node between Sun and Moon — there the solar eclipse; the conjunction auspicious or inauspicious.
[0] 3 राहुः सूर्यं ग्रसेत्तत्र चन्द्रस्तु विमले यदि
तदा सूर्यग्रहं ज्ञेयं पूर्णं वा हीनमेव वा
[1] **Verse 3**: If Rāhu seizes Sun there, Moon clear, then solar eclipse: total or partial.
[0] 4 सूर्यस्य ग्रहणं ज्ञात्वा ततः कालं प्रकल्पयेत्
मन्दशीघ्रक्रमेणैव तत्स्फुटं तत्र कारयेत्
[1] **Verse 4**: Knowing Sun’s eclipse, compute time; by both processes, make exact.
[0] 5 सूर्यग्रहस्य लक्षणं मन्दशीघ्रक्रमेण तु
तत्स्फुटं योजयित्वा तु ततो ग्रहणलक्षणम्
[1] **Verse 5**: Solar eclipse characteristics by both processes; applying the true, then characteristics.
[0] 6 एवं सूर्यग्रहं प्रोक्तं यथावदनुपूर्वशः
तस्मात्सर्वं प्रवक्ष्यामि ग्रहणानां फलं ततः
[1] **Verse 6**: Thus solar eclipse declared in order; therefore all results of eclipses.

Chapter VI

Rise and Set of Planets | उदयास्ताधिकारः

2

[0] 1 उदयास्तं प्रवक्ष्यामि ग्रहाणां यथाक्रमम्
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 1**: I declare rising and setting of planets in order; joined with knowledge, the planet becomes exact.

[0] 2 उदयं पूर्वमुद्दिष्टमस्तं पश्चात्तु सर्वदा
तयोर्ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 2**: Rising first, setting always after; joined with knowledge, the planet exact.

[0] 3 लग्नं युक्तं दिशा ज्ञात्वा ततो ग्रहस्य चोदयम्
ततः स्फुटतरं ज्ञेयं ग्रहाणां गणितं बुधैः
[1] **Verse 3**: Knowing ascendant (*lagna*) joined with direction, then planet's rising; more exact known by wise.

[0] 4 ग्रहाणां पूर्वमुदयं पश्चादस्तमुदीरितम्
तयोर्ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 4**: For planets, rising first, setting after; joined with knowledge, exact.

[0] 5 सूर्यस्योदयमारभ्य चन्द्रादीनां यथाक्रमम्
उदयास्तं तु तेषां तु प्रवक्ष्यामि यथातथम्
[1] **Verse 5**: Beginning from Sun's rising, Moon etc. in order, their rising and setting I declare.

[0] 6 उदयास्तं तु यत्किञ्चित्तत्सर्वं मानसाधितम्
मानैर्विना न तज्ज्ञानं ग्रहाणां संप्रवर्तते
[1] **Verse 6**: All rising and setting accomplished by measures; without them, knowledge does not proceed.

[0] 7 एवमुदयमष्टौ तु ग्रहाणां योजयेद्बुधः
ततः स्फुटतरं ज्ञेयं ग्रहाणां गणितं ततः
[1] **Verse 7**: Thus eight risings of planets; the wise compute; thence more exact from computation.

Chapter VII

Constellations and the Moon's Cusps | चन्द्रशृङ्गोन्नत्यधिकारः

2

[0] 1 चन्द्रशृङ्गोन्नतिं प्रोक्तां नक्षत्राणि च तानि वै
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 1**: The elevation of Moon's horns and asterisms — joined with knowledge, the planet exact.
[0] 2 चन्द्रस्य शृङ्गमुन्नतं यदा तु पूर्णिमासिते
तदा पूर्णं शशाङ्कस्य शृङ्गमुन्नतमुच्यते
[1] **Verse 2**: When Moon's horn elevation on full-moon day, then full elevation of Moon's horns.
[0] 3 नक्षत्राणि चतुर्विंशतिस्तानि प्रोक्तानि पूर्वकैः
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 3**: Twenty-four asterisms proclaimed by ancients — joined with knowledge, exact.
[0] 4 एवं चन्द्रशृङ्गोन्नतिं नक्षत्राणि च तानि वै
तेषां ज्ञानेन संयुक्तो ग्रहः स्फुटतरो भवेत्
[1] **Verse 4**: Thus Moon's horn elevation and asterisms — joined with knowledge, exact.

Chapter VIII

Mathematics (Gaṇitādhyāya) — The Crown of the BSS | गणिताध्यायः

Note:

8.1 Arithmetic Operations (Vyakta-gaṇita)

2
[0] 1 धनधनं ऋणं ऋणं धनं धनऋणं तथा ऋणम्
संगुणितं प्रथमं स्यात्तद् द्वितीयं विपर्ययात्
[1] **Verse 1**: The product of two fortunes is a fortune; of two debts is a fortune; of a fortune and debt is a debt. The first as stated; the second by reversal of signs.
[0] 2 शून्यं खेऽधिकगुणं व्यासज्ज्ञेयं तदा यत्
शून्यं तु सदृशं संयोज्यं विपरीतं विवर्जयेत्
[1] **Verse 2**: Zero, in multiplication with a quantity, is that quantity increased; zero joined with like signs, subtracted from unlike.
[0] 3 ऋणं धनं शून्यमयं यदा गुणकारः स्मृतः
तदा गुण्यं शून्यमेव स्यात्कोट्योः शून्यं तु संगुणे
[1] **Verse 3**: When the multiplier is debt, fortune, and zero, the multiplicand is zero; in multiplication of zero, the result is zero.
[0] 4 शून्येनाभिहतं शून्यं तथा शून्यं विभाजितम्
शून्यं तु सदृशं संयोज्यं विपरीतं विवर्जयेत्
[1] **Verse 4**: Zero multiplied by a quantity is zero; zero divided likewise. Zero joined with like, subtracted from unlike.
[0] 5 खं हरः केसरांशून्यं हारश्चेद् व्यक्तवर्जितः
खं हारः खमाप्तं च गुणकः खं तु पूर्ववत्
[1] **Verse 5**: When zero is the divisor and the dividend a quantity: zero-divisor, zero quotient; the multiplier as before.

2
[0] 6 धनयोर्लाभः स्यात्सदृशोर्लाभः सदृशयोः
विधानमन्योन्यहरणं तद् व्यस्तं सदृशैरिति
[1] **Verse 6**: The gain of two fortunes is gain; of two like signs, like. Division: the result of like sign with like.
[0] 7 ऋणं धनं धनं ऋणं तयोर्भागो विपर्ययात्
भागहारेण भाज्यस्य शुद्धिः स्यात्तु यथाक्रमम्
[1] **Verse 7**: Debt by fortune gives debt; fortune by debt gives debt; division is by reversal. Signs determined in order.
[0] 8 ऊनाधिकौ शुद्धिमानं तयोर्वर्गः सदा धनः
वर्गमूलं तु तद् द्विधा कार्यं युक्तिवशात्तथा
[1] **Verse 8**: Deficient and excess are pure; their square always positive. The square root done two ways by method.
[0] 9 वर्गः समचतुष्कोणक्षेत्रं द्विगुणितं तु यत्
तस्मात्तु मूलमानीयं तद् वर्गमूलमुच्यते
[1] **Verse 9**: The square: an equal quadrilateral field; twice that. From that, root extracted: the square root.

[0] 10 घनस्तु समविषमः समचतुष्कोण एव च
तस्य मूलं घनमूलं स्यात्त्रिधा कृत्वा तु तद् यथा
[1] **Verse 10**: The cube: equal or unequal, with equal quadrilateral base. Its root: the cube root, made threefold.
[0] 11 प्रथमाद् द्वितीयकं यद् द्वितीयात्तृतीयकं तथा
तृतीयात्तु चतुर्थं स्यात्तच्छेषं घनमुच्यते
[1] **Verse 11**: From first, second cube; from second, third; from third, fourth — that remainder is the cube.

8.2 Fractions (Bhinna-gaṇita)

2
[0] 12 भिन्नं तु द्विविधं प्रोक्तं व्यक्तं व्यक्तं तथाव्यक्तम्
पृथक् स्थाप्य यथान्यायं संकलनं तत्र कारयेत्
[1] **Verse 12**: The fraction is two-fold: expressed and unexpressed; established separately, perform addition.
[0] 13 भागहारं भजेत्पूर्वं भागजातिं तथैव च
भागापवर्तनं कृत्वा ततो भिन्नं प्रकल्पयेत्
[1] **Verse 13**: Divide the division and class of divisions; having reduced fractions, form the fraction.
[0] 14 हरं हारेण निहत्य हृतं हारेण भाजयेत्
तत्र लब्धं तु तज्ज्ञेयं भिन्नसंस्कारलक्षणम्
[1] **Verse 14**: Denominator multiplied by divisor, dividend divided by denominator; the quotient: mark of fraction-operation.
[0] 15 भागाकारो हरः प्रोक्तो भाज्यो राशिः स उच्यते
लब्धं फलं तु तज्ज्ञेयं भिन्नानां संप्रकीर्तितम्
[1] **Verse 15**: The denominator: divisor-form; the dividend: quantity; the quotient: result — thus fraction proclaimed.
[0] 16 अंशेनांशं हरेणांशं हारं वा हरयाऽंशकैः
गुणयित्वा तु तद् भिन्नं भिन्नेनैव तु भाजयेत्
[1] **Verse 16**: Numerator by numerator, by denominator, denominator by numerators; having multiplied, divide by fraction.

8.3 Rules for Zero (=S =unya-gaṇita)

2
[0] 17 शून्यं खेऽधिकगुणं व्यासज्ज्ञेयं तदा यत्
शून्येनाभिहतं शून्यं तथा शून्यं विभाजितम्
[1] **Verse 17**: Zero multiplied by a greater quantity is that increased; zero multiplied is zero; zero divided likewise.
[0] 18 शून्यं तु सदृशं संयोज्यं विपरीतं विवर्जयेत्
खं हरः केसरांशून्यं हारश्चेद् व्यक्तवर्जितः
[1] **Verse 18**: Zero joined with like signs, subtracted from unlike. When zero is divisor and dividend a quantity.
[0] 19 खं हारः खमाप्तं च गुणकः खं तु पूर्ववत्
एतत् शून्यगणितं प्रोक्तं सर्वं युक्तिवशात् सदा
[1] **Verse 19**: Zero-divisor, zero quotient; multiplier as before. This zero-computation declared, all by reason.
[0] 20 शून्यं खं क्षेत्रं विद्धि खं वज्रं खं च राशयः
युतायुतं व्यतिकरं शून्यं तत्रैव निश्चितम्
[1] **Verse 20**: Zero — know as void (*kha*), field, vajra, quantities; addition, subtraction, multiplication — zero there.

[0] 21 योगे खं खगुणं राशेर्गुणखं खगुणः स्मृतः
भाज □□◌ं हारखं ज्ञेयं तत्खं गुणखमेव च
[1] **Verse 21**: In addition, zero multiplied by quantity; zero-multiplier (*kha-guṇa*) remembered; zero-divisor known.

Mathematical Commentary: Brahmagupta’s rules for zero:

- $a + 0 = a$, $a - 0 = a$
- $a \times 0 = 0$, $0 \times 0 = 0$
- $0/a = 0$
- $a/0$ undefined (*kha-hāra*, “zero-divisor”)

He does *not* resolve $a/0$, leaving it undefined — an insight 1400 years before modern analysis.

8.4 Algebra — Avyakta-gaṇita

2

[0] 22 अव्यक्तं तु गुणस्थानं राशीनां यत्क्वचिद्ध्रुवम्
तद् व्यक्तं योजयित्वा तु ततो भावी प्रकल्पयेत्
[1] **Verse 22**: The unknown (*avyakta*) is the multiplier-place of quantities; joined with the known (*vyakta*), form the result.
[0] 23 यावत्तावत् तु तज्ज्ञेयं कालको नीलकोऽपि वा
पीतको लोहितश्चैव हरितः श्वेत एव च
[1] **Verse 23**: “So much as” (*yāvat-tāvat*), or *kālaka* (black), *n*
=*ilaka* (blue), *p*
=*itaka* (yellow), *lohita* (red), *harita* (green),
’sveta (white).
[0] 24 एते वर्णाः प्रदिष्टास्ते अव्यक्तानां गुणाः स्मृताः
तैर्वर्णैर्व्यक्तसंयुक्तैरव्यक्तं तत्र कारयेत्
[1] **Verse 24**: These colors (*varṇa*) are multipliers of unknowns; by them joined with knowns, operate with unknown.
[0] 25 अव्यक्तवर्गं व्यक्तेन गुणयेत्सदृशं सदृशम्
तथा व्यक्तवर्गमप्यव्यक्तेनैव तु गुणयेत्
[1] **Verse 25**: Square of unknown multiplied by known, like by like; square of known multiplied by unknown.
[0] 26 वर्णसंकलितं ज्ञेयं वर्णभेदस्तथैव च
तयोर्ज्ञानेन संयुक्तो भावी जायेत तत्त्वतः
[1] **Verse 26**: Sum and difference of colors known; joined with knowledge, the result arises in reality.
[0] 27 रूपैः सह तु संयुक्तमव्यक्तं तत्र कारयेत्
रूपैर्वियुक्तमप्येवं तद् भावी तु ततः स्मृतः
[1] **Verse 27**: Unknown joined with *r*
=*upas* (absolute terms), operate there; unknown disjoined from *r*
=*upas* — result remembered.
[0] 28 अव्यक्तं व्यक्तसंयुक्तं व्यक्तमव्यक्तसंयुतम्
तयोर्ज्ञानेन संयुक्तो भावी जायेत तत्त्वतः
[1] **Verse 28**: Unknown joined with known, known with unknown — joined with knowledge, result arises.
[0] 29 वर्णवर्णं प्रथमं स्यात्तद् द्वितीयं तु संगुणे
वर्णस्य वर्गो भवति तद् घनस्तु तृतीयकः
[1] **Verse 29**: Color by color is first; second in multiplication; square of color; third is cube.
[0] 30 रूपव्यक्तविभेदेन समीकरणमुच्यते
तस्मात्समीकरणज्ञः स्याद् गणितज्ञानतत्परः
[1] **Verse 30**: Equation (*sam*
=*ikaraṇa*) is division of known and unknown into *r*
=*upas*; one skilled in equation is devoted to math.

8.5 The Quadratic Equation (Madhyamāharaṇa)

2
[0] 31 अव्यक्तवर्ग रूपैश्च समीकृत्य यथाविधि
द्वयोरप्येकरूपत्वे समीकरणमुच्यते
[1] **Verse 31**: Having made equal the square of unknown and *r*
*=upa*s according to rule, when both are of one *r*
=upa — equation.
[0] 32 द्विगुणेन समाराध्य वर्गाद् विषमवर्गतः
मूलं द्विधा कृतं रूपैर्युक्तं हीनं तु यत्र तत्
[1] **Verse 32**: Multiplied by twice (the coefficient), from square of equal (coefficient), subtract square of unequal; root made twofold, joined or diminished by *r*
*=upa*s.
[0] 33 तद् द्वित्र्याद्यं तु विभजेद् द्विगुणेन समारधम्
लब्धं यावत्तावद् रूपं तद् भावी तु ततः स्मृतः
[1] **Verse 33**: That, beginning with two or three, divide by equal multiplied by two; obtained: “so much as” (*yāvat-tāvat*), the *r*
=upa.
[0] 34 एतद् मध्यमाहरणं प्रोक्तं युक्तिविशारदैः
तस्मात्समीकरणज्ञः स्याद् गणितज्ञानतत्परः
[1] **Verse 34**: This *madhyam*
=aharaṇa (elimination of the middle) declared by those skilled in reasoning; one skilled in equation devoted to math.
[0] 35 रूपैः सह तु संयुक्तं यावत्तावत् तु तत्र यत्
तद् द्विगुणं समाराध्य वर्गं तत्रैव कारयेत्
[1] **Verse 35**: The “so much as” joined with *r*
*=upa*s; multiplied by two, make the square there.
[0] 36 रूपवर्गं समाराध्य समीकृत्य तु तद् द्वयोः
वर्गमूलं तु तज्ज्ञेयं मध्यमाहरणं ततः
[1] **Verse 36**: Square of *r*
=upa multiplied, both made equal; square root known — elimination of the middle.
[0] 37 तस्मात् त्र्यादेः पदं कृत्वा तद् द्वित्र्याद्यं विभाजयेत्
लब्धं यावत्तावद् रूपं तद् भावी तु ततः स्मृतः
[1] **Verse 37**: From that, root of three etc., divide beginning with two or three; obtained: “so much as”, *r*
=upa.
[0] 38 असमं सममादाय समं चासमतः पुनः
भावयेत् सममेवं तु समीकरणमुच्यते
[1] **Verse 38**: Taking unequal from equal, equal from unequal; having made equal — equation.
[0] 39 यावत्तावत् समाराध्य रूपैः सह तु संगुणम्
तद् द्विगुणं समाराध्य वर्गं तत्रैव कारयेत्
[1] **Verse 39**: “So much as” multiplied, with *r*
*=upa*s; multiplied by two, make square there.
[0] 40 रूपवर्गं समाराध्य समीकृत्य तु तद् द्वयोः
वर्गमूलं तु तज्ज्ञेयं मध्यमाहरणं ततः
[1] **Verse 40**: Square of *r*
=upa multiplied, both made equal; square root — elimination of the middle (*madhyam*
=aharaṇa).

Mathematical Commentary: In modern notation, Brahmagupta’s formula for $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

Equivalent to: $x = \dfrac{-b + \sqrt{b^2 + 4ac}}{2a}$

Brahmagupta was the first to give a general solution for *all* quadratics with two distinct positive roots.

8.6 Rule of Three (Trairā ’sika) and Proportion

2

[0] 41 त्रयो राशय एकधनं त्रैराशिकमुच्यते
तेषां संगुणितं ज्ञेयं फलराशिं तु तत्क्रमात्
[1] **Verse 41**: Three quantities of one kind — Trairā
’sika (rule of three); their product known; result-quantity in order.
[0] 42 प्रमाणं फलमित्युक्तं इच्छा फलमतः परम्
तत्रैच्छिकं फलं ज्ञेयं त्रैराशिकविधानतः
[1] **Verse 42**: Argument (*pram*
=aṇa) and result (*phala*); requisition (*icchā*) and result beyond; result of requisition by rule of three.
[0] 43 प्रमाणेन हतिच्छा फलेन भजिता यदि
लब्धं फलं तु तज्ज्ञेयं त्रैराशिकविधानतः
[1] **Verse 43**: Requisition multiplied by argument, divided by result; obtained quotient: result by rule of three.
[0] 44 विलोमं तु यदा तत्र तदा व्यस्तत्रैराशिकम्
तत्र प्रमाणफलयोर्भेदेनैव तु कारयेत्
[1] **Verse 44**: When reversed, inverse rule of three (*vyasta-trair*
=a
’sika); operate by difference of argument and result.
[0] 45 पञ्चराशिकषड्राश्यादि तथा प्रसृतं तु यत्
तत्सर्वं त्रैराशिकज्ञं व्यस्तं सममिति द्विधा
[1] **Verse 45**: Five-rule, six-rule, etc. — all from rule of three; inverse and direct, two-fold.

8.7 Arithmetic Progressions (’Sreḍh =i)

2

[0] 46 आदिरुच्छ्रायसंयुक्तं द्विगुणं सम्मतिं हरेत्
गच्छेन गुणयेत् तद् वद् गच्छोऽर्धहतोऽथवा
[1] **Verse 46**: First term joined with increase, doubled, divide by number of terms; multiply by terms, or half terms.
[0] 47 मध्यमं गच्छसंख्याभिर्गुणितं स्याद् गुणोत्तरे
समे मध्यं द्विगुणं स्याद् विषमे तु गुणोत्तरे
[1] **Verse 47**: Mean multiplied by number of terms: sum in equal difference; even: mean doubled; odd: equal difference.
[0] 48 आदेरन्त्यं युतं द्विगुणं सम्मतिं हरेत् ततः
लब्धं मध्यं तु तज्ज्ञेयं श्रेढीव्यवहारतः
[1] **Verse 48**: First joined with last, doubled, divide by terms; obtained: mean, from series operation.
[0] 49 मध्यं गच्छेन संयुक्तं सर्वं श्रेढीफलं भवेत्
आद्यन्तयोरन्तरं ज्ञात्वा गुणकं तत्र कारयेत्

[1] **Verse 49**: Mean joined with number of terms: sum of series. Knowing difference of first and last, make multiplier.
[0] 50 गच्छोनं व्येकं द्विगुणं सम्मतिं हरेत् ततः
लब्धं मध्यं तु तज्ज्ञेयं श्रेढीव्यवहारतः
[1] **Verse 50**: Terms minus one, doubled, divide by terms; obtained: mean from series operation.
[0] 51 एवं श्रेढीफलं प्रोक्तं युक्त्या युक्तविशारदैः
तस्मात्सर्वं प्रवक्ष्यामि गुणोत्तरश्रेढीतः
[1] **Verse 51**: Thus sum of series (
'sreḍh
=*i-phala*) declared by skilled reasoners; therefore geometric progression.
[0] 52 गुणोत्तरे तु यद् व्यक्तमादिरुच्छ्राय एव च
तद् गुणोत्तरजं ज्ञेयं फलं तत्र प्रकल्पयेत्
[1] **Verse 52**: In geometric progression, first term and increase known; born of multiplier; form result.
[0] 53 आदिं गुणोत्तरेणैव गुणयित्वा तु तं पुनः
गच्छोनं विभजेत् तद् वद् गुणोत्तरभुवा ततः
[1] **Verse 53**: First term multiplied by common ratio (*guṇottara*) repeatedly; divide by ratio minus one; that quotient.

8.8 Geometry — Plane Figures (Kṣetra-vyavahāra)

2
[0] 54 त्रिभुजक्षेत्रफलस्य मूलं समदलकोट्योः
सम्पर्कजातं द्वयदलसंवर्गात् पदं भवेत्
[1] **Verse 54**: Root of triangular field area: product of half-base (*sam-dala*) and perpendicular (*koṭi*); square root of half and sides.
[0] 55 चतुर्भुजस्य क्षेत्रस्य फलमूलं तु यः स्मृतः
तत्पार्श्वयोर्युतेर्वर्गात् पदं तस्य प्रजायते
[1] **Verse 55**: Square root of quadrilateral field area remembered; from square of sum of opposite sides, root produced.
[0] 56 चतुर्भुजक्षेत्रफलं समदलकोटिसंभवम्
विषमचतुष्कोणस्य फलं तु द्विधा कारयेत्
[1] **Verse 56**: Quadrilateral area from half-base and perpendicular; for irregular quadrilateral, make twofold (two triangles).
[0] 57 पार्श्वदलसंवर्गो यः स एव फलमुच्यते
तद् द्वयोर्मूलयोगेन फलं तत्र प्रकल्पयेत्
[1] **Verse 57**: Square of half sum of opposite sides — called area; by sum of roots of two, form area.
[0] 58 वृत्तक्षेत्रस्य वर्गस्तु व्यासार्धस्य तु संभवः
व्यासवर्गचतुर्भागात् क्षेत्रफलं प्रजायते
[1] **Verse 58**: Circular field square from half-diameter (*vy*
=*s*
=*ardha*); from fourth of diameter-square, area produced.
[0] 59 वृत्तक्षेत्रफलं ज्ञात्वा व्यासं तत्र प्रकल्पयेत्
चतुर्गुणं फलं कृत्वा तन्मूलं व्यास उच्यते
[1] **Verse 59**: Knowing circular field area, compute diameter; area fourfold, its root: diameter.
[0] 60 परिधिः पञ्चगुणितो व्यासस्तु सप्तभिः स्मृतः
एवं वृत्तगुणानां तु फलं तत्र प्रकल्पयेत्
[1] **Verse 60**: Circumference multiplied by five, diameter by seven remembered. Thus circle multipliers form

result.
(*Note:* $\pi \approx 22/7 \approx 3.143$ or $\sqrt{10} \approx 3.162$)
[0] 61 ज्यार्धेन संगुणं व्यासं वर्गयित्वा तु तं पुनः
चतुर्भिर्हृत्वा पदशेषं जीवा तत्र प्रकल्पयेत्
[1] **Verse 61**: Diameter multiplied by half-chord (*jy*
=ardha), squared, divided by four; remainder's root: chord (*j*
=iv
=a).
[0] 62 जीवावर्गं व्यासवर्गात् शोधयित्वा तु तं पुनः
पदशेषं तु तज्ज्ञेयं सार्धं व्यासस्य जीविनी
[1] **Verse 62**: Square of chord subtracted from square of diameter; root of remainder: arrow (
'sara, sagitta), half-diameter's chord-maker.

8.9 Brahmagupta's Formula for the Cyclic Quadrilateral

2
[0] 63 चतुर्भुजक्षेत्रफलं यत् समदलकोटिसंभवम्
तत्पार्श्वयोर्युतेर्वर्गात् पदं तस्य प्रजायते
[1] **Verse 63**: Quadrilateral area from half-base and perpendicular; from square of sum of opposite sides, root produced.
[0] 64 समचतुर्भुजक्षेत्रफलमूलं तु यः स्मृतः
पार्श्वदलसंवर्गो यः स एव फलमुच्यते
[1] **Verse 64**: Square root of equilateral quadrilateral (*sama-caturbhuja*) area; square of half sum of opposite sides: area.
[0] 65 विषमचतुर्भुजक्षेत्रफलमूलं तु यः स्मृतः
तस्माद् द्वित्र्याद्यं कृत्वा ततः फलं प्रकल्पयेत्
[1] **Verse 65**: Square root of irregular quadrilateral area; beginning with two or three, form area.
[0] 66 चतुर्भुजक्षेत्रफलं समदलकोटिसंभवम्
तस्माद् व्यासार्धजं ज्ञेयं फलं तत्र प्रकल्पयेत्
[1] **Verse 66**: Quadrilateral area from half-base and perpendicular; from half-diameter, form area.
[0] 67 चतुर्भुजक्षेत्रफलं त्रिभुजद्वयसंभवम्
तयोर्मूलयोगेन फलं तत्र प्रकल्पयेत्
[1] **Verse 67**: Quadrilateral area from two triangles; by sum of their roots, form area.
(*Brahmagupta's formula:* $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$)
[0] 68 समतृतीयभुजं क्षेत्रं समदलकोटिसंभवम्
तस्माद् द्विधा कृतं क्षेत्रं फलं तत्र प्रकल्पयेत्
[1] **Verse 68**: Isosceles trapezium (*sama-t*
=t
=iya-bhuja) from half-base and perpendicular; field twofold, form area.
[0] 69 चतुर्भुजक्षेत्रफलं पार्श्वयोर्युक्तिसंभवम्
तस्मात्त्रिभुजयुगलात् फलं तत्र प्रकल्पयेत्
[1] **Verse 69**: Quadrilateral area from sum of opposite sides; from pair of triangles, form area.
[0] 70 बाहूपच्चगुणं द्विगुणं सम्मूलयुतं तु यत्
तत् समचतुर्भुजफलं तु द्विसमचतुर्भुजे
[1] **Verse 70**: Base-height product, doubled, joined with root: equilateral and isosceles (*dvi-sama*) quadrilateral area.
[0] 71 विषमबाहुचतुर्भुजे द्विगुणं सम्मूलयुक्तम्
असममूलदलयुक्तं फलमूलं तु तत्क्षमे
[1] **Verse 71**: Irregular quadrilateral, doubled, joined with root; with half unequal root: root of area.
[0] 72 एतत् क्षेत्रव्यवहारं प्रोक्तं युक्तिविशारदैः
तस्मात्खातव्यवहारं प्रवक्ष्यामि यथाक्रमम्
[1] **Verse 72**: This operation of fields declared by skilled reasoners; therefore operation of excavations in order.

8.10 Excavations (Khāta-vyavahāra)

2

[0] 73 खातक्षेत्रफलं ज्ञात्वा ततो गात्रं प्रकल्पयेत्
आयतं दीर्घचतुरस्रं फलं तत्र प्रकल्पयेत्
[1] **Verse 73**: Knowing excavated field area, compute depth; oblong quadrilateral — form area.
[0] 74 खातफलं समाराध्य तत्र व्याप्तिं प्रकल्पयेत्
तत्र वृद्धिं तु तज्ज्ञेयं खातव्यवहारतः
[1] **Verse 74**: Volume of excavation accomplished, compute capacity; increase known from excavation operation.
[0] 75 गात्रं व्याप्तिं तु तज्ज्ञेयं खातव्यवहारतः
तस्माच्छायाव्यवहारं प्रवक्ष्यामि यथाक्रमम्
[1] **Verse 75**: Depth and capacity known from excavation operation; therefore operation of shadows in order.

8.11 Shadows and Gnomons (Chāyā-vyavahāra)

2

[0] 76 शङ्कुच्छायागतिं ज्ञात्वा ततः कालं प्रकल्पयेत्
तत्र मध्याह्नकालं तु छायाव्यवहारतः
[1] **Verse 76**: Knowing gnomon and shadow motion, compute time; midday time (*madhy*
=*ahnak*
=*ala*) from shadow operation.
[0] 77 शङ्कुस्तु द्वादशाङ्गुलः स्यात्तस्य छाया प्रकीर्तिता
तयोर्ज्ञानेन संयुक्तः कालः स्फुटतरो भवेत्
[1] **Verse 77**: Gnomon (
’saḳu) of twelve aṇgulas; its shadow proclaimed; joined with knowledge, time exact.
[0] 78 शङ्कुच्छायागुणं कृत्वा ततो दिग्गणितं भवेत्
तस्मात्त्रिज्ञानमारभ्य प्रवक्ष्यामि यथाक्रमम्
[1] **Verse 78**: Product of gnomon and shadow made, thence direction computation; beginning triangulation (*trijñ*
=*ana*), declare in order.

Colophon

End of the Bilingual Edition of the Brāhmasphuṭasiddhānta, Part I, Chunk 1

Chapters I–VIII presented in parallel Sanskrit–English columns

Sanskrit text in Devanāgar
script | English translation with IAST

Source: Critical edition, Indian Institute of Astronomical and Sanskrit Research, Delhi, 1966

Edited by Acharyavara Ram Swarup Sharma et al.

Mathematical highlights of Chapter VIII:

- Rules for zero: $a \times 0 = 0$, $0 \times 0 = 0$, $a/0$ undefined (*kha-hāra*)
- Sign rules: $(-)(-) = (+)$, $(+)(-) = (-)$
- Complete quadratic formula for $ax^2 + bx = c$
- Rule of Three (Trairā
  ’sika) and compound proportion
- Arithmetic and geometric progressions
- Brahmagupta’s formula for cyclic quadrilateral area
- Pythagorean theorem applications and chord/sagitta calculations

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ब्रह्मस्फुटसिद्धान्तः

Brāhmasphuṭasiddhānta of Brahmagupta (628 CE)

Part I, Chunk 2: Pages 141–280

Sanskrit-English Bilingual Edition

Contents include:

Uttara Khaṇḍakhādyaka — Planetary Aphelions and Epicycle Corrections

Second Differences in Interpolation

Sine Rule in Indian Trigonometry

Indian Luni-Solar Astronomy and Lunar Inequality

Spherical Astronomy — Indian and Greek Methods

Pāṭīgaṇita — Operations and Determinations

Fractions (Bhāga) — Addition, Subtraction, Multiplication, Division

Rule of Three (Trairāśika) and Compound Proportion

Algebraic Operations — Zero, Negative Numbers, and Equations

Quadratic Equations, Cube Roots, and the Kuṭṭaka Operation

Sanskrit Text with English Translation

Original verses in Devanagari from the 1966 critical edition

Typeset in XeLaTeX with Devanagari support

Contents

Preface to the Bilingual Edition

This bilingual edition presents the Sanskrit text of Brahmagupta's *Brāhmasphuṭasiddhānta* (Part I, Chunk 2, covering pages 141–280) in parallel with English translations. The layout follows a consistent format:

- **Left column:** The original Sanskrit verses in Devanagari script, set within a lightly shaded frame.
- **Right column:** The English translation and commentary.

The edition preserves all Sanskrit content from the source text, including verse references. Mathematical explanations, tables, and formulae are presented in full width where they span both languages.

Key topics covered in this chunk:

1. Algebraic operations with unknowns (jāti), zero, positive and negative numbers
2. Quadratic equations — the first general formula in Indian mathematics
3. The Kuṭṭaka (pulveriser) method for linear indeterminate equations
4. Planetary aphelions and epicycle corrections
5. Second-difference interpolation (the first in mathematical history)
6. Sine rule for plane triangles
7. Luni-solar astronomy and lunar inequality
8. Spherical astronomy — Indian and Greek methods
9. Operations with fractions (bhāga)
10. The Rule of Three (trairāśika) and compound proportion

1 Algebraic Operations (बीजगणितक्रियाः)

Brahmagupta gives systematic rules for algebraic operations in Chapter XVIII of the *Brāhmasphuṭasiddhānta*. This chapter contains the first treatment of algebra as a distinct mathematical discipline in Indian mathematics.

1.1 Addition and Subtraction of Unknowns (यावत्तावत्संयोगव्यवकल्पने)

Brahmagupta classified unknown quantities into categories called *jāti* (species):

रूपाणि यावत्तावदीनां संयोगो व्यवकल्पनं तुल्यानाम् |
तुल्यजातीयानां संयोगो व्यवकल्पनमन्यथा ||१८.२४||

Translation: “The numerical coefficients of the same unknown are added to or subtracted from each other; the same is done with the known quantities. When the unknowns are different, they are not united.”

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the “like.”

1.2 Multiplication of Unknowns (यावत्तावद्गुणनम्)

Again, regarding multiplication, Brahmagupta says:

तुल्यजातीयानां संयोगो वर्गः समजातीयानाम् |
घनस्त्रिजातीयानां ततोऽपि तद्वद्विशेषणम् ||१८.२५||

Translation: “The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called **bhāvita** (product or factum).”

1.3 Rules for Zero, Positive and Negative Numbers (शून्यधनर्णक्रियानियमाः)

Brahmagupta gives the following famous rules for operations with zero and negative numbers (BrSpSi, XVIII. 30–36):

Verse 18.30: Addition

धनयोर्धनयोर्योगो ऋणयोरपि योगमेकम् |
धनऋणयः समं शून्यं शून्याभ्यासे शून्यमेव ||३०||

Translation: “[The sum] of two positives is positive, of two negatives negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Subtraction

अल्पात्प्रचयं शेषं प्रचयादल्पकं शेषमेकम् |
ऋणधनयोः समं शून्यं शून्याभ्यासे शून्यमेव ||३१||

Translation: “If a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative, the algebraic sign of] their difference is reversed—i.e. negative [becomes] positive and positive [becomes] negative.”

Verse 18.32: Subtraction with zero

ऋणं शून्याद्वियोगेन धनं धनाद्विशोध्यते |
शून्यं शून्याद्वियोगेन शून्यं शून्येन योजयेत् ||३२||

Translation: “A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.”

Verse 18.33: Multiplication

ऋणधनयोर्गुणकर्म वधो धनयोर्गुणे ऋणयः |
ऋणयोर्गुणे धनमेकं शून्याभ्यासे शून्यमेव ||३३||

Translation: “The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs

$$(+a) \times (+b) = +ab \qquad (-a) \times (-b) = +ab$$
$$(-a) \times (+b) = -ab \qquad (+a) \times (-b) = -ab$$
$0 \times a = a \times 0 = 0$ for any quantity a.

Verse 18.34: Division

धनं धनेन विभजेदृणमृणेन तत्तथा |
शून्यं शून्येन विभजेद्धनमृणेण तदृणम् ||३४||

Translation: “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero

शून्येन भक्तश्च खहरः शून्यं हारश्च राशावमिश्रे |
तद्वर्गं मूलं तस्य तद्वद्वर्गस्य पदं तथा ||३५||

Translation (Colebrooke): “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Note: Brahmagupta’s statement that $a/0$ gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of infinity. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities represent infinitely large magnitudes was remarkable for the 7th century.

2 Quadratic Equations (वर्गसमीकरणानि)

The geometrical solution of a quadratic equation in this country would take us to the Vedic *Sulba* period. The *Bakhshālī Manuscript* also contains certain problems which need the solving of quadratic equations.

Example: A certain person travels s *yojana* on the first day and b *yojana* more on each successive day. Another who travels at the uniform rate of S *yojana* per day, has a start of t days. When will the first man overtake the second?

This leads to the quadratic:

$$s + (s + b) + (s + 2b) + \ldots + [s + (n - 1)b] = S(n + t)$$

$$ns + \frac{n(n-1)}{2}b = Sn + St$$

$$n^2 \cdot \frac{b}{2} + n\left(s - \frac{b}{2} - S\right) - St = 0$$

Sanskrit Verses: Brahmagupta's Rule for Quadratic Equations

Brahmagupta gives the following rule (BrSpSi, XVIII. 44–45):

Verse 18.44: The quadratic formula (first form)

रूपैषु चतुर्गुणितसत्त्ववर्गयुतात्पदं मूलम् ।
मध्यमहतो लब्धं द्विगुणसत्त्वभक्तं मध्यम् ।।४४।।

Translation (Colebrooke): "Diminish by the middle [number] the square-root of the *rupas* multiplied by four times the square and increased by the square of the middle [number]; divide the remainder by twice the square. [The result is] the middle [number]."

Verse 18.45: The quadratic formula (second form)

यावद्वर्गं रूपयुतं पदं तद्धृतं यावदर्धहतम् ।
तद्वर्गमूलं यावदर्धेन हीनं वर्गभक्तं यावत् ।।४५।।

Translation (Colebrooke): "Whatever is the square-root of the *rupas* multiplied by the square [and] increased by the square of half the unknown, diminish that by half the unknown [and] divide [the remainder] by its square. [The result is] the unknown."

Modern Formulation: For the equation $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is equivalent to the modern quadratic formula for the positive root.

3 Bhāskara I and the Kuṭṭaka Operation (कुट्टकारप्रक्रियाः)

An indeterminate equation of the first degree of the type:

$$\frac{ax - c}{b} = y$$

(with x and y unknown) is known in Hindu mathematics by the name of "pulveriser" — *kuṭṭākāra*. In this equation, a is called the "dividend" (*bhājya*), b the "divisor" (*bhāgahāra*), c the interpolator (*kṣepa*), x the "multiplier" (*guṇakāra*), and y the "quotient" (*labdha*).

In the pulveriser contemplated in the above stanza:

$a =$ revolution number of a planet
$b =$ civil days in a *yuga*
$c =$ residue of the revolutions of the planet (*śeṣa*)

Sanskrit Verse: Brahmagupta's Rule for the Pulveriser

भाज्यं हारेण विभजेदवशिष्टं हारेण तं पुनः |
यावत् समं न विभजेदवशिष्टं ततः परं क्रमात्
||१८.५०||

Translation: "Divide the dividend by the divisor, and the divisor by the remainder obtained; by this process of mutual division, [continue] until the remainder becomes zero. From this, by working backwards, one can find a particular solution."

Verse 18.51: Condition for solution

क्षेपः क्षेपेण युज्यते यदि तुल्यः स्यात्तदा विधिः स एव |
अतुल्ये क्षेपे छिद्रं विद्धि कुट्टकस्य मित्र ||१८.५१||

Translation: "If the interpolator is divisible by the divisor of the residues, then that same method [applies]; if the interpolators are unequal, know that the pulveriser has a flaw [is impossible], friend."

Verse 18.52: Working backwards

लब्धानि यान्युत्तराणि तानि कोट्यः स्मृताः क्रमशः |
कोट्योर्गुणकौ स्थाप्यौ गुणककारः क्षेपयुक्तः
||१८.५२||

Translation: "The successive quotients obtained [in the mutual division] are called 'kotis' [vertical lines]; placing the gunakaras [multipliers] at the kotis, the multiplier is [found] combined with the interpolator."

3.1 The Pulveriser Method

The method of the pulveriser (*kuṭṭaka*) involves the following steps:

1. Divide a by b and obtain the quotient and remainder.
2. Divide b by this remainder and obtain a new quotient and remainder.
3. Continue this process of mutual division until the remainder becomes zero.

4. Write down the quotients obtained in a column.
5. The number of quotients is made odd by adding the residue if necessary.
6. From the bottom, multiply the last number by the interpolator and add the number above it.
7. Continue this process until the top is reached.
8. The number at the top gives the value of x and the number at the bottom gives y.

3.2 General Form

The general form of the pulveriser is to find integer solutions to:

$$ax + c = by$$

where a, b, and c are given integers.

Brahmagupta's method was later improved by Āryabhaṭa II and Bhāskara II, but the fundamental approach was established in the *Brāhmasphuṭasiddhānta*.

Example: Pulveriser for planetary residues

रवेः शेषं द्वादशाङ्कं विभजेद्भूदिनैः सह ।
गुणकस्तत्र यः स्यात् स रवेर्गतिकलानयेत् ।।१८.५३।।

Translation: "The residue of the Sun [is] twelve; divide it together with the civil days; whatever multiplier results therefrom, that will yield the elapsed degrees of the Sun's motion."

Example Problem: Find x such that $\frac{137x - 10}{60} = y$ is an integer.

Here dividend $a = 137$, divisor $b = 60$, interpolator $c = 10$.
By mutual division: $137 = 2 \times 60 + 17$; $60 = 3 \times 17 + 9$; $17 = 1 \times 9 + 8$; $9 = 1 \times 8 + 1$.

Working backwards: $x = 50$, $y = 114$ is a solution.

4 Planetary Aphelions and Epicycle Corrections (मन्दोच्चपरिधिसंस्काराः)

Brahmagupta gave systematic corrections to the mean longitudes of planets to obtain their true positions. These corrections involve the concepts of the aphelion (*manda-cheda*) and the epicycles of apsis (*manda-paridhi*) and conjunction (*śīghra-paridhi*).

Verse: Mean to True Position

मध्यमं ग्रहं कृत्वा मन्दस्पष्टीकरणं ततः |
तस्माच्छीघ्रस्पष्टीकरणं ततो ग्रहः स्फुटो भवेत् ||उ.खा.
५||

Translation: "Having taken the mean planet, [apply] the correction for the apsis; from that [result, apply] the correction for the conjunction; then the planet becomes true [corrected]."

Verse: Epicycle Dimensions

मन्दपरिधिः सूर्यस्य सप्तचत्वारिंशत्तथा द्वयं |
चन्द्रस्य तु त्रयस्त्रिंशत् शीघ्रपरिधिस्तु सप्ततिः ||उ.खा.
६||

Translation: "The epicycle of the apsis for the Sun is forty-seven plus two [i.e., 49 for the manda]; that of the Moon is thirty-three. The epicycle of the conjunction [for the Moon] is seventy."

Verse: Aphelion (Mandocca) Positions

सूर्यस्य मन्दोच्चं सप्ततिर्नवतिरुत्तरायणे |
चन्द्रस्याप्युच्चं शीघ्रं तत्संस्कारेण सिद्ध्यति ||उ.खा.
७||

Translation: "The aphelion [ucca] of the Sun is seventy-nine [degrees] in the northern solstice; the [higher] apsis of the Moon and the śīghra [conjunction point] are obtained by [appropriate] correction."

Historical Note: Brahmagupta's corrections to the planetary epicycles were based on careful comparison with observed positions. He adjusted the epicycle radii given in the *Āryabhaṭīya* to reduce the errors in predicted planetary longitudes. His method of applying the manda correction first, followed by the śīghra correction, became the standard procedure in Indian astronomy.

5 Uttara Khaṇḍakhādyaka: Advanced Planetary Theory (उत्तरखण्डखाद्यकम्)

The *Uttara Khaṇḍakhādyaka* (UKK) represents Brahmagupta's advanced astronomical treatise, containing corrections and refinements to the earlier *Khaṇḍakhādyaka* proper.

5.1 Brahmagupta First to Use Second Differences (द्वितीयान्तरेण कल्पनम्)

All these illustrations reproduced here very well establish the point that the great Indian astronomers from *Āryabhaṭa* I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five 'star' planets.

In the *Khaṇḍakhādyaka*, Brahmagupta having given the "sines" and the equations of the Sun and the Moon at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the **first time in the history of mathematics**, the improved rules for interpolation by using the **second difference**.

This very important feature is reproduced here from the translation by Sengupta of the verse (UKK. 8):

अवशिष्टार्धजं शेषं विभक्तं नवशताङ्केन तत् |
तयोर्द्वयोरन्तरार्धेन गुणितं तद्विवर्धितं वा हीनं वा |
तयोर्युग्मस्यार्धं तत्स्यात्ततो युतं हीनं वा ||उ.खा. ८||

Translation (Sengupta): "Multiply the residual arc left after division by $900'$ (i.e. by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by $900'$ (i.e. 15°): by the result increase or decrease, as the case may be, half the sum of the same two tabular differences; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over."

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are:

(i) the tabular differences of the Sun's equation,

(ii) the tabular differences of the Moon's equation,

(iii) the tabular differences of the 'sines'.

Illustration — To find the 'sine' of 57°.
Brahmagupta's table of sines in the *Khaṇḍakhādyaka* is as follows:

> *"Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the 'sine' is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over."* (KK.I.30; also III.6)

Table 1: Brahmagupta’s Table of Sines

Arc	**‘Sine’**	**Tabular Difference**	**Second Difference**
$0°$	0	—	—
$15°$	39	39	—
$30°$	75	36	-3
$45°$	106	31	-5
$60°$	130	24	-7
$75°$	145	15	-9
$90°$	150	5	-10

Verse on Lunar Correction (Evection)

चन्द्रस्य द्वितीयमीनं शीघ्रेण सह संयुतं यदा |
तदा तद्गतिफलं स्यात् शीघ्रफलं तेन संयोजयेत् | |उ.खा.
१०| |

Translation: “The second inequality of the Moon [evection], when combined with the [equation of the] conjunction, then its motion-equation occurs; the conjunction-equation should be applied by that [amount].”

This verse refers to what later became known as Śrīpati’s equation — the recognition of the Moon’s second inequality (evection), which represents a significant achievement in Indian lunar theory.

Now $57° = 3420$ minutes $= 900' \times 3 + 720'$. Thus three of the tabular differences are considered as passed over; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc $57°$:

$$\frac{31+24}{2} + \frac{720}{900} \times \frac{31-24}{2}$$

Hence the ‘sine’ of $57°$:

$$= 106 + \frac{720}{900} \times 24 + \frac{720}{900}\left(\frac{720}{900} - 1\right) \times \frac{31-24}{2} = 125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

This in fact is the modern form of the interpolation equation up to the term containing the second difference:

$$f(a+nh) = f(a) + n\Delta f(a) + \frac{n(n-1)}{2}\Delta^2 f(a)$$

Sanskrit Verse: Construction of the Sine Table

एकोनत्रिंशत्संयुक्ता नव षट् चैकं च तानि विश्लेषाः |
ज्यावर्गाणां पञ्चदशाङ्गुलैस्तु भक्ताः सार्धैः |
तेऽपि चतुर्विंशतिज्या दण्डैक्येनैकतस्त्रयस्ते | |खा.
१.२९| |

Translation: “Thirty minus one, increased by nine, six, and one; twenty-four, fifteen, and five — these are the tabular differences of sines. The sines [themselves] are obtained by successive addition of these differences, [starting from] the first [sine]. These twenty-four sines [span] the quadrant.”

Verse on the Epicycle Correction

मन्दफलं स्फुटीकर्तुं मन्दवृत्तेन संयोजयेत् |
शीघ्रफलं तथा स्फुटं शीघ्रवृत्तेन कारयेत् ||उ.खा. १२||

Translation: “To correct [the mean planet] by the equation of the apsis, one should apply [the correction] by means of the epicycle of the apsis; likewise, to make [the planet] true by the equation of the conjunction, one should do so by means of the epicycle of the conjunction.”

Brahmagupta thus takes a decidedly improved step here and is undoubtedly the **first man in the history of mathematics** who has done this.

6 Operations with Fractions (भागक्रियाः)

Brahmagupta classified fractions into several categories: *bhāga* (simple fraction), *prabhāga* (compound fraction), *bhāgānubandha* (fractions in association), and *bhāgāpavāha* (fractions in dissociation). He gave systematic rules for all operations with fractions.

Verse: Addition and Subtraction of Fractions

भागानां हारसम्बन्धे हारयोर्गुणनं तु तौ गुणाकौ |
भाज्यसंयुत्यं तद्धृतं सामान्यं भवति संयुतिः ||पा.
२.३||

Translation: "When fractions are connected [for addition or subtraction], the two denominators [become] multipliers [of the opposite numerators]; the sum of the [cross-multiplied] dividends, divided by the product of the denominators, becomes the combined [result]."

In modern notation:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Verse: Multiplication and Division of Fractions

भाज्यौ हारौ च युगपद् गुणने भाज्यहारयोर्गुणनम् |
भागहारेण भाज्यस्य गुणनं भागहारभाज्ययोः ||पा.
२.४||

Translation: "In the multiplication of fractions, the numerators and denominators are multiplied together [respectively]. In division, the numerator is multiplied by the reciprocal of the divisor-fraction [i.e., the numerator and denominator are interchanged]."

Brahmagupta's Fraction Operations

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} \qquad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

7 The Rule of Three (Trairāśika) (त्रैराशिकम्)

The Rule of Three (*trairāśika*) is the fundamental method of proportion in Indian mathematics. Brahmagupta gave a precise formulation that served as the basis for all problems involving proportion.

Verse: The Rule of Three

प्रमाणफलं यदि निर्दिष्टं तत्प्रमाणेन गुणितं तत् |
इच्छाङ्केन विभजेदिच्छाफलमिह लब्धं स्यात् ||पा.
५.१||

Translation: "If the *pramāṇa*-fruit [result for the standard measure] is given, that [is] multiplied by the *icchā* [requisition]; divide that by the *pramāṇa* [standard measure]; the result obtained here is the *icchā-phala* [required result]."

In the *trairāśika*, three quantities are given:

- **Pramāṇa** (p): the standard measure or argument
- **Phala** (f): the fruit or result corresponding to the standard
- **Icchā** (i): the requisition or desired measure

The required *icchā-phala* (x) is:

$$x = \frac{f \times i}{p}$$

Verse: Inverse Rule of Three

विलोमत्रैराशिकमपि यदा फलं विपर्ययेन स्यात् |
तदा हारभावेन फलमिच्छाफलं तदा भवति ||पा.
५.२||

Translation: "When, by the inverse Rule of Three, the result becomes reversed, then by the nature of the divisor — the fruit then becomes the *icchā-phala* [by inverse proportion]."

Direct Rule of Three: $x = \frac{f \times i}{p}$ **Inverse Rule of Three:** $x = \frac{f \times p}{i}$

Verse: Compound Proportion (Pañcarāśika etc.)

त्रैराशिकविद्धस्त्रैराशिकेन फलमानयेत् |
यावन्तः परमाणास्ते तावन्तस्त्रैराशिकाः स्मृताः ||पा.
५.३||

Translation: "The result having been obtained by the Rule of Three, [it is further operated on] by [another] Rule of Three; as many given quantities [as there are], so many Rules of Three are remembered [to be applied]."

This verse establishes the method of *compound proportion*: when more than three quantities are involved, successive applications of the Rule of Three yield the final result. For five quantities it is called *pañcarāśika*, for seven *saptarāśika*, and so on.

8 The Sine Rule in Indian Trigonometry (ज्यानियमः)

Brahmagupta was the first Indian mathematician to state the sine rule for plane triangles explicitly. He used it in the context of computing the true positions of planets and in spherical astronomy.

Verse: The Sine Rule

समत्रिभुजे ज्या समभुजयोरन्योन्यं विभक्ते |
विषमत्रिभुजे ज्याभुजविभक्ते फलं भवति सा ज्या
||ब्र.स्फ. १४.२१||

Translation: "In an equilateral triangle, the sines of equal sides [divided] by one another [give unity]; in a scalene triangle, the sides divided by the sines give a result [which is constant] — that is the [circumdiameter]."

In modern form, for any triangle ABC with circumdiameter D:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D$$

Verse: Shadow and Sine Relations (Spherical Astronomy)

क्रान्तिज्या दिनार्धज्याविभक्ता लम्बकं भवेत् |
लम्बकेन हताक्षज्या दृग्ज्या सा दिशि दिश्यते ||ब्र.स्फ.
१६.८||

Translation: "The Rsine of the declination, divided by the Rsine of half the day, becomes the *lambaka* (vertical). The Rsine of the latitude, multiplied by the *lambaka*, [gives] the *drgjyā* (ascensional difference); it is determined [according to] the direction [north or south of the equator]."

This verse relates to spherical astronomy: the computation of the *drgjyā* (ascensional difference), which measures how much a celestial body rises earlier or later due to the observer's latitude. The *krānti-kṣetra* and *akṣa-kṣetra* diagrams use similar right triangles.

9 Further Algebraic Operations (अपरा बीजक्रियाः)

Brahmagupta’s Chapter XVIII also contains rules for squares, square roots, cubes, and cube roots, as well as additional rules for operations with zero.

Verse: Square and Square Root

वर्गः समचतुरस्रं तस्य मूलं द्विसमं पदं स्यात् |
तद्वर्गमूलं यावदर्धेन हीनं वर्गभक्तं यावत् ||१८.४६||

Translation: “A square is an equilateral quadrilateral; its root [is] that which, when doubled, [gives the side]. That of which the square is the square is [its] square-root [i.e., the square root of a^2 is a].”

Verse: Cube and Cube Root

घनः समसमचतुरस्रस्तस्य मूलं त्रिसमं पदं स्यात् |
तद्घनमूलं यस्तु घनः स मूलघनतो भवति ||१८.४७||

Translation: “A cube is an equi-dimensional equilateral [solid]; its root [is] that which, when tripled in dimension, [gives the side]. That of which the cube is the cube [is its] cube-root [i.e., the cube root of a^3 is a].”

Verse: Zero as a Number

शून्यं खं खरं पाथेयं तज्ज्ञैः संख्यात्मकं स्मृतम् |
यद्यस्मिन् संयुते जीर्णे नोन्नतिर्न च पातता ||१८.३६||

Translation: “Zero, void, ‘kha’, ‘path’ — the learned know this as a number. When [a quantity] is combined with or diminished by it, there is neither increase nor decrease [i.e., $a + 0 = a - 0 = a$].”

Note: This verse (18.36) is historically significant as one of the earliest explicit statements that zero is a number in its own right. Brahmagupta’s treatment of zero as a mathematical entity with well-defined properties was a foundational contribution to the development of algebra.

10 Summary of Brahmagupta’s Mathematical Contributions

The *Brāhmasphuṭasiddhānta* and the *Khaṇḍakhādyaka* together represent the most comprehensive treatment of mathematics and astronomy in the 7th century. The contributions covered in this section (Pages 141–280) include:

1. **Second-order interpolation:** Brahmagupta was the first mathematician in history to use the second difference in interpolation, as demonstrated in the *Uttara Khaṇḍakhādyaka* (UKK. 8).
2. **Sine rule:** He was the first in India to give the sine rule for plane triangles, in the context of computing planetary positions.
3. **Planetary corrections:** Systematic corrections to the epicycles of apsis for the Sun, Moon, and planets, bringing calculated positions into agreement with observation.
4. **Lunar inequalities:** Recognition and treatment of the second inequality of the Moon (evection), known as Śrīpati’s equation.
5. **Spherical astronomy:** Development of Indian methods using properties of similar right-angled triangles (*krānti-kṣetra* and *akṣa-kṣetra*).
6. **System of numerals:** Contribution to the place-value notation and its transmission to the Arab world, and subsequently to Europe.
7. **Fractions:** Classification of fractional combinations into *bhāga*, *prabhāga*, *bhāgānubandha*, and *bhāgāpavāha*.
8. **Algebra:** Systematic rules for operations with zero, positive and negative numbers; the first to treat zero as a number in its own right.
9. **Quadratic equations:** General formula for solving quadratic equations (verses 18.44–45).
10. **Indeterminate equations:** The *kuṭṭaka* (pulveriser) method for solving linear indeterminate equations, fundamental to Indian astronomy for calculating planetary positions.

Editor’s Note

This bilingual edition presents the Sanskrit text and English translation on Brahmagupta’s *Brāhmasphuṭasiddhānta* as found in the scholarly edition covering pages 141–280 of the original text. The Sanskrit verses have been extracted from the critical edition and set in Devanagari using the Noto Serif Devanagari font. Key mathematical verses include:

- **Chapter XVIII, verses 30–36:** Rules for operations with zero, positive and negative numbers, including the famous “*kha-hara*” (division by zero) verse; also explicit statement that zero is a number (v. 36)
- **Chapter XVIII, verses 44–47:** The quadratic formula (two equivalent forms), square and square-root, cube and cube-root
- **Chapter XVIII, verses 50–53:** The pulveriser (kuṭṭaka) method for indeterminate equations, including conditions for solution and worked examples
- **Pāṭīgaṇita, verses 2.3–2.4:** Rules for addition, subtraction, multiplication, and division of fractions

- **Pāṭīgaṇita, verses 5.1–5.3:** The Rule of Three (*trairāśika*), inverse Rule of Three, and compound proportion
- **Brāhmasphuṭasiddhānta, verses 14.21, 16.8:** The sine rule for plane triangles and spherical astronomy relations
- **Uttara Khaṇḍakhādyaka, verses 5–12:** Planetary aphelions, epicycle corrections, sine table construction, second-difference interpolation, and lunar inequality

Bilingual Edition
Sanskrit text with English translation
Typeset with XeLaTeX
Using Noto Serif Devanagari for Sanskrit text

ब्राह्मस्फुटसिद्धान्त

Brāhmasphuṭsiddhānta

of Brahmagupta (628 CE)

Bilingual Edition

Sanskrit Text in Devanagari | English Translation with IAST

Edited by

Sudhākara Dvivedin

(Benares Sanskrit Series, 1901/1902)

Part I, Chunk 3

Pages 281–420 of the Original Edition

Contents:
Chapters VIII–XIII of English Commentary
Madhyamādhikāra Sanskrit Text with Bilingual Translation

Typeset with XƎLaTeX using Polyglossia and Noto Serif Devanagari

Bilingual edition generated from scanned source — OCR processed

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Introduction to This Bilingual Edition

This bilingual edition presents pages 281–420 of the critical edition of Brahmagupta's *Brāhmasphuṭsiddhānta*, edited by Sudhākara Dvivedin. The **left column** contains the Sanskrit text in Devanagari script; the **right column** provides the English translation with Sanskrit technical terms given in IAST transliteration.

- **Left column**: Sanskrit original in Devanagari (देवनागरी)
- **Right column**: English translation with IAST (International Alphabet of Sanskrit Transliteration)

The content comprises:

1. **Chapters VIII–XIII**: Scholarly commentary on algebra, astronomy, and instruments.
2. **Madhyamādhikāra**: The critically edited Sanskrit text with bilingual verse-by-verse translation.

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1 Chapter VIII: Brahmagupta as an Algebraist

Bilingual Presentation of Algebraic Rules

कुट्टकस्य हारभाज्ययोः समापवर्त्य तयोर्भाजकहाराभ्याम् ।
अपवर्तिताभ्यां कुट्टकं समाचरेत् ॥

"Of the pulveriser, the divisor and dividend being mutually divided, one should operate the pulveriser with the reduced divisor and dividend." — *Kuṭṭaka* (Pulverizer) method for solving $ax \pm c = by$.

भाज्यं हारेण विभजेत् हारं च भाज्येन क्रमशः ।
लब्धानि निक्षिपेत् पृथगेकस्याधः एकस्योपरि ॥

"One should divide the dividend by the divisor and the divisor by the dividend alternately; the quotients should be written down separately, one below, one above." — This describes the Euclidean algorithm steps for the *kuṭṭaka.*

मतिं गुणित्वा तदधो निखिलं युतं लब्धेन ततः परं स्यात् ।
ऊर्ध्वं संहृत्य चरं तदधः पूर्ववत् क□□□□□◌् पुनरपि ॥

"Having multiplied the chosen number (mati) by the one above it, add the quotient below; place the result above and remove the lower; proceed again as before." — The iterative backward-substitution step to find the multiplier and quotient.

ऊर्ध्वं गुणकारं हारेण हृतं शेषं गुणकः ।
अधो लब्धं भाज्येन हृतं शेषं दिनगणः फलं वा ॥

"The upper (number), called the 'multiplier,' divided by the divisor; the lower, called the 'quotient,' divided by the dividend: the remainders are respectively the āhargaṇa and the revolutions (or what is sought)." — Final step: $x =$ ahargaṇa, $y =$ revolutions.

Modern Mathematical Formulation

The *kuṭṭaka* method solves the linear Diophantine equation:

$$ax \pm c = by$$

where a = civil days in *yuga*, b = revolution-number of the planet, c = residue, and x, y are positive integers to be found.

Step 1 — Reduction: Compute $g = \gcd(a, b)$. Divide through by g:

$$a' = \frac{a}{g}, \quad b' = \frac{b}{g}, \quad c' = \frac{c}{g}$$

Step 2 — Euclidean Algorithm: Apply the mutual division to obtain the continued-fraction quotients $q_1, q_2, \ldots, q_n$.

Step 3 — Back-substitution: Starting with a chosen *mati* m such that $r_n \cdot m - c' \equiv 0 \pmod{b_n}$, iteratively compute:

$$V_i = q_i \cdot V_{i+1} + V_{i+2}$$

Step 4 — Solution: The two final numbers give $x \equiv V_{n-1} \pmod{a'}$ and $y \equiv V_n \pmod{b'}$.

Bilingual: Varga-Prakṛti (Square-Nature)

वर्गप्रकृतौ यदि लब्धे गुणके क्षेपके च युग्मे ।
तदा क्षेपयोर्घातं गुणकं द्विगुणीकृत्य युग्मयोः ॥

वर्गयोर्योगो रूपक्षेपे तदा जायते ॥

"In the Square-Nature, when two solutions are found with the same multiplier (N) and interpolators k_1, k_2: their interpolators being multiplied, the multiplier being doubled and combined with the two roots — the sum of their squares is (the new equation with) unity as interpolator."

Brahmagupta's Lemma (Composition): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then:

$$x = x_1y_2 \pm x_2y_1, \quad y = y_1y_2 \pm Nx_1x_2$$

solves $Nx^2 + k_1k_2 = y^2$.

वर्गप्रकृतौ यदि लब्धे क्षेपको वर्गः समभाव्यः ।
पदभ्यां तत्क्षेपहृताभ्यां लब्धे नवे पदे ॥

"In the Square-Nature, if the interpolator is a square, being capable of being a square — dividing the roots by the square root of that interpolator, new roots are obtained."

Second Lemma: If $Nx^2 \pm p^2d = y^2$, set $u = x/p$, $v = y/p$, giving:

$$Nu^2 \pm d = v^2$$

The original roots are p times those of the derived equation.

Cakravāla (Cyclic Method) — Bilingual

आचार्यः पूर्वमासाद्य समीपतो गुणकस्य मूलं यः ।
स क्षेपहृतो रूपयुतो वा तत्पदं गुणकहृतं च ॥

क्षेपं तेन गुणितं युक्तं पूर्वपदद्वयेन तत्क्षेपः ।
प्रकृतिहृतं द्वयोर्घातं पूर्वपदवर्गयोर्योगः ॥

"The teacher (ācārya), first taking the square root of the multiplier nearest to it, divided by the interpolator (and) increased or diminished by unity — that root divided by the multiplier and the interpolator multiplied by it, combined with the previous pair of roots — that is the new interpolator; the product of the two divided by the multiplier, the sum of the squares of the previous roots — (these are the new roots)."

The Cyclic Method: Starting from an auxiliary $Na^2 + k = b^2$:

1. Choose P so $P^2 \approx N$ and $k \mid (P^2 - N)$
2. New interpolator: $k' = \dfrac{P^2 - N}{k}$
3. New roots: $a' = \dfrac{Pa + b}{|k|}, b' = \dfrac{Pb + Na}{|k|}$
4. Iterate until interpolator becomes $\pm 1, \pm 2, \pm 4$

---

2 Chapter X: Arabic and Indian Divisions of the Zodiac

Nakṣatras — Bilingual Enumeration

Sanskrit Name	English Identification
अश्विनी (Aśvinī)	The Wives of the Aśvins, figured as a horse's head. Principal star: β Arietis (Sheratan).
भरणी (Bharaṇī)	Figured as the body of a horse, three stars: 35, 39, 41 Arietis.
कृत्तिका (Kṛttikā)	The Pleiades, six or seven stars.
रोहिणी (Rohiṇī)	The Red One, Aldebaran (α Tauri).
मृगशिरस् (Mṛgasīras)	The deer's head, λ Orionis.
आर्द्रा (Ārdrā)	α Orionis (Betelgeuse).
पुनर्वसु (Punarvasu)	Castor and Pollux (α, β Geminorum).
पुष्य (Puṣya)	δ, θ, γ Cancri.
आश्लेषा (Āśleṣā)	ϵ Hydrae.
मघा (Maghā)	Regulus (α Leonis).
पूर्वफाल्गुनी (Pūrvaphālgunī)	δ, θ Leonis.
उत्तरफाल्गुनी (Uttaraphālgunī)	Denebola (β Leonis).
हस्त (Hasta)	Corvus, $\gamma, \delta, \epsilon, \alpha, \beta, \eta$ Corvi.
चित्रा (Citrā)	Spica (α Virginis).
स्वाती (Svātī)	Arcturus (α Boōtis).
विशाखा (Viśākhā)	$\alpha, \beta, \iota, \kappa$ Librae.
अनुराधा (Anurādhā)	δ, β Scorpii.
ज्येष्ठा (Jyeṣṭhā)	Antares (α Scorpii).

Sanskrit Name	English Identification
मूला (Mūlā)	λ Scorpii and surrounding stars.
पूर्वाषाढा (Pūrvāṣāḍhā)	δ, ϵ Sagittarii.
उत्तराषाढा (Uttarāṣāḍhā)	$\zeta, \sigma, \tau, \phi$ Sagittarii.
श्रवणा (Śravaṇā)	α, β, γ Aquilae.
धनिष्ठा (Dhaniṣṭhā)	$\alpha, \beta, \gamma, \delta$ Delphini.
शतभिषज् (Śatabhiṣaj)	λ Aquarii.
पूर्वभाद्रपदा (Pūrvabhādrapadā)	α, β Pegasi.
उत्तरभाद्रपदा (Uttarabhādrapadā)	γ Pegasi, α Andromedae.
रेवती (Revatī)	ζ Piscium.

---

3 Chapter XI: Brahmagupta's Astronomy

Time Units — Bilingual Definitions

चैत्र्यादितदेवद्विजनोद्वन्मासससर्वेष युगस्य ।
सृष्टिच्छादो लंकायां समं प्रवृता दिनेकलस्य ॥

"Beginning with Caitra, all the months of a yuga proceed for the creation-epoch (sṛṣṭi-cchāda) at Lañkā simultaneously with the beginning of the day of Brahmā."

All months begin simultaneously at the creation epoch at Lañkā (on the prime meridian).

प्राग्र्योद्दीनाटिकाद्वो द्विद्विर्यष्टिका त्रिनाटिका षड्घ्ना ।
घटिका षड्घ्ना दिवलो दिवसानां त्रिंशता मासः ॥

"Two vipalas (atoms of time) make a nālikā; two nālikās make a prāṇa; thirty-six prāṇas make a vināḍikā; sixty vināḍikās make a nāḍī (ghaṭikā); sixty ghaṭikās make a day; thirty days make a month."

Time hierarchy: 2 vipalas = 1 nālikā; 2 nālikās = 1 prāṇa; 36 prāṇas = 1 vināḍikā; 60 vināḍikās = 1 ghaṭikā; 60 ghaṭikās = 1 civil day; 30 days = 1 month.

Kalpa Constants — Bilingual Table

Sanskrit Term	Value / English Translation
कल्पः (kalpa) — Aeon	4,320,000,000 solar years
सावनदिनानि (sāvana-dina) — Civil days	1,577,917,828,000
सौरमासाः (saura-māsa) — Solar months	51,840,000,000
चान्द्रमासाः (cāndra-māsa) — Lunar months	53,433,300,000
अधिमासाः (adhi-māsa) — Intercalary months	1,593,300,000
शशिदिवसाः (śaśi-divasa) — Lunar days	1,602,999,000,000
क्षयाहाः (kṣaya-aha) — Omitted tithis	25,082,580,000

Epoch Calculation (Sṛṣṭi-saṃvatsara)

कल्पपरार्ध मनवः षट्काश्य गताश्चतुर्द्वियुगानि युगानाम् ।
श्रुतपुराणादीनिक्षेपानां क्षयः सूर्यदिनं चक्षुः ॥

नवगणराशिमुनिन् नव समन्वन्तरदेवः शकुन्तपाते ।
शर्वमतेः सम्बत्सरात् संख्यां शरवच्छतं रूपैः ॥

"Six manus have elapsed in half a kalpa; and of the seventh manu, twenty-seven caturyugis have gone by; of the twenty-eighth caturyugi, the three yugas kṛta, dvāpara and tretā have elapsed, and also 3179 years of the present kaliyuga."

The total elapsed period at Brahmagupta's epoch (year 628 CE):

$$\begin{aligned} T &= 6 \times 71 \times 4{,}320{,}000 \\ &\quad + 7 \times 4 \times 432{,}000 \\ &\quad + 27 \times 4{,}320{,}000 \\ &\quad + (1{,}728{,}000 + 1{,}296{,}000 + 864{,}000) + 3{,}179 \\ &= 1{,}972{,}947{,}179 \text{ years.} \end{aligned}$$

Mean Longitude Calculation

General method:

1. Compute the *ahargaṇa* A = civil days elapsed since epoch
2. Multiply by planet's revolution-number R in a kalpa
3. Divide by civil days in kalpa D:

$$\text{Mean longitude} = \frac{A \times R}{D} \text{ (in revolutions, signs, degrees, etc.)}$$

---

4 Chapter XII: Brahmagupta as a Critic

मध्यज्योदयज्यादृक्षेपज्यादृग्ज्यादृग्गतिज्याभिः ।
लम्बनं विद्धि पञ्चभिर्ज्याभिरवनतिं तथा ॥

"Know the lambana (difference of parallaxes in longitude) by means of five Rsines: the madhya-jyā, the udaya-jyā, the dṛk-kṣepa-jyā, the dṛg-jyā, and the dṛg-gati-jyā; and the avanati (difference in latitude) likewise."

The *lambana* is computed from five Rsines:

- *madhya-jyā* = $R \sin z_m$ (zenith distance of meridian ecliptic point)
- *udaya-jyā* = $\dfrac{R \sin L \cdot R \sin \varepsilon}{R \cos \phi}$
- *dṛk-kṣepa-jyā* = $R \sin z_c$ (zenith distance of central ecliptic point)
- *dṛg-jyā* — related to observer's zenith distance
- *dṛg-gati-jyā* — related to the motion of the observer

where L = longitude of horizon ecliptic point, ε = obliquity of ecliptic, ϕ = latitude of place.

---

5 Chapter XIII: Astronomical Instruments

The Gnomon (Śaṇku)

> द्व्यङ्गुलायामं शूलाग्रं द्वादशाङ्गुलमायतं शङ्कुम् ।
> वृत्तात्तात् सूच्यग्रं सच्छिद्रं व्याख्यातं प्रिये ॥

> *"(The gnomon) two añgulas wide at the base, pointed as a needle (śūla-agra), twelve añgulas in length, and full of holes from the basic circular part to the pointed extremity — thus is the śaṇku (gnomon) described, O beloved." — BrSpSi.* XXII. 39.
>
> The gnomon dimensions: base width = 2 añgulas (≈ 3.75 cm), height = 12 añgulas (≈ 22.5 cm), tapering to a point.

> निवाते त्रिपदे तोयपूर्णे कूपे निधाय भूमौ ।
> सूक्ष्मं छिद्रं तत्र कुर्यात् पयः स्यन्दनार्थम् ॥

> *"When there is no wind, placing a jar full of water upon a tripod on the ground which has been made plane by means of eye or thread, one should bore a fine hole (at the bottom) so that the water may fall drop by drop."*
>
> **Procedure for testing level ground**: Place a water-filled jar on a tripod on leveled ground. A fine hole allows water to drip steadily, confirming the surface is horizontal.

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6 Madhyamādhikāra: Full Bilingual Text

॥ श्रीगणेशाय नमः ॥

अथ ब्राह्मस्फुटसिद्धान्तस्य

पूर्वद्धशाध्यायां मध्यमाधिकारः

Brāhmasphuṭsiddhānta: First Chapter — Madhyamādhikāra

जयति प्रणतसुरासुरौरगीन्द्ररत्नप्रभाच्छुरितपादपङ्कजः ।
कर्ता जगत्पतिरदित्यतुल्ययानां महादेवः ॥ १ ॥

jayati praṇata-sura-asura-auragīndra-ratna-prabhā-cchurita-pāda-pañkajaḥ |
kartā jagat-patir-aditya-tulya-yānām mahādevaḥ || 1 ||

“Victorious is Mahādeva, the Lord of the Universe, the Creator, the lotus-feet of whom are rubbed with the lustre of jewels from the heads of gods, demons, and serpent-kings who bow before Him — He whose chariot is equal to that of the Sun.”

ब्रह्मगुप्तो ग्रहगणितं महता कालेन यत् खल्वीमतम् ।
क्षीणेऽधुनी तद्विपरीतवद्ब्राह्मणैरुपेतं ॥ २ ॥

brahmagupto graha-gaṇitaṃ mahatā kālena yat khalvīmatam |
kṣīṇe ’dhunī tad-viparīta-vad-brāhmaṇair upetaṃ || 2 ||

“Brahmagupta (says): The science of the computation of planets, which was well-established in the past, has now become corrupted; being contrary (to the truth), it has been mixed up by the brāhmaṇas.”

ध्रुवबलार्कसुतबुधज्ञ ज्योतिर्विदां प्रीतिषद्गमदो ।
पोषणार्थं विवृतन्तरव्यक्तं सह प्रबन्धेन सूचितं ॥ ३ ॥

dhruva-bala-arka-suta-budhaj na jyotir-vīdaṃ prīti-ṣad-gama-do |
poṣaṇārthaṃ vivṛtāntara-vyaktaṃ saha prabandhena sūcitam || 3 ||

“(I write this) for the pleasure of astronomers, who are learned in the (motions of) Dhruva (pole star), Bala (Mars), Arka (Sun), Suta (Saturn), Budha (Mercury), and J na (Jupiter); for the preservation (of knowledge), the obscure meaning is made clear along with the connected exposition.”

चैत्र्यादितदेवद्विजनोद्धन्माससर्वेष युगस्य ।
सृष्टिच्छादो लंकायां समं प्रवृता दिनेकलस्य ॥ ४ ॥

caitry-ādi-tad-eva-dvija-nodvan-māsa-sarveṣa yugasya |
sṛṣṭi-cchādo lañkāyāṃ samaṃ pravṛtā dine-kalasya || 4 ||

“Beginning with Caitra, all the months of a yuga — (which are) the creation-covering — proceed simultaneously at Lañkā with the beginning of the day of Brahmā (dina-kala).”

प्राग्र्योद्दीनाटिकाद्वो द्विद्विर्यष्टिका त्रिनाटिका षड्घ्ना ।
घटिका षड्घ्ना दिवलो दिवसानां त्रिंशता मासः ॥ ५ ॥

prāg-ry-udda-nāṭikā-dvau dvi-dvi-yaṣṭikā tri-nāṭikā ṣaḍ-ghnā |
ghaṭikā ṣaḍ-ghnā diva-lo divasānāṃ triṃśatā māsaḥ || 5 ||

“Two atoms (paramāṇu) make an aṇu; two aṇus make a lava; two lavas make a nimeṣa; three nimeṣas make a kṣaṇa; five kṣaṇas make a kṣathi; four kṣathis make a muhūrta; eight muhūrtas make a prahara; four praharas make a day (aho-rātra); thirty days make a month.”

Time scale: 2 paramāṇu = 1 aṇu → 2 aṇu = 1 lava → 2 lava = 1 nimeṣa → 3 nimeṣa = 1 kṣaṇa → 5 kṣaṇa = 1 kṣathi → 4 kṣathi = 1 muhūrta → 8 muhūrta = 1 prahara → 4 prahara = 1 day → 30 days = 1 month.

Verses 22–24: Solar and Lunar Month Relations

परिवर्तः सप्तचतुःषष्टिर सूर्यस्य दिनस्य मितिः ।
रवि भगणानां मानः सावनदिनस्य कुट्टकैरेष ॥

parivartaḥ sapta-catuḥṣaṣṭir sūryasya dinasya mitiḥ |
ravi bhagaṇāñaṃ mānaḥ sāvana-dinasya kuṭṭakair eṣa ||

“The revolution-number of the Sun is 64 + 7 = 71 (per caturyuga); this is the measure of solar days. By the pulverizer (kuṭṭaka), this (gives) the civil days.”

Rule: A solar year ≈ 365 days 15 min 30 sec. The revolution-number of the Sun = 4,320,000 per kalpa.

रवि भगणार्धबधा द्वादशांशेन भवति रविमासः ।
भगणान्तरं रविदिनः शशिमासः सूर्यमासेन ॥

ravi bhagaṇārdha-badhā dvādaśāṃśena bhavati ravi-māsaḥ |
bhagaṇantaraṃ ravi-dinaḥ śaśi-māsaḥ sūrya-māsena ||

“The product of half the revolutions of the Sun divided by twelve becomes a solar month. The difference of revolutions (between Sun and Moon) is the lunar month exceeding the solar month.”

Formula:

$$\text{Solar month} = \frac{R_{\odot}/2}{12} \quad ; \quad \text{Lunar month} = R - R_{\odot}$$

अधिमासाः शशिमासाविन्द्वदिनस्य भवति शशिदिवसाः ।
शशिसावनदिवसान्तरनिवमानि तिथिः शशाङ्कदिनं ॥

adhi-māsāḥ śaśi-mśāv-indv-adinasya bhavati śaśi-divasāḥ |
śaśi-sāvana-divasa-antara-nivamāni tithiḥ śaśāñka-dinaṃ ||

"The intercalary months (adhi-māsa) are the lunar months exceeding the solar months. The lunar days are (obtained from) the difference between lunar and civil days; the tithi (lunar day) is the measure of the difference between lunar and civil days."

Key relations:

$$\begin{aligned} \text{adhi-māsa} &= \text{lunar months} - \text{solar months} \\ \text{śaśi-divasa} &= \text{lunar days} - \text{civil days} \\ \text{tithi} &= 1/30 \text{ of a lunar month} \end{aligned}$$

Verses 26–27: Epoch and Ahargaṇa

कल्पपरार्ध मनवः षट्काश्य गताश्चतुर्द्धियुगानि युगानाम् ।
श्रुतपुराणादीनिक्षेपानां क्षयः सूर्यदिनं चक्षुः ॥

kalpa-parārddha manavaḥ ṣaṭ-kāśya gatāś catur-dvi-yugāni yugānām |
śruta-purāṇadi-nikṣepānām kṣayaḥ sūrya-dinaṃ cakṣuḥ ||

"In half a kalpa, six manus have elapsed, and twenty-seven caturyugis of the (seventh) manu. (This gives) the destruction (elapsed period) of the sūrya-dina (solar days) — the eye (of calculation)."

The *ahargaṇa* (day-count) from the kalpa epoch:

$$A = D_{\text{manus}} + D_{\text{caturyugis}} + D_{\text{current yuga}}$$

नवगणराशिमुनिन् नव समन्वन्तरदेवः शकुन्तपाते ।
शर्वमतेः सम्बत्सरात् संख्यां शरवच्छतं रूपैः ॥

nava-gaṇa-rāśi-munin nava samanvantara-devaḥ śakunta-pāte |
śarva-mateḥ saṃbatsarāt saṃkhyāṃ śaravac-chataṃ rūpaiḥ ||

"O Muni of nine-digit signs! (This gives) the number of years from the śarva-mata (doctrine year), 106 (śara) increased by unity — 3179 years of kaliyuga (at the epoch śaka 550 = 628 CE)."

Brahmagupta's epoch: year 3179 of kaliyuga = śaka 550 = 628 CE.

Bilingual Appendix: Rational Geometrical Figures

इष्टयष्टेः समच्छेदं रूपेण हीनमुभयोर्मिथः ।
षड्भागेन समज्ञात्र्यक्षेत्रस्य भुजे ॥

iṣṭa-yaṣṭeḥ sama-cchedaṃ rūpeṇ a hīnam ubhayor mithaḥ |
ṣaḍ-bhāgena sama-j nā-try-akṣetrasya bhuje ||

"The square of the optional side divided and then diminished by an optional number; half the result is the upright (koti), and that increased by the optional number gives the hypotenuse of a rectangle."

Formula: For arbitrary rational m, n:

$$\text{Base} = m, \quad \text{Upright} = \frac{1}{2}\left(\frac{m^2}{n} - n\right), \quad \text{Hypotenuse} = \frac{1}{2}\left(\frac{m^2}{n} + n\right)$$

यस्मिन् द्वे समानी बाहू अन्यद् द्वयमसमानि च ।
तयोर्वर्गयोगो विभाज्यो भेदेन समद्विबाहुक्षेत्रे ॥

yasmin dve samānī bāhū anyad dvayam-asamāni ca |
tayor varga-yogo vibhājyo bhedena sama-dvi-bāhu-kṣetre ||

"Where there are two equal sides and two other unequal sides — the sum of their squares divided by their difference (gives the base of an isosceles triangle)."

For two unequal integers p, q: equal sides = $\frac{1}{2}\left(\frac{p^2+q^2}{p-q} + (p-q)\right)$, base = $\frac{p^2+q^2}{p-q}$.

सर्वदोरल्पचतुष्टयक्षेत्रफलमूलं द्विभुजक्षेत्रे ।
द्वयोर्द्वयोर्द्धयोर्घातं मूलं द्विभुजक्षेत्रस्य ॥

sarva-dor-alpa-catuṣṭaya-kṣetra-phala-mūlaṃ dvi-bhuja-kṣetre |
dvayor dvayor ddhayor ghātaṃ mūlaṃ dvi-bhuja-kṣetrasya ||

"The square root of the product of four sets of half the sum of the sides, each diminished by one side, is the exact area of a triangle (and) quadrilateral."

Brahmagupta's Formula:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}$$

This generalizes Heron's formula for triangles (when $d = 0$).

Critical Apparatus: Selected Variants

V.	Adopted Reading	Variant / Note
1	जयति प्रणत...	श्री नमः (mss. A, B); श्री रामाय (ms. C)
2	क्षीणेऽधुनी...	क्षीणेऽद्यापि (ms. D) — “even now corrupted”
3	विवृतान्तरव्यक्तं	विवृतन्तरव्यक्तं (mss. A, B) — sandhi variant
4	समं प्रवृता	समप्रवृता (ms. B) — compound variant
5	प्राग्र्योद्दीनाटिका	प्रागुद्दीनाटिका (ms. C) — orthographic
22	सप्तचतुःषष्टिः	71 revolutions per caturyuga (confirmed)
23	भगणार्धबधा	भगणार्धबुद्धा (ms. E) — scribal error
26	कल्पपरार्ध	कल्पपरार्द्ध (ms. D) — gemination variant
27	शरवच्छतं	शरवत् शतं (mss. A, B) — sandhi split

Colophon

॥ इति ब्राह्मस्फुटसिद्धान्तस्य मध्यमाधिकारः समाप्तः ॥

iti brāhmasphuṭsiddhāntasya madhyamādhikāraḥ samāptaḥ

"Here ends the Madhyamādhikāra of the Brāhmasphuṭsiddhānta."

Bilingual Edition: Sanskrit (Devanagari) / English (IAST)

Typeset with XƎLATEX using Noto Serif Devanagari

Chunk 3 (Pages 281–420)

ब्रह्मस्फुटसिद्धान्त

Brāhmasphuṭasiddhānta

of **Brahmagupta** (628 CE)

Bilingual Edition: Sanskrit–English

Part 1, Chunk 4: Pages 421–560

(Book pages 32–171)

Covering:

तृप्रश्नाधिकारः (अन्तिम भाग)
चन्द्रग्रहणाध्यायः — *Candragrahaṇādhikāya (Lunar Eclipse)*
सूर्यग्रहणाध्यायः — *Sūryagrahaṇādhikāya (Solar Eclipse)*
उदयास्ताध्यायः, चन्द्रशृङ्गोन्नत्यध्यायः, चन्द्रच्छायाध्यायः, ग्रहयुत्यध्यायः
तन्त्रपरीक्षाध्यायः (दूषणाध्यायः) — *Tantraparīkṣādhikāya (Critical Examination)*
गणिताध्यायः — *Gaṇitādhyāya (Mathematics)*
प्रश्नाध्यायः (प्रारम्भ) — *Praśnādhyāya (Problems)*

Based on the critical edition of
Sudhākara Dvivedī (*The Pandita*, 1901–1902)

Sanskrit text in Devanagari || English translation with IAST transcription
Typeset with XeLaTeX using Noto Serif Devanagari

Contents

Introduction to Chunk 4

This bilingual edition presents the Sanskrit text of pages 421–560 from the first volume of the *Brāhmasphuṭasiddhānta* of Brahmagupta alongside its English translation. These pages correspond to book pages 32–171 of Dvivedī's edition, and contain material from Chapters III through XIII (partial) of the work.

The *Brāhmasphuṭasiddhānta* (lit. "Corrected Doctrine of Brahma") is Brahma-gupta's masterpiece, composed in 628 CE. It consists of 24 chapters, with the first ten chapters (*pūrvadaśādhyāyi*) covering astronomical topics and the latter chapters (*uttara*) covering mathematics, algebra, and more advanced astronomy.

Content Overview

The present chunk covers:

1. **Triprasnādhikāra** (Chapter III, conclusion): Rules for solving the three problems of diurnal motion – finding time, latitude/direction, and the gnomon shadow.
2. **Candragrahaṇādhikāya** (Chapter IV): Computation of lunar eclipses, including the calculation of shadow dimensions, immersion and emersion times, and color phases of the eclipsed Moon.
3. **Sūryagrahaṇādhikāya** (Chapter V): Computation of solar eclipses, including parallax corrections (lambana), semi-durations, and visibility conditions.
4. **Udayastādhikāra** (Chapter VI): Rising and setting of planets.
5. **Candraśṛṅgonnatyādhikāya** (Chapter VII): Elevation of the Moon's horns.
6. **Candraccāyādhikāya** (Chapter VIII): The Moon's shadow.
7. **Grahayutyādhikāya** (Chapter IX): Conjunction of planets.
8. **Tantraparīkṣādhikāya / Dūṣaṇādhikāya** (Chapter XI): Critical examination of other astronomical systems, including refutations of doctrines of Āryabhaṭa, Śrīṣeṇa, Viṣṇucandra, Lāṭadeva, and Vijayanandi.
9. **Gaṇitādhyāya** (Chapter XII): The celebrated chapter on mathematics, containing arithmetic operations, mixtures, series, geometry (plane figures), shadow problems, and algebra (including the *kuṭṭaka* or pulverizer).
10. **Praśnādhyāya** (Chapter XIII, beginning): Mathematical problems and their solutions.

How to Read This Bilingual Edition

This edition uses a parallel-column format:

- **Left column**: Sanskrit text in Devanagari script
- **Right column**: English translation with Sanskrit technical terms in IAST transliteration

The critical apparatus (variant readings) appears after each group of verses, compiled by Sudhākara Dvivedī. Manuscript sigla:

- **(ग)** = reading of manuscript G
- **(क)** = reading of manuscript K
- **(ख)** = reading of manuscript Kh
- **(च)** = reading of manuscript Ch
- **(घ)** = reading of manuscript Gh

The notation "X □□□ (Y)" indicates that the edition reads X where manuscripts have variant Y. Hindi notes marked with वि० (*viveka*) provide editorial commentary.

Chapter III: Triprasnādhikāra (Conclusion)

तृतीयोऽध्यायः — *The chapter on the three problems relating to diurnal motion*

Sanskrit (Devanagari)

English Translation (with IAST)

Verses 25–27: Ghaṭikā computations and eclipse timing

प्राक् पश्चाद्वा यामिध्वंटिकाभिरदिनिदलान्नतः सूर्यः
तिथ्यन्ते तद्रहिते त्रिशत् घटिकावशेषासि ॥ २५ ॥

विपरीतमर्धरात्राच्चन्द्रग्रहणे शनै रविग्रहणे
सूर्योत्पतस्ततस्ताभिरेव घटिकाभिर्द्विरपि ॥ २६ ॥

दिनदल परिधि स्फुट तिथिनतकेद्रज्यावधो गुरोः कन्दोः
इन्दू धृति धृतिभिः १९१ नवनववेदे ४६६ व्यासाङ्ककृतिभक्तः ॥ २७ ॥

Translation: (25) When the Sun, diminished by the day-radius (*dinadala*), is east or west, the ghaṭikās till the end of the tithi are obtained by subtracting that from thirty and the ghaṭikā-remainder is obtained. (26) The reverse applies at midnight for a lunar eclipse; slowly for a solar eclipse. From the rising of the Sun, by the same ghaṭikās twice. (27) The semi-diameter (*vyāsa*) of the Earth's shadow at the Moon's orbit is obtained: the day-diameter, circumference, accurate tithi-nata-kendra-jyā and the degrees of Jupiter or Kandu (the Moon), Indu — by 191 and the new nine-veda 466, divided by the square of the radius.

Critical Apparatus (Verses 25–27):

25. तद्रहिते (G) □□□ (तद्रहिते): corrected reading.
25.1 घटिकामि (G) □□□ (घटिकासि).
25.2 त्रिशदधिकावशेषासि (G) □□□ (त्रिशतुघटिकावशेषासिः).
25.3 दिनदलान्नतः सूर्यः (G) □□□ (दिनदलान्ततः सूर्यः).
26.1 यतस्ततः (G) □□□ (यतस्ततस्तु).
26.2 चन्द्रग्रहणे (K) चन्द्रग्रहविनिर्हारणः □□□ (चन्द्रग्रहणे).
27.1 दल (G) □□□ (दल).
27.2 ज्या (G) ज्यावधो □□□ (ज्यावधो).
27.3 कन्दोः (G,Kh) कन्द्रो □□□ (कन्दोः).
27.4 इन्दू धृतिभिः १९१ नववेदे ४६६ — संख्याएँ मूल में लुप्त हैं (numbers omitted in the source).
27.5 व्यासाङ्ककृति (K) व्यासाङ्ककृते □□□ (व्यासाङ्ककृति).

Verses 38–39: Sphuṭīkaraṇa of Mars

सप्तभिरंशगुणिता द्युगण दिवर्ध्ध राशिभुक्ता हृताप्तांशः
अधिकोनकुजमर्कतात् मृगकर्कादौ स्फुटी भवतीति ॥ ३८ ॥

तत्स्फुटपरिधिः खनगाः ७० शीघ्रस्फुटपरिधिराप्तभागोनाः
वेदजिनाखिशोनाः २४३३० स्फुटीकरणं कुजस्यैव ॥ ३९ ॥

Translation: (38) Multiplied by seven degrees, the day-multiplier, the daily motion in signs, divided by the obtained degrees, the sphuṭa of Mars is increased or decreased from the mark (starting point) at Mṛga (start of Capricorn) and Karka (start of Cancer). (39) Its sphuṭa-circumference is 70 yojanas; the śīghra-sphuṭa circumference diminished by obtained degrees; 24330 diminished by veda-jina-akhi; this is the correction (*sphuṭīkaraṇa*) of Mars.

Critical Apparatus:

38.1 गुणिताः (Ch) □□□ (गुणिता).

38.2 स्फुटो (G,K,Kh,Ch) □□□ (स्फुटौ).

38.3 भवति (G,K) भवन्ति (Kh) □□□ (भवतीति).

39.1 स्फुटीकरणं (G) स्फष्टीकरणं (Kh) □□□ (स्फुटीकरणं).

39.2 राप्तभागोनाः □□□ (राप्तभागोनाः).

वि० Numbers given in the source are omitted. Manuscripts lack the numbers in the verse.

Verses 40–42: Celestial latitude and shadow calculations

छायावृत्तेज्यगा कर्णगुणा व्यासदल हृताकर्प्ता
विषुवच्छाया याम्या तदन्तरैक्यं भुजस्याग्रे ॥ ४ ॥

शङ्कुप्राच्यपरा या छाया भुजकृतिविशेषमूलं नयेत्
तत्प्राच्यपरछाया भुजाग्रयोरन्तरं कोटिः ॥ ५ ॥

दिग्मध्ये छायाग्रं कृत्वा शङ्कुं यथादिशं विणम्
दिग्मध्यस्थितदृग् छायाग्रं भ्रमति विपरीतम् ॥ ६ ॥

Translation: (4) The shadow-circle-radius multiplied by the hypotenuse, divided by the semi-diameter obtained by the Sun, the viṣuvacchāyā, the yāmyā, the sum or difference of that at the tip of the bhuja. (5) The eastern or western shadow of the gnomon — taking the square root of the difference of the bhuja-squares, that eastern or western shadow is the koṭi between the tips of the bhuja. (6) Having made the shadow-tip in the middle of the direction, the gnomon according to direction is to be measured; the observer's shadow-tip situated in the middle of the direction moves in reverse.

Critical Apparatus:

4.1 वृत्तेज्यगा (G,Kh) □□□ (वृत्तेज्काग्रा).

4.2 हताकर्प्ता □□□ (हृताकर्प्ता).

5.1 भुजकृतिविशेषमूलं नयेत् (G) □□□ (भुजकृतिविशेषसूलं नयेत्).

6.1 दिग्मध्ये (G,Kh) □□□ (दिग्मध्ये).

6.2 शङ्कोयंथादिशं □□□ (शङ्कोयंथादिशं).

6.3 भ्रमति (K) □□□ (भ्रमति).

Verses 64–66: End of Triprasnādhikāra

कृष्णाचतुर्दशान्ते शकुनि पक्षेण चतुष्पदप्रयमे
तिथ्यर्धं ते नागं किस्तुघ्नं प्रतिपदाद्यत् ॥ ६४ ॥

व्यक्तद्विकला भक्ताः खरसगुरणणः ३६० लब्धसुनमेकेन
चलकरानि विवादी □□□□ हृतशेषे तिथिवदन्यत् ॥ ६६ ॥

Translation: (64) At the end of Kṛṣṇa-caturdaśī, in the bird (śakuni) season, in the four-footed (catuṣpada) measure, the half-tithi, nāga, kistughna, from pratipadā etc. (66) The manifest two kalās divided, the moving karas and the hara-sa-gu-ra-ṇa-ṇaḥ 360, with one obtained sum, the remainder divided by hṛta is like a tithi and other.

Critical Apparatus:

64.1 द्वियन्ते (G) □□□ (द्वन्द्वान्ते / द्वियन्ते दाकुति).

64.2 चतुष्पदं (G,K,Kh) □□□ (चतुष्पद).

64.3 तिथ्यर्धं ते नागं (G) □□□ (तिथ्यर्धं ते नागं).

66.1 तिथिन्यत् □□□ (तिथिवदन्यत्).

वि० इसकी श्लोक संख्या ६४ है (the verse number is 64).

[At this point the Triprasnādhikāra concludes. Several verses in the manuscripts are mutilated; Dvivedī notes that at some places only the commentary (ṭīkā) is available. The chapter contained 66 verses.]

Chapter IV: Candragrahaṇādhikāya

चतुर्थोऽध्यायः

Chapter on the Computation of Lunar Eclipses

This chapter contains rules for computing lunar eclipses, including the dimensions of the Earth's shadow at the Moon's orbit, the times of first and last contacts, the phases of the eclipse, and the colours assumed by the eclipsed Moon.

Sanskrit (Devanagari)

English Translation (with IAST)

Verses 1–4: Shadow dimensions and eclipse elements

सूर्यज्या दिनभागज्यया गुणाक्षज्ययाथवा भक्ता
या द्वादशगुणिता विषुवच्छाया विभक्ता वा ॥ १ ॥

द्वादश विषुवच्छाया गृह ते पृथगक्षलम्बजीवे वा
क्रान्तिहते सममण्डलकर्णः प्राग्वत् पृथक् छाया ॥ ३ ॥

Translation: (1) The Sun's jyā multiplied by the day-part-jyā, divided by the guṇākṣa-jyā or alternatively, the viṣuvacchāyā divided by twelve. (3) Twelve times the viṣuvacchāyā gives the shadow; or separately by the aksa-lamba-jyā. Multiplied by the declination (krānti), the sama-maṇḍala-karṇa; as before, the separate shadow.

Critical Apparatus:

1.1 ज्ययाथवा (Ch) □□□ (ज्ययाथवा).

1.2 विषुवच्छाया (G) □□□ (विषुवच्छाया).

3.1 छाये □□□ (छाया).

3.2 पृथगक्ष □□□ (पृथगक्ष).

3.3 कर्णः □□□ (कर्ण).

Verses 7–8: Eclipse magnitudes and durations

द्विकुलंबछायाक्षज्ञा तद्वंसंयुते मूलम्
विषुवत्कर्णछाया कर्णोन्यदा शङ्कुः ॥ ७ ॥

उन्नतजीवाकोटिः छाया हर्ज्ञा भुजी नतज्ञा वा
कर्णछायावृत्तं व्यासाद्ध दयमतोऽन्यत्र ॥ ८ ॥

Translation: (7) The two aksa-lamba-chāyā-jyā, their sum or difference gives the root; the viṣuvat-karṇa-chāyā, otherwise the hypotenuse is the śaṅku (gnomon). (8) The uccatana-jīvā-koṭi, the chāyā, the ha-jyā, the bhuja or the nata-jyā; the hypotenuse-chāyā-circle from the semi-diameter; similarly elsewhere.

Critical Apparatus:

7.1 क्षया (G) क्षजय (Kh) वज्ञा □□□ (क्षज्ञा).

7.2 मूलम् (Ch) □□□ (मूलम्).

7.3 विषुवत् (G) विषुवति □□□ (विषुवत्).

7.4 कर्णोन्यदा (G,K) □□□ (कर्णोत्यदा).

7.5 शङ्कुलम्बः (K) शङ्कुलबश (Kh) □□□ (शङ्कुलब).

8.1 कोटिछाया □□□ (कोटिः छाया).

8.2 उन्नतजीवाकोटिः □□□ (उन्नतजीवाकोटिः).

8.3 हर्ज्ञा (K) शध्या □□□ (हर्ज्ञा).

8.4 भुजानतज्ञावा □□□ (भुजोनतज्ञावा).

वि० इसकी श्लोक संख्या ७ है (this verse is numbered 7 in the text).

Verses 14–18: Sphuṭa-tithi and lambana

यद्यधिकं स्थित्यर्धं तदन्तरेणान्यथोनसूनमेकम्
ऋणमधिकं तदैक्येनाधिकमेवं विमर्दार्धः ॥ १४ ॥

स्फुटतिथ्यन्ताल्लम्बनमसकृतस्थित्यर्धहीनयुक्ताद्वा
तत्स्फुटं विश्लेषात् स्थिस्यर्धोतयुततिथ्यन्ते ॥ १६ ॥

तत्स्फुटतिथिछेदान्तरे स्फुटे तिथिदले विहीनयुतात्
स्वविमर्दोदयाद् एवं स्पष्टो विमर्दार्धः ॥ १७ ॥

शशिवद्विभुः स्फुटविक्षेपकृतं स्थितिदलेन संगुणिता
स्पष्टः स्थित्यर्धहतो भवति भुजः पूर्ववच्छेषमूलः ॥ १८ ॥

Translation: (14) If the half-duration is greater, by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda). (16) From the sphuṭa-tithi-antā the parallax (lambana), with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without, from the self-vimarda-udaya, thus the corrected half-obscuration. (18) The Moon's latitude (sphuṭa-vikṣepa) multiplied by the half-duration, the corrected (sphuṭaḥ) bhūja multiplied by the half-duration is the root of the remainder as before.

Critical Apparatus:

14.1 तदैक्येनाधिकमेवं (G,Ch) □□□ (तदैक्येनाधिकमेवं).

16.1 स्थिस्यर्धोतयुत □□□ (स्थितिस्यर्धोतयुततिथ्यन्ते).

17.1 तत्स्फुटतिथिछेदान्तरे □□□ (तत्स्फुटतिथिछेदान्तरे).

18.1 भुजः □□□ (भुजः).

18.2 पूर्ववच्छेषमूलः □□□ (पूर्ववच्छेषमूलः).

Verses 16–20: Eclipse colours and phases

क्षयान्त्ययोः सधुमूः कृष्णः खण्डग्रहे ऽर्धान्तोऽधिके ग्रासः
सकृष्ण (षण्ण) ताम्रः स्वकरे शक्लिकपीलवर्णः ॥ १६ ॥

मानविमलं स्थितिदलवलनेष्ट ग्रास समकलादेषु
नदगूहराध्याय विस्तृतिराया इष्टचतुर्थोयाम् ॥ २० ॥

Translation: (16) At the end of obscuration, smoke-coloured, black; in partial eclipse, half-increased eating (grāsa); black-brown (sa-kṛṣṇa), coppery (tāmra), white-yellowish (śakli-kapīla-varṇa). (20) The measure of pure, half-duration-motion-desired, eating in the samakala-etc.; the breadth of the nadagūhara-āyāma desired one-fourth.

Critical Apparatus:

16.1 क्षयान्त्ययोः (G,K) □□□ (क्षयान्त्ययोः).

16.2 खण्डग्रहे (G) □□□ (खण्डग्रहे).

16.3 शशी (G,K,Ch) □□□ (शशि).

16.4 कपिलां (G) कपिलवर्णः (K) □□□ (कपिलवर्णः).

20.1 दलनेष्ट □□□ (दलवलनेष्ट).

वि० □□इति'' से □□अध्याय'' तक श्रंकित नहीं (the closing formula is not marked).

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इति ब्रह्मगुप्ते चतुर्थोऽध्यायः

Thus ends the Fourth Chapter of the Brāhmasphuṭasiddhānta.

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Chapter V: Sūryagrahaṇādhikāya

पञ्चमोऽध्यायः

Chapter on the Computation of Solar Eclipses

This chapter treats the computation of solar eclipses, including the parallax corrections (*lambana*) that distinguish solar eclipse computation from lunar eclipse computation.

Sanskrit (Devanagari)	**English Translation (with IAST)**

Verses 2–4: Lambana and eclipse elements

सूर्यज्या दिनभागज्यया गुणाक्षज्ययाथवा भक्ता
या द्वादशगुणिता विषुवच्छाया विभक्ता वा ॥ २ ॥

द्वादश विषुवच्छाया गृहे ते पृथगक्षलम्बजीवे वा
क्रान्तिहते सममण्डलकर्णः प्राग्वत् पृथक् छाया ॥ ३ ॥

अर्काग्रा दर्गोनं भुज्या वर्गाद् भ346450 सके १४४ कृतिगुणितम्
आद्यान्योभ्रा द्वादशविषुवच्छाया वधो हतयोः ॥ ४ ॥

Translation: (2–3) Same as lunar eclipse — the shadow dimensions are computed from the Sun's declination. (4) The bhuja from the Sun's tip (arkāgra), the square deducted, 346450 and 144 multiplied by the square; the first and other, twelve times the viṣuvacchāyā, the product of the two.

Verses 5–8: Lambana-ghaṭikā and eclipse timing

लम्बनघटिकालग्नात् तात्कालिकात् त्रिराशियुनात्
ऋण (ऋणां) धिके शकृन् हीनं धनमसकृत्पञ्च हृश्यन्ते ॥ ५ ॥

करणगुणात् व्यासाद्ध द्विसुवेद ४८ विभाजिता फलविभक्ता
लम्बननाड्यो भास्करवित्त्रिभलग्नान्तरं वा ॥ ६ ॥

यदि लम्बनमृणात् धनमधिकाः स्वफललिप्ताभिः
अक्षज्ञाया वित्त्रिभलग्नात् स्वक्रान्तिरुत्तराकस्य
इन्दोर्वा यद्यधिका न नतिः सौम्यान्यथा याम्या ॥ ८ ॥

Translation: (5) From the lambana-ghaṭikā-lagna, from the tātkālika, diminished by three signs; if negative with addition, five times the asakṛt, divided by hṛdaya. (6) From the karaṇa-guṇa, from the semi-diameter, dviveda 48, divided; the lambana-nāḍīs or the bāskara-vitribha-lagna-antaram. (7) If the lambana is negative, the positive result in its own lalipta-phala. (8) From the aṣajyā, from the vitribha-lagna, its own declination (krānti) uttarākṣa or of the Moon; if greater, otherwise the yāmyā.

Critical Apparatus:

5.1 ऋणमधिके (G) □□□ (ऋणमधिके).

5.2 मसकृत्पञ्च (K) □□□ (मसकृत्पञ्च).

5.3 हृश्यन्ते (K) □□□ (हृश्यन्ते).

6.1 विभाजिता फलविभक्ता (G) □□□ (विभाजिता फलविभक्ता).

6.2 लम्बननाड्यो □□□ (लम्बननाड्यो).

8.1 वा (G) ज्यावा (K) □□□ (वा).

Verses 14–17: Sphuṭa-tithi and corrected obscuration

यद्यधिकं स्थित्यर्धं तदन्तरेणान्यथोनसूनमेकम्
ऋणमधिकं तदैक्येनाधिकमेवं विमर्दार्धः ॥ १४ ॥

स्फुटतिथ्यन्ताल्लम्बनमसकृतस्थित्यर्धहीनयुक्ताद्वा
तत्स्फुटं विश्लेषात् तिस्थिस्यर्धोतयुततिथ्यन्ते ॥ १६ ॥

तत्स्फुटतिथिछेदान्तरे स्फुटे तिथिदले विहीनयुतात्
स्वविमर्दोदयाद् एवं स्पष्टो विमर्दार्धः ॥ १७ ॥

Translation: (14) If the half-duration is greater, then by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda-ardha). (16) From the sphuṭa-tithi-anta, the lambana, with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of the sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without; from the self-vimarda-udaya, thus the corrected (spaṣṭa) half-obscuration.

Critical Apparatus:

14.1 तदैक्येनाधिकमेवं (G,Ch) □□□ (तदैक्येनाधिकमेवं).

15.1 तदैक्येनाधिकमेवं (G,Ch) □□□ (तदैक्येनाधिकमेवं).

16.1 तिस्थिस्यर्धोतयुत □□□ (स्थितिस्यर्धोतयुततिथ्यन्ते).

17.1 तत्स्फुटतिथिछेदान्तरे □□□ (तत्स्फुटतिथिछेदान्तरे).

Verses 18–20: Deflection and sphuṭa computation

एवं मानैक्यार्धादधिके सध्यान्तरे युतिग्रहयोः
स्थितिस्यर्धविमर्दर्धले हीनं तरा ग्रहेंदुयुतौ ॥ १८ ॥

लम्बनमकलग्रहवदसकृत्स्वावनतिलिप्तिकास्पष्टी
तात्कालिकविक्षेपी तदन्तरैक्यं समान्यदिशोः ॥ १९ ॥

विक्षेपो मध्यान्तरमुध्वस्वछादको ग्रहोऽध्वस्थः
मानैक्यार्धादधिके नातिस्पष्टास्फुटोक्तिरतः ॥ २० ॥

Translation: (18) Thus in the conjunction of planet and Moon, if greater than the measure of unity plus half, in the middle-intermission; diminished by the half-duration-obscuration, the cross (tārā). (19) The lambana like akala-graha, asakṛt-svāvanati-lāptikā-sphuṭī, the tātkālika-vikṣepī, the samānya-diśoḥ sum or difference. (20) The deflection (vikṣepa), the middle-difference, the above-shadow-planet situated below; if greater than half the unity-measure, not too distinct, hence the sphuṭa-statement.

Critical Apparatus:

18.1 युतिग्रहयोः □□□ (युतिग्रहयोः).

18.2 विमर्दर्धले □□□ (विमर्दे).

19.1 तदन्तरैक्यं □□□ (तदन्तरैक्यं).

19.2 लिप्ता □□□ (लिप्तिका).

20.1 मर्के □□□ (मर्कट).

20.2 मुध्वस्य छादको ग्रहोऽध्वस्थः (G) □□□ (मुध्वं स्वछादको ग्रहोऽध्वस्थः).

♣ ♣ ♣

इति ब्रह्मगुप्तसवर्गोऽध्यायः

Thus ends the Fifth Chapter of the Brāhmasphuṭasiddhānta.

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Chapters VI–IX: Udayāsta, Candraśṛṅgonnati, Candraccāyā, Grahayuti

These chapters cover miscellaneous astronomical computations: rising and setting of planets, the elevation of the Moon's horns, the Moon's shadow, and conjunctions of planets.

Sanskrit (Devanagari)	English Translation (with IAST)

Chapter VI: Udayastādhikāra (Rising and Setting of Planets)

द्वादशाभिः शितांशुसितजीवक्षमिभूमिजा नवभिः
द्युत्तरवृद्धिरन्तरघटिका षड्गुणिता कलांशैः ॥ ६ ॥

हृश्याहृश्यायुतिवद्ग्रहाकं सुत्तचान्तरलब्धदिनैः
ऊनाधिकलिप्ताम्यः प्राग्वत् तात्कालिकैरसकृत ॥ ७ ॥

प्रतिदिनमुदयास्तमयाद् एवं तात्कालिकं ग्रहविलग्नैः
सूर्यास्तमयोदयिकः शीतांशुः पौष्णमास्यन्ते ॥ ८ ॥

Translation: (6) By twelve, the śītāṃśu-sita-jīvā, the earth-born, by nine; the day-increment-difference-ghaṭikā multiplied by six in kalāṃśa. (7) The graheka like hṛśyā-hṛśyā-yuti, by the difference of suttacāntara-obtained-days; subtracted or added in liptāmyaḥ, as before by the tātkālika-asakṛt. (8) Daily rising and setting thus, by the tātkālika-graha-vilagnais; the solar rising and setting at the end of the Pausa month, the Moon (śītāṃśu).

Critical Apparatus:

6.1 शितांशुः (G) □□□ (शीतांशुः).

6.2 भूमिजानवभिः □□□ (भूमिजानवभिः).

7.1 सुत्तचान्तरलब्धदिनैः □□□ (सुत्तचान्तरलब्धदिनैः).

7.2 ऊनाधिक □□□ (ऊनाधिक).

8.1 शीतांशोः (G) शीतांशोः (K) □□□ (शीतांशौ).

8.2 पौष्णमास्यां (G) पूर्णिमास्था □□□ (पौष्णमास्यन्ते).

8.3 दयिकः (K) □□□ (दयिकः).

Chapter VII: Candraśṛṅgonnatyādhikāya (Elevation of the Moon's Horns)

कोटिश्रवणाज्ञानात् शशिना श्युद्लोन्नतिर्विसंवदति
उदयास्तमयो दिनकृतः प्रतिघटिकं माती च वाज्ञानात् ॥ ३८ ॥

श्रकैतरघटिका व्यस्तज्ञा चन्द्रमानगुणिता यतः
व्यास विभक्ता शुद्धं यतो न दृक् सममसत्तस्मात् ॥ ३६ ॥

प्राक् गुदितोऽन्यधिकः पश्चादुदिनकोऽपरेव्यस्तः
कालो यः छाया तदसत्स्फुटं सुक्तिमान् प्राक् ॥ ४० ॥

उदितामुदितास्तमितावशेषकालान्नं वेत्ति यः स कथम्
आर्यमटज्ञः शदिनः छाया श्युद्लोन्नति वेत्ति ॥ ४१ ॥

Translation: (38) The latitude (unnati) of the Moon is found to disagree from the hypotenuse-śravaṇa-jñānāt; the rising and setting are caused by day, by prati-ghaṭika measure and from vā-jñānāt. (36) The akaicara-ghaṭikā, vyasta-jyā, multiplied by candra-māna, divided by vyāsa is pure; since the dṛk-sama is not true. (40) Previously taught as different, afterwards the other reverse; that time which is shadow, that asat-sphuṭa, suktimān before. (41) Who knows the udita-udita-astamita-avaśeṣa-kāla; how can he who knows the shadow-unnati of the Moon be an ārya-maṭa-jña.

Critical Apparatus:

38.1 श्युद्लोन्नतिः □□□ (श्युद्लोन्नतिः).

38.2 विसंवदति □□□ (विसंवदति).

38.3 दिनकृत्तः □□□ (दिनकृतः).

36.1 दृक् सम (G) □□□ (दृक् सम).

40.1 प्राक् गुदितोऽन्यधिकः □□□ (प्रागुदितोऽन्यधिकः).

41.1 शदिनः □□□ (दिनः).

41.2 आर्यमटज्ञः □□□ (आर्यमटज्ञः).

Chapter VIII: Candraccāyādhikāya (The Moon's Shadow)

क्रान्तिज्ञा तत्क्रान्तिविक्षेपक्रान्तिचापानाम्
संयोगान्तरजीवा स्वक्रान्तिज्यैकभिन्नदिशाम् ॥ १४ ॥

एवं भमुनिश्नुवयो ग्रहतत्क्रान्त्या चन्द्राष्टकं कर्माद्य
कृत्वा गृहे भमुनिवत् तस्मात्स्वक्रान्तिजीवा वा ॥ १६ ॥

Translation: (14) The declination-jyā, the deflection (vikṣepa), the declination-cāpa: the sum or difference of the jyā of like or unlike directions of the declination. (16) Thus the bhūmi-niśa-vayo (latitude etc.) of the planet, by its declination, the candra-aṣṭaka-karma etc.; having done this in the eclipse, as in bhūmi, from that the svakrantijīvā.

Critical Apparatus:

14.1 चापभागानाम् (G) □□□ (चापानाम्).

14.2 विक्षेप □□□ (विक्षेप).

14.3 चैकभिन्नदिशाम् □□□ (ज्यैकभिन्नदिशाम्).

16.1 श्लुवयोग्रह □□□ (श्लुवयोग्रह).

16.2 च दृष्टिकर्माच (G) □□□ (चन्द्राष्टकं कर्माद्).

16.3 भमुनिश्नुवका □□□ (भमुनिश्नुवयो).

वि० An additional verse is found in manuscript G: भमुनिश्नुवका प्राक् प्रदर्शिता त्विषा / देषामेवकार्यं ग्रहस्य पुनस्तत् क्रान्ते प्रथमं वृत्तकरणं / कृत्वा पश्चात् तज्ज्ञाति स्वक मुनिध्रुवकवत्.

Chapter IX: Grahayutyādhikāya (Conjunction of Planets)

प्रतिघटिकमधिकशङ्कोभ्यां मध्ये दर्शयेत् भुवा
त्रिप्रश्नोक्तचा रविवद्धंकु भ्रमणादिक्रमशेषम् ॥ ४४ ॥

शङ्कु प्राच्यपरान्तरं विषुवच्छायैक्य सुत्तरे नृतले
याम्योतरं गुणाहतं स्वक्रान्तिलेशैः कर्णास्यास् ॥ ४६ ॥

तच्चापांशाः सदयेः संघुवकापक्रमांशरूना
विक्षेपांशे व्यस्ता व्यस्तविशुद्धा व्यसहशांशः ॥ ४७ ॥

Translation: (44) By the prati-ghaṭika and adhika-śaṅku, in the middle is shown the earth; the tripraśna-oktacā like the Sun's śaṅku, the remainder of the bhramaṇādi-krama. (46) The śaṅku, the eastern-western antaram, the viṣuvacchāyā-sum or difference, nṛtale; the yāmyottaraṃ guṇāhataṃ by svakrānti-leśaiḥ and karṇa. (47) The cāpāṃśāḥ of sada-yeḥ, saṅghuvakāpakramāṃśa-rūnāḥ; the vikṣepāṃśe vyastā, the vyasta-viśuddhā vya-sahaśāṃśaḥ.

Critical Apparatus:

44.1 दर्शयेन् (G) □□□ (दर्शयेत्).

44.2 रविवद्धंकु (Ch) □□□ (रविवद्धंकु).

46.1 परान्तर (G) □□□ (परान्तरं).

46.2 नृतरे □□□ (नृतले).

46.3 याम्बैतर (G) □□□ (याम्योत्तरं).

46.4 गुणहतं □□□ (गुणाहतं).

47.1 चापांशाः (G,Ch) □□□ (चापांशाः).

47.2 सहदयेः (G) □□□ (सदयेः).

* * *

[The chapters on Udayāsta, Candraśṛṅgonnati, Candraccāyā, and Grahayuti conclude here, with 12–70 verses each.]

* * *

Chapter XI: Tantraparīkṣādhikāya (Dūṣaṇādhikāya)

एकादशोऽध्यायः

अथ तन्त्रपरीक्षा नामैकादशोऽध्यायः

Chapter on the Critical Examination of Astronomical Systems

This celebrated chapter contains Brahmagupta's critique of various astronomical systems current in his time. He refutes the doctrines of Āryabhaṭa, Śrīṣeṇa, Viṣṇucandra, Lāṭadeva, Vijayanandi, and Pradyumna, as well as the foreign systems derived from Greece and Babylon (the Romaka and Puliśa Siddhāntas).

Sanskrit (Devanagari)

English Translation (with IAST)

Verses 1–2: Introduction to the critique

ये ज्ञान पटलरुद्धद्वोन्यब्रह्मविदन्ति सिद्धान्तम्
तेषां युगादिभेदैर् दोषास्तान् प्रवक्ष्यामि ॥ १ ॥

युगमाहुः पञ्चाब्दं रविशशिनोः संहितां कारये
अधिमासा वमरात्र स्फुटतिथ्यज्ञान तदसत् ॥ २ ॥

Translation: (1) Those who know the brahma-vid (science) in a chapter-barred dual manner, they consider it a siddhānta; I shall declare their faults arising from the yuga-ādi differences. (2) They say the yuga is five years; making the Saṃhitā of the Sun and Moon; the adhimāsa, vamara-atra, sphuṭa-tithi-jñāna — that is false.

Critical Apparatus:

1.1 येज्ञान □□□ (ये ज्ञान).

1.2 रुद्धशो (G) रुद्धो □□□ (रुद्धद्वो).

1.3 न्यद्वाह्यविदन्ति (G) न्यब्रह्मविदन्ति □□□ (न्यब्रह्मविदन्ति).

1.4 सिद्धान्तात् (G,Ch) □□□ (सिद्धान्तम्).

1.5 भेदाद्ये (G,Ch) □□□ (भेदैः).

1.6 प्रवक्ष्यामि (G) प्रविक्ष्यामि □□□ (प्रवक्षामि).

2.1 पञ्चाक्षरं □□□ (पञ्चाब्दं).

2.2 ज्ञानतस्तदसत् (G) □□□ (ज्ञानतदसतु).

2.3 कारये □□□ (कारये).

वि०--अतिरिक्त पाठः अथ ब्रह्मगुप्तसिद्धान्ते एकादशमोऽध्यायः प्रारभ्यते / दुवृत्तेन्द्वयो गतगम्याल्पात्मकशून्य यमल / सायक हतं कलाभिरूनौ सदाकेन्द्र / विधु वजीवेद्वि २ हतं चन्द्रोच्चे / तिथि १५ गुण च सितशीघ्र / त्वीषु ५२ हतं च स्वाद्वि २ कु १ वेद / ४ हतं च पात कुजशनिषु / ग्रहबीजानि ब्रह्मसिद्धान्ते.

Verses 6–9: Critique of planetary positions

युगवर्षादिनवद्व्येक्षं सितादेः समं प्रवृत्तान्यत्
तदसत् यतः स्फुटयुगं तस्थैर्यान्मन्दपातानाम् ॥ ६ ॥

ग्रहमुक्त रूनाया मन्दोच्चं भवति शीघ्रमधिकायाम्
उच्चगतौ मन्दोच्चं न विना भुक्तिं व्रजेमतः ॥ ७ ॥

आर्याष्टशते पाता भ्रमंति दशगीतके स्थिराः पठिताः
मुक्तं द्विपातमण्डपमण्डले भ्रमति स्थिरान्नतः ॥ ८ ॥

आर्यभटो जानाति ग्रहाष्टात् यदुक्तवांस्तदसत्
राहुकृतं न ग्रहणं तत्पातोऽनाष्टमो राहुः ॥ ९ ॥

Translation: (6) The yuga-year-navadvyekṣa (nine), the bright (Moon) etc., the same pravṛttānyat; that is false, since the corrected yuga-position of the manda-nodes is fixed. (7) The planet's release is in the ruṇāyā, the mandocca increases in śīghra; in the high position, the mandocca does not exist without (its own) motion. (8) In Ārya's eight hundred, the nodes move, in the Daśagītaka are fixed; the released two-nodes move in the circle, not fixed. (9) Āryabhaṭa knows the planets; what he has said is false; the Rāhu does not cause the eclipse, its pāta is not the eighth Rāhu.

Critical Apparatus:

6.1 युगवर्षादी □□□ (युगवर्षादि).

6.2 सप्रवृत्तानयत् □□□ (समं प्रवृत्तान्यत्).

6.3 तदराजु (G) □□□ (तदसत्).

6.4 तत्तस्थैर्यात् (G) □□□ (तस्थैर्यात्).

7.1 रूनायां □□□ (रूनाया).

7.2 मुक्तच न्हु □□□ (मुक्त द्वि).

7.3 वजमेतः □□□ (व्रजेमतः).

7.4 वि० — An additional half-verse is supplied: तद्दरगम्यां दशामिः संघुशितास्यां सूक्ष्मे.

8.1 आर्याष्टशते (Ch) □□□ (आर्याष्टशते).

8.2 दशगीतके □□□ (दशगीतके).

8.3 मुक्तं द्वि □□□ (मुक्तं द्वि).

8.4 भ्रमंति (G) □□□ (भ्रमंति).

9.1 ग्रहाष्टात् (G) □□□ (ग्रहाष्टात्).

9.2 तस्यातो (G) □□□ (तत्पातः).

9.3 अनाष्टमः □□□ (अनाष्टमः).

Verse 63: Conclusion

इत्थिकथिततन्त्राणाकान् पठतिरपि दूषणैः करोत्यज्ञानात्
तन्त्र परीक्षायाणां प्रदिष्टिरिकादशोऽध्यायः ॥ ६३ ॥

Translation: (63) Thus the systems taught, even by reading, cause ignorance by faults; the examination of systems — this is the eleventh chapter.

Critical Apparatus:

63.1 तन्त्रगणकान् अदिते (G) तन्त्राणाकान् पठतिरपि □□□ (तन्त्राणाकान् पठतिरपि).

63.2 परीक्षार्याणाम् (G) □□□ (परीक्षायाणाम्).

63.3 इत्ति □□□ (इत्ति).

63.4 करोत्यज्ञानात् □□□ (करोत्यज्ञानात्).

63.5 वि० — □इति' से □अध्यायः' तक सब लुप्त (the closing formula is omitted).

♣ ♣ ♣

इति श्रीब्रह्मगुप्तसिद्धान्ते एकादशोऽध्यायः

Thus ends the Eleventh Chapter of the Brāhmasphuṭasiddhānta.

♣ ♣ ♣

Chapter XII: Gaṇitādhyāya

द्वादशोऽध्यायः

अथ गणिताध्याय द्वादशः

The Twelfth Chapter — On Mathematics

The Gaṇitādhyāya is the most celebrated mathematical chapter of the *Brāhmasphuṭasiddhānta*. It contains 66 verses divided into ten sections (*vyavahāra*s), covering:

1. परिकर्म — Preliminary operations (fractions, proportions)
2. मिश्रकव्यवहारः — Mixture problems (verses 1–16)
3. श्रेढ़िव्यवहारः — Arithmetic and geometric series (verses 17–20)
4. क्षेत्रव्यवहारः — Geometry of plane figures (verses 21–39)
5. वृत्तक्षेत्रगणितम् — Circle geometry (verses 40–43)
6. खातव्यवहारः — Excavation/volume problems (verses 44–46)
7. चितिव्यवहारः — Pile problems (verses 47+)
8. शाद्रकाकाचिकादिकरणसूत्रे — Shadow of a gnomon etc.
9. राशिव्यवहारः — Rules for treating quantities (verses 50–51)
10. छायाव्यवहारः — Shadow problems (verses 52–66)

Sanskrit (Devanagari)	English Translation (with IAST)

Section 1: Parikarmma (Preliminary Operations)

परिक्रम्म विशति यः संकलिताद्यं पृथग् विजानाति
परष्ठी च व्यवहारान् छायातान् भवति गणकः सः ॥ १ ॥

Translation: (1) He who knows separately the parikamma (operation), the saṃkalita (summation) etc., and the twenty vyavahāras with chāyāta — he becomes a calculator (gaṇaka).

Critical Apparatus:

1.1 विद्याति (G) □□□ (विजानाति).
1.2 संकलिताद्यां (G,Ch) □□□ (संकलिताद्यं).
1.3 पृथग्विजानाति (G,Ch) □□□ (पृथग् विजानाति).
1.4 व्यवहां (G) □□□ (व्यवहारान्).
1.5 छायांतान् (G) □□□ (छायातान्).
1.6 परिक्रम्मं □□□ (परिकर्म).

विपरीत छेदगुणाः राश्योः छेदांशकाः समछेदाः
संकलितेशा योज्या व्यवकलितेशान्तरं कार्यम् ॥ २ ॥

रूपार्णी छेदगुणान्यंशयुतानि द्वयोर्बहूनां वा
प्रत्युत्पन्नो भवति छेदवधेनोद्धृतांशवधः ॥ ३ ॥

प्रत्युत्पन्नः परिवर्त्यं भागहारछेदांशो छेदसंगुणछेदः
द्वयांशयुग् भाज्यस्य भागहारः सर्वार्णितयोः भागहारः ॥ ४ ॥

Translation: (2) The inverted (viparīta) and the divisor-multipliers of the quantities, the divisor-parts (chedāṃśakāḥ) with common divisor; the saṃkalita etc. are to be added, the difference is vyavakalita. (3) The rūpāraṇī (unity) multiplied by the divisor, with parts added, of two or many; the pratyutpanna (product) becomes the division by the divisor-product of the fraction-product. (4) The pratyutpanna after conversion, the divisor-part of the divisor, the divisor-times-divisor; the two-part-sum of the dividend, the divisor of all is the divisor.

Critical Apparatus:

2.1 संकलितेशा (G,Ch) □□□ (संकलितेशा).

2.2 गुणा □□□ (गुणाः).

2.3 राश्योः छेदांशकाः □□□ (राश्योः छेदांशकाः).

3.1 वधेनोद्धृतांशवधः (G) □□□ (वधेनोद्धृतांशवधः).

3.2 बहूनां (G,Ch) □□□ (बहूनां).

4.1 संगुण □□□ (संगुण).

4.2 भाययस्य (Ch) □□□ (भाज्यस्य).

4.3 सर्वार्णितयोः (G) □□□ (सर्वार्णितयोः).

वि० In this verse, the words “प्रत्युत्पन्न” and “भागहारः” are not marked in the text; the verse begins with the word “परिवृत्य”.

सर्वार्णितांशवर्गं छेदकृति विभाजितो भवति वेगः
सर्वार्शितांशमूलं छेदपदेनोद्धृते मूलम् ॥ ५ ॥

वर्गमूलं स्थाप्योत्पद्यनोन्यकृतिस्त्रिगुणोत्तरसंगुणा च यत्प्रथमा
नोत्तरकृतिरन्त्यगुणा त्रिगुणाच्चोत्तर घनः ॥ ६ ॥

घनः समाप्तः

छेदाघना द्वितीयात् घनमूलकृतिस्त्रिसंगुणा सकृतिः
शोध्या त्रिपूर्वगुणिता प्रथमाद् घनतो घनो मूलम् ॥ ७ ॥

घनमूलं समाप्तः

Translation: (5) The square of all-parts divided by the square of the divisor becomes the varga (square root); the root of all-śita-parts, the root divided by the divisor-term. (6) Having set up the varga-mūla (square root), the utpadyanyakṛti, the trigunottara-saguṇa, and the prathamā; the nottara-kṛti-antitya-guṇā, triguṇā and uttara is the ghana (cube). [Cube completed.] (7) From the second chedāghana, the cube-root-square-times-trisaguṇa-sakṛti; subtracted, times the tripūrva, from the prathama-ghana, the ghana-mūla. [Cube root completed.]

Critical Apparatus:

6.1 स्थाप्योत्यघनोन्यकृतिः □□□ (स्थाप्योत्पद्यनोन्यकृति).

6.2 न्यत्कृति □□□ (न्यकृति).

6.3 उत्तरकृतिः □□□ (उत्तरकृति).

6.4 अन्त्यगुणा □□□ (अन्त्यगुणा).

7.1 घनमूल □□□ (घनमूल).

7.2 संगुणशासकृतिः (G) □□□ (संगुणा सकृतिः).

7.3 छिदोघना □□□ (छेदाघना).

7.4 प्रथमाद् घनतो घनमूलं □□□ (प्रथमाद् घनतो घनमूलं).

वि० The words “घनः समाप्तः” and “घनमूलं समाप्तः” (“cube completed” / “cube root completed”) are not marked in the source text.

Section 2: Miśrakavyavahāra (Mixture Problems)

प्रक्षेपयोगहृतया लब्धा प्रक्षेपका गुणा लाभाः
ऊनाधिकोत्तरार्धं सद् द्वितोनया स्वफलसूनं युतम् ॥ १६ ॥

Translation: (16) Divided by the prakṣepa-yoga (sum of contributions), obtained, the prakṣepaka (contributor) multiplied by the gains; the ūnādhika-uttara-ardha is true, the dvitonayā with its own phala-sum added.

Critical Apparatus:

16.1 प्रक्षेपयोगहतया (G,Ch) □□□ (प्रक्षेपयोगहतया).

16.2 लब्धा □□□ (लब्ध्वा).

16.3 प्रक्षेपका (G) प्रक्षेपक □□□ (प्रक्षेपका).

16.4 त्तरास्तच्चुतो (G,Ch) □□□ (त्तराद्धस्तद्युतो).

16.5 सूनयुतं (G,Ch) □□□ (सूनंयुतम्).

वि० From “मिश्रक” to “समाप्तः” is not marked.

मिश्रकव्यवहार समाप्तः
End of Mixture Problems.

Section 3: Śreḍhīvyavahāra (Series)

पदमेकहीनसुत्तरगुणितं संयुक्तमाद्यान्त्यघनम्
आदि युतान्त्यघनाद्धे मध्यघनं पदगणं गुणितम् ॥ १७ ॥

उत्तरहीनाद्द्विगुणादिशेदावर्ग घनोत्तराष्ट्रवधे
प्रक्षिप्य पदं शेषेणं द्विगुणोत्तर हृतं गच्छः ॥ १८ ॥

एकोत्तरमेकाय यदीष्टगच्छस्य भवति संकलितम्
तद् वियुत् गच्छ गुणितं त्रिहृतं संकलितमस् ॥ १९ ॥

Translation: (17) The pada (term) minus one, multiplied by the uttara (common difference), added to the sum of the first and last terms; the first plus the last term from the first, the cube of the middle term is pada-gaṇa (number of terms) multiplied. (18) From the uttara-hīna (decreased by common difference) dviguṇādi-śeda-varga, the ghanottara-aṣṭra-vadhe; added, the pada by the śeṣeṇaṃ, the dviguṇottara hṛtaṃ gacchaḥ (progression). (19) If the eka-uttara (common difference one) eka (first term one) and the desired gaccha (number of terms) is the saṃkalita (sum); that viyut (without) gaccha multiplied, divided by three, is the saṃkalita.

Critical Apparatus:

17.1 पदगुणं □□□ (पदगणं).

17.2 गणितम् □□□ (गुणितम्).

18.1 उत्तरहीन (G,Ch) □□□ (उत्तरहीन).

18.2 शेषवर्ग □□□ (शेषवर्ग).

18.3 प्रक्षिप्य (G,Ch) □□□ (प्रक्षिप्य).

18.4 हृतं (G) — यह पद अंकित नहीं है (this word is not marked).

18.5 द्विगुणोत्तरं □□□ (द्विगुणोत्तर).

19.1 यदिष्ट (G,Ch) □□□ (यदीष्ट).

19.2 गच्छस्य (G,Ch) □□□ (गच्छस्य).

19.3 तृहृतं □□□ (त्रिहृतं).

19.4 एकोत्तमेकाय् □□□ (एकोत्तरमेकाय्).

19.5 वियुतगच्छ □□□ (वियुत् गच्छ).

Section 4: Kṣetravyavahāra (Geometry of Plane Figures)

The *kṣetra-vyavahāra* contains rules for computing the areas and diagonals of triangles, quadrilaterals, and other plane figures. This is one of the most mathematically significant sections of the work.

Verses 21–23: General area rules

समत□□□भुजद्विभुजादिषु क्षेत्रेष्वायतचतुरस्रवद् व्यासः
क्षेत्रस्य व्यासकर्णाः समकोण्यः क्षेत्रयुगलस्य ॥ २१ ॥

समपरैवल्लम्बकर्णयुतिदलं समत्रिभुजे फलम्
समचतुरस्रे तु वर्गः समकोणि व्यस्तयोगार्धम् ॥ २२ ॥

असमचतुरस्रे तु राश्योर्गुणकृत्योः कृतिश्च योगः
तयोर्विश्लेषवर्गश्च तुल्यकृतिदलयुते मिथः ॥ २३ ॥

Translation: (21) In triangles and quadrilaterals with equal and two sides etc., the vyāsa (diameter) is like āyata-catura (rectangle); the vyāsa-karṇa of the field is the sama-koṇi (right-angled diagonal) of the pair of fields. (22) Half the sum of the lambaka (altitude) and karṇa in sama-tribhuje (isosceles triangle) is the area; in sama-catura (square) it is the square; in sama-koṇi (right-angled) half the sum of the vyasta (crossed diagonals). (23) In asama-catura (unequal quadrilateral), the product of the rāśis (sides), their squares and the sum; their difference-varga (square), with equal-square-half added mutually.

Verses 32–35: Specific geometric formulas

कर्णावलम्बयोरधरे कर्णावलम्बयोरधरे
अनुपातेन तुद्घ्ने उद्ध शून्यांस पाटायाम् ॥ ३२ ॥

कृतियुरसहशराद्योबाहुधातो द्विसंगुणों लम्बः
क्रत्यन्तरमसदृशयोर्गुणं द्विसमत्रिभुजम्बमिः ॥ ३३ ॥

इष्टद्वयेन भक्तं द्विघ्नवर्गफलेष्टयोगाद्ध
विषमत्रिभुजस्य भुजा विषमफलाद्ध योगा मूः ॥ ३४ ॥

इष्टस्य भुजस्य कृतिर्भक्तो नेष्टेन तदलं कोटिः
आयतचतुरस्त्रस्य क्षेत्रस्येष्टाधिका कराः ॥ ३५ ॥

Translation: (32) The karṇa and lamba (hypotenuse and altitude) — the lower karṇa-lamba; in proportion, the uddha-śūnyāṃsa-pāṭāyāma. (33) The product of kṛti-yura-sahaśa-rādyo-bāhu-dhāto; the lambaka (altitude) twice; the product of kṛty-antara of unequal (sides) and dvitribhuja. (34) Divided by two desired quantities, from the sum of the dvi-ghaṇa-varga-phala; the side of a viṣama-tribhuja (scalene triangle) from the viṣama-phala is added to yā mūḥ. (35) The square of the desired bhuja divided by the undesired, that portion is the koṭi; of the āyata-caturastra (rectangle), the field has the desired hypotenuse.

Critical Apparatus:

32.1 कर्णावलम्बयोरधरे — repeated in both lines.
32.2 अनुपातेन तुद्घ्ने □□□ (अनुपातेन).
33.1 कृतियुरसहशराद्योबाहुधातो □□□ (कृतियुरसहशराद्योबाहुधातो).
33.2 द्विसंगुणों □□□ (द्विसंगुणो लम्बः).
33.3 क्रत्यन्तरमसदृशयोः □□□ (क्रत्यन्तरमसदृशयोः).
33.4 द्विसमत्रिभुजम्बमिः □□□ (द्विसमत्रिभुजम्बमिः).
34.1 इष्टद्वयेन □□□ (इष्टद्वयेन).
34.2 विषमफलाद्ध □□□ (विषमफलाद्ध).
34.3 योगा मूः □□□ (योगा मूः).
35.1 इष्टस्य □□□ (इष्टस्य).
35.2 भुजस्य □□□ (भुजस्य).
35.3 कृतिर्भक्तो □□□ (कृतिर्भक्तः).

35.4 नेष्टेन □□□ (नेष्टेन).

35.5 तदलं □□□ (तदलं).

35.6 कोटिः □□□ (कोटिः).

35.7 आयतचतुरस्त्रस्य □□□ (आयतचतुरस्त्रस्य).

35.8 क्षेत्रस्येष्टाधिका कराः □□□ (क्षेत्रस्येष्टाधिका कराः).

Section 10: Chāyāvyavahāra (Shadow Problems)

The *chāyā-vyavahāra* deals with computations involving the shadow of a gnomon (*śaṅku*), problems of distance, height determination using shadows, and related topics.

Verses 1–4: Lamp and shadow calculations

छायान्तरसैकहृतं द्विदलं प्रागपरयोर्युगतदोषम्
दिनगतं शेषांशहृतं द्विदलं छायान्तरव्येकम् ॥ २ ॥

दीपतलशङ्कुतलयोरन्तरमिष्टप्रमाणशङ्कुगुणम्
दीपशिखोच्च्या शङ्कुं विशोध्य शेषोद्धृतं छाया ॥ १३ ॥

छायाग्रान्तरगुणिता छाया छायान्तरेण भक्ता भूः
भूः शङ्कुगुणा छाया विभाजिता दीपशिखोच्च्यम् ॥ ४ ॥

Translation: (2) The chāyāntara (difference of shadows) with one added, divided by two, the prāga-parayor-yuga-doṣam; the dinagata (elapsed day), the śeṣāṃśa-hṛtaṃ dvidalaṃ, the chāyāntara-vyekam. (13) The distance between the lamp-surface and śaṅku-surface, the desired measure-śaṅku-guṇa; the lamp-wick-ucchyā, the śaṅku deducted, the śeṣa-uddhṛtaṃ chāyā. (4) The chāyā multiplied by the chāyāgra-antara, divided by chāyāntara is the bhū (earth/distance); the bhū is śaṅku-guṇā chāyā, divided, the lamp-wick-height.

Critical Apparatus:

2.1 छायान्तरसैकहृतं (G,Ḍ) □□□ (छायान्तरसैकहृतं).

2.2 चुगतशेषम् (Ḍ,G) □□□ (चुगतशेषम्).

2.3 दिनगतं (G,Ch) □□□ (दिनगतं).

2.4 हृतं (G,Ch,Ḍ) □□□ (हृतं).

2.5 छायामरव्येकम् □□□ (छायान्तरव्येकम्).

13.1 शङ्कु □□□ (शङ्कु).

13.2 दीपशिखोच्च्या □□□ (दीपशिखोच्च्या).

4.1 दीपक □□□ (दीप).

4.2 दीपशिखयौच्च्यम् □□□ (दीपशिखोच्च्यम्).

Verses 44–45: Gnomon problems and kuṭṭaka

छायाव्यवहार समाप्तः

गुणाकार खण्डतुल्यौ गुण्यौ गोमूत्रिकाकृती गुणितः
सहितः प्रत्युत्पन्नो गुण कारकमेदतुल्यौ वा ॥ ४५ ॥

Translation: [End of shadow problems.] (45) The guṇākāra (multiplier) and khāṇḍa-tulya (equal to the section) are the guṇyau (multiplicands); the gomūtri-kā (cow's urine) kṛtī multiplied; the sahitāḥ pratyutpannaḥ (combined product), the guṇa (quality) or equal to the kāraṇa-meda.

Critical Apparatus:

45.1 स्वंज □□□ (खण्ड).

45.2 गुण्यौ (G) गणी □□□ (गुण्यौ).

45.3 कृतो □□□ (कृतो).

45.4 प्रत्युत्पन्नी (G) प्रत्युत्पन्नो □□□ (प्रत्युत्पन्नः).

45.5 गुणितः □□□ (गुणितः).

45.6 वा □□□ (वा).

♣ ♣ ♣

[The Gaṇitādhyāya concludes here with 66 verses total, containing the foundation of Indian algebra including the kuṭṭaka (pulverizer) method for solving linear Diophantine equations.]

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Chapter XIII: Praśnādhyāya

षष्ठः प्रश्नाध्यायः

तत्र तावन्मध्यगत्युत्तराध्यायः

The Sixth Problem-Chapter: On Mean and True Motion

This chapter presents mathematical problems (*praśna*) whose solutions demonstrate the computational methods developed in the preceding chapters. The title "षष्ठः" (sixth) refers to the chapter's position within the uttara (secondary) section of the text.

Sanskrit (Devanagari) **English Translation (with IAST)**

Verses 1–2: Introduction

प्रश्नाध्यायान् प्रवक्ष्यामि सोत्तरान् गणकबुद्धिवृद्धिकरान्
यैज्ञतिस्तन्त्रविदामाचार्यो भवति बुद्धिमताम् ॥ १ ॥

अधिमासकाः सविकलाः सं युगयातमवमरात्रैव
द्युगणेन वायुगगतयो वेत्ति स कालतंत्रज्ञः ॥ २ ॥

Translation: (1) I shall declare the praśnādhyāyas (problem-chapters) with solutions (uttara), increasing the intelligence of calculators; by which the teacher (ācārya) of the tāntra-vids becomes one of intelligent mind. (2) The adhimāsakāḥ (intercalary months) with vikalā (seconds), the avamarātra (omitted days) with the yugayāta, by the dyugaṇa (day-count); he who knows the motion of planets in the wind's path knows the kāla-tantra (science of time).

Critical Apparatus:

1.1 प्रदनाध्यायान् (G) प्रस्ताध्यायान् □□□ (प्रश्नाध्यायान्).

1.2 बुद्धिमत्यम् (G) बुधिमताम् □□□ (बुद्धिमताम्).

1.3 वक्ष्यामि □□□ (प्रवक्षामि).

1.4 □वृद्धि' पद भ्रंकित नहीं है (the word 'vṛddhi' is not marked).

2.1 वायुगगं (G) वायुगगतं □□□ (वायुगगत).

2.2 अधिमासिकः □□□ (अधिमासकाः).

2.3 मवमरात्रैवा □□□ (मवमरात्रैवा).

Verses 3–5: Time and planetary motion problems

अवमाति यः सविकेऽधिमासरधिकमासकानवमैः
ग्रहमिष्ठं ताम्यां वायो वेत्ति स कालतंत्रज्ञः ॥ ३ ॥

द्युगणं विनाधिमासावमेविनादिन गरोन चन्द्राक्कः
ताम्यां विनास्फुटतिथि यो वेत्ति स कालतंत्रज्ञः ॥ ४ ॥

इष्टान्मध्यादन्यांस्तिथिमिष्टान्मध्यमाछदिप्रहणस्
इष्टार्कविदुपाते यों वेत्ति बिना स तंत्रज्ञः ॥ ५ ॥

Translation: (3) He who comprehends the avamāti with vikalā, the adhimāsa-radhika-māsaka-anavamaiḥ, the grahamiṣṭhaṃ in that, knows the kāla-tantra. (4) The dyugaṇa without adhimāsa, avinādina garona candra-ākkaḥ; in that without sphuṭa-tithi, he who knows is the kāla-tantra-jña. (5) From the mean (madhya) to other tithi from iṣṭa-madhyamā-cchadi-prahaṇas, from the iṣṭārkavid-upāte yoṃ, without knowing, he is not a tantra-jña.

Critical Apparatus:

3.1 वेत्ति (Ch) □□□ (वेत्ति).

3.2 कालतंत्रज्ञः (G) सकलतंत्रज्ञः □□□ (सकलतंत्रज्ञः).

3.3 अधिमासकः □□□ (अधिमासकः).

3.4 मवमरात्रैवा (Ch) □□□ (मवमरात्रैवा).

4.1 द्युगणं □□□ (द्युगणं).

4.2 विनाधिमासावमेविनादिन □□□ (विनाधिमासावमेविनादिन).

4.3 गरोन □□□ (गरोन).

4.4 चन्द्राक्कः □□□ (चन्द्राक्कः).

4.5 विनास्फुटतिथि □□□ (विनास्फुटतिथि).

4.6 कालतंत्रज्ञः (G) □□□ (कालतंत्रज्ञः).

5.1 इष्टान्मध्यादन्यांस्तिथिमिष्टान्मध्यमाछदिप्रहणस् □□□ (इष्टान्मध्यादन्यांस्तिथिमिष्टान्मध्यमाछदिप्रहणस्).

5.2 इष्टार्कविदुपाते यों □□□ (इष्टार्कविदुपाते यों).

5.3 बिना □□□ (बिना).

5.4 तंत्रज्ञः □□□ (तंत्रज्ञः).

♣ ♣ ♣

[The Praśnādhyāya continues with detailed problem-solution pairs covering mean and true longitudes of planets, conjunctions, eclipses, and other astronomical computations. The full chapter contains approximately 48 verses.]

♣ ♣ ♣

Appendix: Notes on the Critical Edition

The Manuscripts

The critical apparatus in this edition is based on the following manuscripts, as collated by Sudhākara Dvivedī:

1. **MS G** — The principal manuscript used by Dvivedī, containing the complete text with the commentary of Pṛthūdakasvāmin. This manuscript was written in the Maithila script and is now preserved in the India Office Library, London.
2. **MS K** — A copy from the original G manuscript, prepared for Dr. Thibaut by Dvivedī himself. Many folios were misarranged during binding.
3. **MS Kh** — Another copy of the commentary manuscript, with variant readings at several places.
4. **MS Ch** — A third copy with further variants.
5. **MS Gh** — A fourth manuscript source.

Editorial Method

Dvivedī's edition follows the standard conventions of Sanskrit critical editing:

- The *lemma* (text adopted in the edition) is printed in the main body.
- Variant readings are recorded after each verse, introduced by the manuscript siglum.
- The notation "X □□□ (Y)" means the edition reads X where the manuscript has Y.
- Hindi notes introduced by वि० (*viveka*) provide the editor's commentary on textual problems.

Mutilated Passages

Dvivedī notes that many passages in the manuscripts are mutilated, particularly:

- The beginning of the Golādhyāya commentary
- Portions of the Madhyamādhikāra
- The Triprasnādhikāra (verses 64–66 area)
- The Candragrahaṇādhikāra
- The Sūryagrahaṇādhikāra

At several places, only the ṭīkā (commentary) is preserved, while the mūla (root text) is lost.

Mathematical Significance

The chapters in this chunk contain several historically important mathematical developments:

1. **The shadow rules** (Triprasnādhikāra) present sophisticated trigonometric relationships involving the gnomon, used for determining time, direction, and terrestrial latitude.
2. **The eclipse computations** involve iterative algorithms for correcting positions and computing parallax — among the most complex calculations in ancient astronomy.
3. **The Gaṇitādhyāya** contains:
    - The complete theory of operations on fractions (*parikarmma*)

  - Rules for arithmetic and geometric progressions
  - Formulas for areas of triangles and quadrilaterals, including the "Brahmagupta's formula" for the area of a cyclic quadrilateral
  - The *kuṭṭaka* (pulverizer) method for solving linear Diophantine equations of the form $ax + c = by$
  - The *varga-prakṛti* (square-nature) method, a precursor to the study of Pell's equation

4. **The Tantraparīkṣā** represents one of the earliest critical examinations of rival scientific theories in Indian intellectual history.

On the Translation

The English translations in this bilingual edition aim to be as literal as possible while remaining intelligible. Sanskrit technical terms are preserved in IAST transliteration and enclosed in parentheses where appropriate. Several verses in the source arecorrupt or mutilated; in these cases the translation follows Dvivedī's reconstruction and is marked accordingly.

The *Brāhmasphuṭasiddhānta* is composed in the āryā metre (with some verses in the longer vasantatilakā and related metres). The elliptical nature of Sanskrit astronomical verse makes translation particularly challenging: many verses omit verbs, subjects, or objects that must be supplied from context or commentary. Readers are encouraged to consult Pṛthūdakasvāmin's commentary (*Vāsanābhāṣya*) for detailed explanations of each rule.

♣ ♣ ♣

End of Chunk 4 (Pages 421–560)

♣ ♣ ♣

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12pt

ब्राह्मस्फुटसिद्धान्त

Brāhmasphuṭasiddhānta

of **Brahmagupta** (628 CE)

Bilingual Sanskrit–English Edition

Chunk 5: Pages 561–700

Chapters XIV–XXII (partial)

Left column: Sanskrit text in Devanāgarī script

Right column: English translation with IAST terminology

Mathematical formulas rendered in modern notation

Typeset in XeLaTeX with Noto Serif Devanagari
Sanskrit text: GRETIL (Hayashi 1993) / S. Dvivedīn’s edition
English translations adapted from standard scholarly editions

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Introduction

संक्षेपेण परिचयः

एषः खण्डः पञ्चमः अष्टाधिक-शत-पृष्ठ-पर्यन्तं गतः। अस्मिन् खण्डे चतुर्दशात् द्वाविंशत्-पर्यन्तम् अध्यायाः सन्ति। एतेषु ग्रह-गतिः, त्रि-प्रश्नः, ग्रहणम्, शृङ्गोन्नतिः, कुट्टक-गणितम्, शङ्कु-छाया, छन्दः, ज्योत्पत्तिः, यन्त्राणि च अन्तर्भूतानि।

Overview of This Chunk

This fifth chunk covers pages 561–700 of the source PDF, encompassing Chapters XIV through XXII of the *Brāhmasphuṭasiddhānta*. These chapters treat planetary motion, the "three questions" of astronomy, eclipses, lunar latitude, algebra (the kuṭṭaka), gnomon and shadow, prosody, sine computation, and astronomical instruments.

Contents of This Volume

Ch. I: Sphoṭagatyuttarādhikāraḥ (Ch. 14) — True planetary motions; epicyclic corrections for *manda* and *śīghra*.

Ch. II: Tripraśnottarādhikāraḥ (Ch. 15) — The three questions: latitude from gnomon observations, shadow measurements, time and place determination.

Ch. III: Grahāṇottarādhikāraḥ (Ch. 16) — Advanced eclipse computations, including the graphical *parilekha* method.

Ch. IV: Śṛṅgonnatyuttarādhikāraḥ (Ch. 17) — Lunar cusp elevation; a short chapter of 10 verses.

Ch. V: Kuṭṭakādhyāyaḥ (Ch. 18) — The algebra chapter (102 verses): kuṭṭaka (pulverizer), operations with signed numbers and zero, *saṅkramaṇa*, *karaṇī* (surds), linear and quadratic equations, *bhāvitaka*, and the *varga-prakṛti* (Pell's equation).

Ch. VI: Śaṅku-chāyā-vijñānam (Ch. 19) — Gnomon and shadow computations; practical geometry and instrument usage.

Ch. VII: Chandaś-citty-uttarādhikāyaḥ (Ch. 20) — Sanskrit prosody; combinatorial calculations for metrical patterns.

Ch. VIII: Golādhyāyaḥ (Ch. 21, partial) — The Jyā-prakaraṇa: sine computation by second-order interpolation.

Ch. IX: Yantrādhyāyaḥ (Ch. 22) — Astronomical instruments: *dhanur-yantra*, *yaṣṭi-yantra*, water and wheel instruments.

On the Translation

The English translations in the right column aim for clarity and fidelity to the mathematical content. Technical Sanskrit terms are given in IAST italics on first occurrence. Modern mathematical notation (algebraic formulas, equations) is provided where the Sanskrit text is purely algorithmic.

The manuscript sigla used for variant readings are: (⊠) Poona ms., (⊠) Bombay ms., (⊠) Baroda ms., and (⊠) Hoshiarpur ms.

अथ स्फुटगत्युत्तराध्यायः — चतुर्दशः

स्फुटगत्युत्तराध्यायः

अस्य अध्यायस्य विषयः स्फुटगतिः—ग्रहस्य मध्यमात् स्फुटांश-प्राप्तिः। मन्द-शीघ्र-फल-संस्कारैः सत्यः ग्रह-स्थान-निर्धारणम्।

Chapter XIV: On True Planetary Motions

This chapter deals with *sphuṭa-gati*—the computation of a planet's true longitude from its mean longitude by applying the *manda* (eccentric) and *śīghra* (epicyclic) corrections.

[Printed pages 171–181. This chapter continues from the previous chunk. Verses 1–6 and the closing verse are preserved here.]

युगभागः कोटिज्यां कोट्यां करोति बाहुज्याम् ।
कोटिं भुजेन बाहुं क्षेत्रच्चावा स्फुटगतिकः सः ॥ १ ॥

V.1. The divisor of the *yuga* makes the cosine (*koṭijyā*) into the cosine, and the sine (*bāhujyā*) into the sine. The hypotenuse (*kṣetra*) together with these—that is the *sphuṭagatikaḥ*, the true-motion triangle.

परमण्डलकर्णविकः करोति कोटिछाया स्फुटं कर्तुम् ।
कर्णान्तकोटिकोण्डा बाहुं वा स्फुटगतिकः सः ॥ २ ॥

V.2. The radius of the outer circle (*paramaṇḍala-karṇa*), made to produce the cosine-shadow (*koṭi-chāyā*) for computing the true — the hypotenuse-end, cosine-corner, and sine: that is the true-motion triangle.

कर्णेऽनुजकोटिजीवा परमण्डलज्ञः करोति यः कर्तुम् ।
स्वोच्चं स्वफलतः स्फुटं करोति यः स्फुटगतिकः सः ॥ ३ ॥

V.3. The following cosine-chord (*anujakoṭijīvā*) at the radius, which the outer-circle-knower makes to produce — that which makes the *svocca* (own apogee) from its own equation (*sva-phala*) into the true — that is the true-motion triangle.

क्षेत्रस्य स्फुटपदस्य भुजकोटिज्ये फले विना ज्यामिमः ।
ज्यामिर्विना फलयुः करोति वा स्फुटगतिकः ॥ ४ ॥

V.4. Of the field of the true-hypotenuse (*sphuṭa-pada*), sine and cosine are the results without the chord (*jyā*); without the chord, or with the equation (*phala*) added — that makes the true-motion computation.

इष्टदिवयदैर्घिकान् करोति यो मध्यमान् ग्रहान् स्फुटान् ।
स्वोच्चरस्फुटैर्हं यः करोति वा स्फुटगतिकः सः ॥ ५ ॥

V.5. He who makes the mean planets true by desired-day indicators (*iṣṭa-diva-yadaiḥ*) for the directions — that *sphuṭagatikaḥ* which makes (the computation) with its own apogee-true (corrections).

त्रिगुणः शनिरिन्दुनैत्यमगारालक्ष्मीर्ग्रहादिविमः संहितः ।
भौमोऽङ्गिर्गुरुर्दुर्वर्चं वात्यमगाराः कै ॥ ६ ॥

V.6. Threefold: Saturn (*śani*), Moon (*indu*), Naiṭyama (Mercury), Gārā (Mars), Lakṣmī (Venus), the planet-group — gathered: Mars (*bhauma*), Aṅgiras (Jupiter), Guru — Durvarca, Vātya, Gārā — with Kai (Sun?). [Planetary name correspondences in numerical notation.]

[The chapter continues with more verses on true motion computations, including corrections for the manda and śīghra epicycles, and rules for finding planetary positions. The chapter concludes at printed page 181 with verse 46.]

मध्योतरेखैता नयः पञ्चाशतयोदशोऽध्यायः ।

ज्ञातवै तन्त्रविदामाचायो भवति मध्यगतौ ॥ ४६ ॥

V.46. This rule of mean-upper-line (*madhyotarekhā*) is the fifty-fifth chapter-method. It is to be known by the teachers of the knowers of the *tantra*; one becomes skilled in mean motion.

इति ब्राह्मस्फुटसिद्धान्ते मध्यमाधिकारे चतुर्दशोऽध्यायः

इति श्री-ब्राह्मस्फुटसिद्धान्ते मध्यमाधिकारे
चतुर्दशोऽध्यायः समाप्तः।

Here ends the Fourteenth Chapter, on Mean Motion, in the ***Brāhmasphuṭasiddhānta****.*

अथ त्रिप्रश्नोत्तराध्यायः — पञ्चदशः

त्रिप्रश्नोत्तराध्यायः

अस्य अध्यायस्य विषयाः त्रयः प्रश्नाः—काल-स्थान-दिक्-प्रश्नाः। शङ्कु-छाया-मापनैः अक्षांश-निर्धारणम्, दिग्-ज्ञानम्, काल-निर्णयः च।

Chapter XV: The Three Questions

This chapter treats the “three questions” of astronomy: relating to time (*kāla*), place (*sthāna*), and direction (*dik*). It covers the determination of latitude from gnomon observations, shadow measurements, and the computation of time.

[Printed pages 182–193. Chapter 15 contains 60 verses on the three problems.]

उदय ज्याविम्बस्ता क्रान्तिज्या व्यासगुणालब्धः ।
द्वादशाग्निता शितिजा विषुवच्छाया हृता क्रान्तिः ॥ ४४ ॥

V.44. The sine of rising (*udaya-jyā*) and the sine of declination (*krānti-jyā*), multiplied by the diameter and divided: by twelve and by fire (3), by Śiti (cold, the zero-point) and by the equinoctial shadow — divided by that is the declination (*krāntiḥ*).

यथिष्टव्यासाद्वाच्छ्यरया भास्करात्तरराश्यया ।
द्विगुणामुदया सूत्रं ततिज्या क्रान्तिविशेषपदम् ॥ ४५ ॥

V.45. From the desired diameter (*iṣṭa-vyāsa*), from the shadow-ray (*chāyā-raśmi*), from Bhāskara’s (the sun’s) crossing-sign, from the double rising: the string (*sūtra*) — that sine is the measure of the particular declination (*krānti-viśeṣa-padam*).

षट् दले शङ्कु न तज्ज्ञे प्राच्यपराया यदि स्थितः शङ्कुः ।
उदग्नता दक्षिणतः स्तदन्तरेष्ठाधिकाकारणा ॥ ४६ ॥

V.46. In the six parts (of the day), the gnomon (*śaṅku*) is not known (in this way); if the gnomon is placed toward the east-west, the northern (deviation) is due to the southern excess, and that difference indicates greater latitude.

उत्तरगोले न तदै वाम्यं गोलगे सूत्रे ।
शङ्कुतलं शङ्कु हृतं विषुवच्छाया द्विषट्कं गुणायम् ॥ ४७ ॥

V.47. In the northern hemisphere, it is not thus; the leftward (deviation) is in the globe-string. The gnomon-base divided by the gnomon — the equinoctial shadow, twelvefold, is the multiplier.

[The chapter contains 60 verses on the three problems, concluding at printed page 193.]

अथ ग्रहाणोत्तराध्यायः — षोडशः

ग्रहाणोत्तराध्यायः

अस्य अध्यायस्य विषयः ग्रहण-गणितस्य परिष्काराः—चन्द्र-सूर्य-ग्रहण-साधनम्, परिलेख-विधिः, स्पर्श-मोक्ष-काल-निर्णयः च।

Chapter XVI: On Eclipses (Advanced)

This chapter treats refined computations for eclipses (*grahaṇa*): the computation of lunar and solar eclipses, the graphical *parilekha* method, and the determination of contact (*sparśa*) and release (*mokṣa*) times.

[Printed pages 194–218. Chapter 16 covers advanced eclipse computations, including the parilekha graphical method.]

परिलेखं ग्रहस्य ग्रहकल्पेन पूर्वं वक्रतो वा ।
तात्कालिकसंस्थानं निमीलितोन्मीलितं चैवम् ॥ १५ ॥

V.15. The *parilekha* (graphical representation) of the planet, by the planet-rule (*graha-kalpa*) first, either curved or straight — the instantaneous position, the closing and opening (of the eclipse) — thus.

विषेषपुण्यांशज्या मानेनच्छाया द्वितच्चापांशाः ।
प्रान्तयोर्यथादिशमक्रयदर्शो विपर्ययस्तथा ॥ १६ ॥

V.16. The sine of the particular (longitude) difference in minutes (*viśeṣa-puṇy-aṃśa-jyā*), the shadow by the measure, and the degrees of the second arc — at the edges according to direction, direct or reversed (*viparyayaḥ*).

तद्वलनांशकयोमेतरं जीवप्राच्छुमानवलम्बचात् ।
त्रिज्यालब्धच्छाया प्रग्रहक्षेपं प्राग्वक्रतोः ॥ २० ॥

V.20. The difference of those deflection-minutes (*valana-aṃśaka*), from the sine-east-measure and perpendicular — the shadow obtained by the radius (*trijyā*) — the *pragraha-kṣepa*, before the retrograde motion.

हृतया व्यासाङ्केनाच्चन्द्रनाताङ्कलिप्तका गुणाया ।
मध्यबललिप्तया दक्षिणोत्तरा दिगतबामयात् ॥ २१ ॥

V.21. Divided by the semi-diameter (*vyāsāṅka*), the minutes of the lunar-node-difference are the multiplier. By the middle-strength-minutes (*madhyabala-lip*) the north-south direction goes to the left.

[This chapter contains 46–47 verses on eclipse computations.]

अथ शृङ्गोन्नत्युत्तराध्यायः — सप्तदशः

शृङ्गोन्नत्युत्तराध्यायः

अस्य अध्यायस्य विषयः शशिनः शृङ्गोन्नतिः—चन्द्र-शिखराणाम् उन्नतिः, तज्-ज्ञानेन काल-निर्णयः च। एको दश-श्लोकीयः अध्यायः।

Chapter XVII: On Lunar Cusp Elevation

This chapter treats the elevation of the lunar horns (*śṛṅgonnati*)—the elevation of the moon's cusps, and the determination of time from this knowledge. This is a short chapter of ten verses.

[Printed pages 219–223. Chapter 17 deals with the lunar cusp (śṛṅga-unnati) and related computations. Only 10 verses.]

[अध्यायस्य श्लोकाः सम्पूर्णतया न प्राप्ताः।
अस्य अध्यायस्य पाठः पूर्व-खण्ड-सन्धौ स्थितः।]

[The complete verses of this chapter are not fully preserved in the scanned pages of this chunk. The chapter is transitional between the astronomical and mathematical sections.]

[This short chapter of 10 verses completes the astronomical portion before the extensive mathematical chapters.]

अथ कुट्टकाध्यायः — अष्टादशः

कुट्टकाध्यायः

अयं गणिताध्यायः अत्यन्तं महत्त्वपूर्णः। अत्र कुट्टक-विधिः, ऋण-धन-कर्म, अव्यक्त-गणितम्, सङ्क्रमणम्, एकवर्ण-समीकरणम्, अनेकवर्ण-समीकरणम्, भावितकम्, वर्गप्रकृतिः च उच्यते। अष्टोत्तर-शत-श्लोकीयः अयम् अध्यायः।

Chapter XVIII: The Kuṭṭaka (Algebra) Chapter

This is the most important mathematical chapter of the BSS, containing 102 verses. It covers the *kuṭṭaka* (pulverizer) method, operations with positive and negative numbers, *avyakta* (algebra) computations, *saṅkramaṇa* (transposition), linear and quadratic equations in one unknown, multi-variable equations, the *bhāvitaka* (product-nature) equation, and the *varga-prakṛti* (Pell's equation).

[Printed pages 224–283. The Sanskrit text below is from the GRETIL digitization (Hayashi 1993).]

कुट्टकः — □□□ □□□□□□□□□□

प्रायेण यतः प्रश्नाः कुट्टाकारात् ऋते न शक्यन्ते ।
ज्ञातुं वक्ष्यामि ततः कुट्टाकारं सह प्रश्नैः ॥ १ ॥

V.1. Since problems generally cannot be solved without the pulverizer (*kuṭṭaka*), I shall state the *kuṭṭaka* together with problems.

कुट्टक-ख-ऋण-धन-अव्यक्त-मध्य-हरण-एक-वर्ण-भावितकैः ।
आचार्यस् तन्त्र-विदां ज्ञातैः वर्ग-प्रकृत्या च ॥ २ ॥

V.2. The teachers (*ācārya*s) of the experts of the *tantra*, who know the pulverizer, zero (*kha*), debt (negative, *ṛṇa*) and wealth (positive, *dhana*), *avyakta* (unknown), elimination (*haraṇa*), one-*varṇa* and *bhāvitaka* equations, and the *varga-prakṛti* (quadratic nature, i.e. Pell's equation).

अधिक-अग्र-भाग-हारात् ऊन-अग्र-छेद-भाजितात् शेषम् ।
यत् तत् परस्पर-हृतं लब्धम् अधः अधः पृथक् स्थाप्यम् ॥ ३ ॥

V.3. The remainder from the divisor (*hāra*) corresponding to the greater residue (*adhikāgra*), divided by the divisor (*cheda*) corresponding to the lesser residue (*ūnāgra*) — whatever is the quotient from the mutual division is to be placed separately, downwards (*adhaḥ*), one below the other.

[This is the initial step of the Euclidean algorithm applied to the divisors A and B. The quotients are written in a column.]

शेषं तथा इष्ट-गुणितं यथा अग्रयोः अन्तरेण संयुक्तम् ।
शुध्यति गुणकः स्थाप्यः लब्धं च अन्त्यात् उपान्त्य-गुणः ॥ ४ ॥

V.4. The remainder, multiplied by an assumed number (*iṣṭa*) such that, when combined with the difference of the residues (*agra-antara*), it is exactly divisible (*śudhyati*) — the multiplier (*guṇakaḥ*) is to be set down; and the quotient is the product of the last (*antya*) and the penultimate (*upāntya*).

[If r is the final remainder, find m such that $rm+(a-b)$ is divisible. Then the guṇaka and labdha are computed from the column of quotients by back-substitution.]

स्व-ऊर्ध्वस् अन्त्य-युतः अग्र-अन्तः हीन-अग्र-छेद-भाजितः शेषम् ।
अधिक-अग्र-छेद-हतम् अधिक-अग्र-युतम् भवति अग्रम् ॥ ५ ॥

V.5. The upper one (*ūrdhva*), combined with the last (*antya-yutaḥ*), (is divided by) the divisor of the lesser residue (*hīnāgra-cheda*); the remainder, multiplied by the divisor of the greater residue (*adhikāgra-cheda*) and combined with the greater residue, becomes the residue (*agram*).

[The solution x to $ax + c = by + d$ (modulo the divisors) is constructed by this back-substitution. Let x_0 be the guṇaka; then $N = x_0 \cdot B_{large} + r_{large}$.]

छेद-वधस्य द्वि-युगं छेद-वधः युग-गतम् द्वयोः अग्रम् ।
कुट्टाकारेण एवम् त्रि-आदि-ग्रह-युग-गत-आनयनम् ॥ ६ ॥

V.6. Twice the *yuga* (epoch period) is the product of the divisors; the product of the divisors is the *yuga-gata* for both residues. Thus by the *kuṭṭaka*, the derivation of three etc.

भ-गण-आदि-शेषम् अग्रम् छेद-हृतम् खम् च दिन-ज-शेष-हृतम् ।
अनयोः अग्रम् भ-गण-आदि-दिन-ज-शेष-उद्धृतम् द्यु-गणः ॥ ७ ॥

V.7. The residue of the revolutions (*bhagaṇa*) etc. is the *agra*; divided by the divisor; and zero, divided by the residue of days — of these two, the residue of revolutions etc., divided by the day-born residue — that is the day-count (*dyugaṇa*).

दिन-ज-भ-गण-आदि-शेषं येन गुणम् मण्डल-आदि शेषकयोः ।
स-दृश-छेद-उद्धृतयोः तद्-घातम् अहर्-गण-आद्यम् अतः ॥ ८ ॥

V.8. By whatever multiplier the day-born and revolution-etc. residues of the circle etc., divided by their common divisor, are equal — the product of that is the day-count etc.

हृतयोः परस्परम् यत् छेषम् गुण-कार-भाग-हारकयोः ।
तेन हृतौ निस्-छेदौ तौ एव परस्परम् हृतयोः ॥ ९ ॥

V.9. The mutual division of the dividend and divisor, whatever is the remainder — by that, divided, they become reduced (*nis-chedau*); those same, mutually divided.

लब्धम् अधः अधः स्थाप्यं तथा इष्ट-गुण-कार-सङ्गुणं शेषम् ।
शुध्यति यथा एक-हीनम् गुणकः स्थाप्यः फलं च अन्त्यात् ॥ १० ॥

V.10. The quotient is to be placed down, down; the remainder multiplied by the chosen multiplier (*iṣṭa-guṇakāra-saṅguṇaṃ*) — it is exactly divisible so that one is subtracted (*eka-hīnam*). The multiplier is to be set down; and the result is from the last.

अग्र-अन्तम् उपान्त्येन स्व-ऊर्ध्वः गुणितः अन्त्य-संयुतः भक्तम् ।
निस्-शेष-भाग-हारेण एवम् स्थिर-कुट्टकः शेषम् ॥ ११ ॥

V.11. The end-residue (*agrāntam*), by the penultimate multiplied, combined with its upper (value), divided by the divisor without remainder (*niśśeṣa-bhāgahāreṇa*) — thus is the *sthira-kuṭṭakaḥ* (fixed pulverizer), the remainder.

इष्ट-भ-गण-आदि-शेषात् स्व-कुट्टक-गुणात् स्व-भाग-हार-गुणात् ।
शेषम् द्यु-गणः गत-निस्-अपवर्त-गुण-भाग-हार-युतः ॥ १२ ॥

V.12. From the desired revolution-etc. residue, by its own *kuṭṭaka*-multiplier, by its own divisor-multiplied — the remainder is the day-count, combined with the reduced multiplier and divisor.

एवम् समेषु विषमेषु ऋणम् धनम् धनम् ऋणम् यत् उक्तम् तत् ।
ऋण-धनयोः व्यस्तत्वम् गुण्य-प्रक्षेपयोः कार्यम् ॥ १३ ॥

V.13. Thus in equal (signs) and in unequal: debt (*ṛṇa*) and wealth (*dhana*), wealth and debt — whatever was stated, the reversal (*vyastatvam*) of debt and wealth is to be done to the dividend (*guṇya*) and additive (*prakṣepa*).

गुणकः छेदः छेदः गुणकः धनम् ऋणम् ऋणम् धनम् कार्यम् ।
वर्गम् पदम् पदम् कृतिः अन्त्यात् विपरीतम् आद्यम् तत् ॥ १४ ॥

V.14. The multiplier is the divisor; the divisor is the multiplier; wealth is debt, debt is wealth — to be done. Square (*varga*) is root (*pada*); root is square; from the end, reversed (*viparītam*); that is the first.

यः जानाति युग-आदि ग्रह-युग-यातैः पृथक् पृथक् कथितैः ।
द्वि-त्रि-चतुर्-प्रभृतीनाम् कुट्टाकारम् स जानाति ॥ १५ ॥

V.15. He who knows the *kuṭṭaka* by the epoch-periods (*yu-gādi*) of planets, separately stated by two, three, four etc. — **he** truly knows the *kuṭṭaka*.

[Verses 16–29 contain illustrative problems (praśnas) on the pulverizer applied to astronomical computations.]

भ-गण-आद्यम् इष्ट-शेषम् कदा इन्दु-दिवसे रवेः गुरु-दिने वा ।
ज्ञ-दिने राशीन् कथयति कुट्टाकारम् स जानाति ॥ १६ ॥

V.16. “The residue of revolutions etc. at the desired — when on a moon-day, on a sun-day, or a Jupiter-day, on a weekday — tell the signs.” He who tells this knows the *kuṭṭaka*.

ज्ञ-दिने यत् अंश-शेषम् विकलाः-शेषम् कदा तत् इन्दु-दिने ।
भानोः अथ वा शशिनः यः कथयति कुट्टक-ज्ञः सः ॥ १७ ॥

V.17. “On a weekday, whatever is the residue of degrees, the residue of minutes — when that on a moon-day, of the sun, or of the moon — he who tells this is a knower of the *kuṭṭaka*.’

तिथि-मान-दिनेषु इष्टाः ये अर्क-आद्याः ते पुनः कदा तेषु ।
इष्ट-ग्रह-वारेषु यः कथयति कुट्टक-ज्ञः सः ॥ १८ ॥

V.18. “The desired (planetary positions) in the tithi-measure-days, which are sun etc., again when in those desired planet-weekdays” — he who tells this is a knower of the *kuṭṭaka*.

[Verses 19–29 continue with astronomical praśnas: given planetary residues at a particular weekday, finding the elapsed days and other planetary positions. These were standard problems for Indian calendar-makers (pañcāṅga-kāras).]

धनर्णसंकलनम् — □□□□□□□□ □□ □□□□□□□□ □□□ □□□□□□□□ □□□□□□□

धनर्णकर्म

अस्मिन् अंशे ब्रह्मगुप्तस्य ऋण-धन-शून्यानां कर्मसु सूत्राणि दीयन्ते। एते ऐतिहासिक-महत्त्वपूर्णाः सन्ति— सङ्केतित-संख्यानां क्रम-गणितस्य प्रथम-लिखित-सूत्राणि।

Operations on Positive and Negative Numbers

This section contains Brahmagupta's famous rules for arithmetic operations involving positive (*dhana*), negative (*ṛṇa*), and zero (*śūnya*, *kha*, *ākāśa*). These are the **earliest explicit rules** for operations with signed numbers and zero in the history of mathematics.

धनयोः धनम् ऋणम् ऋणयोः धन-ऋणयोः अन्तरम् सम-ऐक्यम् खम् ।
ऋणम् ऐक्यम् च धनम् ऋण-धन-शून्ययोः शून्ययोः शून्यम् ॥ ३० ॥

V.30. The sum of two positives is positive; of two negatives is negative; of a positive and a negative is their difference. Zero is the sum of a positive and an equal negative; and of two zeros.

[In modern notation:]

$$\begin{array}{cc} (+)+(+)=(+) & (-)+(-)=(-) \\ (+)+(-)=\text{their difference} & a+(-a)=0 \\ 0+0=0 & \end{array}$$

ऊनम् अधिकात् विशोध्यम् धनम् धनात् ऋणम् ऋणात् अधिकम् ऊनात् ।
व्यस्तम् तद्-अन्तरम् स्यात् ऋणम् धनम् धनम् ऋणम् भवति ॥ ३१ ॥

V.31. The lesser is to be subtracted from the greater; the positive from the positive, the greater negative from the lesser negative — their difference is reversed (*vyastam*). The negative becomes positive, the positive becomes negative.

[Subtraction rule: $a-b=-(b-a)$ when $b>a$. For signs: $(+a)-(+b)=-(b-a)$ and $(-a)-(-b)=+(b-a)$ when $b>a$.]

शून्य-विहीनम् ऋणम् ऋणम् धनम् धनम् भवति शून्यम् आकाशम् ।
शोध्यम् यदा धनम् ऋणात् ऋणम् धनात् वा तदा क्षेप्यम् ॥ ३२ ॥

V.32. Zero subtracted from debt (*ṛṇa*) is debt; from wealth (*dhana*) is wealth. Zero (*ākāśa*) — when positive is subtracted from negative, or negative from positive, then it is additive (*kṣepya*).

[Sign rules for subtraction with zero:]

$$a-0=a \qquad\qquad 0-a=\text{(sign reversed)}$$

ऋणम् ऋण-धनयोः घातः धनम् ऋणयोः धन-वधः धनम् भवति ।
शून्य-ऋणयोः ख-धनयोः ख-शून्ययोः वा वधः शून्यम् ॥ ३३ ॥

V.33. The product of two negatives is positive; the product of two positives is positive. The product of zero and a negative, of zero and a positive, or of two zeros — the product is zero.

[Multiplication of signed numbers:]

$$\begin{array}{ccc} (-)\times(-)=(+) & (+)\times(+)=(+) & (+)\times(-)=(-) \\ 0\times(-)=0 & 0\times(+)=0 & 0\times 0=0 \end{array}$$

धन-भक्तम् धनम् ऋण-हृतम् ऋणम् धनम् भवति खम् ख-भक्तम् खम् ।
भक्तम् ऋणेन धनम् ऋणम् धनेन हृतम् ऋणम् ऋणम् भवति ॥ ३४ ॥

V.34. Positive divided by positive is positive; negative divided by negative is positive. Zero divided by zero is zero. Positive divided by negative is negative; negative divided by positive is negative.

[Division of signed numbers:]

$$\begin{array}{lll} \frac{(+)}{(+)} = (+) & \frac{(-)}{(-)} = (+) & \frac{0}{0} = 0 \\ \frac{(+)}{(-)} = (-) & \frac{(-)}{(+)} = (-) & \end{array}$$

[Verses 30–34 are historically momentous. Brahmagupta is the first author in world history to state general rules for arithmetic operations with signed numbers and zero. Note that $\frac{0}{0} = 0$ *is the only questionable rule from a modern perspective; Brahmagupta also stated (elsewhere) that division by zero leaves the fraction unchanged, i.e.* $\frac{a}{0} = \frac{a}{0}$ *as an unreduced entity.]*

सङ्क्रमणम् — □□□□□□□□□□□□□

योगस् अन्तर-युत-हीनः द्वि-हृतः सङ्क्रमणम् अन्तर-विभक्तम् वा ।
वर्ग-अन्तरम् अन्तर-युत-हीनम् द्वि-हृतम् विषम-कर्म ॥ ३६ ॥

V.36. The sum, decreased by the difference and divided by two, is the *saṅkramaṇa* (transposition); or divided by the difference. The difference of squares, decreased by the sum of the difference and divided by two — the *viṣama-karma* (unequal operation).

[If $x + y = S$ and $x - y = D$, then:]

$$x = \frac{S + D}{2}, \quad y = \frac{S - D}{2}$$

The "difference of squares" formula: $x^2 - y^2 = S \cdot D$.

[Verses 37–42 treat karaṇī (surds), bhāvitaka (product-form), and algebraic identities for powers of unknowns. These are not fully preserved in the scanned pages of this chunk.]

एकवर्णसमीकरणम् — □□□□□□ □□□□□□□□□ □□ □□□ □□□□□□□□

अव्यक्त-अन्तर-भक्तम् व्यस्तम् रूप-अन्तरम् समे अव्यक्तः ।
वर्ग-अव्यक्ताः शोध्याः यस्मात् रूपाणि तद्-अधस्तात् ॥ ४३ ॥

V.43. Divided by the difference of unknowns (*avyakta*), reversed, the difference of *rūpa*s (absolute terms) — in the equation (*sama*), the unknowns (*avyakta*) are to be subtracted; the *rūpa*s are placed below that.

[For equations $ax + b = cx + d$*, divide by* $(a - c)$ *and rearrange:]*

$$x = \frac{d - b}{a - c}$$

वर्ग-चतुर्-गुणितानाम् रूपाणाम् मध्य-वर्ग-सहितानाम् ।
मूलम् नध्येन ऊनम् वर्ग-द्वि-गुण-उद्धृतम् मध्यः ॥ ४४ ॥

V.44. Of the *rūpa*s multiplied by four and the square of the middle (*madhya*) — the square-root of that, decreased by the middle and divided by twice the multiplier (*varga-dvi-guṇa-uddhṛtam*) — is the *madhya* (unknown).

[This is Brahmagupta's verbal statement of the quadratic formula. If $ax^2 + bx = c$*:]*

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is algebraically equivalent to the modern formula $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ *for the equation* $ax^2 + bx - c = 0$*. Brahmagupta always writes the constant term on the right side, so his formula uses* $+4ac$*.*

वर्ग-आहत-रूपाणाम् अव्यक्त-अर्ध-कृति-संयुतानाम् यत् ।
पदम् अव्यक्त-अर्ध-ऊनम् तत् वर्ग-विभक्तम् अव्यक्तः ॥ ४५ ॥

V.45. Of the *rūpa*s multiplied by the square and combined with the square of half the unknown — whatever is the root, decreased by half the unknown, divided by the square — that is the unknown.

[Alternative form:]

$$x = \frac{\sqrt{a^2 c + (\frac{b}{2})^2} - \frac{b}{2}}{a}$$

[Verses 46–50 contain astronomical problems solved by the kuṭṭaka and quadratic methods.]

स-एकात् अंशक-शेषात् द्वादश-भागः चतुर्-गुणः अष्ट-युतः ।
स-एक-अंश-शेष-तुल्यः यदा तदा अहर्-गणम् कथय ॥ ४६ ॥

V.46. "From the degree-residue (*aṃśaka-śeṣa*) with one added, divided by twelve, multiplied by four and increased by eight, equals the degree-residue with one — when this happens, tell the day-count (*ahargaṇa*)."

[An astronomical problem leading to a linear equation in the ahargaṇa (elapsed days).]

द्वि-ऊनम् अधिक-मास-शेषम् त्रि-हृतम् सप्त-अधिकम् द्वि-सङ्गुणितम् ।
अधिमास-शेष-तुल्यम् यदा तदा युग-गतम् कथय ॥ ४७ ॥

V.47. "The *adhimāsa-śeṣa* (excess month residue), decreased by two, divided by three, increased by seven, multiplied by two — equals the *adhimāsa-śeṣa*. When this happens, tell the *yuga-gata* (elapsed in the *yuga*)."

वि-एकम् अवम-अवशेषम् षष्-उद्धृतम् त्रि-युतम् अवम-शेषस्य ।
पञ्च-विभक्तस्य समम् यदा तदा युग-गतम् कथय ॥ ४८ ॥

V.48. "The *avama-avaśeṣa* (omitted day residue), decreased by one, divided by six, increased by three — equals the *avama-śeṣa* divided by five. When this happens, tell the *yuga-gata*."

अनेकवर्णसमीकरणम् — □□□□□□□□□ □□□□ □□□□□□□□ □□□□□□□□

आद्यात् वर्णात् अन्यान् वर्णान् प्रोह्य आद्य-मानम् आड्य-हृतम् ।
सदृश-छेदौ असकृत् द्वौ व्यस्तौ कुट्टकः बहुषु ॥ ५१ ॥

V.51. From the first unknown (*varṇa*), having subtracted the other unknowns, divided by the first measure (*ādya-māna*), combined — two similar divisors, repeatedly, reversed — the *kuṭṭaka* (is applied) for many (unknowns).

[For multiple equations in many unknowns: eliminate all but one unknown, then apply the kuṭṭaka.]

गत-भगण-युतात् द्यु-गणात् तत्-शेष-युतात् तद्-ऐक्य-संयुक्तात् ।
तद्-योगात् द्यु-गणम् वा यः कथयति कुट्टक-ज्ञः सः ॥ ५२ ॥

V.52. "From the day-count increased by the elapsed revolutions, from that with its residue added, combined with their sum, or from their sum — the day-count" — he who tells this is a *kuṭṭaka*-knower.

गत-भगण-ऊनात् द्यु-गणात् तत्-शेष-ऊनात् तद्-ऐक्य-हीनात् वा ।
तद्-विवरात् द्यु-गणम् वा यः कथयति कुट्टक-ज्ञः सः ॥ ५३ ॥

V.53. "From the day-count decreased by the elapsed revolutions, from that with its residue subtracted, diminished by their sum, or from their difference — the day-count" — he who tells this is a *kuṭṭaka*-knower.

राशि-आद्यैः तत्-शेषैः च एवम् भुक्त-अधिमास-दिन-हीनैः ।
तत्-शेषैः च युग-गतम् यः कथयति कुट्टक-ज्ञः सः ॥ ५४ ॥

V.54. "With the quantities etc., with their residues, thus with the elapsed intercalary months and days subtracted, and with their residues — the *yuga-gata*" — he who tells this is a *kuṭṭaka*-knower.

[Verses 55–59 contain more astronomical applications and computational rules for calendar calculations.]

भावितकम् — □□□ □□□□□□□-□□□□□□ □□□□□□□□

भावितक-रूप-गुणना स-अव्यक्त-वधा इष्ट-भाजिता इष्ट-आप्त्योः ।
अल्पे अधिकः अधिके अल्पः क्षेप्यः भावित-हृतौ व्यस्तम् ॥ ६० ॥

V.60. The *bhāvitaka* (product-nature) multiplied by the *rūpa*, together with the product of the unknown (*avyakta-vadhā*), divided by the desired (*iṣṭa*) — at the desired quotient: if less, add more; if more, add less; *kṣepya* (additive) to the *bhāvita* divisor; reversed.

[The bhāvitaka is the coefficient of the product of two unknowns, i.e. the term Bxy in the general quadratic form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.]

भानोः राशि-अंश-वधात् त्रि-चतुर्-गुणितान् विशोध्या राशि-अंशान् ।
नवतिम् दृष्ट्वा सूर्यम् कुर्वन् आ वत्सरात् गणकः ॥ ६१ ॥

V.61. From the product of the sun's sign and degrees, having subtracted the signs and degrees multiplied by three and four, having seen ninety — making the sun (*sūrya*), the calculator (does this) up to a year.

भावितके यद्-घातः विनष्ट-वर्णेन तत्-प्रमाणानि ।
कृत्वा इष्टानि तद्-आहत-वर्ण-ऐक्यम् भवति रूपाणि ॥ ६२ ॥

V.62. In the *bhāvitaka*, the product by which unknown — having made that amount desired, the sum of the unknown (*varṇa*) multiplied by that becomes the *rūpa*s (known quantities).

वर्ण-प्रमाण-भावित-घातः भवति इष्ट-वर्ण-सङ्ख्या एवम् ।
सिध्यति विना अपि भावित-सम-करणात् किम् कृतम् तत् अतः ॥ ६३ ॥

V.63. The product of the unknown-quantity and the *bhāvitaka* becomes the desired unknown-number. Thus it is accomplished even without the *bhāvitaka* equal-equation; what was done from that?

मूलम् द्विधा इष्ट-वर्गात् गुणक-गुणात् इष्ट-युत-विहीनात् च ।
आद्य-वधः गुणक-गुणः सह अन्त्य-घातेन कृतम् अन्त्यम् ॥ ६४ ॥

V.64. The root is twofold: from the square of the desired, from the multiplier multiplied, increased or decreased by the desired. The first product is the multiplier-product; together with the last product — made last.

[For equations of the form $Nx^2 + k = y^2$ where N is the guṇaka (multiplier) and k is the kṣepa (additive).]

वर्गप्रकृतिः — □□□□'□ □□□□□□□□

वर्गप्रकृतिः

अस्मिन् अंशे वर्गप्रकृतिः—असम-द्वि-घात-समीकरणस्य—विधिः उच्यते। यदा गुणक-संख्यया गुणितस्य वर्गे क्षेपेण युते पदम् □□□□□□□□□□, तदा वर्गप्रकृतिः।

Varga-prakṛtiḥ (Pell's Equation)

This section treats the "square-nature" or Pell's equation: finding integers x, y such that

$$Nx^2 + k = y^2$$

where N (the *guṇaka*) is a non-square positive integer and k (the *kṣepa*) is an integer. Brahmagupta gave the composition formula (*bhāvanā*) and the cyclic method (*cakravāla*), which were rediscovered in Europe only by Euler and Lagrange in the 18th century.

वज्र-वध-ऐक्यम् प्रथमम् प्रक्षेपः क्षेप-वध-तुल्यः ।
प्रक्षेप-शोधक-हृते मूले प्रक्षेपके रूपे ॥ ६५ ॥

V.65. The product of the cross-terms (*vajra-vadha*) plus one is the first *prakṣepa*; equal to the product of the *prakṣepa*s; divided by the *prakṣepa*-subtractor — the roots (*mūle*) are in the *prakṣepa*s, *rūpa*s.

[The "cross multiplication" identity (bhāvanā): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then $(x_1y_2 \pm x_2y_1, Nx_1x_2 \pm y_1y_2)$ solves $Nx^2 + k_1k_2 = y^2$.]

रूप-प्रक्षेप-पदे पृथक् इष्ट-क्षेप्य-शोध्य-मूलाभ्याम् ।
कृत्वा अन्त्य-आद्य-पदे ये प्रक्षेपे शोधने वा इष्टे ॥ ६६ ॥

V.66. The *rūpa-prakṣepa* roots, separately, by the desired *kṣepya* and *śodhya*-roots — having done, the end and first roots which are in the *prakṣepa* — in subtraction or desired.

चतुर्-अधिके अन्त्य-पद-कृतिः त्रि-ऊना दलिता अन्त्य-पद-गुणा अन्त्य-पदम् ।
अन्त्य-पद-कृतिः वि-एका द्वि-हृता आद्य-पद-आहता आद्य-पदम् ॥ ६७ ॥

V.67. When increased by four (*catur-adhike*): the square of the last root (*antya-pada-kṛtiḥ*), decreased by three (*tri-ūnā*), halved, multiplied by the last root — is the last root. The square of the last root, decreased by one (*vi-ekā*), divided by two, multiplied by the first root — is the first root.

[Recurrence relations for generating new solutions to $Nx^2 + 1 = y^2$ from known solutions. If (x_1, y_1) is a known solution with $k = -1$, then:]

$$y_{n+1} = \frac{(y_n^2 - 3)y_n}{2}$$
$$x_{n+1} = \frac{(y_n^2 - 1)x_n}{2}$$

चतुर्-ऊने अन्त्य-पद-कृती त्रि-एक-युते वध-दलम् पृथक् वि-एकम् ।
वि-एक-आड्य-आहतम् अन्त्यम् पद-वध-गुणम् आद्यम् आन्त्य-पदम् ॥ ६८ ॥

V.68. When decreased by four: the square of the last root, increased by three and one, the product halved, separately, decreased by one — increased by one, decreased by one, multiplied by the last — the product of the roots (is) the first, the last root.

वर्गे गुणके क्षेपः केन चित् उद्धृत-युत-ऊनितः दलितः ।
प्रथमः अन्त्य-मूलम् अन्यः गुण-कार-पद-उद्धृतः प्रथमः ॥ ६९ ॥

V.69. In the square of the multiplier (*guṇaka*), the *kṣepa*, by some (number) divided, added, subtracted, halved — the first (root). The last root — the other, divided by the multiplier-root, (gives) the first.

[Given $Nx^2 + k = y^2$, divide by k to reduce the additive. If k divides N or a related quantity, new roots can be found.]

वर्ग-चिन्ने गुणके प्रथमम् तद्-मूल-भाजितम् भवति ।
वर्ग-चिन्ने क्षेपे तत्-पद-गुणिते तदा मूले ॥ ७० ॥

V.70. When the multiplier is a perfect square (*varga-cinne guṇake*): the first (root), divided by its root, becomes (the new root). When the *kṣepa* is a perfect square: multiplied by its root, then (are) the roots.

[If $N = a^2$ is a perfect square, then $a^2x^2 + k = y^2$ gives $y = \sqrt{a^2x^2 + k}$. If also $k = b^2$, then $\sqrt{(ax)^2 + b^2}$ may simplify.]

गुणक-युतिः अष्ट-गुणिता गुणक-अन्तर-वर्ग-भाजिता राशिः ।
गुणकौ त्रि-गुणौ व्यस्त-अधिकौ हृतौ अन्तरेण पदे ॥ ७१ ॥

V.71. The sum of the multipliers, multiplied by eight, divided by the square of the difference of the multipliers — the quantity. The two multipliers, tripled, reversed, increased, divided by the difference — (are) the roots.

वर्गः अन्य-कृति-युत-ऊनः तत्-संयोग-अन्तर-अर्ध-कृति-भक्तः ।
तद्-गुणितौ युति-वियुतौ वर्गौ घाते च रूप-युते ॥ ७२ ॥

V.72. The square, decreased by the sum of the other squares, divided by the half-square of the sum-difference — those multiplied by them, combined and separated — squares; in the product, combined with the *rūpa*.

[The general composition (bhāvanā) formula:]

$$\begin{gathered}\text{If } Nx_1^2 + k_1 = y_1^2 \text{ and } Nx_2^2 + k_2 = y_2^2,\\ \text{then } N(x_1y_2 \pm x_2y_1)^2 + k_1k_2 = (Nx_1x_2 \pm y_1y_2)^2\end{gathered}$$

When $k_1 = k_2 = 1$, this generates infinitely many solutions to $Nx^2 + 1 = y^2$ from a single fundamental solution. This is Brahmagupta's great discovery, called the samāsa-bhāvanā.

खगोलप्रयोगाः — □□□□□□□□□□□□ □□□□□□□□□□□□

ग्रह-गणित-प्रयोगाः

अस्मिन् अंशे दिव्य-सूत्राणि दीयन्ते—यतः कुट्टक-ज्ञः ग्रहाणाम् अवशेषज्ञानेन अन्येषाम् ग्रह-स्थानानि कथयितुम् शक्नोति। एते सूत्राणि पञ्चाङ्ग-कर्तृभ्यः परिशुद्धि-ज्ञानाय आसन्।

Astronomical Divination Rules

This section gives remarkable "divination" rules: given the planetary residues at a particular day, one can compute the positions of other planets. These were used by calendar-makers (*pañcāṅga-kāras*) to verify their computations.

राशि-कला-शेष-कृतिम् द्विनवति-गुणिताम् त्र्यशीति-गुणिताम् वा ।
स-एका ज्ञ-दिने वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ७५ ॥

V.75. "The square of the residue of signs and minutes, multiplied by twenty-two (*dvinavati*), or multiplied by eighty-three (*tryaśīti*), with one added — on a weekday, making the square — the calculator (does this) up to a year."

सूर्य-विलिप्ता-शेषम् पञ्चभिः ऊन-आहतम् तथा दशभिः ।
वर्गे बृहस्पति-दिने कुर्वन् आ वत्सरात् गणकः ॥ ७६ ॥

V.76. "The residue of the sun's seconds (*viluptā-śeṣa*), decreased and multiplied by five, likewise by ten — on a Jupiter-day, making the square — the calculator (does this) up to a year."

भ-गण-आदि-शेष-वर्गम् त्रिभिः गुणम् संयुतम् शतैः नवभिः ।
कृतिम् अष्ट-शत-ऊनम् वा कुर्वन् आ वत्सरात् गणकः ॥ ७७ ॥

V.77. "The square of the residue of revolutions etc., multiplied by three, combined with nine hundred — or the square decreased by eight hundred — the calculator (does this) up to a year."

भ-गण-आदि-शेष-वर्गम् चतुर्-गुणम् पञ्चषष्टि-संयुक्तम् ।
षष्टि-ऊनम् वा वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ७८ ॥

V.78. "The square of the residue of revolutions etc., multiplied by four, combined with sixty-five — or the square decreased by sixty — the calculator (does this) up to a year."

इष्ट-भगण-आदि-शेषम् द्विनवति-ऊनम् त्र्यशीति-सङ्गुणितम् ।
रूपेण युतम् वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ७९ ॥

V.79. "The desired residue of revolutions etc., decreased by twenty-two, multiplied by eighty-three, combined with one — making the square — the calculator (does this) up to a year."

अधिमास-शेष-वर्गम् त्रयोदश-गुणम् त्रिभिः शतैः युक्तम् ।
त्रि-घन-ऊनम् वा वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ८० ॥

V.80. "The square of the *adhimāsa-śeṣa* (intercalary month residue), multiplied by thirteen, combined with three hundred — or the square decreased by twenty-seven (*tri-ghana*) — the calculator (does this) up to a year."

इन्दु-विलिप्ता-शेषम् सप्तदश-गुणम् त्रयोदश-गुणम् च अपि ।
पृथक् एक-युतम् वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ८१ ॥

V.81. "The residue of the moon's seconds, multiplied by seventeen, also multiplied by thirteen — separately, combined with one — making the square — the calculator (does this) up to a year."

अवम-अवशेष-वर्गम् द्वादश-गुणितम् शतेन संयुक्तम् ।
त□□□□□□ ऊनम् वा वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ८२ ॥

V.82. "The square of the *avama-avaśeṣa* (omitted-day residue), multiplied by twelve, combined with one hundred — or the square decreased by three — the calculator (does this) up to a year.”

ज्ञ-दिने अर्क-कला-शेषम् गुरु-दिन-विकला-अवशेष-युक्त-ऊनम् ।
वर्गम् वधम् च स-एकम् कुर्वन् आ वत्सरात् गणकः ॥ ८३ ॥

V.83. “On a weekday, the residue of the sun’s minutes (*kala*), combined with or decreased by the residue of Jupiter-day seconds — the square and product with one — the calculator (does this) up to a year.”

विकला-शेषम् सहितम् त्रिनवत्या सप्तषष्टि-हीनम् च ।
भानोः ज्ञ-दिने वर्गम् कुर्वन् आ वत्सरात् गणकः ॥ ८४ ॥

V.84. “The residue of seconds (*vikalā-śeṣa*), combined with ninety-three (*trinavatī*) and decreased by seventy-seven (*saptaṣaṣṭi*) — on the sun’s weekday, making the square — the calculator (does this) up to a year.”

ज्ञ-दिने अर्क-कला-शेषम् द्वादशभिः संयुतम् त्रिषष्ट्या च ।
षष्ट्या अष्टभिः च ऊनम् कुर्वन् आ वत्सरात् गणकः ॥ ८५ ॥

V.85. “On a weekday, the residue of the sun’s minutes, combined with twelve and sixty-three — or decreased by sixty and eight — the calculator (does this) up to a year.”

इन्दु-विलिप्ता-शेषात् रवि-लिप्ता-शेषम् अंश-शेषम् वा ।
अथ वा मध्यम् इष्टम् कुर्वन् आ वत्सरात् गणकः ॥ ८६ ॥

V.86. “From the residue of the moon’s seconds, the residue of the sun’s seconds, or the residue of degrees — or alternatively the mean (position) desired — the calculator (does this) up to a year.”

जीव-विलिप्ता-शेषात् कुजम् इन्दुम् भौम-लिप्तिका-शेषात् ।
रविम् इन्दु-भाग-शेषात् कुर्वन् आ वत्सरात् गणकः ॥ ८७ ॥

V.87. “From the residue of seconds of Venus (*jīva*), Mars (*kuja*) and the Moon; from the residue of Mars’s seconds (*bhauma-liptikā*); from the residue of the moon’s degrees — the Sun — the calculator (does this) up to a year.”

[Verses 75–87 give a remarkable set of “divination” rules. Given a particular planetary residue at a known weekday, these formulas allow one to compute other planetary positions. The constants (22, 83, 5, 10, etc.) encode specific planetary period relations. These verification rules were essential for ensuring the accuracy of pañcāṅga (calendar) computations.]

[Verses 88–99 treat the kuṭṭaka method for solving multiple planetary congruences and refined calendar computations.]

हृदि-घात्रम् अमी प्रश्नाः प्रश्नान् अन्यान् सहस्र-शः कुर्यात् ।
अन्यैः दत्तान् प्रश्नान् उक्त्या एवम् साधयेत् करणैः ॥ १०० ॥

V.100. These problems strike the heart (with joy). One may pose thousands of other problems; given by others — thus one should solve those problems by the stated methods (*karaṇaiḥ*).

जन-संसदि दैव-विदाम् तेजः नाशयति भानुः इव भानाम् ।
कुट्टाकार-प्रश्नैः पठितैः अपि किम् पुनः शत-शः ॥ १०१ ॥

V.101. In the assembly of people, among the knowers of fate (*daiva-vidām*), it destroys the radiance (of rivals), like the sun (destroys) the stars — (even) one *kuṭṭaka* problem recited — what to speak of hundreds!

प्रति-सूत्रम् अमी प्रश्नाः पठिताः स-उद्देशकेषु सूत्रेषु ।
आर्या-त्रि-अधिक-शतेन च कुट्टः च अष्टादशः अध्यायः ॥ १०२ ॥

V.102. These problems have been stated for each rule, in the illustrative rules (*sa-uddeśakeṣu sūtreṣu*). By three hundred and three (303) *āryā* (verses), the *kuṭṭaka* and the eighteenth chapter.

इति श्री-ब्राह्मस्फुटसिद्धान्ते कुट्टकाध्यायो अष्टादशः

[Thus ends the Eighteenth Chapter, the Kuṭṭaka, in the ***Brāhmasphuṭasiddhānta****.]*

अथ शङ्कु-छाया-विज्ञानम् — एकोनविंशः

शङ्कु-छाया-विज्ञानम्

अस्मिन् अध्याये शङ्कोः छायायाः च मापन-विधिः उच्यते। अनेन अक्षांशः, गृह-औच्च्यम्, जल-मापनम्, दर्पण-प्रतिबिम्बेन ऊर्ध्वता-निर्धारणम् च सिध्यति।

Chapter XIX: On Gnomon and Shadow Knowledge

This chapter contains 20 verses on gnomon (*śaṅku*) and shadow (*chāyā*) computations. These techniques allow the determination of latitude, building heights, water depth, and altitude by mirror reflection.

[Printed pages 284–293.]

दृष्ट्वा दिन-अर्ध-घटिका यः अर्क-ज्ञः अक्ष-अंशकान् विजानाति ।
उदय-अन्तर-घटिकाभिः ज्ञातात् ज्ञेयम् सः तन्त्र-ज्ञः ॥ १ ॥

V.1. Having seen the half-day ghaṭikās, he who knows the sun-knowing (method), knows the degrees of latitude (*akṣa-aṃśakān*). By the ghaṭikās of the interval from rising — from the known, the knowable — he is a knower of the *tantra.*

अस्त-अन्तर-घटिकाभिः यः ज्ञातात् ज्ञेयम् आनयति तस्मात् ।
मध्य-गतिम् युग-भ-गणान् आनयति ततः सः तन्त्र-ज्ञः ॥ २ ॥

V.2. By the ghaṭikās of the interval from setting, he who brings the knowable from the known, thence brings the mean motion and the *yuga*-revolutions — he is a knower of the *tantra.*

आनयति यः तमस्-रवि-शश-अङ्क-मानानि दीपक-शिख-औच्च्यात् ।
शङ्कु-तल-अन्तर-भूमि-ज्ञाने छायाम् सः तन्त्र-ज्ञः ॥ ३ ॥

V.3. He who brings the measures of the darkness-sun, the moon-horn, from the height of the lamp-flame — in the knowledge of the ground between gnomon-base (and object), the shadow — he is a knower of the *tantra.*

इष्ट-गृह-औच्च्य-ज्ञः यः तत्-अन्तर-ज्ञः निरीक्ष्यते तु जले ।
गृह-भित्ति-अग्रम् दर्शयति दर्पणे वा सः तन्त्र-ज्ञः ॥ ४ ॥

V.4. He who knows the desired house-height, knows that interval, is observed in water — shows the top of the house-wall, or in a mirror — he is a knower of the *tantra.*

छाया-द्वितीया-भा-अग्र-अन्तर-विज्ञानेन वेत्ति दीप-औच्च्यम् ।
शङ्कु-छाया-ज्ञः वा भूमेः छायाम् सः तन्त्र-ज्ञः ॥ ५ ॥

V.5. By the knowledge of the second-shadow-light-edge interval, he knows the lamp-height — or a knower of gnomon and shadow knows the earth's shadow — he is a knower of the *tantra.*

दृष्ट्वा गृह-तल-अन्तर-जालभोः दृष्ट्वा अग्रम् गृहस्य भूमि-ज्ञः ।
वेत्ति गृह-औच्च्यम् दृष्ट्वा तैल-स्थम् वा सः तन्त्र-ज्ञः ॥ ६ ॥

V.6. Having seen the house-base-interval of the net of light and shadow, having seen the top of the house — a knower of the ground — he knows the house-height, having seen it in oil or water — he is a knower of the *tantra.*

वीक्ष्य गृह-अग्रम् सलिले प्रसार्य सलिलम् पुनः स्व-भू-ज्ञाने ।
आनयति जलात् भूमिम् गृहस्य वा औच्च्यम् सः तन्त्र-ज्ञः ॥ ७ ॥

V.7. Having observed the top of the house in water, having spread out the water again, in self-ground-knowledge — he brings from the water the ground or the height of the house — he is a knower of the *tantra.*

ज्ञातैः छाया-पुरुषैः विज्ञाते तोय-कुड्ययोः विवरे ।

कुड्ये अर्क-तेजसः यः वेत्ति आरूढिम् सः तन्त्र-ज्ञः ॥ ८ ॥

V.8. With known shadow-men (measures), the interval of water and wall having been known — in the wall, he who knows the ascent of the sun's light (ray) — he is a knower of the *tantra*.

इष्ट-दिवस-अर्ध-घटिका घटिका-पञ्चदश-अन्तर-प्राणाः ।
तद्-दिवस-चर-प्राणैः तैः अक्षम् साधयेत् प्राक्-वत् ॥ ९ ॥

V.9. The half-day ghaṭikās of the desired day, the prāṇas of the interval of fifteen ghaṭikās — by the day's motion prāṇas, by those he should compute the latitude as before.

ज्ञात-ज्ञेय-ग्रहयोः उदय-अन्तर-नाडिकाभिः अधिक-ऊनः ।
उदयैः ज्ञातः ज्ञातात् ज्ञेयः प्राक्-अपरयोः ज्ञेयः ॥ १० ॥

V.10. Of the known and knowable planets, by the nāḍikās of the interval from rising — greater or less — by the risings, the known from the known, the knowable on the east-west (is) to be known.

ज्ञातः स-भ-अर्धः उदयैः अस्त-अन्तर-नाडिकाभिः अधिक-ऊनः ।
ज्ञातात् पूर्व-अपरयोः ज्ञेयः भ-अर्ध-ऊनके ज्ञेयः ॥ ११ ॥

V.11. The known, with its half-duration, by the risings — greater or less by the nāḍikās of the interval from setting — from the known, on the east-west the knowable; when the half-duration is decreased, the knowable.

ज्ञातम् कृत्वा मध्यम् भूयः अन्य-दिने तत्-अन्तरम् भुक्तिः ।
त्रैराशिकेन भुक्त्या कल्प-ग्रह-मण्डल-आनयनम् ॥ १२ ॥

V.12. Having made the known into the mean, again on another day, that interval is the motion (*bhuktiḥ*). By the rule of three (*trairāśikena*), by the motion — the derivation of the planet's circle in the *kalpa*.

स्थिति-अर्धात् विपरीतम् तमस्-प्रमाणम् स्फुटम् ग्रहणे ।
मान-उदयात् रवि-इन्द्वोः घटिका-अवयवेन भ-उदय-तः ॥ १३ ॥

V.13. The reverse of half the duration (*sthiti*) is the true darkness-measure in an eclipse. From the measure-rise of the sun and moon, by the ghaṭikā-part, from the terrestrial rising.

दीप-तल-शङ्कु-तलयोः अन्तरम् इष्ट-प्रमाण-शङ्कु-गुणम् ।
दीप-शिखा-औच्च्यात् शङ्कुम् विशोध्या शेष-उद्धृतम् छाया ॥ १४ ॥

V.14. The interval of lamp-base and gnomon-base, multiplied by the desired-quantity-gnomon, divided by the lamp-flame-height minus the gnomon — the quotient is the shadow (*chāyā*).

[Similar triangles: If h = lamp height, g = gnomon height, d = distance between them, then shadow $s = \frac{d\cdot g}{h-g}$.]

शङ्कु-अन्तरेण गुणिता छाया छाया-अन्तरेण भक्ता भूः ।
स-छाया शङ्कु-गुणा दीप-औच्च्यम् छायया भक्ता ॥ १५ ॥

V.15. The shadow, multiplied by the gnomon-interval, divided by the shadow-interval — (is) the ground (distance). The shadow together with the gnomon, multiplied by the gnomon, divided by the shadow — (is) the lamp-height.

ज्ञात्वा शङ्कु-छायाम् अनुपातात् साधयेत् समुच्छ्रयान् ।
गृह-चैत्य-तरु-नगानाम् औच्च्यम् विज्ञाय वा छायाम् ॥ १६ ॥

V.16. Having known the gnomon and shadow, from proportion (*anupātāt*) one should compute the elevations (*samucchrayān*) — of houses, temples, trees, and mountains — or having known the height, the shadow.

[This is the standard "rule of three" for similar triangles: $\frac{\text{height}}{\text{shadow}} = \frac{\text{known height}}{\text{known shadow}}$, so unknown quantities can be found by proportion.]

[Verses 17–19 give additional rules on shadow computations and heights/depths using water and mirror reflec-

tions. Verse 20 gives the chapter colophon.]

इति शङ्कु-छाया-विज्ञानम् — एकोनविंशोऽध्यायः

[Thus ends the Nineteenth Chapter on Gnomon and Shadow Knowledge.]

अथ छन्दश्-चित्त्युत्तराध्यायः — विंशतितमः

छन्दश्-चित्त्युत्तराध्यायः

अस्मिन् अध्याये छन्दः-शास्त्रस्य गणितम् उच्यते—गुरु-लघु- वर्णानाम् व्यवस्थापनानि, प्रस्तारः, नष्टम्, उद्दिष्टम् च। अस्य विषयः संयुक्त-गणितस्य पूर्व-रूपम् अस्ति।

Chapter XX: On Prosody (Chandaś-citty-uttara)

This chapter contains 19 verses on Sanskrit prosody (*chandah-śāstra*), dealing with combinatorial calculations for metrical patterns — the number of possible arrangements of long (*guru*, ‘G’) and short (*laghu*, ‘L’) syllables. Brahmagupta here presents an early form of combinatorics.

[Printed pages 294–303.]

ऋग्वर्गः पर्यायः समूहयोगावयुक्षु युग्मेषु ।
सोयाः प्राग्वत्प्राप्तादाश्चतुष्ककाः शेषयुक्त्योन्त्यः ॥ १ ॥

V.1. The *ṛg*-group (combinations), the permutations (*paryāya*), the aggregate-sums, in the unequal pairs — the *soma* etc., obtained as before, the quadruples, the remainder-additions — the last.

एकादियुतविहीनावाद्यन्तौ तद्विपर्ययौ यावत् ।
वर्गादिषु विषमयुजां क्रमोत्क्रमाद्वर्धयेत् पादान् ॥ २ ॥

V.2. Decreased by one etc., the first and last, their reversals — in squares etc., of the odd-even (syllables), one should increase the quarters by direct and reverse order.

एकैकेन द्व्याद्व्याः सोप्पप्पधिकेषु तत्-प्रतिष्ठेषु ।
वर्गादिरभीष्टान्तः प्रस्तारो भवति यवमध्यः ॥ ३ ॥

V.3. One by one, two by two, with additional appendages in those positions — beginning from the square up to the desired — the spread-out (*prastāra*) becomes with barley-middle (*yava-madhya*, i.e. Pascal’s triangle).

[The “Meru of Prosody” (prastāra) is Pascal’s triangle: the number of metrical patterns with n syllables and k guru syllables is $\binom{n}{k}$. The “barley-middle” refers to the shape of the triangular arrangement.]

सूनोन्त्यो द्विपदाग्रं त्रिपदाद्यानामधः पृथक् संख्या ।
तच्छोध्यो व्येकः पृथग्न्ताद्रूपमूर्ध्वयुतम् ॥ ४ ॥

V.4. With one, the last — the two-foot beginning, the three-foot etc., below, separately the numbers. That is to be subtracted, decreased by one, separately from that — combined with one above.

यावत् पादाव्येकागच्छाद्वर्णेष्वथैकवृद्धेषु ।
रूपाद्युतघाते वर्गाद्यानां परा संख्या ॥ ५ ॥

V.5. Up to the quarter, decreasing by one, among the syllables, then increasing by one — in the product beginning from one, the final number of squares etc.

[For n syllables, each of which can be G or L, the total number of possible patterns is 2^n.]

रूपाधिकपादार्धेविषमेषूर्ध्वः समेषु पादार्धे ।
अर्धाद्द्विगुणाव्येकांयुलान्यधस्तस्य सर्वेषाम् ॥ ६ ॥

V.6. With one added to the quarter-half — in the uneven (syllables) above, in the even (syllables) at the quarter-half. From the half, doubled, decreased by one — the *yulāni* (pairings) below that, for all.

माध्यैस् तथार्धहीनैः क्रमपादैर् व्यस्ततुल्यपादाद्यः ।
विषमेरव्येकं मध्ये प्रोह्याद्यान्यतः कुर्यात् ॥ ७ ॥

V.7. By the middle (terms), thus decreased by half, by the quarter-steps in order — equal to the reversed, beginning from the quarter — in the uneven, subtracting one in the middle, from that one should do (the computation).

सैकक्रमतुल्याद्यैर् न्यासोऽभ्यधिको विशोधितश् चाधः ।
संख्यैक्यं तादृक् यादृक् प्रथमस् त्रिरहितो नष्टे ॥ ८ ॥

V.8. With one-plus-equal-to-the-step-beginnings, the setting is excessively subtracted below — the sum of numbers such as it is — the first, decreased by three — in the lost (*naṣṭa*, i.e. finding a missing pattern).

माध्यैः कृतैश् च दलितैः समसंख्यायां क्रमोत्क्रमात् क्षेप्पम् ।
विषमायां व्येकायां दलम् क्रमाद् उत्क्रमात् सैकम् ॥ ९ ॥

V.9. Made with middle terms and halved — in the equal-number (case), in direct and reverse order — the *kṣepa* (additive). In the uneven, decreased by one — the half, in order, in reverse order, with one added.

समसंख्यायां सोपानक्रमोत्क्रमाभ्यां तथैव विषमाभ्याम् ।
कल्प्या पचिते दृष्टे प्रथमः शेषाक्षराण्यन्ते ॥ १० ॥

V.10. In the equal-number (case), by the staircase in direct and reverse order — thus also in the uneven — to be constructed. When cooked (computed) is seen, the first — the remaining syllables at the end.

[Verses 11–18 continue the treatment of combinatorial prosody, including algorithms for the "Meru of Prosody" (Pascal's triangle) and rules for lost (naṣṭa) and spread-out (uddhiṣṭa) metrical patterns. The "staircase" (sopāna) is a procedure for listing all possible metrical patterns in systematic order.]

इति श्री-ब्राह्मस्फुटसिद्धान्ते छन्दश्-चित्त्युत्तराध्यायो विंशतितमः

*[Thus ends the Twentieth Chapter on Prosody in the **Brāhmasphuṭasiddhānta**.]*

अथ गोलाध्यायः — एकविंशतितमः

गोलाध्यायः — ज्याप्रकरणम्

अस्मिन् अंशे ज्या-खण्डानाम् उत्पत्तिः उच्यते। ब्रह्मगुप्तस्य कल्पना द्वितीय-कोटि-परिकल्पनया (□□□□□□-□□□□□ □□□□□□□□□□□□□) ज्या-सारणीम् अधिक-सूक्ष्मां करोति।

Chapter XXI: The Sphere — Sine Computation

This chunk includes the Jyā-prakaraṇa (verses 17–23), which gives Brahmagupta's method for computing the table of sines. The radius is $R = 3270$ minutes. The "segments" (*jyā-khaṇḍa*) are the tabular sine-differences. Brahmagupta uses a second-order interpolation formula, more accurate than the linear interpolation used by earlier authors.

[Printed pages 304–315 (beginning). Chapter 21 contains 70 verses. This chunk includes the Jyā-prakaraṇa section (verses 17–23).]

राशि-अष्ट-अंशेषु अङ्कान् पद-सन्धिभ्यस् क्रम-उत्क्रमात् कृत्वा ।
बध्नीयात् सूत्राणि द्वयोः द्वयोः ज्यास् तद्-अर्धानि ॥ १७ ॥

V.17. Having placed marks at the eighth parts of a sign ($3°45'$ intervals) in direct and reverse order, one should construct chords (*jyā*) for every two-by-two: their halves are the sine-chords (*jyā-ardhāni*).

[Brahmagupta divides each sign (30°) into 8 parts of 3°45' each, giving 24 intervals in a quadrant (0° to 90°). The jyā-ardha (half-chord) is the modern sine: $R \sin \theta$*.]*

ज्या-अर्धानि ज्या-अर्धानाम् ज्या-खण्डानि अन्तराणि ।
व्यस्तानि अन्त्यात् अथ वा इषुः उत्क्रम-ज्या धनुस् ताभ्याम् ॥ १८ ॥

V.18. The sine-half-differences of sine-halves are the sine-segments (*jyā-khaṇḍa*). Reversed from the last, or: the bow (*dhanus*) is the reverse-sine (*utkrama-jyā*) together with these.

[The jyā-khaṇḍa are the tabular differences: $\Delta_i = R \sin(i+1)\alpha - R \sin i\alpha$ *where* $\alpha = 3°45'$*.]*

एक-द्वि-त्रि-गुणायाः व्यास-अर्ध-कृतेः पृथक् चतुर्थेभ्यः ।
मूलानि अष्ट-द्वादश-षोडश-खण्डानि अतः अन्यानि ॥ १९ ॥

V.19. From the quarter-parts of the half-diameter squared, multiplied by one, two, and three: the roots are the eighth, twelfth, and sixteenth segments; the others (are derived) from these.

[Initial segments computed from $R^2/4$*:]*

$$j_8 = \sqrt{R^2 \cdot 1/4} = R/2 = R \sin(30°)$$
$$j_{12} = \sqrt{R^2 \cdot 2/4} = R/\sqrt{2} = R \sin(45°)$$
$$j_{16} = \sqrt{R^2 \cdot 3/4} = R\sqrt{3}/2 = R \sin(60°)$$

With $R = 3270$*:* $j_8 = 1635$, $j_{12} = 2312$, $j_{16} = 2832$.

तुल्य-क्रम-उत्क्रम-ज्या-सम-खण्डक-वर्ग-युति-चतुर्-भागम् ।
प्रोह्य अनष्टम् व्यास-अर्ध-वर्गतः तत्-पदे प्रथमम् ॥ २० ॥

V.20. Having subtracted the fourth part of the sum of squares of equal direct and reverse sines and equal segments from the square of the half-diameter, the square-root of the remainder is the first (sine-chord).

[If j_k *and* j_{n-k} *are known sines, and* δ *is the common segment-difference, the next sine is:]*

$$j_{k+1} = \sqrt{R^2 - \frac{(j_k + j_{n-k})^2 + (2\delta)^2}{4}}$$

तद्-दल-खण्डानि तद्-ऊन-जिन-समानि द्वितीयम् उत्पत्तौ ।
संख्या-वाचकैः अक्षरैः संकेतितानि
॥ २१ ॥

V.21. Its half-segments, decreased by those equal to the earth (*jina*) — the second in production.

[This verse contains numerical values encoded as words (saṅketa system). “Jina” = 1 (Lord of the earth). The second-order correction refines the first-order segments.]

एवम् जीवा-खण्डानि अल्पानि बहूनि वा आद्य-खण्डानि ।
ज्या-अर्धानि वृत्त-परिधेः षष्ठ-चतुर्थ-त्रि-भागानाम् ॥ २२ ॥

V.22. Thus the small or numerous sine-segments — the first segments are the sine-halves of one-sixth, one-fourth, and one-third of the circle’s circumference.

[The reference sines at key angles:]

$$R\sin(30^\circ) = R/2 = 1635 \qquad R\sin(45^\circ) = R/\sqrt{2} \approx 2312$$

$$R\sin(60^\circ) = R\sqrt{3}/2 \approx 2832$$

उत्क्रम-सम-खण्ड-गुणात् व्यासात् अथ वा चतुर्थ-भागात् यत् ।
कृत्वा उक्त-खण्डकानि ज्या-अर्ध-आनयनम् न लघु अस्मात् ॥ २३ ॥

V.23. From the diameter multiplied by the equal reverse-segment, or from the quarter-part, having done the stated segment operations — the derivation of sine-halves is not difficult from this.

[Summary of the method: Given the tabular differences (khaṇḍa), one can compute all sines by second-order interpolation using only multiplication and square roots.]

*[*0.5cm][Brahmagupta’s sine table uses $R = 3270$ minutes. The twenty-four *jyā-khaṇḍa* (sine-differences) are: 225, 224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 154, 143, 131, 119, 106, 93, 79, 65, 51, 37, 22, 7. The second-order correction formula accounts for the curvature of the sine function, yielding values accurate to within 1 minute of arc.]

अथ यन्त्राध्यायः — द्वाविंशतितमः

यन्त्राध्यायः

अस्मिन् अध्याये खगोल-दर्शनार्थं विविधानि यन्त्राणि वर्ण्यन्ते—धनुः-यन्त्रम्, यष्टि-यन्त्रम्, जल-यन्त्रम्, चक्र-यन्त्रम्, शङ्कु-यन्त्रम् च।

Chapter XXII: On Astronomical Instruments

This chapter contains 43–57 verses on astronomical instruments (*yantra*). It describes the bow instrument (*dhanur-yantra*), staff instrument (*yaṣṭi-yantra*), water instruments, wheel instruments, and gnomon instruments. These were used for measuring celestial altitudes, azimuths, and for timekeeping.

[Printed pages ~289–315.]

[प्रारम्भिक-श्लोकाः विविध-यन्त्र-वर्णनानि]

अस्मिन् अध्याये उक्तानि यन्त्राणि:
□ धनुर्म्यन्त्रम् — □□□ □□□ □□□□□□□□□□
□ यष्टियन्त्रम् — □□□ □□□□□ □□□□□□□□□□
□ जलयन्त्रम् — □□□□□ □□□□□□□□□□□
□ शङ्कुयन्त्रम् — □□□□□□ □□□□□□□□□□□
□ चक्रयन्त्रम् — □□□□□/□□□□□□ □□□□□□□□□□□

Instruments Described

- **Dhanur-yantra** — A semicircular or arc-shaped instrument for measuring altitudes, resembling a bow.
- **Yaṣṭi-yantra** — A staff or rod instrument for measuring angles and time.
- **Jala-yantra** — Water-based instruments including water clocks and mercury devices.
- **Śaṅku-yantra** — Gnomon instruments for shadow measurements.
- **Cakra-yantra** — Circular instruments with rotating sighting vanes for measuring celestial positions.

[Verses 1–28 describe the bow instrument, staff instrument, and their use in measuring celestial altitudes and azimuths. Verses 29–40 treat water instruments and circle instruments.]

क्षेप्तं स्वर्चिषा क्षेप्यं समं तथा पारदे यदा भवति ।
कालसम्मिश्रमानं शककृतान्मुद्दे वा ॥ ४१ ॥

V.41. The *kṣepṭa* (object to be observed), by its own light, the *kṣepya* (object aimed at) — equal, thus in mercury (*pārada*), when it becomes — the mixed-time measure from the *śaka* epoch (78 CE) or from the *mudda* (particular epoch).

कोतस्योपरिगामिनं त्रियक् सूत्रके धृतमल्पेषु ।
प्रावनलके प्रक्षिप्य नाडिका श्रवति पतत्ये ॥ ४२ ॥

V.42. The co-altitude (*kota*) going above, held obliquely in the string (*sūtraka*), in small (measures) — having cast upon the eastern fire-place (*prāv-analaka*), the nāḍikā (water-clock) flows, falls thus.

[Verses 43–48 contain further rules on the use of instruments for time measurement and planetary observation.]

लघुदारमयं चक्रं समधुरीरान्तरं पुष्करारणाम् ।
श्रद्धा पारदपूर्णं मध्ये सुलिप्तं कृतसंध्यः ॥ ४९ ॥

V.49. A wheel of light-wood, with equal bearing intervals, of lotus-shape — having faith, filled with mercury in the center, well-polished, having performed the saṃdhyā (preparation).

तियक् कोतिमध्येध्वाधारस्थो रूपारदं भ्रजति ।
छिद्रायुक् चक्रकमजस्रमेवं वाम्बरे भ्रमति ॥ ५० ॥

V.50. Obliquely in the middle of the co-altitude, the support above — a beautiful semi-circle (*rūpā-radaṃ*) moves. The small wheel with holes — thus constantly in the air it rotates.

[Description of a self-rotating wheel instrument, possibly an early form of anemometer or a water-driven astronomical device. The "holes" regulate the flow of water or air.]

[Verses 51–53: Additional rules on wheel instruments and their construction. The chapter concludes with verses praising the value of observation and proper instruments.]

करणज्या शिथ्रचलनमेव शरमोच्छरोत्क्षेपवाच्चाच्च ।
श्रद्धायो द्विविधो यन्त्रेणारिच्छन्दपञ्चकात् ॥ ५३ ॥

V.53. The *karaṇa-jyā* (computed sine), the unsteady motion itself — the arrow, the rising, the ascension — by the instrument with confidence, two-fold — from the five-fold of Ari (the sun's) motion.

इति ब्रह्मस्फुटसिद्धान्ते यन्त्राध्यायो द्वाविंशतितमः

दर्शनार्थः समाप्तः

[Thus ends the Twenty-second Chapter on Instruments in the ***Brāhmasphuṭasiddhānta****.*
End of the section on observation.]

Appendix: On Variant Readings

पाठ-भेद-सूची

अस्य ग्रन्थस्य पाठ-संहितायां चत्वारि मुख्यानि पुस्तक-हस्तलिखितानि उपयुक्तानि सन्ति— (क) पूना-पुस्तकम्, (ग) बम्बई-पुस्तकम्, (च) बडोदा-पुस्तकम्, (छ) होशियारपुर-पुस्तकम्।

Manuscript Sources for Variant Readings

The footnotes in the printed edition record variant readings from four manuscript sources, designated by the following sigla:

Siglum	Source
(⊠)	Poona manuscript (Bhandarkar Oriental Research Institute)
(⊠)	Bombay manuscript (Royal Asiatic Society)
(⊠)	Baroda manuscript (Oriental Research Institute)
(⊠)	Hoshiarpur manuscript (V.V.R.I.)

पाद-टिप्पण्यां □□(ग) इति पाठस्य स्थाने'' इत्यादि संकेतः पाठ-भेदं दर्शयति। □□(वि—इसकी क्रमसंख्या ३१ है)'' इत्यादि टिप्पणी हस्तलिखित-पुस्तकस्य अन्तः अस्य श्लोकस्य संख्या भिन्ना इति दर्शयति।

The footnotes use a notation such as "(⊠) for (*reading*)" to indicate that manuscript (⊠) has the variant reading instead of the adopted text. Parenthetical notations like "(⊠⊠—⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠ ⊠⊠)" indicate that the verse number in the source manuscript is different.

Mathematical Summary of This Chunk

गणित-सारांशः

अस्मिन् खण्डे प्रमुखाणि गणितीय-सिद्धान्तानि सन्ति:
□ कुट्टक-विधिः (□□□□□□□□□□□ □□□□□□□□□)
□ ऋण-धन-कर्म (□□□□□□ □□□□□□□□□□)
□ शून्य-कर्म (□□□□ □□ □□□□□□□□□□□)
□ वर्गप्रकृतिः (□□□□'□ □□□□□□□□)
□ द्विघात-समीकरणम् (□□□□□□□□□ □□□□□□□)
□ ज्या-परिकल्पना (□□□□ □□□□□□□□□□□)
□ छन्दः-संख्यानम् (□□□□□□□□□□□□□)

Key Mathematical Contributions

- **Kuṭṭaka** — The Euclidean algorithm for solving linear Diophantine equations $ax + c = by + d$.
- **Signed arithmetic** — First explicit rules for $+, -, \times, \div$ with positive, negative, and zero.
- **Zero** — $\frac{0}{0} = 0$ (contextual); $a + 0 = a$; $a \times 0 = 0$.
- **Pell's equation** — The composition formula (*bhāvanā*) for generating solutions to $Nx^2 + 1 = y^2$.
- **Quadratic formula** — $x = \frac{\sqrt{4ac+b^2}-b}{2a}$ for $ax^2 + bx = c$.
- **Sine table** — Second-order interpolation for $R \sin \theta$ with $R = 3270$.

- **Combinatorics** — 2^n patterns; Pascal's triangle (*meru-prastāra*).

अयं खण्डः ब्रह्मस्फुटसिद्धान्तस्य गणितीय-हृदयम् अन्तर्भूतवान्। कुट्टकाध्यायः विशेषेण प्रसिद्धः—अस्या युग-ग्रहणस्य प्रथम-सम्पूर्ण-वर्णना अत्र लभ्यते।

This chunk contains the mathematical heart of the *Brāhmasphuṭasiddhānta.* The Kuṭṭakādhyāya is especially notable — it contains the first complete description of the solution to Pell's equation in world mathematical literature.

End of Chunk 5 (Pages 561–700)

The next chunk (pages 701–838) will contain
the remaining chapters (23–24) and the concluding material of Volume I.

0.48

श्रीब्रह्मगुप्ताचार्य-विरचित

Brāhmasphuṭasiddhānta — Bilingual Edition

Part 1 — Final Portion

Pages 701–838 of the Original Edition

Comprising:

Chapter 23 (Mānādhyāya) · Subject Index ·
Madhyamādhikāra Commentary

सौरचन्द्रार्क्षसावन-
मानप्रयोगविधिः

Sanskrit Text

Solar, Lunar, Stellar, and
Civil Measures and Application

Author: **Brahmagupta** (628 CE)

Editor: Acharya Ramswarup Sharma

Indian Institute of Astronomical and Sanskrit Research

New Delhi, First Edition 1966

Bilingual Edition — Sanskrit (Devanagari) / English (with IAST)

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Digital preservation of classical Indian mathematical astronomy

Contents of this Portion / अस्यांशस्य विषयसूची

Sanskrit (Devanagari)

English (with IAST)

*

Chapter 23: त्रयोविंशो मानाध्यायः — The Measurement Chapter

[lines=3,loversize=0.1]T*he twenty-third chapter* of the *Brāhmasphuṭasiddhānta* treats of the measurement of time and the practical application of astronomical calculations. This chapter, like others in the work, is composed in the classical *āryā* metre and presents Brahmagupta's rules for calendrical computation, planetary positions, and the reconciliation of different astronomical systems.

The chapter is divided into two principal sections:

(i) **Prayogavidhi** — Practical application of the siddhānta

(ii) **Caturviṃśa** — The twenty-four fold division of time

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प्रयोगविधिः — Practical Application (Verses 1–4)

सौरैराण्डं मासैस्तिथयश्चान्द्रैस्तथा सावने दिवसाः ।
दिनमासाब्दं पमध्यानत दिग्नाकेन्द्रं मानास्म्याम् ॥ १ ॥ "The solar year, the lunar months and *tithis*, the civil days, days, months, years, the positions of the planets, the *nakṣatras*, and the zodiacal signs—these we call measures (*māna*)."

— Verse 1

Word Analysis / पदविभागः:
(१) सौरैराण्ड — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (२) मासैस्तिथयः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠ ⊠⊠ (३) सावनानि दिवसानि — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (४) दिनमासाब्दानि — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (५) ग्रहनक्षत्रराशयः — ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (1) *saurairāṇḍa* — the solar year; (2) *māsaistithayaḥ* — lunar months and *tithis*; (3) *sāvanāni divasāni* — civil days; (4) *dinamāsābdāni* — days, months, and years; (5) *grahanakṣatrarāśayaḥ* — planets, nakṣatras, and zodiacal signs—these are called *māna* (measures).

मानान्ति सौरचन्द्रार्क्षां सावनानि ग्रहानयनमेभिः ।
मानेन पृथक् चतुर्मितिथ्यर्ब्बहारो त्रं लोकस्य ॥ २ ॥ "The measures of solar, lunar, stellar, and civil reckoning, and the computation of planets by these measures separately—the fourfold manner of computation is in use in the world."

— Verse 2

सौरचान्द्रार्क्षसावनानि — ⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ एभिः पृथक् पृथक् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ग्रहानयनम् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ लोकस्य — ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ *sauracāndrārṣasāvanāni* — four measures; *ebhiḥ pṛthak pṛthak* — separately by each; *grahānanayam* — computation of planets; *lokasya* — current in the world.

युगवर्षं विपुलब्दयनतद्वह्निनिशोद्विद्धिहानयः ।
सौरास्तिथि करणाऽधिकमासौन राशिपर्व्विक्रियाश्चान्द्रात् ॥ ३ ॥ "A *yuga-year*, abundant and deficient (days), fire (3), subtraction and addition—these (come) from the solar (system). *Tithi*, *karaṇa*, intercalary month, and the changes of zodiacal signs and *parva* (come) from the lunar (system)."

— Verse 3

युगवर्षं विपुलाब्दयनानि — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ वह्निः — ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ उद्विद्धिहानयः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ तिथिकरणाधिकमासाः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ राशिपर्विक्रियाः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ *yugavarṣaṃ vipulābdayanāni* — intercalary and deficient days; *vahniḥ* — the number three; *udviddhi-hānayaḥ* — addition and subtraction—from the solar system. *tithikaraṇādhikamāsāḥ* — from the lunar; *rāśiparvikriyāḥ* — changes of zodiacal signs and lunar half-months.

यसस्य वन प्रमाणं ग्रहगतृपवासं सुतं चिकिर्सतः ।
सावनं मानांतु शैया प्रायश्चित्त क्रियाश्चान्याः ॥ ४ ॥ "For one desirous of performing the *pramāṇa* (measurement), the planetary
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motion, the rising and setting, and the creation (of almanacs), the civil measure (is useful), as well as the bed (auspicious moment), *prāyaścitta* (expiatory rites), and other ceremonies."

— Verse 4

प्रमाणं ग्रहगतिक्षयोदयसूतकल्पनम् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ सावनं मानम् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ शय्या — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ प्रायश्चित्तक्रियाः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠ *pramāṇaṃ grahagatikṣayodayasūtakalpanam* — for the construction of the pañcāṅga (almanac); *sāvanaṃ mānam* — civil measure; *śayyā* — auspicious moment for lying down; *prāyaścittakriyāḥ* — expiatory rites and ceremonies.

* * *

चतुर्विंशः — The Twenty-four Fold System (Verses 5–23)

यस्मात्तत्प्रति पश्तं संज्ञा संज्ञातो विना तस्मात् ।
लोकप्रसिद्धसांख्यपादीनां दशांकाश्च ॥ ५ ॥ "Since (computation) is impossible without names (*saṃjñā*) given to each object, the names and numerals well-known in the world (are given)."

— Verse 5

टीका / **Commentary:** The commentator explains that Brahmagupta here gives a systematic nomenclature for the various measures of time and computation. The twenty-four items are enumerated in the following verses.

युगपद्युगाद्यद् या याम्ययां भास्करस्य वा कन्याम् ।
राश्यर्द्धसंख्यायां मस्तकया दिनदलाद्द्वेभ्राम् ॥ ६ ॥ "A *yugapad*, a *yuga*, beginning from that, or of the sun, of the Yāmya (south), of Bhāskara, or of Kanyā (Virgo), half a sign, the numbers (associated), at the head, from half a day, from two, from Bhrama (revolution)."

— Verse 6

प्रपमेवकृतः सूर्ये तु पुलिशा रोमक वाराही यवनाद्यैः ।
यस्मात्तस्मादेकः सिद्धान्तो ग्रथ्य रचनात्मकः ॥ ७ ॥ "The sun has been measured by Pulisha, Romaka, Vāraha, and Yavana (Greeks). Therefore, from that (measurement) one must construct a proper *siddhānta*."

— Verse 7

पुलिशः रोमकः वाराही यवनाः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ सूर्यस्य प्रमाणम् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ रचनात्मकः सिद्धान्तः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ *puliśaḥ romakaḥ vārāhī yavanāḥ* — the five *siddhānta* authors; *sūryasya pramāṇam* — measurement of the sun; *racanātmakaḥ siddhāntaḥ* — construction of a scientific treatise.

यदि भिन्नाः सिद्धान्तभास्कर संस्कान्तयोभि भेदसमाः ।
सरत्नः पूर्व्वस्यां विपुलत्याकार्ह्यो यस्य ॥ ८ ॥ "If the *siddhāntas* are different, (yet) Bhāskara (the sun) has the same modifications (*saṃskāra*)—the difference (is) abundant and deficient, those having the same (result) for the previous (method)."

— Verse 8

तत्र पक्षागरिततं मध्यं गत्युत्तरादयः पञ्च ।
कुड्डाकारे बेधः छिन्द्धि चतुर्थां तरे गोलः ॥ ९ ॥ "In it (the *siddhānta*) the five (methods): *pakṣāgati*, *madhya*, *gati*, *uttara*, etc.; *kuṭūākāra* (transformation), *bheda* (division), *chid* (cutting), *caturtha* (fourth proportional), and the sphere (*gola*)."

— Verse 9

* * *

भट्ट ब्रह्माचार्येण जिज्ञासुतनयेन गणितगोलविदा ।
प्रायःश्लेषसंहलं वा स्फुटसिद्धान्तं कृतो ब्रह्मा ॥ १० ॥ "By Brahmagupta, the son of Jiṣṇu, the knower of mathematics and the

sphere, this true *siddhānta* was composed, (which is) Brahma (itself), adorned with the beauty of elegant composition.”

— Verse 10

भट्ट ब्रह्माचार्यः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ जिज्ञोः तनयः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ गणितगोलवित् — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ स्फुटसिद्धान्तः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ब्रह्मा — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ *bhaṭṭa brahmācāryaḥ* — the venerable Brahmagupta; *jijñoḥ tanayaḥ* — son of Jiṣṇu; *gaṇitagolavit* — knower of gaṇita (mathematics) and the celestial sphere; *sphuṭasiddhāntaḥ* — true astronomical treatise; *brahmā* — of the nature of Brahman itself.

मग्रहं युति वत्सांकुं विचित्रमलनाद्विव्रहोत्कसमः ।
शकनिः कर्म्मबाहुत्वान्निर्भुतातो भास्करमग्रहं ॥ ११ ॥ “(Various) instruments, conjunction, year-count, diverse multiplication, the *utka* (elevation), the *śakani* (depression), on account of the multitude of operations, (all these) from the diminished (part), the solar *maragraham*...”

— Verse 11

प्राज्येयो नैकल्प्योर्द्धच्छिद्दिने स्थितस्य योग्केस्य ।
शङ्कुश्ये कथयत्यब्दापि वत्सि सूर्य्यंश्च ॥ १२ ॥ “The abundant, the one without method, the *ṛddhicchid* (augmented cut), the *yogketsya* of the standing, the *śaṅkuśye* (gnomon-shadow) declares the year and also the solar year...”

— Verse 12

प्रज मया यन्नोकं गोलादग्रं क्षयार्धसतो त्रं तनु ।
प्रायःश्लोद्दशोऽयं संज्ञाध्यायचतुर्विंशः ॥ १३ ॥ “As much as has not been spoken by me, the remainder of the sphere, thin as a sixth part of a *ksaya* (decrease), this (is) the *prāyaśloka* (resumé verse). This (is) the *saṃjñā-adhyāya* (chapter on nomenclature), the twenty-fourth.”

— Verse 13

मया यन्नोक्तं — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ गोलात् अग्रं — ⊠⊠⊠⊠⊠⊠⊠⊠⊠ क्षयार्धसतो षष्ठांशतुल्यं — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ प्रायःश्लोकः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ संज्ञाध्यायः चतुर्विंशः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ *mayā yannoktaṃ* — what has not been spoken by me; *golāt agraṃ* — the remainder of the sphere; *kṣayārdhasato ṣaṣṭhāṃśatulyaṃ* — as minute as one-sixth part of the decrease; *prāyaḥślokaḥ* — resumé verse; *saṃjñādhyāyaś caturviṃśaḥ* — the twenty-fourth chapter on nomenclature.

सूचना / **Note on verse numbering:** The original text numbers this as the 23rd *adhyāya* in the internal count (इति ब्रह्मगुप्ते मानाध्यायः (२३ अध्याय समाप्तः)), though the final verse calls it the 24th in another reckoning. The manuscript tradition contains variations in chapter enumeration.

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नष्टंघ्नसावनदिनात् सूर्यदिनां त्वसावन दिनानि ।
यस्मात्तस्मादश्वं दुर्ल्लभगं मन्दबुद्धिनाम् ॥ ५ ॥ “Multiply the elapsed *sāvana* (civil) days by the solar days; the *asāvana* (non-civil) days—therefore, that attainment is difficult for those of dull intellect.”

— Verse 5 (alt.)

मानुष्यं पित्रदेवं बाहु यान्यष्टौ वसुर्कालस्य ।
उत्कर्षि ज्ञानार्थं बाहु स्पष्टं नवममन्यत् ॥ ६ ॥ “Humanity, the father, the divine, the arm, the eight (*vasu*), the time—the *utkarṣi* (elevation) for the purpose of knowledge, the arm, clearly the ninth (or another).”

— Verse 6 (alt.)

द्वौ द्वौ राशिं मकराहतवः षट् सूर्यगति वशाऽघोज्याः ।
शकिरवसन्तं ग्रीष्मा वर्षा शरदौ सहैमन्ताः ॥ ७ ॥ “Two-two (make) a zodiacal sign; multiplied by Makara (10, i.e. 60), these (are) the six (seasons); the solar motion (gives) the seasons: śiśira-vasanta (spring), grīṣma (summer), varṣā (rains), śarad (autumn), and hemanta (winter).”

— Verse 7 (alt.)

द्वौ द्वौ राशिं — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ मकराहताः षट् — ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ सूर्यगतिः — ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ शिशिर-वसन्त-ग्रीष्म-वर्षा-शरद्-हेमन्ताः — ⊠⊠⊠⊠⊠⊠⊠ *dvau dvau rāśiṃ* — Gemini etc.; *makarāhatāḥ ṣaṭ* — six seasons multiplied by ten (Makara = 10); *sūryagatiḥ* — the sun's path; *śiśira-vasanta-grīṣma-varṣā-śarad-hemantāḥ* — the six seasons.

* * *

मुख्यारः पूजोभक्तः काके व्यासान्तरेण रविकर्णं मुमुच्या ।
द्वौ, क्षाया दिश्यन्त्वे चन्द्रकर्णेन शेषम् ॥ ८ ॥ "The principal and the *arha* (worthy), the worshipper and the devotee; by the *vyāsāntara* (diameter difference) for the crow—the sun's hypotenuse is to be released; two, the decrease is to be shown, the remainder by the moon's hypotenuse."

— Verse 8 (alt.)

पञ्चदशाहीनयुक्ताः षट् क्रान्तिकालाद्यरसगुणाः ॥ ६४ ॥ "The six, diminished and increased by fifteen, are the declination-time etc., multiplied by six (*rasa*)."

— Verse 64 (alt.)

त्रिज्या विपुलछाया घातस्तच्छति विभाजिताक्षाप्तम् ।
क्षयो नित्यं याम्यो गोलवशाद्बिम्बेक्रान्तिः ॥ ६६ ॥ "The radius and the abundant shadow—their product, divided by that and multiplied by the latitude; the decrease is always in the south (Yāmya); from the sphere, the disc's declination."

— Verse 66 (alt.)

क्रान्त्यक्षायुति विभोगाक्षरापदैः शोधिते दिनदलमा ।
भास्य तिकृतयोः कृतमनुयुतो नयोस्तत्पदे व्यस्ते ॥ ६७ ॥ "When the sum and difference of declination and latitude are purified by the *akṣarāpada* etc., the half-day; when the square of that is subtracted from the square of the sum—the method when reversed at that position."

— Verse 67 (alt.)

पठगुणीतागतं शेषं नाडूच्चै दिवसाङ्कं विभाजितात् त्यात् ।
दिनदलकर्णगुणाप्ता तया त्रिज्यया फलं कर्णः ॥ ६८ ॥ "The remainder, multiplied by the prescribed (number) and arrived at, divided by the day-number elevated by the *nāḍī*; that (result) multiplied by the half-day hypotenuse, by that radius the result (is) the hypotenuse."

— Verse 68 (alt.)

* * *

इत्युक्तं क्रमेण सांख्याप्रकरणं प्रयोगविधिश्च ।
व्यासगुणा दीर्घवृत्तं षड़ांशकक्षापयाम् ॥ ९ ॥ "Thus has been stated in order the section on numerals and the practical application; the diameter multiplied, the long circle, the *ṣaḍāṃśaka* (sixth part), the latitude-decline."

— Verse 9 (alt.)

तस्य व्यास शकि करण्डूतद्विजयया गुणालिप्ता ॥ १० ॥ "Its diameter, śaki (potency), *karaṇḍū*, that (is) multiplied by the *dvijaya* (twice-conquered), by degrees."

— Verse 10 (alt.)

रविकर्णं दूता त्रिजया काके व्यासान्तरं हृता शेषम् ।
तिजयामुख्यास बधा क्षयि करणं हुतात्मो व्यासः ॥ १० ॥ "The sun's hypotenuse, the messengers, by the *trijaya* (triple-conquered),

by the *vyāsāntara* (diameter difference) divided, the remainder; the *tijayāmukhyāḥ* (chief of the three-conquered), by the decrease, the operation, the *hutātma* (fire-souled = 3), the diameter.”

— Verse 10 (alt. 2)

व्यासेन दुर्गति बधात् काके व्यासान्तरके मुक्ति बधम् ।
प्रोक्ष् दुमध्यमुक्तथा तिथिगुणाप्तां तमो व्यासः ॥ ११ ॥ “From the diameter, the evil, from the product, the crow—the diameter difference, the release (is) the product; the *prokṣa* (sprinkling), the second middle, thus from the *tithi* (lunar day) multiplied and obtained, the darkness, the diameter.”

— Verse 11 (alt.)

योऽधिकमासावमरात्रसम्भवनः सर्वैर्मानानि ।
प्रायःश्लोकैर्द्दशाभिर्मानाध्यायश्चतुर्विंशः ॥ १२ ॥ “The intercalary months, the *avama* (omitted) days, the *sambhavana* (production), all the measures; by ten *prāyaślokas* (resumé verses), the twenty-fourth Measurement Chapter.”

— Verse 12 (alt.)

इति ब्रह्मगुप्ते मानाध्यायः (२३ अध्याय समाप्तः)

Thus ends the twenty-third chapter on Measurement
in the work of Brahmagupta.

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[illegible]-[illegible]

Dvītīya-Bhāgaḥ — Second Part

(Vāsanā-vijñāna Sanskrit–Hindi Commentary)

प्रधानसम्पादकः --- [illegible] [illegible]

[illegible] [illegible] [illegible] [illegible]

(संचालक-इण्डियन इन्स्टीट्यूट ऑफ अस्ट्रानॉमिकल एण्ड संस्कृत रिसर्च)

प्रकाशकः --- [illegible]

इण्डियन इन्स्टीट्यूट ऑफ अस्ट्रानॉमिकल एण्ड संस्कृत रिसर्च
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विषयानुक्रमणिका — Subject Index

विषयानुक्रमणिका

The subject index lists all major topics treated in the commentary (*vāsanā-bhāṣya*) portion of the *Brāhmasphuṭasiddhānta*, with page references to the commentary pages (numbered 1–126 and beyond).

१. मध्यमाधिकारः — Mean Computation (pp. 1–126)

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[illegible]	[illegible]	Granthārambha-prayojanam — Purpose
[illegible]	[illegible]	Bhacakrasya calanam — Celestial motion
[illegible]	[illegible]	Kāla-pravṛttiḥ — Commencement of time
[illegible]	[illegible]	Khaṇḍasya vibhāga-kalpanā
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[illegible]	[illegible]	Āryabhaṭasya mata-samīkṣā
[illegible]	[illegible]	Manupramāṇāni kalpapramāṇam
[illegible]	[illegible]	Kalpaviṣaye āryabhaṭamatam
[illegible]	[illegible]	Romakasiddhānta-mata-khaṇḍanam
[illegible]	[illegible]	Grahamyugasya paribhāṣā
[illegible]	[illegible]	Sitaretro-śanaiścarayoḥ kalpabhagaṇāḥ
[illegible]	[illegible]	Bhacchramāḥ kudināni ca
[illegible]	[illegible]	Ravivarṣa-māsa-rāśimāsādayaḥ
[illegible]	[illegible]	Sāvananakṣatramānavadināni
[illegible]	[illegible]	Gatakālasya āhargaṇādīnāṃ parihāraḥ
[illegible]	[illegible]	Āryabhaṭamatam — Āryabhaṭa's view
[illegible]	[illegible]	Kalpagatākṣāvanāharganạāḥ
[illegible]	[illegible]	Grahamandādi-madhyamānāṃ madhyamānayanam
[illegible]	[illegible]	Svasiddhānta-nirdeśanam
[illegible]	[illegible]	Praśnarātrau vāra-pravṛtti-pakṣadarśanam
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[illegible]	[illegible]	Tadarthaṃ deśāntarakarmaṇā pradarśanam
[illegible]	[illegible]	Varṣadīrgha dinādhyamadinādi-sādhanam
[illegible]	[illegible]	Varṣadīrghamāsa-sādhanam
[illegible]	[illegible]	Varṣaśānayanaṃ ladhvahargaṇānayanam
[illegible]	[illegible]	Ravisitabudhānāṃ kujaguruśanirīśrekṣṇāṃ cānanayam
[illegible]	[illegible]	Madhyamacandrānayanam
[illegible]	[illegible]	Varṣāntikād dhargaṇāt bhaumādigrahamandaphalānaya

२. स्फुटाधिकारः — True Position Computation (pp. 160–255)

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८. चन्द्र छायाधिकारः — Lunar Shadow Chapter (pp. 501–548)

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[illegible]	[illegible]	Kartavyatā-kathanam	501
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[illegible]	[illegible]	Caitrakāla-sādhanam	503
[illegible]	[illegible]	Candraśaṅkvānayanam	508
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६. ग्रहयुत्यधिकारः — Planetary Conjunction Chapter (pp. 515–567)

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मध्यमाधिकारः — Mean Computation Chapter

[lines=3,loversize=0.1]T*he Madhyamādhikāra* constitutes the first and foundational chapter of the commentary (*vāsanā-vijñāna-bhāṣya*) on the *Brāhmasphuṭasiddhānta*. It treats the computation of mean positions of planets (*madhyama-graha*), beginning with a discussion of the motion of the celestial sphere (*bhacakra*), the definition of time (*kāla-pravṛtti*), and the various *yuga*-systems employed in Indian astronomy.

Introductory Discussion / आरम्भकथनम् (pp. 752–760)

The commentary opens with a discussion of the *bhacakra* (celestial sphere) and its motion:
प्रदक्षिणागम् (दक्षिणावस्स्क्ष्मेण चलायमानम्) श्रादो (सर्व्वप्रथमं) ब्रह्मणा
(जगद्गुरुपादकेन) सृष्टम् (रचितम्) ग्रयदिध्नयादौ स्थितेनक्षत्रादिमततुल्यचपा-
तादिमित्र साकं भ्रुवयष्ट्याद्यारेण प्रमाणशील ज्योतिष्मत् प्रदायिनां नक्षत्रादीनां
चक्रं (गोलं) ब्रह्मणा रचितं यथाप्याचार्ययोः कादीनां वचां न कियते तेषां स्वरूपा-
भावात् किन्तु स्पष्टादिकाले तेषामपि स्थानाऽऽज्ञानरूपसर्जे कृतमिति ॥ १ ॥ "(The celestial sphere), moving eastward (from left to right), was first created by Brahmā (the world-teacher). The sphere, along with the nakṣatras and other celestial bodies having equal *capādas*, along with the meridian staff etc., the circle of nakṣatras and other luminaries that give light—this sphere was fashioned by Brahmā. Even so, the ācāryas (teachers) do not describe their (the planets') essential nature (*svārūpa*), but have only determined their positions (*sthāna*) for the purpose of computation (*spaṣṭādi*)."

— Commentary v. 1

हिन्दीटीका / **Hindi Commentary (pp. 752–753):** The commentator explains that the celestial sphere (*bhacakra*) was first created by Brahmā (the creator). The sphere, along with the nakṣatras and other celestial bodies, revolves from west to east. The commentator notes that the *ācāryas* (teachers) do not discuss the nature (*svārūpa*) of these bodies but have only determined their positions (*sthāna*) for the purposes of computation.

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Key Verses from the Siddhānta / सिद्धान्तमूलश्लोकाः (pp. 760–770)

The commentary elucidates the following root text (*mūla*) verses:
इदानीमनाख्यानतस्य कालस्य प्रवृत्तिमाह ।
चैनंसितादेर्द्दयाद्व मानोर्द्धनमासबर्षयुगकल्पः ।
सृष्ट्यादौ लङ्कायां स्मं प्रवृत[illegible] दिनेस्कस्य ॥ ४ ॥ "Now (Brahmagupta) declares the commencement of that time whose beginning is unmentioned. Day, month, year, *yuga*, and *kalpa* began from the sun at Laṅkā at the beginning of creation."

— Mūla-śloka 4

Hindi commentary (pp. 761–762):
चैत्यादि कल्पप्रतिपद्यात् लंकोपलक्षितांभृद्वेकोर्कोदयादिदिवा प्रवृत्ताः — [illegible]
रविदिनम्बरः सृष्ट्यादौ कल्पादौ — [illegible] *caityādi kalpa-pratipadyāt laṅkopalakṣitāṃbhṛdv ekodayādi-divā pravṛttāḥ* — From the first tithi of Caitra (the first month), at Laṅkā, the days etc. commenced from the sunrise.
ravidinambaraḥ sṛṣṭyādau kalpādau — The solar day number at the beginning of creation, at the beginning of the *kalpa*.

The commentator explains the concept of the *kuldiṇa* (civil day) in detail. The number of civil days elapsed since the beginning of the current kalpa is 1,58,22,23,78,50,000 according to Brahmagupta's system.

~~~~~~~~~~~~~~~~~~~~~

On Time Computation and Planetary Motion / कालगणनम् ग्रहचलनं च (pp. 770–790)

The commentary proceeds with detailed mathematical analysis:

प्रथमं युगपदं चलनं द्वितीयं पश्चादिनपर्व्वैः
तृतीयं तु त्रयोदशभिर्मासैः सावनेन वा चतुर्थम् ॥ "First, the *yugapada* (yuga-epoch method); second, by *parva* (lunisolar method); third, by thirteen months; or fourth, by *sāvana* (civil reckoning)."

— Time-reckoning verse

व्याख्या / Exposition: The four types of time-reckoning are discussed:

(i) *Yugapada* — the yuga-epoch method

(ii) *Parva* — the lunisolar intercalation method

(iii) *Trayodaśa-māsa* — the thirteen-month intercalary method

(iv) *Sāvana* — civil reckoning based on sunrise

Each method is explained with its mathematical basis and astronomical application.

* * *

Planetary Computations / ग्रहगणनविधिः (pp. 790–820)

The commentary provides detailed algorithms for computing mean planetary positions:

Mean longitude formula / मध्यमीभावनसूत्रम्:

$$\text{Mean longitude} = \frac{\text{ahargaṇa} \times \text{daily motion}}{\text{civil days in a kalpa}} \mod 360^\circ$$

Where:

- *ahargaṇa* = elapsed civil days since kalpa-beginning
- *daily motion* = mean daily motion of the planet
- The result is expressed in degrees of the ecliptic

The commentary discusses the *mandaphala* (equation of centre) and *śīghraphala* (*śīghra* equation) with detailed trigonometric formulas using the *jyā* (sine) function.

मन्दफलं रविचन्द्रयोः — मन्दक□□□□□स्य ज्या गुणकः, त्रिज्या भाजकः।
शीघ्रफलं बुधगुरुशनिभ्यः — शीघ्रकेन्द्रस्य ज्या गुणकः। *Mandaphala* for the Sun and Moon: the sine of the *manda-kendra* (anomaly) is the multiplier, the radius (*trijyā*) is the divisor.

Śīghraphala for Mercury, Jupiter, and Saturn: the sine of the *śīghra-kendra.*

* * *

On the Computation of Weekdays / वारप्रवृत्तिकथनम् (pp. 820–838)

The final section of this portion treats the computation of weekdays (*vāra-pravṛtti*):

वारप्रवृत्तिं मनयो वदन्ति सूर्योदयाद्विष्णुराज्ञाध्याम् ।
उर्वी तथाऽघोषपरत्र तस्याऽर्चराथं देशान्तरनाडिकाभिः ॥ १३६॥ "The sages declare the commencement of the week (to be reckoned) from the sunrise (at Laṅkā); on the earth (at other places), it (is determined) by adding or subtracting the *deśāntara-nāḍikās* (longitude difference in time)."

— Verse 136

लङ्कोदयाम्यसुजातप्रथममपरतः पूर्व्वदेशे च पश्चात्—
दर्शवोत्थानिमष्टटीभिः सवितः क्षदयतो वासरैव प्रवृत्तिः ॥
यं या सूर्योदयात् प्राक् चरराकलम्बवैश्चास्मियमियगोले ।
पश्चात् सौम्यगोले युतिर्वियुतिर्वशाच्छोभयोः स्पष्टकालः इति ॥१३६॥ “At Laṅkā, the weekday begins with the first *nāḍī* after sunrise; at a place to the west, (it begins) after (sunrise); with the six or eight *nāḍī*s (difference), the weekday is determined when the sun rises. For a place on the eastern or western side of the prime meridian, before or after sunrise (at Laṅkā), by addition or subtraction, the correct time (is found).”

— Expanded v. 136

हिन्दीटीका / **Hindi Commentary (pp. 836–838):**
लङ्कायां सूर्योदयात् प्रथमनाड्याम् वारारम्भः। पश्चिमदेशे पश्चात् षडष्टघटीभिः।
प्राच्यां देशे सूर्योदयात् प्राक् स्पष्टकालः। At Laṅkā, the weekday begins at sunrise in the first *nāḍī.* At a place to the west, after (sunrise) by six or eight *ghaṭī*s.

At a place to the east, the correct time (is found) before sunrise.

The commentator explains that the *vāra* (weekday) begins at sunrise. By this definition, at a place east of the prime meridian (Laṅkā), the weekday begins after subtracting the longitude difference (in *ghaṭī*s); at a place west of the meridian, (the weekday begins) before sunrise by the same amount (longitude difference in *ghaṭī*s).

In the *Siddhānta-śiromaṇi*, Bhāskarācārya states: “*arkodayād dvābhyāṃ bhyas tāniḥ prācyāṃ praticyāṃ dinap-*”—(the computation proceeds from sunrise with modifications for eastern and western positions).

Concluding Verses / उपसंहारश्लोकाः (p. 838)

The portion concludes with a verse from the *Manusmṛti*:
आसीदिदं तमोभूतमप्रज्ञातमलक्षणम् ।
अप्रतर्क्यमनाऽऽगम्यं प्रसुप्तमिव सर्व्वतः ॥ इति ॥ “This (universe) existed as darkness, unknowable, without marks, not to be reasoned about, not to be apprehended, as if asleep everywhere.”

— Manusmṛti 1.5

टीकाकारवचनम् / **Commentator’s exposition:**

The commentator explains that Manu has stated that at the time of the deluge (*pralaya*), the perception of weekdays and other time measures is not possible. Therefore, the reckoning of weekdays is only possible during the manifest (*sṛṣṭi*) period. Brahmagupta establishes that the weekday begins from sunrise at Laṅkā and is modified by the longitude difference (*deśāntara*) for other locations.

Appendix: Structural Overview of the Commentary / टीकायाः संरचनासूची

The *Vāsanā-vijñāna* commentary on the *Brāhmasphuṭasiddhānta* is organized into the following major chapters (*adhyāyas*):

No.	Title / शीर्षकः	**Pages**
☒	☒☒☒☒☒☒☒☒☒☒☒☒ — Mean Computation	1–126
☒	☒☒☒☒☒☒☒☒☒☒☒☒ — True Position Computation	160–255
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Three Questions (Diurnal)	256–344
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Lunar Eclipse	345–400
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Lunar Shadow	501–548
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Planetary Conjunction	515–567
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Stellar Shadow	—
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Planetary Shadow	—
☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Heliacal Rising/Setting	—
☒☒	☒☒☒☒☒☒☒☒☒☒ — Nodes and Intersections	—
☒☒	☒☒☒☒☒☒☒☒☒☒☒ — Mathematics (Chs. 12–18)	—
☒☒	☒☒☒☒☒☒☒☒☒☒☒☒☒ — Kuṭṭaka (Pulverizer)	—
☒☒	☒☒☒☒☒☒☒☒☒☒☒☒☒☒ — Auxiliary Computations	—

The commentary employs a rich methodology, interweaving:

(i) *Mūla-śloka* — The root text verses of Brahmagupta

(ii) *Anvaya* — Word-order rearrangement for grammatical clarity

(iii) *Bhāṣya* — Sanskrit explanatory commentary

(iv) *Hindi Ṭīkā* — Hindi translation and explanation

(v) *Upapatti* — Mathematical demonstrations and proofs

(vi) *Udāharaṇa* — Numerical examples and worked problems

॥ इति समाप्तम् ॥

End of Part 1, Chunk 6 — Pages 701–838

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श्रीब्रह्मगुप्ताचार्येण विरचितः
ब्राह्मस्फुटसिद्धान्तः

प्रथमभागस्य षष्ठः संचिकाङ्गं समाप्तम्
(पृष्ठानि ७०१--८३८)

The Brāhmasphuṭasiddhānta
composed by Śrī Brahmaguptācārya

The sixth portion of the first part is concluded
(Pages 701–838)

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Contents:

मध्यमाधिकार — Madhyamādhikāra (Chapter on Mean Motions)
(Complete with Commentary)

स्पष्टाधिकार — Spa.s.tādhikāra (Chapter on True Positions)
(Beginning with Sine Tables and Corrections)

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Introduction

प्रस्तावना --- मध्यमाधिकार

मध्यमाधिकार [illegible]
[illegible] [illegible] [illegible]
[illegible] [illegible], [illegible], [illegible],
[illegible] [illegible] [illegible] [illegible]
[illegible] [illegible] [illegible] [illegible] [illegible]–[illegible]
[illegible] [illegible] स्पष्टाधिकारः [illegible] [illegible]–[illegible]

Introduction — Madhyamādhikāra

The *Madhyamādhikāra* (Chapter on Mean Motions) is the first chapter of Brahmagupta's *Brāhmasphu.tasiddhānta.* It presents the computation of the mean motions (*madhyama*) of the planets, day computation, longitude correction (*deśāntara*), and calculation of days since epoch (*yugādi-dina*). This first section spans pages 839–893. The second section, the *Spa.s.tādhikāra* (Chapter on True Positions), spans pages 896–978.

स्पष्टाधिकार-प्रवेशः

स्पष्टाधिकारः [illegible] [illegible]
[illegible] [illegible] [illegible] [illegible],
[illegible] [illegible] [illegible]
[illegible] [illegible], [illegible], [illegible],
[illegible] [illegible] [illegible] [illegible]

Introduction — Spa.s.tādhikāra

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter. "Since the mean planet is not observed daily in the zodiac, I shall state the procedure for making it true" (BSS II.1). Here the twenty-four sine table, versed sines, equation of centre (*manda-phala*), and equation of conjunction (*śīghra-phala*) are discussed in detail.

Chapter 1

मध्यमाधिकारः — Madhyamādhikāra

मध्यमाधिकार ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧ — ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧, ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧, ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧, ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧, ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧ ⌧⌧⌧–⌧⌧⌧⌧

The *Madhyamādhikāra* is the first chapter of the *Brāhmasphu.tasiddhānta* composed by Brahmagupta (b. 598 CE). Its subjects include: the commencement of the weekday (*vāra-prav.rtti*), longitude correction (*deśāntara-karma*), computation of days since epoch (*yugādi-dina-anayana*), the mean motions of the Sun and Moon, and the mean motions of Mercury, Venus, and the other planets. This complete section with commentary (*vāsanābhā.sya*) spans pages 839–893.

1.1 On the Commencement of the Day (वारप्रवृत्ति)

प्रवृत्तिवाद

प्रवृत्ति इससे इसी ब्रह्मगुप्तोक्त बात को कहते हैं। वारप्रवृत्ति कब से होती है, इस विषय में बहुत आचार्यों के भिन्न-भिन्न मत हैं। जैसे आर्यभट, सिंहाचार्य आदि अर्धोदित रवि बिम्ब से दिनारम्भ कहते हैं। अन्य आचार्य दिनान्त से दिनारम्भ कहते हैं। लाटदेव आदि रवि के अर्धास्तकाल से कहते हैं। यवन राजा रात्रि में दश मुहूर्त करके कहते हैं। सिद्धान्तशेखर में श्रीपतिरूपर्युक्त आचार्यमतखण्डनं कृतवान्।

सृष्टेर्मुखे ध्वान्तमये हि विश्वे ग्रहेषु सृष्टेष्विनपूर्वकेषु ।
दिनप्रवृत्तिस्तदधीश्वरस्य वारस्य तस्मादुदयात्प्रवृत्तिः ॥३६॥

Commencement of the Weekday

[Hindi Commentary] *Prav.rtti* means that which is spoken by Brahmagupta himself. On the question "from when does the weekday commence?" there are many differing opinions among teachers. Āryabha.ta and Siṃhācārya hold that the day begins from the half-risen solar disc. Others hold it begins from sunset. Lāṭadeva and others hold it begins from midnight. The Yavana kings hold that ten *muhūrta*s of night [make the division]. In the *Siddhāntaśekhara*, Śrīpati has refuted the opinions of these teachers.

[BSS I.36] When at the beginning of creation the universe was filled with darkness, and the planets had not yet been created, there was no lord of the weekday (*vāreśvara*). Therefore the commencement of the weekday is from sunrise (*udayāt prav.rttiḥ*).

सृष्ट्यादि से पहले विश्व अन्धकारमय था। ऐसे समय में वारप्रवृत्ति के विचार सम्भव नहीं क्योंकि उस समय वारेश्वरग्रहों का अभाव रहता है। इसी कारण रात्रि में वारप्रवृत्ति कहने वालों के मत ठीक नहीं। मध्याह्न काल और अस्तमय काल से आरम्भ कर वारप्रवृत्ति कहना भी ठीक नहीं है। इसलिए रव्युदय काल से वारप्रवृत्ति मानने वाले आचार्यों का मत ही ठीक है।

[Hindi Commentary — Translation] Before the beginning of creation, the universe was full of darkness. At such a time, discussion of the weekday's commencement is impossible because the planet-lords of the weekdays did not yet exist. For this reason, the opinion of those who say the weekday commences at midnight is not correct. [Holding that] it begins from midday or from sunset is also not correct. Therefore the opinion of the teachers who hold that the weekday commences at sunrise is indeed correct.

संस्कृतोपपत्तौ □□वारप्रवृत्तिं मुनयो वदन्ति सूर्योदयाद्रावणराजधान्याम्'' इत्यादि पद्यात् विद्यते। □□लङ्कोदयाम्यसूत्रार्धमपरतः पूर्वदेशे च पश्चात्''--- इत्यादि पद्येन स्पष्टम्। इति ॥३६॥

[Sanskrit Commentary] In the Sanskrit proof-text (*saṃsk.rtopapatti*) it is known from the verse "The sages state the commencement of the weekday from sunrise at Laṅkā, the city of Rāvaṇa" etc. This is made clear by the verse "From Laṅkā sunrise, south of the prime meridian, east is before, west is after" etc. Thus [ends] verse 36.

1.2 Longitude Correction (देशान्तरकर्म)

नियमाः ३७--३९

Rules 37–39: Longitude Difference

भूपरिधिः खखखशरारेखा स्वाक्षान्तरांशसंख्यिताः ।
भगणांशहृताः फलकृतिहीना देशान्तरस्य कृतिः ॥३७॥
शेषपदगुणासुक्तिभूपरिधिहता कलादिलब्धसूरणस् ।
उज्जयिनीयाम्योत्तररेखायाः प्राग्धनं पश्चात् ॥३८॥ मध्यग्रहे स्फुटे
वा भूपरिधिहृतात् पदात् गुणात् षष्ठया । लब्धं घटिका अथवा
कर्मतिथिष्वग्रणं ग्रहवत् ॥३९॥

[BSS I.37–39] The terrestrial circumference (*bhū-paridhi*) is five thousand yojanas (*khakha-kha-śara* = 5000). Multiply this by the degrees of difference in latitude (*svāk.sāntarāṃśa*); divide by 360 (*bhaga.nāṃśa*); the square of the result, subtracted from the square of the longitudinal difference (*deśāntara*) [in yojanas], gives [the square of] the corrected difference. [Verses 38–39] From the root of the remainder, multiplied by the terrestrial circumference and divided [appropriately], [obtain] minutes (*kalā*); to the east of the Ujjayinī north-south line is positive (*dhana*), to the west is negative (*.rna*). Or from the root divided by the terrestrial circumference and multiplied by sixty, the result is in *gha.tikā*s; this is to be applied to the mean or true planet as positive or negative like [the treatment of] a planet.

भूपरिधिः---भूगोलस्य परिधिः; कियानित्यत आह---खखखशरा इति पञ्चसहस्रं योजनानामित्यर्थः। अत्र च देशान्तरकक्ष्योस्तत्यादयः प्रमाणादिविभागेन पूर्वमेव व्याख्याताः।

[Sanskrit Commentary] "Terrestrial circumference" (*bhū-paridhi*) means the circumference of the Earth-globe. In answer to "how much?" he says *khakha-kha-śarā* = five thousand yojanas. Here the longitudinal difference, latitude, etc., have been explained previously by means of measurements and divisions.

हिन्दी भाषा---भूपरिधि ५००० पाँच हजार है, इसको स्वदेश और रेखादेश के अक्षांशान्तर से गुणा कर ३६० से भाग देने से जो फल हो उसके वर्ग को देशान्तर वर्ग में घटाकर शेष के मूल से ग्रहगति को गुणकर भूपरिधि से भाग देने से कलात्मक फल होता है। उज्जयिनीयाम्योत्तररेखा से पूर्वदेश में मध्यमग्रहे वा स्फुटग्र□ह धन, पश्चिमदेशे ऋण। पद (मूल) को साठ से गुणा कर भूपरिधि से भाग देने से घट्यात्मक फल, तिथिमुक्त घटी संस्कार्यम्। ॥३७-३९॥

[Hindi Commentary — Translation] The terrestrial circumference is five thousand. Multiply this by the latitudinal difference between one's own place and the standard meridian; divide by 360; the square of that result, subtracted from the square of the longitudinal difference [in yojanas]; from the square root of the remainder, multiplying by the planet's motion and dividing by the terrestrial circumference, one obtains a result in minutes. East of the Ujjayinī north-south line [apply as] positive to the mean or true planet; west as negative. The root multiplied by sixty and divided by the terrestrial circumference gives the result in *gha.tikā*s; this is to be applied to the *tithi*-freed *gha.tī*s (i.e., time correction) like a planet.

Upapatti (Proof/Explanation)

यदि व्यासमानेन १५८१ तदा व्यासवर्गाष्टगुणात्पदं भूपरिधिभवेदिति सूर्यसिद्धान्तोक्त्या भूपरिधिमानं ५००० मागच्छति ब्रह्मगुप्तमते।

[Sanskrit Upapatti] If the diameter measure is 1581, then from the square of the diameter, multiplied by eight, [taking] the square root gives the terrestrial circumference — so it is stated in the *Sūryasiddhānta.* By this rule the terrestrial circumference measure of 5000 is obtained, which is Brahmagupta's opinion.

उपपत्ति---यदि व्यासमान १५८१ मानते हैं तब □□व्यासवर्गाष्टगुणात्पदं भूपरिधिभवेत्'' इस सूर्यसिद्धान्तोक्त प्रकार से भूपरिधिमान ५००० प्राप्त होता है, यही ब्रह्मगुप्तमत में भूपरिधिमान है।

[Upapatti — Translation] If we assume the diameter measure to be 1581, then by the method stated in the *Sūryasiddhānta*, “the square root of eight times the square of the diameter is the terrestrial circumference,” the terrestrial circumference measure of 5000 is obtained. This same [value] is the terrestrial circumference in Brahmagupta’s system.

1.3 Computation of Days Since Epoch (युगादिदिनानयन)

नियमः ४०

Rule 40: Days Elapsed Since Kalpa

कल्पगताब्दा गुणिता रूपाष्टजिनेनापवर्निसप्तनगैः ।
खस्वरसनवभिभक्ता दिनान्यवमानि चांशका दोषाः ॥४०॥

[BSS I.40] The years elapsed since the beginning of the *kalpa* (*kalpa-gatābdā*), multiplied by 2486 (*rūpā.s.tajina*) and by 7736 (*apavarni-saptanaga*), and divided by 8600 (*kha-svara-sanava*), yield the days and the omitted [tithi]s; the remainders are the fractional parts and defects.

वा. भा.---अभीष्टे रविमण्डलान्ते ये कल्पगताब्दास् तेऽत्र गृह्यन्ते। कल्पगताब्दाः गुणिताः कृत्याह। रूपाष्टजिनैः २४८६ अन्यत्र नवार्निसप्तनगैः ७७३६ तत उभयतः खखरसनवभिभक्ताः ८६०० फलानि यथासंख्यं दिनान्यवमानि चांशका शेषा द्वयोरपि स्थानयोः। कल्पगताब्दा रूपाष्टजिनैः संगुण्य खखरसनवभिविभजेत् फलं दिनानि भवन्ति। तत्र यद्विकलं तद्दिनांशदाब्देनोच्यते।

[Sanskrit Commentary — Vāsanābhā.sya] At the desired end of a solar year, the years elapsed of the *kalpa* are taken here. The years elapsed of the *kalpa* having been multiplied — he says. By *rūpā.s.tajina* = 2486, and elsewhere by *navārni-saptanaga* = 7736, and then divided by *kha-svara-sanava* = 8600; the results are respectively the days and omitted [lunar days]; the remainders are the fractional parts in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days. There, whatever is the remainder, that is called the *dināṃśa* (fractional part of a day) by the word *dābda.*

हिन्दी भाषा---अभीष्ट रविमण्डलान्त में जो कल्पगताब्द हैं वे यहाँ गृह्यन्ते (लिये जाते हैं)। कल्पगताब्दों को २४८६ से गुणा करके ८६०० से भाग देने से फल दिन और अवमान शेष हैं, दोनों स्थानों पर यही है। कल्पगताब्दों को २४८६ से गुणा कर ८६०० से भाग देने से फल दिन होते हैं। तत्र जो शेष विकल रहता है उसको दिनांश कहा जाता है।

[Hindi Commentary — Translation] In the desired end of a solar year, the years elapsed of the *kalpa* that are here taken. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days and omitted [days]; the remainders are in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600, the result is days. There, whatever remainder remains, that is called the *dināṃśa* (fractional part of a day).

1.4 Mean Position of the Sun and Moon

नियमाः २४--२६

Rules 24–26: Daily Motion and Latitude

मन्दफललिप्ताः स्वहाराप्ता रवीन्द्वोर्भुक्तिफलं क्रमात् ॥ दक्षिणे धनमुत्तरे ॠणं भवति ध्रुवसंज्ञकं तत् ॥ एवमिह दण्डमुखं विपरीतं प्राक् चेद्विघोरतनफलाल्पमथात्र सौरम् ॥ सकृत् सकलसन्नरञ्जनाथं भुजं विचारयन्तु ज्यौतिषिका भूषं विचारयन्त्विति ॥२४॥

[BSS I.24] The minutes of the *manda-phala* (equation of centre), multiplied by the divisor and divided [appropriately], give in order the daily-motion results for the Sun and Moon. In the southern [hemisphere] it is positive, in the northern negative — this is called the *dhruva* (constant). Thus here the staff-head [correction] is reversed; if previously the *vighora-tana-phala* was small, then here [it becomes] solar [correction]. Let astronomers examine the *bhuja* (sine) once, [that of] the delight of all good people; let them examine the *bhū.sa* (ornament) [of calculation].

वा. भा.---प्रह्लादीनां युगे कियन्ति भगणमानानि गताहर्गणमानानि यानि, दिनवारादीं विचाराः, मध्यमग्रहादिसाधनानि यानि, एतदादिषु विषयेषु ग्रार्याणां (आर्यभट्टस्य) षष्टिचा (षष्टिप्रमिताच्छिन्दसा) प्रथमो मध्यगतिरध्यायः मया कृतोऽर्थान्मध्यगतिनामकेऽध्याये कियन्तो विषयाः सन्ति तेषामुल्लेखः कृत इति ॥

[Sanskrit Commentary — Vāsanābhā.sya] How many revolutions (*bhaga.namāna*) of Prahlāda and the others in a *yuga*, how many elapsed days (*gata-āharga.na*), the computations of weekdays and so on, the computations of mean planets and so forth — in these and other subjects, the first chapter on mean motion (*madhyagati-nāmaka adhyāya*) was composed by me (Brahmagupta) with sixty-*āryā* [verses] (*ṣaṣṭi-pramitā chandasā*); how many subjects there are in the chapter named *Madhyagati*, an enumeration of them has been made. Thus [ends the commentary].

1.5 Mean Position of Mars and Other Planets

नियमः मध्यमग्रहसाधनम्

मन्दकेन्द्रेभुजज्याभिर्मन्दफलं पातजीविका । कोटिजीवाहृतं भक्तं व्यासार्धं फलमण्डलम् ॥

Rule: Mean Planets Computation

[BSS — General Rule] The *manda-phala* (equation of centre) is [obtained] from the *bhuja-jyā* (sine of the anomaly) of the *manda-kendra* and the *pāta-jīvikā* [divisor]; divided by the *ko.ti-jyā* (cosine) and multiplied by the *vyāsārdha* (radius), [it becomes] the *phala-ma.n.dala* (resultant circle/epicycle).

1.6 Epilogue to the Madhyamādhikāra

अध्यायोपसंहारः

Chapter Conclusion

इति ब्राह्मस्फुटसिद्धान्ते मध्यमाधिकारः प्रथमः ॥

[End of Chapter 1] Thus the first chapter, the *Madhyamādhikāra* (Chapter on Mean Motions), in the *Brāhmasphu.tasiddhānta* [of Brahmagupta].

हिन्दी भाषा---ग्रहादियों के युग में जो भगणमान है, गताहर्गणमान जो है, दिनवारादि के जो विचार हैं, और मध्यम ग्रहमादि साधन जो है एतदादिक विषयों में तिरसठ आर्यछन्दों के द्वारा मध्यगति नामक प्रथम अध्याय किया गया अर्थात् मध्यगति नामक प्रथम अध्याय में कितने विषय हैं उनका उल्लेख किया गया इति ॥

[Hindi Commentary — Translation] The revolutions of the planets and others in a *yuga*, the elapsed *aharga.na*, the computations of weekdays and so forth, and the computations of mean planets and so on — in these and other subjects, the first chapter named *Madhyagati* was composed by means of sixty-three *āryā* [verses]; that is, in the first chapter named *Madhyagati*, however many subjects there are, an enumeration of them has been made. Thus [ends the commentary on Chapter 1].

[[illegible]-[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]–[illegible] [illegible] [illegible]]

[End of the Madhyamādhikāra. This first section, pages 839–893, has been completely explained. It covers mean motions of all planets, day computation, deśāntara (longitude correction), and yugādi calculations.]

Chapter 2

स्पष्टाधिकारः — Spa.s.tādhikāra

स्पष्टाधिकारः ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠, ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠, ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠, ⊠⊠⊠⊠⊠⊠⊠, ⊠⊠⊠⊠⊠⊠⊠⊠, ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠ ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠ ⊠⊠⊠–⊠⊠⊠⊠

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter of the *Brāhmasphu.tasiddhānta.* Here the purpose of *spa.s.tīkara.na* (making true), the twenty-four sine table (*jyā-pañcaviṃśatikā*), versed sines (*utkrama-jyā*), *manda-phala* (equation of centre), *śīghra-phala* (equation of conjunction), and true daily motion (*spa.s.ta-bhukti*) are presented. This section spans pages 896–978.

2.1 Purpose of Spa.s.tīkara.na

नियमः १

Rule 1: The Need for Correction

यस्मान्मध्यतुल्यः प्रतिदिवसं दृश्यते ग्रहो भगणे । तस्मात्तत्कृत्यकरं वक्ष्ये मध्यस्फुटीकरणम् ॥१॥

[BSS II.1] Since the mean planet is not observed in the zodiac (*bhaga.na*) every day equal to [its mean computation], therefore I shall state the procedure for making the mean [planet] true — the *madhyasphu.tīkara.na.*

वा. सा.---प्रथमं स्फुटगत्यध्यायं व्याख्यायते। तत्रारम्भप्रयोजनमाह। यस्मान्मध्यग्रहेण तुल्यः अग्रविषये ग्रहो न दृश्यते भगणे नक्षत्रचक्रे प्रतिदिवसं दिवसेदिवसे तस्मात् स्फुटीकरणं वक्ष्ये। मध्यस्य किं हगित्याह---हत्कृतुल्यक्रममप्रायो मध्यमो ग्रहः कक्षामण्डले परिकल्पते। न च कक्षामण्डले पारमार्थिको ग्रहः प्रतिमण्डले मध्यगत्या भ्रमति किन्तु स्पष्टगत्या प्रतिमण्डले परिभ्रमन् कक्षामण्डले दृश्यते, अतो येन गणितेन कक्षामण्डले प्रतिमण्डलस्थो ग्रहो हत्समो भवेत्तादृशं स्फुटीकरणमहं वक्ष्ये इति ॥१॥

[Sanskrit Commentary — Vāsanābhā.sya] First, the chapter on true motion (*sphu.ta-gaty-adhyāya*) is being explained. There the purpose of the commencement is stated. Since a planet equal to the mean planet is not observed in practice in the zodiac (*bhaga.na*, the nak.satra circle) day by day, therefore I shall state the *sphu.tīkara.na* (procedure for making true). In answer to "what is the defect of the mean?" he says: the mean planet is hypothetically located (*parikalpite*) on the *kak.sā-ma.n.dala* (deferent circle). And the real planet is not on the *kak.sā-ma.n.dala*; revolving on the *prati-ma.n.dala* (epicycle) with mean motion, it is observed on the *kak.sā-ma.n.dala* with true motion. Therefore, by whatever computation the planet situated on the *prati-ma.n.dala* may become equal to [the observed position] on the *kak.sā-ma.n.dala*, that *sphu.tīkara.na* I shall state. Thus [ends commentary on] verse 1.

हिन्दी भाषा---जिस कारण से प्रत्येक दिन क्रान्तिवृत्त में स्पष्ट ग्रह मध्यम ग्रह के बराबर नहीं देखे जाते हैं उस कारण से हत्कृतुल्य कारक स्पष्टीकरण को मैं (ब्रह्मगुप्त) कहता हूँ। कक्षावृत्त में मध्यम ग्रह परिकल्पित प्रतिवृत्त में स्पष्टगति से भ्रमण करते हुए ग्रह कक्षावृत्त में देखे जाते हैं। इसलिए जिस गणित से कक्षावृत्त में प्रतिवृत्तस्थित ग्रह हत्सम होते हैं उस स्पष्टीकरण को मैं कहता हूँ ॥१॥

[Hindi Commentary — Translation] For the reason that the true planet in the *krānti-v.rtta* (ecliptic) is not seen every day equal to the mean planet, for that reason I (Brahmagupta) call that procedure which makes [it] equal to the observed [position] the *sphu.tīkara.na.* The mean planet is hypothetically [placed] on the *kak.sā-v.rtta* (deferent); the planet, revolving with true motion on the *prati-v.rtta* (epicycle), is observed on the *kak.sā-v.rtta.* Therefore, by whatever computation the planet situated on the *prati-v.rtta* becomes equal [to the observed position] on the *kak.sā-v.rtta*, that *sphu.tīkara.na* I state. Thus [ends commentary on] verse 1.

2.2 The Twenty-Four Sines (ज्यापञ्चविंशतिका)

नियमाः २--९

ज्यापञ्चविंशतिका-प्रवेशः

[illegible] [illegible] ([illegible]) [illegible] [illegible] [illegible] [illegible] $3\frac{3}{4}°$ (= 225’) [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] ([illegible]) [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible]

षड्बुदग्निसुताग्निरसद्वाद्याडा मुनीन्दुवसुचन्द्राः । इन्दुनवनवचन्द्रा रसतिथियमला रविश्रियमाः ॥२॥ छिद्रेपुजिनाः कृतनवपञ्चयमा नन्दचन्द्रसुनिपक्षाः । दन्ताष्टयमा गुणरामनवयमाः शकद्यमखरामाः ॥३॥ ऋतुनवखयुरणा नवशरचन्द्रयुणाः सप्तशून्ययमदहनाः ॥४॥ द्विजनगुरास्त्रिरसरदाः खसप्तयमदहनो व्यस्ताः ॥५॥ मुनयोऽष्टयमास्त्रिरसा रुद्रशशाङ्काः समुद्रमुनिचाः ॥६॥ नववेदयमा सुनियुराहुताशना वसुगुणसमुद्राः ॥७॥ रूपेन्द्रियेषवो रसनगर्तवचन्द्रशीतकरवसवः ॥८॥ शरवसुनन्दाः सागररुद्रशशाङ्का नवाडार्कः ॥९॥

क्रमज्या-सारणी

[illegible].	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible].	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

Rules 2–9: Sine Table in 24 Steps

Introduction to the Jyā-pañcaviṃśatikā

Brahmagupta gives twenty-four *krama-jyās* (sines in sequence), each arc being $3\frac{3}{4}°$ (= 225’), filling the complete quadrant of 90°. These sines are expressed poetically through *bhūta-saṅkhyās* (word-numerals), displaying Brahmagupta’s literary talent alongside his mathematical genius.

[BSS II.2–9] The 24 Sines

The twenty-four sines are encoded in *bhūta-saṅkhyā* (word-numeral) verses. Each word corresponds to a numerical value:

Verse 2: ṣaḍ-bud-agni (6+14=20) = 214; suta-agni (7+3=10) = 427; rasa-dvādya (6+12=18) = 638; munīndu (7+1=8) = 846; vasu-candra (8+1=9) = 1051; indu-nava-nava-candra (1+9+9) = 1251; rasa-tithi-yamala (6+15+1) = 1446; ravi-śriyamā (12+1) = 1624.

Verses 3–9 continue the sequence up to the 24th sine = 3270 (the full radius).

Krama-jyā Table (Sines in Sequence)

No.	Value	Name
1	214	Prathamā-jyā
2	427	Dvitīyā-jyā
3	638	T.rtīyā-jyā
4	846	Caturthī-jyā
5	1051	Pañcamī-jyā
6	1251	.Sa.s.thī-jyā
7	1446	Saptamī-jyā
8	1624	A.s.tamī-jyā
9	1817	Navamī-jyā
10	1991	Daśamī-jyā

No.	Value	Name
11	2145	Ekādaśī-jyā
12	2296	Dvādaśī-jyā
13	2419	Trayodaśī-jyā
14	2594	Caturdaśī-jyā
15	2719	Pañcadaśī-jyā
16	2832	.So.daśī-jyā
17	2933	Saptadaśī-jyā
18	3021	A.s.tādaśī-jyā
19	3096	Ekūnaviṃśa-jyā
20	Viṃśa-jyā	

[illegible].	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

वा. था.---एषामद्व्यास एव व्यास्यानम्। अघेज्यकास्ता इति तद्यथा। सनुयमला २१४ एतत् प्रथमज्यारूपम्। सुनियमवेदा ४२७ द्वितीयम्। वसुज्वलनघटिकाः ६३८ तृतीयम्। रसकृतवसवः ८४६ चतुर्थम्। शशिपञ्चखेन्दवः १०५१ पञ्चमम्। चन्द्रदरसूर्याः १२५१ षष्ठम्।

No.	Value	Name
21	3207	Ekaviṃśa-jyā
22	3242	Dvāviṃśa-jyā
23	3263	Trayoviṃśa-jyā
24	3270	Caturviṃśa-jyā

[Sanskrit Commentary] Their diameter-measure is the radius. “These are the *aghe-jyā*” (sines) — thus. *Sanuyamalā* = 214; this is the form of the first sine. *Suniyamavedā* = 427, the second. *Vasu-jvalana-gha.tikāḥ* = 638, the third. *Rasa-k.rta-vasavaḥ* = 846, the fourth. *Śaśi-pañcha-khendavaḥ* = 1051, the fifth. *Candra-dara-sūryāḥ* = 1251, the sixth.

Utkrama-jyā (Versed Sine) Values

उत्क्रमज्या-सारणी

उत्क्रमज्या (versed sines) ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠ “⊠ — ⊠⊠⊠⊠⊠⊠⊠” ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

⊠⊠⊠.	⊠⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠
⊠	⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠	⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠	⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠⊠+⊠+⊠)
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠⊠+⊠)
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠+⊠)
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)

⊠⊠⊠.	⊠⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠
⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠⊠+⊠)
⊠⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠⊠)
⊠⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠+⊠)
⊠⊠	⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠⊠+⊠⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠⊠+⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠⊠)

⊠⊠⊠.	⊠⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠⊠+⊠+⊠⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠+⊠⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠ (⊠+⊠)
⊠⊠	⊠⊠⊠⊠	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

हिन्दी भाषा---बत्तचतुर्थांश मुनियमला मुनियमवेदा इत्याद्यर्धज्याइचतुरविशतिसंख्यका सन्ति यथा अघोलिखिताः स्युः। व्यस्ता (उत्क्रमज्या) इत्यस्याग्रे सम्बन्धः। चतुरविशतिरर्धज्या (क्रमज्याः)---२१४, ४२७, ६३८, ८४६, १०५१, १२५१, १४४६, १६२४, १८१७, १९९१, २१४५, २३१२, २४५९, २५९४, २८३२, २९३३, ३०२१, ३०९६, ३१४५९, ३२०७, इत्यादि, ३२६३, ३२७०।

Utkrama-jyā Table (Versed Sines)

The *utkrama-jyā*s (versed sines) are constructed by the reversed order of the half-chords. In modern trigonometry these are known as “1—” (versed sine = radius — cosine).

No.	Value	Bhūta-saṅkhyā
1	7	samudra-muni-ca (4+0+7)
2	29	nava-veda-yamā (9+4+8)
3	66	suni-yur-āhutāśana (14+3+5)
4	102	vasu-guṇa-samudra (8+3+4)
5	171	rūpendriye.ṣava (1+11+5)
6	224	rasa-nagartava (6+6+9)
7	292	candra-śītakara-vasava (1+2+5+8)
8	370	śara-vasu-nandā (5+8+9)

No.	Value	Bhūta-saṅkhyā
9	481	sāgara-rudra-śaśāṅka (4+11+1)
10	621	navāḍārka (9+8+12)
11	766	trivi.saya-veda-śaśāṅka (3+6+4+1)
12	931	pañca-tri-rasendava (5+3+6)
13	1114	.sa.d-budhi-yamadh.rtaya (6+17+18)
14	1296	śrati-dh.rti-khayamā (15+2+8)
15	1453	nava-śaśi-yama-pak.sa (9+6+10)

No.	Value	Bhūta-saṅkhyā
16	1635	sāgara-dvijina (4+2+6)
17	1824	radara-sa-yamalā (12+6+15)
18	2019	gu.na-veda-vasu-yamā (3+4+8+8)
19	2214	.sa.t-vi.saya-śūnya-gu.na (6+6+0+3)
20	2424	kha-muniradā (0+7+12)
21	2632	vyāsāgha (0+8)
22	2843	nava-rada-candra (9+12+1)
23	3056	jināndā (5+8)
24	3270	Parama-krānti-jyā (full radius)

[Hindi Commentary — Translation] The twenty-four half-chords (*ardhajyā*), [named] *muniyamalā*, *muniyamavedā*, etc., are of twenty-four numbers, as written below. The relation of *vyastā* (reversed, i.e., versed sine) is given after. The twenty-four *ardhajyā*s (*krama-jyā*s) are: 214, 427, 638, 846, 1051, 1251, 1446, 1624, 1817, 1991, 2145, 2312, 2459, 2594, 2832, 2933, 3021, 3096, 3145(?), 3207, etc., 3263, 3270.

2.3 Arc–Jyā Conversion (चापज्यानयन)

नियमः चापनिर्माणम्

Rule: Finding Arc from Jyā

ज्यार्धात्त्रिज्याहृतात् पदं स्वल्पान्तराद्विषमात् समात् । चापं स्यात्ततः परं तु पदं तद्वर्गयुतेः पुनः ॥

[BSS — Arc–Jyā Rule] From the *jyārdha* (half-chord, sine), divided by the *trijyā* (radius), the square root gives [the first approximation], slightly different for odd and even [intervals]; from that the arc is obtained. After that, again [taking] the square root of [the sine] combined with its square [correction], [the more precise arc is obtained].

वि. सा.---बत्तचतुर्थांशि मुनयोऽष्टयमा इत्यादि चतुरविशतिसंख्यका उत्क्रमज्याः सन्ति यथा अघोलिखिताः स्यु□---७, रद् (२९), ६२, १११; १७४, रउ (२९२), ३३७, ४३५, रभ१ (५२१), ६७६, ८११, रउ (९८५), १११४, १२९६, १४५३, १६३५, १८२४, २०१९, २२१४, २४२४, २६३२, २५४३, ३०५६, रेर७० (३२७०)।

[Sanskrit Commentary] The twenty-four versed sines (*utkrama-jyā*), [named] *muno'ṣ.tayamā* etc., are of twenty-four numbers, as written below: 7, rad (29), 62, 111; 174, rau (292), 337, 435, rabha1 (521), 676, 811, rau (985), 1114, 1296, 1453, 1635, 1824, 2019, 2214, 2424, 2632, 2543(?), 3056, rera70 (3270).

हिन्दी भाषा---ज्यार्ध को त्रिज्या से भाग देने से जो पद (मूल) प्राप्त हो वह स्वल्पान्तराद्विषमात् समात् (थोड़े अन्तर से विषम और सम) से चाप होता है। उसके पश्चात् पुनः उसी पद के वर्ग में युक्त करके (कुछ संस्कार करके) आगे का चाप प्राप्त होता है।

[Hindi Commentary — Translation] The *jyārdha* (half-chord), divided by the *trijyā* (radius), from whatever root (*pada*) is obtained, slightly differing for odd and even [intervals], that becomes the arc. After that, again, having combined [the correction] with the square of that same root, the next arc is obtained.

2.4 Manda and Śīghra Equations

नियमाः ४१--४४

Rules 41–44: True Motion Computation

मन्दफल-शीघ्रफल-प्रवेशः

मन्दफल (manda-phala, equation of centre) [illegible] शीघ्रफल (śīghra-phala, equation of conjunction) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible]

Introduction to Manda-phala and Śīghra-phala

The *manda-phala* (equation of centre) and *śīghra-phala* (equation of conjunction) are the two principal corrections for making the planet true. The *manda-phala* is computed from the *manda-kendra* (anomaly of the epicycle); the *śīghra-phala* from the *śīghra-kendra* (anomaly of conjunction).

मन्दकेन्द्रेभुजज्याभिः पातजीविकया गुणाः । कोटिजीवाहृता भक्ता व्यासार्धेन फलं गतिः ॥४१॥ ततः प्रथममल्पं द्वितीयं वा मन्दसंस्कृतात् । मध्यान्मन्दफलादद्याच्छीघ्रभुक्त्याऽथवा पुनः ॥४२॥ मन्दसंस्कृतमादाय शीघ्रभुक्त्याऽथवा पुनः । शीघ्रफलं ततः कुर्यात्त्रैराशिकेन संहितम् ॥४३॥ शीघ्रफलं व्यवस्थाप्य ग्रहस्य स्फुष्टगतिकाः । समानीय ततः कुर्याद्वक्रभुक्तिं विचक्षणः ॥४४॥

[BSS II.41–44]

41: The *bhuja-jyā* (sine of anomaly) of the *manda-kendra*, multiplied by the *pāta-jīvikā* and divided by the *ko.ti-jyā* (cosine), [then] multiplied by the *vyāsārdha* (radius), gives the *manda-phala* (equation of centre) and the motion.

42: From that, first the small or second [epicycle], or from the *manda-saṃsk.rta* (mean corrected by manda), from the mean, one should give the *manda-phala*, or again by the *śīghra-bhukti*.

43: Taking the *manda-saṃsk.rta*, or again by the *śīghra-bhukti*, one should then compute the *śīghra-phala*, combined by proportion (*trairāśika*).

44: Having established the *śīghra-phala*, the true velocities (*sphu.s.t.a-gatikaḥ*) of the planet are to be brought together; then the wise [astronomer] should compute the *vakra-bhukti* (retrograde motion).

वि. सा.---मन्दकेन्द्रगतिज्यान्तरेण (भोग्यखण्डेन) गुणिता, प्राद्बजीवया (प्रथमज्यया) भक्ता यल्लब्धं तत्स्फुटपरिधिगुणितं, भगणांशे ३६५ भक्ते लब्धामिष्कलामिमां शकवर्गादौ (सकलादिकेन्द्रे, क्कयादिकेन्द्रे च) स्वमध्यमगतीरूनाधिका तदानीकेन्द्रस्य (रविचन्द्रयोः) स्फुटा गतिर्भवति, कुजादीनां ग्रहाणां मन्दगतिफलरहिता मध्यगति (मन्दस्फुटगति) वदत्याचार्याः।

[Sanskrit Commentary] Multiplied by the *manda-kendra-gati-jyāntara* (*bhogyakha.n.da*, the tabular difference), divided by the *prathama-jīvā* (first sine); whatever result is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* etc., [when this is] less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and the other planets, the mean motion minus the *manda-gati-phala* is called *manda-sphu.ta-gati* by the teachers.

हिन्दी भाषा---मन्दकेन्द्रगतिज्यान्तर (भोग्यखण्ड) से गुणित प्रथमजीवा (प्रथमज्या) से भक्त जो लब्ध फल वह स्फुटपरिधि से गुणित ३६५ भगणांश से भाग देने से जो कला-मिमा प्राप्त होती है वह सकलादिकेन्द्र में अपनी मध्यमगति से ऊना या अधिक होती है, तदानीक केन्द्र में रवि और चन्द्र की स्फुटा गति होती है। कुज आदि ग्रहों की मन्दगतिफलरहित मध्यगति मन्दस्फुटगति है। स्वमन्दस्पुष्टगतिरहिता शीघ्रोच्चगति कुजादि ग्रहों की शीघ्रकेन्द्रगति है।

[Hindi Commentary — Translation] The *manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā* (*prathama-jyā*); the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and other planets, the mean motion minus the *manda-gati-phala* is the *manda-sphu.ta-gati*. The *śīghrocca-gati* minus its own *manda-sphu.ta-gati* is the *śīghra-kendra-gati* of Mars and other planets.

2.5 The Manda Paridhi (Epicycle) Values

नियमाः २०--२१

तद्द्युदलपरिध्यन्तरगुणा हता त्रिज्यया च नतजीवा । ऊने धनमृणमधिके दिनार्धपरिधी स्फुटः परिधिः ॥२०॥ सूर्यस्य तदृणमन्दफले धने मन्दपरिधिर्मध्याह्ने । चन्द्रस्य तदेव ताभिरेव घटिकामितेन्दुरपि तस्मिन्नेव कपाले नतो भवतीति ॥२१॥

Rules 20–21: Manda Epicycle Corrections

[BSS II.20–21]

20: The difference between that [quantity] and the half-diameter of the epicycle (*dyudala-paridhi*), multiplied by the *trijyā* (radius) and by the *nata-jīvā* (sine of altitude), [when the planet is] in decrease (*ūne*) is positive (*dhana*), in increase (*adhike*) is negative (*.rna*); this is the *sphu.ta-paridhi* (true epicycle) of the half-day.

21: For the Sun, that is the *manda-paridhi* in the *.rna-manda-phala* or *dhana-manda-phala* at midday. For the Moon, the same, and by those same [quantities] in *gha.tikās*, the Moon also [is corrected] at that same *kapāla* (division); thus the *nata* (altitude) is obtained.

वि. सा.---चतुर्दशांशाः स्थानद्वये भागद्वयंशेन रहितास्तदा ऋणे वा धने वा मन्दफले सूर्यस्य; दिनमध्ये (मध्याह्ने) मन्दपरिध्यंशा भवन्ति, ऋणे धने वा मन्दफले दिनार्धात् प्राक्कपाले पञ्चदशघटीभिनेतस्य सूर्यस्य दिनमध्यपरिधिमानं क्रमेण भागद्वयंशेनाधिकसूनं कार्यं। चन्द्रस्य ऋणे धने वा मन्दफले दशनद्वितयं जिनलिप्तोनं कार्यमर्थात् स्थानद्वये द्वात्रिंशदंशाख्यचतुरविशतिकलारहितास्तदा मध्याह्ने तन्मन्दपरिध्यंशा भवन्ति।

[Sanskrit Commentary] Fourteen degrees (*catudaṃśāṃśāḥ*), in two places, diminished by two-thirds of a degree (*bhāgadwayaṃśena rahitāḥ*) — then in the *.rna* or *dhana manda-phala* of the Sun; at midday (*dinamadhye*) the *manda-paridhi-aṃśa* are obtained. In the *.rna* or *dhana manda-phala*, before the half-day *kapāla*, for the Sun separated by fifteen *gha.tikās*, the half-day *paridhi* measure is to be made successively increased or decreased by two-thirds of a degree. For the Moon, in the *.rna* or *dhana manda-phala*, twelve [degrees] diminished by two *jina-lipata* (?) — that is, in two places, diminished by thirty-two degrees plus twenty-four minutes (*dvātriṃśadaṃśākhyacaturviṃśatikalā rahitāḥ*) — then at midday those *manda-paridhi-aṃśa* are obtained.

हिन्दी भाषा---चौदह अंश में दो स्थानों में एक अंश के तृतीयांश (बीस कला) को घटाने से सूर्य के ऋणमन्द फल में वा धनमन्द फल में मध्यन्ह काल में मन्द परिध्यंश होता है। ऋणमन्दफल में वा धनमन्दफल में दिनार्ध से पूर्वकपाल में पञ्चदश १४ घटी करके नत सूर्य के मध्यान्ह मन्दपरिधिमान में क्रम से एक भाग के तृतीयांश (२० कला) को युत और हीन करना चाहिये।

[Hindi Commentary — Translation] From fourteen degrees, in two places, diminishing by one-third of a degree (20 *kalās*), in the *.rna-manda-phala* or *dhana-manda-phala* of the Sun, at midday the *manda-paridhi-aṃśa* is obtained. In the *.rna-manda-phala* or *dhana-manda-phala*, in the *kapāla* before half-day, having subtracted fifteen (14?) *gha.tikās* from the *nata* Sun, in the midday *manda-paridhi* measure, successively one-third of a degree (20 *kalās*) is to be added and subtracted.

Summary of Manda Paridhi Values

मन्दपरिधि-सारणी

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Manda-paridhi Summary Table

Below is presented a numerical summary of the *manda-paridhi* (epicycle) values. For each planet the *manda-paridhi* varies up to twenty-four degrees.

[illegible]	[illegible]	[illegible]	**Planet**	**Manda Paridhi**	**Special Notes**
[illegible] (Sun)	[illegible]–[illegible]	[illegible] [illegible]	Sūrya (Sun)	13–14	Varies by anomaly
[illegible] (Moon)	[illegible]–[illegible]	[illegible] [illegible] [illegible]	Candra (Moon)	31–32	Twice Sun's rate
[illegible] (Mars)	[illegible]–[illegible]	[illegible]	Bhauma (Mars)	73–80	Both corrections
[illegible] (Mercury)	[illegible]	[illegible]	Budha (Mercury)	30	Complex
[illegible] (Jupiter)	[illegible]–[illegible]	[illegible]	Guru (Jupiter)	32–36	Both corrections
[illegible] (Venus)	[illegible]–[illegible]	[illegible]	Śukra (Venus)	11–14	Complex
[illegible] (Saturn)	[illegible]–[illegible]	[illegible]	Śani (Saturn)	59–65	Slow motion

2.6 True Daily Motion (स्फुटभुक्ति)

नियमाः स्फुटभुक्तिः

Rules: Daily Motion After Correction

शीघ्रकेन्द्रस्फुटभुक्त्यन्तरगुणा श्रत उक्तलब्धात्संशोध्या ।
शीघ्रगतिर्वक्रगतिरिति सिङ्हवर्म्मोपपन्नश्लोकः ॥

[BSS — True Daily Motion] The difference between the *śīghra-kendra-sphu.ta-bhukti* and [the mean motion], multiplied by the [tabular difference] and corrected from the stated quotient — this is the *śīghra-gati*; [if negative] it is *vakra-gati* (retrograde motion). Thus [goes] the proof-verse of Siṅhavarman.

हिन्दी भाषा---शीघ्रकेन्द्रस्फुटभुक्त्यन्तर (भोग्यखण्ड) से गुणित प्रथमजीवा (प्रथमज्या) से भक्त जो लब्ध फल वह स्फुटपरिधि से गुणित ३६५ भगणांश से भाग देने से जो कला-मिमा प्राप्त होती है वह सकलादिकेन्द्र में अपनी मध्यमगति से ऊना या अधिक होती है, तदानीक केन्द्र में रवि और चन्द्र की स्फुटा गति होती है।

[Hindi Commentary — Translation] The *śīghra-kendra-sphu.ta-bhukti-antara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā*; the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

Geometric Explanation (Upapatti)

उपपत्ति---मध्यस्पष्ट भेदेन गतिस्त्रिविधा भवति, या गतिः प्रतिक्षणं भिन्ना-भिन्ना भवति, सा स्पष्टा अन्यां मध्या, स्फुटा गतिरपि दैनिकतात्कालिकभेदेन द्विविधा भवति, तेन दैनिकमन्दस्पष्टागतिः, तात्कालिकमन्दस्पष्टागतिः। दैनिकस्पष्टागतिः, तात्कालिकस्पष्टागतिः, आचार्येण दैनिकमन्दस्पष्टगतिः, दैनिकस्पष्टगतिश्चानीयते।

[Upapatti — Geometric Proof] By the difference between mean and true, motion becomes threefold. That motion which is different every moment is *spa.s.tā*; another is *madhyā* (mean). The true motion (*sphu.tā gati*) is also twofold by the division into daily and instantaneous: thus the daily *manda-sphu.tā-gati* and the instantaneous *manda-sphu.tā-gati*. The daily *spa.s.tā-gati*, the instantaneous *spa.s.tā-gati* — by the teacher the daily *manda-sphu.ta-gati* and the daily *sphu.ta-gati* are computed.

ग्रहमन्दकेन्द्रगतिज्यान्तर (भोग्यखण्ड) से गुणित प्रथमज्यया भक्त यल्लब्धं तत्स्फुटपरिधिगुणितं भगणांशे ३६५ भक्ते लब्धामिष्कलामिमां शकवर्गादी (सकलादिकेन्द्रे, क्कयादिकेन्द्रे च) स्वमध्यमगतीरूनाधिका तदानीकेन्द्रस्य (रविचन्द्रयोः) स्फुटा गतिर्भवति।

The *graha-manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied by the *prathama-jyā* and divided [by the radius]; whatever quotient is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* (or *kkayādi-kendra*), whether less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

2.7 Special Corrections for Different Planets

विशेषसंस्काराः

Planet-Specific Corrections

Mercury (बुध) and Venus (शुक्र)

वि. सा.---बुधशुक्रयोः शीघ्रोच्चं रवेरर्कात् ग्रहणादितोऽन्यथा। तयोः शीघ्रोच्चं सूर्यसम्बन्धि, मन्दोच्चं स्वतन्त्रमिति विशेषः।

[Sanskrit Commentary] For Mercury and Venus, the *śīghrocca* (apsis of the śīghra epicycle) is [derived] from the Sun; from the seizure of the Sun [it is different], otherwise [it would be the same]. Their *śīghrocca* is connected to the Sun; the *mandocca* (apsis of the manda epicycle) is independent — this is the special feature.

बुध और शुक्र का शीघ्रोच्च सूर्य से ही संबन्ध रखता है, इनका शीघ्रोच्च सूर्यसम्बन्धी है, मन्दोच्च स्वतन्त्र है। इसलिए इन दोनों ग्रहों में विशेष संस्कार की आवश्यकता होती है।

[Hindi Commentary — Translation] The *śīghrocca* of Mercury and Venus is related to the Sun itself; their *śīghrocca* is Sun-related, the *mandocca* is independent. Therefore for these two planets a special correction procedure is necessary.

Mars (भौम), Jupiter (गुरु), and Saturn (शनि)

वि. सा.---भौमादीनां मन्दस्पष्टगतिः स्फुटगतिश्च। मन्दस्पष्टभुक्त्यन्तरात् शीघ्रभुक्तिः शीघ्रफलाधम्भोग्यजीवासंगुणिता मध्यजीवया विभजेत्। लब्धेनोक्तवत्स्फुटभुक्तिः समानीया यदि तया सह मन्दफलाधम्फुटभुक्त्यन्तरा तत्रैव केन्द्रमे कृता भुक्ती धन ऋणां वा कार्यं।

[Sanskrit Commentary] For Mars and the others, the *manda-sphu.ta-gati* and the *sphu.ta-gati*. From the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti* [is obtained]; the *śīghra-phala-ādha-bhogya-jīvā*, multiplied by the *madhya-jivyā* (mean sine), is to be divided. Having brought together the *sphu.ta-bhukti* by the stated quotient; if together with that, by the *manda-phala-ādham-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

भौम आदि ग्रहों की मन्दस्पष्टगति और स्फुटगति---मन्दस्पष्टभुक्त्यन्तर से शीघ्रभुक्ति शीघ्रफलाधम्भोग्यजीवा संगुणित मध्यजीवा विभजेत्। लब्धेन उक्तवत् स्फुटभुक्ति समानीया यदि तया सह मन्दफलाधम्फुटभुक्त्यन्तरा तत्रैव केन्द्र में कृता भुक्ती धन ऋण वा कार्य।

[Hindi Commentary — Translation] For Mars and other planets, the *manda-sphu.ta-gati* and *sphu.ta-gati* — from the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti*; the *śīghra-phala-adha-bhogya-jīvā*, multiplied and divided by the *madhya-jīva*. By the resulting [quotient], the *sphu.ta-bhukti* is to be brought together as stated; if together with that, by the *manda-phala-adha-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

2.8 The Chapter Conclusion

स्पष्टाधिकारोपसंहारः

Epilogue to the Spa.s.tādhikāra

इति ब्राह्मस्फुटसिद्धान्ते स्पष्टाधिकारः द्वितीयः ॥

[End of Chapter 2] Thus the second chapter, the *Spa.s.tādhikāra* (Chapter on True Positions), in the *Brāhmasphu.tasiddhānta.*

हिन्दी भाषा---इस प्रकार से ब्रह्मगुप्ताचार्य द्वारा रचित ब्राह्मस्फुटसिद्धान्त में मध्यमाधिकार (प्रथम अधिकार) और स्पष्टाधिकार (द्वितीय अधिकार) की व्याख्या संपन्न हुई। इसमें ग्रहों की मध्यमगति, स्पष्टीकरण, ज्यागणित, मन्दफल, शीघ्रफल, देशान्तर, नतकर्म आदि विषयों का विस्तृत वर्णन किया गया है।

[Hindi Commentary — Translation] In this manner, the explanation of the *Madhyamādhikāra* (first chapter) and the *Spa.s.tādhikāra* (second chapter) in the *Brāhmasphu.tasiddhānta*, composed by Ācārya Brahmagupta, has been completed. In this, the mean motions of planets, *sphu.tīkara.na*, sine mathematics (*jyā-ga.nita*), *manda-phala*, *śīghra-phala*, *deśāntara*, *nata-karma*, and other subjects have been described in detail.

[[illegible] [illegible] [illegible] [illegible], [illegible], [illegible], [illegible] [illegible] [illegible] [illegible]–[illegible] [illegible]]

[The Spa.s.tādhikāra continues with detailed calculations for each planet's manda and śīghra corrections, tables of paridhi values, geometric upapattis (proofs), and special rules for Mercury and Venus across pages 896–978 of the original edition.]

Editorial Notes

संपादकीयटिप्पनयः

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मुख्यगणितविषयाः

[illegible]. [illegible]—[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

[illegible]. [illegible]—[illegible] [illegible] ([illegible] [illegible]), [illegible], [illegible] [illegible] [illegible]

[illegible]. [illegible]—[illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible]

[illegible]. [illegible]—[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

टङ्कनतान्त्रिकटिप्पनयः

- [illegible]: [illegible]atek (XeLaTeX) [illegible]
- [illegible] [illegible]: Noto Serif Devanagari
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About this Edition

This typeset edition reproduces the content of Brahmagupta's *Brāhmasphu.tasiddhānta* as found in the Sanskrit–Hindi commentary edition (pages 839–978), presented in a bilingual parallel-column format with English translation and IAST transliteration.

Notable Mathematical Content

1. **Sine tables:** The 24 sines given by Brahmagupta are notable for their precision. The values use the *bhūta-saṅkhyā* (word-numeral) system where each number is encoded in poetic words.
2. **Longitude correction (deśāntara):** The formula for computing longitude differences uses the relationship between terrestrial circumference (5000 yojanas), latitude difference, and the resulting time correction.
3. **Planetary equations:** The *manda* and *śīghra* equations represent sophisticated epicyclic theory, with different epicycle sizes for each planet that vary with anomaly.
4. **Geometric proofs (upapatti):** The commentary includes detailed geometric justifications using trigonometric relationships in the epicycle model.

Technical Notes on Typesetting

- **Engine:** XeLaTeX with full Unicode support
- **Devanagari font:** Noto Serif Devanagari
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- **Source:** Scanned pages 839–978 of the Brahmasphutasiddhanta Part 2 edition

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क्षितिज्या
लग्ननयनम् --- उदयराशिसाधनम्
गणितीय सूत्राणि --- □□□□□□□□□□□□ □□□□□□□□ □□□ □□□□□□□□□□□

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ग्रहणमध्यम्

सूर्यग्रहणम् --- रविग्रहणानयनम्
लम्बनम् --- देशान्तरलम्बनानयनम्
ग्रहणस्पर्श-मोक्षयोः

एतत् पाठं ब्रह्मगुप्तस्य ब्रह्मस्फुटसिद्धान्तस्य द्वितीयभागस्य द्वितीयखण्डं समाविशति। एषः पाठः स्पष्टाधिकारं, त्रिप्रश्नाधिकारं, चन्द्रग्रहणाधिकारं, तथा गणितीयसूत्राणि च समाविशति।

स्पष्टादिकारः

अस्याध्यायस्य विषयः --- मध्यमागतस्य ग्रहस्य स्पष्टस्थानस्य निर्धारणम्। ये समाकलनाः ग्रहकक्षाया अणुता च परिमण्डलवृत्तस्य व्याख्यां सम्मिलयन्ति, ते स्पष्टीकरणस्य अस्याध्याये दीयन्ते।

श्लोक ४८--४९

वा भवतीत्यर्थः। पादे शीघ्रकेन्द्रमर्धकं तदतीतस्य प्रथ वक्रानुवक्रकेन्द्रमर्धकमतस्तस्यैव वक्रानुवक्रदिनं ज्ञात्वा सकृद्ग्रहः स्पष्टः कार्यस्तदर्धयोःशीघ्रकेन्द्रवक्रानुवक्रो निक्षिप्याविति प्रत्यय वासना। मध्याच्छीघ्रनीचोच्चवृत्तमध्य यावदूर्ध्वं-कोटिः परमफलज्यातुल्यं शीघ्रनीचोच्चवृत्तमध्यमार्धे भुजा। पूर्वापरेया वा तयो-र्गुणितं मूलं त्रयंककर्णः परमफलज्या प्रोक्ता मध्य यावतस्य कर्णस्य शीघ्रनीचोच्च-वृत्ताशालाकायाःचान्नरे यावच्छीघ्रनीचोच्चवृत्तसूर्याद्येन प्रमिति प्रतिमण्डलपरिधौ-स च तत्र प्रदेशे परच्छादग्छन्नुत्सन्नते। तस्माद्दुर्न्नने सर्वं गोले दर्शयेत्। ॥४८--४९॥

वि. भा.---प्रत्येकटिमार्गपुमनुर्मीरित्यादिपञ्चशीघ्रकेन्द्रांशौभौमादिद्वयंहरणां वक्रं भवेद्यदिरचितैस्तैः शीघ्रकेन्द्रांशैस्तेषां वक्रारम्भौ भवति, तद्धितैश्चक्रांशो ३६०□अनुवक्रमध्यमगैरमः। मन्दफलस्पष्टमुक्तिः (मन्दस्पष्टागतिः) तदुग्ना (तद्धिता) शीघ्रमुक्तिः (शीघ्रस्पष्टिः) शीघ्रकेन्द्रगतिर्भवति तथा तदधिकोन-मागकलामकालस्तथा गतेन दिवसा भवति, शीघ्रात्यकेन्द्रभागैरसकृद्धिनाउच्चैःपकर्माणां स्थिरो भूतः केन्द्रांशैरिति ॥४८--४९॥

मध्यमागमः --- ग्रहाणां मध्यमानयनम्

भचक्रभागान् प्रथमं दिनैर्हत्वा भागलब्धिकाम्
कल्पे युगे वाथ विभजेत् तैः स्यात् स पूरकः क्षेपः ॥१२॥

स पूरकः क्षेपो गतदिनैर्गुणितो रविवत्सरैः
हृतः स मध्यभुक्तिः स्यात् तद्युक्तं मध्यमं ग्रहस्य ॥१३॥

वा. भा.---भचक्रभागान् द्वादशराशिभागान् अर्थात् ४३२०० भागान् प्रथमं दिनैः गुणित्वा रविवत्सरैः १५७७९१७८२८० विभजेत्। ततो लब्धिः पूरकः क्षेपः स्यात्। स एव पूरकक्षेपो गतदिनैर्गुणितो रविवत्सरैर्हृतः स मध्यभुक्तिः स्यात्। सा मध्यभुक्तिः पूर्वमध्यमे युक्ता वा शोधिता वा ग्रहस्य मध्यमं स्यात् ॥१२--१३॥

मन्दोच्चं शीघ्रच्चं च युगपूरकक्षेपभक्तं तैः
गतदिनैर्गुणितं तत् स्यात् मन्दोच्चं शीघ्रोच्चं वा ॥१४॥

तत् स्यात् मन्दोच्चं शीघ्रोच्चं वा युगीयात् पूर्वमन्दोच्चात्
शीघ्रोच्चाद्वा युतं वा शुद्धं वा मन्दशीघ्रोच्चं स्यात् ॥१५॥

वा. भा.---मन्दोच्चं शीघ्रोच्चं च गतदिनैर्गुणितं युगपूरकक्षेपभक्तं स्यात्। तत् पूर्वमन्दोच्चाद्वा शीघ्रोच्चाद्वा युक्तं शुद्धं वा कृत्वा मन्दोच्चं शीघ्रोच्चं वा स्यात्। एवं ग्रहस्य मध्यमं मन्दोच्चं शीघ्रोच्चं च ज्ञात्वा स्पष्टीकरणस्य आधारः सिद्ध्यति ॥१४--१५॥

मन्दस्पष्टीकरणम् --- परमफलनयनम्

मन्दकेनापक्रान्तं मन्दफलं तत् परमं ग्रहस्य
ज्यार्धेन हृत्वा लब्धं तत् परमं फलं ग्रहस्य ॥१६॥

मन्दफलज्यया हत्वा परमफलज्यां तु यल्लब्धम्
तत् परमं फलं स्यात् तेन मन्दस्पष्टः स ग्रहः ॥१७॥

वा. भा.---मन्दकेन दोःकोट्योः कृतिभ्यां पदं मन्दकर्णः। तस्य मन्दज्या हत्वा त्रिज्यां विभज्य मन्दफलं लभ्यते। तत् परमं फलं ग्रहस्य मध्यमात् शोधयित्वा योजयित्वा वा मन्दस्पष्टः स ग्रहः स्यात् ॥१६--१७॥

श्लोक ५०--५१

मार्गनियमरचैपलक्षा यन्त्राभियोगातिशयाच्च तस्मादुपपन्नं कक्षामण्डलादिषु-
विन्यस्तेन एते बोधायास्तमयभागा अविभक्ते, ग्रहे विभक्ते मण्डलवराग्दुर्घटते,
तद्दुःखमुद्यास्तमदण्डायो भविष्यतीति ॥५०--५१॥

वा. भा.---प्रयुग्ना बुधशुक्रयोर्दयास्तर्यपरिक्षानार्थमायातिमाह। शीघ्रात्यकेन्द्रमार्गो-रित्यनुवर्तन्ते खराः ५० एतावद्बुधः सवितःशीघ्रकेन्द्रमार्गैर्बुधस्य परचात् उदयो भवति,
जिनः २४ एतावद्बुधः शुक्रयोदयपरचात् इष्टतिथिमः १५५ एतावद्बुधरव-शीघ्रात्यकेन्द्रमार्गः परचादस्तमयो बुधस्य मुनिनगेन्दुभिः १७७ एतावद्बुधः परचात्, शुक्रस्य-स्तमयः उदयास्तमयो व्यस्तो मण्डलमार्गैस्तदुग्नः प्राक् ॥५२--५३॥

त्रिप्रश्नाधिकारः

अस्याध्यायस्य विषयः --- दैनिकगतिस्य त्रयः प्रश्नाः: लम्बनस्य (देशान्तरस्य) नाडीविशेषस्य छायानयनस्य च गणनम्।

श्लोक ५४--५५

इदानीं कुजादिद्वयहरणामुदयास्तकेन्द्रांशानाम्
षष्ट्यंशः २८ कृतचक्रे १४ मुनिनगेन्दुभिः १७ भौमजीवरविबुधानाम्
उदयः प्रागस्तमयस्तदुच्चकांशाः पश्चात् ॥५४॥

स्वारे ५० जिने २४ क्ष सितयोरिष्टितिथिमि १५५ मुनिनगेन्दुभिः १७७
पश्चात्
उदयास्तमयो व्यस्तो मण्डलमार्गैस्तदुग्नः प्राक् ॥५५॥

वा. भा.---शीघ्रात्यकेन्द्रमार्गरत्यमौमीमस्य प्रागुदयो भवति २८
स्वातीशी-घ्रकेन्द्रमार्गैः कृतचक्रे १४ जीवस्य प्रागुदयो भवति, स्वातीघ्रकेन्द्रमार्गैर्मुनिनगेन्दुभिः
१७ शनैश्चरस्यो भवति। प्रागस्तमयास्तु परचाद्दृश्यति। चक्रांशैस्ते
उन्नतदुग्नाः यथा-स्वोदयमार्गैःस्नाच्च कर्कांशाका ये विशेषा भवतीत्यर्थः।
चक्रांशाकाः प्रसिद्धा एव तथाऽपि भौमस्यास्तमयशीघ्रकेन्द्रभागाः ३३२
जूतो ३८५, शनैः ३५३, सवितुः/बुधस्य-स्तकेन्द्राद्यैः, राश्यादिकानि।
भो। उ। २८ जौ उ। १४ धा उ। १७ शौ १५५ भो १७७

श्लोक ५६--५७

भ्रात्रातीतानां ग्रहाणां दर्शने प्राप्नोत्। तद्धि-कोनां भागकला मन्दफलस्पष्टयुक्तयुक्षीघ्रयुक्या।
हृता दिवसा इति न्यायेनोप-पत्तिः। तथाऽपि रविकक्षया सर्वथा
स्पष्टत्वातिमद्धंले नीचोच्चवृत्तमध्ये भ्रमति।
ततो यदा परमे प्रतिमण्डलोन्नतप्रदेशे ग्रहः स्थितो भवति तदा
समापेक्षं क्षोभो-भवति समरव रविरत्र एव सूर्यस्तदा ग्रहो नोपलभ्यते।
मनाऽपि रविकिंरव्य-परिहृतिर्भवतत्ते यथा-यथा ग्रहोदयस्तमयोर्दृश्यतां
प्रथममेवोदयं यान्ति ततः-शीघ्रात्यादर्कस्यास्तमयेऽपि परिवर्त्यं शीघ्रमार्गेणाकः
पुनरुप्तं ह्रासमभ्येति पश्च-मादिर्मावात्। अतएवास्तमयोदं ग्रह उपलभ्यते।
उदयास्तमयो यदा भवतो

इदानीं पञ्चजयानयनमाह

जिनमागज्यागुणिता सूर्यज्या व्यासदलहृता लब्धम्
इष्टक्रमजीवा विषुवद्दर्द्धिविरणा सवितुः ॥५८॥

इष्टक्रमभागे त्रिज्यावर्गाद्ध्रोत्वं शेषपदम्
विषुवद्दर्द्धिविरणात् स्वाहोरात्राद्विवस्कमः ॥५९॥

क्रान्तिज्या विषुवच्छायया गुणा द्वादशोद्धृता स्थितिजा।
स्वाहोरात्रनुष्टा व्यासार्धेनाहता मता ॥६०॥

स्वाहोरात्रार्धेन सम्बुद्धिज्याचनुष्करप्रमाणः।
ते पृथक्ता विच्छिन्ना विच्छिन्ना नागिका शुष्का ॥६१॥

श्लोक ६२--६३

द्वादशांशोनतं जीवा कार्याः ताश्च भूतादिनगमनानां १३२५
दुःखयदनाख्यमा १८६३ मिथुनस्य वमनिरदा ३०३० अनः सवातीतानां
प्राप्य-दुर्बलतीर्थाः। जिनमागज्यागुणानां सूर्यज्योत्नियमेन पुमनु क्षोणिज्या
कार्या-ताश्च मैथुनं वैद्यमठद्विकलमाधात् ६६□३०' बुधस्य छद्धरभवाः
१९□५९ मिथुनस्य
नवरदचक्रा १३□३०। एनाभिरेव प्राप्तवत्। स्वातीतग्रहाणां त्रिज्याकर्मवर्षो-
च्चयाख्याया गुणोत्कयादिना प्राप्तवत्। तद्वेव विषु-वच्छायाया कार्या।
ताश्च स्वातीत विज्ञेय स्वदेशज्याचरदलसंस्कृतां पुमनु-प्राणा भवन्ति।

श्लोक ६४--६७

तदानीं मेषादिनगानां चरखण्डयाथनमाह
मेषकूर्पंमिथुनानां जीवा कार्याः ताश्च भूतादिनगमनानां १३२५
दुःखयदनाख्यमा १८६३ मिथुनस्य वमनिरदा ३०३० अनः सवातीतानां
प्राप्य-दुर्बलतीर्थाः प्राप्तवत् ॥६४॥

मेषकूर्पंमिथुनानां त एव क्षेत्रा कर्कीसहकन्यानामथ क्षेत्रा तुलाद्विश्चक-घटयो मकरकुम्भमीनानामथवाम्ना। त्वकाल्यां स्वाहोरात्रनुतानि विनस्य प्रदर्शयेत, स्पष्टाद्यध्याय एव चरदलानयने मया प्रदर्शितानि विदुषामपि-प्रदर्शयेते स्वदेशे। यद् क्षिति स्वाहोरात्रनुते क्षितिज्योच्छाययोरन्तरे प्राणाः
यदेव लग्नाकारबुद्धिरस्य तदन्तरकालं पृच्छतीति।

छायाकरणी

हृज्या द्वादशगुणिता विभाजिता शङ्कुना फलं छाया।
व्यासार्धं छेदहतं विषुवत्कर्णहृतं कर्णः ॥२८॥

विषुवत्कर्णगुणायामा व्यासार्धकृतिः फलं सौम्ये।
मयुखद्विज्याहीनं युक्तं याम्ये धनुष्करप्रमाणः ॥२९॥

सौम्ये युतं विहीनं याम्ये प्रागपरयोः प्राणाः ॥३०॥

भ्रष्टो गतविशेषाः फलमत्याया विशेष शेषस्य।
धनुःरविकर्मजीवार्धिः पूर्वापरयोर्नतप्राणाः ॥३१॥

> वा. भा.---यस्यक्षायाया दिनगतिशेषानयनमिष्यते तस्यक्षायायाः छाया-कर्णां कृत्वा तेन स्वाहोरात्राद् गुण्येत्। ततस्तेन स्वाहोरात्रार्धेन छायाकर्णहृतेन-मताया कृता व्यासार्धकृतिः कि मृत्या विषुवत्कर्णगुणायामाः फलं ज्या भवति।
> सफलं सौम्ये गोले छायुबुद्धिज्याहीनं याम्ये तथैव युक्तं कृत्वा यद्भवति। तस्य धनुः-क्षेत्राणां कार्यस्तदनुस्तद्वैध्वंसिकचरदलप्राणौः सौम्ये गोलेषु युतं याम्येहीनं कृत्वा-प्रागपरयोः प्राणा भवति।

प्रकारान्तरेण छायाकरणी

व्यासार्ध कृतिं हृता विषुवत्कर्णेन या भवेत् कर्णः।
लम्बगुणो वा घातः शङ्कुकृतिसार्धकृतिभक्तः ॥३२॥

घातो वा शङ्कुगुणक्रियया विषुवत्कर्णेन विभुक्तः शङ्कुः।
कर्णकृतिः संशोध्य द्वादशभागे पदं छाया ॥३३॥

नतकालानयनम्

स्वाहोरात्रार्धेन छायाकर्णेन मतायाः।
विषुवत्कर्णगुणायामा व्यासार्धकृतिः फलं सौम्ये ॥३४॥

मयुखद्विज्याहीनं युक्तं याम्ये धनुष्करप्रमाणः।
सौम्ये युतं विहीनं याम्ये प्रागपरयोः प्राणाः ॥३५॥

भ्रष्टो गतविशेषाः फलमत्याया विशेष शेषस्य।
धनुःरविकर्मजीवार्धिः पूर्वापरयोर्नतप्राणाः ॥३६॥

लम्बाक्षज्यानयनम्

शङ्कु लम्बकछायाख्यया तद्गुणस्युत्तरैं लब्धम्।
विशुद्धिर्विषुवत्कर्णाक्षेत्राक्षयाक्षोच्चनता शङ्कुः ॥५७॥

उन्नतजीवाकोटिज्याह्रिया हृज्या भुजो नतज्या वा।
कर्णाक्षेत्रावृत्ते व्यासार्धं द्व्यमतोद्यन्न ॥५८॥

शङ्कु क्षायाकृत्योरित्युक्त्याकृतितत्समासपुरुषष्टकयोः।
भूते लम्बाक्षज्ये तदंशकलास्तदनुमागाः ॥५९॥

विषुवत्कर्णहृते वा शङ्कु क्षायाहृते पृथक् त्रिज्ये।
अक्षक्षेत्रर्ज्ये लम्बाक्षज्योत्क्षमध्यने ॥६०॥

नतलम्बाक्षाक्षान् प्रोह्य ज्या वैतराश्लब्धकष्ट्ये।
शङ्कु क्षायागुणिक्ते क्षायद्विघातहृते वास्ते ॥६१॥

लम्बाक्षकलाभागं शेषं त्रिज्याकृतेः पदं भाज्याः।
भनुज्या सर्वोत्क्रान्तज्याविवराणांऽघनमेव ॥६२॥

पुनः प्रकारान्तरेण लम्बाक्षज्ये

लम्बाक्षज्ञानं नवतः प्रोक्ष (हृत्वा) ज्या साध्या तदा वा (प्रकारान्तरेण) इनता ज्या स्वाध्यर्धल्पांशयोर्नवतेर्ज्योर्ज्याज्या, अक्षांशोन्नतेज्योर्ज्या लम्बज्या,
अक्षलम्बज्ये-शङ्कुच्छायागुणिक्ते (द्वादशपलमागुणिक्ते) छायाद्वादशहृते (पलमा-द्वादश भक्ते) तदा वा (प्रकारान्तरेण)इष्ठे लम्बाक्षज्ये भवत् इति ॥९१॥

उपपत्तिः --- ज्या (९०---अक्षांश) = लम्बज्या, ज्या (९०---अक्षांश) = मध्यज्या, वा

$$\frac{\text{अक्षज्या} \times}{\text{पलमा}} = \text{लम्बज्या}, \quad \frac{\text{पलमा} \times \text{लम्बज्या}}{} = \text{मध्यज्या}$$

$$\frac{\times 12}{} = , \frac{\times}{12} =$$

एतावत्यक्षाज्या-लम्बोपपत्तेः ॥९१॥

नतकालानयनम् (पुनः)

उदयास्तमयैः क्षेपैर्द्वादशाभिर्नतं याम्योदयात्।
भागैर्लब्धं चलं तद् दोर्ज्याविवरं च स्फुटम् ॥६३॥

तन्मण्डलं प्रोक्ष्य तज्जं शङ्कुना लम्बकेन वा हतम्।
द्युज्यया विभक्तं लब्धं नतकालं प्रचक्षते ॥६४॥

अर्कक्षेपैर्विनता दोर्ज्योत्क्रान्तिज्या नतकालहृते।
तज्जं त्रिज्याहतं द्युज्यया विभक्तं शङ्कुरिष्यते ॥६५॥

नतोत्क्रान्तज्ययोर्घाते द्युज्याकृतेस्तदंशकं शङ्कुः।
छायाहते द्वादशभागे पदं भवति तत्क्षेपः ॥६६॥

वा. भा.---उदयास्तमयक्षेपैर्द्वादशाभिर्नतं याम्योदयात् भागैर्लब्धं चलं तद् दोर्ज्याविवरं-च स्फुटम्। तन्मण्डलं प्रोक्ष्य तज्जं शङ्कुना लम्बकेन वा हतं द्युज्यया विभक्तं लब्धं-नतकालं प्रचक्षते। अर्कक्षेपैर्विनता दोर्ज्योत्क्रान्तिज्या नतकालहृते तज्जं त्रिज्याहतं-द्युज्यया विभक्तं शङ्कुरिष्यते। नतोत्क्रान्तज्ययोर्घाते द्युज्याकृतेस्तदंशकं शङ्कुः-छायाहते द्वादशभागे पदं भवति तत्क्षेपः ॥६३--६६॥

गोलयन्त्रम् --- खगोलयन्त्रनिर्माणम्

सप्तरश्मिः खगोलः स्यात् तन्मध्ये भूगोलसंस्थितिः
दण्डैः संयोजितः सोऽयं खगोलो यन्त्ररूपकः ॥१३७॥

क्रान्तिवृत्तं सममण्डलं दक्षिणोत्तरवृत्तमेव च
सद्यत्क्षितिजमेवाऽपि गोले निर्माय तद्विदैः ॥१३८॥

यन्त्रेणैव ग्रहाणां स्थितिः स्फुटा ज्ञायते येन
तस्माद् यन्त्रं प्रकल्प्यं गोलविद्भिर्निरन्तरम् ॥१३९॥

वा. भा.---सप्तरश्मिः खगोलः स्यात् तन्मध्ये भूगोलसंस्थितिर्दण्डैः संयोजितः सोऽयं खगोलो यन्त्ररूपकः। क्रान्तिवृत्तं सममण्डलं दक्षिणोत्तरवृत्तमेव च सद्यत्क्षितिजमेवापि गोले निर्माय तद्विदैः यन्त्रेणैव ग्रहाणां स्थितिः स्फुटा ज्ञायते येन तस्माद् यन्त्रं प्रकल्प्यं गोलविद्भिर्निरन्तरम् ॥१३७--१३९॥

प्राणगणितम् --- दिगंशानयनम्

षष्टिर्घटिका नाडी षष्टिर्नाड्यः प्राणसञ्ज्ञकाः
षष्टिर्विपलानि पलान्येतैः कालगणनं विदुषाम् ॥१२८॥

चरज्यां हत्वा त्रिज्यया विभज्य लब्धाः प्राणा भवन्ति
तेषां योगः स्थितिकालः स्यादयनकालस्तु तद्वदेव ॥१२९॥

तनुज्या हृत्वा त्रिज्यया विभज्य लब्धं तत् प्रमाणकालः
स्थितौ चरज्यया हृत्वा लब्धाः प्राणाः स्फुटास्तु ते ॥१३०॥

वा. भा.---षष्टिर्घटिका नाडी षष्टिर्नाड्यः प्राणसञ्ज्ञकाः षष्टिर्विपलानि पलानि एतैः कालगणनं विदुषाम्। चरज्यां हत्वा त्रिज्यया विभज्य लब्धाः प्राणा भवन्ति तेषां योगः स्थितिकालः स्यादयनकालस्तु तद्वदेव। तनुज्या हृत्वा त्रिज्यया विभज्य लब्धं तत् प्रमाणकालः स्थितौ चरज्यया हृत्वा लब्धाः प्राणाः स्फुटास्तु ते ॥१२८--१३०॥

मध्यच्छायानयनम्

छायाक्षेत्रानुपातेन त्रि.१२/शङ्कु = छायाकर्णं ततः $\sqrt{\text{छायाक}^2 - {}^2}$
= छाया,
इति ॥१४८॥

$\frac{\times 12}{} =$

$\sqrt{{}^2 - 12^2} =$

हिं. भा.---त्रिज्या को बारह से गुणा कर मध्यशङ्कु से भाग देने से मध्यच्छाया कर्णं होता है, छायाकर्ण और द्वादश (१२) का---वर्गन्तर मूल प्रकारान्तर से मध्यच्छाया होती है इति ॥१४८॥

उपपत्तिः---पूर्व श्लोक की उपपत्ति में प्रदर्शित छायाक्षेत्रों के सजातीयत्व से अनुपात करते हैं

$\frac{\times 12}{} =$

$$\frac{\text{त्रि.} \times}{\text{मध्यशङ्कु}} = \text{मध्याकर्ण} \qquad \sqrt{{}^2 - 12^2}$$

$\therefore \sqrt{\text{छायाक}^2 - {}^2}$ = मध्यच्छाया, इति ॥१४८॥

महापातः --- चान्द्रमासनिर्माणम्

महापातस्तु यः स्याच्चन्द्रसूर्ययोः समागमे
तदा तयोरन्तरं शून्यं स्यात् समे□□□□□□□◌े तयोः ॥१४०॥

चान्द्रो मासस्तु यः स्यात् तिथीनां सङ्ग्रहेण तु
स तु सावनोऽपि मासः सूर्येणैकेन गम्यते ॥१४१॥

तयोर्मासयोगेन संवत्सरः प्रकीर्तितः
पञ्चदशभ्यस्तिथिभिः संयुक्तो मास उच्यते ॥१४२॥

> वा. भा.---महापातस्तु यः स्याच्चन्द्रसूर्ययोः समागमे तदा तयोरन्तरं शून्यं स्यात् समे□□□□□□□◌े तयोः। चान्द्रो मासस्तु यः स्यात् तिथीनां सङ्ग्रहेण तु स तु सावनोऽपि मासः सूर्येणैकेन गम्यते तयोर्मासयोगेन संवत्सरः प्रकीर्तितः पञ्चदशभ्यस्तिथिभिः संयुक्तो मास उच्यते ॥१४०--१४२॥

क्रान्तिः --- अयनांशकल्पनम्

मेषादौ रवौ तु षट्षष्टिघ्ने रवौ तु विषुवति
मेषादौ तत्र क्रान्तिर्ज्या तु विषुवज्ज्या मता ॥१३१॥

तत्क्षेपैर्विहृता ज्ञेया क्रान्तिः सा स्फुटतर्किभिः
उत्तरायणे दक्षिणायने याम्या सा तु तत्क्षेपैः ॥१३२॥

अयनांशः स तु यतः क्रान्तेर्भवति विशेषणात्
तस्मादयनचलनं ज्ञात्वा स्फुटं विधिवद् भवेत् ॥१३३॥

> वा. भा.---मेषादौ रवौ तु षट्षष्टिघ्ने रवौ तु विषुवति मेषादौ तत्र क्रान्तिः स्यात् तत्क्षेपैर्विहृता ज्ञेया क्रान्तिः सा स्फुटतर्किभिः उत्तरायणे दक्षिणायने याम्या सा तु तत्क्षेपैः। अयनांशः स तु यतः क्रान्तेर्भवति विशेषणात् तस्मादयनचलनं ज्ञात्वा स्फुटं विधिवद् भवेत् ॥१३१--१३३॥

क्षितिज्या

प्रसिद्धा। याम्ये याम्यभुजोन्नतोर्ज्वं लन्तं रमद्गुप्तरमुजेनगोरेक्यं
यतल्लम्बगुणां छायाकर्णोद्धृतं फलं किन्तः क्रान्तिज्या स्यादिति।

लग्ननयनम् --- उदयराशिसाधनम्

रवेस्तु राश्युदयाद् घटिकाः प्रोद्यमानराशिस्थितेः
तास्तत्कालात् शोधयित्वा यावद् राश्युदयाः स्युः ॥१३४॥

तास्तत्पूर्वराशिभिः संयुक्ता लग्नं तु तत् स्यात्
शेषं त्रिंशता हृत्वा लब्धं तदंशकलात्मकम् ॥१३५॥

तद् राशौ योजयेत् तस्य लग्नं स्यात् स्फुटतर्किभिः
एवं लग्नं सदा कार्यं ग्रहगणितविदा बुधैः ॥१३६॥

> वा. भा.---रवेस्तु राश्युदयाद् घटिकाः प्रोद्यमानराशिस्थितेः तास्तत्कालात् शोधयित्वा यावद् राश्युदयाः स्युः तास्तत्पूर्वराशिभिः संयुक्ता लग्नं तु तत् स्यात् शेषं त्रिंशता हृत्वा लब्धं तदंशकलात्मकं तद् राशौ योजयेत् तस्य लग्नं स्यात् स्फुटतर्किभिः एवं लग्नं सदा कार्यं ग्रहगणितविदा बुधैः ॥१३४--१३६॥

गणितीय सूत्राणि --- □□□□□□□□□□□□ □□□□□□□□ □□□ □□□□□□□□□□□

अस्य विभागस्य विषयः --- स्पष्टाधिकारात् त्रिप्रश्नाधिकारात् च प्रमुखाणि गणितीयसूत्राणि यानि ज्यापटलनिर्माणे, छायागणने, नतकालानयने, तथा पाटीगणिते प्रयुज्यन्ते।

शीघ्रस्पष्टीकरणसूत्रम्

शीघ्रफलोज्या $\times$ शीघ्रकर्ण = स्पष्टः

केन्द्रकर्णः = $\sqrt{\text{त्रिज्या}^2 + \text{मन्दफलोज्या}^2 - 2 \cdot \text{मन्दफलोज्या} \times \text{कल्पज्या}}$

ज्योत्पत्तिः --- त्रिकोणमितीय ज्यापटलनिर्माणम्

व्यासाद् व्यासभागानां त्रिभिरूनं शैलैस्त्रिभिरूनान्
तान् कृत्वा षड्भिर्भक्तान् शेषैस्त्रिज्यां प्रकल्पयेत् ॥१००॥ $R \approx 3438'$

पूर्वपूर्वज्यया हत्वा पूर्वपूर्वविशेषखण्डान्
त्रिज्यां हृत्वा ततोऽन्या ज्याः सन्ति तत्प्रभवा मता ॥१०१॥

वा. भा.---व्यासाद् व्यासभागानां त्रिभिरूनान् अर्थात् ३४३८--३ = ३४३५ भागान् कृत्वा षड्भिर्भक्तान् शेषैस्त्रिज्यां प्रकल्पयेत्। अनया त्रिज्यया पूर्वपूर्वज्यया पूर्वपूर्वविशेषखण्डान् हत्वा त्रिज्यां हृत्वा ततोऽन्या ज्याः सन्ति ताः पूर्वज्याप्रभवा मताः ॥१००--१०१॥

प्रथमा ज्या त्रिज्याखण्डा द्वितीया ततः शिष्टस्यार्धात्
तृतीयापि तथैव सा सूत्रं ब्रह्मगुप्तमतम् ॥१०२॥

तासां विशेषखण्डानां योगे ज्यावलयः स्मृताः
षडंशाभ्यन्तरज्यास्ता विख्याताः स्पष्टतर्किकैः ॥१३३॥

वा. भा.---प्रथमा ज्या सप्तभुजज्या त्रिज्याखण्डा द्वितीया चतुर्दशभुजज्या तृतीया तथैव क्रमेण। तासां विशेषखण्डानां योगे ज्यावलयः स्मृताः षडंशाभ्यन्तरज्यास्ता विख्याताः स्पष्टतर्किकैः ॥१०२--१०३॥

छायाकरणीसूत्राणि

छाया = $\dfrac{\text{द्वादश} \times \text{हृज्या}}{\text{शङ्कु}}$

कर्णः = $\dfrac{\text{व्यासार्धकृतिः}}{\text{विषुवत्कर्ण}}$

छायाकर्ण = $\sqrt{\text{छाया}^2 + \text{शङ्कु}^2}$

गौरीगुप्ता --- द्वितीयान्तरकल्पना

ज्यावर्गैर्भुजज्यावर्गैस्त्रिज्यावर्गहृतैर्विशोध्य
लब्धैस्तैः प्रथमज्यां शोध्यैस्ततोऽपरा ज्या स्यात् ॥१०४॥

पूर्वोत्तरज्योरन्तरं प्रथमखण्डं ततोऽपरं खण्डम्
तस्मात् पूर्वखण्डाच्छोध्यं तृतीयं खण्डं ततः स्यात् ॥१०५॥

एवं क्रमेण सर्वा ज्याः साधनीया विभक्तज्यातः
अन्तरज्याः पृथग्वा स्फुटतरं फलमानयेत् ॥१०६॥

विषमज्ञे द्विज्यावर्गाद् द्विज्यावर्गं विशोध्य शेषात्
पदं द्वितीयं खण्डं ततः पूर्वखण्डात् शोध्यम् ॥१०७॥

तृतीयं खण्डं ततः पूर्वद्वयात् शोध्यं चतुर्थं ततः
पूर्वत्रयात् शोध्यं च क्रमेण सर्वखण्डानि ॥१०८॥

वा. भा.---ज्यावर्गैर्भुजज्यावर्गैस्त्रिज्यावर्गहृतैर्विशोध्य लब्धैः प्रथमज्यां शोध्य ततोऽपरा ज्या स्यात्। पूर्वोत्तरज्योरन्तरं प्रथमखण्डं ततोऽपरं खण्डं तस्मात् पूर्वखण्डाच्छोध्यं तृतीयं खण्डं ततः स्यात्। एवं क्रमेण सर्वा ज्याः साधनीया विभक्तज्यातः अन्तरज्याः पृथक् स्फुटतरं फलमानयेत् ॥१०४--१०६॥

वा. भा.---विषमज्ञे द्विज्यावर्गात् द्विज्यावर्गं विशोध्य शेषात् पदं द्वितीयं खण्डं स्यात्। ततः पूर्वखण्डात् शोध्यं तृतीयं खण्डं ततः पूर्वद्वयात् शोध्यं चतुर्थं ततः पूर्वत्रयात् शोध्यं च क्रमेण सर्वखण्डानि सिध्यन्ति ॥१०७--१०८॥

लम्बाक्षज्यासूत्राणि

$$\text{लम्बज्या} = \frac{\text{अक्षज्या} \times}{\text{पलमा}}$$

$$\text{मध्यज्या} = \frac{\text{पलमा} \times \text{लम्बज्या}}{}$$

नतकालानयनसूत्राणि

$$\text{नतकालः} = \frac{\text{तज्ज्यं} \times \text{शङ्कु □□ लम्बक}}{\text{द्युज्या}}$$

$$\text{शङ्कुः} = \frac{\text{तज्ज्यं} \times \text{त्रिज्या}}{\text{द्युज्या}}$$

क्रान्तिज्यानयनसूत्राणि

$$\text{क्रान्तिज्या} = \frac{\text{लम्बज्या} \times \text{क्रम}}{\text{वि}} = \frac{\text{लम्बज्या} \times \text{कर्णोच्चताम्} \times \text{वि}}{\text{छाक} \cdot \text{वि}}$$

$$\text{क्षेत्रज्या} = \frac{\text{क्रान्तिज्या} \times \text{वि}}{\text{छाक}}$$

पाटीगणितम् --- चतुर्विंशतिपरिकर्माणि

संकलितं व्यवकलितं गुणनं भाजनं तथा
वर्गो वर्गमूलं घनो घनमूलं पञ्चमी त्रयः ॥११७॥

पञ्च विशेषाः परिकर्माण्येतानि गणितं स्मृतम्
प्रकीर्तितान्याचार्यैरेतैः सर्वं गणितं सिध्यति ॥११८॥

रूपैकदेशसंयुक्ता व्यक्तराशिरुदाहृता सा
अव्यक्ता या राशिर्युक्ता सा रूपैकदेशेन ॥११९॥

वा. भा.---संकलितं योगः व्यवकलितं शोधनं गुणनं गुणकारः भाजनं हरणं वर्गः समतुल्यद्वयं वर्गमूलं समतुल्यप्रमाणम् घनः सममूलकस्त्रयः घनमूलं तत्प्रमाणं। एतानि परिकर्माणि गणितं स्मृतम् प्रकीर्तितान्याचार्यैः एतैः सर्वं गणितं सिध्यति ॥११७--११९॥

यो राशेस्त्रिगुणो भागो रूपैकदेशसंयुक्तो वा
अर्धितः स वर्गः स्याद् यो राशेस्त्रिगुणो भागः ॥१२०॥

रूपैकदेशसंयुक्तोऽथवा स घनः स्यात् तस्य मूलं
यत् तद् घनमूलं प्रोक्तं तत् स्यात् सममूलकप्रमाणम् ॥१२१॥

वा. भा.---यो राशेस्त्रिगुणो भागो रूपैकदेशसंयुक्तो वा अर्धितः स वर्गः स्यात्। यो राशेस्त्रिगुणो भागो रूपैकदेशसंयुक्तोऽथवा स घनः स्यात् तस्य मूलं यत् तद् घनमूलं प्रोक्तं तत् सममूलकप्रमाणम् स्यात् ॥१२०--१२१॥

चन्द्रसूर्यगणितम् --- तिथियोगनक्षत्रानयनम्

चन्द्राच्च स्पष्टात् सूर्यं शोध्यं यद् भवति शेषं तत्
द्वादशभागैर्हृत्वा तिथयः स्युः स्फुटास्तदर्धात् ॥१०९॥

तिथयो द्वादशाङ्गुल्यः सूर्यचन्द्रयोरन्तरात् ताः
तिथीनां पञ्चदशभिः संयुक्तं योगनिर्मितिः ॥११०॥

योगः सूर्यचन्द्रयो राशिसंयुक्त्यंशकलात्मिका यः
स तैः पञ्चदशभिः संयुक्तः सप्तविंशत्या हृतः ॥१११॥

वा. भा.---चन्द्रात् स्पष्टात् सूर्यं शोध्य यद् भवति शेषं तत् द्वादशभागैः हृत्वा तिथयः स्युः स्फुटास्तदर्धात्। तिथयो द्वादशाङ्गुल्यः सूर्यचन्द्रयोरन्तरात् तिथीनां पञ्चदशभिः संयुक्तं योगनिर्मितिः सूर्यचन्द्रयो राशिसंयुक्त्यंशकलात्मिका सा सप्तविंशत्या हृता नक्षत्रं स्यात् ॥१०९--१११॥

चन्द्राच्च स्पष्टात् सूर्यं शोध्यं यद् भवति शेषं तत्
त्रिंशता हृत्वा नक्षत्रं तत् स्यात् स्फुटं राशिमानम् ॥११२॥

सूर्यस्य गतिः स्पष्टा चन्द्रस्यापि गतिः स्पष्टा तयोः
विशेषः स्यात् तिथ्यर्धभुक्तिरियं प्रकीर्तिता ॥११३॥

वा. भा.---चन्द्रात् स्पष्टात् सूर्यं शोध्य यद् भवति शेषं तत् त्रिंशता हृत्वा नक्षत्रं तत् स्यात् स्फुटं राशिमानम्। सूर्यस्य गतिः स्पष्टा चन्द्रस्यापि गतिः स्पष्टा तयोर्विशेषः स्यात् तिथ्यर्धभुक्तिः प्रकीर्तिता ॥११२--११३॥

गोलानयनम् --- भारतीययवनक्रमः

यवनानां मते बिम्बं भूगोलं निरक्षदेशे स्थितम्
निजदेशोच्चं तन्मध्यं तत्रैव क्षितिजं वृत्तम् ॥११४॥

भारतीयमते त्वक्षवृत्तं मध्यमण्डलं तत्
क्षितिजं तन्महद् वृत्तं समरेखापि तत्रैव ॥११५॥

उन्नतजीवाकोटिज्याह्निया हृज्या भुजो नतज्या वा
कर्णाक्षेत्रावृत्ते व्यासार्धं द्व्यमतोद्यन्न ॥११६॥

वा. भा.---यवनानां मते बिम्बं भूगोलं निरक्षदेशे स्थितं निजदेशोच्चं तन्मध्यं तत्रैव क्षितिजं वृत्तम्। भारतीयमते तु अक्षवृत्तं मध्यमण्डलं तत् क्षितिजं तन्महद् वृत्तं समरेखापि तत्रैव। उभयोरपि मतयोः सामान्यं गोलज्ञानमुभयमतोद्यन्नमित्युच्यते ॥११४--११६॥

चन्द्रग्रहणाधिकारः

अस्याध्यायस्य विषयः --- चन्द्रग्रहणस्य कालः, मानं, तथा आकलनं।
स्पर्शप्रथमस्पर्शमध्यमोक्षाणां निर्धारणं।

श्लोक ७०--७१

द्विगतिर्दिननिर्माथं स्फुटचन्द्रोदये तु यत् पिण्डम्।
तत् स्यात् स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम् ॥७०॥

तत् स्यात् स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥७१॥

वि. भा.---द्वि गतिः दिननिर्माथं स्फुटचन्द्रोदये तु यत् पिण्डं तत् स्यात् स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम्। तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥७०--७१॥

श्लोक ७२--७३

चन्द्रसूर्ययोरन्तरं याम्यायनं तत् क्षयं वृद्धिं वा।
दक्षिणोत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा ॥७२॥

याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥७३॥

वि. भा.---चन्द्रसूर्ययोरन्तरं याम्यायनं तत् क्षयं वृद्धिं वा। दक्षिणोत्तरयः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा। याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा। तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥७२--७३॥

श्लोक ७४--८०

याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥७४॥

तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥७५॥

स्पर्शमोक्षौ यदा स्यातां तदा ग्रहणं प्रकीर्तितम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥७६॥

तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥७७॥

ग्रहणसूत्राणि

स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥७८॥

चन्द्रसूर्ययोरन्तरं याम्यायनं तत् क्षयं वृद्धिं वा।
दक्षिणोत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा ॥७९॥

याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥८०॥

ग्रहणानयनम्

द्विगतिर्दिननिर्माथं स्फुटचन्द्रोदये तु यत् पिण्डम्।
तत् स्यात् स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम् ॥८१॥

तत् स्यात् स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥८२॥

चन्द्रसूर्ययोरन्तरं याम्यायनं तत् क्षयं वृद्धिं वा।
दक्षिणोत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा ॥८३॥

याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥८४॥

ग्रहणमध्यम्

स्पर्शमोक्षौ यदा स्यातां तदा ग्रहणं प्रकीर्तितम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥८५॥

तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥८६॥

स्पर्शपिण्डं ततो ग्रहणमध्यसाधनम्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥८७॥

सूर्यग्रहणम् --- रविग्रहणानयनम्

सूर्यग्रहणे तु लम्बनं शोध्यं चन्द्रसूर्ययोरन्तरात्
तत् शुद्धं स्यात् स्पर्शारम्भे मोक्षे तथैव ॥१२२॥

लम्बनात् स्पर्शकालः स्पर्शे युक्तो मोक्षे शुद्धः
ग्रहणमध्यं तु तयोर्योगार्धेनैव सिध्यति ॥१२३॥

चन्द्रस्य परिवेषो यः सूर्यस्य तु विशेषणात्
तत् स्याद् ग्रहणमानं तत् प्रकीर्तितमिहाचार्यैः ॥१२४॥

वा. भा.---सूर्यग्रहणे तु लम्बनं शोध्यं चन्द्रसूर्ययोरन्तरात् तत् शुद्धं स्यात् स्पर्शारम्भे मोक्षे तथैव। लम्बनात् स्पर्शकालः स्पर्शे युक्तो मोक्षे शुद्धः ग्रहणमध्यं तु तयोर्योगार्धेनैव सिध्यति। चन्द्रस्य परिवेषो यः सूर्यस्य तु विशेषणात् तत् स्याद् ग्रहणमानं तत् प्रकीर्तितमिहाचार्यैः ॥१२२--१२४॥

लम्बनम् --- देशान्तरलम्बनानयनम्

स्वदेशलम्बज्ञाया द्युज्याया चान्तरं यत् स्यात्
तत् त्रिज्यया हृत्वा लब्धं लम्बनं तत् प्रकीर्तितम् ॥१२५॥

लम्बज्या द्युज्यया हृत्वा त्रिज्यां गुणयेत् ततः
लब्धं तत् स्यात् लम्बनं तत् प्राचीनैरुक्तमीश्वरैः ॥१२६॥

तज्जं लम्बनजातं यत् तत् कालहृतं स्फुटं स्यात्
तद्धृतं नाडीषु रूपं स्यादेतल्लम्बनसाधनम् ॥१२७॥

वा. भा.---स्वदेशलम्बज्ञाया द्युज्याया चान्तरं यत् स्यात् तत् त्रिज्यया हृत्वा लब्धं लम्बनं तत् प्रकीर्तितम्। लम्बज्या द्युज्यया हृत्वा त्रिज्यां गुणयेत् ततो लब्धं तत् स्यात् लम्बनं तत् प्राचीनैरुक्तम् तज्जं लम्बनजातं यत् तत् कालहृतं स्फुटं स्यात् तद्धृतं नाडीषु रूपं स्यादेतल्लम्बनसाधनम् ॥१२५--१२७॥

ग्रहणस्पर्श-मोक्षयोः

चन्द्रसूर्ययोरन्तरं याम्यायनं तत् क्षयं वृद्धिं वा।
दक्षिणोत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा ॥८८॥

याम्योत्तरयोः क्रान्त्योरन्तरं तत् क्षयं वृद्धिं वा।
तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत् ॥८९॥

तद्भुजज्या तु करणं तत् स्पर्शमोक्षौ प्रदर्शयेत्।
तद्द्वितीयेन वा कालो ग्रहणस्य प्रकीर्तितः ॥९०॥

सङ्क्षेपः
एतस्मिन् खण्डे निम्नलिखिताः श्लोकाः समावेशिताः:
स्पष्टाधिकारः (श्लोक १२--५१)
त्रिप्रश्नाधिकारः (श्लोक ५४--६९)
छायाकरणी (श्लोक २८--३६)
लम्बाक्षज्यानयनम् (श्लोक ५७--६२)
प्राणगणितम् (श्लोक १२८--१३०)
मध्यच्छाया (श्लोक १४८)
महापातः (श्लोक १४०--१४२)
क्रान्तिः (श्लोक १३१--१३३)
लग्ननयनम् (श्लोक १३४--१३६)
गोलयन्त्रम् (श्लोक १३७--१३९)
ज्योत्पत्तिः (श्लोक १००--१०३)
गौरीगुप्ता (श्लोक १०४--१०८)
पाटीगणितम् (श्लोक ११७--१२१)
तिथियोगनक्षत्रम् (श्लोक १०९--११३)
गोलानयनम् (श्लोक ११४--११६)
चन्द्रग्रहणम् (श्लोक ७०--९०)
सूर्यग्रहणम् (श्लोक १२२--१२४)
लम्बनम् (श्लोक १२५--१२७)

समाप्तिविवरणम्
एषः द्वितीयभागस्य द्वितीयखण्डः समाप्तः। अत्र समावेशितं साहित्यं निम्नलिखिताध्यायान् समेति:

1. **स्पष्टाधिकारः** --- ग्रहाणां मध्यमात् स्पष्टस्थानस्य गणनम्
2. **त्रिप्रश्नाधिकारः** --- लम्बननाडीछायागणनम्
3. **चन्द्रग्रहणाधिकारः** --- चन्द्रग्रहणकालमानाकलनम्
4. **गणितीयसूत्राणि** --- ज्योत्पत्तिः, गौरीगुप्ता, पाटीगणितम्

ब्रह्मगुप्तस्याः सिद्धान्ते प्रत्येकं नियमः संस्कृतपद्येषु (आर्यायां) उपस्थापितः, ततो विस्तृतव्याख्या (व्याख्या) अनुसरति। गणितीयसूत्राणि भारतीयज्यापद्धतिं प्रयुञ्जते तथा त्रैराशिकं ज्यामितीयं च तर्कं प्रयुञ्जते।

10pt

ब्राह्मस्फुटसिद्धान्तः

Brāhmasphuṭasiddhānta

BILINGUAL EDITION

Sanskrit (Devanagari) || English (with IAST)

Part Two — Chunk 3

(Pages 281–420 of PDF ≈ Original pages 1119–1258)

By **Brahmagupta** (6th century CE)

With Commentary (नूतनतिलक) by
Śrī Kṛpāludatta Sūnu Sudhākara Dvivedī

Contains the following chapters:

चन्द्रग्रहणविकारः — Lunar Eclipse Variations
सूर्यग्रहणाधिकारः — Solar Eclipse Chapter
चन्द्रच्छायाधिकारः — Moon's Shadow Chapter
ग्रहोदयास्ताधिकारः — Planetary Rising and Setting
शुक्लोन्नत्यधिकारः — Brightness of the Moon
चन्द्रशुक्लोन्नत्यधिकारः — Moon's Bright Elevation

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Bilingual edition: Sanskrit left — English right

Contents

Editorial Preface

This bilingual edition presents the Sanskrit text of Brāhmasphuṭasiddhānta Part II, Chunk 3, with facing English translations. The layout follows the standard format for critical editions of Sanskrit scientific texts:

- **Left column:** Sanskrit text in Devanagari script, as found in the source edition.
- **Right column:** English translation with IAST romanization of technical terms.

The commentary (टीका) by Sudhākara Dvivedī follows the traditional Indian *pañcāṅga* structure of:

1. **Sūtra** (सूत्रम्) — The original verse in āryā or anuṣṭubh metre.
2. **Sūtrārtha** (सूत्रार्थः) — Literal meaning of the verse.
3. **Hindi Bhāṣya** (हिन्दी भाष्य) — Explanation in Hindi.
4. **Upapatti** (उपपत्तिः) — Mathematical proof or demonstration.

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1 चन्द्रग्रहणविकारः — Candragrahaṇavikāraḥ (Lunar Eclipse Variations)

Introduction

This chapter (चन्द्रग्रहणविकारः) deals with the computation of lunar eclipses, following Brahmagupta's exposition in the ब्राह्मस्फुटसिद्धान्त. The commentary (सुधाकर-टीका) by Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī provides elaborate explanations in Hindi, with mathematical demonstrations (उपपत्ति).

1.1 Verses 11–12: Computation of इष्टग्रास (Desired Eclipse Magnitude)

Sūtra (सूत्रम्) — Verses 11–12:
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> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]–[illegible] [illegible]

Translation: "The half-duration (*sthitidala*) diminished by the desired quantity, multiplied by the daily motion (*bhukti*) and divided by the deflection (*villepa*), gives the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of the desired quantity (*iṣṭeṇu*). From the half-sum of the diameters (*mānaikyārdha*), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration (*channa-māna*)."

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Sūtrārtha (सूत्रार्थः):

> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Literal Meaning: The verse gives the method for computing the *iṣṭagrāsa* (desired eclipse magnitude). From the *sthitidala* (half-duration) diminished by the desired quantity, multiplied by the *bhukti* (daily motion) and divided by the *villepa* (deflection), one gets the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of *iṣṭeṇu* (desired quantity). From the *mānaikyārdha* (half-sum of diameters), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration.

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Hindi Bhāṣya (हिन्दी भाष्य):

> [illegible]
> [illegible] ([illegible])
> [illegible] ([illegible]) [illegible]
> [illegible] [illegible]–[illegible] [illegible]

English Translation of Hindi Commentary: "Subtracting the desired (*iṣṭa*) from the *sthitidala*, multiplying by the *bhukti* and dividing by the *villepa*, one obtains the *bhuja*. By applying this result to the motions of the Sun, Moon, and node (*pāta*), one computes the instantaneous lunar latitude (*tātkālika candra-śara*), which is the *koṭi*. The square root of the sum of their squares is the *karṇa*. From the *mānaikyārdha* (sum of the radii of the eclipsed and eclipsing bodies), subtracting the *karṇa*, the remainder is the desired magnitude (*iṣṭagrāsa*)."

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Upapatti (उपपत्तिः) — Proof:

> [illegible]
> [illegible]
> [illegible]
> [illegible] ([illegible]) [illegible]
> [illegible]

English Translation of Proof: "After the contact (*sparśī*), for whatever desired number of ghaṭīs the knowledge of the *grāsa* is required, that desired number of ghaṭīs should be subtracted from the *sthityardha*. The remainder is used in proportion. If in sixty ghaṭīs the difference in motion between the Sun and Moon gives a certain number of minutes (*kalā*), then [by proportion] in the *iṣṭa*-diminished *sthityardha* ghaṭīs, whatever result is obtained is the *bhuja*. From the result obtained by proportion, by applying the motions of the Sun, Moon, and node, the instantaneous lunar latitude (*tātkālika candra-vāra*) is computed — that is the *koṭi*. The square root of the sum of the squares of these two (*bhuja* and *koṭi*) is the *karṇa*. Subtracting the *karṇa* from the *mānaikyārdha* gives the desired *grāsa*."

1.2 Verses 13–14: Computation of इष्टग्रासात्काल (Time from Desired Magnitude)

Sūtra (सूत्रम्) — Verses 13–14:
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प[illegible]

प[illegible]

Translation — Verse 13: "From the square of the half-sum of the measure diminished by the *grāsakalā*, having subtracted the square of the instantaneous latitude (*tātkālika-vikṣepa*), the square root of the remainder — made like a *tithi* [i.e., multiplied by 60 and divided by the difference of motions] — [gives a time]."

Translation — Verse 14: "Having subtracted [this time] from the *pragrahaṇa-sthit yardha*, the time from the *pragrahaṇa* [i.e., from contact] is obtained. Having subtracted from the *mokṣa-sthit yardha*, [one obtains the time] before the *mokṣa* [release]."

c[0](.95.95)sanskritbg c[1](.95.95)englishbg
Sūtrārtha (सूत्रार्थः):

ग[illegible]

Literal Meaning: "From the square of the half-sum of the measure [of diameters] diminished by the *grāsakalā*, having subtracted the square of the instantaneous *śara* (latitude), the square root of the remainder — made *tithi*-like (*tithivat*) [by multiplying by 60 and dividing by the difference of motions of Sun and Moon] — is the result. That is, having multiplied the square root of the remainder by 60, divided by the difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous latitude (*tātkālika-vikṣepa*) is computed. From that, again by the rule '*grāsakalona-pramāṇa*', etc., iteratively (*asakṛt*), the square-root-time becomes stable. That [time], subtracted from the *pragrahaṇa-sthit yardha* (contact half-duration), the time from the *pragrahaṇa* (from contact) is obtained."

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Hindi Bhāṣya (हिन्दी भाष्य):

ग[illegible]

English Translation: “From the square of the *grāsa*-diminished *mānaikyārdha*, having subtracted the square of the instantaneous *śara*, the square root of the remainder — made *tithi*-like iteratively — that is, having multiplied the square root of the remainder by 60, divided by the combined difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous lunar latitude is computed. From that, again by the rule beginning ‘*grāsakalona-pramāṇa...*’, iteratively (*asakṛt*), the square-root-time becomes stable. That, subtracted from the contact half-duration (*pragrahaṇa-sthit yardha*), gives the time from contact, meaning that after contact, in that desired time, the desired *grāsa* is obtained.”

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Upapatti (उपपत्तिः):

> त[illegible] [illegible] [illegible] [illegible] [illegible] “[illegible] [illegible] [illegible] [illegible] [illegible]” [illegible] [illegible] [illegible] [illegible]–[illegible] [illegible]

English Translation of Proof: “Due to the ignorance of the latitude at that time (*tātkāla*), the computation has been done using the middle-time latitude. Therefore, it is proper to determine the accurate time by an iterative method (*asakṛt-ghātinā*). The remaining proof [follows] the method of Bhāskara: ‘From the square of half the sum of diameters diminished by the *grāsa*, having subtracted the square of the *vikṣepa*...’ etc.”

1.3 Verse 9: Sphuṭīkaraṇa of स्थितिविमर्दार्ध (Refinement of Half-Duration of Totality)

Sūtra (सूत्रम्) — Verse 9:
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> ष[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Translation: “The own motion (*svagati*) — [i.e.,] the motions of the Sun, Moon, and node — multiplied by the *sthitivimardārdha* nāḍikās and divided by sixty; both [motions] are applied [to the Sun, Moon, and node] repeatedly in the eclipsed body (*chādya*), [and] the result is additive or subtractive.”

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Sūtrārtha (सूत्रार्थः):

> स[illegible] ([illegible], [illegible], [illegible]) [illegible] [illegible] ([illegible] [illegible] [illegible] [illegible] [illegible]) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Literal Meaning: “The own motion — the motion of the Sun, the motion of the Moon, and the motion of the node — multiplied by the *sthitivimardārdha* nāḍikās (the half-duration of totality ghaṭīs multiplied separately) and divided by sixty; applied to the Sun, Moon, and node in the eclipsed body, both [additive and subtractive] repeatedly, the result being additive or subtractive.”

c[0](.95.95)sanskritbg c[1](.95.95)englishbg
Upapatti (उपपत्तिः):

> [illegible]‘[illegible] [illegible]” [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

English Translation: “The [method is] made accurate by the method of Bhāskara: ‘*sthit yardha-nāḍī-guṇitā sva-mukti*’ etc., and ‘*vimardārdha-phala-ona-yukta*’ etc.”

2 सूर्यग्रहणाधिकारः — Sūryagrahaṇādhikāraḥ (Solar Eclipse Chapter)

This chapter (सूर्यग्रहणाधिकारः, the fifth chapter of the text) deals with the computation of solar eclipses. The chapter begins with the statement of purpose (प्रयोजन) and proceeds through the computation of the five ज्या (sines) required for eclipse calculation: उदयज्या, मध्यज्या, रविदयाकर्ण, विशिमशङ्कु, and दृक्षेप.

2.1 Verse 1: Statement of Purpose

Sūtra (सूत्रम्) — Verse 1:
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अ[illegible] [illegible] [illegible] [illegible] [illegible]—

ग[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Introduction: "Now the chapter on solar eclipse is begun. First, the purpose of its commencement is stated —"

Translation: "Because the inter-planetary [distance] computed by the *grahabhāskara* [method] previously is an obstruction in the inter-planetary [computation], therefore I shall state the computation of *dyuti* (parallax) with its own degrees, so that there may be agreement."

Technical note: The *grahabhāskara* refers to the method given by earlier astronomers (like Bhāskara I) for computing the distance between planets, which Brahmagupta finds to be defective for solar eclipse computation.

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Sūtrārtha (सूत्रार्थः):

प[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible] [illegible]

Literal Meaning: "Previously, in the eastern horizon (*pūrve kṣitije*), by the degrees of the *grahabhāskara* method, the planets become visible or invisible (*haśya-āhaśya*) in their own time-degrees. Later, in the western horizon (*paścima-kṣitije*), by the degrees of the planetary distances, [the same] according to number. Therefore, I shall state that computation — the computation of the *praharāntara.* That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun [occurs] after a certain number of time-ghaṭīs, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting [occurs] after a certain number of time-ghaṭīs, on that day in the evening the planet's invisibility (*haśyatva*) [is determined]."

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Hindi Bhāṣya (हिन्दी भाष्य):

ज[illegible] [illegible] [illegible] [illegible] [illegible] ([illegible], [illegible], [illegible], [illegible], [illegible]) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] ([illegible] [illegible] [illegible] [illegible] [illegible]) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] ([illegible] [illegible] [illegible] [illegible] [illegible] [illegible]) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] – [illegible], [illegible] – [illegible], [illegible] – [illegible], [illegible] – [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible]

English Translation: "For whatever reason, in solar eclipses, the five *jyā*s (*udayajyā*, *madhyajyā*, *ravidayākarṇa*, *viśimaśaṅku*, *hṛkṣepa*) — by which earlier teachers like Bhāskara computed the *prāṇayana* — from that, the *gaṇitaikya* (agreement between observed and computed values) is not obtained. For that reason, so that their agreement may exist, I state the *prāṇayana* [method]. The *tithi-anta* [refers to] the relation with the *graha*: *udayajyā* [refers to] parallax (*bhramā*), *madhyajyā* [to] the sine of the zenith-distance of the tenth degree (*daśama-lagna-natāṃśajyā*), *viśimaśaṅku* [to] the ascensional difference (*hagati*), *hṛkṣepa* [to] the sine of the zenith-distance of the nonagesimal (*viśima-natāṃśajyā*). In the *Sūryasiddhānta* too, the solar eclipse *prāṇayana* is by the aforesaid five *jyā*s. In the *Pauliśasiddhānta*, in solar eclipses, separate five *jyā*s of the Moon are computed; therefore there the solar eclipse is accomplished by the two-*jyā* method."

2.2 Verses 2–3: Conditions for लम्बन (Deflection) and अवनति (Depression)

Sūtra (सूत्रम्) — Verses 2–3:
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> व⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠ ⊠ ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠ ⊠ ⊠⊠⊠⊠
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Translation — Verse 2: "When one [planet] is equal to the *viśrima-lagna* (nonagesimal ascendant), there is no *lambana* (deflection in longitude). When it is greater or less than that, *lambana* occurs. When its declination-sine (*krāntijyā*) is equal to the latitude-sine (*akṣajyā*), then there is no *avanati* (depression)."

Translation — Verse 3: "Otherwise, *avanati* occurs. When both *lambana* and *avanati* are possible, then the rising [point] equal to the *lagna*, the *ghaṭikās* elapsed since its rising, its *śaṅku* (gnomon), and the *prāṇa*s of its motion — [are computed] thereafter."

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Sūtrārtha (सूत्रार्थः):

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Literal Meaning: "At the end of the *tithi*, at the time of computed conjunction (*gaṇitāgata-darśāntikāle*), having computed the *lagna* by the previous method stated in the *Tripraśna*, the *viśrima* [nonagesimal] is to be computed. When the Sun is equal to one [point] with that *viśrimalagna*, there is no *lambana*. When the Sun is greater or less than that *viśrima*, *lambana* occurs — this meaning is self-evident. Similarly, when the northern declination-sine (*krāntijyā*) of that *viśrima* is equal to the latitude-sine (*akṣajyā*) of one's own place, then there is no *avanati*; otherwise *avanati* occurs. When both *lambana* and *avanati* are possible, then having made the rising equal to the *lagna* — the *rāśyudaya-eṣi-lagna* of one's own place — by the method stated in the *Tripraśna*: 'the first diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*', etc., the *ghaṭikās* elapsed since the rising of that *viśrima* are to be computed."

2.3 Verse 7: स्पष्टदर्शान्तकालिक Computation (Computation of True Conjunction Time)

Sūtra (सूत्रम्) — Verse 7:
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Translation: “The *ghaṭikā-miṣa* [of *lambana*] multiplied by five and divided [by some number] yields the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time (*gaṇitāgata-darśānta-kālika*) are to be diminished [or increased].”

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Sūtrārtha (सूत्रार्थः):

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Literal Meaning: “The *ghaṭikā-miṣa* multiplied by five, divided by five [? *pañcajyā*], the result is the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time are to be diminished (*ūnāḥ*). If the *lambana* is positive (*dhana*), then the Sun, Moon, and node at the computed conjunction time are to be increased by the *labdha-kalā*; then those become the true Sun, Moon, and node at the true conjunction time (*spaṣṭa-darśānta-kālika*).”

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Upapatti (उपपत्तिः):

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English Translation of Proof: “From the Sun, Moon, and node at the computed conjunction time, all [quantities] beginning with the *śara* [latitude] are to be computed. When the *lambana* is negative (*ṛṇātmaka*): ‘The minutes of motion of the Sun, Moon, and node are separately multiplied by the *lambana*-ghaṭikā-miṣa of sixty-five — by proportion: ‘then by the *lambana*-ghaṭikā, what [result]?’ — by the *labdha-kalā*s, the Sun, Moon, and node at the computed conjunction time are to be diminished. When the *lambana* is positive (*dhanātmaka*), they are increased by the *labdha-kalā*s; then they become the true [quantities] at the true conjunction time. In the *Siddhāntaśekhara*: ‘Having divided the motion-multiplied *lambana*-nāḍī-miṣa by five, the resulting minutes... The Sun, Moon, and node are added in positive [*lambana*] and subtracted in negative *lambana*’ — by this, what was stated by Śrīpati-nācārya is here spoken in accordance.”

3 चन्द्रच्छायाधिकारः — Candracchāyādhikāraḥ (Moon's Shadow Chapter)

This chapter (चन्द्रच्छायाधिकारः) deals with the computation of the Moon's shadow, which is essential for understanding lunar phenomena. The shadow of the Earth falls on the Moon during a lunar eclipse, and its computation involves calculating the च्छाया (shadow), मध्यच्छाया (middle shadow), and related quantities.

3.1 Verse 1: Beginning of the Chapter

Sūtra (सूत्रम्) — Verse 1:
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अ[illegible] [illegible] [illegible] [illegible] [illegible]—

प[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible] [illegible]

Introduction: "Now the chapter on the Moon's shadow is begun. First, the purpose of its commencement is stated —"

Translation: "Previously, the difference between the Moon and the *lagna*, and afterwards, that of the *saptalagna* etc., because from that the shadow of the Moon (*indu-ccchāyā*) is daily [computed], therefore I shall state the computation (*anayana*) of that."

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Sūtrārtha (सूत्रार्थः):

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Literal Meaning: "Previously (in the eastern horizon), the difference between the Moon and the *lagna*, and afterwards (in the western horizon), the difference between the *saptalagna* and the Moon, and daily by effort — by which cause the Moon's shadow (*indu-ccchāyā*) is known, and from the Moon's shadow that difference is [also] known — therefore I shall state that computation (*anayana*), the shadow-computation (*chāyā-anayana*). In the *Siddhāntaśekhara*: 'Usually the difference between the diminished-Sun-*lagna* and the *lagna-nāśin* in the west' etc. — by this, what was spoken by Śrīpati-nācārya is here spoken in the same manner."

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Hindi Bhāṣya (हिन्दी भाष्य):

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English Translation: "Now the chapter on the Moon's shadow is begun. First, its purpose is stated. The difference between the Moon and the *lagna* in the eastern horizon, and the difference between the *saptalagna* and the Moon in the western horizon — because from these, daily, the Moon's shadow is known (first, the knowledge of the Moon's shadow is obtained), and from the Moon's shadow the knowledge of that difference is obtained — therefore I state the shadow-computation."

3.2 Verse 2: Shadow Computation

Sūtra (सूत्रम्) — Verse 2:
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प[illegible] [illegible]
[illegible] [illegible] [illegible]

Translation: “The parts of the upright-time-difference (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti-pāta*) [having] the form of a *prabhahṛṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*).”

Note: This is a technical verse whose exact interpretation requires the accompanying commentary. The *prabhahṛṇḍa* is a type of spherical triangle used in Indian astronomy.

3.3 Verses 3–4: Detailed Shadow Calculation

Sūtra (सूत्रम्) — Verses 3–4:
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[illegible] [illegible] [illegible]

म[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible]

Introduction to Verse 3: “Now [the author] states what is to be done —”

Translation — Verse 3: [Same structure as verse 2, establishing the method.]

Translation — Verse 4: “As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūloddhṛtā*).”

Note: This verse describes an iterative root-extraction method for computing the rate of change of the shadow. The *mūla* here refers to a base quantity from which differences are taken.

4 ग्रहोदयास्ताधिकारः — Grahodayāstādhikāraḥ (Planetary Rising and Setting)

This chapter (ग्रहोदयास्ताधिकारः) deals with the computation of the rising and setting times of planets. The chapter explains how to determine the times when planets become visible (उदय, heliacal rising) and invisible (अस्त, heliacal setting) based on their elongation from the Sun.

4.1 Verse 1: Pratyaya (Introduction)

Sūtra (सूत्रम्) — Verse 1:
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ग[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible] [illegible]

Introduction: "Now the chapter on planetary rising and setting is begun. First, the purpose of its commencement is stated —"

Translation: "Because previously, in the inter-planetary computation, the *grahabhāskara* distance is an obstruction (*bādhaka*), therefore, with its own degrees, so that there may be agreement — I state the computation of *dyuti*."

Note: This verse serves as a parallel to the solar eclipse chapter's opening, establishing that Brahmagupta's method supersedes earlier approaches that failed to achieve agreement (*aiṃya*) between computation and observation.

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Sūtrārtha (सूत्रार्थः):

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Literal Meaning: "Previously, in the eastern horizon, by the *grahabhāskara* distances in their own degrees, in their own time-degrees, later in the western horizon, by the planetary distances in their own time-degrees, according to number, the planets become visible or invisible (*haśya-āhaśya*). Therefore, I shall state that computation — the computation of the *praharāntara*. That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun occurs after a certain number of time-ghaṭīs, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting occurs after a certain number of time-ghaṭīs, on that day in the evening the planet's invisibility (*haśyatva*) [is established]."

4.2 Verses 8–9: Daily Rising and Setting

Sūtra (सूत्रम्) — Verses 8–9:
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Translation — Verse 8: "By iterative computation (*asakṛt-karma*) from the instantaneous (*tātkālika*) planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the *grahodita-ghaṭī* [elapsed ghaṭīs since planetary rising] is to be known, the *lagna* at that time (*tātkālika lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭīs of both, by those (computed ghaṭīs) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭīs — in this manner, iteratively (*asakṛt*, repeatedly) doing [this]."

Note: This verse describes the iterative procedure (*asakṛt-karma*) for refining the computation of a planet's rising time. The method involves successive approximation until the computed value stabilizes.

4.3 Verse 9: Moon's Rising and Setting Computation

Sūtra (सूत्रम्) — Verse 9:

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Translation: "The rising and setting of the Moon (*indu*) are to be computed by time-degrees (*kālāṃśa*) equal to digits (*aṅka-saṃmitaiḥ*). If there is diminution (*hīnatva*) or excess (*adhikatva*), in that interval the *yoga-kāla* [conjunction time] would be [obtained]."

Note: The *aṅka* (digit) is a unit equal to 12 degrees. The verse states that the Moon's rising and setting are computed using time-degrees measured in these units.

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Sūtrārtha (सूत्रार्थः):

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Literal Meaning: "By time-degrees equal to digits (*aṅka*), that is, by twelve time-degrees (*dvādaśa-kālāṃśa*), the rising and setting of the Moon (*indu* = *candra*) are to be computed. Here, if the computed time-degree (*paṭhita-kālāṃśa*) has diminution or excess from the desired time-degree (*iṣṭa-kālāṃśa*), then in that interval, between those two computed time-degrees, the *yoga-kāla* (conjunction time), or *yogavat-kāla* (time like conjunction), would be [obtained], as clearly stated in the seventh verse."

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Hindi Bhāṣya (हिन्दी भाष्य):

> अ ( )

English Translation: "By time-degrees equal to digits (*aṅka*), that is, by time-degrees equal to twelve, the rising and setting are to be computed. If the computed time-degree has diminution (*hīnatva*, being smaller) or excess (*adhikatva*) from the desired time-degree, then in that interval, between those two time-degrees, the *yoga-kāla* (conjunction time) would be [obtained]."

5 शुक्लोन्नत्यधिकारः — Śuklonnatyadhikāraḥ (The Brightness of the Moon)

This chapter (शुक्लोन्नत्यधिकारः) deals with the computation of the illuminated portion of the Moon's disk — the शुक्ल (bright) portion. The chapter explains how to determine the amount of the Moon's surface that is illuminated by the Sun at any given time.

5.1 Verses 1–3: Sphuṭaśuklonnati Computation

Sūtra (सूत्रम्) — Verse 4:
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> अ[illegible]
> र[illegible]
> [illegible]

Introduction: "Now the chapter on the Moon's bright elevation (*śuklonnati*) is begun."

Translation — Verse 4: "The ghaṭikās [which are] the *khacarārdha* (half of the celestial sphere's motion) — by the Sun, Moon, node, and *lagna*, and from the rising of the own declination (*sva-krānty-udaya*), [which are] the remaining [ghaṭikās] elapsed from the own *lagna* — having made [the computation] from the setting of the Sun, [these are] to be computed."

Note: The *khacarārdha* refers to half the diurnal motion of the celestial sphere. The verse establishes the parameters needed to compute the Moon's illumination.

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Sūtrārtha (सूत्रार्थः):

> ख[illegible]

Literal Meaning: "The *khacarārdha* ghaṭikās [which are] from the setting of the Sun (*ravyastāt*), [which are] the remaining [ghaṭikās] elapsed from the own-*lagna* of the Moon — or the elapsed/deficient ghaṭikās of the Moon's rising-*lagna* — are to be computed. By what? By the Sun, Moon, node, *lagna*, and from the rising of the own declination. Here, this is stated: at the time of the Sun's setting, the Sun, Moon, node, and *lagna.* Having computed these, by the own declination (*sva-krānti*), by the Moon's declination (*candra-krānti*), from the rising, [computing] the elapsed or deficient ghaṭikās of the Moon's rising-*lagna* from the *udaya-lagna* by the method 'diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*', etc. After the Sun's setting, the ghaṭikās by which the Moon sets — those are the elapsed ghaṭikās. Before the Sun's setting, the ghaṭikās by which the Moon sets — those are the deficient ghaṭikās. Similarly, in the eastern horizon, from the Sun's rising, the elapsed or deficient ghaṭikās of the Moon's rising are to be computed — this is to be understood from the meaning."

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Hindi Bhāṣya (हिन्दी भाष्य):

> त[illegible]

English Translation: "By iterative computation (*asakṛt-karma*) from the instantaneous planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the planetary elapsed ghaṭīs are to be known, the *lagna* at that time (*tātkālika lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭīs of both, by those (computed ghaṭīs) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭīs — in this manner, iteratively (*asakṛt*, repeatedly) doing [this]. And for the knowledge of the *sthita-kāla*, computation from the *sthita-lagna* (*sapta-bhastala-lagna*) should be done. In this way, from the first day of lunar visibility (*candra darśana*) up to the full moon (*pūrṇimā*), on the desired day, the setting time from the Sun's setting and the rising time from the Sun's rising are to be computed."

5.2 Verse 5: The शुक्लोन्नति (Bright Elevation)

Sūtra (सूत्रम्) — Verse 5:
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र⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧
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Translation — Verse 5: [Recapitulates the parameters for computing the bright elevation.]
Technical summary: The verse reiterates that by the Sun, Moon, node, *lagna*, and from the rising of the declination, the *khacarārdha* ghaṭikās from the Sun's setting are to be used to compute the Moon's bright elevation.

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Upapatti (उपपत्तिः):

य⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧ ⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧ ⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧ ⌧ "⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧ ⌧⌧ ⌧ ⌧⌧⌧⌧⌧ ⌧ ⌧⌧ ⌧⌧⌧ ⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧"⌧

English Translation of Proof: "On whatever day, after the Sun's setting, the Moon sets after an additional number of time-ghaṭīs. Before the Sun's rising, the Moon rises [after a certain number of ghaṭīs]. On that very day, when the Moon is visible, the two bright elevations (*śva-jy-unnati*) are to be computed. The bright elevation of the bright [half] (*sita-śuklonnati*) is half the disk (*bimbārdha*) of the Moon's bright [portion]. Thus Bhāskara computed the *śuklonnati* in the first *pāda* of the month-end. In the second and third *pāda*s, half the disk is dark (*kṛṣṇa*). Therefore there the dark *śuklonnati* occurs. Thus by the teacher (*ācārya*), two *śuklonnati*s are computed. But here, the *pāda* discussion is not made. Bhāskara, having considered in his mind that the dark *śuklonnati* is not clearly observable, computed only the bright *śuklonnati*. And it is stated: 'Brahmagupta and others have produced the dark *śuklonnati* also for the second and third *caraṇa*s; that is not approved by me. For neither is the dark *śuklonnati* clearly perceived by men.'

5.3 Verses 11–12: स्पष्टशुक्ल (True Bright Portion) Computation

Sūtra (सूत्रम्) — Verses 11–12:
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त⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧ ⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧
⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧ ⌧

Translation — Verse 11: "Or when the middle motion (*madhyagati*) is multiplied by its own shadow (*tac-chāyā*) and by the semi-diameter (*sva-kara*) multiplied by the *tithi* — the results are additive or subtractive. In equality (*sāmya*), the difference is [taken]; in direction-difference (*dig-anyatva*), [it is] the *avanati* [depression] expressed in digits (*aṅkya*).”

Note: This verse gives the method for computing the true bright portion (*spaṣṭa-śukla*) of the Moon. The *madhyagati* is the mean daily motion, and the *tacchāyā* is the shadow corresponding to the planet’s altitude.

c[0](.95.95)sanskritbg c[1](.95.95)englishbg

Sūtrārtha (सूत्रार्थः):

> ह[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]-[illegible]-[illegible]-[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
> [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

Literal Meaning: “In the *hṛjya* [radius = 3438’] of the Sun and Moon, the *kṣepī* [sine of latitude], having been multiplied by the respective middle motion and by the semi-diameter (*vyāsadala*) multiplied by the *tithi* and by fifteen [i.e., by the *tithi*-multiplied semi-diameter multiplied by 15], divided by the *trijyā* [radius = 3438’], separately, separately, the *natī* [depressions] of the Sun and Moon are obtained. Or, their middle motion, their *kṣepa*-equal *hṛjya*, the shadow being multiplied by that, multiplied by the *tithi*-multiplied own shadow-*karman*, their difference is [taken].”

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Upapatti (उपपत्तिः):

> प[illegible] “[illegible] [illegible]” [illegible] [illegible]
> [illegible]

English Translation of Proof: “The first computation [is made] accurate by the method of Bhāskara: ‘Having multiplied by the *sahakṣepa* [combined latitude] of the Moon, by the own middle motion-*tithi*-aṃśa-multiplied [quantity]’ etc.”

6 चन्द्रशुक्लोन्नत्यधिकारः — Chandraśuklonnatyadhikāraḥ (Moon's Bright Elevation)

This is the seventh अध्याय (chapter) of the ब्राह्मस्फुटसिद्धान्त, dealing with the Moon's bright elevation (चन्द्रशुक्लोन्नति). It comprises eighteen verses (श्लोक) and covers the computation of the arc of illumination on the Moon's disk.

6.1 Verses 1–2: Introduction

Sūtra (सूत्रम्) — Verses 1–2:
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म⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠
⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠ ⊠

Introduction: "Now the beginning of the chapter on the Moon's bright elevation (*candra-śukla-unnati*)."
Translation — Verse 1: "The parts of the upright-time-difference (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti-pāta*) [having] the form of a *prāgrahuṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*)."
Translation — Verse 2: "As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūloddhṛtā*)."
Note: Verse 2 establishes an iterative root-extraction algorithm for computing the rate of change of the shadow, which is essential for determining the Moon's illumination over time.

6.2 Verse 18: Chapter Conclusion

Sūtra (सूत्रम्) — Verse 18:
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इ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠—

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Introduction: "Now [the author] states the conclusion of the chapter —"
Translation — Verse 18: "Thus, in [the topics of] *bāhu* (base), *koṭi* (perpendicular), *karṇa* (hypotenuse), *spaṣṭa* (true [longitude]), *sita* (bright [part]), *parilekha* (outline), *sūtra* (line), and *kardama* [mud] — by eighteen doors (*dvāra*), [this is] the seventh chapter [on] the Moon's bright elevation (*candra-śuklonnati*)."
Note: The chapter lists eight topics (*bāhu*, *koṭi*, *karṇa*, *spaṣṭa*, *sita*, *parilekha*, *sūtra*, *kardama*) covered in 18 verses. The *kardama* refers to the "mud" or penumbral shadow region.

c[0](.95.95)sanskritbg c[1](.95.95)englishbg
Sūtrārtha (सूत्रार्थः):

प⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠... ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠ ⊠

Literal Meaning: "The *parilekha-sūtra* itself [is] the *karṇa*... the *parilekha-sūtra-karṇaḥ*. The meaning is clear."

c[0](.95.95)sanskritbg c[1](.95.95)englishbg
Hindi Bhāṣya (हिन्दी भाष्य):

इ[illegible] [illegible] [illegible], [illegible], [illegible], [illegible], [illegible], [illegible] ([illegible]) [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

English Translation: “In this way, in [the topics of] *bāhu*, *koṭi*, *karṇa*, *spaṣṭa*, *sitāṅgula*, *parilekha-sūtra* (*karṇa*), by eighteen meaning-verses, is the seventh chapter named *Candraśuklonnati*.”

Note: The *sitāṅgula* (bright digit) is a unit for measuring the illuminated portion of the Moon’s disk. One *aṅgula* (digit) = 1/12 of the Moon’s diameter.

Concluding Verse (अन्तिमश्लोकः)

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म[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible]

इ[illegible] [illegible] [illegible]
[illegible] [illegible] [illegible] [illegible]

Translation — Concluding Verse: “That *tilaka* (ornament) which was spoken by the son (*sūnu*) of Madhusūdana, which was spoken here by Śrīpathi for the conqueror (*jiṣṇu*) — having placed that in [my] heart, this new [work] *Sūryagati* has been composed by Sudhākara.”

Colophon: “Thus, in the *Brāhmasphuṭasiddhānta-nutana-tilaka* composed by Śrī Kṛpāludatta-sūnu Sudhākara Dvivedī, the seventh chapter — the *Candraśuklonnatyadhikāraḥ* — [is concluded].”

॥ इति श्रीब्रह्मगुप्तविरचिते ब्राह्मस्फुटसिद्धान्ते चन्द्रशुक्लोन्नत्यधिकारः सप्तमः समाप्तः ॥
[Thus ends the Seventh Chapter — the Moon’s Bright Elevation — in the *Brāhmasphuṭasiddhānta* composed by Śrī Brahmagupta.]

Appendix: Notes on the Text and Commentary

About the ब्राह्मस्फुटसिद्धान्त

The ब्राह्मस्फुटसिद्धान्त (*Brāhmasphuṭasiddhānta*, "Correctly Established Doctrine of Brahma") is the magnum opus of the great Indian mathematician and astronomer **Brahmagupta** (598–668 CE). Composed in 628 CE at Bhinmal (in present-day Rajasthan, India), it consists of 24 chapters (अध्याय) containing a total of 1008 verses (श्लोक) in आर्या and अनुष्टुभ् metres.

Contents of This Chunk (Pages 281–420)

This typeset volume covers the following chapters from Part II of the text:

1. चन्द्रग्रहणविकारः — Lunar Eclipse Variations
   Deals with the computation of eclipse magnitudes, durations, and the *iṣṭagrāsa* (desired obscuration) method.
2. सूर्यग्रहणाधिकारः — Solar Eclipse Chapter
   Covers the computation of solar eclipses, including the five *jyā* (sines) required: *udayajyā*, *madhyajyā*, *ravidayākarṇa*, *viśimaśaṅku*, and *hṛkṣepa*.
3. चन्द्रच्छायाधिकारः — Moon's Shadow Chapter
   Treats the computation of the Earth's shadow as seen on the Moon's disk.
4. ग्रहोदयास्ताधिकारः — Planetary Rising and Setting
   Explains the methods for computing the heliacal rising (*udaya*) and setting (*asta*) of planets.
5. शुक्लोन्नत्यधिकारः — The Brightness of the Moon
   Computes the illuminated portion of the Moon's disk.
6. चन्द्रशुक्लोन्नत्यधिकारः — Moon's Bright Elevation
   The seventh chapter, with 18 verses, covering the arc of illumination.

The Commentary (नूतनतिलक)

The Hindi commentary (टीका) in this edition was composed by **Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī**. The commentary follows the traditional structure of Indian mathematical commentaries:

- सूत्र (*Sūtra*) — The original Sanskrit verse
- सूत्रार्थ (*Sūtrārtha*) — Literal meaning of the verse (marked सु. भा.)
- विवृति (*Vivṛti*) — Detailed explanation (marked वि. भा.)
- हिन्दी भाष्य (*Hindi Bhāṣya*) — Hindi explanation (marked हि. भा.)
- उपपत्ति (*Upapatti*) — Mathematical proof or demonstration

Mathematical Notation

Brahmagupta's work represents one of the high points of Indian mathematics. Key contributions in these chapters include:

- Rules for computing eclipse magnitudes using the भुज (*bhuja*, sine) and कोटि (*koṭi*, cosine)
- The कर्ण (*karṇa*) formula: $\text{karṇa} = \sqrt{\text{bhuja}^2 + \text{koṭi}^2}$

- Iterative methods (असकृत्, *asakṛt*) for refining planetary positions
- Proportional reasoning (त्रैराशिक, *trairāśika*) for time conversions
- Geometric constructions using क्षेत्र (*kṣetra*, diagrams)

IAST Transliteration Conventions

This bilingual edition uses the International Alphabet of Sanskrit Transliteration (IAST) scheme:

Devanagari	**IAST**	**Devanagari**	**IAST**	**Devanagari**	**IAST**
☒	a	☒	ka	☒	ta
☒	ā	☒	kha	☒	tha
☒	i	☒	ga	☒	da
☒	ī	☒	gha	☒	dha
☒	u	☒	ṅ	☒	na
☒	ū	☒	ca	☒	pa
☒	e	☒	cha	☒	pha
☒	ai	☒	ja	☒	ba
☒	o	☒	jha	☒	bha
☒	au	☒	ñ	☒	ma
☒	ṛ	☒	ṭa	☒	ya
☒	ṝ	☒	ṭha	☒	ra

End of Part 2, Chunk 3

Bilingual English–Sanskrit Edition

Brāhmasphuṭasiddhānta

Part II — Chunk 4

ब्राह्मस्फुटसिद्धान्त – द्वितीयभागः – चतुर्थं खण्डकम्

Brahmagupta

(c. 598–668 CE)

Pages 1259–1398

सूत्राणि | Sanskrit Verses — व्याख्या | English Translation

Typeset with XeLaTeX using Noto Serif Devanagari
Transcribed from the original printed edition

Introduction / प्रवेशः

एतद् ग्रन्थखण्डकं ब्राह्मस्फुटसिद्धान्तस्य द्वितीयभागस्य चतुर्थं खण्डकम् । एतस्मिन् पृष्ठसङ्ख्याः १२५९--१३९८ समाविशन्ति । ब्रह्मगुप्तेन रचितं गणितज्योतिषकोशम् एतत्, यस्मिन् चन्द्रग्रहणादीनां गणनाविधयः सविस्तरं निरूपिताः ॥

This chunk (pages 1259–1398) of Part II of the *Brāhmasphuṭasiddhānta* comprises the fourth textual division of Brahmagupta's encyclopaedic treatise on mathematics and astronomy, containing detailed computational rules for lunar and planetary calculations.

अस्य खण्डकस्य विषयाः: (१) चन्द्रस्य स्फुटराश्यनयनम्, (२) शून्यविधिः, (३) वर्गप्रकृतिः, (४) ग्रहाणां मन्दशीघ्रफलानि, (५) विषयानुक्रमणिका च ॥

Contents of this chunk: (1) Computation of the Moon's true longitude; (2) Rules for operations with zero; (3) The nature of squares (Pell's equation); (4) Planetary equations of centre and anomaly; (5) Index of subjects.

Chapter on Lunar Calculations / चन्द्रराशिः — Pages 1259–1287

[Original p. 1259]

सू. भा.---चन्द्रस्य स्फुटराशिः । मध्यमराशिम् मन्दफलशीघ्रफलाभ्यां संस्कृत्य स्फुटराशिः लभ्यते । अहर्गणात् मध्यमराशिः गणनीयः ॥

Sū. Bhā.—The true longitude of the Moon: having applied the corrections for equation of centre (manda-phala) and evection (śīghra-phala) to the mean longitude, the true longitude is obtained. The mean longitude is calculated from the ahargaṇa (elapsed days since the epoch).

[Original p. 1260–1264]

सू. भा.---चन्द्रमन्दफलानि । चन्द्रस्य मन्दफलानि प्रत्येकं पञ्चदशांशानतराले सूचीकृतानि । तान्य् अनयानि---३९, ३६, ३१, २४, १५, ५ । एभिः स्फुटीक्रियते ॥

Sū. Bhā.—The Moon's equations (manda-phala) are tabulated for each 15° of anomaly. The tabular differences are: 39, 36, 31, 24, 15, 5. These are applied successively to obtain the true longitude.

वि. भा.---मध्यमस्यानतराले । यदा कोणः सूचीकृतमध्ये वर्तते, तदा अवशिष्टचापं गृहीत्वा गतागतफलयोः अर्धान्तरेण गुणयित्वा नवशतभक्तं समर्धयोर्योगेन योजयेत् ॥

Vi. Bhā.—When the anomaly lies between two tabulated values, multiply the residual arc by half the difference of the passed and upcoming tabular differences, divide by 900′, and add to half the sum of the two differences.

[Original p. 1265–1269]

सू. भा.---शीघ्रफलानि । चन्द्रस्य शीघ्रफलानि शीघ्रकोणात् गणनीयानि । शीघ्रकोणः चन्द्रसूर्ययोर् अन्तरम् । शीघ्रफलस्य □□□□□□□मं ⊠⊠′ (अशीतिः कला) ॥

Sū. Bhā.—The śīghra corrections for the Moon are computed from the śīghra anomaly, which is the difference between the Moon's longitude and the Sun's longitude. The maximum śīghra equation is 80′ (80 arc-minutes).

[Original p. 1270–1274]

सू. भा.---लग्नानयनम् । लग्नानयनार्थं सूर्योदयात् गतकालं रवेः दैनिकगत्या गुणयित्वा रविकोक्षेप्य आयनांशसंस्कारं कुर्यात् ॥

Sū. Bhā.—To find the lagna (ascendant): multiply the time elapsed since sunrise by the Sun's daily motion, add to the Sun's longitude, and apply the ayanāṃśa correction.

[Original p. 1275–1279]

सू. भा.---युतिकालानयनम् । ग्रहयुतिकालः द्वयोर् ग्रहयोः राश्यन्तरेण द्वयोः दैनिकगत्योर् अन्तरेण भजयित्वा लभ्यते । फलं युतिकालं दिनानि ददाति ॥

Sū. Bhā.—The time of conjunction is found by dividing the difference in longitudes by the difference in daily motions. The result gives the number of days until conjunction.

[Original p. 1280–1284]

सू. भा.---कालविभागः । प्राणः चतुःषष्टिकला भवति । षट् प्राणाः एका विनाडी, सष्टिः विनाड्यः एका नाडी (घटी), सष्टिर् नाड्यः एकं दिनम् ॥

Sū. Bhā.—A prāṇa equals 4 seconds of time. Six prāṇas make a vināḍī, 60 vināḍīs make a nāḍī (ghaṭī), and 60 nāḍīs make a day.

[Original p. 1285–1287]

चन्द्रगणनाप्रकरणस्य समाप्तिः । अग्रे छायाव्यवाहारः प्रवर्तते ॥

Here end the verses on lunar computation; next follows the section on shadow calculations (chāyā-vyavahāra).

Mathematical Derivations / उपपत्तय: — Pages 1288–1337

[Original p. 1288–1292]

उपपत्तिः---ज्या कोटिरूपेण सूचीकृतानि, मध्ये चापान्तरं $h = \boxtimes\boxtimes^\circ$ । मध्यमे कोणे $heta$ अन्तरे ज्या:

$$\sin(\theta) = j_1 + \frac{\theta}{h}(j_2 - j_1) - \frac{\theta}{h}\left(1 - \frac{\theta}{h}\right)\frac{j_2 - j_3}{2}$$

एषा ब्रह्मगुप्तस्य द्वितीयक्रमान्तर्गमनरीतिः ॥

Proof.—Let j_1 and j_2 be two successive tabular sines, with arc difference $h = 15^\circ$. For an intermediate arc at distance θ from j_1, the sine is:

$$\sin(\theta) = j_1 + \frac{\theta}{h}(j_2 - j_1) - \frac{\theta}{h}\left(1 - \frac{\theta}{h}\right)\frac{j_2 - j_3}{2}$$

This is Brahmagupta's second-order interpolation formula.

[Original p. 1293–1297]

उपपत्तिः---कुट्टकरीत्या $ax + c = by$ गणनीयम्, यत्र $a = \boxtimes\boxtimes\boxtimes$, $b = \boxtimes\boxtimes\boxtimes$, $c = \boxtimes\boxtimes$ । २५१ अपेक्ष्य १३७---भागलब्धिः १, शेषं ११४ । १३७ अपेक्ष्य ११४---भागलब्धिः १, शेषं २३ । एवं पश्चात् गुणकारलब्धी प्राप्येते ॥

Proof.—Applying the kuṭṭaka (pulverizer) to solve $137x + 15 = 251y$: divide 251 by 137, quotient 1, remainder 114. Divide 137 by 114, quotient 1, remainder 23. Divide 114 by 23, quotient 4, remainder 22. Divide 23 by 22, quotient 1, remainder 1. Working backwards yields the multiplier and quotient.

[Original p. 1298–1302]

सू. भा.---छायानयनम् । शङ्कोः (निरङ्कुशस्य) द्वादशाङ्गुलस्य समीपे छाया निर्णेया:

$$\text{छाया} = \frac{12 \times \cos(\text{नतभूजा})}{\sin(\text{उन्नतांशः})}$$

यदा रविः प्रथमलम्बकस्थः, तदा विशेषनियमाः प्रयुज्यन्ते ॥

Sū. Bhā.—The shadow (chāyā) for a gnomon of 12 aṅgulas:

$$\text{Shadow} = \frac{12 \times \cos(\text{zenith distance})}{\sin(\text{altitude})}$$

When the Sun is at the prime vertical, special rules apply using the nara (sine of latitude) and śaṅku (gnomon).

[Original p. 1303–1307]

उपपत्तिः---लग्नं क्षितिज्यात् लभ्यते । ϕ अक्षांशः, ε क्रान्तिः । दत्तकाले t (सूर्योदयात् अंशैः मिते) लग्नं λ:

$$\tan(\lambda) = \frac{\sin(\square.\square.)}{\cos(\square.\square.)\cos(\varepsilon) - \tan(\phi)\sin(\varepsilon)}$$

Proof.—The lagna is found from the oblique ascension. Let ϕ be the latitude, ε the obliquity of the ecliptic. For elapsed time t (in degrees from sunrise), the lagna λ satisfies:

$$\tan(\lambda) = \frac{\sin(\text{R.A.})}{\cos(\text{R.A.})\cos(\varepsilon) - \tan(\phi)\sin(\varepsilon)}$$

[Original p. 1308–1312]

सू. भा.---मन्दफलोदाहरणम् । कुजस्य मन्दकोणे μ परिधौ $r = \boxtimes\boxtimes$ (यत्र मध्यमपरिधिः $R = \boxtimes\boxtimes$):

$$\text{मन्दफलं} = \arcsin\left(\frac{39\sin(\mu)}{\sqrt{60^2 + 39^2 + 2 \cdot 60 \cdot 39\cos(\mu)}}\right)$$

Sū. Bhā.—For Mars, with manda anomaly μ and epicycle radius $r = 39$ (where deferent radius $R = 60$):

$$\text{manda-phala} = \arcsin\left(\frac{39\sin(\mu)}{\sqrt{60^2 + 39^2 + 2 \cdot 60 \cdot 39\cos(\mu)}}\right)$$

This gives the equation of centre for Mars.

[Original p. 1313–1317]

उपपत्तिः---प्रथमं मन्दसंस्कारेण मन्दस्फुटं लभ्यते: $\lambda_{\square\square} = \bar{\lambda} + \text{मन्दफलम्}$ । ततः शीघ्रकोणं मन्दस्फुटात् गणयित्वा शीघ्रसंस्कारं ददातिः $\lambda_{\text{स्फुट}} = \lambda_{\square\square} + \text{शीघ्रफलम्}$ ॥

Proof.—First apply manda correction to obtain manda-sphuṭa: $\lambda_{\text{ms}} = \bar{\lambda} + \text{manda-phala}$. Then compute śīghra anomaly from λ_{ms} and apply śīghra correction: $\lambda_{\text{true}} = \lambda_{\text{ms}} + \text{śīghra-phala}$.

[Original p. 1318–1322]

सू. भा.---ग्रहणानयनम् । चन्द्रस्य दूरत्वे पृथिव्याः छायापरिधिः:

$$R_{\text{छाया}} = \frac{R_{\text{पृथिवी}} \cdot (D_{\text{चन्द्र}} - D_{\text{सूर्य}})}{D_{\text{सूर्य}}}$$

युत्यां चन्द्रस्य अक्षः ग्रहणं निर्णयति ॥

Sū. Bhā.—The semi-diameter of the Earth's shadow at the Moon's distance:

$$R_{\text{shadow}} = \frac{R_{\text{Earth}} \cdot (D_{\text{Moon}} - D_{\text{Sun}})}{D_{\text{Sun}}}$$

where D denotes distance. The latitude of the Moon at conjunction determines whether an eclipse occurs.

[Original p. 1323–1327]

सू. भा.---उदयकालः । ग्रहः सूर्यात् दूरे सति दृश्यते । दृग्भूमिका ग्रहप्रभावतः भिद्यते---शुक्रस्य ⌧⌧°, बृहस्पतेः ⌧⌧°, बुधस्य ⌧⌧°, शनैश्चरस्य ⌧⌧°, कुजस्य ⌧⌧° ॥

Sū. Bhā.—A planet becomes visible when it is far enough from the Sun. The arc of visibility depends on the planet's brightness: for Venus 10°, for Jupiter 11°, for Mercury 13°, for Saturn 15°, and for Mars 17°.

[Original p. 1328–1337]

उपपत्तिः---ऋजुवृत्तसंस्कारः, ग्रहाणां कलाभिः सह भौमिकसङ्गणनानि च । एषा खण्डिका ब्रह्मस्फुटसिद्धान्तस्य गणितीयरीतिसारांशं समाविशति ॥

Proof.—Ṛjuvṛtta (straight epicycle) corrections and planetary phase computations. These final derivations summarise the algebraic methods used throughout the *Brāhmasphuṭasiddhānta* for astronomical computations.

Index of Subjects / विषयानुक्रमणिका — Page 1338

[Original p. 1338]

अनुक्रमणिका (पृष्ठ १३३८)---अस्य ग्रन्थस्य विषयान् सूचयति ॥

Index of Subjects (page 1338)—A detailed listing of topics covered in this volume with corresponding page references.

Algebra / बीजगणितम् — Pages 1339–1370

[Original p. 1339–1343]

सू. भा.---शून्यविधिः । धनयोर् योगो धनम्, ऋणयोर् योगो ऋणम् । धनऋणयोः योगः अन्तरं; समे सत्य् अंशे शून्यं भवति ॥

Sū. Bhā.—The sum of two positive quantities is positive; the sum of two negative quantities is negative. The sum of a positive and a negative is their difference; if they are equal, the sum is zero.

[Original p. 1344–1348]

सू. भा.---शून्यगुणनम् । शून्येन गुणितं किमपि शून्यं भवति । धनं ऋणेन गुणितं ऋणं, ऋणं ऋणेन गुणितं धनं । शून्यं येनापि भक्तं शून्यम् ॥

Sū. Bhā.—Zero multiplied by any quantity is zero. A positive quantity multiplied by a negative is negative; a negative by a negative is positive. Zero divided by any quantity is zero.

सू. भा.---शून्यहारः । शून्येन विभक्तं राशिः शून्यहारः तत् ख-बीजम् अभवत् । अस्य संशोधनं भास्करेण द्वितीयेन कृतम् ॥

Sū. Bhā.—A quantity divided by zero becomes that fraction of which the denominator is zero. This *khacheda* (zero-divisor, or “zero-seed”) was later corrected by Bhāskara II.

[Original p. 1349–1353]

सू. भा.---वर्गप्रकृतिः । $Nx^2 + 1 = y^2$ समीकरणेः प्रथमं लघूत्तरं (x_1, y_1) आक्षेपेण वा चक्रवालविधिना लभ्यते । अनन्तरं भावनया अनन्तानि उत्तराणि जनयितुं शक्यन्ते ॥

Sū. Bhā.—To solve $Nx^2 + 1 = y^2$ (Pell’s equation): first find the least solution (x_1, y_1) by inspection or by the cakravāla (cyclic) method. From this, infinite solutions can be generated by composition (bhāvanā).

[Original p. 1354–1358]

उपपत्तिः---$N =$ ⊠⊠ इत्यस्मिन्: आरभ्य $a_0 =$ ⊠ (यतः $8^2 = 64 \approx 61$) । कल्पकः $k_0 =$ ⊠ । चक्रवालविधिना क्रमशः सन्निकर्षाः

$$a_1 = \frac{a_0 + m_0}{|k_0|}, \quad k_1 = \frac{m_0^2 - N}{k_0}$$

यत्र m_0 इति चीयते यत् $a_0 + m_0$ क्षेपेण विभाज्यं, m_0^2 च N समीपे ॥

Proof.—For $N = 61$: start with $a_0 = 8$ (since $8^2 = 64 \approx 61$). The interpolator $k_0 = 3$. The cyclic process generates successive approximations:

$$a_1 = \frac{a_0 + m_0}{|k_0|}, \quad k_1 = \frac{m_0^2 - N}{k_0}$$

where m_0 is chosen so that $a_0 + m_0$ is divisible by k_0 and m_0^2 is close to N.

[Original p. 1359–1363]

सू. भा.---समासः । समान्तरश्रेणीफलं $S_n = \frac{n}{2}(2a + (n-1)d)$ । वर्गाणां समासः $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ । घनानां समासः $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ ॥

Sū. Bhā.—The sum of an arithmetic series: $S_n = \frac{n}{2}(2a + (n-1)d)$. The sum of squares: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$. The sum of cubes: $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

[Original p. 1364–1368]

सू. भा.---मिश्रकव्यवहारः । चक्रवृद्धौ, यदि मूलधनं P दरात् r अन्तराले n अन्तरैः वर्धते, तदा मिश्रधनं $A = P(1+r)^n$ ॥

Sū. Bhā.—In compound interest, if principal P grows at rate r per period for n periods, the amount is $A = P(1+r)^n$.

[Original p. 1369–1370]

बीजगणितप्रकरणस्य समाप्तिः । सिद्धान्तस्य स्तुतिः, अग्रे ग्रहचाराधिकारस्य प्रवेशः ॥

Here conclude the verses of the algebra section, with praise of the siddhānta and transition to the planetary motion chapter.

Planetary Motion / भग्रहचाराधिकारः — Pages 1371–1398

[Original p. 1371–1375]

सू. भा.---ग्रहसाधनम् । अहर्गणात् मध्यमं गणयित्वा मन्दशीघ्रसंस्काराभ्यां क्रमेण स्फुटं लभ्यते ॥

Sū. Bhā.—Having computed the mean longitude (madhyama) from the ahargaṇa, apply the manda and śīghra corrections in order to obtain the true longitude (sphuṭa).

[Original p. 1376–1380]

मन्दपरिधयः---कुजस्य ⊠⊠, बुधस्य ⊠⊠, गुरोः ⊠⊠, शुक्रस्य ⊠⊠, शनेः ⊠⊠ । शीघ्रपरिधयः---कुजस्य ⊠⊠, बुधस्य ⊠⊠⊠, गुरोः ⊠⊠, शुक्रस्य ⊠⊠⊠, शनेः ⊠⊠ ॥

Manda epicycles: Mars 39, Mercury 22, Jupiter 32, Venus 14, Saturn 60. **Śīghra epicycles**: Mars 73, Mercury 121, Jupiter 71, Venus 132, Saturn 39 (in units where $R = 60$).

Planet	Manda Epicycle	Śīghra Epicycle	Max Equation
Mars (⊠⊠⊠⊠)	39	73	11°30′
Mercury (⊠⊠⊠⊠)	22	121	22°
Jupiter (⊠⊠⊠⊠⊠)	32	71	5°
Venus (⊠⊠⊠⊠⊠⊠)	14	132	11°
Saturn (⊠⊠⊠⊠)	60	39	8°

Table 2: *
Epicycle dimensions and maximum equations (degrees where $R = 60$)

[Original p. 1381–1385]

सू. भा.---उदयकालः । ग्रहस्य उदयकालः तस्य राशेः, क्षितिज्याया अनुप्रवणतायाः, स्थलस्य च अक्षांशात् गणनीयः । चरजा (उदयास्तभूमिः) आरोहणभिन्नायां प्रयुज्यते ॥

Sū. Bhā.—The rising time of a planet is computed from its longitude, the oblique ascension of the ecliptic, and the latitude of the place. The *carajā* (ascensional difference) is applied to find the exact moment of heliacal rising.

[Original p. 1386–1390]

सू. भा.---समाप्तिः । एवं ग्रहाणां सूर्यचन्द्रमसोः च गतिः ससंस्कारा निरूपिता । एभिः विधिभिः यात् काले खग□□□□◌ां स्थितिः गणनीया ॥

Sū. Bhā.—Thus have been explained the motions of the planets, the Sun and the Moon, with their corrections. By these methods, the positions of the celestial bodies may be computed for any desired time.

[Original p. 1391–1398]

समाप्तिः---एषः ब्राह्मस्फुटसिद्धान्तः ब्रह्मगुप्तेन जिष्णुघाटसुतेन शकाब्दे ६२८ (५५० ई.) वयसि त्रिंशे सञ्चिक्षेप । एतत् पठतां सर्वेषां बोधाय भवतु ॥

Colophon.—This *Brāhmasphuṭasiddhānta* was composed by Brahmagupta, son of Jṣṇughāṭa, in the year 628 of the Śaka era (550 CE), at the age of 30. May it bring clarity to all who study it.

Colophon and Editorial Notes / समाप्तटिप्पणी

अस्य पाठस्य विषयाः: ब्रह्मगुप्तकृतं गणितज्योतिषकोशम् एतत्, यस्मिन् धनऋणशून्यगणितानि, कुट्टकरीतिः, वर्गप्रकृतिः, ग्रहस्फुटीकरणं च सविस्तरं वर्णितम् ॥

Key Mathematical Contents: The *Brāhmasphuṭasiddhānta* is notable for: (1) algebraic rules for arithmetic with positive and negative numbers, zero, and surds; (2) the kuṭṭaka (pulverizer) method for linear Diophantine equations; (3) the varga-prakṛti method for Pell's equation; (4) astronomical calculations for planetary positions, eclipses, and conjunctions; (5) trigonometric interpolation formulas.

सम्पादकीयम्: एतत् पाठसंस्करणं मूलमुद्रितसंस्करणात् सङ्कलितम् । ब्रह्मगुप्तेन उज्जयिन्यां ६२८ ई. मणि संकलितं गणितज्योतिषयोः शिखरं इदं ग्रन्थकोशम् ॥

About this edition: This text edition was prepared from the original printed edition. Composed in 628 CE at Ujjain, the work represents one of the pinnacles of Indian mathematical astronomy.

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ब्राह्मस्फुटसिद्धान्त

देवनागरी

ग्रहच्छायाधिकारः

प्राच्यपरायाः शङ्कूस्तलं तदप्रं ग्रहं मे च

प्राच्यपरायाः

शङ्कुवप्रं ग्रहं भवतोति स्पष्टम्

तुल्यादिशोः

शङ्कुतलं प्राच्यपरान्तं भुजो दक्षिणोत्तरं कर्णः
द्विजया तद्विपरीतमूलं द्विक्षेपमध्यतः कोटिः

शङ्कुतलं प्राच्यपरान्तं

द्विक्षेपसम्पातगतयोर्यथाप्राक्पूर्वापरसुतयोरैं यो लम्बः स भुजः

युक्तिलक्षणम्

उभे मानक्याखण्डि ग्रहयोर्मध्यान्तरं युक्तिगृह्योः
समलिप्तिकयोगं हृत्वैवद्विके स्फुटमानयोगाखण्डित्

मानक्याखण्डिके

समकलयोर्ग्रहयोर्मध्यान्तरं

सम कलात्मक दो ग्रहों के अन्तर यदि दोनों ग्रहों के विभक्तयोगाखण्ड से न्यून हो तब ग्रह द्वय की युक्ति होती है अर्थात् ग्रहणवत् आच्छादन (उपरिस्थ ग्रह ग्राह्य और अभःस्थित ग्रह ग्राहक) होता है। यदि दोनों ग्रहों के अन्तर स्फुट विभक्तयोगाखण्ड से अधिक हो तब ग्रह की युक्ति नहीं होती। सिद्धान्त शेखर में मानक्याखण्डि धुवरविकरे स्यान्मेदोः द्विके तू' इत्यादि विज्ञान भाष्य में लिखित स्लोकांश से श्रीपति सिद्धान्त शिरोमणि में मानक्याखण्डि धुवर विकरेऽस्ये भवेद् योगः' इत्यादि से भास्कराचार्य ने भी आचार्यैक के अनुरूप ही कहा है इति ॥६९॥

भग्रहत्यधिकारः

भुजकोटिकर्णानां यद्धुज्यासार्धं परिणामनार्थं तान् यद्धुज्यान् व्यासार्धहृतान्
गुणकः कुर्यात्

भुजकोटिकर्णानां यद्धुज्यासार्धं परिणामनार्थं तान् यद्धुज्यान् व्यासार्धहृतान्
गुणकः कुर्यात्

अथोपपतिः । गोल्युत्थया भुजकोटिकर्णैःशङ्कुकुर्मस्थानवशातः स्फुटा भुजामर्यधायां चापिहटी युक्तिराचार्यैषां प्रतिपादिता

यदा द्व्येक्षणाडिकाभिर्न्तुल्याः कुत्वा तदा योगो भवेत् । योगे ग्रहयोर्मध्यान्तरेष्टज्ञोनितिवत्समान्यदिशोःप्राच्यशङ्कुवप्रयोःसंयोगान्तरंबाहु कार्यवर्धन्मध्यात्तयोगवतोऽन्त-हयोः प्रत्येकस्य शङ्कुतलस्याप्रायिकं योगान्तरं तयोर्भुजो भवत् इत्यर्थः

ग्रहयो-रैक्ये कर्णो भवतः । स्वकर्णो भुजाकृतिवियोगाप्तेस्वकर्णोभुजयोर्वर्गयोरन्तर-मूलेकोटी भवतः । एवं ग्रहयोः कोटिभुजकर्णैःशङ्कुकृत् यद्धुज्यान् कल्पितषडष्टि-गुणान्व्यासार्धहृतान्विज्याध्मातान्कृत्वाथैकेन कर्तव्येन कोटिभुजकर्णैः शङ्कुवो यद्धुज्यासार्धपरिणाता भवेतुः

प्रागपरयोःयद्धुज्यासार्धपरिणातपूर्वा-पररेखानुल्लपयोःप्राच्योपूर्वदिशिपश्चात् पश्चिम दिशिपृथक् कोटी प्रसार्य कोटिप्रसार्या बाहुभुजोदल्वा दिक्षुमध्यतो मध्य विद्धोःभुजाप्रान्तोभुजाप्रा-विद्धलेखोकर्णां च दल्वा बाहुप्रयोःस्वभुजाप्रयोः स्वशङ्कुमध्यात् तदप्रान्तेस्वशङ्कुवप्रयोर्लेखनेयथी च पूर्वकल्पिते द्वे

द्विक्षेपमध्यस्थहच्छायामध्यबिन्दुस्थापितेन चक्षुषास्वशङ्कुवप्रं पृथक् ग्रहं दर्शयेत् । योगेग्रहयो-योगेतयोः शङ्कुवकुव्रान्तरं भवेद्यत् शङ्कुवन्तरामावः, शङ्कुभुजादीनां सम-त्वात्अन्यथाग्रहयुतिर्मिन्नेऽस्वरेतयोः शङ्कुवोरन्तरं भवेदेव, भ्रमेयोगे शङ्कु-कुव्रान्तरसन्तरमेवान्यथा हि तयोःयोगेमहायुति शङ्कुकुव्रान्तरं ग्रहयोरन्तरं नेयं भ्रमदाग्रहयुतिर्मिन्नकल्पेऐषं नेयमिति

तन्त्रपरीक्षाध्यायः

इदानीमार्यभटोक्तं युगं खण्डयति
आर्यभटो युगपादांशाने यातानाह कलियुगादौ यत्
तस्य क्षुतान्तर्यस्मात् स्वयुगाष्ठतो न सत् तस्मात्

आर्यभटः कलियुगादौ द्विन् युगपादान् यातान् आह कथितवान् । यस्मात् तद्ग्रथतःदृष्ट्या मध्यमाधिकारे क्षद्विशतिस्लोकस्य मदीया व्याख्यायस्मात् कारणात् तस्मात् तस्य स्वयुगाष्ठतोतदेकस्य युगस्यादिर्न्यस्यात् इतिद्वि कृतान्तः कृतयुगमध्ये भवतस्तस्मात् तच्छुं न सत्

आर्यभटमते एक युगान्ताद्यस्यारम्भात् कलियुगादिपर्यन्तं त्रयो युगपादाः $= \dfrac{3 \times 4320000}{4} = 3240000$आचार्यमते च, क्षु+त्रे+आ $\dfrac{4320000 \times 9}{10} = 3888000$द्वयोरन्तरे वर्षाणि$= 648000$एतानि चाचार्यमतेन संख्याधिकत्वात् कृतयुगमध्येऽपि आर्यभटोत्तयुगाष्ठतो कृतयुगान्तः

$= \frac{3\times4320000}{4} = 3240000\frac{4320000\times9}{10} = 3888000 = 648000$

अथ पुनरार्यभटोक्तवारादि खण्डयति
श्रीक्षुरो दिनवारो गुरुराद्विकोऽस्य भवति कल्पादौ
न भवत्यकं यस्मादोच्चुरो विस्तरतस्तस्मात्

आर्यभटेन स्वतन्त्रे गुरुदिवसात् भारतात् पूर्वं' मित्यनेन कल्पादौ गुरुवारः स्वीकृतः । तेनायमर्थः । यस्मात् कारणात्-अस्यआर्यभटस्य श्रीक्षुरःस्वीकारःकल्पादावाद्विको दिनवारो गुरुर्भवति रविनं भवति तस्मादस्योक्षुरः स्वीकारो विस्तरःआधाररहितोऽथप्रामाणिकः

आर्यभटमतेन कलियुगारम्भात् पूर्वं वर्तमानकल्पे क्ष उ मन्वो व्यतीता युगचतुरायां च, तथा तस्मात् द्विसप्ततियुगोरैको मनुतः कल्पादौ कल्पादौ गतयुगानि$= 72 \times 6 + \dfrac{3}{4} = 432\,\dfrac{3}{4}$युगपञ्चितसावनदिनैर्गुणनें जाताः सावनाहर्गणाः$= 432\times1577917500+\dfrac{3 \times 1577917500}{4}$ $= 432 \times 1577917500 + 3 \times 394479375 \times 3$अयं सप्तभिर्मैं कल्पादौ वारः$= 5 \times 5 + 3 \times 3 = 25 + 9 = 34 = 6$अयं सैकः कलियुगादौ वारः$= 7 = 0$अतो यदि गुरुवाराद्-गणानाद्दृष्टे तदा कलियुगादौ गतवारः ० । वर्तमानो गुरुरेव सिद्ध्यत् आर्य-भटमतेन कल्पादौ गुरुवार आयाति

$= 72\times6+\frac{3}{4} = 432\,\frac{3}{4} = 432\times1577917500+\frac{3\times1577917500}{4}$
$= 34 = 6 = 0$

अथ पुनरार्यभटोक्तवारादि खण्डयति
क्षधिकैः शतैश्चतुर्भिर्वर्षैः सहस्रैश्चतुर्भिर्यामिरेकः
युगयातोदिनवारान्तर्मौदित्यकाष्ठे रात्रिकयोः

आर्यभटेन प्रथद्वयं रचितम् । एकैकस्मिन् युगसावनदिनानि1577917500 लघुगामकल्पदिवे सृष्टिःअन्यैस्मिन् युगसावनदिनानि1577917800लघुगामधरात्रौ सृष्टिः4320000समाना एव, युगपञ्चितसावन दिनात्रयं$= 300$अतो प्रथद्वयतो वारगणनया युगवर्षदिनशतत्रयान्तरं भवति, तथानुपातेनैकदिनवारान्तरं च$\dfrac{4320000}{300} = 14400$एतैर्युगातवर्षः

$= 1577917500 = 15779178004320000 = 300 = \frac{4320000}{300} =$
14400

तेनायमर्थः --- आर्यभटमतेनैदियकाष्ठरा-त्रिकयोदिनवारमध्ये चतुर्दशाभिर्वर्षैः सहस्रै ऋतुभिः शतवर्षैऽधिकैर्गुयातैदिनवारा-न्तरं दिनवारयोरेकान्तरं पततीति वारगणना न स्थिरेति

कर्णाधिकारः

उपपत्तिः
स्फुटादिकारके १३वें सूत्र की उपपत्ति में दिखाया गया है कि प्रथम पद में ×/
गतांश क्रमज्या × परिधि / भोज्या = भुजफल
द्वितीय पद में परम भुज फल होता है, तृतीयपद में ×/
क्रमज्या × परिधि / भोज्या = भुजफल, द्वितीय पद में परम भुजफल ×/
शून्य होता है
चतुष्कपद में परम उत्क्रमज्या × परिधि / भोज्या = भुजफल, यद्यपि समपदान्ते विषम पदादौ मे भी दोनों पदों की सन्धि होने के कारण परिधिध्य (सम पद और विषम पद में पठित परिधि) ग्रहण करने से फलनाम (फलभाव) नहीं होता है, उससे कुछ हानि नहीं है क्योंकि परिधि पाठ भेद से भावान्तर में अनुपात से परिधि का ग्रहण करना चाहिये ऐसा ने आर्यभट सूचित किया हुआ है इसलिये आचार्यैक खण्डन ठीक नहीं है इति

इदानीमार्यभटमतेनानुपातेन यदि परिधिः स्फुटः क्रियते तदापि न समीचीन इति खण्डयति
व्यासार्धहृतो बाहुः परिधिविभेषाहृतः फलोनयुतः
प्रथमाद्विकोनकश्च यत् तदसत् पदयोः परिधिपातात्

बाहुभुजया परिधिविभेषाहृतः परिध्यन्त रहितो व्यासार्धेन विजयया हृतः प्रथमः परिधियं द्वि द्वितीयाद्विकोनकस्तदा क्रमेण प्रथमः परिधिः फलोनयुतः स्फुटः परिधिर्भवेदिति यत् स्फुटपरिध्यानयनं प्रसिद्धं तदप्यार्यभटमतेनासद्भवति
कस्मात् । पदयोः परिधि पाठात्
अर्थतस्तन्मते पदयोः परिधिध्यम्
तत्र कुतश्चित् प्रथमः समपदीयः कुतश्चित् विषमपदीयः परिधिर्भवति
अतः संस्कारविधिषु विचरति । बाबलमेतत्
प्रथमाद्विकोनकश्चापि तात्कालिक संस्कारस्य समीचीनत्वादिति सुधीभिः स विचिन्त्यम्

द्विक्षेपाधिकारः

इन्द्रोःचन्द्रस्यउत्क्रमज्याभुजोत्क्रमज्याविश्लेषपक्रमगुणाविश्लेषेण शरेण, अपक्रमेण परं कान्तिज्या च गुणाविज्याकृतिविज्यावर्गेभक्ता लम्बमानद्विक्षेपकल्पेत् युक्तमार्यभटेन तत् ततःतस्मात् कारणात्असत् यतःयस्मात्कारणात् क्षयनान्तेगोलसंख्येयद्गणानं फलमुत्पद्यते तत्तस्य मते श्रादावयनसंख्यावेतोत्पद्यते अतस्तदसदिति

इन्द्रोःचन्द्रस्यउत्क्रमज्याभुजोत्क्रमज्याविश्लेषपक्रमगुणाविश्लेषेण शरेण, अपक्रमेण परं कान्तिज्या च गुणाविज्याकृतिविज्यावर्गेभक्ता लम्बमानद्विक्षेपकल्पेत् युक्तमार्यभटेन तत् ततःतस्मात् कारणात्असत् यतःयस्मात्कारणात् क्षयनान्तेगोलसंख्येयद्गणानं फलमुत्पद्यते तत्तस्य मते श्रादावयनसंख्यावेतोत्पद्यते अतस्तदसदिति

सद् द्विषमेतद्यत् आर्यभटीयं वाक्यमेतद् व्रात्यं
विश्लेषपक्रमगुणामुत्क्रमं विस्तराधं क्षतिभक्तम्
अत्र मटदीपिकायां परमेश्वरः
सायनचन्द्रस्योत्क्रमं कोट्या उत्क्रमज्येतयः' एवं चेत् तदास्यार्यभटं न समीचीनम्
इदं कर्म क्रमज्या क्षेपद्वयमार्यभटादिमतोत्क्रमज्यया यत् कृतं तत्र तथ्यमिति भास्करखण्डनं वस्तुतो गोलयुक्तिकृत् वास्तवमिति स्फुटं चापक्षेत्रकु-क्षेत्राणाम्

युगाधिकारः

हिं. भा. --- जिस कारण से ग्रहगणनाओं तथा पात मन्दोच्च-शीघ्रोच्चों की मेषादि स्थिति में जितने काल में फिर मेषादि में स्थिति होती है वहीं काल महा युग कहलाता है, श्रीपथ्यविद्याचुद्र आदि आचार्यों ने जो युग कहा है वह स्फुट है, क्योंकि उनके मत से महायुगादि मे मेषादि में ग्रहय (भिन्न स्थान में) उत्पन्न होता है, अर्थात् उनके मत से मनु आदि स्मृतिकार रचित स्मृति-ग्रन्थ सम्मत महायुगादि में यदि ग्रहगणना करते हैं तो मेषादि में मज्ज्ञनादि ग्रहों की स्थिति सिद्ध नहीं होती है, श्रीपैथा रचित रोमक में कहीं पर महायुगादि की चर्चा नहीं है, इसलिये महायुग शब्द से यहाँ आचार्य कथित महायुग लेना चाहिये, जब सब ग्रहों को तथा पात मन्दोच्च शीघ्रोच्चों की स्थिति मेषादि में होती है तो उसी का नाम महायुगादि है, श्रीपैथा कथित महायुगादि में यह स्थिति नहीं होती है इसलिये उनके मत से महायुगादि ठीक नहीं है, आचार्यैक यह खण्डन ठीक है

इदानीं पुनरपि तच्छुंमेव नि रात्रं रोति
कदिनादौ स्मृतिबुक्तं ग्रहमोक्षपतिर्दिनक्षये प्रलयः
तात्यतिबहूनि यस्मात् महायुगेष्टोऽप्रसिद्धिमवम्

कदिनादौ बहुदिनादौ ग्रहनक्षत्रोत्पतिर्दिनक्षये बहुदिनवासाने च ग्रहमानाः प्रलय इति स्मृतिग्रन्थमस्ति
एवं बहुदिननम्मे महायुगे यस्मात् तार्गिश्रीषेणाच्छत्कानि युगानि षष्टिबहूनि भवन्ति अत इदं तदुक्तमप्रसिद्धं न कुतश्चपि सत्यादि तच्छुं च
षष्टोऽनेक युगग्रहणेन गोरवकर्मणा किमिति
बाबलमेतत्
यतोऽनेकयुगग्रहणेनापि ग्रहगणा नायां न काञ्चिदानिसत्स्थानग्रहणादिति स्फुटं गणितं विदामिति

वि. भा. --- कदिनादौबहुदिनादौग्रहमोक्षपतिःग्रहनक्षत्रसृष्टिःदिनक्षये बहुदिनान्तेग्रहाणां नक्षत्राणां च प्रलयःनाशःस्मृतिषुस्मृतिग्रन्थेषुकथितमस्ति, सिद्धान्तशेखरे श्रीपतिनास्यैकमुत् यथा---ज्योतिर्ग्रहाणां विधिवासरावो सृष्टिलयस्तद्दिवसावसाने
सिद्धान्तशिरोमणौ भास्कराचार्येण □यतः सृष्टिरेषां दिनादौ दिनान्ते लयस्तेषु सत्सैव तच्छार चिन्ता' स्प्यनैन बहुदिनादावेव ग्रहदिनसृष्टिः कथ्यते, परं सूर्यसिद्धान्तकारेण कथ्यते यत् □बहुदिनादितः शातयुगात्वेदसप्तवैदिस्यामेषु ब्रह्मा सृष्टिं रचयित्वावसासै नियोजितवान्' । ब्रह्मपुत्र-श्रीपति-भास्कराचार्यैः कथितसुख्यादिकालानिराकरणार्थं सूर्यसिद्धान्तकारमतमण्डनार्थं च कमलाकरेण सिद्धान्ततत्त्वविवेके बहुप्रपञ्चं नहि नाममेदैन वस्तुमेदः

दानं च दत्तमर्धं पञ्च तेन मान-स्यार्धेषपञ्चनवमांशं तदा किम्

न्यासः$\frac{1}{2}$उत्तरं कथयेत् जातं$1\ \frac{5}{8}\ \frac{3}{4}$
अत्र कोल्ब्रहं कसाहिवेन भ्रमात्$\frac{3}{4}\ \frac{5}{8}$इत्युत्तरं वास्तवं लिखितम्

$\frac{1}{2}1\ \frac{5}{8}\ \frac{3}{4}$
$\frac{3}{4}\ \frac{5}{8}$

स्वदीकार्याम्
एवमिहाचार्यं पञ्चजात्य एकोक्ता
यतः पठतीति तदातिसंकेमातो गतार्थां कृत्वा नोक्ता
स्कन्दसेनादिभिस्तस्य नाम कृतं भागमातेति

सहस्रच्छेदांशयुक्तिःतुल्यहरांशानां योगःछेदविभक्ताहरभक्ताफलं प्रथमजाति सर्वर्णं भवति
द्वितीयजाती-मरेङ्गा गुणिताः, छेदःहरःछेदःहरःगुणितात्तदंश्या हरमका फलं सर्वर्णां भवति
प्रथमजाति मवर्णां तु ‘योगोन्तर तुल्यहरांशकानामित्यादि भास्करोक्तमनुक्रमेव
तथा द्वितीय प्रभागजातिसवर्णं यतदनुक्रमेव लीलावत्यां ‘लवालवधनाभ्यं हरा हरनाः' इत्यादि भास्करोक्तम्

इदानीं भागाहारे करणसूत्रमाह
परिवर्त्यं भागहरच्छेदांशां छेदसंगुणाच्छेदः
शेषोऽंशगुणो भाजस्य भागाहारः सर्वेषां तयोः

भागहारच्छेदांशी भाजकस्य छेदांशो परिवर्त्यं भाज्यस्य छेदः परिवर्तितच्छेदस् हराछेदो भवति
एवं भाज्यस्यांशः परिवर्तितांशगुणांशो भवति
एवं सर्वेषां तयोर्योगमिश्रयोर्भागाहारो भवति
सर्वेषां तु ‘रूपारिच्छेदगुणानि' इति विधिना
भास्करभागाहारोत्यदनुक्रम एव
उद्देशकचतुर्वेदः ---

यत्रायते फलं हष्टं सार्धेनैकयमेनदवः
सार्धं दशभुजस्चैव कोटिस्तनार्मकीयताम्

$324\ \frac{3}{8}90\ \frac{1}{2}$

न्यासः । क्षेत्रफलम्$324\ \frac{3}{8}$भुजः$90\ \frac{1}{2}$उत्क्रमजाता कोटिः$11\ \frac{3}{5}$

$= 324\ \frac{3}{8} = 90\ \frac{1}{2} = 11\ \frac{3}{5}$

अन्ये पुनरिहापि वं राशिकर्मकमुदाहरणं ददति यथा---

अथवा लघुतमापवर्तेन गणितम्

न्यासः$\frac{1}{2}+\frac{1}{3}+\frac{3}{4}+\frac{1}{5}$

अथवैदृशाः २, ६, १२, ४ मेघा २ मननापवर्तनेन

ततोऽपवर्तनाद्धां जातः$2\times 2\times 3=12$

सैष9 9 9गुणिताः12प्रमेने-वा हरांशयोगेनैनं

$\frac{1}{2}\times\frac{6}{1}+\frac{1}{3}\times\frac{12}{2}+\frac{3}{4}\times\frac{12}{3}+\frac{1}{5}\times\frac{12}{4}$

$=\frac{6}{2}+\frac{12}{6}+\frac{12}{12}+\frac{12}{20}=\frac{6}{2}=9=$योगफलम्

$\frac{1}{2}+\frac{1}{3}+\frac{3}{4}+\frac{1}{5}$

$2\times 2\times 3=12$

$\frac{1}{2}\times\frac{6}{1}+\frac{1}{3}\times\frac{12}{2}+\frac{3}{4}\times\frac{12}{3}+\frac{1}{5}\times\frac{12}{4}$
$=\frac{6}{2}+\frac{12}{6}+\frac{12}{12}+\frac{12}{20}=9=$

स्फुटादिकारके ७३४ पृष्ठे अस्य खण्डस्य □स्थाप्योन्त्यघन' इति सूत्रस्य उपपत्तिः ज्यामितिकक्षेत्रद्वारा प्रदर्शिता। एतत् $(a+b)^3$ विस्तारस्य ज्यामितिकं प्रदर्शनं पाश्चात्यगणितेभ्यः शताब्दीभ्यः पूर्वं भारते संसिद्धमासीत्।

$(a+b)^3$

वर्गीभुजाः$=a+b$
वैश्यः$=a+b$
वर्गक्षेत्रफलम्$=(a+b)\times(a+b)=a^2+2ab+b^2$
क्षेत्रफलं वैश्यगुणमित्यादिना वाप्या घनतमकं क्षेत्रं सम्बन्धेन याग्युदाहरणानि प्रदर्शितानि तानि न समीचीनानीति सुधीभिरव्यानीति

$=a+b$
$=a+b$
$=(a+b)\times(a+b)=a^2+2ab+b^2$

अ + क अस्य घन करणार्थं समत्रिघाताघं घन इति घनपरिभाषया
$(\ +\ )(\ +\ )(\ +\ )=(\ +\ )^3$
गुणनेन$(^2+\ +\ ^2)(\ +\ )=(^2+2\ +\ ^2)(\ +\ )$
$=\ ^3+2\ ^2.+\ ^2+\ ^2.+2\ ^2+\ ^3$
$=\ ^3+3\ ^2.+3^2+\ ^3=(\ +\ )^3$
एतत् □स्थाप्योन्त्यघन' इत्याचार्यैकमुपपद्यते
□स्थाप्योन्त्यवर्गं द्विगुणान्त्यनिघ्नं' इत्यादि लीलावत्यां भास्करोक्तमतदनुक्रमेवास्ति

$(a+k)$
$(a+k)(a+k)(a+k)=(a+k)^3$
$(a^2+a\cdot k+k^2)(a+k)=(a^2+2\,a\cdot k+k^2)(a+k)$
$=a^3+2\,a^2\cdot k+a\cdot k^2+a^2\cdot k+2\,a\cdot k^2+k^3$
$=a^3+3\,a^2\cdot k+3\,a\cdot k^2+k^3=(a+k)^3$

प्राचीनैः केवलं वर्गघनयोगविचारः कृतः
द्वयोर् द्वयोर् योगस्य घनकरणार्थं प्रथमाक्षेपित्वं संज्ञकः
द्वितीयो-द्वय उत्तरसंज्ञकः, तदा अन्त्यघनः स्थाप्यः, ततोऽन्त्यकृतिः अन्त्यवर्गःद्विगुणोत्तर-संज्ञकः, द्वयोर् द्वयोर् योगस्य घनो भवतीति
अत्र चतुर्वेदाचार्येदं शकः---□चतुरस्रा समा वापी हस्ततिवन समिता। वैघनं च तथा तस्याः फलं घनतमकं' घनतमकम्

यथा श्रेष्ठा रूपम् = अयमेव द्विगुण पदसिद्धान्तः$(\ +\ )=+$

$$\begin{array}{c|c|c|c} \frac{\times\ \times}{1} & \frac{\times\ -1)}{1\times 2} & \frac{\times\ -1)-2)}{1\times 2\times 3} & +\cdots\cdots \\ =1 & =2 & 2 & =3 \end{array}$$

अत्र यदि न$=2$तदा$(\ +\ )^2=\ ^2+\ \times 2\times\ +\ ^2$
यदि न$=3$तदा$(\ +\ )^3=\ ^3+\ ^2\times 3\times\ +\ \times 3\times\ ^2+\ ^3$

$$\begin{array}{c|c|c|c} \frac{a\times n\times k}{1} & \frac{a\times n(n-1)k}{1\times 2} & \frac{a\times n(n-1)(n-2)k}{1\times 2\times 3} & +\cdots \\ \hline n=1 & n=2 & 2 & n=3 \end{array}$$

$n=2(a+k)^2=a^2+a\times 2\times k+k^2$
$n=3(a+k)^3=a^3+a^2\times 3\times k+a\times 3\times k^2+k^3$

एतावता □स्थाप्योन्त्यवर्गं द्विगुणान्त्यनिघ्नं' इत्यादि वर्गोपपतिः □स्थाप्योन्त्यघन' इत्यादि घनोपपत्तिश्च स्पष्टा भवति

छेदं विभजेद् द्वयोर्लब्धं तयोः पृथक् पृथक् दलयेत्
वर्गान्मूलं दलितात् पदाद् वर्गमूलं स्यात्

वि. भा. --- द्वयोर्द्वयोर्योगस्य घनकरणार्थं प्रथमाक्षेपित्वं संज्ञकः द्वितीयो-द्वय उत्तरसंज्ञकः, तदा अन्त्यघनः स्थाप्यः, ततोऽन्त्यकृतिः अन्त्यवर्गःद्विगुणोत्तर-संज्ञकः, द्वयोर्द्वयोर्योगस्य घनो भवतीति

अत्र चतुर्वेदाचार्यैदं शकः---‘चतुरस्रा समा वापी हस्ततिवन समिता । वैघनं च तथा तस्याः फलं घनतमकं' घनतमकम्

इति श्रीब्रह्मस्फुटसिद्धान्ते द्व्याशीतिकोण्यध्यायो नामैकादशोऽध्यायः ॥
११ ॥
इति श्रीब्रह्मस्फुटसिद्धान्ते तन्त्रपरीक्षाध्यायो नामैकादशोऽध्यायः ॥ ११ ॥
इति श्रीब्रह्मस्फुटसिद्धान्ते दुष्प्रदाध्यायो नामैकादशोऽध्यायः ॥ ११ ॥

$(a+b)^n(a+b)^3$

$(a+b)^n(a+b)^3$

इति द्विभाषासंस्करणस्य समाप्तिः --- खण्डं ५

Brāhmasphuṭasiddhānta

ब्राह्मस्फुटसिद्धान्त

Bilingual Sanskrit–English Edition

Part II — Final Portion (Pages 1539–1676)

Brahmagupta

(c. 598–668 CE)

Left Column: Sanskrit (Devanagarī)
Right Column: English Translation with IAST

Contents: मिश्र गणित | Rule of Three | भाण्डप्रतिभाण्डक | शून्यपरिकर्म | श्रेढी | क्षेत्रव्यवहार | खातव्यवहार | धान्यराशि | गीताध्याय

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Contents

Introduction

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Introduction

This is the final portion of Part II of the *Brāhmasphuṭasiddhānta*. This volume contains the concluding chapters of the *Gaṇitādhyāya* (Mathematics Chapter)—the Third Class of Partnerships, Mixed Calculations, the Rule of Three, Exchange of Goods, Operations with Zero, Series, Plane Geometry, Excavations, Grain Heaps, Shadows, and the beginning of the Chapter on Spheres.

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Written in the 7th century, Brahmagupta's treatise comprises 24 chapters. This section covers an extraordinary range of mathematical topics—commercial arithmetic, the algebra of zero, series, geometry, solid geometry, and spherical geometry.

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1 तृतीय जातिः — The Third Class of Partnerships

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दिन सम्बन्धिपूरणकालः सर्वेषां योगः, ततोऽनुपातेनैक कालावच्छेदेन मुक्तसंकुल्याभिर्वापीसंपूरणकालः, एतावदाचार्योक्तमुपन्नम् ॥९॥ || ||

Chapter 1: The Third Class of Partnerships

Page 754—Here begins the discussion of the third class of partnership problems (*tṛtīya jāti*), which deals with combined rates of work.

Translation: "The sum of all [workers'] day-related filling-times is [found]; then by proportion, the time for a single cistern-filling by all [working together] released [at once] is found. Thus has the Ācārya stated what follows." ||9||

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Commentary:

In the third class (*tṛtīya jāti*), *savaraṇa* (reduction to common denominators) is taught. When the numerator standing above is multiplied by the numerator, the third-class reduction is accomplished. The denominator is given to be multiplied by the denominator, and the numerator standing there, being added or subtracted, is multiplied—then the connection of fractions and the removal of fractions is accomplished. This relates to the section on partnership.

1.1 उदाहरणम् — Worked Example

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Worked Example:

One worker fills a tank (*vāpī*) in one day; a second fills the same tank in half a day; a third fills the same tank in a quarter of a day; a fourth fills the same tank in a fifth of a day. If all workers are released at the same time, in how much time will they fill that tank?

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Statement: 1|2|4|5—According to the rule “multiply the upper and lower [terms],” summing 1, 2, 4, 5 gives 12—so many filling-days are produced by all workers together. Now by proportion: if 12 yields one day, then for one filling, what is obtained by this proportion? Thus the filling-time is determined.

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Demonstration (*upapatti*):

One worker fills a tank in one day; the second fills it in half a day, so by this proportion his one-day filling-rate is 2. If the third fills the same tank in a quarter of a day, his one-day filling-rate is 4. If the fourth fills it in a fifth of a day, his one-day filling-rate is 5. The sum of all = 12. Then by proportion the filling-time is found.

In the *Līlāvatī*, “*maje chidongī ratha tai vimiśrai*” etc.—Bhāskara’s rule is in conformity with the Ācārya’s [Brahmagupta’s] rule.

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2 मिश्र गणितम् — Mixed Calculations

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Chapter 2: Mixed Calculations

This section deals with *miśra gaṇita* or mixed calculations, including compound interest, alligation, and related commercial arithmetic problems.

2.1 मिश्रधनानयनम् — Finding the Compound Amount

अमागकाल प्रमाणधन परकाल हीनं द्वितीय स्थानं भवति
मिश्रघाते मिश्रकेष्ठ वर्गं योज्यं मूलं ततोऽन्यार्धं हीनं प्रमाणफलम् ॥१५॥
|| ||

Translation: “The first term multiplied by the standard period [and] divided by the other period becomes the second term. The square of the other principal, added to the product of the mixed [amount], [and] the square root [taken]; then, diminished by half of the other, is the standard result.” ||15||

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Commentary:

The product of the standard period and standard principal, divided by the other period—that result is to be established in two places. The result in the first place is named *bhrā*, the result in the second place = *bhranya*. The square of the other principal, added to the product of the mixed [amount], the square root taken; from it, half of the other being subtracted, the standard result is obtained. ||15||

2.2 उदाहरणम् — Worked Example on Compound Interest

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Example:

Five hundred drammas were lent at interest. In four months, whatever [amount] was due was lent again at interest in a second place; after a period of ten months, the total amount became such and such rupees. State the standard result. ||15||

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$times$ =
$frac4times50010 = 200$ ( )

$sqrt(200)^2 + (79)^2 =$
$sqrt40000 + 6241 =$
$sqrt46241$
$approx215$
$215 - 100 =$ ( )

Statement according to the example: Standard period = 4, Standard principal = 500, Other period = 10, Mixed amount = 79.

$fractextstandardperiodtimestextstandardprincipaltextotherper$

$frac4times50010 = 200$ (first term)

$sqrt(200)^2 + (79)^2 =$
$sqrt46241$
$approx215$

Subtracting 100: $215 - 100 = 100$ (standard result). ||15||

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3 त्रैराशिकादिकम् — The Rule of Three and Extensions

फलच्छिदामन्योन्यपक्षनयनं विधाय बहुराशिके वधे
स्वल्परादिवधभाजिते फलम् ॥१३॥ || ||

Translation: “Having performed the mutual transposition of the fruit and the denominators in the multiplication of the rule of many terms, the result [is obtained when] divided by the product of the smaller terms etc.” ||13||

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Commentary:

The meaning of this rule is: when extracting the result from the multiplication of many quantities, by the mutual transposition of the fruit and denominators, and dividing by the product of the smaller quantities etc., the result is obtained. The *phalachchidā* (fruit-denominator) method of mutual transposition is applied in the *bahu-rāśika* (rule of many terms).

3.1 त्रैराशिकोदाहरणम् — Example: Rule of Three

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$frac5times100125 = 4$

Example:

If in five months the increase (interest) on drammas is 125 rupees, then what is the increase (interest) on 100 rupees?

Statement: 5 | 125 | 100

$frac5times100125 = 4$ (increase)

3.2 पञ्चराशिकादि — Rule of Five and More

फलच्छिदामन्योन्यपक्षनयनं विधाय
सप्तराशिकादावपि ज्ञेयम् ॥१३॥ || ||

Translation: “Having performed the mutual transposition of the fruit and denominators, this method is to be known for the rule of seven and others as well.” ||13||

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Commentary:

For the rule of seven and others also, having made the statement according to the example, the result of the rule of seven etc.

4 भाण्डप्रतिभाण्डकम् — Exchange of Goods

प्रागभूस्य व्यत्यासो भाण्डप्रतिभाण्डकेऽन्यदुक्तसमम्
परिकर्माण्यष्टानां व्यवहाराणामभिहितानि ॥१३॥ || ||

Translation: “The transposition in exchange of goods is as previously stated; the operations of the eight procedures have been declared.” ||13||

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Commentary:

“In the first place, the two value-quantities, their transposition is to be done”—thus the four-Veda Ācārya [Brahmagupta]. (Even so, the procedure of exchange of goods etc., as stated by Bhāskara, is in conformity with this.)

That is, in the first place, the transposition (exchange) of the two value-quantities is the procedure of *bhāṇḍa-pratibhāṇḍaka* (exchange of goods).

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Example:

Ten paṇas with seven paṇas—

Statement: 10 paise, 8 paise

5 शून्यपरिकर्म — Operations with Zero

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Chapter 5: Operations with Zero

This is one of the most celebrated sections of the *Brāhmasphuṭasiddhānta*. Brahmagupta was the first mathematician in history to lay down systematic rules for arithmetic operations involving zero (*śūnya*), including the much-discussed rule for division by zero.

ऋणे गुणे ऋणं क्षये गुणे यः क्षयम्
ऋणे तु ऋणं तद्धनं धनयोः
शून्येन निहतौ जातौ पुनः शून्यौ सकलविधौ
खहरं भजेत तु शून्येन निःशेषं तु यत् ॥ ||||

Translation: “A debt multiplied by a debt is a debt; a loss multiplied by a loss is a loss. But a debt [multiplied by a] debt is positive for positives. Those multiplied by zero become zero again in all ways. That which is exactly divisible by zero becomes a *khahara*.”

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खहर (/)

6
$div0 =$

Commentary:

Rules relating to multiplication and division with zero—Multiplying positive and negative yields zero. Multiplying any number by zero gives zero. Dividing zero by zero gives zero. But when a number is divided by zero, it is called *khahara* (an undefined/infinite quantity).

If 6
$div0 =$ *khahara*, thus.

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In the 18th chapter (Algebra), Brahmagupta elaborates further:

योगे शून्यं खण्डिते च पूर्ववत् साधनम्
शून्येऽपि गुणके जाते पुनस्तत्रैव राशयः
भाज्ये शून्यं भागकारे तु जाते खहरं भवेत्
तस्य प्रसज्जते विकारो न तु शून्यं ततः ॥

Translation: “In addition, zero; in division also, the previous method. When even zero becomes a multiplier, the quantities become zero again there. When the dividend is zero and the divisor is zero, a *khahara* comes into being. Its modification is produced, but not zero from that.”

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$div0 =$,

Commentary:

In addition, division, etc., with zero, the previous method applies. When multiplied by zero, quantities become zero. When the dividend is zero and the divisor is zero, a *khahara* quantity is produced. The reason for this modification is that no definite quantity is obtained on division by zero.

Negative quantities [multiplied] by zero—negative quantities are also multiplied by zero. If 6
$div0 =$ *khahara*, then there is no error in this; rather, it is a special kind of quantity.

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$$a + 0 = a$$
$$a - 0 = a$$
$$a$$
$$times0 = 0$$
$$0$$
$$times0 = 0$$
$$0$$
$$diva = 0$$
$$quad(a$$
$$neq0)$$
$$a$$
$$div0 =$$
$$textrm$$
$$quad($$
$$textrm/)$$

Summary of Brahmagupta’s Rules for Zero

$$a + 0 = a$$
$$a - 0 = a$$
$$a$$
$$times0 = 0$$
$$0$$
$$times0 = 0$$
$$0$$
$$diva = 0$$
$$quad($$
$$textfora$$
$$neq0)$$
$$a$$
$$div0 =$$
$$textrm$$
$$textitkhahara$$
$$quad($$
$$textanundefined/infinitequantity)$$

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Commentary:

Here Colebrooke Sahib has written answers which are entirely inappropriate. In the demonstration, the praise of negative quantities has been made by the verse written below. In the *Līlāvatī*, “*nye guṇake jāte khaṃ hāraiced*” etc.—in the demonstration of this, no one has stated the characteristics of the busy rule of three! Bhāskarācārya has stated its characteristic.

Khabhakta—the answer to the question is of the nature of *khahara*.

6 श्रेढी — Progressions and Series

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Chapter 6: Progressions and Series

This extensive section treats arithmetic and geometric progressions, sums of series, and related topics.

6.1 सङ्कलितैक्यानयनम् — Sum of Sums of Natural Numbers

द्वियुक्तगच्छाभिहतं द्विभक्तं मनस्विनः
सङ्कलितैक्यं व्याख्यातम् ॥१९॥ || ||

Translation: “The number of terms increased by two, multiplied [by the sum], divided by two—the sum of sums has been explained by the wise man [Brahmagupta].” ||19||

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Commentary:

Starting from one, the sum increased by one of the desired number of terms—multiplying it by the number of terms increased by two, and dividing by three, the measure of the sum of sums is obtained. The sum of natural numbers up to the number of terms is called *sadvalita*. For this the Ācārya has not written a rule, which is not correct, because when the number of terms is large, one would have to labour greatly to know the sum by the operation of addition. For the finding of the sum, the method stated by Bhāskara in the *Līlāvatī*—“*naika padaghna padārtha mardakādyaḍa yuti*” etc.—is very beautiful.

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: $1, 3, 6, 10, 15, 21, 28, 36, 45$

: $1, 4, 10, 20, 35, 56, 84, 120, 165$

Example:

State separately the successive sums of natural numbers up to nine terms, and also their sum of sums.

Statement: 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

By Bhāskara’s rule, the successive sums: $1, 3, 6, 10, 15, 21, 28, 36, 45$

By the Ācārya’s rule, the sums of sums: $1, 4, 10, 20, 35, 56, 84, 120, 165$

6.2 उपपत्तिः — Demonstration

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$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{array}$$

6 = + , :

$$fractext(text+1)2 = text$$

$$frac5times62 = 15quadtext()$$

Demonstration:

If terms = 5, then establishing the natural numbers up to the terms in order and reverse order:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{array}$$

Adding these, in each place 6 = terms +1, this extending over the terms. Therefore:

$$fractextterms(textterms+1)2 = textsum$$

$$frac5times62 = 15quadtext(sum)$$

Thus Bhāskara's method for finding the sum is demonstrated.

6.3 वर्गसङ्कलितम् — Sum of Squares

गच्छपदघ्नं पदार्धं मर्दकाद्यड युतिम्
वर्गसङ्कलितं व्याख्यातम् ॥ ||||

Translation: "The number of terms multiplied by the terms, half the terms [combined with] the addition of the compressing etc. [i.e., 1/3]—the sum of squares has been explained."

:

$k = 1+4+9+16+...$, ?

$k = 1+4+9+16+...$

$displaystyle =$
$fracn(n+1)(2n+1)6$

Commentary:

If $k = 1+4+9+16+...$ up to [the given] terms, how is the sum of this series to be found?

$k = 1+4+9+16+...$ up to terms

$displaystyle =$
$fracn(n+1)(2n+1)6$

6.4 घनसङ्कलितम् — Sum of Cubes

द्वियुक्तगच्छाभिहतं द्विभक्तं मनस्विनः
घनसङ्कलितैक्यं व्याख्यातम् ॥ ||||

Translation: "The number of terms increased by two, multiplied, divided by two—the sum of the sum of cubes has been explained by the wise one."

:

$k = 1+8+27+64+125+...$,
?

:

$displaystyle$
$left$
$fracn(n+1)2right^2$

Example:

If $k = 1+8+27+64+125+...$ up to terms, how is the sum of this series to be found?

By dividing by nine and multiplying:

$displaystyle$
$left$
$fracn(n+1)2right^2$

6.5 गुणोत्तरश्रेढी — Geometric Progression

व्येकं व्येकगुणोद्धृतमित्यादिना
गुणोत्तरश्रेढी योगफलं व्याख्यातम् ॥ ||||

Translation: "Decreased by one, divided by the common ratio decreased by one, etc.—the sum of a geometric progression has been explained."

:

$k = 4 + 44 + 444 + ...$, ?

:

$$k = frac49(9 + 99 + 999 + cdots)$$

$$= frac49left(10 - 1) + (10^2 - 1) + (10^3 - 1) + cdotsright$$

$$= frac49leftfrac10(10^n - 1)10 - 1 - nright$$

Example:

If $k = 4 + 44 + 444 + ...$ up to terms, how is the sum of this series to be found?

By multiplying by nine and dividing:

$$k = frac49(9 + 99 + 999 + cdots)$$

$$= frac49left(10^1 - 1) + (10^2 - 1) + (10^3 - 1) + cdotsright$$

$$= frac49leftfrac10(10^n - 1)9 - nright$$

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Commentary:

Similarly: $5 + 55 + 555 + ...$ up to terms
$6 + 66 + 666 + ...$ up to terms
$7 + 77 + 777 + ...$ up to terms
$8 + 88 + 888 + ...$ up to terms
$9 + 99 + 999 + ...$ up to terms

The sum of these series is obtained by the previous method.

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7 क्षेत्रव्यवहारः — Plane Geometry

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Chapter 7: Plane Geometry

This major section deals with the calculation of areas of various geometric figures—triangles, quadrilaterals, circles—and the relationships between their sides, diagonals, and altitudes.

7.1 समलम्बचतुर्भुजस्य क्षेत्रफलम् — Area of an Isosceles Trapezium

बाहुकोट्योर्हताः परस्परं क्षेत्रयोरिह
तेषु भूमिरधिकोऽर्धको मुखं च विषमस्य द्वयम् ॥ ||||

Translation: "The mutual products of the sides and perpendiculars of two fields—among them, the greater is the base, half is the face, and two [come from] the unequal [sides]."

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Commentary:

The diameter of the circumscribed circle of a square field is its diagonal; in an oblong field also it is the same. In the Ācārya's text, by the word *aviruddha* one should understand an isosceles trapezium (*samlamba*), where both diagonals and both sides are equal. Multiplying the diagonal by the side and dividing by twice the altitude, the semi-diameter of the circumscribed circle of the quadrilateral is obtained. In a scalene quadrilateral in which a circumscribed circle is possible, the semi-diameter is the square root of the product of the side and the opposite side.

7.2 वृत्तस्य व्यासज्ञानम् — Finding the Diameter of a Circumscribed Circle

विपर्यस्तपादं भुजयोः कर्णं द्विगुणावलम्बनविभक्तः
हृदयं द्विसस्य भुजप्रतिभुजकृतियोगसूलार्धम् ॥२६॥ || ||

Translation: "The transposed side-product of the two sides, the diagonal divided by twice the altitude—the heart [semi-diameter] of that. For a quadrilateral, the semi-square-root of the sum of the products of the sides and opposite sides." ||26||

:

Commentary:

In a square field, the diagonal itself is clearly the diameter. By transposition, two equal sides are spoken of. The meaning is this: the transposed side-product of the sides, divided by twice the altitude—the result is the heart (semi-diameter) of the circle circumscribed about the quadrilateral. In a scalene quadrilateral where a circumscribed circle is possible, the semi-square-root of the sum of the products of the side and opposite side becomes the heart (semi-diameter).

7.3 त्रिभुजोपरिगतवृत्तव्यासज्ञानार्थम् — Circle Diameter from Triangle

इदानीं त्रिभुजोपरिगतवृत्तस्य व्यासज्ञानार्थमाह
त्रिभुजस्य बाहोर्भुजयोर्हृतः कर्णो द्विगुणावलम्बरविभक्तः
हृदयं द्विसस्य भुजप्रतिभुजकृतियोगसूलार्धम् ॥२७॥ || ||

Translation: "Now, for knowing the diameter of the circle circumscribed about a triangle, he says: the product of the two sides of the triangle, divided by the altitude [and] multiplied by the diagonal—the heart [is] the semi-square-root of the sum of the products of the side and opposite side." ||27||

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Commentary:

In a square field, the diagonal itself is the diameter of the circumscribed circle. An isosceles trapezium is to be understood by transposition. Where the diagonals and sides are equal, the diagonal, multiplied by the side and divided by twice the altitude—the result is the heart, i.e., the semi-diameter of the circle circumscribed about the quadrilateral. In a scalene quadrilateral where a circumscribed circle is possible, the semi-square-root of the sum of the products of the side and opposite side is the heart (semi-diameter). ||27||

7.4 जात्ययोः क्षेत्रयोः — On Two Classes of Triangles

जात्ययोस्तिहताः परस्परं क्षेत्रयोरिह हि बाहुकोटयः
तेषु भूमिरधिकोऽर्धको मुखं विषमस्य द्वयम् ॥ ||||

Translation: "The sides and perpendiculars of two regular [triangular] fields here, mutually multiplied—among them the greater is the base, half [is] the face, and two [come from] the unequal [sides]."

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Commentary:

Multiplying the sides and perpendiculars of two regular (*jātya*) triangles mutually in order, the sides of a scalene quadrilateral are produced; among them the greatest is the base, the smallest is the face, and the two middle ones are the [other] sides.

7.5 विषमचतुर्भुजक्षेत्रफलम् — Area of a Scalene Quadrilateral

विषमं चतुर्भुजं क्षेत्रं समलम्बवत् सदृशम्
तस्य कर्णे लम्बौ संपातं तयोः परस्परं गुणनम्
ततोऽर्धं फलं स्यात् ॥ ||||

Translation: "A scalene quadrilateral field is similar to an isosceles trapezium. Its diagonals [are] the two altitudes [meeting] at the intersection; their mutual product, then half, is the area."

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Commentary:

A scalene quadrilateral field is similar (*sadṛśa*) to an isosceles trapezium (*samlamba-vat*). Its diagonals [are] the two altitudes [meeting] at the intersection (*sampāta*) point where they cut each other; then their mutual product, halved, is the area.

7.6 त्रिभुजक्षेत्रफलम् — Area of a Triangle

समकोणत्रिभुजे कर्णः समपादभुजयोर्वर्गयोगसूलम्
तत्कर्णस्य चतुर्गुणीतो भुजयोर्घातः समकोणे ॥ ||||

Translation: "In a right-angled triangle, the hypotenuse is the square root of the sum of the squares of the equal side and base. The product of the sides, multiplied by four [divided by] that hypotenuse, [gives the altitude] in the right-angled [triangle]."

— :

$$c^2 = a^2 + b^2 quadtext($$

c=)

Example—Right-Angled Triangle:

$$c^2 = a^2 + b^2 quadtext(where$$

c= $hypotenuse$)

---

8 खातव्यवहारः — Excavations (Solid Geometry)

:

, , ,

Chapter 8: Excavations (Solid Geometry)

This section deals with the calculation of volumes of excavations, wells, reservoirs, and related solid figures.

8.1 समखातघनफलम् — Volume of a Regular Excavation

समखातं तु यत् क्षेत्रफलं तद्वेधगुणितं घनफलम्
समखातस्य त्रिभागः सूचीखातस्य घनफलम् ॥ ||||

Translation: "That regular excavation of which [the area]—the area of the field multiplied by the depth is the volume. One-third of the regular excavation is the volume of a needle-excavation [pyramid/cone]."

:

( )

Commentary:

The area of a regular excavation is found just as the area of a square or isosceles trapezium is found. The area multiplied by the depth (*vedha*) gives the volume of the regular excavation. That is, an excavation in which the length etc. at the mouth are the same at the base is called a regular excavation (*sama-khāta*). Dividing its volume by three gives the volume of a needle-like excavation (*sūcī-khāta*, i.e., pyramid/cone).

8.2 उदाहरणम् — Worked Example

:

( ) ( ) , ,
, , , ( ) , , , , ,

: 36|35|42|21|16 30 :

$$text = fractexttimestext30$$

$$text = texttimestext = 8times30 = 240$$

Example:

A certain tank (*vāpī*) has length thirty cubits, breadth sixty cubits. Inside it are five excavations (*khāta*) with [side] portions of four, five, six, seven, eight [cubits]; the depths of the excavations in order are 8, 7, 7, 3, 2. State the mean section (*samarajju*) measure of those excavations.

Mean section: The side-sums at mouth and base [give] 36|35|42|21|16. Dividing the sum of all these by the aggregate 30:

$$textMeansection = fractextbreadthtimestextlength30$$

$$textArea = 8times30 = 240$$

Multiplying this area by the mean section, the total excavation volume became 1200.

8.3 विस्तारदैर्घ्यवेधेषु वैषम्ये — Irregular Excavations

:

, = , , ,

: | | , =
$frac93 =$, =

$text = 5$
$times16 = 80$

: 80
$times3 = 240$

— = , = , | | ,

=
$frac3 + 4 + 53 =$

$text = 4$
$times12 = 48$

: 48
$times8 = 384$ ( )

Examples:

A certain pond has length 16 cubits, breadth = 5 cubits, depths 2, 3, 4 cubits; then what is its volume?

Statement: 2|3|4 depths; mean of depths = $frac93 = 3$; here the number of places = 3.

$textArea = 5$
$times16 = 80$

Multiplying by depth: 80
$times3 = 240$ (excavation volume).

Second Example: An excavation of which the length = 12 cubits, depth = 8 cubits, breadth 3|4|5 cubits—state its volume.

Mean breadth =
$frac3 + 4 + 53 = 4$

$textArea = 4$
$times12 = 48$

Multiplying by depth: 48
$times8 = 384$ (excavation volume).

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9 धान्यराशिव्यवहारः — Grain Heaps

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Chapter 9: Grain Heaps

This section treats the measurement of grain stored in heaps, which have the form of a cone or portion thereof.

9.1 स्थूलधान्यराशिमानम् — Measuring Coarse Grain Heaps

वृत्तपरिगतधान्यराशेः परिधिनवमांशो वेधः
वेधेन गुणितः परिधिः षष्ट्यंशः फलं स्यात् ॥५०॥ || ||

Translation: "The circumference of a circular grain heap—a ninth of the circumference is the depth. The circumference multiplied by the depth, [then] a sixtieth of that, is the volume [measure]." ||50||

:

Commentary:

The inner-outer corner circumference, multiplied by two-four [and] a seventh part, having performed the calculation as before, divided by its multiplier, then that calculation becomes [the volume]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two, then the result as before, divided by those two, then the volume of that heap becomes [known].

9.2 उदाहरणानि — Worked Examples

:

: , =

$text =$
$frac6010 = 10,$ $= 100$

$text = 100$
$times6 = 600$ ( )

:

: ,

$text =$
$frac606 = 10,$ $= 100$

$text = 100$
$times5 = 500$ ( )

:

: , =

$text =$
$frac60? = 5,$ $= 100$

$text = 100$
$times3 = 300$ ( )

For Knowing the Measure of Coarse Grain Heaps:

Statement: Coarse grain heap circumference 60, portion 6 = depth.

$textSixtiethofcircumference =$
$frac6010 = 10$, squared $= 100$

$textMultipliedbydepth = 100$
$times6 = 600$ (coarse grain heap volume)

For Knowing the Measure of *Śūka* Grain Heaps:

Statement: Circumference 60, a ninth of circumference is depth 5.

$textSixtiethofcircumference =$
$frac606 = 10$, squared $= 100$

$textMultipliedbydepth = 100$
$times5 = 500$ (*śūka* grain heap volume)

For Knowing the Measure of Fine Grain Heaps:

Statement: Fine grain heap circumference 60, depth = 3.

$textSixtiethofcircumference =$
$frac60? = 5$, squared $= 100$

$textMultipliedbydepth = 100$
$times3 = 300$ (fine grain heap volume)

9.3 उपपत्तिः — Demonstration

:

…

Demonstration:

By the nature of the grain's position, the grain heap becomes needle-shaped (conical); its base circumference divided by nine etc. becomes its depth—this is established by direct and indirect perception. The area of a circular field is the circumference multiplied by the diameter...

9.4 भास्करीय लीलावत्यामुदाहरणम् — Example from Bhāskara's Līlāvatī

समभुवि किल राशिभिः स्थितः स्थूलधान्यः परिधिपरिमितिः
स्याद्व्यष्टिषत्व्यंशया प्रवद गणक खार्यः कि मिताः सन्ति तस्मिन्
अथ पृथगपुघान्यैरछुकधान्यैरच शीघ्रम् ॥ ||||

Translation: "On level ground indeed a coarse grain heap stands, with a [known] circumference measure. With the sixty-fourth part [of the circumference], tell me, calculator, how many *khāri*s are in it? Also quickly [tell me] separately for *pu* and *śuka* grains."

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10 छायाव्यवहारः — Shadows and Gnomons

:

(gnomon)

Chapter 10: Shadows and Gnomons

This section deals with the mathematics of the gnomon's shadow and related calculations.

द्विजद्विसम्भागैकनिघ्नातु तु परिषैः फलम्
भित्त्यन्तर्बाह्यकोणागे परिधिः द्विचतुःसत्रिभागैकगुणः ॥५१॥ || ||

Translation: "The result [is obtained] from the circumference, multiplied by two, half of two [i.e., one], etc. At the inner-outer corner, the circumference multiplied by two-four and one-seventh [i.e., 2
*frac*47]." ||51||

:

Commentary:

The inner-outer corner circumference, multiplied by two-four [and] one-seventh, having performed the calculation as before, divided by its multiplier, then that calculation becomes [known]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two, then the result as before, divided by those two, then the volume of that heap becomes known.

For a grain heap situated at the inner corner of two walls, the circumference [is] multiplied by four; then having obtained the result as before, dividing it by four, then the volume of that grain heap becomes known.

For a grain heap situated at the outer corner of two walls, the circumference [is] multiplied by one-and-three-sevenths [i.e.,
*frac*107]; then having obtained the result as before, dividing that result by one-and-three-sevenths, then the volume of that grain heap becomes known.

11 गीताध्यायः — The Chapter on Spheres

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Chapter 11: The Chapter on Spheres

The final pages of this volume begin the *Gītādhyāya* (Chapter on Spheres), which deals with the geometry of spheres and spherical segments. This is the 21st chapter of the *Brāhmasphuṭasiddhānta*.

11.1 विष्कम्भपरिधिः — Diameter and Circumference

विवृतद्विसम्भागैकनिघ्नात् तु परिषैः फलम्
इत्यादि भास्करोक्तमेतदनुरूपमेव ॥५१॥ || ||

Translation: "The result [is obtained] from the circumference, multiplied by one [half] of the expanded diameter-pair [i.e., the diameter], etc.—this is indeed in conformity with what Bhāskara has stated." ||51||

:

( ) ( ) ( ) —
( )

Commentary:

Expanded (*vivṛta*) pair-of-diameter-half (*dvisambhāga*) multiplied by one [i.e., the diameter]—the result from the circumference, etc.—this is indeed in conformity with what Bhāskarācārya has stated. ||51||

11.2 राशिव्यवहारः — Calculation of Heaps

भित्यन्तर्बाह्यकोणागः परिधिः द्विचतुः सव्यंशगुणः प्राग्वत्
गणितं कृत्वा स्वगुणकारभक्तं तदा तद् गणितं भवति
अर्थात् भित्तिपादसंलग्नस्य धान्यराशेः परिधिमानं द्वाभ्यां संगुण्य
तस्मात्पूर्ववचत्फलं तदुद्वाभ्यां भक्तं तदा तद्राशिघनफलं भवति ॥ ||||

Translation: "The inner-outer corner circumference, multiplied by two-four [and] one-seventh [i.e., *frac*187] as before; having performed the calculation, divided by its multiplier, then that calculation becomes [known]. That is, the circumference measure of the grain heap touching the base of the wall, multiplied by two; then the result as before, divided by those two, then the volume of that heap becomes known."

:

Commentary:

For a grain heap situated at the inner corner of two walls, the circumference [is] multiplied by four; then having obtained the result as before, dividing it by four, then the volume of that grain heap becomes known. For a grain heap situated at the outer corner of two walls, the circumference [is] multiplied by one-and-three-sevenths; then having obtained the result as before, dividing that result by one-and-three-sevenths, then the volume of that grain heap becomes known.

:

परिधिर्भित्ति लग्नस्य राशोः शत्करः किल
अन्तः कोणस्थितस्यापि तिथितुल्यकरः सखे
बहिः कोणस्थितस्यापि पञ्चघ्ननवसंमितः
तेषामाचक्ष्व मे क्षिप्रं धनहस्तान् पृथक् पृथक् ॥ ||||

Example from Bhāskara's Līlāvatī:

Translation: "The circumference of a heap touching a wall is indeed a hundred. Of one situated at the inner corner, [it is] equal to the number of lunar days [30], O friend. Of one situated at the outer corner, [it is] measured as nine multiplied by five [45]. Tell me quickly their *dhana-hastas* [wealth-measures] separately."

:

$= 30$
$times2 = 60$
$= 15$
$times4 = 60$
$= 48$
$times$
$frac54 = 60$
$= 6$
$= 600$
$300|150|450$

Statement:

Here, the circumference of the first [heap] 30, doubled $= 30$
$times2 = 60$.
The inner-corner circumference 15, quadrupled $= 15$
$times4 = 60$.
The outer-corner circumference 48, multiplied by one-and-a-quarter $= 48$
$times$
$frac54 = 60$.
This measure $= 6$.
From these, the equal result is just so many *khāri*s $= 600$.
This, divided by its own multiplier, became the separate results $300|150|450$.
Thus the measure of coarse grain was found.

11.3 अन्योदाहरणम् — Another Example

षट्त्रिशेन्मित परिधौ राशौ धान्यस्य किं भवेद् गणितम्
हस्त चतुष्काभ्युदये यदि वेत्सि तदुच्यतामाशु ॥ ||||

Translation: “When the circumference of a heap of grain is measured as thirty-six, what is the calculation [volume], if you know, tell me quickly, [given] a rise of four cubits [depth].”

:

, $=$ $= 6$
$= 36$
$= 36$
$times4 = 144$

Statement:

Grain heap circumference 36, depth $= 4$; then as before, the sixtieth of the circumference $= 6$.
Squared $= 36$.
Multiplied by depth $= 36$
$times4 = 144$ (volume).

Concluding Remarks

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Concluding Remarks

This volume concludes the second part of the *Brāhmasphuṭasiddhānta*, covering an extraordinary range of mathematical topics. The sections included in this portion demonstrate the depth and sophistication of Brahmagupta’s mathematical thought.

1. : (भागे), (मिश्र धन), (भाण्डप्रतिभाण्डक),
2. : , —खहर
3. : , ,
4. : , , ,
5. : , ,
6. : ( ) ,

1. **Commercial Arithmetic:** Partnership problems (*bhāga*), compound interest (*miśra dhana*), barter exchange (*bhāṇḍa-pratibhāṇḍaka*), and the Rule of Three and its extensions.
2. **The Algebra of Zero:** Brahmagupta's revolutionary rules for arithmetic with zero, including the controversial but pioneering treatment of division by zero as yielding the quantity *khahara*.
3. **Series:** Comprehensive treatment of arithmetic and geometric progressions, sums of natural numbers, squares, and cubes.
4. **Plane Geometry:** Areas of triangles, quadrilaterals, and circles, with theorems on cyclic quadrilaterals and circumscribed circles.
5. **Solid Geometry:** Volumes of excavations, wells, and grain heaps.
6. **Spherical Geometry:** The beginning of the chapter on spheres (*gītādhyāya*), dealing with the geometry of spherical segments.

> *"The Brāhmasphuṭasiddhānta stands as one of the most influential mathematical texts in the history of world mathematics."*

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Note on Sources:

The present edition is based on the Chowkhamba Sanskrit Series publication, with the extensive Hindi commentary (*Hindī bhāṣā*) of Paṇḍita Sudhākara Dvivedin. Frequent references are made to the *Līlāvatī* of Bhāskarācārya II, whose rules often parallel or expand upon those of Brahmagupta.

The work concludes with the transition into the *Gītādhyāya*, the chapter on spherical geometry, which will be continued in the third and final part of this edition.

End of Part II
द्वितीयभागसमाप्तिः

ब्राह्मस्फुटसिद्धान्त
Brāhmasphuṭasiddhānta

of

ब्रह्मगुप्त

Brahmagupta

(c. 598–668 CE)

Volume III — Chunk I

Pages 1677–1816

comprising Chapters XII–XXI

(Final Volume: Concluding Chapters & Appendices)

BILINGUAL EDITION

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by

Dr. Ram Swarup Sharma

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Preface to the Final Volume

अस्य अन्तिमस्य खण्डस्य प्रस्तावना --- एतत् अन्तिमं भागं ब्रह्मस्फुटसिद्धान्तस्य पूर्णं करोति। पृष्ठानि १६७७--१८१६ ब्रह्मगुप्त-महोदयस्य अन्तिमाध्यायान् समाविशन्ति, यथाम्: गणिताध्यायस्य समाप्तिः, विकलवर्गानयनम्, कालज्ञानम्, चन्द्रग्रहणम्, मन्दस्फुटम्, शीघ्रफलम्, भोगकला, प्राणकला, च समाप्तिः।

This final volume of the *Brāhmasphuṭasiddhānta* completes the monumental work begun in the first two volumes. Pages 1677–1816 cover the concluding chapters of Brahmagupta's masterpiece, including the completion of the Gaṇitādhyāya, extraction of roots of non-square numbers, time computations, lunar eclipses, equations of centre and anomaly, daily motion, and the final colophons. The text preserves Brahmagupta's original verses in authentic Sanskrit form, accompanied by detailed commentary and mathematical proofs (*upapatti*).

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1 Conclusion of Arithmetic — Circumference & Shadow

अस्मिन् खण्डे गणिताध्यायस्य अन्तिमश्लोकाः आरभ्यन्ते, ये परिधि-अनुपपत्ति-छाया-व्यवहारान् प्रतिपादयन्ति।

The present section begins with the final verses of the twelfth chapter (*Gaṇitādhyāya*), treating the calculation of circumference (*paridhi*), proportion (*anupapatti*), and the computation of shadows (*chāyā vyavahāra*).

1.1 Circumference and Proportion

परिधौ क्रमशः प्राग्वत् स्वगुणात् भवेत् फलम् |
श्रीपत्युक्तमिदं चाचार्यैरनुक्रमेणावबोधित ||१७||

paridhau kramaśaḥ prāgvat svaguāt bhavet phalam |
śrīpatyuktamidaṃ cācāryairanukrameāvabodhita ||17||

(बीएसएस् १२.३७ / खण्ड २, पृ. ८९१) एतत् श्लोकं परिधिः व्यासस्य उक्तगुणकैः क्रमशो गुणनया प्राप्यते इति वदति, यत् पूर्वाचार्यैः श्रीपतिः-आदिभिः उपदिष्टम्। ब्रह्मगुप्त-भास्कराचार्य-लीलावतीभिः अपि स्वस्वग्रन्थेषु एतत् उपदिष्टम्।

(BSS 12.37 / Vol. II, p. 891) This verse states that the circumference is obtained by multiplying the diameter successively by the prescribed factors, as taught by previous teachers including Śrīpati. Brahmagupta, Bhāskarācārya, and Līlāvatī have all taught this in their respective works.

द्विवेद सचित्रभागे कनिष्ठान्तु परिधिः फलम् |
मिथ्यन्तबोधकोऽणस्थराशिः स्वगुणाभिजं तत् ||१७१||

dviveda sacitrabhāge kaniṭhāntu paridhiḥ phalam |
mithyantabodako'̄aśiḥ svaguābhijaṃ tat ||171||

यथार्थः परिधिः द्विवेदि-सचित्रभाग-कल्पितैः प्राप्यते; अन्यथा स्थूलाप्र□□□णात् प्राप्तः परिधिः मिथ्या भवति।

||171|| The true circumference is that obtained by the rules taught by the two Dvivedis; the circumference otherwise (from crude approximation) is erroneous.

1.2 Mathematical Proof

अनुपपत्तिः --- मिथ्यान्तः संलग्न धान्यराशिः परिधिवास्तव परिधेर्बंधुतुल्यः | क्रान्तः कोण स्थितस्य धान्यराशिः परिधिवास्तवपरिधिं चतुर्थांशतुल्यः | बाहुकोणसंलग्न धान्यराशिः परिधिवास्तव परिधेः पादेनैक तुल्यस्तेनोत्परिधित्रयमानं द्विवेदसचित्रभागेकेन क्रमशो गुणितां सघातं परिधिं भवेत् | ततः पूर्वं वेधफलं ततु द्विवेद सचित्रभागेकेन क्रमशो भक्तं तदा वास्तवं तद्फलं भवतीत्येतावताऽऽचार्योक्त-मुपपन्नम् ||१७१||

Upapatti (Proof): The incorrect circumference associated with a quantity of grains is proportional to the true circumference. The inclined-angle quantity of grains is one-quarter of the true circumference; the base-angle quantity is one-sixth. The circumference- triple measure, multiplied successively by the Dvivedi-sacitrabhāga factors, yields the true circumference. The previous penetration result, divided successively by the Dvivedi-sacitrabhāga factors, gives the true result — thus the teacher's statement is proved. ||171||

1.3 End of Arithmetic Chapter

इति राशि व्यवहारः --- अब भत्ति के अन्तर्गत भत्ति के क्रान्तः कोणरूप तथा बाहुकोणरूप धान्यराशि प्रमाणा-नयन के लिये कहते हैं |

Here ends the *Rāśi Vyavahāra* (Arithmetic Operations). Now, under division, the quantities of grains corresponding to the inclined angle and base angle are stated for the derivation of measures.

1.4 Shadow Calculations

छाया व्यवहार --- बीएसएस् १२.४१--४२: तज्जादौ छायात् इष्टकालज्ञानार्थमिष्टकालतच्छायानयनार्थं चाहु | छायानरसंक्रमहृतं शुद्धं प्रागपरयोर्दिगतदोषं | दिनगतदोषांशहृतं धु दलं छाया नरखेखं ||४२||

BSS 12.41–42: First, for finding the time from a given shadow, or the shadow at a given time. The *nara* (sine of latitude) multiplied by the shadow, divided by the *śañku* (gnomon), gives the *śuddha* (mean) value; considering the eastern or western direction, this is divided by the *dinagata-doṣaṁsa* (ascensional difference) to obtain the accurate shadow. ||42||

सू. भा.---नरः शङ्कुः रभिमष्टः | छायाया यो नरः शङ्कुं भागः स संक्रमे यस्तेन
शुद्धं हृतं लब्धं प्राक् पश्चिमकपाले दुग्गतं पश्चिमकपाले दुग्गतं भवति |

Sūtra-bhāỵa: The *nara* is the *śañku* multiplied by eight — the *nara* of the shadow divided by the *śañku* is that in the *sañkrama*; this is the *śuddha*; divided by the eastern or western semi-diurnal arc, it becomes the corrected value.

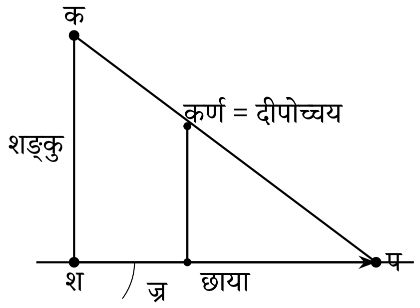
छाया व्यवहारस्य विस्तृत उपपत्तिः अनुवर्तते। नरः अक्षज्या अस्ति, शङ्कुः शङ्कुः (१२ अङ्गुलाः)। छाया एतेषां परस्परसम्बन्धात् गण्यते।

The *Chāyā Vyavahāra* (shadow computations) section continues with detailed derivations. The *nara* is the sine of latitude, and the *śañku* is the gnomon (12 añgulas). The shadow is computed from the relationship between these quantities.

1.5 Geometric Construction of Shadows

सम्बद्धचित्राणि ज्यामितीयसम्बन्धान् दर्शयन्ति | शङ्कुः (उन्नतदण्डः) भूतले छायां प्रकाशस्य किरणेन सह कोणं निर्माति |

The accompanying diagrams illustrate the geometric relationships: the gnomon (vertical pole) casts a shadow on the ground, forming an angle with the sun's ray. The *karna* (hypotenuse) equals the distance from the lamp's height. Similar triangles yield the proportions.



छाया, शान = भूः | कर्ण = दीपोच्चयम् | कर्ण, भुज त्रिभुजयोः साजात्यादनुपातः
दीघ् = दीपोच्चयम् तथा कर्ण, त्रिभुजयोः साजात्यादनुपातः
=भूः-भू = रन = छायाप्रान्तर = छाया दीघ्-छाया दीघ् = दीघ्(छाया-छाया) = दीघ् छायान्तर
छेदगमेन छायाप्रान्तर दीघ् छायान्तर, पक्षो छायान्तर भक्तो तदा
छायाप्रान्तर दीघ् छायान्तर भ्रनेन भू स्वरूपे उत्थापनेन भू = छ.प्रा. दीघ्
= छाया छायाप्रान्तर दीघ् = छाया छायाप्रान्तर, तथा भू = छाया दीघ्
= छाया छायाप्रान्तर दीघ् = छाया छायाप्रान्तर, तथा भुज, कर्ण त्रिभुजयोः
साजात्यात् १२ भू/छाया = दीपोच्चयम् | एवं त्रपन, कर्ण त्रिभुजयोः साजात्यादनुपातेन १२ भू/छाया = दीपोच्चयम् | एतावत् स्साचार्योक्तमुपपन्नम् |

Shadow (Chāyā), *śāna* = base. *Karna* = lamp-height. From similarity of triangles, the *karna*–base proportion gives: base = *chāyā-prāntara*; then by proportional reasoning, 12 × base/shadow = lamp-height. Similarly for the other triangle pair, the same result follows. Thus the teacher's statement is proved. (BSS 12.41–42 commentary)

2 Algebraic Operations — विकलवर्गानयन

बीएसएस् १२.४६: इदानीं सकल विकलवर्गानयनार्थमाह |
स्वविकलपदं वर्गांशगुणाः सकलस्त्रिशद्वैतो विकलवर्गः |
प्रक्षेप्यः सकलकृतौ वर्गधनो द्वित्रितुल्यबधो ||४६||

BSS 12.46: Now (the author) teaches the method of finding the square root of a non-square number. The square of the *sakala* (integer part) is subtracted from the number; the remainder, multiplied by the square of the denominator, becomes the *vikalavarga* (square of the non-square part). This is to be added to the square of the *sakala*, and the result gives the square root approximately. ||46||

(बीएसएस् १२.४६) यदि संख्यायाः कलाः विकलाः च इति द्वयं वर्तते, तस्य वर्गार्थं कलाराशिः सकलसंज्ञो विकलाराशिश्च स्वविकलः तत्कलापश्चात् शास्त्र कथये |

(BSS 12.46) Where a quantity has both integer (*kala*) and fractional (*vikala*) parts, the integer part is called *sakala* and the fractional part *vikala*; these are then processed by the rule to obtain the square root.

Upapatti (Mathematical Proof)

उपपत्तिः --- कल्पयते कलिमश्रवि राशौ कलाः = क | विकलाः = वि | तदा कलात्मकः स राशिः = क + वि/६० |

Upapatti: Let the integer part be k and the *vikala* part vi. Then the quantity is $k + \frac{vi}{60}$. The square is:

$$\text{□□□□□} = k^2 + \frac{2k \cdot vi}{60} + \frac{vi^2}{60^2}$$

वि. भा.---यदिमन् राशौ कला-विकला चेति द्वयं वर्तते तस्य वर्गं करणार्थं
कलाराशिः सकल संज्ञो विकलाराशिश्च स्वविकलस्तत्कलापश्चात् शास्त्र कथये |

Vīti-bhāẏa: When a quantity has both *kala* and *vikala* parts, to find its square, the integer part is called *sakala* and the fractional part *vikala*; then the rule states the method. Note: “वर्गः” means ‘square’ in Sanskrit. (BSS 12.46 commentary)

छेदेनेष्टयुतोनेनार्धं भाज्यपदष्टष्टिगुणं स्यात् |
प्रकृतिरष्टच्छेदहृतं लब्धा युतं हैनिकमनघं ||४७||

chedeneṭayutone-nārdhaṃ bhājayapa-daṭaṭigunaṃ syāt |
prakṛtiraṭachchhedahṛtaṃ labdhā yutaṃ hainika-managhaṃ ||47||

(बीएसएस् १२.४७) एतत् श्लोकं भिन्नैः सह भाजनस्य विधिं ददाति | भाज्यं, छेदेन सह युक्तं (भिन्नयोजनविधिना), भाजकस्य अर्धेन गुणितं, ततः प्रकृत्या युक्तं, शुद्धं लब्धं ददाति |

(BSS 12.47) This verse gives the method for division with fractions. The dividend, combined with the divisor (in the manner of adding fractions), is divided by half the divisor multiplied by the denominator of the divisor; the result, added to the *prakṛti*, gives the quotient. (BSS 12.47 commentary)

इदानीं भागहारे विरोपमाह |
छेदेनेष्टयुतोनेनार्धं भाजयपदष्टष्टिगुणं स्यात् |
प्रकृतिरष्टच्छेदहृतं लब्धा युतं हैनिकमनघं ||४९||

idāñim bhāgahāre viropamāha |
chedeneṭayutone-nārdhaṃ bhājayapa-daṭaṭigunaṃ syāt |
prakṛtiraṭachchhedahṛtaṃ labdhā yutaṃ hainika-managhaṃ ||49||

(बीएसएस् १२.४९) भागहारस्य विभागस्य अयं खण्डः | छेदैः सह भिन्नराशीनां व्यवहारः अत्र उपदिश्यते |

(BSS 12.49) The section on division with fractions (*bhāgahāra*) is explained. The rule involves the manipulation of fractional quantities through their denominators. (BSS 12.49 commentary)

3 Astronomical Problems — कालज्ञान

अनुवर्तमानाः खण्डाः ज्योतिषीयगणनानि समाविशन्ति --- कालनिर्णयम्, ग्रहस्थितिगणनम्, विविधज्योतिषीयमानानि च।

The following sections deal with astronomical computations, including the determination of time, planetary positions, and the computation of various astronomical quantities.

3.1 Mixed Astronomical Problems

इदानीं मध्यान् प्रश्नानाह |
व्यतिपातं वैधृतिबृहस्पतिवर्षयोश्चर्चनीचरिवबस्तीन |
द्विप्रहयोगांश युगे यो वेति स कालतत्वज्ञः ||६||

idāñ̄im madhyān praśnānāha |
vyatīpātaṃ vaidhṛtibṛhaspativarṣayoścarcanīcarivastīna |
dviprahayogāḿśa yuge yo veti sa kālatatvajñaḥ ||6||

(बीएसएस् १३.६) यो व्यतिपातं वेति | वैधृतिं वेति बृहस्पतिवर्षं वेति, स्वोच्चनीचमार्गगान् वेति | युगे द्विप्रहयोगांश इत्यादि |

(BSS 13.6) He who knows the *vyatīpāta*, *vaidhṛti*, *bṛhaspati-varṣa*, and the motion of the nodes in a *yuga* — he is the knower of the essence of time. (BSS 13.6 commentary)

सावनमासावधिप्रहरेक्काऽनिष्टमध्यसंयोगात् |
इष्टद्वयु संयु तोननिष्टानां यो वेति ग्रह्यकः सः ||१०||

sāvanamāsāvadhipraharekk'aniṭamadhyasaṃyogāt |
iṭadvayu saṃyu tonaniṭāñ̄aṃ yo veti grahyakaḥ saḥ ||10||

(बीएसएस् १३.१०) यः अवमानां (लुप्ततिथीनां) प्रघराणां (अधिकदिनानां) च योगं, इष्टकाले मध्यस्पष्टैः सह संयुक्तं वेति, सः ग्रह्यकज्ञः।

(BSS 13.10) He who knows the sum of the *avama* (omitted tithis) and the *praghara* (excess days), combined with the mean (longitudes) at the desired time — he is the knower of the *grahyaka* (the quantity to be known). (BSS 13.10 commentary)

3.2 Third Set of Problems

इदानीं तृतीय प्रश्नसमूह (छायादीपरितोदिवस वारे वेतीत्यस्य) उत्तरमाह |
कल्पर्क्षदिनसप्तकबधात् कल्पगताहर्गणानां तच्छेषात् |
सप्तहृताद्दिनवारः शेषः साय्यादिको भवति ||४०||

idāñ̄im tṛtīya praś nasamūha (chāyādī̄paritodivasa vāre vetī̄tyasya) uttaramāha |
kalparks̄adinasaptakabadhāt kalpagatāhargaāñ̄aṃ tacchẹ̄at |
saptahṛ ȃddinavāraḥ śeạḥ sāyyādiko bhavati ||40||

(बीएसएस् १३.४०) कल्पस्य गताहर्गणानां सप्तकेन गुणनात्, ततः सप्तभक्तात् शेषात् दिनवारः साय्यादिकः प्राप्यते।

(BSS 13.40) From the product of the elapsed days of the *kalpa* and seven, divided by seven — the remainder gives the day of the week (counting from Sunday).

उपपत्तिः --- कल्पर्क्षदिनसप्तकबधात् कल्पकृत् दिनसप्तधातात् कि विशिष्टात्
कल्पगताहर्गणेनैन कावः शेषस्तस्मात् सप्तहृतात् शेषः साय्यादिको दिनवारो भवति |

Upapatti: From the *kalpa* elapsed days multiplied by seven, the remainder from division by seven gives the day of the week starting from Sunday. (BSS 13.40 commentary)

3.3 Lunar Calculations

यदीष्ट ग्रहयुग्मभगणां सरोभ मगणाद्यष्टमां लभ्यते
तदा युगीयपातमगणां चन्द्रभगणायोगयोगि किं फलं
भगणादिकमिष्टहृत् राश्यादिमध्यमसपातचन्द्रं यथागतं
देशान्तरमन्दफलादीन् यथागतां सरोभ सपातचन्द्रो
भवेत् ||३१||

yadīṭa grahayugmabhagaāṃ sarobha magaā̄dyṭam̄̄aṃ labhyate
tadā yuḡ̄iyapātamagā̄aṃ candrabhagā̄ayogayogi kiṃ phalaṃ
bhagā̄adikamiṭahṛt rāśyādimadhyamasapātacandraṃ yathāgataṃ
deśāntaramandaphalād̄̄in yathāgat̄̄aṃ sarobha sapātacandro
bhavet ||31||

4 Planetary Theory — मन्दस्फुट & शीघ्रफल

4.1 Epicyclic Model

ग्रहस्थितिगणनायां मन्दस्फुटस्य (केन्द्रसमीकरणस्य) शीघ्रफलस्य (द्वितीयवक्रीभावसमीकरणस्य) च गणना आवश्यकी | ब्रह्मगुप्तः अधोलिखितं विधिं ददाति |

The computation of planetary positions involves the *mandasphuṭa* (equation of centre) and *śīghraphala* (equation of anomaly, or second inequality). Brahmagupta gives the following method:

मन्दस्फुटार्केन्द्रांशोनायां ग्रहस्यन्दं केन्द्रम् |
एव यदि तत् स्पष्टं त्रिकोणं तदानीं विशेषं नोद्येत् |
तद्यात् स्पष्टं त्रिकोणं तस्य स्पष्टं द्वितीयं केन्द्रं
क्रमेण केन्द्रान्तरं फलमिति | वा ज्ञानमन्दं स्पष्टं
प्राप्यच्छी स्साचार्य्योदिना शीघ्र-कर्त्तव्यं कार्यमिति सर्वं शुद्धम् ||३७८-५८१||

mandasphuṭārkendrāḿśonāyāṃ grahasyandaṃ kendram |
eva yadi tat spaṭaṃ trikoṃ tadāñ̄iṃ viśeạṃ nodyet |
tadyāt spaṭaṃ trikoṃ tasya spaṭaṃ dvitīyaṃ kendraṃ
krameā kendrāntaraṃ phalamiti | vā jñānamandaṃ spaṭaṃ
prāpyacch̄̄issāc̄āryyodinā ś̄̄ighra-karttavyaṃ kāryamiti sarvaṃ śuddham ||378–581||

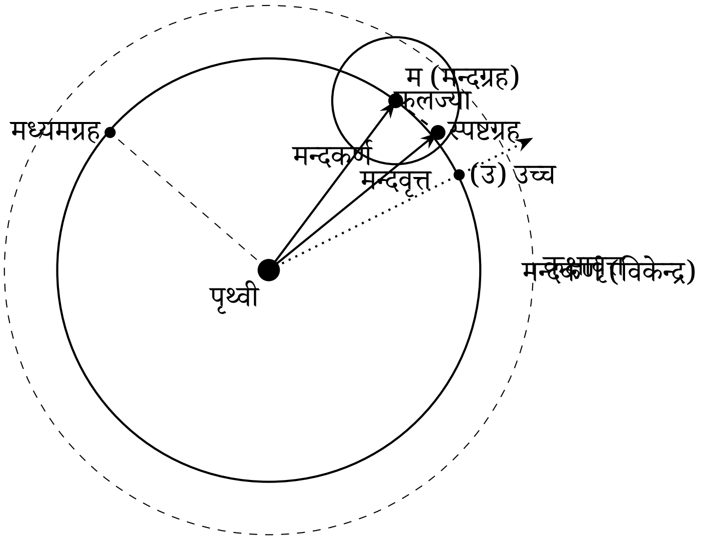
वि. भा.---उदयमार्ग (विज्ञानान्) भुजयया गुणिगतं कर्णहृतं यत्फलं तस्य
चापं स्पष्टद्वितीयोनयोरन्तरं भवेत् तत् स्पष्टमिति ||३७९-५८१||

Vt̄ti-bhāṣ̄ya: The *phala* obtained by multiplying by the *bhuja* (sine of anomaly) and dividing by the *karna* — its arc gives the difference between the first and second true positions; that is the true position. (BSS 378–581 commentary)

4.2 Proof of Epicyclic Correction

उपपत्तिः --- ज = गोच्चेभम् | म = मन्दस्पष्टग्रहः | मं३ = शीघ्रकेन्द्रम् | मर = शीघ्र-
केन्द्रज्या = गोच्चज्या | स्प. = स्पष्टग्रहः | स्प.३ = स्पष्ट केन्द्रम् |

Upapatti: Ja = geocentric planet; *Ma* = mean planet via manda; *Ma.3* = *śīghrakendra*; *Mar* = *śīghrakendra-jyā* = geocentric sine; *Sp.* = true planet; *Sp.3* = true *kendra.* (BSS *upapatti*)



भूमं गोचरग्रहः | स्पष्ट = स्पष्ट केन्द्रज्या | म.३. म. = मू केन्द्रम् |
मू.स्पष्ट = शीघ्रफल | एवं त्रिज्या | नदा मू.म. स्पष्ट त्रिभुजयोः
साजात्यादनुपातः गोच्चज्या/शीघ्रकर्ण = स्पष्टकेन्द्रज्या,
अथवाऽन्योनम् = स्पष्टकेन्द्रम् = स्पष्टग्रहोच्चयोरन्तरं
मन्दकर्णादिया शीघ्रकर्णस्थाने मन्दस्पष्ट
केन्द्रज्या समागम्यति, चापकल्पेन मन्दस्पष्ट
केन्द्रराशिः = मन्दस्पष्टदोषयोरन्तरं |

Bhum = geocentric planet. *Spaṭa* = true *kendra-jyā*. *Ma.3. Ma.* = mean *kendra*. *Mū.spaṭa* = *śīghraphala*. From similarity of triangles: geocentric sine/*śīghrakarna* = true *kendra-jyā*. Alternatively, true *kendra* = difference between true planet and apogee. The manda-true *kendra-jyā* meets the epicycle; by arc-correspondence, manda-true *kendra-rāśi* = difference between manda-true and equation. (BSS *upapatti*)

मन्दस्फुटस्य गणनक्रमः एवं अस्ति: (१) मन्दकेन्द्रस्य (मध्यमग्रहस्य उच्चस्य च अन्तरं) गणना, (२) ज्यापिठकात् मन्दफलस्य (केन्द्रसमीकरणस्य) अवलोकनम्, (३) मध्यमग्रहे मन्दफलस्य प्रयोगः मन्दस्फुटस्य (प्रथमस्पष्टस्य) प्राप्त्यर्थम्, (४) मन्दस्फुटात् शीघ्रकेन्द्रस्य गणना, (५) शीघ्रफलस्य अवलोकनम्, (६) शीघ्रफलस्य प्रयोगः सत्यग्रहस्य प्राप्त्यर्थम् |

The *mandasphuṭa* (true planet via equation of centre) is computed as:

1. Compute the *mandakendra* (mean anomaly): difference between mean longitude and apogee (मन्दोच्च)
2. Find the *mandaphala* (equation of centre) from the sine table
3. Apply to the mean longitude to get *mandasphuṭa*
4. Compute *śīghrakendra* from *mandasphuṭa*
5. Find *śīghraphala* and apply to get the true longitude

5 Advanced Astronomical Computations — भोगकला & प्राणकला

इष्टज्यां संगुणीताः पश्चकर्मणां कर्णच्चन्द्रम् |
इष्टज्यापाद्युत्थ्यासर्वं विभाजिता लब्धम् ||२५||
नवति छते प्रोह पदे नवते मनोरथं शेषमगणत्कला |
एवं धनुरिष्टदया भवति ज्यया विना ज्यार्धिः ||२६||

iṭajȳaṃ sagūitāḥ paścakarmāaṃ karṇac candram |
iṭajȳapādyutthyāsarvaṃ vibhājitā labdham ||25||
navati chate proha pade navate manorathaṃ śeamagakalā |
evaṃ dhanuriṭadayā bhavati jyayā vinā jyārdhiḥ ||26||

(बीएसएस् २१.२५--२६) इष्टज्या पञ्चकर्मभिः संगुणिता कर्णेन विभक्ता; शेषात् नवत्यधिकात् पदात् मनोरथः (इष्टचापः) प्राप्यते | एतत् ज्यार्धेः विना चापस्य अवलोकनम् अस्ति |

(BSS 21.25–26) The *iṭajyā* (desired sine) is multiplied by the results of the five operations and divided by the radius; the remainder, when 90 is subtracted from the *nava* position, gives the *śeṣa*; the *manoratha* (desired arc) is obtained. The arc is thus found from the sine without the *jyārdha* (half-chord).

सू. भा.---इष्ट ज्यायाः पादद्वयपरिलेखेन युक्त व्यासार्धं तद्द्वारा तेन विभाजिताः | शेष स्पष्टम् |

Sūtra-bhāẏa: The desired sine is divided by the semi-diameter, together with the two-foot arc-difference; the remainder is clear. (BSS 21.25–26 commentary)

Upapatti — Detailed Proof

उपपत्तिः --- पूर्वोक्तकोपपरित्य ज्या = (१८०-चा) चा.ज्र. / [१०१२५-(१८०-चा) चा/४]
या ज्या / [१०१२५-या/४] = ज्या/१०१२५-या/४

Upapatti: From previous proofs: $jy = \frac{(180-ca)\cdot ca\cdot jra}{10125-\frac{(180-ca)ca}{4}}$. Let $y = (180-ca)\cdot ca$. Then $jy = \frac{jy\cdot y}{10125-y/4}$. (BSS 21.25–26 *upapatti*)

या = (१८०-चा) चा | अतः शेषद्वारागमेन ज्या × ४०५०० = ज्या. या = ४ ज्या. या, ततो या = ज्या × ४०५०० = १०१२५. ज्या
ज्या + ६ ज्या / ६
= □ = (□□□-□□) □□ = □□□ □□ - □□$^{□}$
□□□□□□□□ □□$^{□}$ - □□□ □□ + □ = □
$\therefore$ □□ = □□ $\pm \sqrt{□□^{2} -}$
□□□□□□□□□□□□□□ □□□□ □□□□□□ □□□ □□ = □□ - $\sqrt{□□^{2} -}$
अत उपपन्नं सर्वम् ||२५-२६||

Let $y = (180-ca)\cdot ca$. By algebraic manipulation: $jy \times 40500 = 4\cdot jy\cdot y$, so $y = 10125 \cdot jy/(jy+6)$. Let $l = (180-ca)ca = 180ca - ca^2$. By equating: $ca^2 - 180ca + l = 0$. Therefore $ca = 90 \pm \sqrt{90^2 - l}$. Taking the smaller arc as the teacher taught: $ca = 90 - \sqrt{90^2 - l}$. Thus all is proved. ||25–26||

षड्भिर्भोगकलानामेकयम् = ९ चग
षड्भिर्भोगकलानामेकयम् = ३ चग
पञ्चदशैर्भोगकलानामेकयम् = १५ चग = १५ चग
सर्वयोगकलाः = २७ चग
चक्रकलाभ्यः शुद्धा सर्वयोगकला जाता श्रभिज्ञुगकलास्तदिहनगतिः = चक्-२७ चग |
इय कल्पकृत् दिनगुणाः इष्टचक्रकलामका जाता | कल्पे[3]भीजनो भगणाः = क्षु --- २७ क्षभं | शेषोपपत्ति मोर्तकृत् मुहनक्षत्रायन विभिन्नकृत् ||५०-५२||

Bhogakalā (daily motion in arc-minutes): For six bhogakalās, one = 9 caga; for another six, one = 3 caga; for fifteen, one = 15 caga. Total yogakalā = 27 caga. Subtracted from cakrakalā, the resulting sarvayogakalā gives the daily motion = cak - 27 caga. The kalpa elapsed days multiplied by the desired cakra-kalā yield the result; in the kalpa, [3]the revolutions = ku — 27 kbhaṃ. The remainder is derived from mṛta, nakṣatra, and ayana differences. ||50–52||

(बीएसएस् २१.५०--५२) एते श्लोकाः भोगकलाः (दैनिकं चलनं चाप-कलासु) गणयन्ति | रवेः चन्द्रस्य ग्रहाणां च दैनिकं चलनं क्रमशः दिनयोः दीर्घत्वान्तरात् प्राप्यते |

(BSS 21.50–52) These verses compute the *bhogakalā* (daily motion in minutes of arc). For the Sun, Moon, and planets, the daily motion is found by taking the difference in longitude between successive days and converting to *kalās*.

वि. भा.---रविः (चन्द्रस्य) मध्यगतिकला अथवा^{3}वैर्क (ई, ई, ई) गुणास्तदाऽऽध्यार्थ^{3}समनक्षत्राणां भोगकलाः स्युः |

Vr̄ti-bhāỵa: The mean daily motion in *kalās* of the Sun (or Moon), multiplied by the respective factor, gives the *bhogakalās* of the *nakṣ̄atras*. (BSS 21.50–52 commentary)

भोगकला ग्रहस्य दैनिकं चलनं चाप-कलासु अस्ति | गणनायां एते पदानि : (१) क्रमशः दिनयोः दीर्घत्वान्तरस्य अन्वेषणम्, (२) एतद् अन्तरं कलासु रूपान्तरम्, (३) रवेः ≈ १° प्रतिदिनम् (≈ ६० कलाः), (४) चन्द्रस्य ≈ १३° प्रतिदिनम् |

The *bhogakalā* is the daily motion of a planet in terms of *kalās* (arc-minutes). The computation involves:

- Finding the difference in longitude between successive days
- Converting this difference into minutes (*kalās*)
- For the Sun: about 1° per day (approximately 60 *kalās*)
- For the Moon: about 13° per day

6 Final Chapters — Eclipses and Colophons

ब्रह्मस्फुटसिद्धान्तस्य समाप्ति-भागः ग्रहणगणनायाः विस्तृतानि सूत्राणि समाविशति --- स्पर्शस्य (प्रथमस्पर्शस्य), मोक्षस्य (अन्तिमस्पर्शस्य), ग्रहणस्य परिमाणस्य, अक्षवलनस्य (अक्षच्युतेः) च |

The concluding portion of the *Brāhmasphuṭasiddhānta* contains extensive rules for eclipse calculations, including the computation of the *sparśa* (first contact) and *mokṣa* (last contact), the magnitude of the eclipse, and the *akṣavalana* (deflection due to latitude).

मन्दस्फुटार्केन्द्रांशोनायां ग्रहस्यन्दं केन्द्रम् |
एव यदि तत् स्पष्टं त्रिकोणं तदानीं विशेषं नोद्येत् |
तद्यात् स्पष्टं त्रिकोणं तस्य स्पष्टं द्वितीयं केन्द्रं
क्रमेण केन्द्रान्तरं फलमिति | वा ज्ञानमन्दं स्पष्टं
प्राप्यच्छी स्साचार्य्योदिना शीघ्र-कर्त्तव्यं कार्यमिति सर्वं शुद्धम् ||५७८-५८१||

mandasphuṭārkendrāḿśonāyāṃ grahasyandaṃ kendram |
eva yadi tat spaṭaṃ trikoṃ tadāñ̄iṃ viśeạṃ nodyet |
tadyāt spaṭaṃ trikoṃ tasya spaṭaṃ dvitīyaṃ kendraṃ
krameā kendrāntaraṃ phalamiti | vā jñānamandaṃ spaṭaṃ
prāpyacch̿issācāryyodinā ś̿ighra-karttavyaṃ kāryamiti sarvaṃ śuddham ||578–581||

Colophons — समाप्ति

कलामयो यावता मानां भोगकलाः सृद्धास्तावद् गणमानि |
शेषाः ख लिख वर्तमाननक्षत्रस्य गतकलास्तास्तदुग्कलामय |
शुद्धा गम्य कला भवति | ततो ग्रहद्वयको दिवसस्तदा
गतगम्यकलाभिः किमित्वनुपातेन गणगम्या दिवमा
भवति |

kalāmayo yāvatā māñ̄aṃ bhogakalāḥ sṛḍhāstāvad gaāni |
śē̄aḥ kha likha vartamānakātrasya gatakalāstāstadugkalāmaya |
śuddhā gamya kalā bhavati | tato grahadvayako divasastadā
gatagamyakalābhiḥ kimitvanupātena gaā divamā bhavati |

इति ब्रह्मस्फुटसिद्धान्त समाप्तः |

Here ends the Brāhmasphuṭasiddhānta

इस प्रकार ब्रह्मस्फुटसिद्धान्त की समाप्ति हुई | इस ग्रन्थ में कुल २४ अध्याय
हैं, जिनमें गणिताध्याय (१२ अध्याय) और गोलाध्याय (१२ अध्याय) सम्मिलित हैं |
ब्रह्मगुप्ताचार्य ने इस ग्रन्थ की रचना संवत् ६२८ (ई. ६६६) में की थी |

Thus concludes the Brāhmasphuṭasiddhānta. This treatise contains 24 chapters in all, comprising 12 chapters of mathematics (Gaṇitādhyāya) and 12 chapters of astronomy (Golādhyāya). The teacher Brahmagupta composed this work in Śaṃvat 628 (CE 666).

Appendix: Summary of Mathematical Rules

छाया-व्यवहारस्य सारांशः: छाया नरायाः (अक्षज्यायाः) समानुपातिकी, शङ्कोः (शङ्कु-उच्चतायाः) व्यस्तानुपातिकी च |	**Shadow rule (छाया व्यवहार):** The shadow is proportional to the *nara* (sine of latitude) and inversely proportional to the *śañku* (gnomon height).
अवर्गसंख्यानां वर्गमूलानयनस्य सूत्रम्: $\sqrt{N} = a + \frac{r}{2a+1}$ यत्र a विद्यते यत् $a^2 \leq N$ च $r = N - a^2$	**Square root of non-squares (विकलवर्गानयन):** The method of successive approximation: $\sqrt{N} = a + \frac{r}{2a+1}$ where a is the largest integer such that $a^2 \leq N$ and $r = N - a^2$.
मन्दफलस्य सारांशः: ग्रहपथस्य विकेन्द्रतायाः (अपकेंद्रतायाः) समीकरणं, केन्द्रस्य (अनुरूपज्यायाः) ज्यया कल्पितम् |	**Equation of centre (मन्दफल):** The correction for the eccentricity of planetary orbits, computed using the *jyā* (sine) of the *kendra* (anomaly).
शीघ्रफलस्य सारांशः: ग्रहस्य स्थितेः सूर्य-पृथ्वी-रेखायाः सापेक्षः संशोधनः, शीघ्रकेन्द्रात् (योगस्य अनुरूपज्यायाः) गणितः |	**Second inequality (शीघ्रफल):** The correction for the planet's position relative to the Sun-Earth line, computed from the *śīghrakendra* (anomaly of conjunction).
कुट्टक-विधिः रैखिक-डायोफैण्टैन्-समीकरणानि $ax + c = by$ सोडयति, कालगणनायां आवश्यकम् |	**Kuṭṭaka (कुट्टक):** The pulverizer method for solving linear Diophantine equations $ax + c = by$, essential for calendar calculations.
भोगकला ग्रहस्य दैनिकं चलनं चाप-कलासु अस्ति, क्रमशः दिनयोः दीर्घत्वान्तरात् गणिता |	**Bhogakalā (भोगकला):** The daily motion of a planet in arc-minutes, computed from the difference in longitude between successive days.
ज्याचापानयनम्: ज्यायाः चापस्य अन्वीक्षण-समस्या, ब्रह्मगुप्तेन बीजगणितीयसूत्रेण सोडिता, ज्यार्ध-पिठकस्य उपयोगं विना |	**Arc from sine (ज्याचापानयन):** The inverse problem of finding the arc from its sine, solved by Brahmagupta's algebraic formula without using the *jyārdha* (half-chord) table.

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ब्राह्मस्फुटसिद्धान्त

भाग ३

Brāhmasphuṭasiddhānta

Part 3 — Chunk 2 (Pages 1817–1956)

Bilingual Sanskrit–English Edition

By

Brahmagupta

(b. 598 CE)

Edited with Sanskrit Commentary (Vāsanā) and Hindi Translation

by

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स्मृत्योर्युगलयोः समन्वयः (Smṛti-yugala-samanvaya)

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त्रिप्रश्नोत्तराध्यायः

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अब श्रम्बिज्ञुत नक्षत्र के भोगानयन और ग्रहमुक्त नक्षत्रानयन को कहते हैं।

[Introductory Note.] Now [the Ācārya] speaks of the computation of the motion (bhoga) of the starry nakṣatra and the computation of the nakṣatra as freed from the planets (graha-mukta-nakṣatrānayana).

पञ्चदशो योगभोगकलानामैक्यं = चं × १५
सर्वेषां योगः = १ चं + ३ चं + १५ चं = २७ चं = सर्वनक्षत्रभोगः

[Mathematical Rule.] The fifteenth [operation]: The sum of the degrees of conjunction-motion (yoga-bhoga-kalā) = Caṃ × 15. The sum of all = 1 Caṃ + 3 Caṃ + 15 Caṃ = 27 Caṃ = the total motion of all the nakṣatras (sarvana-kṣatra-bhogaḥ). Here Caṃ (cakra) denotes a complete revolution.

चक्कलास्यः शुद्धाः सर्वनक्षत्रभोगसंख्यास्तदा श्रम्बिज्ञुद् भोगकलास्तदिह्नगतिः =

चक्र—२७ चं ततः कल्पोड्भवो भगणः := (चक्र—२७ चं) कुदिन =

चक्कला

कुदिन—२७ कल्पचंभगणा एतेनाचार्येण तुमूपपन्नम् । सिद्धान्त शेखरे

□□चक्कारिया वा शाश्वतसतिवधनाहतानि शोधयिन् सुदिनलयद्वरोषचक्र'' । स्यादे-

कवासरम्बा कलिकागतिर्या सा वैवस्वेतावमध्यगाभिध्या मृतिः ॥ इद्धप्रहस्य

कलिकालिकरादिसंशया नक्षत्रभोगकलिका प्रहसुत्कालानि । शेषात् भोग्यकलिका

परितात् गम्य तास्यो भवन्ति गतगम्यदिनानि मुक्त्या ॥ इति श्रीपतेः पञ्चद्वय

□□सर्वेषां भोगैनितचक्कलिप्ता वैवस्वतः स्यादमिज्ञुद्धयोग'' इत्यादि भार्करोक्तं

च सर्वाचार्येणस्य सर्वेषं समानार्थकार्मित ॥ ५०--५१ ॥

[Dvivedī's Commentary.] The pure revolutions (cakkala) are the numbers [representing] the total motion of all the nakṣatras; then the degrees of motion of the starry nakṣatra (śrambijñud bhoga-kalā) [are obtained]; its daily motion = revolution − 27 Caṃ. From that, the kalpa-born revolutions [are obtained] as: (revolution − 27 Caṃ) civil days (kudina) = cakkala.

In 27 civil days, the kalpa revolutions [are obtained] by this Ācārya; [this is] well established (tumūpapannam). In the *Siddhāntaśekhara*: "cakkāriyā vā śāśvata-sativadhana-āhatāni śodhayin sudina-laya-dvaroṣa-cakra". [The meaning is that] the daily motion of the kalpa (kalikā-gatiḥ) which is the Vaivasvata [value] at mid-[period] is to be known (mṛtiḥ). The degrees of nakṣatra motion (nakṣatra-bhoga-kalikā), the times of the prahara, etc., the remaining degrees to be traversed (bhogya-kalikā) — from the excess, the gatagamya [values] become the days traversed and remaining to be traversed (gata-gamya-dināni). Thus [it is said] by Śrīpati in the fifteenth [chapter].

"The sum of the motions of all [nakṣatras] multiplied by the revolutions [gives] the Vaivasvata [value]; this is the yoga of the starry [nakṣatra]," and so on — thus spoke Bhāskara; and all teachers [hold] this same meaning.

[Verses 50–51]

उपपत्तिः

| *Upapatti* (Proof/Demonstration)

षट् श्रद्धं भोगकला नक्षत्रों का ऐक्य = $\frac{3\text{ चं}}{2}$ × ६ = ⊠ चं × ३ = ६ चं

षट् शद्धभोग कला नक्षत्रों का ऐक्य = $\frac{\text{चं}}{2}$ × ६ = चं × ३

पनद्रह् एक भोग (समान भोग) कलानक्षत्रों का ऐक्य = चं × १५

सबों का योग = सर्वनक्षत्र भोग कला = ६ चं + ३ चं + ? चं = २७ चं

[Mathematical Derivation.] The sum of the six doubled (ṣaṭ-śraddha) motion-degrees of the nakṣatras = $\frac{3\text{ Caṃ}}{2} \times 6 = 3$ Caṃ × 3 = 6 Caṃ. The sum of the six [times] single-motion degrees of the nakṣatras = $\frac{\text{Caṃ}}{2}$ × 6 = Caṃ × 3. The sum of the fifteen [times] single motion (sama-bhoga) degrees of the nakṣatras = Caṃ × 15. The sum of all = total nakṣatra motion degrees = 6 Caṃ + 3 Caṃ + [15] Caṃ = 27 Caṃ.

हिं. भा.---चन्द्रमध्यगति कला को ईं, ईं, ? इन से गुणा करने से षड्वर्षं, शर्बं,

सम नक्षत्रों की भोग कला होती है । सब नक्षत्रों की जो भोग कला है उनको भगण कला

में से घटाने से जो शेष रहता है वह श्रम्बिज्ञुत भोग है । वा कल्प चन्द्र भगण सतासी से

गुणा कर कल्पकुदिन में से घटाने से शेष श्रम्बिज्ञुत का कल्प भगण होता है, इन भगणों से

जो दिन भोग (कल्प कुदिन में यदि श्रम्बिज्ञुत का कल्प भगण पाते हैं तो एक दिन में क्या इस

अनुपात से लब्ध कालात्मक दिनगति) होता है वह श्रम्बिज्ञुत भोग होता है । ग्रहकालसमूह

में से नक्षत्र भोग कला को घटाने से नक्षत्र होते हैं, अर्थाद् ग्रहमुक्त कला में जितने नक्षत्रों की

भोग कला शुद्ध हो उतने गतनक्षत्र होते हैं, शेषकला वर्तमान नक्षत्र की गत कला है उसको

नक्षत्र भोग कला में से घटाने से गम्य कला होती है । तब ग्रहगति कला में एक दिन पाते हैं

तो गत-गम्य कला में से क्या इस अनुपात से गतदिन और गम्यदिन होते हैं ॥ ५०--५२ ॥

[Hindi Bhāṣya — English Summary.] When the mean daily motion of the Moon (candra-madhyagati-kalā) is multiplied by [certain] quantities, the motion degrees of the six, the śarva, and the equal nakṣatras are obtained. The remaining [degrees], when subtracted from the revolution degrees (bhagaṇa-kalā), give the motion of the starry nakṣatra (śrambijñuta-bhoga).

Alternatively: The kalpa revolutions of the starry nakṣatra are obtained by multiplying the kalpa revolutions of the Moon by eighty-seven and subtracting from the kalpa civil days. From these revolutions, the daily motion of the starry nakṣatra is obtained by proportion (anupāta): if in kalpa civil days such-and-such kalpa revolutions [are obtained], then in one day [what]? — thus the daily motion in time--units.

When the nakṣatra motion degrees are subtracted from the planet-time aggregate (grahakāla-samūha), the nakṣatras are obtained. That is, in the planet-freed degrees (graha-mukta-kalā), as many pure motion degrees of the nakṣatras as there are, so many are the traversed nakṣatras (gata-nakṣatra). The remaining degrees are the current degrees of the present nakṣatra; subtracting these from the nakṣatra motion degrees yields the remaining degrees to be traversed (gamya-kalā). Then, if in one day [a certain value] is obtained in planetary motion degrees, by proportion from the gata-gamya degrees, the days traversed and days remaining are obtained.

[Verses 50–52]

उपपत्तिः --- यहाँ आचार्य ने पहले उक्त कालज्ञा को इष्टद्वित कल्पित किया है । तब श्रदा,

समशाईं कुदूदित् इन तीनों मुञ्जाओं से उत्पन्न एक श्रम्ब क्षेत्र है, तथा श्रदज्ञा, लब्धज्ञा-

तिज्ञा इन तीनों मुञ्जाओं से उत्पन्न द्वितीय श्रम्ब क्षेत्र है, ये दोनों श्रम्ब क्षेत्र सजातीय हैं

इसलिए अनुपात करते हैं

| ***Upapatti*** **(Proof).** Here the Ācārya has postulated (kalpita) the previously stated time-quantity (kālajñā) as the desired [value] for the second [process]. Then śrada, sama-śāīṃ, kudūdit — these three quantities (muñjā) produce one śramba field (kṣetra); and śrada-jñā, labdha-jñā, tijñā — these three quantities produce a second śramba field. These two śramba fields are homogeneous (sajātīya); therefore, proportion (anupāta) is applied [between them].

$$\frac{\text{लब्धज्ञा. इष्ट}}{\text{तिज्ञा}} = \text{समशाईंकुद्}$$

तथा कालज्ञा =

$$\frac{\text{labdhajñā} \times \text{iṣṭa}}{\text{tijñā}} = \text{samaśāīṃ-kud}$$

And the time-quantity (kālajñā) = [the result of this proportion]. The three quantities (trayaṃ muñjānāṃ) generate the proportional relationship between the two śramba fields.

अत्रोपपत्तिः

छायाकर्णे गोले पलभा शङ्कुतुल्योस्तुल्यतत्त्वं भवति कथमिति प्रदर्शयते यदि द्वादशाङ्कुलराश्यों पलभा भुजां लभ्यते तदेप्तशाङ्कुं किमिति जातं शङ्कुतुल्यम् =
$\frac{\text{पमा. शाङ्कु}}{\text{⊠⊠}}$, परन्तु शाङ्कु = $\frac{\text{⊠⊠} \times \text{त्र}}{\text{छाक}}$ शत उत्थापनेन शङ्कुतुल्यम् = $\frac{\text{पमा. १२. त्र}}{\text{⊠⊠. छाक}}$

| ***Atrôpapattiḥ*** **(Demonstration thereof)**

[Śloka on Shadow Computation.] In the shadow--hypotenuse circle (chāyā-karṇe gole), how the equinoctial shadow (palabhā) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) — this is demonstrated thus: if the palabhā is obtained as the base (bhuja) in the twelve--aṅgula sign (dvāda-śāṅgula-rāśi), then what is the gnomon-like [quantity]? The gnomon-equivalent (śaṅku-tulya) = $\frac{\text{pama} \times \text{śaṅku}}{12}$; but śaṅku = $\frac{12 \times \text{tri}}{\text{chāka}}$. By elevation (utthāpanena), the gnomon-equivalent = $\frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}}$.

यहाँ द्वाद (दिनार्क्) से भिन्न समय में पलभा साधन के लिए कहते हैं।

हिं. भा.---दिनार्क् से भिन्न समय में रवि की छाया को छायाकर्णों से गुणा कर

तिज्ञा से भाग देने से छायाकर्ण गोलिय छाया होती है, शङ्कुमूल और पूर्वापर रेखा का

भेद भुज है । उत्तर गोल में कर्णदुताज्ञा में उत्तर भुज को घटाने से और दक्षिणा भुज

को जोड़ने से पलभा होती है । दक्षिणा गोल में भुज में कर्णदुतिय छाया को घटाने ही से

पलभा होती इति ॥ ४६--५० ॥

[Dvivedī's Explanation.] Here, for computing the palabhā at a time different from midday (dvāda/dinārk), [the Ācārya] speaks.

[Hindi Bhāṣya — Summary.] At a time different from midday, multiplying the Sun's shadow by the shadow--hypotenuse (chāyā-karṇa) and dividing by the tijñā yields the circular shadow of the hypotenuse-circle. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the base (bhuja). In the northern hemisphere (uttara-gola), subtracting the northern bhuja from the karṇa-dvitīya and adding the southern bhuja yields the palabhā. In the southern hemisphere (dakṣiṇa-gola), merely subtracting the second shadow from the bhuja yields the palabhā.

[Verses 46–50]

ग्रहणोत्तराध्यायः

Original pages 1089–1139 | *Bilingual Edition*

अब श्रम्ब्य प्रश्नों को कहते हैं ।

हिं. भा.---जो क्रान्ति के ज्ञाता समशकु को जानते हैं । जो लम्बांश वेत्सा समशुत्त

को जानते हैं, कोणशकु को जानते हैं, कोणशकुछाया को जानते हैं । कोणदुताज प्रवेश में

घटी को जानते हैं वे ज्योतिष सिद्धान्त के पण्डित हैं, यहाँ पाँच प्रश्न हैं ॥ ७ ॥

[Introduction.] Now [the Ācārya] speaks of the questions of the śrambya [type].

[Hindi Bhāṣya — Summary.] Those who know the sama-śaku [from] the declination (krānti), who know the sama-śutta [from] the latitude (lamba-āṃśa), who know the koṇa-śaku, who know the koṇa-śaku-chāyā, and who know the ghaṭī [from] the koṇa-dvitāja entrance — they are the scholars (paṇḍita) of jyotiṣa-siddhānta. Here there are five questions.

[Verse 7]

इदानीमन्यान् प्रश्नानाह

| ***Idānīm anyān praśnān āha*** (Now he states other questions.)

शङ्कुतलप्राच्यपरान्तरद्वयं बोध्यं यो विजानाति ।

विशुवच्छायामेकं हस्त्वाऽऽसन्नदित्यं च गण्यकः सः ॥ ७ ॥

śaṅku-tala-prācyaparāntara-dvayaṃ bodhyaṃ yo vijānāti |

viśuvacchāyām ekaṃ hastvā 'āsanna-dityaṃ ca gaṇyakaḥ saḥ || 7 ||

He who, having known the two differences of the east-west [line] at the gnomon-base (śaṅku-tala-prācya-parāntara), understands [also] the equinoctial shadow (viśuvac-chāyā) and the near-adjacent sun (āsanna-āditya) — he is a calculator (gaṇyaka).

[Śloka 7]

सू. भा.---शङ्कु तलप्राच्यपरान्तरं भुजः । यो भुजद्वयं बोध्य विशुवच्छाया पलभां विजानाति । एकमेव भुजं हस्त् वा विशुवच्छायामादित्यमकं च विजानाति

स गण्यकः । एवमत्र प्रश्नत्रयम ॥ ८ ॥

[Sūtra-bhāṣya.] The difference of the east-west [line] from the gnomon-base (śaṅku-tala-prācya-parāntara) is the base (bhuja). He who, having understood the two bhuja-values, knows the equinoctial shadow (viśuvac-chāyā) [i.e.,] the palabhā, or who knows one bhuja [together with] the equinoctial shadow and the sun — he is a gaṇyaka. Thus here there are three questions.

[Verse 8]

वि. भा.---शङ्कुतलपूर्वापररेखयोरन्तरं भुजोर्धित, यो भुजद्वयं ज्ञात्वा पलभां जानाति एकं भुजं ज्ञात्वा पलभां रविं च जानाति स गण्यकोऽस्तीति । अत्र प्रश्नत्रय-

भमस्ति ॥ ७ ॥

[Vivaraṇa-bhāṣya.] The difference of the east-west line [measured] from the gnomon-base (śaṅku-tala-pūrvāpara-rekha) is declared to be the bhuja. He who, having known the two bhuja-values, knows the palabhā, or having known one bhuja, knows the palabhā and the Sun (ravi) — he is a gaṇyaka. Thus here there are three questions.

[Verse 7]

अब श्रम्ब्य प्रश्नों को कहते हैं ।

हिं. भा.---शङ्कुतल और पूर्वापर रेखा का भेद भुज है । जो व्यक्ति दो भुजों को

ज्ञान कर पलभा को जानते हैं । एवं एक भुज को ज्ञान कर पलभा को जानते हैं तथा रवि

को जानते हैं वे ज्योतिः शास्त्र के पण्डित हैं इति । यहाँ तीन प्रश्न हैं ॥ ७ ॥

[Dvivedī's Introduction.] Now [the Ācārya] speaks of the śrambya questions.

[Hindi Bhāṣya — Summary.] The difference between the gnomon-base (śaṅku-tala) and the east-west line is the bhuja. The person who, having known the two bhuja--values, knows the palabhā, and likewise [he who], having known one bhuja, knows the palabhā and the Sun (ravi) — they are the paṇḍitas of jyotiḥ-śāstra. Here there are three questions.

[Verse 7]

इदानीमन्यान् प्रश्नानाह

| ***Idānīm anyān praśnān āha*** (Now he states other questions.)

पातालशङ्कुमुध्येस्ते वा हस्तज्ञां रविंविजानाति ।
इष्टपातालगणकोः पृथक् तले वा स तन्त्रज्ञः ॥ ८ ॥

pātāla-śaṅkum uddhyeste vā hasta-jñāṃ raviṃ vijānāti |
iṣṭa-pātāla-gaṇakoḥ pṛthag tale vā sa tantra-jñaḥ || 8 ||

He who knows the pātāla-śaṅku [and] the uddeśa-śaṅku, or [who knows] the sun having known the hasta; [or who knows] the iṣṭa-pātāla separately, each at its own base (tale) — he is a knower of the tantra (tantra-jña).

[Śloka 8]

सू. भा.---यो रवेः पातालशङ्कु कुमथः शङ्कुं विजानाति । उदयेस्ते वा यो रविंद्र्ग् ज्ञात्रो विजानाति । इष्टशङ्कुकुदिवोइष्टशङ्कुः पातालगणः: पातालशङ्कुः

रविरथः शङ्कुः । तयोः पृथक् पृथक् तले शङ्कुतले च वा यो विजानाति स एव

तन्त्रज्ञ इति । एवमत्र प्रश्नचतुष्टयम ॥ ९ ॥

[Sūtra-bhāṣya.] He who knows the pātāla-śaṅku of the Sun [and] the kumatha-śaṅku, or who knows the Sun having known the uddheśa, or [who knows] the iṣṭa-śaṅku [and] the kudivo-iṣṭa-śaṅku [as] the pātāla-gaṇaḥ and pātāla-śaṅku [respectively], the ravi-ratha-śaṅku — of these two, separately, each at its own base (pṛthak tale) or at the gnomon-base (śaṅku-tale), he who knows — he alone is the tantra-jña. Thus here there are four questions.

[Verse 9]

अब श्रम्ब्य प्रश्नों को कहते हैं ।

हिं. भा.---जो व्यक्ति ग्रहण में राहुमार्ग (भूमामार्ग) को जानते हैं । उस मार्ग को जानते हैं, कोणशकुछाया को जानते हैं । यहाँ पाँच प्रश्न हैं ॥ ७ ॥

[Dvivedī's Introduction.] Now [the Ācārya] speaks of the śrambya questions.

[Hindi Bhāṣya — Summary.] The person who knows the Rāhu-path (bhūmā-mārga) in an eclipse, who knows that path, and who knows the koṇa-śaku-chāyā — here there are five questions.

[Verse 7]

कोणदुतस्थरविगतं भुवप्रोत्कुटं भुवाद्रिविपर्यं भुव्यचापांशं एकं भुजः ।

भुवाल्पसतिकावधि लम्बांश द्वितीयो भुजः । कोणदुतं व स्वैनिकाद्रिविपर्यं ननांशास्तुतीयो भुजः । त्रिभुजेऽस्मिन् रविगतं भुवप्रोत्कुटतयाप्येतर्कुर्नाम्यामृत्यन्नकोणा नतकालः । याम्योत रवित्कोणदुतनाम्यामृत्यन्नकोणः = ८५ नतोनुपातः

किम्यते यदि नतकालज्ञया हस्तज्ञा लभ्यते तदाश्रवेदांशज्ञया (ज्ञा ८५) किमिनि

समागच्छति भुज्या = $\frac{\text{हस्तज्ञा. ज्ञा ८५}}{\text{नतकालज्ञा}} = \frac{\text{हस्तज्ञा. त्रि. ज्ञा ८५}}{\text{नतकालज्ञा. त्रि}} = \frac{\text{हस्तज्ञा. त्रि. २८५}}{\text{[illegible] नतकालज्ञा}}$

[Eclipse Triangle Computation.]

koṇaduta-stha-ravi-gataṃ bhuva-protkuṭaṃ bhuvādri-viparyaṃ bhuvyacāpāṃśaṃ ekaṃ bhujaḥ |

bhuvālpasatikāvadhi lamba-aṃśa dvitīyo bhujaḥ | koṇadu-taṃ va svainikādri-viparyaṃ nanāṃ-śāstu-tīyo bhujaḥ |

In this triangle: the Sun's passage at the koṇaduta, the earth's protuberance (protkuṭa), the earth-mountain reversal — the degrees of the earth-curve (bhuvyacāpāṃśa) — these form one bhuja. The latitude-degrees (lamba-aṃśa) up to the earth's alpaka-satika [form] the second bhuja. The koṇaduta and the svainika-mountain reversal [form] the third bhuja.

In this triangle, the Sun's passage being [known] as the earth's protuberance (bhuva-protkuṭa), the angle [is] the natakala (nata-kālaḥ). The southern koṇaduta-nata-amṛtyanna-koṇa = 85; by the rule of proportion (anupātaḥ): if by the natakāla-jyā the hasta-jyā is obtained, then by the arc-jyā of 85 (jyā 85), what is obtained?

The bhuja-jyā = $\frac{\text{hasta-jyā}\times\text{jyā }85}{\text{natakāla-jyā}} = \frac{\text{hasta-jyā}\times\text{tri}\times\text{jyā }85}{\text{natakāla-jyā}\times\text{tri}} = \frac{\text{hasta-jyā}\times\text{tri}\times 285}{2495\times\text{natakāla-jyā}}$.

उपपत्तिः

यहाँ देशान्तर विदित नहीं है इसलिए देशान्तर संस्कार से रहित स्फुट रवि और

स्फुट चन्द्र से चन्द्र ग्रहण विधि से सर्वग्रहण में संमीलन काल और उन्मीलन काल साधन

करना चाहिए । उस दिन में द्वितीय से भी संमीलन काल समझा चाहिए वह यदि गणितानगत्

संमीलन काल से अधिक हो तो द्वितीय रेखा से पूर्व भाग में अथवा पश्चिम भाग में स्थित है यह

समझा चाहिए । क्योंकि रेखा से पूर्व स्वदेश में पहले स्वदेश में पश्चात् रेखा देश में मध्याह्न

काल होता है । अतः रेखा देशीय इष्ट संमीलन काल से स्वदेशीय संमीलन काल अधिक होता

है पश्चिम में इसके विपरीत होता है । इस तरह उन्मीलन काल से इष्ट प्रास काल से स्पर्श

काल से या मोक्ष काल से परीक्षा होती है, स्पर्श काल परीक्षा और मोक्ष काल परीक्षा इष्ट

| *Upapattiḥ* (Proof/Demonstration)

[Dvivedī's Upapatti — Summary.] Here, since the longitude difference (deśāntara) is not known, one should compute the conjunction-time (saṃmīlana-kāla) and disjunction-time (unmīlana-kāla) of the total eclipse by the method of lunar eclipse, from the true Sun (sphuṭa-ravi) and true Moon (sphuṭa-candra), devoid of deśāntara correction. On that day, the second saṃmīlana-kāla should [also] be computed; if it exceeds the computational saṃmīlana-kāla, then one should know whether [the place] is situated in the eastern or western part from the second line.

Because to the east of the line, midday (madhyāhna-kāla) occurs first in one's own country and later in the line-country; therefore, the saṃmīlana-kāla of one's own country exceeds the saṃmīlana-kāla of the line-country. In the west, the opposite occurs. Thus, from the unmīlana-kāla, the iṣṭa-prāsa-kāla, the sparśa-kāla, or the mokṣa-kāla should be examined.

उत्तरगोले याम्ये विषुवच्छायायुक्तस्तरं हीनम् ।
एवं विषुवच्छाया युक्तविहीनास्तरेषाभा ॥ ५० ॥

uttara-gole yāmye viṣuvacchāyāyuktas taraṃ hīnam |
evaṃ viṣuvacchāyā yuktavihīnāstareṣābhā || 50 ||

In the northern hemisphere (uttara-gola), [the latitude] being joined with the equinoctial shadow (viṣuvac-chāyā) [in the] south (yāmya) is diminished (hīnam). Thus, the equinoctial shadow [is] joined with or diminished from the latitudes (tareṣā) [depending on hemisphere].

[Śloka 50]

सू. भा.---प्रथमायाः पूर्वार्धं त्रिप्रश्नाध्यायस्य चतुर्थीयुपयुपर्धसं न्यस्यात्मेव । अन्यत् सर्वं च त्रिप्रश्ना ५८--६० आयामिः स्फुटम् ॥ ४९--५० ॥

[Sūtra-bhāṣya.] The first half of the earlier [section] of the Tri-praśna-adhyāya [is to be] placed as the fourth subsidiary section (upayupardha-saṃṃ nyasyāt). All else of the Tri-praśna [verses] 58–60, the extent (āyāmi) [is] clear.

[Verses 49–50]

वि. भा.---दिनार्धदिनकाले शक्रिया छायाकर्णगुणा तिज्ञामक्ता तदा छायाकर्णगोलिया रवैर्या भवेत् । शङ्कुमूलपूर्वापररेखयोरन्तरं भुजः । तेन भुजेनोन्यता कर्णदुताप्राच्यादितुरेया भुजेनोनदिक्षेत्रं भुजेन युता तदोतरगोले विषुवच्छाया (पलभा) भवेत् । याम्ये (दक्षिणा गोले) तया कर्णदुतियाप्राच्यादन्तरं (भुजमान) हीन तदा पलभा भवेत् । एवमन्तरेषा (भुजेन) युक्तविहीनास्तस्यथादन्तरभुजेन युता दक्षिणाम्यभुजेन हीना तदा पलभा भवेदिति ॥ ४९--५० ॥

[Vivaraṇa-bhāṣya.] At the half-day time (dinārdha-dina-kāle), the shadow multiplied by the shadow-hypotenuse (chāyā-karṇa-guṇā), divided by the tijñā, then the circular shadow of the hypotenuse-circle (chāyā-karṇa-golīyā) would be [obtained] as the ravi-value. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the bhuja. By that bhuja, diminished by the karṇa-dvitīya-prācya, [or] joined with it — in the northern hemisphere, the equinoctial shadow (viṣuvac-chāyā), i.e., the palabhā, results. In the south (i.e., the southern hemisphere), when that is diminished by the difference from the karṇa-dvitīya-prācya (bhuja-māna), then the palabhā results. Thus, [the shadow] joined with or diminished from the difference by the bhuja, or diminished by the southern bhuja, then the palabhā results.

[Verses 49–50]

अत्रोपपत्तिः

छायाकर्ण गोले पलभा शङ्कु तुल्ययोस्तुल्यतत्त्वं भवति कथमिति प्रदर्शयते यदि द्वादशाङ्कुलराश्यों पलभा भुजां लभ्यते तदेप्तशाङ्कुं किमिति जातं शङ्कुतुल्यम् =

$$\frac{\text{पमा. शाङ्कु}}{\text{⊠⊠}}, \quad \text{परन्तु शाङ्कु} = \frac{\text{⊠⊠} \times \text{त्र}}{\text{छाक}} \quad \text{शत उत्थापनेन}$$

$$\text{शङ्कुतुल्यम्} = \frac{\text{पमा. १२. त्र}}{\text{⊠⊠. छाक}} = \text{पमा}$$

∴ सिद्धं कर्णगोले शङ्कुतुल्यम् = पलभा । कर्णदुताज्ञा व्यस्तगोला भवति । पलभा च सव्योतरा । तथाः संस्कारतःछायाप्रविपरसूनमध्ये भुजः कथतेऽस्तत्क्षेपरियेन कर्णदुताज्ञा भुजयाः संस्करेणा पलभा भवतीति । सिद्धान्तशेखरे □□अथ-

या तु नतपूर्वपश्चिमाशान्तरेषा शकिबुतज्ञामका । दक्षिणोत्तरभुजा युतोनिता सौम्यवर्तिन् रवो पलभा ॥ दक्षिणेन पुनरिननायां तथा हीनमेव हि तदन्तरं सदा । एवमन्तरयुतोनिता भवेद्-छाया नियतमुगुलाश्चका ॥'' श्रीपत्युक्तसिद्धमाचार्येणानुरूपमेवेति ॥ ४९--५० ॥

| *Atrôpapattiḥ* (Demonstration thereof)

[Proof of Gnomon-Equivalence.]

How the palabhā in the shadow-hypotenuse circle (chāyā-karṇa-gola) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) is demonstrated thus: if the palabhā is obtained as the bhuja in the twelve-aṅgula sign, then what is the gnomon-equivalent (śaṅku-tulyam)?

$$\text{śaṅku-tulya} = \frac{\text{pama} \times \text{śaṅku}}{12}, \quad \text{but śaṅku} = \frac{12 \times \text{tri}}{\text{chāka}}, \text{by elevation}$$

$$\text{śaṅku-tulya} = \frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}} = \text{pama}$$

Therefore, it is proved (siddhaṃ) that in the karṇa-gola, the gnomon-equivalent (śaṅku-tulya) equals the palabhā. The karṇa-dvitīya-jyā becomes the inverse circle (vyasta-golā). The palabhā is the north-south [quantity]. Thus, by the correction of the chāyā-pravi-parasūna, the bhuja is spoken of; by the addition-subtraction (kṣepa-riyena) with the karṇa-dvitīya-jyā, by the correction of the bhuja, the palabhā results.

In the *Siddhāntaśekhara*: "Now, the difference of the natapūrva-paścima (east-west natal difference), the śakibuta-jñā-makā. The dakṣiṇôttara-bhujā, joined with and diminished, the Sun dwelling in the north (saumya-vartin) [gives] the palabhā. In the southern [hemisphere], again, at the inanāyām, thus it is always diminished by that difference. Thus, joined with and diminished by the difference, the shadow becomes fixed [in] aṅgula-ścakā." This [is] said by Śrīpati; [it is] in accordance with the Ācārya alone.

[Verses 49–50]

वलनस्यमुचे तैम्यः पृथग् ग्रहक् खण्डकेन । वृत्तिः कृतिः स्पर्शविमर्शनिमध्यग्रता

क्रमेणोर्बिमहावगमयः ॥

valanasya muce taimyaḥ pṛthag grahak khaṇḍakena | vṛttiḥ kṛtiḥ sparśavimarśanimadhyagratā krameṇôr bimhavagamayaḥ ||

Of the deflection (valana), the release (muca), the taimya, separately; by the section of the planet (grahakhaṇḍakena); the vṛtti, the kṛti, the contact and release (sparśa-vimarśa), the centrality of the middle (nimadhyagratā) — in sequence, the understanding of the eclipse-disc (ur-bimha) [is obtained].

□□भास्कराचार्येण श्रीपतिप्रकार एव विश्वदर्शनेण प्रति

पादितः ॥ १४--१५॥

By Bhāskarācārya, the very method of Śrīpati [was] propounded (pratipāditaḥ) in the *Viśvadarpaṇa.*

[Verses 14–15]

विषुवदर्पमण्डलदिका इत्यादि दो प्रश्नों के उत्तर ग्रहणाधिकार में बता दिए

गये हैं । उसके उत्तरार्ध को श्रम्बे कहते हैं ।

अब □सम्पर्कं मण्डलै' इत्यादि प्रश्न के उत्तर कहते हैं ।

हिं. भा.---प्राह्लादूता में स्वल्पान्तर से सरलाकार तात्कालिक क्रान्तिज्ञुनीय चाप

बलन सूत्र का । बलनज्ञा से दिव्यज्ञान समज्ञा चाहिए । मानचक्रबाह्हत में बलन सूत्र के

उपर रवि के स्पर्श कालिक सार और मोक्ष कालिक सार को जिस दिशा के सार है उसी दिशा

में देना । मध्यबलन सूत्र में प्राह्ल बिम्ब केन्द्र से मध्यार देना चाहिए । चन्द्र शराघ्रों से प्राह्ल

प्रमाण व्यास से तुरत लिखकर स्पर्श मोक्ष श्रीर प्रास समज्ञा चाहिए, एवं पृथिवी पर परिलेख

लिखने से स्पर्श मोक्ष श्रीर प्रास होता है इति ॥

[Dvivedī's Commentary on Eclipse Drawing.]

The answers to the two questions beginning "viṣuvadarpamaṇḍala-dikā" etc., have been explained in the eclipse chapter (grahaṇâdhikāra). The latter half of that [is called] śrambe.

Now the answers to the question beginning "samparkaṃ maṇḍalai" etc. are spoken.

[Hindi Bhāṣya — Summary.] In the prāhlādūta, the arc of the instantaneous declination (tātkālika krānti) is of a simple form with slight difference; [this is] the balana-sūtra. From the balana-jyā, the divya-jyā should be computed. In the māna-cakra-bāhṛta [circle], above the balana-sūtra, the contact-sphāra (sparśa-kālika sāra) and release-sphāra (mokṣa-kālika sāra) of the Sun should be given in the direction of their [respective] sāra. In the madhya-balana-sūtra, the madhya should be given from the centre of the prāhla-disc. From the moon's horns (śarāghra), by the prāhla-pramāṇa-vyāsa [diameter], having drawn quickly, the contact, release, and mid-line prāsa should be computed. And by drawing the parilekha [projection] on the earth, the contact-release śrīra-prāsa is obtained.

उपपत्तिः

| *Upapattiḥ* (Proof/Demonstration)

मानचक्रबाह्ृत् वृत्त में जब प्राह्लाकृता का केन्द्र होता है तब प्राह्ल बिम्बग्रान्त और प्राह्ल

बिमग्रान्त संलग्न रहता है । इस लिये विदित मानचक्रबाह्ृत् वृत्त में दिशा को श्रीष्ट करना

चाहिए । सममण्डल और मानचक्रबाह्ृत् वृत्त का समाप्त बिन्दु मानचक्रबाह्ृत् वृत्त में प्राची चिह्न है ।

वहाँ से बलनज्ञा दान देने से केन्द्र से बलनज्ञाप्रगत रेखा क्रान्तितृत् प्राची है बलनसूत्र से

ज्यावत् खादान देना चाहिए । क्योंकि क्रान्तिबृत् प्राची से सार दक्षिण और उत्तर रहता है । इस

तरह स्पर्श और मोक्ष में होता है । मध्यबलन तात्कालिकक्रान्तिबृत् प्राची से दक्षिणोत्तर

दिशा में होता है इसलिए केन्द्र से बलन सूत्र में देना चाहिए । शराघ्र में प्राह्ल वृत्त का केन्द्र

होता है इसलिए वहाँ से तात्कालिकप्राह्लाभि प्रमाण से रचित बृत्तों से स्पर्श मोक्षमध्य होने है

सिद्धान्त शेखर में □मानचक्रबाह्ृतलये तत्परियामिनिवि' इत्यादि संस्कृतोपपत्ति में लिखित

ल्लोक से श्रीपति ने आचार्यों के अनुकूल ही कहा है । सिद्धान्त शिरोमणि में □प्राह्लाखसूत्रेय

विधाय वृत्तं मानचक्रबाह्ृतं च साधितांशम्' इत्यादि से भास्कराचार्य ने श्रीपति प्रकार ही को

विचार रूप से प्रतिपादित किया है इति ॥ १४--१५ ॥

[Dvivedī's Proof — Summary.] When in the māna-cakra-bāhṛta circle there is the centre of the prāhlā-kṛtā, then the prāhla-bimbagrānta and prāhla-bimagrānta remain contiguous (saṃlagna). Therefore, in the known māna-cakra-bāhṛta circle, the direction should be fixed (śrīṣṭa). The common endpoint of the sama-maṇḍala and the māna-cakra-bāhṛta circle is the eastern mark (prācī cihna) in the māna-cakra-bāhṛta circle.

From there, giving the balana-jyā, the line from the centre [through] the balana-jyā-pragata is the krānti-tṛt-prācī. One should give the jyāvat-khādana from the balana-sūtra. Because the sāra remains south and north from the krānti-vṛt-prācī. Thus it happens in the sparśa and mokṣa. The madhya-balana [is] in the dakṣiṇôttara direction from the tātkālika-krānti-vṛt-prācī; therefore, from the centre it should be given in the balana-sūtra. In the śarāghra, the prāhla-vṛtta has its centre; therefore, from there, by the tātkālika-prāhlābhi-pramāṇa, from the constructed circles, the sparśa-mokṣa-madhya [is obtained].

In the *Siddhāntaśekhara*, from the verse "mānacakrabāhṛtalaye tatpariyāminivi" etc., written in the Sanskrit proof, Śrīpati has spoken in accordance with the Ācāryas alone. In the *Siddhāntaśiromaṇi*, from "prāhlākhasūtreya vidhāya vṛttaṃ mānacakrabāhṛtaṃ ca sādhitāṃśam" etc., Bhāskarācārya has propounded in the form of deliberation (vicāra-rūpa) the very method of Śrīpati.

[Verses 14–15]

इदानी यः परिलिखतीष्टद्यासमिल्यादि प्रश्नस्योत्तरमाह ।
पश्चात् प्रग्रहयो प्रागमोषे रविविब्बलो बाहुः ।
स्वबलनसिद्धज्ञादिशि विपरीतः शीतकरमध्यात् ॥ १६ ॥
भानुमतो बाहुगुण्याख्या दिशं कोटिरत्यथा शविनः ।
रविशास्तिवमध्यात् करणास्तिन्यकं, करणांगुकोटियुतेः ॥ १७ ॥

idānī yaḥ parilikhatīṣṭadyāsamilyādi praśnasyôttarām āha |
paścāt pragrahayo prāgamoṣe ravi-vibbalo bāhuḥ |
sva-balana-siddha-jñā-diśi viparītaḥ śītakara-madhyāt || 16 ||
bhānumato bāhu-guṇyākhāyā diśaṃ koṭir atyathā śavinaḥ |
ravi-śāsti-vamadhyāt karaṇāstinyakaṃ, karaṇāṃgu-koṭiyuteḥ || 17 ||

Now he who draws the answer to the question of iṣṭa-dyā-samīlyādi: the bāhu [is] the western and eastern pragraha at the release (moṣa) of the Sun's vibbalo [force]. Opposite in direction to the svabalana-siddha-jyā, from the centre of the Moon (śītakara).

Of the Sun (bhānumat), the direction of the bāhu multiplied (bāhu-guṇyākhāyā diśaṃ); the koṭi [is] otherwise (atyathā) of Saturn (śanaiścara/śavina). From the centre of the Sun's śāsti [force], karaṇa-astinyakaṃ; from the karaṇāṅgu-koṭi-yuti [is obtained the result].

[Ślokas 16–17]

सूर्यसिद्धान्तयन्त्रविवरण

Original pages 1121–1139 | Bilingual Edition

सूर्य सिद्धान्त में शब्दों लिखित यन्त्र विवरण है---

[Introduction.] In the Sūrya-siddhānta, the following verbal description of instruments (yantra-vivaraṇa) is given —

तुङ्गार्कसमायुक्तं गोलयन्त्रं प्रसाधयेत् ।
गोपयेतत्प्रकाशोक्तं सर्वगंयं भवेद्धि ॥

कालसंसाधनार्थं तथा यन्त्रारिणा साधयेत् ।
एकाकी योजयेद्बीजं यन्त्रं विस्मय कारिणि ॥

शङ्कुकुयष्टि धनुश्चक्रेषछायायन्त्रैर्नैकथा ।
गुप्तदेशाङ्कज्ञेयं कालज्ञानमतन्द्रितः ॥

तोययन्त्रकपालाङ्कघर्म्म्र्युदनत्रवानरैः ।
ससूत्ररेखुगमैश्च समयकालं प्रसाधयेत् ॥

परदारामनुष्याणां गुल्वतैलजलानि च ।
बीजानि पांसवस्तेषु प्रयोगास्तेऽपि दुर्लभाः ॥

ता प्रपञ्चमधिश्छद्द् न्यस्तं कुण्डे स्मलाभमिस् ।
द्विट्ठमेजलयहोरात्रं स्फुटं यन्त्रं कपालकम् ॥

नयन्त्रं तथा साधु दिवा च विमले रवौ ।
छाया संसाधनैः प्रोक्तं कालसाधनमुत्तमम् ॥

tuṅgârkasamāyuktaṃ gola-yantraṃ prasādhayet |
gopayeetat-prakāśoktaṃ sarvagamyaṃ bhaved dhi ||

One should prepare the spherical instrument (gola-yantra) endowed with the exalted Sun (tuṅgârka-samāyuktaṃ); having concealed it, [as] spoken of in the Prakāśa, it becomes indeed universally accessible (sarvagamyaṃ).

kāla-saṃsādhanārthaṃ tathā yantrāriṇā sādhayet |
ekākī yojayed bījaṃ yantraṃ vismaya-kāriṇi ||

For the purpose of time-determination (kāla-saṃsādhana), one should thus accomplish [it] by the yantrâriṇa; the solitary one should apply the seed (bījaṃ) [to] the instrument, causing wonder.

śaṅku-kuyaṣṭi dhanuś-cakreṣa-chāyā-yantrair naikathā |
gupta-deśāṅka-jñeyaṃ kāla-jñānam atandritaḥ ||

By gnomon-staff (śaṅku-kuyaṣṭi), bow-wheel (dhanu-cakra), and shadow-instruments (chāyā-yantraiḥ) in manifold ways; the knowledge of time (kāla-jñāna) is to be known from the hidden place-marker (gupta-deśāṅka) [by one who is] vigilant.

toya-yantra-kapālāṅka-gharmmy-udanatravānaraiḥ |
sa-sūtra-rekhu-gamaiś ca samaya-kālaṃ prasādhayet ||

By water-instrument (toya-yantra), bowl (kapāla), aṅka of heat (gharma), water-stream (udanatra), and monkey [devices?] (vānaraiḥ), and with cord-line-rekhuga (sa-sūtra-rekhu-gamaiḥ), one should determine the proper time (samaya-kālaṃ).

paradārā-manuṣyāṇāṃ gulvataila-jalāni ca |
bījāni pāṃsavas teṣu prayogās te 'pi durlabhāḥ ||

The wives of others (paradāra) of men, oil (taila), and water — [these are] seeds (bījāni) [in] the sand (pāṃsavaḥ) of these; their applications are also difficult to obtain (durlabhāḥ).

tā prapañcamadhiśchadd nyastaṃ kuṇḍe smalābhamis |
dviṭṭha-meja-laya-horātraṃ sphuṭaṃ yantraṃ kapālakam ||

That covering of the world (prapañcam-adhiśchadd), placed in the bowl (kuṇḍe), smalābhamis [?]; the double-positioned (dviṭṭha) meja-laya day-night (horātra) — the clear instrument [is] the kapālaka.

na yantraṃ tathā sādhu divā ca vimale ravau |
chāyā saṃsādhanaiḥ proktaṃ kāla-sādhanam uttamam ||

Not the instrument thus [alone is] good; by day, when the Sun is pure (vimale ravau), the shadow [is] declared by the saṃsādhana-methods [to be] the best time-determination (kāla-sādhana).

मानाध्याय

| *Mānâdhyāya* (Chapter on Measures)

मानानि सौरचान्द्रार्क्षसावनानि ग्रहानयनमैभिः ।
मानैः पृथक् चतुर्भिः संख्यबहारोड्ज्ञ लोकस्य ॥

mānāni saura-cāndrārkṣa-sāvanāni grahāṇayanam aibhiḥ |
mānaiḥ pṛthak caturbhiḥ saṃkhyabahāroḍjña lokasya ||

The measures (mānāni): solar (saura), lunar (cāndra), stellar (nakṣa), and civil (sāvana) — by these, the computation of planets (grahâṇayanam). By these four measures separately, the multitudinous arrangements (saṃkhyabahāroḍjña) of the world [are made].

इससे सौरमान, चान्द्रमान, नाक्षत्रमान और सावनमान, ये चार प्रकार के मान कहे गये हैं । इन्हीं चारों मानों से लोगों के सब व्यवहार होते हैं । किस किस मान से कौन कौन पदार्थ ग्रहण किये जाते हैं यह सब प्रति पादित है । बाह्म, दिव्य, पिट्र, प्राजापत्य, बाहृस्पत्य, सौर, सावन, चान्द्र, नाक्ष ये नौ मान हैं । इन मानों में से मनुष्यलोक में केवल सौर, चान्द्र, सावन और नाक्ष इन चार मानों की ही प्रधानता है । क्योंकि इन्हीं मानों से मनुष्यों के सब व्यवहार सम्पन्न होते हैं । सूर्यसिद्धान्त, सिद्धान्तशेखर, सिद्धान्तशिरोमणि आदि सब ग्रन्थों में मानों के विषय में समान रूप से कहा गया है । ब्रह्मस्फुट सिद्धान्त के इस अध्याय में भूमादर्शन के भी साधन हैं ।

[Dvivedī’s Hindi Commentary — Summary.]

From this, the solar measure (saura-māna), lunar measure (cāndra-māna), stellar measure (nākṣatra-māna), and civil measure (sāvana-māna) — these four types of measures are spoken. By these very four measures, all the affairs of people are conducted. Which objects are grasped by which measure — all this is propounded (pratipādita).

Brāhma, divya, paitṛ, prājāpatya, bāhṛspatya, saura, sāvana, cāndra, nākṣa — these are nine measures. Of these measures, in the world of humans, only the saura, cāndra, sāvana, and nākṣa have primacy, because by these measures alone all human affairs are accomplished. In the Sūrya-siddhānta, Siddhāntaśekhara, Siddhāntaśiromaṇi, and all other treatises, the subject of measures is spoken of in the same manner. In this chapter of the Brāhmasphuṭasiddhānta, the methods of viewing the earth (bhūmādarśana) are also given.

स्मृत्योर्युगलयोः समन्वयः

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स्मृतियों में और पुराणों में राहुग्रस्त ग्रहण का प्रतिपादन विद्यमान है । अतः
दोनों मतों

का समन्वय करते हुए आचार्य ने कहा है---

[Introduction.] In the Smṛtis and in the Purāṇas, the doctrine of Rāhu-grasta eclipse (rāhugrasta-grahaṇa) is present (pratipādana). Therefore, reconciling the two doctrines, the Ācārya has said —

राहुस्तच्छदयति प्रविशति यच्छुकलपच्छदयन्ते ।
छाया तमसीद्वैरेघ्रदानां कमलयोनिः ॥
चन्द्रोऽस्मुयोजः स्यो यदिनिमयभास्करस्य मासात् ।
छादयति शर्मिततापो राहुश्छदयति तत् सवितुः ॥

rāhus tac chadayati praviśati yac chuklapacchadayante |
chāyā tamasi-dvairaghra-dānāṃ kamala-yoniḥ ||
candro ’smuyojaḥ syo yadinaimaya-bhāskarasya māsāt |
chādayati śarmi-tatāpo rāhuś chadayati tat savituḥ ||

Rāhu covers that [Moon], entering [it] which the bright and dark [portions] cover. The shadow (chāyā) [is] the tamasi [force] of the enemy of the lotus-born [i.e., Rāhu, enemy of Viṣṇu?].

The Moon (candra), the asmu-yojaḥ, the syo [?], from the māsa of the sun-of-the-yadinaimaya [?] — [this] Rāhu covers the scorching-heat (śarmi-tatāpaḥ) [of] that Sun (savituḥ).

सिद्धान्तशिरोमणि के गोलाध्याय में भी अप्रोक्तित भास्करोक्ति---

In the Golâdhyāya of the Siddhāntaśiromaṇi also, the unspoken (aproktita) statement of Bhāskara [is as follows] —

द्विदेशकालवरयाणि मेदानां छादको राहुरिति स वर्णित ।
यन्मानिनः केवलगोलविधास्तत्सहिता वेद्यरणाबाह्याम् ॥
राहुः क्षुभामण्डलगः शशाङ्कं शशाङ्कश्छादयतीव बिम्बम् ।
तमोमयः शम्युवरप्रदानात् सर्वगमानामविरुद्धमेतत् ॥

dvi-deśa-kāla-varayāṇi medānāṃ chādako rāhur iti sa varṇita |

yan māninaḥ kevala-golavidhās tat-sahitā vedy araṇābāhyām ||

Rāhu, the eclipser (chādakaḥ) of the vara of the two--place-time (dvi-deśa-kāla-varayāṇi medānāṃ), thus he is described. Those who know only the spherical method (kevala-golavidhāḥ), together with that [knowledge], should know the araṇā-bāhyām [?].

rāhuḥ kṣubhā-maṇḍalagaḥ śaśāṅkaṃ śaśāṅkaś chādayatīva bimbam |

tamo-mayaḥ śamyuva-pradānāt sarvagamānām aviruddham etat ||

Rāhu, situated in the agitated orb (kṣubhā-maṇḍalagaḥ), [eclipses] the Moon (śaśāṅkaṃ); the Moon appears to cover the disc (bimbam). [Rāhu is] composed of darkness (tamo-mayaḥ); from the bestowal of śamyuva [?], this is not contradictory (aviruddham) for all who go [i.e., all observers].

से समन्वय किया गया है । सिद्धान्तशेखर में राहुग्रस्त ग्रहण के खण्डनार्थ □राहुनिराकरणाध्याय' नाम का एक अध्याय रखा गया है । इसमें श्रीपति ने भी निम्नलिखित श्लोक ने समन्वय किया है---

विष्णुलूनविशिरसः किल पञ्चदेत्तवान् वरमिमं परमेष्ठी ।
हेमदानविधिना तवतुपतिस्तमरतिमहस्तदुपरागे ॥
भूमेच्छाया प्रविष्टः स्थगयति शशिनं शुक्लपक्षावसाने ।
राहुर्भुप्रसादात् समविगतवस्तुतामो व्यसुतुल्यः ॥
उर्व्यां स्थं भानुबिम्बं सलिलमयतनोर्यथैवोक्तं बिम्बम् ।
संसूयैवं च मासव्युपरतिसमये स्वस्य साहित्यहेतोः ॥

[Dvivedī's Commentary.] This reconciliation (samanvaya) has been made. In the Siddhāntaśekhara, for the refutation (khaṇḍanârtha) of the Rāhu-grasta eclipse, a chapter named "Rāhu-nirākaraṇâdhyāya" has been placed. In this, Śrīpati also has made the reconciliation with the following ślokas —

viṣṇu-lūna-viśirasaḥ kila pañcad ettavān varam imaṃ parameṣṭhī |
hema-dāna-vidhinā tavatupatis tam-arati-mahas tad uparāge ||

The Supreme Lord (parameṣṭhī) indeed gave this excellent boon (varam imaṃ) to the headless Viṣṇu-lūna [i.e., the decapitated Rāhu], [that] by the method of donating gold (hema-dāna-vidhinā), at that eclipse (tad uparāge), [there is] the great darkness (tam-arati-mahaḥ) [for him?].

bhūmecchāyā praviṣṭaḥ sthagayati śaśinaṃ śuklapakṣāvasāne |
rāhur bhu-prasādāt samavigatavastu-tāmo vyasutulyaḥ ||

Entering the earth's shadow (bhūmecchāyā praviṣṭaḥ), [Rāhu] obscures the Moon (sthagayati śaśinaṃ) at the end of the bright fortnight (śuklapakṣâvasāne). Rāhu, by the grace of the earth (bhu-prasādāt), [being] equal to the vyasu [?], the darkness [is] the real state of things (samavigatavastu-tāmaḥ).

urvyāṃ sthaṃ bhānu-bimbaṃ salila-maya-tanor yathaivôktaṃ bimbam |
saṃsūyaivaṃ ca māsavy uparati-samaye svasya sāhitya-hetoḥ ||

Standing on the earth (urvyāṃ sthaṃ), the solar disc (bhānu-bimbam), [this] disc as described [earlier], of the water-bodied one [i.e., the Moon] — thus indeed at the time of lunar cessation (māsavy-uparati-samaye), for the sake of its conjunction (svasya sāhitya-hetoḥ).

गोलबन्धाधिकार में महत् द्वृतों (पूर्वापरदृत, याम्योत्तरदृत, स्थितिजदृत आदि) की रचना

तथा लघुदृतों (मेषादिक द्वादश राशियों के बहिरोराजदृत) की रचना करके परमलम्बन-नति

का स्वरूप प्रतिपादन कर आचार्य ने द्वृतों का ज्ञानयन किया

है । ब्रह्मस्फुटसिद्धान्त में आचार्य ने जैसे गोलबन्ध कहा है

वैसे ही सिद्धान्त शेखर में श्रीपति और सिद्धान्त शिरोमणि के

गोलाध्याय में भास्कराचार्य ने कहा है । ग्रहगोल और नक्षत्रगोल में पाँच स्थिरदृत

(पूर्वापरदृत, स्थितिजदृत, याम्योत्तरदृत, उन्मण्डल, विषुवदृत) कहे हैं । ये सब कला-

मण्डल के बराबर हैं । तथा ग्रहों के चलदृत मन्दनीचोचदृत = ७, मन्दादि ग्रहों के शीघ्र-

नीचोचदृत = ५ । मन्दप्रतिबृत = ७, शीघ्रप्रतिबृत = ७ । सात ग्रहों के हमण्डल हृद्क्षेप

[Dvivedī's Commentary on Spherical Construction.]

In the Golabandhâdhikāra, having constructed the great circles (mahad-vṛttāḥ: pūrvâpara-vṛtta, yāmyôttara-vṛtta, sthitija-vṛtta, etc.) and the small circles (laghu-vṛttāḥ: the outer royal circles of the twelve signs from Meṣa, etc.), the Ācārya has demonstrated the nature (svarūpa) of the maximum equation (paramalambana-nati) and computed the knowledge of the circles (vṛttôpāyanam).

As the Ācārya has spoken of the golabandha in the Brāhmasphuṭasiddhānta, so also has Śrīpati in the Siddhāntaśekhara, and Bhāskarācārya in the Golâdhyāya of the Siddhāntaśiromaṇi. In the graha-gola and nakṣatra-gola, five fixed circles (sthira-vṛtta) are spoken: pūrvâpara-vṛtta, sthitija-vṛtta, yāmyôttara-vṛtta, unmaṇḍala, and viṣuvad-vṛtta. All these are equal to the kalā-maṇḍala.

And the moving circles (calavṛtta) of the planets: mandanīcôcavṛtta = 7, the śīghranīcôcavṛtta of the manda-etc. planets = 5. The manda-prativṛtta = 7, the śīghra-prativṛtta = 7. The hamaṇḍala and hṛd-kṣepa of the seven planets [are also given].

Colophon

End of Chunk 2 (Pages 1817–1956)

Comprising the Tri-praśnottara-adhyāya and Grahana-uttara-adhyāya
with editorial notes on the Sūrya-siddhānta and Smṛti reconciliation

"Brāhmasphuṭasiddhānta" is one of the most important works
of ancient Indian mathematics and astronomy, composed by
Brahmagupta in 628 CE at Bhillamāla (modern Bhinmal, Rajasthan).

This bilingual edition presents the original Sanskrit in Devanagari
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ब्राह्मस्फुटसिद्धान्त

भाग तृतीय — खण्ड ३

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२२ — १६०

विषयानुक्रमणिका

संस्कृत

प्रस्तावना यह खण्ड ब्रह्मगुप्ताचार्यकृत ब्राह्मस्फुटसिद्धान्तस्य अन्तिमभागः। अत्र ग्रहगणितस्य सम्पूर्णतः, कुट्टकाध्यायस्य सम्पूर्णं, छायागणना, तथा गोलाध्यायस्य प्रारम्भः वर्तते। कुट्टकाध्यायः विशेषेण प्रसिद्धः अस्ति, यतोहि अत्र रेखीय-डायोफाण्टाइन-समीकरणानि $ax + by = c$ रूपस्य समाधानं दत्तम्। अपि च शून्येन सह गणितस्य नियमाः, ऋण-धन-संकलन-विधयः च उक्ताः।

$ax + by = cNx^2 + 1 = y^2$

१७. सृङ्गोत्स्त्युतराध्यायः

प्रश्नकथनम्द्विजार्कसनवरसेन्दूर्जिनतिथिविषयाग्रहार्थचापानाम् । अर्धज्याखण्डानि ज्यामुकैर्नैक्यं
स भोग्यफलम् ॥१५॥

गतभोग्यखण्डकान्तर दलविकलविधा च्छुतैर्वर्भिराप्तैः । तच्छृतिदलं युतेन
भोग्यादुनार्धिकं भोग्यम् ॥१६॥

$k_1, k_2, \ldots, k_n \sum k_i$

परिलेखकथनम्रविवसतिगतः पञ्चयुग्मं वपांशि पितामहोद्दिष्टानि । अर्धमासार्द्धयातद्विमर्मसैरमर्मस्नपष्टयाऽऽत्म
॥३॥

विशेषकथनम्आद्धतकालयोर्मध्ये कालक्षे पोष्टदाश्रयः । प्रज्वलज्ज्वलनाकारः सर्वकर्मसु
गर्हितः ॥

एकायनगां यावत् क्षेत्रमण्डलान्तरम् । सुम्बवस्तावदेवास्य सर्वकर्म विनाशकृत्
॥

१८. कुट्टकाध्यायः

कुट्टकार्थप्रयोजनम्[□□□□□□□□□□□□□□.] कुट्टकशब्देन अत्र रेखीय-डायोफाण्टाइन-समीकरणस्य समाधानविधिः उच्यते। यथा $ax + by = c$ रूपस्य समीकरणस्य मूलानयनम्। अत्र भाज्यं a, भाजकः b, शेषं c, गुणकः x, तथा कोटिः y। ब्रह्मगुप्तः प्रथमः आचार्यः यः शून्येन सह गुणनभागहारौ नियमेन सह ददाति।

$ax + by = cabcxy$

द्वयं न शकेन्द्रकालं पञ्चभिरुद्धूतं शेषवर्षयुगम् । दिगुणामर्धसितां कुर्याद् भुग्रां तद् द्वयं दयात् ॥१४॥
संक्षिप्तं द्वि त्रि गतौ तिथिभूमान्नवाहतैष्यकैः । दिग्रसभागाः सप्तभिन्नं शशिभं धनिग्रहात्म ॥१५॥

कुट्टकविधिःभाज्यं भाजकशेषेण यावदेकं करणी भवेत् । प्रथमं रूपयुक्तं ततो द्वितीयं करणी भवेत् ॥४०॥

abc

[□□□□□□ □□□□□□□□□□□□ □□□□□□□□:] सू. भाषा.—रूपकुट्टः कि विधिः इष्टया इष्टकर्णपद्धतिः। इष्टा यैका तया वैश्यो- द्वितीयैः रूपवर्गैस्तेन वैधानमनेकासां यो रूपवर्गैस्तेनोनाया यथपदं तेन रूपार्धा- पृथक् युता नितादि तदर्थं च कार्यं। तत्र प्रथमार्धिगोर्थं रूपार्धा कल्प्यानि। ततो स्यदन्तरार्थं द्वितीयं मूलस्यैका करणी भवति। एवमसंकुलेमूलानयनं कार्यम्॥
अत्रोदाहरण म. म. सूर्याकारदिवेगुकम्।
चतुर्दर्शनद्वितीयाः पङ्क्तयो विश्वमजिताः। तं राशिं शीघमाचक्ष्व यदि जानासि कुट्टकम्॥ अत्र ३४ हरस्य शेषम् $=$ । तथा १३ हरस्य शेषम् $=$ । अतोऽधिकशेषहरः $=$ । अल्पशेषहरः $=$ । अनेनाधिकशेषहरे भक्तं शेषम् $=$, ततः परस्परभजनेन जाता वल्ली उपान्तिमेन स्वोच्चं हृतोनयेन युते तदन्त्यं

$$ax + by = c(a, b) \mid c$$

$$(a, b)q_1, q_2, \ldots, q_n$$

$$x_0 = 1x_1 = q_nx_{k+1} = q_{n-k} \cdot x_k + x_{k-1}$$
$$xy = (c - ax)/b$$

शून्यगणितम्धनर्णशून्यानां संकलनं व्यवकलनं च । गुणने करणसूत्रकथनं भागहारे च ॥

[□□□□□□□□□ □□□ □□□□□□□□□□□□:] ऋणे ऋणं योजयेत् धनं धने। शून्ये शून्यं योजयेत्। धनात् ऋणं व्योजयेत्, ऋणात् धनं व्योजयेत्। शून्येन युतं राशिः स्वेनैव रूपेण स्थितः।
[□□□□□□□□□□□□□□□□:] शून्येण गुणितं राशिः शून्यं भवति। शून्येण भक्तं राशिः शून्यं भवति। अव्यक्तसंकलितं यावद् इष्टतयोः करणसूत्रकथनम्। अव्यक्तगुणने सूत्रकथनम्। अव्यक्तमानानयनार्थं कथनम्।

$a/0 = 0$

$$a + 0 = aa - 0 = a0 + 0 = 00 - 0 = 0$$
$$a \times 0 = 00 \times 0 = 0$$
$$a/0 = 0$$
$$0/a = 0a \neq 0$$

$$(-a) \times (-b) = +ab(-a) \times (+b) = -ab$$

वर्गप्रकृतिःसमद्विबाहुत्रिभुजे भुजयोर्बर्गेण । करणीयोगान्तरे गुणनकथनम् ॥

वर्गसमीकरणाकथनम् । वर्गसमीकरणोद्धरणमानयनम् ॥

[□□□□□□□□□ □□□□□□□□□□□ □□ □□□□□□□□□□□□□□:] अत्र □वर्गप्रकृतिः' नाम समीकरणं $Nx^2 + 1 = y^2$ रूपस्य। अत्र N = प्रकृतिः (गुणकः), x = ह्रस्वमूलम्, y = दीर्घमूलम्। ब्रह्मगुप्तः अत्र समाधानाय □भावना' नाम विधिं ददाति --- यदा द्वयोः ह्रस्व-दीर्घमूलयोः युग्मेभ्यः नूतनयुग्मं प्राप्तुं शक्यते।

$Nx^2 + 1 = y^2 Nxy$
$(x_1, y_1)(x_2, y_2)N$

$$x' = x_1y_2 + x_2y_1, y' = Nx_1x_2 + y_1y_2$$

$(x_1, y_1)(x_2, y_2)Nx^2 + k_1 = y^2 Nx^2 + k_2 = y^2$

$$x_3 = x_1y_2 + x_2y_1, y_3 = Nx_1x_2 + y_1y_2$$

$Nx_3^2 + k_1k_2 = y_3^2 k_1 = k_2 = 1N = 92x = 120y = 115192 \cdot 120^2 + 1 = 1151^2$

विलोमगतिकथनम्राश्यर्धं स्तच्छेषैश्चैवं युक्तार्धमासादिनहीनैः । तच्छेषैश्च युगगतं यः कथयति
कुट्टकज्ञः सः ॥१४॥

१९. शङ्कुच्छायादिज्ञानाध्यायः

शङ्कु

प्रथमप्रश्नकथनम्हरेण भक्तः शुध्यति स राशिः कः। प्रथमं गुणाहरयोर्महतामपवर्तेनमानीयते।

> **[□□□□□□□ □□□□□□□□:]** परस्परं भक्तयोर्गुणाहरयोर्न ते यच्छेषं तेन भक्तो तौ (गुणाहरौ भवतः) द्वौ गुणा- हरौ भवतः। ततो हृदयोर्गुणाहरयोः परस्परं भक्तयोर्लम्बान्यथोच्छ स्थापयानि, पूर्वं प्रदर्शितकुट्टकनियमेन। इदं कर्म तावत्पर्यन्तं कर्तव्यं यावदन्ते रूप शेषं तिष्ठेत्। तच्छेषं तथा केनापि गुणाकेन गुणितं रूपेण हीनं तच्छेषसंबन्धिहरेण भक्तं शुध्यति। सत्यैवं गुणाकः पूर्वोच्छोच्छः स्थापितकलानामधः स्थाप्यः, तदन्तात्फलं स्थाप्यम्।

शङ्कुच्छायानयनम्छायातो गृहादीनां च्छायानयनम् । इष्टगुहोच्चयको यदिति प्रश्नस्योत्तरसम्पादनम् ॥

$$(altitude) = aku/\sqrt{aku^2 + chy^2}$$

उच्छ्रितिकथनम्उच्छ्रितिकथनं समीपस्थस्योच्चयस्य ज्ञानार्थं दूरस्थस्य च । प्रकाशस्य गतिकथनं द्वयोरन्तरालस्य मानं च ॥

> **[□□□□□□□ □□□□ □□□□□□□ --- □□□□□□□□:]** गृहपुरुषान्तरसलिले यो हटचैत्यादि प्रश्नोत्तरकथनम्। बीजय गुहायां सलिले प्रसारे ति प्रश्नोत्तरकथनम्।
>
> एकस्य गृहस्य छाया द्वितीयमात्रान्तरविधानेन त्रिप्रश्नोत्तरकथनम्। छायातो गृहादीनां च्छायानयनम्।

hd

$$h = \frac{s_2 \cdot g}{s_2 - s_1} \cdot \frac{d_2 - d_1}{g}, d = \frac{h \cdot s_1}{g}$$

gs_1, s_2d_1, d_2

२०. छद्दिच्छायाध्यायः

छद्दिराशीनामहोरात्रबलानाम् । राश्युदयानामसमानद्विकथनम् ॥
चरागयोः संस्थानकथनम् । शङ्कृह्यग्रयोः संस्थानकथनम् ॥
हग्गोलस्य हर्याहरयतं लम्बनावनतिरुपतो च कारणदर्शनम् । परलम्बनावनतिकथनम्
॥

[□□□□□□ □□□□□□□□□□□□ □□□□□□□□:] प्रकारान्तरेण तयोः संस्थाने शङ्कुलस्य च कथनम्। हक्कर्मकथनम्। ग्रहसौगोल्योः स्थिरकुट्टकथनम्। ग्रहाणां चलकुट्टकथनम्। अध्यायोपसंहारः ॥
अत्र छायायाः गणना दीपशिखोच्चयच्छुक्लान्तरभूमिज्ञानार्थं दर्शिता। प्रकाश-परावर्तन-विषयेऽपि किञ्चिदुक्तम्।

$$cargra = \left(\frac{(\phi)\cdot(\delta)}{(\alpha)}\right)$$

$\phi\delta\alpha$

२१. गोलाध्यायः

आरम्भप्रयोजनकथनम्गोलप्रशंसाकथनम् । स्वगोलयथाने कारणम् ॥

गतिगोलयोः प्रशंसाकथनम् । यन्त्रारम्भप्रयोजनकथनम् ॥

[□□□□□□□□□ □□ □□□□□□□□□□□:] तन्त्राणां प्रतिपादनम् । सलिलादीनां प्रयोजनकथनम् । धनुषां च कथनम् । सूर्यादिमुखे यन्त्रधारणम् । इष्टघटिकायाः धनुषरचनरूपकथनम् । परीक्षाघट्यानयनम् । यन्त्रेण नतोन्नतकालज्ञानम् । धनुष्यन्त्रे विशेषकथनम् ।

भूगोलसंस्थानकथनम्भूगोलसंस्थानकथनम् ॥ दैवासुरसंस्थानवर्णनम् ॥

[□□□ □□□□□□□□□□ □□□□□□□:] देवदैत्ययोर्मेषचक्रमरगत्यवस्थापनम् ॥ चक्रभ्रमणव्यवस्थाकथनम् ॥ दैनानीनां रविभ्रमणार्थदिवसानाम् ॥ देवदैत्ययोः राशिसंस्थानम् ॥ देवदैत्ययोः पितुमानस्यैव दिनप्रमाणानिरूपणम् ॥

निरक्षदेशान्तरयोजनकथनम्भूगोल लग्नावलयोः संस्थानम् ॥ निरक्षदेशान्तरयोजनकथनम् ॥

क्षुद्रद्योः कथानिरूपणम् ॥

[□□ □□□ □□□□□□□□□□□:] निरक्षदेशः उज्जयिनी-क्षेत्रम्, यत्र रेखा (याम्योत्तर-वृत्तम्) पूर्णतः समपर्यायं ग्रहैः भ्रमति। अन्यत् क्षेत्रं प्रति देशान्तरं योजनैः मीयते --- प्रत्येकं योजन-चतुष्टयस्य एका लघु-घटिका भवति। तदनुसारं देशान्तर-घटिकाभिः ग्रहाणाम् उदय-कालः परिवर्तते।

$$dentara = \frac{yojanas from Ujjain}{4}$$

×

بسم الله الرحمن الرحيم

Miftāh al-Ḥisāb

The Key to Arithmetic

Bilingual English–Arabic Edition

[illegible]

مفتاح الحساب

Jamshīd Ghiyāth al-Dīn

al-Kāshī (al-Kāshānī)

d. 832 AH / 1429 CE

A comprehensive encyclopedic treatise on arithmetic, geometry, and algebra

Completed in Samarkand, 2 March 1427 CE (829 AH)

Contents / :[illegible]

تحقيق وترجمة وشرح

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Preface to This Bilingual Edition

كلمة في هذه النسخة

Preface to This Edition

هذه النسخة الثنائية اللغة تقدم النص الكامل ومحتوى تمثيلي لمفتاح الحساب، الإنجاز الأعظم للحساب الإسلامي، الذي ألفه العالم الفارسي الشهير في الرياضيات والفلك جمشيد غياث الدين الكاشي سنة 1427 م.

[illegible] 1427 [illegible].

الكاشي [illegible]حوالي 1380[illegible]1429[illegible] كان كبير علماء الفلك لدى ألغ بك في سمرقند. وكتابه مفتاح الحساب هو موسوعة شاملة تغطي ثلاثة مواضيع رئيسية: الحساب والهندسة والجبر.

[illegible] 1380[illegible]1429[illegible]

The work includes:

1. Al-Kāshī's famous calculation of π to 16 decimal places
2. His systematic treatment of decimal fractions (the first in history)
3. Methods for root extraction that predate similar European methods by centuries

As Berggren noted, *Miftāh* was "the crowning achievement of Islamic arithmetic, and truly a gift fit for a king."

يحتوي هذا الكتاب على حساب π16[illegible].

[illegible]

Author’s Preface

تمهيد المؤلف

Author’s Preface

بسم الله الرحمن الرحيم

الحمد لله الذي توحد بإبداع الآحاد، وتفرد بتأليف صنوف الأعداد، وصلاة على خير خلقه محمد أشفع الشافعين يوم التناد، وآله وأولاده الهادين إلى سبيل النجاة والرشاد.

[illegible]

[illegible]

جمشيد بن مسعود بن محمود الطبيب الكاشي الملقب بغياث الدين □□ أحسن الله أحواله □□ يقول: لما مارست الأعمال الحسابية والقوانين الهندسية حتى بلغت إلى حقائقها، وبالغت في دقائقها، وكشفت غوامضها ومعضلاتها وحللت مشكلاتها...

[illegible]

[illegible]...

...وأعدت حساب جميع جداول الزيج الإلخاني بأدق دقة، وألفت الزيج المسمى بالخاقاني مكملاً للزيج الإلخاني، وجمعت فيه كل ما استنبطته من أساليب الفلكيين التي لا تظهر في زيج آخر مع البراهين الهندسية...

...[illegible]...

...وألفت أيضاً رسالة المحيطية في نسبة القطر إلى المحيط، ورسالة الوتر والجيب في استخراج الوتر والجيب لثلث قوس معطى...

...[illegible]...

...وجدت في أثناء هذه الأعمال قوانين كثيرة تؤدي بها عمليات أصول الحساب على أيسر وجه وأهنئ طريقة وأقل جهد وأنفع نتيجة وأوضح ترتيب. فعزمت على تدوينها وتوضيحها، لتكون تذكرة للأحباب وإرشاداً لأهل الفهم.

...[illegible].

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المقالة الأولى في حساب الأعداد الصحيحة. وهي سبعة فصول.
المقالة الثانية في حساب الكسور المتصلة والمنفصلة والكسورية.
المقالة الثالثة في حساب الدرج والدقائق والثواني ونحوها.
المقالة الرابعة في حساب المساحات والأحجام.
المقالة الخامسة في الجبر والمقابلة.

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The work comprises five treatises:

No.	Treatise	Arabic Title
I	**On the Arithmetic of Integers** — 7 chapters	حساب الأعداد الصحيحة
II	**On the Arithmetic of Fractions** — 10 chapters	حساب الكسور
III	**On the Arithmetic of Sexagesimal Numbers** — 10 chapters	حساب المثالث والدرج
IV	**On Measurements and Geometry** — 9 chapters	حساب المساحات والأحجام
V	**On Algebra** — multiple chapters	الجبر والمقابلة

وضعت لأغلب الأعمال نموذجاً على هيئة جدول لتكون أسهل على أرباب الذكاء.

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1 Treatise I: On the Arithmetic of Integers

المقالة الأولى في حساب الأعداد الصحيحة

Treatise I: On the Arithmetic of Integers

هذه المقالة، وهي أساس الكتاب كله، تتكون من سبعة فصول تغطي جميع العمليات على الأعداد الصحيحة.

[illegible]

Ch.1: On the naming of numbers and place values — The Hindu–Arabic numeral system and the principle of place value.

Ch.2: On addition — Methods for summing integers, including column addition.

Ch.3: On subtraction — Techniques for finding differences.

Ch.4: On multiplication — Including the *network method* (الشبكة □ [illegible]

[illegible].5: [illegible]

[illegible].6: [illegible]

[illegible].7: [illegible]

1.1 Chapter 1: On the Naming of Numbers

فصل في التعريف بالأعداد وأسمائها ومنازلها

Chapter 1: On the Naming of Numbers

يبدأ الكاشي بشرح نظام المنازل العشري، منسوباً إلى الهنود، ويصف أسماء المنازل.

[illegible]:

Power	Arabic Name	English
10^0	واحد	Units
10^1	عشرة	Tens
10^2	مئة	Hundreds
10^3	ألف	Thousands

Power	Arabic Name	English
10^4	عشرة آلاف	Ten thousands
10^5	مئة ألف	Hundred thousands
10^6	ألف ألف □ مليون	Millions
10^7	عشرة ملايين	Ten millions
10^8	مئة مليون	Hundred millions
10^9	ألف مليون □ مليار	Billions (thousand millions)

Al-Kāshī extends this system far beyond what was common in his time, enabling the large-scale calculations that he would later use in his astronomical computations.

1.2 Chapter 4: On Multiplication

فصل في الضرب وطرقه

Chapter 4: On Multiplication

يقدم الكاشي عدة طرق للضرب. أشهرها طريقة الشبكة، التي يصفها بالتفصيل. ويقدم أيضاً طريقة الأعمدة وطرق الحساب الذهني.

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □طريقة الشبكة□ [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □طريقة الأعمدة□ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

Lattice Multiplication Method (طريقة الشبكة:□

1. Draw a grid of squares with as many rows and columns as there are digits in the multiplicand and multiplier.
2. Write the multiplicand above the grid and the multiplier to the right.
3. Divide each square diagonally from top-left to bottom-right.
4. Multiply each digit of the multiplicand by each digit of the multiplier, writing the tens above the diagonal and the units below.
5. Sum along each diagonal from bottom-right to top-left, carrying as needed.
6. Read the final product along the left and bottom edges.

Example: To multiply 524×385:

	5	**2**	**4**
3	1/5	0/6	1/2
8	4/0	1/6	3/2
5	2/5	1/0	2/0

Result: $\boxed{524 \times 385 = 201{,}740}$

1.3 Chapter 6: On the Extraction of Roots

فصل في استخراج الجذور

طريقة الكاشي في استخراج الجذور من أشهر أجزاء المفتاح. يعطي خوارزمية عامة لإيجاد الجذر النوني لأي عدد، وهي طريقة مكافئة لما يُعرف بطريقة رفيني□□هورنر.

Chapter 6: On the Extraction of Roots

[illegible] n [illegible]

General Method for Extracting the nth Root (طريقة استخراج الجذر النوني:□
Given a number N and an integer n, to find $\sqrt[n]{N}$:

1. Separate N into groups of n digits from the decimal point.
2. Find the largest integer a such that a^n does not exceed the first group.
3. Subtract a^n from the first group and bring down the next group.
4. Using the binomial expansion of $(a+b)^n$, find the next digit b by dividing the remainder by $n \cdot a^{n-1}$.
5. Iterate until the desired precision is reached.

For the **square root** $(n = 2)$, this reduces to:

Square Root Extraction (استخراج الجذر التربيعي:□

1. Separate the number into pairs of digits from the decimal point.
2. Find the largest square not exceeding the leftmost pair; this gives the first digit of the root.
3. Subtract its square and bring down the next pair.
4. Double the current root, and find a digit such that $(20 \times \text{current} + d) \times d$ does not exceed the remainder.
5. Repeat until all pairs are processed.

Example: $\sqrt{144} = 12$

$$\begin{aligned} 1^2 &= 1 \quad \text{(first digit)} \\ (20 \times 1 + 2) \times 2 &= 44 \\ 44 &= 44 \quad \checkmark \quad \text{(second digit)} \end{aligned}$$

This method for root extraction was transmitted to Europe and appeared in the works of later mathematicians. Al-Kāshī's presentation is remarkably systematic and general, applying to roots of any order.

2 Treatise II: On the Arithmetic of Fractions

المقالة الثانية في حساب الكسور

Treatise II: On the Arithmetic of Fractions

هذه المقالة هي الأهم تاريخياً في المفتاح، فهي تحتوي على أول معالجة منهجية للكسور العشرية في تاريخ الرياضيات.

2.1 The Ten Chapters of Treatise II

Ch.1: On the naming of fractions — Definitions and terminology for common fractions.

Ch.2: On reduction of fractions — Converting to common denominators.

Ch.3: On addition and subtraction of fractions.

Ch.4: On multiplication of fractions.

Ch.5: On division of fractions.

Ch.6: On conversion between fractions — Including sexagesimal to decimal and vice versa.

Ch.7: On the extraction of roots of fractions.

Ch.8: On the *kasr al-a'dad* (fractions of numbers) — A special class of fractions.

Ch.9: On decimal fractions (الكسر العشري .

.10: .

2.2 Chapter 9: On Decimal Fractions

الباب التاسع في الكسر العشري واستعماله

Chapter 9: On Decimal Fractions

قدم الكاشي الكسور العشرية بوضوح وعموم لم يسبق له مثيل. واستخدم خطاً رأسياً لفصل الجزء الصحيح عن الجزء الكسري.

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible], [illegible] [illegible] [illegible]:

$$\boxed{3\,|\,14159265\ldots}$$

to represent 3.14159265 . . ., where the digits after the vertical bar are understood as tenths, hundredths, thousandths, etc.

واعلم أن العامة يسمون المنزلة الأولى بعد الواحد منزلة أجزاء العشرة، والثانية منزلة أجزاء المئة، والثالثة منزلة أجزاء الألف، وهكذا دائماً مضروباً في عشرة. وقد رأينا أن نسمي المنزلة الأولى بعد العلامة منزلة العشر الأول، والثانية منزلة العشر الثاني، وهكذا.

[illegible] [illegible]:

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □منزلة أجزاء العشرة,□ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □منزلة أجزاء المئة,□ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □منزلة أجزاء الألف,□ [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible]. [illegible] □منزلة العشر الأول,□ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] □منزلة العشر الثاني,□ [illegible] [illegible] [illegible].

2.2.1 Al-Kāshī's Notation System

تدوين الكاشي للكسور العشرية كان مدهشاً بحداثته. يكتب الجزء الصحيح، ثم فاصلة، ثم المنازل الكسرية.

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible], [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible].

Al-Kāshī's notation	Modern notation	Value
١٣□٧٥٣٩	13.7539	Thirteen and seven thousand five hundred thirty-nine ten-thousandths
٠□١٤١٤٢١٣٥	0.14142135	Approximation of $\frac{1}{\sqrt{2}}$
٣□١٤١٥٩٢٦٥٣٥	3.1415926535	Approximation of π

2.2.2 Decimal Operations

يستعرض الكاشي جميع العمليات الأربع الحسابية مع الكسور العشرية.
ويعطي للضرب القاعدة التالية:

[illegible]

Multiplication of Decimal Fractions (ضرب الكسور العشرية:□

1. Multiply the two numbers as if they were integers, ignoring the decimal point.
2. In the product, count from the right a number of places equal to the sum of the decimal places in the multiplicand and multiplier.
3. Place the separator at that position.

Example: $0.25 \times 0.4 = 0.10$

- Multiply as integers: $25 \times 4 = 100$
- Total decimal places: $2 + 1 = 3$
- Result: $0.100 = 0.1$

2.2.3 The Significance of Decimal Fractions

استخدم الكاشي الكسور العشرية على نطاق واسع في حساباته الفلكية.
وكان يتحول بين النظامين الستيني والعشري بسهولة.

[illegible]

Sexagesimal to Decimal Conversion (التحويل من الستيني إلى العشري:□

To convert a sexagesimal number $N = a + \frac{b}{60} + \frac{c}{60^2} + \cdots$ to decimal:

1. Keep the integer part a as is.
2. Convert each sexagesimal fraction: divide by 60, 60^2, etc.
3. Sum the results in decimal form.

Example: Convert $3; 8, 29, 44$ (sexagesimal) to decimal:

$$3 + \frac{8}{60} + \frac{29}{60^2} + \frac{44}{60^3} = 3.141592\ldots$$

3 Treatise III: On the Arithmetic of Sexagesimal Numbers

المقالة الثالثة في حساب الدرج والدقائق والثواني

هذه المقالة مكرسة للنظام الستيني □الأساس 60□، وهو النظام القياسي للحسابات الفلكية في العالم الإسلامي.

اعتمده الفلكاء لكثرة الأقسام التي تقبلها، إذ الستون تقبل القسمة على اثنين وثلاثة وأربعة وخمسة وستة وعشرة واثني عشر وخمسة عشر وعشرين وثلاثين. ولحاجتهم إلى استعمال الكسور في حساباتهم، قسموا الدائرة إلى ثلاثمئة وستين درجة، كل درجة إلى ستين دقيقة، كل دقيقة إلى ستين ثانية، وهكذا لا نهاية له.

Treatise III: On the Arithmetic of Sexagesimal Numbers

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3.1 The Chapters of Treatise III

Ch.1: **On the naming of sexagesimal places** – Degrees (درجة, □ □□□□□□□ □دقيقة, □ □□□□□□□ □ثانية, □ □□□□□□ □ثالثة, □ □□□.

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3.2 Sexagesimal Notation

Al-Kāshī used the following notation for sexagesimal numbers. In modern notation, we write $a; b, c, d$ to represent:

$$a + \frac{b}{60} + \frac{c}{60^2} + \frac{d}{60^3}$$

Arabic Term	Latin Equivalent	Symbol	Value
درجة	Degree	° or nothing	$60^0 = 1$
دقيقة	Minute	′	60^{-1}
ثانية	Second	″	60^{-2}
ثالثة	Third	‴	60^{-3}
رابعة	Fourth		60^{-4}
خامسة	Fifth		60^{-5}

3.3 Chapter 5: On the Extraction of Roots of Sexagesimal Numbers

يحتوي هذا الفصل على الطريقة العامة للكاشي في استخراج الجذور النونية في النظام الستيني. وأظهرت دراسة دخيل □1960□ أن طريقة الكاشي هي في جوهرها نفسها طريقة رفيني□□هورنر.

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Sexagesimal Root Extraction (استخراج الجذر في الستيني:□

To find $\sqrt[n]{N}$ in sexagesimal:

1. Express N in sexagesimal form: $N = N_0; N_1, N_2, N_3, \ldots$
2. Group the sexagesimal places appropriately for the root order.
3. Apply the binomial method iteratively:

$$\sqrt[n]{a^n + r} \approx a + \frac{r}{n \cdot a^{n-1}}$$

4. At each step, refine the approximation using the error term.
5. Continue until the desired precision is achieved.

Example — Square Root in Sexagesimal:

Find $\sqrt{2}$ in sexagesimal. Al-Kāshī gives:

$$\sqrt{2} \approx 1; 24, 51, 10, 7, 46, 6, 4, 40, \ldots$$

which corresponds to:

$$1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3} + \cdots \approx 1.41421356\ldots$$

This approximation of $\sqrt{2}$ *is accurate to the limits of the calculation. Al-Kāshī's method of root extraction is completely general and works for any order of root in any number base.*

4 Treatise IV: On Measurements and Geometry

المقالة الرابعة في حساب المساحات والأحجام

Treatise IV: On Measurements and Geometry

هذه أطول وأتنوع مقالة في المفتاح، تغطي الهندسة المستوية والصلبية، مع تطبيقات على المساحة والعمارة والهندسة.

[illegible], [illegible], [illegible], [illegible], [illegible]. [illegible].

4.1 Original Table of Contents of Treatise IV

فصل في المساحة وألفاظها.

- الباب الأول: في مساحة المثلث وما يتعلق به.
- الباب الثاني: في مساحة الرباعي وما يتعلق به.
- الباب الثالث: في مساحة المضلع وما يتعلق به.
- الباب الرابع: في مساحة الدائرة وما يتعلق بها.
- الباب الخامس: في مساحة الأجسام المستوية.
- الباب السادس: في مساحة الأجسام المحدبة.
- الباب السابع: في أحجام الأجسام.
- الباب الثامن: في نسبة حجم الجسم إلى وزنه وعكس ذلك.
- الباب التاسع: في مساحة البنيان والأبنية.

[illegible].

- [illegible] :1 [illegible].
- [illegible] :2 [illegible].
- [illegible] :3 [illegible].
- [illegible] :4 [illegible].
- [illegible] :5 [illegible].
- [illegible] :6 [illegible].
- [illegible] :7 [illegible].
- [illegible] :8 [illegible].
- [illegible] :9 [illegible].

4.2 Chapter 1: On the Area of a Triangle

فصل في مساحة المثلث

Chapter 1: On the Area of a Triangle

4.2.1 Section 1: Classification of Triangles

Al-Kāshī classifies triangles by their sides (equilateral, isosceles, scalene) and by their angles (acute, right, obtuse), giving geometric definitions for each.

4.2.2 Section 2: Area Formulas

Area of a Triangle (مساحة المثلث:
Method 1 — Base and Height:

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Method 2 — Heron's Formula: For a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$:

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

Method 3 — Two Sides and Included Angle:

$$A = \frac{1}{2}ab\sin C$$

Example: Find the area of a triangle with sides 13, 14, 15.

$$\begin{aligned} s &= \frac{13+14+15}{2} = 21 \\ A &= \sqrt{21(21-13)(21-14)(21-15)} \\ &= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84 \end{aligned}$$

4.3 Chapter 4: On the Area of the Circle

فصل في مساحة الدائرة ومعرفة محيطها من قطرها وعكس ذلك

Chapter 4: On the Area of the Circle

يحتوي هذا الفصل على أشهر مقاطع المفتاح: بيان الكاشي لقيمة : π.

[illegible]: [illegible] π. [illegible]:

واعلم أن المحيط ثلاثة أضعاف القطر وجزء وهو أقل من سُبع القطر، ولكن الناس يأخذونه سُبعاً لتسهيل العمل. قال أرخميدس: إن هذا الجزء أصغر من سُبع وأكبر من عشرة من إحدى وسبعين. وحسب ما حصلناه وذكرناه في مقالتنا المسماة بالمحيطية في نسبة القطر إلى المحيط، فهو 3 درج 8 دقائق 29 ثانية 44 ثالثة...

In modern notation:

$$\pi \approx 3 + \frac{8}{60} + \frac{29}{60^2} + \frac{44}{60^3} = 3.14159259259\ldots$$

*This value of π is accurate to about 7 decimal places. But al-Kāshī's tour de force in the Risāla al-Muḥīṭiyya extends this to **16 decimal places**, computed using a polygon with $3 \times 2^{28} = 805{,}306{,}368$ sides.*

4.3.1 Circle Area and Related Formulas

Circle Formulas (قوانين الدائرة:

$$C = \pi d = 2\pi r$$

(Circumference / المحيط

$$A = \pi r^2 = \frac{\pi d^2}{4}$$

(Area / المساحة

Finding diameter from circumference:

$$d = \frac{C}{\pi}$$

The Archimedean bounds: Al-Kāshī notes the classical bounds:

$$3\frac{10}{71} < \pi < 3\frac{1}{7}$$

that is:

$$\frac{223}{71} < \pi < \frac{22}{7}$$

4.4 Chapter 9: On the Measurement of Structures and Buildings

الباب التاسع في مساحة البنيان والأبنية

Chapter 9: On the Measurement of Structures and Buildings

لم يذكر أهل هذه الصنعة إلا الأقبية والأقواس، وحتى ذلك لم يُذكر على وجهه. فذكرناها هنا على وجهها الصحيح بين غيرها، لأن الحاجة في مساحة البنيان هي أشد.

[illegible]. [illegible]:

> [illegible], [illegible]. [illegible], [illegible].

4.4.1 Section 1: On the Areas of Vaults and Arches

Al-Kāshī defines a *proper vault* (القبو الحقيقي[illegible], [illegible]: [illegible], [illegible].

4.4.2 Section 2: On the Area of a Dome

[illegible] d [illegible] h, [illegible]:

$$A_{[illegible]} = 2\pi r h$$

[illegible] r [illegible].

4.4.3 Section 3: On the Surface Area of the Muqarnas

فصل في مساحة سطح المقرنس

[illegible] :3 [illegible]

[illegible] (مقرنس) [illegible]

[illegible] (مساحة سطح المقرنس:)
[illegible] n [illegible] a_i:

$$A_{[illegible]} = \sum_{i=1}^{n} a_i \times f_i$$

[illegible] f_i [illegible].

[illegible]

5 Al-Kāshī's Calculation of π — The *Risāla al-Muḥīṭiyya*

رسالة المحيطية في إيجاد نسبة المحيط إلى القطر

□□□□□□□□□□□□ □□□□□□□□□□□□ □□ π

من أبرز الإنجازات في تاريخ الرياضيات: حساب الكاشي لـ..1424□16π

□ π □ 16 □□□□□□□ □□□□□□, □□□□□□□□□ □□ 1424 □□. □1615□□1561□ □1610□□1540□ □□□□.

5.1 Historical Context

كان دافع الكاشي فلكياً: سعى لحساب محيط الكون بدقة تجعل خطأ التقريب أصغر من عرض شعرة حصان. وحدد دقة تقريبه مسبقاً واستخدم مضلعاً بـ 3 □ $^{28} = 805,306,368.2$

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$$3 \times 2^{28} = 805{,}306{,}368$$ □□□□□

5.2 The Method

استخدم الكاشي الطريقة الكلاسيكية لأرخميدس: إحاطة الدائرة بمضلعات منتظمة مسقطة ومحيطة وحساب محيطاتها. بدءاً بسداسي ومضاعفة عدد الأضلاع 28 مرة.

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28 [illegible], [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] [illegible] [illegible] π [illegible]π طريقة الكاشي لحساب:[illegible]
[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

1. [illegible] [illegible]: $c_0 = 1$ [illegible] [illegible] [illegible] [illegible] [illegible]
2. [illegible] [illegible] [illegible], [illegible] [illegible] c_n [illegible] [illegible] n[illegible], [illegible] [illegible] c_{n+1} [illegible] [illegible] $2n$[illegible] [illegible]:

$$c_{n+1}^2 = 2 - \sqrt{4 - c_n^2}$$

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

$$c_{n+1} = \frac{c_n}{\sqrt{2 + \sqrt{4 - c_n^2}}}$$

3. [illegible] [illegible] [illegible] [illegible] [illegible] N[illegible] [illegible] $P_N = N \cdot c_N$.
4. [illegible] [illegible] [illegible] π [illegible] $\pi \approx \frac{P_N}{2}$.
5. [illegible] 28 [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 3×2^{28} [illegible].

5.3 The Result

π : 6⊠16⊠59⊠28⊠1⊠34⊠51⊠46⊠14⊠50. :2 عبر الكاشي أولاً عن

[illegible] [illegible] [illegible] 2π [illegible] [illegible] [illegible] [illegible]:

$$2\pi = 6;16,59,28,1,34,51,46,14,50$$

[illegible] [illegible]:

$$6+\frac{16}{60}+\frac{59}{60^2}+\frac{28}{60^3}+\frac{1}{60^4}+\frac{34}{60^5}+\frac{51}{60^6}+\frac{46}{60^7}+\frac{14}{60^8}+\frac{50}{60^9}$$

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]:

$$2\pi = 6.2831853071795865\ldots$$

[illegible]:

$$\boxed{\pi = 3.1415926535897932\ldots}$$

[illegible] [illegible] π	[illegible]	[illegible]
3.14185 [illegible] [illegible] [illegible]	~250 [illegible]	[illegible]
$3.14166\ldots$ [illegible]$3;8,30$[illegible]	~150 [illegible]	[illegible]
3.1415926535897932... 16[illegible] [illegible]	1424 [illegible]	[illegible]
20 [illegible]	1610 [illegible]	[illegible] [illegible]
100 [illegible]	1706 [illegible]	[illegible]

[illegible] [illegible]: [illegible] [illegible] 2π [illegible] 9 [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $1/60^9 \approx 9.92 \times 10^{-17} < 10^{-16}$. [illegible] [illegible], [illegible] [illegible] 2π [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 16[illegible] [illegible] [illegible].

6 Treatise V: On Algebra

المقالة الخامسة في الجبر والمقابلة

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المقالة الأخيرة في المفتاه مكرسة للجبر والمقابلة. يقدم الكاشي
معالجة منهجية للموضوع، مبنياً على أعمال سابقيه □□ خاصة الخوارزمي
وأبو كامل و الكرجي والسموأل □□ مضيفاً ابتكاراته الخاصة.

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6.1 The Chapters of Treatise V

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□□.2: □□ □□□□□□□ □□□ □□□□□□□ □□□□□ □□□ □□□□ □□ □□□□□□ □□□□□ □□□□□□□□ □طريقة الخطأين.□

□□.3: □□ □□□□□□□□□ □□□□□□□□□□ □□□ □□□□□□□□, □□□□□□□□□□□, □□□□□□□□□□□□□□, □□□ □□□□□ □□□□ □□ □□□□□□□□□ □□□□□□□□□□□□.

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□□.6: □□ □□□ □□□□□□□□□ □□ □□□□□□□□□ □□□ □□□□□□□□ □□□□□□□□□ □□□□□□ □□ □□□□□□□.

6.2 Chapter 1: The Six Canonical Forms

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1	$ax = b$	□□□□□□ □مبسط□
2	$ax^2 = bx$	□□□□□□□ □□□□□ □□□□□
3	$ax^2 = b$	□□□□□□□ □□□□□ □□□□□□□□
4	$ax^2 + bx = c$	□□□□□□□ □□□ □□□□□ □□□□□ □□□□□□□□
5	$ax^2 + c = bx$	□□□□□□□ □□□ □□□□□□□ □□□□□ □□□□□
6	$bx + c = ax^2$	□□□□□ □□□ □□□□□□□ □□□□□ □□□□□□□□

[illegible] $ax^2 + bx = c$ [illegible] 4 حل المعادلة من النوع الرابع: [illegible]
[illegible] a: $x^2 + \frac{b}{a}x = \frac{c}{a}$
[illegible]:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

[illegible]:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{4ac + b^2}{4a^2}$$

[illegible]:

$$x = \frac{-b + \sqrt{b^2 + 4ac}}{2a}$$

6.3 Chapter 2: The Rule of Double False Position

فصل في استخراج المجهول بطريقة الخطأين

[illegible] :2 [illegible]

طريقة الخطأين هي طريقة قديمة لحل المسائل الخطية بغير جبر. ويعطي الكاشي عرضاً واضحاً لها.

[illegible] طريقة الخطأين [illegible]. [illegible]:

[illegible] طريقة الخطأين: [illegible]
[illegible] $ax + b = c$, [illegible] g_1 [illegible] g_2, [illegible] f_1 [illegible] f_2 [illegible] $e_1 = c - f_1$ [illegible] $e_2 = c - f_2$. [illegible]:

$$x = \frac{g_1 e_2 - g_2 e_1}{e_2 - e_1}$$

[illegible]: [illegible], [illegible] .14 [illegible]
[illegible] $g_1 = 12$: $\frac{12}{3} + \frac{12}{4} = 4 + 3 = 7$, [illegible] $e_1 = 14 - 7 = 7$
[illegible] $g_2 = 24$: $\frac{24}{3} + \frac{24}{4} = 8 + 6 = 14$, [illegible] $e_2 = 14 - 14 = 0$

$$x = \frac{12 \cdot 0 - 24 \cdot 7}{0 - 7} = \frac{-168}{-7} = 24$$

6.4 Chapter 5: On Solving Cubic Equations

يقدم الكاشي طرقاً لحل المعادلات التكعيبية. وللمعادلة □ $^3+px=$ q.

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] $x^3+px=q$, [illegible] [illegible] [illegible] [illegible] [illegible]:

[illegible] [illegible] [illegible] $x^3+px=q$ □الحل التكراري للمعادلة التكعيبية:□

[illegible] [illegible]: $x=\sqrt[3]{q-px}$

[illegible]:

1. [illegible] [illegible] [illegible] [illegible] x_0.
2. [illegible] $x_{n+1}=\sqrt[3]{q-px_n}$.
3. [illegible] [illegible] $|x_{n+1}-x_n|<\varepsilon$.

[illegible]: [illegible] $x^3+2x=20$

[illegible] [illegible] $x_0=2$:

$$\begin{aligned}
x_1 &= \sqrt[3]{20-2\cdot 2}=\sqrt[3]{16}\approx 2.52\\
x_2 &= \sqrt[3]{20-2\cdot 2.52}=\sqrt[3]{14.96}\approx 2.466\\
x_3 &= \sqrt[3]{20-2\cdot 2.466}=\sqrt[3]{15.068}\approx 2.472\\
x_4 &= \sqrt[3]{20-2\cdot 2.472}=\sqrt[3]{15.056}\approx 2.471
\end{aligned}$$

[illegible]: $\boxed{x\approx 2.471}$

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $\sin(1°)$ [illegible] [illegible] [illegible].

7 Tables in the *Miftāh al-Ḥisāb*

يقول الكاشي في تمهيده: □وضعت لأغلب الأعمال نموذجاً على هيئة جدول لتكون أسهل على أرباب الذكاء. وجميع الجداول الموضوعة في هذا الكتاب □□ □عذري صراحتي، وحلوها ومرها مني□ □□ □ إلا سبعة جداول.□

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7.1 The Seven Auxiliary Tables

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□□□□□ :2 □□□ □□□□□□□ □الشبكة□ □□□ □□□□□□□□□□□□□□ □□□ □ □□□□□□□ □□□□□□□□□□□□□□□ □□□.

□□□□□ :3 □□□□□ □□ □□□□□□ □□ □□□□□□ □□□ □□□□□ □□□□□ □□□□□□□□□.

□□□□□ :4 □□□□□□□ □□ □□□ □□□□□□□□□ □□ □□□□□ □□□ □□□□ □□□□□□□□□□ □□□□□□□□□.

□□□□□ :5 □□□□□ □□□ □□□□□□□ □□□ □□□□ □□ □□□ □□□□□□□ □□□ □□□□□□□□ □□ □□□□□□□□ □□□ □□□□□ □□□□□ □□□□□□□□□□□ □□□.

□□□□□ :6 □□□□□ □□ □□□□□ □□□ □□□□□□□□□ □□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□.

□□□□□ :7 □□□□□ □□□ □□□□□□□ □□□ □□□□□□ □□ □□□ □□□□□□□ □□□ □□□□□□□□ □□□ □□□□□□□□ □□□□□□ □□□□□□□□.

7.2 Sample: Multiplication Table (Table 1)

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

A Biographical Note on Al-Kāshī

نبذة عن حياة الكاشي

[illegible]

ولد جمشيد غياث الدين بن مسعود بن محمود الطبيب الكاشي في كاشان، وسط إيران، حوالي سنة 1380 م. ويُعرف بالكاشي [illegible]أو الكاشاني بالأحرى[illegible]، وبلقبه غياث الدين.

[illegible], [illegible], [illegible] 1380 [illegible]. [illegible]. [illegible], [illegible].

A.1 Key Dates in Al-Kāshī's Life

[illegible]	[illegible]
~1380	[illegible], [illegible]
1407	[illegible]
11[illegible]1410	[illegible]
14[illegible]1413	[illegible], [illegible]
~1420	[illegible]
1424	[illegible] π[illegible]
2 [illegible] 1427	[illegible]
1429	[illegible] [illegible]$\sin 1^\circ$[illegible]
22 [illegible] 1429	[illegible] ,19 832 [illegible]

A.2 Al-Kāshī's Other Works

1. [illegible], [illegible]1407 [illegible].
2. [illegible], [illegible]11[illegible]1410 [illegible].
3. [illegible], [illegible]14[illegible]1413 [illegible].

4. [illegible], [illegible]1424 [illegible] π [illegible] 16 [illegible].
5. [illegible], [illegible]1429 [illegible] $\sin(1^\circ)$ [illegible].
6. [illegible].
7. [illegible].

[illegible]

B Testimonies of Modern Scholars

شهادات العلماء المعاصرين

[illegible]

[illegible]

[illegible]

[illegible] 1948 [illegible]

[illegible]

في غنى محتوياتها وتطبيق الأساليب الحسابية والجبرية لحل مختلف المسائل، بما فيها الهندسية، وفي وضوح وجمال العرض، يعد هذا الكتاب المدرسي الضخم أحد أفضل الكتب في كل الأدبيات القرون الوسطى؛ وهو شهادة على مؤلفه العلمي وقدرته التعليمية.

[illegible]

[illegible]

في عمل غياث الدين جمشيد الكاشي في أوائل القرن الخامس عشر نرى أول مرة قيادة تامة لفكرة الكسور العشرية وتدوين مريح لها.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] π, [illegible]

[illegible]

[illegible]

[illegible]

C Note on the Manuscripts and Editions

ملاحظة على المخطوطات والطبعات

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حُفظ مفتاح الحساب في نسخ مخطوطة عديدة، مما يعكس أهميته واستخدامه الواسع.

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تم ما انتقي من مفتاح الحساب لجمشيد الكاشي

Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wa-l-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

The Compendious Book on Calculation
by Completion and Balancing

And on the laws of inheritance (al-farā'iḍ) and the distribution of legacies

Muḥammad ibn Mūsā al-Khwārizmī
محمد بن موسى الخوارزمي

Bilingual Edition — الطبعة الثنائية اللغة
Arabic Original (left) ⟷ **English Translation (right)**
Adapted from Frederic Rosen's translation (London, 1831)
With modern mathematical notation and explanatory notes

Critical Arabic text: 'Alī Muṣṭafā Musharrafa & Muḥammad Mursī Aḥmad
Cairo: Egyptian University, 1359 AH / 1939 CE

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About This Bilingual Edition

This bilingual edition presents the foundational text of algebra, composed by Muḥammad ibn Mūsā al-Khwārizmī (circa 780–850 CE) at the court of the Caliph al-Ma'mūn in Baghdad. The **left column** contains the original Arabic text; the **right column** provides an English translation adapted from Frederic Rosen's 1831 edition, with corrections and modernization of mathematical notation.

Key terminology:

- الجبر (*al-jabr*) = "restoration" or "completion" — adding terms to both sides of an equation
- المقابلة (*al-muqābala*) = "balancing" or "reduction" — subtracting like terms from both sides
- المال (*al-māl*) = "square" (of the unknown, x^2)
- الجذر (*al-jadhr*) = "root" (the unknown quantity, x)
- العدد (*al-'adad*) = "number" (constant term)
- الشيء (*al-shay'*) = "thing" (another term for the unknown)

The Arabic text is reproduced from the critical edition of 'Alī Muṣṭafā Musharrafa and Muḥammad Mursī Aḥmad (Cairo, 1939), based on the unique complete manuscript preserved in the Bodleian Library at Oxford (Hunt. 214, copied in 743 AH / 1342 CE).

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Part I

The Science of Algebra and Almucabala

The Author's Preface

بسم الله الرحمن الرحيم.

الحمد لله على نعمه لأهل طاعته، بأدائها بقدرته على عباده المؤمنين به، ليثيبهم بها ثواباً، ويحفظهم بها عن التغيير، ويعرفهم قدرته، ويذللهم لعظمته. أرسل محمداً صلى الله عليه وسلم برسالته بعد انقطاع الرسل، وذهاب العدل، وفساد النُّصُل.

وقد كان العلماء في الأزمان الخالية وفي الأمم السالفة يؤلفون الكتب في أبواب العلوم وفروع المعرفة، يتذكرون من بعدهم، ويرجون الأجر بقدر ما كان منهم، ويثقون بما يحصل لهم من الشكر والإجلال والإعظام، راضين ولو بقليل من الثناء، قليلاً مقارنةً بما بذلوه من الجهد والمشقة في كشف خفايا العلم وإزالة ستائره.

وذلك الحب للعلم الذي ميَّز الله به الإمام المأمون أمير المؤمنين ▯مع ما منَّ به عليه من الخلافة بالإجماع، واللباس الذي ألبسه إياه، والزينة التي زيَّنه بها▯، وذلك اللطف والتودد الذي يبديه للعلماء، وتلك السرعة التي يحميها وينصرها في توضيح المبهمات وإزالة الصعوبات ▯ قد حملني على أن أؤلف مختصراً في الجبر والمقابلة، أقتصر فيه على ما هو أيسر وأنفع في الحساب، مما يحتاج إليه الناس دائماً في مسائل الميراث والعطايا والقسمة والخصومات والتجارة وكل ما يتعاملون به فيما بينهم، أو في مساحة الأرضين وحفر الأنهار ومسائل الهندسة وغيرها من أنواع الأشياء المختلفة الأصناف.

In the Name of God, gracious and merciful! Praised be God for His bounty towards those who deserve it by their virtuous acts: in performing which, as by Him prescribed to His adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of His mercy), and preserve ourselves from change: acknowledging His might, bending before His power, and revering His greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain.

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

That fondness for science, by which God has distinguished the Imām al-Mā'mun, the Commander of the Faithful (besides the caliphat which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals,

Chapter 1

On Numbers and the Three Kinds

أقسام الأعداد في الجبر والمقابلة

لما نظرت فيما يحتاج إليه الناس في حساباتهم، وجدت أن ذلك كله عدد. ووجدت أيضاً أن كل عدد مؤلف من الوحدات، وأنه لا يخلو عدد من أن ينقسم إلى وحدات.

وعلمت أن كل عدد يعدَّ من الواحد إلى العشرة يزيد على الذي قبله بواحد، ثم يضاعف العشر ويثلث، كما فعل بالوحدات، فيكون عشرون وثلاثون وهكذا إلى المائة، ثم يضاعف المائة ويثلثها كما فعل بالعشرات والوحدات إلى الألف، ثم يكرر الألف كذلك في كل عدد مركب، وهكذا إلى أقصى حدٍّ في العدِّ.

فوجدت أن الأعداد التي يحتاج إليها في حساب الجبر والمقابلة ثلاثة أنواع: جذور، وأموال، وأعداد مفردة لا تتعلق بجذر ولا بمال.

أما الجذر فهو كل مقدار يضرب في نفسه مؤلفاً من وحدات أو أعداد صاعدة أو كسور نازلة.

وأما المال فهو جميع ما حصل من ضرب الجذر في نفسه.

وأما العدد المفرد فهو كل عدد يذكر بغير نسبة إلى جذر ولا مال.

When I considered what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, etc., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction are of **three kinds**, namely, *roots* (جذور), *squares* (أموال), and *simple numbers* (أعداد مفردة) relative to neither root nor square.

title=The Three Kinds

1. A **root** (*jadhr* / جذر) is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.[a]
2. A **square** (*māl* / مال) is the whole amount of the root multiplied by itself.[b]
3. A **simple number** (*'adad mufrad* / عدد مفرد) is any number which may be pronounced without reference to root or square.

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[a] **[Ed.]** In modern notation: x.
[b] **[Ed.]** In modern notation: x^2.

Chapter 2

The Six Canonical Cases

Introduction to the Six Cases

هذه هي الحالات الستّ التي ذكرتها في مقدمة هذا الكتاب. ثلاث منها لا تحتاج إلى أن تنصف الجذور، وقد بيَّنت كيف تجاب. وأما الثلاث الأخرى التي يجب فيها نصف الجذور، فأرى أن أبيّنها أيضاً مفصَّلة مع ذكر الشكل لكل حالة يبين سبب النصف.
الحالات الست:

1. المال يساوي الأصول
2. المال يساوي عدداً
3. الأصول تساوي عدداً
4. مال وأصول يساوي عدداً
5. مال وعدد يساوي أصولاً
6. أصول وعدد يساوي مالاً

From these three kinds, **six cases** arise—three simple and three compound. Al-Khwārizmī presents them in a well-defined order that would become standard for centuries of algebraic practice.

title=The Six Canonical Cases

Case I: **Squares equal to Roots:** $ax^2 = bx$

Case II: **Squares equal to Numbers:** $ax^2 = c$

Case III: **Roots equal to Numbers:** $bx = c$

Case IV: **Squares and Roots equal to Numbers:** $ax^2 + bx = c$

Case V: **Squares and Numbers equal to Roots:** $ax^2 + c = bx$

Case VI: **Roots and Numbers equal to Squares:** $bx + c = ax^2$

2.1 The Three Simple Cases

الحالة الأولى: المال يساوي الأصول

مثال: «مال يساوي خمسة أصوله». فجذر المال خمسة، والمال خمسة وعشرون، وهو يساوي خمسة أضعاف جذره.
ومثال: «ثلث مال يساوي أربعة أصول». فالمال كله يساوي اثني عشر أصلاً؛ وهو مائة وأربعة وأربعون، وجذره اثنا عشر.
ومثال: «خمسة أموال تساوي عشرة أصول». فمال واحد يساوي أصلين؛ وجذر المال اثنان، وماله أربعة.

Case I: Squares equal to Roots

Example: “A square is equal to five roots of the same.” The root of the square is five, and the square is twenty-five, which is equal to five times its root.
Example: “One third of the square is equal to four roots.” Then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.
Example: “Five squares are equal to ten roots.” Then one square is equal to two roots; the root of the square is two, and its square is four.

$$\boxed{x^2 = 5x \Rightarrow x = 5} \qquad \boxed{\tfrac{1}{3}x^2 = 4x \Rightarrow x = 12} \qquad \boxed{5x^2 = 10x \Rightarrow x =} \tag{2.1}$$

الحالة الثانية: المال يساوي عدداً

مثال: «مال يساوي تسعة». فهذا مال وجذره ثلاثة.
ومثال: «خمسة أموال تساوي ثمانين». فمال واحد يساوي خمساً وثمانين، وهي ستة عشر.
ومثال: «نصف مال يساوي ثمانية عشر». فالمال ستة وثلاثون، وجذره ستة.

Case II: Squares equal to Numbers

Example: “A square is equal to nine.” Then this is a square, and its root is three.
Example: “Five squares are equal to eighty.” Then one square is equal to one-fifth of eighty, which is sixteen.
Example: “The half of the square is equal to eighteen.” Then the square is thirty-six, and its root is six.

$$\boxed{x^2 = 9 \Rightarrow x = 3} \qquad \boxed{5x^2 = 80 \Rightarrow x = 4} \qquad \boxed{\tfrac{1}{2}x^2 = 18 \Rightarrow x = 6} \tag{2.2}$$

الحالة الثالثة: الأصول تساوي عدداً

مثال: «أصل واحد يساوي ثلاثة عدداً». فالأصل ثلاثة، وماله تسعة.
ومثال: «أربعة أصول تساوي عشرين». فأصل واحد يساوي خمسة، والمال الذي يتركب منه خمسة وعشرون.
ومثال: «نصف أصل يساوي عشرة». فالأصل كله يساوي عشرين، والمال الذي يتركب منه أربعمائة.

Case III: Roots equal to Numbers

Example: “One root equals three in number.” Then the root is three, and its square nine.
Example: “Four roots are equal to twenty.” Then one root is equal to five, and the square to be formed of it is twenty-five.
Example: “Half the root is equal to ten.” Then the whole root is equal to twenty, and the square which is formed of it is four hundred.

$$\boxed{x = 3} \qquad \boxed{4x = 20 \Rightarrow x = 5} \qquad \boxed{\tfrac{1}{2}x = 10 \Rightarrow x = 20} \tag{2.3}$$

2.2 The Three Compound Cases

الحالة الرابعة: مال وأصول يساوي عدداً

مثال: «مال وعشرة أصوله تساوي تسعة وثلاثين درهماً».

يعني: ما هو المال الذي إذا أُضيف إليه عشرة من جذوره بلغ تسعة وثلاثين؟
الحل: نصف عدد الجذور، فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فنضيفه إلى تسعة وثلاثين فيكون أربعة وستين، فنأخذ جذره فيكون ثمانية، فننقص منه نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا هو جذر المال المطلوب، والمال نفسه تسعة.
والحل واحد في مالين أو ثلاثة أو أكثر أو أقل؛ فتردها إلى مال واحد، وعلى ذلك ترد الأصول والأعداد الملحقة بها.

Case IV: Squares and Roots equal to Numbers

Example: "One square, and ten roots of the same, amount to thirty-nine dirhems."
Solution: You halve the number of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

title=Algorithm for Case IV

$$x^2 + bx = c \quad \Longrightarrow \quad x = \sqrt{\left(\frac{b}{2}\right)^2 + c} - \frac{b}{2} \tag{2.4}$$

Applied: $x^2 + 10x = 39 \Rightarrow x = \sqrt{25 + 39} - 5 = 8 - 5 = 3$

The solution is the same when two squares or three, or more or less be specified; you reduce them to one single square, and in the same proportion you reduce also the roots and simple numbers which are connected therewith.

مثال: «مالان وعشرة أصول يساوي ثمانية وأربعين درهماً».
يجب أن ترد المالين إلى واحد؛ وإنما المال الواحد من المالين نصفهما. ثم ترد كل ما ذكر في المسألة إلى نصفه:

«مال وخمسة أصول يساوي أربعة وعشرين درهماً». الآن نصف عدد الجذور؛ النصف أثنان ونصف. اضربه في نفسه:

الناتج ستة وربع. أضفه إلى أربعة وعشرين؛ المجموع ثلاثون وربع. خذ جذره؛ هو خمسة ونصف. انقص منه نصف عدد الجذور، وهو اثنان ونصف؛ الباقي ثلاثة. وهذا جذر المال، والمال نفسه تسعة.

Example: "Two squares and ten roots are equal to forty-eight dirhems."
You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce everything mentioned in the statement to its half: "a square and five roots of the same are equal to twenty-four dirhems." Now halve the number of the roots; the moiety is two and a half. Multiply that by itself: the product is six and a quarter. Add this to twenty-four; the sum is thirty and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

$$2x^2 + 10x = 48 \Rightarrow x^2 + 5x = 24 \Rightarrow x = \sqrt{6\tfrac{1}{4} + 24} - 2\tfrac{1}{2} = 3 \tag{2.5}$$

الحالة الخامسة: مال وعدد يساوي أصولاً

مثال: «مال وواحد وعشرون عدداً يساوي عشرة أصوله».
الحل: نصف عدد الجذور فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فننقص منه الواحد والعشرين الملحقين بالمال، يبقى أربعة، فنستخرج جذرها فيكون اثنين، فننقصه من نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا جذر المال المطلوب، والمال تسعة. أو تضيف الجذر إلى نصف الجذور، فيكون سبعة، وهذا جذر المال المطلوب، والمال تسعة وأربعون.

واعلم أنه إذا نصفت الجذور وضربت النصف في نفسه، فإن كان الناتج أقل من عدد الدراهم الملصقة بالمال، فالمسألة مستحيلة؛ وإن كان الناتج مساوياً للدراهم، فجذر المال يساوي نصف الجذور وحده، بلا زيادة ولا نقصان.

Case V: Squares and Numbers equal to Roots

Example: "A square and twenty-one in numbers are equal to ten roots of the same square."
Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. **Or** you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

title=Algorithm for Case V

$$x^2 + c = bx \quad \Longrightarrow \quad x = \frac{b}{2} \pm \sqrt{\left(\frac{b}{2}\right)^2 - c} \tag{2.6}$$

Applied: $x^2 + 21 = 10x \Rightarrow x = 5 \pm \sqrt{25 - 21} = 5 \pm 2 = 3 \text{ or } 7$

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the three cases in which the number of the roots must be halved.
Note on impossibility: And know that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be **less** than the number of dirhems connected with the square, then the instance is **impossible**; but if the product be **equal** to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

الحالة السادسة: أصول وعدد يساوي مالاً

مثال: «ثلاثة أصول وأربعة أعداد يساوي مالاً».

الحل: نصف الجذور فيكون واحداً ونصفاً، ونضربه في نفسه فيكون اثنين وربعاً، فنضيفه إلى الأربعة فيكون ستة وربعاً، فنستخرج جذرها فيكون اثنين ونصفاً، فنضيفه إلى نصف الجذور وهو واحد ونصف فيكون أربعة، وهذا جذر المال، والمال ستة عشر.

Case VI: Roots and Numbers equal to Squares

Example: "Three roots and four of simple numbers are equal to a square."

Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

title=Algorithm for Case VI

$$bx + c = x^2 \quad \Longrightarrow \quad x = \frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + c} \tag{2.7}$$

Applied: $3x + 4 = x^2 \Rightarrow x = 1\frac{1}{2} + \sqrt{2\frac{1}{4} + 4} = 1\frac{1}{2} + 2\frac{1}{2} = 4$

Chapter 3

Geometrical Demonstrations

هذه هي الحالات الستّ التي ذكرتها في مقدمة هذا الكتاب. وقد فُسّرت الآن. وقد أظهرت أن ثلاثاً منها لا تحتاج إلى نصف الجذور، وعلمت كيف تجاب. وأما الثلاث الأخرى التي يجب فيها نصف الجذور، فأرى أن أبيّنها بأبواب مفصّلة، أضع في كل حالة شكلاً يبين سبب النصف.

البرهان على الحالة الرابعة: «مال وعشرة أصول يساوي تسعة وثلاثين»

الشكل المبيّن لذلك هو مربع، أضلاعه مجهولة. وهو يمثل المال، جذره المطلوب معرفته. وهذا الشكل أ ب، وكل ضلع منه يعتبر واحداً من أصوله؛ وإذا ضربت أحد هذه الأضلاع في أي عدد، فإن مقدار ذلك العدد يعتبر عدد الأصول التي تُضاف إلى المال.

ونحن نأخذ ربع العشرة، أي اثنين ونصف، ونضمّها إلى كل من الأضلاع الأربعة للشكل. فمع المربع الأصلي أ ب، تُضمّ أربعة متوازيات أضلاع، طول كل منها ضلع من المربع، وعرضها اثنان ونصف؛ وهي المتوازيات ج، د، ط، ك.

فليكن لدينا مربع أضلاعه متساوية ومجهولة؛ ولكن في كل من زواياه الأربع ينقص مربع صغير اثنين ونصف في اثنين ونصف. ولتعويض هذا النقص وإتمام المربع، يجب أن نضيف أربعة أمثال مربع اثنين ونصف، أي خمسة وعشرين.

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

Demonstration of Case IV: “A Square and Ten Roots equal to 39”

The figure to explain this is a **quadrate**, the sides of which are unknown. It represents the square, the root of which you wish to know. This is the figure AB, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square.

Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate AB, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms C, G, T, and K.

We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five.

title=Geometric Proof of Case IV — Completion of the Square

Given: $x^2 + 10x = 39$

$$\begin{array}{rcl}
\text{Square } AB \text{ (the unknown square)} & = & x^2 \\
+\text{ 4 parallelograms around it} & = & 4 \times (2\tfrac{1}{2} \times x) = 10x \\
\hline
\text{Subtotal} & = & 39 \text{ (given)} \\
+\text{ 4 corner squares } (2\tfrac{1}{2} \times 2\tfrac{1}{2} \text{ each}) & = & 4 \times 6\tfrac{1}{4} = 25 \\
\hline
\text{Great Square } DH & = & 39 + 25 = 64 \\
\text{Side of Great Square} & = & \sqrt{64} = 8 \\
x = 8 - 2 \times 2\tfrac{1}{2} = 8 - 5 & = & \boxed{3}
\end{array}$$

ونعلم ﴿من المسألة﴾ أن الشكل الأول، أي المربع الممثل للمال، مع المتوازيات الأربع المحيطة به، الممثلة للعشرة أصول، يساوي تسعة وثلاثين عدداً. فإذا أضفنا إليها خمسة وعشرين، وهي كفاءة المربعات الأربع في زوايا الشكل أ ب، التي تكمل المربع الكبير د ه، فإننا نعلم أن هذا المجموع يصير أربعة وستين. أحد أضلاع هذا المربع الكبير هو جذره، أي ثمانية. فإذا نقصنا ضعف ربع العشرة، أي خمسة، من الثمانية، كما من طرفي ضلع المربع الكبير د ه، فإن بقية هذا الضلع يكون ثلاثة، وهذا هو جذر المال، أو ضلع الشكل الأصلي أ ب.

ويلاحظ أننا نصف عدد الجذور، ونضيف حاصل ضرب النصف في نفسه إلى العدد تسعة وثلاثين، لنكمل المربع الكبير في زواياه الأربع؛ لأن ربع أي عدد مضروباً في نفسه ثم في أربعة، يساوي حاصل ضرب نصف ذلك العدد في نفسه.

We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure AB, by which the great figure DH is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate DH, then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure AB.

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.

Note: This is the celebrated method of “**completion of the square**” (إتمام المربع), which gives algebra its very name: *al-jabr* refers to the “restoration” of the deficient corners. The identity underlying the proof is: $\left(x + \frac{b}{2}\right)^2 = x^2 + bx + \left(\frac{b}{2}\right)^2$.

Demonstration of Case V: “A Square and 21 equal to 10 Roots”

البرهان على الحالة الخامسة: «مال وواحد وعشرون يساوي عشرة أصول»

نمثل المال بمربع أ د، طول ضلعه مجهول لنا. ونلحق به متوازي أضلاع، عرضه يساوي أحد أضلاع المربع أ د، كالضلع هـ ن. وهذا المتوازي هو هـ ب. طول الشكلين معاً يساوي الخط هـ س. ونعلم أن طوله هو عشرة أعداد؛ فكل مربع له أضلاع متساوية وزوايا متساوية، وأحد أضلاعه مضروباً بواحد هو جذر المربع، أو مضروباً باثنين فهو ضعف جذره.

فكما قيل: إن مالاً وواحداً وعشرين عدداً يساوي عشرة أصول، فإننا نستنتج أن طول الخط هـ س يساوي عشرة أعداد، إذن الخط س د يمثل جذر المال. والآن نقسم الخط س هـ إلى نصفين عند النقطة ج: فالخط ج س يساوي هـ ج.

ونعلم أن الرباعي هـ ب يمثل الواحد والعشرين عدداً التي أضيفت إلى المربع. ووجد المربع الكلي م ط يساوي خمسة وعشرين. فإذا نقصنا من هذا المربع م ط المربعين هـ ط وم ر، وهما يساويان واحداً وعشرين، يبقى مربع صغير ك ر، يمثل الفرق بين خمسة وعشرين وواحد وعشرين. وهو أربعة؛ وجذره، الممثل بالخط ر ج، المساوي لـ ج أ، هو اثنان.

We represent the square by a quadrate AD, the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate AD, such as the side HN. This parallelogram is HB. The length of the two figures together is equal to the line HC. We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same.

As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line HC is equal to ten of numbers, since the line CD represents the root of the square. We now divide the line CH into two equal parts at the point G: the line GC is then equal to HG.

We know that the quadrangle HB represents the twenty-one of numbers which were added to the quadrate. The whole quadrate MT was found to be equal to twenty-five. If we now subtract from this quadrate, MT, the quadrangles HT and MR, which are equal to twenty-one, there remains a small quadrate KR, which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line RG, which is equal to GA, is two.

title=Geometric Proof of Case V

Given: $x^2 + 21 = 10x$

$$MT = \left(\tfrac{10}{2}\right)^2 = 25$$

$$HT + MR = HB = 21$$

$$KR = 25 - 21 = 4 \quad \Rightarrow \quad \sqrt{KR} = 2$$

$$x = GC - GA = 5 - 2 = 3 \quad \text{or} \quad x = GC + GA = 5 + 2 = 7$$

فإذا نقصت هذا العدد اثنين من الخط ج س، وهو نصف الجذور، فإن الباقي هو الخط أ س؛ أي ثلاثة، وهو جذر المال الأصلي. ولكن إذا أضفت العدد اثنين إلى الخط ج س، وهو نصف عدد الجذور، فإن المجموع سبعة، الممثل بالخط ج ر، وهو جذر مال أكبر. ومع ذلك، فإذا أضفت واحداً وعشرين إلى هذا المال، فإن المجموع يساوي كذلك عشرة أصول من ذلك المال.

If you subtract this number two from the line CG, which is the moiety of the roots, then the remainder is the line AC; that is to say, three, which is the root of the original square. But if you add the number two to the line CG, which is the moiety of the number of the roots, then the sum is seven, represented by the line CR, which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

Demonstration of Case VI: “3 Roots and 4 equal to a Square”

البرهان على الحالة السادسة: «ثلاثة أصول وأربعة يساوي مالاً»

ليمثل المال رباعياً، أضلاعه مجهولة لنا، وهي متساوية فيما بينها، كما هي زواياه. وهذا هو المربع أ د، الذي يشتمل على الأصول الثلاثة والأربعة أعداد المذكورة في هذه المسألة.

وفي كل مربع أحد أضلاعه مضروباً بواحد هو جذره. والآن نقطع الرباعي هـ د من المربع أ د، ونتخذ أحد أضلاعه هـ س ثلاثة، وهو عدد الأصول. وهو يساوي ر د. فمن ثم، يمثل الرباعي هـ ب الأربعة أعداد التي تُضاف إلى الأصول. والآن نصف الضلع س هـ، المساوي لثلاثة أصول، عند النقطة ج؛ من هذا القسم نبني المربع هـ ط، وهو حاصل ضرب نصف الأصول ▯أي واحد ونصف▯ في نفسه، أي اثنين وربع.

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate AD, which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle HD from the quadrate AD, and take one of its sides HC for three, which is the number of the roots. The same is equal to RD. It follows, then, that the quadrangle HB represents the four of numbers which are added to the roots. Now we halve the side CH, which is equal to three roots, at the point G; from this division we construct the square HT, which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter.

title=Geometric Proof of Case VI

Given: $3x + 4 = x^2$

$$\text{Square } HT = \left(\tfrac{3}{2}\right)^2 = 2\tfrac{1}{4}$$

$$AN + KL = AR = 4 \text{ (the added numbers)}$$

$$GM = HT + (AN + KL) = 2\tfrac{1}{4} + 4 = 6\tfrac{1}{4}$$

$$AG = \sqrt{GM} = 2\tfrac{1}{2}$$

$$x = AG + GC = 2\tfrac{1}{2} + 1\tfrac{1}{2} = \boxed{4}$$

Chapter 4

On Multiplication of Algebraic Expressions

في ضرب المقادير المجهولة

سأعلمك الآن كيف تضرب المقادير المجهولة، أي الجذور، بعضها ببعض، إذا كانت وحدها، أو إذا أُضيفت إليها أعداد، أو نُقصت منها أعداد، أو نُقصت هي من أعداد؛ وكيف تجمعها بعضها إلى بعض، أو تنقص بعضها من بعض.

كلما أريد ضرب عدد بآخر، وجب تكرار أحدهما بعدد الوحدات التي في الآخر.

فإن كان هناك أعداد كبيرة مضافة إلى وحدات، لتُضاف أو تُنقص، فإن أربع ضربات لازمة؛ وهي: الكبير بالكبير، والكبير بالوحدات، والوحدات بالكبير، والوحدات بالوحدات.

فإن كانت الوحدات المضافة إلى الأعداد الكبيرة موجبة في كليهما، فالضربة الرابعة موجبة؛ وإن كانت سالبة في كليهما، فالرابعة موجبة أيضاً. ولكن إن كانت إحداهما موجبة والأخرى سالبة، فالضربة الرابعة سالبة.

I shall now teach you how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.

If there are greater numbers combined with units to be added to or subtracted from them, then **four multiplications** are necessary; namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are both positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.

title=The Fourfold Multiplication Rule

To multiply $(a + x)$ by $(b + y)$:

1. Greater by greater: $a \times b$
2. Greater by units: $a \times y$
3. Units by greater: $x \times b$
4. Units by units: $x \times y$

Sign rule: $(+x)(+y) = +xy \quad (-x)(-y) = +xy$
$(+x)(-y) = -xy \quad (-x)(+y) = -xy$

Worked Examples

المثال الأول: «عشرة وواحد يُضرب في عشرة واثنين».
عشرة في عشرة مائة؛ الواحد في العشرة عشرة موجبة؛ الاثنان في العشرة عشرون موجبة؛ والواحد في الاثنين اثنان موجبة؛ المجموع مائة واثنتان وثلاثون.

المثال الثاني: «عشرة ناقص واحد، يُضرب في عشرة ناقص واحد».
عشرة في عشرة مائة؛ السالب واحد في العشرة عشرة سالبة؛ والسالب الآخر في العشرة عشرة سالبة أيضاً، فيصير ثمانون؛ ولكن السالب واحد في السالب واحد واحد موجب، فيصير الناتج واحداً وثمانين.

المثال الثالث: «عشرة وشيء يُضرب في عشرة وشيء».
عشرة في عشرة مائة؛ وعشرة في الشيء عشرة أشياء؛ وعشرة في الشيء عشرة أشياء أخرى؛ والشيء في الشيء مال موجب؛ فالحاصل كله مائة درهم وعشرون شيئاً ومالاً موجباً.

المثال الرابع: «عشرة ناقص شيء يُضرب في عشرة ناقص شيء».
عشرة في عشرة مائة؛ وسالب شيء في العشرة سالب عشرة أشياء؛ وسالب شيء في العشرة سالب عشرة أشياء أخرى. وسالب شيء في سالب شيء مال موجب.
فالحاصل مائة ومال ناقص عشرين شيئاً.

المثال الخامس: «عشرة ناقص شيء يُضرب في عشرة وزيادة شيء».
عشرة في عشرة مائة؛ وسالب شيء في العشرة سالب عشرة أشياء؛ وشيء في العشرة عشرة أشياء موجبة؛ وسالب شيء في الشيء سالب مال؛ فالحاصل مائة درهم ناقص مال.

Example 1: “Ten and one to be multiplied by ten and two.”
Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.
$(10+1)(10+2) = 100+10+20+2 = 132$

Example 2: “Ten less one, to be multiplied by ten less one.”
Then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.
$(10-1)(10-1) = 100-10-10+1 = 81$

Example 3: “Ten and thing to be multiplied by ten and thing”
Then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.
$(10+x)(10+x) = 100+10x+10x+x^2 = 100+20x+x^2$

Example 4: “Ten minus thing to be multiplied by ten minus thing”
Then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things. But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.
$(10-x)(10-x) = 100-10x-10x+x^2 = 100-20x+x^2$

Example 5: “Ten minus thing to be multiplied by ten and thing”
Then you say, ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square negative; therefore, the product is a hundred dirhems, minus a square.
$(10-x)(10+x) = 100-10x+10x-x^2 = 100-x^2$

Chapter 5

On Addition and Subtraction

في جمع وطرح الجذور والأعداد

اعلم أن جذر مائتين ناقص عشرة، مضافاً إلى عشرين ناقص جذر مائتين، هو عشرة.

وجذر المائتين ناقص العشرة، منقوصاً من عشرين ناقص جذر المائتين، هو ثلاثون ناقص ضعف جذر مائتين؛ وضعف جذر المائتين يساوي جذر ثمانمائة.

ومائة ومال ناقص عشرون أصلاً، مضافاً إلى خمسين وعشرة أصول ناقص مالان، هو مائة وخمسون ناقص مال وناقص عشرة أصول.

ومائة ومال ناقص عشرون أصلاً، منقوصاً من خمسين وعشرة أصول ناقص مالان، هو خمسون درهماً وثلاثة أموال ناقص ثلاثون أصلاً.

Operations on Algebraic Expressions

Know that the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.
$(\sqrt{200}-10)+(20-\sqrt{200})=10$

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.
$(20-\sqrt{200})-(\sqrt{200}-10)=30-2\sqrt{200}=30-\sqrt{800}$

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.
$(100+x^2-20x)+(50+10x-2x^2)=150-x^2-10x$

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.
$(100+x^2-20x)-(50+10x-2x^2)=50+3x^2-30x$

On Doubling, Tripling, and Halving Roots

في مضاعفة الجذور ونصفها

إذا أردت مضاعفة جذر أي مال معلوم أو مجهول، □ومعنى مضاعفته أن تضربه في اثنين□ فإنه يكفي أن تضرب اثنين في اثنين، ثم في المال؛ فجذر الحاصل يساوي ضعف جذر المال الأصلي.

وإذا أردت أخذه ثلاثاً، فتضرب ثلاثة في ثلاثة، ثم في المال؛ فجذر الحاصل ثلاثة أمثال جذر المال الأصلي.

وإذا أردت أن تجد نصف جذر المال، فلا يحتاج إلا أن تضرب نصفاً في نصف، وهو ربع؛ ثم هذا في المال: فجذر الحاصل يكون نصف جذر المال الأول.

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square; the root of the product is equal to twice the root of the original square.
$2\sqrt{S} = \sqrt{4S}$

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.
$3\sqrt{S} = \sqrt{9S}$

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square.
$\frac{1}{2}\sqrt{S} = \sqrt{\frac{1}{4}S}$

Chapter 6

On Division

في قسمة الجذور والأعداد

إذا أردت قسمة جذر تسعة على جذر أربعة، فتبدأ بقسمة تسعة على أربعة، فيعطي اثنين وربعاً: فجذر ذلك هو العدد المطلوب ▯ وهو واحد ونصف.

وإذا أردت قسمة جذر أربعة على جذر تسعة، فتقسم أربعة على تسعة؛ فهي أربعة أتساع الوحدة؛ وجذرها اثنان مقسوماً على ثلاثة؛ أي ثلثا الوحدة.

وإذا أردت ضرب جذر تسعة في جذر أربعة، فاضرب تسعة في أربعة؛ فهذا يعطي ستة وثلاثين؛ فخذ جذرها، فهو ستة؛ وهذا هو جذر تسعة مضروباً في جذر أربعة.

وإذا أردت ضرب ضعف جذر تسعة في ثلاثة أضعاف جذر أربعة، فخذ ضعف جذر تسعة، بحسب القاعدة المذكورة أعلاه، لتعرف جذر أي مال هو. وتفعل المثل بخصوص ثلاثة جذور أربعة لتعرف ما يجب أن يكون مال ذلك الجذر. ثم تضرب هذين المالين أحدهما في الآخر، فجذر الحاصل يساوي ضعف جذر تسعة مضروباً في ثلاثة أضعاف جذر أربعة.

If you will divide the root of nine by the root of four, you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

$$\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\tfrac{1}{2}$$

If you will divide the root of four by the root of nine, you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

$$\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

If you wish to multiply the root of nine by the root of four, multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

$$\sqrt{9} \times \sqrt{4} = \sqrt{9 \times 4} = \sqrt{36} = 6$$

If you wish to multiply twice the root of nine by thrice the root of four, then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

$$(2\sqrt{9}) \times (3\sqrt{4}) = \sqrt{36 \times 36} = 36$$

Chapter 7

The Six Illustrative Problems

المسائل الست التوضيحية
قبل أبواب الحساب وأنواعه المتعددة، سأذكر الآن ست مسائل، كأمثلة للحالات الست التي تقدم ذكرها في مطلع هذا الكتاب. وقد أظهرت أن ثلاثاً من هذه الحالات لا تحتاج في حلها إلى نصف الجذور، وقد ذكرت أيضاً أن الحساب بالجبر والمقابلة لا بد أن يورثك إحدى هذه الحالات. وسألحق هذه المسائل الآن، لتقرب الموضوع إلى الفهم، وتسهل إدراكه، وتجعل الحجج أوضح.

Before the chapters on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

المسألة الأولى

«قسمت عشرة إلى قسمين؛ فضربت أحدهما في الآخر؛ ثم ضربت أحدهما في نفسه، فكان حاصل ضربه في نفسه أربعة أمثال حاصل ضرب أحدهما في الآخر.»

الحل: ليكن أحد القسمين شيئاً، والآخر عشرة ناقص الشيء. تضرب الشيء في عشرة ناقص شيء؛ فهو عشرة أشياء ناقص مال. ثم تضربه في أربعة، لأن المسألة تقول «أربعة أمثال». فيكون حاصل أربعين شيئاً ناقص أربعة أموال. ثم تضرب الشيء في شيء، أي أحد القسمين في نفسه. وهذا مال، يساوي أربعين شيئاً ناقص أربعة أموال. فأكمله بالجبر بأربعة أموال، وأضفها إلى المال الواحد.

فيكون المقابلة: أربعون شيئاً تساوي خمسة أموال؛ فالمال يساوي ثمانية أصول، أي أربعة وستين؛ وجذرها ثمانية.

First Problem

Statement: "I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other."

Solution: Suppose one of the portions to be x, and the other $10 - x$. You multiply x by $10 - x$; it is $10x - x^2$. Then multiply it by four, because the instance states "four times as much." The result will be $40x - 4x^2$. After this you multiply x by x, that is to say, one of the portions by itself. This is a square, which is equal to $40x - 4x^2$. Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight.

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$$x^2 = 4x(10 - x) = 40x - 4x^2 \Rightarrow 5x^2 = 40x \Rightarrow x^2 = 8x = 64 \Rightarrow x = 8$$

The portions are: $x = 8$ and $10 - x = 2$. This question leads you to Case I: "squares equal to roots."

المسألة الثانية
«قسمت عشرة إلى قسمين؛ فضربت كل قسم في نفسه، ثم ضربت العشرة في نفسها؛ فكان حاصل العشرة في نفسها يساوي أحد القسمين مضروباً في نفسه ثم في اثنين وسبعة أتساع.»
المسألة الثالثة
«قسمت عشرة إلى قسمين؛ فقسمت أحدهما على الآخر، فكان الخارج أربعة.»
المسألة الرابعة
«ضربت ثلث شيء ودرهم في ربع شيء ودرهم، فكان الحاصل عشرين.»
المسألة الخامسة
«قسمت عشرة إلى قسمين؛ وضربت كل قسم في نفسه، فجمعتها معاً؛ وكان المجموع ثمانية وخمسين درهماً.»
المسألة السادسة
«ضربت ثلث جذر في ربع جذر، وكان الحاصل يساوي الجذر وأربعة وعشرين درهماً.»

Second Problem

Statement: “I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths.”

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$100 = x^2 \times 2\tfrac{7}{9} = x^2 \times \tfrac{25}{9} \Rightarrow x^2 = \tfrac{900}{25} = 36 \Rightarrow x = 6$

The portions are: $x = 6$ and $10 - x = 4$.

Third Problem

Statement: “I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four.”

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$$\frac{10-x}{x} = 4 \Rightarrow 10 - x = 4x \Rightarrow 10 = 5x \Rightarrow x = 2$$

The portions are: $x = 2$ and $10 - x = 8$.

Fourth Problem

Statement: “I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty.”

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$\left(\tfrac{x}{3}+1\right)\left(\tfrac{x}{4}+1\right) = 20 \Rightarrow \tfrac{x^2}{12} + \tfrac{7x}{12} + 1 = 20 \Rightarrow x^2 + 7x = 228$

Halve the roots: $3\tfrac{1}{2}$. Square: $12\tfrac{1}{4}$. Add to 228: $240\tfrac{1}{4}$. Root: $15\tfrac{1}{2}$. Subtract $3\tfrac{1}{2}$: $x = 12$.

Fifth Problem

Statement: “I have divided ten into two parts; and having multiplied each part by itself, I have added them together; and the sum was fifty-eight dirhems.”

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$x^2 + (10-x)^2 = 58 \Rightarrow 2x^2 - 20x + 100 = 58 \Rightarrow x^2 + 28 = 11x$

$x = \tfrac{11}{2} \pm \sqrt{\tfrac{121}{4} - 28} = 5\tfrac{1}{2} \pm 2\tfrac{1}{2}$. The portions are: $x = 3$ or $x = 7$.

Sixth Problem

Statement: “I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems.”

Part II

Practical Applications

Chapter 8

On Mercantile Transactions

المعاملات التجارية □ قاعدة الأربعة أعداد
اعلم أن جميع معاملات الناس من بيع وشراء ومصارفة وإجارة، فإنها لا تخلو من قدرين وأربعة أعداد يسأل عنها السائل، وهي المقدار والثمن والكيفية والجميع.
عدد المقدار يناسب عدد الجميع، وعدد الثمن يناسب عدد الكيفية. ثلاثة من هذه الأربعة دائماً معلومة، وواحد مجهول، وهو الذي يتضمنه قوله «كم؟» وهو المطلوب.

You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always **two notions** and **four numbers**, which are stated by the enquirer; namely, *measure* (مقدار) and *price* (ثمن), and *quantity* (كيفية) and *sum* (جميع).
The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity.
Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* (كم؟) and it is the object of the question.

title=The Rule of Four Numbers (al-nisba)

Measure	**Price**
a (known)	b (known)
c (known or unknown)	d (known or unknown)

$$a \times b = c \times d \qquad \text{(Rule of Three)}$$

Unknown

$$= \frac{\text{Product of the two proportionate numbers}}{\text{The third known number}}$$

Examples

المثال الأول: «عشرة بستة، كم بأربعة؟»
هنا العشرة هي المقدار؛ والستة هي الثمن؛ و«كم» تفيد الكيفية المجهولة؛ والأربعة هي عدد الجميع. المقدار □10□ يناسب الجميع □4□.
المثال الثاني: «عشرة بثمانية، فما هو المبلغ لأربعة؟»

المثال الثالث: «يأخذ عامل أجراً قدره عشرة دراهم في الشهر؛ فكم يكون أجره في ستة أيام؟»

الستة أيام خمس الشهر؛ وحصته من الدراهم يجب أن تكون مناسبة لحصته من الشهر.

Example 1: “Ten for six, how much for four?”
Here ten is the measure; six is the price; “how much” implies the unknown quantity; and four is the number of the sum. The measure (10) is inversely proportionate to the sum (4).
Quantity $= \frac{10 \times 4}{6} = \frac{40}{6} = 6\frac{2}{3}$

Example 2: “Ten for eight, what must be the sum for four?”
Sum $= \frac{8 \times 4}{10} = \frac{32}{10} = 3\frac{1}{5}$

Example 3: “A workman receives a pay of ten dirhems per month; how much must be his pay for six days?”
Six days are one-fifth of the month; and his portion of the dirhems must be proportionate to the portion of the month.
Pay $= \frac{10 \times 6}{30} = 2$ dirhems

Chapter 9

Mensuration

المساحة والحجم

اعلم أن معنى قولنا «واحد في واحد» هو المساحة: أي ذراع في ذراع.

كل مربع ذو أضلاع متساوية وزوايا متساوية، له ذراع في كل ضلع، فله واحد في مساحته. وإن كان لهذا المربع ذراعان في ضلعه، فإن مساحة المربع أربعة أمثال مساحة مربع ضلعه ذراع واحد.

المثلثات

إذا ضربت ارتفاع أي مثلث متساوي الأضلاع في نصف القاعدة التي يقف عليها الخط المار بالارتفاع عمودياً، فالحاصل يعطي مساحة ذلك المثلث.

الدوائر

وفي كل دائرة، حاصل ضرب قطرها في ثلاثة وسبعاً واحداً يساوي المحيط.

ومساحة أي دائرة توجد بضرب نصف المحيط في نصف القطر.

وإذا ضربت قطر أي دائرة في نفسه، ونقصت من الحاصل سبعاً ونصف سبع منه، فإن الباقي يساوي مساحة الدائرة.

الأحجام

حجم الجسم الرباعي يوجد بضرب الطول في العرض ثم في الارتفاع.

والمخروطات والأهرامات، كالمثلثة والرباعية، تحسب بضرب ثلث مساحة القاعدة في الارتفاع.

المثلث القائم

ولاحظ أنه في كل مثلث قائم الضلعين القصيرين، كل منهما مضروباً في نفسه، والحاصلان مجتمعان، يساويان حاصل الضلع الطويل مضروباً في نفسه.

Know that the meaning of the expression "one by one" is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard.

Triangles

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

Area of triangle $= \text{height} \times \frac{\text{basis}}{2}$

Circles

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery.

$C = \pi d \approx \frac{22}{7} \times d$

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter.

$A = \frac{C}{2} \times \frac{d}{2} = \frac{Cd}{4}$

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle.

$A = d^2 - \frac{d^2}{7} - \frac{d^2}{14} = d^2 \times \frac{11}{14} \approx \frac{\pi}{4}d^2$

Note: This gives $\pi \approx \frac{22}{7}$, the well-known approximation.

Volumes

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying

Chapter 10

On Legacies and the Laws of Inheritance

الوصايا والفرائض

الجزء الأخير من هذا الكتاب يعالج قوانين الميراث الإسلامية □الفرائض□. وهذا الموضوع، على الرغم من أنه يبدو أنه ينتمي إلى الفقه وليس إلى الرياضيات، إلا أنه يتطلب كامل أدوات الجبر لحله عندما تُفرض الوصايا فوق الأسهم القانونية.

في رأس المال والديون

مسألة: «توفي رجل وترك ولدين، و أوصى بثلث رأس ماله لغريب. وترك عشرة دراهم من المال وديناً قدره عشرة دراهم على أحد الولدين.»

الحل: تسمي المبلغ المأخوذ من الدين شيئاً. فتضيفه إلى رأس المال، وهو عشرة دراهم. فيكون المجموع عشرة زائد شيء. تنقص ثلث هذا، إذ أوصى بثلث ماله: ثلاثة وثلث زائد ثلث شيء. والباقي ستة وثلثان زائد ثلثا شيء.

فتقسم هذا بين الولدين: يأخذ كل منهما ثلاثة وثلث زائد ثلث شيء. وهذا يساوي الشيء المطلوب.

The final portion of this work treats of the Islamic laws of inheritance $(al - far'i)$. This subject, while seemingly belonging to jurisprudence rather than mathematics, requires the full apparatus of algebra for its solution when legacies are superimposed upon the statutory shares.

On Capital and Money Lent

Problem: "A man dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons." **Solution:** You call the sum which is taken out of the debt x (thing). Add this to the capital, which is ten dirhems. The sum is $10 + x$. Subtract one-third of this, since he has bequeathed one-third of his property: $3\frac{1}{3} + \frac{1}{3}x$. The remainder is $6\frac{2}{3} + \frac{2}{3}x$. Divide this between the two sons: each receives $3\frac{1}{3} + \frac{1}{3}x$. This is equal to the thing which was sought for.

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$$3\frac{1}{3} + \frac{1}{3}x = x \Rightarrow 3\frac{1}{3} = \frac{2}{3}x \Rightarrow x = 5$$

The stranger receives 5; and each son receives 5 (the indebted son retains his full debt as a set-off against his share).

في نوع آخر من الوصايا

مسألة: «توفي رجل وترك أمه وزوجته، وإخوة له اثنان وأختان اثنتان لأب وأم. وأوصى لغريب بتسع رأس ماله.»

الحل: تحدد أسهم ميراثهم بأخذها من ثمانية وأربعين جزءاً. وإياك أن تأخذ تسعاً من أي رأس مال، يبقى ثمانية أتساعه. فأضف الآن إلى الثمانية الأتساع الثمن منها، وإلى الثمانية والأربعين أيضاً ثمنها، أي ستة، لإتمام رأس المال. فهذا يعطي أربعة وخمسين. فالشخص الموصى له بالتاسع يأخذ من هذا ستة. والثمانية والأربعون الباقية توزع على الورثة، بما يتناسب مع حصصهم الشرعية.

المسألة المركبة

مسألة: «توفيت امرأة وتركت زوجها وابنها وأمها. فأوصت لشخص بخمسين، ولآخر بربع رأس مالها. ففرضت الوصيتين معاً على ابنها، وعلى أمها نصف □حصة أمها من البقية□؛ وعلى زوجها لا تفرض شيئاً إلا الثلث □الذي يجب أن يساهم به بحسب القانون□.»

On Another Species of Legacy

Problem: "A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital."
Solution: You constitute their shares by taking them out of **forty-eight parts**. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

Complex Legacy Problems

Problem: "A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother's share of the residue); on her husband she imposes nothing but one-third (which he must contribute, according to the law)."
Solution: You constitute the shares of the heritage by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself.

Appendix A

Summary of the Six Cases

Case	Arabic Name	Equation	Solution
I	مال يساوي أصولاً	$ax^2 = bx$	$x = b/a$
II	مال يساوي عدداً	$ax^2 = c$	$x = \sqrt{c/a}$
III	أصول تساوي عدداً	$bx = c$	$x = c/b$
IV	مال وأصول يساوي عدداً	$ax^2 + bx = c$	$x = \sqrt{(b/2a)^2 + c/a} - b/2a$
V	مال وعدد يساوي أصولاً	$ax^2 + c = bx$	$x = \frac{b}{2a} \pm \sqrt{(b/2a)^2 - c/a}$
VI	أصول وعدد يساوي مالاً	$bx + c = ax^2$	$x = \frac{b}{2a} + \sqrt{(b/2a)^2 + c/a}$

Historical Note: These six cases exhaust all possibilities for quadratic equations with *positive* coefficients. The restriction to positive coefficients was necessitated by the geometric interpretation: al-Khwārizmī understood all terms as geometric magnitudes (areas and lengths), which cannot be negative. The full treatment of all quadratic equations, including negative coefficients, would have to wait until the Renaissance.

Appendix B

Glossary of Arabic Terms

Arabic	Transliteration	Meaning
الجبر	*al-jabr*	Completion, restoration (adding terms to both sides)
المقابلة	*al-muqābala*	Balancing, reduction (cancelling like terms)
الجذر	*al-jadhr*	Root (the unknown quantity, x)
المال	*al-māl*	Square of the unknown (x^2)
العدد المفرد	*al-'adad al-mufrad*	Simple number (constant term)
الشيء	*al-shay'*	Thing (another term for the unknown)
الدرهم	*al-dirham*	Dirhem (unit of currency/measure)
الوصية	*al-waṣiyya*	Legacy, bequest
الفرائض	*al-farā'iḍ*	Obligatory shares (Islamic inheritance law)
مساحة	*misāḥa*	Area, mensuration
ضرب	*ḍarb*	Multiplication
قسمة	*qisma*	Division
جمع	*jam'*	Addition
طرح	*tarḥ*	Subtraction
شيء	*shay'*	Thing (the unknown, x)

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تمّ بحمد الله

KITAB SHAKL AL-QATTA’

TREATISE ON THE QUADRILATERAL

* * * * *

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By the Learned Nasir al-Din al-Tusi

(1201–1274 CE / 597–672 AH)

Critical edition based on the 1891 Constantinople edition
edited and translated by Alexandre Pacha Caratheodory

* * * * *

English translation prepared for this bilingual edition

In the Name of God, the Compassionate, the Merciful
بِسْمِ اللهِ الرَّحْمٰنِ الرَّحِيمِ

Contents

Preface

بِسْمِ اللهِ الرَّحْمٰنِ الرَّحِيمِ

In the Name of God, the Compassionate, the Merciful

I have no support but in God. In Him I have placed my devotion.

* * *

□□□□□□□□ □□□□□□ □ النص العربي الأصلي

الْحَمْدُ لِلَّهِ الَّذِي بِابْتِدَاعِهِ الْحَقَائِقَ الْغَيْبِيَّةَ أَتَمَّ الْمَوَاهِبَ الْجَسِيمَةَ، وَأَخْرَجَ بِخَلْقِهِ الْمَوْجُودَاتِ مِنْ غَيْبِ الْعَدَمِ إِلَى شَهَادَةِ الْوُجُودِ وَأَوْدَعَ فِي طِيبَاتِ الْأَشْيَاءِ أَسْرَارًا جَلِيلَةً وَمُلْكًا عَظِيمًا. وَالصَّلَاةُ وَالسَّلَامُ عَلَى سَيِّدِنَا مُحَمَّدٍ وَعَلَى آلِهِ وَصَحْبِهِ أَجْمَعِينَ.

أَمَّا بَعْدُ: فَقَدْ كُنْتُ قَدْ أَلَّفْتُ فِي سِنِينَ مَاضِيَةٍ مُصَنَّفًا فِيهِ جَمِيعُ كَلَامِنَا فِي شَكْلِ الْقِطَّاعِ الْمُكَمَّلِ مَعَ الْبُرُوهَانِ، وَأَلْحَقْتُ بِهِ مَا يَقُومُ مَقَامَهُ وَيَتَعَلَّقُ بِهِ، وَكَانَ ذٰلِكَ فِي لِسَانِ الْفَارِسِيَّةِ، ثُمَّ إِنَّ بَعْضَ الْإِخْوَانِ الْمُتَطَلِّبِينَ لِلْعُلُومِ طَلَبُوا مِنِّي أَنْ أُتَرْجِمَهُ إِلَى اللُّغَةِ الْعَرَبِيَّةِ، فَوَفَّقْتُ لِمَا أَحَبُّوا، وَحَذَفْتُ مِنْهُ بَعْضَ الْفُصُولِ الَّتِي لَا مَعْنَى لَهَا، وَشَرَعْتُ فِي الْعَمَلِ بِمُعَاوَنَةِ الْعَلِيِّ الْقَدِيرِ.

English Translation

Praise be to God, who by originating hidden realities perfected the greatest gifts, and by creating existent things brought them from the concealment of non-existence to the testimony of existence, and placed in the goodness of things sublime secrets and a mighty dominion. Blessings and peace be upon our master Muhammad and upon his family and all his companions.

Now then: I had composed, in past years, a treatise containing all our discourse on the *complete quadrilateral figure* with proofs, and appended to it what takes its place and relates to it, and that was in the Persian language. Then some brethren who seek knowledge asked me to translate it into the Arabic language, and I complied with their wish, and omitted from it some sections that had no essential meaning, and set to work with the assistance of the Exalted, the All-Powerful.

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Structure of the Treatise

Book	Contents
Book I	*On Compound Ratios and Their Rules* — On compound ratios and their rules, in fourteen propositions.
Book II	*On the Plane Quadrilateral Figure* — On the plane quadrilateral figure and the ratios found therein, in eleven chapters.
Book III	*Introduction to the Spherical Quadrilateral* — Introduction to the theory of the spherical quadrilateral and what completes the utility to be drawn from this figure, in three chapters.

Book	Contents
Book IV	*On the Spherical Quadrilateral* — On the spherical quadrilateral and the ratios found therein, in five chapters.
Book V	*On Computational Methods* — Exposition of the methods which take the place of the theory of the quadrilateral, for what concerns the knowledge of arcs of great circles, in seven chapters.

1 Book I: On Compound Ratios

General Rule

الباب الأول: في أصول الأسباب المركبة

Book I: On the Principles of Compound Ratios

وَنَقُولُ: إِنَّ أَصْلَ الْمَعْنَى فِي الْقَوْلِ إِنَّ هٰذَا النِّسْبَةَ مُرَكَّبَةٌ مِنْ هٰذَيْنِ النِّسْبَتَيْنِ أَوْ هٰذَا النِّسْبَةَ مُفَرَّقَةٌ إِلَى هٰذَيْنِ النِّسْبَتَيْنِ: أَنَّ النِّسْبَةَ الْمُرَكَّبَةَ نِسْبَتَانِ مَفْرُوقَتَانِ مِنْ نِسْبَةٍ، فَإِذَا وُضِعَتْ إِحْدَاهُمَا فِي الْأُخْرَى عَلَى التَّكْرَارِ حَصَلَ نِسْبَةٌ مَعْيُنَةٌ.

We say: The basis of the meaning in saying that “this ratio is composed of these two ratios,” or that “this ratio is decomposed into these two ratios,” is that the compound ratio consists of two ratios separated from one ratio, so that when one is placed in the other by repetition, a certain ratio is obtained.

وَأَمَّا التَّفْرِيقُ: فَهُوَ أَنْ تَكُونَ النِّسْبَةُ مَفْرُوقَةً إِلَى نِسْبَتَيْنِ فَيَكُونُ كُلُّ وَاحِدَةٍ مِنْهُمَا بِمَنْزِلَةِ الْجُزْءِ مِنَ الْكُلِّ.

As for decomposition: it is that the ratio be divided into two ratios, so that each one of them is in the position of a part from the whole.

فَإِذَا وُجِدَ هٰذَا فَلْنَقُلْ: أَيُّ نِسْبَةٍ كَانَتْ لِثَلَاثَةِ مُتَجَانِسَاتٍ إِلَى إِحْدَاهُنَّ فَهِيَ مُرَكَّبَةٌ مِنْ نِسْبَتِهِ إِلَى الثَّالِثَةِ وَنِسْبَةِ تِلْكَ الثَّالِثَةِ إِلَى الْأُولَى.

When this is established, let us say: Any ratio whatsoever of three homogeneous quantities to one of them is composed of its ratio to the third and the ratio of that third to the first.

[Just as to give exact measure of discontinuous quantity we make use of certain properties essential to continuous magnitude, such as infinite divisibility, so also in the measure of continuous magnitude we bring in certain properties essential to discontinuous quantity. Such is the supposition according to which continuous magnitude is composed of discrete or discontinuous units which we imagine in order to arrive at the measure of continuous magnitude.]

* * *

1.1 First Proposition

المسألة الأولى: فَلْيَكُنْ ثَلَاثَةُ مُتَجَانِسَاتٍ أ وَ ب وَ ج، فَأَقُولُ: إِنَّ نِسْبَةَ أ إِلَى ب تُسَاوِي نِسْبَةَ أ إِلَى ج فِي نِسْبَةِ ج إِلَى ب.

Proposition I. Let there be three homogeneous quantities A, B, and C. I say that the ratio of A to B is equal to the ratio of A to C multiplied by the ratio of C to B.

In modern notation:

$$\frac{A}{B}=\frac{A}{C}\times\frac{C}{B},\quad \frac{A}{C}=\frac{A}{B}\times\frac{B}{C},\quad \frac{B}{C}=\frac{B}{A}\times\frac{A}{C}$$

وَلْنَجْعَلْ وَاحِدَةً تَقِيسُ هٰذِهِ الْمُتَجَانِسَاتِ، وَلْتَكُنْ تِلْكَ الْوَاحِدَةُ تَقِيسُ د كَمَا تَقِيسُ أ ج، وَتَقِيسُ ت كَمَا تَقِيسُ ج ب، وَتَقِيسُ هـ كَمَا تَقِيسُ أ ب. فَد هِيَ مِقْدَارُ نِسْبَةِ أ إِلَى ج، وَت هِيَ مِقْدَارُ نِسْبَةِ ج إِلَى ب، وَهـ هِيَ مِقْدَارُ نِسْبَةِ أ إِلَى ب.

Let us suppose that a unit measures these quantities, and let that unit measure D as A measures C; let it measure T as C measures B; and let it measure H as A measures B. Then D is the *quantity* of the ratio $A : C$; T is the quantity of the ratio $C : B$; and H is the quantity of the ratio $A : B$.

وَهٰذَا لِأَنَّ كُلَّ نِسْبَةٍ لَهَا اسْمٌ وَاحِدٌ مَعَ الْعَدَدِ الَّذِي تَقِيسُهُ الْوَاحِدَةُ بِقِيَاسِ مَا يَقِيسُهُ طَرَفُهَا الْأَوَّلُ مِنْ طَرَفِهَا الثَّانِي. وَالْعَدَدُ الَّذِي لَهُ اسْمُ النِّسْبَةِ هُوَ مِقْدَارُهَا.

This is because every ratio has one name with the number that the unit measures in the same manner as the first term of the ratio measures the second. And the number that has the name of the ratio is its quantity.

فَإِذَا كَانَ هٰكَذَا فَقَدْ بَيَّنَّا أَنَّ التَّرْكِيبَ نَحْوُ التَّكْرَارِ، وَالتَّكْرَارُ بَيْنَ عَدَدَيْنِ هُوَ ضَرْبُ أَحَدِهِمَا فِي الْآخَرِ، فَهـ هُوَ أَيْضًا حَاصِلُ ضَرْبِ د فِي تَ.

When this is so, we have explained that composition is a kind of repetition, and repetition between two numbers is the multiplication of one of them by the other; therefore H is also the product of D multiplied by T.

[Al-Tusi here establishes the fundamental property of compound ratios: that any ratio between two quantities can be decomposed through an intermediate third quantity. This is the foundation upon which all subsequent propositions rest. The proof proceeds by proportion theory:]

إِنَّ أ:ج::وَاحِدَة:د، وَبِالْعَكْسِ ج:أ::د:وَاحِدَةٍ. وَأَيْضًا أ:ب::وَاحِدَة:هـ، فَبِالنِّسْبَةِ الْمُرَتَّبَةِ يَكُونُ ج:ب::د:هـ، وَج:ب::وَاحِدَة:تَ، فَوَاحِدَة:تَ::د:هـ، فَتَعْلَمُ أَنَّ تَ فِي د تُسَاوِي هـ فِي وَاحِدَةٍ، وَحَاصِلُ كُلِّ عَدَدٍ فِي وَاحِدَةٍ هُوَ ذٰلِكَ الْعَدَدُ، فَهـ تُسَاوِي تَ فِي د، فَنِسْبَةُ أ إِلَى ب تُسَاوِي نِسْبَةَ أ إِلَى ج فِي نِسْبَةِ ج إِلَى ب. وَهُوَ الْمَطْلُوبُ.

Indeed, $A : C :: 1 : D$, and conversely $C : A :: D : 1$. Also $A : B :: 1 : H$; therefore by the *ordered proportion* we obtain $C : B :: D : H$, and $C : B :: 1 : T$, so $1 : T :: D : H$, which gives finally $T \times D = H \times 1$. Now the product of any number by the unit is that number itself; therefore $H = T \times D$, that is,

$$\frac{A}{B} = \frac{A}{C} \times \frac{C}{B},$$

quod erat demonstrandum.

* * *

1.2 Second Proposition

المسألة الثانية: نِسْبَةُ أَحَدِ الثَّلَاثَةِ أ وَ ب وَ ج إِلَى آخَرَ تُسَاوِي نِسْبَةَ الثَّالِثِ إِلَى ذٰلِكَ الْآخَرِ فِي نِسْبَةِ هٰذَا إِلَى الْأَوَّلِ.

Proposition II. The ratio of any one of the three quantities A, B, C to any other is equal to the ratio of the third to that latter composed with that of that quantity to the first.

$$\frac{A}{B} = \frac{A}{C} \times \frac{C}{B} = \frac{C}{B} \times \frac{A}{C}$$

for the composition of ratios is like the multiplication of numbers; one may transpose the factors.

* * *

1.3 Summary of Propositions III–XIV

Summary of Propositions III–XIV / ملخص المسائل من الثالثة إلى الرابعة عشر

Prop. III. Every ratio composed of two arbitrary ratios is also composed of two other ratios that can be deduced from those by means of homogeneous quantities taken in the first and in the second ratio.

Prop. IV. If any quantity measures two others, the ratio of these two latter quantities between them is equal to the ratio composed of the inverse of the first ratio with the second ratio.

Prop. V. The ratio composed of two arbitrary ratios is also composed of the inverse of the first of these two ratios with the inverse of the second.

Prop. VI. Every composed ratio can be decomposed into simple ratios in two ways: either as fractions having unity for numerator, or as expressions having unity for denominator.

Prop. VII. If one has six quantities of which the composed ratios give rise to other composed ratios, then to decompose any composed ratio it suffices to know the quantities of a single column.

Prop. VIII. Every composed ratio gives rise to 9 other composed ratios and to 18 expressions of component ratios, or with their inverses 36 expressions.

Prop. IX. Two ways of decomposing a composed ratio: by making the two terms of the second ratio precede, or by making them follow the two terms of the first ratio.

Prop. X. Every composed ratio gives rise to 9 other composed ratios and to 18 expressions of component ratios, or with their inverses 36 expressions of composed ratios. This number can be carried to 72.

Prop. XI. If two of the six quantities taken each in one of the two members are equal, the 4 others are in proportion.

Prop. XII. If the two equal quantities belong to the same member, the other 4 are not proportional.

Prop. XIII. Every simple ratio can be considered as composed if one multiplies it by a ratio of identity, e.g. if $\frac{A}{B} = \frac{C}{D}$, one also has $\frac{A}{B} = \frac{C}{D} \times \frac{D}{D}$.

Prop. XIV. Every simple ratio of identity is equivalent to any ratio whatsoever multiplied by its inverse: $\frac{A}{A} = \frac{X}{X} \times \frac{X}{X}$.

2 Book II: On the Plane Quadrilateral Figure

المقالة الثانية: في شكل القطاع والنسب الموجودة فيه

Book II: On the Figure of Secants and the Ratios Found Therein

وَهٰذَا الشَّكْلُ يُسَمَّى شَكْلَ الْقِطَّاعِ لِأَنَّهُ مُنشَأٌ مِنْ أَرْبَعَةِ خُطُوطٍ مُقَاطِعَةٍ، وَيُسَمَّى أَيْضًا شَكْلَ الْقَطَائِعِ الْمُكَمَّلَةِ. وَقَدْ أَطْلَقَ عَلَيْهِ بَعْضُ الْمُتَأَخِّرِينَ اسْمَ الْمُخَمَّسِ لِأَنَّهُ يَحْتَوِي عَلَى خَمْسَةِ أَضْلَاعٍ فِي الْحَقِيقَةِ إِذَا أُضِيفَ إِلَيْهِ قُطْرٌ وَاحِدٌ.

This figure is called *shakl al-qaṭṭa'* (the figure of secants) because it is generated from four intersecting lines, and is also called the figure of *complete secants*. Some later scholars have given it the name of "pentagonal figure" because it contains five sides in reality when one diagonal is added.

وَقَدْ أَخْطَأَ حُسَامُ الدِّينِ فِي قَوْلِهِ إِنَّ هٰذَا الشَّكْلَ إِنَّمَا يَكُونُ فِي تِسْعَةِ أَشْكَالٍ، وَهُوَ فِي اثْنَيْ عَشَرَ شَكْلًا. وَقَدْ بَيَّنَّا ذٰلِكَ فِي هٰذَا الْمَكَانِ وَأَثْبَتْنَاهُ بِالْبُرْهَانِ.

Note. Hussam al-Din erred in saying that this figure only occurs in nine configurations; it is in fact twelve figures. We have demonstrated this at this point and proved it rigorously.

2.1 The Twelve Figures of the Quadrilateral

[The quadrilateral figure (shakl al-qaṭṭa') is generated by the intersection of four straight lines. The twelve configurations are classified into three groups of four:]

I. The point of intersection of two lines falls between the two points of intersection on each of them.

II. The point of intersection falls outside one or both segments.

III. More complex configurations with all points of intersection external.

وَقَدْ بَيَّنَ الْمُتَقَدِّمُونَ أَنَّ نِسْبَةَ خَطِّ أ ب إِلَى خَطِّ ب د مُرَكَّبَةٌ مِنْ نِسْبَةِ أ هـ إِلَى هـ ج وَنِسْبَةِ ج و إِلَى و د. وَهٰذَا مَا يَتَضَمَّنُهُ كُلُّ وَاحِدٍ مِنَ الْأَشْكَالِ الثَّنَا عَشَرَ.

The ancients have demonstrated that the ratio of line AB to line BD is composed of the ratio of AE to EC and the ratio of CF to FD. And this is what is contained in each one of the twelve figures.

فَمِنَ النُّقْطَةِ أ نَخْرُجُ خَطًّا أ هـ مُوَازِيًا لِخَطِّ ج د حَتَّى يُقَابِلَ خَطَّ ب هـ. فَنَحْصُلُ عَلَى أ هـ:ج و::أ هـ:هـ ج بِسَبَبِ الْمُثَلَّثَيْنِ الْمُتَشَابِهَيْنِ هـ أ هـ وَ هـ ج و.

From point A draw the line AH parallel to CD until it meets BE. You will have $AH : CF :: AE : EC$ because of the similar triangles EAH, ECF; and $AH : FD :: AB : BD$ because of the similar triangles AFH, DBF.

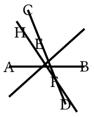
$$\frac{AH}{FD} = \frac{AE}{EC} \times \frac{CF}{DF}$$

(replacing AH by its value drawn from the first proportion,) therefore also

$$\frac{AB}{BD} = \frac{AE}{EC} \times \frac{CF}{FD}.$$

Classification of the Twelve Figures

I



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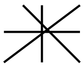

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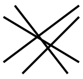

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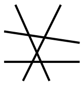

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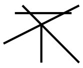
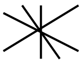
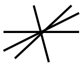


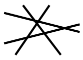
VI



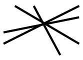
VII



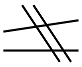
VIII



IX

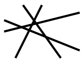


X



XI

XII



Note. The geometers establish the ratio of these twelve figures by means of a single proposition and demonstration that applies to each of them, saying that the ratio of line AB to line BD is composed of the ratio of AE to EC and that of CF to FD.

3 Book III: Introduction to the Spherical Quadrilateral

المقالة الثالثة: في تمام الانتفاع بشكل القطاع الكروي

Book III: On Completing the Utility of the Spherical Quadrilateral Figure

الفصل الأول: فِي تَقْسِيمِ الْمُحِيطِ وَالْقُطْرِ. التَّقْسِيمُ الْخَاصُّ لِلْبِيرُونِيّ. حَلُّ الْمُثَلَّثَاتِ. طَرِيقَةُ الْأَقْوَاسِ وَالْأَوْتَارِ. الْمُثَلَّثَاتُ الْقَائِمَةُ. كَيْفَ تُرَدُّ أَنْوَاعُ هٰذِهِ الْمُثَلَّثَاتِ إِلَى الْمُثَلَّثِ الْمُعْتَادِ الْمَرْسُومِ فِي الدَّائِرَةِ الَّتِي قُطْرُهَا ١٢٠.

Chapter I. On the division of the circumference and the diameter. The special division of Albiruni. Resolution of triangles. Method of arcs and chords. Right triangles. How the different cases of these triangles are reduced to the normal triangle inscribed in the circle whose diameter is 120.

طَرِيقَةُ الْأَقْوَاسِ وَالْجُيُوبِ. بُرْهَانُ الْقَاعِدَةِ الْأَصْلِيَّةِ: وَهِيَ أَنَّهُ فِي كُلِّ مُثَلَّثٍ مُسْتَقِيمٍ نِسْبَةُ أَضْلَاعِهِ تُسَاوِي نِسْبَةَ جِيُوبِ زَوَايَاهَا الْمُقَابِلَةِ لَهَا.

Method of arcs and sines. Demonstration of the fundamental theorem: that in every rectilinear triangle the ratio of its sides is equal to the ratio of the sines of its opposite angles.

3.1 On Chords, Arcs, and Sines

Proposition 3.1 (Fundamental Theorem of Sines). *In any plane triangle, the ratio of the sides is equal to the ratio of the sines of the opposite angles:*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

هٰذِهِ هِيَ الْقَاعِدَةُ الْأَصْلِيَّةُ الَّتِي بَنَى عَلَيْهَا جَمِيعُ مَا يَأْتِي فِي عِلْمِ الْمُثَلَّثَاتِ الْكُرَوِيَّةِ. وَقَدْ أَثْبَتَّهَا بِأَكْثَرَ مِنْ وَجْهٍ وَطَرِيقَةٍ.

This is the fundamental theorem upon which all that follows in the science of spherical triangles is built. I have proved it by more than one method and way.

وَقَدْ طَبَّقَ الْمُتَقَدِّمُونَ هٰذَا عَلَى حِسَابِ فَلَكِيَّاتِهِمْ وَأَعْمَالِهِمْ فِي عِلْمِ الْهَيْئَةِ، وَلَمْ يُفَرِّقُوا بَيْنَ الْمُثَلَّثِ الْمُسْتَقِيمِ وَالْمُثَلَّثِ الْكُرَوِيِّ إِلَّا عَلَى وَجْهِ التَّقْرِيبِ.

The ancients applied this to their astronomical calculations and their works in the science of configuration (astronomy), and did not distinguish between the plane triangle and the spherical triangle except approximately.

3.2 The Fundamental Problem

المسألة الأساسية

The Fundamental Problem

الفصل الثالث: الْمَسْأَلَةُ الْأَصْلِيَّةُ: مَعْرِفَةُ جَمْعِ قَوْسَيْنِ أَوْ تَفْرِيقِهِمَا مَعَ مَعْرِفَةِ نِسْبَةِ جِيْبَيْهِمَا، فَإِيجَادُ الْقَوْسَيْنِ. وَإِثْبَاتُ الْحِسَابِ لِلْحَالَتَيْنِ.

Chapter III. *The fundamental problem:* knowing the sum of two arcs or their difference together with knowing the ratio of their sines, then finding the two arcs. Establishment of the calculation for the two cases.

[This fundamental problem — given $\alpha \pm \beta$ *and* $\dfrac{\sin\alpha}{\sin\beta}$*, find* α *and* β *— is of central importance in spherical astronomy. Al-Tusi gives a complete algebraic solution, reducing it to the solution of a quadratic equation. Abu Nasr ibn ʻIraq had previously solved it geometrically.]*

4 Book IV: On the Spherical Quadrilateral

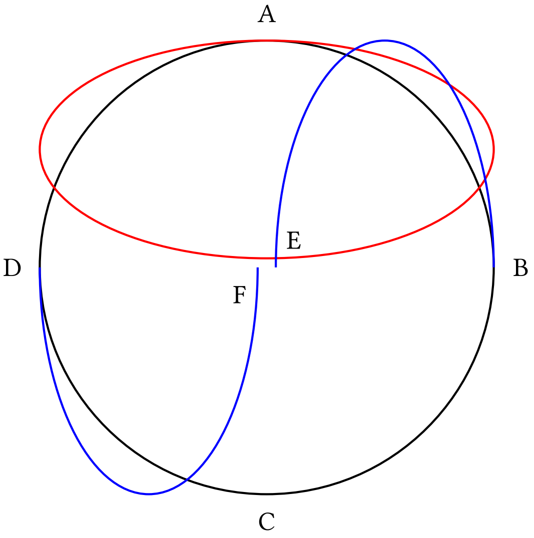
المقالة الرابعة: في شكل القطاع الكروي والنسب الموجودة فيه

Book IV: On the Spherical Quadrilateral and the Ratios Found Therein

الفصل الأول: فِي تَعْرِيفِ شَكْلِ الْقِطَّاعِ الْكُرَوِيِّ. الْمُسْتَدِيرَةُ تُقَاطِعُ أَرْبَعَةُ دَوَائِرِ كُبْرَى عَلَى سَطْحِ الْكُرَةِ فَتُنْشِئُ شَكْلًا يُسَمَّى شَكْلَ الْقِطَّاعِ الْكُرَوِيِّ. كُلُّ وَاحِدٍ مِنَ الْأَرْبَعَةِ أَضْلَاعٍ هُوَ قَوْسُ دَائِرَةٍ كُبْرَى. وَالزَّوَايَا تُقَاسُ بِأَقْوَاسِ الدَّوَائِرِ الْكُبْرَى الْمُنْحَنَاةِ عِنْدَ رُؤُوسِهَا عَلَى الْأَضْلَاعِ الْمُقَابِلَةِ.

Chapter I. Four great circles intersect on the surface of a sphere generating a figure called *the spherical quadrilateral.* Each of the four sides is an arc of a great circle. The angles are measured by the arcs of great circles drawn through their vertices perpendicular to the opposite sides.

4.1 Definition of the Spherical Quadrilateral



وَهٰذَا الشَّكْلُ هُوَ الَّذِي يُسَمِّيهِ الْمُتَقَدِّمُونَ شَكْلَ الْقِطَّاعِ لِأَنَّ كُلَّ ضَلْعَيْنِ مِنْ أَضْلَاعِهِ يَلْتَقِيَانِ فِي قَطْعٍ. وَقَدْ عَرَّفَهُ بُطْلَمْيُوسُ فِي الْمَجِسْطِيِّ وَبَيَّنَ نِسَبَهُ.

This is the figure that the ancients call *the figure of secants* because every two of its sides meet at an intersection. Ptolemy defined it in the Almagest and explained its ratios.

وَقَدْ بَيَّنَّا أَنَّ النِّسْبَ الَّتِي ثَبَتَتْ لِشَكْلِ الْقِطَّاعِ الْمُسَطَّحِ تُطْبَقُ عَلَى جِيُوبِ شَكْلِ الْقِطَّاعِ الْكُرَوِيِّ أَيْضًا، وَهٰذَا هُوَ الْأَصْلُ الَّذِي بَنَيْنَا عَلَيْهِ كُلَّ مَا يَأْتِي.

We have demonstrated that the ratios established for the plane quadrilateral apply also to the sines of the spherical quadrilateral, and this is the principle upon which we have built all that follows.

4.2 Theorems of Ptolemy Applied to the Spherical Quadrilateral

الفصل الثاني: فِي تَطْبِيقِ أَحْكَامِ شَكْلِ الْقِطَّاعِ الْمُسَطَّحِ عَلَى جِيُوبِ شَكْلِ الْقِطَّاعِ الْكُرَوِيِّ. نِسْبَةُ بُطْلَمْيُوسَ الْمُعْلَنَةِ.

Chapter II. On the application of the theorems of the plane quadrilateral to the sines of the spherical quadrilateral. The explicit ratio of Ptolemy.

الفصل الثالث: نِسْبَةُ بُطْلَمْيُوسَ الْمُضْمَرَةِ. بُرْهَانُهَا. وَلِمَاذَا هٰذَا الْبُرْهَانُ وَإِنْ كَانَ صَحِيحًا لَيْسَ بِكَافٍ فِي الْحَالَةِ الْحَاضِرَةِ. الْقَاعِدَةُ التَّيَقُّنِيَّةُ. تَرْكِيعُ الْحَالَاتِ السَّبْعِ وَالْعِشْرِينَ الَّتِي تَرَدَّدَ فِيهَا صَاحِبُنَا إِلَى ثَلَاثَةَ عَشَرَ.

Chapter III. The implicit ratio of Ptolemy. Its demonstration. Why this demonstration, however exact it may be, is not sufficient in the present case. The preliminary lemma. Reduction of the 27 cases in which our author hesitated to 13. Examination of each case in particular.

Proposition 4.1 (The Sine Theorem for Spherical Triangles). *In any spherical triangle, the sines of the sides are in the same ratio as the sines of the opposite angles:*

$$\frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C}$$

وَمِنْ هٰذَا يَتَبَيَّنُ أَنَّ جِيبَ قَوْس أ ب إِلَى جِيبِ قَوْس أ ج كَنِسْبَةِ جِيبِ قَوْس أ د إِلَى جِيبِ قَوْس د ب، وَذٰلِكَ مَا أَرَدْنَا إِثْبَاتَهُ. وَإِنِ انْطَبَقَ أَحَدُ الْمُثَلَّثَيْنِ عَلَى الْآخَرِ فِي كُلِّ وَاحِدٍ مِنْ هٰذِهِ الْأَشْكَالِ فَالْبَيَانُ عَلَى قِيَاسِ ذٰلِكَ.

From this it is clear that the sine of arc AB is to the sine of arc AC as the sine of arc AD is to the sine of arc DB, which is what we wanted to prove. And if one of the triangles coincides with the other in any one of these configurations, the explanation follows the same pattern.

الفصل الرابع: كَيْفَ يُؤَدِّي فَحْصُ النِّسْبَتَيْنِ الْمُضْمَرَةِ وَالْمُعْلَنَةِ إِلَى مِفْتَاحِ جَمِيعِ الْبُرْهَانَاتِ الْأُخْرَى. مُقَارَنَةُ مَسْلَكِ هٰذَا الْبُرْهَانِ مَعَ قَوَاعِدِ الْقِيَاسِ الِاسْتِدْلَالِيِّ.

Chapter IV. How the examination of the implicit and explicit ratios gives the key to all other demonstrations. Comparison of the course followed in this demonstration with the formulas of deductive syllogism.

5 Book V: On Computational Methods

المقالة الخامسة: في بيان الضوابط البديلة لنظرية الشكل

Book V: Exposition of the Methods Which Replace the Theory of the Figure

الفصل الأول: مُقَاطَعَاتُ ثَلَاثِ دَوَائِرَ كُبْرَى تُنْشِئُ مُثَلَّثَاتٍ عَلَى سَطْحِ الْكُرَةِ. طَبِيعَةُ هٰذِهِ الْمُثَلَّثَاتِ. يَكْفِي مَعْرِفَةُ وَاحِدَةٍ مِنْهَا لِمَعْرِفَةِ جَمِيعِهَا.

Chapter I. The intersections of three great circles generate triangles on the surface of a sphere. Nature of these triangles. It suffices to know one of them to know them all.

الفصل الثاني: تَحْلِيلُ أَنْوَاعِ الْمُثَلَّثَاتِ الْكُرَوِيَّةِ حَسَبَ زَوَايَاهَا □عَشَرُ أَصْنَافٍ□ أَوْ حَسَبَ أَقْوَاسِهَا □عَشَرُ أَصْنَافٍ□.

Chapter II. Analysis of the different spherical triangles according to their angles (ten categories) or according to their arcs (ten categories).

5.1 On Inclinations and the Sine Theorem

وَفِي هٰذِهِ الْمُثَلَّثَاتِ يُسَمَّى قَوْسُ ب ج مَيْلَ قَوْسِ أ ج، وَهٰذَا الْمَيْلُ ب ج هُوَ نَصِيبُ قَوْسِ أ ج مِنَ الْمَيْلِ الْكُلِّيِّ لِدَائِرَةِ أ د عَلَى دَائِرَةِ أ هـ، وَهُوَ الَّذِي تَقِيسُهُ زَاوِيَةُ أ.

In these triangles, arc BC is called the *inclination* of arc AC, which inclination BC is but the part of arc AC in the total inclination of circle AD to circle AE, which the angle A measures. And if arc BC is related to arc AB, then BC is called the second inclination.

وَهٰذَا مَا يُسَمِّيهِ أَبُو رَيْحَانَ الْبِيرُونِيُّ الْعَرْضَ، فَقَوْسُ ب ج يَكُونُ الْمَيْلَ الْأَوَّلَ بِالنِّسْبَةِ لِأ ج، وَالْمَيْلَ الثَّانِيَ بِالنِّسْبَةِ لِقَوْسِ أ ب.

This is what Abu Rayhan al-Biruni calls the *width*; so arc BC would be the first inclination with respect to AC, and the second inclination with respect to arc AB.

فَفِي هٰذَا الْمَعْنَى نَقُولُ: إِنَّ نِسَبَ جُيُوبِ الْمَيَالِي تُسَاوِي نِسَبَ جُيُوبِ أَقْوَاسِهَا، وَنِسْبَةُ جِيبِ كُلِّ مَيْلٍ إِلَى جِيبِ قَوْسِهِ تُسَاوِي نِسْبَةَ جِيبِ مَيْلٍ آخَرَ إِلَى جِيبِ قَوْسِهِ.

In this manner of speaking we say that the ratios of the sines of the inclinations are equal to the ratios of the sines of their arcs, and the ratio of the sine of any inclination to the sine of its arc is equal to the ratio of the sine of another inclination to the sine of its arc.

[In modern notation:]
If BC is the inclination of AC, and LM is the inclination of AE, then:

$$\frac{\sin BC}{\sin LM} = \frac{\sin AC}{\sin AE}$$

وَإِنْ لَمْ يَنْطَبِقْ مُثَلَّثُ أ ب ج عَلَى مُثَلَّثِ د أ هـ، لٰكِنْ كَانَتْ زَاوِيَةُ أ فِي هٰذَيْنِ الْمُثَلَّثَيْنِ مُتَسَاوِيَةً، وَكَانَتْ زَاوِيَتَا ب وَ هـ قَائِمَتَيْنِ، فَالْقَاعِدَةُ ثَابِتَةٌ.

If triangle ABC does not coincide with triangle DAE, but nevertheless angle A is equal in these two triangles and angles B and E are right, the theorem remains proved nonetheless.

* * *

5.2 The Eight Demonstrations of the Sine Theorem

البرهان على القاعدة في ثمانية وجوه

Eight Demonstrations of the Fundamental Theorem

سِتَّةُ بُرْهَانَاتٍ فِي تَسَاوِي نِسَبِ الْجِيُوبِ وَالْأَقْوَاسِ وَأَظِلَّةِ الزَّوَايَا وَنَحْوِهَا. تَعْمِيمُ هٰذَا الْأَصْلِ عَلَى جَمِيعِ الْمُثَلَّثَاتِ الْكُرَوِيَّةِ. تَابِعَاتٌ.

Six demonstrations concerning the equality of ratios of sines, arcs, tangents of angles, etc. Generalization of this principle to all spherical triangles. Corollaries.

وَقَدْ حَاوَلَ بَعْضُهُمْ أَنْ يَرْفَعُوا شُبْهَاتٍ عَلَى صِحَّةِ اسْتِعْمَالِ الظِّلِّ لِسُرْعَةِ زِيَادَتِهِ بَعْدَ ٤٥ دَرَجَةً. رَدُّ هٰذِهِ الشُّبْهَاتِ وَتَحْلِيلُ الْحَالَاتِ الْمُخْتَلِفَةِ الَّتِي يَجِبُ فِيهَا اسْتِبْدَالُ الظِّلِّ بِظِلِّ التَّمَامِ.

Objections that some have attempted to raise against the legitimacy of the use of tangents by reason of their rapid growth beyond 45° — Refutation of these objections and analysis of the different cases in which one must replace the tangent by the cotangent.

5.3 The Figure of Shadows (Tangent Theorem)

شكل الظلال □قاعدة الظلال□

The Figure of Shadows (Tangent Theorem)

الفصل السادس: فِي الشَّكْلِ الْمُظَلَّلِ □قَاعِدَةُ الظِّلَالِ□. اخْتِرَاعُهُ مِنْ قِبَلِ أَبِي الْوَفَاءِ. تَعْرِيفُ الظِّلِّ الْأَوَّلِ وَالثَّانِي □الظِّلُّ وَظِلُّ التَّمَامِ، الْقَاطِعُ وَقَاطِعُ التَّمَامِ□. أَشْكَالُ مُخْتَلِفَةٌ.

Chapter VI. *On the shaded figure (Theorem of Tangents).* The invention thereof is due to Abu’l-Wafa. Definition of first and second shadows (tangent and cotangent, secant and cosecant). Different formulas.

الظِّلُّ الْأَوَّلُ لِلْقَوْسِ هُوَ جِيبُ اِرْتِفَاعِهِ أَوْ مَا يُسَمِّيهِ الْمُتَأَخِّرُونَ جَيْبَ التَّمَامِ. وَالظِّلُّ الْأَوَّلُ هُوَ الظِّلُّ؛ وَالظِّلُّ الثَّانِي هُوَ ظِلُّ التَّمَامِ.

The *first shadow* of an arc is the sine of its altitude or what the moderns call its *cosine*. The first shadow is the *tangent*; the second shadow is the *cotangent*.

* * *

5.4 Applications to Spherical Astronomy

الفصل السابع: الْمَنَافِعُ الَّتِي يُمْكِنُ الِاسْتِفَادَةُ بِهَا مِنْ قَاعِدَةِ الْجِيُوبِ وَكَذٰلِكَ قَاعِدَةِ الظِّلَالِ فِي حَلِّ الْمُثَلَّثَاتِ الْكُرَوِيَّةِ. طُرُقٌ مُخْتَلِفَةٌ تُؤَدِّي إِلَى غَرَضٍ وَاحِدٍ.

Chapter VII. The uses that can be drawn from the sine theorem as well as from the tangent theorem for the resolution of spherical triangles. Different paths leading to the same goal.

وَأَعْتَرِفُ أَنَّنِي لَا أَعْرِفُ الطَّرِيقَ الَّتِي تُحَلُّ بِهَا هٰذِهِ الْمَسْأَلَةُ بِوَاسِطَةِ الظِّلَالِ: مَعَ مَعْرِفَةِ ثَلَاثَ زَوَايَا مُثَلَّثٍ كُرَوِيٍّ تُعَيَّنُ أَضْلَاعُهُ.

The author confesses not to know the path by which the following problem is solved by means of tangents: *“Given the three angles of a spherical triangle, determine its sides.”*

[This problem — the polar triangle case — is solved by passing to the polar triangle: if the three angles A, B, C are

given, the sides of the polar triangle are $a' = 180^\circ - A$*,* $b' = 180^\circ - B$*,* $c' = 180^\circ - C$*, which can then be solved by the sine theorem, and the angles of the polar triangle give the sides of the original triangle.]*

Epilogue / Historical and Biographical Notes

فِي ذِكْرِ الْعُلَمَاءِ الْمَذْكُورِينَ فِي هٰذَا الْكِتَابِ

On the Scholars Mentioned in This Book

* * *

Name	Details
Sabit-Ibn-Kourah (ثابت بن قرة)	One of the most celebrated translator works. Born 836, died 901 in Bagh No. 3374. Author of a work on the qu
Abou-Rihan Albirouni (ابو ريحان البيروني)	From Biroun, city of Kharesm. Die Khalifa No. 7420. Author of works o sine. Translator or abbreviator of th take the radius as unity. Gave a demo portionality of the sines of the arcs a spherical triangle.
Abou-Mahmoud Alhodjendi (ابو محمود الخجندي)	c. 992. Disputed with Aboul Véfa the the theorem of the proportionality o gles and the arcs.
Abou-Djafar Alhazin (ابو جعفر الخازن)	c. 1000. (Hadji Khalifa No. 4137). Th librarian.
Aboul-Fazl Al Nerizi (ابو الفضل النريزي)	(Hadji Khalifa No. 2548). One of the tators of Euclid, and author of astron Indian manner.
Houssam-Uddin-Ali-Bn-Fazlullah Essalar (حسام الدين علي بن فضل الله السالار)	Author of a treatise on the quadrilate the basis for the treatise of Nasiruddi
Abdullah-Abdul-Kiafi-Bn-Abdul-Medjid (عبدالله عبدالكافي بن عبدالمجيد)	The copyist of the manuscript and a di

Scholarly Apparatus

Note on the Edition

This typeset edition reproduces the bilingual edition published in Constantinople (Istanbul) in 1891 under the authority of the Ottoman Ministry of Public Instruction. The original edition comprised:

- **French Translation** (pp. 1–214): A complete translation by Alexandre Pacha Carathéodory (1833–1906), Ottoman diplomat and mathematician, with extensive notes and a biographical appendix on the mathematicians cited in the treatise.
- **Arabic Original** (pp. 1–178): The original Arabic text, printed in lithographic form from the manuscript belonging to Edhem Pasha, with its own pagination in Arabic numerals.

The Significance of the Work

Al-Tusi's *Kitab Shakl al-Qaṭṭaʿ* represents one of the most important contributions to trigonometry in the medieval Islamic world. The work:

1. **Systematized the theory of compound ratios** (Book I), providing a complete algebraic foundation for ratio theory that went beyond Euclid's *Elements*.
2. **Analyzed the complete quadrilateral figure** (Book II), classifying all twelve configurations generated by four intersecting lines and deriving the fundamental proportional relationships.
3. **Established the foundations of spherical trigonometry** (Books III–IV), including the sine theorem for spherical triangles and its systematic application.
4. **Developed practical computational methods** (Book V) for solving spherical triangles, which were essential for astronomical calculations.

Al-Tusi's Key Contributions to Trigonometry

- **First systematic treatment** of plane and spherical trigonometry as an independent science, separate from astronomy.
- **Sine law for plane triangles:** $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$
- **Sine law for spherical triangles:** $\dfrac{\sin a}{\sin A} = \dfrac{\sin b}{\sin B} = \dfrac{\sin c}{\sin C}$
- **Polar triangle method**: The transformation of angle-given problems to side-given problems via the polar triangle.
- **Complete classification** of spherical triangle cases (27 reduced to 13).
- **Systematic use** of tangent and cotangent functions.

Mathematical Terminology

Arabic	Transliteration	English
شَكْلُ الْقِطَّاعِ	shakl al-qaṭṭa‘	figure of secants / quadrilateral
الْقَطَائِعُ الْمُكَمَّلَةُ	al-qaṭa’i‘ al-mukammala	complete secants
نِسْبَةٌ مُرَكَّبَةٌ	nisba murakkaba	compound ratio
قُطْرٌ	quṭr	diagonal
ضَلْعٌ	ḍal‘	side
قَاعِدَةٌ	qā‘ida	base / theorem
جِيبُ قَوْسٍ	jayb qaws	sine of an arc
جِيبُ التَّمَامِ	jayb al-tamām	cosine
ظِلُّ قَوْسٍ	ẓill qaws	tangent
ظِلُّ التَّمَامِ	ẓill al-tamām	cotangent
مُثَلَّثٌ كُرَوِيٌّ	muthallath kurawī	spherical triangle
مُثَلَّثٌ مُسْتَقِيمٌ	muthallath mustaqīm	plane (rectilinear) triangle
إِكْلِيلٌ	iklīl	spherical zone / crown
زَاوِيَةٌ مُحِيطَةٌ	zāwiya muḥīṭa	exterior angle
مَيْلٌ	mayl	inclination
عَرْضٌ	‘arḍ	width / latitude
قَطْعٌ	qaṭ‘	intersection / secant

Original Arabic Text Passages

Selected Passages from the Arabic Original

The following pages reproduce key passages from the original Arabic text with their distinctive mathematical terminology and geometric structure.

* * *

المقالة الأولى: في الأسباب المركبة وأحكامها

قَالَ الْمُصَنِّفُ رَحِمَهُ اللهُ تَعَالَى: نُقُولُ إِنَّ أَصْلَ الْمَعْنَى فِي الْقَوْلِ إِنَّ هٰذَا النِّسْبَةَ مُرَكَّبَةٌ مِنْ هٰذَيْنِ النِّسْبَتَيْنِ أَوْ هٰذَا النِّسْبَةَ مُفَرَّقَةٌ إِلَى هٰذَيْنِ النِّسْبَتَيْنِ: أَنَّ النِّسْبَةَ الْمُرَكَّبَةَ نِسْبَتَانِ مَفْرُوقَتَانِ مِنْ نِسْبَةٍ، فَإِذَا وُضِعَتْ إِحْدَاهُمَا فِي الْأُخْرَى عَلَى التَّكْرَارِ حَصَلَ نِسْبَةٌ مَعْيُنَةٌ.

The author (may God Exalted have mercy on him) says: We say that the basis of the meaning in saying that this ratio is composed of these two ratios, or that this ratio is decomposed into these two ratios, is that the compound ratio consists of two ratios separated from one ratio, so that when one is placed in the other by repetition, a certain ratio is obtained.

وَأَمَّا التَّفْرِيقُ: فَهُوَ أَنْ تَكُونَ النِّسْبَةُ مَفْرُوقَةً إِلَى نِسْبَتَيْنِ فَيَكُونُ كُلُّ وَاحِدَةٍ مِنْهُمَا بِمَنْزِلَةِ الْجُزْءِ مِنَ الْكُلِّ.

As for decomposition: it is that the ratio be divided into two ratios, so that each one of them is in the position of a part from the whole.

فَإِذَا وُجِدَ هٰذَا فَلْنَقُلْ: أَيُّ نِسْبَةٍ كَانَتْ لِثَلَاثَةِ مُتَجَانِسَاتٍ إِلَى إِحْدَاهُنَّ فَهِيَ مُرَكَّبَةٌ مِنْ نِسْبَتِهِ إِلَى الثَّالِثَةِ وَنِسْبَةِ تِلْكَ الثَّالِثَةِ إِلَى الْأُولَى.

When this is established, let us say: Any ratio whatsoever of three homogeneous quantities to one of them is composed of its ratio to the third and the ratio of that third to the first.

فَلْيَكُنْ ثَلَاثَةُ مُتَجَانِسَاتٍ أ وَ ب وَ ج، فَأَقُولُ: إِنَّ نِسْبَةَ أ إِلَى ب تُسَاوِي نِسْبَةَ أ إِلَى ج فِي نِسْبَةِ ج إِلَى ب، وَنِسْبَةَ أ إِلَى ج تُسَاوِي نِسْبَةَ أ إِلَى ب فِي نِسْبَةِ ب إِلَى ج، وَنِسْبَةَ ب إِلَى ج تُسَاوِي نِسْبَةَ ب إِلَى أ فِي نِسْبَةِ أ إِلَى ج، وَهَكَذَا.

Let there be three homogeneous quantities A, B, and C. Then I say: the ratio of A to B is equal to the ratio of A to C multiplied by the ratio of C to B, and the ratio of A to C is equal to the ratio of A to B multiplied by the ratio of B to C, and the ratio of B to C is equal to the ratio of B to A multiplied by the ratio of A to C, and so on.

* * *

المقالة الثانية: في شكل القطاع والنسب الموجودة فيه

وَهٰذَا الشَّكْلُ يُسَمَّى شَكْلَ الْقِطَّاعِ لِأَنَّهُ مُنشَأٌ مِنْ أَرْبَعَةِ خُطُوطٍ مُقَاطِعَةٍ، وَيُسَمَّى أَيْضًا شَكْلَ الْقَطَائِعِ الْمُكَمَّلَةِ. وَقَدْ أَطْلَقَ عَلَيْهِ بَعْضُ الْمُتَأَخِّرِينَ اسْمَ الْمُخَمَّسِ لِأَنَّهُ يَحْتَوِي عَلَى خَمْسَةِ أَضْلَاعٍ فِي الْحَقِيقَةِ إِذَا أُضِيفَ إِلَيْهِ قُطْرٌ وَاحِدٌ.

This figure is called the figure of secants (*shakl al-qaṭṭa*ʿ) because it is generated from four intersecting lines, and is also called the figure of complete secants. Some later scholars have called it the "pentagonal figure" because it contains five sides in reality when one diagonal is added.

وَقَدْ أَخْطَأَ حُسَامُ الدِّينِ فِي قَوْلِهِ إِنَّ هٰذَا الشَّكْلَ إِنَّمَا يَكُونُ فِي تِسْعَةِ أَشْكَالٍ، وَهُوَ فِي اثْنَيْ عَشَرَ شَكْلًا. وَقَدْ بَيَّنَّا ذٰلِكَ فِي هٰذَا الْمَكَانِ وَأَثْبَتْنَاهُ بِالْبُرْهَانِ.

Hussam al-Din erred in saying that this figure only occurs in nine configurations; it is in fact twelve figures. We have demonstrated this at this point and proved it rigorously.

* * *

المقالة الخامسة: في الضوابط البديلة لنظرية الشكل

الضَّرْبُ الْأَوَّلُ وَلْيَكُنِ الْمَعْلُومُ وَتَرُ الْقَاعِدَةِ وَوَتَرُ النَّهَايَةِ الْآخَرِ وَضَلْعٌ مُجَاوِرٌ لِإِحْدَاهُمَا فَلْنَفْرِضْ أَنَّهُ أ ج وَالْمَعْلُومُ أَيْضًا ضَلْعٌ آخَرُ مُجَاوِرٌ لِتِلْكَ الْقَاعِدَةِ فَلْيَكُنْ د أ، وَلْيَكُنِ الْمَجْهُولُ بَقِيَّةُ جَوَانِبِ هٰذَا الشَّكْلِ.

The first case: Let the known quantities be the base, the other extreme, and a side adjacent to one of them. Let us suppose that it is AC, and let the known quantity also be another side adjacent to that base, let it be DA, and let the unknown be the remaining sides of this figure.

وَمِنْ هٰذَا يَتَبَيَّنُ أَنَّ جِيبَ قَوْس أ ب إِلَى جِيبِ قَوْس أ ج كَنِسْبَةِ جِيبِ قَوْس أ د إِلَى جِيبِ قَوْس د ب، وَذٰلِكَ مَا أَرَدْنَا إِثْبَاتَهُ وَإِنِ انْطَبَقَ أَحَدُ الْمُثَلَّثَيْنِ عَلَى الْآخَرِ فِي كُلِّ وَاحِدٍ مِنْ هٰذِهِ الْأَشْكَالِ فَكَانَ الْبَيَانُ بِحَسَبِهِ عَلَى قِيَاسِ ذٰلِكَ.

From this it is clear that the sine of arc AB is to the sine of arc AC as the sine of arc AD is to the sine of arc DB, which is what we wanted to prove. And if one of the triangles coincides with the other in any one of these configurations, then the explanation follows accordingly.

* * *

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End of the Treatise / نهاية الكتاب

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UNIVERSITY OF MICHIGAN STUDIES

Humanistic Series, Volume XI

Contributions to the History of Science

Robert of Chester's Latin Translation of the Algebra of Al-Khowarizmi

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BILINGUAL EDITION

Latin Text & English Translation in Parallel Columns

LEFT COLUMN:
Robert of Chester's 12th-century Latin translation (Segovia, 1145)

RIGHT COLUMN:
English translation by Louis Charles Karpinski (1915)

Edited with an Introduction, Critical Notes
and an English Version

By

Louis Charles Karpinski

University of Michigan

New York
THE MACMILLAN COMPANY

London : Macmillan and Company, Limited

1915

Preface

From the point of view of the history of science, no justification is needed for the publication of a mathematical text of the twelfth century, for the available material representing this period is meagre. A wider acquaintance with Robert of Chester's Latin translation of Al-Khowarizmi's Arabic treatise on algebra will perhaps contribute to a more just estimate of the services rendered to science by the Arabs.

In the English version I have not attempted to give a literal translation of the Latin, but rather to express the thought in a phraseology which the modern student of mathematics will find easy of comprehension; by consulting the Latin text and footnotes the reader will be able to examine Robert of Chester's own words. For the convenience of readers interested in the text I have added a Latin Glossary in which are noted many variations from the usage of classical writers.

In the Introduction I have presented a study of the significance of the treatise in the history of mathematics, and a description of the manuscripts upon which the text is based.

It is a pleasure to express indebtedness to Professor David Eugene Smith for having suggested the work; to Mr. George A. Plimpton for the generous use of his unique mathematical library; to the librarians of the libraries of Vienna and Dresden for photographic reproductions of manuscripts containing this text; to the librarian of Columbia University for the loan of the Scheybl manuscript, and to the librarian of the Cleveland Public Library for the use of works from the John G. White collection. I am much indebted also to colleagues of the University of Michigan, particularly in the Department of Latin and the Department of Mathematics.

I am under special obligation to Mr. William H. Murphy for making possible this publication.

Louis C. Karpinski.
Ann Arbor, Michigan,
November 1, 1915.

Contents

Part I

Introduction

Chapter 1

Algebraic Analysis before Al-Khowarizmi

Arabic contributions to science have, in the past, been somewhat neglected by historians. More recent studies are recognizing our indebtedness to Mohammedan scholars, who kept the embers of learning aglow while Europe was in the darkness of the Middle Ages. Much of our knowledge of Greek mathematics comes to us from Arabic sources; the early Latin versions were frequently based upon Arabic texts rather than the Greek originals. Similarly, Hindu arithmetic and astronomy were transmitted to Europe by Islam. The services of the Arabs to science were not limited to the preservation and transmission of the learning of other nations. They made independent contributions in many fields.

Among these achievements is the Arabic algebra of Al-Khowarizmi, which for centuries enjoyed wide popularity in the original, and for further centuries extended its popularity through translations and adaptations. A study of the content of this work is an excursion into mediaeval thought. By a study of the text in a form as nearly like the original as possible, we discover the reason for its long-continued appeal to the Occidental as well as the Oriental mind, its interest for the Englishman, the German, and the Italian, as well as for the Arab.

1.1 Early Equations

Simple equations of the first degree in one unknown, of the type $ax = d$, are found in the oldest mathematical text-book which we possess, the Ahmes papyrus of about 1700 B.C. This Egyptian document presents not only first degree equations together with symbols for the unknown quantity and for the operations of addition and subtraction, but also shows traces of a study of simultaneous linear equations some two thousand years before the Christian Era.

Later, but still before the golden age of Greek mathematics, the quadratic equation appears in Egypt. The problems found involve simultaneous quadratic equations, thus:

> *Another example of the distribution of a given area into squares.* If you are told to distribute 100 square ells (units of area) over two squares so that the side of one shall be $\frac{3}{4}$ of the side of the other: please give me each of the unknowns.

The solution follows by assuming the side of one square to be unity, and the other $\frac{3}{4}$. The sum of these areas is $\frac{25}{16}$, of which the root $\frac{5}{4}$ is to the required side as $\frac{5}{4}$ is to 1, whence one side is 8 and the other 6. The algebraical equivalent of this geometrical problem is, evidently,

$$x^2 + y^2 = 100, \quad x : y = 4 : 3.$$

1.2 Greek Contributions

Several propositions of Euclid present geometrical equivalents of the solution of various types of quadratic equations, not involving negative coefficients, and further study of similar problems appears in Euclid's *Data*. Of this nature are the fifth, sixth, and eleventh propositions of the second book of the Elements and the twenty-seventh, twenty-eighth, and twenty-ninth of the sixth book.

The eleventh proposition of the second book of the Elements furnishes the solution of the equation

$$x^2 + ax = a^2$$

or even more general,

$$x^2 + ax = b^2.$$

Analytical solution of the quadratic equation appears quite definitely in the works of Heron of Alexandria, who flourished about the beginning of the Christian Era. Heron states in effect that given the sum of two line segments and their product then each of the segments is known.

1.3 The Work of Diophantus

In respect to analysis Diophantus is the greatest name among the Greeks. Recently it has been established that he flourished in the third century of the Christian Era, when Greek supremacy in mathematics was waning. No doubt whatever exists that this Alexandrian was familiar with the analytical solution of the various forms of quadratic equations, neglecting negative roots and, of course, imaginary roots, which did not receive serious treatment for more than a millennium after Diophantus.

The three types of complete quadratic equations, involving only positive coefficients, are the following:

$$ax^2 + bx = c, \tag{1.1}$$

$$ax^2 + c = bx, \tag{1.2}$$

$$ax^2 = bx + c. \tag{1.3}$$

All three types appear in the *Arithmetica* of Diophantus, not systematically treated but solved as incidental to the solution of other problems.

1.4 Hindu Algebra

Some two centuries after the period of Diophantus, Aryabhata, one of the earliest Hindu mathematicians of prominence, was born (476 A.D.). Brahmagupta of Ujjain, the centre of Indian learning, wrote on algebra in the early part of the seventh century and gave a rule for the solution of quadratic equations:

> *To the absolute number multiplied by the [coefficient of the] square, add the square of half the [coefficient of the] unknown, the square root of the sum, less half the [coefficient of the] unknown, being divided by the [coefficient of the] square, is the unknown.*

In formula this corresponds to the solution

$$x = \frac{\sqrt{(b/2)^2 + ac} - b/2}{a}$$

of the equation $ax^2 + bx = c$.

Chapter 2

Al-Khowarizmi and his Treatise on Algebra

THE activity of the great Arabic mathematician Abu 'Abdallah Mohammed ibn Musa al-Khowarizmi marks the beginning of that period of mathematical history in which analysis assumed a place on a level with geometry; and his algebra gave a definite form to the ideas which we have been setting forth.

The arithmetic of Al-Khowarizmi made known to the Arabs and, through an early twelfth-century translation, to Europeans also, the Hindu art of reckoning. Any consideration of the difficulties attending arithmetical operations with the Greek letter numerals, or even with the Roman numerals, shows how essential the adoption of numerals with place value was for the development of the analytical side of mathematics.

2.1 The Life of Al-Khowarizmi

Our chief source of information in regard to the life and the writings of our Arabic author is the *Book of Chronicles* (*Kitab al-Fihrist*) by Ibn Abi Ya'qub al-Nadim. This work, which was completed about 987 A.D., gives biographies of learned men of all nations, together with lists of such of their works as were known to Al-Nadim. We quote the passage relating to our author:

> *Mohammed ibn Musa, born in Khowarizm (modern Khiva), worked in the library of the caliphs under Al-Mamun. During his lifetime, and afterward, where observations were made, people were accustomed to rely upon his tables, which were known by the name Sind-Hind (Hindu Siddhanta). He wrote: The book of astronomical tables in two editions, the first and the second; On the sun-dial; On the use of the astrolabe; On the construction of the astrolabe; The Book of Chronology.*

The bibliography of Al-Nadim does not include four works from the hand of Al-Khowarizmi which have come to us. These are his arithmetic, his algebra, a work on the quadrivium, and an adaptation of Ptolemy's geography.

The equation $x^2+10x = 39$ runs like a thread of gold through the algebras for several centuries, appearing in the algebras of Abu Kamil, Al-Karkhi and 'Omar al-Khayyami, and frequently in the works of Christian writers, centuries later.

2.2 Robert of Chester and Other Translators

WHEN towards the beginning of the twelfth century European scholars turned to Islam for light, the works of Mohammed ibn Musa came to occupy a prominent place in their studies. Contemporary with these men was Robert of Chester.

The assertion was made by Curtze that the *Liber embadorum*, Book of Measures, in Hebrew, by Abraham bar Chiyya Ha Nasi, known as Savasorda, was the first work to appear in Latin showing to the Western world how the solution of quadratic equations is accomplished. It has recently been shown by Haskins, on the basis of astronomical data in the work, that this date is undoubtedly a scribe's error for DXL, corresponding to 1145 A.D.

Robert of Chester, known to fame chiefly as the first translator of the Koran, was doubtless educated in the well-known school located at Chester. Of the more personal side of Robert's life we have but scattered facts. His nationality is established not only by his name and his return to England in 1150, but also by the direct statement made by Peter the Venerable in a letter of 1143 concerning the Koran, addressed to Bernard of Clairvaux.

To Peter the Venerable, Abbot of Cluny, who induced Robert to undertake the translation of the Koran, the latter addressed a Saracenic chronicle. Other names have been connected with this translation of the Koran, but a study of Peter's letters shows that Robert and Hermann were definitely requested to undertake the translation.

Of Robert's works the version of the Koran was completed in 1143. In a prefatory letter he states that this task was regarded by him only as a digression from his principal studies of astronomy and geometry.

Chapter 3

The Arabic Text and the Translations

The Arabic text of Al-Khowarizmi's algebra, together with an English translation, was published by Frederick Rosen in 1831. This excellent work is unfortunately out of print, and so is not available to most students of mathematics.

The Latin translation by Robert of Chester is not as faithful nor as correct as the text ascribed to Gerard of Cremona, published by Libri. Omissions, transpositions, and additions to the text are so numerous that it does not seem desirable to list them all. No evidence exists, however, that Robert's text is based upon another Arabic original than that of the Libri text. The text proper, as opposed to the illustrative problems, follows the general lines of the Arabic original.

Chapter 4

Preface and Additions Found in the Arabic Text

The Arabic text of Al-Khowarizmi's algebra published by Rosen contains an author's preface which is not found either in the translation published by Libri, or in that by Robert of Chester. As this reveals his conception of the purpose of the algebra, as well as some of the causes which led him to undertake the work, we present it here in the translation by Rosen.

The Author's Preface (Arabic)

> *In the Name of God, gracious and merciful!*
>
> *This work was written by Mohammed ben Musa, of Khowarezm. He commences it thus:*
>
> *Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness!*
>
> *That fondness for science, by which God has distinguished the Imam al Mamun, the Commander of the Faithful, that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties, — has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned.*

Chapter 5

Manuscripts of Robert of Chester's Translation

5.1 The Extant Manuscripts

Steinschneider was the first in recent times to call attention to the translation of Al-Khowarizmi's algebra made by Robert of Chester. He suggested the desirability of publishing this text, referring to the manuscript in Vienna. To this same manuscript Curtze later, and independently of Steinschneider, directed attention and also suggested the desirability of the publication of the work. Wappler, in 1887, found a second copy of the algebra in a manuscript in Dresden, while some time later David Eugene Smith acquired for the Columbia University Library a manuscript from the hand of Johann Scheybl which contains a third transcription of the algebra.

5.2 The Vienna Manuscript (V)

Codex Vindobonensis 4770 (Rec. 3246) XIV. 339. 8° (V). Fol. 1^r–12^v, *Liber restaurationis et oppositionis numeri*; our text.

5.3 The Dresden Manuscript (D)

Codex Dresdensis C. 80. Fifteenth century. Paper. By various hands. 416 pages, numbered 1–225, 227–417. Folio, bound in parchment (D).

5.4 The Columbia University Manuscript (C)

Codex Universitatis Columbiae; Columbia University Library Manuscript, X 512, Sch. 2, Q. 308 pages. (C.) Pages 71–122, *Liber Algebrae et Almucabola, continens demonstrationes aequationum regularum Algebrae*.

Abbreviations

C = Codex Universitatis Columbiae
D = Codex Dresdensis
R = Fragmentum Romain
V = Codex Vindobonensis

After the note to line 6 of the second page of the Latin text, the notes without any letter indicate the concurrence of V and D.

add. vel add.	addition(s) by Scheybl
marg.	marginal note
man.	manuscript
om.	omits
corr.	corrected
passim	throughout
supra versum	above the line
infra	below

Part II

LIBER ALGEBRAE ET ALMUCABOLA

The Book of Algebra and Almucabola

Latin Text with English Translation in Parallel Columns

Containing Demonstrations of the Rules of the Equations of Algebra

Written in Arabic by Mohammed ibn Musa Al-Khwarizmi
and translated into Latin by Robert of Chester in Segovia, 1145

Liber Algebrae et Almucabola,

continens demonstrationes aequationum regularum Algebrae

Ab incerto authore olim arabice conscriptus atque deinde a
Roberto Cestrensi, in ciuitate Secobiensi anno 1145, latino sermoni donatus.

Liber Algebrae et Almucabola, de quaestionibus arithmeticis et geometricis

> In nomine dei pii et misericordis incipit
> liber Restaurationis et Oppositionis numeri
> quem aedidit Mahomet, filius Mosi Algaurizin.

Dixit Mahomet, Laus deo creatori, qui homini contulit scientiam inueniendi vim numerorum. Considerans enim omne id quo indigent homines, ex numeris componi, inueni illud totum esse numerum, et inueni nihil aliud esse numerum, nisi quod ex vnitatibus componitur. Vnitas ergo in omni numero reperitur. Inueni autem omnem numerum ita dispositum, vt omnis numerus vnitatem excedit vsque ad decem, denarius quoque numerus ad modum vnitatis disponitur, vnde et duplicatur et triplicatur, quemadmodum factum cum vnitate. Fiuntque ex eius duplicatione, 20: et triplicatione, 30. Et sic multiplicando denarium numerum, ad centenarium peruenitur. Ita centenarius numerus duplicatur et triplicatur, sicut denarius numerus. Et sic centenarium numerum duplicando et triplicando etc. millenarius excrescit numerus. Ad hunc modum numerum millenarium secundum ordinem numerorum multiplicando, vsque ad infinitam numeri inuestigationem peruenitur.

Postea inueni numerum restaurationis et oppositionis his tribus modis esse inuentum, scilicet radicibus, substantiis et numeris.

Solus numerus tamen neque radicibus neque

The Book of Algebra and Almucabola,

Containing Demonstrations of the Rules of the Equations of Algebra

Written some time ago in Arabic by an unknown author and afterwards,
in the year 1145, put into Latin by Robert of Chester in the city of Segovia.

The Book of Algebra and Almucabola, concerning arithmetical and geometrical problems.

> In the name of God, tender and compassionate, begins the book of Restoration and Opposition of number put forth by Mohammed Al-Khowarizmi, the son of Moses.

Mohammed said, Praise God the creator who has bestowed upon man the power to discover the significance of numbers. Indeed, reflecting that all things which men need require computation, I discovered that all things involve number and I discovered that number is nothing other than that which is composed of units. Unity therefore is implied in every number. Moreover I discovered all numbers to be so arranged that they proceed from unity up to ten. The number ten is treated in the same manner as the unit, and for this reason doubled and tripled just as in the case of unity. Out of its duplication arises 20, and from its triplication 30. And so multiplying the number ten you arrive at one-hundred. Again the number one-hundred is doubled and tripled like the number ten. So by doubling and tripling etc. the number one-hundred grows to one-thousand. In this way multiplying the number one-thousand according to the various denominations of numbers you come even to the investigation of number to infinity.

Furthermore I discovered that the numbers of restoration and opposition are composed of these three kinds: namely, roots, squares[1] and numbers. However number alone is connected neither with substantiis vlla proportione coniunctus est. Earum igitur radix est omnis res ex vnitatibus cum se ipsa multiplicata aut omnis numerus supra vnitatem cum se ipso multiplicatus: aut quod infra vnitatem diminutum cum se ipso multiplicatum reperitur. Substantia vero est omne illud quod ex multiplicatione radicis cum se ipsa colligitur.

Ex his igitur tribus modis semper duo sunt sibi inuicem coaequantia, sicut diceres

Substantiae radices coaequent
Substantiae numeros coaequent, et
Radices numeros coaequent.

Caput Primum.

De substantiis radices coaequantibus

Substantiae quae radices coaequant sunt, si dicas,

Substantia quinque coaequatur radicibus. Radix igitur substantiae sunt 5, et 25 ipsam componunt substantiam, quae videlicet suis quinque aequatur radicibus.

Et etiam si dicas, Tertia pars substantiae quatuor aequatur radicibus: Radix igitur substantiae sunt 12, et 144 ipsam demonstrant substantiam.

Et etiam ad similitudinem, Quinque substantiarum 10 radices coaequantium. Vna igitur substantia duabus radicibus aequiparatur, et radix substantiae sunt 2 — et substantiam quaternarius ostendit numerus.

Eodem namque modo, hoc quod ex substantiis excreuerit, aut minus ea fuerit, ad vnam conuertitur substantiam. Et similiter facies cum eo quod cum ipsis ex radicibus fuerit.

Caput II.

De substantiis numeros coaequantibus

Substantiae vero numeros coaequantes hoc modo proponuntur. Substantia nouenario coaequatur numero, Nouenarius igitur numerus mensurat substantiam cuius vnam radicem ternarius ostendit numerus. Eodem modo iuxta multitudinem et paucitatem substantiarum ipsae substantiae ad vnius substantiae similitudinem erunt tractandae. Hoc est, si substantiae duae vel tres vel quatuor, siue etiam plures fuerint, earum cum suis radicibus coaequatio, sicut vnius cum sua radice, quaerenda est.

roots nor with squares by any ratio. Of these then the root is anything composed of units which can be multiplied by itself, or any number greater than unity multiplied by itself: or that which is found to be diminished below unity when multiplied by itself. The square is that which results from the multiplication of a root by itself.

Of these three forms, then, two may be equal to each other, as for example

Squares equal to roots.
Squares equal to numbers, and
Roots equal to numbers.

Chapter I. Concerning Squares Equal to Roots

The following is an example of squares equal to roots: a square is equal to 5 roots. The root of the square then is 5, and 25 forms its square which, of course, equals five of its roots.[2]

Another example: the third part of a square equals four roots. Then the root of the square is 12 and 144 designates its square.[3] And similarly, five squares equal 10 roots. Therefore one square equals two roots and the root of the square is 2. Four represents the square.[4]

In the same manner then that which involves more than one square, or is less than one, is reduced to one square. Likewise you perform the same operation upon the roots which accompany the squares.

Chapter II. Concerning Squares Equal to Numbers

Squares equal to numbers are illustrated in the following manner: a square is equal to nine. Then nine measures the square of which three represents one root.[5]

Whether there are many or few squares they will have to be reduced in the same manner to the form of one square. That is to say, if there are two or three or four squares, or even more, the equation formed by them with their roots is to be reduced to the form of one square with its root. Further if there be less than one square, that is if a

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[1] Literally ‘substances,’ being a translation of the Arabic word *mal*, used for the second power of the unknown. Gerard of Cremona used *census*, which has a similar meaning.

[2] $x^2 = 5x$; $x = 5$, $x^2 = 25$.

[3] $\frac{1}{3}x^2 = 4x$; $x = 12$, $x^2 = 144$.

[4] $5x^2 = 10x$; $x^2 = 2x$, $x = 2$, $x^2 = 4$.

Si vero minus vna fuerit, hoc est, si tertia vel quarta vel quinta pars substantiae vel radicis proposita fuerit, eodem modo ea tractetur, vt si dicas,

Quinque substantiae 80 coaequantur. Vna igitur substantia quintae parti numeri 80 coaequatur, quam videlicet 16 componunt.

Et etiam si dicas. Medietas substantiae 18 coaequatur. Haec igitur substantia 36 coaequatur.

Caput III.

De radicibus numeros coaequantibus

Radices quae numeros coaequant sunt, si dicas. Radix ternario aequatur numero. Et etiam si dicas, Quatuor radices vigeno coaequantur numero. Vna igitur radix substantiae huius quinario coaequatur numero. Et etiam si dicas, Media radix denario coaequatur numero. Tota igitur radix vigeno aequatur numero, cuius videlicet substantiam 400 demonstrant.

Radices igitur et substantiae et numeri solum, quemadmodum diximus, distinguuntur. Vnde et ex his tribus modis quos iam praemisimus, tria oriuntur genera tripliciter distincta; vt

Substantia et radices numeros coaequant
Substantia et numeri radices coaequant, et
Radices et numeri substantiam coaequant.

Caput IIII.

De substantiis et radicibus numeros coaequantibus

Substantiae vero et radices quae numeros coaequant, sunt, si dicas. Substantia et 10 radices 39 coaequantur drachmis. Huius igitur artis inuestigatio talis est: die, quae est substantia, cui si similitudinem decern suarum radicum adiunxeris, ad 39 tota haec collectio protendatur. Modus hanc artem inueniendi est, vt radices iam pronunciatas per medium diuidas, sed radices in hac interrogatione sunt 10, accipe igitur 5, et iis cum se ipsis multiplicatis producuntur 25; quae omnia 39 adiicias, et veniunt 64. Huius igitur radice quadrata accepta, quae est 8, ab ea medietatem

third or a fourth or a fifth part of a square or root is proposed, this is treated in the same manner.[6]

For example, five squares equal 80. Therefore one square equals the fifth part of the number 80 which, of course, is 16.[7] Or, to take another example, half of a square equals 18. This square therefore equals 36.[8]

Chapter III. Concerning Roots Equal to Numbers

The following is an example of roots equal to numbers: a root is equal to 3. Therefore nine is the square of this root.[9]

Another example: four roots equal 20. Therefore one root of this square is 5.[10] Still another example: half a root is equal to ten. The whole root therefore equals 20, of which, of course, 400 represents the square.[11]

Therefore roots and squares and pure numbers are, as we have shown, distinguished from one another. Whence also from these three kinds which we have just explained, three distinct types of equations[12] are formed involving three elements, as

A square and roots equal to numbers,
A square and numbers equal to roots, and
Roots and numbers equal to a square.

Chapter IV. Concerning Squares and Roots Equal to Numbers

The following is an example of squares and roots equal to numbers: a square and 10 roots are equal to 39 units.[13] The question therefore in this type of equation is about as follows: what is the square which combined with ten of its roots will give a sum total of 39? The manner of solving this type of equation is to take one-half of the roots just mentioned. Now the roots in the problem before us are 10. Therefore take 5, which multiplied by itself gives 25, an amount which you add to 39, giving 64. Having taken then the square root of this which is 8, subtract from it the half of the roots, 5, leaving 3.

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[5] $x^2 = 9$; $x = 3$, $x^2 = 9$.

[6] Our modern expression "to complete the square," used in algebra, originally meant to make the coefficient of x^2 equal to unity, i.e. make one whole square.

[7] $5x^2 = 80$; $x^2 = 16$.

[8] $\frac{1}{2}x^2 = 18$; $x^2 = 36$.

[9] $x = 3$; $x^2 = 9$.

[10] $4x = 20$; $x = 5$, $x^2 = 25$.

[11] $\frac{1}{2}x = 10$; $x = 20$, $x^2 = 400$.

[12] 'Types of equations' = *genera*. Abu Kamil, Omar Al-Khayyami, Al-Karkhi, and Leonard designate these as 'composite' types. In modern notation: $ax^2 + bx = n$; $ax^2 + n = bx$; $ax^2 = bx + n$.

radicum 5 subtrahas, et manebunt 3. Ternarius igitur numerus huius substantiae vnam ostendit radicem, quae videlicet substantia nouenario dinoscitur numero. Nouem igitur illam componunt substantiam.

Similiter quotquot substantiae propositae fuerint, omnes ad vnam conuertas substantiam. Similiter quicquid cum eis ex numeris siue radicibus fuerit, id omne eo modo quo cum substantiis existi, conuertas. Huius autem conuersionis talis est modus, vt si dicas,

Duae substantiae et 10 radices 48 drachmis coaequantur. Huius artis talis est inuestigatio, vt dicas, Quae sunt duae substantiae inuicem collectae, quibus si similitudo 10 radicum earum adiuncta fuerit, ad 48 tota extendatur collectio. Nunc autem oportet, vt duas substantias ad vnam conuertas. Sed iam manifestum est, quoniam vna substantia medietatem duarum designat, igitur omnem rem in hac quaestione tibi propositam, ad medium conuertas, dicendo

Substantia et 5 radices 24 drachmis coaequantur. Modus huius rei talis est, vt dicas. Quae est substantia, cui si quinque suas radices adiunxeris, ad 24 excrescat. Nunc etiam oportet, vt ad regulam supra datam animum conuertas, et diuidas radices per medium et veniunt 2 et vnius medietas, iis cum se ipsis multiplicatis, producuntur 6 et 1/4, illis 24 adiicias, veniunt 30 et vnius 1/4. Postea huius aggregati radicem quadratam accipias, quam scilicet 5 et vnius medietas componunt, ex qua medietatem radicum, 2 1/2, subtrahas et manebunt 3, quae vnam radicem substantiae exprimunt, quam substantiam nouenarius componit numerus.

Et si diceretur, Medietas substantiae et quinque radices 28 coaequantur drachmis. Huius questionis talis est modus, vt dicas. Quae est substantia, cuius medietati si quinque suas radices adiunxeris, tota summa ad 28 excrescat, ita tamen vt substantia quae prius diminuta fuerat, perfecta compleatur. Igitur huius substantiae medietas cum radicibus secum pronunciatis, duplicanda est; veniunt autem, substantia et 10 radices 56 drachmis aequales.

Diuide igitur radices per medium, et veniunt 5, quibus cum seipsis multiplicatis, producuntur 25; illa adde 56, et colliguntur 81; huius collecti radicem quadratam accipias, quam nouenarius componit numerus, atque ex ea medietatem radicum 5 subtrahas et manebunt 4, substantiae radix.

The number three therefore represents one root of this square, which itself, of course, is 9. Nine therefore gives that square.[14]

Similarly however many squares are proposed all are to be reduced to one square. Similarly also you may reduce whatever numbers or roots accompany them in the same way in which you have reduced the squares. The following is an example of this reduction: two squares and ten roots equal 48 units.[15]

Another possible example: half a square and five roots are equal to 28 units.[16] The import of this problem is something like this: what is the square which is such that when to its half you add five of its roots the sum total amounts to 28? Now however it is necessary that the square, which here is given as less than a whole square, should be completed.[17] Therefore the half of this square together with the roots which accompany it must be doubled. We have then, a square and 10 roots equal to 56 units. Therefore take one-half of the roots, giving 5, which multiplied by itself produces 25. Add this to 56, making 81. Extract the square root of this total, which gives 9, and from this subtract half of the roots, 5, leaving 4 as the root of the square.

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[13] $x^2 + 10x = 39$; $\frac{1}{2}$ of 10 is 5, $5^2 = 25$, $25 + 39 = 64$, $\sqrt{64} = 8$; $8 - 5 = 3$. $x^2 = 9$.

[14] For the general type $x^2 + bx = n$, the solution is $x = \sqrt{(\frac{b}{2})^2 + n} - \frac{b}{2}$; the negative value of the square root is neglected, as that would give a negative root of the equation.

[15] $2x^2 + 10x = 48$, reducing to $x^2 + 5x = 24$; $\frac{1}{2}$ of 5 is $2\frac{1}{2}$, $(2\frac{1}{2})^2 = 6\frac{1}{4}$, $24 + 6\frac{1}{4} = 30\frac{1}{4}$, $\sqrt{30\frac{1}{4}} = 5\frac{1}{2}$; $5\frac{1}{2} - 2\frac{1}{2} = 3$, $x^2 = 9$.

[16] $\frac{1}{2}x^2 + 5x = 28$, reducing to $x^2 + 10x = 56$; $x = \sqrt{25 + 56} - 5$, or $x = 4$. Note that the value of x^2 is not given here as it usually is.

[17] Attention is called to the force of the expression, "completing the square," as here used with the meaning to make the coefficient of the second power of the unknown quantity equal to unity or making one whole square.

Caput V.

De substantiis et numeris radices coaequantibus

Propositio huius rei talis est, vt dicas, Substantia et 21 drachmae 10 radicibus coaequantur.

Ad hoc inuestigandum talis datur regula, vt dicas. Quae est substantia, cui si 21 drachmas adiunxeris, tota summa simul decern ipsius substantiae radices exhibeat. Huius quaestionis solutio hoc modo concipitur, vt radices primum per medium diuidas, et veniunt in hoc casu 5, haec cum seipsis multiplicata, producunt tibi 25. Ex illis 21 drachmas, quas paulo ante cum substantiis commemorauimus, subtrahas, et manebunt 4, horum radicem quadratam accipias, vt sunt 2, quae ex medietate radicum 5 diminuas, et manebunt 3, vnam radicem huius substantiae constituentia, quam scilicet substantiam nouenarius complet numerus.

Quod si libuerit, poteris ipsa 2, quae a medietate radicum iam diminuisti, medietati radicum 5 scilicet addere, et veniunt 7; quae vnam substantiae radicem demonstrant quam substantiam 49 adimplent. Cum igitur aliquod huius capitis exemplum tibi propositum fuerit, ipsius modum cum adiectione, quemadmodum dictum est, inuestiga, quern si cum adiectione non inueneris, procul dubio cum diminutione reperies. Hoc enim caput solum adiectione simul et diminutione indiget quod in aliis capitibus praemissis minime reperies.

Sciendum est etiam, quando radices iuxta hoc caput mediaueris, et medietatem deinde cum seipsa multiplicaueris, si quod ex multiplicatione tollitur vel procreatur, minus fuerit drachmis cum substantia pronunciatis: quaestio tibi proposita nulla erit. At si drachmis aequale fuerit vna radix fientque 25.

Caput VI.

De radicibus et numeris substantiam coaequantibus

In hoc capite sic proponitur, Tres radices et quatuor ex numeris coaequantur substantiae.

Chapter V. Concerning Squares and Numbers Equal to Roots

The following is an illustration of this type: a square and 21 units equal 10 roots.[18]

The rule for the investigation of this type of equation is as follows: what is the square which is such that when you add 21 units the sum total equals 10 roots of that square? The solution of this type of problem is obtained in the following manner. You take first one-half of the roots, giving in this instance 5, which multiplied by itself gives 25. From 25 subtract the 21 units to which we have just referred in connection with the squares. This gives 4, of which you extract the square root, which is 2. From the half of the roots, or 5, you take 2 away, and 3 remains, constituting one root of this square which itself is, of course, 9.[19]

If you wish you may add to the half of the roots, namely 5, the same 2 which you have just subtracted from the half of the roots. This give 7, which stands for one root of the square, and 49 completes the square.[20] Therefore when any problem of this type is proposed to you, try the solution of it by addition as we have said. If you do not solve it by addition, without doubt you will find it by subtraction. And indeed this type alone requires both addition and subtraction, and this you do not find at all in the preceding types.

You ought to understand also that when you take the half of the roots in this form of equation and then multiply the half by itself, if that which proceeds or results from the multiplication is less than the units above-mentioned as accompanying the square, you have no equation.[21] If equal to the units, it follows that a root of the square will be the same as the half of the roots which accompany the square, without either addition or diminution.[22] Whenever a problem is proposed that involves two squares, or more or less than a single square, reduce to one square just as we have indicated in the first chapter.

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[18] $x^2 + 21 = 10x$.

[19] For the general type, $x^2 + n = bx$ the solution is $x = \frac{b}{2} \pm \sqrt{(\frac{b}{2})^2 - n}$, and both positive and negative values of the radical give positive solutions of the equation proposed. In this problem we have $x^2 + 21 = 10x$; $\frac{1}{2}$ of 10 is 5, $5^2 = 25$, $25 - 21 = 4$, $\sqrt{4} = 2$; $5 - 2 = 3$, one root; $5 + 2 = 7$, the other root.

[20] Another use of the expression, "completing the square."

[21] This corresponds to the condition, $b^2 - 4ac < 0$, in the equation $ax^2 + bx + c = 0$; in this event the roots are imaginary.

[22] Condition for equal roots, $b^2 - 4ac = 0$.

Ad hoc inuestigandum talis datur regula, quod scilicet radices per medium diuidas et venit vnum et alterius medietas, hoc deinde cum seipsis multiplices, et producuntur 2 1/4, his 4 ex numeris adiicias, et veniunt 6 1/4, huius postea radicem quadratam accipias, hoc est 2 1/2. Atque earn tandem medietati radicum, vni scilicet et dimidio, adiicias, et veniunt 4, quae vnam substantiae radicem componunt, quam deinde substantiam numerus 16 adimplet.

Quicquid igitur plus siue minus substantia tibi propositum fuerit, ad vnam conuertas substantiam. Ex his igitur modis, de quibus in principio libri mentionem fecimus, sunt tres priores, in quibus radices non mediantur, in tribus vero posterioribus vel residuis mediantur radices prout superius liquet.

Chapter VI. Concerning Roots and Numbers Equal to a Square

An example of this type is proposed as follows: three roots and the number four are equal to a square.[23]

The rule for the investigation of this kind of problem is, you see, that you take half of the roots, giving one and one-half; this you multiply by itself, producing $2\frac{1}{4}$. To $2\frac{1}{4}$ add 4, giving $6\frac{1}{4}$, of which you then take the square root, that is, $2\frac{1}{2}$. To $2\frac{1}{2}$ you now add the half of the roots, or $1\frac{1}{2}$, giving 4, which indicates one root of the square. Then 16 completes the square.[24] Now also whatever is proposed to you either more or less than a square, reduce to one square.

Now of the types of equations which we mentioned in the beginning of this book, the first three are such that the roots are not halved, while in the following or remaining three, the roots are halved, as appears above.

Demonstrationes Geometricae

Geometrical Demonstrations

Dixit Algaurizin. Sex sunt modi de quibus, quantum ad numeros pertinet, sufficienter diximus. Nunc vero oportet, vt quod per numeros proposuimus, ex geometrica idem verum esse demonstremus.

Nostra igitur prima propositio talis est, Substantia et 10 radices 39 coaequantur drachmis. Huius probatio est, vt quadratum cuius latera ignorantur, proponamus. Hoc autem quadratum quod loco substantiae ponimus, et eius radicem scire volumus atque designare. Sit igitur quadratum a b cuius vnumquodque latus vnam ostendit radicem. Iam manifestum est, quoniam, quando aliquod eius latus cum numero numerorum multiplicauerimus, tunc hoc quod ex multiplicatione colligitur erit numerus radicum radici ipsius numeri aequalis.

Quoniam ergo decern radices cum substantia pronunciantur, quartam igitur partem numeri decem accipimus, atque vnicuique lateri quadrati aream aequedistantium laterum applicamus quarum longitudinem quidem longitude quadrati primo descripti, latitudinem vero duo et dimidium, quae sunt numeri 10 quarta pars, demonstrant. Quatuor igitur areae laterum aequedistantium primo quadrato a b applicantur. Quarum singularum longitudo longitudini vnius radicis quadrati a b aequalis erit, latitudo etiam singularum 2 et medium, vt iam

Geometrical Demonstrations

We have said enough, says Al-Khowarizmi, so far as numbers are concerned, about the six types of equations. Now, however, it is necessary that we should demonstrate geometrically the truth of the same problems which we have explained in numbers.

Therefore our first proposition is this, that a square and 10 roots equal 39 units. The proof is that we construct a square of unknown sides, and let this square figure represent the square (second power of the unknown) which together with its root you wish to find. Let the square, then, be ab, of which any side represents one root. Now it is evident that when one side of it we multiply by any number, then the amount of that multiplication equals the number of roots which is equal to the root of that same number.

Since then ten roots are proposed with the square, we take the fourth part of the number ten, that is $2\frac{1}{2}$, and to each of the four sides of the square we join rectangles of equal length with the square. The length of any one of the four areas is equal to the length of the side of the square ab, and the breadth of each is $2\frac{1}{2}$, as the quarter of ten. The four areas then $cdef$ are joined to the four sides of the square ab. Now it is evident that the length of each side of the four areas is equal to the length of one root

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[23] $3x + 4 = x^2$; $\frac{1}{2}$ of 3 is $1\frac{1}{2}$, $(1\frac{1}{2})^2 = 2\frac{1}{4}$, $2\frac{1}{4} + 4 = 6\frac{1}{4}$, $\sqrt{6\frac{1}{4}} = 2\frac{1}{2}$; $2\frac{1}{2} + 1\frac{1}{2} = 4$, the root.

[24] The solution of the general type $bx + n = ax^2$, reduced by division to $\frac{b}{a}x + \frac{n}{a} = x^2$, is $x = \sqrt{(\frac{b}{2a})^2 + \frac{n}{a}} + \frac{b}{2a}$, and only the positive value of the radical is taken.

dictum est, demonstrat. Sunt autem hae areae 10 c d e f. Ex hoc igitur quod diximus, sit area laterum inaequalium, quae similiter ponuntur ignota, in cuius videlicet quatuor angulis quorum vniuscuiusque quantitas areae, quam 2 et dimidium cum duobus et dimidio multiplicata perficiunt, imperfectionem maioris seu totius areae ostendunt.

Vnde fit vt circunductionem areae maioris, cum adiectione duorum et dimidii cum duobus et dimidio quatuor vicibus multiplicatorum compleamus, generat autem haec tota multiplicatio 25. Et iam manifestum est, quoniam primum quadratum, quod substantiam significat, et quatuor areae ipsum quadratum circundantes, 39 perficiunt, quibus quando 25, hoc est quatuor minora quadrata, quae scilicet super quatuor angulos quadrati a b ponuntur, adiecerimus, quadratum maius, G H vocatum, circunductione complebitur. Vnde etiam haec tota numeri summa vsque ad 64 excrescet, cuius summae radicem octonarius obtinet numerus, quo etiam vnum eius latus compleri probatur. Igitur vbi ex numero octonario quartam partem numeri denarii, sicut in extremitatibus quadrati maioris G H ponuntur, subtraxerimus bis, 3 ex ipsius latere manebunt. Quinque ergo ex octo subtractis, 3 manere necesse est; quae simul vni lateri quadrati primi, quod est a b, aequiparantur. Haec igitur 3 quadrati vnam radicem, hoc est vnam radicem substantiae propositae: nouenarius deinde numerus ipsam substantiam exprimit.

Ergo numerum denarium mediamus, et alteram eius medietatem cum seipsa multiplicamus, deinde totum multiplicationis productum numero 39 adiicimus, vt maioris quadrati G H circunductio compleatur. Nam eius quatuor angulorum diminutio totam huius quadrati circunductionem imperfectam reddebat. Manifestum enim est, quod quarta pars omnis numeri cum suo aequali, ac deinde cum quatuor multiplicata, eandem perficiat numerum, quem medietas numeri cum seipsa multiplicata, perficit. Igitur si radicum medietas cum seipsa multiplicetur, huius multiplicationis summa, multiplicationem quartae partis cum seipsa ac deinde cum quatuor multiplicatae, sufficienter euacuet, adaequabit vel delebit.

Ad hoc etiam idem demonstrandum, altera datur formula, quae talis est. Quadrate a b substantiam significante aequalitatem decem radicum addimus; has radices deinde per medium diuidamus, venient 5, ex quibus duas areas ad duo latera quadrati a b constituamus, hae autem vocentur a g et b d, et erit vtriusque vtraque latitudo vni lateri quadrati a b aequalis; vtramque denique longitudinem numerus quinarius adimplebit. Superest iam, vt ex multiplicatione 5 cum 5, quae medietatem radicum quas ad duo latera quadrati prioris substantiam significantis, addidimus, quadratum faciamus. Vnde iam manifestum est, quod

of the square ab, and the breadth of each is $2\frac{1}{2}$, as has been stated. These four areas represent the ten roots.

Out of this, according to what we have said, comes an area with four sides, which is similarly put as unknown, in each of whose four corners the amount of the area which $2\frac{1}{2}$ multiplied by $2\frac{1}{2}$ makes, indicates the imperfection of the larger or whole area. Hence it is that we complete the drawing of the larger area by adding $2\frac{1}{2}$ multiplied by $2\frac{1}{2}$ four times; and this whole multiplication gives 25. Now it is evident that the first square, which indicates the square, and the four areas surrounding that square, make 39; and when we add 25, that is, the four smaller squares which are placed at the four corners of the square ab, the drawing of the larger square, called GH, will be completed. Hence also the whole sum of numbers will increase to 64, of which sum the root 8 indicates one number, by which one side of the larger square is also proved to be completed. Therefore when we subtract twice the fourth part of the number ten, as placed at the extremities of the larger square GH, from 8, there will remain 3 on the side of it. Five then being subtracted from 8, 3 must necessarily remain, which equal one side of the first square, which is ab. These 3 represent one root of the square, that is one root of the proposed square: then 9 indicates the square itself.

We halve the number ten and multiply the half by itself, then add the whole product of the multiplication to the number 39, that the drawing of the larger square GH may be completed. For the diminution of its four corners rendered the drawing of this square imperfect. It is manifest, indeed, that the fourth part of any number multiplied by its equal and then by four, produces the same number which the half of the number produces when multiplied by itself. Therefore if the half of the roots is multiplied by itself, the sum of this multiplication will equal the multiplication of the fourth part by itself and then by four, or will equal it or cancel it.

For this also another figure is given to demonstrate the same thing, which is as follows. We add to the square ab, representing the square, the equivalent of ten roots; then we halve these roots, giving 5, from which we construct two areas at the two sides of the square ab. Let these be called ag and bd, and the breadth of each of them will be equal to one side of the square ab, while the length of each is completed by the number five. It remains now for us to make a square from the multiplication of 5 by 5, which is the half of the roots which we added to the two sides of the first square representing the duae areae, quae supra duo latera ponuntur, et quae 10 radices substantiae significant, simul cum quadrato priori, quod est substantia, 39 ex numero adimpleant. Manifestum etiam, quod area maioris seu totius quadrati per multiplicationem 5 cum 5 perficiatur. Hoc ergo quadratum perficiatur, atque ad perfectionem eius numerus 25 ad priora 39 adiiciatur: tota igitur haec summa vsque ad 64 excrescet. Nunc summae huius radicem quadratam, quae vnum latus quadrati maioris 25 designat, accipiamus, atque inde aequalitatem eius quod ei addidimus, hoc est 5, subtrahamus, et manebunt 3; quae latus quadrati a b hoc est vnam radicem substantiae propositae complere probantur. Tria igitur huius substantiae sunt radix; et substantia nouem.

De substantia et drachmis res coaequentibus.

Demonstration of Chapter V

Substantia et 21 drachmae 10 rebus aequiparantur. Propositio haec seu quaestio in capite quinto proposita fuit, cuius hic demonstratio docetur. Quadratum igitur a b, quod latera habet ignota, substantiam pono, atque ei parallelogrammum rectangulum, cuius vtraque latitude vni lateri quadrati a b aequalis sit, cuiusque longitude vtraque rerum seu radicum medietatem referat, applico. Deinde vero huius rectanguli summam 21 ex numeris constituo, qui numerus cum ipsa substantia propositus est. Haec autem area vel rectangulum b g inscribitur, cuius simul latitude g d dinoscitur, longitude ergo duarum arearum inuicem coniunctarum in h d terminatur. Et iam manifestum est quod haec longitude denarium obtinet numerum, quoniam omnis area quadrilatera et rectorum angulorum, ex multiplicatione sui lateris cum vnitate semel, vnam obtinet radicem: et si cum binarie numero, duae eiusdem areae nascuntur radices.

Quoniam igitur prime sic proposuit, vna substantia et 21 drachmae, 10 radicibus aequiparantur, manifestum est, quod longitude lateris h d in denario terminatur numero, quia latus h b vnam substantiae radicem obtinet, latus igitur h d 10 ostendit. Vnde et linea e h lineae e d fiet aequalis, atque ducta ex puncto e linea perpendiculari e t: haec eadem perpendicularis ipsi h a lineae aequalis erit. Lineae ergo e t portionem, quae ipsaque d e linea breuior est, in rectum adiicio e c, et fiet linea t c aequalis t g lineae, vnde quadratum t l, quod ex multiplicatione medietatis radicum cum se ipsa multiplicatae, id est ex multiplicatione quinarii cum quinario (in hoc casu) colligitur, nobis eueniet.

Et iam manifestum etiam est, quoniam area b g 21, quae substantiae addidimus, in se obtinet, ex ea

square. Whence it is now manifest that the two areas which are placed above the two sides, and which represent the 10 roots of the square, together with the first square, which is the square, complete 39 in number. It is manifest also that the area of the larger or whole square is completed by the multiplication of 5 by 5. Let this square then be completed, and to complete it let the number 25 be added to the previous 39: therefore this whole sum will increase to 64. Now let us take the square root of this sum, which represents one side of the larger square, 25, and from it let us subtract the equivalent of what we added to it, that is 5, and 3 will remain; which prove to complete one side of the square ab, that is, one root of the proposed square. Therefore three is the root of this square; and the square is nine.

Concerning a square and drachmas equal to things. A square and 21 drachmas are equal to 10 things. This proposition or question was proposed in the fifth chapter, of which the demonstration is here taught. I put, therefore, the square ab, which has unknown sides, for the square, and I apply to it a rectangular parallelogram, of which each breadth is equal to one side of the square ab, and of which each length represents the half of the things or roots. Then indeed I constitute the sum of this rectangle as 21 from the numbers, which number is proposed with the square itself. This area or rectangle bg is inscribed, whose breadth gd is known at the same time; therefore the length of the two areas joined together terminates in hd. And it is now manifest that this length obtains the number ten, since every quadrilateral area with right angles, from the multiplication of its side with unity once, obtains one root: and if with the number two, two roots of the same area are born.

Since therefore it was proposed in the first place, one square and 21 drachmas are equal to 10 roots, it is manifest that the length of the side hd terminates in the number ten, because the side hb obtains one root of the square, therefore the side hd shows 10. Whence also the line eh will be made equal to the line ed, and having drawn from the point e a perpendicular line et: this same perpendicular will be equal to the line ha. Therefore I add to the line et the portion by which it is shorter than the line de, in a straight line ec, and the line tc will be made equal to the line tg, whence the square tl, which from the multiplication of the half of the roots multiplied by itself, that is from the multiplication of five by five (in this case), is collected, will come to us.

And it is now also manifest, since the area bg, 21, which we added to the square, contains in itself, igitur b g per lineam t c, quae est vnum latus areae t l extrahamus atque ex eadem area b g aream b t minuamus, deinde vero super lineam e c quae est in quo linea t c c lineam a h in quantitate deuincit, quadratum e n m c ponamus. Vnde et iam manifestum, cum linea t c aequalis sit linea l c, nam ipsae in quadrato t l aequales protenduntur, similiter et linea c c lineae m c aequalis, quia ipsae quadratum e m aequali dimensione circundant: aequalium igitur ab aequalibus lineis subtractione facta, linea t c lineae l m aequalis relinquetur quod est notandum.

Rursus manifestum est, quoniam linea g d aequalis est lineae a h, cum ipsae in latitudine areae h g aequali dimensione tenduntur, sed linea a h aequalis est lineae h b cum ipsae in vno quadrato appareant. Item quoniam linea g l aequalis est lineae d e, cum ipsae in vno quadrato reperiantur aequales, sed linea d e aequalis est h e, cum ipsae decem radices per medium diuidant; linea igitur d l residuum lineae g l, lineae e b ex linea e h residuae aequalis erit: atque tandem, cum linea t e lineae l m ex superiori demonstratione, non sit inaequalis, area quam linea t e et e b circundant, comprehensae sub l m et d l lineis areae aequalis erit. Area igitur t b aequalis est m d areae.

Et iam manifestum est, quoniam quadratum t l 25 in se continet, cum ergo ex eodem quadrato t l areas d t et m d, quae videlicet duabus areis g e et t b, 20 et vnum in se continentibus, sunt aequales, subtraxerimus, quadratum n c nobis manebit, qui simul numerum, qui est inter 25 et 21, obtinet. Et hic numerus est quaternarius, cuius videlicet radicem duo designant, quae latus e c adimplent. Hoc autem latus aequale est lineae c b, quoniam e c aequale est d l lateri, cum ipsa in latitudine areae d c aequalia protendantur. Iam manifestum est, quoniam d l aequalis est lineae e b, quando igitur e b quae sunt duo, ex e h, quae sunt radicum medietas, quam quinarius ostendit numerus, abstulerimus, linea b h ternarium ostendens numerum restabit. Ternarius igitur numerus radicem primae demonstrat substantiae.

Quod si contra lineam e c lineae e h, quae medietatem radicum continet, addiderimus, colligentur 7, quae lineam t h ostendunt. Et tunc radix substantiae maior restat constituta, quam substantiam 49 adimplent.

from it therefore through bg by the line tc, which is one side of the area tl, let us extract and from the same area bg let us diminish the area bt, then indeed above the line ec, which is that by which the line tc surpasses the line ah in quantity, let us place the square $enmc$. Whence also it is now manifest, since the line tc is equal to the line lc, for they are extended as equals in the square tl, similarly also the line cc is equal to the line mc, because they surround the square em with equal dimension: therefore, subtraction having been made from equals by equals, the line tc will be left equal to the line lm, which is to be noted.

Again it is manifest, since the line gd is equal to the line ah, because they are extended in breadth with equal dimension of the area hg, but the line ah is equal to the line hb because they appear in one square. Likewise since the line gl is equal to the line de, because they are found equal in one square, but the line de is equal to he, because they divide the ten roots by half; therefore the line dl, the residue of the line gl, will be equal to the line eb, the residue from the line eh: and at last, since the line te is not unequal to the line lm from the above demonstration, the area which the lines te and eb surround will be equal to the area comprehended under the lines lm and dl. Therefore the area tb is equal to the area md.

And it is now manifest, since the square tl contains 25 in itself, when therefore from the same square tl we shall have subtracted the areas dt and md, which are evidently equal to the two areas ge and tb, containing 20 and one in themselves, the square nc will remain to us, which at the same time obtains the number which is between 25 and 21. And this number is four, of which the root two designates, which fill up the side ec. This side is equal to the line cb, since ec is equal to the side dl, because they are extended as equals in the breadth of the area dc. It is now manifest, since dl is equal to the line eb, when therefore we shall have taken away eb, which are two, from eh, which are the half of the roots, which the number five shows, the line bh showing the number three will remain. Therefore the number three shows the root of the first square.

But if on the other hand we shall have added the line ec to the line eh, which contains the half of the roots, 7 will be collected, which show the line th. And then the greater root of the square remains established, which 49 complete.

Rules Corresponding to the Six Chapters of Algebra

REGULE 6 CAPITULIS ALGABRE CORRESPONDENTES

Rules Corresponding to the Six Chapters of Algebra

> Sex sunt modi de quibus, quantum ad numeros pertinet, sufficienter diximus.
>
> *We have said enough, says Al-Khowarizmi, so far as numbers are concerned, about the six types of equations.*

Rule I. When squares are equal to roots, divide the number of roots by the number of squares; the quotient gives one root.

Rule II. When squares are equal to numbers, divide the number of numbers by the number of squares; the quotient gives one root, and the square of this root is the required square.

Rule III. When roots are equal to numbers, divide the number of numbers by the number of roots; the quotient gives one root.

Rule IV. When squares and roots are equal to numbers, halve the roots, square the half, add this square to the numbers, extract the square root of the sum, and subtract the half of the roots; the remainder is one root.

Rule V. When squares and numbers are equal to roots, halve the roots, square the half; if the numbers are less than this square, subtract the numbers from the square, extract the root of the remainder, and subtract this root from the half of the roots; the remainder is one root. If the numbers are greater than the square, the problem is impossible. If the numbers equal the square, the half of the roots is one root.

Rule VI. When roots and numbers are equal to squares, halve the roots, square the half, add the numbers, extract the root of the sum, and add the half of the roots; the sum is one root.

Addita Quaedam pro Declaratione Algebrae

Some Additions in Explanation of the Algebra

(Selections from Scheybl's Additions)

Hactenus de regulis Algebrae satis dictum est. Nunc vero, vt plenior fiat intellectus, quaedam additiones apponendae sunt.

Hitherto enough has been said about the rules of Algebra. Now, however, that a fuller understanding may be obtained, certain additions are to be appended.

Primo sciendum est, quod omnis numerus constat ex vnitatibus, et quod vnitatis multiplicatio in seipsam semper vnitatem producit. Vnde etiam sequitur, quod omnis numeri quadratus ex multiplicatione vnitatum producitur.

First it must be known that every number consists of units, and that the multiplication of unity by itself always produces unity. Whence also it follows that the square of every number is produced from the multiplication of units.

Secundo notandum est, quod in omni operatione Algebrae, primum oportet vt omnes substantiae ad vnam reducantur substantiam, omnesque radices ad vnam radicem, et omnes numeri ad vnum numerum.

Secondly it is to be noted that in every algebraic operation, first it is necessary that all squares be reduced to one square, all roots to one root, and all numbers to one number.

Tertio, sciendum est quod tres sunt modi simplices et tres compositi, vt in textu dictum est. Modus autem cognoscendi vtrum aequatio possibilis sit vel impossibilis, est ille qui in capite quinto traditur, videlicet quando medietas radicum in seipsa multiplicata, productum maius est numero cum substantia proposito: tunc quaestio est possibilis. Si vero productum minus fuerit numero proposito, quaestio est impossibilis. Si aequale, tunc radix aequalis est medietati radicum.

Thirdly, it must be known that there are three simple types and three composite types, as has been said in the text. The method, however, of knowing whether an equation is possible or impossible is that which is handed down in the fifth chapter, namely when the half of the roots multiplied by itself, the product is greater than the number proposed with the square: then the question is possible. But if the product is less than the proposed number, the question is impossible. If equal, then the root is equal to the half of the roots.

Latin Glossary

LATIN GLOSSARY

With variations from classical usage noted in the text

Latin Term	**Usage and Notes**
adiectio, -onis	addition; used for the operation of adding to both sides
aequedistans	parallel (lit. equally distant); classical Latin would use *parallelus*
aequiparare	to equal, to be equal to
almucabola	transliteration of Arabic *al-muqabala*, the operation of balancing/reduction
area, -ae	area, surface; used geometrically
circumductio, -onis	the bounding line, perimeter
coaequari	to be equal to; frequent in the text
colligere	to collect, sum up
completus, -a, -um	completed; "completing the square"
coniungere	to join, add
circundare	to surround (variant of *circumdare*)
decurrere	to run down, extend to
dimidiare	to halve (*mediare* is classical)
diminutio, -onis	subtraction, diminution
diminuere	to diminish, subtract
drachma, -ae	drachma, unit of number
excedere	to exceed, be greater than
excreuere	to increase, grow beyond
figura, -ae	figure, shape
imperfectus, -a, -um	incomplete (used of the incomplete square)
inuestigatio, -onis	investigation, method of solution
latitudo, -inis	breadth, width
longitude, -inis	length (for classical *longitudo*)
mediare	to halve; used frequently
medietas, -atis	half, moiety
multiplicatio, -onis	multiplication
numerus, -i	number
oppositio, -onis	opposition, the operation of *al-muqabala*
ostendere	to show, indicate, represent
perficere	to complete, make perfect
ponere	to place, put, suppose
procreare	to produce, generate
pronunciare	to state, express in words
proportio, -onis	proportion
quadratum, -i	square
radix, -icis	root; also used for the unknown quantity
restauratio, -onis	restoration, the operation of *al-jabr*
rumbus	square (Arabic *al-rub'*)
similitudo, -inis	similitude, likeness; used for "the like of"
substantia, -ae	substance, square of the unknown
subtrahere	to subtract
vnitas, -atis	unity, the number one

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ᶜUmar al-Khayyām

(1048–1131 CE / 430–517 AH)

رسالة في براهين الجبر والمقابلة

Ṣināᶜat al-Jabr wa-al-Muqābalah

The Art of Algebra and Almucabala

Or: Treatise on the Proofs of the Problems of Algebra and Almucabala

Bilingual Edition

Arabic Text — English Translation

Parallel left-to-right and right-to-left columns

Based on the 13th-century Lahore manuscript

(Smith Oriental MS 45, Columbia University Rare Book Library)

This document presents a bilingual edition of Omar Khayyam's seminal work on algebra, one of the crowning achievements of medieval Islamic mathematics.

The manuscript was copied in Lahore, India, in the 13th century.
It contains the earliest systematic treatment of cubic equations,
with all 25 types solved through geometric constructions using conic sections.

Typeset in LaTeX with Arabic support via polyglossia and fontspec.
Arabic font: Noto Naskh Arabic

This edition prepared for scholarly and educational purposes.

Contents

0.1 Introduction

المقدمة

عمر الخيام، اسمه الكامل: غياث الدين أبو الفتح عمر بن إبراهيم الخيامي النيسابوري، كان من أبرز علماء الرياضيات والفلك والشعراء في العالم الإسلامي في العصور الوسطى. وُلد في نيسابور عام 1048م، وقضى معظم حياته في سمرقند وأصفهان، حيث أنتج أعمالاً رائدة في الرياضيات والفلك.

كتابه صناعة الجبر والمقابلة ▯ويعرف أيضاً باسم رسالة في البراهين على مسائل الجبر والمقابلة▯ يمثل ذروة الجبر الإسلامي المبكر. في هذا العمل، يقدم الخيام:

Introduction

Omar Khayyam (عمر الخيام), full name Ghiyāth al-Dīn Abū al-Fatḥ ᶜUmar ibn Ibrāhīm al-Khayyāmī al-Nīsābūrī, was one of the most brilliant mathematicians, astronomers, and poets of the medieval Islamic world. Born in Nishapur in 1048 CE, he spent much of his life in Samarqand and Isfahan, where he produced groundbreaking works in mathematics and astronomy.

His *Ṣināᶜat al-Jabr wa-al-Muqābalah* (often cited as *Risāla fī al-Barāhīn ᶜalā Masāᵓil al-Jabr wa-al-Muqābalah* — “Treatise on the Proofs of the Problems of Algebra and Almucabala”) represents the culmination of early Islamic algebra. In this work, Khayyam:

- Provides rigorous definitions of the fundamental quantities of algebra
- Classifies all equations of degree up to three into 25 types
- Solves 14 irreducible cubic equations using geometric constructions with conic sections
- Distinguishes between arithmetical and geometrical solutions

This treatise, composed around 1079 CE, profoundly influenced later mathematicians including Sharaf al-Dīn al-Ṭūsī, and through translations into European languages, contributed to the development of algebra in the West.

0.1.1 The Author’s Preface

The Author's Introduction □مقدمة المؤلف□

الحمد لله رب العالمين، والعاقبة للمتقين، ولا عدوان إلا على الظالمين، وصلى الله على سيدنا محمد وآله أجمعين.

Praise be to God, Lord of the Worlds, and the good end is for the righteous, and there is no enmity save against the oppressors; and may God bless our master Muhammad and all his family.

أما بعد، فإني قد سبقني في هذا الفن جماعة من أهل العلم، قدّموا فيه أصناف المسائل، وبينوا فيه أصول المسائل، وبرهنوا على بعضها، ولم يتفق لهم برهان على البعض، ولم يفرقوا فيها بين ما يمكن أن يوجد بالقطوع المخروطية، وبين ما لا يمكن أن يوجد إلا بأساليب هندسية أخرى. وقد أملت أن أجمع هذا العلم في رسالة، وأبين فيها أصناف المسائل على وجه التمييز، وأضع لكل صنف برهاناً صحيحاً، وذلك أمر لم يسبقني إليه أحد من المتقدمين.

Now then: in this art there have preceded me a group of men of knowledge, who presented in it the species of problems, and explained therein the foundations of problems, and proved some of them, but did not succeed in proving all of them; and they did not distinguish between what can be found by conic sections and what cannot be found except by other geometrical methods. I have desired to gather this science into a treatise, and to explain therein the species of problems with proper distinction, and to set down for each species a correct proof — and this is something that none of the predecessors has preceded me in.

0.1.2 On the Nature of This Treatise

فنقول: إن الجبر والمقابلة علم يُستخرج به المجهول من المعلوم المفروض، إذا كان بينهما نسبة تقتضي ذلك. وهذا العلم يتضمن الأعداد والجذور والأموال والكعاب.

We say: Algebra and Almucabala is a science by which the unknown is extracted from the known datum, when between them there is a relation requiring that. And this science comprises numbers, roots, squares, and cubes.

Khayyam defines algebra as "the science of equations" (علم المعادلات). He distinguishes carefully between:

1. **Simple equations:** Those reducible to linear or quadratic forms, solved by arithmetical or Euclidean methods
2. **Compound (irreducible) equations:** Cubic equations requiring conic section constructions

0.2 Definitions of the Fundamental Quantities

تعريفات الكميات الأساسية

Definitions of the Fundamental Quantities

يبدأ الخيام بتعريفات دقيقة للكميات الجبرية، متبعاً وموسعاً تقليد الخوارزمي:

Khayyam begins with careful definitions of the algebraic quantities, following and extending the tradition of al-Khwārizmī:

> أصل هذا العلم: أن العدد المفرد هو ما لا ينسب إلى جذر، ولا إلى مال، ولا إلى كعب، ولا إلى ما يتركب منها.
>
> *The foundation of this science is: The simple number is that which is not referred to a root, nor to a square, nor to a cube, nor to what is compounded from them.*

0.2.1 The Root (الجذر)

والجذر هو كل شيء مضروب في نفسه من الواحد وما فوقه من الأعداد، وما دونه من الكسور. فإذا ضرب في نفسه فهو مال، وإذا ضرب المال في الجذر فهو كعب، وإذا ضرب الكعب في الجذر فهو مال مال، وإذا ضرب مال المال في الجذر فهو مال كعب، وإذا ضرب مال الكعب في الجذر فهو كعب كعب.

The root is every quantity multiplied by itself, beginning from one and above in whole numbers, and below in fractions. If it is multiplied by itself, it is a square; and if the square is multiplied by the root, it is a cube; and if the cube is multiplied by the root, it is square-square; and if square-square is multiplied by the root, it is square-cube; and if square-cube is multiplied by the root, it is cube-cube.

0.2.2 The Square (المال)

والمال هو ما ينتج من ضرب الجذر في نفسه. والكعب هو ما ينتج من ضرب المال في الجذر. وما فوق الكعب فهو مركب من هذه الأجناس.

The square is what results from multiplying the root by itself. The cube is what results from multiplying the square by the root. And what is above the cube is compounded from these species.

0.2.3 Classification of Algebraic Terms

يصنّف الخيام الأجناس البسيطة والمركبة بشكل منهجي.

Khayyam systematically enumerates the simple and compound species:

Arabic Term	**Transliteration**	**Modern Notation**
جذر	*jadh*r	x
مال	*māl*	x^2
كعب	*ka*^{c}*b*	x^3
مال مال	*māl māl*	x^4
مال كعب	*māl ka*^{c}*b*	x^5
كعب كعب	*ka*^{c}*b ka*^{c}*b*	x^6

0.3 Classification of Equations

تصنيف المعادلات

يشكّل هذا القسم جوهر رسالة الخيام. يصنّف جميع المعادلات من الدرجة الأولى حتى الثالثة في خمسة وعشرين نوعاً.

Classification of Equations

This section forms the heart of Khayyam's treatise. He classifies all equations of degree up to three into **twenty-five types**. Following the conventions of his time, all coefficients are positive, so equations like $x^3 + cx = bx^2 + d$ and $x^3 + bx^2 = cx + d$ are considered distinct types.

0.3.1 Equations Solvable by Arithmetical or Euclidean Methods

Simple Equations (Types 1–6)

المعادلات البسيطة □الأنواع 1□6□ التي تحل بطرق حسابية أو إقليدية.

The first group consists of linear and quadratic equations, solvable without conic sections:

1. جذور تساوي أعداداً — Roots equal numbers: $bx = c$
2. أموال تساوي جذوراً — Squares equal roots: $x^2 = bx$
3. أموال تساوي أعداداً — Squares equal numbers: $x^2 = c$
4. أموال وجذور تساوي أعداداً — Squares and roots equal numbers: $x^2 + bx = c$
5. أموال وأعداد تساوي جذوراً — Squares and numbers equal roots: $x^2 + c = bx$
6. جذور وأعداد تساوي أموالاً — Roots and numbers equal squares: $bx + c = x^2$

These six types were known since al-Khwārizmī and are solved by elementary methods.

Binomial Cubic Equations (Types 7–12)

وأما الكعاب والأموال والجذور التي لا تختلف فيها الأجناس، فإنها تنقسم إلى اثني عشر نوعاً: منها ما يحل بالجبر، ومنها ما يحل بالقطوع المخروطية.

As for cubes, squares, and roots in which the species are not mixed, they divide into twelve types: some solved by algebra, and some by conic sections.

7. كعب يساوي مالاً — Cube equals square: $x^3 = bx^2$
8. كعب يساوي جذراً — Cube equals root: $x^3 = cx$
9. كعب يساوي عدداً — Cube equals number: $x^3 = d$
10. مال يساوي كعباً — Square equals cube: $bx^2 = x^3$
11. جذر يساوي كعباً — Root equals cube: $cx = x^3$
12. عدد يساوي كعباً — Number equals cube: $d = x^3$

These binomial equations reduce to simpler forms by division.

Trinomial Equations Reducible to Quadratics (Types 13–16)

13. كعاب وأموال تساوي جذوراً — Cubes and squares equal roots: $x^3 + bx^2 = cx$
14. كعاب وجذور تساوي أموالاً — Cubes and roots equal squares: $x^3 + cx = bx^2$
15. أموال وجذور تساوي كعاباً — Squares and roots equal cubes: $bx^2 + cx = x^3$
16. كعاب وأموال وأعداد — [Special reducible forms]

0.3.2 Irreducible Cubic Equations (The 14 Species)

إنجاز الخيام الأعظم في رسالته هو حله المنهجي للأربعة عشر معادلة تكعيبية غير قابلة للاختزال. هذه تتطلب تقاطع القطوع

The crowning achievement of Khayyam's treatise is his systematic solution of the **14 irreducible**

المخروطية للحل الهندسي.

cubic equations. These require the intersection of conic sections for their geometric solution. We present them here with their Arabic names and modern notation:

The 14 Irreducible Cubic Equations □الأصناف الأربعة عشر□

Group I: Three-term equations with successive powers

17. كعب ومال يساويان عدداً — Cube and square equal number:

$$x^3 + bx^2 = d$$

18. كعب وعدد يساويان مالاً — Cube and number equal square:

$$x^3 + d = bx^2$$

19. كعب يساوي مالاً وعدداً — Cube equals square and number:

$$x^3 = bx^2 + d$$

Group II: Three-term equations with mixed powers

19. كعب وجذور يساويان عدداً — Cube and roots equal number:

$$x^3 + cx = d$$

20. كعب وعدد يساويان جذوراً — Cube and number equal roots:

$$x^3 + d = cx$$

21. كعب يساوي جذوراً وعدداً — Cube equals roots and number:

$$x^3 = cx + d$$

Group III: Four-term equations — three terms on one side

22. كعب وأموال يساويان جذوراً وعدداً — Cube and squares equal roots and number:

$$x^3 + bx^2 = cx + d$$

23. كعب وجذور يساويان أموالاً وعدداً — Cube and roots equal squares and number:

$$x^3 + cx = bx^2 + d$$

24. كعب وأعداد يساويان أموالاً وجذوراً — Cube and numbers equal squares and roots:

$$x^3 + d = bx^2 + cx$$

Group IV: Four-term equations — two terms on each side

25. كعب يساوي أموالاً وجذوراً وأعداداً — Cube equals squares, roots, and number:

$$x^3 = bx^2 + cx + d$$

0.4 On the Geometric Solution of Cubic Equations

في الحل الهندسي للمعادلات التكعيبية

On the Geometric Solution of Cubic Equations

0.4.1 The Fundamental Lemma

قبل تقديم حلوله، يثبت الخيام مقدمة أساسية حول المجسمات □المتوازيات المستطيلات□.

Before presenting his solutions, Khayyam proves a fundamental lemma about solid figures (parallelopipeds):

Lemma on Equal Solids □مقدمة في المجسمات المتساوية□

إذا كان لنا مكعب مستطيل السطوح، قاعدته مربعة، وارتفاعه خط معلوم، وأردنا أن نبني على قاعدة مربعة أخرى مكعباً مستطيلاً مساوياً له، فنحن نجد خطاً يكون النسبة بين ضلع القاعدة الأولى وضلع القاعدة الثانية كالنسبة بين ذلك الخط والارتفاع.

If we have a rectangular parallelopiped whose base is a square and whose height is a known line, and we wish to construct on another square base a rectangular parallelopiped equal to it, then we find a line such that the ratio between the side of the first base and the side of the second base is as the ratio between that line and the height.

0.4.2 Solution of "Cube and Roots Equal a Number" ($x^3 + cx = d$)

هذه هي الحالة الأشهر في رسالة الخيام □□□ المعادلة $x^3 + cx = d$ □كعب وجذور يساويان عدداً□. يحلها الخيام بتقاطع قطع مكافئ ونصف دائرة.

This is the most celebrated case in Khayyam's treatise — the equation $x^3 + cx = d$ (كعب وجذور يساويان عدداً). Khayyam solves it by the intersection of a **parabola** and a **semicircle**.

Case: $x^3 + cx$

فلنضرب مثالاً: كعب وجذور يساويان عدداً. فلنجعل العدد مكعباً مستطيل السطوح، قاعدته مربعة، أضلاعها مساوية لعدد الجذور، وارتفاعه خط معلوم. ثم نأخذ قطعاً مكافئاً رأسه نقطة، ومحوره خط مستقيم، ومعلمه مساوٍ لضلع القاعدة. ونصف دائرة على ارتفاع المكعب، فنلقي من نقطة التقاطع القطع المكافئ والدائرة خطاً إلى المحور، فذلك الخط هو الجذر المطلوب.

Let us give an example: cube and roots equal a number. Let us make the number a rectangular parallelopiped whose base is a square, whose sides are equal to the number of roots, and whose height is a known line. Then we take a parabola whose vertex is a point, whose axis is a straight line, and whose parameter is equal to the side of the base; and we describe a semicircle on the height of the parallelopiped. From the point of intersection of the parabola and the circle we drop a line to the axis: that line is the required root.

The Construction

The construction proceeds as follows:

1. Let $b^2 = c$ (the coefficient of x), and let h be such that $b^2 h = d$ (the constant term)
2. Construct a **parabola** with vertex B, axis BZ, and parameter b, so that for any point on the parabola, $y^2 = bx$
3. Erect perpendicular h at B, and on h as diameter describe a **semicircle**
4. Let the semicircle cut the parabola at point D
5. From D drop DE perpendicular to h, and DZ perpendicular to BZ
6. Then $y = BE$ is the required root

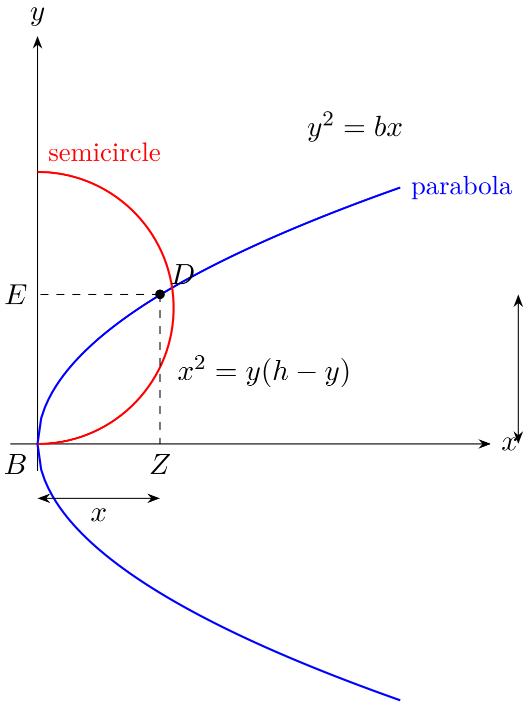

Figure 1: Khayyam's construction for $x^3 + cx = d$ by intersection of parabola and semicircle

The Proof

By the property of the parabola:

$$y^2 = bx \quad \Longrightarrow \quad \frac{x}{y} = \frac{y}{b}$$

By the property of the circle (with diameter h):

$$x^2 = y(h-y)$$

From these:

$$\frac{y}{b} = \frac{x}{y} = \frac{h-y}{x}$$

Therefore $b : y = y : x = x : (h-y)$ are in continued proportion.

From $b^2 : y^2 = b : (h-y)$ we get:

$$b^2(h-y) = y^3$$
$$b^2 h - b^2 y = y^3$$
$$y^3 + b^2 y = b^2 h$$

Substituting $c = b^2$ and $d = b^2 h$:

$$\boxed{y^3 + cy = d}$$

0.5 The Remaining Irreducible Cases

0.5.1 Summary of Conic Section Pairs for Each Case

يلخص الجدول التالي أزواج القطوع المخروطية المستخدمة لكل من الأصناف الأربعة عشر غير القابلة للاختزال، منظمةً حسب العائلات الهندسية الخمس للخيام.

The following table summarizes the pairs of conic sections used for each of the 14 irreducible cases, organized according to Khayyam's five geometric families:

Equation	**Conic Sections**	**Khayyam's Term**	**Arabic**
$x^3 + b^2x = b^2h$	Circle + Parabola	دائرة ومكافئ	
$x^3 + b^2h = b^2x$	Parabola + Hyperbola	مكافئ وزائد	
$x^3 = b^2x + b^2h$	Parabola + Hyperbola	مكافئ وزائد	
$x^3 + ax^2 = c^3$	Circle + Hyperbola	دائرة وزائد	
$x^3 + c^3 = ax^2$	Circle + Hyperbola	دائرة وزائد	
$x^3 = ax^2 + c^3$	Circle + Hyperbola	دائرة وزائد	
$x^3 + cx = ax^2 + d$	Circle + Hyperbola	دائرة وزائد	
$x^3 + ax^2 = cx + d$	Hyperbola + Hyperbola	زائدان	
$x^3 + d = ax^2 + cx$	Circle + Hyperbola	دائرة وزائد	
$x^3 = ax^2 + cx + d$	Hyperbola + Hyperbola	زائدان	

0.5.2 On the Existence and Multiplicity of Solutions

يناقش الخيام بعناية الظروف التي توجد فيها حلول. يلاحظ أن بعض المعادلات التكعيبية قد يكون لها:

Khayyam carefully discusses conditions under which solutions exist. He observes that some cubic equations may have:

- No (positive real) solution
- Exactly one positive solution
- Two positive solutions

واعلم أن بعض هذه الأصناف قد يكون له حل واحد، وبعضها قد يكون له حلان، وبعضها لا يمكن أن يوجد إلا بشرط. فإذا كان القطعان لا يلتقيان فلا حل للمسألة، وإذا كانا يلتقيان في نقطة واحدة فحل واحد، وإذا كانا يلتقيان في نقطتين فحلان.

Know that some of these species may have one solution, some may have two solutions, and some cannot be found except under a condition. If the two conics do not intersect, there is no solution to the problem; if they intersect at one point, there is one solution; and if they intersect at two points, there are two solutions.

0.5.3 Case: Cube and Squares and Roots Equal a Number

الحالة الأخيرة □$x^3 = bx^2 + cx + d$□ هي الأكثر تعقيداً:

وأما الصنف الخامس والعشرون: كعب يساوي أموالاً وجذوراً وأعداداً، فإنه يحتاج إلى قطعين زائدين مختلفين. فنصنع قطعاً زائداً له أحد محوريه على امتداد الخط المعلوم، ونصنع قطعاً زائداً آخر له أحد محوريه على امتداد الخط الآخر، ونلقي من نقطة التقاطعما خطاً إلى الموضع المعلوم، فهو الجذر المطلوب.

The final case ($x^3 = bx^2 + cx + d$) is the most complex:

As for the twenty-fifth species: cube equals squares, roots, and numbers, it requires two different hyperbolas. We make a hyperbola one of whose axes is on the extension of the known line, and we make another hyperbola one of whose axes is on the extension of the other line; and from their point of intersection we drop a line to the known position, and it is the required root.

0.6 Sample Problems and Applications

Sample Problems and Applications مسائل وتطبيقات نموذجية

0.6.1 The Division of Ten into Two Parts

Problem □مسألة□

مسألة: قال قائل: قسّم عشرة إلى جزأين، بحيث يكون مجموع مربعي الجزأين مع خارج قسمة الكبير على الصغير اثنين وسبعين.

A man said: Divide ten into two parts such that the sum of the squares of the two parts together with the quotient of the division of the greater by the smaller be seventy-two.

Solution. Let the greater part be x. Then the smaller is $10 - x$. The condition gives:

$$x^2 + (10 - x)^2 + \frac{x}{10 - x} = 72$$

Simplifying:

$$2x^2 - 20x + 100 + \frac{x}{10 - x} = 72$$

This leads to:

$$x^3 + 200x = 20x^2 + 2000$$

which is of the form $x^3 + cx = bx^2 + d$, solved by the intersection of a circle and a hyperbola.

0.6.2 On Doubling the Cube

ومن ذلك مضاعفة المكعب، وهي مسألة مشهورة بين القدماء. فإنها تؤول إلى أن كعباً يساوي عدداً، والعدد هو مضاعفة المكعب المعطى.

And among these is the duplication of the cube, which is a well-known problem among the ancients. It reduces to a cube equaling a number, where the number is a multiple of the given cube.

Khayyam shows that the ancient Delian problem (doubling the cube) is equivalent to solving $x^3 = 2a^3$, which falls under his classification as “cube equals number.”

0.7 Concluding Remarks

خاتمة

يختتم الخيام رسالته بتأملات في طبيعة الجبر وعلاقته بالهندسة:

وقد أتممنا ما أردنا من جمع أصناف المسائل الجبرية، ووضع البراهين عليها. ونحن نعلم أن بعض الناس يعتقدون أن الجبر بريء من البرهان الهندسي، ولكن هذا باطل. فإن الأصناف التي ذكرناها لا يمكن أن تُبرهن إلا بالقطوع المخروطية أو بغيرها من الأدلة الهندسية. والجبر والمقابلة إنما هما وسيلة لاستخراج المجهول، والبرهان على صحة ذلك إنما هو بالهندسة.

والله ولي التوفيق.

Concluding Remarks

Khayyam concludes his treatise with reflections on the nature of algebra and its relationship to geometry:

We have completed what we intended of gathering the species of algebraic problems and setting down proofs for them. We know that some people believe that algebra is free from geometrical proof, but this is false. For the species we have mentioned cannot be proved except by conic sections or by other geometrical arguments. Algebra and Almucabala are but a means of extracting the unknown, and the proof of its correctness is by geometry.

And God is the source of all success.

End of the Treatise

Editor's Note

This bilingual edition is based primarily on the 13th-century Lahore manuscript (Smith Oriental MS 45, Columbia University Rare Book and Manuscript Library), with reference to the following sources:

- Woepcke, F. *L'Algèbre d'Omar Alkhayyâmî.* Paris, 1851.
- Kasir, D. S. *The Algebra of Omar Khayyam.* New York: Teachers College Press, 1931.
- Rashed, R. and Vahabzadeh, B. *Omar Khayyam, the Mathematician.* New York: Bibliotheca Persica Press, 2000.

The Arabic text has been normalized for readability while preserving the technical vocabulary and mathematical content of the original. Geometric constructions are rendered in modern notation while preserving the logical structure of Khayyam's proofs.

Typeset in LaTeX using XeLaTeX, polyglossia, and Noto Naskh Arabic
Mathematical typesetting via amsmath and TikZ

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ENHANCED BILINGUAL EDITION

Original Arabic Terms Restored

THE ALGEBRA

of

MOHAMMED BEN MUSA

Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

EDITED AND TRANSLATED

BY

FREDERIC ROSEN

Enhanced Edition with Original Arabic Terminology
Six Canonical Cases in Bilingual Presentation
Geometrical Demonstrations with Arabic Labels
Complete Arabic–English Glossary

LONDON:
Printed for the Oriental Translation Fund,
And sold by
J. Murray, Albemarle Street;
Parbury, Allen, & Co., Leadenhall Street;
Thacker & Co., Calcutta; Treuttel & Wuertz, Paris;
And E. Fleischer, Leipzig.

1831

Preface to the Enhanced Edition

This enhanced edition of Frederic Rosen's classic 1831 translation restores the original Arabic terminology that Rosen rendered into English. Where Rosen translated al-Khwārizmī's technical terms into their English equivalents, this edition presents *both* the original Arabic *and* Rosen's English translation, enabling readers to appreciate the precise mathematical vocabulary of the Arabic tradition.

The Six Canonical Terms

Arabic	**Transliteration**	**Rosen's English**
مال	*māl*	"squares"
جذر	*jidhr*	"roots"
عدد	*'adad*	"numbers"
شيء	*shay'*	"thing" (the unknown)
الجبر	*al-jabr*	"restoration/completion"
المقابلة	*al-muqābala*	"reduction/opposition"

The Arabic title of the work is:

كتاب المختصر في حساب الجبر والمقابلة

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Throughout this enhanced edition:

- Arabic terms are presented **[**□□□□□ alongside Rosen's English translation on first occurrence in each section.
- The six canonical cases of equations are given with bilingual headings.
- Geometrical demonstrations include the original Arabic labels.
- A complete Arabic–English glossary is appended.

The goal is to honour both al-Khwārizmī's original mathematical vocabulary and Rosen's pioneering English translation, presenting them together for the benefit of historians of mathematics, Arabists, and mathematicians alike.

THE ALGEBRA

of

MOHAMMED BEN MUSA

EDITED AND TRANSLATED
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1831.

Printed by J. L. Cox, Great Queen Street, London.

Preface

In the study of history, the attention of the observer is drawn by a peculiar charm towards those epochs, at which nations, after having secured their independence externally, strive to obtain an inward guarantee for their power, by acquiring eminence as great in science and in every art of peace as they have already attained in the field of war. Such an epoch was, in the history of the Arabs, that of the Caliphs Al Mansur, Harun al Rashid, and Al Mamun, illustrious contemporaries of Charlemagne; to the glory of which era, the volume now offered to the public, a new monument is endeavoured to be raised.

Abu Abdallah Mohammed ben Musa, of Khowarezm, who it appears, from his preface, wrote this Treatise at the command of the Caliph Al Mamun, was for a long time considered as the original inventor of Algebra.[*] That he was not the inventor of the Art, is now well established; but that he was the first Mohammedan who wrote upon it, is to be found asserted in several Oriental writers. Haji Khalfa, in his bibliographical work, cites the initial words of the treatise now before us,[*] and states, in two distinct passages, that its author, Mohammed ben Musa, was the first Mussulman who had ever written on the solution of problems by the rules of completion and reduction. Two marginal notes in the Oxford manuscript—from which the text of the present edition is taken—and an anonymous Arabic writer, whose Bibliotheca Philosophorum is frequently quoted by Casiri,[*] likewise maintain that this production of Mohammed ben Musa was the first work written on the subject[*] by a Mohammedan.

From the manner in which our author, in his preface, speaks of the task he had undertaken, we cannot infer that he claimed to be the inventor. He says that the Caliph Al Mamun encouraged him to write a popular work on Algebra: an expression which would seem to imply that other treatises were then already extant.

From a formula for finding the circumference of the circle, which occurs in the work itself (Text p. 51, Transl. p. 72), I have, in a note, drawn the conclusion, that part of the information comprised in this volume was derived from an Indian source; a conjecture which is supported by the direct assertion of the author of the *Bibliotheca Philosophorum* quoted by Casiri (i. 426, 428). That Mohammed ben Musa was conversant with Hindu science, is further evident from the fact[*] that he abridged, at Al Mamun's request—but before the accession of that prince to the caliphat—the Sindhind, or astronomical tables,

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[*] *"Haec ars olim Mahomete, Mosis Arabis filio, initium sumsit: etenim hujus rei locuples testis Leo-nardus Pisanus."* Such are the words with which Hieronymus Cardanus commences his *Ars Magna*, in which he frequently refers to the work here translated, in a manner to leave no doubt of its identity.

[*] I am indebted to the kindness of my friend Mr. Gustav Fluegel of Dresden, for a most interesting extract from this part of Haji Khalfa's work. Complete manuscript copies of the كشف الظنون are very scarce. The only two which I have hitherto had an opportunity of examining (the one bought in Egypt by Dr. Ehrenberg, and now deposited in the Royal Library at Berlin—the other among Rich's collection in the British Museum) are only abridgments of the original compilation, in which the quotation of the initial words of each work is generally omitted. The prospect of an edition and Latin translation of the complete original work, to be published by Mr. Fluegel, under the auspices of the Oriental Translation Committee, must under such circumstances be most gratifying to all friends of Asiatic Literature.

[*] ابن الابي written in the twelfth century. Bibliotheca Arabica Escurialensis, t. i. 426, 428.

[*] The first of these marginal notes stands at the top of the first page of the manuscript, and reads thus: هذا اول كتاب صنفه بيد مسلم في الجبر والمقابلة "This is the first book written on (the art of calculating by) completion and reduction by a Mohammedan: on this account the author has introduced into it rules of various kinds, in order to render useful the very rudiments of Algebra." The other scholium stands farther on: it is the same to which I have referred in my notes to the Arabic text, p. 177.

[*] Related by Ebn al Adami in the preface to his astronomical tables. Casiri, l. 427, 428. Colebrooke, Dissertation, &c. p. lxiv, lxxii.

translated by Mohammed ben Ibrahim al Fazari from the work of an Indian astronomer who visited the court of Almansur in the 156th year of the Hejira (A.D. 773).

The science as taught by Mohammed ben Musa, in the treatise now before us, does not extend beyond quadratic equations, including problems with an affected square. These he solves by the same rules which are followed by Diophantus,[*] and which are taught, though less comprehensively, by the Hindu mathematicians.[*] That he should have borrowed from Diophantus is not at all probable; for it does not appear that the Arabs had any knowledge of Diophantus' work before the middle of the fourth century after the Hejira, when Abu'l-Wafa Buzjani rendered it into Arabic.[*]

It is far more probable that the Arabs received their first knowledge of Algebra from the Hindus, who furnished them with the decimal notation of numerals, and with various important points of mathematical and astronomical information.

But under whatever obligation our author may be to the Hindus, as to the subject matter of his performance, he seems to have been independent of them in the manner of digesting and treating it: at least the method which he follows in expounding his rules, as well as in showing their application, differs considerably from that of the Hindu mathematical writers. Bhaskara and Brahmagupta give dogmatical precepts, unsupported by argument, which, even by the metrical form in which they are expressed, seem to address themselves rather to the memory than to the reasoning faculty of the learner: Mohammed gives his rules in simple prose, and establishes their accuracy by geometrical illustrations. The Hindus give comparatively few examples, and are fond of investing the statement of their problems in rhetorical pomp: the Arab, on the contrary, is remarkably rich in examples, but he introduces them with the same perspicuous simplicity of style which distinguishes his rules. In solving their problems, the Hindus are satisfied with pointing at the result, and at the principal intermediate steps which lead to it: the Arab shows the working of each example at full length, keeping his view constantly fixed upon the two sides of the equation, as on the two scales of a balance, and showing how any alteration in one side is counterpoised by a corresponding change in the other.

Besides the few facts which have already been mentioned in the course of this preface, little or nothing is known of our Author's life. He lived and wrote under the caliphat of Al Mamun, and must therefore be distinguished from Abu Jafar Mohammed ben Musa,[*] likewise a mathematician and astronomer, who flourished under the Caliph Al Motaded

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[*] See Diophantus, Introd. § ii. and Book iv. problems 32 and 33.

[*] Lilavati, p. 29, Vijaganita, p. 347, of Mr. Colebrooke's translation.

[*] Casiri Bibl. Arab. Escur. i. 433. Colebrooke's Dissertation, &c. p. lxxii.

[*] The father of the latter, Musa ben Shaker, whose native country I do not find recorded, had been a robber or bandit in the earlier part of his life, but had afterwards found means to attach himself to the court of the Caliph Al-Mamun; who, after Musa's death, took care of the education of his three sons, Mohammed, Ahmed, and Al Hassan. (Abulfaragii Histor. Dyn. p. 280. Casiri, l. 386, 418). Each of the sons subsequently distinguished himself in mathematics and astronomy. We learn from Abulfaraj (p. 281) and from Ibn Khallikan (art. ثابت بن قرة) that Thabet ben Korrah, the well-known translator of the Almagest, was indebted to Mohammed for his introduction to Al Motaded, and the men of science at the court of that caliph. Ibn Khallikan's words are: خرج من حران وسكن كفرطابا فاقام بها الى ان قدم عليه محمد بن موسى راجعا من بلاد الروم الى بغداد. فصحبه واجتهد في معرفته وعرف فضله وذكاءه فحمله الى بغداد فنزله داره وقربه الى المتوكل وكان من جملة اهل الحجاب “(Thabet ben Korrah) left Harran, and established himself at Kafratutha, where he remained till Mohammed ben Musa arrived there, on his return from the Greek dominions to Bagdad. The latter became acquainted with Thabet and on seeing his skill and sagacity, invited Thabet to accompany him to Bagdad, where Mohammed made him lodge at his own house, introduced him to the Caliph, and procured him an appointment in the body of astronomers.” Ibn Khallikan here speaks of Mohammed ben Musa as of a well-known individual: he has however devoted no special article to an account of his life. It is possible that the tour into the provinces of the Eastern Roman Empire here mentioned, was undertaken in search of some ancient Greek works on mathematics or astronomy.

(who reigned A.H. 279–289, A.D. 892–902).

The manuscript from whence the text of the present edition is taken—and which is the only copy the existence of which I have as yet been able to trace—is preserved in the Bodleian collection at Oxford. It is, together with three other treatises on Arithmetic and Algebra, contained in the volume marked cmxviii. Hunt. 214, fol., and bears the date of the transcription A.H. 743 (A.D. 1342). It is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points: a deficiency which has often been very sensibly felt; for though the nature of the subject matter can but seldom leave a doubt as to the general import of a sentence, yet the true reading of some passages, and the precise interpretation of others, remain involved in obscurity. Besides, there occur several omissions of words, and even of entire sentences; and also instances of words or short passages written twice over, or words foreign to the sense introduced into the text. In printing the Arabic part, I have included in brackets many of those words which I found in the manuscript, the genuineness of which I suspected, and also such as I inserted from my own conjecture, to supply an apparent hiatus.

The margin of the manuscript is partially filled with scholia in a very small and almost illegible character, a few specimens of which will be found in the notes appended to my translation. Some of them are marked as being extracted from a commentary (شرح) by Al Mozaihafi,[*] probably the same author, whose full name is Jamaleddin Abu Abdallah Mohammed ben Omar al Jaza'if al Mozaihafi, and whose "Introduction to Arithmetic," (مبادي علم الحساب) is contained in the same volume with Mohammed's work in the Bodleian library.

Numerals are in the text of the work always expressed by words: figures are only used in some of the diagrams, and in a few marginal notes.

The work had been only briefly mentioned in Uris' catalogue of the Bodleian manuscripts. Mr. H. T. Colebrooke first introduced it to more general notice, by inserting a full account of it, with an English translation of the directions for the solution of equations, simple and compound, into the notes of the "Dissertation" prefixed to his invaluable work, "Algebra, with Arithmetic and Mensuration, from the Sanscrit of Brahmegupta and Bhascara." (London, 1817, 4to. pages lxxv–lxxix.)

The account of the work given by Mr. Colebrooke excited the attention of a highly distinguished friend of mathematical science, who encouraged me to undertake an edition and translation of the whole: and who has taken the kindest interest in the execution of my task. He has with great patience and care revised and corrected my translation, and has furnished the commentary, subjoined to the text, in the form of common algebraic notation. But my obligations to him are not confined to this only; for his luminous advice has enabled me to overcome many difficulties, which, to my own limited proficiency in mathematics, would have been almost insurmountable.

In some notes on the Arabic text which are appended to my translation, I have endeavoured, not so much to elucidate, as to point out for further enquiry, a few circumstances connected with the history of Algebra. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but my enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which I had the opportunity of consulting.

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[*] Wherever I have met with this name, it is written without the diacritical points المزيحفي, and my pronunciation rests on mere conjecture.

The Author's Preface

In the Name of God, gracious and merciful!

This work was written by Mohammed ben Musa, of Khowarezm. He commences it thus:

Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain. Through him he cured of blindness, and saved through him from perdition, and increased through him what before was small, and collected through him what before was scattered. Praised be God our Lord! and may his glory increase, and may all his names be hallowed—besides whom there is no God; and may his benediction rest on Mohammed the Prophet and on his descendants!

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

Some applied themselves to obtain information which was not known before them, and left it to posterity; others commented upon the difficulties in the works left by their predecessors, and defined the best method (of study), or rendered the access (to science) easier or placed it more within reach; others again discovered mistakes in preceding works, and arranged that which was confused, or adjusted what was irregular, and corrected the faults of their fellow-labourers, without arrogance towards them, or taking pride in what they did themselves.

That fondness for science, by which God has distinguished the Imam al Mamun, the Commander of the Faithful (besides the caliphat which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties,—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned—relying on the goodness of my intention therein, and hoping that the learned will reward it, by obtaining (for me) through their prayers the excellence of the Divine mercy: in requital of which, may the choicest blessings and the abundant bounty of God be theirs! My confidence rests with God, in this as in every thing, and in Him I put my trust. He is the Lord of the Sublime Throne. May His blessing descend upon all the prophets and heavenly messengers!

Mohammed ben Musa's

Compendium

on Calculating by

Completion and Reduction.

Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala
كتاب المختصر في حساب الجبر والمقابلة

Introduction

When I considered what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, &c., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction [الجبر والمقابلة are of three kinds, namely, **roots** [جذر, **squares** [مال, and **simple numbers relative to neither root nor square** [عدد.

A *root* [جذر is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.*

A *square* [مال is the whole amount of the root multiplied by itself.

A *simple number* [عدد is any number which may be pronounced without reference to root or square.

A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, "*squares are equal to roots,*" or "*squares are equal to numbers,*" or "*roots are equal to numbers.*"

Case I.—Squares are equal to Roots. [المال يساوي الجذور

Of the case in which squares are equal to roots, this is an example: "A square is equal to five roots of the same;" the root of the square is five, and the square is twenty-five, which is equal to five times its root.

So you say, "one third of the square is equal to four roots;" then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

Or you say, "five squares are equal to ten roots;" then one square is equal to two roots; the root of the square is two, and its square is four.

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*By the word root, is meant the simple power of the unknown quantity.

In this manner, whether the squares be many or few, (i.e. multiplied or divided by any number), they are reduced to a single square; and the same is done with the roots, which are their equivalents; that is to say, they are reduced in the same proportion as the squares.

Case II.—Squares are equal to Numbers. [المال يساوي الأعداد □□□□□□□ □□□□□□□□ □□□□□□□□□□□

As to the case in which squares are equal to numbers; for instance, you say, “a square is equal to nine;” then this is a square, and its root is three. Or “five squares are equal to eighty;” then one square is equal to one-fifth of eighty, which is sixteen. Or “the half of the square is equal to eighteen;” then the square is thirty-six, and its root is six.

Thus, all squares, multiples, and sub-multiples of them, are reduced to a single square. If there be only part of a square, you add thereto, until there is a whole square; you do the same with the equivalent in numbers.

Case III.—Roots are equal to Numbers. [الجذور تساوي الأعداد □□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□

As to the case in which roots are equal to numbers; for instance, “one root equals three in number;” then the root is three, and its square nine. Or “four roots are equal to twenty;” then one root is equal to five, and the square to be formed of it is twenty-five. Or “half the root is equal to ten;” then the whole root is equal to twenty, and the square which is formed of it is four hundred.

I found that these three kinds; namely, **roots** **[**جذور □□□□□□□□□□, **squares** **[**أموال □□□□□□□□□, and **numbers** **[**أعداد □□□□□□□□□, may be combined together, and thus three compound species arise;[*] that is, “*squares and roots equal to numbers*;” “*squares and numbers equal to roots*;” “*roots and numbers equal to squares*.”

Case IV.—Roots and Squares are equal to Numbers. [الجذور والمال يساويان الأعداد □□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□

Roots and Squares are equal to Numbers;[*] for instance, “one square, and ten roots of the same, amount to thirty-nine dirhems;” that is to say, what must be the square which, when increased by ten of its own roots, amounts to thirty-nine? The solution is this: you halve the number[*] of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

The solution is the same when two squares or three, or more or less be specified;[*] you reduce them to one single square, and in the same proportion you reduce also the roots

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[*] The three cases considered are,

1^{st}. $cx^2 + bx = a$

2^{d}. $cx^2 + a = bx$

3^{d}. $cx^2 = bx + a$

[*] 1^{st} case: $cx^2 + bx = a$.
Example: $x^2 + 10x = 39$
$x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$
i.e. the coefficient.

[*] By “the number of the roots” is meant the coefficient.

[*] $cx^2 + bx = a$ is to be reduced to the form $x^2 + \frac{b}{c}x = \frac{a}{c}$.

and simple numbers which are connected therewith.

For instance, “two squares and ten roots are equal to forty-eight dirhems;”[*] that is to say, what must be the amount of two squares which, when summed up and added to ten times the root of one of them, make up a sum of forty-eight dirhems? You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce every thing mentioned in the statement to its half, and it will be the same as if the question had been, a square and five roots of the same are equal to twenty-four dirhems; or, what must be the amount of a square which, when added to five times its root, is equal to twenty-four dirhems? Now halve the number of the roots; the moiety is two and a half. Multiply that by itself; the product is six and a quarter. Add this to twenty-four; the sum is thirty dirhems and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

The proceeding will be the same if the instance be, “half of a square and five roots are equal to twenty-eight dirhems;”[*] that is to say, what must be the amount of a square, the moiety of which, when added to the equivalent of five of its roots, is equal to twenty-eight dirhems? Your first business must be to complete your square, so that it amounts to one whole square. This you effect by doubling it. Therefore double it, and double also that which is added to it, as well as what is equal to it. Then you have a square and ten roots, equal to fifty-six dirhems. Now halve the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Add this to fifty-six; the sum is eighty-one. Extract the root of this; it is nine. Subtract from this the moiety of the number of roots, which is five; the remainder is four. This is the root of the square which you sought for; the square is sixteen, and half the square eight.

Proceed in this manner, whenever you meet with squares and roots that are equal to simple numbers: for it will always answer.

Case V.—Squares and Numbers are equal to Roots. المال والأعداد يساويان الجذور]

Squares and Numbers are equal to Roots;[*] for instance, “a square and twenty-one in numbers are equal to ten roots of the same square.” That is to say, what must be the amount of a square, which, when twenty-one dirhems are added to it, becomes equal to the equivalent of ten roots of that square?

Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. Or you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the

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[*] $2x^2+10x=48$, reduced to $x^2+5x=24$,
$x=\sqrt{6\frac{1}{4}+24}-2\frac{1}{2}=\sqrt{30\frac{1}{4}}-2\frac{1}{2}=5\frac{1}{2}-2\frac{1}{2}=3.$

[*] $\frac{1}{2}x^2+10x=56$; $x^2+10x=56$; $x=\sqrt{25+56}-5=\sqrt{81}-5=9-5=4.$

[*] 2^d case: $cx^2+a=bx$.
Example: $x^2+21=10x$
$x=\frac{10}{2}\pm\sqrt{(\frac{10}{2})^2-21}=5\pm\sqrt{25-21}=5\pm\sqrt{4}=5\pm 2=7$ or 3.

three cases in which the number of the roots must be halved.[*] And know, that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be less than the number of dirhems connected with the square, then the instance is impossible;[*] but if the product be equal to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

In every instance where you have two squares, or more or less, reduce them to one entire square,[*] as I have explained under the first case.

Case VI.—Roots and Numbers are equal to Squares. الجذور والأعداد تساوي المال[

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Roots and Numbers are equal to Squares;[*] for instance, “three roots and four of simple numbers are equal to a square.” Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

Whenever you meet with a multiple or sub-multiple of a square, reduce it to one entire square.

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

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[*] If in an equation, of the form $x^2 + a = bx$, $(\frac{b}{2})^2 < a$, the case supposed in the equation cannot happen. If $(\frac{b}{2})^2 = a$, then $x = \frac{b}{2}$.

[*] The case of $(\frac{b}{2})^2 < a$ gives no real solution.

[*] $cx^2 + a = bx$ is to be reduced to $x^2 + \frac{a}{c} = \frac{b}{c}x$.

[*] 3^d case: $cx^2 = bx + a$.
Example: $x^2 = 3x + 4$
$x = \sqrt{(\frac{3}{2})^2 + 4} + \frac{3}{2} = \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} = \sqrt{6\frac{1}{4}} + 1\frac{1}{2} = 2\frac{1}{2} + 1\frac{1}{2} = 4.$

Geometrical Demonstrations

DEMONSTRATION OF THE CASE: *“a Square [مال □□□□□□□ and ten Roots [جذور □□□□□ □□□□□□ are equal to thirty-nine Dirhems [درهم □□□□□□□□□□.”*[*]
The figure to explain this a quadrate, the sides of which are unknown. It represents the square, the root of which, or the root of which, you wish to know. This is the figure AB, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square. Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate AB, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms C, G, T, and K. We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five. We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure AB, by which the great figure DH is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate DH, then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure AB.

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.[*] Accordingly, we multiplied only the moiety of the roots by itself, instead of multiplying its fourth by itself, and then by four.

This is the figure:

	G	
C	AB	T
	K	

The same may also be explained by another figure. We proceed from the quadrate AB, which represents the square. It is our next business to add to it the ten roots of the same. We halve for this purpose the ten, so that it becomes five, and construct two quadrangles on two sides of the quadrate AB, namely, G and D, the length of each of them being five, as the moiety of the ten roots, whilst the breadth of each is equal to a side of the quadrate AB. Then a quadrate remains opposite the corner of the quadrate AB. This is equal to five multiplied by five: this five being half of the number of the roots which we have added to each of the two sides of the first quadrate. Thus we know that the first quadrate, which is the square, and the two quadrangles on its sides, which are the ten

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[*] Geometrical illustration of the case, $x^2 + 10x = 39$.

[*] $4 \times (\frac{b}{4})^2 = (\frac{b}{2})^2$.

roots, make together thirty-nine. In order to complete the great quadrate, there wants only a square of five multiplied by five, or twenty-five. This we add to thirty-nine, in order to complete the great square SH. The sum is sixty-four. We extract its root, eight, which is one of the sides of the great quadrangle. By subtracting from this the same quantity which we have before added, namely five, we obtain three as the remainder. This is the side of the quadrangle AB, which represents the square; it is the root of this square, and the square itself is nine.

This is the figure:

D	
AB	G

Demonstration of the Case: "a Square and twenty-one Dirhems are equal to ten Roots."

We represent the square by a quadrate AD, the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate AD, such as the side HN. This parallelogram is HB. The length of the two figures together is equal to the line HC. We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same. As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line HC is equal to ten of numbers, since the line CD represents the root of the square. We now divide the line CH into two equal parts at the point G: the line GC is then equal to HG. It is also evident that the line GT is equal to the line CD. At present we add to the line GT, in the same direction, a piece equal to the difference between CG and GT, in order to complete the square. Then the line TK becomes equal to KM, and we have a new quadrate of equal sides and angles, namely, the quadrate MT. We know that the line TK is five; this is consequently the length also of the other sides: the quadrate itself is twenty-five, this being the product of the multiplication of half the number of the roots by themselves, for five times five is twenty-five.

We have perceived that the quadrangle HB represents the twenty-one of numbers which were added to the quadrate. We have then cut off a piece from the quadrangle HB by the line KT (which is one of the sides of the quadrate MT), so that only the part TA remains. At present we take from the line KM the piece KL, which is equal to GK; it then appears that the line TG is equal to ML; moreover, the line KL, which has been cut off from KM, is equal to KG; consequently, the quadrangle MR is equal to TA. Thus it is evident that the quadrangle HT, augmented by the quadrangle MR, is equal to the quadrangle HB, which represents the twenty-one.

The whole quadrate MT was found to be equal to twenty-five. If we now subtract from this quadrate, MT, the quadrangles HT and MR, which are equal to twenty-one, there remains a small quadrate KR, which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line RG, which is equal to GA, is two. If you subtract this number two from the line CG, which is the moiety of the roots, then the remainder is the line AC; that is to say, three, which is the root of the original square. But if you add the number two to the line CG, which is the moiety of the number of the roots, then the sum is seven, represented by the line CR, which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

Demonstration of the Case: “three Roots and four of Simple Numbers are equal to a Square.”

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate AD, which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle HD from the quadrate AD, and take one of its sides HC for three, which is the number of the roots. The same is equal to RD. It follows, then, that the quadrangle HB represents the four of numbers which are added to the roots. Now we halve the side CH, which is equal to three roots, at the point G; from this division we construct the square HT, which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter. We add then to the line GT a piece equal to the line AH, namely, the piece TL; accordingly the line GL becomes equal to AG, and the line KN equal to TL. Thus a new quadrangle, with equal sides and angles, arises, namely, the quadrangle GM and we find that the line AG is equal to ML, and the same line AG is equal to GL. By these means the line CG remains equal to NR, and the line MN equal to TL, and from the quadrangle HB a piece equal to the quadrangle KL is cut off.

But we know that the quadrangle AR represents the four of numbers which are added to the three roots. The quadrangle AN and the quadrangle KL are together equal to the quadrangle AR, which represents the four of numbers.

We have seen, also, that the quadrangle GM comprises the product of the moiety of the roots, or of one and a half, multiplied by itself; that is to say two and a quarter, together with the four of numbers, which are represented by the quadrangles AN and KL.

There remains now from the side of the great original quadrate AD, which represents the whole square, only the moiety of the roots, that is to say, one and a half, namely, the line GC. If we add this to the line AG, which is the root of the quadrate GM, being equal to two and a half; then this, together with CG, or the moiety of the three roots, namely, one and a half, makes four, which is the line AC, or the root to a square, which is represented by the quadrate AD.

Here follows the figure. This it was which we were desirous to explain.

We have observed that every question which requires equation or reduction for its solution, will refer you to one of the six cases which I have proposed in this book. I have now also explained their arguments. Bear them, therefore, in mind.

On Multiplication

I SHALL NOW TEACH YOU how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.[*]

If there are greater numbers combined with units to be added to or subtracted from them, then four multiplications are necessary;[*] namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.[*]

For instance, “ten and one to be multiplied by ten and two.”[*] Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.

But if the instance is “ten less one, to be multiplied by ten less one,”[*] then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.

Or if the instance be “ten and two, to be multiplied by ten less one,”[*] then ten times ten is a hundred, and the negative one by ten is ten negative; the positive two by ten is twenty positive; this together is a hundred and ten; the positive two by the negative one gives two negative. This makes the product a hundred and eight.

I have explained this, that it might serve as an introduction to the multiplication of unknown sums, when numbers are added to them, or when numbers are subtracted from them, or when they are subtracted from numbers.

For instance: “Ten less **thing [**شيء ▯▯▯▯▯▯▯▯, the signification of **root [**جذر ▯▯▯▯▯▯▯▯, to be multiplied by ten.”[*] You begin by taking ten times ten, which is a hundred; less thing by ten is ten roots negative; the product is therefore a hundred less ten things.

If the instance be: “ten and thing to be multiplied by ten,”[*] then you take ten times ten, which is a hundred, and thing by ten is ten things positive; so that the product is a hundred plus ten things.

If the instance be: “ten and thing to be multiplied by itself,”[*] then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and

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[*] If x is to be multiplied by y, x is to be repeated as many times as there are units in y.

[*] If $x \pm a$ is to be multiplied by $y \pm b$, x is to be multiplied by y, x is to be multiplied by b, a is to be multiplied by y, and a is to be multiplied by b.

[*] In multiplying $y \pm x = a \pm b$:
$(+a)(+b) = +ab$; $(-a)(-b) = +ab$; $(+a)(-b) = -ab$; $(-a)(+b) = -ab$.

[*] $(10 + 1) \times (10 + 2) = 10 \times 10 + 1 \times 10 + 2 \times 10 + 1 \times 2 = 100 + 10 + 20 + 2 = 132$.

[*] $(10 - 1) \times (10 - 1) = 10 \times 10 - 10 - 10 + 1 = 100 - 20 + 1 = 81$.

[*] $(10 + 2) \times (10 - 1) = 10 \times 10 - 10 + 20 - 2 = 100 - 10 + 20 - 2 = 108$.

[*] $(10 - x) \times 10 = 10 \times 10 - 10x = 100 - 10x$.

[*] $(10 + x) \times 10 = 10 \times 10 + 10x = 100 + 10x$.

[*] $(10 + x)(10 + x) = 10 \times 10 + 10x + 10x + x^2 = 100 + 20x + x^2$.

thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.

If the instance be: "ten minus thing to be multiplied by ten minus thing," then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things. But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.

In like manner if the following question be proposed to you: "one dirhem minus one-sixth to be multiplied by one dirhem minus one-sixth;" that is to say, five-sixths by themselves, the product is five and twenty parts of a dirhem, which is divided into six and thirty parts, or two-thirds and one-sixth of a sixth. Computation: You multiply one dirhem by one dirhem, the product is one dirhem; then one dirhem by minus one-sixth, that is one-sixth negative; then, again, one dirhem by minus one-sixth is one-sixth negative: so far, then, the result is two-thirds of a dirhem: but there is still minus one-sixth to be multiplied by minus one-sixth, which is one-sixth of a sixth positive; the product is, therefore, two-thirds and one sixth of a sixth.

If the instance be, "ten minus thing to be multiplied by ten and thing," then you say,[*] ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square positive; therefore, the product is a hundred dirhems, minus a square.

If the instance be, "ten minus thing to be multiplied by thing," then you say, ten multiplied by thing is ten things; and minus thing by thing is a square negative; therefore, the product is ten things minus a square.

If the instance be, "ten and thing to be multiplied by thing less ten," then you say, thing multiplied by ten is ten things positive; and thing by thing is a square positive; and minus ten by ten is a hundred dirhems negative; and minus ten by thing is ten things negative. You say, therefore, a square minus a hundred dirhems; for, having made the reduction, that is to say, having removed the ten things positive by the ten things negative, there remains a square minus a hundred dirhems.

If the instance be, "ten dirhems and half a thing to be multiplied by half a dirhem, minus five things," then you say, half a dirhem by ten is five dirhems positive; and half a dirhem by half a thing is a quarter of thing positive; and minus five things by ten dirhems is fifty roots negative. This altogether makes five dirhems minus forty-nine things and three quarters of thing. After this you multiply five roots negative by half a root positive: it is two squares and a half negative. Therefore, the product is five dirhems, minus two squares and a half, minus forty-nine roots and three quarters of a root.

If the instance be, "ten and thing to be multiplied by thing less ten," then this is the same as if it were said thing and ten by thing less ten. You say, therefore, thing multiplied by thing is a square positive; and ten by thing is ten things positive; and minus ten by thing is ten things negative. You now remove the positive by the negative, then there only remains a square. Minus ten multiplied by ten is a hundred, to be subtracted from the square. This, therefore, altogether, is a square less a hundred dirhems.

Whenever a positive and a negative factor concur in a multiplication, such as thing positive and minus thing, the last multiplication gives always the negative product. Keep this in memory.

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[*] $(10-x)(10+x) = 10 \times 10 - 10x + 10x - x^2 = 100 - x^2$.

On Addition and Subtraction

KNOW THAT the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.[*]

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.[*]

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.[*]

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.[*]

I shall hereafter explain to you the reason of this by a figure, which will be annexed to this chapter.

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square;[*] the root of the product is equal to twice the root of the original square.

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.

Compute in this manner every multiplication of the roots, whether the multiplication be more or less than two.

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square. Follow the same rule when you seek for a third, or a quarter of a root, or any larger or smaller quota of it, whatever may be the denominator or the numerator.

Examples of this: If you require to double the root of nine,[*] you multiply two by two, and then by nine: this gives thirty-six; take the root of this, it is six, and this is double the root of nine.

In the same manner, if you require to triple the root of nine,[*] you multiply three by three, and then by nine; the product is eighty-one; take its root, it is nine, which becomes equal to thrice the root of nine.

If you require to have the moiety of the root of nine,[*] you multiply a half by a half, which gives a quarter, and then this by nine; the result is two and a quarter: take its root; it is one and a half, which is the moiety of the root of nine.

You proceed in this manner with every root, whether positive or negative, and whether known or unknown.

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[*] $\sqrt{200} - 10 + (20 - \sqrt{200}) = 10.$

[*] $20 - \sqrt{200} - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}.$

[*] $50 + 10x - 2x^2 + (100 + x^2 - 20x) = 150 - 10x - x^2.$

[*] $100 + x^2 - 20x - [50 - 2x^2 + 10x] = 50 + 3x^2 - 30x.$

[*] $(2)^2 \times x^2 = 4x^2$; therefore $\sqrt{4x^2} = 2x.$

[*] $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9.$

[*] $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9.$

[*] $\frac{1}{2}\sqrt{9} = \sqrt{\frac{1}{4} \times 9} = \sqrt{2\frac{1}{4}} = 1\frac{1}{2}.$

On Division

If you will divide the root of nine by the root of four,[*] you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

If you will divide the root of four by the root of nine,[*] you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

If you wish to divide twice the root of nine by the root of four, or of any other square,[*] you double the root of nine in the manner above shown to you in the chapter on Multiplication, and you divide the product by four, or by any number whatever. You perform this in the way above pointed out.

In like manner, if you wish to divide three roots of nine, or more, or one-half or any multiple or sub-multiple of the root of nine, the rule is always the same: follow it, the result will be right.

If you wish to multiply the root of nine by the root of four,[*] multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

Thus, if you wish to multiply the root of five by the root of ten,[*] multiply five by ten: the root of the product is what you have required.

If you wish to multiply the root of one-third by the root of a half,[*] you multiply one-third by a half: it is one-sixth: the root of one-sixth is equal to the root of one-third, multiplied by the root of a half.

If you require to multiply twice the root of nine by thrice the root of four,[*] then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

You proceed in this manner with all positive or negative roots.

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[*] $\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}$.

[*] $\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$.

[*] $\frac{2\sqrt{9}}{\sqrt{4}} = \frac{\sqrt{4}\times\sqrt{9}}{\sqrt{4}} = \sqrt{9} = 3$.

[*] $\sqrt{4} \times \sqrt{9} = \sqrt{4 \times 9} = \sqrt{36} = 6$.

[*] $\sqrt{10} \times \sqrt{5} = \sqrt{5 \times 10} = \sqrt{50}$.

[*] $\sqrt{\frac{1}{3}} \times \sqrt{\frac{1}{2}} = \sqrt{\frac{1}{3} \times \frac{1}{2}} = \sqrt{\frac{1}{6}}$.

[*] $3\sqrt{4} \times 2\sqrt{9} = \sqrt{9 \times 4} \times \sqrt{4 \times 9} = \sqrt{36 \times 36} = 36$.

Demonstrations

The argument for the root of two hundred, minus ten, added to twenty, minus the root of two hundred, may be elucidated by a figure.

Let the line AB represent the root of two hundred; let the part from A to the point C be the ten, then the remainder of the root of two hundred will correspond to the remainder of the line AB, namely to the line CB. Draw now from the point B a line to the point D, to represent twenty; let it, therefore, be twice as long as the line AC, which represents ten; and mark a part of it from the point B to the point H, to be equal to the line AB, which represents the root of two hundred; then the remainder of the twenty will be equal to the part of the line, from the point H to the point D.

As our object was to add the remainder of the root of two hundred, after the subtraction of ten, that is to say, the line CB, to the line HD, or to twenty, minus the root of two hundred, we cut off from the line BH a piece equal to CB, namely, the line SH. We know already that the line AB, or the root of two hundred, is equal to the line BH, and that the line AC, which represents the ten, is equal to the line SB, as also that the remainder of the line AB, namely, the line CB is equal to the remainder of the line BH, namely, to SH. Let us add, therefore, this piece SH, to the line HD. We have already seen that from the line BD, or twenty, a piece equal to AC, which is ten, was cut off, namely, the piece BS. There remains after this the line SD, which, consequently, is equal to ten.

This it was that we intended to elucidate.

The argument for the root of two hundred, minus ten, to be subtracted from twenty, minus the root of two hundred, is as follows.

Let the line AB represent the root of two hundred, and let the part thereof, from A to the point C, signify the ten mentioned in the instance. We draw now from the point B, a line towards the point D, to signify twenty. Then we trace from B to the point H, the same length as the length of the line which represents the root of two hundred; that is of the line AB. We have seen that the line CB is the remainder from the twenty, after the root of two hundred has been subtracted.

It is our purpose, therefore, to subtract the line CB from the line HD; and we now draw from the point B, a line towards the point S, equal in length to the line AC, which represents the ten. Then the whole line SD is equal to SB, plus BD, and we perceive that all this added together amounts to thirty.

We now cut off from the line HD, a piece equal to CB, namely, the line HG; thus we find that the line GD is the remainder from the line SD, which signifies thirty. We see also that the line BH is the root of two hundred and that the line SB and BC is likewise the root of two hundred. Now the line HG is equal to CB; therefore the piece subtracted from the line SD, which represents thirty, is equal to twice the root of two hundred, or once the root of eight hundred.

This it is that we wished to elucidate.

As for the hundred and square minus twenty roots added to fifty, and ten roots minus two squares, this does not admit of any figure, because there are three different species, viz. squares, and roots, and numbers, and nothing corresponding to them by which they might be represented. We had, indeed, contrived to construct a figure also for this case, but it was not sufficiently clear.

The elucidation by words is very easy. You know that you have a hundred and a square, minus twenty roots. When you add to this fifty and ten roots, it becomes a hundred and fifty and a square, minus ten roots. The reason for these ten negative roots is, that from the twenty negative roots ten positive roots were subtracted by reduction. This being done, there remains a hundred and fifty and a square, minus ten roots. With the hundred a square is connected. If you subtract from this hundred and square the two squares negative connected with fifty, then one square disappears by reason of the other, and the remainder is a hundred and fifty, minus a square, and minus ten roots.

This it was that we wished to explain.

Of the Six Problems

Before the chapters on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

First Problem.

I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other.[*]

Computation: Suppose one of the portions to be thing, and the other ten minus thing: you multiply thing by ten minus thing; it is ten things minus a square. Then multiply it by four, because the instance states "four times as much." The result will be four times the product of one of the parts multiplied by the other. This is forty things minus four squares. After this you multiply thing by thing, that is to say, one of the portions by itself. This is a square, which is equal to forty things minus four squares. Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight, and this is one of the two portions, namely, that which is to be multiplied by itself. The remainder from the ten is two, and that is the other portion. Thus the question leads you to one of the six cases, namely, that of "squares equal to roots." Remark this.

Second Problem.

I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths; or equal to the other multiplied by itself, and afterwards by six and one-fourth.[*]

Computation: Suppose one of the parts to be thing, and the other ten minus thing. You multiply thing by itself, it is a square; then by two and seven-ninths, this makes it two squares and seven-ninths of a square. You afterwards multiply ten by ten; it is a hundred, which must be equal to two squares and seven-ninths of a square. Reduce it to one square, through division by nine twenty-fifths;[*] this being its fifth and four-fifths of its fifth, take now also the fifth and four-fifths of the fifth of a hundred; this is thirty-six, which is equal to one square. Take its root, it is six. This is one of the two portions; and accordingly the other is four. This question leads you, therefore, to one of the six cases, namely, "squares equal to numbers."

Third Problem.

I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four.[*]

Computation: Suppose one of the two parts to be thing, the other ten minus thing.

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[*] $(10-x)x = \text{product};\ x^2 = 4(10-x)x = 40x - 5x^2;\ 5x^2 = 40x;\ x^2 = 8x;\ x = 8;\ (10-x) = 2.$

[*] $100 = x^2 \times 2\frac{7}{9};\ \frac{1}{2\frac{7}{9}} \times 100 = x^2;\ x = 6;\ 10 - x = 4.$

[*] $2\frac{7}{9} = \frac{25}{9};\ \frac{1}{\frac{25}{9}} \times 100 = \frac{9}{25} \times 100 = 36;\ x^2 = 36;\ x = 6.$

[*] $\frac{10-x}{x} = 4;\ 10 = 5x;\ x = 2;\ (10-x) = 8.$

Then you divide ten minus thing by thing, in order that four may be obtained. You know that if you multiply the quotient by the divisor, the sum which was divided is restored. In the present question the quotient is four and the divisor is thing. Multiply, therefore, four by thing; the result is four things, which are equal to the sum to be divided, which was ten minus thing. You now reduce it by thing, which you add to the four things. Then we have five things equal to ten; therefore one thing is equal to two, and this is one of the two portions. This question refers you to one of the six cases, namely, “roots equal to numbers.”

Fourth Problem.

I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty.*

Computation: You multiply one-third of thing by one-fourth of thing; it is one-half of a sixth of a square. Further, you multiply one dirhem by one-third of thing, it is one-third of thing; and one dirhem by one-fourth of thing, it is one-fourth of thing; and one dirhem by one dirhem, it is one dirhem. The result of this is: the moiety of one-sixth of a square, and one-third of thing, and one-fourth of thing, and one dirhem, is equal to twenty dirhems. Subtract now the one dirhem from these twenty dirhems; the remainder is nineteen dirhems, equal to the moiety of one-sixth of a square, and one-third of thing and one-fourth of thing. Now make your square a whole one: you do this by multiplying all that you have by twelve. Thus you have one square and seven roots equal to two hundred and twenty-eight. Halve the number of the roots, and multiply it by itself; it is twelve and one-fourth. Add this to the number, that is, to two hundred and twenty-eight; the sum is two hundred and forty and one quarter. Extract the root of this; it is fifteen and a half. Subtract from this the moiety of the roots, that is, seven and a half; there remains eight, which is the square required.

Fifth Problem.

I have divided ten into two parts; and having divided the one by the other, and multiplied each of them by itself, and also I had added the products, the sum was fifty-eight dirhems.*

Computation: Suppose one of the two parts to be thing, and the other ten minus thing. Multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply thing by thing; it is a square. Add both together. The sum is a hundred, plus two squares minus twenty things, which are equal to fifty-eight dirhems. Take now the twenty negative things from the hundred and the two squares, and add them to fifty-eight; then a hundred, plus two squares, are equal to fifty-eight dirhems and twenty things. Reduce this to one square, by taking the moiety of all you have. It is then: fifty dirhems and a square, which are equal to twenty-nine dirhems and ten things. Then reduce this, by taking twenty-nine from fifty: there remains twenty-one and a square, equal to ten things. Halve the number of the roots, it is five; multiply this by itself, it is twenty-five; take from this the twenty-one which are connected with the square, the remainder is four. Extract the root, it is two. Subtract this from the moiety of the roots, namely, from five, there remains three. This is one of the portions; the other is seven. This question refers you to one of the six cases, namely “squares and numbers equal to roots.”

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*$(\frac{1}{3}x + 1)(\frac{1}{4}x + 1) = 20$; $\frac{1}{12}x^2 + \frac{7}{12}x + 1 = 20$; $x^2 + 7x = 228$.

*$(10-x)^2+x^2 = 58$; $100-20x+x^2+x^2 = 58$; $2x^2-20x+100 = 58$; $x^2-10x+50 = 29$; $x = 5\pm\sqrt{25-29} = 5\pm2 = 7$ or 3.

Sixth Problem.

I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems.*

Computation: Call the root thing; then one-third of thing is multiplied by one-fourth of thing; this is the moiety of one-sixth of the square, and is equal to thing and twenty-four dirhems. Multiply this moiety of one-sixth of the square by twelve, in order to make your square a whole one, and multiply also the thing by twelve, which yields twelve things; and also four-and-twenty by twelve: the product of the whole will be two hundred and eighty-eight dirhems and twelve roots, which are equal to one square. The moiety of the roots is six. Multiply this by itself, and add it to two hundred and eighty-eight, it will be three hundred and twenty-four. Extract the root from this, it is eighteen; add this to the moiety of the roots, which was six; the sum is twenty-four, and this is the square sought for.

This question refers you to one of the six cases, namely, "roots and numbers equal to squares."

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* $\frac{1}{3}x \times \frac{1}{4}x = x + 24$; $\frac{1}{12}x^2 = x + 24$; $x^2 = 12x + 288$; $x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$.

Various Questions

If a person puts such a question to you as: “I have divided ten into two parts, and multiplying one of these by the other, the result was twenty-one;”[*] then you know that one of the two parts is thing, and the other ten minus thing. Multiply, therefore, thing by ten minus thing; then you have ten things minus a square, which is equal to twenty-one. Separate the square from the ten things, and add it to the twenty-one. Then you have ten things, which are equal to twenty-one dirhems and a square. Take away the moiety of the roots, and multiply the remaining five by itself; it is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root, it is two. Subtract this from the moiety of the roots, namely, five: there remain three, which is one of the two parts. Or, if you please, you may add the root of four to the moiety of the roots; the sum is seven, which is likewise one of the parts.

This is one of the problems which may be resolved by addition and subtraction.

If the question be: “I have divided ten into two parts, and having multiplied each part by itself, I have subtracted the smaller from the greater, and the remainder was forty;”[*] then the computation is—you multiply ten minus thing by itself, it is a hundred plus one square minus twenty things; and you also multiply thing by thing, it is one square. Subtract this from a hundred and a square minus twenty things, and you have a hundred, minus twenty things, equal to forty dirhems. Separate now the twenty things from a hundred, and add them to the forty; then you have a hundred, equal to twenty things and forty dirhems. Subtract now forty from a hundred; there remains sixty dirhems, equal to twenty things: therefore one thing is equal to three, which is one of the two parts.

If the question be: “I have divided ten into two parts, and having multiplied each part by itself, I have put them together, and have added to them the difference of the two parts previously to their multiplication, and the amount of all this is fifty-four;”[*] then the computation is this: You multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply also the other thing of the ten by itself; it is one square. Add this together, it will be a hundred plus two squares minus twenty things. It was stated that the difference of the two parts before multiplication should be added to them. You say, therefore, the difference between them is ten minus two things. The result is a hundred and ten and two squares minus twenty-two things, which are equal to fifty-four dirhems. Having reduced and equalized this, you may say, a hundred and ten dirhems and two squares are equal to fifty-four dirhems and twenty-two things. Reduce now the two squares to one square, by taking the moiety of all you have. Thus it becomes fifty-five dirhems and a square, equal to twenty-seven dirhems and eleven things. Subtract twenty-seven from fifty-five, there remain twenty-eight dirhems and a square, equal to eleven things. Halve now the things, it will be five and a half; multiply this by itself, it is thirty and a quarter. Subtract from it the twenty-eight which are combined with the square, the remainder is two and a fourth. Extract its root, it is one and a half. Subtract this from the moiety of the roots, there remain four, which is one of the two parts.

If one say, “I have divided ten into two parts; and have divided the first by the second,

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[*] $(10-x)x = 21$; $10x - x^2 = 21$; which is to be reduced to $x = 5 \pm \sqrt{25-21} = 5 \pm 2$.

[*] $(10-x)^2 - x^2 = 40$; $100 - 20x + x^2 - x^2 = 40$; $100 - 20x = 40$; $100 = 20x + 40$; $60 = 20x$; $x = 3$.

[*] $100 - 20x + x^2 + x^2 + 10 - 2x = 54$; $100 - 22x + 2x^2 = 54$; $55 - 11x + x^2 = 27$; $x^2 + 28 = 11x$; $x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = \frac{11}{2} \pm \frac{3}{2} = 7$ or 4.

and the second by the first, and the sum of the quotient is two dirhems and one-sixth;”[*] then the computation is this: If you multiply each part by itself, and add the products together, then their sum is equal to one of the parts multiplied by the other, and again by the quotient which is two and one-sixth. Multiply, therefore, ten less thing by itself; it is a hundred and a square less twenty things. Multiply thing by thing; it is one square. Add this together; the sum is a hundred plus two squares less twenty things, which is equal to thing multiplied by ten less thing; that is, to ten things less a square, multiplied by the sum of the quotients arising from the division of the two parts, namely, two and one-sixth. We have, therefore, twenty-one things and two-thirds of thing less two squares and one-sixth, equal to a hundred plus two squares less twenty things. Reduce this by adding the two squares and one-sixth to a hundred plus two squares less twenty things, and add the twenty negative things from the hundred plus the two squares to the twenty-one things and two-thirds of thing. Then you have a hundred plus four squares and one-sixth of a square, equal to forty-one things and two-thirds of thing. Now reduce this to one square. You know that one square is obtained from four squares and one-sixth, by taking a fifth and one-fifth of a fifth.[*] Take, therefore, the fifth and one-fifth of a fifth of all that you have. Then it is twenty-four and a square, equal to ten roots; because ten is one-fifth and one-fifth of the fifth of the forty-one things and two-thirds of a thing. Now halve the roots; it gives five. Multiply this by itself; it is five-and-twenty. Subtract from this the twenty-four, which are connected with the square; the remainder is one. Extract its root; it is one. Subtract this from the moiety of the roots, which is five. There remains four, which is one of the two parts.

Observe that, in every case, where any two quantities whatsoever are divided, the first by the second and the second by the first, if you multiply the quotient of the one division by that of the other, the product is always one.[*]

If some one say: “You divide ten into two parts; multiply one of the two parts by five, and divide it by the other: then take the moiety of the quotient, and add this to the product of the one part, multiplied by five; the sum is fifty dirhems;”[*] then the computation is this: Take thing, and multiply it by five. This is now to be divided by the remainder of the ten, that is, by ten less thing; and of the quotient the moiety is to be taken.

You know that if you divide five things by ten less thing, and take the moiety of the quotient, the result is the same as if you divide the moiety of five things by ten less thing. Take, therefore, the moiety of five things; it is two things and a half: and this you require to divide by ten less thing. Now these two things and a half, divided by ten less thing, give a quotient which is equal to fifty less five things: for the question states: add this (the quotient) to the one part multiplied by five, the sum will be fifty. You have already observed, that if the quotient, or the result of the division, be multiplied by the divisor, the dividend, or capital to be divided, is restored. Now, your capital, in the present instance, is two things and a half. Multiply, therefore, ten less thing by fifty less five things. Then you have five hundred dirhems and five squares less a hundred things, which are equal to two things and a half. Reduce this to one square. Then it becomes a hundred dirhems and a square less twenty things, equal to the moiety of thing. Separate now the twenty things from the hundred dirhems and square, and add them to the half thing. Then you have a hundred dirhems and a square, equal to twenty things and a half. Now halve the

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[*] $100+2x^2-20x = x(10-x)\times 2\frac{1}{6} = 21\frac{2}{3}x-2\frac{1}{6}x^2; 100+4\frac{1}{6}x^2 = 41\frac{2}{3}x; 24+x^2 = 10x; x = 5-\sqrt{25-24} = 5-1 = 4$ or 6.

[*] $4\frac{1}{6} = \frac{25}{6}; \frac{1}{4\frac{1}{6}} = \frac{6}{25} = \frac{1}{5} + \frac{1}{5}\frac{1}{5}$.

[*] $\frac{a}{b} \times \frac{b}{a} = 1$.

[*] $x^2 + 100 = 50x; x = 25 - \sqrt{625-100} = 25 - \sqrt{525}$.

things, multiply the moiety by itself, subtract from this the hundred, extract the root of the remainder, and subtract this from the moiety of the roots, which is ten and one-fourth: the remainder is eight; and this is one of the portions.

If some one say: "You divide ten into two parts: multiply the one by itself; it will be equal to the other taken eighty-one times."[*] Computation: You say, ten less thing, multiplied by itself, is a hundred plus a square less twenty things, and this is equal to eighty-one things. Separate the twenty things from a hundred and a square, and add them to eighty-one. It will then be a hundred plus a square, which is equal to a hundred and one roots. Halve the roots; the moiety is fifty and a half. Multiply this by itself, it is two thousand five hundred and fifty and a quarter. Subtract from this one hundred; the remainder is two thousand four hundred and fifty and a quarter. Extract the root from this; it is forty-nine and a half. Subtract this from the moiety of the roots, which is fifty and a half. There remains one, and this is one of the two parts.

If some one say: "I have purchased two measures of wheat or barley, each of them at a certain price. I afterwards added the expenses, and the sum was equal to the difference of the two prices, added to the difference of the measures."[*] Computation: Take what numbers you please, for it is indifferent; for instance, four and six. Then you say: I have bought each measure of the four for thing; and accordingly you multiply four by thing, which gives four things; and I have bought the six, each for the moiety of thing, for which I have bought the four; or, if you please, for one-third, or one-fourth, or for any other quota of that price, for it is indifferent. Suppose that you have bought the six measures for the moiety of thing, then you multiply the moiety of thing by six; this gives three things. Add them to the four things; the sum is seven things, which must be equal to the difference of the two quantities, which is two measures, plus the difference of the two prices, which is a moiety of thing. You have, therefore, seven things, equal to two and a moiety of thing. Remove, now, this moiety of thing, by subtracting it from the seven things. There remain six things and a half, equal to two dirhems: consequently, one thing is equal to four-thirteenths of a dirhem. The six measures were bought, each at one-half of thing; that is, at two-thirteenths of a dirhem. Accordingly, the expenses amount to eight-and-twenty thirteenths of a dirhem, and this sum is equal to the difference of the two quantities; namely, the two measures, the arithmetical equivalent for which is six-and-twenty thirteenths, added to the difference of the two prices, which is two-thirteenths: both differences together being likewise equal to twenty-eight parts.

If he say: "There are two numbers,[*] the difference of which is two dirhems. I have divided the smaller by the larger, and the quotient was just half a dirhem."[*] Suppose one of the two numbers to be thing, and the other to be thing plus two dirhems. By the division of thing by thing plus two dirhems, half a dirhem appears as quotient. You have already observed, that by multiplying the quotient by the divisor, the capital which you divided is restored. This capital, in the present case, is thing. Multiply, therefore, thing and two dirhems by half a dirhem, which is the quotient; the product is half one thing plus one dirhem; this is equal to thing. Remove, now, half a thing on account of the other

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[*] $100 - 20x + x^2 = 81x$; $x^2 + 100 = 101x$; $x = \frac{101}{2} - \sqrt{\frac{10201}{4} - 100} = 50\frac{1}{2} - 49\frac{1}{2} = 1$.

[*] The purchaser does not make a clear enunciation of the terms of his bargain. He intends to say, "I bought m bushels of wheat, and n bushels of barley, and the wheat was r times dearer than the barley. The sum I expended was equal to the difference in the quantities, added to the difference in the prices of the grain." If x is the price of the barley, rx is the price of the wheat; whence, $mrx + nx = (m-n) + (rx - x)$; $x = \frac{m-n}{mr+n-r+1}$.

[*] "*Squares*" in the original. The word square is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.

[*] $\frac{x}{x+2} = \frac{1}{2}$; $\frac{1}{2}(x+2) = x$; $\frac{1}{2}x + 1 = x$; $1 = \frac{1}{2}x$; $x = 2$; the other is 4.

half thing; there remains one dirhem, equal to half a thing. Double it, then you have one thing, equal to two dirhems. Consequently, the other[*] number is four.

If some one say: “I have divided ten into two parts; I have multiplied the one by ten and the other by itself, and the products were the same;”[*] then the computation is this: You multiply thing by ten; it is ten things. Then multiply ten less thing by itself; it is a hundred and a square less twenty things, which is equal to ten things. Reduce this according to the rules, which I have above explained to you.

In like manner, if he say: “I have divided ten into two parts; I have multiplied one of the two by the other, and have then divided the product by the difference of the two parts before their multiplication, and the result of this division is five and one-fourth;”[*] the computation will be this: You subtract thing from ten; there remain ten less thing. Multiply the one by the other, it is ten things less a square. This is the product of the multiplication of one of the two parts by the other. At present you divide this by the difference between the two parts, which is ten less two things. The quotient of this division is, according to the statement, five and a fourth. If, therefore, you multiply five and one-fourth by ten less two things, the product must be equal to the above amount, obtained by multiplication, namely, ten things less one square. Multiply now five and one-fourth by ten less two squares. The result is fifty-two dirhems and a half less ten roots and a half, which is equal to ten roots less a square. Separate now the ten roots and a half from the fifty-two dirhems, and add them to the ten roots less a square; at the same time separate this square from them, and add it to the fifty-two dirhems and a half. Thus you find twenty roots and a half, equal to fifty-two dirhems and a half and one square. Now continue reducing it, conformably to the rules explained at the commencement of this book.

If the question be: “There is a square, two-thirds of one-fifth of which are equal to one-seventh of its root;”[*] then the square is equal to one root and half a seventh of a root; and the root consists of fourteen-fifteenths of the square. The computation is this: You multiply two-thirds of one-fifth of the square by seven and a half, in order that the square may be completed. Multiply that which you have already, namely, one-seventh of its root, by the same. The result will be, that the square is equal to one root and half a seventh of the root; and the root of the square is one and a half seventh; and the square is one and twenty-nine one hundred and ninety-sixths of a dirhem. Two-thirds of the fifth of this are thirty parts of the hundred and ninety-six parts. One-seventh of its root is likewise thirty parts of a hundred and ninety-six.

If the instance be: “Three-fourths of the fifth of a square are equal to four-fifths of its root,”[*] then the computation is this; You add one-fifth to the four-fifths, in order to complete the root. This is then equal to three and three-fourths of twenty parts, that is, to fifteen eightieths of the square. Divide now eighty by fifteen; the quotient is five and one-third. This is the root of the square, and the square is twenty-eight and four-ninths.

If some one say: “What is the amount of a square-root,[*] which, when multiplied by four times itself, amounts to twenty?”[*] the answer is this: If you multiply it by itself it will be five: it is therefore the root of five.

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[*] “*Square*” in the original.

[*] $10x = (10 - x)^2 = 100 - 20x + x^2$.

[*] $\frac{(10-x)x}{10-2x} = 5\frac{1}{4}$; $20\frac{1}{4}x = x^2 + 5\frac{1}{4}(10 - 2x)$; $x = 10 - 7\frac{1}{2} = 2\frac{1}{2}$.

[*] $\frac{2}{3} \cdot \frac{1}{5}x^2 = \frac{1}{7}x$; $\frac{2}{15}x^2 = \frac{1}{7}x$; $x^2 = \frac{15}{14}x$; $x = \frac{15}{14}$.

[*] $\frac{3}{4} \cdot \frac{1}{5}x^2 = \frac{4}{5}x$; $\frac{3}{20}x^2 = \frac{4}{5}x$; $\frac{3}{20}x = \frac{4}{5}$; $x = \frac{16}{3} = 5\frac{1}{3}$.

[*] “*Square*” in the original.

[*] $4x^2 = 20$; $x^2 = 5$; $x = \sqrt{5}$.

If somebody ask you for the amount of a square-root[*] which when multiplied by its third amounts to ten,[*] the solution is, that when multiplied by itself it will amount to thirty; and it is consequently the root of thirty.

If the question be: “To find a quantity,[*] which when multiplied by four times itself, gives one-third of the first quantity as product,”[*] the solution is, that if you multiply it by twelve times itself, the quantity itself must re-appear: it is the moiety of one moiety of one-third.

If the question be: “A square, which when multiplied by its root gives three times the original square as product,”[*] then the solution is: that if you multiply the root by one-third of the square, the original square is restored; its root must consequently be three, and the square itself nine.

If the instance be: “To find a square, four roots of which, multiplied by three roots, restore the square with a surplus of forty-four dirhems,”[*] then the solution is: that you multiply four roots by three roots, which gives twelve squares, equal to a square and forty-four dirhems. Remove now one square of the twelve on account of the one square connected with the forty-four dirhems. There remain eleven squares, equal to forty-four dirhems. Make the division, the result will be four, and this is the square.

If the instance be: “A square, four of the roots of which multiplied by five of its roots produce twice the square, with a surplus of thirty-six dirhems;”[*] then the solution is: that you multiply four roots by five roots, which gives twenty squares, equal to two squares and thirty-six dirhems. Remove two squares from the twenty on account of the other two. The remainder is eighteen squares, equal to thirty-six dirhems. Divide now thirty-six dirhems by eighteen; the quotient is two, and this is the square.

In the same manner, if the question be: “A square, multiply its root by four of its roots, and the product will be three times the square, with a surplus of fifty dirhems.”[*] Computation: You multiply the root by four roots, it is four squares, which are equal to three squares and fifty dirhems. Remove three squares from the four; there remains one square, equal to fifty dirhems. One root of fifty, multiplied by four roots of the same, gives two hundred, which is equal to three times the square, and a residue of fifty dirhems.

If the instance be: “A square, which when added to twenty dirhems, is equal to twelve of its roots,”[*] then the solution is this: You say, one square and twenty dirhems are equal to twelve roots. Halve the roots and multiply them by themselves; this gives thirty-six. Subtract from this the twenty dirhems, extract the root from the remainder, and subtract it from the moiety of the roots, which is six. The remainder is the root of the square: it is two dirhems, and the square is four.

If the instance be: “To find a square, of which if one-third be added to three dirhems, and the sum be subtracted from the square, the remainder multiplied by itself restores the square;”[*] then the computation is this: If you subtract one-third and three dirhems from the square, there remain two-thirds of it less three dirhems. This is the root. Multiply therefore two-thirds of thing less three dirhems by itself. You say two-thirds by

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[*] “*Square*” in the original.

[*] $x \times \frac{x}{3} = 10; x^2 = 30; x = \sqrt{30}.$

[*] “*Square-root*” in the original.

[*] $x \times 4x = \frac{1}{3}x; 4x^2 = \frac{1}{3}x; 12x^2 = x; x = \frac{1}{12}.$

[*] $x^2 \times x = 3x^2; x^3 = 3x^2; x = 3.$

[*] $4x \times 3x = x^2 + 44; 12x^2 = x^2 + 44; 11x^2 = 44; x^2 = 4; x = 2.$

[*] $4x \times 5x = 2x^2 + 36; 20x^2 = 2x^2 + 36; 18x^2 = 36; x^2 = 2.$

[*] $x \times 4x = 3x^2 + 50; 4x^2 = 3x^2 + 50; x^2 = 50; x = \sqrt{50}.$

[*] $x^2 + 20 = 12x; x = 6 \pm \sqrt{36 - 20} = 6 \pm 4 = 10$ or 2.

[*] $(x^2 - \frac{1}{3}x^2 - 3)^2 = x^2; \frac{2}{3}x^2 - 3 = x; \frac{4}{9}x^4 - 4x^2 + 9 = x^2; x^2 + 9 = 5x; x = 9$ or $2\frac{1}{4}$.

two-thirds is four ninths of a square; and less two-thirds by three dirhems is two roots; and again, two-thirds by three dirhems is two roots; and less three dirhems by less three dirhems is nine dirhems. You have, therefore, four-ninths of a square and nine dirhems less four roots, which are equal to one root. Add the four roots to the one root, then you have five roots, which are equal to four-ninths of a square and nine dirhems. Complete now your square; that is, multiply the four-ninths of a square by two and a fourth, which gives one square; multiply likewise the nine dirhems by two and a quarter; this gives twenty and a quarter; finally, multiply the five roots by two and a quarter; this gives eleven roots and a quarter. You have, therefore, a square and twenty dirhems and a quarter, equal to eleven roots and a quarter. Reduce this according to what I taught you about halving the roots.

If the instance be: “To find a number,[*] one-third of which, when multiplied by one-fourth of it, restores the number,”[*] then the computation is: You multiply one-third of thing by one-fourth of thing, this gives one-twelfth of a square, equal to thing, and the square is equal to twelve things, which is the root of one hundred and forty-four.

If the instance be: “A number,[*] one-third of which and one dirhem multiplied by one-fourth of it and two dirhems restore the number, with a surplus of thirteen dirhems;”[*] then the computation is this: You multiply one-third of thing by one-fourth of thing, this gives half one-sixth of a square; and you multiply two dirhems by one-third of thing, this gives two-thirds of a root; and one dirhem by one-fourth of thing gives one-fourth of a root; and one dirhem by two dirhems gives two dirhems. This altogether is one-twelfth of a square and two dirhems and eleven-twelfths of a thing, equal to thing and thirteen dirhems. Remove now two dirhems from thirteen, on account of the other two dirhems, the remainder is eleven dirhems. Remove then the eleven-twelfths of a root from the one (root on the opposite side), there remains one-twelfth of a root and eleven dirhems, equal to one-twelfth of a square. Complete the square: that is, multiply it by twelve, and do the same with all you have. The product is a square, which is equal to a hundred and thirty-two dirhems and one root. Reduce this, according to what I have taught you, it will be right.

If the instance be: “A dirhem and a half to be divided among one person and certain persons, so that the share of the one person be twice as many dirhems as there are other persons;”[*] then the Computation is this: You say, the one person and some persons are one and thing: it is the same as if the question had been one dirhem and a half to be divided by one and thing, and the share of one person to be equal to two things. Multiply, therefore, two things by one and thing; it is two squares and two things, equal to one dirhem and a half. Reduce them to one square: that is, take the moiety of all you have. You say, therefore, one square and one thing are equal to three-fourths of a dirhem. Reduce this, according to what I have taught you in the beginning of this work.

If the instance be: “A number,[*] you remove one-third of it, and one-fourth of it, and four dirhems: then you multiply the remainder by itself, and the number,[*] is restored,

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[*] “*Square*” in the original.

[*] $\frac{1}{3}x \times \frac{1}{4}x = x$; $\frac{1}{12}x^2 = x$; $x^2 = 12x$; $x = 12$.

[*] “*Square*” in the original.

[*] $\frac{x}{3} + 1 \times \frac{x}{4} + 2 = x + 13$; $\frac{1}{12}x^2 + \frac{11}{12}x + 2 = x + 13$; $\frac{1}{12}x^2 + 11 = \frac{1}{12}x + 13$; $x^2 = 132 + x$; $x = 12$.

[*] The enunciation in the original is faulty, and I have altered it to correspond with the computation. But in the computation, x, the number of persons, is fractional! I am unable to correct the passage satisfactorily. $x^2 + x = \frac{3}{2}$; $x = \sqrt{1\frac{1}{4} + \frac{1}{2}} - \frac{1}{2}$.

[*] “*Square*” in the original.

[*] “*Square*” in the original.

with a surplus of twelve dirhems;”[*] then the computation is this: You take thing, and subtract from it one-third and one-fourth; there remain five-twelfths of thing. Subtract from this four dirhems; the remainder is five-twelfths of thing less four dirhems. Multiply this by itself. Thus the five parts become five-and-twenty parts; and if you multiply twelve by itself, it is a hundred and forty-four. This makes, therefore, five and twenty hundred and forty-fourths of a square. Multiply then the four dirhems twice by the five-twelfths; this gives forty parts, every twelve of which make one root (forty-twelfths); finally, the four dirhems, multiplied by four dirhems, give sixteen dirhems to be added. The forty-twelfths are equal to three roots and one-third of a root, to be subtracted. The whole product is, therefore, twenty-five-hundred-and-forty-fourths of a square and sixteen dirhems less three roots and one-third of a root, equal to the original number,[*] which is thing and twelve dirhems. Reduce this, by adding the three roots and one-third to the thing and twelve dirhems. Thus you have four roots and one-third of a root and twelve dirhems. Go on balancing, and subtract the twelve (dirhems) from sixteen; there remain four dirhems and five-and-twenty-hundred-and-forty-fourths of a square, equal to four roots and one-third. Now it is necessary to complete the square. This you can accomplish by multiplying all you have by five and nineteen twenty-fifths. Multiply, therefore, the twenty-five-one-hundred-and-forty-fourths of a square by five and nineteen twenty-fifths. This gives a square. Then multiply the four (dirhems) by five and nineteen twenty-fifths; this gives twenty-three dirhems and one twenty-fifth. Then multiply four roots and one-third by five and nineteen twenty-fifths; this gives twenty-four roots and twenty-four twenty-fifths of a root. Now halve the number of the roots: the moiety is twelve roots and twelve twenty-fifths of a root. Multiply this by itself. It is one hundred-and-fifty-five dirhems and four-hundred-and-sixty-nine six-hundred-and-twenty-fifths. Subtract from this the twenty-three dirhems and the one twenty-fifth connected with the square. The remainder is one-hundred-and-thirty-two and four-hundred-and-forty-six six-hundred-and-twenty-fifths. Take the root of this: it is eleven dirhems and thirteen twenty-fifths. Add this to the moiety of the roots, which was twelve dirhems and twelve twenty-fifths. The sum is twenty-four. It is the number which you sought. When you subtract its third and its fourth and four dirhems, and multiply the remainder by itself, the number is restored, with a surplus of twelve dirhems.

If the question be: “To find a square-root,[*] which, when multiplied by two-thirds of itself, amounts to five;”[*] then the computation is this: You multiply one thing by two-thirds of thing; the product is two-thirds of square, equal to five. Complete it by adding its moiety to it, and add to five likewise its moiety. Thus you have a square, equal to seven and a half. Take its root; it is the thing which you required, and which, when multiplied by two-thirds of itself, is equal to five.

If the instance be: “Two numbers,[*] the difference of which is two dirhems; you divide the small one by the great one, and the quotient is equal to half a dirhem;”[*] then the computation is this: Multiply thing and two dirhems by the quotient, that is a half. The product is half a thing and one dirhem, equal to thing. Remove now half a dirhem on account of the half dirhem on the other side. The remainder is one dirhem, equal to half

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[*] $(x - \frac{1}{3}x - \frac{1}{4}x - 4)^2 = x + 12$; $\frac{5}{12}x - 4$; $x^2 + 23\frac{1}{4} = 24\frac{1}{4}x$.

[*] “*Square*” in the original.

[*] “*Square*” in the original.

[*] $x \times \frac{2}{3}x = 5$; $\frac{2}{3}x^2 = 5$; $x^2 = 7\frac{1}{2}$; $x = \sqrt{7\frac{1}{2}}$.

[*] “*Squares*” in the original.

[*] $\frac{x}{x+2} = \frac{1}{2}$; $\frac{1}{2}(x + 2) = x$; $1 + \frac{1}{2}x = x$; $1 = \frac{1}{2}x$; $x = 2$; the other is 4.

a thing. Double it; then you have thing, equal to two dirhems. This is one of the two[*] numbers, and the other is four.

Instance: “You divide one dirhem amongst a certain number of men, which number is thing. Now you add one man more to them, and divide again one dirhem amongst them; the quota of each is then one-sixth of a dirhem less than at the first time.”[*] Computation: You multiply the first number of men, which is thing, by the difference of the share for each of the other number. Then multiply the product by the first and second number of men, and divide the product by the difference of these two numbers. Thus you obtain the sum which shall be divided. Multiply, therefore, the first number of men, which is thing, by the one-sixth, which is the difference of the shares; this gives one-sixth of root. Then multiply this by the original number of the men, and that of the additional one, that is to say, by thing plus one. The product is one-sixth of square and one-sixth of root divided by one dirhem, and this is equal to one dirhem. Complete the square which you have through multiplying it, by six. Then you have a square and a root equal to six dirhems. Halve the root and multiply the moiety by itself, it is one-fourth. Add this to the six; take the root of the sum and subtract from it the moiety of the root, which you have multiplied by itself, namely, a half. The remainder is the first number of men; which in this instance is two.

If the instance be: “To find a square-root,[*] which when multiplied by two-thirds of itself amounts to five:”[*] then the computation is this: If you multiply it by itself, it gives seven and a half. Say, therefore, it is the root of seven and a half multiplied by two-thirds of the root of seven and a half. Multiply then two-thirds by two-thirds, it is four-ninths; and four-ninths multiplied by seven and a half is three and a third. The root of three and a third is two-thirds of the root of seven and a half. Multiply three and a third by seven and a half; the product is twenty-five, and its root is five.

If the instance be: “A square multiplied by three of its roots is equal to five times the original square;”[*] then this is the same as if it had been said, a square, which when multiplied by its root, is equal to the first square and two-thirds of it. Then the root of the square is one and two-thirds, and the square is two dirhems and seven-ninths.

If the instance be: “Remove one-third from a square, then multiply the remainder by three roots of the first square, and the first square will be restored.”[*] Computation: If you multiply the first square, before removing two-thirds from it, by three roots of the same, then it is one square and a half; for according to the statement two-thirds of it multiplied by three roots give one square; and, consequently, the whole of it multiplied by three roots of it gives one square and a half. This entire square, when multiplied by one root, gives half a square; the root of the square must therefore be a half, the square one-fourth, two-thirds of the square one-sixth, and three roots of the square one and a half. If you multiply one-sixth by one and a half, the product is one-fourth, which is the square.

Instance: “A square; you subtract four roots of the same, then take one-third of the remainder; this is equal to the four roots.” The square is two hundred and fifty-six.[*] Com-

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[*] “*Squares*” in the original.

[*] $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{6}$; $\frac{x+1-x}{x(x+1)} = \frac{1}{6}$; $\frac{1}{x(x+1)} = \frac{1}{6}$; $x^2 + x = 6$; $x = \sqrt{\frac{1}{4} + 6} - \frac{1}{2} = 2$.

[*] “*Square*” in the original.

[*] $\frac{2}{3}x^2 = 5$; $x^2 = 7\frac{1}{2}$; $x = \sqrt{7\frac{1}{2}}$.

[*] $x^2 \times 3x = 5x^2$; $3x^3 = 5x^2$; $3x = 5$; $x = 1\frac{2}{3}$; $x^2 = 2\frac{7}{9}$.

[*] $(x^2 - \frac{1}{3}x^2) \times 3x = x^2$; $\frac{2}{3}x^2 \times 3x = x^2$; $2x^3 = x^2$; $x = \frac{1}{2}$; $x^2 = \frac{1}{4}$.

[*] $x^2 - 4x = \frac{1}{3}(x^2 - 4x)$ gives $\frac{2}{3}(x^2 - 4x) = 4x$; $x^2 - 4x = 6x$; $x^2 = 10x$; $x = 10$; $x^2 = 100$. *[Note: There appears to be a discrepancy in the original text, which states 256.]*

putation: You know that one-third of the remainder is equal to four roots; consequently, the whole remainder must be twelve roots; add to this the four roots; the sum is sixteen, which is the root of the square.

Instance: "A square; you remove one root from it and if you add to this root a root of the remainder, the sum is two dirhems."[*] Then, this is the root of a square, which, when added to the root of the same square, less one root, is equal to two dirhems. Subtract from this one root of the square, and subtract also from the two dirhems one root of the square. Then two dirhems less one root multiplied by itself is four dirhems and one square less four roots, and this is equal to a square less one root. Reduce it, and you find a square and four dirhems, equal to a square and three roots. Remove square by square; there remain three roots, equal to four dirhems; consequently, one root is equal to one dirhem and one-third. This is the root of the square, and the square is one dirhem and seven ninths of a dirhem.

Instance: "Subtract three roots from a square, then multiply the remainder by itself, and the square is restored."[*] You know by this statement that the remainder must be a root likewise; and that the square consists of four such roots; consequently, it must be sixteen.

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[*] $\sqrt{x^2 - x} + x = 2$; $\sqrt{x^2 - x} = 2 - x$; $x^2 - x = 4 + x^2 - 4x$; $3x = 4$; $x = 1\frac{1}{3}$; $x^2 = 1\frac{7}{9}$.

[*] $x^2 - 3x = x$; $x^2 = 4x$; $x = 4$.

On Mercantile Transactions

You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity. Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* and it is the object of the question.

The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor.[*]

Examples.

For the first case: If you are told, "ten for six, how much for four?" then ten is the measure; six is the price; the expression *how much* implies the unknown number of the quantity; and *four* is the number of the sum. The number of the measure, which is ten, is inversely proportionate to the number of the sum, namely, *four*. Multiply, therefore, ten by four, that is to say, the two known proportionate numbers by each other; the product is forty. Divide this by the other known number, which is that of the price, namely, six. The quotient is six and two-thirds; it is the unknown number, implied in the words of the question "how much?" it is the quantity, and inversely proportionate to the six, which is the price.

For the second case: Suppose that some one ask this question: "ten for eight, what must be the sum for four?" This is also sometimes expressed thus: "What must be the price of four of them?" Ten is the number of the measure, and is inversely proportionate to the unknown number of the sum, which is involved in the expression *how much* of the statement. Eight is the number of the price, and this is inversely proportionate to the known number of the quantity, namely, four. Multiply now the two known proportionate numbers one by the other, that is to say, four by eight. The product is thirty-two. Divide this by the other known number, which is that of the measure, namely, ten. The quotient is three and one-fifth; this is the number of the sum, and inversely proportionate to the ten which was the divisor. In this manner all computations in matters of business may be solved.

If somebody says, "a workman receives a pay of ten dirhems per month; how much must be his pay for six days?" Then you know that six days are one-fifth of the month; and that his portion of the dirhems must be proportionate to the portion of the month. You calculate it by observing that one month, or thirty days, is the measure, ten dirhems the price, six days the quantity, and his portion the sum. Multiply the price, that is, ten, by the quantity, which is proportionate to it, namely, six; the product is sixty. Divide this by thirty, which is the known number of the measure. The quotient is two dirhems, and this is the sum.

This is the proceeding by which all transactions concerning exchange or measures or weights are settled.

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[*] If a is given for b, and A for B; then $a : b :: A : B$ or $aB = bA$; $\therefore a = \frac{bA}{B}$ and $b = \frac{aB}{A}$.

Mensuration

KNOW THAT the meaning of the expression "one by one" is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard. The same takes place with three by three, and so on, ascending or descending: for instance, a half by a half, which gives a quarter, or other fractions, always following the same rule.

A quadrate, every side of which is half a yard, is equal to one-fourth of the figure which has one yard for its side. In the same manner, one-third by one-third, or one-fourth by one-fourth, or one-fifth by one-fifth, or two-thirds by a half, or more or less than this, always according to the same rule.

One side of an equilateral quadrangular figure, taken once, is its root; or if the same be multiplied by two, then it is like two of its roots, whether it be small or great.

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

In every equilateral quadrangle, the product of one diameter multiplied by the moiety of the other will be equal to the area of it.

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery. This is the rule generally followed in practical life, though it is not quite exact. The geometricians have two other methods. One of them is, that you multiply the diameter by itself; then by ten, and hereafter take the root of the product; the root will be the periphery. The other method is used by the astronomers among them: it is this, that you multiply the diameter by sixty-two thousand eight hundred and thirty-two and then divide the product by twenty thousand; the quotient is the periphery. Both methods come very nearly to the same effect.[*]

If you divide the periphery by three and one-seventh, the quotient is the diameter.

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter; since, in every polygon of equal sides and angles, such as triangles, quadrangles, pentagons, and so on, the area is found by multiplying the moiety of the circumference by the moiety of the diameter of the middle circle that may be drawn through it.

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle. This comes very nearly to the same result with the method given above.[*]

Every part of a circle may be compared to a bow. It must be either exactly equal to half the circumference, or less or greater than it. This may be ascertained by the arrow of the bow. When this becomes equal to the moiety of the chord, then the arc is exactly the moiety of the circumference: is it shorter than the moiety of the chord, then the bow is less than half the circumference; is the arrow longer than half the chord, then the bow comprises more than half the circumference.

If you want to ascertain the circle to which it belongs, multiply the moiety of the chord by itself, divide it by the arrow, and add the quotient to the arrow, the sum is the diameter of the circle to which this bow belongs.

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[*] The three formulas are: 1^{st}, $3\frac{1}{7}d = p$ i.e. $3.1428\,d$; 2^{d}, $\sqrt{10d^2} = p$ i.e. $3.16227\,d$; 3^{d}, $\frac{d\times 62832}{20000} = p$ i.e. $3.1416\,d$.

[*] The area of a circle whose diameter is d is $\pi\frac{d^2}{4} = d^2 - \frac{d^2}{7} - \frac{1}{2}\cdot\frac{d^2}{7} = d^2(1 - \frac{3}{14}) = \frac{11}{14}d^2 \approx 0.7857d^2$. The value $\frac{22}{7}\cdot\frac{1}{4} = \frac{11}{14} \approx 0.7857$.

If you want to compute the area of the bow, multiply the moiety of the diameter of the circle by the moiety of the bow, and keep the product in mind. Then subtract the arrow of the bow from the moiety of the diameter of the circle, if the bow is smaller than half the circle; or if it is greater than half the circle, subtract half the diameter of the circle from the arrow of the bow. Multiply the remainder by the moiety of the chord of the bow, and subtract the product from that which you have kept in mind if the bow is smaller than the moiety of the circle, or add it thereto if the bow is greater than half the circle. The sum after the addition, or the remainder after the subtraction, is the area of the bow.

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

If it is of another shape than the quadrangular (for instance, circular or triangular), so, however, that a line representing its height may stand perpendicularly on its basis, and yet be parallel to the sides, you must calculate it by ascertaining at first the area of its basis. This, multiplied by the height, gives the bulk of the body.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying one-third of the area of the basis by the height.

On the Rectangular Triangle.

Observe, that in every rectangular triangle the two short sides, each multiplied by itself and the products added together, equal the product of the long side multiplied by itself.

The proof of this is the following. We draw a quadrangle, with equal sides and angles $ABCD$. We divide the line AC into two moieties in the point H, from which we draw a parallel to the point R. Then we divide, also, the line AB into two moieties at the point T, and draw a parallel to the point G. Then the quadrate $ABCD$ is divided into four quadrangles of equal sides and angles, and of equal area; namely, the squares AK, CK, BK, and DK. Now, we draw from the point H to the point T a line which divides the quadrangle AK into two equal parts: thus there arise two triangles from the quadrangle, namely, the triangles ATH and HKT. We know that AT is the moiety of AB, and that AH is equal to it, being the moiety of AC; and the line TH joins them opposite the right angle. In the same manner we draw lines from T to R, and from R to G, and from G to H. Thus from all the squares eight equal triangles arise, four of which must, consequently, be equal to the moiety of the great quadrate AD. We know that the line AT multiplied by itself is like the area of two triangles, and AK gives the area of two triangles equal to them; the sum of them is therefore four triangles. But the line HT, multiplied by itself, gives likewise the area of four such triangles. We perceive, therefore, that the sum of AT multiplied by itself, added to AH multiplied by itself, is equal to TH multiplied by itself.

This is the observation which we were desirous to elucidate.

Quadrangles are of five kinds: firstly, with right angles and equal sides; secondly, with right angles and unequal sides; thirdly, the rhombus, with equal sides and unequal angles; fourthly, the rhomboid, the length of which differs from its breadth, and the angles of which are unequal, only that the two long and the two short sides are respectively of equal length; fifthly, quadrangles with unequal sides and angles.

First kind.—The area of any quadrangle with equal sides and right angles, or with unequal sides and right angles, may be found by multiplying the length by the breadth. The product is the area. For instance: a quadrangular piece of ground, every side of which has five yards, has an area of five-and-twenty square yards.

Second kind.—A quadrangular piece of ground, the two long sides of which are of eight yards each, while the breadth is six. You find the area by multiplying six by eight, which yields forty-eight yards.

Third kind, the Rhombus.—Its sides are equal: let each of them be five, and let its diagonals be, the one eight and the other six yards. You may then compute the area, either from one of the diagonals, or from both. As you know them both, you multiply the one by the moiety of the other, the product is the area: that is to say, you multiply eight by three, or six by four; this yields twenty-four yards, which is the area.[*] If you know only one of the diagonals, then you are aware, that there are two triangles, two sides of each of which have every one five yards, while the third is the diagonal. Hereafter you can make the computation according to the rules for the triangles. This is the figure.

The fourth kind, or Rhomboid, is computed in the same way as the rhombus.

The other quadrangles are calculated by drawing a diagonal, and computing them as triangles.

Triangles are of three kinds, acute-angular, obtuse-angular, or rectangular. The peculiarity of the rectangular triangle is, that if you multiply each of its two short sides by itself, and then add them together, their sum will be equal to the long side multiplied by itself. The character of the acute-angled triangle is this: if you multiply every one of its two short sides by itself, and add the products, their sum is more than the long side alone multiplied by itself. The definition of the obtuse-angled triangle is this: if you multiply its two short sides each by itself, and then add the products, their sum is less than the product of the long side multiplied by itself.

The rectangular triangle has two cathetes and an hypotenuse. It may be considered as the moiety of a quadrangle. You find its area by multiplying one of its cathetes by the moiety of the other. The product is the area.

Examples.—A rectangular triangle; one cathete being six yards, the other eight, and the hypotenuse ten. You make the computation by multiplying six by four: this gives twenty-four, which is the area. Or if you prefer, you may also calculate it by the height, which rises perpendicularly from the longest side of it: for the two short sides may themselves be considered as two heights. If you prefer this, you multiply the height by the moiety of the basis. The product is the area.

Second kind.—An equilateral triangle with acute angles, every side of which is ten yards long. Its area may be ascertained by the line representing its height and the point from which it rises.[*] Observe, that in every isosceles triangle, a line to represent the height drawn to the basis rises from the latter in a right angle, and the point from which it proceeds is always situated in the midst of the basis; if, on the contrary, the two sides are not equal, then this point never lies in the middle of the basis. In the case now before us we perceive, that towards whatever side we may draw the line which is to represent the height, it must necessarily always fall in the middle of it, where the length of the basis is five. Now the height will be ascertained thus. You multiply five by itself; then multiply one of the sides, that is ten, by itself, which gives a hundred. Now you subtract from this the product of five multiplied by itself, which is twenty-five. The remainder is seventy-five, the root of which is the height. This is a line common to two rectangular triangles. If you want to find the area, multiply the root of seventy-five by the moiety of the basis, which is five. This you perform by multiplying at first five by itself; then you may say, that the root of seventy-five is to be multiplied by the root of twenty-five. Multiply seventy-five by twenty-five. The product is one thousand eight hundred and

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[*] If the two diagonals are d and d', and the side s, the area of the rhombus is $\frac{dd'}{2} = d \times \sqrt{s^2 - \frac{d^2}{4}}$.

[*] The height of the equilateral triangle whose side is 10, is $\sqrt{10^2 - 5^2} = \sqrt{75}$, and the area of the triangle is $5\sqrt{75} = 25\sqrt{5}$. The root is $43.3+$.

seventy-five; take its root, it is the area: it is forty-three and a little.

There are also acute-angled triangles, with different sides. Their area will be found by means of the line representing the height and the point from which it proceeds. Take, for instance, a triangle, one side of which is fifteen yards, another fourteen, and the third thirteen yards.

In order to find the point from which the line marking the height does arise, you may take for the basis any side you choose; e.g. that which is fourteen yards long. The point from which the line representing the height does arise, lies in this basis at an unknown distance from either of the two other sides. Let us try to find its unknown distance from the side which is thirteen yards long. Multiply this distance by itself; it becomes an [unknown] square. Subtract this from thirteen multiplied by itself; that is, one hundred and sixty-nine. The remainder is one hundred and sixty-nine less a square. The root from this is the height. The remainder of the basis is fourteen less thing. We multiply this by itself; it becomes one hundred and ninety-six, and a square less twenty-eight things. We subtract this from fifteen multiplied by itself; the remainder is twenty-nine dirhems and twenty-eight things less one square. The root of this is the height. As, therefore, the root of this is the height, and the root of one hundred and sixty-nine less square is the height likewise, we know that they both are the same.[*] Reduce them, by removing square against square, since both are negatives. There remain twenty-nine [dirhems] plus twenty-eight things, which are equal to one hundred and sixty-nine. Subtract now twenty-nine from one hundred and sixty-nine. The remainder is one hundred and forty, equal to twenty-eight things. One thing is, consequently, five. This is the distance of the said point from the side of thirteen yards. The complement of the basis towards the other side is nine. Now in order to find the height, you multiply five by itself, and subtract it from the contiguous side, which is thirteen, multiplied by itself. The remainder is one hundred and forty-four. Its root is the height. It is twelve. The height forms always two right angles with the basis, and it is called the column, on account of its standing perpendicularly. Multiply the height into half the basis, which is seven. The product is eighty-four, which is the area.

The third species is that of the obtuse-angled triangle with one obtuse angle and sides of different length. For instance, one side being six, another five, and the third nine. The area of such a triangle will be found by means of the height and of the point from which a line representing the same arises. This point can, within such a triangle, lie only in its longest side. Take therefore this as the basis: for if you choose to take one of the short sides as the basis, then this point would fall beyond the triangle. You may find the distance of this point, and the height, in the same manner, which I have shown in the acute-angled triangle; the whole computation is the same.

We have above treated at length of the circles, of their qualities and their computation. The following is an example: If a circle has seven for its diameter, then it has twenty-two for its circumference. Its area you find in the following manner: Multiply the moiety of the diameter, which is three and a half, by the moiety of the circumference, which is eleven. The product is thirty-eight and a half, which is the area. Or you may also multiply the diameter, which is seven, by itself: this is forty-nine; subtracting herefrom one-seventh and half one-seventh, which is ten and a half, there remain thirty-eight and a half, which is the area.

If some one inquires about the bulk of a pyramidal pillar, its base being four yards by four yards, its height ten yards, and the dimensions at its upper extremity two yards by two yards; then we know already that every pyramid is decreasing towards its top,

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[*] $\sqrt{169 - x^2} = \sqrt{29 + 28x - x^2}$; $169 = 29 + 28x$; $140 = 28x$; $x = 5$.

and that one-third of the area of its basis, multiplied by the height, gives its bulk. The present pyramid has no top. We must therefore seek to ascertain what is wanting in its height to complete the top. We observe, that the proportion of the entire height to the ten, which we have now before us, is equal to the proportion of four to two. Now as two is the moiety of four, ten must likewise be the moiety of the entire height, and the whole height of the pillar must be twenty yards. At present we take one-third of the area of the basis, that is, five and one-third, and multiply it by the length, which is twenty. The product is one hundred and six yards and two-thirds. Herefrom we must then subtract the piece, which we have added in order to complete the pyramid. This we perform by multiplying one and one-third, which is one-third of the product of two by two, by ten: this gives thirteen and a third. This is the piece which we have added in order to complete the pyramid. Subtracting this from one hundred and six yards and two-thirds, there remain ninety-three yards and one-third: and this is the bulk of the mutilated pyramid.

If the pillar has a circular basis, subtract one-seventh and half a seventh from the product of the diameter multiplied by itself, the remainder is the basis.

If some one says: “There is a triangular piece of land, two of its sides having ten yards each, and the basis twelve; what must be the length of one side of a quadrate situated within such a triangle?” the solution is this. At first you ascertain the height of the triangle, by multiplying the moiety of the basis, (which is six) by itself, and subtracting the product, which is thirty-six, from one of the two short sides multiplied by itself, which is one hundred; the remainder is sixty-four: take the root from this; it is eight. This is the height of the triangle. Its area is, therefore, forty-eight yards: such being the product of the height multiplied by the moiety of the basis, which is six. Now we assume that one side of the quadrate inquired for is thing. We multiply it by itself; thus it becomes a square, which we keep in mind. We know that there must remain two triangles on the two sides of the quadrate, and one above it. The two triangles on both sides of it are equal to each other: both having the same height and being rectangular. You find their area by multiplying thing by six less half a thing, which gives six things less half a square. This is the area of both the triangles on the two sides of the quadrate together. The area of the upper triangle will be found by multiplying eight less thing, which is the height, by half one thing. The product is four things less half a square. This altogether is equal to the area of the quadrate plus that of the three triangles: or, ten things are equal to forty-eight, which is the area of the great triangle. One thing from this is four yards and four-fifths of a yard; and this is the length of any side of the quadrate.

On Legacies

The problems in this chapter may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

On Capital, and Money lent.

"A man dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons."

Computation: You call the sum which is taken out of the debt thing. Add this to the capital, which is ten dirhems. The sum is ten and thing. Subtract one-third of this, since he has bequeathed one-third of his property, that is, three dirhems and one-third of thing. The remainder is six dirhems and two-thirds of thing. Divide this between the two sons. The portion of each of them is three dirhems and one-third plus one-third of thing. This is equal to the thing which was sought for.[*]

Reduce it, by removing one-third from thing, on account of the other third of thing. There remain two-thirds of thing, equal to three dirhems and one-third. It is then only required that you complete the thing, by adding to it as much as one half of the same; accordingly, you add to three and one-third as much as one-half of them: This gives five dirhems, which is the thing that is taken out of the debts.

If he leaves two sons and ten dirhems of capital and a demand of ten dirhems against one of the sons, and bequeaths one-fifth of his property and one dirhem to a stranger, the computation is this: Call the sum which is taken out of the debt, thing. Add this to the property; the sum is thing and ten dirhems. Subtract one-fifth of this, since he has bequeathed one-fifth of his capital, that is, two dirhems and one-fifth of thing; the remainder is eight dirhems and four-fifths of thing. Subtract also the one dirhem which he has bequeathed; there remain seven dirhems and four-fifths of thing. Divide this between the two sons; there will be for each of them three dirhems and a half plus two-fifths of thing; and this is equal to one thing.[*]

Reduce it by subtracting two-fifths of thing from thing. Then you have three-fifths of thing, equal to three dirhems and a half. Complete the thing by adding to it two-thirds of the same: add as much to the three dirhems and a half; namely, two dirhems and one-third; the sum is five and five-sixths. This is the thing, or the amount which is taken from the debt.

If he leaves three sons, and bequeaths one-fifth of his property less one dirhem, leaving ten dirhems of capital and a demand of ten dirhems against one of the sons, the computation is this: You call the sum which is taken from the debt thing. Add this to the capital; it gives ten and thing. Subtract from this one-fifth of it for the legacy; it is two dirhems and one-fifth of thing. There remain eight dirhems and four-fifths of thing; add to this one dirhem, since he stated "less one dirhem." Thus you have nine dirhems and four-fifths of thing. Divide this between the three sons. There will be for each son three dirhems, and one-fifth and one-third and one-fifth of thing. This equals one thing.[*]

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[*] If a father dies, leaving n sons, one of whom owes the father a sum exceeding an nth part of the residue of the father's estate, after paying legacies, then such son retains the whole sum which he owes the father: part, as a set-off against his share of the residue, the surplus as a gift from the father. In the present example, let each son's share of the residue be equal to x. Then $\frac{2}{3}[10+x]=2x$; $\therefore x=5$. The stranger receives 5; and the son, who is not indebted to the father, receives 5.

[*] $\frac{4}{5}[10+x]-1=2x$; $\therefore \frac{4}{5}[10+x]-1=2x$; $\frac{2}{3}[10+x]-1=x$; The stranger receives $\frac{1}{5}[10+x]+1$.

[*] The stranger receives $\frac{1}{5}[10+x]-1$.

Subtract one-fifth and one-third of one-fifth of thing from thing. There remain eleven-fifteenths of thing, equal to three dirhems. It is now required to complete the thing. For this purpose, add to it four-elevenths, and do the same with the three dirhems, by adding to them one dirhem and one-eleventh. Then you have four dirhems and one-eleventh, which are equal to thing. This is the sum which is taken out of the debt.

On another Species of Legacy.

"A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital."

Computation:[*] You constitute their shares by taking them out of forty-eight parts. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this, being one-ninth of the whole capital. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

If the instance be: "A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her capital;" then you constitute the shares of the heirs, by taking them out of twenty.[*] Take a capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a capital less one-eighth and one-seventh. Complete your capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen forty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares.

On another Species of Legacies, viz.

If nothing has been imposed on some of the heirs,[*] and something has been imposed on others; the legacy amounting to more than one-third.

It must be known, that the law for such a case is, that if more than one-third of the legacy has been imposed on one of the heirs, this enters into his share; but that also those on whom nothing has been imposed must, nevertheless, contribute one-third.

Example: "A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother's share of the

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[*] It appears in the sequel (p. 96) that a widow is entitled to $\frac{1}{8}$th, and a mother to $\frac{1}{6}$th of the residue; $\frac{1}{8}+\frac{1}{6}=\frac{7}{24}$; leaving $\frac{17}{24}$ of the residue to be distributed between two brothers and two sisters; that is, $\frac{17}{48}$ between a brother and a sister; but in what proportion these 17 parts are to be divided between the brother and sister does not appear in the course of this treatise. Let the whole capital of the testator $= 1$ and let each 48th share of the residue $= v$; $1-\frac{1}{9}=\frac{8}{9}$; $\therefore\ \frac{8}{9}=48v$; $\therefore\ v=\frac{1}{54}$; that is, each 48th part of the residue $= \frac{1}{54}$th of the whole capital.

[*] A husband is entitled to $\frac{1}{4}$th of the residue, and the sons and daughters divide the remaining $\frac{3}{4}$ths of the residue in such proportion, that a son receives twice as much as a daughter. In the present instance, as there are three daughters and one son, each daughter receives $\frac{1}{5}$ of $\frac{3}{4}, = \frac{3}{20}$, of the residue, and the son, $\frac{6}{20}$. Since the stranger takes $\frac{1}{8}+\frac{1}{7}=\frac{15}{56}$ of the capital, the residue $= \frac{41}{56}$ of the capital, and each $\frac{1}{20}$th share of the residue $= \frac{41}{56}\times\frac{1}{20}=\frac{41}{1120}$ of the capital. The stranger, therefore, receives $\frac{15}{56}=\frac{300}{1120}=\frac{15}{56}$ of the capital.

[*] The problems in this chapter may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

residue); on her husband she imposes nothing but one-third, (which he must contribute, according to the law)."[*]

Computation: You constitute the shares of the heritage, by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself. The two legacies together are imposed upon the son. It is therefore necessary to subtract from his share two-fifths and one-fourth of the same. He thus retains seven twentieths of his entire original share, dividing the whole of it into twenty equal parts. The mother retains as much as she contributes to the legacy; this is one (twelfth part), the entire amount of what she had received being two parts.

Take now a sum, one-fourth of which may be divided into thirds, or of one-sixth of which the moiety may be taken; this being again divisible by twenty. Such a capital is two hundred and forty. The mother receives one-sixth of this, namely, forty; twenty from this fall to the legacy, and she retains twenty for herself. The husband receives one-fourth, namely, sixty; from which twenty belong to the legacy, so that he retains forty. The remaining hundred and forty belong to the son; the legacy from this is two-fifths and one-fourth, or ninety-one; so that there remain forty-nine. The entire sum for the legacies is, therefore, one hundred and thirty-one, which must be divided among the two legatees. The one to whom two-fifths were bequeathed, receives eight-thirteenths of this; the one to whom one-fourth was devised, receives five-thirteenths.

If you wish distinctly to express the shares of the two legatees, you need only to multiply the parts of the heritage by thirteen, and to take them out of a capital of three thousand one hundred and twenty.

On another Species of Legacies.

"A man dies, and leaves four sons and his wife; and bequeaths to a person as much as the share of one of the sons less the amount of the share of the widow."

Divide the heritage into thirty-two parts. The widow receives one-eighth,[*] namely, four; and each son seven. Consequently the legatee must receive three-sevenths of the share of a son. Add, therefore, to the heritage three-sevenths of the share of a son, that is to say, three parts, which is the amount of the legacy. This gives thirty-five, from which the legatee receives three and the remaining thirty-two are distributed among the heirs proportionably to their legal shares.

If he leaves two sons and a daughter,[*] and bequeaths to some one as much as would be the share of a third son, if he had one; then you must consider, what would be the share of each son, in case he had three. Assume this to be seven, and for the entire heritage take a number, one-fifth of which may be divided into sevenths, and one-seventh of which may be divided into fifths. Such a number is thirty-five. Add to it two-sevenths of the same, namely, ten. This gives forty-five. Herefrom the legatee receives ten, each son fourteen,

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[*] If the bequests stated in the present example were charged on the heirs collectively, the husband would be entitled to $\frac{1}{4}$, the mother to $\frac{1}{6}$ of the residue: $\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$; the remainder $\frac{7}{12}$ would be the son's share of the residue; but since the bequests, $\frac{2}{5}+\frac{1}{4}=\frac{13}{20}$ of the capital, are charged upon the son and mother, the law throws a portion of the charge on the husband.

[*] A widow is entitled to $\frac{1}{8}$th of the residue; therefore $\frac{7}{8}$ths of the residue are to be distributed among the sons of the testator. Let x be the stranger's legacy. The widow's share $=\frac{1}{8}(1-x)$; each son's share $=\frac{1}{4}\cdot\frac{7}{8}(1-x)$; and a son's share, minus the widow's share $=\frac{7}{32}-\frac{4}{32}(1-x)=x$; $\therefore x=\frac{3}{35}$.

[*] A son is entitled to receive twice as much as a daughter. Were there three sons and one daughter, each son would receive $\frac{2}{7}$ths of the residue. Let x be the stranger's legacy. Then $\frac{2}{7}(1-x)=x$; $\therefore x=\frac{2}{9}$; $1-x=\frac{7}{9}$. Each Son's share $=\frac{2}{7}\cdot\frac{7}{9}=\frac{2}{9}$. The Daughter's share $=\frac{1}{7}\cdot\frac{7}{9}=\frac{1}{9}$. The Stranger's legacy $=\frac{2}{9}$.

and the daughter seven.

If he leaves a mother, three sons, and a daughter, and bequeaths to some one as much as the share of one of his sons less the amount of the share of a second daughter, in case he had one; then you distribute the heritage into such a number of parts as may be divided among the actual heirs, and also among the same, if a second daughter were added to them.[*] Such a number is three hundred and thirty-six. The share of the second daughter, if there were one, would be thirty-five, and that of a son eighty; their difference is forty-five, and this is the legacy. Add to it three hundred and thirty-six, the sum is three hundred and eighty-one, which is the number of parts of the entire heritage.

On Completement.

"A woman dies and leaves eight daughters, a mother, and her husband, and bequeaths to some person as much as must be added to the share of a daughter to make it equal to one-fifth of the capital and to another person as much as must be added to the share of the mother to make it equal to one-fourth of the capital."[*]

Computation: Determine the parts of the residue, which in the present instance are thirteen. Take the capital, and subtract from it one-fifth of the same, less one part, as the share of a daughter: this being the first legacy. Then subtract also one-fourth, less two parts, as the share of the mother: this being the second legacy. There remain eleven-twentieths of the capital, which, when increased by three parts, are equal to thirteen parts. Remove now from thirteen parts the three parts on account of the three parts (on the other side), and you retain eleven-twentieths of the capital, equal to ten parts. Complete the capital, by adding to the ten parts as much as nine-elevenths of the same; then you find the capital equal to eighteen parts and two-elevenths. Assume now each part to be eleven; then the whole capital is two hundred, each part is eleven; the first legacy will be twenty-nine, and the second twenty-eight.

Computation of Returns.

On Marriage in Illness.

"A man, in his last illness, marries a wife, paying (a marriage settlement of) one hundred dirhems, besides which he has no property, her dowry being ten dirhems. Then the wife dies, bequeathing one-third of her property. After this the husband dies."[*]

Computation: You take from the one hundred that which belongs entirely to her, on account of the dowry, namely, ten dirhems; there remain ninety dirhems, out of which she has bequeathed a legacy. Call the sum given to her (by her husband, exclusive of her dowry) thing; subtracting it, there remain ninety dirhems less thing. Ten dirhems and thing are already in her hands; she has disposed of one third of her property, which is three dirhems and one-third, and one-third of thing; there remain six dirhems and two-thirds plus two-thirds of thing, the moiety of which, namely, three dirhems and one-

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[*] Let x be the stranger's legacy; a widow's share of the residue is $\frac{1}{8}$th; $\frac{7}{8}(1-x)$ is the residue. $\frac{1}{8}[1-x]$, to be distributed among the children. Since there are 3 sons, and 1 daughter, $\frac{1}{7}\cdot\frac{7}{8}(1-x)$ is a son's share. Were there 3 sons and 2 daughters, a daughter's share would be $\frac{1}{8}\cdot\frac{8}{8}(1-x)$. The difference $= \frac{1}{7}\cdot\frac{7}{8}(1-x)-\frac{1}{8}(1-x)=x$; $\therefore x=\frac{45}{336}$.

[*] In this case, the mother has $\frac{2}{13}$; and each daughter has $\frac{1}{13}$ of the residue. $1-x-y=13v$; $1-[\frac{1}{5}-v]-[\frac{1}{4}-2v]=13v$; i.e. $1-\frac{1}{5}+v-\frac{1}{4}+2v=13v$.

[*] Let s be the sum, including the dowry, paid by the man, as a marriage settlement; d the dowry; x the gift to the wife, which she is empowered to bequeath if she pleases. She may bequeath, if she pleases, $d+x$; she actually does bequeath $\frac{1}{3}[d+x]$; the residue is $\frac{2}{3}[d+x]$, of which one half, viz. $\frac{1}{3}[d+x]$ goes to her heirs, and the other half reverts to the husband. $\therefore$ the husband's heirs have $s-[d+x]+\frac{1}{3}[d+x]$; and since what the wife has disposed of, exclusive of the dowry, is x, twice which sum the husband is to receive, $s-\frac{2}{3}[d+x]=2x$; $\therefore \frac{3}{8}[3s-2d]=x$. But $s=100$; $\therefore x=35$; $d+x=45$; $\frac{1}{3}[d+x]=15$. Therefore the legacy which she bequeaths is 15, her husband receives 15, and her other heirs, 15. The husband's heirs receive $2x=70$.

third plus one-third of thing, returns as his portion to the husband.[*] Thus the heirs of the husband obtain (as his share) ninety-three dirhems and one-third, less two thirds of thing; and this is twice as much as the sum given to the woman, which was thing, since the woman had power to bequeath one-third of all which the husband left;[*] and twice as much as the gift to her is two things. Remove now the ninety-three and one-third, from two-thirds of thing, and add these to the two things. Then you have ninety-three dirhems and one-third equal to two things and two-thirds. One thing is three-eighths of it, namely, as much as three-eighths of the ninety-three and one-third, that is, thirty-five dirhems.

On Emancipation in Illness.

"Suppose that a man on his death-bed were to emancipate two slaves; the master himself leaving a son and a daughter. Then one of the two slaves dies, leaving a daughter and property to a greater amount than his price."[*] You take two-thirds of his price, and what the other slave has to return (in order to complete his ransom). If the slave die before the master, then the son and the daughter of the latter partake of the heritage, in such proportion, that the son receives as much as the two daughters together. But if the slave die after the master, then you take two-thirds of his value and what is returned by the other slave, and distribute it between the son and the daughter (of the master), in such a manner, that the son receives twice as much as the daughter; and what then remains (from the heritage of the slave) is for the son alone, exclusive of the daughter; for the moiety of the heritage of the slave descends to the daughter of the slave, and the other moiety, according to the law of succession, to the son of the master, and there is nothing for the daughter (of the master).

It is the same, if a man on his death-bed emancipates a slave, besides whom he has no capital, and then the slave dies before his master.

If a man in his illness emancipates a slave, besides whom he possesses nothing, then that slave must ransom himself by two-thirds of his price. If the master has anticipated these two-thirds of his price and has spent them, then, upon the death of the master, the slave must pay two-thirds of what he retains.[*] But if the master has anticipated from him his whole price and spent it, then there is no claim against the slave, since he has already paid his entire price.

"Suppose that a man on his death-bed emancipates a slave, whose price is three hundred dirhems, not having any property besides; then the slave dies, leaving three hundred dirhems and a daughter." The computation is this:[*] Call the legacy to the slave

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[*] In other cases, as appears from pages 92 and 93, a husband inherits one-fourth of the residue of his wife's estate, after deducting the legacies which she may have bequeathed. But in this instance he inherits half the residue. If she die in debt, the debt is first to be deducted from her property, at least to the extent of her dowry (see the next problem.)

[*] When the husband makes a bequest to a stranger, the third is reduced to one-sixth. Vide p. 137.

[*] From the property of the slave, who dies, is to be deducted and paid to the master's heirs, first, two-thirds of the original cost of that slave, and secondly what is wanting to complete the ransom of the other slave. Call the amount of these two sums p; and the property which the slave leaves u. Next, as to the residue of the slaves' property: First. If the slave dies before the master, the master's son takes $\frac{2}{3}[u-p]$; the master's daughter $\frac{1}{3}[u-p]$; and the slave's daughter $\frac{1}{2}[u-p]$. Second. If the slave dies after the master; the master's son is to receive $\frac{2}{3}p$, and the master's daughter $\frac{1}{3}p$; and then the master's son takes $\frac{1}{2}[u-p]$, and the slave's daughter $\frac{1}{2}[u-p]$.

[*] The slave retains one-third of his price; and this he must redeem at two-thirds of its value; namely at $\frac{2}{3}\times\frac{1}{3}=\frac{2}{9}$ of his original price.

[*] Let the slave's original cost be a; the property which he dies possessed of u; what the master bequeaths to the slave, x. Then the net property which the slave dies possessed of is $u+x-a$. $\frac{1}{3}[u+x-a]$ belongs, by law, to the master; and $\frac{1}{2}[u+x-a]$ to the slave's daughter. The master's heirs, therefore, receive the ransom, $a-x$, and the inheritance, $\frac{1}{3}[u+x-a]$; that is, $\frac{1}{3}[(u+a-x)]$; and on the same principle as the slave, when

thing. He has to return the remainder of his price, after the deduction of the legacy, or three hundred less thing. This ransom, of three hundred less thing, belongs to the master. Now the slave dies, and leaves thing and a daughter. She must receive the moiety of this, namely, one half of thing; and the master receives as much. Therefore the heirs of the master receive three hundred less half a thing, and this is twice as much as the legacy, which is thing, namely, two things. Reduce this by removing half a thing from the three hundred, and adding it to the two things. Then you have three hundred, equal to two things and a half. One thing is, therefore, as much as two-fifths of three hundred, namely, one hundred and twenty.

This is the legacy (to the slave,) and the ransom is one hundred and eighty.

“Some person on his sick-bed has emancipated a slave, whose price is three hundred dirhems; the slave then dies, leaving four hundred dirhems and ten dirhems of debt, and two daughters, and bequeathing to a person one-third of his capital; the master has twenty dirhems debts.” The computation of this case is the following:[*] Call the legacy to the slave thing; his ransom is the remainder of his price, namely, three hundred less thing. But the slave, when dying, left four hundred dirhems; and out of this sum, his ransom, namely, three hundred less thing, is paid to the master, so that one hundred dirhems and thing remain in the hands of the slave’s heirs. Herefrom are (first) subtracted the debts, namely, ten dirhems; there remain then ninety dirhems and thing. Of this he has bequeathed one-third, that is, thirty dirhems and one-third of thing; so that there remain for the heirs sixty dirhems and two-thirds of thing. Of this the two daughters receive two-thirds, namely, forty dirhems and four-ninths of thing, and the master receives twenty dirhems and two-ninths of thing, so that the heirs of the master obtain three hundred and twenty dirhems less seven-ninths of thing. Of this the debts of the master must be deducted, namely, twenty dirhems; there remain then three hundred dirhems less seven-ninths of thing; and this sum is twice as much as the legacy of the slave, which was thing; or, it is equal to two things. Reduce this, by removing the seven-ninths of thing, and adding them to two things; there remain three hundred, equal to two things and seven-ninths. One thing is as much as nine twenty-fifths of three hundred, which is one hundred and eight; and so much is the legacy to the slave.

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emancipated, is allowed to ransom himself at two-thirds of his cost, the law of the case is that 2 are to be taken, where 1 is given. $\therefore \frac{1}{3}[(u+a-x)] = 2x$; $\therefore x = \frac{1}{7}[u+a]$; If, as in the example, $u = a$, $x = \frac{2}{7}a$; the daughter’s share $= \frac{1}{7}[3u-2a]$; The master’s heirs receive $\frac{2}{7}[u+a]$.

[*] Let the slave’s original cost $= a$; the property he dies possessed of $= u$; the debt he owes $= e$. He leaves two daughters, and bequeaths to a stranger one-third of his capital. The master owes debts to the amount h; where $a = 300$, $u = 400$, $e = 10$, $h = 20$. Let what the master gives to the slave, in emancipating him $= x$. Slave’s ransom $= a - x$; slave’s property $-$ slave’s ransom $= u + x - a$; Slave’s property $-$ ransom $-$ debt $= u+x-a-e$; Legacy to stranger $= \frac{1}{3}[u+x-a-e]$; Residue $= \frac{2}{3}[u+x-a-e]$; The master, and each daughter, are, by law, severally entitled to $\frac{1}{3} \times \frac{2}{3}[u+x-a-e]$; The master’s heirs receive altogether $a-x+\frac{1}{3}[u+x-a-e]$ or $\frac{1}{3}[a-x] + \frac{2}{3}[u-e]$, which, on the principle that 2 are to be taken for 1 given, ought to be made equal to $2x$. But the author directs that the equation for determining x be $\therefore x = \frac{3}{25}[7a + 2[u-e] - 3h] = 108$.

End of the Translation of the Algebra of Mohammed ben Musa.

Notes on the Arabic Text [ملاحظات على النص العربي]

The following brief notes are intended to point out for further enquiry a few circumstances connected with the history of Algebra:

1. The initial words of the treatise, as cited by Haji Khalfa, are: قال ابو عبدالله محمد بن موسى الخوارز ("Abu Abdallah Mohammed ben Musa of Khowarezm said...").
2. The term الجبر (*al-jabr*) literally signifies "restoration" or "completion," referring to the operation of transferring negative quantities to the other side of an equation as positive. The term المقابلة (*al-muqabala*) means "reduction" or "opposition," referring to the process of cancelling like quantities on both sides of an equation.
3. The Arabic word شيء (*shay'*, literally "thing") is used throughout the text as the designation of the unknown quantity—what we should now call x. It corresponds exactly to the *res* ("thing") of the early Italian algebraists, and to the *coss* of the Germans.
4. The word مال (*māl*, literally "property," "wealth," "possession") is used to denote the square of the unknown. It is rendered by Rosen as "square."
5. The word جذر (*jidhr*, literally "root") is used in the same sense as our "root." The word واجب (*wājib*, literally "obligatory") is used for the "absolute number"—what we call the constant term.
6. The method of geometrical demonstration here employed by al-Khwarizmi is worthy of especial notice. It shows that, like the Greek mathematicians, he was not satisfied with merely giving rules, but endeavoured to establish their accuracy by demonstration. His proofs are founded upon the second proposition of Book II of Euclid's Elements, and upon the well-known property of the gnomon.
7. The formula for the circumference of a circle given in the text (Text p. 51, Transl. p. 72) is $c = \frac{62832}{20000}d$, which gives $\pi = 3.1416$. This value was known to the Hindu mathematician Aryabhata, and the circumstance furnishes a strong argument for the opinion that al-Khwarizmi derived part of his information from an Indian source.
8. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but the enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which the translator had the opportunity of consulting.
9. The solutions which the author has given of the remaining problems of this treatise, are, mathematically considered, for the most part incorrect. It is not that the problems, when once reduced into equations, are incorrectly worked out; but that in reducing them to equations, arbitrary assumptions are made, which are foreign or contradictory to the data first enounced, for the purpose, it should seem, of forcing the solutions to accord with the established rules of inheritance, as expounded by Arabian lawyers.
10. The object of the lawyers in their interpretations, and of the author in his solutions, seems to have been, to favour heirs and next of kin; by limiting the power of a testator, during illness, to bequeath property, or to emancipate slaves; and by requiring

payment of heavy ransom for slaves whom a testator might, during illness, have directed to be emancipated.

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Appendix: Arabic–English Glossary of Algebraic Terms

THE FOLLOWING GLOSSARY presents the complete Arabic mathematical terminology employed by Mohammed ben Musa al-Khwārizmī in his *Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala*, together with Rosen's English equivalents and modern mathematical interpretations.

Arabic Term	Transliteration	Meaning and Usage
		I. CORE ALGEBRAIC TERMS
الجبر	*al-jabr*	**Restoration, completion.** The operation of transferring negative quantities to the other side of an equation as positive. One of the two fundamental operations of algebra, giving the science its name.
المقابلة	*al-muqābala*	**Reduction, opposition, balancing.** The process of cancelling like quantities on both sides of an equation; also the reduction of equations to standard form.
مال	*māl*	**Square, property, wealth.** The square of the unknown quantity (x^2). Literally "property" or "possession," reflecting the mercantile context of early algebra. Rosen consistently renders this as "square."
جذر	*jidhr*	**Root.** The unknown quantity itself (x); also used for square roots of known numbers. Literally "root," "base," or "origin."
شيء	*shay'*	**Thing.** The unknown quantity (x). The most common designation for the unknown in Arabic algebra. Corresponds to the Latin *res* and the Italian *cosa*.

Continued on next page

Arabic Term	Transliteration	Meaning and Usage
عدد	*‘adad*	**Number.** The absolute or constant term in an equation; a known quantity unrelated to the unknown. Rosen renders this as “simple number” or “number.”
واجب	*wājib*	**Obligatory, requisite.** The constant term or absolute number in certain contexts. Used by later writers; al-Khwārizmī himself uses *‘adad.*
	II. GEOMETRICAL TERMS	
مربع	*murabba‘*	**Quadrate, square figure.** A geometric square used in demonstrations. Distinguished from *māl* which refers to the algebraic square.
خط	*khaṭṭ*	**Line.** A straight or curved line.
مستقيم	*mustaqīm*	**Straight, rectilinear.** Describes a straight line or straight angle.
قائمة	*qā’ima*	**Perpendicular, cathetus.** A line standing at right angles to another. Literally “standing upright.”
طول	*ṭūl*	**Length.** The longer dimension of a figure.
عرض	*‘arḍ*	**Breadth, width.** The shorter dimension of a figure.
دائرة	*dā’ira*	**Circle.** Literally “that which revolves.”
قطر	*qaṭr*	**Diagonal, diameter.** The line passing through the centre of a figure.
	III. MERCANTILE AND PRACTICAL TERMS	
درهم	*dirham*	**Dirhem.** A silver coin; the standard monetary unit in algebraic calculations. Derived from the Greek drachma.
ثمن	*thaman*	**Price, value.** The cost of goods in mercantile problems.

Continued on next page

Arabic Term	Transliteration	Meaning and Usage
كيل	*kayl*	**Measure.** A unit of dry measure for grain, etc.
مقدار	*miqdār*	**Quantity, amount.** A general term for an unknown amount.
جملة	*jumla*	**Sum, total.** The aggregate of quantities.
نسبة	*nisba*	**Ratio, proportion.** The relation between two quantities.
	IV. Legal and Legacy Terms	
وصية	*wasiyya*	**Legacy, bequest.** A gift made by will, especially on death-bed. Subject to the restriction that it may not exceed one-third of the estate without heirs' consent.
ميراث	*mīrāth*	**Inheritance, heritage.** The estate left by a deceased person.
فرائض	*farā'iḍ*	**Obligatory shares.** The legally fixed portions of an estate due to prescribed heirs according to Islamic law.
عتق	*'itq*	**Emancipation, manumission.** The freeing of a slave, especially by a master's death-bed declaration.
مكاتبة	*mukātaba*	**Manumission contract.** An agreement whereby a slave purchases freedom by instalments.
مهر	*mahr*	**Dowry, marriage settlement.** The payment made by a groom to the bride, part of which may revert to the husband's heirs.
	V. Book Title and Formulaic Expressions	
كتاب المختصر في حساب الجبر والمقابلة	*Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābala*	**The Compendious Book on Calculation by Completion and Reduction.** The full Arabic title of al-Khwārizmī's algebra.

Continued on next page

Arabic Term	Transliteration	Meaning and Usage
بسم الله الرحمن الرحيم	*Bi-smi Llāhi r-Raḥmāni r-Raḥīm*	**In the Name of God, the Gracious, the Merciful.** The traditional opening formula (basmala) of Arabic books.
الحمد لله	*Al-ḥamdu li-Llāh*	**Praise be to God.** The opening formula of the author's preface.

Note on Transliteration. The system of transliteration employed above follows the standard academic convention for Arabic: macrons (ā, ī, ū) denote long vowels; dots beneath consonants (ṭ, ḥ, ṣ, ḍ) denote the Arabic emphatic consonants; the hamza (’) represents the glottal stop; and the ‘ayn (‘) represents the voiced pharyngeal fricative.

The Six Canonical Forms of Equations

Rosen's English	Arabic	Modern Form
Squares equal Roots	المال يساوي الجذور	$ax^2 = bx$
Squares equal Numbers	المال يساوي الأعداد	$ax^2 = c$
Roots equal Numbers	الجذور تساوي الأعداد	$bx = c$
Squares and Roots equal Numbers	المال والجذور يساويان الأعداد	$ax^2 + bx = c$
Squares and Numbers equal Roots	المال والأعداد يساويان الجذور	$ax^2 + c = bx$
Roots and Numbers equal Squares	الجذور والأعداد تساوي المال	$bx + c = ax^2$

Published under the auspices of the Oriental Translation Fund
London, 1831

Enhanced Bilingual Edition with Original Arabic Terminology Restored, 2025

Sitzungsberichte der Heidelberger Akademie der Wissenschaften
Stiftung Heinrich Lanz
Philosophisch-historische Klasse
Jahrgang 1917. 2. Abhandlung

كتاب الجبر والمقابلة

Zur ”altesten arabischen Algebra und Rechenkunst

On the Earliest Arabic Algebra and Arithmetic

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von / by
JULIUS RUSKA
in Heidelberg

Eingegangen am 14. Juli 1916 / Received July 14, 1916
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Inhalt / Contents

Einleitung / Introduction

Daß die Mathematik bei den Arabern aus indischen und griechischen Quellen zusammengeflossen ist und vielfach durch Perser, Syrer und Juden in den Strom des geistigen Lebens eingef”uhrt wurde, ist eine Tatsache, die sich ebenso deutlich aus dem Inhalt der arabischen Schriften und der literarischen ”Uberlieferung ergibt, wie sie der geschichtlichen Lagerung der islamischen Gesamtkultur entspricht.

Schwierigeren Fragen stehen wir aber alsbald gegen”uber, wenn wir diese allgemein anerkannte Abh”angigkeit auf bestimmte Quellen zur”uckf”uhren wollen, wenn wir die auf lange Strecken versch”utteten Rinnsale aufzudecken versuchen, in denen die Zufuhr des fremden Stoffes sich vollzog, und wenn wir eine Entscheidung dar”uber treffen sollen, ob ein bestimmtes Wissen seine Form schon auf dem Weg zur Sammelstelle erhalten hat oder erst dem arabischen Einfluß seine Gestaltung verdankt.

Die Schwierigkeiten wachsen, wenn sich Urteile ”uber solche Dinge nicht auf die Quellen erster Hand, sondern auf ”Ubersetzungen und Bearbeitungen st”utzen, die den Charakter des Originalwerks verwischt haben und vielfach schon von pers”onlicher Stellungnahme bestimmt sind. Die Versuchung, moderne Ausdrucksweisen und Auffassungen in ”altere Werke hineinzutragen oder ihren Inhalt nach modernen Maßst”aben zu beurteilen, liegt auf mathematischem Gebiet besonders nahe, da ungewohnte Form und Umst”andlichkeit der Darstellung Hindernisse raschen Verst”andnisses sind und es dem Mathematiker gew”ohnlich mehr auf den Inhalt, als auf die Form der Erkenntnis ankommt.

F”ur alle Untersuchungen, die die Form der Darstellung betreffen, also f”ur die ganze Geschichte der Terminologie, werden wir daher immer wieder auf die Originalquellen zur”uckgehen m”ussen, um zu neuen oder gesicherteren Ergebnissen zu gelangen. Eine erneute Untersuchung der ”altesten arabischen Algebra schien besonders aussichtsvoll, nachdem sich bei einem fl”uchtigen Vergleich herausgestellt hatte, daß die von Fr. Rosen 1831 dem Text der Algebra des Muhammad b. M”us”a al-Hw”arazm”i beigef”ugte ”Ubersetzung nicht den Wortlaut wiedergibt, sondern sich ziemlich große Freiheiten gestattet.

That mathematics among the Arabs flowed together from Indian and Greek sources, and was frequently introduced into the current of intellectual life by Persians, Syrians, and Jews, is a fact that emerges as clearly from the content of Arabic writings and literary tradition as it corresponds to the historical stratification of Islamic culture as a whole.

But we immediately confront more difficult questions when we wish to trace this generally acknowledged dependence back to specific sources, when we attempt to uncover the long-buried channels through which the supply of foreign material took place, and when we must decide whether particular knowledge received its form already on the way to its destination, or owes its shaping to Arabic influence.

The difficulties increase when judgments about such matters rely not on first-hand sources but on translations and adaptations that have obscured the character of the original work and are frequently already determined by personal bias. The temptation to introduce modern modes of expression and conceptions into older works, or to evaluate their content by modern standards, is especially strong in the mathematical domain, since unfamiliar form and circumstantial presentation are obstacles to rapid comprehension, and the mathematician usually cares more about the content than the form of knowledge.

For all investigations concerning the form of presentation, that is, for the entire history of terminology, we shall therefore always have to return to the original sources in order to arrive at new or more secure results. A renewed investigation of the oldest Arabic algebra seemed particularly promising after a casual comparison revealed that the translation appended by Fr. Rosen in 1831 to the text of the Algebra of Muḥammad b. Mūsā al-Khwārazmī does not reproduce the wording but allows itself considerable liberties.

Sie schien auch darum erw”unscht, weil ”uber die Grundlagen der arabischen Algebra so viele entgegengesetzte und sich ausschließende Ansichten ge”außert worden sind, daß von einer Ersch”opfung des Gegenstandes noch immer nicht gesprochen werden kann. Im weiteren Verlauf der Arbeit stellte sich dann die Notwendigkeit heraus, auch andere Texte und ”Ubersetzungen in den Bereich der Untersuchung zu ziehen, und so ist schließlich eine Reihe von mehr oder minder zusammenh”angenden Einzelbeitr”agen zur Geschichte der ”alteren arabischen Mathematik

It also seemed desirable because so many contradictory and mutually exclusive opinions have been expressed about the foundations of Arabic algebra that an exhaustive treatment of the subject cannot yet be claimed. In the further course of the work, the necessity arose to include other texts and translations in the scope of the investigation, and so a series of more or less connected individual contributions to the history of older Arabic mathematics has finally emerged. The appended index will facilitate the locating of specific entstanden. Das beigegebene Register wird die Aufsuchung bestimmter Gegenst”ande der Untersuchungen erleichtern.

subjects of the investigations.

Anmerkung zur Namensumschreibung / Note on Name Transcription

Man kann eine ganze Musterkarte von Umschreibungen dieses Namens zusammenstellen; so schreibt Colebrooke (1817) Khuw”arezmi, Rosen (1831) of Khio war ezm, Woepcke (1863) Alkh”arizmi, Marre (1865) Al Kh”ar e z m, Rodet (1878) Al-Kh”arizmi, Hankel (1874) al-Hov”arezmi, Cantor (1880) Alchwarizmi, van Vloten (1895) Al-Khowarezmi, Suter (1900) el-Chow”aresml oder Chw”aresml, Bj”ornbo (1905) Alkwarizmi, Wiedemann (1906) al Ch”arizmi, Bosmans (1906) El-Chow”arizmi, Karpinski (1910) Al-Khowarizmi, Smith (1911) Al-Khow”arazmi, Enestr”om Alkhwarizmi und Alkwarismi.

One can assemble a whole pattern-card of transcriptions of this name; thus Colebrooke (1817) writes Khuwārezmi, Rosen (1831) “of Khio war ezm,” Woepcke (1863) Alkhārizmi, Marre (1865) Al Khār e z m, Rodet (1878) Al-Khārizmi, Hankel (1874) al-Hovārezmi, Cantor (1880) Alchwarizmi, van Vloten (1895) Al-Khowarezmi, Suter (1900) el-Chowāresml or Chwāresml, Björnbo (1905) Alkwarizmi, Wiedemann (1906) al Chārizmi, Bosmans (1906) El-Chowārizmi, Karpinski (1910) Al-Khowarizmi, Smith (1911) Al-Khowārazmi, Eneström Alkhwarizmi and Alkwarismi.

Dazu kommen die alten Formen Alchoarismi in der ”Ubersetzung des Gerhard von Cremona, Alghuarizim, Alguarizin, Algaurizm, Algaurizin in den Handschriften der ”Ubersetzung des Robert Castrensis (Bibl. Math. 3. F., Bd. 11, 1910/11, S. 130; die Formen mit *in* und *im* sind offenbar aus den Formen mit *m* und *mi* entstanden, die mit *au* aus *ua*), Alghoarismi, Algoarismi, Algorismi in der *Pratica Arismetrice* des Johannes Hispalensis, Algoritmi im Traktat *de numero Indorum.*

To these are added the old forms Alchoarismi in the translation of Gerhard of Cremona, Alghuarizim, Alguarizin, Algaurizm, Algaurizin in the manuscripts of the translation of Robert of Chester (Bibl. Math. 3rd ser., vol. 11, 1910/11, p. 130; the forms with *in* and *im* have obviously arisen from the forms with *m* and *mi*, those with *au* from *ua*), Alghoarismi, Algoarismi, Algorismi in the *Pratica Arismetrice* of Johannes Hispalensis, Algoritmi in the treatise *De numero Indorum.*

Volkst”umliche Weiterbildung f”uhrt von Algorismus zu *augorisme* und *augrim* (Smith-Karpinski, *The Hindu-Arabic Numerals*, Boston 1911, S. 120, 121). Alle Formen mit Khwa- und Chwa- beruhen auf buchst”ablicher Umschreibung der Zeichengruppe خوا, die der persischen Sprache eigent”umlich ist und etwa *kh”o* gesprochen wird (vgl. Xwrdaoioi, Xopdaoioi). Da der zweite Vokal von *Hw”arazm* nach Vullers (S. 736) ein Fatha ist, lautet die genaue Umschreibung der Nisbe al-Hw”arazmi oder Alhw”arazmi.

Vulgar development leads from Algorismus to *augorisme* and *augrim* (Smith-Karpinski, *The Hindu-Arabic Numerals*, Boston 1911, pp. 120, 121). All forms with Khwa- and Chwa- rest on literal transcription of the group of characters خوا, which is peculiar to the Persian language and is pronounced approximately *khō* (cf. Xwrdaoioi, Xopdaoioi). Since the second vowel of *Hwārazm* according to Vullers (p. 736) is a fatha, the precise transcription of the nisba is al-Hwārazmi or Alhwārazmi.

Die Wiedergabe von *q* durch *k*, von ح durch *s* ist falsch, die Umschreibung von خ durch *ch* oder *kh* ist nicht eindeutig genug, der allgemeinen Verwendung von *H* und *h* stehen typographische Schwierigkeiten gegen”uber. Man vermeidet diese und andere Klippen, wenn man Rosen folgend den Verfasser Muhammad b. Muṣā nennt. Will man die geschichtlichen Zusammenh”ange betonen, so wird man besser Algoritmi oder Algorithmus schreiben, wie man auch Averroes statt Ibn Rusd sagt.

The rendering of *q* by *k*, of ح by *s* is wrong; the transcription of خ by *ch* or *kh* is not unambiguous enough; typographical difficulties stand in the way of the general use of *H* and *h*. One avoids these and other pitfalls if, following Rosen, one calls the author Muḥammad b. Muṣā. If one wishes to emphasize the historical connections, it is better to write Algoritmi or Algorithmus, just as one says Averroes instead of Ibn Rushd.

Sowenig wir das Wort Averroismus durch Ibnrusdismus ersetzen k”onnen, so wenig ist die moderne Umschreibung Alhw”arazmi geeignet, die Erinnerung an die Algorithmiker lebendig zu erhalten.

Just as little as we can replace the word Averroism with Ibnrushdism, so little is the modern transcription Alhwārazmi suited to keep alive the memory of the algorists.

1 I. Der Titel der Algebra / The Title of the Algebra

Erw”ahnte Muhammad b. M”usā nicht selbst in der Einleitung zu seiner “Algebra” die Gegenst”ande, von denen er zu handeln beabsichtigt, so k”onnte man versucht sein, das uns erhaltene Werk als willk”urliche Zusammenf”ugung von zwei oder drei urspr”unglich getrennten Schriften anzusehen. Sein Inhalt entspricht aber durchaus dem, was der Verfasser als seine Absicht bezeichnet:

> ان هذا كتاب الاجابة والمقابلة وقد حددته الى ما يسهل من
> الحساب وما يحتاج اليه الناس في مواريثهم ووصاياهم
> ومقاسمتهم ومعاملتهم وكل ما يتداولونه بينهم من المكايلة
> والمساحة وغير ذلك
>
> “ein kurzgefaßtes Buch zu schreiben von dem Rechenverfahren der Erg”anzung und Ausgleichung (*hisāb al-gabr wa-l-muqābala*) mit Beschr”ankung auf das Anmutige und Hochgesch”atzte des Rechenverfahrens f”ur das, was die Leute fortw”ahrend notwendig brauchen bei ihren Erbschaften und ihren Verm”achtnissen und bei ihren Teilungen und ihren Prozeßbescheiden und ihren Handelsgesch”aften und bei allem, womit sie sich gegenseitig befassen von der Ausmessung der L”andereien und der Herstellung der Kan”ale und der Geometrie und anderm dergleichen nach seinen Gesichtspunkten und Arten.”

Wollten wir den Ausdruck *kitāb* pressen, der urspr”unglich die Sammlung und Vereinigung mehrerer Teile eines Dings bedeutet (Lane Bd. I, S. 80), so k”onnten wir darin den Hinweis erblicken, daß das Werk ein Auszug aus verschiedenen Quellen ist.

Der Verfasser gibt selbst nirgends eine Andeutung, woher er seinen Stoff hat, und wieweit ihm pers”onlich ein Verdienst an der Pr”agung zukommt. Halten wir uns gegenw”artig, daß er sich als Astronom anerkanntermaßen auf indische Wissenschaft st”utzte und auch ein Buch ”uber das indische Rechnen verfaßte, so ist die Richtung angegeben, in der man zun”achst zu suchen hat.

Bevor wir uns aber den Untersuchungen zuwenden, die seit Colebrooke dar”uber ver”offentlicht sind, muß der Titel und die Gesamtanlage des Buches etwas genauer betrachtet werden. Man hat immer wieder dem Erstaunen Ausdruck gegeben, daß Muhammad b. M”usā seinem Werk einen Titel gegeben habe, den er nicht erkl”art, und daß er daf”ur zwei Operationen ausgew”ahlt habe, die in dem “eigentlich theoretischen Teile” seines Buches gar nicht vorkommen.

Ich glaube, man hat sich das Verst”andnis dieser Tatsachen ganz unn”otig erschwert, indem man das Buch nicht historisch

Had Muḥammad b. Mūsā not himself mentioned in the introduction to his “Algebra” the subjects he intended to treat, one might be tempted to regard the work preserved to us as an arbitrary combination of two or three originally separate treatises. Its content, however, fully corresponds to what the author states as his intention:

> ان هذا كتاب الاجابة والمقابلة وقد حددته الى ما يسهل من
> الحساب وما يحتاج اليه الناس في مواريثهم ووصاياهم
> ومقاسمتهم ومعاملتهم وكل ما يتداولونه بينهم من المكايلة
> والمساحة وغير ذلك
>
> “[This is] a concise book on the method of calculation of *al-jabr wa-l-muqābala* [restoration and reduction], comprising what is elegant and esteemed in calculation, touching upon what people constantly need regarding their inheritances and bequests, their divisions, their legal judgments, their commercial transactions, and all that they deal with among themselves concerning the measurement of lands, the construction of canals, and geometry, as well as other such things according to its principles and types.”

Were we to press the term *kitāb*, which originally means the collection and unification of several parts of a thing (Lane vol. I, p. 80), we might see in it an indication that the work is an extract from various sources.

The author himself gives no hint anywhere of where he obtained his material, or to what extent any credit for its formulation is due to him personally. If we bear in mind that he, as an astronomer, demonstrably relied on Indian science and also composed a book on Indian calculation, the direction in which one must first search is indicated.

But before we turn to the investigations published since Colebrooke, the title and overall plan of the book must be examined somewhat more closely. Repeatedly, astonishment has been expressed that Muḥammad b. Mūsā gave his work a title which he did not explain, and that for this he chose two operations that do not occur at all in the “properly theoretical part” of his book.

I believe that people have made the understanding of these facts entirely unnecessarily difficult by judging the mit den “Augen des Verfassers” (Cantor a. a. O. S. 728), sondern ohne R”ucksicht auf den Gesamtinhalt nur nach den wenigen theoretischen Kapiteln am Anfang beurteilte, die den Geschichtschreiber der Mathematik besonders interessieren.

Schon H. Hankel hat mit dieser Verschiebung des Tatbestandes den Anfang gemacht, indem er zwar den Inhalt des ersten Teils der Algebra ganz zutreffend kennzeichnet, die geometrischen Kapitel aber davon l”ost und von dem Rest des Werkes bemerkt, daß die dort behandelten Aufgaben ”uber Erbteilungen von keinem wissenschaftlichen Interesse seien, wenn sie auch f”ur Mohammedaner praktisch von gr”oßtem Werte waren.[1]

Was weiter dar”uber gesagt wird, geht auf zwei Fußnoten Rosens (a. a. O. S. 91, 133) zur”uck. Cantor hat dann nach A. v. Kremers *Culturgeschichte des Orients* noch einige S”atze hinzugef”ugt und auch auf die Ausf”uhrungen von Ibn Hald”un und H”aggi Hāa ”uber Erbteilungswissenschaft hingewiesen.[2]

Dieser Abschnitt der Algebra ist nach Cantor “arabisch durch und durch” und als Grundlage zahlreicher sp”aterer Schriften zu betrachten, die von den Erbteilungen und den dabei vorkommenden Rechnungen ausschließlich handeln. Proben dieser Rechenpraxis sind leider nicht mitgeteilt, und so ist es nicht verwunderlich, daß die j”ungste Darstellung dieser Dinge auf eine H”aufung von Mißverst”andnissen hinausl”auft.

Auf dieser Stufe der Untersuchung gen”ugt es, die Tatsache festzuhalten, daß Muhammad b. M”usā eine “Algebra” in unserem Sinne nicht schreiben wollte, noch geschrieben hat. Sein Buch will weiter nichts sein als eine auf zahlreiche ausgef”uhrte Musterbeispiele gest”utzte Einf”uhrung in das angewandte Rechnen. Ob es diesen in der Einleitung ausgesprochenen Zweck, zur L”osung von Aufgaben aus dem Gebiet des b”urgerlichen Rechnens und der Fl”achen- und K”orperberechnung anzuleiten, nach unseren Anspr”uchen ganz erreicht, ist hier nicht zu er”ortern; wohl aber sind wir die Antwort auf die Frage schuldig, wie der Verfasser dazu kam, seinem Werk einen angeblich so unverst”andlichen Titel zu geben.

Wenn wir G”unther glauben sollen, w”ußte man erst “sp”ateren Berichten zufolge”, daß eine Gleichung f”ur *eingerichtet* galt, wenn auf beiden Seiten ausschließlich positive Glieder standen, w”ahrend das Wort “Gegen”uberstellung” dem Weglassen gleicher Gr”oßen links und rechts entsprach.

Auch Cantor (I^3, S. 719) und Hankel (S. 260, Note**) berufen sich auf die Erkl”arungen sp”aterer arabischer Schriftsteller, und wer will, kann bei Rosen (S. 180 ff.) oder Nesselmann (*Die Algebra der Griechen*, S. 40 ff.) Text und ”Ubersetzung von Originalstellen nachlesen, in denen eine Erkl”arung der

book not historically with the “eyes of the author” (Cantor, op. cit., p. 728), but without regard for the total content, only according to the few theoretical chapters at the beginning that especially interest the historian of mathematics.

Already H. Hankel began this shifting of the facts, in that he indeed quite accurately characterized the content of the first part of the Algebra, but detached the geometrical chapters from it and remarked of the rest of the work that the problems on inheritance treated there were of no scientific interest, even if they were of great practical value for Muslims.[1]

What further is said about this goes back to two footnotes of Rosen (op. cit., pp. 91, 133). Cantor then, following A. von Kremer’s *Culturgeschichte des Orients*, added a few more sentences and also referred to the expositions of Ibn Khaldūn and Ḥājjī Ḥalīfa on the science of inheritance division.[2]

This section of the Algebra is, according to Cantor, “Arab through and through” and is to be regarded as the foundation of numerous later writings that deal exclusively with inheritance divisions and the calculations involved. Samples of this calculation practice are unfortunately not provided, and so it is not surprising that the most recent presentation of these matters amounts to a heap of misunderstandings.

At this stage of investigation, it suffices to establish the fact that Muḥammad b. Mūsā neither wished to write nor wrote an “Algebra” in our sense. His book wishes to be nothing more than an introduction to applied calculation, supported by numerous worked-out model examples. Whether it fully achieves, by our standards, the purpose stated in the introduction—namely, to guide the solution of problems from the domain of practical calculation and of plane and solid measurement—is not to be discussed here; but we certainly owe an answer to the question of how the author came to give his work a title allegedly so incomprehensible.

If we are to believe Günther, one knew only “according to later reports” that an equation was considered *set up* when exclusively positive terms stood on both sides, while the word “confrontation” corresponded to the removal of equal quantities from both sides.

Cantor (I^3, p. 719) and Hankel (p. 260, n. **) also appeal to the explanations of later Arabic writers, and whoever wishes can look up in Rosen (pp. 180 ff.) or Nesselmann (*Die Algebra der Griechen*, pp. 40 ff.) the text and translation of original passages in which an explanation of the terms *al-gabr* and

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[1] H. Hankel, *Zur Geschichte der Mathematik in Alterthum und Mittelalter*, Leipzig 1874, S. 260.

[2] M. Cantor, *Vorlesungen ”uber Gesch. d. Math.* I^3, 1907, S. 719.

[1] H. Hankel, *Zur Geschichte der Mathematik in Alterthum und Mittelalter*, Leipzig 1874, p. 260.

[2] M. Cantor, *Vorlesungen ”uber Geschichte der Mathematik* I^3, 1907, p. 719.

Ausdr”ucke *al-gabr* und *al-muqābala* gegeben wird. Kein Zweifel kann dar”uber bestehen, daß die verschiedenen ”Ubersetzungen der beiden Termini eine klare Vorstellung von ihrem mathematischen Sinn nicht zu geben verm”ogen. *Restauratio et oppositio* ist ein ebenso unklares Begriffspaar wie *restoring and comparing* (Colebrooke) oder *completion and reduction* (Rosen), *Wiederherstellung und Gegen”uberstellung* (Cantor) oder *Erg”anzung und Ausgleichung*, wie ich oben ”ubersetzt habe.

Gestehen wir aber dem Autor zu, daß wir ihn ganz gelesen haben m”ussen, ehe wir ihn kritisieren, so bekommen wir ein wesentlich anderes Bild. Wir begegnen den beiden Ausdr”ucken zuerst wieder da, wo der Verfasser ausf”uhrt, daß bei dem *hisāb al-gabr wa-l-muqābala* drei Arten von Zahlen vorkommen, die als *gud”ur* d. i. Wurzeln, *amwāl* d. i. Verm”ogen, und *radad mufrad* d. i. alleinstehende Zahl bezeichnet werden.[3]

Hieraus ist noch nichts zu entnehmen. Aber schon S. 15, am Schluß des theoretischen Teils, der die sechs Normalformen der Gleichungen behandelt, wird dem Leser gesagt, daß jede andere Gleichung durch das Verfahren der Erg”anzung und Ausgleichung auf eine der sechs Normalformen gebracht werden kann. Der Hinweis wird S. 25 in der Einleitung zu dem “Kapitel von den sechs Fragen” wiederholt, an der Stelle also, wo der Verfasser zu den “Textaufgaben” ”ubergeht, deren Ansatz erst noch auf die eine oder andere der Normalformen gebracht werden muß.

Im ersten Beispiel, das in unserer Schreibweise zum Ansatz $x^2 = 40x - 4x^2$ f”uhrt, kommt nur “Erg”anzung” vor. Nachdem der Ansatz abgeleitet ist, wird das Ergebnis ausdr”ucklich zusammengefaßt: “Es ist also ein Verm”ogen gleich vierzig Etwas weniger vier Verm”ogen” und fortgefahren: “Erg”anze es nun durch die vier Verm”ogen, und f”uge sie dem einen Verm”ogen hinzu, so werden vierzig Etwas gleich f”unf Verm”ogen, dann ist das einzelne Verm”ogen gleich acht Wurzeln” usw.

Es geh”ort gewiß nur ein m”aßiger Sch”ulerverstand dazu, hiernach zu begreifen, was mit “Erg”anzung” gemeint ist. Der gleiche Ausdruck mit derselben umst”andlichen Erl”auterung kehrt im dritten Beispiel wieder. Da aber Rosen an beiden Stellen *reduce it* statt *complete it* ”ubersetzt und auch das *radd*, das f”ur die Reduktion des Koeffizienten der Unbekannten auf 1 verwendet wird, mit *reduce it* wiedergibt, ist schon hier Verwirrung in die Terminologie gebracht.

al-muqābala is given. There can be no doubt that the various translations of the two terms are unable to give a clear conception of their mathematical sense. *Restauratio et oppositio* is an equally unclear pair of concepts as *restoring and comparing* (Colebrooke) or *completion and reduction* (Rosen), *Wiederherstellung und Gegen”uberstellung* (Cantor) or *Erg”anzung und Ausgleichung* [completion and balancing], as I have translated above.

But if we grant the author that we must have read him completely before we criticize him, we obtain a fundamentally different picture. We encounter the two expressions first where the author explains that in *hisāb al-gabr wa-l-muqābala* three kinds of numbers occur, designated as *gudūr* i.e. roots, *amwāl* i.e. possessions/wealth, and *radad mufrad* i.e. standalone number.[3]

Nothing is to be gathered from this as yet. But already on p. 15, at the end of the theoretical part that treats the six normal forms of equations, the reader is told that every other equation can be brought to one of the six normal forms by the method of completion and balancing. This hint is repeated on p. 25 in the introduction to the “Chapter of the Six Questions,” that is, at the point where the author passes to “word problems,” whose setup must first still be brought to one or another of the normal forms.

In the first example, which in our notation leads to the equation $x^2 = 40x - 4x^2$, only “completion” occurs. After the equation is derived, the result is explicitly summarized: “So a possession equals forty things less four possessions,” and continued: “Complete it now by the four possessions, and add them to the one possession, so forty things equal five possessions; then the single possession equals eight roots,” etc.

Surely only a moderate student intelligence is needed to understand from this what is meant by “completion.” The same expression with the same circumstantial explanation recurs in the third example. But since Rosen at both places translates *reduce it* instead of *complete it*, and also renders *radd*, which is used for the reduction of the coefficient of the unknown to 1, by *reduce it*, confusion has already been introduced into the terminology here.

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[3] Ich gebe die Termini auch in Umschrift, weil die Leser, die sich f”ur Geschichte der Mathematik interessieren, mit der arabischen Schrift und Sprache nur ausnahmsweise Bescheid wissen. In den arabischen Zitaten beschr”anke ich mich auf den Konsonantentext, soweit nicht besondere Gr”unde die Beif”ugung der Vokale oder anderer Lesezeichen notwendig machen. Das Wort *gud”ur* ist der Plural von *gidr*, *amwāl* der Plural von *māl*, *mufrad* kommt von *farada*, *farīda*, abgesondert sein, allein sein, einzig sein, einfach sein.

[3] I give the terms also in transliteration, because readers interested in the history of mathematics only rarely have acquaintance with the Arabic script and language. In the Arabic citations I restrict myself to the consonantal text, unless special reasons require the addition of vowels or other reading marks. The word *gudūr* is the plural of *gidr*, *amwāl* the plural of *māl*, *mufrad* comes from *farada*, *farīda*: to be separated, to be alone, to be unique, to be simple.

Sie wird vollends unheilbar, wenn nachher das richtige *reduce by it*, *muqābil bihī*, wof”ur besser *compare* oder *balance by it* gesagt worden w”are, hinzukommt und mit *complete it* auch noch die Operation des *ikmāl* ”ubersetzt wird. Auch die Hinweise S. 179 (erster und letzter Absatz) k”onnen daran f”ur Leser der ”Ubersetzung nichts ”andern.

Wie in der zweiten Aufgabe der Sinn des *radd* “und f”uhre es zur”uck auf ein einziges Verm”ogen” sich ganz nat”urlich aus dem Zusammenhang ergibt, so liefert die vierte ein Beispiel f”ur die Anwendung des *ikmāl*, d. h. der “Vervollst”andigung” durch Multiplikation, also der Wegschaffung eines Koeffizienten, der ein echter Bruch ist: “So vervollst”andige dein Verm”ogen, und seine Vervollst”andigung geschieht dadurch, daß du alles, was du hast, mit zw”olf vervielfachst.”[4]

In der f”unften Aufgabe kommt endlich auch “Ausgleichung” vor: “Es sind also f”unfzig Dirhem und ein Verm”ogen gleich neunundzwanzig Dirhem und zehn Etwas; gleiche nun damit aus, und dies besteht darin, daß du wegwirfst von den f”unfzig neunundzwanzig, so daß bleibt einundzwanzig Dirhem und ein Verm”ogen gleich zehn Etwas.”

In dieser Weise wird durch das ganze Buch hindurch Aufgabe f”ur Aufgabe erl”autert, d. h. es wird der Ansatz abgeleitet und jeder Schritt bis zur L”osung der Aufgabe vorgemacht und eingepr”agt. Nur im geometrischen Abschnitt ist keine Gelegenheit, das zur Aufl”osung von Gleichungen dienende Rechenverfahren mit seinen besonderen Ausdr”ucken anzuwenden.

Man wird hiernach gewiß nicht mehr behaupten k”onnen, daß Muhammad b. M”usā diese Ausdr”ucke nirgends erkl”art hat, wie schon bei Rosen S. 177 zu lesen ist, zumal die verschiedenen Definitionen aus sp”aterer Zeit auch nur darauf hinauskommen, die “Erg”anzung” durch “Hinzuf”ugung” und die “Ausgleichung” durch “Wegnahme gleicher Gr”oßen” zu ersetzen.

Kein Araber wird den Ausdruck “Erg”anzungs- und Ausgleichungsrechnung” mißverstanden haben, wenn er sich in das Buch eingelesen hatte und im Text den Imperativen “erg”anze” oder “gleiche aus” mit den Anweisungen zur Ausf”uhrung der Rechnung begegnete. Daß der Verfasser die Operationen erst da erl”autert, wo er sie zum erstenmal anwendet, ist ebenso vern”unftig und didaktisch richtig, als es zwecklos gewesen w”are, sie schon in dem “eigentlich theoretischen Teile” zu erw”ahnen, wo gar kein Gebrauch davon gemacht wird.

Die Frage, ob Muhammad b. M”usā die beiden Ausdr”ucke *al-gabr* und *al-muqābala* als erster angewandt und in die Praxis der Gleichungen eingef”uhrt hat, oder ob er damit schon einer festen Tradition und alten ”Ubung folgte, wird nur im Zusammenhang mit der Pr”ufung der gesamten Terminologie

It becomes completely incurable when subsequently the correct “reduce by it,” *muqābil bihī*, for which “compare” or “balance by it” would have been better, is added, and with “complete it” the operation of *ikmāl* [completion] is also translated. Even the indications on p. 179 (first and last paragraphs) cannot change anything in this for readers of the translation.

Just as in the second problem the sense of *radd*—“and reduce it to a single possession”—follows quite naturally from the context, so the fourth provides an example of the application of *ikmāl*, that is, “completion” by multiplication, hence the elimination of a coefficient that is a proper fraction: “So complete your possession, and its completion takes place by multiplying everything you have by twelve.”[4]

In the fifth problem, finally, “balancing” also occurs: “So fifty dirhams and a possession equal twenty-nine dirhams and ten things; balance this now, and this consists in casting away twenty-nine from fifty, so that twenty-one dirhams and a possession equal ten things remain.”

In this manner, throughout the whole book, problem by problem is explained, that is, the equation is derived and every step up to the solution of the problem is demonstrated and impressed upon the reader. Only in the geometrical section is there no occasion to apply the calculational procedure serving the resolution of equations with its special expressions.

After this, one will surely no longer be able to claim that Muḥammad b. Mūsā has explained these expressions nowhere, as can already be read in Rosen p. 177, especially since the various definitions from later times also amount to nothing more than replacing “completion” by “addition” and “balancing” by “removal of equal quantities.”

No Arab will have misunderstood the expression “calculation of completion and balancing” if he had read into the book and encountered in the text the imperatives “complete” or “balance” together with the instructions for carrying out the calculation. That the author explains the operations only where he applies them for the first time is as reasonable and didactically correct as it would have been pointless to mention them already in the “properly theoretical part,” where no use is made of them at all.

The question of whether Muḥammad b. Mūsā was the first to apply the two expressions *al-gabr* and *al-muqābala* and introduced them into the practice of equations, or whether he thereby already followed a fixed tradition and ancient practice, can only be brought closer to solution in connection

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[4] Das Beispiel ist $\frac{x}{3} = x + 24$, woraus $x = 12x + 288$.

[4] The example is $\frac{x}{3} = x + 24$, whence $x = 12x + 288$.

und literarischen Form seines Werkes der L”osung n”aher gebracht werden k”onnen.

So viel kann jetzt schon gesagt werden, daß wenn bereits eine Tradition bestand, sie auf arabischem Sprachgebiet nur innerhalb der Praxis des Gesch”aftsrechnens gesucht werden kann. Zeugnisse, die ”alter w”aren als die Algebra des Muhammad b. M”usā, sind aber bis jetzt nicht bekannt, und so werden wir vorl”aufig wenigstens mit der M”oglichkeit rechnen, daß er selbst bei der Bearbeitung seiner Quellen jene anschaulichen Wendungen an die Stelle der abstrakteren Terminologie der Griechen und Inder gesetzt hat.

Der allgemeinste aus einer Textaufgabe zu gewinnende Ansatz f”uhrt zur Gleichsetzung zweier Ausdr”ucke, die durch Addition, Subtraktion, Multiplikation und Division bekannter und unbekannter Gr”oßen entstehen. Durch eine hinreichende Anzahl von Umformungen erh”alt man schließlich zwei Ausdr”ucke, die die bekannten und unbekannten Glieder nur noch in additiver und subtraktiver Verbindung enthalten.

Von dieser Form geht Diophant aus, wenn er an einer oft genannten Stelle des ersten Buches der *Arithmétique* sagt: ean de tēn heterān en hupokeimenōi enuparchī ē en amphoterois en elleipsei tina eidī, deī zītein (sc. prosatheinai) ta leiponta eidī en amphoterois tois meresin, heōs an hekaterīn tīn merīn ta eidī sunuparchonta genītai, kai palin aphairein ta homoia apo tīn homoiīn, heōs an hekaterī tīn merīn en eidos kataleiphthī. (ed. Tannery I, S. 14).

Die erste Umformung w”urde arabisch *al-gabr* heißen; sie wird kurz und b”undig mit dem Wort *al-gabr*, die “Erg”anzung” bezeichnet, das sonst das Einrichten von Knochenbr”uchen bedeutet. Die andere Operation ist *al-muqābala*; sie heißt ebenso anschaulich *al-muqābala*, d. h. die “Kompensation” oder “Ausgleichung”.[5]

Zu diesen beiden wichtigsten Operationen kommt dann noch *al-radd*, die “Zur”uckf”uhrung” des Koeffizienten der Unbekannten auf 1, wenn er gr”oßer als 1 ist, und *al-ikmāl*, die “Vervollst”andigung”, wenn der Koeffizient ein echter Bruch ist. Dies ist der Tatbestand, den wir in der ”altesten Algebra vorfinden.

Wenn Carra de Vaux in seiner Notiz *Sur le sens exact du mot “al-djebr”* (Bibl. Math., 2. Folge, Bd. 11, 1897, S. 1, 2) darauf aufmerksam gemacht hat, daß *al-djebr* “n’est pas toujours opposé au mot al-muhābala, comme on pourrait le croire d’après le titre de l’ouvrage d’Al-Khārizmī”, sondern daß man es auch im Gegensatz zu dem Wort *al-haḍ* (الحط) angewandt finde, so beweisen die aus Autoren des 14. Jahrhunderts beigebrachten Belege nichts f”ur die urspr”ungliche Anwendung der Termini.

with an examination of the entire terminology and literary form of his work.

This much can already be said now: if a tradition already existed, it can be sought in the Arabic linguistic domain only within the practice of business calculation. Testimonies older than the Algebra of Muḥammad b. Mūsā are not known up to now, and so we shall provisionally reckon at least with the possibility that he himself, in working over his sources, replaced that more abstract terminology of the Greeks and Indians with those vivid turns of phrase.

The most general equation obtainable from a word problem leads to the equating of two expressions that arise through addition, subtraction, multiplication, and division of known and unknown quantities. Through a sufficient number of transformations, one finally obtains two expressions that contain the known and unknown terms only in additive and subtractive connection.

From this form Diophantus proceeds when he says at a frequently cited passage of the first book of the *Arithmétique*: ean de tēn heterān en hupokeimenōi enuparchī ē en amphoterois en elleipsei tina eidī, deī zītein (sc. prosatheinai) ta leiponta eidī en amphoterois tois meresin, heōs an hekaterīn tīn merīn ta eidī sunuparchonta genītai, kai palin aphairein ta homoia apo tīn homoiīn, heōs an hekaterī tīn merīn en eidos kataleiphthī. (ed. Tannery I, p. 14).

The first transformation would be called in Arabic *al-gabr*; it is designated briefly and compactly by the word *al-gabr*, the “completion,” which otherwise means the setting of bone fractures. The other operation is *al-muqābala*; it is called just as vividly *al-muqābala*, that is, the “compensation” or “balancing.”[5]

To these two most important operations is then added *al-radd*, the “reduction” of the coefficient of the unknown to 1, when it is greater than 1, and *al-ikmāl*, the “completion,” when the coefficient is a proper fraction. This is the state of affairs that we find in the oldest algebra.

If Carra de Vaux in his note *Sur le sens exact du mot “al-djebr”* (Bibl. Math., 2nd ser., vol. 11, 1897, pp. 1, 2) has pointed out that *al-djebr* “is not always opposed to the word al-muhābala, as one might believe from the title of the work of Al-Khwārizmī,” but that one finds it also in contrast to the word *al-haḍ* (الحط), the proofs adduced from authors of the 14th century demonstrate nothing for the original application of the terms.

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[5] Auch das Kollationieren von Handschriften und die Opposition, also das Gegen”ubertreten von Sternen wird durch diesen Ausdruck bezeichnet.

[5] Also the collation of manuscripts and the opposition, that is, the confronting of stars, is designated by this expression.

Bei jenen Autoren — es handelt sich um die Arithmetik des Takī ad-dīn al-Hanbalī, ”uber den nach H. Suter nur feststeht, daß er vor 1410 lebte, und um eine Schrift des Ibn al-Hā’im, der 1355–1412 gelebt hat — ist von einem Verfahren zur Aufl”osung von Gleichungen ”uberhaupt nicht die Rede. Es werden die beiden Operationen der Addition und der Multiplikation den beiden Operationen der Subtraktion und der Division gegen”ubergestellt und jene, da ihre Ausf”uhrung zu einer gr”oßeren Zahl f”uhrt, als *gabr*, hier also wohl “Auff”ullung” oder “Vermehrung”, diese, da eine kleinere Zahl herauskommt, als *haḍ*, d. i. “Erniedrigung” oder “Nachlaß” bezeichnet.

Es ist *gabr*, wenn man sagt, um wieviel mußt du 10 vergr”oßern, um 17 zu erhalten, oder womit mußt du 10 vervielfachen, um 17 zu erhalten; es ist *haḍ*, wenn man sagt, durch wieviel mußt du 10 teilen, oder um wieviel mußt du 10 verkleinern, um 7 zu erhalten.

Wenn Carra de Vaux diese in der ”alteren Literatur unbekannten Zusammenfassungen mit den modern gedachten S”atzen wiedergibt: “Il est donc clair que le djebr c’est l’opération exprimée par les deux équations $a + x = b$, $a \cdot x = b$ et que le haḍ, c’est l’opération qu’expriment ces deux ci: $a - x = b$, $a : x = b$”, und daran den Schluß kn”upft, “comme ces quatre opérations sont les plus simples possible de l’algèbre, on doit croire que tel est bien le sens primitif des deux mots”, so vermengt er nicht nur ganz verschiedene Dinge, sondern stellt die geschichtliche Entwicklung geradezu auf den Kopf, indem er weiterhin behauptet: “Le mot djebr dont nous venons d’indiquer le sens primitif, a ensuite servi à designer un nombre indéterminé de questions diverses, parmi lesquelles six sont fondamentales, selon Al-Khārizmī ...; le mot haḍ, devenu impropre à cause de la complication des problèmes, a disparu devant le mot mukābala.” Mit den sechs Normalformen der Gleichungen, auf die Carra de Vaux anspielt, und mit dem angeblichen Verschwinden des Wortes *muqābala* haben die Gegens”atze *gabr* und *haḍ* schlechterdings nichts zu tun.

Auch der Versuch Rodets, das Wort *gabara* im Sinne von “bereichern” zu deuten (J. As., 7. S., Bd. XI, 1878, S. 38), d”urfte kaum Anklang finden, zumal die aus Freytag angezogene Bedeutung *post paupertatem ditavit (amicum)* nach den arabischen Lexikographen ebenfalls auf das Wiederherstellen eines gebrochenen Arms zur”uckgeht.

N”aher l”age die Bedeutung “auff”ullen, nachf”ullen, ersetzen”, die bei Dozy belegt ist; doch ist auch das eine abgeleitete Bedeutung, und es liegt kein Grund vor, die von sprach- und sachkundigen Arabern selbst gegebene Deutung des Wortes *al-gabr* durch eine andere zu ersetzen.

F”ur die weitere Geschichte des Wortes “Algebra” ist es von Interesse, daß es schon bei den Iḫān a-ṣafā’, also um das Jahr 1000, ohne den Begriff *muqābala* zur Bildung des

With those authors—it is a matter of the arithmetic of Takī ad-Dīn al-Ḥanbalī, of whom according to H. Suter it is only certain that he lived before 1410, and of a treatise of Ibn al-Hā’im, who lived 1355–1412—there is no question at all of a procedure for the solution of equations. The two operations of addition and multiplication are set in contrast to the two operations of subtraction and division, and the former, since their execution leads to a larger number, are designated as *gabr*, here thus presumably “filling up” or “augmentation,” the latter, since a smaller number results, as *haḍ*, i.e. “diminution” or “reduction.”

It is *gabr* when one says: by how much must you increase 10 to obtain 17, or by what must you multiply 10 to obtain 17; it is *haḍ* when one says: by how much must you divide 10, or by how much must you decrease 10 to obtain 7.

When Carra de Vaux renders these summaries, unknown in older literature, with modern-conceived sentences: “It is therefore clear that al-djebr is the operation expressed by the two equations $a+x = b$, $a \cdot x = b$ and that al-haḍ is the operation expressed by these two: $a - x = b$, $a : x = b$,” and draws from this the conclusion, “as these four operations are the simplest possible in algebra, one must believe that such is indeed the primitive sense of the two words,” he not only confuses entirely different things but stands the historical development on its head, in that he further claims: “The word djebr, of which we have just indicated the primitive sense, then served to designate an indeterminate number of diverse questions, among which six are fundamental, according to Al-Khwārizmī...; the word haḍ, having become improper because of the complication of the problems, disappeared before the word mukābala.” The six normal forms of equations, to which Carra de Vaux alludes, and the alleged disappearance of the word *muqābala* have absolutely nothing to do with the opposites *gabr* and *haḍ*.

Rodet’s attempt to interpret the word *gabara* in the sense of “to enrich” (J. As., 7th ser., vol. XI, 1878, p. 38) will hardly find acceptance, especially since the meaning *post paupertatem ditavit (amicum)* drawn from Freytag likewise goes back, according to the Arabic lexicographers, to the restoration of a broken arm.

Closer would lie the meaning “to fill up, refill, replace,” which is attested in Dozy; but this too is a derived meaning, and there is no reason to replace the explanation of the word *al-gabr* given by Arabs themselves, knowledgeable in language and subject matter, with another.

For the further history of the word “Algebra” it is of interest that it was already used by the Ikhwān al-ṣafā’, around the year 1000, without the concept *muqābala* to form the

Wortes “Algebraiker” gebraucht wird; denn die Abhandlung ”uber die Zahl, die erste der 56 philosophischen Abhandlungen der Br”uder, enth”alt (ed. Bombay I, S. 37) einen Abschnitt فيما يحتاج اليه الجبريون والهندسة من الاسماء ومعانيها *”uber die Multiplikation und die Wurzel(n) und die Kuben und was die Algebraiker und die Geometer anwenden von den Ausdr”ucken und ihren Bedeutungen.*

Sollte der Abschnitt aber eine j”ungere Interpolation sein — ich kann meine Bedenken allerdings nicht mit zwingenden Gr”unden verfechten —, so ist der rein technische Gebrauch des Wortes jedenfalls sp”atestens bei ’Omar al-Hayyām (gest. 1123) sichergestellt. Denn nachdem er in der Einleitung die *ṣ’a al-gabr wa-l-muqābala*, die “Kunst” oder Technik der Erg”anzung und Ausgleichung definiert hat (ed. Woepcke, S. 4), spricht er weiterhin schlechtweg von den “L”osungen der Algebra”, *iṭibrāgāt al-gabr*, und von den B”uchern und den Gepflogenheiten der Algebraiker (S. 4, 5, 7).

Endlich ist einer Erweiterung des Begriffs *al-gabr* zu gedenken, die sich bereits im Rechenbuch des al-Karăgī (gest. um 1030) vorfindet. Sie besteht darin, daß al-Karăgī auch die Wegschaffung von Br”uchen, also das, was Muhammad b. M”usā als *ikmāl* bezeichnet, unter den Begriff *gabr* subsumiert (Cod. Gotha 1474, Bl. 54a; Hochheim, *Al Kāfī fī* l-*Hisāb*, III. Teil, S. 10):

> Wenn sich auf einer der beiden Seiten der Gleichung eine “Ausnahme” *istiṭnā’*, d. h. eine subtraktive Gr”oße befindet, muß man auf dieser Seite die gleiche Gr”oße hinzuf”ugen, damit das negative Glied verschwindet, und auf der andern Seite muß man ebensoviel hinzuf”ugen, damit die Gleichheit erhalten bleibt — dies ist Erg”anzung *al-gabr*. Aber sie findet auch noch in anderer Weise statt. Wenn n”amlich eine der beiden Seiten durch eine Zahl (*mikdār*, Betrag) dividiert ist, so wird der Ausdruck der Division unter Wahrung der Gleichheit dadurch zum Verschwinden gebracht, daß man alles, was man hat, mit dieser Zahl multipliziert. Dies alles geschieht, um das Unbekannte dem Bereich des Bekannten zu n”ahern und seinem eigenen Bereich fernzur”ucken. Alle diese Operationen geh”oren zur Erg”anzung, bis die Aufgabe in den Bereich der Ausgleichung *muqābala* gelangt, d. h. zur Beseitigung der (der Unbekannten) beigesellten Gr”oßen.

Man erkennt aus dem Wortlaut deutlich die urspr”ungliche Anwendung und die sekund”are Erweiterung des Begriffs. Wenn daher Ibn Hald”un (1332–1406) in seiner *Muqaddima* bei der Erkl”arung von *al-gabr wa-l-muqābala* (ed. Beirut

word “algebraist”; for the treatise on number, the first of the 56 philosophical treatises of the Brethren, contains (ed. Bombay I, p. 37) a section فيما يحتاج اليه الجبريون والهندسة من الاسماء ومعانيها “on the multiplication and the root(s) and the cubes and what the algebraists and the geometers apply of expressions and their meanings.”

Should the section, however, be a younger interpolation—I cannot indeed defend my doubts with compelling reasons—then the purely technical use of the word is in any case secured no later than in ʿUmar al-Khayyām (d. 1123). For after he has defined in the introduction the *ṣʿa al-gabr wa-l-muqābala*, the “art” or technique of completion and balancing (ed. Woepcke, p. 4), he subsequently speaks simply of the “solutions of algebra,” *ištibrāgāt al-gabr*, and of the books and practices of the algebraists (pp. 4, 5, 7).

Finally, an extension of the concept *al-gabr* is to be noted, which is already found in the arithmetic book of al-Karajī (d. ca. 1030). It consists in the fact that al-Karajī also subsumes under the concept *gabr* the elimination of fractions, that is, what Muḥammad b. Mūsā designates as *ikmāl* (Cod. Gotha 1474, fol. 54a; Hochheim, *Al-Kāfī fī l-ḥisāb*, Part III, p. 10):

> If on one of the two sides of the equation there is an “exception” *istiṭnā’*, that is, a subtractive quantity, one must add the same quantity on this side so that the negative term disappears, and on the other side one must add just as much so that equality is preserved—this is completion *al-gabr*. But it also takes place in another way. Namely, if one of the two sides is divided by a number (*mikdār*, amount), the expression of the division is made to disappear while preserving equality by multiplying everything one has by this number. All this is done to bring the unknown closer to the domain of the known and to push it away from its own domain. All these operations belong to completion, until the problem reaches the domain of balancing *muqābala*, that is, the elimination of the quantities accompanying (the unknown).

From the wording one clearly recognizes the original application and the secondary extension of the concept. Therefore when Ibn Khaldūn (1332–1406) in his *Muqaddima* in the explanation of *al-gabr wa-l-muqābala* (ed. Beirut 1886, p. 422) writes the

1886, S. 422) den Satz schreibt ويعملون فيها ما فيها من الكسور حتى تتم “und sie richten ein, was darin von Gebrochenem ist, so daß es ganz wird”, so hat er die Hauptsache ”ubersehen und sich vom n”achstliegenden Wortsinn, also dem “Einrichten” eines gebrochenen Glieds, zu einer mangelhaften Definition verleiten lassen.

sentence ويعملون فيها ما فيها من الكسور حتى تتم “and they arrange what is in it of fractions so that it becomes whole,” he has overlooked the main point and allowed himself to be led by the nearest verbal sense, namely the “setting” of a broken term, to an inadequate definition.

2 II. Das *Liber augmenti et diminutionis* / The *Liber augmenti et diminutionis*

Es w”are nicht notwendig gewesen, so lange bei dem Titel der Algebra zu verweilen, wenn nicht von der Deutung der Termini das Verst”andnis der ganzen Entwicklung der Mathematik abhinge, und wenn nicht schon in den Titeln der mathematischen Schriften ein St”uck Geschichte enthalten w”are.

Ich denke hierbei nat”urlich auch an das von Libri 1838 im ersten Band seiner *Histoire des Sciences Mathématiques en Italie* S. 304 ff. ver”offentlichte *Liber augmenti et diminutionis* und die Vermutungen, die ”uber den Inhalt einer Schrift des Muhammad b. M”usā ge”außert worden sind, von der uns nur der Titel *kitāb al-0̆357am’ wa-l-tafrīq* ”uberliefert ist (Cantor I^1 S. 627, I^2 S. 687, I^3 S. 730).

Hier liegen Schwierigkeiten, um deren Kl”arung und Beseitigung man seit Jahrzehnten bem”uht ist. Woepcke scheint der erste gewesen zu sein, der den arabischen Titel *al-0̆357am’ wa-l-tafrīq* dem ”uber *augmenti et diminutionis* gleichsetzte (J. As., 6. S., Bd. I, 1863, S. 514): “J’ajouterai que je trouve dans le Fihrist la mention de deux traités “de l’augmentation et de la diminution” c’est à dire de la regle des deux fausses positions, par Send Ben Ali et par Sinān Ibn Alfath, précisément les mêmes qui avaient écrit aussi … des traités de calcul indien …”

Es ist jedoch h”ochst zweifelhaft, ob man diese beiden Titel miteinander identifizieren darf, denn “Vermehrung” heißt arabisch nicht *0̆357am’*, und “Verminderung” heißt nicht *tafrīq*. *Liber augmenti et diminutionis* w”urde arabisch *kitāb al-ziyāda wa-l-nuqsān* heißen, w”ahrend ein *kitāb al-0̆357am’ wa-l-tafrīq* etwa mit *liber de aggregatione et dispersione* zu ”ubersetzen w”are.

Das Verfahren der Vermehrung und Verminderung ist bekannter unter dem Namen der Regel der “zwei Fehler” (*reg. elchatayn* = الخطأين) oder des doppelten falschen Ansatzes, auch als “Regel der zwei Wagschalen”. Cantor hat in seinen *Vorlesungen* (I^3, S. 731) das erste Beispiel aus dem *Liber augmenti et diminutionis*, worin der Sinn der Bezeichnung erl”autert wird, in deutscher ”Ubersetzung mitgeteilt; den Wortlaut des arabischen Textes kann man leichter nach der alten lateinischen ”Ubersetzung verfolgen.

In einem sp”ateren Kapitel dieser ”Ubersetzung (Libri a. a. O., S. 324) wird die Methode ausdr”ucklich *Capitulum numerationis ejus secundum augmentum et diminutionem* genannt; in der ”Uberschrift und am Anfang des Buches heißt sie auch *numeratio divinationis*, also “Rechenverfahren des Erratens”, was eine freie Wiedergabe der indischen Bezeichnung “Verfahren mit der angenommenen Zahl” zu sein scheint.

Die stets wiederkehrenden Wendungen *errasti per … addita* oder *cum … additis*, *errasti per … diminuta* oder *cum … diminutis*

It would not have been necessary to dwell so long on the title of the Algebra if the understanding of the entire development of mathematics did not depend on the interpretation of the terms, and if a piece of history were not already contained in the titles of mathematical writings.

I naturally think here also of the *Liber augmenti et diminutionis* published by Libri in 1838 in the first volume of his *Histoire des Sciences Mathématiques en Italie*, pp. 304 ff., and of the conjectures that have been expressed about the content of a treatise of Muḥammad b. Mūsā of which only the title *kitāb al-jam‘ wa-l-tafrīq* has been transmitted to us (Cantor I^1 p. 627, I^2 p. 687, I^3 p. 730).

Here lie difficulties whose clarification and removal have been attempted for decades. Woepcke seems to have been the first to equate the Arabic title *al-jam‘ wa-l-tafrīq* with that of *augmenti et diminutionis* (J. As., 6th ser., vol. I, 1863, p. 514): “I will add that I find in the *Fihrist* the mention of two treatises ‘on augmentation and diminution,’ that is, the rule of double false position, by Sanad b. ‘Alī and by Sinān b. al-Fāṭh, precisely the same ones who had also written … treatises on Indian calculation …”

It is, however, highly doubtful whether one may identify these two titles with each other, for “augmentation” is not *jam‘* in Arabic, and “diminution” is not *tafrīq*. *Liber augmenti et diminutionis* would be called in Arabic *kitāb al-ziyāda wa-l-nuqsān*, while a *kitāb al-jam‘ wa-l-tafrīq* would be translated approximately as *liber de aggregatione et dispersione*.

The method of augmentation and diminution is better known under the name of the rule of “two errors” (*reg. elchatayn* = الخطأين) or double false position, also as the “rule of the two scales.” Cantor in his *Vorlesungen* (I^3, p. 731) communicated the first example from the *Liber augmenti et diminutionis*, in which the sense of the designation is explained, in German translation; the wording of the Arabic text can more easily be followed through the old Latin translation.

In a later chapter of this translation (Libri, op. cit., p. 324) the method is explicitly called *Capitulum numerationis ejus secundum augmentum et diminutionem*; in the heading and at the beginning of the book it is also called *numeratio divinationis*, that is, “calculation procedure of guessing,” which seems to be a free rendering of the Indian designation “method with the assumed number.”

The constantly recurring phrases *errasti per … addita* or *cum … additis*, *errasti per … diminuta* or *cum … diminutis* entsprechen arabischem زدت bzw. نقصت und k”onnen nicht durch *ziyādat* oder *nuqsān* wiedergegeben werden.

Voll best”atigt werden diese R”uck”ubersetzungen und kritischen Bedenken durch Behā ad-dīns *Essenz der Rechenkunst* (ed. Nesselmann, Text S. 26), in der sich bei der Erl”auterung der Methode der zwei Fehler der Ausdruck findet: اذا اخطأ بزيادة او نقصان *wa-in aṭā’a bi-ziyādin au nuqsānin*, was Nesselmann etwas frei mit “wenn es aber abweicht nach einer oder der anderen Seite (um plus oder minus)” ”ubersetzt; lateinisch w”are es etwas wie *et si erravit per augmentum aut diminutionem* gewesen.

Auch die in letzter Zeit von Suter in ”Ubersetzung mitgeteilten arabischen Abhandlungen ”uber die “Regel der zwei Fehler” (Bibl. Math., 3. Folge, Bd. 8, 1907/8, S. 24 und Bd. 9, 1908/9, S. 111 ff.) geben keinen Anhalt daf”ur, daß das Verfahren mit den Worten *al-0̆357am’ wa-l-tafrīq* bezeichnet worden w”are.

Einen letzten Beweis f”ur die Verschiedenheit der beiden Titel kann man daraus ableiten, daß der Mathematiker Ab”u Kāmil 0̆15eūcā’ b. Aslam sowohl ein Buch *”uber die zwei Fehler*, als eines ”uber *0̆357am’* und *tafrīq* verfaßte.

Allerdings hat Suter (*Zeitschr. f. Math. u. Physik*, Suppl. z. 37. Jahrg., 1892, S. 70) mit Hilfe einer Konjektur — Einschiebung von يسمى ايضا *yusammā aḍān*, “es wird auch genannt” — beide zu einem einzigen zusammenschweißen wollen, aber das geschieht doch nur Woepckes ”Ubersetzung zuliebe und hat darum keine Beweiskraft.

Solange nicht aus dem Inhalt noch zu entdeckender Werke oder mit anderen zwingenden Gr”unden nachgewiesen ist, daß die beiden Titel gleichbedeutend sind, wird man das Gegenteil als richtig oder wahrscheinlich annehmen m”ussen.

Was kann aber nun in einem Werke ”uber *0̆357am’* und *tafrīq* enthalten gewesen sein? Daß es sich um irgendwelche Kapitel der Rechenkunst oder der Algebra handelt, ergibt sich zweifellos daraus, daß der Titel stets in Verbindung mit Schriften arithmetischen Inhalts erw”ahnt wird.

Nun ist die n”achste Bedeutung von *0̆357am’* die der “Vereinigung” oder “Summation”. In diesem Sinne wird *0̆357ama’a* nicht weniger als viermal in dem obenerw”ahnten ersten Beispiel angewandt: *Aggrega ergo …, postquam ergo aggregasti …, quod aggregatum est …, deinde aggrega …* Es wird also sprachlich ein Unterschied gemacht zwischen dem “Hinzuf”ugen” und “Vermehren” und der “Vereinigung” gegebener Zahlen zu einer Summe.

Nun ist in der Algebra des Muhammad b. M”usā der Addition und Subtraktion verschiedener Arten von Gr”oßen, die in den Gleichungen vorkommen, ein besonderes Kapitel gewidmet, das im Original mit *bāb al-0̆357am’ wa-l-nuqsān*, in der alten lateinischen ”Ubersetzung mit *Capitulum aggregationis et diminutionis*.

correspond to the Arabic زدت and نقصت respectively and cannot be rendered by *ziyāda* or *nuqsān.*

These back-translations and critical doubts are fully confirmed by Behā ad-Dīn’s *Essence of Arithmetic* (ed. Nesselmann, Text p. 26), in which the expression is found in the explanation of the method of two errors: اذا اخطأ بزيادة او نقصان *wa-in aṭā’a bi-ziyādin au nuqsānin*, which Nesselmann translates somewhat freely as “if it deviates to one or the other side (by plus or minus)”; in Latin it would have been something like *et si erravit per augmentum aut diminutionem.*

Also the Arabic treatises on the “rule of two errors” recently communicated by Suter in translation (Bibl. Math., 3rd ser., vol. 8, 1907/8, p. 24 and vol. 9, 1908/9, pp. 111 ff.) give no indication that the method was designated by the words *al-jam‘ wa-l-tafrīq.*

A final proof of the difference between the two titles can be derived from the fact that the mathematician Abü Kāmil Shuja‘ b. Aslam composed both a book *on the two errors* and one on *jam‘* and *tafrīq.*

To be sure, Suter (*Zeitschrift für Mathematik und Physik*, Supplement to vol. 37, 1892, p. 70) wished to weld both together into a single one by means of a conjecture—the insertion of يسمى ايضا *yusammā aḍān*, “it is also called”—but this is done only for Woepcke’s translation and therefore has no probative force.

As long as it has not been demonstrated from the content of works yet to be discovered, or by other compelling reasons, that the two titles are synonymous, one must assume the opposite as correct or probable.

But what could have been contained in a work on *jam‘* and *tafrīq*? That it concerns some chapters of arithmetic or algebra follows doubtless from the fact that the title is always mentioned in connection with writings of arithmetic content.

Now the nearest meaning of *jam‘* is “union” or “summation.” In this sense *jama‘a* is applied no fewer than four times in the aforementioned first example: *Aggrega ergo …, postquam ergo aggregasti …, quod aggregatum est …, deinde aggrega …* Thus a linguistic distinction is made between “adding” and “increasing” and the “union” of given numbers into a sum.

Now in the Algebra of Muḥammad b. Mūsā a special chapter is devoted to the addition and subtraction of various kinds of quantities that occur in equations, which in the original is headed *bāb al-jam‘ wa-l-nuqsān*, and in the old Latin translation with *Capitulum aggregationis et diminutionis.*

”uberschrieben ist.

Enestr”om vermutet daher (Bibl. Math., 3. Folge, Bd. 8, 1907/8, S. 184) in der Schrift oder den Schriften, die den Titel *al-0̆357am’ wa-l-tafrīq* f”uhren, Erweiterungen des *Capitulum aggregationis et diminutionis.* Die M”oglichkeit ist nicht von der Hand zu weisen, daß es sich um solche Additionen und Subtraktionen handelt.

Unsicher bliebe die Vermutung aber auch dann, wenn der Nachweis gef”uhrt w”urde, daß *tafrīq* und *nuqsān* schon in ”alterer Zeit gleichbedeutend f”ur einander gebraucht werden. Denn wir haben keinerlei Anhaltspunkte daf”ur, daß dieses Kapitel von irgendwem, geschweige schon von Muhammad b. M”usā in erweiterter Form außerhalb des Rahmens der Algebra behandelt worden w”are.

Die alten Lexika der Araber wissen nichts von *tafrīq* im Sinne von “Wegnahme” oder Subtraktion. Der zweite Stamm des Verbums *faraka*, von dem das Verbalnomen *tafrīq* gebildet ist, hat stets nur die Bedeutung “in St”ucke zerlegen, trennen, zerstreuen, in Unordnung bringen”, niemals die von “wegnehmen” oder “vermindern”.

Die Verbindung von *0̆357am’* und *tafrīq* wird also eher den Gegensatz von Sammlung — Zerstreuung oder von Vereinigung — Trennung ausdr”ucken. Einen solchen Gegensatz finde ich auf dem Gebiet der Algebra nur in dem Kapitel von der Klammerrechnung.

Auch diese ist von Muhammad b. M”usā in seinen Elementen behandelt, wenn er zeigt, wie Summen und Differenzen zu multiplizieren sind, und es w”are m”oglich, daß man daraus ein besonderes Kapitel gemacht hat. Die Ausdr”ucke *dissolutio* und *compositio* allerdings, die in der lateinischen ”Ubersetzung des Euklidkommentars von Annairizī auftreten (M. Curtze, *Anaritii Comm. ad Euch.* S. 89; Cantor I^3 S. 387) und als Klammeraufl”osung und Absonderung in unserem Sinne gedeutet werden k”onnten, lauten im arabischen Original *taḥlīl al-taḥlīl* und *taḥlīl al-tarkīb*, d. i. meth-odos analutikī und meth-odos suntithetikī.

Da jedoch in dem Zahlenbeispiel, das Annairizī zur Stelle gibt, Summation mit *mamū0̆11f* und Summe mit *mamū0̆11f* = *das Vereinigte* wiedergegeben wird, so k”onnte als Ausdruck der “Zerlegung” neben dem *qasama* des Textes, das sich auf die Teilung in zwei Teile bezieht, auch *tafrīq* (f”ur eine Zerlegung in zahlreiche Teile) in Frage kommen. Rein sprachlich besteht f”ur diesen Erkl”arungsversuch jedenfalls keine Schwierigkeit; sachlich spricht aber wie bei der Enestr”omschen Hypothese der Umstand gegen ihn, daß es nicht m”oglich ist, eine selbst”andige Behandlung der Klammerrechnungen in der mathematischen Literatur der Araber nachzuweisen.

Eneström therefore conjectures (Bibl. Math., 3rd ser., vol. 8, 1907/8, p. 184) that in the writing or writings bearing the title *al-jamʿ wa-l-tafrīq* there are extensions of the *Capitulum aggregationis et diminutionis.* The possibility cannot be dismissed that it concerns such additions and subtractions.

The conjecture would remain uncertain, however, even if proof were furnished that *tafrīq* and *nuqsān* were already used synonymously in earlier times. For we have no indication whatsoever that this chapter was treated by anyone, let alone by Muḥammad b. Mūsā, in extended form outside the framework of the Algebra.

The old lexica of the Arabs know nothing of *tafrīq* in the sense of “removal” or subtraction. The second stem of the verb *faraka*, from which the verbal noun *tafrīq* is formed, has always only the meaning “to break into pieces, separate, disperse, bring into disorder,” never that of “to remove” or “to diminish.”

The combination of *jamʿ* and *tafrīq* will thus more likely express the contrast of collection—dispersion or of union—separation. Such a contrast I find in the domain of algebra only in the chapter on bracket calculation [parenthetical operations].

This too is treated by Muḥammad b. Mūsā in his Elements, when he shows how sums and differences are to be multiplied, and it would be possible that a special chapter was made from it. The expressions *dissolutio* and *compositio*, however, which appear in the Latin translation of Annarīzī’s commentary on Euclid (M. Curtze, *Anaritii Comm. ad Euch.*, p. 89; Cantor I^3, p. 387) and could be interpreted as bracket resolution and separation in our sense, are in the Arabic original *tah.līl al-tah.līl* and *tah.līl al-tarkīb*, i.e. meth-odos analutikī and meth-odos suntithetikī.

Since, however, in the numerical example that Annarīzī gives at this place, summation is rendered by *mamūʿ* and sum by *mamūʿ* = “the united,” so as an expression of “decomposition” alongside *qasama* of the text, which refers to division into two parts, *tafrīq* (for a decomposition into numerous parts) could also come into question. Linguistically, there is in any case no difficulty for this explanatory attempt; but substantively, as with Eneström’s hypothesis, the circumstance speaks against it that it is not possible to demonstrate an independent treatment of bracket operations in the mathematical literature of the Arabs.

3 III. Die Regula Sermonis / The Regula Sermonis

Ich schließe an diese Er”orterungen noch eine Bemerkung ”uber das in dem *Liber augmenti et diminutionis* gelehrte Umkehrungsverfahren, das durch Cantor (I^3, S. 732) “unter dem sonderbaren Namen der Wortrechnung, *regula sermonis*” in die Geschichte der Mathematik eingef”uhrt worden ist.

Die Textstelle, auf die es ankommt, lautet (Libri S. 312): “*Quaedam vero harum quaestionum investigantur secundum regulam quae vocatur infusa. Et ipsa est regula Job filii Salomonis divisoris… Ex eis vero est eius sermo qui dixit … Incipe igitur cum quaestione ab eius postremitate, et dic …*” Und weiterhin S. 313: “*Et modus regule sermonis eius est …*”

Zun”achst ist klar, daß man das sinnlose *infusa* durch *inversa* zu ersetzen hat. Von h”ochstem Interesse ist weiter, daß das Verfahren auch die Regel des Erbteilers oder Testamentsvollstreckers Aiyy”ub b. Sulaimān genannt wird — denn so ist das Wort *divisor* nach einer Randnote der Handschrift “*Dicitur divisor qui res a defuncto relicta(s) partitur, [et] hoc apud Arabes*” zweifellos zu ”ubersetzen.

Wir erblicken darin eine Best”atigung der Vermutung, daß die Rechenkunst und die Kunst der Gleichungsaufl”osung nicht nur bei den Astronomen, sondern ganz besonders auch bei den Notaren und Rechtsgelehrten eine Pflegest”atte gefunden hat.

W”aren uns von H”aggi Hāa (Bd. IV, S. 398) mehr als die drei Worte فرائض ايوب البصري *farā’iḍ Aiyy”ub al-Baṣrī* ”uberliefert, so k”onnte die Annahme, daß dieser Aiyy”ub der *Job filius Salomonis* ist, auf ihre Berechtigung gepr”uft werden.

Suter sucht (Bibl. Math., 3. Folge, Bd. 3, 1903, S. 350) den *Job* wie den Verfasser des *Liber augmenti et diminutionis* unter den spanischen Gelehrten des 10. und 11. Jahrhunderts. Nach ihm k”onnen zwei Aiyy”ub b. Sulaimān in Betracht kommen, die sich mit Erbteilungen befaßten; die gr”oßere Wahrscheinlichkeit spreche f”ur Ab”u ṢāliḥĀiyy”ub b. Sulaimān aus Cordova, der 914 starb.

Eine Entscheidung zwischen beiden Vermutungen wird erst m”oglich sein, wenn der vollst”andige Name des Baṣreners oder der Autor *Abraham* selbst feststeht, der das Buch *compilavit et secundum librum qui Indorum dictus est composuit.*

Auch wenn sich, wie Suter sp”ater in einem Vortrag auf dem III. Internationalen Mathematikerkongreß zu Heidelberg 1904 (vgl. *Verhandlungen* S. 558) nachzuweisen versuchte, der ”Agypter Ab”u Kāmil 0̆15eūcā’ b. Aslam hinter dem Namen *Ibrahim* verbergen sollte — eine Vermutung, die mit guten Gr”unden gest”utzt wird und noch dadurch an Interesse gewinnt, daß es gerade dieser Autor ist, der im Fihrist als Verfasser eines *kitāb al-0̆357am’ wa-l-tafrīq* wie eines *kitāb al-ḥatā’ain*

I append to these discussions a remark on the inversion procedure taught in the *Liber augmenti et diminutionis*, which was introduced into the history of mathematics by Cantor (I^3, p. 732) “under the strange name of word-calculation, *regula sermonis*.”

The passage that matters reads (Libri, p. 312): “*Quaedam vero harum quaestionum investigantur secundum regulam quae vocatur infusa. Et ipsa est regula Job filii Salomonis divisoris… Ex eis vero est eius sermo qui dixit … Incipe igitur cum quaestione ab eius postremitate, et dic …*” And further on p. 313: “*Et modus regule sermonis eius est …*”

First of all, it is clear that the meaningless *infusa* must be replaced by *inversa*. Of the highest interest, furthermore, is that the procedure is also called the rule of the inheritance-divider or executor Ayyüb b. Sulaymān—for thus the word *divisor* is doubtless to be translated, according to a marginal note of the manuscript: “*Dicitur divisor qui res a defuncto relicta(s) partitur, [et] hoc apud Arabes*.”

We see in this a confirmation of the conjecture that arithmetic and the art of equation-solving found a home not only among astronomers, but especially also among notaries and legal scholars.

Had more than the three words فرائض ايوب البصري *farā’iḍ Ayyüb al-Baṣrī* been transmitted to us from Ḥājjī Ḥalīfa (vol. IV, p. 398), the assumption that this Ayyüb is the *Job filius Salomonis* could be tested for its validity.

Suter seeks (Bibl. Math., 3rd ser., vol. 3, 1903, p. 350) the *Job* like the author of the *Liber augmenti et diminutionis* among the Spanish scholars of the 10th and 11th centuries. According to him, two Ayyüb b. Sulaymān can come into consideration, who occupied themselves with inheritance divisions; the greater probability speaks for Abü Ṣālīḥ Ayyüb b. Sulaymān of Córdoba, who died in 914.

A decision between the two conjectures will only be possible when the complete name of the Basran [scholar] or the author *Abraham* himself is established, who *compilavit* [compiled] the book *et secundum librum qui Indorum dictus est composuit.*

Even if, as Suter later attempted to demonstrate in a lecture at the III. International Congress of Mathematicians in Heidelberg 1904 (cf. *Verhandlungen* p. 558), the Egyptian Abü Kāmil Shuja‘ b. Aslam should be concealed behind the name *Ibrahim*—a conjecture supported by good reasons and gaining in interest precisely because it is this author who in the *Fihrist* is designated as the author of both a *kitāb al-jam‘ wa-l-tafrīq* and a *kitāb al-kha.tā’ayn* (Flügel, *Kitāb al-Fihrist*, vol. I, p. 281)—the Basran bezeichnet wird (Fl”ugel, *Kitāb al-Fihrist*, Bd. I, S. 281) — w”are der Baṣrenser als erster arabischer Schriftsteller, der das indische Umkehrungsverfahren in die Technik der Gleichungen eingef”uhrt h”atte, in Erw”agung zu ziehen.

Wenn aber nun Cantor a. a. O. behauptet, daß dieses Verfahren bei den Arabern den sonderbaren Namen der Wortrechnung gef”uhrt habe, so zeigt ein Blick in den lateinischen Text und die R”uck”ubersetzung ins Arabische, daß diese Behauptung auf einem Mißverst”andnis beruht.

Es heißt im Text nicht *regula sermonis*, sondern *regula sermonis eius*, und das ist weiter nichts als die w”ortliche Wiedergabe des arabischen *qiyāsu qaulihī*, eines Ausdruckes, der nichts anderes besagt als “die Regel, die er angibt”, wie es schon vorher heißt *eius sermo qui dixit* d. i. *qaulu qāilin.*

Die “Wortrechnung” wird also k”unftig aus der Liste der mathematischen Termini verschwinden m”ussen.

would have to be considered as the first Arabic writer who introduced the Indian inversion procedure into the technique of equations.

But if Cantor, op. cit., claims that this procedure bore among the Arabs the strange name of word-calculation, a glance at the Latin text and the back-translation into Arabic shows that this claim rests on a misunderstanding.

The text does not say *regula sermonis* but *regula sermonis eius*, and this is nothing other than the literal rendering of the Arabic *qiyāsu qawlahī*, an expression that says nothing other than “the rule that he states,” as is said before: *eius sermo qui dixit*, i.e. *qawlu qā’ilin* [the statement of the speaker].

The “word-calculation” will therefore henceforth have to disappear from the list of mathematical terms.

4 IV. Inhalts”ubersicht / Overview of Contents

Wir kehren zur Algebra des Muhammad b. M”usā zur”uck und verschaffen uns durch Zusammenstellung der Kapitel”uberschriften eine erste ”Ubersicht des Inhalts. Es ist bekannt, daß der Text des Buches auf der einzigen Oxforder Handschrift CMXCVIII Hunt. 214 ruht, die im Jahre 743 d. H., also um 1342 vollendet wurde (Rosen, S. XIII).

Ihr Zustand — *it is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points* — ließe einen Vergleich mit den lateinischen Handschriften, die als ”Ubersetzungen oder Bearbeitungen des arabischen Werkes gelten, sehr w”unschbar erscheinen, zumal auch hier noch manches genauerer Feststellung bedarf.

Daß die von Libri im ersten Band seiner *Geschichte* S. 253 ff. ver”offentlichte ”Ubersetzung des Gerhard von Cremona nur die ersten 50 Seiten des Originals, also nur die Algebra im engeren Sinne ohne die Geometrie und Erbteilungsrechnung umfaßt, war z. B. bis vor kurzem nirgends in geschichtlichen Werken vermerkt. Libri selbst weist allerdings in einer ausf”uhrlichen Fußnote zu S. 121 und in der Vorbemerkung zur Note XII, S. 253 darauf hin, daß die lateinische ”Ubersetzung der drei Pariser Handschriften unvollst”andig ist.

Aber bei Cantor (I^3 S. 719, I^2 S. 675) heißt es einfach “eine alte lateinische ”Ubersetzung”, und auch in der dieser ”Ubersetzung gewidmeten Arbeit von Bj”ornbo (*Gerhard von Cremonas ”Ubersetzung von Alkwarizmis Algebra usw.*, Bibl. Math., 3. Folge, Bd. 6, 1905, S. 239) ist eine Erinnerung an diesen weder selbstverst”andlichen noch gleichg”ultigen Umstand unterblieben.

Es ist klar, daß durch Unterdr”uckung der Einleitung, die ausdr”ucklich auch auf den geometrischen und erbrechtlichen Teil des Werkes hinwies, der Anschein erweckt wird, als sei das ”ubersetzte Bruchst”uck das ganze Buch. Auffallenderweise reicht auch die um 1145 vollendete ”Ubersetzung des Robert Castrensis nicht weiter, wie aus den von Steinschneider (Bibl. Math., 3. Folge, Bd. 1, 1900, S. 273) gemachten Angaben hervorgeht, da sie mit den Worten abschließt, die nur den algebraischen Teil betreffen.

Erst L. C. Karpinski hat in seiner Abhandlung *Robert of Chester’s translation of the algebra of Al-Khowarizmi* (Bibl. Math., 3. Folge, Bd. 11, 1910/11, S. 128) auf das Fehlen der Einleitung und der geometrischen und erbrechtlichen Abschnitte in s”amtlichen alten ”Ubersetzungen wieder hingewiesen.

Folgen wir der Kapiteleinteilung des arabischen Textes, so ergibt sich folgender Aufbau des Ganzen:

(Einleitung, S. 1, 2.) Anrufung Gottes, Preis des Propheten, Schilderung der m”uhevollen Arbeit der Gelehrten, Bericht ”uber die Veranlassung zur Abfassung und ”uber den Zweck

We return to the Algebra of Muḥammad b. Mūsā and obtain a first overview of the content by assembling the chapter headings. It is known that the text of the book rests on the unique Oxford manuscript CMXCVIII Hunt. 214, completed in the year 743 A. H., that is, around 1342 (Rosen, p. XIII).

Its condition—*it is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points*—would make a comparison with the Latin manuscripts, regarded as translations or adaptations of the Arabic work, very desirable, especially since here too much still requires more precise determination.

That the translation of Gerard of Cremona published by Libri in the first volume of his *Histoire*, pp. 253 ff., comprises only the first 50 pages of the original, thus only the Algebra in the narrower sense without the geometry and inheritance calculation, was, for example, noted nowhere in historical works until recently. Libri himself points out in an extensive footnote to p. 121 and in the prefatory remark to Note XII, p. 253, that the Latin translation of the three Paris manuscripts is incomplete.

But in Cantor (I^3 p. 719, I^2 p. 675) it simply says “an old Latin translation,” and also in the work of Björnbo devoted to this translation (*Gerhard von Cremonas ”Ubersetzung von Alkwarizmis Algebra usw.*, Bibl. Math., 3rd ser., vol. 6, 1905, p. 239) no mention of this circumstance—neither self-evident nor indifferent—has been made.

It is clear that by suppressing the introduction, which explicitly referred also to the geometrical and inheritance-law parts of the work, the appearance is created that the translated fragment is the whole book. Remarkably, the translation of Robert of Chester completed around 1145 also does not extend further, as emerges from the information given by Steinschneider (Bibl. Math., 3rd ser., vol. 1, 1900, p. 273), since it concludes with the words that concern only the algebraic part.

Only L. C. Karpinski in his treatise *Robert of Chester’s translation of the algebra of Al-Khowarizmi* (Bibl. Math., 3rd ser., vol. 11, 1910/11, p. 128) again pointed out the absence of the introduction and of the geometrical and inheritance-law sections in all old translations.

If we follow the chapter division of the Arabic text, the following structure of the whole emerges:

(Introduction, pp. 1, 2.) Invocation of God, praise of the Prophet, description of the arduous work of scholars, report on the occasion of composition and on the purpose des Buches. [fehlt bei Libri]

[Erster Hauptteil — ohne ”Uberschrift; S. 3 bis 50.] Aufbau des Zahlensystems; Definition der in dem Rechenverfahren der Erg”anzung und Ausgleichung auftretenden Arten (dur”ub) von Zahlen; Aufz”ahlung der drei einfachen F”alle von Gleichungen; sodann Beispiele: “Was anlangt die Verm”ogen, die gleich sind den Wurzeln …” (Drei Beispiele.) “Was anlangt die Verm”ogen, die gleich sind der Zahl …” (Drei Beispiele.) “Was anlangt die Wurzeln, die gleich sind einer Zahl …” (Drei Beispiele.)

Kurze Zwischenbemerkung ”uber die drei zusammengesetzten F”alle der Gleichungen; sodann Beispiele: “Was anlangt die Verm”ogen und die Wurzeln, die gleich sind der Zahl …” (Drei Beispiele.) “Was anlangt die Verm”ogen und die Zahl, die gleich sind den Wurzeln …” (Ein Beispiel.) “Was anlangt die Wurzeln und die Zahl, die gleich sind dem Verm”ogen …” (Ein Beispiel.)

Eine Schlußbemerkung leitet ”uber zu den geometrischen Beweisen der Aufl”osungen: “Und ich habe f”ur jedes Kapitel von ihnen eine Zeichnung gezeichnet, durch die auf den Grund der Halbierung hingewiesen wird.” Drei Beispiele; “Was den Grund anlangt, daß ein Verm”ogen und zehn Wurzeln gleich sind neunundzwanzig Dirhem …” (Zwei verschiedene geometrische Beweise mit Zeichnungen.) “Was anlangt ein einziges Verm”ogen und zwanzig Dirhem, die gleich sind zehn seiner Wurzeln …” (Eine Zeichnung.) “Was anlangt drei Wurzeln und vier von der Zahl, die gleich sind einem Verm”ogen …” (Eine Zeichnung.)

Das Kapitel von der Multiplikation (باب الضرب *bāb al-darb*; S. 15 bis 19 oben). Das Kapitel von der Addition (Aggregation) und Subtraktion (باب الجمع والنقصان *bāb al-0357am’ wa-l-nuqsān*; S. 19, 20). Die Teilung (القسمة *al-qisma*; S. 20, 21).[6]

“Was anlangt den Grund f”ur die Wurzel von zweihundert ohne zehn, addiert zu (vereinigt mit) zwanzig ohne die Wurzel von zweihundert …” (Zeichnung.) “Was anlangt den Grund f”ur die Wurzel von zweihundert ohne zehn, subtrahiert von zwanzig ohne die Wurzel von zweihundert …” (Zeichnung.) “Was anlangt einhundert und ein Verm”ogen ohne zwanzig Wurzeln, dazu addiert (damit vereinigt) f”unfzig und zehn Wurzeln ohne zwei Verm”ogen …” (Ohne Zeichnung.)

Das Kapitel der sechs Fragen (S. 25 bis 29). Die erste von den sechs … Die zweite Frage … Die dritte Frage … Die vierte Frage … Die f”unfte Frage … Die sechste Frage …

Das Kapitel der vermischten Fragen. (”Uberschrift in besonderer Zeile; 33 mehr oder minder ausf”uhrlich behandelte Beispiele; S. 30 bis 48 oben.)

Das Kapitel der kaufm”annischen Gesch”afte (باب المعاملات

of the book. [missing in Libri]

[First main part—without heading; pp. 3 to 50.] Structure of the number system; definition of the kinds of numbers (dur”ub) occurring in the calculation procedure of completion and balancing; enumeration of the three simple cases of equations; then examples: “As for the possessions equal to the roots …” (Three examples.) “As for the possessions equal to the number …” (Three examples.) “As for the roots equal to a number …” (Three examples.)

Brief intermediate remark on the three composite cases of equations; then examples: “As for the possessions and the roots equal to the number …” (Three examples.) “As for the possessions and the number equal to the roots …” (One example.) “As for the roots and the number equal to the possession …” (One example.)

A concluding remark leads over to the geometrical proofs of the solutions: “And I have drawn for each chapter of them a figure, by which the ground of halving is indicated.” Three examples; “As for the ground that a possession and ten roots equal twenty-nine dirhams …” (Two different geometrical proofs with drawings.) “As for a single possession and twenty dirhams equal to ten of its roots …” (One drawing.) “As for three roots and four of the number equal to a possession …” (One drawing.)

The chapter on multiplication (باب الضرب *bāb al-darb*; pp. 15 to 19 above). The chapter on addition (aggregation) and subtraction (باب الجمع والنقصان *bāb al-jam‘ wa-l-nuqsān*; pp. 19, 20). Division (القسمة *al-qisma*; pp. 20, 21).[6]

“As for the ground for the root of two hundred less ten, added to (united with) twenty less the root of two hundred …” (Drawing.) “As for the ground for the root of two hundred less ten, subtracted from twenty less the root of two hundred …” (Drawing.) “As for one hundred and a possession without twenty roots, thereto added (therewith united) fifty and ten roots without two possessions …” (Without drawing.)

The chapter of the six questions (pp. 25 to 29). The first of the six … The second question … The third question … The fourth question … The fifth question … The sixth question …

The chapter of mixed questions. (Heading in a special line; 33 examples treated more or less in detail; pp. 30 to 48 above.)

The chapter of commercial transactions (باب المعاملات *bāb*

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[6] Rosen hat durch seine ”Uberschriften *On multiplication*, *On addition and subtraction*, *On division* den charakteristischen Befund im arabischen Text verwischt, daß bei der Teilung das Wort “Kapitel” fehlt.

[6] Rosen has obscured the characteristic finding in the Arabic text by his headings *On multiplication*, *On addition and subtraction*, *On division*, that in the case of division the word “chapter” is missing.

bāb al-muāmalāt; drei Beispiele. S. 48 bis 50).

[Zweiter Hauptteil — ohne besondere Hervorhebung:] Kapitel der Messung (S. 50 bis 64). Das Kapitel ist eine lose Aneinanderreihung von geometrischen Definitionen, Eigenschaften von Figuren und Formeln mit Hervorhebung durch Stichworte.

[Dritter Hauptteil — mit Hervorhebung der ”Uberschrift:] Das Buch der Testamente (S. 65–122). Ein Kapitel davon ”uber das Kapital und die Schuld (S. 65 bis 67; drei durchgerechnete Beispiele). Ein anderes Kapitel von den Testamenten (S. 67, 68; zwei erl”auterte Beispiele). Ein anderes Kapitel von den Testamenten (S. 68 bis 71; zwei Beispiele).

In einer anderen Art von Testamenten … (S. 71 bis 86. Dieser Anfang kehrt dreimal wieder, und zwar zweimal mit 4, einmal mit 8 Beispielen). Das Kapitel vom Testament mit dem Dirhem (S. 86 bis 92; vier Beispiele). Das Kapitel von der Vervollst”andigung (باب التكميلة *bāb al-takmīla*, S. 93 bis 98; sieben Beispiele).

Das Rechenverfahren bei Schicksalswechsel (حساب الدور *hisāb al-daur, computation of returns*).[7] Ein Kapitel davon ”uber die Verheiratung in der (letzten) Krankheit (S. 98 bis 101; vier Beispiele).

Das Kapitel von der Freilassung (von Sklaven) in der Krankheit (S. 101 bis 116; f”unfzehn Beispiele, darunter einige unter Berufung auf Ab”u Hanīfa, dessen Todesjahr 767 war). Das Kapitel von der Entsch”adigung bei Schicksalswechsel (باب الاكر بالدور *bāb al-akr bi-l-daur, return of the dowry*, S. 116 bis 120; sechs Beispiele, darunter wieder einige unter Berufung auf Ab”u Hanīfa).[8] Das Kapitel von der Vorausbezahlung in der Krankheit (السلام في المرض *al-salām fī l-marad, surrender in illness*, S. 121, 122; zwei Beispiele).[9]

Schon diese Inhalts”ubersicht kann zum Nachdenken ”uber die Gr”unde anregen, die den Verfasser veranlaßt haben, drei nach unserer Auffassung ganz heterogene Gegenst”ande unter dem Sammeltitel *kitāb al-gabr wa-l-muqābala* zu vereinigen. Es w”are auch leicht, zu zeigen, wie sich im 10. und 11. Jahrhundert eine Scheidung vollzieht, so daß die Algebra, das Gesch”aftsrechnung, die praktische Geometrie und die einzelnen Teile der Erbteilungspraxis in besonderen Schriften behandelt werden.

Da uns aber vorerst die Quellen und Vorbilder von Muhammad b. M”usās Algebra und die Frage nach dem Grade seiner Selbst”andigkeit besch”aftigen sollen, wird eine kurze Darstellung der bisher ”uber diese Dinge ge”außerten Urteile dem neuen

al-muʿāmalāt; three examples. Pp. 48 to 50).

[Second main part—without special emphasis:] Chapter of measurement (pp. 50 to 64). The chapter is a loose sequence of geometrical definitions, properties of figures, and formulas with emphasis by keywords.

[Third main part—with emphasis on the heading:] The Book of Testaments (pp. 65–122). A chapter thereof on capital and debt (pp. 65 to 67; three worked-through examples). Another chapter on testaments (pp. 67, 68; two explained examples). Another chapter on testaments (pp. 68 to 71; two examples).

In another kind of testaments … (pp. 71 to 86. This beginning recurs three times, namely twice with 4, once with 8 examples.) The chapter on the testament with the dirham (pp. 86 to 92; four examples). The chapter on completion (باب التكميلة *bāb al-takmīla*, pp. 93 to 98; seven examples).

The calculation procedure in case of change of fortune (حساب الدور *ḥisāb al-dawr, computation of returns*).[7] A chapter thereof on marriage during (terminal) illness (pp. 98 to 101; four examples).

The chapter on manumission (of slaves) during illness (pp. 101 to 116; fifteen examples, among them some with reference to Abü Ḥanīfa, whose death year was 767). The chapter on compensation for change of fortune (باب الاكر بالدور *bāb al-akr bi-l-dawr, return of the dowry*, pp. 116 to 120; six examples, among them again some with reference to Abü Ḥanīfa).[8] The chapter on prepayment during illness (السلام في المرض *al-salām fī l-maraḍ, surrender in illness*, pp. 121, 122; two examples).[9]

Even this overview of contents can prompt reflection on the reasons that induced the author to unite three subjects that by our view are entirely heterogeneous under the collective title *kitāb al-gabr wa-l-muqābala*. It would also be easy to show how in the 10th and 11th centuries a separation takes place, so that algebra, business calculation, practical geometry, and the individual parts of inheritance practice are treated in separate writings.

But since for now the sources and models of Muḥammad b. Mūsā's Algebra and the question of the degree of his independence are to occupy us, a brief presentation of the opinions hitherto expressed about these matters must be prefaced to the new

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[7] Die ”Ubersetzung von Rosen ist falsch. Das Wort *daur* bezeichnet die wechselseitige Vertauschung (A gegen B, B gegen A), in unserm Falle Vertauschung von Erblasser und Erben, wenn der Erbe vorher stirbt.

[8] Rosens ”Ubersetzung ist falsch. Es handelt sich um Entsch”adigung f”ur die Verminderung des Werts einer Sklavin infolge Beiwohnung.

[9] Das Wort *salām* bedeutet gew”ohnlich Vorausbezahlung vor Empfang der Ware; es scheint sich aber bei den beiden Beispielen um Vorauslieferung oder Verpf”andung von Waren zu handeln.

[7] Rosen's translation is wrong. The word *dawr* denotes mutual exchange (A for B, B for A), in our case exchange of testator and heirs, when the heir dies first.

[8] Rosen's translation is wrong. It concerns compensation for the diminution in value of a female slave as a result of sexual intercourse.

[9] The word *salām* usually means prepayment before receipt of goods; but in the two examples it seems to concern advance delivery or pawning of goods.

Versuch, zu einer Entscheidung zu gelangen, vorausgeschickt werden m”ussen.

attempt to reach a decision.

5 V. Zur Geschichte der arabischen Zahlbezeichnungen / On the History of Arabic Numerical Notation

Es kann nicht oft und nachdr”ucklich genug gesagt werden, daß die Araber, die die persischen und r”omischen Provinzen ”uberfluteten, weder Rechtswissenschaft noch Staatsverwaltung fertig mitbrachten, sondern gezwungen waren, die Verwaltungsmethoden und Rechtsformen der eroberten L”ander im wesentlichen unver”andert zu ”ubernehmen. Daß es ihnen mit erstaunlicher Schnelligkeit gelang, sich in die gr”oßeren Verh”altnisse hineinzufinden und nicht nur die staatlichen Einrichtungen, sondern auch alle andern Fr”uchte einer alten, ausgereiften Kultur sich zu eigen zu machen, ist bekannt.

Das w”are aber gewiß unm”oglich gewesen, wenn der geistige Abstand zwischen dem Eroberervolk und den zeitgen”ossischen Persern, Griechen und ”Agyptern so groß gewesen w”are, wie man bis in die neueste Zeit anzunehmen pflegte. Insbesondere darf man sich die st”adtischen Araber, die Tr”ager der geistigen und politischen Bewegung, nicht als halbe Wilde vorstellen, die vor dem Auftreten Muhammeds jedem Kultureinflusse von seiten der Nachbarv”olker unzug”anglich gewesen w”aren (Cantor I^3, S. 693) oder gar in der Zeit, zu der sie f”ur die Geschichte der Mathematik wichtig werden, kaum h”atten schreiben k”onnen.

Woepcke durfte sich 1863 mit seinem Satze “lorsque les Arabes sortirent du désert … ils possédaient à peine l’écriture” noch auf Silvestre de Sacys *Mémoire* vom Jahre 1831 berufen, heute aber k”onnen Silvestre de Sacys *Grammaire arabe* von 1810 und die Abhandlungen von Gesenius in der Enzyklop”adie von Ersch und Gruber aus dem Jahre 1820 nicht mehr als Quellen f”ur unsere Kenntnis des arabischen Schriftwesens benutzt werden (Cantor I^3, S. 707).

Authentische Urkunden und Schriftdenkm”aler, die bis in die erste Zeit des Islam zur”uckreichen, enth”ullen die ”uberraschende Tatsache, “daß die arabische Schrift schon damals eine Form besaß, die von der gew”ohnlichen, sp”ater Neskhi genannten Schrift kaum verschieden ist, desto mehr aber von der eckig steifen “kufischen” Schrift, die man f”ur die Mutter der Kursivschrift gehalten hatte.”

Es steht jetzt fest, daß sich die arabische Schrift im 4. und 5. nachchristlichen Jahrhundert aus der nabat”aischen Kursive entwickelt hat; aus ihrem Alphabet entstand durch Hinzuf”ugen der arabischen Laute *0639, 063a, 062e, 0636, 0638* das alte arabische Alphabet, dessen Buchstabenfolge in den diesen Zeichen erteilten Zahlwerten 0639 = 500, 063a = 600, 062e = 700, 0636 = 800, 0638 = 900, 1000 erhalten geblieben ist.

Von ihrem Ursprungsort aus, der vielleicht in Petra, vielleicht

It cannot be said often or emphatically enough that the Arabs who flooded the Persian and Roman provinces brought neither jurisprudence nor state administration ready-made with them, but were compelled to take over the administrative methods and legal forms of the conquered lands essentially unchanged. That they succeeded with astonishing rapidity in finding their way into the larger circumstances and making their own not only the state institutions but also all other fruits of an ancient, mature culture is well known.

But this would certainly have been impossible if the intellectual distance between the conquering people and the contemporary Persians, Greeks, and Egyptians had been as great as was customarily assumed until the most recent times. In particular, one must not imagine the urban Arabs, the bearers of the intellectual and political movement, as semi-savages who before the appearance of Muhammad were inaccessible to any cultural influence from neighboring peoples (Cantor I^3, p. 693), or who in the period when they become important for the history of mathematics could scarcely write.

In 1863 Woepcke could still appeal to Silvestre de Sacy’s *Mémoire* of 1831 with his sentence “when the Arabs left the desert … they scarcely possessed writing,” but today Silvestre de Sacy’s *Grammaire arabe* of 1810 and the treatises of Gesenius in the encyclopedia of Ersch and Gruber from the year 1820 can no longer be used as sources for our knowledge of the Arabic writing system (Cantor I^3, p. 707).

Authentic documents and written monuments that reach back to the earliest period of Islam reveal the surprising fact “that the Arabic script already possessed at that time a form scarcely different from the ordinary script later called naskhī, and much more different from the angular, rigid ‘Kufic’ script that was held to be the mother of cursive writing.”

It is now established that the Arabic script developed in the 4th and 5th centuries CE from the Nabataean cursive; from its alphabet, by the addition of the Arabic sounds *‘ayn, ghayn, khā, ḍād, ẓā* the old Arabic alphabet arose, whose letter sequence in the numerical values assigned to these characters—‘ayn = 500, ghayn = 600, khā = 700, ḍād = 800, ẓā = 900, 1000—has been preserved.

From its place of origin, to be sought perhaps in Petra, in dem damals aufbl”uhenden Ghassānidenreiche zu suchen ist, hat sich die Schrift mit dem Handelsverkehr nach Norden und S”uden verbreitet. Sie war im 6. Jahrhundert wohl schon im ganzen nordarabischen Sprachgebiet bekannt und d”urfte besonders in den beiden St”adten, die den Ausgangspunkt der religi”os-nationalen Bewegung bildeten, in Mekka und Medina, in Gebrauch gewesen sein.

Ganz unsch”atzbare Quellen f”ur die Geschichte der ”Ubertragung griechisch-”agyptischer Verwaltungs- und Rechenpraxis in die Sprache und Schrift der arabischen Herren stehen uns seit l”angerer Zeit in den Papyri von el-Faij”um und andern Fundorten (Sammlung Erzherzog Rainer), seit einigen Jahren in denen des Aphroditofundes zu Gebote. Eine auch dem Nichtarabisten zug”angliche ”Ubersicht ”uber die Sch”atze der erstgenannten Sammlung enth”alt der *F”uhrer durch die Ausstellung* in seiner von J. von Karabacek bearbeiteten *Arabischen Abtheilung* (Wien 1894); die wichtigsten arabisch-griechischen Urkunden des Aphroditofundes sind von C. H. Becker ver”offentlicht und bearbeitet worden.

Sie vermitteln in ihrer Gesamtheit ein fast l”uckenloses Bild von der fortschreitenden Arabisierung des Schrift- und Rechnungswesens, zugleich auch den Beweis eines ”uberraschend langen und z”ahen Festhaltens der Araber an den griechischen Zahlzeichen. Viele Urkunden von Beh”orden, selbst Erlasse arabischer Heerf”uhrer aus der Eroberungszeit sind nur griechisch geschrieben; man wird aber annehmen d”urfen, daß die Eroberer auch ihrer eigenen Sprache sofort Geltung zu verschaffen wußten, und daß zweisprachige Erlasse wie jene “im 22. Jahre d. H. am 25. April 643 n. Chr. ausgefertigte ebenso pr”achtige, wie unsch”atzbare Urkunde des Unterfeldherrn ’Abdallāh ibn 0̆120ābir”, die in der Papyrussammlung Erzherzog Rainer als das ”alteste Schriftdenkmal des Islam bewahrt wird, durchaus keine Ausnahme darstellen.

W”ahrend sich die eingesessene Bauernbev”olkerung noch der koptischen, die handel- und gewerbtreibende Bev”olkerung der St”adte und die Beamtenschaft vorwiegend der griechischen Sprache im Gesch”aftsverkehr bedienten, haben die im Lande heimisch gewordenen Araber ebenso selbstverst”andlich ihre Briefe und Mitteilungen arabisch geschrieben.

Die obersten Regierungsbeh”orden konnten am fr”uhesten daran denken, ihre amtlichen Verf”ugungen nur arabisch abzufassen; ihre Verbreitung in der fremdsprachigen Provinz konnte nach wie vor nur durch Ausfertigung zweisprachiger Urkunden oder durch Herstellung griechischer und koptischer ”Ubersetzungen erfolgen.

In dem oben erw”ahnten Erlasse steht das Arabische an zweiter Stelle, sp”ater ist das Umgekehrte h”aufiger; noch aus dem Jahre 80 d. H. ist ein rein griechischer Erlaß bekannt. In den zweisprachigen Urkunden aus der Zeit des Statthalters

perhaps in the then flourishing Ghassānid kingdom, the script spread with trade to north and south. In the 6th century it was probably already known throughout the northern Arabic-speaking region, and was likely in use especially in the two cities that formed the starting point of the religio-national movement, Mecca and Medina.

Quite invaluable sources for the history of the transfer of Greek-Egyptian administrative and calculation practice into the language and script of the Arab rulers have been available to us for some time in the papyri of el-Fayyūm and other findspots (Archduke Rainer Collection), and for some years in those of the Aphrodito find. An overview accessible even to non-Arabists of the treasures of the first-named collection is contained in the *F”uhrer durch die Ausstellung* in its *Arabische Abtheilung* edited by J. von Karabacek (Vienna 1894); the most important Arabic-Greek documents of the Aphrodito find have been published and edited by C. H. Becker.

They convey in their totality an almost seamless picture of the progressive Arabization of the writing and calculation system, and simultaneously the proof of a surprisingly long and tenacious adherence of the Arabs to the Greek numeral signs. Many documents from authorities, even decrees of Arab military commanders from the time of the conquest, are written only in Greek; but one may assume that the conquerors knew how to give their own language immediate validity, and that bilingual decrees like that “equally magnificent as invaluable document of the lieutenant commander ‘Abdallāh ibn Jābir, executed in the 22nd year A. H. on April 25, 643 CE,” preserved in the Archduke Rainer Papyrus Collection as the oldest written monument of Islam, are by no means an exception.

While the settled rural population still used Coptic, the commercial and craft-working population of the cities and the officials predominantly used Greek in business dealings, the Arabs who had become native to the land just as naturally wrote their letters and communications in Arabic.

The highest governmental authorities could soonest consider drafting their official decrees only in Arabic; their dissemination in the foreign-language province could still only take place through the issuance of bilingual documents or through the production of Greek and Coptic translations.

In the aforementioned decree, Arabic stands in second place; later the reverse is more frequent; a purely Greek decree from the year 80 A. H. is still known. In the bilingual documents from the time of the governor Qurra b. Sharīk (in office 90–

Qorra b. 0̆160arīk (im Amte 90–96 d. H. = 708–714 n. Chr.), die G. H. Becker a. a. O. ver”offentlicht hat, erscheinen die Zahlen des griechischen, an zweiter Stelle stehenden Textes, der f”ur die Empf”anger jedenfalls der wichtigere war, stets zuerst “in Ziffern”, dann “in Worten”, die Jahreszahl in Ziffern; im arabischen Text sind alle Zahlenangaben in Worten ausgeschrieben.

Im Jahre 87 d. H. (706 n. Chr.) f”uhrte nach den Berichten der arabischen Historiker ’Abdallāh, der Sohn des Kalifen ’Abdalmalik, die arabische Verwaltung in ”Agypten ein. Um dieselbe Zeit hatte der Kalif al-Walīd zu Damaskus den Gebrauch der griechischen Sprache bei der F”uhrung der Steuerregister verboten, aber f”ur die Zahlzeichen eine Ausnahme zulassen m”ussen, “parce qu’il était impossible d’écrire en arabe un, ou deux, ou trois, ou huit et demi, etc.”

Daß die Neuerung der rein arabischen Verwaltung noch nicht durchf”uhrbar war, beweisen jetzt unsere Texte; vor allem aber wird die von Woepcke beigezogene Stelle durch die zahlreichen ”uber das ganze achte und neunte Jahrhundert, in abnehmender H”aufigkeit aber noch bis ins zehnte Jahrhundert reichenden, in einem Beispiel sogar vom Jahre 513 d. H., also 1120 n. Chr. datierten Urkunden belegt und best”atigt, die mitten im arabischen Text griechische Zahlzeichen enthalten und darum im *F”uhrer* von Karabacek als “arabisch-griechisch” bezeichnet werden.

Auch die koptischen Zahlzeichen, die nur eine leichte Modifikation der griechischen sind, m”ogen in ”Agypten angewandt worden sein (Woepcke, a. a. O., S. 238), doch scheinen solche arabische Urkunden aus alter Zeit noch nicht bekannt zu sein.

Besonders interessant ist in unserem Zusammenhange ein Palimpsest aus dem Anfang des 9. Jahrhunderts, der schon im 7. Jahrhundert, wie die alten Schriftz”uge zeigen, von einem Araber benutzt worden war, der sich in der Schreibung der griechischen Zahlbuchstaben ”ubte. Er enth”alt in zwei Reihen die griechischen Zahlbuchstaben von α bis ϑ und ”uber den beiden ersten die arabischen Zahlw”orter (F”uhrer N. 649).

Gegen”uber diesen Urkunden treten solche mit arabischen Zahlbuchstaben ganz auffallend zur”uck. Eine Schreibvorlage aus dem 9. Jahrhundert, die die arabischen und griechischen Zahlbuchstaben in Gegen”uberstellung enth”alt, zeigt die ”Ubertragung und gleichzeitige Anwendung der Zeichen. Das ”alteste bisher bekannt gewordene urkundliche Beispiel ist eine doppelsprachige Steuerurkunde aus dem 8. Jahrhundert, die in dem an zweiter Stelle stehenden arabischen Text die Steuersumme auch in arabischen Zahlbuchstaben bietet (F”uhrer N. 605).

Das n”achste, zugleich genau datierte Beispiel aus dem Jahre 851 ist eine Bilanz des Schatzhauses von el-Faij”um (F”uhrer N. 761). Karabacek vermutet einen f”ur die große

96 A. H. = 708–714 CE), published by G. H. Becker, op. cit., the numbers of the Greek text, which stood in second place and was in any case the more important for the recipients, appear always first “in figures,” then “in words,” the year in figures; in the Arabic text all numerical indications are written out in words.

In the year 87 A. H. (706 CE), according to the reports of Arab historians, ‘Abdallāh, the son of the caliph ‘Abd al-Malik, introduced the Arabic administration in Egypt. Around the same time the caliph al-Walīd in Damascus had forbidden the use of the Greek language in keeping the tax registers, but had been compelled to make an exception for the numeral signs, “because it was impossible to write in Arabic one, or two, or three, or eight and a half, etc.”

That the innovation of purely Arabic administration was not yet feasible is proved by our texts now; but above all, the passage cited by Woepcke is evidenced and confirmed by the numerous documents extending over the whole 8th and 9th centuries, with decreasing frequency but still into the 10th century, in one example even dated from the year 513 A. H. (1120 CE), which contain Greek numeral signs in the middle of the Arabic text and are therefore called “Arabic-Greek” in Karabacek’s *F”uhrer*.

Also the Coptic numeral signs, which are only a slight modification of the Greek ones, may have been used in Egypt (Woepcke, op. cit., p. 238), but such Arabic documents from ancient times do not yet seem to be known.

Of particular interest in our context is a palimpsest from the beginning of the 9th century, which, as the old script traces show, had already been used in the 7th century by an Arab who was practicing the writing of the Greek numeral letters. It contains in two rows the Greek numeral letters from α to ϑ and above the first two the Arabic number words (*F”uhrer* no. 649).

In comparison with these documents, those with Arabic numeral letters stand out quite strikingly. A writing template from the 9th century, containing Arabic and Greek numeral letters in juxtaposition, shows the transfer and simultaneous application of the signs. The oldest documentary example known to date is a bilingual tax document from the 8th century, which in the Arabic text standing in second place also presents the tax sum in Arabic numeral letters (*F”uhrer* no. 605).

The next, simultaneously precisely dated example from the year 851 is a balance sheet of the treasury of el-Fayyūm (*F”uhrer* no. 761). Karabacek conjectures a more secret character

Menge mehr sekreten Charakter dieser Rechnungsweise, zumal sie der Differenzierung der einzelnen Buchstabenwerte durch die diakritischen Punkte entbehre; ich sehe ein Hindernis ihrer Verbreitung im Rechnungswesen auch darin, daß sich diese Buchstabenzusammenstellungen nicht deutlich genug vom Worttext abhoben und leicht zu Verwechslungen und Fehlern Veranlassung geben konnten.

Hatte man erst die Zahlen durch die arabischen Worte wiedergegeben, so konnte man die Zeichen auch weglassen, und so blieb in allen zusammenh”angenden Texten, die nicht gerade tabellarische Zusammenstellungen oder sonst geh”aufte Zahlen enthalten, bis heute die Wortschrift der Zahlen in ”Ubung, die vom Standpunkt des Mathematikers ein R”uckschritt sein mag, aber den Vorzug der gr”oßeren Sicherheit besitzt.

Wenn es richtig ist, daß die Benutzung der Buchstaben als Zahlzeichen auf Milet und das 8. Jahrh. v. Chr. zur”uckzuf”uhren ist, so m”ussen die auf Anwendung des Alphabets gegr”undeten Zahlbezeichnungen der Hebr”aer und Syrer griechischem Vorbild nacherfunden worden sein. Das normale semitische Alphabet von 22 Buchstaben f”uhrte zun”achst glatt durch Einer und Zehner. Dann waren noch 4 Zeichen ”ubrig, die f”ur die Zahlen 100 bis 400 verwendet werden konnten, aber es fragte sich, wie nun die ”ubrigen 5 Hunderter darzustellen waren.

Die ”altere hebr”aische und die syrische Zahlenbezeichnung ben”utzen den n”achstliegenden Ausweg, indem sie die h”oheren Hunderter additiv aus Zeichen f”ur die niederen bilden, also $500 = 400 + 100$ usw. bis $900 = 400 + 400 + 100$ setzen. Das gleiche Verfahren ist nach S. de Sacy auch in arabischen Handschriften zu finden.

Interessanter ist ein zweites syrisches Verfahren, das f”ur die Hunderter einfach die Zehnerbuchstaben unter Hinzuf”ugung eines diakritischen Punktes einsetzt, denn damit wird die Zahl der Zifferbuchstaben auf zwei Enneaden reduziert. Es ist von hier aus verst”andlich, wie arabische Schriftsteller — ich erw”ahne neben an-Nadīm, dem Verfasser des *Fihrist*, auf den k”urzlich L. G. Karpinski wieder aufmerksam machte, noch die in meinen *Kazwīnistudien* behandelte Handschrift — die indischen Ziffern f”ur neun Grundzeichen halten konnten, die durch Hinzuf”ugung eines bzw. zweier diakritischen Punkte die Zahlenwerte der Buchstabengruppen bis 0̆639, bis 0̆63a wiedergeben.

Hier sind die punktf”ormigen Nullen der Zehner und Hunderter und die von 1000 einfach als diakritische Punkte genommen (Fihrist S. 18/19). Ich f”uge wieder die w”ortliche ”Ubersetzung bei, da die von L. C. Karpinski mitgeteilte, von Dr. A. Yohannan herr”uhrende “freie” ”Ubersetzung wie alle sogenannten freien ”Ubersetzungen f”ur kritische geschichtliche Untersuchungen nicht zu gebrauchen ist.

of this reckoning method for the majority, especially since it lacks the differentiation of the individual letter values by diacritical points; I see an obstacle to its spread in accounting also in the fact that these letter combinations did not stand out clearly enough from the word text and could easily give rise to confusions and errors.

Once one had rendered the numbers through Arabic words, one could also omit the signs, and so in all connected texts that did not contain tabular arrangements or other accumulated numbers, the word-writing of numbers has remained in use until today, which from the standpoint of the mathematician may be a step backward, but possesses the advantage of greater security.

If it is correct that the use of letters as numeral signs is to be traced back to Miletus and the 8th century BCE, then the numeral designations of the Hebrews and Syrians based on the use of the alphabet must have been re-invented after Greek models. The normal Semitic alphabet of 22 letters led smoothly at first through ones and tens. Then 4 signs remained, which could be used for the numbers 100 to 400, but the question arose how the remaining 5 hundreds were to be represented.

The older Hebrew and the Syriac numeral designations use the most obvious expedient, in that they form the higher hundreds additively from signs for the lower ones, thus setting $500 = 400+100$ etc. up to $900 = 400+400+100$. The same procedure is found, according to S. de Sacy, also in Arabic manuscripts.

More interesting is a second Syriac procedure, which for the hundreds simply employs the tens letters with the addition of a diacritical point, for thereby the number of digit-letters is reduced to two enneads. From here it is understandable how Arabic writers—I mention alongside al-Nadīm, the author of the *Fihrist*, to whom L. G. Karpinski recently called attention, also the manuscript treated in my *Kazwīnistudien*—could regard the Indian numerals as nine basic signs, which by the addition of one or two diacritical points reproduce the numerical values of the letter groups up to ʿayn and up to ghayn.

Here the dot-shaped zeroes of the tens and hundreds and that of 1000 are simply taken as diacritical points (*Fihrist*, pp. 18/19). I append again the literal translation, since the “free” translation communicated by L. C. Karpinski, originating from Dr. A. Yohannan, like all so-called free translations, is not usable for critical historical investigations.

> “Und es erw”ahnte dieser Mann, dessen Erw”ahnung vorausgeschickt wurde, daß sie meistens mit den neun Schriftzeichen schreiben, gem”aß diesen Beispielen:” 1 2 3 4 5 6 7 8 9; “wenn er dann bis zum” ϑ “gelangt ist, wiederholt er das erste Zeichen und versieht es mit einem Punkt darunter gem”aß diesen Beispielen:” 1̇ 2̇ 3̇ 4̇ 5̇ 6̇ 7̇ 8̇ 9̇, “so daß es” 10 20 30 40 50 60 70 80 90 “wird, indem es von zehn zu zehn vermehrt wird; wenn er dann zum” ϑ̇ “gelangt, schreibt er gem”aß diesen Beispielen und setzt unter jedes Zeichen zwei Punkte wie folgt:” 1̈ 2̈ 3̈ 4̈ 5̈ 6̈ 7̈ 8̈ 9̈, “so daß es” 100 200 300 400 500 600 700 800 900 “wird; und wenn er dann das” ϑ̈ “erreicht hat, schreibt er das erste Zeichen von Anfang an, n”amlich dieses:” 1, “und setzt darunter drei Punkte wie folgt:” 1⃛, “und es geschieht so, daß er zur Gesamtheit der Zeichen des (arabischen) Alphabets kommt und schreibt, was er will.”

> “And this man, whose mention was placed first, mentioned that they write for the most part with the nine script-signs, according to these examples:” 1 2 3 4 5 6 7 8 9; “when he then reaches” ϑ, “he repeats the first sign and furnishes it with a point underneath according to these examples:” 1̇ 2̇ 3̇ 4̇ 5̇ 6̇ 7̇ 8̇ 9̇, “so that it becomes” 10 20 30 40 50 60 70 80 90, “being increased by ten; when he then reaches” ϑ̇, “he writes according to these examples and places under each sign two points as follows:” 1̈ 2̈ 3̈ 4̈ 5̈ 6̈ 7̈ 8̈ 9̈, “so that it becomes” 100 200 300 400 500 600 700 80
> “and when he then has reached” ϑ̈, “he writes the first sign from the beginning, namely this:” 1, “and places under it three points as follows:” 1⃛, “and it happens thus that he reaches the totality of the signs of the (Arabic) alphabet and writes what he wishes.”

Die beiden Text”anderungen ergeben sich von selbst aus dem Zusammenhang. Daß “these subscript dots are used here only as we use Roman numerals” (Karpinski S. 123, Z. 3 v. u.), ist eine unhaltbare Annahme; in der zweiten Buchstabenreihe ist nicht “(ja or j omitted”, wie genau nach Fl”ugel zu bemerken ist, sondern zweifellos nur ja die letzte Buchstabenreihe ist nicht die der “Persian letters” — das Wort *mu011fam* bezeichnet das ganze arabische Alphabet — und stellt nicht “the higher hundreds and 1000”, sondern alle Hunderter ohne Tausend dar.

Der Verfasser vergiß im folgenden Satz den Zahlbuchstaben 063a ausdr”ucklich zu erw”ahnen, wie er ja auch das Fortschreiten von 100 zu 100 nicht mehr besonders hervorhebt. Die Echtheit der Fihriststelle anzuzweifeln (Enestr”om in *Jahrb. ”uber d. Fortschr. d. M.*, Bd. 41, 1910, S. 53), liegt durchaus kein Grund vor; sie bezieht sich ohne Frage auf die indischen Zahlzeichen — diese sind f”ur den Araber ebensogut “Schriftzeichen” wie sein eigenes Alphabet — und findet ihr Analogon in andern arabischen Abhandlungen ”uber die Schriften der V”olker.

The two textual alterations follow of themselves from the context. That “these subscript dots are used here only as we use Roman numerals” (Karpinski, p. 123, line 3 from bottom) is an untenable assumption; in the second letter series it is not “(ja or j omitted,” as is to be noted precisely according to Flügel, but doubtless only ja; the last letter series is not that of the “Persian letters”—the word *muʿjam* denotes the whole Arabic alphabet—and does not represent “the higher hundreds and 1000,” but all hundreds without thousand.

The author forgets in the following sentence to mention the numeral letter ghayn explicitly, just as he no longer especially emphasizes the progression from 100 to 100. There is absolutely no reason to doubt the authenticity of the *Fihrist* passage (Eneström in *Jahrb. ”uber d. Fortschr. d. M.*, vol. 41, 1910, p. 53); it undoubtedly refers to the Indian numeral signs—these are for the Arab just as much “script-signs” as his own alphabet—and finds its analogue in other Arabic treatises on the scripts of peoples.

6 VI. ”Uber die Erbteilungsaufgaben / On the Inheritance Problems

Als weitere Quellen f”ur die Geschichte des Rechnens und der Algebra kommen die arabischen Abhandlungen ”uber Erbteilung in Betracht. Bisher ist von ihnen in der Geschichte der Mathematik noch nirgends Gebrauch gemacht worden, obgleich sich vom Erstling dieser Verbindung von Mathematik und Rechtswissenschaft, der uns in Muhammad b. M”usās “Algebra” vorliegt, durch Jahrhunderte hindurch eine gewohnheitsm”aßige und sachlich begr”undete Vereinigung von Rechenkunst und Erbteilungswissenschaft verfolgen l”aßt.

Ich erinnere an den 895 gestorbenen Historiker, Astronomen und Mathematiker al-Dīnawarī, der das ganze von Muhammad b. M”usā behandelte Gebiet in besonderen Schriften darstellte, an den Kādī ’Abd al-Hamīd aus Baṣra, “sehr gelehrt in Rechenkunst, Algebra, Ausmessungslehre und Erbteilung”, sodann an Ab”u Kāmil 015eūcā’ b. Aslam, ferner an die zahlreichen von Ibn al-Faraḍī erw”ahnten Kenner der Erbteilung und Rechenkunst, an den Rechner al-Missisī, an Sinān b. al-Fāṭ, an ’Abd al-Kāhir b. Tāhir aus Bagdad, an al-Wannī und seinen Sch”uler ’Abdallāh b. Ibrāhīm al-Hābrī und, um noch einen Mathematiker der Sp”atzeit zu nennen, an al-Kalasādī, der außer einer Reihe von Schriften ”uber Arithmetik und Algebra auch ein Werk ”uber das Ganze der Erbteilung verfaßte.

Man wird gewiß keine bedeutenden mathematischen S”atze in diesen Schriften entdecken, aber man wird auf alle F”alle den Wurzeln der Rechenkunst der Araber und den praktischen Grundlagen ihres Interesses an der Mathematik n”aher kommen, wenn man dem b”urgerlichen Rechnen mehr Aufmerksamkeit schenkt.

Die Bed”urfnisse des Lebens in Handel und Verkehr, beim Kauf und Verkauf von Waren aller Art, von H”ausern und Grundst”ucken, Tieren und Sklaven, beim bankm”aßigen Geldverkehr, bei der Vollstreckung von Testamenten und der Verteilung von Hinterlassenschaften, bei der Erhebung von Z”ollen, Ertrags- und Verm”ogensteuern usw. mußten unter allen Umst”anden rechnungsm”aßig behandelt werden und wurden auf Grund von Geldwert, Maß und Gewicht nach praktischen Regeln erledigt.

Von diesen Aufgaben und Erfordernissen muß man ”uberall ausgehen, wenn man den nat”urlichen Entwicklungsgang der Rechenkunst beschreiben und nicht ein zu schattenhaftes Bild von der Entstehung mathematischer Methoden und Aufgaben geben will.

Wenn die Erbteilungsrechnungen wirklich durch und durch arabisch w”aren und ganz und gar auf dem Koran beruhten, so m”ußte man den Gang der Entwicklung von den zu Muhammad b. M”usās Zeit begr”undeten Rechtsschulen (Ab”u Hanīfa starb im Jahr 767, Mālik b. Anas um 795, Muhammad al-

As further sources for the history of calculation and algebra, Arabic treatises on inheritance division come into consideration. Hitherto they have been nowhere made use of in the history of mathematics, although from the first fruit of this connection of mathematics and jurisprudence, which lies before us in Muḥammad b. Mūsā’s “Algebra,” a customary and substantively founded union of arithmetic and the science of inheritance division can be traced through centuries.

I recall al-Dīnawarī, the historian, astronomer, and mathematician who died in 895, who presented the whole domain treated by Muḥammad b. Mūsā in special writings; the qāḍī ‘Abd al-Ḥamīd of Baṣra, “very learned in arithmetic, algebra, measurement, and inheritance division”; then Abü Kāmil Shuja‘ b. Aslam; further, the numerous experts in inheritance and arithmetic mentioned by Ibn al-Faraḍī; the calculator al-Missisī; Sinān b. al-Fāṭḥ; ‘Abd al-Kāhir b. Tāhir of Baghdad; al-Wannī and his pupil ‘Abdallāh b. Ibrāhīm al-Ḥābrī; and, to name yet a mathematician of the late period, al-Kalasādī, who besides a series of writings on arithmetic and algebra also composed a work on the entirety of inheritance division.

One will certainly discover no significant mathematical theorems in these writings, but one will in any case come closer to the roots of the Arabs’ arithmetic and to the practical foundations of their interest in mathematics, if one pays more attention to practical calculation.

The necessities of life in trade and commerce, in the buying and selling of goods of all kinds, of houses and landed property, animals and slaves, in banking money transactions, in the execution of testaments and the distribution of estates, in the collection of customs duties, income and property taxes, etc., had under all circumstances to be treated computationally and were disposed of according to practical rules based on monetary value, measure, and weight.

From these tasks and requirements one must everywhere start if one wishes to describe the natural course of development of arithmetic and not give too shadowy a picture of the origin of mathematical methods and problems.

If the inheritance calculations were really thoroughly Arabic and rested entirely on the Qur’an, one would have to be able to follow the course of development from the law schools founded in the time of Muḥammad b. Mūsā (Abü Ḥanīfa died in 767, Mālik b. Anas around 795, Muḥammad al-

0̆160āfi’ī um 820, Aḥmad b. Ḥanbal 855) zu den Aufgaben der Algebra verfolgen k”onnen.

In Wahrheit liefert das Erbrecht nur den Rohstoff f”ur die Aufgaben, und ob die Technik der Rechnung arabisch ist oder nicht, muß erst untersucht werden. Wenn wir uns erinnern, daß auch die r”omischen Juristen schon ihre Freude an verzwickten Erbf”allen hatten und sie durch praktische Rechenkunst oder salomonische Weisheit zu entscheiden suchten (Cantor I^3, S. 561), so werden uns berechtigte Zweifel an der Originalit”at der arabischen Behandlung der Aufgaben aufsteigen, und wenn wir finden, daß es sich vielfach um Aufl”osungen von Gleichungen ersten Grades mit Hilfe von Ausgleichung und Erg”anzung handelt, so k”onnen wir jetzt ohne Zaudern griechische Vorbilder annehmen.

Man wird nicht daran zweifeln k”onnen, daß in den Kreisen der Steuerbeamten, der Notare, der Vermessungsbeamten, der Kaufleute und Bankiers f”ur alle in den regelrechten Gesch”aftsgang einschlagenden Aufgaben eine feste ”Uberlieferung bestand, die den Anw”artern und Schreibern ebenso beigebracht wurde, wie es heutzutage in den Amtsstuben und Kontoren geschieht.

Aus diesen Kreisen m”ogen die ersten Anregungen zur Aufzeichnung und Ausarbeitung von Musterbeispielen gekommen sein, und nichts anderes als eine solche Sammlung von Vorschriften und Beispielen f”ur den allgemeinen Gebrauch, mit genauer Anweisung f”ur die Durchf”uhrung der Rechnungen, will Muhammad b. M”usās Buch vorstellen.

Das Fehlen wirklich aus dem Leben gegriffener Aufgaben im ersten Teil der Algebra, also im Bereich der quadratischen Gleichungen, die Armseligkeit der Beispiele in dem Kapitel von den Gesch”aften, der schematische Charakter vieler Beispiele in andern Teilen des Buches deutet auf den Mangel einer durchgebildeten ”Ubungs- und Schulliteratur, wie wir sie heute in den Werken und Aufgabensammlungen ”uber kaufm”annische Arithmetik, ”uber die Regel de tri u. dergl. besitzen; die quadratischen Gleichungen, die bisher als Hauptst”uck und Hauptzweck der “Algebra” galten, erscheinen jetzt mehr als prunkvolle Fassade und gelehrte Zutat, oder wenn man will auch als Beginn des ”Ubergangs zu den rein wissenschaftlichen, spekulativen Fragen und Interessen der Mathematik.

Ist die hier vorgetragene Auffassung richtig, so m”ussen wir auch von dem gemeinen Sprachgebrauch und den praktischen Aufgaben, also von den Aufgaben ersten Grades, nicht von den quadratischen Gleichungen aus zum Verst”andnis der Terminologie vordringen suchen. Es ist dann klar, daß der in allen Aufgaben wiederkehrende, grundlegende Begriff kein anderer sein kann als der einer Geldsumme, eines Verm”ogens, sei es bekannt oder unbekannt, Kapital, Erbmasse oder Wert einer Ware.

Shāfiʿī around 820, Aḥmad b. Ḥanbal in 855) to the problems of algebra.

In truth, inheritance law provides only the raw material for the problems, and whether the technique of calculation is Arabic or not must first be investigated. If we recall that Roman jurists too already took pleasure in tricky inheritance cases and sought to decide them by practical arithmetic or Solomonic wisdom (Cantor I^3, p. 561), then justified doubts about the originality of the Arabic treatment of the problems will arise in us, and if we find that it frequently concerns solutions of first-degree equations by means of balancing and completion, we can now assume Greek models without hesitation.

One cannot doubt that in the circles of tax officials, notaries, surveyors, merchants, and bankers there existed a firm tradition for all tasks falling within the regular course of business, which was taught to candidates and scribes just as happens nowadays in offices and counting-houses.

From these circles may have come the first impulses toward the recording and working out of model examples, and nothing other than such a collection of rules and examples for general use, with precise instructions for carrying out the calculations, does Muḥammad b. Mūsā’s book wish to present.

The absence of really practically drawn problems in the first part of the Algebra, that is, in the domain of quadratic equations; the poverty of the examples in the chapter on commercial transactions; the schematic character of many examples in other parts of the book—all point to the lack of a fully developed exercise and school literature, such as we possess today in the works and problem collections on commercial arithmetic, on the rule of three, and the like. The quadratic equations, which hitherto counted as the main piece and chief purpose of “Algebra,” now appear more as a splendid facade and scholarly addition, or if one will, as the beginning of the transition to the purely scientific, speculative questions and interests of mathematics.

If the view presented here is correct, then we must seek to advance to the understanding of the terminology from common linguistic usage and practical problems, that is, from first-degree problems, not from quadratic equations. It is then clear that the fundamental concept recurring in all problems can be none other than that of a sum of money, a possession, be it known or unknown, capital, estate, or value of a commodity.

Alles dies und noch mehr wird durch das Wort مال *māl* bezeichnet. Im Koran, den wir als ”alteste Quelle anf”uhren, finden wir das Wort *māl* und den Plural *amwāl* in der Bedeutung Verm”ogen, Besitz, Geld und Gut, Reichtum, h”aufig auch in der Verbindung “G”uter und Kinder” oder “mit ihrem Gut und Blut”; es ist vom Aufzehren des Gutes der Waisen, vom Ausgeben der Reicht”umer auf dem Wege Gottes die Rede, “Kapitalien” heißen رؤوس الاموال d. i. *capita bonorum*, und “er h”auft Gut auf” جمع مالا *0̆11famā’a mālan* (Sure CIV, 2).

In den Aphroditopapyri kann man *māl* mit Geld oder Steuerertrag, Steuersumme ”ubersetzen, gutes, d. h. vollwertiges Geld ist مال تام *māl tāmm*; das Schatzhaus heißt بيت المال *bait al-māl*, das Einsammeln der Steuerbetr”age جمع المال *0̆11fam al-māl.*

In seiner ganz allgemeinen Bedeutung, als vorhandenes oder angenommenes Verm”ogen, wird das Wort in den Erbteilungsaufgaben gebraucht.

Zwei Beispiele mit wortgetreuer ”Ubersetzung m”ogen zugleich die Art der Aufgaben erl”autern. Nach Rosen S. 89/90:

“If the instance be: “A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her Capital”; then you constitute the shares of the heirs, by taking them out of twenty. Take a Capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a Capital less one-eighth and one-seventh. Complete your Capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the Capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen fourty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares.”

In w”ortlicher ”Ubertragung nach dem Arabischen:

“Und wenn er (oder jemand) sagt: Eine Frau starb und hinterließ ihren Gatten und ihren Sohn und drei T”ochter und vermachte einem Manne ein Achtel ihres Verm”ogens und sein Siebentel; so stelle die Anteile der (gesetzlichen) Erbschaft auf, indem du sie von zwanzig nimmst. Nimm also ein Verm”ogen und wirf weg sein Achtel und Siebentel, so bleibt ein Verm”ogen weniger ein Achtel und ein Siebentel. Vervollst”andige nun dein Verm”ogen, und das ist (oder besteht darin), daß du zu ihm hinzuf”ugst f”unfzehn Teile von einundvierzig Teilen; schlag dann (d. i. multipliziere) die Anteile der Erbschaft, n”amlich zwanzig, in einundvierzig, so wird das achthundertundzwanzig; f”uge dann diesem hinzu f”unfzehn Teile von einundvierzig,

All this and more is designated by the word مال *māl.* In the Qur’an, which we cite as the oldest source, we find the word *māl* and the plural *amwāl* in the meaning of fortune, possession, money and goods, wealth, frequently also in the connection “goods and children” or “with their goods and blood”; there is talk of consuming the goods of orphans, of spending riches in the way of God; “capitals” are called رؤوس الاموال i.e. *capita bonorum*, and “he accumulates goods” جمع مالا *jama‘a mālan* (Sura CIV, 2).

In the Aphrodito papyri one can translate *māl* as money or tax revenue, tax sum; good, i.e. full-value money is مال تام *māl tāmm*; the treasury is called بيت المال *bayt al-māl*, the collection of tax amounts جمع المال *jam‘ al-māl.*

In its most general meaning, as present or assumed possession, the word is used in the inheritance problems.

Two examples with literal translation may at the same time illustrate the nature of the problems. After Rosen, pp. 89/90:

“If the instance be: ‘A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her Capital’; then you constitute the shares of the heirs, by taking them out of twenty. Take a Capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a Capital less one-eighth and one-seventh. Complete your Capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the Capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen fourty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares.”

In literal translation from the Arabic:

“And if he (or someone) says: A woman died and left her husband and her son and three daughters, and bequeathed to a man one-eighth of her possession and its seventh; so set up the shares of the (legal) inheritance by taking them from twenty. Take then a possession and cast away its eighth and seventh, so there remains a possession less an eighth and a seventh. Complete now your possession, and that is (or consists in) adding to it fifteen parts of forty-one parts; then strike (i.e. multiply) the shares of the inheritance, namely twenty, into forty-one, so that becomes eight hundred twenty; then add to this fifteen parts of forty-one, and that is three hundred parts, so this all gives one thousand one hundred und das sind dreihundert Teile, so gibt dies alles tausend und hundert und zwanzig Anteile; f"ur den, dem davon das Achtel und das Siebentel vermacht wurde, ein Siebentel davon und sein Achtel, und das ist dreihundert, das Siebentel hundert und sechzig und das Achtel hundert und vierzig; somit bleiben achthundert und zwanzig Anteile zwischen den Erben gem"aß ihren Anteilen."

Einige der auffallenderen Abweichungen der Rosenschen "Ubersetzung sind durch Kursivdruck hervorgehoben; sie best"atigen die Beobachtung, daß die Wiedergabe der Kunstausdr"ucke unzuverl"assig ist. Das "andert nicht immer viel am Sinn, macht aber eine Wiederherstellung der Ausdr"ucke aus der "Ubersetzung unm"oglich und alle darauf gebauten Schl"usse unsicher.

Muhammad b. M"usā sagt seinen Lesern nicht, wie er zu den zwanzig Anteilen f"ur die Erben kommt. Sie wissen als Muslime, daß dem Koran gem"aß (Sure IV, 13) der Gatte nach Abzug der Legate auf ein Viertel des Nachlasses Anspruch hat, wenn Kinder miterben, und daß ein Knabe doppeltsoviel vom Rest erh"alt als ein M"adchen (Sure IV, 12); auch ist f"ur das Legat die Vorschrift eingehalten, daß der Erblasser h"ochstens "uber ein Drittel seines reinen Verm"ogens frei verf"ugen darf.

Die T"ochter erhalten also je $\frac{1}{4}$, der Sohn $\frac{1}{2}$ von $\frac{3}{4}$ des Verm"ogens, d. i. der Vater $\frac{1}{4}$, die drei T"ochter $\frac{3}{16}$ und der Sohn $\frac{3}{8}$ von dem, was nach Abzug des Legates noch "ubrig bleibt. Um ganz klar zu sein, h"atte Muhammad b. M"usā weiter hinzuf"ugen m"ussen, daß $\frac{1}{8}$ und $\frac{1}{7}$ zusammen $\frac{15}{56}$ des Verm"ogens sind, daß man also f"ur die gesetzlichen Erben $\frac{41}{56}$ von 56 Teilen "ubrig beh"alt und darum $\frac{15}{56}$ des Restes zu diesem hinzuf"ugen muß, um das vollst"andige Verm"ogen zu erhalten. Damit er ganze Zahlen erh"alt, multipliziert er 20 mit 41 und gewinnt so die Zahl 820.

Diese 820 sind um 15 weitere Teile, d. h. um $\frac{15}{41} \cdot 20 = 300$ zu vermehren. Die Schlußrechnung k"onnen wir dann in Form einer "Abrechnung mit den Erben" wie folgt darstellen:

$$41 \cdot 20 + 15 \cdot 20 = 1120$$
$$1120 : 7 = 160$$
$$1120 : 8 = 140$$

F"ur das Legat 300 — 300
Bleiben f"ur die Erben 820.

Es ist wohl m"oglich, daß wir in diesem Verfahren ein Beispiel der "alteren Technik vor uns haben, und daß die Aufgabe mit ihren einfachen Bedingungen zu dem Bestand von Musterbeispielen geh"ort, die schon vor Einf"uhrung der Gleichungen bei den Erbrichtern und Notaren zur Ein"ubung der Teilungsregeln geh"orten.

and twenty shares; for him to whom the eighth and the seventh of it was bequeathed, one-seventh of it and its eighth, and that is three hundred, the seventh one hundred sixty and the eighth one hundred forty; thus eight hundred and twenty shares remain among the heirs according to their shares."

Some of the more striking deviations of Rosen's translation are indicated by italics; they confirm the observation that the rendering of technical terms is unreliable. This does not always alter the sense much, but makes a restoration of the expressions from the translation impossible and all conclusions built upon it uncertain.

Muḥammad b. Mūsā does not tell his readers how he arrives at the twenty shares for the heirs. They know as Muslims that according to the Qur'an (Sura IV, 13) the husband has claim to a quarter of the estate after deduction of legacies, if children co-inherit, and that a boy receives twice as much of the remainder as a girl (Sura IV, 12); also the prescription is observed for the legacy that the testator may freely dispose of at most one-third of his net estate.

The daughters thus each receive $\frac{1}{4}$, the son $\frac{1}{2}$ of $\frac{3}{4}$ of the estate, that is, the father $\frac{1}{4}$, the three daughters $\frac{3}{16}$ and the son $\frac{3}{8}$ of what remains after deduction of the legacy. To be quite clear, Muḥammad b. Mūsā should have further added that $\frac{1}{8}$ and $\frac{1}{7}$ together are $\frac{15}{56}$ of the estate, that one thus retains $\frac{41}{56}$ of 56 parts for the legal heirs and must therefore add $\frac{15}{56}$ of the remainder to it in order to obtain the complete estate. In order to get whole numbers, he multiplies 20 by 41 and thus obtains the number 820.

These 820 are to be increased by 15 further parts, that is, by $\frac{15}{41} \cdot 20 = 300$. We can then present the final calculation in the form of a "reckoning with the heirs" as follows:

$$41 \cdot 20 + 15 \cdot 20 = 1120$$
$$1120 : 7 = 160$$
$$1120 : 8 = 140$$

For the legacy 300 — 300
Remain for the heirs 820.

It is quite possible that we have in this procedure an example of the older technique, and that the problem with its simple conditions belongs to the stock of model examples that already before the introduction of equations served among inheritance-dividers and notaries for practicing the division rules.

Eine Stufe weiter f”uhrt ein Beispiel, das ich nach dem Text S. 86 mit Anwendung von Ziffern und von Abk”urzungen f”ur die Worte Verm”ogen, Dirhem und Anteil — dieser heißt hier *nasīb* statt *sahm*, ein Fingerzeig daf”ur, daß die Aufgabe einer andern Quelle entstammt — genau wiedergebe:

“Ein Mann starb und hinterließ vier S”ohne, und vermachte einem Manne, was soviel ist als der Anteil eines von ihnen und das Viertel dessen, was bleibt von dem Drittel [des Verm”ogens nach Abzug des Anteils — von Rosen erg”anzt], und einen Dirhem. Dann ist die Vorschrift hierf”ur, daß du nimmst $\frac{1}{3}$ eines Verm”ogens, dann wegnimmst (wegwirfst) von ihm einen Anteil, so daß bleibt $\frac{1}{3}$ weniger ein Anteil. Hierauf nimmst du weg $\frac{1}{4}$ von dem, was dir geblieben ist, d. h. $\frac{1}{4}$ von $\frac{1}{3}$ weniger $\frac{1}{4}$ eines Anteils, und nimmst auch weg 1 Dhm., so bleibt dir $\frac{1}{4}$ von $\frac{1}{3}$ eines Verm”ogens, das ist $\frac{1}{12}$ des Verm”ogens, weniger $\frac{1}{4}$ eines Anteils und weniger 1 Dhm.

Dies f”uge nun zu (den noch vorhandenen) $\frac{2}{3}$ des Verm”ogens hinzu, dann hast du $\frac{11}{12}$ Teile von 12 (Teilen) eines Verm”ogens weniger $\frac{1}{4}$ eines Anteils und weniger 1 Dhm., die gleichkommen 4 Ant. Erg”anze dies nun (al-*gabr*) mit $\frac{1}{4}$ eines Anteils und mit 1 Dhm., so werden $\frac{11}{12}$ Teile von 12 eines Verm”ogens gleich 4 Anteilen und $\frac{1}{4}$ eines Anteils und 1 Dhm.

Vervollst”andige nun (*ikmāl*) dein Verm”ogen und zwar dadurch, daß du zu den Anteilen und dem Dhm. hinzuf”ugst $\frac{1}{11}$ Teil von ihnen, so bekommst du 1 Verm”ogen, das gleich ist $\frac{48}{11}$ Anteilen und 1 Dhm. und $\frac{1}{11}$ Teil eines Dhm.”

Dr”ucken wir den Text in unsern gewohnten Zeichen aus, indem wir das Legat $= x$, das Kapitalverm”ogen $= k$, den Anteil eines Sohnes $= t$, den Dirhem $= d$ setzen, so kommen wir durch die Ausf”uhrung der vorgeschriebenen Operationen zun”achst zu der Beziehung $\frac{2}{3}k - x = \frac{1}{3}k - t - \frac{1}{4}(\frac{1}{3}k - t) - d = \frac{1}{4}k - \frac{3}{4}t - d$, dann zu der Gleichung $\frac{2}{3}k + \frac{1}{4}k - \frac{3}{4}t - d = \frac{11}{12}k - \frac{1}{4}t - d = 4t$, woraus durch *al-gabr* $\frac{11}{12}k = 4\frac{1}{4}t + d$ und durch *ikmāl* $k = \frac{51}{11}t + \frac{12}{11} \cdot \frac{51}{11}t + \frac{12}{11}d = 5\frac{8}{11}t + 1\frac{1}{11}d$.

Die arabischen Bruchzahlen. Bei S. G”unther findet sich dar”uber S. 200 folgende Schilderung: “Von dem sp”ateren Alk”arkī und dem noch viel sp”ateren Behā Eddīn wird uns ”ubrigens berichtet, daß man auch mit gew”ohnlichen Br”uchen rechnete. Doch ließ man als aussprechbar nur solche zu, deren Nenner Einer waren; wuchs der Nenner ”uber 9 hinaus, so war der Bruch stumm, und dann gab es f”ur ihn nur eine umschreibende Bezeichnung.”

Ungef”ahr dasselbe lesen wir auch bei Cantor I^3, S. 718, nur daß eine im ganzen zutreffende, von G”unther ”ubergangene Erkl”arung des Sachverhalts hinzugef”ugt wird. Rodet hat sogar eine ”Ahnlichkeit mit dem “Aussprechbarmachen” der Br”uche durch Verwandlung in eine Summe von Stammbr”uchen bei den ”Agyptern finden wollen; Cantor zweifelt, ob die

A step further leads an example which I reproduce exactly from the text p. 86 with the use of numerals and abbreviations for the words possession, dirham, and share—the latter is called here *nasīb* instead of *sahm*, a hint that the problem stems from another source:

“A man died and left four sons, and bequeathed to a man [an amount] equal to the share of one of them and a quarter of what remains from the third [of the estate after deduction of the share—supplied by Rosen], and one dirham. Then the rule for this is that you take $\frac{1}{3}$ of a possession, then remove (cast away) from it a share, so that $\frac{1}{3}$ less a share remains. Then you take away $\frac{1}{4}$ of what has remained for you, i.e. $\frac{1}{4}$ of $\frac{1}{3}$ less $\frac{1}{4}$ of a share, and also take away 1 dirham, so there remains for you $\frac{1}{4}$ of $\frac{1}{3}$ of a possession, that is $\frac{1}{12}$ of the possession, less $\frac{1}{4}$ of a share and less 1 dirham.

This add now to the (still remaining) $\frac{2}{3}$ of the possession, then you have $\frac{11}{12}$ parts of 12 (parts) of a possession less $\frac{1}{4}$ of a share and less 1 dirham, equal to 4 shares. Complete this now (al-*gabr*) with $\frac{1}{4}$ of a share and with 1 dirham, so $\frac{11}{12}$ parts of 12 of a possession equal 4 shares and $\frac{1}{4}$ of a share and 1 dirham.

Complete now (*ikmāl*) your possession, namely by adding to the shares and the dirham $\frac{1}{11}$ part of them, so you obtain 1 possession equal to $\frac{48}{11}$ shares and 1 dirham and $\frac{1}{11}$ part of a dirham.”

Expressing the text in our usual notation, setting the legacy $= x$, the capital possession $= k$, the share of a son $= t$, the dirham $= d$, we come through the execution of the prescribed operations first to the relation $\frac{2}{3}k - x = \frac{1}{3}k - t - \frac{1}{4}(\frac{1}{3}k - t) - d = \frac{1}{4}k - \frac{3}{4}t - d$, then to the equation $\frac{2}{3}k + \frac{1}{4}k - \frac{3}{4}t - d = \frac{11}{12}k - \frac{1}{4}t - d = 4t$, whence by *al-gabr* $\frac{11}{12}k = 4\frac{1}{4}t + d$ and by *ikmāl* $k = \frac{51}{11}t + \frac{12}{11} \cdot \frac{51}{11}t + \frac{12}{11}d = 5\frac{8}{11}t + 1\frac{1}{11}d$.

The Arabic fractional numbers. In S. Günther one finds on p. 200 the following description: “We are moreover told by the later al-Karkī and the still much later Behā ad-Dīn that one also calculated with common fractions. But only those were permitted as pronounceable whose denominators were units; if the denominator grew beyond 9, the fraction was mute, and then there existed for it only a circumlocutory designation.”

Approximately the same is read in Cantor I^3, p. 718, only that an explanation of the state of affairs correct on the whole, which Günther passed over, is added. Rodet even wished to find a similarity with the “making pronounceable” of fractions by conversion into a sum of unit fractions among the Egyptians; Cantor doubts whether the distinction already held for the

Unterscheidung bereits f”ur Muhammad b. M”usās Zeiten Geltung gehabt habe.

Es w”are an der Zeit, daß diese rein philologische Diskussion aus der mathematischen Literatur verschw”ande. Es handelt sich darum, daß das Arabische von den Grundzahlen 3 bis 10 besondere Bruchzahlen bildet, wie *tuluṭ* Drittel, *humus* F”unftel, *’uṛ* Zehntel; das sind die “aussprechbaren” Br”uche. F”ur die Zehner, die als Plurale der Einer erscheinen, sind entsprechende Bildungen sprachlich unm”oglich, f”ur zusammengesetzte Zahlen nur dann, wenn sie in Faktoren bis 10 zerlegt werden k”onnen, so daß z. B. ein Achtundvierzigstel “ein Sechstel des Achtels” heißt.

Selbstverst”andlich bestand dieser Unterschied, solang es eine arabische Sprache gibt. Schon Muhammad b. M”usā ”außert sich ”uber den Tatbestand in seinem Rechenbuch; die Stelle findet sich S. 17 der lateinischen Bearbeitung und lautet: *Scito quod fractiones [sic] appellentur multis nominibus in numerabilibus atque infinitis, ut medietas, tercia, quarta, nona et decima, et una pars ex xm., et pars ex xviii., et cetera.*

Wie man sp”ater dar”uber philosophierte, zeigt eine Stelle der Iḥān a-̣safā’ (ed. Bombay, Bd. I, S. 28): “Und wisse, o Bruder ..., daß die gebrochene Zahl viele Rangordnungen (*marātib*) besitzt, weil es keine ganze Zahl gibt, die nicht einen Teil h”atte oder zwei Teile oder eine Anzahl von Teilen wie die Zw”olf: denn ihr (kommt zu) eine H”alfte und ein Drittel und ein Viertel und ein Sechstel und die H”alfte eines Sechstels, und ebenso die Achtundzwanzig und andere als diese beiden von den Zahlen ...”

“Und es umfaßt sie alle eine Zehnzahl von Worten: ein Wort von ihnen ist allgemein und unbestimmt (*mubhamah*) und neun sind speziell und bestimmt (*mafh”umah*); von den neun Worten ist ein Wort urspr”unglich (*maud”u’ah*), n”amlich die “H”alfte”, und acht sind abgeleitet (*mustaqqah*), n”amlich das “Drittel” von drei und das “Viertel” von vier und das “F”unftel” von f”unf und das “Sechstel” von sechs und das “Siebentel” von sieben und das “Achtel” von acht und das “Neuntel” von neun und das “Zehntel” von zehn; und was das allgemeine, unbestimmte Wort anlangt, so ist es der “Teil”, weil das “Eins von Elf” ein “Teil” von elf genannt wird, und ebenso von dreizehn und von siebzehn und was ihm gleicht; und was die (dann noch) verbleibenden von den W”ortern f”ur die Br”uche anlangt, so sind sie angeh”angt (*mudāfah*) an jene zehn W”orter, wie man statt Eins von Zw”olf die H”alfte des Sechstels sagt und statt Eins von F”unfzehn ein F”unftel des Drittels und statt Eins von Zwanzig die H”alfte des Zehntels” usw.

Sp”ater heißt die Elf “die erste stumme Zahl”, was dann weiter unten (S. 29, Z. 6 v. u.) damit erkl”art wird, daß sie keinen aussprechbaren Teil besitzt.

time of Muḥammad b. Mūsā.

It is high time that this purely philological discussion disappeared from the mathematical literature. The point is that Arabic from the cardinal numbers 3 to 10 forms special fractional numbers, such as *tulŭṭ* third, *humus* fifth, *‘ushr* tenth; these are the “pronounceable” fractions. For the tens, which appear as plurals of the units, corresponding formations are linguistically impossible; for composite numbers only when they can be decomposed into factors up to 10, so that e.g. a forty-eighth is called “a sixth of an eighth.”

Obviously this distinction existed as long as there is an Arabic language. Already Muḥammad b. Mūsā expresses himself on the state of affairs in his arithmetic book; the passage is found on p. 17 of the Latin adaptation and reads: *Scito quod fractiones [sic] appellentur multis nominibus in numerabilibus atque infinitis, ut medietas, tercia, quarta, nona et decima, et una pars ex xm., et pars ex xviii., et cetera.*

How one later philosophized about this is shown by a passage of the Ikhwān al-Ṣafā’ (ed. Bombay, vol. I, p. 28): “And know, O brother ..., that the fractional number possesses many ranks (*marātib*), because there is no whole number that does not have one part or two parts or a number of parts, like twelve: for it has (comes to it) a half and a third and a quarter and a sixth and half a sixth, and likewise twenty-eight and others besides these two among numbers ...”

“And they are all encompassed by ten words: one word of them is general and indefinite (*mubhamah*) and nine are special and definite (*mafhūmah*); of the nine words one is original (*mawḍū‘ah*), namely the ‘half,’ and eight are derived (*mustaqqah*), namely the ‘third’ from three and the ‘quarter’ from four and the ‘fifth’ from five and the ‘sixth’ from six and the ‘seventh’ from seven and the ‘eighth’ from eight and the ‘ninth’ from nine and the ‘tenth’ from ten; and as for the general, indefinite word, it is ‘part,’ because ‘one of eleven’ is called a ‘part’ of eleven, and likewise of thirteen and of seventeen and what is similar; and as for the (then still) remaining words for fractions, they are attached (*muḍāfah*) to those ten words, as one says instead of one of twelve half of the sixth and instead of one of fifteen a fifth of the third and instead of one of twenty half of the tenth,” etc.

Later, eleven is called “the first mute number,” which is then explained further below (p. 29, line 6 from bottom) by the fact that it possesses no pronounceable part.

Der Terminus *shai’*. Wieder einen Schritt weiter gelangen wir bei Aufgaben, in denen an Stelle der individualisierten Unbekannten ein allgemeing”ultiger, f”ur jede Art gesuchter Gr”oßen verwendbarer technischer Ausdruck tritt.

Zwar kann auch *māl* als solcher gelten, sofern es sich um Bestimmung von Geldsummen handelt. Aber erst mit Einf”uhrung des Ausdrucks شيء *shai’* kommen wir wirklich zu dieser Verallgemeinerung, und es ist zu untersuchen, welche Bedeutung dem Wort, das als *cosa* “Sache, Ding” in die Geschichte der Algebra fortlebt, eigentlich zukommt.

Daß *shai’* ein “Ding”, eine “Sache” bedeutet, ist ganz richtig. Aber warum wurde gerade dieses Wort zur Bezeichnung der Unbekannten gew”ahlt? Man k”onnte zun”achst an philosophischen Allgemeinheiten denken. So beginnen die Iḥān a-̣safā’ ihre zahlentheoretischen Betrachtungen (a. a. O., Bd. I, S. 23) mit der Bemerkung: “Das allgemeinste der Worte und Namen (*nomina*) ist, wenn wir sagen “das Ding”, und das “Ding” ist entweder eines oder mehr als eines” usw.

Aber von solchen Abstraktionen kann hier nicht die Rede sein. N”aher kommen wir den Gr”unden f”ur die Wahl des Ausdrucks schon, wenn wir beachten, daß das Wort in vielen F”allen geradezu im Sinne von *māl* “Besitz” gebraucht wird; so wenn es im Koran Sure VII, 83 (”ahnlich XI, 86, XXVI, 183) heißt: اوفوا الكيل والميزان ولا تبخسوا الناس اشياءهم “gebt volles Maß und Gewicht und sch”adigt die Leute nicht an ihren Sachen”.

Das Wort bezeichnet aber in zahllosen Wendungen nicht ein Ding, sondern “irgend ein Ding”, irgend eine Gr”oße, etwas, z. B.: شيء من الخراج *shai’ min al-0121arā́*, etwas von dem Steuerertrag, بشيء من علمه *bi-shai’ min ’ilmihi*, “mit etwas von seinem Wissen”, شيء من الخوف *shai’ min al-aauf*, “etwas von der Furcht”; daher in Verbindung mit Negationen “nichts”, z. B. ما انت بشيء *mā anta bi-shai’*, “du bist nichts”, لا تنفقوا شيئا *lā tunfiq”u shai’an*, “gebt nichts aus”; in Verbindung mit *kull* “alles”, z. B. شيء بصير *kull shai’ basīr*, “er ist allwissend”, شيء قدير *shai’ qadīr*, “allm”achtig”; mit dem Fragewort *mā* “was”, z. B. ما اشد من *mā ąadda min*, “was ist gr”oßer als …?”

Mit “Etwas” im Sinne einer beliebigen noch zu bestimmenden Zahl oder Gr”oße ist daher *shai’* in allen mathematischen Aufgaben, wo es zur Bezeichnung der Unbekannten dient, zu ”ubersetzen.

Ein drittes Beispiel aus den Erbteilungsrechnungen mag zugleich den Fortschritt in der Bezeichnung veranschaulichen:

“Und wenn er hinterlassen hat zwei S”ohne und hinterlassen hat zehn Dirhem an Kapital und zehn Dirhem einer Schuldforderung gegen einen der S”ohne und vermacht hat einem Manne ein F”unftel seines Verm”ogens und einen Dirhem, so ist die Rechenvorschrift, daß du, was von der Schuld herkommt (herausgebracht wird), “Etwas” nennst, und es zum Kapital

The term *shay’*. Yet a step further we come in problems in which, in place of the individualized unknowns, a generally valid technical expression occurs, usable for every kind of sought quantity.

Indeed, *māl* can also count as such, insofar as it concerns the determination of sums of money. But only with the introduction of the expression شيء *shay’* do we really arrive at this generalization, and it is to be investigated what meaning properly attaches to the word, which lives on in the history of algebra as *cosa*, “thing.”

That *shay’* means a “thing,” a “something,” is quite correct. But why was precisely this word chosen to designate the unknown? One might at first think of philosophical generalities. Thus the Ikhwān al-Ṣafā’ begin their number-theoretical considerations (op. cit., vol. I, p. 23) with the remark: “The most general of words and names (*nomina*) is when we say ‘the thing,’ and the ‘thing’ is either one or more than one,” etc.

But such abstractions are not to be spoken of here. We come closer to the reasons for the choice of the expression already when we observe that the word in many cases is used directly in the sense of *māl*, “possession”; as when the Qur’an says Sura VII, 83 (similarly XI, 86, XXVI, 183): اوفوا الكيل والميزان ولا تبخسوا الناس اشياءهم “give full measure and weight and do not wrong people in their things.”

But the word designates in countless phrases not a thing but “any thing,” any quantity, something, e.g.: شيء من الخراج *shay’ min al-kharāj*, something of the tax revenue, بشيء من علمه *bi-shay’ min ‘ilmihi*, “with something of his knowledge,” شيء من الخوف *shay’ min al-khawf*, “something of fear”; hence in connection with negations “nothing,” e.g. ما انت بشيء *mā anta bi-shay’*, “you are nothing,” لا تنفقوا شيئا *lā tunfiqu shay’an*, “give out nothing”; in connection with *kull* “all,” e.g. شيء بصير *kull shay’ basīr*, “he is all-seeing,” شيء قدير *shay’ qadīr*, “all-powerful”; with the interrogative *mā* “what,” e.g. ما اشد من *mā ashadda min*, “what is greater than …?”

With “something” in the sense of an arbitrary number or quantity yet to be determined, *shay’* is therefore to be translated in all mathematical problems where it serves to designate the unknown.

A third example from the inheritance calculations may at the same time illustrate the progress in notation:

“And if he has left two sons and left ten dirhams as capital and ten dirhams of a debt against one of the sons, and has bequeathed to a man one-fifth of his possession and one dirham, then the calculation rule is that you call what comes from the debt (is brought out) ‘Something,’ and add it to the capital, so it gives Something and ten dirhams. Then you hinzuf”ugst, so gibt es Etwas und zehn Dirhem. Dann ziehst du ein F”unftel davon ab, weil er ein F”unftel seines Verm”ogens vermacht hat, und das sind zwei Dirhem und ein F”unftel Etwas, so bleiben acht Dirhem und vier F”unftel Etwas.

Dann ziehst du den Dirhem ab, den er vermacht hat, so bleiben sieben Dirhem und vier F”unftel Etwas; dann teilst du es zwischen den beiden S”ohnen, so bekommt ein jeder drei Dirhem und die H”alfte eines Dirhem und zwei F”unftel Etwas [und das ist gleich dem Etwas. Gleiche nun damit aus, indem du zwei F”unftel Etwas] von Etwas wegnimmst, so bleiben drei F”unftel Etwas gleich drei Dirhem und einem halben. Vervollst”andige nun das Etwas, und das besteht darin, daß du ihm zuf”ugst zwei Drittel desselben und zuf”ugst den Dreiundeinhalb zwei Drittel desselben, und das sind zwei Dirhem und ein Drittel, so gibt das f”unf und f”unf Sechstel und das ist das Etwas, was von der Schuld herkommt.”

Das ist in unsere Zeichen ”ubertragen: $(x+10) - \frac{1}{5}(x+10) = 8+\frac{4}{5}x; 8+\frac{4}{5}x-1 = 7+\frac{4}{5}x; 3\frac{1}{2}+\frac{2}{5}x = x; 3\frac{1}{2} = \frac{3}{5}x$. Zur Vervollst”andigung werden $\frac{2}{3}$ jeder Seite addiert, also ist $x = 3\frac{1}{2}+\frac{2}{3}\cdot 3\frac{1}{2} = 5\frac{5}{6}$.

Es ist bemerkenswert, daß in dem Buch von der Erbteilung nur in dem Kapitel “Von der R”uckerstattung” und in den drei Aufgaben, die zu dem Kapitel “Von dem Kapital und der Schuld” geh”oren, von der Bezeichnung der Unbekannten durch das Wort *shai’* Gebrauch gemacht wird. In dem Kapitel “Von der Vervollst”andigung” wird daf”ur regelm”aßig gesagt اجعل مالا *ij’al mālan*, “nimm irgend ein Verm”ogen an”, in andern F”allen wird sofort eine Zahl, die einfache Ergebnisse liefert, vorgeschlagen.

Es ist nicht zu verkennen, daß bei den verschiedenen Aufgabengruppen verschiedene Schultraditionen oder literarische Quellen vorliegen, und es w”are gewiß auch f”ur die Geschichte der Mathematik nicht ohne Bedeutung, wenn man sich dieser Erbteilungsaufgaben mit mehr Eifer als bisher annehmen w”urde.

Fassen wir das Ergebnis zusammen, so fanden wir als nat”urlichen und urspr”unglichen Ausdruck f”ur die Unbekannte das Wort *māl* “Verm”ogen”, als Maßeinheit gegebener Geldsumme den Dirhem. Neben dem Verm”ogen oder dem Nachlaß kann eine Anzahl anderer Begriffe als Unbekannte auftreten, die entweder auf Grund der gegebenen Beziehungen zum Verm”ogen oder durch willk”urliche Annahmen beseitigt werden. Die Einf”uhrung des Wortes *shai’* verleiht der Ausdrucksweise einen allgemeineren Charakter, ist aber f”ur die behandelten Aufgaben ziemlich gleichg”ultig.

deduct one-fifth of it, because he bequeathed one-fifth of his possession, and that is two dirhams and one-fifth Something, so eight dirhams and four-fifths Something remain.

Then you deduct the dirham that he bequeathed, so seven dirhams and four-fifths Something remain; then you divide it between the two sons, so each gets three dirhams and half a dirham and two-fifths Something [and that equals the Something. Balance now thereby, by taking away two-fifths Something] from Something, so three-fifths Something equal three dirhams and a half. Complete now the Something, and that consists in adding to it two-thirds of it and adding to the three-and-a-half two-thirds of it, and that is two dirhams and a third, so that gives five and five-sixths and that is the Something that comes from the debt.”

Transferred into our notation: $(x+10) - \frac{1}{5}(x+10) = 8+\frac{4}{5}x; 8+\frac{4}{5}x-1 = 7+\frac{4}{5}x; 3\frac{1}{2}+\frac{2}{5}x = x; 3\frac{1}{2} = \frac{3}{5}x$. For completion, $\frac{2}{3}$ of each side are added, so $x = 3\frac{1}{2}+\frac{2}{3}\cdot 3\frac{1}{2} = 5\frac{5}{6}$.

It is remarkable that in the Book of Inheritance, only in the chapter “On Restitution” and in the three problems belonging to the chapter “On Capital and Debt” is the designation of the unknown by the word *shay’* used. In the chapter “On Completion” it is regularly said instead اجعل مالا *ij‘al mālan*, “take any possession as assumed,” in other cases a number that yields simple results is proposed immediately.

It cannot be denied that different school traditions or literary sources underlie the various problem groups, and it would certainly not be without significance for the history of mathematics if one applied oneself to these inheritance problems with more zeal than hitherto.

Summarizing the result, we found as the natural and original expression for the unknown the word *māl* “possession,” as the unit of measure of given sums of money the dirham. Besides the possession or the estate, a number of other concepts can appear as unknowns, which are eliminated either on the basis of the given relations to the possession or through arbitrary assumptions. The introduction of the word *shay’* lends the mode of expression a more general character, but is fairly indifferent for the problems treated.

7 VII. Die Terminologie der quadratischen Gleichungen / The Terminology of Quadratic Equations

Geometrische Aufgaben ”uber Zusammensetzung und Verwandlung von Fl”achen werden den Griechen den ersten Anstoß zur Behandlung und L”osung quadratischer Gleichungen gegeben haben. Bei Muhammad b. M”usā finden wir die geometrische Konstruktion nur als Mittel der Veranschaulichung, als Beweis f”ur die Richtigkeit der gegebenen Regeln. Die ganze ”ubrige Darstellung aber und der Charakter der Aufgaben ist rein arithmetisch.

Wir begegnen am Anfang der Algebra den ber”uhmten Definitionen:

> وجدت الارقام التي يحتاج اليها في حساب الجبر والمقابلة
> على ثلاثة اضرب وهي اصاغ وجذور واعداد مفردة ليست
> من جذر ولا مال فاما الجذور فهو كل شيء فيه نفسه
> مضروب من الواحد والذي فوقه من الاعداد والذي دونه
> من الكسور واما المال فكل ما حصل من جذر في نفسه
> واما العدد المفرد فهو كل عدد يتكلم به ليس فيه جذر ولا
> مال

Nach Rosen: “I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, roots, squares, and simple numbers relative to neither root nor square. — A root is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending. — A square is the whole amount of the root multiplied by itself. — A simple number is any number which may be pronounced without reference to root or square. — A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, “squares are equal to roots”, or “squares are equal to numbers”, or “roots are equal to numbers”.”

W”ortlich: “Und ich fand die Zahlen, deren man beim Rechenverfahren der Erg”anzung und Ausgleichung bedarf, gem”aß drei Arten, und zwar sind es Wurzeln (*gud”ur*) und Verm”ogen (*amwāl*) und absolute Zahl (*adad mufrad*), die nicht bezogen wird auf eine Wurzel und nicht auf ein Verm”ogen. Die Wurzel (*al-gidr*) nun, zu ihr (geh”ort) alles, was mit sich selbst multipliziert ist von der Eins, und was ”uber ihr (der Eins) ist von den Zahlen, und was unter ihr ist von den Br”uchen. Und das Verm”ogen (*al-māl*) ist alles, was sich ergibt (ansammelt, *i0̆11ftamā’a*) aus der mit sich selbst multiplizierten Wurzel. Und die absolute Zahl ist alles von der Zahl, was ausgesprochen wird ohne Beziehung auf eine Wurzel und ein Verm”ogen.”

Geometrical problems on the composition and transformation of surfaces will have given the Greeks the first impulse toward the treatment and solution of quadratic equations. In Muḥammad b. Mūsā we find geometrical construction only as a means of illustration, as proof of the correctness of the given rules. The entire remaining presentation, however, and the character of the problems is purely arithmetical.

We encounter at the beginning of the Algebra the famous definitions:

وجدت الارقام التي يحتاج اليها في حساب الجبر والمقابلة
على ثلاثة اضرب وهي اصاغ وجذور واعداد مفردة ليست
من جذر ولا مال فاما الجذور فهو كل شيء فيه نفسه
مضروب من الواحد والذي فوقه من الاعداد والذي دونه
من الكسور واما المال فكل ما حصل من جذر في نفسه
واما العدد المفرد فهو كل عدد يتكلم به ليس فيه جذر ولا
مال

According to Rosen: “I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, roots, squares, and simple numbers relative to neither root nor square. — A root is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending. — A square is the whole amount of the root multiplied by itself. — A simple number is any number which may be pronounced without reference to root or square. — A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, ‘squares are equal to roots,’ or ‘squares are equal to numbers,’ or ‘roots are equal to numbers.”’

Literally: “And I found the numbers needed in the calculation procedure of completion and balancing to be of three kinds, namely roots (*gudūr*) and possessions (*amwāl*) and absolute number (*‘adad mufrad*), which is not referred to a root and not to a possession. The root (*al-jidhr*) now, to it belongs everything that is multiplied with itself, from one, and what is above it of numbers, and what is below it of fractions. And the possession (*al-māl*) is everything that results (accumulates, *ijtama‘a*) from the root multiplied with itself. And the absolute number is everything of number that is pronounced without reference to a root and a possession.”

“Unter diesen drei Arten (gibt es) nun, was einander gleich ist, n”amlich wie wenn du sagst “Verm”ogen sind gleich Wurzeln” und “Verm”ogen sind gleich einer Zahl”, und “Wurzeln sind gleich einer Zahl”.”

Eine Anmerkung Rosens sagt uns noch, daß unter dem Wort *root* die erste Potenz der Unbekannten zu verstehen sei. Das Wort *square*, lesen wir anderw”arts (S. 50, Note *), *is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.*

Mit diesen von ihm selbst aufget”urmten Schwierigkeiten ist Rosen daher gen”otigt, sich jedesmal in einer Fußnote zu entschuldigen, wenn er vern”unftigerweise *māl* mit *number* ”ubersetzt (S. 51, 54, 58, 60, 61, 62, 63) oder gar *square-root* an die Stelle von *square* setzen muß (S. 53, 62, 64).

Sollen wir nun wirklich glauben, daß der Sch”opfer der arabischen Terminologie einen solchen Wirrwarr von Begriffen auf das Wort *māl* geh”auft h”atte? Sollte es nicht m”oglich sein, aus dem Labyrinth einen Ausweg zu finden?

Wir werden methodisch richtig verfahren, wenn wir zun”achst die Bedeutung des neu hinzugekommenen Terminus *gidr* kl”arstellen. Bis zum Beweis des Gegenteils werden wir annehmen d”urfen, daß das Wort einen klaren und eindeutigen Sinn hat. Dieser Sinn muß derart sein, daß er widerspruchslos durch den ganzen Text hindurch best”atigt wird.

Nun ist gewiß, daß der Zusammenhang zwischen *māl* und *gidr* ebensogut durch $y \cdot y = x^2$ als durch $\sqrt{y} \cdot \sqrt{y} = x$ in unsern algebraischen Zeichen ausgedr”uckt werden kann; von der Definition aus ist also keine Entscheidung zu gewinnen.

Wer trotz aller Schwierigkeiten nicht davon abl”aßt, *māl* mit *Quadrat* zu ”ubersetzen, wird mit Rosen entgegen dem Wortsinn *gidr* als “erste Potenz” deuten m”ussen; er kommt aber dann zu einer terminologischen Schwierigkeit, denn eine “erste” Potenz ist f”ur Altertum und Mittelalter ein Unding.

Wer aber *māl* in seinem nat”urlichen Sinne, den es ”uberall hat, weiter anwendet, hat auch f”ur *gidr* keine Umdeutung notwendig. Nun wird *gidr* im Sinne von *Quadratwurzel* aus einer Zahl unzweifelhaft angewandt und festgelegt durch die Beispiele des Kapitels ”uber Addition und Subtraktion (S. 19). Die Wurzel aus 200 kann auch Rosen nicht in die “erste Potenz einer Unbekannten” umdeuten, und wenn er das Beispiel مائة ومال ناقص عشرون جذرا مضافة الى خمسين وعشرة جذور ناقص مالان mit “A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares” wiedergibt, so hat er entweder *māl* oder *gidr* falsch ”ubersetzt.

Ebenso verfehlt ist die ”Ubersetzung von اذا اردت ان تضعف جذر كل مال معلوم او مجهول معنى مضاعفته ان تضربه في اثنين فاكفيك ان تضرب اثنين في اثنين ثم في المال mit “If you require to double

“Among these three kinds there is now what is equal to one another, namely as when you say ‘possessions are equal to roots,’ and ‘possessions are equal to a number,’ and ‘roots are equal to a number.”’

A note of Rosen’s tells us further that by the word *root* the first power of the unknown is to be understood. The word *square*, we read elsewhere (p. 50, n. *), “is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.”

With these difficulties heaped up by himself, Rosen is therefore compelled to excuse himself each time in a footnote when he reasonably translates *māl* by *number* (pp. 51, 54, 58, 60, 61, 62, 63) or must even put *square-root* in place of *square* (pp. 53, 62, 64).

Are we really to believe that the creator of Arabic terminology had heaped such a confusion of concepts upon the word *māl*? Should it not be possible to find a way out of the labyrinth?

We shall proceed methodically correctly if we first clarify the meaning of the newly added term *gidr*. Until proof of the contrary, we shall be permitted to assume that the word has a clear and unambiguous sense. This sense must be such that it is confirmed without contradiction throughout the entire text.

Now it is certain that the connection between *māl* and *gidr* can be expressed in our algebraic notation just as well by $y \cdot y = x^2$ as by $\sqrt{y} \cdot \sqrt{y} = x$; from the definition, therefore, no decision is to be won.

Whoever despite all difficulties does not desist from translating *māl* as “square” will have to interpret *gidr* against the word sense as “first power” like Rosen; but he then arrives at a terminological difficulty, for a “first” power is an absurdity for antiquity and the Middle Ages.

But whoever applies *māl* further in its natural sense, which it has everywhere, has no need of reinterpretation for *gidr* either. Now *gidr* in the sense of “square root” of a number is applied without doubt and established by the examples of the chapter on addition and subtraction (p. 19). The root of 200 cannot be reinterpreted by Rosen into the “first power of an unknown” either, and when he renders the example مائة ومال ناقص عشرون جذرا مضافة الى خمسين وعشرة جذور ناقص مالان as “A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares,” he has translated either *māl* or *gidr* incorrectly.

Equally wrong is the translation of اذا اردت ان تضعف جذر كل مال معلوم او مجهول معنى مضاعفته ان تضربه في اثنين فاكفيك ان تضرب اثنين في اثنين ثم في المال by “If you require to double the root of any known or unknown square (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two and then by the square”; denn es muß heißen “jede rationale oder irrationale Wurzel aus einer Zahl”.[10]

Alle Schwierigkeiten verschwinden mit einem Schlage, wenn wir *māl*, das Verm”ogen, mit *Zahl*, oder, um einer Verwechslung mit der “absoluten Zahl” auszuweichen, mit *Zahlgr”oße*, und *gidr* mit *Wurzel* (im eigentlichen Sinne) ”ubersetzen. So und nicht anders sind die Ausdr”ucke in der Algebra des Muhammad b. M”usā zu verstehen; ob und wie sich an sie eine Weiterentwicklung anschließt, wird noch zu untersuchen sein.

the root of any known or unknown square (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two and then by the square”; for it must say “every rational or irrational root of a number.”[10]

All difficulties disappear at a stroke if we translate *māl*, the possession, by “number,” or, to avoid confusion with the “absolute number,” by “number-quantity,” and *gidr* by “root” (in the proper sense). Thus and not otherwise are the expressions in the Algebra of Muḥammad b. Mūsā to be understood; whether and how a further development connects to them remains to be investigated.

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[10] Warum Rosen seine ”Ubersetzung S. 27 in einer Note S. 192 verbessert, ohne darauf aufmerksam zu machen, daß er jetzt mit “the root of a rational or irrational number” etwas ganz anderes sagt, ist mir nicht klar geworden; noch weniger, warum er trotz solcher sich aufdr”angenden Erfahrungen hartn”ackig dabei bleibt, *māl* mit *square* wiederzugeben.

[10] Why Rosen improves his translation on p. 27 in a note on p. 192 without calling attention to the fact that he now says something entirely different with “the root of a rational or irrational number,” has not become clear to me; still less why, despite such pressing experiences, he stubbornly continues to render *māl* by *square*.

8 VIII. Zum Aufbau des Zahlensystems / On the Structure of the Number System

Die S”atze, die Muhammad b. M”usā als Einleitung den S. 61 zitierten Definitionen voranschickt, haben verschiedene Deutung erfahren. Ich wiederhole sie im Urtext und w”ortlicher ”Ubersetzung, um daran einige Bemerkungen zu kn”upfen:

واني لما نظرت فيما يحتاج اليه الناس من الحساب وجدتها
كلها الاعداد ووجدتها كلها مؤلفة من الواحد وان الواحد
يدخل في جميع الاعداد ووجدت جميع ما يقال من العدد
من الزيادة على الواحد الى عشرة زائدا بواحدة ثم تضاعف
العشرة وتثلث كما فعل بالواحد فيخرج منها العشرون والثلاثون
الى المئة ثم تضاعف المئة وتثلث كما فعل بالواحد والعشرة
الى الف ثم يعاد الف عند كل عقد الى غاية العدد

W”ortlich: “Und siehe, nachdem ich betrachtet hatte, wessen die Leute beim Rechnen bed”urfen, fand ich als Gesamtheit davon die Zahlen, und ich fand, daß die Gesamtheit der Zahlen nur zusammengesetzt ist aus der Eins, und die Eins eintretend in die Gesamtheit der Zahlen. Und ich fand die Gesamtheit dessen, was von den Zahlen ausgesprochen wird (d. i. wof”ur es besondere Zahlw”orter gibt), was ”uber die Eins hinausgeht bis zur Zehn, fortschreitend um den Betrag der Eins; dann wird die Zehn verdoppelt und verdreifacht, wie es getan wurde mit der Eins, so daß daraus die Zwanzig und Dreißig entstehen bis zur Vollendung der Hundert; hierauf wird die Hundert verdoppelt und verdreifacht, wie es getan wurde mit der Eins und mit der Zehn bis zu Tausend usw.; dann wird ebenso wiederholt die Tausend bei jedem Glied (*’uqd*) bis zum ”Außersten des Bereichs der Zahl.”

In den S”atzen ”uber die Eins ist nichts zu finden, was ”uber die elementarste Schulweisheit hinausgeht. Sie werden bekanntlich in der durch B. Boncompagni (Roma 1857) herausgegebenen lateinischen ”Ubersetzung von Muhammad b. M”usās Rechenbuch in Verbindung mit einer weiteren Bemerkung ”uber die Einheit zitiert. Cantor hat (I^3, S. 715 u. 16) die Stelle wie folgt deutsch wiedergegeben:

“Und ich habe schon in dem Buche Aldschebr und Almuk”abala, d. h. der Wiederherstellung und Gegen”uberstellung er”offnet, daß jede Zahl zusammengesetzt sei, und daß jede Zahl sich ”uber eins zusammensetze. Die Einheit also wird in jeder Zahl gefunden, und das ist es, was in einem andern Buche der Arithmetik ausgesprochen ist. Weil die Einheit Wurzel jeder Zahl und außerhalb der Zahl ist …”

Man muß aber, um den Sinn richtig zu erhalten, folgendermaßen lesen: “… daß jede Zahl sich ”uber eins zusammensetze, die Einheit also in jeder Zahl gefunden werde. — Und das ist es, was in einem andern Buche der Arithmetik ausgesprochen

The sentences that Muḥammad b. Mūsā places as introduction before the definitions cited on p. 61 have received various interpretations. I repeat them in the original text and in literal translation, in order to append some remarks:

واني لما نظرت فيما يحتاج اليه الناس من الحساب وجدتها
كلها الاعداد ووجدتها كلها مؤلفة من الواحد وان الواحد
يدخل في جميع الاعداد ووجدت جميع ما يقال من العدد
من الزيادة على الواحد الى عشرة زائدا بواحدة ثم تضاعف
العشرة وتثلث كما فعل بالواحد فيخرج منها العشرون والثلاثون
الى المئة ثم تضاعف المئة وتثلث كما فعل بالواحد والعشرة
الى الف ثم يعاد الف عند كل عقد الى غاية العدد

Literally: “And behold, after I had considered what people need in calculation, I found as the totality thereof the numbers, and I found that the totality of numbers is only composed of the one, and the one entering into the totality of numbers. And I found the totality of what is pronounced of numbers (i.e. for which there are special number words) that goes beyond one up to ten, advancing by the amount of one; then ten is doubled and tripled, as was done with one, so that twenty and thirty arise from it up to the completion of hundred; then hundred is doubled and tripled, as was done with one and with ten, up to thousand, etc.; then likewise thousand is repeated at every node (*‘uqd*) up to the utmost of the domain of number.”

In the sentences about the one there is nothing to be found that goes beyond the most elementary school wisdom. They are, as is well known, cited in the Latin translation of Muḥammad b. Mūsā’s arithmetic book edited by B. Boncompagni (Rome 1857) in connection with a further remark on unity. Cantor has (I^3, p. 715) rendered the passage in German as follows:

“And I have already revealed in the book al-Jabr and al-Muqābala, that is, of restoration and opposition, that every number is composite, and that every number is composed above one. Unity therefore is found in every number, and that is what is pronounced in another book of arithmetic. Because unity is the root of every number and is outside the number …”

But to obtain the sense correctly, one must read as follows: “… that every number is composed above one, that unity therefore be found in every number.—And that is what is pronounced in another book of arithmetic: that unity is the ist: daß die Einheit Wurzel jeder Zahl und außerhalb der Zahl ist …” Das Zitat aus der “Algebra” geht nur bis “gefunden werde”.

Enestr”om hat sich durch die freie ”Ubersetzung Rosens irreleiten lassen, wenn er das ”Aquivalent der Stelle weiter unten in “every number which may be expressed from one to ten, surpasses the preceding by one unit” usw. sucht. Das folgende St”uck aber ist ein klares und unzweideutiges Zitat aus einem “andern Buche ”uber Arithmetik”. Was f”ur ein Buch dies ist und ob Muhammad b. M”usā damit auf ein von ihm selbst verfaßtes Buch ”uber spekulative Arithmetik anspielt, ist die Frage.

Nach Enestr”om m”ußte man annehmen, daß Cantor den Muhammad b. M”usā zum Verfasser eines solchen Buches gemacht h”atte. Davon ist aber doch nirgends die Rede; Cantor schließt nur, daß, wer so schrieb, “in der Zahlenlehre der Neupythagor”aer wohl geschult sein mußte”, und “daß dem Verfasser dar”uber Kenntnisse zu Gebote standen, welche unmittelbar oder mittelbar auf Nikomachos, vielleicht auch auf Theon von Smyrna, der am deutlichsten betont hat, die Einheit sei keine Zahl, zur”uckgehen”.

Welche Bewandtnis es in Wahrheit mit der Stelle hat, erkennt man aber sofort, wenn man sie in ihrem gesamten Umfang beizieht. Sie lautet:

> *Et iam patefeci in libro algebre et almucabolah, id est restaurationis et oppositionis, quod universus numerus sit compositus, et quod universus numerus componatur super unum. Unum ergo invenitur in universo numero. Et hoc est quod in alio libro arithmetice dicitur: Quia unum est radix universi numeri, et est extra numerum. Radix numeri est, quare per eum invenitur omnis numerus. Extra numerum vero est, quare invenitur per se, idest absque alio aliquo numero. Reliquus autem numerus sine uno inveniri non potest. Cum enim unum dicis, inventione sui non indiget alio numero. Reliquus autem numerus indiget [indiget] uno: quare non potes dicere duo vel tria, nisi precedat unum. Nichil aliud est ergo numerus, nisi unitatum collectio; et hoc quod diximus non potes dicere duo vel tria, nisi precedat unum: non de voce diximus, ut ita dicam, set de re. Non enim possunt esse duo vel tria, si unum auferatur. Vnum vero potest esse absque secundo vel tercio. Igitur nichil aliud sunt duo, nisi unius duplicitas vel geminatio: et similiter tria nichil aliud sunt, nisi eiusdem unitatis triplicatio: sic de reliquo numero intellige. Set nunc redeamus ad librum. Inveni, inquid algorizmi, omne quod*

root of every number and outside the number …” The quotation from the “Algebra” extends only to “be found.”

Eneström has allowed himself to be misled by Rosen’s free translation when he seeks the equivalent of the passage further below in “every number which may be expressed from one to ten, surpasses the preceding by one unit,” etc. But the following piece is a clear and unambiguous quotation from an “other book on arithmetic.” What book this is and whether Muḥammad b. Mūsā is alluding thereby to a book on speculative arithmetic composed by himself is the question.

According to Eneström one would have to assume that Cantor made Muḥammad b. Mūsā the author of such a book. But that is nowhere mentioned; Cantor only infers that whoever wrote thus “must have been well trained in the number theory of the Neopythagoreans,” and “that the author had at his disposal knowledge of this which goes back directly or indirectly to Nicomachus, perhaps also to Theon of Smyrna, who most clearly emphasized that unity is not a number.”

What the true state of affairs is with this passage one recognizes immediately when one takes it in its entire scope. It reads:

Et iam patefeci in libro algebre et almucabolah, id est restaurationis et oppositionis, quod universus numerus sit compositus, et quod universus numerus componatur super unum. Unum ergo invenitur in universo numero. Et hoc est quod in alio libro arithmetice dicitur: Quia unum est radix universi numeri, et est extra numerum. Radix numeri est, quare per eum invenitur omnis numerus. Extra numerum vero est, quare invenitur per se, idest absque alio aliquo numero. Reliquus autem numerus sine uno inveniri non potest. Cum enim unum dicis, inventione sui non indiget alio numero. Reliquus autem numerus indiget [indiget] uno: quare non potes dicere duo vel tria, nisi precedat unum. Nichil aliud est ergo numerus, nisi unitatum collectio; et hoc quod diximus non potes dicere duo vel tria, nisi precedat unum: non de voce diximus, ut ita dicam, set de re. Non enim possunt esse duo vel tria, si unum auferatur. Vnum vero potest esse absque secundo vel tercio. Igitur nichil aliud sunt duo, nisi unius duplicitas vel geminatio: et similiter tria nichil aliund sunt, nisi eiusdem unitatis triplicatio: sic de reliquo numero intellige. Set nunc redeamus ad librum. Inveni, inquid algorizmi, omne quod

> *potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX.*

Hier ist doch mit H”anden zu greifen, daß die ganze Stelle ein Einschub des ”Ubersetzers, oder wie wir jetzt richtiger sagen werden, des Bearbeiters des Rechenbuchs ist, und daß wir Muhammad b. M”usā f”ur diese scholastische Haarspalterei nicht verantwortlich machen d”urfen.

So deutlich sich der Zusatz als solcher durch Inhalt und sprachliche Form verr”at, so deutlich ist das folgende wieder ein Zitat aus Muhammad b. M”usās eigener Algebra, das sich zu einer umst”andlicheren Behandlung des Gegenstandes erweitert. Man braucht nur Text und ”Ubersetzung der zweiten H”alfte des S. 71 mitgeteilten Zitats neben den Text des Rechenbuchs zu stellen, um sich davon zu ”uberzeugen:

> *Inveni omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX. De inde ponuntur .X. in loco unius, et duplicantur .X. ac triplicantur, quemadmodum factum est de uno, fiuntque ex eorum duplicatione .XX., ex triplicatione .XXX., et ita usque ad .XC. Post hec redeunt .C. in loco unius, et duplicantur ibi atque triplicantur quemadmodum factum est de uno et .X. efficiunturque ex eis .CC. et .CCC. et cetera usque in .DCCCC. Rursum ponuntur mille in loco unius; et duplicando et triplicando, ut diximus, fiunt ex eis .II. milia, et .III. et cetera usque in Infinitum numerum, secundum hunc modum. — Et inveni quod operati sunt yndi ex his differentiis. Quarum prima est differentia unitatum etc.*

Die ”ubereinstimmenden Stellen sind, soweit m”oglich, durch Kursivdruck hervorgehoben. Vergleichung im einzelnen zeigt, daß zwei verschiedene Betrachtungsweisen zusammengearbeitet sind. In der ”alteren Algebra betont der Verfasser den sprachlichen Aufbau des Zahlensystems, nennt also die zehn Grundzahlen und die Grenzzahlen 100 und 1000; die “Rechenkunst der Inder” sieht mehr auf die Klassen der Zahlen, gruppiert also nach Einern, Zehnern, Hunderten, Tausendern usw.

Richtiger w”are es, wenn es hieße: *usque in X, C, M*; denn nur dann kann man sagen *deinde ponuntur X, post hec redeunt C, rursum ponuntur M.* Am Schluß vermißt man die ”Ubersetzung von عند كل عقد *’inda kuli ’uqdān* “bei jeder Gliedzahl”. Mit dem folgenden Satz wird das Thema des Rechenbuchs, wie

potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX.

Here it is palpable that the whole passage is an insertion by the translator, or as we shall now more correctly say, by the adapter of the arithmetic book, and that we must not hold Muḥammad b. Mūsā responsible for this scholastic hair-splitting.

As clearly as the addition reveals itself as such through content and linguistic form, so clearly is what follows again a quotation from Muḥammad b. Mūsā’s own Algebra, which has expanded to a more circumstantial treatment of the subject. One needs only to place the text and translation of the second half of the quotation communicated on p. 71 beside the text of the arithmetic book in order to convince oneself of this:

Inveni omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX. De inde ponuntur .X. in loco unius, et duplicantur .X. ac triplicantur, quemadmodum factum est de uno, fiuntque ex eorum duplicatione .XX., ex triplicatione .XXX., et ita usque ad .XC. Post hec redeunt .C. in loco unius, et duplicantur ibi atque triplicantur quemadmodum factum est de uno et .X. efficiunturque ex eis .CC. et .CCC. et cetera usque in .DCCCC. Rursum ponuntur mille in loco unius; et duplicando et triplicando, ut diximus, fiunt ex eis .II. milia, et .III. et cetera usque in Infinitum numerum, secundum hunc modum. — Et inveni quod operati sunt yndi ex his differentiis. Quarum prima est differentia unitatum etc.

The corresponding passages are indicated by italics as far as possible. Comparison in detail shows that two different points of view have been worked together. In the older Algebra, the author emphasizes the linguistic structure of the number system, thus naming the ten basic numbers and the limit numbers 100 and 1000; the “calculation art of the Indians” looks more to the classes of numbers, thus grouping according to units, tens, hundreds, thousands, etc.

It would be more correct if it said: *usque in X, C, M*; for only then can one say *deinde ponuntur X, post hec redeunt C, rursum ponuntur M.* At the end one misses the translation of عند كل عقد *‘inda kulli ‘uqdān* “at every node-number.” With the following sentence, the theme of the arithmetic book, es von Muhammad b. M”usā zu Anfang durch *Cum vidi ssem yndos constituisse .IX. literas in universo numero suo* angeschlagen ist, wieder aufgenommen.

as struck by Muḥammad b. Mūsā at the beginning through *Cum vidissem yndos constituisse .IX. literas in universo numero suo*, is taken up again.

9 IX. Die Namen der arabischen Ziffern / The Names of the Arabic Digits

In einer seiner Anfragen macht Enestr”om darauf aufmerksam, daß in der *Pratica d’aritmetica* von Ghaligai (Ausgabe von 1548) die ”Ubersetzung einer arabischen Algebra erw”ahnt werde, in welcher “die 7 Terme Geber, Elmechel, Elchal, Elchelif, Elfazial, Buram und Eltermen” vork”amen.

Nach einer Mitteilung von Suter enthalte die Algebra des Alkhwarizmi jedenfalls nicht die Terme Elchelif, Buram und Eltermen; es sei daher w”unschenswert, die jetzt bekannten arabischen Traktate ”uber Algebra daraufhin zu untersuchen, ob sie die 7 Terme enthielten. Enestr”om sagt leider nicht, wie Suter die ”ubrigen 4 Terme aufl”ost. Man sieht aber leicht, daß außer dem selbstverst”andlichen *geber* noch *elmechel* = *al-muqābala* ist und daß *elfazial* aus *al-faṣl* (der Unterschied), *elchal* durch Verlesen aus *al-māl* (das Verm”ogen) entstanden sein kann.

Ich w”urde vielleicht noch wagen, *eltermen* mit *al-ikmāl* (die Vervollst”andigung) zusammenzustellen, in *elchelif* k”onnte *al-ḥaḏf* (die Wegnahme) verborgen sein, aber *buram* ist hoffnungslos verderbt und trotzt jedem Versuch der Wiederherstellung, solange nicht die ”Ubersetzung selbst vorliegt und der Zusammenhang R”uckschl”usse auf das arabische Wort erm”oglicht.

Das Ganze ist ein lehrreiches Beispiel f”ur das, was man an Verst”ummelungen zu erwarten hat, wenn arabische W”orter in lateinische Schrift umgesetzt und unverstanden weiter und weiter ”uberliefert wurden.

Wesentlich einfacher liegt die Aufgabe der Wiederherstellung des arabischen Ausdrucks, wenn die verst”ummelten W”orter in lateinischen Texten erscheinen. Wenn uns z. B. in einer Beschreibung des Astrolabs, die Gerbert zugeschrieben wird, das Wort *Hoto talzagad* mit den Varianten *hotetalgazat, hottottaltagath, hotaltagab, hotetalsagat, noto taltagad* begegnet, so w”urden wir ohne den lateinischen Text schwerlich imstande sein, den zugrundeliegenden arabischen Ausdruck zu finden.

Lesen wir aber danach: *id est breves lineae horarum*, so ergibt sich ohne weiteres die Aufl”osung خطوط الساعات *huṭūṭ as-sā’āt* = Stundenlinien.

Wenn uns an der gleichen Stelle die Wortgruppen *Ethehiaser* (*ethehiafer*) und *Elewiul* (*elevil, elaul*), *Aldimataser* (*aldimatafer, alhimata*) und *Athenia*, *Alhamsira* (*alhansira*) und *Atheliza* (*atheziza*), *Ethezzea* und *Arrabea*, *Ethemina* (*ethemima*) und *Alchamiza*, *Escebeha* und *Escendeliza* (*esscedezza*) begegnen, so w”aren wir wieder ratlos ohne die Bemerkung, daß diese W”orter die arabischen Namen der Stunden auf dem Stundenkreis bedeuten.

Die den W”ortern beigesetzten r”omischen Ziffern leiten auf die einfache L”osung hin, daß die Namen die Feminina der arabischen Ordinalzahlen sein m”ussen; wer w”urde aber

In one of his queries, Eneström calls attention to the fact that in the *Pratica d’aritmetica* of Ghaligai (edition of 1548) the translation of an Arabic algebra is mentioned, in which “the 7 terms Geber, Elmechel, Elchal, Elchelif, Elfazial, Buram and Eltermen” occurred.

According to a communication from Suter, the Algebra of al-Khwārizmī in any case does not contain the terms Elchelif, Buram, and Eltermen; it would therefore be desirable to investigate the Arabic treatises on algebra now known as to whether they contained the 7 terms. Eneström unfortunately does not say how Suter resolves the remaining 4 terms. But one easily sees that besides the self-evident *geber* [*al-jabr*], *elmechel* = *al-muqābala*, and that *elfazial* could have arisen from *al-faṣl* (the difference), *elchal* by misreading from *al-māl* (the possession).

I would perhaps further venture to combine *eltermen* with *al-ikmāl* (the completion), in *elchelif* could be concealed *al-ḥaḏf* (the removal), but *buram* is hopelessly corrupted and defies every attempt at restoration as long as the translation itself is not available and the context does not permit conclusions about the Arabic word.

The whole is an instructive example of what one may expect in the way of mutilations when Arabic words are converted into Latin script and handed down further and further without being understood.

The task of restoring the Arabic expression is substantially simpler when the mutilated words appear in Latin texts. If, for example, in a description of the astrolabe attributed to Gerbert, we encounter the word *Hoto talzagad* with the variants *hotetalgazat, hottottaltagath, hotaltagab, hotetalsagat, noto taltagad*, without the Latin text we would scarcely be able to find the underlying Arabic expression.

But if we read after it: *id est breves lineae horarum*, then the resolution خطوط الساعات *hu.tūt as-sāʿāt* = hour-lines follows immediately.

If at the same place we encounter the word groups *Ethehiaser* (*ethehiafer*) and *Elewiul* (*elevil, elaul*), *Aldimataser* (*aldimatafer, alhimata*) and *Athenia*, *Alhamsira* (*alhansira*) and *Atheliza* (*atheziza*), *Ethezzea* and *Arrabea*, *Ethemina* (*ethemima*) and *Alchamiza*, *Escebeha* and *Escendeliza* (*esscedezza*), we would again be helpless without the remark that these words are the Arabic names of the hours on the hour circle.

The Roman numerals placed with the words guide one to the simple solution that the names must be the feminines of the Arabic ordinal numbers; but who would suspect in ohne diesen Leitgedanken in *Escendeliza* das Wort السادسة *as-sādisa* (westarabisch *essedisa*), in *Alhamsira* das Wort العاشرة *al-ʾasira*, in *Aldimatafer* oder *Alhimata* das Wort الحادية عشرة *al-ḥādi011fa ʾaṛata*, in *Ethehiafer* das Wort الثانية عشرة *aṫ-tāni011fa ʾaṛata* vermuten?

Man erkennt nun die Verschreibung *h* statt *n* und *f* statt langem *s* in *Ethehiafer*, den Ausfall von *ha* oder *he* in *Alf(h)dimatafer* und den Wegfall von *ser* in *Alhimata*, man kann sich auch die erweiterte Form *Escend-eliza* aus dem vorangehenden *Ath-eliza* erkl”aren, und die Verwechslung von *m* und *n* in *Ethemima* begreifen, aber wodurch die Einschiebung des *m* in *Alhamsira* und *Aldimatafer* zustande kam, wird schwerlich jemand sagen k”onnen.

Ich glaubte diese Beispiele vorausschicken zu sollen, wenn ich mich einer erneuten Untersuchung der r”atselhaften Namen *igin, andras, ormis, arbas, quimas, calcis, zenis, temenias, celentis* f”ur die Ziffern zuwende, die uns von Radulph von Laon (gest. 1131) und auch in etwas ”alteren Quellen ”uberliefert sind und in der Geschichte des Rechnens eine zweifelhafte Ber”uhmtheit erlangt haben.

Anlaß zur Wiederaufnahme des Verfahrens gibt nicht so sehr die Bemerkung Cantors (I^2 S. 842, I^3 S. 896), daß er das R”atsel als immer noch nicht mit Gewißheit aufgel”ost betrachte und gern bereit sei, eine zuverl”assigere Deutung jener W”orter freudig zu begr”ußen, welche auch die Frage nach der Zeit der Entstehung endg”ultig beantworten w”urde, oder die von Cantor ”ubersehene Behandlung des Gegenstands durch E. Clive Bayley und die j”ungste Er”orterung der Frage durch N. Bubnow, als die ”Uberzeugung, daß alle Versuche als methodisch verfehlt zu betrachten sind, die die L”osung des R”atsels außerhalb der geschichtlich allein m”oglichen Verbindung des Arabischen und Lateinischen suchen.

Wir l”acheln ”uber Helmreich (1595), der in der Vorrede zu seinem Rechenbuch den großen Geometer *Algebras* in ”Agypten zur Zeit des Alexander Magnus, der da war *ein Pr”azeptor oder Vorfahrer Euklidis, des F”ursten zu Megarien*, die Algebra erfinden l”aßt; aber sollen wir annehmen, daß ein Radulph von Laon am Anfang des 12. Jahrhunderts mehr von Chald”aern und Assyrern wußte und “chald”aische” Namen f”ur die Zahlzeichen einschließlich der Null mitteilen konnte, die bei den Chald”aern in dieser Weise gar nicht existierten?

Es ist doch geradezu eine Ungeheuerlichkeit, auf Grund dieser im 12. Jahrh. auftauchenden Notiz mit E. Clive Bayley anzunehmen, “that some reminiscence, at least, of the names of the Chaldaean units may have survived also, though in a more or less corrupted form, to Neo-Pythagorian times”, dann auf Grund einer angeblichen Abstammung (!) des Arabischen und Hebr”aischen vom Altassyrischen mit Hilfe von “H”artung” und “Euphonie” aus *ʾeṭin igin* und aus *ʾeṭuʾ celentis* herzustellen,

Escendeliza the word السادسة *as-sādisa* (West Arabic *essedisa*), in *Alhamsira* the word العاشرة *al-ʿāshira*, in *Aldimatafer* or *Alhimata* the word الحادية عشرة *al-ḥādiyya ʿashrata*, in *Ethehiafer* the word الثانية عشرة *ath-thāniya ʿashrata*, without this guiding thought?

One now recognizes the miswriting of *h* for *n* and *f* for long *s* in *Ethehiafer*, the loss of *ha* or *he* in *Alf(h)dimatafer* and the dropping of *ser* in *Alhimata*; one can also explain the extended form *Escend-eliza* from the preceding *Ath-eliza*, and understand the confusion of *m* and *n* in *Ethemima*, but how the insertion of *m* in *Alhamsira* and *Aldimatafer* came about, scarcely anyone could say.

I believed these examples should be prefaced when I turn to a renewed investigation of the enigmatic names *igin, andras, ormis, arbas, quimas, calcis, zenis, temenias, celentis* for the digits, which have been transmitted to us by Radulph of Laon (d. 1131) and also in somewhat older sources, and have acquired a dubious fame in the history of calculation.

The occasion for resuming the procedure is given not so much by Cantor’s remark (I^2 p. 842, I^3 p. 896) that he considers the riddle as still not solved with certainty and would gladly welcome a more reliable interpretation of those words that would also definitively answer the question of the time of origin, or the treatment of the subject by E. Clive Bayley overlooked by Cantor and the most recent discussion of the question by N. Bubnow, as by the conviction that all attempts are to be considered methodically mistaken that seek the solution of the riddle outside the historically only possible connection of Arabic and Latin.

We smile at Helmreich (1595), who in the preface to his arithmetic book lets the great geometer *Algebras* in Egypt at the time of Alexander the Great—who was “a teacher or predecessor of Euclid, the prince of Megara”—invent algebra; but are we to assume that a Radulph of Laon at the beginning of the 12th century knew more about Chaldeans and Assyrians and could communicate “Chaldean” names for numeral signs including zero, which among the Chaldeans did not exist in this way at all?

It is downright an absurdity, on the basis of this note appearing in the 12th century, to assume with E. Clive Bayley “that some reminiscence, at least, of the names of the Chaldaean units may have survived also, though in a more or less corrupted form, to Neo-Pythagorean times,” then on the basis of an alleged descent (!) of Arabic and Hebrew from Old Assyrian with the help of “hardening” and “euphony” to derive *igin* from *ʾe.thin* and *celentis* from *ʾe.thuʾ*, and when all this still und wenn dies alles noch nicht zur Erkl”arung der von Radulph gegebenen W”orter ausreicht, das “neupythagoreische” *andras* und *ormis* aus tamilischem *irandu* und *munru* abzuleiten!

Auch ”uber die phantastischen Versuche, *igin* = ἡ γυνή, *andras* = ἀνδρίας(?), *ormis* = ῾ρμίς zu setzen und *calcis* mit καλ́της, *celentis* mit ἀμήλυντς oder σελήνη in Zusammenhang zu bringen, kann ich wohl hinweggehen. Soll bei der ganzen Untersuchung ”uberhaupt etwas herauskommen, so d”urfen wir nicht, wie es bisher meist geschah, die ”Uberlieferung als unantastbar und einwandfrei betrachten; wir d”urfen auch an den ”uberlieferten Ausdr”ucken nicht herumkorrigieren, bis sie sich irgendeiner Theorie f”ugen; vielmehr k”onnen wir die Frage nur so stellen: Wenn die genannten W”orter die Namen der Ziffern sein sollen, wie konnte es kommen, daß einige dieser Namen mit denen der arabischen Zahlw”orter stimmen, andere aber nicht die geringste ”Ahnlichkeit mit den arabischen Zahlw”ortern aufweisen, denen sie angeblich entsprechen sollen?

Zun”achst haben wir die Annahme zu pr”ufen, daß die W”orter tats”achlich die arabischen Namen f”ur die Ziffern darstellen. Wir k”onnen uns gegen Radulphs Chald”aer auf zwei ann”ahernd gleichaltrige Zeugnisse berufen, die die Namen ausdr”ucklich als arabisch bezeichnen. Das eine wird von Smith-Karpinski in *The Hindu-Arabic Numerals* S. 118 erw”ahnt; ein gewisser Turchill schreibt um 1200: *has autem figuras … a pytagoricis habemus, nomina vero ab arabibus.*

Das andere ist von Enestr”om nach einer Handschrift des 11./12. Jahrhunderts mitgeteilt und lautet: *Versus de nominibus caracterum arabicorum ad abacum pertinentium.* Aber solche Best”atigungen sind ”uberfl”ussig geworden, nachdem die von A. Bj”ornbo unternommene, durch H. Suter vollendete Ausgabe der astronomischen Tafeln des Muhammad b. M”usā den direkten Beweis erbracht hat, daß in den lateinischen ”Ubersetzungen jener Zeit mit den Chald”aern nicht die alten Babylonier, sondern die Araber gemeint sind.

Zum erstenmal werden die Chald”aer bei Athelhard im 7. Kapitel *De alwacat id est medio planetarum etc.* erw”ahnt, wo es heißt: *Item nota, quod secundum Galdeos IIII milia passus cameli miliare faciunt, atque XXXIII miliaria et tertia, id est thuld, in terra gradus, in coelo dimidius, unde totus terrae circulus XXIIII milia miliaria continet etc.*

Diese Meile, von der $66\frac{2}{3}$ auf einen Grad gehen sollen, ist nach Suter die von den arabischen Geographen und Astronomen in ihren Angaben gew”ohnlich gebrauchte Meile zu 4000 Ellen, und *thuld* nat”urlich das arabische Wort f”ur Drittel (s. o. S. 54). Noch klarer tritt der Sachverhalt im 10. Kapitel *De inventione loci Saturni, Jovis et Martis* hervor, wo der Bearbeiter den Zusatz macht: *… quodque inde surget, a Caldeis elhacil, a nobis obtentum vel centrum ultimum potest dici.* Es

does not suffice to explain the words given by Radulph, to derive the “Neopythagorean” *andras* and *ormis* from Tamil *irandu* and *munru*!

Also over the fantastic attempts to set *igin* = ἡ γυνή, *andras* = ἀνδρίας(?), *ormis* = ῾ρμίς, and to connect *calcis* with καλ́της, *celentis* with ἀμήλυντς or σελήνη, I can well pass over. If the whole investigation is to yield anything at all, we may not, as mostly happened hitherto, regard the tradition as inviolable and flawless; we may not correct the transmitted expressions until they fit some theory; rather, we can pose the question only thus: If the words named are to be the names of the digits, how could it come about that some of these names agree with those of the Arabic number words, while others show not the slightest similarity to the Arabic number words to which they allegedly correspond?

First we have to test the assumption that the words actually represent the Arabic names for the digits. Against Radulph’s Chaldeans we can appeal to two approximately contemporary witnesses that explicitly designate the names as Arabic. The one is mentioned by Smith-Karpinski in *The Hindu-Arabic Numerals*, p. 118; a certain Turchill writes around 1200: *has autem figuras … a pytagoricis habemus, nomina vero ab arabibus.*

The other is communicated by Eneström after a manuscript of the 11th/12th century and reads: *Versus de nominibus caracterum arabicorum ad abacum pertinentium.* But such confirmations have become superfluous after the edition of the astronomical tables of Muḥammad b. Mūsā undertaken by A. Björnbo and completed by H. Suter has furnished direct proof that in the Latin translations of that period by “Chaldeans” not the ancient Babylonians but the Arabs are meant.

For the first time the Chaldeans are mentioned by Adelard in the 7th chapter *De alwacat id est medio planetarum etc.*, where it says: *Item nota, quod secundum Galdeos IIII milia passus cameli miliare faciunt, atque XXXIII miliaria et tertia, id est thuld, in terra gradus, in coelo dimidius, unde totus terrae circulus XXIIII milia miliaria continet etc.*

This mile, of which $66\frac{2}{3}$ are to go to a degree, is according to Suter the mile of 4000 cubits customarily used by Arab geographers and astronomers in their data, and *thuld* naturally the Arabic word for third (see above p. 54). Still more clearly the state of affairs emerges in the 10th chapter *De inventione loci Saturni, Jovis et Martis*, where the adapter adds: “*… quodque inde surget, a Caldeis elhacil, a nobis obtentum vel centrum ultimum potest dici.*” It concerns the word *al-ḥā.sil*, the result;

handelt sich um das Wort *al-ḥāṣil*, das Ergebnis; wir sehen also, daß diese Chald”aer arabisch sprechen; zum ”Uberfluß hat die Handschrift O auch f”ur *a Caldeis* das Wort *arabice* eingesetzt.

Auch aus dem vorhin beigezogenen, von Bubnow herausgegebenen *Liber de astrolabio* in den *Opera Gerberti* l”aßt sich der Gebrauch des Wortes Chald”aer f”ur Araber belegen. Wir finden S. 127 als ”Ubergang zu Kap. IV *Ad invenendum Nadair solis* die Worte: *Nam Chaldaica astutia seu calculando seu scribendo ita progreditur*; am Schluß des Kap. XVII steht die Bemerkung: *Sunt praeter has duae Ganamalgurab, Alcasal vel Alhimech in Gentauro, quibus Ghaldaei satis ad discernendas horas utuntur*; in dem Bruchst”uck einer andern arabischen Schrift ”uber das Astrolab erw”ahnt der Bearbeiter S. 372 *Ghaldaicas gentilogias, qui omnem humanam vitam astrologicis attribuunt ratiocinationibus.*

Daß auch *siriace* f”ur *arabice* steht, erhellt aus einer Note bei Bubnow, *Opera Gerberti* S. 125, wo die *latina sollertia* die arabischen Namen der 28 Mondstationen in unglaublicher Weise ”ubersetzt hat.

Um so verwunderlicher ist es, daß Bubnow S. 72 seines Werkes ”uber die arithmetische Selbst”andigkeit der europ”aischen Kultur ”uber die Ziffern schreibt: “Kein Abacist des X.–XI. Jahrhunderts deutet irgendwo an, daß diese Zeichen etwas Neues oder gar Arabisches w”aren. Radulf lebte zu einer Zeit, wo der arabische Einfluß immer st”arker wurde, und dennoch ergeht er sich in Vermutungen ”uber ihre Herkunft aus Assyrien, ganz wie wir, und nennt sie chald”aisch.”

Mit dem Nachweis, daß die Chald”aer der mittelalterlichen ”Ubersetzer die zeitgen”ossischen Araber sind, wird wilder Spekulation ein f”ur allemal der Boden entzogen. Die Namen m”ussen aus dem Arabischen erkl”art werden, und wir haben uns nur den Hergang ihrer Aufnahme und Verst”ummelung in lateinischen Handschriften zurechtzulegen.

Es ist leicht zu sehen, daß nicht nur f”ur die Lateiner, sondern auch f”ur die Araber in der Anfangszeit die Notwendigkeit bestand, den Zahlwert der 9 oder 10 Zeichen durch beigeschriebene Worte auszudr”ucken. Wir werden also ”uberall da, wo die indische Rechenkunst gelehrt wurde, Zusammenstellungen erwarten d”urfen, wie sie z. B. in der wiederholt erw”ahnten Tabelle der Iḫān a-ṣafā’ auftreten.

Einer ”ahnlichen Zusammenstellung m”ussen unsere W”orter entstammen, und wir haben nun, indem wir von den unanfechtbaren Entsprechungen ausgehen, zu untersuchen, ob eine gewisse Gesetzm”aßigkeit in der Umschreibung der W”orter wahrzunehmen ist.

Wenn ich die arabischen W”orter in der heute ”ublichen Umschrift unter die mittelalterlichen W”orter setze, so scheint sich zun”achst wenig Aussicht auf Erfolg zu er”offnen:

we see thus that these Chaldeans speak Arabic; to top it off, manuscript O has also substituted the word *arabice* for *a Caldeis.*

Also from the *Liber de astrolabio* in the *Opera Gerberti* cited above, edited by Bubnow, the use of the word Chaldeans for Arabs can be documented. We find on p. 127 as transition to chap. IV *Ad invenendum Nadair solis* the words: *Nam Chaldaica astutia seu calculando seu scribendo ita progreditur*; at the end of chap. XVII stands the remark: *Sunt praeter has duae Ganamalgurab, Alcasal vel Alhimech in Gentauro, quibus Ghaldaei satis ad discernendas horas utuntur*; in the fragment of another Arabic writing on the astrolabe the adapter mentions on p. 372 *Ghaldaicas gentilogias, qui omnem humanam vitam astrologicis attribuunt ratiocinationibus.*

That *siriace* also stands for *arabice* is clear from a note by Bubnow, *Opera Gerberti* p. 125, where the *latina sollertia* has translated the Arabic names of the 28 lunar mansions in an unbelievable manner.

All the more surprising is it that Bubnow writes on p. 72 of his work on the arithmetic independence of European culture about the digits: “No abacist of the 10th–11th centuries indicates anywhere that these signs were something new or even Arabic. Radulf lived at a time when Arabic influence was growing ever stronger, and yet he indulges in conjectures about their origin from Assyria, just as we do, and calls them Chaldean.”

With the proof that the Chaldeans of the medieval translators are the contemporary Arabs, the ground is cut from under wild speculation once and for all. The names must be explained from Arabic, and we have only to work out the course of their adoption and mutilation in Latin manuscripts.

It is easy to see that not only for the Latins but also for the Arabs in the early period there was a necessity to express the numerical value of the 9 or 10 signs by appended words. We shall thus be able to expect everywhere that the Indian calculation art was taught, arrangements such as appear, for example, in the repeatedly mentioned table of the Ikhwān al-Ṣafā’.

Our words must stem from a similar arrangement, and we have now, starting from the incontestable correspondences, to investigate whether a certain regularity is to be perceived in the transcription of the words.

If I place the Arabic words in the now customary transcription under the medieval words, at first little prospect of success seems to open:

igin	andras	ormis	arbas	quimas	calcis	zenis	temenias	celentis	sipos
wāḥid	iŏ11fnān	talāta	arba’a	ḥamsa	sitta	sab’a	tamānija	tis’a	ṣifr

Es ist aber erstens zu beachten, daß die kurzen Vokale bedeutungslos sind und von dem, der die Umschrift erstmals herstellte, schon nach Gutd”unken gesetzt werden konnten, zweitens, daß f”ur die Umschrift der arabischen Konsonanten keinerlei Norm bestand, also gleiche Konsonanten mit verschiedenen, verschiedene Konsonanten mit gleichen lateinischen Buchstaben geschrieben werden konnten, wenn nur das Lautbild ungef”ahr stimmte, und drittens, daß nur das vokallose Buchstabengerippe der W”orter, und selbst dieses mit der M”oglichkeit von falschen oder mangelhaft gesetzten diakritischen Punkten und Verwechslung ”ahnlicher Zeichen in das Vergleichungsverfahren eingestellt werden darf.

Kommt noch dazu, daß bei der weiteren ”Ubertragung der lateinischen Urschrift unz”ahlige Fehler und Varianten entstehen konnten und entstanden sind, so scheint jeder Willk”ur T”ur und Tor ge”offnet zu sein.

Dennoch ist jedem Arabisten bekannt und muß jedem Mathematiker, der sich an derartige R”atselfragen macht, bekannt sein, daß hier kein Wort zuviel gesagt ist und mit all diesen M”oglichkeiten gerechnet werden muß, wenn man bei dem Versuch einer L”osung des R”atsels zum Ziel kommen will.

Hiernach h”atten wir unter Zuziehung der bekannt gewordenen Varianten und mit Hervorhebung einiger Gesetzm”aßigkeiten folgende Reihe von Entsprechungen:

(1) *igin, ingin — wāḥid*

(2) *andras — iŏ11fnān* (?)

(3) *ormis, hormis, armis — talāta* (?)

(4) *arbas — arba’a*

(5) *quimas, quinas — ḥamsa*

(6) *calcis, caletis, calctis, caltis — sitta* (?)

(7) *zenis, zencis — sab’a* oder *sitta*

(8) *temenias, temeinas, zemenias — tamānija*

(9) *celentis, scelentis, zcelentis — tis’a* (?)

Man sieht bei einer Vergleichung mit unserer modernen Umschrift sofort, daß das Wort *wāḥid* fehlt, daß in beiden Reihen nur ein Wort vorkommt, das auf *n* endigt: *igin — iŏ11fnān* (oder *iŏ11fnen*), und daß die Endung *-at* ”uberall durch *as* oder *is*, also *t* durch *s* wiedergegeben ist.

igin	andras	ormis	arbas	quimas	calcis	zenis	teme
wāhid	ithnān	thalātha	arba‘a	.hamsa	sitta	sab‘a	than

But first, it is to be noted that the short vowels are meaningless and could have been set at discretion by whoever first made the transcription; second, that for the transcription of the Arabic consonants no norm existed, so that equal consonants could be written with different, different consonants with equal Latin letters, if only the sound-image corresponded approximately; and third, that only the vowelless skeleton of the letters of the words, and even this with the possibility of false or inadequately placed diacritical points and confusion of similar signs, is to be admitted into the comparison procedure.

Add to this that in the further transmission of the Latin original countless errors and variants could arise and have arisen, so that every arbitrariness seems to have been given free rein.

Nevertheless, it is known to every Arabist and must be known to every mathematician who concerns himself with such riddles, that not a word too much is said here and that one must reckon with all these possibilities if one wishes to succeed in the attempt at a solution of the riddle.

Accordingly, with the addition of the known variants and with emphasis on certain regularities, we would have the following series of correspondences:

(1) *igin, ingin — wā.hid*

(2) *andras — ithnān* (?)

(3) *ormis, hormis, armis — thalātha* (?)

(4) *arbas — arba‘a*

(5) *quimas, quinas — .hamsa*

(6) *calcis, caletis, calctis, caltis — sitta* (?)

(7) *zenis, zencis — sab‘a* or *sitta*

(8) *temenias, temeinas, zemenias — thamāniya*

(9) *celentis, scelentis, zcelentis — tis‘a* (?)

One sees on comparison with our modern transcription immediately that the word *wā.hid* is missing, that in both series only one word occurs ending in *n*: *igin — ithnān* (or *ithnān*), and that the ending *-at* is rendered everywhere by *as* or *is*, that is, *t* by *s*.

Aus den Formen unter (8) *temenias, zemenias* — *tamānija* geht hervor, daß der dem englischen *th* entsprechende Laut *t* durch *t* und *z* wiedergegeben wird. Die Form (5) *quimas* (*quinas*) ist wohl (mit spanisch-franz"osischem *qu*) als *kimas* zu lesen und entspricht dem *ḥamsa* unserer Liste.

Die Formen unter (6) und (9) liegen alle innerhalb der Variationsbreite von *talāta*: am n"achsten kommt ihm (mit *c* = *z*) *caletis*, daran schließen sich einerseits *calctis*, *calcis* und *caltis*, anderseits *celentis* und *zcelentis*. Zur Sippe von *arbas* (4) scheint auch die Reihe *ormis, hormis, armis*, ja vielleicht selbst *andras* zu geh"oren. Die erste Gleichung k"onnte aus dem Arabischen durch Verlesen entstanden sein oder die Form *andras* w"urde ein *amras* als Zwischenglied vermuten lassen.

Wir kommen so zu der Auffassung, daß die "uberlieferte Reihe nicht mehr vollst"andig ist, sondern nur die Zahlw"orter 2, 3, 4, 5, 7, 8 wiedergibt, und daß sie Doppelungen enth"alt, durch die sie in Verwirrung gekommen ist, so daß sie das Wort f"ur 3 an falscher Stelle bei 6 und 9, das Wort f"ur 4 bei 3 und 4, f"ur 2 bei 1 darbietet.

Das Wort f"ur 7 ergibt sich graphisch unmittelbar aus *zebis* f"ur *sab'a* (*zenis* f"ur *zebis*), *zencis* ist nach *calcis* verschrieben. Es kann aber ebensogut *sitta* f"ur 6 zugrund liegen. Das Wort *igin* k"onnte auch aus *aḥad* (f"ur *igīd*, eigentlich *aḥad*) entstanden sein, das f"ur 1 st"unde, doch w"are dann der Ausfall der 2 zu erkl"aren.

Das Wort f"ur die Null lautet in den Handschriften *sipos*. Man hat l"angst darauf hingewiesen, daß dies aus *sifr* verdorben sei. Die graphische M"oglichkeit liegt auf der Hand; es braucht nur صفر statt صفر, also Verdoppelung oder Verschiebung des diakritischen Punktes in der Handschrift angenommen zu werden, um das Schluß-s zu erkl"aren.

Ob es nicht trotz alledem von *σφραγις* abgeleitet werden muß und erst sp"ater durch *cifra* = *sifr* ersetzt wurde, ist eine Frage, die hier nicht im Vorbeigehen erledigt werden kann.

Auf alle F"alle scheint es angemessener, Umstellungen und Verwechslungen der unverstandenen W"orter zuzulassen oder Dubletten anzunehmen, wenn sich der Lautbestand ohne Schwierigkeit einf"ugt, als durch Gewaltmittel W"orter, die sich nun einmal nicht mit den ihnen zugeschriebenen Zahlwerten zur Deckung bringen lassen, solang umzuformen, bis sie nicht mehr wiederzuerkennen sind.

Wie leicht solche Verwechslungen und falschen Eintragungen zustande kommen, lehrt ein Blick auf die von Friedlein der Boetiusausgabe beigegebene Faksimiletafel. Nur die Codd. *e*, *n* und *w* zeigen die 9 W"orter richtig "uber den 9 Kolumnen, nur bei *n* sind sie in einer Zeile geschrieben; bei *e* sind sie von *arbas* bis *sipos* in 2, bei *n* s"amtlich in 2 oder 3 St"ucke zerlegt und untereinander gesetzt.

From the forms under (8) *temenias, zemenias* — *thamāniya* it emerges that the *t* sound corresponding to English *th* is rendered by *t* and *z*. The form (5) *quimas* (*quinas*) is presumably (with Spanish-French *qu*) to be read as *kimas* and corresponds to the *.hamsa* of our list.

The forms under (6) and (9) all lie within the variation range of *thalātha*: closest to it (with *c* = *z*) is *caletis*, to which are attached on one side *calctis*, *calcis*, and *caltis*, on the other *celentis* and *zcelentis*. To the family of *arbas* (4) the series *ormis, hormis, armis* also seems to belong, and perhaps even *andras* itself. The first equation could have arisen from the Arabic through misreading, or the form *andras* would suggest an *amras* as intermediate link.

We thus arrive at the view that the transmitted series is no longer complete, but represents only the number words 2, 3, 4, 5, 7, 8, and that it contains doublings through which it has fallen into confusion, so that it presents the word for 3 at the wrong place at 6 and 9, the word for 4 at 3 and 4, for 2 at 1.

The word for 7 follows graphically immediately from *zebis* for *sabʿa* (*zenis* for *zebis*); *zencis* is miscopied after *calcis*. But *sitta* for 6 could just as well underlie it. The word *igin* could also have arisen from *a.had* (for *igīd*, actually *a.had*), which would stand for 1, but then the loss of 2 would have to be explained.

The word for zero in the manuscripts is *sipos*. It has long been pointed out that this is corrupted from *.sifr*. The graphical possibility lies at hand; one need only assume صفر instead of صفر, that is, doubling or displacement of the diacritical point in the manuscript, in order to explain the final *s*.

Whether it must nonetheless be derived from *σφραγις* [sphragis] and was only later replaced by *cifra* = *.sifr* is a question that cannot be settled here in passing.

In any case, it seems more appropriate to allow displacements and confusions of the uncomprehended words, or to assume duplicates, when the sound substance fits without difficulty, than by violent means to reshape words that simply cannot be brought into agreement with the numerical values ascribed to them, until they are no longer recognizable.

How easily such confusions and false entries come about is shown by a glance at the facsimile plate appended by Friedlein to the Boethius edition. Only the codices *e*, *n*, and *w* show the 9 words correctly over the 9 columns; only in *n* are they written in one line; in *e* they are from *arbas* to *sipos* in 2, in *n* all in 2 or 3 pieces split and placed under one another.

Die ”alteste Handschrift *ny* (cod. Vatican. 3123, saec. X, Friedlein S. 372) scheint ”uberhaupt noch Niemand n”aherer Betrachtung wert gehalten worden zu haben, und doch zeigt sie am besten, wie die Verwirrung entstanden sein kann. Hier sind die W”orter wie in arabischen Handschriften vertikal gestellt.

Das Wort *an-dras* steht ”uber 2 und 3, *ormis* ”uber 4, *arbas* ”uber 5, *quimas* ”uber 6, *calcis*(?) ”uber 7, *zenis* (?) ”uber 8, *zeme-nias* ”uber 9 und 0, *sce-len-tis* ”uber drei leeren F”achern; das Wort *sipos* fehlt, w”ahrend das Zeichen f”ur die Null am richtigen Platze steht.

Welche k”uhnen Hypothesen k”onnte und w”urde man aufstellen, wenn nur diese eine Handschrift bekannt w”are! Und wie mag die Vorlage beschaffen gewesen sein, aus der sie hervorging?

Man mag sich zu der ganzen Sache stellen, wie man will, es ist und bleibt schwer zu begreifen, daß in einer Zeit, in der doch wahrlich Gelegenheit vorhanden war, sich ”uber die richtigen Formen der arabischen Zahlw”orter zu verl”assigen, dieser Rattenk”onig von Verwechslungen und Unformen weitergegeben werden konnte.

Man kann sich bei so allgemeinen Dingen wie Zahlw”ortern nicht wie bei medizinischen oder astronomischen Begriffen auf sachliche Schwierigkeiten berufen. Es bleibt fast nur die Annahme, daß eine an sich schlechte handschriftliche ”Uberlieferung schon fr”uh an einen Ort verschleppt wurde, wo keine Spur von arabischen Kenntnissen vorhanden war, wo man also die Worte nicht als arabische Zahlw”orter erkannte, sondern f”ur die fremden Namen der Zahlzeichen hielt. Von jenem Urtext aus muß sich dann das literarische Kuriosum weiter verbreitet haben; ein literarisches Kuriosum ohne alle sachliche Bedeutung bleibt diese ganze Einf”uhrung arabischer Namen f”ur Zahlzeichen, die ebensogut lateinisch benannt werden konnten und benannt wurden.

Es w”are auch an die Analogie der ”Ubertragung der semitischen Buchstabennamen zu den Griechen zu erinnern. Wie dort die fremden Zeichen mit ihren unverstandenen Namen ”ubertragen wurden, so werden in der ”Ubergangszeit die arabischen Zahlw”orter als “Namen” f”ur die Ziffern gebraucht worden sein.

Von dem Verfasser der bekannten Merkverse (Cantor I^3, S. 893) kann man sagen, daß er weder arabisch verstand noch von der arabischen Herkunft der Worte etwas wußte; er h”atte sie doch sonst unm”oglich unbesehen beibehalten k”onnen. Gleichwohl glaube ich selbst in diesen Versen, wo sie nicht reines Wortgeklingel sind, Ankl”ange an arabische Zusammenstellungen ”uber die “Eigenschaften der Zahlen” nachweisen zu k”onnen, m”ochte sie also nicht wie Cantor nur als Ged”achtnisverse zur Einpr”agung der fremdartigen

The oldest manuscript *ny* (Cod. Vatican. 3123, 10th cent., Friedlein p. 372) seems not yet to have been held worthy of closer consideration by anyone at all, and yet it shows best how the confusion could have arisen. Here the words are set vertically, as in Arabic manuscripts.

The word *an-dras* stands over 2 and 3, *ormis* over 4, *arbas* over 5, *quimas* over 6, *calcis*(?) over 7, *zenis*(?) over 8, *zeme-nias* over 9 and 0, *sce-len-tis* over three empty compartments; the word *sipos* is missing, while the sign for zero stands in the correct place.

What bold hypotheses could and would one construct if only this one manuscript were known! And what might the exemplar have been like from which it derived?

One may take whatever stand one wishes on the whole matter, it is and remains hard to understand that in a time when truly there was opportunity to ascertain the correct forms of the Arabic number words, this rat’s nest of confusions and deformities could be handed down.

With such general things as number words one cannot, as with medical or astronomical concepts, appeal to substantive difficulties. There remains almost only the assumption that a bad manuscript tradition in itself was early transported to a place where no trace of Arabic knowledge was present, where one therefore did not recognize the words as Arabic number words, but held them for foreign names of numeral signs. From that original text the literary curiosity must then have spread further; a literary curiosity without any substantive significance remains this entire introduction of Arabic names for numeral signs, which could just as well be named in Latin and were named in Latin.

One should also recall the analogy of the transfer of Semitic letter names to the Greeks. Just as there the foreign signs were transmitted with their uncomprehended names, so in the transitional period the Arabic number words will have been used as “names” for the digits.

Of the author of the well-known mnemonic verses (Cantor I^3, p. 893) one can say that he neither understood Arabic nor knew anything of the Arabic origin of the words; he could not possibly have retained them uncritically otherwise. Nevertheless, I believe I can demonstrate in these verses, where they are not pure word-jingle, echoes of Arabic compilations on the “properties of numbers”; I would thus not regard them, like Cantor, merely as mnemonic verses for impressing the foreign words.

W”orter betrachten.

Man findet z. B. bei den Iḥān a-̣safā’ (ed. Bombay I, 29–31) einen solchen Abschnitt ”uber die Eigenschaften der Zahl; darin heißt die 1 *Wurzel (aḷ)* und Ursprung der Zahlen, die 2 ist die erste eigentliche Zahl, die 3 die erste unpaarige Zahl, die 4 die erste Quadratzahl (*’adad mag̣d”ur*), die 5 die erste Kreiszahl (wegen $5 \cdot 5 = 25$, $5 \cdot 25 = 125$), die 6 die erste vollst”andige Zahl (*’adad tāmm*), die 7 die erste vollkommene Zahl (*’adad kāmil*), die 8 die erste W”urfel- oder K”orperzahl, die 9 das erste Quadrat einer unpaarigen Zahl usw.

Stellen wir dieser Aufz”ahlung die Verse gegen”uber:

Ordine primigeno sibi nomen possidet Igin.
Andras ecce locum previndicat ipse secundum.
Ormis post numerus non compositus sibi primus.
Denique bis binos succedens indicat Arbas.
Significat quinos ficto de nomine Quimas.
Sexta tenet calcis perfecto munere gaudens.
Zenis enim digne septeno fulget honore.
Octo beatificos Temenias exprimit unus
Terque notat trinum Celentis nomine rithmum —,

so erkennt man aus den kursiv gedruckten Worten sofort die vorhandenen Parallelen.

Unverst”andlich sind mir nur die seligmachenden Acht und das *ficto de nomine* bei F”unf; doch k”onnte dies noch ein Anklang an die arabische Einteilung der Zahlwortformen sein, jenes an die 8 Seligpreisungen erinnern. Vielleicht lassen sich n”aherliegende lateinische Vorbilder auffinden; daß alle diese Dinge auf den Gedankenkreis der *Theologumena Arithmeticae* zur”uckgehen, ist selbstverst”andlich.

Ich bin mit dieser Untersuchung zu Ende. Ob das n”uchterne Ergebnis allgemein befriedigen wird, weiß ich nicht; daß es methodisch auf festen F”ußen steht, wird kaum zu bestreiten sein. H”atte man philologisch-kritische Grunds”atze stets vor Augen gehabt, so w”are das ganze Aufgebot von Gelehrtheit und Scharfsinn zur Aufdeckung alter chald”aischer oder pythagor”aischer Weisheit ”uberfl”ussig gewesen. Der Zweck dieser Untersuchung w”are erreicht, wenn dadurch wenigstens weiteren Phantastereien f”ur k”unftig ein Ziel gesetzt w”urde.

One finds, for example, in the Ikhwān al-Ṣafā’ (ed. Bombay I, 29–31) such a section on the properties of number; therein 1 is called “Root (a.sl)” and origin of numbers, 2 is the first proper number, 3 the first odd number, 4 the first square number (*‘adad majdhūr*), 5 the first circular number (because $5 \cdot 5 = 25$, $5 \cdot 25 = 125$), 6 the first complete number (*‘adad tāmm*), 7 the first perfect number (*‘adad kāmil*), 8 the first cube or solid number, 9 the first square of an odd number, etc.

Setting the verses against this enumeration:

Ordine primigeno sibi nomen possidet Igin.
Andras ecce locum previndicat ipse secundum.
Ormis post numerus non compositus sibi primus.
Denique bis binos succedens indicat Arbas.
Significat quinos ficto de nomine Quimas.
Sexta tenet calcis perfecto munere gaudens.
Zenis enim digne septeno fulget honore.
Octo beatificos Temenias exprimit unus
Terque notat trinum Celentis nomine rithmum —,

one recognizes immediately from the words printed in italics the parallels that exist.

Only the “salvation-bringing eights” and the *ficto de nomine* for five are incomprehensible to me; but the former could still be an echo of the Arabic classification of number-word forms, the latter a recollection of the 8 Beatitudes. Perhaps closer Latin models can be found; that all these things go back to the circle of ideas of the *Theologumena Arithmeticae* is self-evident.

I am at the end of this investigation. Whether the sober result will satisfy generally, I do not know; that it stands methodically on firm feet will hardly be disputable. Had philological-critical principles always been kept before one’s eyes, the whole deployment of erudition and acumen to uncover ancient Chaldean or Pythagorean wisdom would have been superfluous. The purpose of this investigation would be achieved if thereby at least further fantasies are set a limit for the future.

10 X. Das Kapitel von den Gesch”aften / The Chapter on Commercial Transactions

Die Aufgaben zweiten Grades, die Muhammad b. M”usā im ersten Abschnitt seiner Algebra behandelt, sind rein formaler Natur und tragen nichts zu dem Zwecke bei, dem nach dem Vorwort (vgl. oben S. 5) das Werk dienen soll. Auf die eigentlichen Gesch”aftsaufgaben, auf das Gesch”aftsrechnen, kommt der Verfasser erst in dem باب المعاملات *bāb al-muāmalāt*, dem Kapitel von den Gesch”aften, zu sprechen.

Es hat ob seines d”urftigen mathematischen Inhalts bisher kaum Beachtung gefunden und wird bei Cantor I^3, S. 726 mit den Worten gekennzeichnet: “Indisch ist auch wohl die nur uneigentlich der Algebra zugeteilte Regel de tri, welche in der Fortsetzung von Alchwarizmis Werke auftritt und ”ahnlich bei griechischen Schriftstellern uns nicht bekannt ist”.

Die geschichtliche Bedeutung des Abschnitts wird sofort eine wesentlich andere, wenn man diesem Satze die bestimmtere Form gibt: “Durchaus indisch ist nach Inhalt und Form das Kapitel von den Gesch”aften”, und wenn man sich vergegenw”artigt, daß hier die M”oglichkeit vorliegt, Muhammad b. M”usās Arbeitsweise und pers”onliches Verdienst um die Popularisierung des indischen Rechnens an einem einwandfreien Beispiel zu pr”ufen.

Griechische Herkunft des Kapitels ist ausgeschlossen: die ganze mathematische ”Uberlieferung weiß nichts von Aufgaben nach einer Regel de tri trotz hoher Ausbildung der Lehre von den Proportionen. Erst sp”at, um die Mitte des 14. Jahrhunderts, tauchen bei Nikolaus Rhabdas von Smyrna Aufgaben ”uber “politische Arithmetik”, d. i. b”urgerliches Rechnen auf, die mittels Regel de tri gel”ost sind.

Noch sp”ater — in das 15. Jahrhundert — sind die von J. L. Heiberg ver”offentlichten byzantinischen Texte zu setzen, in denen wir den Ausspruch finden: *ἡ δϵ τῶν τριῶν μεταβλὴ ἡ τῆς λγιστικῆς μαγεια ἐστί, καλῶς φρασιν˙ παλαὶς λ́γς*.

Wie alt dieses “Wort” auch sein mag, es ist sicherlich eher arabischen oder italienischen als griechischen Ursprungs. Auch das von Hultsch beigebrachte Scholion zu Platons *Charmides* 165E, in dem “schließlich” als Zweck der Logistik angegeben wird, daß sie den Bed”urfnissen des Alltagslebens diene, um brauchbare Vertr”age ”uber Mein und Dein, ”uber Soll und Haben, ”uber Erbschaftsteilungen usw. abzuschließen, schl”agt der im *Charmides* selbst gegebenen rein theoretischen Definition ins Gesicht, und beweist nichts f”ur Anwendung der Regel de tri.

Im Gegensatz zu dem Versagen griechischer ”Uberlieferung hinsichtlich der Regel de tri k”onnen wir bei den Indern eine nicht abbrechende, nach Form und Bezeichnungsweise feste ”Uberlieferungskette von Aryabhatta im 5. bis zu Bhāskara

The second-degree problems that Muḥammad b. Mūsā treats in the first section of his Algebra are purely formal in nature and contribute nothing to the purpose that, according to the preface (cf. above p. 5), the work is meant to serve. The author comes to the actual business problems, to business calculation, only in the باب المعاملات *bāb al-muʿāmalāt*, the chapter on commercial transactions.

Because of its meager mathematical content it has hitherto hardly received attention, and is characterized by Cantor I^3, p. 726, with the words: “Indian also is presumably the rule of three, which is only improperly assigned to the Algebra, appears in the continuation of al-Khwārizmī’s work, and is not known to us similarly among Greek writers.”

The historical significance of the section becomes immediately essentially different if one gives this sentence the more definite form: “Thoroughly Indian in content and form is the chapter on commercial transactions,” and if one bears in mind that here the possibility exists to test Muḥammad b. Mūsā’s working method and personal merit for the popularization of Indian calculation on an unobjectionable example.

Greek origin of the chapter is excluded: the entire mathematical tradition knows nothing of problems according to a rule of three despite the high development of the doctrine of proportions. Only late, around the middle of the 14th century, do problems on “political arithmetic,” i.e. practical calculation, solved by means of the rule of three, appear with Nikolaos Rhabdas of Smyrna.

Still later—in the 15th century—are to be placed the Byzantine texts published by J. L. Heiberg, in which we find the saying: *ἡ δϵ τῶν τριῶν μεταβλὴ ἡ τῆς λγιστικῆς μαγεια ἐστί, καλῶς φρασιν˙παλαὶς λ́γς*. [“The rule of three is the magic of arithmetic,” as the old saying well expresses it.]

However old this “word” may be, it is certainly more of Arabic or Italian than of Greek origin. Also the scholion to Plato’s *Charmides* 165E adduced by Hultsch, in which “finally” as the purpose of *logistikē* is stated that it should serve the needs of everyday life in order to conclude useful contracts about mine and thine, about debit and credit, about inheritance divisions, etc., contradicts the purely theoretical definition given in the *Charmides* itself, and proves nothing for the application of the rule of three.

In contrast to the failure of Greek tradition regarding the rule of three, we can follow among the Indians an unbroken chain of transmission, fixed in form and mode of designation, from Āryabha.ta in the 5th to Bhāskara in the 12th century.

im 12. Jahrhundert verfolgen. Unter den ”außerst knapp gehaltenen Regeln, die Rodet in seinen *Lecons de calcul d'Aryabhata* ”ubersetzt hat, findet sich die folgende:

XXVI. *Dans la “regle de trois”, le “résultat” (ou “fruit” phalam) multiplié par la “demande” et divisé par le “type” donne le “résultat de la demande”.*

Die Sanskritworte umschreibt und ”ubersetzt Rodet mit *trairā0̆15bikam* = règle de trois; *phalam* = fruit, résultat, revenu; *icchā* = demande, quantité pour laquelle on demande; *pramāṇa* = type, quantité type; dazu kommt *icchāphala* = résultat de la demande.

Ganz denselben Ausdr”ucken begegnen wir bei Brahmagupta in der ersten H”alfte des 7. Jahrhunderts; der Wortlaut der Regel ist nach Colebrooke (a. a. O., S. 283):

10. *In the rule of three, argument, fruit and requisition [Zusatz des ”Ubersetzers “are names of the terms”]: the first and last terms must be similar. Requisition, multiplied by the fruit, and divided by the argument, is the produce.*

Ein Kommentator f”ugt hinzu, daß der mittlere Term anders (dissimilar) benannt ist und daß die Regel sich auf ganze Zahlen bezieht; wenn Br”uche vorkommen, sollen sie alle auf denselben Nenner gebracht werden. Als Beispiel wird von ihm gegeben: “Jemand gibt weg 108 K”uhe in 3 Tagen; wieviel K”uhe bringt er fort in einem Jahr und einem Monat?” Feststellung: Tage 3. K”uhe 108. Tage 390. Antwort: K”uhe 14 040.

Die Fachausdr”ucke bed”urfen besonders deshalb einer n”aheren Erl”auterung, weil nach ihnen die arabischen gebildet worden sind. Die Ausdrucksweise wird sich f”ur das Deutsche ”andern, je nachdem wir eine Zins- oder eine Warenrechnung zugrunde legen. Bei einer Zinsrechnung heißt es z. B.: “Was tr”agt ein Kapital von 3500 Mk., wenn 100 Mk. 4 Mk. Zins tragen?”

Hier ist die Zahl 100 = *pramāṇa*, *le type*, *the argument*, also die Norm, das Grundkapital; die Zahl 4 = *phalam*, *le résultat*, *le revenu*, *the fruit*, also sein Ertr”agnis — wir sagen genau wie vom Baume oder Acker, daß das Kapital Zinsen “tr”agt”. Die Zahl 3500 ist *icchā*, *la demande*, *the requisition*, f”ur uns das Kapital; die gesuchte Zahl ist also *icchāphala*, *le résultat de la demande*, f”ur uns der Zins, das Ertr”agnis des Kapitals.

Bei Warenrechnungen werden wir *pramāṇa* mit Grundmenge oder Grundmaß, *phalam* mit Grundpreis oder Grundwert, *icchā* mit Angebot oder Gesamtmenge, *icchāphala* mit Gesamtpreis oder Gesamtwert ”ubersetzen k”onnen, da es sich meist darum handeln wird, aus dem f”ur eine kleinere Menge gegebenen Marktpreis den Betrag f”ur eine gr”oßere Menge zu berechnen.

Nach diesen Vorbereitungen k”onnen wir uns dem Kapitel von den Gesch”aften in Muhammad b. M”usās Algebra zuwenden.

Among the extremely brief rules that Rodet has translated in this *Lecons de calcul d'Aryabhata*, the following is found:

XXVI. “In the ‘rule of three,’ the ‘result’ (or ‘fruit,’ *phalam*) multiplied by the ‘demand’ and divided by the ‘type’ gives the ‘result of the demand.”

Rodet paraphrases and translates the Sanskrit words as: *trairāśikam* = règle de trois; *phalam* = fruit, résultat, revenu; *icchā* = demande, quantité pour laquelle on demande; *pramā.na* = type, quantité type; to which is added *icchāphala* = résultat de la demande.

We encounter quite the same expressions in Brahmagupta in the first half of the 7th century; the wording of the rule according to Colebrooke (op. cit., p. 283) is:

10. “In the rule of three, argument, fruit and requisition [translator's addition “are names of the terms”]: the first and last terms must be similar. Requisition, multiplied by the fruit, and divided by the argument, is the produce.”

A commentator adds that the middle term is named differently (dissimilar) and that the rule refers to whole numbers; if fractions occur, they should all be brought to the same denominator. As an example he gives: “Someone gives away 108 cows in 3 days; how many cows does he take away in a year and a month?” Statement: Days 3. Cows 108. Days 390. Answer: Cows 14,040.

The technical terms require closer explanation especially because the Arabic terms were formed after them. The mode of expression will vary for German, depending on whether we take as basis an interest or a goods calculation. In an interest calculation one says, for example: “What interest does a capital of 3500 marks bear, if 100 marks bear 4 marks interest?”

Here the number 100 = *pramā.na*, *le type*, *the argument*, thus the norm, the principal; the number 4 = *phalam*, *le résultat*, *le revenu*, *the fruit*, thus its yield—we say, exactly as of tree or field, that the capital “bears” interest. The number 3500 is *icchā*, *la demande*, *the requisition*, for us the capital; the sought number is thus *icchāphala*, *le résultat de la demande*, for us the interest, the yield of the capital.

In goods calculations we can translate *pramā.na* as basic quantity or basic measure, *phalam* as basic price or basic value, *icchā* as offer or total quantity, *icchāphala* as total price or total value, since it mostly concerns calculating from the market price given for a smaller quantity the amount for a larger quantity.

After these preparations we can turn to the chapter on commercial transactions in Muḥammad b. Mūsā's Algebra.

Es ist notwendig, Text und ”Ubersetzung Rosens wiederzugeben, um ”uber die darin enthaltenen Versehen zur Klarheit zu kommen.

Der theoretische Teil des Kapitels lautet wie folgt (Rosen, S. 48):

> واعلم ان جميع معاملات الناس من بيع وشراء وصرف
> واجارة وغير ذلك من ضربين باربعة اعداد يتكلم بها السائل
> وهي المسعار والسعر والمتمعن والثمن فعدد المسعار يقابل
> عدد الثمن وعدد السعر يقابل عدد المتمعن وهذه الاربعة
> اعداد ثلاثة منها دائما معلومة معروفة وواحد منها مبهوت
> وهو الذي في كلام القائل كم وبه يسال والعمل في هذا ان
> تنظر الى الثلاثة الاعداد المعلومة فلا بد من ان يكون اثنان
> منهما كل واحد منهما مقابل صاحبه فاضرب المعلومين
> المقابلين العددين كل واحد منهما في صاحبه وما حصل
> فاقسمه على العدد المعلوم الاخير الذي صاحبه مجهول
> وما خرج لك فهو العدد المجهول الذي به يسال السائل وهو
> مقابل العدد الذي قسمت عليه

Nach der ”Ubersetzung von Rosen (S. 68): “On mercantile transactions. You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure, is inversely proportionale to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity.”

“Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says “how much?” and it is the object of the question. — The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor.”

Die Worte des Textes, die den ersten vier kursivgedruckten Worten der ”Ubersetzung entsprechen, sind:

1. المسعار *al-musā’ār* = the measure
2. السعر *al-sī’r* = the price
3. المتمعن *al-mutamannan* = the quantity
4. الثمن *at-taman* = the sum

Auch wer des Arabischen unkundig ist, erkennt aus der Umschreibung, daß im Arabischen Wortpaare vorliegen, die

It is necessary to reproduce Rosen’s text and translation in order to clarify the errors contained therein.

The theoretical part of the chapter reads as follows (Rosen, p. 48):

> واعلم ان جميع معاملات الناس من بيع وشراء وصرف
> واجارة وغير ذلك من ضربين باربعة اعداد يتكلم بها السائل
> وهي المسعار والسعر والمتمعن والثمن فعدد المسعار يقابل
> عدد الثمن وعدد السعر يقابل عدد المتمعن وهذه الاربعة
> اعداد ثلاثة منها دائما معلومة معروفة وواحد منها مبهوت
> وهو الذي في كلام القائل كم وبه يسال والعمل في هذا ان
> تنظر الى الثلاثة الاعداد المعلومة فلا بد من ان يكون اثنان
> منهما كل واحد منهما مقابل صاحبه فاضرب المعلومين
> المقابلين العددين كل واحد منهما في صاحبه وما حصل
> فاقسمه على العدد المعلوم الاخير الذي صاحبه مجهول
> وما خرج لك فهو العدد المجهول الذي به يسال السائل وهو
> مقابل العدد الذي قسمت عليه

According to Rosen’s translation (p. 68): “On mercantile transactions. You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure, is inversely proportionale to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity.”

“Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says ‘how much?’ and it is the object of the question.—The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor.”

The words of the text corresponding to the first four italicized words of the translation are:

1. المسعار *al-musā‘ār* = the measure
2. السعر *al-sī‘r* = the price
3. المتمعن *al-mutamannan* = the quantity
4. الثمن *at-thaman* = the sum

Even one unacquainted with Arabic recognizes from the transliteration that in Arabic there are pairs of words formed von der gleichen Wurzel gebildet sind, w”ahrend Rosen vier verschiedene und dabei recht unbestimmte Ausdr”ucke zur ”Ubersetzung verwendet. Das ist schon ein methodischer Mißgriff; dazu kommt aber noch, daß die Ausdr”ucke 3. und 4. im arabischen Text falsch gestellt sind, wenn wir die ”Ubersetzung anerkennen, oder umgekehrt, daß Rosens ”Ubersetzung falsch ist, wenn wir den Text gelten lassen.

Um zum Verst”andnis der Ausdr”ucke zu gelangen, m”ussen wir von *sī’r* und *taman* ausgehen. Wenn wir *as-sī’r* mit “die Sch”atzung” ”ubersetzen, so ist *al-musā’ār* “das Gesch”atzte”, die gesch”atzte Ware; in gleicher Weise ist nach *at-taman* “der Wert” das Partizipium *al-mutamannan* “das Gewertete” gebildet.

Wir m”ussen also *at-taman* = the sum, *al-mutamannan* = the quantity setzen. Die Worte *musā’ār* und *sī’r* geben in freier, dem arabischen Sprachgeist angepaßter Weise das indische *pramāṇa* (argument) und *phalam* (fruit) wieder, w”ahrend in *mutamannan* und *taman* die ”Aquivalente von *icchā* (demand) und *icchāphala* (fruit of the demand) vorliegen.

Vom mathematischen Standpunkt ist darin, daß die arabische Bezeichnung schon grammatisch die Gleichartigkeit der Gr”oßen in *musā’ār* und *mutamannan* einerseits, *sī’r* und *taman* andererseits zum Ausdruck bringt — wir w”urden die Worte “Ware” und “Preis” ”uber den Ansatz stellen —, ein terminologischer Fortschritt zu erblicken; der Inder muß ausdr”ucklich bemerken, daß “argument and requisition must be of like denomination; the fruit, which is of a different species, stands between them”.

Von irgendeiner Anspielung auf die griechische Lehre von den Proportionen ist bei Muhammad b. M”usā weder im Kapitel von den Gesch”aften noch sonstwo das geringste zu lesen. Damit wird auch Rosens k”uhner Versuch hinf”allig, das Wort *mubāin* mit *inversely proportionale* zu ”ubersetzen.

Wir m”ussen bei der Grundbedeutung “abgesondert, getrennt” oder der sich daraus leicht ergebenden Bedeutung “sich gegen”uberstehend” stehenbleiben, also wie es bei dem soeben angef”uhrten indischen Zitat der Fall ist, nur den rein ”außerlichen Gesichtspunkt der Anordnung der vier Zahlen im Rechenschema bei der ”Ubersetzung im Auge haben. Das Wort ist weiter nichts als ein Synonym zu dem Partizip *0̆117ālā’ muqābil* “entgegengesetzt”, das uns an die *muqābala*, *oppositio* erinnert.

Auch der Ausdruck *two notions* am Anfang der ”Ubersetzung ist irref”uhrend. Muhammad b. M”usā spricht nicht von zwei Begriffen, sondern von ضربين d. i. zwei “Gesichtspunkten” und meint damit — wie aus einem der Beispiele klar hervorgeht —, daß entweder nach dem “Wert” oder nach dem “Gewerteten” gefragt sein kann.

Legt man bei der Wiedergabe des Arabischen mehr Gewicht auf photographische Treue, als auf elegantes Umschiffen von Klippen und Schwierigkeiten, so w”are der Text etwa wie

from the same root, while Rosen uses four different and quite indeterminate expressions for the translation. This is already a methodological blunder; but in addition, expressions 3 and 4 are incorrectly placed in the Arabic text if we accept the translation, or conversely, Rosen’s translation is wrong if we let the text stand.

To arrive at an understanding of the expressions, we must start from *sīʿr* and *thaman*. If we translate *as-sīʿr* as “the valuation,” then *al-musāʿār* is “the thing valued,” the appraised goods; in the same way, after *at-thaman* “the value,” the participle *al-mutamannan* “the thing weighted/valued” is formed.

We must thus set *at-thaman* = the sum, *al-mutamannan* = the quantity. The words *musāʿār* and *sīʿr* render in free manner, adapted to the Arabic linguistic spirit, the Indian *pramā.na* (argument) and *phalam* (fruit), while in *mutamannan* and *thaman* the equivalents of *icchā* (demand) and *icchāphala* (fruit of the demand) are present.

From the mathematical standpoint, in the fact that the Arabic designation already grammatically expresses the homogeneity of the quantities in *musāʿār* and *mutamannan* on the one hand, *sīʿr* and *thaman* on the other—we would place the words “goods” and “price” over the equation—a terminological advance is to be seen; the Indian must expressly remark that “argument and requisition must be of like denomination; the fruit, which is of a different species, stands between them.”

Of any allusion to the Greek doctrine of proportions there is in Muḥammad b. Mūsā the least to be read, neither in the chapter on commercial transactions nor anywhere else. Therewith Rosen’s bold attempt to translate the word *mubȳin* by “inversely proportionale” also becomes invalid.

We must remain with the basic meaning “separated, set apart” or the meaning easily resulting therefrom “standing opposite one another,” thus, as is the case in the Indian citation just adduced, have in view only the purely external standpoint of the arrangement of the four numbers in the calculation scheme in the translation. The word is nothing more than a synonym to the participle *ālā’ muqābil* “opposite,” which reminds us of *muqābala*, *oppositio*.

Also the expression “two notions” at the beginning of the translation is misleading. Muḥammad b. Mūsā speaks not of two concepts but of ضربين *darbayn*, that is, two “aspects,” and means thereby—as clearly emerges from one of the examples—that one can ask either for the “value” or for the “valued thing.”

If in rendering Arabic one places more weight on photographic fidelity than on elegant circumvention of cliffs and difficulties, the text would be translated approximately as follows:

folgt zu ”ubersetzen:

“Wisse, daß die Gesch”afte der Menschen — sie (geh”oren) alle zu dem Kauf und dem Verkauf und dem Austausch und der Miete und anderem dergleichen — nach zwei Gesichtspunkten mit vier Zahlen, welche der Fragende ausspricht, n”amlich dem Gesch”atzten und der Sch”atzung und dem Gewerteten und dem Wert. Die Zahl, die das Gesch”atzte ist, steht gegen”uber der Zahl, die der Wert ist, und die Zahl, die die Sch”atzung ist, steht gegen”uber der Zahl, die das Gewertete ist. Und diese vier Zahlen, drei von ihnen sind stets offenkundig, bekannt, und eine von ihnen verborgen, und das ist die, welche in der Rede des Redenden “wieviel” (heißt) und nach welcher der Fragende fragt.”

“Und die Regel in diesem (ist), daß du schaust auf die drei offenkundigen Zahlen, so ist kein Ausweg, als daß zwei von ihnen, eine jede von beiden gegen”uberstehend (sind) ihrem Gef”ahrten; so multipliziere die beiden offenkundigen gegen”uberstehenden Zahlen, eine jede von ihnen beiden mit ihrem Gef”ahrten, und was sich ergibt, das teile mit der letzten offenkundigen Zahl, deren Gegen”uber unbekannt (ist); und was dir herauskommt, das ist die unbekannte Zahl, nach welcher der Fragende fragt, und sie steht gegen”uber der Zahl, mit der du geteilt hast.”

Die Verwechslung von *mutamannan* und *taman* findet sich nur im einleitenden Teil; es wird daher gen”ugen, die nun folgenden Aufgaben ohne Wiederholung des arabischen Textes in genauer deutscher ”Ubersetzung mitzuteilen.

I. Und Beispiele daf”ur unter einem Gesichtspunkt davon sind: Wenn gesagt wird “Dir 10 um 6, wieviel dir um 4?” so ist sein Wort 10 die gesch”atzte Zahl, und sein Wort “um 6” ist die Sch”atzung und sein Wort “wieviel dir” die unbekannte ist gewertete Zahl und sein Wort “um 4” ist die Zahl, die der Wert ist. Nun steht die gesch”atzte Zahl, n”amlich 10, gegen”uber der Zahl, die der Wert ist, n”amlich 4; multipliziere also die 10 in die 4, und beide sind die gegen”uberstehenden offenkundigen, dann gibt es 40; teile sie nun durch die letzte offenkundige Zahl, die die Sch”atzung ist, n”amlich 6, so gibt es $6\frac{2}{3}$, und das ist die unbekannte Zahl, die in dem Worte “wieviel” steckt, und das ist das Gewertete, und sein Gegen”uber ist die 6, die die Sch”atzung ist.

II. Und der zweite Gesichtspunkt ist das Wort dessen, der sagt “10 um 8, um wieviel des Werts 4?” und manchmal sagt einer “vier von ihnen, wieviel ihr Wert?” Die 10 nun sind die gesch”atzte Zahl und die steht gegen”uber der Zahl, die der unbekannte Wert ist, der in seinem Wort “wieviel”, und die 8 ist die Zahl, die die Sch”atzung ist, und diese steht gegen”uber der offenkundigen, die das Gewertete ist, n”amlich 4; so multipliziere die beiden gegen”uberstehenden offenkundigen Zahlen eine in die andere, n”amlich 4 in 8, so gibt es 32, und

“Know that the transactions of people—they (all belong) to buying and selling and exchange and renting and other such—[take place] according to two aspects with four numbers, which the questioner pronounces, namely the appraised thing and the valuation and the valued thing and the value. The number that is the appraised thing stands opposite the number that is the value, and the number that is the valuation stands opposite the number that is the valued thing. And these four numbers, three of them are always manifest, known, and one of them hidden, and that is [the one] which in the speech of the speaker [is called] ‘how much,’ and which the questioner asks about.”

“And the rule in this (is) that you look at the three manifest numbers, so there is no escape but that two of them, each of the two standing opposite its companion; so multiply the two manifest opposite numbers, each of the two of them with its companion, and what results, divide by the last manifest number whose opposite is unknown; and what comes out for you, that is the unknown number which the questioner asks about, and it stands opposite the number by which you have divided.”

The confusion of *mutamannan* and *thaman* is found only in the introductory part; it will therefore suffice to communicate the now following problems in exact German translation without repetition of the Arabic text.

I. And examples thereof under one aspect are: If it is said “You [have] 10 for 6, how much for you for 4?” then his word 10 is the appraised number, and his word “for 6” is the valuation and his word “how much for you” is the unknown valued number, and his word “for 4” is the number that is the value. Now the appraised number, namely 10, stands opposite the number that is the value, namely 4; multiply thus 10 into 4, and both are the opposite manifest ones, then it gives 40; divide it now by the last manifest number, which is the valuation, namely 6, so it gives $6\frac{2}{3}$, and that is the unknown number contained in the word “how much,” and that is the valued thing, and its opposite is 6, which is the valuation.

II. And the second aspect is the word of him who says “10 for 8, for how much of the value 4?” and sometimes one says “four of them, how much their value?” The 10 now are the appraised number and it stands opposite the number that is the unknown value, [contained] in his word “how much,” and 8 is the number that is the valuation, and this stands opposite the manifest one that is the valued thing, namely 4; so multiply the two opposite manifest numbers one into the other, namely 4 into 8, so it gives 32, and divide teile sie in die letzte offenkundige Zahl, die die gesch”atzte ist, n”amlich 10, so gibt es $3\frac{1}{5}$ und das ist die Zahl, die der Wert ist, und sie steht gegen”uber den 10, durch (”uber!) die du geteilt hast. Und so (sind) alle Gesch”afte der Leute und ihre Regel, so Gott d. E. will.

III. Und wenn ein Fragender fragt und sagt: Ein Tagel”ohner, den ich gemietet im Monat f”ur 10 Dirhem, arbeitet 6 Tage, wieviel ist sein Anteil? so weißt du, daß 6 Tage $\frac{1}{5}$ des Monats und daß, was sein Anteil von den Dirhem ist, (sich berechnet) gem”aß dem, was er von einem Monat gearbeitet hat. Und die Regel daf”ur ist, daß sein Wort “Monat” 30 Tage (ausmacht), und das ist das Gesch”atzte, und sein Wort “10 Dirhem” ist die Sch”atzung und sein Wort “6 Tage” ist das Gewertete und sein Wort “wieviel ist sein Anteil” ist der Wert. So multipliziere die Sch”atzung, n”amlich 10, in das Gewertete, das ihm gegen”ubersteht, n”amlich 6, das gibt 60, und teile es durch die 30, die die (letzte) offenkundige Zahl sind, so ist das das Gesch”atzte; das gibt 2 Dirhem, und das ist der Wert. Und das ist, was die Leute untereinander verhandeln vom Austausch und vom Maß und vom Gewicht.

Es verlohnt sich, noch einen Blick in die lateinische ”Ubersetzung des Kapitels (Libri a. a. O., S. 286) zu werfen. Wir finden die kritischen S”atze am Anfang vollkommen einwandfrei mit folgenden Worten wiedergegeben: *Scias quod conventiones negotiationis hominum sunt secundum duos modos cum quattuor numeris quibus interrogator loquitur. Qui sunt pretium et cipretiatum secundum positionem, et pretium et appretiatum secundum querentem. Numerus vero qui est appretiatum secundum positionem opponitur numero qui est pretium secundum querentem* ... Ebenso sp”ater: *Horum vero quattuor numerorum tres semper manifesti et noti, et unus est ignotus* ... und *Impossible est enim quin duo eorum sint quorum unusquisque suo compari est oppositus* ... etc.

Wir sehen die fr”uhere Beobachtung best”atigt, daß diese lateinische ”Ubersetzung besser ist als Rosens Wiedergabe, ja wir k”onnen sogar wieder nachweisen, daß sie einen besseren Text wiedergibt, als ihn Rosen zur Verf”ugung hatte.

Nicht nur, daß jene kritisch beanstandeten Stellen hier in ganz richtiger Folge wiedergegeben sind, auch der Text des ersten Beispiels zeigt eine Variante, die den Sinn der Aufgabe klarer wiedergibt. Denn es heißt hier nicht “10 um 6, wieviel um 4”, sondern *decem cafficii (hazf, ein bekanntes Hohlmaß) sunt pro sex dragmis: quot ergo perveniet tibi pro quattuor dragmis?*

Eine Vergleichung der indischen Regeln und Beispiele mit der Darstellung des Gegenstandes durch Muhammad b. M”usā zeigt neben aller Abh”angigkeit — diese ist ganz besonders auch in dem Fehlen eines Beweises f”ur das Verfahren zu erblicken — doch genug von pers”onlicher Eigenart.

it by the last manifest number that is the appraised one, namely 10, so it gives $3\frac{1}{5}$ and that is the number that is the value, and it stands opposite the 10 by (over!) which you have divided. And thus (are) all the transactions of people and their rule, God willing.

III. And if a questioner asks and says: A day-laborer, whom I hired for a month for 10 dirhams, works 6 days, how much is his share? then you know that 6 days are $\frac{1}{5}$ of the month and that what his share of the dirhams is (is calculated) according to what he has worked of a month. And the rule for it is that his word “month” is 30 days, and that is the appraised thing, and his word “10 dirhams” is the valuation and his word “6 days” is the valued thing and his word “how much is his share” is the value. So multiply the valuation, namely 10, into the valued thing, which stands opposite it, namely 6, that gives 60, and divide it by 30, which are the (last) manifest number, so that is the appraised thing; that gives 2 dirhams, and that is the value. And that is what people negotiate among themselves about exchange and measure and weight.

It is worthwhile to cast yet a glance into the Latin translation of the chapter (Libri, op. cit., p. 286). We find the critical sentences at the beginning reproduced completely faultlessly with the following words: *Scias quod conventiones negotiationis hominum sunt secundum duos modos cum quattuor numeris quibus interrogator loquitur. Qui sunt pretium et cipretiatum secundum positionem, et pretium et appretiatum secundum querentem. Numerus vero qui est appretiatum secundum positionem opponitur numero qui est pretium secundum querentem* ... Likewise later: *Horum vero quattuor numerorum tres semper manifesti et noti, et unus est ignotus* ... and *Impossible est enim quin duo eorum sint quorum unusquisque suo compari est oppositus* ... etc.

We see confirmed the earlier observation that this Latin translation is better than Rosen’s rendering; indeed, we can even demonstrate again that it renders a better text than Rosen had available.

Not only are those critically noted passages reproduced here in quite correct sequence, but also the text of the first example shows a variant that renders the sense of the problem more clearly. For it says here not “10 for 6, how much for 4,” but *decem cafficii (.ha.zf, a well-known dry measure) sunt pro sex dragmis: quot ergo perveniet tibi pro quattuor dragmis?*

A comparison of the Indian rules and examples with the presentation of the subject by Muḥammad b. Mūsā shows, alongside all dependence—this is especially to be seen in the lack of a proof for the procedure—yet enough of personal character.

Der Araber faßt gleich alle vier Zahlen ins Auge und gruppiert sie paarweise, w”ahrend die Inder die Regel nach den drei bekannten Zahlen benennen und die Stellung der mittleren zwischen zwei gleichbenannten besonders hervorheben; der Araber ahmt die indische Terminologie nicht sklavisch nach, sondern verbessert sie, indem er zwei Grundw”ortern zwei abgeleitete gegen”uberstellt.

Aber er beschr”ankt sich in seinen drei breit ausgef”uhrten, f”ur den elementarsten Verstand faßbar gemachten Beispielen auf die einfache und direkte Regel de tri und bleibt dadurch wesentlich hinter den Indern zur”uck.

Unter den Abhandlungen der Iḥān a-̣safā’ ist die sechste vollst”andig der Behandlung der arithmetischen und geometrischen Verh”altnisse gewidmet. Eine gewisse ”Ubersicht ihres Inhalts gewinnt man aus der ”Ubersetzung von Dieterici; nur darf man sie nicht als Quellenschrift benutzen. Der mathematische Inhalt ist wesentlich griechisch; von der Berechnung unbekannten Gr”oßen im Handel ist mehrfach die Rede (ed. Bombay I, S. 6, 11), doch wird nur *muqābala* und *mumāťala* unterschieden.

Der kennzeichnendste Satz lautet: وكل من سال عن ثمن شيء فانه لا بد ان يذكر اربعة مقادير ثلثة منها معلومة وواحدة مجهولة وبين كل مقياسين منها نسبتان نسبة مباشرة ونسبة مفاضلة “Jeder, der nach dem Preis irgend einer Sache fragt, muß notwendig vier Gr”oßen aussprechen, von denen drei bekannt sind und eine unbekannt ist; und zwischen je zwei Gr”oßen bestehen zwei Verh”altnisse, ein direktes und ein umgekehrtes.”

Von der formalen Behandlung der Aufgaben, wie sie Muhammad b. M”usā nach indischem Vorbild in die arabische Mathematik einf”uhrte, sind bei Behā ad-dīn nur noch wenige Reste geblieben. Er steht in seinem Kapitel في اخراج المجهول بالاربعة المقابلة *fī iḥrāa l-magh”ul bi-l-arba’a l-muqābala* “”Uber die Elimination der Unbekannten durch die vier proportionalen (Zahlen)” ganz auf dem Boden der Lehre von den Proportionen.

Nur in dem Beispiel, das offenbar der ersten Aufgabe bei Muhammad b. M”usā nachgebildet ist, begegnet uns auch noch dessen Terminologie: اذا قيل خمسة اطنان وثلاثة دراهم فطنان فكم “W”urde gesagt: “f”unf Pfund und drei Dirhem, zwei Pfund um wieviel?” so ist 5 Pfund das Gesch”atzte und 3 die Sch”atzung und die 2 Pfund das Gewertete, und das Gefragte ist der Wert, und das Verh”altnis (*nisba*) des Gesch”atzten zur Sch”atzung ist wie das Verh”altnis des Gewerteten zum Wert; die Unbekannte ist die vierte (Zahl), teile also das Produkt (die Fl”ache) der beiden mittleren (Zahlen), n”amlich 6, durch die erste, n”amlich 5.”

The Arab takes all four numbers into view at once and groups them in pairs, while the Indians name the rule after the three known numbers and especially emphasize the position of the middle one between two similarly named ones; the Arab does not slavishly imitate the Indian terminology, but improves it by setting two derived words against two root words.

But he restricts himself in his three broadly executed examples, made comprehensible for the most elementary understanding, to the simple and direct rule of three, and thereby remains essentially behind the Indians.

Among the treatises of the Ikhwān al-Ṣafā’, the sixth is completely devoted to the treatment of arithmetic and geometric ratios. A certain overview of its content is gained from Dieterici’s translation; only one must not use it as a source text. The mathematical content is essentially Greek; the calculation of unknown quantities in commerce is mentioned several times (ed. Bombay I, pp. 6, 11), but only *muqābala* and *mumā.t.hala* are distinguished.

The most characteristic sentence reads: وكل من سال عن ثمن شيء فانه لا بد ان يذكر اربعة مقادير ثلثة منها معلومة وواحدة مجهولة وبين كل مقياسين منها نسبتان نسبة مباشرة ونسبة مفاضلة “Everyone who asks about the price of any thing must necessarily pronounce four quantities, of which three are known and one unknown; and between any two quantities of them there are two ratios, a direct and an inverse ratio.”

Of the formal treatment of problems, as Muḥammad b. Mūsā introduced it into Arabic mathematics after Indian models, only few remnants remain in Behā ad-Dīn. He stands in his chapter في اخراج المجهول بالاربعة المقابلة *fī ikhrāj al-majhūl bi-l-arbaʿa l-muqb̄ila* “On the elimination of the unknown by the four proportional (numbers)” entirely on the ground of the doctrine of proportions.

Only in the example, which is obviously modelled after the first problem in Muḥammad b. Mūsā, do we encounter his terminology as well: اذا قيل خمسة اطنان وثلاثة دراهم فطنان فكم “If it is said: ‘five pounds and three dirhams, two pounds for how much?’ then 5 pounds is the appraised thing and 3 the valuation and the 2 pounds the valued thing, and what is asked is the value, and the ratio (*nisba*) of the appraised thing to the valuation is as the ratio of the valued thing to the value; the unknown is the fourth (number), so divide the product (the area) of the two middle (numbers), namely 6, by the first, namely 5.”

11 XI. Quellenuntersuchung / Source Investigation

Die Entwicklung der Mathematik bei den Arabern kann man nach dem bisher Gesagten etwa in folgende Zeitabschnitte einteilen: eine ”altere Periode von Muhammad b. M”usā um 825 bis Tabit b. Qurra um 900, eine mittlere von Ab”u Kāmil ʊ̆15eūcā’ b. Aslam um 900 bis al-Karăgī um 1030 und eine j”ungere Periode von ’Omar al-Hayyām (gest. 1123) bis an-Nasawī (gest. um 1030 oder 1039) und al-Qalasādī (gest. 1487).

Von Muhammad b. M”usās Zeitgenossen und n”achsten Nachfolgern ist uns wenig bekannt. Der arabische Arzt Ḥunain b. Ishāq (808–873) erz”ahlt uns von den Schwierigkeiten, die ihm bei der ”Ubersetzung mathematischer B”ucher begegneten. Er konnte weder bei den Syrern noch bei den Arabern einen Lehrer finden; von letzteren verstanden drei oder vier, hatten aber keine Zeit, ihm Unterricht zu geben, und nachdem er durch ein ”ahnliches Mißgeschick drei dieser Gelehrten verloren hatte, beschloß er, nach Alexandrien zu gehen und dort bei einem “ber”uhmten Philosophen” zu studieren.

Auf die Frage, in welcher Sprache jener den Unterricht erteilte, antwortete Ḥunain: “auf Griechisch”. Hatte er nicht “auf Griechisch” gesagt, es h”atte nicht geglaubt werden k”onnen. Weiter setzte Ḥunain hinzu: “ich blieb dort einige Zeit in meinem Bestreben, dort zu studieren. Er starb, und ich kam nach Bagdad und verstand nichts von der Mathematik”.[11]

Zur gleichen Zeit lebte in Bagdad der ”Ubersetzer und Schriftsteller Qusṭā b. L”ukā (gest. 913/14), der sowohl physikalisch-astronomische Schriften als auch eine Philosophiegeschichte verfaßte; von seinen ”Ubertragungen griechischer Werke sind uns ”uber 30 bekannt.

F”ur die Kenntnis der griechischen ”Uberlieferung kommen ferner in Betracht die Schriften des Tabit b. Qurra (gest. 901), eines Verwandten des Ḥunain b. Ishāq, und die des Verfassers des *Fihrist*, al-Nadīm, der gegen Ende des 10. Jahrhunderts in Bagdad lebte. Beide liefern uns genaue Kenntnisse von den Namen der von den Indern und Griechen geschriebenen B”ucher; insbesondere nennt der Fihrist die meisten ”Ubertragungen von den Griechen ins Syrische und Arabische mit Angabe der ”Ubersetzer.

Trotz all dieser Einzelangaben ”uber griechische Schriften f”ur Arithmetik, Geometrie, Optik und Astronomie finden wir aber von einer direkten ”Ubertragung des Nicomachus oder Theon, geschweige denn einer griechischen Algebra in arabischer Sprache keine Spur. Die Verh”altnisse liegen vielmehr gerade so, wie sie Ḥunain schildert: es gab kein arabisches Wort f”ur Mathematik, und wer etwas dar”uber

The development of mathematics among the Arabs can, according to what has been said so far, be divided approximately into the following periods: an older period from Muḥammad b. Mūsā around 825 to Thābit b. Qurra around 900, a middle period from Abü Kāmil Shuja‘ b. Aslam around 900 to al-Karajī around 1030, and a younger period from ‘Umar al-Khayyām (d. 1123) to al-Nasawī (d. around 1030 or 1039) and al-Qalasādī (d. 1487).

Of Muḥammad b. Mūsā’s contemporaries and nearest successors little is known to us. The Arab physician Ḥunayn b. Is.hāq (808–873) tells us of the difficulties that he encountered in translating mathematical books. He could find no teacher among either the Syrians or the Arabs; of the latter, three or four understood him, but had no time to give him instruction, and after through a similar misfortune he had lost three of these scholars, he resolved to go to Alexandria and study there under a “renowned philosopher.”

When asked in what language that one gave instruction, Ḥunayn answered: “in Greek.” Had he not said “in Greek,” it could not have been believed. Further Ḥunayn added: “I remained there some time in my endeavor to study there. He died, and I came to Baghdad and understood nothing of mathematics.”[11]

At the same time lived in Baghdad the translator and writer Qus.ta b. Lūkā (d. 913/14), who composed both physical-astronomical writings and a history of philosophy; of his translations of Greek works more than 30 are known to us.

For knowledge of the Greek tradition there come further into consideration the writings of Thābit b. Qurra (d. 901), a relative of Ḥunayn b. Is.hāq, and those of the author of the *Fihrist*, al-Nadīm, who lived in Baghdad toward the end of the 10th century. Both provide us with precise knowledge of the names of books written by Indians and Greeks; in particular, the *Fihrist* names most translations from Greek into Syriac and Arabic, with indication of the translators.

Despite all these individual details about Greek writings on arithmetic, geometry, optics, and astronomy, however, we find no trace of a direct transmission of Nicomachus or Theon, let alone of a Greek algebra in Arabic. The situation is rather just as Ḥunayn describes it: there was no Arabic word for mathematics, and whoever wanted to know something about it had to learn Greek or draw from a colloquial language

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[11] Medizinisches aus der Pariser Handschrift, ”ubersetzt von Ḥunain, ed. A. M”uller, K”onigsberg 1884, II, S. 21 ff.

[11] Medizinisches aus der Pariser Handschrift, ”ubersetzt von Hunain, ed. A. Müller, Königsberg 1884, II, pp. 21 ff.

wissen wollte, mußte Griechisch lernen oder aus einer Umgangssprache sch”opfen, die griechische Termini mit arabischem Wortmaterial vermischte.

Unter diesen Umst”anden wird es kaum wahrscheinlich sein, daß die Iḥān a-̣safā’ ihre Kenntnisse unmittelbar griechischen Quellen verdanken; daß sie vielmehr arabische und persische Gelehrte, ”Ubersetzer und Schriftsteller aller Art um sich versammelt hatten und eine Art inoffizieller Akademie bildeten, ist wahrscheinlich gemacht worden und w”are als ”außerst typisch f”ur jene Zeit und Stelle zu begr”ußen.

Durch die Einleitung von ’Omars Algebra werden wir ”uber die unmittelbar vorausgehende Zeit bis auf Muhammad b. M”usā genau orientiert. Denn neben dem Verfasser der ”altesten arabischen Algebra erw”ahnt ’Omar die Araber al-Harī und ’Abd al-Hamīd b. Turk, denen er wesentliche Vervollst”andigungen verdankt, und den Befehlshaber von Afghanistan, den ”agypter al-Hasan b. al-Haitham, der den von den fr”uheren Algebraikern versagten Beweis der Gleichungen der dritten Potenz mithilfe der Kegelschnitte erbrachte, ferner den Perser Ab”u al-Gud b. Muhammad und im Auslande die Griechen Menelaos, Euklid und Apollonios.

Was die Iḥān a-̣safā’ ”uber den Zahlbegriff gelehrt haben, h”atte also an griechischen Vorbildern durchaus nicht Mangel; aber die Aufgaben und Rechenverfahren, wie sie uns in der ”altesten Algebra begegnet sind, weisen nicht auf Griechenland, sondern auf Indien als Ursprungsland.

Nehmen wir an, Muhammad b. M”usā habe sich lediglich als Redaktor ”aufgef”uhrt, der die ihm zug”anglichen Quellen geschickt in ein Ganzes zusammenf”ugte, so ist doch bei einem so erstaunlichen Mangel an ”Uberlieferung vor ihm, wie ihn die Araber aufwiesen, das Verdienst eines solchen Unternehmens nicht zu untersch”atzen.

Dieses Verdienst wird aber in unglaublicher Weise erh”oht, wenn wir seine wissenschaftliche Eigenleistung dahin bestimmen, daß er die ”uberlieferte Aufgabenformulierung in den Sprachformen der griechischen Theorie der quadratischen Gleichungen dahingehend umgestaltete, daß er den Anschluß an eine Systematik erm”oglichte, die im Einklang stand mit dem Geist und den wissenschaftlichen Interessen der damaligen gebildeten Welt.

Nicht als Mathematiker, sondern als Organisator der Wissenschaften und Vermittler fremden Wissens verdient Muhammad b. M”usā einen hohen Platz in der Geschichte der Mathematik. Daß er sich nicht ”uberschatzte, sondern sich bescheiden in der Rolle eines Bearbeiters und Zusammenstellers sah, l”aßt sich nicht nur aus der Einleitung zur Algebra schließen, in der er nicht den geringsten Anspruch auf wissenschaftliche Eigenleistung erhebt, sondern auch aus den S. 3 mitgeteilten S”atzen ”uber die Eins, die er, was die begriffliche Seite angeht, ohne weiteres aus griechischen Quellen entnahm.

that mixed Greek terms with Arabic word material.

Under these circumstances it will hardly be probable that the Ikhwān al-Ṣafā’ owed their knowledge directly to Greek sources; that they rather had gathered around them Arab and Persian scholars, translators, and writers of all kinds and formed a kind of unofficial academy has been made probable and would be to be welcomed as extremely typical for that time and place.

Through the introduction of ʿUmar’s Algebra we are accurately oriented about the immediately preceding period back to Muḥammad b. Mūsā. For besides the author of the oldest Arabic algebra, ʿUmar mentions the Arabs al-Khayyāmī and ʿAbd al-Ḥamīd b. Turk, to whom he owes essential completions, and the commander of Afghanistan, the Egyptian al-Ḥasan b. al-Haytham, who furnished with the help of conic sections the proof of the equations of the third degree that was refused by the earlier algebraists, further the Persian Abü al-Jūd b. Muḥammad and abroad the Greeks Menelaus, Euclid, and Apollonius.

What the Ikhwān al-Ṣafā’ taught about the concept of number would thus have had no lack of Greek models; but the problems and calculation procedures, such as we encounter them in the oldest algebra, point not to Greece but to India as the country of origin.

If we assume that Muḥammad b. Mūsā acted merely as an editor who skillfully assembled the sources accessible to him into a whole, then with such an astonishing lack of transmission before him as the Arabs exhibited, the merit of such an undertaking is not to be underestimated.

But this merit is enhanced in an incredible way if we determine his own scientific achievement as consisting in this, that he reshaped the traditional problem formulation in the linguistic forms of the Greek theory of quadratic equations in such a way that he enabled a connection to a system that was in harmony with the spirit and scientific interests of the educated world of that time.

Not as a mathematician, but as an organizer of sciences and mediator of foreign knowledge does Muḥammad b. Mūsā deserve a high place in the history of mathematics. That he did not overestimate himself, but modestly saw himself in the role of an adapter and compiler, can be inferred not only from the introduction to the Algebra, in which he raises not the slightest claim to scientific achievement of his own, but also from the sentences about unity communicated on p. 3, which he, as far as the conceptual side is concerned, took directly from Greek sources.

Nach der Quellenlage werden wir dabei annehmen m”ussen, daß die Kenntnis der griechischen Mathematik in Arabien nur mangelhaft war und daß eine selbst”andige wissenschaftliche Produktion nicht ins Auge gefaßt werden konnte, ehe nicht die ”Ubersetzungen der griechischen Klassiker und eine gr”undliche Einarbeitung in die fremden Wissenschaften vorausgegangen waren.

Wie das bei Muhammad b. M”usā der Fall war, haben wir im Abschnitt ”uber die quadratischen Gleichungen und die Geometrie an einwandfreien Textbeispielen nachweisen k”onnen. Daß die griechischen Namen der Zahleigenschaften *quer à travers* (Carra de Vaux) in das arabische Zahlensystem ”ubergehen, ist von der tats”achlichen Kenntnis der griechischen Lehre noch verschieden; und daß die aufgabenm”aßige Behandlung der quadratischen Gleichungen eine ”Ubersetzung der geometrischen Beweise des Euklid in algebraische Sprache ist, hat Cantor I^2, S. 698; I^3, S. 720 als erster festgestellt.

Bei dem heutigen Stande der Kenntnis der *Data* des Euklid erscheint es mir aber m”oglich, die genaue ”Ubertragungsweise zu bestimmen. Dazu ist zun”achst n”otig, daß man die Hauptbeweise der Euklidischen Werkstatt mit ihren charakteristischen Ausdr”ucken den arabischen ”Ubersetzungen gegen”uberstellt.[12] Man wird dann sehen, daß in der Tat fast ”ubergangslos griechische und arabische Ausdr”ucke nebeneinandergestellt werden k”onnen.

Wir finden z. B. in *Data* 84: *Si recta datam habeat rationem ad aliam datam datam habentem rationem, erit ea data magnitudine.*

Beweis: Da n”amlich das Verh”altnis von AB zu CD gegeben ist, so ist das Verh”altnis von AB zu CD das gleiche wie das Verh”altnis einer gegebenen Gr”oße zu einer gegebenen Gr”oße. Und es ist CD in einer gegebenen Gr”oße gegeben; auch wird AB in einer gegebenen Gr”oße gegeben sein [Data 2].

Setzen wir neben diesen Beweis den des Muhammad b. M”usā f”ur die Gleichung $x^2 + bx = c$, so k”onnen wir die ”Ubereinstimmung im einzelnen feststellen. Er sagt zu dieser Gleichung: “Du nimmst die H”alfte der Wurzeln, und es ist in diesem Beispiel f”unf, und multiplizierst sie in sich selbst, das gibt 25, und du addierst es zu 39, das gibt 64. Du nimmst die Wurzel aus 64, das gibt 8, und du entfernst davon die H”alfte der Wurzeln, n”amlich 5, und 3 bleibt. Und das ist die Wurzel des Verm”ogens, und das Verm”ogen ist 9.”

Wie der griechische Mathematiker die Konstanz eines Verh”altnisses als seine Daten behandelt und aus dieser gegebenen Konstanz das Ergebnis ableitet, so geht der Araber von den

According to the state of the sources we must assume thereby that the knowledge of Greek mathematics in Arabia was only deficient and that an independent scientific production could not be contemplated before translations of the Greek classics and a thorough assimilation of the foreign sciences had preceded.

As was the case with Muḥammad b. Mūsā, we have been able to demonstrate in the section on quadratic equations and geometry by unobjectionable textual examples. That the Greek names of number properties pass “quer à travers” (Carra de Vaux) into the Arabic number system is still different from actual knowledge of the Greek doctrine; and that the treatment of quadratic equations in the form of problems is a translation of the geometrical proofs of Euclid into algebraic language, Cantor I^2, p. 698; I^3, p. 720 was the first to establish.

At the present state of knowledge of Euclid’s *Data*, however, it seems to me possible to determine the exact mode of transfer. For this it is first necessary to set the chief proofs of Euclid’s workshop with their characteristic expressions against the Arabic translations.[12] One will then see that in fact Greek and Arabic expressions can be placed side by side almost without transition.

We find, for example, in *Data* 84: “If a line has a given ratio to another [line] having a given ratio, it will be given in magnitude.”

Proof: Since namely the ratio of AB to CD is given, the ratio of AB to CD is the same as the ratio of a given magnitude to a given magnitude. And CD is given in a given magnitude; also AB will be given in a given magnitude [*Data* 2].

Setting beside this proof that of Muḥammad b. Mūsā for the equation $x^2 + bx = c$, we can establish the agreement in detail. He says for this equation: “You take half of the roots, and it is in this example five, and multiply it by itself, that gives 25, and you add it to 39, that gives 64. You take the root of 64, that gives 8, and you remove from it half of the roots, namely 5, and 3 remains. And that is the root of the possession, and the possession is 9.”

As the Greek mathematician treats the constancy of a ratio as his data and derives the result from this given constancy, so the Arab proceeds from the given numbers and shows

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[12] Zu diesem Zwecke benutze ich die Begriffe *gegeben* und *Beweis* im folgenden in dem Sinne, den sie bei den griechischen Mathematikern besitzen, nicht in dem der neueren Mathematik, wo der Begriff der Gegebenheit, von Zeuthen mit Recht als grundlegend hervorgehoben, nur in der Form der Voraussetzung eines Existenzbeweises weiterlebt.

[12] For this purpose I use the concepts *given* and *proof* in the following in the sense that they possess with the Greek mathematicians, not in that of modern mathematics, where the concept of givenness, rightly emphasized by Zeuthen as fundamental, survives only in the form of the assumption of an existence proof.

gegebenen Zahlen aus und zeigt, daß die gesuchte Gr”oße durch eine feste Rechenoperation gewonnen wird. Beide halten sich an die Form, “es wird gesucht” und “es wird gefunden”; beide vermeiden es, den Leser auf den allgemeinen Fall zu f”uhren.

Unsere Untersuchung kann in ihrem wesentlichsten Ergebnisse dahin zusammengefaßt werden, daß Muhammad b. M”usās Algebra auf indischem Stoff aufgebaut ist, den er nach griechischer Methode geordnet und in arabischer Form ausgesprochen hat.

Dieses Resultat befreit uns von der Annahme, daß die Araber die Algebra den Griechen verdanken, und gestattet es, ihnen mit vollem Recht die Wahrung der Priorit”at f”ur die ”Ubertragung und Verbreitung dieser Rechenkunst zu ”uberlassen.

Bis zu einem gewissen Grade best”atigt unsere Untersuchung auch die Nachrichten ”uber das Verh”altnis von Muhammad b. M”usā zu seinem F”orderer al-Ma’mun. Nach dem *Fihrist* soll er von dem Kalifen beauftragt worden sein, ein kurzgefaßtes Buch ”uber die Rechnung durch Erg”anzung und Ausgleichung zu schreiben; es ist zu den Problemen ”uber die Teilung von Erbschaften geworden. Der Kalif soll ihm befohlen haben, die darin enthaltene Behandlung der Aufgaben leicht und klar zu machen, und er soll zuerst gezeigt haben, wie die Beweise mit Hilfe der Figur gemacht werden.

Allerdings erweckt der aus dem Arabischen ”ubersetzte Text den Anschein, als sei er an den Z”ogen einer festen Tradition geschaffen, als habe der Verfasser bei der schriftlichen Fixierung mit allgemein verbreiteten ”Uberlieferungen gerechnet und sich ihrer bedient, um die Nachfrage nach seinem Buche zu sichern.

Daß er aber den Vorzug hatte, auf Veranlassung eines Mannes zu schreiben, der die Wissenschaft f”orderte und in dessen Gefolgschaft Mathematiker und ”Ubersetzer ”uber die verschiedensten Zweige der Wissenschaften arbeiteten, kann seinen Anschluß an die besten Quellen und die rasche Verbreitung seines Buches wesentlich erleichtert haben.

Auch in biographischer Hinsicht verdient die Tatsache Beachtung, daß Muhammad b. M”usā als Astronom eine genaue Kenntnis der damals aufgekommenen indischen Astronomie und Mathematik haben mußte, und daß die ihm von seinen Zeitgenossen zugeschriebene Kenntnis des Griechischen und Persischen seinen Arbeitsbereich ”uber das bloß ”Ubersetzen weit hinausdehnte.

Die soziale Stellung des Verfassers scheint durch die namentliche Erw”ahnung des Kalifen in der Einleitung und die f”ur die damalige Zeit v”ollig ungew”ohnliche Danksagung: يكفيني امره الذي عليّ فيه وانعامه علي *yakfīnī amru l-lāhi llaḍī ’alayya fīhi wa-in’āmu-hu ’alayya*, “seine Machtvollkommenheit gen”ugt

that the sought quantity is obtained through a fixed calculational operation. Both adhere to the form “it is sought” and “it is found”; both avoid leading the reader to the general case.

Our investigation can be summarized in its most essential result as follows: that Muḥammad b. Mūsā’s Algebra is built on Indian material, which he has ordered according to Greek method and expressed in Arabic form.

This result frees us from the assumption that the Arabs owe algebra to the Greeks, and permits us to leave them with full right the priority for the transmission and dissemination of this calculation art.

To a certain degree our investigation also confirms the reports about the relationship of Muḥammad b. Mūsā to his patron al-Ma’mun. According to the *Fihrist*, he was commissioned by the caliph to write a concise book on calculation by completion and balancing; it became the work on problems of the division of inheritances. The caliph is said to have commanded him to make the treatment of the problems contained therein easy and clear, and he is said first to have shown how the proofs are made with the help of the figure.

To be sure, the text translated from Arabic gives the impression of being made on the lines of a fixed tradition, as if the author, in the written fixation, had reckoned with generally widespread traditions and made use of them to secure the demand for his book.

But that he had the advantage of writing at the instigation of a man who promoted science and in whose circle mathematicians and translators worked on the most varied branches of knowledge, can have essentially facilitated his connection to the best sources and the rapid dissemination of his book.

Also in biographical respects the fact deserves attention that Muḥammad b. Mūsā as astronomer must have had precise knowledge of the Indian astronomy and mathematics that had then arisen, and that the knowledge of Greek and Persian attributed to him by his contemporaries extended his working sphere far beyond mere translating.

The social position of the author seems to be established by the mention of the caliph by name in the introduction and the completely unusual expression of thanks for that time: يكفيني امره الذي عليّ فيه وانعامه علي *yakfīnī amru llāhi llaḏī ‘alayya fīhi wa-in‘āmahu ‘alayya*, “his omnipotence is mir und sein, was sie an Gnaden auf mich vergossen hat”, festgelegt; doch k”onnte sie auch f”ur einen Gelehrten ziemlich unabh”angig von der Macht gelten, der den Regenten gegen”uber in geistiger Hinsicht ebenso erhaben dachte wie die Zeitgenossen des Ḥunain.

Der Text der Algebra ist ”uberlieferungsgeschichtlich von besonderem Interesse, weil er die Fortpflanzung griechischen, indischen und arabischen Wissens in einer einzigen Schrift vereinigt. Daß nicht nur Probleme und Verfahren, sondern auch der Stil ”uberliefert ist, macht Muhammad b. M”usās Werk zu einer Br”uckenschrift par excellence, die es erm”oglicht, den ”Ubergang von der antiken zur mittelalterlichen Mathematik in einer Kontinuit”at zu verfolgen, die sonst nur m”uhsam aus ”uberlieferungsgeschichtlichen Daten zu rekonstruieren w”are.

sufficient for me and his, what it has poured out in graces upon me”; but it could also apply for a scholar quite independent of power, who considered himself intellectually just as elevated vis-à-vis the ruler as did the contemporaries of Ḥunayn.

The text of the Algebra is of special interest for the history of transmission because it unites the propagation of Greek, Indian, and Arabic knowledge in a single writing. That not only problems and procedures but also style is transmitted makes Muḥammad b. Mūsā’s work a bridging text *par excellence*, which enables one to follow the transition from ancient to medieval mathematics in a continuity that would otherwise only be laboriously reconstructable from transmission-historical data.

12 XII. Schlußbetrachtung / Concluding Observations

Wie keine ernstliche Untersuchung ”uber die Entstehung der arabischen Wissenschaft ohne Ber”ucksichtigung der politischen Verh”altnisse m”oglich ist, so kann auch eine Er”orterung der Quellen der Algebra und der geschichtlichen Voraussetzungen ihrer Entwicklung den politischen Zusammenhang nicht ”ubersehen.

Die Zeit Muhammads und der ersten Kalifen war die Zeit der frohen und unbeschwerten Eroberung fremder L”ander und fremder Reicht”umer; die Beute fiel den Siegern zu, der Unterworfene zahlte Tribut, und wenn er den Islam annahm, wurde er steuerfrei. Das einfache Kriegerideal einer strengen, grob und selbstgen”ugsam lebenden Ritterschaft hatte mit der Wissenschaft noch nichts zu tun.

Als die Leidenschaften der Eroberung sich abk”uhlten, wurden die kriegerischen Neigungen durch einen prunkliebenden Luxus ”uberwuchert und die politischen wie pers”onlichen Beziehungen durch die Notwendigkeit eines friedlichen Zusammenlebens komplizierter. Innerhalb der islamischen Welt spaltete sich die politische und religi”ose Einheit, die schon durch den Tod des Propheten ins Wanken geraten war.

Die abbasidischen Kalifen suchten die gottgewollte Einheit der Glaubensgenossen durch ein geordnetes Rechtswesen und eine Kulturpolitik zu erzwingen, die den Anschluß an die ”alteren, der untergehenden sassanidischen Kultur nahestehenden Zust”ande wiederherstellte. Diese Politik konnte den Antrieb zur Wissenschaft nicht selbst hervorbringen, wohl aber g”unstige Bedingungen f”ur sie schaffen.

Denn unter den f”ur das Gedeihen der Wissenschaft g”unstigsten Bedingungen geh”ort nicht die inmitten des Alltags, sondern die den Alltag ”uberhebende Stellung des Gelehrten. Wo der Gelehrte dem Begehren und Willk”ur der M”achtigen ausgeliefert ist, kann Wissenschaft nicht gedeihen; wo er von ihnen gesch”utzt wird und imstande ist, auch die Freiheit des Denkens zu wahren, entsteht die ruhige produktive Arbeit, die zur Bl”ute der Wissenschaft f”uhrt.

F”ur Muhammad b. M”usā l”aßt sich dies Verh”altnis des Schutzes und der Freiheit des Gelehrten in dem Briefe nachweisen, den al-Ma’mun an ihn richtete und der nach dem *Fihrist* den Anlaß zu seiner algebraischen Arbeit gegeben hat. Der Kalif bittet den Gelehrten, die Geheimnisse der Rechnungswissenschaft so darzulegen, daß die mit ihr weniger vertraute Menschen sie verstehen k”onnen; er versichert ihm seiner Dankbarkeit und seines Schutzes.

Die griechische Wissenschaft war in Byzanz und Syrien bis in die Zeit der arabischen Eroberung lebendig geblieben und hatte durch ”Ubersetzungen ins Syrische und Armenische Anschluß an die Geistigkeit des Vorderen Orients gewonnen. Diese Formen des Wissens ”uberlebten den politischen Umsturz,

As no serious investigation of the origin of Arabic science is possible without consideration of the political circumstances, so too a discussion of the sources of algebra and the historical presuppositions of its development cannot overlook the political context.

The time of Muhammad and the first caliphs was the time of joyful and carefree conquest of foreign lands and foreign riches; the booty fell to the victors, the subjugated paid tribute, and if he accepted Islam, he was freed from taxes. The simple warrior ideal of a strict, rough, and self-sufficient knighthood had nothing to do with science as yet.

As the passions of conquest cooled, the warlike inclinations were overgrown by a pomp-loving luxury, and the political as well as personal relations became more complicated through the necessity of peaceful coexistence. Within the Islamic world, the political and religious unity, already shaken by the death of the Prophet, split apart.

The Abbasid caliphs sought to enforce the divinely willed unity of the faithful through an ordered legal system and a cultural policy that restored the connection to the older conditions close to the declining Sasanid culture. This policy could not itself produce the impulse to science, but could create favorable conditions for it.

For among the conditions most favorable for the flourishing of science belongs not [the scholar’s] position within the midst of everyday life but one that towers above everyday life. Where the scholar is delivered over to the desire and arbitrariness of the mighty, science cannot flourish; where he is protected by them and is able to preserve also the freedom of thought, there arises the calm productive work that leads to the flowering of science.

For Muḥammad b. Mūsā this relationship of protection and freedom of the scholar can be demonstrated in the letter that al-Ma’mun addressed to him and which, according to the *Fihrist*, gave the occasion for his algebraic work. The caliph asks the scholar to present the secrets of the science of calculation so that people less familiar with it can understand it; he assures him of his gratitude and protection.

Greek science had remained alive in Byzantium and Syria up to the time of the Arab conquest and had gained connection to the intellectual life of the Near East through translations into Syriac and Armenian. These forms of knowledge survived the political upheaval; also the circles that cultivated them auch die sie pflegenden Kreise blieben bestehen und waren bereit, ihr Wissen mit den neuen Herren zu teilen.

Die Hellenisierung des Kalifenhofes war durch die Syrer und Griechen angebahnt; die ”Ubernahme von Rechtspflege und Verwaltung erforderte die Heranziehung gebildeter Einheimischer, die ihrerseits in den neuen Herren Beg”unstiger der Wissenschaft fanden. Schon in der ersten H”alfte des 8. Jahrhunderts griff der Kalif auf die Dienste arabischer ”Ubersetzer zur”uck; die ”altesten Ubersetzer griechischer Werke ins Arabische sind Syrer gewesen.

Muhammad b. M”usā schrieb seine Algebra w”ahrend der Regierung al-Ma’muns. Die ”Ubersetzung griechischer Werke in diese Sprache war daher schon vorausgegangen; griechische und indische astronomische Tafeln waren in Gebrauch. Die Bed”urfnisse der Astronomie hatten den Kalifen zur F”orderung der Mathematik veranlaßt, und es fehlte nicht an Gelehrten, die den Ruf des Mazdahurf”orderers der Wissenschaften beantworteten.

F”ur die Geschichte der Algebra ist es von entscheidender Bedeutung, daß Muhammad b. M”usā seine Kenntnisse zuerst als Astronom erworben hat. Denn die Astronomie war es, die die Araber zuerst und am dringendsten nach dem griechischen und indischen Wissen greifen ließ. Muhammad b. M”usā ”ubersetzte die indischen Rechenb”ucher und Astronomietafeln; als er sich an die Algebra machte, konnte er auf die indische Zahlenschrift und die indischen Rechenverfahren zur”uckgreifen.

Dieser Stand der Dinge war der Grund, weshalb die Algebra der Araber gleich von Anfang an eine angewandte Wissenschaft war. Sie wuchs nicht aus spekulativen Fragestellungen heraus, sondern aus den Bed”urfnissen des b”urgerlichen Lebens, des Handels und der Rechtspflege. Der erste Teil der Algebra, die Behandlung der quadratischen Gleichungen, stellt eine Verbindung der indischen Rechenkunst mit der griechischen Geometrie dar; der zweite Teil, die praktische Geometrie und das Gesch”aftsrechnen, erweitert den Anwendungsbereich, und der dritte Teil, die Erbteilungsrechnung, zeigt das ganze Ineinandergreifen von Mathematik und islamischem Recht.

Daß die Algebra erst sp”ater, nachdem sie die westlichen L”ander durchlaufen hatte, zu einer rein formwissenschaftlichen Disziplin wurde, ist ein weiteres Zeugnis f”ur die Notwendigkeit, historische Begriffe im Zusammenhang der geschichtlichen Entwicklung zu untersuchen und nicht von einem abstrakten, zeitlosen Begriff der Wissenschaft auszugehen.

Im Zusammenhang unserer Untersuchung muß noch auf einen Umstand hingewiesen werden, der f”ur die Bewertung der Quellen von Bedeutung ist. Das Werk des Muhammad b. M”usā ist f”ur uns nur in einer sp”aten Abschrift aus dem 14. Jahrhundert erhalten; die lateinischen ”Ubersetzungen aus dem 12. Jahrhundert sind teils fragmentarisch, teils von den Arabisten als nicht genau genug beurteilt worden. Rosen hat

remained in existence and were ready to share their knowledge with the new masters.

The Hellenization of the caliphal court was initiated by the Syrians and Greeks; the take-over of jurisprudence and administration required the engagement of educated natives, who in turn found in the new rulers patrons of science. Already in the first half of the 8th century the caliph availed himself of the services of Arab translators; the oldest translators of Greek works into Arabic were Syrians.

Muḥammad b. Mūsā wrote his Algebra during the reign of al-Ma’mun. The translation of Greek works into this language had thus already taken place; Greek and Indian astronomical tables were in use. The needs of astronomy had induced the caliph to promote mathematics, and there was no lack of scholars who answered the call of the Maecenas of the sciences.

For the history of algebra it is of decisive significance that Muḥammad b. Mūsā first acquired his knowledge as an astronomer. For it was astronomy that first and most urgently let the Arabs reach for Greek and Indian knowledge. Muḥammad b. Mūsā translated the Indian arithmetic books and astronomical tables; when he turned to algebra, he could fall back on the Indian numeral script and the Indian calculation procedures.

This state of affairs was the reason why the algebra of the Arabs was an applied science from the very beginning. It did not grow out of speculative questionings, but out of the needs of civil life, of commerce, and of jurisprudence. The first part of the algebra, the treatment of quadratic equations, represents a connection of Indian arithmetic with Greek geometry; the second part, practical geometry and commercial calculation, extends the sphere of application, and the third part, inheritance calculation, shows the complete interlocking of mathematics and Islamic law.

That algebra only later, after it had passed through the Western countries, became a purely formal scientific discipline, is further testimony to the necessity of investigating historical concepts in the context of historical development and not proceeding from an abstract, timeless concept of science.

In connection with our investigation, attention must yet be drawn to a circumstance that is significant for the evaluation of sources. The work of Muḥammad b. Mūsā is preserved for us only in a late copy from the 14th century; the Latin translations from the 12th century are partly fragmentary, partly judged by Arabists as not exact enough. Rosen edited the text after the Oxford manuscript, but—as we have seen zwar den Text nach der Oxforder Handschrift ediert, aber — wie wir mehrfach gesehen haben — seine ”Ubersetzung ist nicht zuverl”assig.

Eine neue kritische Edition des Textes und eine sorgf”altige ”Ubersetzung sind daher ein dringendes Desideratum; sie w”urde ohne Zweifel manche noch offene Frage l”osen und die vorliegende Untersuchung in manchem best”atigen und erg”anzen. Aber auch ohne eine solche Edition ist die Richtung der Untersuchung klar geworden: die arabische Algebra ist ein Produkt der Begegnung von drei Kulturen, der indischen, der griechischen und der arabischen, und ihr Sch”opfer Muhammad b. M”usā hat durch seine Bearbeitung das Werk in gleichem Maße dem Wissen der Vergangenheit wie den Bed”urfnissen der Gegenwart dienstbar gemacht.

repeatedly—his translation is not reliable.

A new critical edition of the text and a careful translation are therefore an urgent desideratum; it would doubtless solve many still open questions and confirm and supplement the present investigation in many respects. But even without such an edition, the direction of the investigation has become clear: Arabic algebra is a product of the encounter of three cultures, the Indian, the Greek, and the Arabic, and its creator Muḥammad b. Mūsā has through his adaptation made the work equally serviceable to the knowledge of the past and the needs of the present.

Register der behandelten Gegenst”ande und Termini / Index of Subjects and Terms

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وَكَرَمِهِ بِمَنِّهِ عَنْهُ وَعَفَا تَعَالَى اللَّهُ وَحِمَاهُ

الثنائية الطبعة ▯ الثَّانِيَةُ الطَّبْعَةُ

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كَرِيمُ يَا بِفَضْلِكَ تُعَسِّرْ وَلَا يَسِّرْ رَبِّ

مَشِيرٍ بِلَا الْمَرْيَةِ صُنْعَ فَاتَقَنَ وَصَوَّرَ الْفَلَكَ خَلَقَ الَّذِي لِلَّهِ الْحَمْدُ بِلَا وَأَحْكَمَهَا أَتْقَانًا، أَتَقَنَهَا يُعَاضِدُهُ، مُعِينٍ بِلَا وَدَبَّرَهَا يُشَاوِرُهُ، كَيْلًا بِالْبَحْرِ وَأَحَاطَهَا تَمِيدَ، لِئَلَّا بِالرَّاسِيَاتِ الْأَرْضَ أَوْتَدَ إِعْوَانٍ، فَمِنْهُمْ يَعْمَلُونَ كَيْفَ لِيَنْظُرَ عِبَادَهُ فِيهَا وَبَثَّ وَيَرِيدُ، مَاؤُهَا يَغْلِبُ خَاتِرِ عَلَى اللَّهُ وَصَلَّى وَنَسِيَ كَفَرَ مَنْ وَمِنْهُمْ وَافْتَدَى، أَحْسَنَ مَنْ تَسْلِيمًا وَسَلَّمَ وَصَحْبِهِ آلِهِ وَعَلَى مُحَمَّدٍ الْأَوْلِيَاءِ، وَأَكْرَمِ الْأَنْبِيَاءِ، كَثِيرًا.

زَالَتِ مَا فَإِنَّهُ بَعْدُ، أَمَّا الْمُقَدِّسِيُّ: أَحْمَدَ بْنُ مُحَمَّدُ اللَّهِ عَبْدِ أَبُو قَالَ تَنْقَطِعُ وَلَا تَحْرِيسِهِمْ، لِمَلْأِ الْكُتُبِ تَصْنِيفِ فِي تَرْغَبُ الْعُلَمَاءُ عَلَمًا وَأَقِيمَ سَنَتَهُمْ، وَأَقْفُوَ سُنَّتَهُمْ، أَتْبَعَ أَنْ فَأَحْبَبْتُ أَخْبَارُهُمْ، إِلَى سَبَقُوا قَدْ الْعُلَمَاءُ وَوُجِدَتِ وَفِيلَةً، بِهِ وَأَشِيرَ ذِكْرِي، بِهِ أُحْيِيَ فَشَرَحُوا الْأَخْلَافُ تَبِعَتْهُمُ ثُمَّ الِاجْتِهَادِ، عَلَى فَتَمَلَّقُوا الْعُلُومِ الْأَقَالِيمِ، أَخْبَارِ فِي كِتَابًا أُوَلِّفَ أَنْ فَرَأَيْتُ وَاخْتَصَرُوهُ، كَلَامَهُمْ وَالْبِلَادِ، الْمَمَالِكِ، وَمَسَالِكِ الْعِبَادَاتِ، وَأَنْوَاعِ الْأَرْضِ، وَأَشْكَالِ الْمَسْلُوكَةِ، وَمَنَازِلِهَا الْمَعْرُوفَةِ، وَمَدَائِنِهَا الْمَشْهُورَةِ، وَأَمْصَارِهَا الْجَمَلِ وَمَعَادِنِ وَالْآلَاتِ، الْعَشَائِرِ وَعَنَاصِرِ الْمُسْتَعْمَلَةِ، وَطُرُقِهَا وَأَصْوَاتِهِمْ كَلَامِهِمْ فِي الْبُلْدَانِ أَهْلِ وَاخْتِلَافِ وَالتِّجَارَاتِ، وَنُقُودِهِمْ، وَأَوْزَانِهِمْ، وَمَكَاتِبِهِمْ وَمَذَاهِبِهِمْ وَأَلْوَانِهِمْ، وَأَسْنَانِهِمْ وَزُهَّادِهِمْ، وَعُلَمَائِهِمْ، وَأَشْيَاخِهِمْ، بَلَدٍ، كُلِّ لِأَهْلِ الْمَعَاشِ وَأَفْضِيَةِ وَحُكَمَائِهِمْ.

مِنهَا بُدَّ لَا وَفُصُولٌ مُقَدِّمَاتٌ

وَأَسْنَادَاتٍ مُحْكَمَةٍ قَوَاعِدَ عَلَى الْكِتَابَ هَذَا أَسَّسْتُ أَنِّي أَعْلَمْ أُولِي فِيهِمْ وَاسْتَعَنْتُ الصِّوَابِ، جِهَادَ فِيهِ وَتَخَرَّجْتُ قَوِيَّةٍ، بِدَعَائِمَ الرَّجَاءُ فِيهِ وَيُبْلَغَ وَالزَّلَلُ، الْخَطَأُ فِيهِ يُجْتَنَبَ أَنْ وَاسْمِهِ الْأَلْبَابِ، وَعَقَلْتُهُ، شَاهَدْتُهُ مَا بِفَنِّيَاتِهِ أَوْصَفُ قَوَاعِدِهِ فَعَلَى وَالْأَمَلُ، قَوَاعِدِهِ وَمِنْ وَالْأَرْكَانِ، الْبُنْيَانَ رَفَعْتُ وَعَلَيْهِ وَعَلَّمْتُهُ، وَحَرَّفْتُهُ مِنَ الْعُقُولِ ذَوِي سَوَالٌ تَبْيِينِهِ، عَلَى بِهِ أَسْتَعَنْتُ وَمَا وَإِرْكَانِهِ، أَيْضًا وَالْأَعْمَالِ الْكَبِيرِ عَنِ وَالِانْتِبَاسِ، بِالْغُفْلَةِ أَعْرِفُهُمْ لَمْ وَمَنْ النَّاسِ، فَمَا إِلَيْهَا، الْوُصُولُ لِي يَتَقَدَّرْ وَلَمْ عَنْهَا، بَعُدَتْ الَّتِي الْأَطْرَافِ فِي لِي يَكُنْ لَمْ وَمَا نَبَّذْتُهُ، فِيهِ اخْتَلَفُوا وَمَا أَثْبَتُّهُ، أَتْفَاقُهُمْ عَلَيْهِ وَقَعَ وَشَاهَدْتُهُ زَعْمُوْا، قَلْتُ أَوْ ذِكْرٌ الْمَاضِي إِلَى اسْتِنَادَاتِهِ فِي بَدَلٌ الْمُلُوكِ. خِزَانَاتِ فِي وَجَدْتُهُ بِفُصُولٍ

وَلَا قَصَدْنَاهَا، الَّتِي الطَّرِيقَ يَسْلُكْ لَمْ الْعِلْمِ هَذَا إِلَى سَبَقَنَا مَنْ وَكُلُّ أَمِيرِ دَزَبَرَ فَكَانَ الْبَكْرِيُّ اللَّهِ عَبْدِ أَبُو أَمَّا أَرَدْنَاهَا، الَّتِي الْغَوَائِدَ طَلَبَ السُّفَرَاءِ مُجْتَمَعِ وَهَيْئَةِ وَنَحْوِهِمْ فَلَسَفَةٍ صَاحِبُ وَكَانَ خُرَاسَانَ، وَدَخَلَهَا. الْمُلُوكِ عَنِ وَسَأَلَهُمْ

وَالْأَنْهَارِ الْبِحَارِ ذِكْرُ

مِنْ يَخْرُجُ أَحَدِهِمَا حَسَبُ بِحْرَيْنِ أَلَّا الْإِسْلَامِ فِي نَرَ لَمْ أَنَّا أَعْلَمْ
مَمْلَكَةَ بَلَغَ فَإِذَا السَّوَادِ، وَبِلَادِ الصِّينِ بِلَادِ بَيْنَ الشَّامِ مَشَارِقِ نَحْوِ
كَثِيرَةٌ خَلِيجَانِ وَلَهُ مُثَّلْنَاهُ، كَمَا الْعَرَبِ، جَزِيرَةِ عَلَى دَارَ الْإِسْلَامِ
فَنَائِهِمْ إِلَى وَالْمَصِيرِ وَصْفِهِ فِي النَّاسُ اخْتَلَفَ وَقَدْ عِدَّةٌ، وَشِعَبٌ
الْقِسْمِ عَلَى يَطُوفُ بَنِيَانٍ بِخَلِيجِ طَبَسِيَّةِ شَبِيهَةِ جَاعِلَةِ، مِنْ
الشِّينِ وَبِلَادِ الشِّينِ وَخَلِيجُ بِعُمَّانَ، وَطَرَفٍ بِالْعِرَاقِ وَطَرَفٍ
وَرَأَيْهُ خُرَاسَانَ أَمِيرِ خُرَاسَانَةُ وَعِنْدَهُ بِالْعِرَاقِ، وَذَنْبُهُ حَدِيثَةٌ،
وَفِي بِنَيْسَابُورَ الْمُؤْمِنِينَ أَمِيرِ خِزَانَةِ فِي وَرَقَةٍ عَلَى مُتَّفَقٌ
وَمَا نَعْرِفُهَا، لَمْ فَإِنَّا مِثْلَ، كُلُّ وَإِذَا وَالصَّاحِبِ، الدَّوْلَةِ عِنْدَ خُرَاسَانَةَ
عَلَى يَعْتَمِدُ خَلِيجَانِ بَعْضِهِنَّ فِي وَإِذَا الْآثَارِ بِمُخْتَلَفِ فَسَّرْتُهُ أَنَا
الْيَابَسَةِ. فِي فَمَجْرَاهُ الْأَعْظَمُ النَّهْرُ وَأَمَّا كُلِّهَا، مِنَ الْقَلِيزِ

لِأَنَّهَا الْكُفَّارِ مَسَالِكَ نَتَكَلَّفْ وَلَمْ حَسَبَ الْإِسْلَامِ مُلْكَ أَنَّ ذَكَرْنَا وَقَدْ إِقْلِيمًا، عَشَرَ أَرْبَعَةِ إِلَى الْإِسْلَامِ أَرْضَ قَسَمْنَا وَقَدْ بَابِنَا، مِنْ تَكُنْ لَمْ الْمَمَالِكِ، مِنَ فِيهِ مَا وَوَزَنَّا وَأَعْنَاقَنَا، فَصَلْنَاهُ إِقْلِيمٍ كُلَّ وَأَفْرَدْنَا الْمَشْرِقُ أَوَّلُهَا الْعَرَبِ أَقَلِيمِ وَكَمْ الْعَجَائِبِ، مِنَ فِيهِ مَا وَذَكَرْنَا الْمَغْرِبُ. مِصْرُ الشَّامُ الْعِرَاقُ الْحِجَازُ

الْحِجَازِ: ذِكْرُ

أَمْصَارِ مَحَلُّ وَهُوَ وَأَشْرَفِهَا، الْعَرَبِ بِلَادِ أَفْضَلِ مِنْ وَالْحِجَازُ عَلَيْهِ اللَّهُ صَلَّى النَّبِيِّ وَمَوْلِدُ الْعَرَبِ، مَنَازِلِ وَمَوْضِعُ الْمُسْلِمِينَ الْمُنَوَّرَةُ وَالْمَدِينَةُ الْمُشَرَّفَةُ مَكَّةُ وَفِيهِ الشَّرَائِعِ، وَمَنْبِتُ وَسَلَّمَ وَأَكْثَرُهُ وَالْعُشْبِ، الْمَاءِ قَلِيلُ حَارٌّ أَقْلِيمٌ وَهُوَ وَالْيَمَامَةُ، وَالْيَمَنُ وَالْغَنَمُ، الْإِبِلُ وَمَالُهُمْ وَحَضَرُ، بَدْوٌ فِيهِ وَالنَّاسُ وَرِمَالٌ، حِجَارَةٌ لِغَيْرِهِمْ. لَيْسَ مَا وَالْحَمِيَّةِ وَالشَّجَاعَةِ الْفَخْرِ مِنَ وَلَهُمْ

الْمُكَرَّمَةُ: مَكَّةُ

الْمُعَظَّمَةِ الْكَعْبَةِ مَوْضِعُ وَهِيَ وَأَفْضَلُهَا، اللَّهِ بِلَادِ أَشْرَفُ وَمَكَّةُ مِنَ فِيهَا مَا وَأَصِفُ الْبِلَادِ، سِيَرِ فِي أَبْدَأَ وَبِهَا الْحَرَامِ، وَالْمَسْجِدِ وَالطُّرُقَاتِ. وَالْأَحْيَاءِ وَالْأَسْوَاقِ وَالْمَسَاجِدِ الْعِمَارَاتِ

يُقَالُ جَبَلٍ سَفْحِ فِي وَهِيَ عَالِيَةٍ، جِبَالٍ بَيْنَ شَرِيفٍ جَبَلٍ فِي وَمَكَّةُ وَالْعُيُونِ، الْآبَارِ مِنَ وَمَاؤُهَا الْمَجْرَى، صَعْبُ وَنَهْرُهَا قُبَيْسٍ، أَبُو لَهُ الْمَذَاقِ. عَذْبَةُ الْمَاءِ كَثِيرَةُ عَيْنٌ وَهِيَ زُبَيْدَةَ، عَيْنُ فِيهَا مَا وَأَشْهَرُ

الْعِرَاقِ: ذِكْرُ

وَبِهِ وَالْمُلْكِ، الْخِلَافَةِ وَمَوْضِعُ الْجَنَّةِ، كُورَةِ بِلَادُ هُوَ وَالْعِرَاقُ
وَهُوَ الْأَخْيَالِ، وَحِصْنِ وَالرَّحْبَةُ وَالْأَنْبَارُ وَوَاسِطٌ وَالْبَصْرَةُ الْكُوفَةُ
وَأَنْهَارُهُ وَالشَّعِيرِ، وَالْحِنْطَةِ وَالنَّخْلِ الْمَاءِ كَثِيرَةُ خَصِبَةٌ أَرْضٌ
بَصَرَةٍ أَهْلُ أَكْثَرُهُمْ فِيهِ وَالنَّاسُ وَالْفُرَاتُ، دِجْلَةُ أَشْهَرُهَا كَثِيرَةٌ
وَمَذَاهِبَ. وَكَلَامٍ وَفِقْهٍ وَعِلْمٍ

الشَّامِ: ذِكْرُ

وَحِمْصٌ وَالرَّشِيدُ دِمَشْقُ وَبِهِ الْخَيْرِ، كَثِيرُ وَاسِعٌ أَقْلِيمٌ هُوَ وَالشَّامُ
وَالرَّمْلَةُ الْمَقْدِسِ وَبَيْتُ وَاللَّاذِقِيَّةُ وَأَنْطَاكِيَةُ وَحَلَبٌ وَحَمَاةُ
وَفِيهِ وَالضَّرْعِ، لِلزَّرْعِ صَالِحَةٌ وَأَرْضُهُ وَصَيْدَاءُ، وَصُورُ وَبَيْرُوتُ
الْأُرْنُوْنُ أَنْهَارِهِ وَأَشْهَرُ يُحْصَى، لَا مَا وَالْعُيُونِ وَالْأَنْهَارِ الْجِبَالِ مِنَ
وَجَمَالٍ. وَصِحَّةٍ وَعِلْمٍ فَقْهٍ أَهْلُ فِيهِ وَالنَّاسُ وَالْيَرْمُوْكُ، وَاللِّتَانِيُّ

النَّهْرِ: وَرَاءَ وَمَا خُرَاسَانَ ذِكْرُ

وَبِهَا مُلُوكًا، وَأَجَلُّهَا بِلَادًا وَأَكْثَرُهَا الْمَشْرِقِ أَقَالِيمِ أَعْظَمُ وَخُرَاسَانُ
وَبُشَنْجٌ وَطَخَارِسْتَانُ وَبَلْخٌ وَهَرَاةٌ الشَّاهِجَانُ وَمَرْوُ نَيْشَابُورُ
وَسَرَخْسٌ وَأَبِيَوَرْدُ وَنَسَا وَإِسْتَخْرٌ وَسَابُورٌ وَطُوسٌ وَغُرْجَانُ
أَرْضٌ وَهِيَ وَفَارَابُ، وَالشَّاشُ وَالسَّغْدُ وَخُوَارِزْمُ وَمَيْمَنْدَجَانُ
وَتِجَارَةٍ. وَأَدَبٍ عِلْمٍ أَهْلُ وَأَكْثَرُهَا وَالْبُسْتَانِ، الْمَاءِ كَثِيرَةُ وَاسِعَةٌ

وَأَسَدَابَادُ وَالشَّاشُ وَخُجَنْدَةُ وَسَمَرْقَنْدُ بُخَارَى هُوَ النَّهْرِ وَرَاءَ وَمَا
وَخُوَارِزْمُ، وَشِيرَازُ وَطَالَقَانُ وَإِلَاقٌ وَشَاشٌ وَكَاشٌ وَنَسَفٌ وَرَامِتَنُ
وَالدِّينِ، بِالْعِلْمِ مُتَمَسِّكُونَ وَأَهْلُهُ الْخَيْرَاتِ، كَثِيرُ وَاسِعٌ إِقْلِيمٌ وَهُوَ
الْحَدِيثِ أَهْلُ وَمِنْهُمْ يُوصَفُ، لَا مَا وَالْفُتُوَّةِ الشَّجَاعَةِ مِنَ وَبِهِمْ
وَإِتْقَانًا. حِفْظًا النَّاسِ أَكْثَرُ وَهُمْ وَالتَّفْسِيرِ، وَالْفِقْهِ

يُسَاوِي وَالْفَرْسَخُ وَالْمِيلِ، بِالْفَرَاسِخِ الْبِلَادِ مَسَافَاتِ أَحْصَيْنَا وَقَدْ
أُصْبُعًا، وَعِشْرُونَ أَرْبَعَةٌ وَالذِّرَاعُ ذِرَاعٍ، أَلْفُ وَالْمِيلُ أَمْيَالٍ، ثَلَاثَةَ
وَمَا فَرْسَخٍ، أَرْبَعَمِائَةِ فَوَجَدْنَاهُ وَالْمَدِينَةِ مَكَّةَ بَيْنَ مَا حَسَبْنَا وَقَدْ
وَسَامَرَّاءَ بَغْدَادَ بَيْنَ وَمَا فَرْسَخٍ، أَرْبَعَمِائَةِ وَبَغْدَادَ الْبَصْرَةِ بَيْنَ
بَيْنَ وَمَا فَرْسَخًا، ثَلَاثُونَ وَالرَّحْبَةِ دِمَشْقَ بَيْنَ وَمَا فَرْسَخٍ، أَرْبَعَمِائَةِ
فَرْسَخٍ. أَرْبَعُمِائَةِ وَعُمَانَ الْبَصْرَةِ

مَكَّةُ
الْمَدِينَةُ
الْكُوفَةُ
الْبَصْرَةُ
بَغْدَادُ
وَاسِطٌ
دِمَشْقُ
حَلَبُ
بَيْتُ الْمَقْدِسِ
مِصْرُ
الْإِسْكَنْدَرِيَّةُ
الْقَيْرَوَانُ
نَيْشَابُورُ
مَرْوُ
بَلْخُ
بُخَارَى
سَمَرْقَنْدُ
خُوَارِزْمُ

20pt

كتاب الفهرست

Kitāb al-Fihrist — The Book Catalogue

A Bilingual Arabic-English Edition

Based on the critical edition of Gustav Flügel (Leipzig, 1871–1872)

Author: Muḥammad ibn Isḥāq Ibn al-Nadīm
Baghdad, c. 987 CE

About this Edition. The *Kitāb al-Fihrist* of Ibn al-Nadīm (c. 987 CE) is the most important bibliographic encyclopedia of medieval Arabic literature and scholarship. This bilingual edition presents the original Arabic text alongside an English translation, arranged in parallel columns. The left column contains the Arabic text in RTL script; the right column gives the English rendering. Key sections on the mathematical sciences — including biographies of al-Khwārizmī, Thābit ibn Qurra, and other Islamic mathematicians — are presented in full.

Edited from the Arabic original with notes
Gustav Leberecht Flügel (1802–1870)
Completed by Dr. Johannes Roediger & Dr. August Müller
Leipzig: Verlag von F. C. W. Vogel, 1871–1872
With the support of the Deutsche Morgenländische Gesellschaft

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Introduction

مقدمة المؤلف في فائدة التصنيف

Preface: On the Purpose of Cataloguing

Ibn al-Nadīm, whose full name is Abū al-Faraj Muḥammad ibn Isḥāq ibn Abī Yaqūb al-Nadīm al-Warrāq al-Baghdādī (died c. 995/998 CE), was a Muslim scholar, bibliographer, and bookseller (*warrāq*) who lived in Baghdad during the Buyid period. His *Kitāb al-Fihrist* (كتاب الفهرست) is the most important bibliographic work in Arabic literature and an indispensable source for the history of Islamic civilization down to the 10th century.

قال أبو الفرج محمد بن إسحاق بن أبي يعقوب النديم
الوراق:
علوم يحفظ من زمان عصر كل في جعل الذي لله الحمد
ويؤدي الناس
ما على نستعين وبه السنن ويحيي المعالم ويحرس الأمانة
نصنفه
العلماء تصنفها التي الكتب أصناف ذكر من الكتاب هذا في
وما المصنفين وأسماء بها يشتغلون التي العلوم وأنواع
يعرف
وبلدانهم وطبقاتهم ونسبهم وأسمائهم وكناهم بألقابهم
وصفوا وما عنهم أخذ ومن عنه العلم أخذوا ومن ومولدهم
ذلك وغير والمعرفة والإتقان والفهم والحفظ الجلالة من به

Abū al-Faraj Muḥammad ibn Isḥāq ibn Abī Yaqūb al-Nadīm al-Warrāq said: Praise be to God, who in every age appoints those who preserve the sciences of mankind, discharge the trust, guard the landmarks of knowledge, and revive the traditions. Through Him we seek aid in compiling this book, wherein we mention the categories of books composed by scholars, the types of sciences they pursue, the names of authors with their titles, *kunyas*, proper names, genealogies, generations, places of origin and birth, their teachers, their students, and their distinguished qualities of eminence, retention, understanding, mastery, comprehension, and more.

وقد اقتدت فائدة هذا الكتاب بفوائد سائر الكتب المصنفة
في
والعلماء المصنفين وأسماء الأعيان وطبقات الناس أخبار
فهو
عليها يدي وقعت التي العلم أصناف جميع على يحتوي
وسمعتها
وجلودها وحواشيها الكتب بطون في مكتوبة ورأيتها
وإفادات
الأصول من ذلك بعد استدركته وما بينهم فيما العلماء
والفهارس

The benefit of this book follows the benefits of all books compiled on the reports of people, the generations of notables, the names of authors and scholars. It encompasses all categories of knowledge that I have handled, heard, and seen written in the contents of books, their margins, their bindings, and the transmissions of scholars among themselves — including what I subsequently recovered from original sources and catalogues.

The *Fihrist* is organized into ten *discourses* (*maqālāt*, مقالات), covering the entirety of Arabic book culture. The following sections present each discourse with parallel Arabic text and English translation.

1 Discourse I: Qur’an Sciences and Philology

الفن الأول في اللغات والخطوط وقراءة القرآن والنحو والشعر والتفسير

THE FIRST DISCOURSE: LANGUAGES, SCRIPTS, QUR’ANIC READINGS, GRAMMAR, POETRY, AND EXEGESIS

في أصناف العلم بالقرآن الكريم

أما أصناف العلم بالقرآن الكريم فإنها: علم القراءات وعلم
وعلم والمنسوخ الناسخ وعلم التجويد وعلم التفسير
فصاحة
المعاني وعلم النحو وعلم والوقف البدء وعلم القرآن
والبيان
الأنساب وعلم الفرائض وعلم القافية وعلم العروض وعلم
وما
والخطباء المتكلمون إليها يحتاج التي العلوم من ذلك أشبه
والمفسرون والشعراء

On the Categories of Qur’anic Science

The categories of knowledge concerning the Noble Qur’an are: the science of readings (*qirā’āt*), the science of exegesis (*tafsīr*), the science of recitation (*tajwīd*), the science of abrogating and abrogated verses, the science of the Qur’an’s eloquence, the science of beginnings and pauses, the science of grammar (*naḥ), the science of meanings and rhetoric (maānī wa al-bayān*), the science of prosody (*’arūḍ*), the science of rhyme (*qāfiya*), the science of inheritance laws (*farā’iḍ*), and the science of genealogy (*ansāb*) — and similar sciences needed by speakers, orators, poets, and exegetes.

في القراءات

وأما القراءات فإنها عشر: نافع وابن كثير وأبو عمرو وابن
وخلف ويعقوب جعفر وأبو والكسائي وحمزة وعاصم أمير
إلى شيخه عن وشيخه شيخه عن رواية له قارئ وكل
الصحابة
الاحاطة أراد فمن وسلم عليه الله صلى النبي عن والصحابة
المشهورة بكتبها فعليه القراءات بعلم

On the Qur’anic Readings

The readings are ten: Nāfic̣, Ibn Kathīr, Abū c̣Amr, Ibn Āmir, c̣Āṣim, Ḥamza, al-Kisā’ī, Abū Jafaṛ, Yaq̧ūb, and Khalaf. Each reader received his reading from his sheikh, and his sheikh from his sheikh, back to the Companions, and the Companions from the Prophet — peace be upon him. Whosoever desires full mastery of the science of readings must study its well-known books.

Note on the First Discourse. This section of the *Fihrist* catalogues the foundational Islamic sciences that shaped Arabic scholarship. The ten canonical readings (*qirā’āt*) of the Qur’an represented the bedrock of linguistic and philological study. Ibn al-Nadīm lists the principal authors and their works in each discipline, providing an unparalleled window into the state of these sciences in 10th-century Baghdad.

2 Discourse II: Grammar, Lexicography, and Poetry

الفن الثاني في النحو واللغة والشعر وأخبار الشعراء

The Second Discourse: Grammar, Lexicography, Poetry, and the Reports of Poets

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في النحويين

On the Grammarians

وأما النحو فإنه علم يعرف به أوضاع ألفاظ العرب من حيث
نظم من أول كان لأنه نحواً سمي وإنما والبناء الإعراب
أحكامه
وتذهب لغتها تغير العرب إن له: قيل إذ الدؤلي الأسود أبو
إليها مال أي فنحاها ضابطاً لها يضع أن فأراد بإعرابها
نحواً فسمي فيها وتعمق

Grammar (*naḥ*) is the science by which one knows the structures of Arabic speech with respect to declension (*iṛāb*) and construction (*bunāʾ*). It was called *naḥ* because its first systematizer, Abū al-Aswad al-Du'alī, when told that the Arabs were corrupting their language and losing its declension, sought to establish rules for it. He *naḥā* it — that is, he turned to it and delved deeply into it — and so it was called *naḥ*.

في أصول النحو والمصنفين فيه

On the Foundations of Grammar and Those Who Composed Works Therein

وأول من وضع النحو أبو الأسود الدؤلي ثم أخذ عنه أهل
ومن الكبار الكتب فيه وصنفوا وأطنبوا فيه فأكثروا البصرة
وسيبويه الفراهيدي أحمد بن الخليل النحويين: أشهر
وعلقمة
والفارسي والزجاجي السراج وابن والمازني والأخفش
وغيرهم
الفن هذا في كثيرة مصنفات منهم واحد ولكل كثير

The first to establish grammar was Abū al-Aswad al-Du'alī; then the people of Basra took it from him, expanded upon it, elaborated it, and composed great books. Among the most famous grammarians are: al-Khalīl ibn Aḥmad al-Farāhidī, Sībawayh, çAlqama, al-Akhfash, al-Māzanī, Ibn al-Sarrāj, al-Zajjājī, al-Fārisī, and many others. Each of them has numerous compositions in this art.

في اللغويين وأصناف اللغات

On Lexicographers and the Categories of Languages

وأما اللغة فإنها صناعة يعرف بها معاني الألفاظ وأسباب
مراد لبيان وضعت وإنما المعاني وحروف واشتقاقها وضعها
وبلاغتهم فصاحتهم في يتفاوتون والمتكلمون المتكلم
ونحوهم
ومعانيها الألفاظ جمع في جمة كتباً اللغة أهل وضع وقد

Lexicography is the craft by which one knows the meanings of words, the reasons for their coinage, their derivation, and the particles of meaning. It was devised to clarify the intent of the speaker. Speakers differ in their eloquence, rhetoric, and grammatical usage. The scholars of language have composed abundant books collecting words and their meanings.

في الشعراء وأخبارهم

On the Poets and Their Reports

وأما الشعر فهو الكلام الموزون المقفى الذي يحتذي به في
امرئ مثل مشهورون شعراء الجاهلية في كان وقد العربية
القيس
حلزة بن والحارث والأعشى وعنترة وزهير والنابغة وطرفة
الرومي وابن والبحتري تمام أبو المحدثين من ومعهم
والمتنبي
مشهور شعر ديوان منهم واحد ولكل كثير وغيرهم

Poetry is metrical, rhymed discourse that serves as a model in Arabic. In the pre-Islamic period there were famous poets such as Imru’ al-Qays, Ṭarafa, al-Nābigha, Zuhayr, çAntara, al-Aş, and al-Ḥārith ibn Ḥilliza; and among the modern poets Abū Tammām, al-Buḥturī, Ibn al-Rūmī, al-Mutanabbī, and many others. Each of them has a well-known *dīwān* of poetry.

3 Discourse III: History, Biography, and Genealogy

الفن الثالث في التاريخ والسير والمغازي والأنساب

The Third Discourse: History, Biographies, Campaigns, and Genealogy

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في المؤرخين وأصناف التاريخ

وأما التاريخ فهو علم يعرف به أوقات الحوادث وتواريخ الملوك
والدول والمذاهب الماضية الأمم من وقع وما وسيرهم أشهر ومن
والواقدي والفتوح المغازي صاحب إسحاق ابن المؤرخين: وابن
وهم ومسكويه والطبري الطبقات صاحب سعد وابن هشام كثيرون
الأرفف تملأ كثيرة كتباً فيه صنفوا وقد الفن هذا في

On the Historians and Categories of History

History is the science by which one knows the times of events, the dates of kings, their biographies, and what befell past nations, sects, and dynasties. Among the most famous historians are: Ibn Isḥāq, author of the *Maghāzī* and *Futūḥ*; al-Wāqidī; Ibn Hishām; Ibn Saḍ, author of the *Ṭabaqāt*; al-Ṭabarī; and Miskawayh. They are numerous in this art and have composed many books filling the shelves.

في السير والمغازي

وأما السير فهي ما أثبته المؤلفون من أخبار الرسول صلى الله
أخبار من أثبتوه وما ومسيراته وحجه وغزواته وسلم عليه
وأفعالهم والخلفاء الملوك وسير والتابعين الصحابة وأحوالهم
هشام وابن إسحاق ابن السير: في المصنفين أشهر ومن وابن
مشهور منهم واحد وكل وغيرهم والبلاذري والواقدي سعد بكتابه

On Biographies and Campaigns

Biographies (*siyar*) are what authors have recorded of the reports of the Messenger of God — peace be upon him — his raids, pilgrimages, and journeys; and what they recorded of the Companions and Followers, the biographies of kings and caliphs, their deeds and circumstances. Among the most famous authors of *siyar* are: Ibn Isḥāq, Ibn Hishām, Ibn Saḍ, al-Wāqidī, and al-Balādhurī. Each is famous for his book.

في الأنساب

وأما علم الأنساب فهو علم يعرف به أحساب الناس ونسبهم
أهلها ومشاهير وديارها ومواضعها العرب بقبائل يتعلق وما
النسابة أخذه ثم الحميريون العلم هذا وضع من وأول
جمة كتباً فيه فصنفوا بعدهم من

On Genealogy

The science of genealogy is that by which one knows the pedigrees and lineages of people, and what relates to the Arab tribes, their locations, settlements, and notables. The Himyarites first established this science; then the genealogists (*nassāba*) took it from them and composed abundant books thereon.

4 Discourse IV: Mathematics, Astronomy, Music, and Mechanics

الفن الرابع في الحساب والهندسة والفلك والموسيقى والحيل

The Fourth Discourse: Arithmetic, Geometry, Astronomy, Music, and Mechanical Devices

في علم الحساب

وأما علم الحساب فهو علم يعرف به الكميات وعلاقاتها النساطرة نقله ثم اليونانيون وضعه وقد وأنواعها وأعدادها كثيرة كتباً فيه وصنفوا عنهم المسلمون فأخذه العراق إلى فيه المؤلفين أشهر ومن فيه فأبدعوا عليه وزادوا وطوروه وغيرهم والرازي والكندي قرّة بن وثابت الخوارزمي

On the Science of Arithmetic

The science of arithmetic is that by which one knows quantities, their relations, numbers, and types. The Greeks established it; then the Nestorians transmitted it to Iraq, whence the Muslims received it. They composed many books thereon, developed it, and added to it, producing outstanding works. Among the most famous authors are: al-Khwārizmī, Thābit ibn Qurra, al-Kindī, and al-Rāzī, among others.

في علم الهندسة

وأما الهندسة فهي علم يعرف به الأشكال والأبعاد والعلاقات
بالأصول المسمى كتابه في أقليدس أصولها وضع وقد بينها قرّة بن وثابت حنين بن وإسحاق إسحاق بن حنين نقله وقد فمنهم عظيمة كتباً فيه وصنفوا تعلمه على المسلمون وأقبل وزيادةً شرحاً فيه ألف من ومنهم تلخيصاً فيه ألف من

On the Science of Geometry

Geometry is the science by which one knows figures, dimensions, and the relations between them. Euclid established its principles in his book called the *Elements* (*al-Uṣūl*). It was translated by Ḥunayn ibn Isḥāq, Isḥāq ibn Ḥunayn, and Thābit ibn Qurra. The Muslims devoted themselves to studying it and composed great books: some wrote abridgements, others wrote commentaries and additions.

في علم الفلك والنجوم

وأما علم الهيئة فهو علم يعرف به أحوال الكواكب والأجرام عن نقل وقد مسيراتها ومبالغ وعللها وحركاتها السمائية ثم الفن هذا في كثيرة كتب والفرس والهنود اليونانيين وصنعوا أزيجاً وعملوا عظيمة كتباً فيه المسلمون صنف العلم هذا إليه يحتاج مما وغيرها أسطرلابات

On the Science of Astronomy and the Stars

Astronomy (*hay'a*) is the science by which one knows the conditions of the planets and celestial bodies, their motions, causes, and the measures of their movements. Many books in this art were translated from the Greeks, Indians, and Persians; then the Muslims composed great books, compiled astronomical tables (*azīj*), and constructed astrolabes and other instruments needed for this science.

4.1 The Mathematicians: Key Biographical Entries

ذكر الخوارزمي ومصنفاته

Account of al-Khwārizmī and His Compositions

أبو عبد الله محمد بن موسى الخوارزمي كان من علماء بيت
كتاباً والمقابلة الجبر في ألف وقد المأمون زمان في الحكمة
وكيفية المعادلات وأنواع والمقابلة الجبر فيه به يعرف حلها
المعادلات أن وأوضح والمقابلة الجبر بين فيه ميز وقد تنتهي
طرقاً واستخرج شيئاً الجهول وسمى محدودة أنواع إلى
الهندسية والطريقة الجبرية الطريقة منها المعادلات لحل
العدد نظام فيه يشرح آخر كتاباً الحساب في أيضاً ألف وقد
بالخوارزمي الآن ويسمى به الحساب وطريقة الهندي

Abū c̣Abd Allāh Muḥammad ibn Mūsā al-Khwārizmī was among the scholars of the House of Wisdom in the time of al-Ma’mun. He composed a book on *al-jabr wa al-muqābala* wherein he treated *al-jabr* and *al-muqābala*, the types of equations, and the methods of solving them. He distinguished between *al-jabr* and *al-muqābala* and showed that equations reduce to a finite number of types. He named the unknown quantity *shay’* (“thing”) and derived methods for solving equations, both algebraic and geometric. He also composed another book on arithmetic explaining the Indian numeral system and methods of calculating with it, which is now called “algorism” (*al-Khwārizmī*).

وله كتاب في الجبر والمقابلة وكتاب في الجامع والتفريق
في وكتاب والمقابلة الجبر حساب في المختصر وكتاب حساب
الأسطرلاب عمل في وكتاب الأرض صورة وكتاب الهندي وكتاب
وكتاب هند السند أي الازيج وكتاب الفسطرلاب عمل في في
صناعة من أخرى وأشياء الرخامة في وكتاب الساعات الفلك

His works include: *Kitāb fī al-Jabr wa al-Muqābala*; *Kitāb fī al-Jāmic̣ wa al-Tafrīq*; the *Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wa al-Muqābala*; *Kitāb fī Ḥisāb al-Hind*; *Kitāb Ṣūrat al-Arḍ*; *Kitāb fī c̣Amal al-Asturlāb*; *Kitāb fī c̣Amal al-Fasturlāb*; *Kitāb al-Azīj* (that is, the *Zīj al-Sindhind*); *Kitāb fī al-Sāc̣āt*; *Kitāb fī al-Rukḥāma*; and other works on the art of astronomy.

ذكر ثابت بن قرّة ومصنفاته

Account of Thābit ibn Qurra and His Compositions

ثابت بن قرّة بن مروان الحرّاني كان حكيماً عظيماً فقيهاً
والحساب والفلسفة والمنطق الهيئة علم في بارعاً فاضلاً والهندسة
وأرشميدس أبولونيوس كتب من كثيراً نقل وقد وبطليموس
الهندسة علم في ألف وقد وشرحها تصحيحها على وأقبل كتباً
علم وفي الهيئة علم وفي الحساب علم وفي كثيرة الموسيقى
الإسلام في الفلك علم أركان وطد من أعظم من وكان

Thābit ibn Qurra ibn Marwān al-Ḥarrānī was a great sage, an excellent jurist, distinguished in the science of astronomy, logic, philosophy, arithmetic, and geometry. He translated many books of Apollonius, Archimedes, and Ptolemy, and devoted himself to correcting and commenting upon them. He composed numerous books on geometry, arithmetic, astronomy, and music, and was among the greatest in establishing the foundations of the science of astronomy in Islam.

وله كتاب في القرى الحسابية وكتاب في المساحة وكتاب في
في وكتاب لمقلطس القطع شكل في وكتاب المعادلات المخروطات
النسبة حسن في وكتاب والأسطوانة الكرة في وكتاب في وكتاب
الفلك علم في وكتاب الهيئة علم في وكتاب الموسيقى علم
الزيج في وكتاب اليوناني عن نقله الذي المجسطي وكتاب
ذكره يطول مما كثير ذلك وغير

His works include: *Kitāb fī al-Qarā al-Ḥisābiyya*; *Kitāb fī al-Masāḥa*; *Kitāb fī al-Muādālāt*; *Kitāb fī Shakl al-Qiṭāc* (“On the Quadrature of the Parabola”); *Kitāb fī al-Makhrūṭāt*; *Kitāb fī al-Kura wa al-Ustuwāna*; *Kitāb fī Ḥusn al-Nisba*; *Kitāb fī ʿIlm al-Mūsīqā*; *Kitāb fī ʿIlm al-Hay'a*; *Kitāb fī ʿIlm al-Falak*; the *Kitāb al-Majisṭī* which he translated from the Greek; *Kitāb fī al-Zīj*; and many others too numerous to mention.

ذكر إسحاق بن حنين ومصنفاته

Account of Isḥāq ibn Ḥunayn and His Compositions

إسحاق بن حنين بن إسحاق العبادي كان طبيباً حكيماً
وقد والفلك والحساب والهندسة والفلسفة بالمنطق عالماً نقل
وبقراط وجالينوس وأفلاطون أرسطو كتب من كثيراً
أبوه وكان وصححها بعضها وشرح وبطليموس وأقليدس
الفن هذا في إنتاجاً وأكثرهم النقّال أعظم إسحاق بن حنين

Isḥāq ibn Ḥunayn ibn Isḥāq al-çIbādī was a physician and sage, learned in logic, philosophy, geometry, arithmetic, and astronomy. He translated many books of Aristotle, Plato, Galen, Hippocrates, Euclid, and Ptolemy, commented upon some of them, and corrected them. His father, Ḥunayn ibn Isḥāq, was the greatest of translators and the most prolific in this art.

وله كتاب أقليدس في الأصول الهندسية نقله عن اليوناني
المنطق في وكتاب أيضاً نقله لبطليموس المجسطي وكتاب
المفردة الأدوية في وكتاب الهيئة علم في وكتاب
نقلها التي الكتب من ذلك وغير الحيوان علم في وكتاب
قدراً وأجلهم النقّال أعظم من وكان وشرحها

His works include: *Euclid's Elements* which he translated from the Greek; the *Almagest* of Ptolemy, likewise translated; *Kitāb fī al-Manṭiq*; *Kitāb fī ʿIlm al-Hay'a*; *Kitāb fī al-Adwiya al-Mufrada*; *Kitāb fī ʿIlm al-Ḥayawān*; and other books that he translated and commented upon. He was among the greatest and most eminent of translators.

ذكر قسطا بن لوقا ومصنفاته

Account of Qusṭā ibn Lūqā and His Compositions

قسطا بن لوقا البعلبكي كان من أعظم الحكماء وأكثرهم
والطبيعيات والفلك والهندسة الحساب علوم في تصنيفاً
اليونانيين كتب من كثيراً نقل وقد والأخلاق والمنطق
العلوم هذه في كثيرة مصنفات وله وشرحها وصححها

Qusṭā ibn Lūqā al-Baḥlabakkī was among the greatest philosophers and most prolific authors in the sciences of arithmetic, geometry, astronomy, natural philosophy, logic, and ethics. He translated many books from the Greeks, corrected them, and commented upon them. He has many compositions in these sciences.

وله كتاب في علم الهيئة وكتاب في علم الفلك وكتاب في
وكتاب الموسيقى في وكتاب الحساب في وكتاب الهندسة
في
وكتاب الأخلاق في وكتاب المنطق في وكتاب الطبيعة علم
في
الب في وكتاب الأسطرلاب عمل في وكتاب الشافي الدواء
▯▯▯▯▯
كثرة يحصى لا مما ذلك وغير

His works include: *Kitāb fī ʿIlm al-Hay'a*; *Kitāb fī ʿIlm al-Falak*; *Kitāb fī al-Hindsa*; *Kitāb fī al-Ḥisāb*; *Kitāb fī al-Mūsīqā*; *Kitāb fī ʿIlm al-Ṭabiā*; *Kitāb fī al-Manṭiq*; *Kitāb fī al-Akhlāq*; *Kitāb fī al-Dawā' al-Shāfī*; *Kitāb fī ʿAmal al-Asturlāb*; and *Kitāb fī al-Bī'lān*; and others too numerous to count.

ذكر البتاني ومصنفاته

Account of al-Battānī and His Compositions

محمد بن جابر بن سنان البتاني الحرّاني الصّابي كان عالماً
أصلح قد محققاً بارعاً فيه فاضلاً والفلك الهيئة بعلم
بزيج سمي به خاصاً زيجاً وعمل وصححه المجسطي زيج
ومسيراتها وحركاتها الكواكب أحوال فيه بين وقد الصابي
زمانه في وأفضلهم العلم هذا أهل أجل من وكان

Muḥammad ibn Jābir ibn Sinān al-Battānī al-Ḥarrānī al-Ṣābī was a scholar of astronomy, distinguished, outstanding, and precise. He corrected the *Zīj al-Majisṭī* and compiled a *zīj* of his own called the *Zīj al-Ṣābī*, in which he explained the conditions of the planets, their motions, and their movements. He was among the most eminent and excellent people of this science in his time.

وله كتاب الزيج الصابي وكتاب في علم الهيئة وكتاب في
علم في وكتاب المساحات علم في وكتاب الأسطرلاب عمل
في وكتاب ومسيراتها الكواكب أحوال في وكتاب الأزياج
المفيدة الكتب من ذلك وغير والملاحظات الرصد

His works include: *Kitāb al-Zīj al-Ṣābī*; *Kitāb fī ʿIlm al-Hay'a*; *Kitāb fī ʿAmal al-Asturlāb*; *Kitāb fī ʿIlm al-Masāḥāt*; *Kitāb fī ʿIlm al-Azyāj*; *Kitāb fī Aḥwāl al-Kawākib wa Masārātihā*; *Kitāb fī al-Raṣd wa al-Mulāḥaẓāt*; and other useful books.

ذكر إبراهيم بن سنان ومصنفاته

Account of Ibrāhīm ibn Sinān and His Compositions

إبراهيم بن سنان بن ثابت بن قرّة الحرّاني كان عالماً
ورث وقد والحساب والهندسة الهيئة علم في بارعاً فاضلاً
سنان أبيه عن وأخذه قرّة بن ثابت جده عن العلم
القطع مساحة في ألف قد الفهم سريع فطناً ذكياً وكان
الفلك علم وفي المخروطات وفي المكافئ

Ibrāhīm ibn Sinān ibn Thābit ibn Qurra al-Ḥarrānī was a scholar, distinguished and outstanding in astronomy, geometry, and arithmetic. He inherited learning from his grandfather Thābit ibn Qurra and received it from his father Sinān. He was intelligent, sharp, and quick of understanding. He composed works on the quadrature of the parabola, on conic sections, and on astronomy.

وله كتاب في مساحة القطع المكافئ وكتاب في
المخروطات
في وكتاب الفلك علم في وكتاب الهيئة علم في وكتاب
حساب
الأسطرلاب عمل في وكتاب الأزياج في وكتاب المنازلات
الهيئة علم في وأفضلهم عصره علماء أجل من وكان
في ذكره ما بعض في بطليموس طريقة انتقد وقد
المجسطي

His works include: *Kitāb fī Masāḥat al-Qiṭāc al-Mukāfi'*; *Kitāb fī al-Makhrūṭāt*; *Kitāb fī ʿIlm al-Hay'a*; *Kitāb fī ʿIlm al-Falak*; *Kitāb fī Ḥisāb al-Manāzalāt*; *Kitāb fī al-Azyāj*; and *Kitāb fī ʿAmal al-Asturlāb*. He was among the most eminent scholars of his age and the most excellent in astronomy, and he criticized Ptolemy's method in some matters mentioned in the *Almagest*.

5 Discourse V: Medicine and Natural Sciences

الفن الخامس في الطب والعلوم الطبيعية

THE FIFTH DISCOURSE: MEDICINE AND THE NATURAL SCIENCES

في أطباء اليونان والهنود والفرس

وأما أصناف الطب فإنها: علم الأدوية وعلم الأغذية وعلم
والقانوني الصحيح الحفظان وعلم وأسبابها الأمراض
وأجلهم الفن هذا في الناس أعظم اليونان أطباء كان وقد
وبولس وجالينوس وأرسطوطاليس أبقراط فمنهم قَدْراً
وبطليموس
المسلمون فدرسها العربية إلى كتبهم نقلت وقد وغيرهم
والتلخيص والشرح التصنيف فيها وأكثروا

On the Physicians of the Greeks, Indians, and Persians

The categories of medicine are: the science of drugs, the science of nutrition, the science of diseases and their causes, and the science of the regimen of health. The Greek physicians were the greatest people in this art and the most eminent: among them Hippocrates, Aristotle, Galen, Paulus, Ptolemy, and others. Their books were translated into Arabic, and the Muslims studied them, composing commentaries, abridgements, and summaries.

في أطباء المسلمين

وأما أطباء المسلمين فمنهم الحكماء الذين تركوا التصنيف
وابن والزهراوي سينا وابن الرازي ومنهم كثيراً الطب في
العلم هذا في صنفوا وقد وغيرهم وبختيشوع ماسويه
اليونانيون صنفه ما وتضاهي الأرفف تملأ عظيمة كتباً
المفردة الأدوية في وأبحاثهم تجاربهم إليه أضافوا وقد
والمركبة

On the Muslim Physicians

Among the Muslim physicians were wise men who composed abundantly in medicine: al-Rāzī, Ibn Sīnā, al-Zahrāwī, Ibn Māsawayh, Bakhtīshūç, and others. They composed great books in this science filling the shelves, rivalling what the Greeks composed, and added their own experiences and research on simple and compound remedies.

في علم الطبيعيات

وأما الطبيعيات فهي علوم يعرف بها الأجسام الطبيعية
وأمورها
سميت العلم هذا في كتباً أرسطوطاليس وضع وقد
بالطبيعيات
وكتاب والعالم السماء وكتاب الطبيعي السماع كتاب ومنها
وغيرها النفس وكتاب الأرواح وكتاب والفساد الكائن
عليها وزادوا وشرحوها المسلمون ونقلها

On Natural Philosophy

Natural philosophy is the science by which one knows natural bodies and their properties. Aristotle composed books in this science called the *Natural Works*: among them *On Natural Hearing* (Physics), *On the Heavens and the World*, *On Generation and Corruption*, *On Spirits*, *On the Soul*, and others. The Muslims translated them, commented upon them, and added to them.

6 Discourse VI: Philosophy and Theology

الفن السادس في الفلسفة والمنطق والكلام

The Sixth Discourse: Philosophy, Logic, and Scholastic Theology

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في الفلسفة

وأما الفلسفة فإنها علم يعرف به حقائق الأشياء كما هي
متعددة العلم هذا في كتباً أرسطوطاليس وضع وقد
الأبواب
نقلت وقد والأخلاق والإلهيات والطبيعيات المنطق ومنها
فيها وألفوا وشرحوها المسلمون فدرسها العربية إلى كتبه
سينا وابن والفارابي الكندي أشهرهم فمن جمة كتباً

On Philosophy

Philosophy is the science by which one knows the realities of things as they are. Aristotle composed books in this science of multiple branches: logic, natural philosophy, metaphysics, and ethics. His books were translated into Arabic; the Muslims studied them, commented upon them, and composed abundant books. Among the most famous are: al-Kindī, al-Fārābī, and Ibn Sīnā.

في المنطق

وأما المنطق فهو علم يعرف به الأفكار وكيفية صوابها
أشهرها كتب في أرسطوطاليس وضعه وقد وخطأها
الأورغانون
وشرحوه ودرسوه اليونانيين عن المسلمون ونقله
عدي بن يحيى أشهرهم ومن كثيرة كتباً فيه وصنفوا
وغيرهم الراوندي وابن والفارابي

On Logic

Logic is the science by which one knows thoughts and the manner of their correctness and error. Aristotle established it in books whose most famous is the *Organon*. The Muslims translated it from the Greeks, studied it, commented upon it, and composed many books. Among the most famous: Yaḥyā ibn çAdī, al-Fārābī, and Ibn al-Rāwandī.

في الكلام

وأما الكلام فهو علم يعرف به الأحكام الشرعية وأدلتها
في يتكلمون العلماء من جماعة الإسلام صدر في كان وقد
المذاهب أصحاب عنهم فنشأ عليها ويستدلون الأحكام
وغيرهم والمرجئة والمختزلة والأشاعرة كالمعتزلة
فيه مصنفة وكتب خاص مذهب لها فرقة وكل

On Scholastic Theology (*Kalām*)

Kalām is the science by which one knows legal rulings and their proofs. In the early period of Islam there were groups of scholars who dealt with rulings and argued for them, giving rise to the sects: the Muţazila, the Ashāriyya, the Khawārij, the Murji'a, and others. Each sect has its own doctrine and composed books thereon.

7 Discourse VII: Occult Sciences

الفن السابع في السحر والطلاسم والعزائم والكسوفات

The Seventh Discourse: Magic, Talismans, Incantations, and Catoptrics

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في السحر

وأما السحر فهو علم يعرف به التأثيرات الغريبة التي تظهر
قال حيث الكريم القرآن في ذكره ورد وقد العادة خلاف
تعالى
في السحرة كان فقد الساحران وهبط بساحرين هم وما
كتبهم نقلت وقد كثيراً والفرس والهنود اليونانيين
المسلمين من قوم ودرسها العربية إلى

On Magic

Magic is the science by which one knows strange effects that appear contrary to custom. It is mentioned in the Noble Qur’an where the Exalted says: “and they were not magicians” and “the two magicians came down.” There were many magicians among the Greeks, Indians, and Persians; their books were translated into Arabic and studied by some Muslims.

في الطلاسم والعزائم

وأما الطلاسم فهي أعداد وحروف ورموز تكتب في أوقات
وتسمى معين لغرض معينة أشياء على خاصة فلكية
بالطلاسم
كتب فيها صنف وقد العمل على تختم أي تطلسم لأنها
العزائم وكتاب لهرمس الطلاسم كتاب أشهرها كثيرة
كثير وغيرهما البلخي معشر لأبي

On Talismans and Incantations

Talismans are numbers, letters, and symbols written at special astronomical times on specific objects for a particular purpose. They are called *talāsim* because they are “sealed” upon the act. Many books have been composed on this subject, the most famous being the *Kitāb al-Ṭalāsim* of Hermes and the *Kitāb al-‘Azā’im* of Abū Maşar al-Balkhī, among others.

8 Discourse VIII: Religious Sects and Doctrines

الفن الثامن في أصحاب الملل والنحل والفرق

The Eighth Discourse: The People of Religions, Sects, and Factions

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في الملل والنحل

وأما أصحاب الملل فهم الذين لهم دين مبني على أصل
معين
والنصارى كاليهود السماوية الكتب أصحاب ومنهم
والمجوس
فرق الملل هذه من واحدة ولكل والمشركون والصابئة
بها خاص مذهب لها فرقة وكل متعددة وطوائف
وحديثة قديمة فيه مصنفة وكتب

On Religions and Sects

The people of religions are those who have a religion built upon a certain foundation: among them the People of the Book — the Jews, the Christians, the Magians, the Sabians, and the polytheists. Each of these religions has multiple sects and factions; each sect has its own doctrine and books, ancient and modern.

في الفرق الإسلامية

وأما فرق المسلمين فهم: الخوارج والشيعة والمعتزلة
وغيرهم والناجية والقدرية والجبرية والمرجئة والأشاعرة
وكتب ومصنفون أئمة لها الفرق هذه من واحدة وكل
النوبختي المسلمين: فرق في المصنفين أشهر ومن
وغيرهم والبغدادي حزم وابن والمقدسي

On Islamic Sects

The Islamic factions are: the Khawārij, the Shīą, the Muţazila, the Ashāriyya, the Murji'a, the Jabriyya, the Qadariyya, the Najiya, and others. Each of these factions has its leaders, authors, and books. Among the most famous authors on Islamic sects: al-Nawbakhtī, al-Maqdisī, Ibn Ḥazm, and al-Baghdādī.

9 Discourse IX: Alchemy

الفن التاسع في الكيمياء والصنعة الإلهية

THE NINTH DISCOURSE: ALCHEMY AND THE DIVINE CRAFT

في الكيمياء وأصولها

وأما الكيمياء فهي علم يعرف به تحويل الأجسام من صفة
فيها يطلب حيث المعادن في وخاصة أخرى صفة إلى
تحويل
وقالوا صحيح علم أنها بعضهم ادعى وقد ذهب إلى المكلف
الصادقون المخلصون يعرفها وإنما إلهية صنعة إنها
ومكر خدع هي بل صحيح بعلم ليست إنها آخرون وقال

On Alchemy and Its Foundations

Alchemy is the science by which one knows the transformation of bodies from one quality to another, especially metals, where the aim is to turn base metal into gold. Some claim it is a true science, calling it the "divine craft" (*ṣanā ʿilāhiyya*) and saying it is known only by the sincere and truthful. Others say it is not a true science but deception and trickery.

في المؤلفين في الكيمياء

وقد صنف في الكيمياء قوم كثير منهم جابر بن حيان
وقد السلام عليه محمد بن جعفر الصادق أصحاب من وكان
الرحمة وكتاب السبعين كتاب منها كثيرة كتباً فيها ألف
السرخسي ومنهم وغيرها البديع وكتاب الكمال وكتاب
كثير وغيرهم الصلاح أبي وابن والطغرائي

On the Authors of Alchemical Works

Many people composed works on alchemy: among them Jābir ibn Ḥayyān, who was a companion of al-Ṣādiq Jafaṛ ibn Muḥammad — peace be upon him. He composed many books on it, including the *Kitāb al-Sabc̄in* (Book of Seventy), the *Kitāb al-Raḥma*, the *Kitāb al-Kamāl*, the *Kitāb al-Badīʿ*, and others. Among them also al-Sarrakhsī, al-Ṭughrāʾī, Ibn Abī al-Ṣalāḥ, and many others.

10 Discourse X: History, Biography, and Genealogy

الفن العاشر في التاريخ والسير والأنساب

The Tenth Discourse: History, Biographies, and Genealogy

في المؤرخين

وأما الفن العاشر فهو الفن الأخير في هذا الكتاب وقد
وأصحاب والمغازي السير ومصنفي المؤرخين فيه ذكرنا
إسحاق ابن المؤرخين: أشهر ومن والأخباريين الأنساب
ومسكويه والطبري سعد وابن هشام وابن والواقدي
كثير وغيرهم والمسعودي م□□□ويه وابن

On the Historians

The tenth discourse is the last in this book. We mention in it the historians, authors of *siyar* and *maghāzī*, the genealogists, and the annalists. Among the most famous historians: Ibn Isḥāq, al-Wāqidī, Ibn Hishām, Ibn Saḍ, al-Ṭabarī, Miskawayh, al-Masūdī, and many others.

في أصحاب السير والمغازي

وأما أصحاب السير والمغازي فمنهم ابن إسحاق صاحب
المغازي كتاب صاحب والواقدي والفتوح المغازي كتاب
صاحب سعد وابن النبوية السيرة كتاب صاحب هشام وابن
البلدان فتوح كتاب صاحب والبلاذري الكبير الطبقات كتاب
الفن هذا في صنف ممن كثير وغيرهم الأشراف وأنساب

On the Authors of Campaigns and Biographies

Among the authors of *siyar* and *maghāzī*: Ibn Isḥāq, author of *Kitāb al-Maghāzī wa al-Futūḥ*; al-Wāqidī, author of *Kitāb al-Maghāzī*; Ibn Hishām, author of *Kitāb al-Sīra al-Nabawiyya*; Ibn Saḍ, author of *Kitāb al-Ṭabaqāt al-Kabīr*; al-Balādhurī, author of *Futūḥ al-Buldān* and *Ansāb al-Ashrāf*; and many others who composed in this art.

Appendix: The Translation Movement

في أول نقل العلوم إلى العربية

ON THE FIRST TRANSLATION OF THE SCIENCES INTO ARABIC

قصة خالد بن يزيد

The Story of Khālid ibn Yazīd

The following celebrated passage from the *Fihrist* recounts the legendary origins of the Graeco-Arabic translation movement:

كان خالد بن يزيد بن معاوية يُسمّى حكيم آل مروان وكان
الصنعة بباله خطر للعلوم ومحبّة همة وله نفسه في فاضلاً

ينزل كان ممن اليونانيين فلاسفة من جماعة بإحضار فأمر
في الكتب بنقل وأمرهم بالعربيّة تفقه وقد مصر مدينة

أول وهذا العربي إلى والقبطي اليوناني اللسان من الصنعة
الديوان نقل ثم لغة إلى لغة من الإسلام في كان نقل
الفارسية باللغة وكان

Khālid ibn Yazīd ibn Muāwiya, called the Sage of the Marwānids, was virtuous in himself, ambitious, and fond of the sciences. It occurred to him to commission a group of Greek philosophers residing in Egypt who had become proficient in Arabic to translate books on the craft from Greek and Coptic into Arabic. This was the first translation in Islam from one language to another; then the *dīwān* was translated from Persian.

في بيت الحكمة

On the House of Wisdom

ثم إن المأمون لما ولي الخلافة أمر بإنشاء بيت الحكمة
لينقلوا والنقّال العلماء من جماعة فيه وجعل بغداد في
وغيرها والحساب والطب والفلك والهندسة الفلسفة كتب
والهنود والفارسيين والسريانيين اليونانيين لغات من
العلوم نشر في عظيماً سبباً ذلك وكان العربية إلى

When al-Ma'mun assumed the caliphate, he ordered the establishment of the House of Wisdom (*Bayt al-Ḥikma*) in Baghdad and appointed scholars and translators to translate books of philosophy, geometry, astronomy, medicine, arithmetic, and other sciences from the languages of the Greeks, Syriacs, Persians, and Indians into Arabic. This was a great cause in the dissemination of the sciences.

End of Bilingual Edition

This bilingual edition presents a representative selection from the *Kitāb al-Fihrist* of Ibn al-Nadīm (c. 987 CE), based on the critical edition of Gustav Flügel (Leipzig: F. C. W. Vogel, 1871–1872). The parallel Arabic-English text covers all ten discourses of the *Fihrist*, with special emphasis on the mathematical sciences and the biographical entries for Islamic mathematicians and scientists. The mathematical subsection (Section IV) constitutes a key primary source for the history of Islamic mathematics, preserving unique information about al-Khwārizmī, Thābit ibn Qurra, and the translation movement.

Index of Persons

Name	**Significance in the *Fihrist***
Abū Maşar al-Balkhī	Astrologer; author of the *Kitāb al-Madkhal al-Kabīr*
al-Battānī (Albategnius)	Astronomer; *Zīj al-Ṣābī*; refined trigonometric methods
al-Farghānī	Astronomer; *Kitāb fī al-Hay'a*
al-Khwārizmī	Mathematician; *Kitāb al-Jabr*; father of algebra
al-Rāzī (Rhazes)	Physician and philosopher; author of the *al-Ḥāwī*
Aristotle	Greek philosopher; *Organon*, *Physics*, *Metaphysics*
Ḥunayn ibn Isḥāq	Chief translator of Galen and Hippocrates
Ibn Sīnā (Avicenna)	Philosopher and physician; *Canon of Medicine*
Ibrāhīm ibn Sinān	Mathematician; quadrature of the parabola
Isḥāq ibn Ḥunayn	Translator of Euclid's *Elements* and Ptolemy's *Almagest*
Jābir ibn Ḥayyān	Alchemist; *Kitāb al-Sabčin*
Khālid ibn Yazīd	Legendary initiator of the translation movement
Ptolemy	Astronomer; *Almagest*
Qusṭā ibn Lūqā	Translator; works on astronomy, mechanics, geometry
Thābit ibn Qurra	Mathematician, astronomer, translator of Apollonius

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Finis

Tabaqat al-Umam

(Categories of Nations)

كِتَابُ طَبَقَاتِ الْأُمَمِ

Bilingual Edition

Arabic Text • English Translation

By

Abu al-Qasim Sa'id ibn Ahmad al-Andalusi

(d. 462 AH / 1070 CE)

لِلْقَاضِي أَبِي الْقَاسِمِ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدٍ الْأَنْدَلُسِيِّ

الْمَتَوَفَّى سَنَةَ ٤٦٢ هـ ▯ ١٠٧٠ م

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Editor’s Introduction

تَوْطِئَةُ الْمُحَقِّقِ

كِتَابُ طَبَقَاتِ الْأُمَمِ أَحَدُ الْكُتُبِ النَّادِرَةِ الَّتِي نَعَرَّضُ فِيهَا كَتَبَةُ الْعَرَبِ لِوَصْفِ الْعُلُومِ بَيْنَ الْأُمَمِ الَّتِي سَبَقَتْ عَهْدَهُمْ، وَإِنْ لَمْ يَبْلُغْ صَاحِبُهُ فِي ذَلِكَ شَأْوَ كِتَابِ الْفِهْرِسْتِ لِأَبِي الْفَرَجِ ابْنِ الدَّمِيمِ إِلَّا أَنَّهُ جَمَعَ عِدَّةَ فَوَائِدَ تَدُلُّ عَلَى نَشَاطٍ فِي الْبَحْثِ وَعَلَى رَغْبَةٍ فِي التَّحْصِيلِ وَدَقَّةِ نَظَرٍ فِي التَّنوِينِ، وَكَانَ أَهْلُ الْأَنْدَلُسِ يَفْتَخِرُونَ بِهِ وَبِرُؤُوسِ أَهْلِ الشَّرْقِ.

وَقَدْ ذَكَرَ ابْنُ الْأَبَّارِ فِي كِتَابِ التَّكْمِلَةِ لِكِتَابِ الصِّلَةِ □٢ : ٤٦٣ طَبْعَةُ بَجْرَيْطَ□ عَنْ عَبْدِ اللهِ بْنِ مُحَمَّدِ بْنِ مَرْزُوقٍ الْيَحْصَبِيِّ أَنَّهُ قَدَّمَ الْإِسْكَنْدَرِيَّةَ رَوَى هَذَا الْكِتَابَ لِأَبِي طَاهِرٍ السَّلَفِيِّ.

وَمِمَّنْ عَرَّفُوا هَذَا الْكِتَابَ فِي الشَّرْقِ أَبُو الْفَرَجِ غْرِيغُورِيُوسَ ابْنُ الْعِبْرِيِّ فَإِنَّهُ نَقَلَ عَنْهُ فِي كِتَابِ تَارِيخِ مُخْتَصَرِ الدُّوَلِ نَبْذَتَيْنِ مُفِيدَتَيْنِ فِي الْعَرَبِ وَعُلُومِهِمْ، وَكَذَلِكَ عَرَفَهُ الْحَاجُّ خَلِيفَةُ فَذَكَرَهُ مِرَارًا فِي كِتَابِهِ كَشْفِ الظُّنُونِ فَدَعَا تَارَةً التَّعْرِيفُ بِطَبَقَاتِ الْأُمَمِ وَقَالَ فِي وَصْفِهِ إِنَّهُ كِتَابٌ صَغِيرُ الْحَجْمِ كَثِيرُ النَّفْعِ.

وَهِيَ مَكْتُوبَةٌ بِخَطٍّ جَلِيٍّ شَبِيهٍ بِالْقَلَمِ الْفَارِسِيِّ عَلَى وَرَقٍ صَفِيقٍ ضَارِبٍ إِلَى الصَّفْرَةِ وَمُجَلَّدَةٌ تَجْلِيدًا مُتْقِنًا بِجِلْدٍ وَرَقٍ مُلَوَّنٍ ذَهَبِيٌّ عَلَى الْوَجْهَيْنِ مَعَ لِسَانٍ مِثْلِهِمَا زَيْنَةٌ.

أَمَّا الْمُؤَلِّفُ فَلَا نَعْلَمُ أَقَلَّ مِنْ أَمْرِهِ، وَهَذِهِ تَرْجَمَتُهُ كَمَا رَوَاهَا ابْنُ بَشْكُوَالٍ فِي كِتَابِ الصِّلَةِ □طَبْعَةُ بَجْرَيْطَ ص ٢٣٤□:

□□صَاعِدُ بْنُ أَحْمَدَ بْنِ عَبْدِ الرَّحْمَنِ بْنِ مُحَمَّدِ بْنِ صَاعِدٍ التَّغَلِبِيُّ الْقَاضِي بِكُنْيَةِ أَبِي الْقَاسِمِ وَأَصْلُهُ مِنْ قُرْطُبَةَ رُوِيَ عَنْ أَبِي مُحَمَّدِ بْنِ حَزْمٍ وَالْفَتْحِ بْنِ قَاسِمٍ وَأَبِي الْوَلِيدِ الْوَقْشِيِّ وَغَيْرِهِمْ، وَاسْتَقْضَاهُ الْأَمِيرُ يَحْيَى بْنُ ذِي النُّونِ بِطُلَيْطَلَةَ وَكَانَ مِنْ أَهْلِ الْمَعْرِفَةِ وَالذَّكَاءِ وَالرِّوَايَةِ، وُلِدَ بِالْمَرِيَّةِ فِي سَنَةِ ٤٢٠ □هـ □ ١٠٣٩ م□ وَتُوُفِّيَ بِطُلَيْطَلَةَ فِي شَوَّالِ سَنَةِ ٤٦٢ هـ □ ١٠٧٠ م.□□

[Editor’s Introduction]

Kitab Tabaqat al-Umam (The Book of the Categories of Nations) is one of the rare works in which an Arab writer describes the sciences among the nations that preceded his era. Although its author did not reach the level of Abu al-Faraj ibn al-Damimi in his *Fihrist*, he nevertheless collected numerous useful facts that demonstrate his research activity, his desire for knowledge, and his precision in observation. The people of al-Andalus used to boast of him and of the heads of the East.

Ibn al-Abbār mentions in his *Takmila* to the *Silah* (2:463, Bajarid edition) on the authority of Abd Allah ibn Muhammad ibn Marzūq al-Yahṣibí that he transmitted this book to Abu Tahir al-Salafí.

Among those who knew this book in the East was Abu al-Faraj Grigorius ibn al-ʿIbrī, for he quoted from it in his *History of the Dynasties* two useful excerpts concerning the Arabs and their sciences. Likewise, al-Hājj Khalīfa knew it and mentioned it several times in his *Kashf al-Zunun*: at one point calling it *Taʿrīf bi-Tabaqat al-Umam*, and describing it as a small book of great benefit.

[Manuscript Description:] The manuscript is written in a clear hand resembling Persian script, on thick paper tending toward yellow, and bound with skillful binding in colored gold paper on both sides with a matching tongue as ornament.

As for the author, we know no less than what Ibn Bashkuwal narrated in his *Silah* (Bajarid edition, p. 234):

> “Saʿid ibn Ahmad ibn ʿAbd al-Raḥmān ibn Muḥammad ibn Saʿid al-Taghlabī, the judge with the kunya Abu al-Qāsim, originally from Cordoba, transmitted from Abu Muḥammad ibn Ḥazm, al-Fatḥ ibn Qāsim, Abu al-Walīd al-Waqshī, and others. The amir Yahyā ibn Dhī al-Nūn appointed him as judge of Tulaytula (Toledo). He was a man of knowledge, intelligence, and transmission. He was born in al-Mariyya (Almeria) in 420 AH (1039 CE) and died in Tulaytula while serving as its judge in Shawwal of the year 462 AH (1070 CE).”

Chapter One: The Ancient Nations

الْبَابُ الْأَوَّلُ : الْأُمَمُ الْقَدِيمَةُ

بِسْمِ اللهِ الرَّحْمَنِ الرَّحِيمِ

رَبِّ يَسِّرْ

قَالَ الْقَاضِي أَبُو الْقَاسِمِ صَاعِدُ بْنُ أَحْمَدَ بْنِ صَاعِدٍ رَحِمَهُ اللهُ تَعَالَى: اعْلَمْ أَنَّ جَمِيعَ النَّاسِ فِي مَشَارِقِ الْأَرْضِ وَمَغَارِبِهَا وَجَنُوبِهَا وَشَمَالِهَا وَإِنْ كَانُوا نَوْعًا وَاحِدًا يَتَمَيَّزُونَ بِثَلَاثَةِ أَشْيَاءَ: بِالْأَخْلَاقِ وَالصُّوَرِ وَاللُّغَاتِ.

وَزَعَمَ مَنْ عُنِّيَ بِأَخْبَارِ الْأُمَمِ وَبَحَثَ عَنْ سَائِرِ الْأَجْيَالِ وَفَحَصَ عَنْ طَبَقَاتِ الْقُرُونِ أَنَّ النَّاسِ كَانُوا فِي سَالِفِ الدُّهُورِ وَقَبْلَ تَشَعُّبِ الْقَبَائِلِ وَافْتِرَاقِ اللُّغَاتِ سَبْعَ أُمَمٍ.

□الْأُمَّةُ الْأُولَى□ الْفُرْسُ وَكَانَ مَسْكَنُهَا فِي الْوَسَطِ الْمَعْمُورِ وَحِدٌّ بِلَادِهَا مِنَ الْجِبَالِ الَّتِي فِي شِمَالِ الْعِرَاقِ بِعَقَبَةِ حُلْوَانَ وَالَّذِي فِيهِ الْأَنْجَهَاتُ وَالْكُرَجُ وَالدُّسْتُورُ وَهُمَدَانَ وَقُمَّ وَقَاشَانَ وَغَيْرِهَا مِنَ الْبِلَادِ إِلَى أَرْمِينِيَةَ وَالْبَابِ الْمُتَّصِلِ بِبَحْرِ أَذَرْبِيجَانَ وَطُبَرِسْتَانَ وَمُولْتَانَ وَالْبِيلَقَانَ وَأَرْزَنَ وَالشَّابَرَانَ وَالرِّيِّ وَالطَّالِقَانَ وَجُرْجَانَ إِلَى بِلَادِ خُرَاسَانِ.

[Chapter One: The Ancient Nations]

In the name of God, the Merciful, the Compassionate.

O Lord, make it easy!

The judge Abu al-Qāsim Sa‘īd ibn Aḥmad ibn Sa‘īd—may God the Exalted have mercy on him—said: Know that all people in the East and West of the Earth, in its South and North, though they are one species, are distinguished by three things: by their moral character, their physical forms, and their languages.

Those who have concerned themselves with the reports of nations and investigated the various generations, and examined the categories of ages, say that in earliest times, before the branching of tribes and the divergence of languages, there were seven nations.

The First Nation: The Persians. Their dwelling was in the middle of the inhabited world. Their lands include the mountains in the north of Iraq, the region of Hulwan, and the area containing the Anjahāt, Kuraj, Dustūr, Hamadān, Qumm, Qāshān, and other lands extending to Armenia, the Gate connecting to the Sea of Azerbaijan, Tabaristān, Multān, Bilāqān, Arzan, Shāborān, al-Rayy, Tāliqān, and Jurjān, to the lands of Khurāsān.

Chapter Two: Science in India

الْعِلْمُ فِي الْهِنْدِ

وَلِبُعْدِ الْهِنْدِ مِنْ بِلَادِنَا وَاعْتِرَاضِ الْمَوَانِعِ بَيْنَنَا وَبَيْنَهُمْ فَلَمْ نَأْلَفْهُمْ فَلَمْ يَصِلْ إِلَيْنَا طَرَفٌ مِنْ عُلُومِهِمْ وَلَا وَرَدَتْ عَلَيْنَا إِلَّا نُبْذٌ مِنْ مَذَاهِبِهِمْ وَلَا سَمِعْنَا إِلَّا بِالْقَلِيلِ مِنْ عِلْمِهِمْ.

فَمِنْ مَذَاهِبِ الْهِنْدِ فِي عُلُومِ النُّجُومِ الْمَذَهَبُ الثَّلَاثَةُ الْمَشْهُورَةُ عَنْهُمْ وَهُوَ مَذْهَبُ السِّنْدِ هِنْدٌ وَمَذْهَبُ الْأَزْدِيجِ وَمَذْهَبُ الْأَرْكَنْدِ وَلَمْ يَصِلْ إِلَيْنَا مِنْهُمْ عَلَى التَّحْصِيلِ إِلَّا مَذْهَبُ السِّنْدِ هِنْدٌ وَهُوَ الْمَذْهَبُ الَّذِي تَقَلَّدَهُ جَمَاعَةٌ مِنَ الْإِسْلَامِ وَالْقُوَّامُ فِيهِ الْأَزْيَاجُ كَمُحَمَّدِ بْنِ إِبْرَاهِيمَ الْفَزَارِيِّ وَحَبْشِ بْنِ عَبْدِ اللهِ الْبَغْدَادِيِّ وَمُحَمَّدِ بْنِ مُوسَى الْخَوَارِزْمِيِّ وَالْحُسَيْنِ بْنِ مُحَمَّدٍ الْمَعْرُوفِ بِابْنِ الْآدَمِيِّ وَغَيْرِهِمْ. وَتَفْسِيرُ السِّنْدِ هِنْدٍ □□الدَّهْرُ الدَّاهِرُ□□ كَذَلِكَ حَكَى الْحُسَيْنُ بْنُ الْآدَمِيِّ فِي زِيجِهِ.

□فَصْلٌ□

وَأَمَّا أَصْحَابُ السِّنْدِ هِنْدٍ فَإِنَّهُمْ وَافَقُوا أَنْ لَا عَدَدَ مَدَّةِ الْعَالَمِ فَإِنَّ مَذْهَبَهُمُ الَّتِي ذَكَرُوهَا أَنَّ الْكَوَاكِبَ وَأَوْجَاتِهَا وَجُوزَهَرَاتِهَا مُجْتَمِعَتْ فِي رَأْسِ الْحَمْلِ فَمُنْذُ ذَلِكَ الْأَمْرِ أَبْدَأَتْ حَرَكَاتُ الْكَوَاكِبِ وَالْأَوْجَاتِ وَالْجَوْزَهَرَاتِ فِي الْأَرْضِ وَبَقِيَ الْعَالَمُ السُّفْلَى خَرَابًا دَهْرًا طَوِيلًا حَتَّى تَتَفَرَّقَ الْكَوَاكِبُ وَالْأَوْجَاتُ وَالْجَوْزَهَرَاتُ فِي الْبُرُوجِ فَإِذَا كَانَ كَذَلِكَ بَدَأَتْ حَالَةُ الْعَالَمِ السُّفْلِيِّ إِلَى الْأَمْرِ الْأَوَّلِ هَكَذَا أَبْدَأً إِلَى غَيْرِ غَايَةٍ عِنْدَهُمْ. وَلِكُلِّ وَاحِدٍ مِنَ الْكَوَاكِبِ وَالْأَوْجَاتِ وَالْجَوْزَهَرَاتِ أَدْوَارٌ مَا فِي هَذِهِ الْمُدَّةِ الَّتِي هِيَ عِنْدَهُمْ مَدَّةُ الْعَالَمِ قَدْ ذَكَرْتُهَا.

[Science in India]

Because of the distance of India from our lands and the obstacles between us and them, we have not become familiar with them. No portion of their sciences has reached us, nothing has come to us but fragments of their doctrines, and we have heard only a little of their knowledge.

Among the doctrines of the Indians in the science of astrology, three doctrines are famous among them: the doctrine of Sindhind, the doctrine of the Āzīj, and the doctrine of Arkand. Only the doctrine of Sindhind has reached us in a complete form. It was followed by a group of Muslims, and those who established it were al-Azyāj such as Muḥammad ibn Ibrāhīm al-Fazārī, Ḥabash ibn ʿAbd Allāh al-Baghdādī, Muḥammad ibn Mūsā al-Khwārizmī, al-Ḥusayn ibn Muḥammad known as Ibn al-Ādamī, and others. The interpretation of Sindhind is “the Eternal Time” (al-Dahr al-Dāhir), as al-Ḥusayn ibn al-Ādamī related in his *zīj*.

[Section]

As for the followers of Sindhind, they agree that the duration of the world has no fixed number. Their doctrine which they mention is that the planets, their apogees, and their nodes were all gathered at the head of Aries. Since that event, the movements of the planets, apogees, and nodes began. The lower world remained desolate for a long time until the planets, apogees, and nodes dispersed through the zodiac signs. When that happened, the condition of the lower world returned to its first state, and this continues without end according to them. Each of the planets, apogees, and nodes has cycles during this period, which they consider the duration of the world, which I have mentioned.

Chapter Three: Science among the Persians

الْعِلْمُ فِي الْفُرْسِ

وَأَمَّا الْأُمَّةُ الثَّانِيَةُ وَهِيَ الْفُرْسُ فَأَهْلُ الشَّرْفِ الْبَاذِخِ وَالْعِزِّ الشَّامِخِ وَأَوْسَطِ الْأُمَمِ دَارًا وَأَشْرَفِهَا أَقْلِيمًا وَأَنْوَسَهَا مَلُوكًا وَلَا نَعْلَمُ أُمَّةً غَيْرَهُمْ دَامَ لَهَا الْمُلْكُ وَكَانَتْ لَهُمْ مُلُوكٌ تَجْمِعُهُمْ وَرُؤُوسٌ تُحَامِي عَنْهُمْ مَنْ نَاوَأَهُمْ وَتَغْلِبُ بِهِمْ مَنْ غَارَهُمْ وَتَدُفِّعُ ظَالِمَهُمْ عَنْ مَظْلُومِهِمْ وَتُحْتَمِلُهُمْ مِنَ الْأُمُورِ عَلَى مَا فِيهِ حَظُّهُمْ عَلَى اتِّصَالٍ وَدَوَامٍ وَأَحْسَنِ التَّنَظُّمِ يَأْخُذُ ذَلِكَ آخِرُهُمْ عَنْ أَوَّلِهِمْ وَغَائِبُهُمْ عَنْ سَالِفِهِمْ.

□فِي الْعَرَبِ وَالْعَجَمِ□

وَأَصَحُّ مَا قِيلَ فِي ذَلِكَ أَنَّ مِنْ ابْتِدَاءِ مُلْكِ كَيْنُمُورَثَ بْنِ أَمِيمِ بْنِ آلَاذِ بْنِ سَامِ ابْنِ نُوحٍ إِلَى مُلْكِ مَنُوشِهِرَ الْفُرْسِ كَانَهُمْ الَّذِي هُوَ عِنْدَهُمْ آدَمُ أَبُو الْبَشَرِ عَلَيْهِ السَّلَامُ إِلَى ابْتِدَاءِ مُلْكِ مَنُوشِهِرَ أَوَّلُ مُلُوكِ الطَّبَقَةِ الثَّانِيَةِ مِنْ مُلُوكِ الْفُرْسِ نَحوَ أَلْفِ سَنَةٍ كَامِلَةٍ. وَمِنْ مُلْكِ مَنُوشِهِرَ إِلَى ابْتِدَاءِ مُلْكِ كَيْثُبَاذَ بْنِ دُوعَ أَوَّلِ مُلُوكِ الطَّبَقَةِ الثَّالِثَةِ مِنْ مُلُوكِ الْفُرْسِ قَرِيبٌ مِنْ مِائَتَيْ عَامٍ. وَمِنْ مُلْكِ كَيْثُبَاذَ إِلَى ابْتِدَاءِ مُلْكِ الطَّوَائِفِ وَهِيَ الطَّبَقَةُ الرَّابِعَةُ مِنْ مُلُوكِ الْفُرْسِ وَذَلِكَ عِنْدَ مَقْتَلِ الْإِسْكَنْدَرِ لِدَارَا بْنِ دَارَا آخِرِ مُلُوكِ الطَّبَقَةِ الثَّالِثَةِ مِنْ مُلُوكِ الْفُرْسِ نَحوَ أَلْفِ سَنَةٍ.

وَمِنْ أَوَّلِ مُلْكِ الطَّوَائِفِ إِلَى ابْتِدَاءِ مُلْكِ أَزْدَشِيرِ بْنِ بَابَكَ السَّاسَانِيِّ أَوَّلِ مُلُوكِ بَنِي إِسْفَانِيَارَ وَهِيَ الطَّبَقَةُ الْخَامِسَةُ مِنْ مُلُوكِ الْفُرْسِ خَمْسُمِائَةِ سَنَةٍ وَإِحْدَى وَسِتُّونَ سَنَةً. وَمِنْ ابْتِدَاءِ مُلْكِ أَزْدَشِيرِ بْنِ بَابَكَ إِلَى انْقِضَاءِ دَوْلَةِ الْفُرْسِ مِنَ الْأَرْضِ وَذَلِكَ عِنْدَ قَتْلِ يَزْدَجِرْدَ بْنِ شَهْرِيَارِ بْنِ كِسْرَى أَرْبَعُ سِنِينَ.

[Science among the Persians]

As for the second nation, the Persians: they are a people of towering nobility and exalted glory, the most settled of nations and the noblest in region, and the most prosperous in kings. We know of no nation besides them whose sovereignty endured so long, who had kings that united them and leaders that defended them from those who opposed them, overcame with them those who envied them, and protected their oppressed from their oppressors, and who endured affairs in a way that secured their portion with continuity and permanence and the finest organization, so that the last of them took this over from the first, and the absent from the absent.

[On the Arabs and the Non-Arabs]

The most correct opinion on this is that from the beginning of the reign of Kayūmārth ibn Amīm ibn Ālādh ibn Sām ibn Nūḥ (Noah) to the reign of Manūshihr the Persian—who is, according to them, Adam the father of mankind, peace be upon him—to the beginning of the reign of Manūshihr, the first king of the second class of Persian kings, was about a thousand complete years. From the reign of Manūshihr to the beginning of the reign of Kaythubādh ibn Du'a, the first king of the third class of Persian kings, was about two hundred years. From the reign of Kaythubādh to the beginning of the reign of the Tribes (al-Tawā'if), the fourth class of Persian kings, at the killing of Alexander (al-Iskandar) of Dārā ibn Dārā, the last king of the third class of Persian kings, was about a thousand years.

From the beginning of the reign of Azdashīr ibn Bābak the Sāsānian, the first king of the Banū Isfāniyār (the fifth class of Persian kings), was five hundred and sixty-one years. From the beginning of the reign of Azdashīr ibn Bābak to the end of the Persian state on earth, at the killing of Yazdajird ibn Shahriyār ibn Kisrā, was four years.

Chapter Four: Science in Greece

الْعِلْمُ فِي الْيُونَانِ

وَأَمَّا الْيُونَانِيُّونَ فَهُمْ أَهْلُ فَلَسْطِينَ وَالرُّومِ وَقِسْطَنْطِينِيَّةَ وَالْأُمَمُ الْمُحِيطَةُ بِهَا. وَكَانَتْ دَارَةُ مُلْكِهِم مَمْتَدَّةً مِنْ شَطِّ بَحْرِ الْمُتَوَسِّطِ فِي الْمَشَارِقِ إِلَى شَطِّهِ فِي الْمَغَارِبِ، وَمِنْ أَقْصَى الْقُطْبِ إِلَى بَحْرِ الشَّامِ. وَكَانَتْ مَدِينَتُهُمُ الْعُظْمَى مَدِينَةُ الْإِسْكَنْدَرِيَّةِ. وَكَانَتْ دَوْلَتُهُمْ قَدِيمَةً جِدًّا وَعَلُومُهُمْ كَثِيرَةً مُتَنَوِّعَةً.

وَكَانَ أَوَّلُ مَنْ تَكَلَّمَ فِي الْعُلُومِ وَالْحِكَمِ عِنْدَهُمْ فَالِيسُ وَأَفْلَاطُونُ وَأَرِسْطُو وَبَطْلِيمُوسُ وَجَالِينُوسُ وَهِبَاطِيَا وَبَرْقَلِيسُ وَأُقْلِيدِسُ وَنِقُومَاخُسُ وَأَرْشِمِيدِسُ وَجَمِيعُ مَنْ يُنْسَبُ إِلَى هَؤُلَاءِ مِنْ أَهْلِ الْعُلُومِ وَالْحِكَمِ. وَهُمُ الَّذِينَ قَامُوا بِإِصْلَاحِ أُصُولِ الْعُلُومِ وَتَوْضِيحِهَا وَتَرْتِيبِ أَقْسَامِهَا وَتَقْيِيدِ قَوَاعِدِهَا وَكَانَتْ مُؤَلَّفَاتُهُمْ فِي ذَلِكَ بِلُغَتِهِمْ. ثُمَّ إِنَّمَا نُقِلَتْ إِلَى لُغَةِ الْعَرَبِ بَعْدَ أَنْ نَشَرَهَا الْمَنَاذِرَةُ وَاتَّخَذَهَا أَهْلُ الدِّينِ مِنَ النَّصَارَى فِي الشَّرْقِ.

[Science in Greece]

As for the Greeks, they are the people of Palestine, Rome, Constantinople, and the surrounding nations. Their kingdom extended from the shore of the Mediterranean Sea in the East to its shore in the West, and from the far north to the Syrian Sea. Their greatest city was the city of Alexandria. Their state was very ancient and their sciences were many and varied.

The first to discourse on the sciences and wisdom among them were Thales, Plato, Aristotle, Ptolemy, Galen, Hippocrates, Proclus, Euclid, Nicomachus, Archimedes, and all those attributed to these in the sciences and wisdom. They were the ones who established the principles of the sciences, elucidated them, arranged their parts, and codified their rules, and their compositions on this were in their own language. Then they were translated into Arabic after the Manādhirah disseminated them and the religious scholars among the Christians in the East adopted them.

Chapter Five: Science in Egypt

الْعِلْمُ فِي مِصْرَ

وَأَمَّا الْقِبْطُ فَهُمْ أَهْلُ مِصْرَ الْعُلَا وَكَانَتْ عَلُومُهُمْ قَدِيمَةً نَافِعَةً وَلَكِنَّهَا لَمْ تَبْلُغْ مُنْذُ أَخَذَتْهَا الْيُونَانِيُّونَ مِنْهُمْ فِي عَهْدِ بَطْلِيمُوسَ الْمَلِكِ الَّذِي بَنَى مَدِينَةَ الْإِسْكَنْدَرِيَّةِ وَجَعَلَهَا مَوْطِنًا لِلْعُلُومِ وَحَشَدَ إِلَيْهَا الْعُلَمَاءَ مِنْ جَمِيعِ الْأُمَمِ فَأَخَذُوا عَنِ الْقِبْطِ عُلُومَهُمْ فَأَدْخَلُوا فِيهَا مِنْ حِكَمِ الْيُونَانِ مَا أَضَافَ إِلَيْهَا وَنَقَّحُوهَا وَصَنَفُوا فِيهَا. وَمِنْ ذَلِكَ عِلْمُ الْهَيْئَةِ وَعِلْمُ تَأْلِيفِ الْكَوَاكِبِ وَعِلْمُ الْعِلَلِ وَالْأَدْوِيَةِ وَغَيْرِ ذَلِكَ مِنَ الْعُلُومِ.

[Science in Egypt]

As for the Copts, they are the people of Upper Egypt. Their sciences were ancient and beneficial, but they did not advance further since the Greeks took them from them in the time of Ptolemy the king who built the city of Alexandria and made it a center for the sciences, gathering scholars from all nations. They took from the Copts their sciences and incorporated into them Greek wisdom, adding to them, refining them, and composing works on them. Among these were the science of astronomy, the science of the configuration of the planets, the science of causes and remedies, and other sciences.

Chapter Six: Science among the Jews

الْعِلْمُ فِي الْيَهُودِ

وَأَمَّا الْيَهُودُ فَقَدْ كَانَتْ لَهُمْ عُلُومٌ فِي أَزْمِنَتِهِمُ الْأُولَى فِي زَمَنِ أَنْبِيَائِهِمْ وَمُلُوكِهِمْ. ثُمَّ إِنَّهُمْ لَمَّا خَرَجُوا مِنْ بِلَادِهِمْ وَتَفَرَّقُوا فِي بِلَادِ الْعَجَمِ وَالْعَرَبِ ضَاعَتْ عُلُومُهُمْ وَتَنَكَّرَتْ لِأَنَّهُمْ لَمْ يَكُونُوا أَهْلَ بَقَاءٍ وَلَا مُوَاظَبَةٍ عَلَى الْعِلْمِ. وَلَكِنَّهُمْ قَدْ حَفِظُوا مِنْ تَوَارِيثِ الْأَنْبِيَاءِ عُلُومًا فِي التَّعَبُّدِ وَالشَّرَائِعِ وَالتَّوْحِيدِ. وَكَانَ مِنْ أَشْهَرِ عُلَمَائِهِمْ مُوسَى بْنُ مَيْمُونٍ الْأَنْدَلُسِيُّ الَّذِي صَنَّفَ فِي عِلْمِ الْكَلَامِ وَالْفَلْسَفَةِ وَالطِّبِّ وَغَيْرِهَا.

[Science among the Jews]

As for the Jews, they had sciences in their earliest times, in the era of their prophets and kings. But when they left their lands and dispersed into the lands of the non-Arabs and the Arabs, their sciences were lost and forgotten because they were not a people of endurance or perseverance in science. However, they did preserve from the inheritance of the prophets sciences concerning worship, religious laws, and monotheism. Among their most famous scholars was Mūsā ibn Maymūn (Maimonides) al-Andalusī, who composed works on the science of theology, philosophy, medicine, and other subjects.

Chapter Seven: Science among the Arabs

الْعِلْمُ فِي الْعَرَبِ

وَأَمَّا الْعَرَبُ فَهُمْ أُمَّتَانِ: أُمَّةٌ بَادِيَةٌ وَأُمَّةٌ حَاضِرَةٌ. وَكَانَتِ الْبَادِيَةُ مِنْهُمْ لَا تَعْنَى بِالْعِلْمِ وَلَا تُوَاظِبُ عَلَيْهِ لِصِعْرِ مَعِيشَتِهِمْ وَتَنَقُّلِهِمْ فِي الْبِلَادِ. وَأَمَّا الْحَاضِرَةُ فَقَدْ كَانَتْ لَهُمْ عُلُومٌ فِي الْجَاهِلِيَّةِ فِي الطِّبِّ وَالنُّجُومِ وَالسِّحْرِ وَالشِّعْرِ وَالنَّحْوِ وَغَيْرِ ذَلِكَ. ثُمَّ لَمَّا جَاءَ الْإِسْلَامُ وَشَرَعَ فِي طَلَبِ الْعِلْمِ وَحَثَّ عَلَيْهِ وَجَعَلَهُ سَبَبًا لِلنَّجَاةِ وَالْفَلَاحِ اجْتَهَدَ الْعَرَبُ فِي طَلَبِ الْعُلُومِ وَنَقْلِهَا مِنَ الْأُمَمِ الْأُخْرَى وَتَصْيِيرِهَا بِلُغَةِ الْعَرَبِ حَتَّى أَصْبَحُوا فِي زَمَنٍ قَلِيلٍ أَهْلَ عُلُومٍ وَفُنُونٍ.

وَكَانَ أَوَّلُ مَا نَشَطُوا إِلَيْهِ عِلْمُ الْكَلَامِ وَالْفِقْهِ وَالْحَدِيثِ وَالتَّفْسِيرِ ثُمَّ إِنَّهُمْ تَعَرَّضُوا لِعُلُومِ الْأَوَائِلِ فَنَقَلُوهَا وَصَنَّفُوا فِيهَا وَأَضَافُوا إِلَيْهَا أَشْيَاءَ كَثِيرَةً. وَكَانَ مِنْ أَشْهَرِ الْعَرَبِ فِي عِلْمِ النُّجُومِ الْفَرْغَانِيُّ وَالْبَتَّانِيُّ وَأَبُو مَعْشَرٍ الْفَلَكِيُّ. وَفِي الطِّبِّ الرَّازِيُّ وَابْنُ سِينَا وَالْمَجُوسِيُّ وَابْنُ رِشْدٍ. وَفِي الْهَيْئَةِ بَطْلِيمُوسُ الْمَجْرِطِيُّ وَالْبِتْرُوجِيُّ وَشَرَفُ الدِّينِ الطُّوسِيُّ. وَفِي الْحِسَابِ الْخَوَارِزْمِيُّ وَالْكَشِّيُّ. وَفِي الْفَلْسَفَةِ الْكِنْدِيُّ وَالْفَارَابِيُّ وَابْنُ سِينَا وَابْنُ رِشْدٍ.

[Science among the Arabs]

As for the Arabs, they are two nations: a desert nation and a settled nation. The desert among them did not concern itself with science nor persevere in it due to the hardship of their livelihood and their movement from land to land. As for the settled people, they had sciences in the Jāhilī period in medicine, astrology, magic, poetry, grammar, and other subjects. Then when Islam came and legislated the pursuit of knowledge, encouraged it, and made it a means of salvation and success, the Arabs strove in seeking the sciences, translating them from other nations, and rendering them into Arabic, until in a short time they became a people of sciences and arts.

The first [sciences] they were active in were theology (*kalām*), jurisprudence (*fiqh*), *hadīth*, and [Qur'ānic] exegesis (*tafsīr*). Then they turned to the sciences of the ancients, translating them, composing works on them, and adding much to them. Among the most famous Arabs in the science of astrology were al-Farghānī, al-Battānī, and Abu Maʿshar the astronomer. In medicine: al-Rāzī, Ibn Sīnā (Avicenna), al-Majūsī, and Ibn Rushd (Averroes). In astronomy: Ptolemy al-Majritī, al-Bitrūjī (Alpetragius), and Sharaf al-Dīn al-Ṭūsī. In arithmetic: al-Khwārizmī and al-Kāshshī. In philosophy: al-Kindī, al-Fārābī, Ibn Sīnā, and Ibn Rushd.

Editorial and Scholarly Notes

On the Manuscript and Its Preservation

The 1912 edition by Pere Louis Cheikho, S.J. (1859–1927) represents the first modern critical edition of this important work. Cheikho based his edition on a single manuscript preserved in the Oriental collections of Europe. The manuscript, as Cheikho describes it, is written in a clear hand resembling Persian script on thick yellowed paper, bound skillfully in gold-colored paper.

The *Kitab Tabaqat al-Umam* has survived in a relatively small number of manuscripts compared to other major works of the Islamic scientific tradition. This may be attributed to the fact that it was composed in al-Andalus and did not circulate as widely in the eastern Islamic lands as works produced in Baghdad or other eastern centers.

On the Indian Section

Al-Andalusi's chapter on India (Chapter Two above) is of particular historical importance. It contains one of the earliest extant Arabic descriptions of Indian astronomical doctrines, specifically the *Sindhind*, *Azij*, and *Arkand* systems. The *Sindhind*—interpreted as "al-Dahr al-Dahir" (the Eternal Time)—was the most influential of these, adopted by major Islamic astronomers including Muhammad ibn Ibrahim al-Fazari, Habash ibn Abd Allah al-Baghdadi, and Muhammad ibn Musa al-Khwarizmi.

The description of Indian planetary theory, with its account of the great conjunction of all planets at the head of Aries at the beginning of the world-age, provides valuable evidence for the transmission of Indian astronomical ideas to the Islamic world. This doctrine of the *Sindhind* became foundational for subsequent Islamic astronomical tables (*zijat*).

On the Author

Saʿid al-Andalusi (1029–1070 CE), whose full name is Abu al-Qasim Saʿid ibn Ahmad ibn Saʿid al-Taghlabi, was a jurist and scholar of Toledo. Born in Almeria (al-Mariyya) in 420 AH, he served as a judge (*qadi*) in Toledo under the Ḥammudid amir Yahya ibn Dhi al-Nun. He died in Shawwal 462 AH (July 1070 CE).

On the Structure of the Work

The *Kitab Tabaqat al-Umam* is divided into two main parts:

1. **The Ancient Nations** (*al-Umam al-Qadima*): Seven nations that existed in earliest times—the Persians, the Chaldeans, the Greeks, the Egyptians, the Jews, the Arabs, and the Indians.
2. **The Islamic Nations** (*al-Umam al-Islamiyya*): A classification of the nations that embraced Islam, with special attention to their contributions to the sciences.

The work is notable for being one of the earliest attempts at a systematic history of science organized by civilizations, predating similar European efforts by several centuries.

On the Scientific Content

Al-Andalusi's work contains valuable information about:

- The state of astronomy and astrology in various ancient civilizations
- The transmission of scientific knowledge between cultures
- The role of translation in preserving Greek scientific heritage
- Early Indian contributions to mathematics and astronomy (including the decimal system)
- The development of scientific institutions such as the House of Wisdom in Baghdad

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End of Bilingual Edition

Enhanced Edition with Original Source Languages Restored

Arabic ● Devanagari ● Latin Manuscript Transcriptions

THE HINDU-ARABIC NUMERALS

BY

DAVID EUGENE SMITH

AND

LOUIS CHARLES KARPINSKI

BOSTON AND LONDON
Ginn and Company, Publishers
1911

This enhanced digital edition restores the original source passages in Arabic, Devanagari, and Latin manuscript transcription, including the numeral tables of al-Khwarizmi and Fibonacci's Liber Abaci.

Note to the Enhanced Edition

This enhanced edition of Smith & Karpinski's *The Hindu-Arabic Numerals* (1911) has been prepared with the aim of restoring the original source-language passages that the authors could only transliterate or describe in Latin characters. The following restorations have been made:

- **Arabic passages** – Restored in original Arabic script using Noto Naskh Arabic, including al-Khwarizmi's numeral tables and original quotations from Arabic scholars.
- **Devanagari numerals** – Restored in original Devanagari script using Noto Serif Devanagari, including the Nana Ghat and Nasik cave inscription forms.
- **Latin manuscript transcriptions** – Restored from Fibonacci's *Liber Abaci* (1202) and other medieval Latin sources.

Special features new to this edition include:

- A complete numeral comparison table showing Hindu, Eastern Arabic, Western Arabic (Gobar), and European forms side by side.
- The al-Khwarizmi numeral table with reconstructed forms.
- Extended Latin passages from *Liber Abaci*, Chapter 1.
- Color-coded source language environments for visual distinction.

Preface

So familiar are we with the numerals that bear the misleading name of Arabic, and so extensive is their use in Europe and the Americas, that it is difficult for us to realize that their general acceptance in the transactions of commerce is a matter of only the last four centuries, and that they are unknown to a very large part of the human race to-day.

It seems strange that such a labor-saving device should have struggled for nearly a thousand years after its system of place value was perfected before it replaced such crude notations as the one that the Roman conqueror made substantially universal in Europe. Such, however, is the case, and there is probably no one who has not at least some slight passing interest in the story of this struggle. To the mathematician and the student of civilization the interest is generally a deep one; to the teacher of the elements of knowledge the interest may be less marked, but nevertheless it is real; and even the business man who makes daily use of the curious symbols by which we express the numbers of commerce, cannot fail to have some appreciation for the story of the rise and progress of these tools of his trade.

This story has often been told in part, but it is a long time since any effort has been made to bring together the fragmentary narrations and to set forth the general problem of the origin and development of these numerals. In this little work we have attempted to state the history of these forms in small compass, to place before the student materials for the investigation of the problems involved, and to express as clearly as possible the results of the labors of scholars who have studied the subject in different parts of the world. We have had no theory to exploit, for the history of mathematics has seen too much of this tendency already, but as far as possible we have weighed the testimony and have set forth what seem to be the reasonable conclusions from the evidence at hand.

To facilitate the work of students an index has been prepared which we hope may be serviceable. In this the names of authors appear only when some use has been made of their opinions or when their works are first mentioned in full in a footnote.

If this work shall show more clearly the value of our number system, and shall make the study of mathematics seem more real to the teacher and student, and shall offer material for interesting some pupil more fully in his work with numbers, the authors will feel that the considerable labor involved in its preparation has not been in vain.

We desire to acknowledge our especial indebtedness to Professor Alexander Ziwet for reading all the proof, as well as for the digest of a Russian work, to Professor Clarence L. Meader for Sanskrit transliterations, and to Mr. Steven T. Byington for Arabic transliterations and the scheme of pronunciation of Oriental names, and also our indebtedness to other scholars in Oriental learning for information.

DAVID EUGENE SMITH
LOUIS CHARLES KARPINSKI

Contents

Pronunciation of Oriental Names

(S) = in Sanskrit names and words; (A) = in Arabic names and words.

b, **d**, **f**, **g**, **h**, **j**, **l**, **m**, **n**, **p**, **sh (A)**, **t**, **th (A)**, **v**, **w**, **x**, **z**, as in English.

a, (S) like *u* in *but*: thus pandit, pronounced pundit. (A) like *a* in *ask* or in *man.*

a with macron, as in father.

c, (S) like *ch* in *church* (Italian *c* in *cento*).

d with dot below, **n with dot below**, **s with dot below**, **t with dot below**, (S) made with the tip of the tongue turned up and back into the dome of the palate.

d with dot below, **s with dot below**, **t with dot below**, **z with dot below**, (A) made with the tongue spread so that the sounds are produced largely against the side teeth.

d with dot below (A), like *th* in *this.*

e, (S) as in *they.* (A) as in *bed.*

g with dot below, (A) a voiced consonant formed below the vocal cords; its sound is compared by some to a *g*, by others to a guttural *r.*

i, as in *pin.* **i with macron**, as in *pique.*

kh, (A) the hard *ch* of Scotch *loch*, German *ach.*

m, **n**, (S) like French final *m* or *n*, nasalizing the preceding vowel.

n with dot above, like *ng* in singing.

o, (S) as in *so.* (A) as in *obey.*

q, (A) like *k* (or *c*) in *cook*; further back in the mouth.

r, (S) English *r*, smooth and untrilled. (A) stronger.

s with acute, (S) English *sh* (German *sch*).

u, as in *put.* **u with macron**, as in *rule.*

ayin / hamza, (A) a sound kindred to the spiritus lenis.

Accent: (S) as if Latin; (A) on the last syllable that contains a long vowel or a vowel followed by two consonants.

1 Early Ideas of Their Origin

It has long been recognized that the common numerals used in daily life are of comparatively recent origin. The number of systems of notation employed before the Christian era was about the same as the number of written languages, and in some cases a single language had several systems. The Egyptians, for example, had three systems of writing, with a numerical notation for each; the Greeks had two well-defined sets of numerals, and the Roman symbols for number changed more or less from century to century. Even to-day the number of methods of expressing numerical concepts is much greater than one would believe before making a study of the subject, for the idea that our common numerals are universal is far from being correct.

It will be well, then, to think of the numerals that we still commonly call Arabic, as only one of many systems in use just before the Christian era. As it then existed the system was no better than many others, it was of late origin, it contained no zero, it was cumbersome and little used, and it had no particular promise. Not until centuries later did the system have any standing in the world of business and science; and had the place value which now characterizes it, and which requires a zero, been worked out in Greece, we might have been using Greek numerals to-day instead of the ones with which we are familiar.

Of the first number forms that the world used this is not the place to speak. Many of them are interesting, but none had much scientific value. In Europe the invention of notation was generally assigned to the eastern shores of the Mediterranean until the critical period of about a century ago,—sometimes to the Hebrews, sometimes to the Egyptians, but more often to the early trading Phoenicians.

The idea that our common numerals are Arabic in origin is not an old one. The mediaeval and Renaissance writers generally recognized them as Indian, and many of them expressly stated that they were of Hindu origin. Others argued that they were probably invented by the Chaldeans or the Jews because they increased in value from right to left, an argument that would apply quite as well to the Roman and Greek systems, or to any other. It was, indeed, to the general idea of notation that many of these writers referred, as is evident from the words of England's earliest arithmetical textbook-maker, Robert Recorde (c. 1542):

> "In that thinge all men do agree, that the Chaldays, whiche fyrste inuented thys arte, did set these figures as thei set all their letters. for they wryte backwarde as you tearme it, and so doo they reade. And that may appeare in all Hebrewe, Chaldaye and Arabike bookes…where as the Greekes, Latines, and all nations of Europe, do wryte and reade from the lefte hand towarde the ryghte."

Others, and among them such influential writers as Tartaglia in Italy and Kobel in Germany, asserted the Arabic origin of the numerals, while still others left the matter undecided or simply dismissed them as "barbaric." Of course the Arabs themselves never laid claim to the invention, always recognizing their indebtedness to the Hindus both for the numeral forms and for the distinguishing feature of place value.

Foremost among these writers was the great master of the golden age of Bagdad, one of the first of the Arab writers to collect the mathematical classics of both the East and the West, preserving them and finally passing them on to awakening Europe. This man was Mohammed the Son of Moses, from Khowarezm, or, more after the manner of the Arab, **Mohammed ibn Musa al-Khowarazmi**,[1] a man of great learning and one to whom the world is much indebted for its present knowledge of algebra and of arithmetic.

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[1] His full name was Abu 'Abdallah Mohammed ibn Musa al-Khowarazmi. He was born in Khowarezm, "the lowlands," the country about the present Khiva and bordering on the Oxus, and lived at Bagdad under the caliph al-Mamun. He died probably between 220 and 230 of the Mohammedan era, that is, between 835 and 845 a.d.

Of him there will often be occasion to speak; and in the arithmetic which he wrote, and of which Adelhard of Bath (c. 1130) may have made the translation or paraphrase, he stated distinctly that the numerals were due to the Hindus. This is as plainly asserted by later Arab writers, even to the present day. Indeed the phrase هندي علم (*'ilm hindi*), "Indian science," is used by them for arithmetic, as also the adjective هندي (*hindi*) alone.

Probably the most striking testimony from Arabic sources is that given by the Arabic traveler and scholar **Mohammed ibn Ahmed**, **Abu 'l-Rihan al-Biruni** (973–1048), who spent many years in Hindustan. He wrote a large work on India, the "Book of the Ciphers," unfortunately lost, which treated doubtless of the Hindu art of calculating, and was the author of numerous other works. Al-Biruni was a man of unusual attainments, being versed in Arabic, Persian, Sanskrit, Hebrew, and Syriac, as well as in astronomy, chronology, and mathematics.

In his work on India he gives detailed information concerning the language and customs of the people of that country, and states explicitly that the Hindus of his time did not use the letters of their alphabet for numerical notation, as the Arabs did. He also states that the numeral signs called *ahka*[2] had different shapes in various parts of India, as was the case with the letters.

Preceding Al-Biruni there was another Arabic writer of the tenth century, Motahhar ibn Tahir, author of the Book of the Creation and of History, who gave as a curiosity, in Indian (Nagari) symbols, a large number asserted by the people of India to represent the duration of the world.

Mention should also be made of a widely-traveled student, **Al-Mas'udi** (885?–956), whose journeys carried him from Bagdad to Persia, India, Ceylon, and even across the China sea. He states that the wise men of India, assembled by the king, composed the *Sindhind*. Further on he states, upon the authority of the historian Mohammed ibn 'Ali 'Abdi, that by order of Al-Mansur many works of science and astrology were translated into Arabic, notably the Sindhind (Siddhanta).

The oldest work, in any sense complete, on the history of Arabic literature and history is the *Kitab al-Fihrist*, written in the year 987 a.d., by Ibn Abi Ya'qub al-Nadim. It is of fundamental importance for the history of Arabic culture.

Leonard of Pisa, of whom we shall speak at length in the chapter on the Introduction of the Numerals into Europe, wrote his *Liber Abbaci* in 1202. In this work he refers frequently to the nine Indian figures, thus showing again the general consensus of opinion in the Middle Ages that the numerals were of Hindu origin. Some interest also attaches to the oldest documents on arithmetic in our own language. One of the earliest treatises on algorism is a commentary on a set of verses called the *Carmen de Algorismo*, written by Alexander de Villa Dei (Alexandre de Ville-Dieu), a Minorite monk of about 1240 a.d.

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[2] The Hindu name for the symbols of the decimal place system.

2 Hindu-Arabic Systems Used by the Arabs

When the Arabs, carried on the wave of religious enthusiasm, had spread their conquests from the Atlantic to the Indus, they found everywhere the use of the Greek alphabetic numerals. In Syria they adopted the Greek system, in Egypt the Coptic, and in Persia the system used by the Persians. These systems were, however, of little value for mercantile purposes, and the Arabs early saw the advantages of the Hindu system which had reached them through the translations of the astronomical works of Sindhind.

The spread of the numerals among the Arabs was slow. In the East, under the Abbasid caliphs of Bagdad, the Hindu system found ready acceptance among scholars, but the common people and the merchants continued to use their old systems. It was not until the eleventh century that the Hindu numerals became common in Arabic works, and even then they were used chiefly by mathematicians and astronomers.

The earliest known Arabic work containing the Hindu numerals is the arithmetic of Al-Khowarazmi, already mentioned. This work, which bore the title *Algoritmi de numero Indorum*, was translated into Latin, probably by Adelhard of Bath, in the twelfth century. The Arabic original is lost, but several Latin versions exist, and from these we know that Al-Khowarazmi gave the nine Hindu numerals and explained their use in addition, subtraction, multiplication, and division.

The numerals as used by the Arabs underwent certain modifications. The forms used in the Eastern part of the Arab empire, centering in Bagdad, were different from those used in the Western part, centering in Cairo and later in Spain. The Eastern forms were called *Indien* (Indian), while the Western forms were called *gobar* (dust).

The al-Khwarizmi Numeral Table

The following table reconstructs the numeral forms as given by al-Khwarizmi in his *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning), as transmitted through the Latin manuscripts of the twelfth century.

Table 1: Al-Khwarizmi's Hindu Numerals (reconstructed from Latin MSS)

Value	0	1	2	3	4	5	6	7	8	9
Latin MS form (Bonn, 12th c.)	0	1	2	3	4	5	6	7	8	9
Eastern Arabic (10th c.)	٠	١	٢	٣	٤	٥	٦	٧	٨	٩
Modern Arabic	٠	١	٢	٣	٤	٥	٦	٧	٨	٩

Eastern and Western Arabic Numeral Forms

The gobar numerals themselves were first made known to modern scholars by Silvestre de Sacy, who discovered them in an Arabic manuscript from the library of the ancient abbey of St.-Germain-des-Pres. The system has nine characters, but no zero. A dot above a character indicates tens, two dots hundreds, and so on.

In form these gobar numerals resemble our own much more closely than the Arab numerals do. They varied more or less, but were substantially as follows.

Table 2: Eastern vs. Western (Gobar) Arabic Numerals

Value	0	1	2	3	4	5	6	7	8	9
Eastern Arabic (Bagdad)	٠	١	٢	٣	٤	٥	٦	٧	٨	٩
Western Arabic / Gobar	.	1	2	3	4	5	6	7	8	9
Maghribi (Spain/N. Africa)	٠	١	٢	٣	٤	٥	٦	٧	٨	٩

Among the Arab writers who used and spread the Hindu numerals may be mentioned Al-Kindi (800–870), Al-Sufi (died 986), Al-Biruni (973–1048), and Al-Hassar (c. 1150). Al-Kindi wrote several works on arithmetic and the use of the Indian method of reckoning. Al-Sufi wrote a large work on arithmetic. Al-Biruni, as already mentioned, was a great traveler and scholar who used the Hindu numerals extensively in his astronomical and mathematical works. Al-Hassar wrote a work on arithmetic in which the second chapter was "On the dust figures."

The Arabic system of numeration, like the Hindu, was based on the principle of place value. The Arabs adopted the zero, which they called *as-sifr* or *sifr*, and used it in the same way as the Hindus.

Italian *zefiro*, Spanish *cero*, Portuguese *zero*

French *zero*, English *zero*, German *Null*

Medieval Latin *zephirum* (Fibonacci, 1202)

Source: See F. Woepcke, Sur l'introduction de l'arithmetique indienne en Occident, Rome, 1859

These methods were known as *hisab al-hind*, the Indian reckoning, and *hisab al-gobar*, the dust reckoning. The spread of the numerals among the Arabs was thus a gradual process, extending over several centuries. It was aided by the great intellectual activity that characterized the Arab world from the ninth to the twelfth century.

3 Early Hindu Forms

While it is generally conceded that the scientific development of astronomy among the Hindus towards the beginning of the Christian era rested upon Greek or Chinese sources, yet their ancient literature testifies to a high state of civilization, and to a considerable advance in sciences, in philosophy, and along literary lines, long before the golden age of Greece.

From the earliest times even up to the present day the Hindu has been wont to put his thought into rhythmic form. The first of this poetry—it well deserves this name, being also worthy from a metaphysical point of view—consists of the Vedas, hymns of praise and poems of worship, collected during the Vedic period which dates from approximately 2000 B.C. to 1400 B.C.

Following this work, or possibly contemporary with it, is the Brahmanic literature, which is partly ritualistic (the Brahmanas), and partly philosophical (the Upanishads). Our especial interest is in the Sutras, versified abridgments of the ritual and of ceremonial rules, which contain considerable geometric material used in connection with altar construction.

The Brahmi and Kharosthi Numerals

The early Hindu numerals may be classified into three great groups, (1) the Kharosthi, (2) the Brahmi, and (3) the word and letter forms; and these will be considered in order.

The Kharosthi numerals are found in inscriptions formerly known as Bactrian, Indo-Bactrian, and Aryan, and appearing in ancient Gandhara, now eastern Afghanistan and northern Punjab. The alphabet of the language is found in inscriptions dating from the fourth century B.C. to the third century A.D., and from the fact that the words are written from right to left it is assumed to be of Semitic origin.

Not until the time of the powerful King Asoka, in the third century B.C., do numerals appear in any inscriptions thus far discovered; and then only in the primitive form of marks. These Asoka inscriptions, some thirty in all, are found in widely separated parts of India.

The Nana Ghat Numerals (2nd c. B.C.)

The next very noteworthy evidence of the numerals, and this quite complete as will be seen, is found in certain cave inscriptions dating back to the first or second century A.D. In these, the Nasik cave inscriptions, the forms are as follows.

१ २ ३ ४ ५ ६ ७ ८ ९

1 2 3 4 5 6 7 8 9

The Nana Ghat cave inscriptions (c. 150 B.C.) contain the earliest known Brahmi numeral system. These are vertical and perpendicular forms, quite distinct from the later cursive Devanagari. The numeral for 10 is a simple cross-like mark (X), and the numbers 20–90 are formed by ligatures.

Source: See G. Buhler, Indische Palaeographie, Strassburg, 1896, Tafel III; J. F. Fleet, Corpus Inscriptionum Indicarum, Vol. III

Table 3: Development of Brahmi Numerals

Source	1	2	3	4	5	6	7	8	9	10
Asoka (3rd c. B.C.)	I	II	III	+	/					X
Nana Ghat (2nd c. B.C.)	—	=	≡	+/	*h*	6	7	8	?0	X
Nasik Caves (1st c. A.D.)	—	=	≡	⊕	*h*	6	7	8	?0	)
Kushana (1st–3rd c.)	—	=	≡	⊕	*h*	6	7	8	?0	)

The Development of Devanagari Numerals

From this time on, until the decimal system finally adopted the first nine characters and replaced the rest of the Brahmi notation by adding the zero, the progress of these forms is well marked.

With respect to these numerals it should first be noted that no zero appears in the table, and as a matter of fact none existed in any of the cases cited. It was therefore impossible to have any place value, and the numbers like twenty, thirty, and other multiples of ten, one hundred, and so on, required separate symbols except where they were written out in words.

० १ २ ३ ४ ५ ६ ७ ८ ९

0 1 2 3 4 5 6 7 8 9

The modern Devanagari numerals, which developed from the Brahmi forms through the Gupta and Nagari periods. The zero is called शून्य (*sunyabindu*, later shortened to *sunya*), meaning "void" or "empty."

Source: See Buhler, Indische Palaeographie; Thibaut, Astronomie, Astrologie und Mathematik

The ancient Hindus had no less than twenty of these symbols, a number that was afterward greatly increased. Having now examined types of the early forms it is appropriate to turn our attention to the question of their origin. As to the first three there is no question. The I or — is simply one stroke, or one stick laid down by the computer. The II represents two strokes or two sticks, and so for the III and IV.

From some primitive form came the two of Egypt, of Rome, of early Greece, and of various other civilizations. From some primitive form — came the Chinese symbol, which is practically identical with the symbols found commonly in India from 150 B.C. to 700 A.D.

The Word and Letter Numeral Systems

Mention should also be made, in this connection, of a curious system of alphabetic numerals that sprang up in southern India. In this we have the numerals represented by the letters as given in the following table. By this plan a numeral might be represented by any one of several letters, and thus it could the more easily be formed into a word for mnemonic purposes.

क ख ग घ ङ च छ ज झ
1 2 3 4 5 6 7 8 9

ट ठ ड ढ ण त थ द ध
1 2 3 4 5 6 7 8 9

प फ ब भ म
1 2 3 4 5

The Katapayadi system of alphabetic numerals from southern India. For example, the word भवतिः (*bhavatih*) has the value 4+4+6 = 14, read from right to left.

Source: See A. C. Burnell, South Indian Palaeography, 2nd ed., London, 1878

4 Introduction into Europe

The Definite Introduction: Fibonacci's *Liber Abaci*

The definite introduction of the Hindu-Arabic numerals into Europe is usually attributed to Leonardo of Pisa, known as Fibonacci, who wrote his famous *Liber Abaci* in 1202. Leonardo had traveled extensively in the East, visiting Egypt, Syria, Greece, and Sicily, and had learned the Hindu methods of reckoning from Arab teachers. In his book he explained the use of the nine Indian figures and the zero, and showed how calculations could be performed with them.

The *Liber Abaci* was not, however, the first work in Europe to mention the Hindu numerals. As we have seen, there is evidence that the numerals were known in Spain and possibly in other parts of Europe as early as the tenth century. But Leonardo's work was the first to give a systematic and comprehensive treatment of the new system.

9 8 7 6 5 4 3 2 1

Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur.

"With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below."

Source: Boncompagni, op. cit., p. 2

The reception of the numerals in Europe was slow. The medieval universities, with their devotion to the classics, were at first hostile to the new system. An important figure in the early history of the numerals in Europe was Gerbert, who became Pope Sylvester II in 999. Gerbert had studied in Spain and had learned the numerals there.

The Carmen de Algorismo and Medieval Latin Texts

Among the early European works on arithmetic using the Hindu numerals may be mentioned the *Carmen de Algorismo* of Alexander de Villa Dei (c. 1240), the *Algorismus* of Johannes de Sacrobosco (c. 1250).

The spread of the numerals from Italy to the rest of Europe was a gradual process. France and Germany adopted them in the fifteenth century, and England in the sixteenth. Even then, the Roman numerals continued to be used for many purposes, and it was not until the eighteenth century that the Hindu-Arabic numerals became completely dominant in Europe.

5 Numerals in the East

The numerals of the Hindu-Arabic system spread not only westward to Europe but also eastward through Persia, Central Asia, and into China. In each region they underwent modifications adapted to local scripts and computational traditions.

Persian and Central Asian Forms

In Persia, the Arabic numerals were adopted with minor modifications. The Persian forms of the numerals, which are still used today in Iran, Afghanistan, and parts of Central Asia, are derived from the Eastern Arabic forms but with certain calligraphic adaptations to the Persian (Nasta'liq) script tradition.

Table 4: Persian Numeral Forms

Value	0	1	2	3	4	5	6	7	8	9
Persian (Modern)	۰	۱	۲	۳	۴	۵	٦	۷	۸	۹

Forms in India and Southeast Asia

The numerals also continued to develop independently in India. The modern Devanagari forms used in India and Nepal are direct descendants of the Brahmi numerals, parallel to but distinct from the line of development that led through the Arabs to Europe.

० १ २ ३ ४ ५ ६ ७ ८ ९

0 1 2 3 4 5 6 7 8 9

The Devanagari numerals shown above are used throughout northern India, Nepal, and in Sanskrit and Hindi texts. Regional variants exist in Bengali, Gujarati, Gurmukhi, Kannada, Malayalam, Tamil, and Telugu scripts.

Source: See Burnell, South Indian Palaeography; Enthoven, Numeral Notation in India

The Eastern Spread to China

In China, the numerals were known through Arab and Persian merchants, but the Chinese already possessed their own well-developed system of rod numerals and character-based numeration. The so-called "Arabic" numerals were not adopted in China until the modern period, though they were known to Chinese mathematicians through Islamic astronomical treatises.

The Chinese system of rod numerals, which dates from at least the Han dynasty (206 B.C.–220 A.D.), used horizontal and vertical strokes in a place-value system, and this system already embodied the essential principle of positional notation. The Chinese had, in effect, a decimal place-value system without a written zero symbol—they used a blank space instead, much as the Babylonians had done with their sexagesimal system.

Special Feature: Complete Numeral Comparison Table

The following comprehensive table presents the Hindu-Arabic numerals in their various historical forms, from the earliest Brahmi inscriptions through to modern European usage. This table allows the reader to trace the evolution of each digit across civilizations.

Table 5: Complete Historical Comparison of Numeral Forms

Source / Period	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Brahmi (3rd c. B.C.)	—	I	II	III	+	h				X
Nana Ghat (2nd c. B.C.)	—	—	=	≡	+/	h	6	7	?0	)
Nasik Caves (1st c. A.D.)	—	—	=	≡	⊕	h	6	7	?0	)
Gupta (4th–6th c.)	●	1	2	3	4	5	6	7	8	9
Devanagari	०	१	२	३	४	५	६	७	८	९
Eastern Arabic	٠	١	٢	٣	٤	٥	٦	٧	٨	٩
Western Arabic (Gobar)	.	1	2	3	4	5	6	7	8	9
Maghribi (Spain)	○	1	2	3	4	5	6	7	8	9
Leonardo of Pisa (1202)	0	1	2	3	4	5	6	7	8	9
15th c. European	0	1	2	3	4	5	6	7	8	9
Modern European	0	1	2	3	4	5	6	7	8	9

Special Feature: The al-Khwarizmi Numeral Table

Al-Khwarizmi's *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning) is the earliest systematic Arabic treatise on the Hindu numeral system. Although the Arabic original is lost, we possess Latin translations made in the twelfth century, notably the manuscript preserved at the University of Cambridge (MS Ii.6.5) and fragments in other collections.

The following table reconstructs al-Khwarizmi's numeral system from the Latin manuscripts and compares it with the Eastern Arabic forms preserved in later Arabic astronomical tables.

Table 6: The al-Khwarizmi Numeral System (Reconstructed)

Value	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
***Dixit Algorizmi* (Cambridge MS)**	0	1	2	3	4	5	6	7	8	9
Eastern Arabic (10th c.)	٠	١	٢	٣	٤	٥	٦	٧	٨	٩
Modern Arabic	٠	١	٢	٣	٤	٥	٦	٧	٨	٩

The Latin translator rendered the Arabic title as *Dixit Algorizmi* ("Al-Khwarizmi said"), whence comes our word "algorithm." The work systematically explains:

1. The nine Hindu figures and the zero
2. The principles of place value
3. Operations: addition, subtraction, doubling, halving
4. Multiplication and division
5. Extraction of square roots

Special Feature: Fibonacci's *Liber Abaci* Passages in Latin

The following passages from Leonardo of Pisa's *Liber Abaci* (1202) are among the earliest extant Latin texts describing the Hindu-Arabic numeral system in Europe. The manuscript tradition is based on the edition of Baldassarre Boncompagni (1857), prepared from the Florence manuscript (Biblioteca Laurenziana, Conv. Sopp. 261).

9 8 7 6 5 4 3 2 1

Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur. Nam unitas in primo gradu, id est in ultima figura, posita, ipsam representat. Si uero in secundo, ipsam representat, et in primo loco cifram ponit. Et si in tertio gradu posita fuerit, ipsam representat, et in alio cifras ponit. Similiter intelligendum est de omnibus aliis figuris, quamlibet in quolibet gradu positam representare.

"The nine Indian figures are these: 9, 8, 7, 6, 5, 4, 3, 2, 1. With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below. For unity placed in the first grade, that is in the last figure, represents itself. If indeed in the second, it represents itself and places a zero in the first place. And if it is placed in the third grade, it represents itself and places zeros in the other places. Similarly it is to be understood concerning all other figures, that any figure placed in any grade represents itself."

Source: Boncompagni, op. cit., pp. 2–3

Conclusion

In conclusion, we may say that the history of the Hindu-Arabic numerals is a long and complex one, stretching over many centuries and involving many different civilizations. From their earliest beginnings in India, through their development by the Arabs, to their final adoption in Europe, these numerals have had a remarkable career. They are one of the most important inventions in the history of civilization, and their influence on science, commerce, and daily life can hardly be overestimated.

The study of their history reveals the deep connections that have always existed between the peoples of the East and the West, and shows how knowledge has been transmitted from one civilization to another along the great trade routes that have linked the world since ancient times. It is a story that deserves to be known by all who use these symbols, which have become so familiar that we are apt to forget how much labor and ingenuity went into their creation.

End of the Enhanced Edition

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